A FIELD REFERENCE FOR THE DATA-MINDED

Mathematics

A Comprehensive Review | Second Edition

Formulas, definitions, context, and cut-out flashcards
spanning high-school and university mathematics, physics, engineering,
finance, and machine learning.

01	**Math Signs & Operations**	M
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19	**Glossary & Reference Cards**	GL

Includes: concept-indexed cross references • data-science context for every formula • named formulas with explanations • step-by-step methods • key-topic deep dives • duplex-aligned flashcards

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How to use this guide

Every idea is a **concept block** with a **concept ID** built from the chapter code and a number (**C1.7** is the seventh concept of Calculus I). Every block has the same anatomy, shown at right: a one-line **summary**; a box of **named formulas**, each with a one-line explanation of how it works; a read-aloud *stated equation*; a memory aid; where a result is reached by a procedure, numbered **method** steps with a small worked example; and, where you must make a choice, a blue **decision point** on how to think about it. Every symbol is defined under *Terms*.

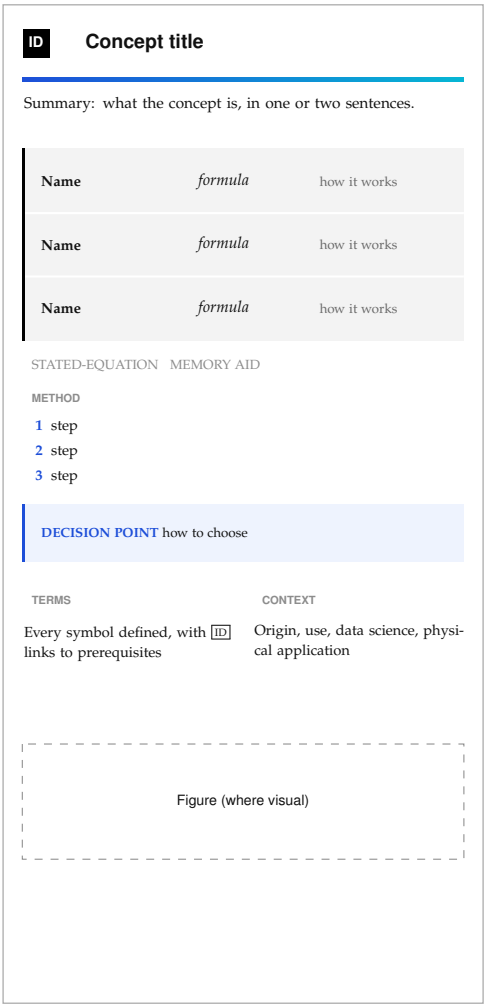
Key topics. Fundamentals (the unit circle, limits, vectors, partial derivatives, . . .) get full-page **key topic** spreads, titled *Title | Key Topic* with their own ID (e.g. T.K1). Each chapter opener lists the chapter's concepts and key topics.

Cross references. A small box such as A1.15 §03 Algebra I links to the concept it names; when the concept is in another chapter the chapter number and name follow it, e.g. C1.9 §07 Calculus I. In the PDF every box is clickable.

Printing. Every chapter starts on an odd (right-hand) page, so a printed chapter can be removed on its own; a left-over even page is left intentionally blank.

Flashcards. Cards are grouped by chapter at the back, each chapter starting on a fresh sheet. The front shows the concept and the *names* of the formulas to recall; the back shows the formulas, key terms and memory aid. Busy concepts are split into Part 1, Part 2, ... Print double-sided at **100%**, **flip on long edge**, and cut along the dashed lines.

Chapter codes. M Math Signs • GR Greek • A1 A2 Algebra I, II • G Geometry • T Trigonometry • C1 C2 C3 Calculus I, II, III • LA Linear Algebra • V Vector Calculus • P Physics • S Statistics • O Optimization • ML Machine Learning • EE Electrical Eng. • FN Finance • GG Graph Gallery • GL Glossary.



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01
Math Signs & Operations

The vocabulary of mathematics: every sign used in this book, how to read it aloud, what it means, and a tiny example with small numbers. Each row is also a flashcard: the front shows the symbol, the back gives its name, meaning and example.

In this chapter

M.K1 **Key topic:** Logic: Implication, Converse, Contrapositive & Truth Tables 22

M.K2 **Key topic:** Reading Mathematical Notation 24

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SYMBOL	NAME / READ AS	MEANING	EXAMPLE
Arithmetic			
$+$	**Plus** M.1	Addition; as a prefix, marks a positive number.	$3+4=7$
$-$	**Minus** M.2	Subtraction; as a prefix, the negative (additive inverse) of a number.	$9-5=4$; $-(-2)=2$
$\times$	**Times** M.3	Multiplication of numbers. Between vectors it is the cross product (see Vectors below).	$3\times 4=12$
$\cdot$	**Dot, "times"** M.4	Multiplication where $\times$ could be confused with the letter x; between vectors, the dot product.	$2\cdot 5=10$
ab	**Juxtaposition, "a b"** M.5	Writing symbols side by side means multiplication. It binds tighter than $\div$ in most texts, so add parentheses when in doubt.	$3x$ at $x=2$ is 6; $2(3+1)=8$
$\div$	**Divided by** M.6	Division (obelus). Rare beyond arithmetic; replaced by a fraction bar or slash.	$8\div 2=4$
a/b	**Slash, "a over b"** M.7	In-line division. Everything to the right of / is *not* automatically in the denominator.	$6/3=2$; write $1/(2x)$, not $1/2x$
$\pm$	**Plus or minus** M.8	Two values at once (both signs), or a tolerance around a central value.	$\pm\sqrt{9}=\pm 3$; 5 ± 1 means 4 to 6
$\mp$	**Minus or plus** M.9	Used together with $\pm$: take the opposite sign (top sign with top sign).	$\cos(a\pm b)=\cos a\cos b\mp\sin a\sin b$
$\bmod$	**a mod n (modulo)** M.10	Remainder after dividing a by n (with $0\le r<n$); clock arithmetic.	$10 \bmod 3=1$; $9 \bmod 3=0$
$a \mid b$	**a divides b** M.11	b is an integer multiple of a; $\nmid$ means "does not divide".	$3\mid 9$; $3\nmid 10$
$\gcd$	**GCD (and lcm)** M.12	Greatest common divisor: largest integer dividing both. lcm: least common multiple. $\gcd\cdot\operatorname{lcm}=ab$.	$\gcd(6,9)=3$; $\operatorname{lcm}(4,6)=12$

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Equality & Comparison

$=$	**Equals** M.13	Both sides have the same value.	$2+3=5$
$\neq$	**Not equal to** M.14	The two sides have different values.	$3 \neq 4$
$\approx$	**Approximately equal** M.15	Close in value; the result of rounding or an approximation (linearization, series truncation C1.21 §07 Calculus I).	$\pi \approx 3.14$; $\sqrt{2} \approx 1.41$
$\equiv$	**Identical to; congruent mod n** M.16	True for *every* value of the variables (an identity), or logically equivalent; with $\pmod n$, same remainder.	$(x+1)^2 \equiv x^2+2x+1$; $10 \equiv 1 \pmod 3$
$\cong$	**Congruent / isomorphic to** M.17	Geometry: same shape *and* size G.9 §05 Geometry. Algebra: same structure.	$\triangle ABC \cong \triangle DEF$ if all sides match: $3,4,5$ and $3,4,5$
$\sim$	**Similar / equivalent / asymptotic** M.18	Geometry: same shape, any size G.9 §05 Geometry. Analysis: $f \sim g$ means $f/g \to 1$. Loosely: "on the order of". (In statistics $X \sim$ means "distributed as".)	Triangles $3,4,5 \sim 6,8,10$; $n^2+n \sim n^2$
$\propto$	**Proportional to** M.19	$y = kx$ for some constant k: doubling x doubles y A1.18 §03 Algebra I.	$y \propto x$ with $y = 3x$: $x=2 \to y=6$
$<$ $>$	**Less than / greater than** M.20	Strict order; the open end faces the larger number.	$3<7$; $9>2$
$\leq$ $\geq$	**At most / at least** M.21	Less than or equal; greater than or equal. Equality allowed.	$x \leq 5$ includes 5; $4 \geq 4$
$\ll$ $\gg$	**Much less / greater** M.22	Smaller (larger) by orders of magnitude, so the smaller term can often be neglected.	$x \ll 1 \Rightarrow x^2 \approx 0$; $1000 \gg 1$
$:=$	**Is defined as** M.23	Defines the left side by the right side (assignment, not a claim). Also written $\triangleq$ or $\stackrel{\text{def}}{=}$.	$f(x) := x^2$, so $f(3)=9$

Grouping & Order

$(\;)$	**Parentheses** M.24	Group: evaluate inside first. Also function arguments, ordered pairs, open intervals.	$2(3+4)=14$; $f(2)$; point $(1,2)$
$[\;]$	**Brackets** M.25	Outer grouping around parentheses; closed intervals; matrices; $\mathbb{E}[X]$.	$2[1+(3+1)]=10$; $[1,3]$
$\{ \; \}$	**Braces** M.26	Outermost grouping; sets; piecewise cases.	$\{1,2,3\}$; $3\{2+[1+(1+1)]\}=15$
$\overline{}$	**Vinculum (overbar)** M.27	A bar groups like parentheses (fraction bar, root). Also marks repeating decimals, complex conjugates, complements and means.	$0.\overline{3}=\frac{1}{3}$; $\overline{3+2i}=3-2i$

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Powers, Roots, Exponentials & Logarithms

x^n	**x to the n (power)** M.28	n factors of x multiplied together A1.3 §03 Algebra I. x^2: "x squared"; x^3: "x cubed".	$2^3 = 2 \cdot 2 \cdot 2 = 8$
x^{-n}	**x to the minus n** M.29	Reciprocal of the power: $1/x^n$. Not a negative number.	$2^{-3} = \frac{1}{8}$
$x^{1/n}$	**Fractional exponent** M.30	The n-th root; $x^{m/n} = (\sqrt[n]{x})^m$ A1.4 §03 Algebra I.	$8^{1/3} = 2$; $4^{3/2} = 8$
$\sqrt{x}$	**Square root** M.31	The *non-negative* number whose square is x (principal root).	$\sqrt{9} = 3$, not ± 3
$\sqrt[n]{x}$	**n-th root** M.32	The number whose n-th power is x; for even n, the non-negative one.	$\sqrt[3]{8} = 2$; $\sqrt[4]{16} = 2$
$\exp(x)$	**Exponential, "exp of x"** M.33	e^x, written as a function when the exponent is long A2.13 §04 Algebra II.	$\exp(0) = 1$; $\exp(\ln 5) = 5$
$\log_b x$	**Log base b of x** M.34	The exponent b must be raised to in order to give x: $b^y = x$ A2.14 §04 Algebra II.	$\log_2 8 = 3$; $\log_{10} 100 = 2$
$\ln x$	**Natural log, "L-N of x"** M.35	Logarithm base e; the inverse of e^x.	$\ln e = 1$; $\ln 1 = 0$
$\log x$	**Log of x (no base)** M.36	Base depends on the field: 10 in school and engineering, e in math and ML, 2 in computer science. Check the context.	$\log 1000 = 3$ (base 10); $\log_2 8 = 3$ (base 2)
`1e-3`	**E-notation** M.37	Computer scientific notation: `aEk` means $a \times 10^k$ A1.5 §03 Algebra I.	`2.5e3` $= 2500$; `1e-3` $= 0.001$

Constants

π	**Pi** M.38	Circumference ÷ diameter of any circle, ≈ 3.14159 G.17 §05 Geometry.	Circle of radius 2: $C = 2\pi r = 4\pi \approx 12.57$
e	**Euler's number** M.39	≈ 2.71828, the base of natural growth: $\lim_{n\to\infty}(1+\frac{1}{n})^n$ C1.6 §07 Calculus I.	$e^2 \approx 7.39$; $(1+\frac{1}{10})^{10} \approx 2.59$
i	**Imaginary unit** M.40	$i^2 = -1$ A2.6 §04 Algebra II. Engineers write j (since i is current).	$(1+i)(1-i) = 1 - i^2 = 2$
φ	**Golden ratio, "phi"** M.41	$\varphi = \frac{1+\sqrt{5}}{2} \approx 1.618$; satisfies $\varphi^2 = \varphi + 1$.	Fibonacci ratios $\frac{8}{5} = 1.6$, $\frac{13}{8} \approx 1.625 \to \varphi$
∞	**Infinity** M.42	Unbounded growth; not a real number. Appears in limits, intervals and sums.	$\frac{1}{x} \to 0$ as $x \to \infty$; $[1, \infty)$

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Absolute Value, Floor, Ceiling & Rounding

$\lvert x \rvert$	**Absolute value** M.43	Distance from x to 0, never negative A1.11 §03 Algebra I. Same bars mean size of a set, determinant, or modulus of a complex number.	$\lvert -3 \rvert = 3$; $\lvert 4 \rvert = 4$; $\lvert 3+4i \rvert = 5$
$\lfloor x \rfloor$	**Floor of x** M.44	Greatest integer $\leq x$ (round down) F.18 §18 Graph Gallery.	$\lfloor 3.7 \rfloor = 3$; $\lfloor -2.5 \rfloor = -3$
$\lceil x \rceil$	**Ceiling of x** M.45	Least integer $\geq x$ (round up).	$\lceil 3.2 \rceil = 4$; $\lceil -2.5 \rceil = -2$
$\lfloor x \rceil$	**Round (nearest integer)** M.46	Round to the closest integer. Ties: half up in school, half to even in most software.	$\lfloor 2.6 \rceil = 3$; `round(2.5)` $= 2$ in Python
$\operatorname{sgn} x$	**Sign of x, "signum"** M.47	$+1$ if $x > 0$, 0 if $x = 0$, -1 if $x < 0$; $x = \operatorname{sgn}(x)\,\lvert x \rvert$.	$\operatorname{sgn}(-5) = -1$; $\operatorname{sgn}(3) = 1$

Factorials & Counting

$n!$	**n factorial** M.48	Product $1 \cdot 2 \cdots n$; counts orderings of n items. By convention $0! = 1$ S.8 §13 Statistics.	$4! = 24$; $5! = 120$
$\binom{n}{k}$	**n choose k (binomial coefficient)** M.49	Number of ways to choose k of n items, order ignored: $\frac{n!}{k!(n-k)!}$ S.8 §13 Statistics, A2.23 §04 Algebra II.	$\binom{5}{2} = 10$; $\binom{4}{2} = 6$
${}_nC_r$	**n C r (combinations)** M.50	Same as $\binom{n}{r}$; also written $C(n,r)$ or ${}^nC_r$.	${}_5C_2 = 10$
${}_nP_r$	**n P r (permutations)** M.51	Ordered arrangements of r of n items: $\frac{n!}{(n-r)!}$ S.8 §13 Statistics.	${}_5P_2 = 5 \cdot 4 = 20$
$n!!$	**Double factorial** M.52	Product of every second integer down from n; not $(n!)!$.	$7!! = 7 \cdot 5 \cdot 3 \cdot 1 = 105$; $6!! = 48$

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Sums, Products & Index Notation

$\sum_{i=1}^{n} a_i$	**Sigma, "sum from i = 1 to n"** M.53	Add the terms a_i for every integer i from the lower limit to the upper limit A2.21 §04 Algebra II.	$\sum_{i=1}^{4} i = 1+2+3+4 = 10$
$\prod_{i=1}^{n} a_i$	**Pi, "product from i = 1 to n"** M.54	Multiply the terms a_i for $i = 1, \ldots, n$; logs turn it into a sum.	$\prod_{i=1}^{4} i = 24 = 4!$
a_i	**a sub i** M.55	The i-th element of a sequence or vector; a_{ij} is row i, column j of a matrix.	$a = (5,8,2)$: $a_2 = 8$; $A = \left[\begin{smallmatrix} 1 & 2 \\ 3 & 4 \end{smallmatrix}\right]$: $a_{21} = 3$
$\sum_{x \in S}$	**Sum over a set** M.56	Add over every element of S (no counting index needed). $\sum_{i,j}$ sums over all pairs.	$\sum_{x \in \{1,3,5\}} x = 9$
$a_i b_i$	**Einstein summation** M.57	A repeated index implies a sum (physics, `einsum`): $a_i b_i = \sum_i a_i b_i$.	$a = (1,2)$, $b = (3,4)$: $a_i b_i = 3 + 8 = 11$

Fractions, Ratios & Percent

$\frac{a}{b}$	**Fraction, "a over b"** M.58	Numerator divided by denominator $b \neq 0$; the bar groups both parts A1.2 §03 Algebra I.	$\frac{6+2}{4} = 2$; $\frac{1}{2} + \frac{1}{3} = \frac{5}{6}$
$a : b$	**Ratio, "a to b"** M.59	Compares two quantities; equals $\frac{a}{b}$. Odds $a : b$ mean probability $\frac{a}{a+b}$.	$6 : 4 = 3 : 2$; odds $1 : 3 \Rightarrow P = \frac{1}{4}$
$::$	**Proportion, "as"** M.60	$a : b :: c : d$ reads "a is to b as c is to d": two equal ratios; solve by cross-multiplying $ad = bc$ A1.19 §03 Algebra I.	$2 : 3 :: 4 : 6$ since $2 \cdot 6 = 3 \cdot 4$
$\%$	**Percent** M.61	Per hundred: $p\% = \frac{p}{100}$ A1.19 §03 Algebra I.	25% of 8 is 2; $3 \to 4$ is $+33.\overline{3}\%$

Sets

$\{a, b, c\}$	**Set (roster)** M.62	An unordered collection of distinct elements; order and repetition do not matter.	$\{a,b,c\} = \{c,a,b\} = \{a,a,b,c\}$
$\{x \mid P\}$	**Set-builder, "the set of x such that"** M.63	All x for which the condition P holds; also written $\{x : P\}$.	$\{x \in \mathbb{Z} \mid 1 \leq x \leq 3\} = \{1,2,3\}$
$\in$	**Is an element of** M.64	Membership of a single element.	$a \in \{a,b,c\}$
$\notin$	**Is not an element of** M.65	Non-membership.	$d \notin \{a,b,c\}$
$\subseteq$	**Is a subset of** M.66	Every element of the left set is in the right set (equality allowed).	$\{a,b\} \subseteq \{a,b,c\}$; $\{a,b\} \subseteq \{a,b\}$
$\subset$	**Proper subset** M.67	Subset but not equal ($\subsetneq$). Some authors use $\subset$ for $\subseteq$: check.	$\{a\} \subset \{a,b,c\}$

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$\supseteq$	**Is a superset of** M.68	The left set contains the right ($\supset$: proper superset).	$\{a,b,c\} \supseteq \{c\}$
$\cup$	**Union, "or"** M.69	Elements in either set (or both) A1.21 §03 Algebra I. $\bigcup_i A_i$: union of many sets.	$\{a,b,c\} \cup \{d,e,f\} = \{a,b,c,d,e,f\}$
$\cap$	**Intersection, "and"** M.70	Elements in both sets.	$\{a,b,c\} \cap \{b,c,d\} = \{b,c\}$
$\setminus$	**Set difference, "minus"** M.71	Elements of the first set that are not in the second.	$\{a,b,c\} \setminus \{b\} = \{a,c\}$
A^c	**Complement** M.72	Everything in the universal set U not in A; also $\overline{A}$ or A'.	$U = \{a,\dots,f\}$, $A = \{a,b,c\}$: $A^c = \{d,e,f\}$
$A \triangle B$	**Symmetric difference** M.73	In exactly one of the two sets: $(A \setminus B) \cup (B \setminus A)$; the set version of XOR.	$\{a,b,c\} \triangle \{b,c,d\} = \{a,d\}$
$\varnothing$	**Empty set** M.74	The set with no elements, $\{\}$; a subset of every set. Not the Greek ϕ.	$\{a,b,c\} \cap \{d,e,f\} = \varnothing$
$\lvert A \rvert$	**Cardinality, "size of A"** M.75	Number of elements; also $\#A$ or $n(A)$.	$\lvert\{a,b,c\}\rvert = 3$; $\lvert\varnothing\rvert = 0$
$A \times B$	**Cartesian product** M.76	All ordered pairs (x,y) with $x \in A$, $y \in B$; $\lvert A \times B\rvert = \lvert A\rvert\,\lvert B\rvert$.	$\{a,b\} \times \{d,e\} = \{(a,d),(a,e),(b,d),(b,e)\}$
$\mathcal{P}(A)$	**Power set** M.77	The set of all subsets of A (also 2^A); it has $2^{\lvert A\rvert}$ elements.	$\mathcal{P}(\{a,b\}) = \{\varnothing,\{a\},\{b\},\{a,b\}\}$
$\mathbb{N}\,\mathbb{Z}\,\mathbb{Q}\,\mathbb{R}\,\mathbb{C}$	**Number sets** M.78	Naturals $\{1,2,\dots\}$ (some include 0), integers, rationals $\frac{p}{q}$, reals, complex numbers: $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$. A superscript $^+$ keeps the positives.	$3 \in \mathbb{N}$; $-3 \in \mathbb{Z}$; $\frac{1}{2} \in \mathbb{Q}$; $\sqrt{2} \in \mathbb{R}$; $2+i \in \mathbb{C}$
$\mathbb{R}^n$	**R n, real n-space** M.79	All ordered lists of n real numbers: vectors with n components LA.2 §09 Linear Algebra.	$(1,2,3) \in \mathbb{R}^3$
(a,b)	**Open interval** M.80	All reals strictly between a and b; parentheses exclude the endpoints A1.21 §03 Algebra I. (Same notation as a point: read the context.)	$(2,5) = \{x \mid 2 < x < 5\}$; $2 \notin (2,5)$
$[a,b]$	**Closed interval** M.81	All reals from a to b, endpoints included.	$[2,5] = \{x \mid 2 \le x \le 5\}$; $2 \in [2,5]$
$[a,b)$	**Half-open interval** M.82	Includes a, excludes b; ∞ always takes a parenthesis.	$[2,5) = \{x \mid 2 \le x < 5\}$; $[1,\infty)$

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Logic

$\neg p$	**Not p (negation)** M.83	Flips the truth value; also written $\sim p$, $\overline{p}$ or $!p$. See the truth tables M.K1.	$\neg(3 > 5)$ is true
$p \wedge q$	**p and q (conjunction)** M.84	True only when both are true; like $\cap$.	$(2 < 3) \wedge (4 < 5)$: true
$p \vee q$	**p or q (disjunction)** M.85	True when at least one is true (inclusive or); like $\cup$.	$(2 > 3) \vee (4 < 5)$: true
$p \oplus q$	**Exclusive or (XOR)** M.86	True when exactly one is true; on bits, addition mod 2 EE.36 §16 Electrical Eng.. ($\oplus$ also denotes a direct sum.)	$1 \oplus 1 = 0$; $5 \oplus 3 = 101_2 \oplus 011_2 = 110_2 = 6$
$\top$	**Top, "true"** M.87	The always-true statement (tautology).	$p \vee \neg p \equiv \top$
$\bot$	**Bottom, "false"** M.88	The always-false statement (contradiction). In geometry the same shape means perpendicular.	$p \wedge \neg p \equiv \bot$
$\forall$	**For all** M.89	Universal quantifier: the statement holds for every element. Negation: $\neg\forall x\, P \equiv \exists x\, \neg P$.	$\forall x \in \mathbb{R},\ x^2 \geq 0$
$\exists$	**There exists** M.90	Existential quantifier: at least one element works. $\nexists$: there is none.	$\exists x \in \mathbb{N} :\ x + 2 = 5$ (namely $x = 3$)
$\exists!$	**Exists exactly one** M.91	Existence *and* uniqueness.	$\exists!\, x :\ 2x = 6$ (only $x = 3$)
$\therefore$	**Therefore** M.92	Introduces a conclusion.	$x + 2 = 5\ \therefore\ x = 3$
$\because$	**Because** M.93	Introduces a reason.	6 is even $\because\ 6 = 2 \cdot 3$
$p \to q$	**If p then q (conditional)** M.94	The logical connective: false only when p is true and q is false. Equivalent to $\neg p \vee q$.	$(n = 4) \to (n \text{ even})$: true for $n = 1, \ldots, 10$
$\Rightarrow$	**Implies** M.95	Asserts that whenever the left side holds, so does the right (p is *sufficient* for q; q is *necessary* for p). The converse need not hold.	$x = 2 \Rightarrow x^2 = 4$, but $x^2 = 4 \not\Rightarrow x = 2$ ($x = -2$)
$\Leftarrow$	**Is implied by, "if"** M.96	The reverse direction: $q \Leftarrow p$ means $p \Rightarrow q$.	$x^2 = 4 \Leftarrow x = 2$
$\leftarrow$	**Gets (assignment)** M.97	In algorithms: overwrite the variable on the left with the value on the right.	$x \leftarrow x + 1$ turns 3 into 4
$p \leftrightarrow q$	**p if and only if q (biconditional)** M.98	The connective that is true when p and q have the same truth value; $(p \to q) \wedge (q \to p)$.	$(x + 1 = 3) \leftrightarrow (x = 2)$
$\Leftrightarrow$	**If and only if (iff)** M.99	Implication in both directions: each side is necessary and sufficient for the other; the statements are equivalent.	$2x = 6 \Leftrightarrow x = 3$

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iff	**If and only if** M.100	Written abbreviation of $\Leftrightarrow$; used in definitions and theorems.	n is even iff $n \bmod 2 = 0$
$q \to p$	**Converse** M.101	Swap hypothesis and conclusion. *Not* equivalent to $p \to q$.	"If n is even then $n = 4$": false ($n = 6$)
$\neg p \to \neg q$	**Inverse** M.102	Negate both parts. Equivalent to the converse, not to the original.	"If $n \neq 4$ then n is odd": false ($n = 2$)
$\neg q \to \neg p$	**Contrapositive** M.103	Swap *and* negate. Always equivalent to $p \to q$; proving it proves the original.	"If n is odd then $n \neq 4$": true
$\vdash$	**Proves, "turnstile"** M.104	Syntactic consequence: the right side can be derived from the left by proof rules.	$p,\ p \to q \vdash q$ (modus ponens)
$\models$	**Entails / satisfies** M.105	Semantic consequence: every interpretation making the left true makes the right true.	$p \wedge q \models p$

Functions & Mappings

$f\colon A \to B$	**f maps A to B** M.106	A function with domain A and codomain B A1.9 §03 Algebra I.	$f\colon \mathbb{R} \to \mathbb{R},\ f(x) = 2x$
$\mapsto$	**Maps to** M.107	Where one element goes (element level, while $\to$ is set level).	$x \mapsto x^2$ sends $3 \mapsto 9$
$f(x)$	**f of x** M.108	The output of f at input x; not multiplication.	$f(x) = x + 1$: $f(2) = 3$
$g \circ f$	**g composed with f, "g after f"** M.109	Apply f first, then g: $(g \circ f)(x) = g(f(x))$ A2.1 §04 Algebra II.	$f(x) = x + 1,\ g(x) = 2x$: $(g \circ f)(3) = 8$
f^{-1}	**f inverse** M.110	The function that undoes f A2.1 §04 Algebra II. Not $\frac{1}{f}$ (that is $f(x)^{-1}$).	$f(x) = 2x$: $f^{-1}(6) = 3$
$f'(x)$	**f prime of x** M.111	The derivative of f (Lagrange notation); f'': second derivative C1.9 §07 Calculus I.	$f(x) = x^2$: $f'(3) = 6$, $f''(3) = 2$
max	**Maximum (and min)** M.112	Largest (smallest) value of a list or of a function over a set.	$\max(3,7,5) = 7;\quad \min_x (x-2)^2 = 0$
arg max	**Argmax (argmin)** M.113	The *input* at which the maximum (minimum) is reached, not the value itself. Training a model is an argmin ML.1 §15 Machine Learning.	$\arg\max_{x \in \{1,2,3\}} (4 - (x-2)^2) = 2$
$\mathbf{1}_A$	**Indicator function** M.114	Equals 1 when the condition or membership holds, 0 otherwise; $\mathbb{E}[\mathbf{1}_A] = P(A)$.	$\mathbf{1}_{x>3}(5) = 1;\quad \mathbf{1}_{x>3}(2) = 0$
$O(g)$	**Big-O, "order g"** M.115	$f = O(g)$: f grows no faster than a constant times g O.10 §14 Optimization. Ω: at least; Θ: exactly.	$3n^2 + 5n = O(n^2)$
$f * g$	**Convolution** M.116	Slide one function over the flipped other and sum the products; used in signals and CNNs EE.26 §16 Electrical Eng..	$(1,2) * (1,1) = (1,3,2)$

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Calculus

$\lim_{x\to a}$	**Limit** M.117	The value $f(x)$ approaches as x gets close to a (from both sides; a^+, a^- for one side) C1.1 §07 Calculus I.	$\lim_{x\to 2}(x+3)=5$; $\lim_{x\to 0^+}\frac{1}{x}=\infty$
$x \to a$	**x tends to a** M.118	x approaches a (in limits); in logic $\to$ is "implies", between sets it is "maps".	$\frac{1}{x}\to 0$ as $x\to\infty$
Δx	**Delta x, "change in x"** M.119	A finite change: final minus initial A1.6 §03 Algebra I.	x from 2 to 5: $\Delta x = 3$
dx	**Differential, "d x"** M.120	An infinitesimal change in x; in an integral it names the variable of integration.	$dy = f'(x)\,dx$: $y=x^2$, $x=3$, $dx=0.1 \Rightarrow dy=0.6$
$\frac{dy}{dx}$	**d y d x (Leibniz)** M.121	Derivative of y with respect to x: instantaneous rate of change. $\frac{d}{dx}$ is the operator C1.9 §07 Calculus I.	$\frac{d}{dx}x^3=3x^2$; at $x=2$: 12
$\frac{d^2y}{dx^2}$	**Second derivative** M.122	Derivative of the derivative: concavity, acceleration C1.23 §07 Calculus I.	$y=x^3$: $\frac{d^2y}{dx^2}=6x$; at $x=1$: 6
$\dot{x}\ \ \ddot{x}$	**x dot, x double-dot** M.123	Newton's notation for time derivatives (velocity, acceleration) P.2 §12 Physics.	$x=t^2$: $\dot{x}=2t$, $\ddot{x}=2$
∂	**Partial, "del" or "dee"** M.124	Partial derivative: differentiate in one variable, hold the others fixed C3.9 §10 Calculus III.	$f=x^2y$: $\frac{\partial f}{\partial x}=2xy$; at $(1,3)$: 6
∇f	**Gradient, "del f"** M.125	Vector of all partial derivatives; points uphill fastest C3.12 §10 Calculus III.	$f=x^2+y^2$: $\nabla f=(2x,2y)$; at $(1,2)$: $(2,4)$
$\nabla \cdot \mathbf{F}$	**Divergence** M.126	Net outflow per unit volume of a vector field V.5 §11 Vector Calculus.	$\mathbf{F}=(x,y)$: $\nabla\cdot\mathbf{F}=1+1=2$
$\nabla \times \mathbf{F}$	**Curl** M.127	Local rotation (circulation density) of a vector field V.5 §11 Vector Calculus.	$\mathbf{F}=(-y,x,0)$: $\nabla\times\mathbf{F}=(0,0,2)$
∇^2	**Laplacian (or Hessian)** M.128	Physics: $\nabla^2 f=\sum \partial^2 f/\partial x_i^2$, also Δf V.13 §11 Vector Calculus. Optimization texts use $\nabla^2 f$ for the Hessian matrix C3.16 §10 Calculus III.	$f=x^2+y^2$: Laplacian 4; Hessian $\left[\begin{smallmatrix}2&0\\0&2\end{smallmatrix}\right]$
$\int f\,dx$	**Integral (indefinite)** M.129	Antiderivative: a function whose derivative is f; add $+C$ C1.29 §07 Calculus I.	$\int 2x\,dx = x^2+C$
$\int_a^b$	**Definite integral** M.130	Signed area under f between a and b C1.30 §07 Calculus I.	$\int_0^3 2x\,dx = 9$
$\Big\vert_a^b$	**Evaluated from a to b** M.131	$F(x)\big\vert_a^b = F(b)-F(a)$, the last step of the Fundamental Theorem C1.32 §07 Calculus I.	$x^2\big\vert_1^3 = 9-1=8$
$\iint$	**Double integral** M.132	Integrate over a 2-D region (triple: $\iiint$) C3.19 §10 Calculus III.	$\int_0^2\int_0^3 1\,dy\,dx = 6$ (area of a 2×3 rectangle)
$\oint$	**Contour (closed-loop) integral** M.133	Integral around a closed curve: circulation or flux V.7 §11 Vector Calculus.	$\oint_C dx = 0$ for any closed curve C

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$\sum_{n=1}^{\infty}$	**Infinite series** M.134	The limit of the partial sums, if it exists C2.16 §08 Calculus II.	$\sum_{n=1}^{\infty} \frac{1}{2^n} = \frac{1}{2} + \frac{1}{4} + \cdots = 1$

Vectors & Matrices

$\vec{v}$ $\mathbf{v}$	**Vector v** M.135	A quantity with magnitude and direction; arrow (handwriting) or bold (print) LA.2 §09 Linear Algebra.	$\mathbf{v} = (3, 4)$
$\hat{\mathbf{u}}$	**u hat (unit vector)** M.136	A vector of length 1: $\hat{\mathbf{u}} = \mathbf{u}/\lVert\mathbf{u}\rVert$; $\hat{\imath}, \hat{\jmath}, \hat{k}$ are the axis directions.	$(3, 4) \to (0.6, 0.8)$
$\lVert\mathbf{v}\rVert$	**Norm, "length of v"** M.137	Euclidean length $\sqrt{\sum v_i^2}$ LA.29 §09 Linear Algebra.	$\lVert(3, 4)\rVert = 5$
$\lVert\mathbf{v}\rVert_p$	**p-norm** M.138	$\left(\sum \lvert v_i\rvert^p\right)^{1/p}$; $p = 1$: taxicab, $p = 2$: Euclidean, $p = \infty$: largest entry ML.11 §15 Machine Learning.	$\mathbf{v} = (3, -4)$: $\lVert\mathbf{v}\rVert_1 = 7$, $\lVert\mathbf{v}\rVert_\infty = 4$
$\mathbf{u} \cdot \mathbf{v}$	**Dot product** M.139	$\sum u_i v_i$, a scalar; $= \lVert\mathbf{u}\rVert\lVert\mathbf{v}\rVert \cos\theta$ T.21 §06 Trigonometry.	$(1, 2) \cdot (3, 4) = 3 + 8 = 11$
$\mathbf{u} \times \mathbf{v}$	**Cross product** M.140	A vector perpendicular to both (3-D only), length = parallelogram area.	$(1, 0, 0) \times (0, 1, 0) = (0, 0, 1)$
$\langle\mathbf{u}, \mathbf{v}\rangle$	**Inner product** M.141	General dot product; also $\langle 1, 2\rangle$ for a vector in some texts.	$\langle(1, 2), (3, 4)\rangle = 11$
$\otimes$	**Outer (tensor) product** M.142	Also the Kronecker product of matrices. $\mathbf{u} \otimes \mathbf{v} = \mathbf{u}\mathbf{v}^{\mathsf{T}}$: every product $u_i v_j$ LA.12 §09 Linear Algebra.	$(1, 2) \otimes (3, 4) = \left[\begin{smallmatrix} 3 & 4 \\ 6 & 8 \end{smallmatrix}\right]$
$\odot$	**Hadamard product** M.143	Multiply matching entries; same shape in and out LA.12 §09 Linear Algebra.	$(1, 2) \odot (3, 4) = (3, 8)$
A^{T}	**A transpose** M.144	Rows become columns: $(A^{\mathsf{T}})_{ij} = a_{ji}$ LA.11 §09 Linear Algebra. Also A' or $A^{\top}$.	$\left[\begin{smallmatrix} 1 & 2 \\ 3 & 4 \end{smallmatrix}\right]^{\mathsf{T}} = \left[\begin{smallmatrix} 1 & 3 \\ 2 & 4 \end{smallmatrix}\right]$
A^{-1}	**A inverse** M.145	The matrix with $AA^{-1} = I$; exists iff $\det A \neq 0$ A2.25 §04 Algebra II.	$\left[\begin{smallmatrix} 2 & 0 \\ 0 & 4 \end{smallmatrix}\right]^{-1} = \left[\begin{smallmatrix} 1/2 & 0 \\ 0 & 1/4 \end{smallmatrix}\right]$
A^{+}	**Pseudoinverse** M.146	Moore–Penrose inverse; gives least-squares solutions when A^{-1} does not exist LA.31 §09 Linear Algebra.	$\left[\begin{smallmatrix} 2 & 0 \\ 0 & 0 \end{smallmatrix}\right]^{+} = \left[\begin{smallmatrix} 1/2 & 0 \\ 0 & 0 \end{smallmatrix}\right]$
$\det A$	**Determinant** M.147	Scalar giving the volume scaling of A; also $\lvert A\rvert$. Zero means singular A2.25 §04 Algebra II.	$\det \left[\begin{smallmatrix} 1 & 2 \\ 3 & 4 \end{smallmatrix}\right] = 4 - 6 = -2$
$\operatorname{tr} A$	**Trace** M.148	Sum of the diagonal entries; equals the sum of the eigenvalues LA.29 §09 Linear Algebra.	$\operatorname{tr} \left[\begin{smallmatrix} 1 & 2 \\ 3 & 4 \end{smallmatrix}\right] = 5$
I $\mathbf{0}$	**Identity / zero matrix** M.149	I: 1s on the diagonal, $AI = A$; $\mathbf{0}$: all zeros, $A + \mathbf{0} = A$ LA.16 §09 Linear Algebra.	$I_2 = \left[\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right]$
$\operatorname{rank} A$	**Rank** M.150	Number of linearly independent rows (or columns) LA.19 §09 Linear Algebra.	$\operatorname{rank} \left[\begin{smallmatrix} 1 & 2 \\ 2 & 4 \end{smallmatrix}\right] = 1$

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δ_{ij}	**Kronecker delta** M.151	1 if $i = j$, 0 otherwise; the entries of I.	$\delta_{11} = 1; \quad \delta_{12} = 0$
$A \succ 0$	**Positive definite** M.152	$\mathbf{x}^{\mathsf{T}} A\mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0}$ ($\succeq$: semidefinite) LA.27 §09 Linear Algebra.	$\left[\begin{smallmatrix} 2 & 0 \\ 0 & 3 \end{smallmatrix}\right] \succ 0$

Probability & Statistics

$P(A)$	**Probability of A** M.153	A number in $[0,1]$; $P(A^c) = 1 - P(A)$ S.7 §13 Statistics. Also $\Pr(A)$.	Fair die: $P(\text{even}) = \frac{3}{6} = \frac{1}{2}$
$P(A \mid B)$	**Probability of A given B** M.154	Probability of A once B is known to have happened: $\frac{P(A \cap B)}{P(B)}$ S.9 §13 Statistics.	Die: $P(6 \mid \text{even}) = \frac{1}{3}$
$P(A, B)$	**Joint probability** M.155	Probability that both happen, $P(A \cap B)$.	Die: $P(\text{even}, > 3) = P(\{4,6\}) = \frac{1}{3}$
$\mathbb{E}[X]$	**Expected value of X** M.156	Probability-weighted average of X S.13 §13 Statistics; also $\mathrm{E}(X)$, $\langle X \rangle$, μ.	Fair die: $\mathbb{E}[X] = 3.5$
$\mathrm{Var}(X)$	**Variance of X** M.157	$\mathbb{E}[(X - \mu)^2]$: average squared spread S.2 §13 Statistics. $\mathrm{Cov}(X, Y)$ is the two-variable version S.38 §13 Statistics.	Coin $X \in \{0,1\}$: $\mathrm{Var}(X) = \frac{1}{4}$
μ	**Mu (population mean)** M.158	The true mean of a population or distribution S.1 §13 Statistics.	Fair die: $\mu = 3.5$
$\bar{x}$	**x bar (sample mean)** M.159	Average of the observed data: $\frac{1}{n}\sum x_i$ S.1 §13 Statistics.	Data $2, 4, 9$: $\bar{x} = 5$
$\sigma \quad s$	**Standard deviation** M.160	σ: population SD (σ^2 is the variance); s: sample SD, divides by $n - 1$ S.2 §13 Statistics.	Data $2, 4, 4, 4, 5, 5, 7, 9$: $\sigma = 2$
$r \quad \rho$	**Correlation** M.161	Strength of linear relationship, from -1 to 1; r for a sample, ρ for a population S.38 §13 Statistics.	Points $(1,2), (2,4), (3,6)$: $r = 1$
$X \sim$	**Distributed as** M.162	Names the distribution a random variable follows S.20 §13 Statistics.	$X \sim \mathrm{Bin}(10, 0.5)$: 10 fair coin flips
$\overset{\text{iid}}{\sim}$	**i.i.d.** M.163	Independent and identically distributed: every draw from the same distribution, no influence between them S.24 §13 Statistics.	$X_1, \ldots, X_5 \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$
$\mathcal{N}(\mu, \sigma^2)$	**Normal distribution** M.164	Bell curve with mean μ and *variance* σ^2 (check: some software takes σ) S.20 §13 Statistics.	$\mathcal{N}(5,4)$: mean 5, SD 2
$X \perp\!\!\!\perp Y$	**Independent** M.165	$P(X,Y) = P(X)P(Y)$: knowing one tells nothing about the other S.9 §13 Statistics.	Two dice: $P(6,6) = \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36}$
$\hat{\theta}$	**Theta hat (estimate)** M.166	A hat marks an estimate or prediction of the un-hatted quantity: $\hat{y}$, $\hat{\beta}$, $\hat{p}$ S.26 §13 Statistics.	3 heads in 10 flips: $\hat{p} = 0.3$
$\mathcal{L}(\theta)$	**Likelihood** M.167	Probability of the observed data viewed as a function of the parameter S.26 §13 Statistics.	2 heads in 3 flips: $\mathcal{L}(p) = 3p^2(1-p)$

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H_0 H_1	**Null / alternative hypothesis** M.168	H_0: the no-effect claim being tested; H_1 (or H_a): what you accept if H_0 is rejected S.28 §13 Statistics.	$H_0: \mu = 5$ vs $H_1: \mu \neq 5$

Geometry

$\angle$	**Angle** M.169	The angle at the middle vertex; $m\angle$ is its measure G.3 §05 Geometry.	$\angle ABC = 90^\circ$ in a square corner
$^\circ$	**Degree** M.170	$\frac{1}{360}$ of a full turn; $1^\circ = 60'$ (minutes) $= 3600''$ (seconds) T.2 §06 Trigonometry.	$180^\circ = \pi$ rad; $\quad 30' = 0.5^\circ$
rad	**Radian** M.171	Angle whose arc equals the radius; 2π rad $= 360^\circ$ T.2 §06 Trigonometry. Angles with no unit are radians.	$\frac{\pi}{2}$ rad $= 90^\circ$; $\quad 1$ rad $\approx 57.3^\circ$
$\triangle ABC$	**Triangle ABC** M.172	The triangle with vertices A, B, C G.5 §05 Geometry.	$\triangle ABC$ with sides $3, 4, 5$ is a right triangle
$\parallel$	**Is parallel to** M.173	Same direction, never meet; equal slopes A1.8 §03 Algebra I.	$y = 2x + 1 \parallel y = 2x + 5$
$\perp$	**Is perpendicular to** M.174	Meet at a right angle; slopes multiply to -1 A1.8 §03 Algebra I. For vectors: orthogonal.	$y = 2x \perp y = -\frac{1}{2}x$; $\quad (1,2) \perp (2,-1)$
$\overline{AB}$	**Segment AB** M.175	The segment from A to B; often its length AB too G.12 §05 Geometry.	$A = (0,0)$, $B = (3,4)$: $AB = 5$
$\overrightarrow{AB}$	**Ray / vector AB** M.176	Starts at A through B; as a vector, $B - A$. ($\overleftrightarrow{AB}$: the whole line.)	$A = (1,1)$, $B = (4,5)$: $\overrightarrow{AB} = (3,4)$
$\overset{\frown}{AB}$	**Arc AB** M.177	The arc of a circle from A to B; length $r\theta$ G.17 §05 Geometry.	90° arc, $r = 2$: length $\pi \approx 3.14$

Miscellaneous

$\ldots$	**Ellipsis (low dots)** M.178	"And so on" in a list separated by commas.	$1, 2, \ldots, 10$
$\cdots$	**Centered dots** M.179	"And so on" between operators. Vertical and diagonal dots continue the columns and diagonal of a matrix.	$1 + 2 + \cdots + 10 = 55$
$\triangleq$	**Equal by definition** M.180	Same as $:=$; also written $\overset{\text{def}}{=}$.	$\sigma \triangleq \sqrt{\text{Var}(X)}$
$\blacksquare$	**QED (end of proof)** M.181	*Quod erat demonstrandum*, "which was to be shown"; also $\square$.	"…so n^2 is even. $\blacksquare$"
s.t.	**Such that / subject to** M.182	Introduces a condition; in optimization, the constraints O.1 §14 Optimization.	$\min x^2$ s.t. $x \geq 1$: minimum 1 at $x = 1$

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Logic: Implication, Converse, Contrapositive & Truth Tables |

Key Topic M.K1

Every theorem is a statement built from a few connectives. A truth table lists every combination of truth values (T = true, F = false) and the result. The conditional $p \to q$ is the one most often misread: it says nothing when p is false, and it does *not* run backwards.

Truth tables

p	q	$\neg p$ not	$p \wedge q$ and	$p \vee q$ or	$p \oplus q$ xor	$p \to q$ if…then	$p \leftrightarrow q$ iff
T	T	F	T	T	F	T	T
T	F	F	F	T	T	**F**	F
F	T	T	F	T	T	T	F
F	F	T	F	F	F	T	T

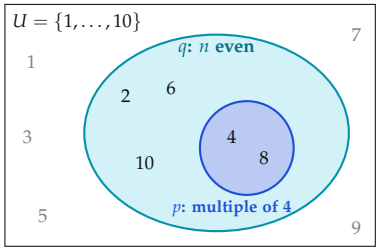
$\wedge$ is true only in the first row; $\vee$ is false only in the last; $\oplus$ is true when $p \neq q$; $\leftrightarrow$ is true when $p = q$ (so $p \oplus q \equiv \neg(p \leftrightarrow q)$).

$p \to q$ **is false in exactly one row**: p true, q false. When p is false the conditional is (vacuously) true: "if $3 > 5$ then $1 = 2$" is true. Hence $p \to q \equiv \neg p \vee q$ and $\neg(p \to q) \equiv p \wedge \neg q$.

The four related conditionals ($n \in \{1, \dots, 10\}$)

Let p: "n is a multiple of 4" and q: "n is even".

Name	Form	Statement	Truth
Original	$p \to q$	If n is a multiple of 4, n is even.	T
Converse	$q \to p$	If n is even, n is a multiple of 4.	F ($n = 6$)
Inverse	$\neg p \to \neg q$	If n is not a multiple of 4, n is odd.	F ($n = 2$)
Contrapositive	$\neg q \to \neg p$	If n is odd, n is not a multiple of 4.	T



$p \Rightarrow q$: the p-region lies inside the q-region.
Converse fails at 2, 6, 10; contrapositive: outside $q \Rightarrow$ outside p.

Equivalent pairs: original $\equiv$ contrapositive; converse $\equiv$ inverse (each is the contrapositive of the other). Checking all ten values: p is true only for $n = 4, 8$, and both are even, so $p \to q$ never fails.

Implication vocabulary

$p \Rightarrow q$ M.95 reads "p implies q", "if p then q", "p only if q", "q whenever p". p is **sufficient** for q; q is **necessary** for p.

$p \Leftrightarrow q$ M.99 ("iff") asserts *both* $p \Rightarrow q$ and $q \Rightarrow p$: each is necessary and sufficient for the other. Example: n is even $\Leftrightarrow$ $n \bmod 2 = 0$.

Negating quantifiers and connectives

Quantifiers: swap and push the $\neg$ inside

$\neg\forall x\, P(x) \;\equiv\; \exists x\, \neg P(x)$
$\neg\exists x\, P(x) \;\equiv\; \forall x\, \neg P(x)$

"Every $n \in \{1, \dots, 10\}$ is less than 10" is false: its negation "$\exists n$ with $n \geq 10$" is true ($n = 10$). One counterexample disproves a $\forall$.

Order matters: $\forall x\, \exists y\, (y > x)$ is true in $\mathbb{N}$ (take $y = x + 1$), but $\exists y\, \forall x\, (y > x)$ is false (no largest number).

De Morgan's laws (logic and sets)

$\neg(p \wedge q) \equiv \neg p \vee \neg q \qquad (A \cap B)^c = A^c \cup B^c$
$\neg(p \vee q) \equiv \neg p \wedge \neg q \qquad (A \cup B)^c = A^c \cap B^c$

The negation of "$2 < x$ and $x < 5$" is "$x \leq 2$ or $x \geq 5$". Sets: A1.21 §03 Algebra I.

Pattern: negating swaps $\wedge \leftrightarrow \vee$ and $\forall \leftrightarrow \exists$, and pushes $\neg$ onto every piece.

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Proof strategies

Direct: assume p, derive q. **Contrapositive:** assume $\neg q$, derive $\neg p$. **Contradiction:** assume $p \wedge \neg q$, derive $\bot$. **Counterexample:** one case kills a $\forall$.

Standard equivalences (check any of them with a truth table)

Double negation	$\neg\neg p \equiv p$	Implication as "or"	$p \to q \equiv \neg p \vee q$
Commutative	$p \wedge q \equiv q \wedge p, \quad p \vee q \equiv q \vee p$	Contrapositive	$p \to q \equiv \neg q \to \neg p$
Associative	$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	Negated implication	$\neg(p \to q) \equiv p \wedge \neg q$
Distributive	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	Biconditional	$p \leftrightarrow q \equiv (p \to q) \wedge (q \to p)$
Idempotent	$p \wedge p \equiv p \vee p \equiv p$	XOR	$p \oplus q \equiv (p \vee q) \wedge \neg(p \wedge q)$
Absorption	$p \vee (p \wedge q) \equiv p$	Exportation	$(p \wedge q) \to r \equiv p \to (q \to r)$
Identity / domination	$p \wedge \top \equiv p, \quad p \vee \top \equiv \top$	Excluded middle	$p \vee \neg p \equiv \top$

Translating English into logic

English	Logic	English	Logic
p only if q	$p \to q$	p unless q	$\neg q \to p \ \ (\equiv p \vee q)$
p is sufficient for q	$p \to q$	neither p nor q	$\neg p \wedge \neg q$
p is necessary for q	$q \to p$	not all x are P	$\exists x\, \neg P(x)$
p if q	$q \to p$	no x is P	$\forall x\, \neg P(x)$
p exactly when q	$p \leftrightarrow q$	every P is a Q	$\forall x\, (P(x) \to Q(x))$
either p or q (not both)	$p \oplus q$	some P is a Q	$\exists x\, (P(x) \wedge Q(x))$

Check with $n \in \{1, \ldots, 10\}$: "every multiple of 4 is even" is $\forall n\, (4 \mid n \to 2 \mid n)$, true. "Some even number is a multiple of 4" is $\exists n\, (2 \mid n \wedge 4 \mid n)$, true ($n = 4$). Note $\forall$ pairs with $\to$ and $\exists$ pairs with $\wedge$: $\exists n\, (2 \mid n \to 4 \mid n)$ is trivially true (take $n = 1$) and says nothing useful.

Common mistakes

DECISION POINT **Affirming the consequent**: from $p \to q$ and q, concluding p ("n is even, so it is a multiple of 4": $n = 6$). **Denying the antecedent**: from $p \to q$ and $\neg p$, concluding $\neg q$ ("n is not a multiple of 4, so it is odd": $n = 2$). Both confuse a statement with its converse or inverse. Valid: *modus ponens* (p, $p \to q \vdash q$) and *modus tollens* ($\neg q$, $p \to q \vdash \neg p$).

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Reading Mathematical Notation | *Key Topic M.K2*

A formula is a sentence written in symbols. Read it from the outside in: identify the main relation ($=$, $<$, $\Rightarrow$), then the outermost operator on each side, then work inward, naming every index and what it runs over. Precedence rules fix the order of operations, and context fixes the meaning of overloaded symbols.

Decoding a dense formula: the sample variance

$$\underbrace{s^2}_{\text{sample variance}} = \underbrace{\frac{1}{n-1}}_{\text{divide by } n-1} \overbrace{\sum_{i=1}^{n}}^{\text{for each data point } i} \Big(\underbrace{x_i}_{i\text{-th value}} - \underbrace{\bar{x}}_{\text{mean}} \Big) \overbrace{{}^{2}}^{\text{square}}$$

Read it aloud

"s squared equals one over n minus one times the sum, for i from 1 to n, of x_i minus x-bar, squared." S.2 §13 Statistics

Evaluate inside out (data 2, 4, 6)

$\bar{x} = 4$; deviations $-2, 0, 2$; squares $4, 0, 4$; sum 8; divide by $n - 1 = 2$: $s^2 = 4$, $s = 2$.

A second example: the ε–δ limit

$$\forall \varepsilon > 0 \;\; \exists \delta > 0 : \quad 0 < |x - a| < \delta \;\Rightarrow\; |f(x) - L| < \varepsilon$$

"For every tolerance ε there is a closeness δ such that whenever x is within δ of a (but not equal), $f(x)$ is within ε of L." That is the definition of $\lim_{x \to a} f(x) = L$ C1.1 §07 Calculus I. Quantifier order matters: δ may depend on ε. Example: $f(x) = 2x$, $a = 3$, $L = 6$: choose $\delta = \varepsilon/2$.

Operator precedence (highest first)

	Level	Example
1	Grouping ( ), bars, roots, $\lvert\,\rvert$	$2(3+1) = 8$
2	Function application, subscripts, !	$2 \cdot 3! = 12$; $\sin 2x = \sin(2x)$
3	Exponents (right to left)	$2^{3^2} = 2^9 = 512$
4	Unary minus	$-3^2 = -9$, but $(-3)^2 = 9$
5	$\times$, $\div$, juxtaposition (left to right)	$8 \div 2 \cdot 4 = 16$
6	$\sum$, $\prod$, $\int$ (scope: the following product)	$\sum_i a_i b_i + c = \left(\sum_i a_i b_i\right) + c$
7	$+$, $-$ (left to right)	$9 - 4 + 1 = 6$
8	Relations $=, <, \leq, \in, \subseteq$	$1 + 2 < 4$
9	$\neg$, then $\wedge$, then $\vee$	$\neg p \wedge q = (\neg p) \wedge q$
10	$\Rightarrow$, then $\Leftrightarrow$; quantifiers widest	$p \wedge q \Rightarrow r$

Index notation

a_i: i-th entry; a_{ij}: row i, column j.
$x^{(i)}$: i-th *sample* (ML), not a power; $x_j^{(i)}$: feature j of sample i.
a^i (superscript index): a component in physics, a power elsewhere.
Bound index: summed over, its name is irrelevant: $\sum_i a_i = \sum_k a_k$. **Free index**: appears on both sides: $y_i = 2x_i$ holds for each i.
Einstein: a repeated index means sum: $A_{ij}x_j = \sum_j A_{ij}x_j = (A\mathbf{x})_i$.

Overloaded symbols: read the context

$|\cdot|$: absolute value, set size, determinant. $\sim$: similar, distributed as, asymptotic. $\perp$: perpendicular, false, independent ($\perp\!\!\!\perp$). $\to$: tends to, maps to, implies. Bar: mean, complement, conjugate, segment. $'$: derivative, transpose, complement.

Classic misreadings

DECISION POINT $\sin^2 x = (\sin x)^2$, but $\sin^{-1} x = \arcsin x \neq \frac{1}{\sin x}$. $f^{-1}(x)$ is the inverse function, $f(x)^{-1} = \frac{1}{f(x)}$. $1/2x$ is ambiguous: write $\frac{1}{2x}$ or $\frac{x}{2}$. $\log x$ has no fixed base. $\sum_{i=1}^{3} i^2 = 14$ but $\left(\sum_{i=1}^{3} i\right)^2 = 36$. $\mathcal{N}(\mu, \sigma^2)$ takes a variance; `numpy` takes a standard deviation. $x \in A$ (element) vs $\{x\} \subseteq A$ (subset).

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02
Greek Alphabet

The 24 letters that name the constants, parameters and operators of science and engineering. Each row gives the capital and lower-case forms (plus variant forms), a common English pronunciation, and what the letter conventionally means in mathematics, statistics, machine learning, physics, electrical engineering and finance. Many capitals (A, B, E, Z, H, I, K, M, N, O, P, T, X) look exactly like Latin letters, so only the distinct ones appear in formulas.

In this chapter

GR.K1 **Key topic:** Greek Letters by Field 28

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KEY	LETTER	NAME / SAY IT	COMMON USES
GR.1	A α	**Alpha** *AL-fuh*	**Stats:** significance level / Type I error rate S.28 §13 Statistics. **ML:** learning rate; elastic-net mix ML.8 §15 Machine Learning; Beta/Dirichlet shape D.10 §18 Graph Gallery. **Physics:** angle, angular acceleration, thermal diffusivity P.31 §12 Physics. **Finance:** excess return (Jensen's alpha) FN.14 §17 Finance.
GR.2	B β	**Beta** *BAY-tuh (UK: BEE-tuh)*	**Stats:** regression coefficients β_0, β_1 S.40 §13 Statistics; Type II error rate, power $= 1 - \beta$ S.28 §13 Statistics; Beta distribution D.10 §18 Graph Gallery. **ML:** Adam momentum β_1, β_2 ML.27 §15 Machine Learning. **Finance:** market beta FN.14 §17 Finance. **EE:** transistor current gain.
GR.3	Γ γ	**Gamma** *GAM-uh*	**Math:** $\Gamma(n) = (n-1)!$, the gamma function D.9 §18 Graph Gallery; Euler's constant $\gamma \approx 0.577$. **ML:** RL discount factor O.21 §14 Optimization; RBF kernel width ML.15 §15 Machine Learning. **Physics:** Lorentz factor, heat-capacity ratio, photon. **Finance:** option gamma FN.28 §17 Finance.
GR.4	Δ δ	**Delta** *DEL-tuh*	Δ: a finite change Δx A1.6 §03 Algebra I; discriminant A1.15 §03 Algebra I; Laplacian $\Delta = \nabla^2$. δ: a small change; ε–δ limits C1.1 §07 Calculus I; Kronecker δ_{ij} and Dirac $\delta(x)$; backprop error signal ML.22 §15 Machine Learning. **Finance:** option delta $\partial V / \partial S$ FN.28 §17 Finance.
GR.5	E ϵ ε	**Epsilon** *EP-sih-lon*	Both forms (ϵ, ε) are the same letter. **Math:** an arbitrarily small positive number C1.1 §07 Calculus I. **Stats:** error term $\varepsilon \sim \mathcal{N}(0, \sigma^2)$ S.40 §13 Statistics. **ML:** stability constant (`1e-8`) ML.27 §15 Machine Learning; ε-greedy. **Physics/EE:** permittivity ε_0 EE.18 §16 Electrical Eng.; emissivity P.28 §12 Physics; strain.
GR.6	Z ζ	**Zeta** *ZAY-tuh (UK: ZEE-tuh)*	**Math:** Riemann zeta function $\zeta(s) = \sum n^{-s}$, with $\zeta(2) = \frac{\pi^2}{6}$ C2.16 §08 Calculus II. **Engineering:** damping ratio of a second-order system: $\zeta < 1$ underdamped, $\zeta = 1$ critical, $\zeta > 1$ overdamped EE.16 §16 Electrical Eng., P.15 §12 Physics.
GR.7	H η	**Eta** *AY-tuh (UK: EE-tuh)*	**ML:** learning rate (step size) of gradient descent C1.26 §07 Calculus I, ML.27 §15 Machine Learning. **Engineering:** efficiency $\eta = P_{\text{out}} / P_{\text{in}}$ P.6 §12 Physics. **Physics:** dynamic viscosity. **Stats:** linear predictor $\eta = X\beta$ in GLMs; effect size η^2.
GR.8	Θ θ ϑ	**Theta** *THAY-tuh, THEE-tuh*	θ: an angle T.1 §06 Trigonometry, polar angle T.19 §06 Trigonometry; generic model parameters, estimate $\hat{\theta}$ S.26 §13 Statistics. Θ: parameter space; tight bound $\Theta(n)$ O.10 §14 Optimization. **Finance:** option theta (time decay). ϑ: variant, used for a second angle.
GR.9	I ι	**Iota** *eye-OH-tuh*	Rarely used because it looks like i (and has no dot). Inclusion map $\iota\colon A \hookrightarrow B$; index generator in APL. "Not one iota": not the smallest amount.
GR.10	K κ $\varkappa$	**Kappa** *KAP-uh*	**Linear algebra:** condition number $\kappa(A)$ LA.29 §09 Linear Algebra. **Geometry:** curvature. **Stats/ML:** Cohen's κ, agreement beyond chance ML.17 §15 Machine Learning. **Physics:** thermal conductivity (k or κ) P.24 §12 Physics; dielectric constant. $\varkappa$: variant form.
GR.11	Λ λ	**Lambda** *LAM-duh*	λ: eigenvalue LA.23 §09 Linear Algebra (Λ = diagonal eigenvalue matrix); Poisson / exponential rate S.18 §13 Statistics; regularization strength ML.8 §15 Machine Learning; Lagrange multiplier O.17 §14 Optimization; wavelength $\lambda = v/f$ P.15 §12 Physics. Programming: anonymous function.
GR.12	M μ	**Mu** *MYOO*	**Stats:** population mean $\mu = \mathbb{E}[X]$ S.1 §13 Statistics. **Units:** micro-, 10^{-6} (μm, μF) P.1 §12 Physics. **Physics:** friction coefficient P.4 §12 Physics; dynamic viscosity P.21 §12 Physics; permeability μ_0 EE.18 §16 Electrical Eng.. **Finance:** drift of a price process FN.27 §17 Finance.
GR.13	N ν	**Nu** *NOO (or NYOO)*	Looks like a Latin v. **Stats:** degrees of freedom S.21 §13 Statistics of the t and χ^2 distributions S.30 §13 Statistics. **Physics:** frequency ($\omega = 2\pi\nu$) P.15 §12 Physics; kinematic viscosity $\nu = \mu/\rho$ P.21 §12 Physics; Poisson's ratio.

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GR.14 Ξ ξ **Xi**
KSY, ZY or KSEE

ML: SVM slack variables ξ_i ML.15 §15 Machine Learning. **Math:** an unspecified intermediate point (mean value theorem, Taylor remainder) C1.27 §07 Calculus I; a generic random variable. **Physics:** correlation length; damping ratio in some texts.

GR.15 O o **Omicron**
OM-ih-kron

Identical to Latin o, so almost never used as a symbol. The asymptotic symbols big-O and little-o O.10 §14 Optimization are Latin letters ("order of").

GR.16 Π π ϖ **Pi**
PIE

$\pi \approx 3.14159$, circumference over diameter G.17 §05 Geometry. Π: product operator M.54 §01 Math Signs. **ML/Stats:** mixing weights π_k ML.37 §15 Machine Learning; RL policy $\pi(a \mid s)$; prior $\pi(\theta)$. **Number theory:** prime-counting $\pi(x)$. ϖ: rare variant.

GR.17 P ρ ϱ **Rho**
ROH

Stats: population correlation S.38 §13 Statistics. **Physics:** density (mass per volume) P.19 §12 Physics; resistivity $R = \rho L/A$ P.33 §12 Physics. **Math:** spherical radius C3.21 §10 Calculus III; spectral radius $\rho(A)$. **Finance:** option rho (rate sensitivity). ϱ: variant form.

GR.18 Σ σ ς **Sigma**
SIG-muh

Σ: summation M.53 §01 Math Signs; covariance matrix. σ: standard deviation, σ^2 variance S.2 §13 Statistics; sigmoid $\sigma(z)$ ML.9 §15 Machine Learning; singular values LA.28 §09 Linear Algebra; Stefan–Boltzmann constant P.28 §12 Physics; volatility (finance). ς: word-final form, not used in math.

GR.19 T τ **Tau**
TOW (rhymes with cow)

EE: time constant $\tau = RC$ EE.11 §16 Electrical Eng.. **Physics:** torque $\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}$; shear stress P.21 §12 Physics. **Stats:** Kendall's rank correlation τ. **ML:** softmax temperature ML.24 §15 Machine Learning; a threshold. **Math:** $\tau = 2\pi$ (one full turn).

GR.20 Υ υ **Upsilon**
OOP-sih-lon

Rare because υ looks like u and v; occasionally a velocity. Particle physics: the Υ meson.

GR.21 Φ ϕ φ **Phi**
FY (or FEE)

Φ: standard normal CDF S.20 §13 Statistics; magnetic or electric flux EE.18 §16 Electrical Eng.. ϕ: standard normal pdf; golden ratio $\varphi \approx 1.618$; phase angle EE.14 §16 Electrical Eng.; azimuth C3.21 §10 Calculus III; feature map $\phi(\mathbf{x})$ ML.15 §15 Machine Learning; Euler's totient. ϕ, φ: same letter. Not $\varnothing$.

GR.22 X χ **Chi**
KY (rhymes with sky)

Stats: χ^2 distribution and tests of fit and independence S.32 §13 Statistics, D.11 §18 Graph Gallery. **Math:** indicator (characteristic) function χ_A; Euler characteristic $\chi = V - E + F$. **Physics:** magnetic or electric susceptibility. Not the Latin x.

GR.23 Ψ ψ **Psi**
SY (or PSY)

Physics: quantum wave function ψ ($|\psi|^2$ = probability density); stream function in fluid flow. **Math:** digamma function $\psi = \Gamma'/\Gamma$ D.9 §18 Graph Gallery. **Engineering:** yaw angle; magnetic flux linkage Ψ.

GR.24 Ω ω **Omega**
oh-MAY-guh

Ω: the ohm, unit of resistance EE.1 §16 Electrical Eng.; sample space S.7 §13 Statistics; asymptotic lower bound $\Omega(n)$ O.10 §14 Optimization; solid angle. ω: angular frequency $\omega = 2\pi f$ EE.13 §16 Electrical Eng.; angular velocity T.2 §06 Trigonometry; an outcome $\omega \in \Omega$.

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Greek Letters by Field | *Key Topic GR.K1*

The same letter means different things in different fields, and each field has strong conventions. When you read a formula, first ask which field it comes from. The table lists the conventional meaning of the most-used letters; a dash means no common convention.

	Statistics	Machine learning	Physics	Electrical eng.	Finance
α	significance level	learning rate; L1/L2 mix	angular acceleration; diffusivity	temperature coefficient	excess return (alpha)
β	regression coefficient; Type II error	momentum (Adam β_1, β_2)	v/c; expansion coefficient	transistor current gain	market sensitivity (beta)
γ	skewness	discount factor; kernel width	Lorentz factor	propagation constant	option gamma
Δ, δ	difference; effect size	error signal (backprop)	change; Dirac delta	small-signal change	option delta
ε	error / noise term	numerical-stability constant	strain; emissivity	permittivity	residual shock
η	effect size η^2	learning rate	viscosity	efficiency	—
θ	parameter (estimate $\hat{\theta}$)	model weights	angle	phase angle	option theta
κ	Cohen's agreement	—	thermal conductivity	dielectric constant	—
λ	Poisson / exponential rate	regularization strength; eigenvalue	wavelength	failure rate	jump intensity
μ	population mean	mean of a Gaussian	friction; viscosity; micro-	permeability; micro-	drift (expected return)
ν	degrees of freedom	—	frequency	—	—
π	—	policy; mixing weights	3.14159 . . .	3.14159 . . .	profit
ρ	correlation	—	density	resistivity	option rho; correlation
σ, Σ	SD; covariance matrix	sigmoid; singular value	Stefan–Boltzmann; stress	conductivity	volatility
τ	Kendall's rank correlation	softmax temperature	torque; shear stress	time constant RC	time to expiry
ϕ, Φ	normal pdf / CDF	feature map; activation	potential; flux	phase; magnetic flux	normal CDF in Black–Scholes
χ	χ^2 test	—	susceptibility	—	—
ω, Ω	outcome; sample space	weights (also w)	angular frequency	angular frequency; ohm	portfolio weights

Look-alikes to watch

Lower case vs. Latin

ν (nu) vs. v; ω (omega) vs. w; ρ (rho) vs. p;
ι (iota) vs. i; κ (kappa) vs. k; χ (chi) vs. x;
υ (upsilon) vs. u; o (omicron) vs. o; η (eta) vs. n.

Handwriting tip: give ν a sharp point, ρ a tail that starts in the bowl, and χ a curved stroke.

Variant forms are the same letter

$\epsilon = \varepsilon$, $\theta = \vartheta$, $\phi = \varphi$, $\rho = \varrho$, $\pi = \varpi$, $\kappa = \varkappa$.

ς is the Greek word-final sigma (never a math symbol). $\varnothing$ (empty set) and $\in$ (element of) are not Greek letters, though $\in$ descends from ε. Capitals that equal Latin letters (A, B, E, . . .) are not used as Greek symbols.

General mathematics (any field)

Σ sum, Π product M.53 §01 Math Signs; Δ finite change; δ, ε small positive quantities C1.1 §07 Calculus I; θ, ϕ, α, β angles T.1 §06 Trigonometry; λ eigenvalue LA.23 §09 Linear Algebra; π the circle constant; Γ gamma function; ζ zeta function; χ_A indicator. **Not Greek:** ∂ (a rounded d), ∇ (nabla, an inverted Δ), $\in$ (element of), $\varnothing$ (empty set), $\aleph$ (Hebrew aleph).

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03
Algebra I

The grammar of mathematics: operating on real numbers, manipulating expressions, and modeling with lines and parabolas.

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A1.1 Properties of Real Numbers & Order of Operations

The rules that let you rearrange and regroup any arithmetic expression without changing its value, and the fixed order in which operations are carried out.

Commutative	$a+b=b+a, \quad ab=ba$	Order of the operands does not matter for $+$ and $\times$ (it does for $-$ and $\div$).
Associative	$(a+b)+c=a+(b+c), \quad (ab)c=a(bc)$	Grouping does not matter for $+$ and $\times$.
Distributive	$a(b+c)=ab+ac$	Multiplication spreads over each term of a sum; read right-to-left it is factoring.
Identities	$a+0=a, \quad a\cdot 1=a$	0 and 1 leave a number unchanged.
Inverses	$a+(-a)=0, \quad a\cdot\frac{1}{a}=1\ (a\neq 0)$	Every number can be "undone"; there is no inverse of 0 under $\times$.
Zero property	$a\cdot 0=0; \quad ab=0\Rightarrow a=0$ or $b=0$	The basis of solving by factoring A1.14.
Order of operations	Multiplication does not beat division (and $+$ does not beat $-$): equal-rank operations go left to right. $\underbrace{(\,)}_{\text{Parentheses}} \to \underbrace{x^n}_{\text{Exponents}} \to \underbrace{\times\ \div}_{\text{left to right}} \to \underbrace{+\ -}_{\text{left to right}}$	

STATED-EQUATION *a plus b equals b plus a; a b equals b a; the quantity a plus b, plus c, equals a plus the quantity b plus c; a times the quantity b plus c equals a b plus a c. Order: parentheses, exponents, multiplication and division left to right, then addition and subtraction left to right. a plus zero equals a, and a times one equals a. a plus minus a equals zero, and a times one over a equals one when a is not zero. a times zero equals zero, and if a b equals zero then a equals zero or b equals zero.*

MEMORY AID PEMDAS: "Please Excuse My Dear Aunt Sally" (Parentheses, Exponents, Multiply/Divide, Add/Subtract).

METHOD

1. Evaluate innermost parentheses (and fraction bars, roots, absolute values, which group like parentheses).
2. Apply exponents.
3. Multiply and divide from left to right.
4. Add and subtract from left to right. Example: $8-6\div 2\cdot 3=8-3\cdot 3=8-9=-1$.

TERMS

a, b, c – any real numbers
Operand – a number an operation acts on
Identity element – 0 for addition, 1 for multiplication
Additive inverse $-a$ – the opposite of a
Reciprocal $\frac{1}{a}$ – the multiplicative inverse of a
Exponent – repeated multiplication A1.3

CONTEXT

- **Origin.** The field axioms, formalized in the 19th century by Peacock and Hamilton.
- **Use.** Simplifying any expression unambiguously.
- **Data science.** Floating-point addition is *not* associative, so the order of a large sum (e.g. a GPU reduction) can change the last digits of a result.
- **Physical application.** The cost of 3 bags each holding a \$4 and a \$2 item: $3(4+2)=3\cdot 4+3\cdot 2=18$.

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Canceling & Common Mistakes | *Key Topic A1.K1*

Canceling is not deleting symbols: it is replacing an inverse pair by its identity. A factor over the same factor is 1 (multiplicative inverse); a term plus its opposite is 0 (additive inverse) A1.1. Everything that goes wrong comes from canceling something that is not a whole factor, or from forgetting the values the canceled factor excluded.

Factors cancel, terms do not

Factors cancel

$$\frac{x+3}{x+3} = 1 \quad (x \neq -3)$$

$$\frac{\cancel{(x+3)}\,(x-2)}{\cancel{(x+3)}\,(x+1)} = \frac{x-2}{x+1}$$

each box is *multiplied*: a whole factor

Terms do not

the 3 is *added*: a term, not a factor

$$\frac{x+3}{3} \neq x \qquad = \frac{x}{3}+1$$

cancel only after factoring top and bottom completely

What may be canceled

A common factor, top and bottom: $\frac{ac}{bc} = \frac{a}{b}$ $(c \neq 0)$.

Opposite terms in a sum: $x + \cancel{3} - \cancel{3} = x$.

An operation and its inverse: $\sqrt{x}^2 = x$ $(x \geq 0)$, $\ln e^x = x$, $e^{\ln x} = x$ $(x > 0)$, $\sqrt{x^2} = |x|$ A1.4.

Opposite factors give -1: $\frac{2-x}{x-2} = -1$.

Before you cancel

1. Factor numerator and denominator *completely* A1.14.
2. Write the restrictions: every value that makes an original denominator 0.
3. Cancel only factors that appear, whole, on both sides of the bar.
4. Keep the restrictions: the simplified form has a *hole* there A2.10 §04 Algebra II.

DECISION POINT **Can I cancel it?** Ask whether the piece is *multiplied* into everything above the bar and everything below it. If you cannot draw a single box around it on each side, with only multiplication connecting that box to the rest, it is a term: factor first or leave it. When you divide both sides of an equation by an expression, set that expression equal to 0 separately so no solution is lost.

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Worked cancellations, simple to advanced

Type	Cancellation	Why it works
Monomials	$\dfrac{6x^2y^3}{9xy^5} = \dfrac{2\cdot\cancel{3}\cdot\cancel{x}\cdot x\cdot\cancel{y^3}}{\cancel{3}\cdot\cancel{3}\cdot\cancel{x}\cdot y^2\cdot\cancel{y^3}} = \dfrac{2x}{3y^2}$	Split every number into prime factors and every power into repeated factors; $x, y \neq 0$.
Trinomials	$\dfrac{x^2-9}{x^2+x-6} = \dfrac{(x-3)\cancel{(x+3)}}{\cancel{(x+3)}(x-2)} = \dfrac{x-3}{x-2}$	Factor both first; restrictions $x \neq -3$, 2 stay.
Opposite factor	$\dfrac{x^2-4x}{12-3x} = \dfrac{x\cancel{(x-4)}}{-3\cancel{(x-4)}} = -\dfrac{x}{3}$	Rewrite $12-3x = -3(x-4)$ to expose the match.
Cubes	$\dfrac{x^3-8}{x-2} = \dfrac{\cancel{(x-2)}(x^2+2x+4)}{\cancel{x-2}} = x^2+2x+4$	Difference of cubes (SOAP) A1.14.
Complex fraction	$\dfrac{\frac{1}{x}-\frac{1}{a}}{x-a} = \dfrac{\frac{a-x}{ax}}{x-a} = \dfrac{-\cancel{(x-a)}}{ax\cancel{(x-a)}} = -\dfrac{1}{ax}$	Combine the small fractions first; then $a-x = -(x-a)$.
Difference quotient	$\dfrac{(x+h)^2-x^2}{h} = \dfrac{\cancel{x^2}+2xh+h^2-\cancel{x^2}}{h} = \dfrac{\cancel{h}(2x+h)}{\cancel{h}} = 2x+h$	Expand, cancel opposite terms, *then* factor out h C1.7 §07 Calculus I.
Conjugate	$\dfrac{\sqrt{x+h}-\sqrt{x}}{h}\cdot\dfrac{\sqrt{x+h}+\sqrt{x}}{\sqrt{x+h}+\sqrt{x}} = \dfrac{\cancel{h}}{\cancel{h}(\sqrt{x+h}+\sqrt{x})}$	$(\sqrt{u}-\sqrt{v})(\sqrt{u}+\sqrt{v}) = u-v$ removes the roots A1.4.
Reciprocals	$\left(a^{-1}+b^{-1}\right)^{-1} = \left(\dfrac{a+b}{ab}\right)^{-1} = \dfrac{ab}{a+b}$	Negative exponents are reciprocals; combine before inverting.
Exponent and log	$e^{2\ln x} = e^{\ln x^2} = x^2, \qquad \log_2 8^x = 3x$	Move the coefficient inside as a power, then the inverse pair cancels A2.14 §04 Algebra II.
Telescoping	$\left(1-\frac{1}{2}\right)+\left(\frac{1}{2}-\frac{1}{3}\right)+\cdots+\left(\frac{1}{n}-\frac{1}{n+1}\right) = 1-\dfrac{1}{n+1}$	Each middle term meets its opposite; only the ends survive C2.16 §08 Calculus II.

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Common mistakes

Wrong	Right	What went wrong
$\frac{x+5}{5} = x$	$\frac{x+5}{5} = \frac{x}{5} + 1$	5 is a term of the top, not a factor of it.
$\frac{a+b}{a+c} = \frac{b}{c}$	does not simplify	Only whole, multiplied factors cancel; a is added.
$\frac{x^2+y^2}{x+y} = x+y$	does not simplify	x^2+y^2 does not factor over the reals; x^2-y^2 would.
$\frac{1}{a+b} = \frac{1}{a} + \frac{1}{b}$	$\frac{1}{a} + \frac{1}{b} = \frac{a+b}{ab}$	Fractions split over a sum in the *numerator* only.
$\sqrt{a^2+b^2} = a+b$	$\sqrt{a^2b^2} = \lvert ab \rvert$	Roots distribute over products, not sums ($\sqrt{9+16} = 5 \neq 7$).
$(a+b)^2 = a^2+b^2$	$(a+b)^2 = a^2+2ab+b^2$	Exponents distribute over products, not sums A1.13.
$\ln(a+b) = \ln a + \ln b$	$\ln(ab) = \ln a + \ln b$	Logs turn products into sums, never sums into products A2.14 §04 Algebra II.
$\frac{x^6}{x^2} = x^3$	$\frac{x^6}{x^2} = x^4$	Divide powers by *subtracting* exponents A1.3.
$-3^2 = 9$	$-3^2 = -9,\ (-3)^2 = 9$	The exponent binds before the minus sign A1.1.
$\frac{\sin x}{x} = \sin$	$\lim_{x\to 0} \frac{\sin x}{x} = 1$	A function name is not a factor; $\sin x$ is one object.
$x^2 = x \Rightarrow x = 1$	$x(x-1) = 0 \Rightarrow x = 0, 1$	Dividing by x silently assumes $x \neq 0$ and loses a root.
$\frac{x^2-x}{x} = x-1$ for all x	$x-1,\ x \neq 0$	The canceled factor leaves a hole; the domain does not change.

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A1.2 Fractions & Rational Arithmetic

A fraction $\frac{a}{b}$ is division of a by b. Adding needs a common denominator; multiplying and dividing do not.

Add / subtract	$\frac{a}{b} \pm \frac{c}{d} = \frac{ad \pm bc}{bd}$	Rewrite both over a common denominator, then combine numerators.
Multiply	$\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}$	Tops times tops, bottoms times bottoms.
Divide	$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \cdot \frac{d}{c}$	Multiply by the reciprocal ("keep, change, flip").
Simplify	$\frac{ka}{kb} = \frac{a}{b}\ (k \neq 0)$	Cancel common *factors*, never common terms: $\frac{2+3}{2} \neq 3$.
Mixed number	$3\frac{1}{2} = \frac{3 \cdot 2 + 1}{2} = \frac{7}{2}$	Convert to an improper fraction before computing.

STATED-EQUATION *a over b plus or minus c over d equals a d plus or minus b c, over b d. a over b times c over d equals a c over b d. Dividing by c over d is multiplying by d over c. Common nonzero factors cancel. k a over k b equals a over b for nonzero k. Three and one half equals three times two plus one, over two, which is seven halves.*

MEMORY AID Adding: "same bottom first". Dividing: "keep, change, flip".

METHOD

1. Factor every numerator and denominator.
2. For $\pm$: find the least common denominator (LCD), rewrite each fraction over it.
3. Combine, then cancel common factors. Example: $\frac{1}{2} + \frac{1}{3} = \frac{3}{6} + \frac{2}{6} = \frac{5}{6}$.

TERMS

Numerator a – top; *denominator* $b \neq 0$: bottom
LCD – least common multiple of the denominators
Reciprocal – $\frac{d}{c}$, the flipped fraction A1.1
Rational number – any $\frac{a}{b}$ with integers a, b, $b \neq 0$; the set $\mathbb{Q}$ A1.21
c, d – numerator and denominator of the second fraction, $d \neq 0$
k – a common nonzero factor of top and bottom
Mixed number – whole number plus a proper fraction, e.g. $3\frac{1}{2}$; an *improper fraction* has numerator $\geq$ denominator

CONTEXT

- **Origin.** Egyptian unit fractions (c. 1650 BCE); the bar notation is from al-Hassar (12th c.).
- **Use.** Probabilities, rates, proportions, simplifying rational expressions A2.10 §04 Algebra II.
- **Data science.** Probabilities multiply as fractions: $P(A)P(B) = \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{6}$ for independent events S.9 §13 Statistics.
- **Physical application.** A recipe needing $\frac{3}{4}$ cup, halved, needs $\frac{3}{4} \cdot \frac{1}{2} = \frac{3}{8}$ cup.

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A1.3 Exponent Rules

An exponent counts repeated multiplication. Every rule follows from counting how many copies of the base are being multiplied.

Product of powers	$a^m \cdot a^n = a^{m+n}$	Same base multiplied: add the exponents.
Quotient of powers	$\dfrac{a^m}{a^n} = a^{m-n}$	Same base divided: subtract the exponents.
Power of a power	$(a^m)^n = a^{mn}$	Repeating a^m n times multiplies the exponents.
Power of a product	$(ab)^n = a^n b^n$	An exponent distributes over multiplication (not over addition).
Power of a quotient	$\left(\dfrac{a}{b}\right)^n = \dfrac{a^n}{b^n}$	An exponent distributes over division.
Zero exponent	$a^0 = 1 \ (a \neq 0)$	Follows from $\frac{a^n}{a^n} = a^{n-n} = 1$.
Negative exponent	$a^{-n} = \dfrac{1}{a^n}$	A negative exponent means reciprocal, not a negative number: $2^{-3} = \frac{1}{8}$.

STATED-EQUATION *a to the m times a to the n equals a to the m plus n. a to the m over a to the n equals a to the m minus n. a to the m, all to the n, equals a to the m n. a b to the n equals a to the n b to the n. a over b to the n equals a to the n over b to the n. a to the zero equals one. a to the minus n equals one over a to the n.*

MEMORY AID Multiply → add exponents; power of a power → multiply exponents.

TERMS

a, b – bases; nonzero when the exponent is 0 or negative
m, n – exponents (integers here; fractions in A1.4)
a^n – a multiplied by itself n times
Pitfall – $(a+b)^2 \neq a^2 + b^2$; see A1.13

CONTEXT

- **Origin.** Modern superscript notation comes from Descartes, *La Géométrie* (1637).
- **Use.** Scientific notation A1.5, compound growth A2.13 §04 Algebra II, simplifying rational expressions.
- **Data science.** Products of many probabilities underflow; taking logs A2.14 §04 Algebra II turns $\prod p_i$ into $\sum \log p_i$, which is why models maximize the *log*-likelihood.
- **Physical application.** $(3 \times 10^8)(2 \times 10^3) = 6 \times 10^{11}$ multiplies by adding exponents.

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A1.4 Radicals & Rational Exponents

A root undoes a power. Writing roots as fractional exponents lets every exponent rule [A1.3] apply to them.

Rational exponent	$a^{m/n} = \sqrt[n]{a^m} = \left(\sqrt[n]{a}\right)^m$	Denominator is the root, numerator is the power: $8^{2/3} = (\sqrt[3]{8})^2 = 4$.
Product rule	$\sqrt{ab} = \sqrt{a}\,\sqrt{b}$	Split or merge roots over multiplication $(a, b \geq 0)$.
Quotient rule	$\sqrt{\dfrac{a}{b}} = \dfrac{\sqrt{a}}{\sqrt{b}}$	Split or merge roots over division.
Rationalize	$\dfrac{c}{\sqrt{a}} = \dfrac{c\sqrt{a}}{a}$	Multiply top and bottom by $\sqrt{a}$ to clear the root from the denominator.
Conjugate	$\dfrac{1}{\sqrt{a}+\sqrt{b}} = \dfrac{\sqrt{a}-\sqrt{b}}{a-b}$	Multiply by the conjugate to use $(x+y)(x-y) = x^2 - y^2$.
Even root of a square	$\sqrt{x^2} = \lvert x \rvert$	Even roots return the non-negative value [A1.11].

STATED-EQUATION *a to the m over n equals the n-th root of a to the m, which equals the n-th root of a, raised to the m. The square root of a b equals root a times root b. The square root of a over b equals root a over root b. c over root a equals c root a over a. One over root a plus root b equals root a minus root b, over a minus b. The square root of x squared equals the absolute value of x.*

MEMORY AID "Power over root": in $a^{m/n}$ the top is the power, the bottom is the root.

TERMS

n – index (root); $\sqrt{\ }$ means $n = 2$
a – radicand; $a \geq 0$ when n is even (for real results)
m – power, obeying the exponent rules [A1.3]
Conjugate – $\sqrt{a} - \sqrt{b}$ for $\sqrt{a} + \sqrt{b}$
Pitfall – $\sqrt{a+b} \neq \sqrt{a} + \sqrt{b}$: $\sqrt{9+16} = 5 \neq 7$
b – second radicand, $b > 0$ in a denominator
c – any real numerator
Rationalize – rewrite so no root remains in the denominator
$|x|$ – absolute value [A1.11]

CONTEXT

- **Origin.** The radical sign first appeared in Christoff Rudolff's *Coss* (1525).
- **Use.** Undoing powers; geometric lengths [G.7] §05 Geometry.
- **Data science.** Root-mean-square error $\sqrt{\frac{1}{n}\sum e_i^2}$ and the standard deviation [S.2] §13 Statistics both use a square root to return to the original units.
- **Physical application.** The side of a square garden with area $50\,\text{m}^2$ is $\sqrt{50} = 5\sqrt{2} \approx 7.07\,\text{m}$.

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A1.5 Scientific Notation & Significant Figures

Write any number as a coefficient between 1 and 10 times a power of ten; the number of digits kept in the coefficient records the precision.

Scientific notation	$x = c \times 10^k, \quad 1 \le \lvert c \rvert < 10$	Move the decimal point k places; $k > 0$ for large, $k < 0$ for small numbers.
Multiply / divide	$(c_1 \times 10^a)(c_2 \times 10^b) = c_1 c_2 \times 10^{a+b}$	Multiply the coefficients, add the exponents A1.3, then renormalize c.
Add / subtract	$c_1 \times 10^k + c_2 \times 10^k = (c_1 + c_2) \times 10^k$	Exponents must match first.
Significant figures ($\times, \div$)	keep the fewest sig. figs. of the inputs	$2.0 \times 3.14 = 6.3$ (2 sig. figs.).
Decimal places ($+, -$)	keep the fewest decimal places	$1.2 + 3.45 = 4.7$.

STATED-EQUATION *x equals c times ten to the k, where c is at least one and less than ten in size. When multiplying, multiply the coefficients and add the exponents. Results keep the precision of the least precise input. c one times ten to the a, times c two times ten to the b, equals c one c two times ten to the a plus b. With the same power of ten, c one times ten to the k plus c two times ten to the k equals c one plus c two, times ten to the k. Multiplying or dividing keeps the fewest significant figures; adding or subtracting keeps the fewest decimal places.*

MEMORY AID Big number → positive exponent; tiny number → negative exponent.

METHOD

1. Place the decimal point after the first nonzero digit to get c.
2. Count the places moved: that count is $|k|$; k is negative if the original number was less than 1. Example: $0.00052 = 5.2 \times 10^{-4}$.

TERMS

c – coefficient (mantissa), $1 \le |c| < 10$
k – integer exponent of 10 A1.3
Significant figure – a digit that carries measured precision; leading zeros do not count
Order of magnitude – the exponent k, i.e. $\approx \log_{10} x$ A2.14 §04 Algebra II
c_1, c_2; a, b – coefficients and exponents of the two numbers
Decimal places – digits to the right of the decimal point

CONTEXT

- **Origin.** Archimedes' *Sand Reckoner* counted with powers of a myriad; modern E-notation comes from computing (1950s).
- **Use.** Physics, chemistry, astronomy, engineering tolerances.
- **Data science.** Floats are binary scientific notation ($c \times 2^k$); `1e-8` learning rates and epsilon constants are written this way.
- **Physical application.** Avogadro's number 6.022×10^{23}; a red blood cell is about 8×10^{-6} m wide.

A1.6 Slope

Slope measures steepness: how much y changes for each one-unit change in x. A line has the same slope everywhere.

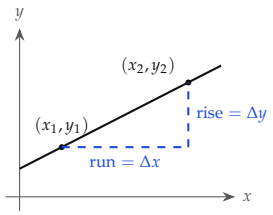
Slope (two points)	$m = \dfrac{y_2 - y_1}{x_2 - x_1} = \dfrac{\Delta y}{\Delta x}$	Subtract in the same order top and bottom; any two points on the line give the same m.
Sign of slope	$m > 0$ ↗, $m < 0$ ↘, $m = 0$ →	Positive rises left-to-right, negative falls, zero is horizontal; vertical lines have undefined slope.

STATED-EQUATION *m equals y two minus y one over x two minus x one, which is delta y over delta x: rise over run. m greater than zero rises, m less than zero falls, and m equal to zero is horizontal.*

MEMORY AID "Rise over run" — y changes on top, like rising up.

TERMS

$(x_1, y_1), (x_2, y_2)$ – two distinct points on the line
m – slope: change in y per unit change in x
Δ – "change in"; m undefined when $\Delta x = 0$
Rate of change – slope with units, e.g. km per hour



CONTEXT

- **Origin.** Coordinate geometry of Descartes and Fermat (17th c.).
- **Use.** Any constant rate: speed, cost per unit, growth per month.
- **Data science.** A regression coefficient S.40 §13 Statistics is a slope: the expected change in the target per one-unit change in a feature. The derivative C1.9 §07 Calculus I is a slope in the limit.
- **Physical application.** Points $(1, 2)$ and $(3, 8)$: $m = \frac{8-2}{3-1} = 3$. A road rising 6 m over 100 m has slope 0.06 (6% grade).

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A1.7 Equations of a Line

Three interchangeable ways to write the same straight line; pick the one that matches the information you have.

Slope–intercept	$y = mx + b$	Use when you know the slope and the y-intercept; easiest to graph.
Point–slope	$y - y_1 = m(x - x_1)$	Use when you know the slope and any one point.
Standard form	$Ax + By = C$	Integer coefficients; intercepts are $x = \frac{C}{A}$, $y = \frac{C}{B}$.

STATED-EQUATION *Slope-intercept form: y equals m x plus b. Point-slope form: y minus y one equals m times x minus x one. Standard form: A x plus B y equals C.*

DECISION POINT Which form? Given slope + intercept → slope–intercept. Given a point + slope, or two points (compute m first A1.6) → point–slope, then rearrange if needed. Need intercepts or a system A1.12 → standard form. All three describe the same line, so the choice is only about convenience.

TERMS

m – slope A1.6
b – y-intercept (value of y when $x = 0$)
(x_1, y_1) – any known point on the line
A, B, C – constants, usually integers with $A \geq 0$
Vertical line – $x = c$ (no slope–intercept form)

CONTEXT

- **Origin.** Descartes and Fermat's analytic geometry.
- **Use.** Modeling anything that changes at a constant rate.
- **Data science.** A linear model $\hat{y} = \beta_0 + \beta_1 x$ is slope–intercept form; a single neuron computes $w \cdot x + b$ before its activation ML.22 §15 Machine Learning.
- **Physical application.** A taxi fare of \$3 plus \$2 per mile is the line $y = 2x + 3$: intercept is the flag drop, slope is the per-mile rate.

A1.8 Parallel & Perpendicular Lines

Slopes tell you how two lines meet: equal slopes never meet; slopes that are negative reciprocals meet at a right angle.

Parallel	$m_1 = m_2 \ (b_1 \neq b_2)$	Same steepness, different intercepts: the lines never meet.
Perpendicular	$m_1 \cdot m_2 = -1 \iff m_2 = -\dfrac{1}{m_1}$	Flip the slope and change its sign: $m = 2 \Rightarrow m_\perp = -\frac{1}{2}$.
Distance point to line	$d = \dfrac{\lvert Ax_0 + By_0 - C\rvert}{\sqrt{A^2 + B^2}}$	Shortest (perpendicular) distance from (x_0, y_0) to $Ax + By = C$.

STATED-EQUATION *Parallel lines: m one equals m two. Perpendicular lines: m one times m two equals minus one, so m two equals the negative reciprocal of m one. Parallel lines have different intercepts, b one not equal to b two. The distance d from the point x zero, y zero to the line A x plus B y equals C is the absolute value of A x zero plus B y zero minus C, over the square root of A squared plus B squared.*

MEMORY AID Perpendicular slopes: "flip it and switch the sign."

TERMS

m_1, m_2 – slopes of the two lines A1.6
Negative reciprocal – flip the fraction and change the sign
Exception – a vertical line ($x = c$) is perpendicular to a horizontal one ($y = c$)
(x_0, y_0) – the outside point; A, B, C from standard form A1.7
b_1, b_2 – y-intercepts; if equal, the lines coincide
d – shortest (perpendicular) distance from the point to the line

CONTEXT

- **Origin.** Euclid's parallel postulate; the slope test follows from the Pythagorean theorem G.7 §05 Geometry.
- **Use.** Constructing right angles, shortest-distance problems.
- **Data science.** Perpendicular (orthogonal) directions carry non-overlapping information: PCA axes are orthogonal, and OLS residuals are orthogonal to fitted values S.40 §13 Statistics. See the dot product T.21 §06 Trigonometry.
- **Physical application.** Distance from $(1, 2)$ to $3x + 4y = 1$: $\frac{|3+8-1|}{5} = 2$.

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A1.9 Functions, Domain & Range

A function assigns exactly one output to each allowed input. Its domain is the set of allowed inputs; its range is the set of outputs it actually produces.

Function notation	$y = f(x)$	$f(3)$ means "substitute 3 for x": if $f(x) = 2x + 1$ then $f(3) = 7$.
Vertical line test	each $x \mapsto$ exactly one y	A graph is a function if no vertical line hits it twice.
Domain restrictions	$\frac{1}{g(x)}:\ g \neq 0; \quad \sqrt{g(x)}:\ g \geq 0; \quad \ln g(x):\ g > 0$	Exclude inputs that divide by zero, take even roots of negatives, or log non-positives.

STATED-EQUATION *y equals f of x: the function f maps each input x to one output y. Denominators cannot be zero, even roots need non-negative inputs, and logarithms need positive inputs. Vertical line test: a graph is a function when each x maps to exactly one y, so no vertical line meets it twice.*

MEMORY AID Domain = "what can go in"; range = "what comes out".

METHOD

1. Start from all real numbers.
2. Remove values that make a denominator 0, an even root's radicand negative, or a log's argument ≤ 0.
3. Write the result in interval notation A1.21. Example: $f(x) = \frac{1}{x-2}$ has domain $(-\infty, 2) \cup (2, \infty)$.

TERMS

f – the rule (name of the function)
x – input / independent variable
$f(x)$ – output / dependent variable
Domain – all allowed inputs
Range – all outputs produced
Composition, inverse – see A2.1 §04 Algebra II
$g(x)$ – the inner expression being restricted
ln – natural logarithm A2.14 §04 Algebra II
$\mapsto$ – "maps to"

CONTEXT

- **Origin.** Leibniz coined "function" (1673); Dirichlet gave the modern definition (1837).
- **Use.** Every model, formula and program maps inputs to outputs.
- **Data science.** A trained model is a function $\hat{y} = f(\mathbf{x})$; invalid inputs (e.g. $\log 0$ on a zero-valued feature) produce `-inf` or `NaN`.
- **Physical application.** Height of a ball as a function of time, $h(t)$, has domain $0 \leq t \leq t_{\text{land}}$.

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A1.10 Solving Linear & Literal Equations

Solve by doing the same operation to both sides until the variable stands alone. A literal equation is the same process applied to a formula with several variables.

Balance principle	$a = b \Rightarrow a + c = b + c, \;\; ac = bc$	Any operation applied to both sides keeps the equation true (never divide by 0).
Linear solution	$ax + b = c \Rightarrow x = \dfrac{c - b}{a}$	Undo addition first, then multiplication: reverse PEMDAS A1.1.
Literal equation	$A = \frac{1}{2}bh \Rightarrow h = \dfrac{2A}{b}$	Treat the other letters as constants and isolate the target letter.

STATED-EQUATION *If a equals b, then a plus c equals b plus c and a c equals b c. a x plus b equals c gives x equals c minus b, over a. A equals one half b h gives h equals two A over b.*

MEMORY AID "Undo in reverse order": last thing done to x is the first thing undone.

METHOD

1. Clear fractions (multiply by the LCD A1.2) and expand parentheses.
2. Collect variable terms on one side and constants on the other.
3. Divide by the coefficient of the variable.
4. Check by substituting back. Example: $3x + 4 = 10 \Rightarrow 3x = 6 \Rightarrow x = 2$; check $3(2) + 4 = 10$.

TERMS

a, b, c – constants, $a \neq 0$
x – unknown to solve for
Identity – equation true for all x (e.g. $2x = 2x$): infinitely many solutions
Contradiction – equation true for no x (e.g. $0 = 5$): no solution
Literal equation – a formula solved for one of its letters, e.g. triangle area A, base b, height h

CONTEXT

- **Origin.** Al-Khwarizmi's *al-jabr* ("restoring") and *al-muqabala* ("balancing"), c. 820, gave algebra its name.
- **Use.** Rearranging any formula for the quantity you need.
- **Data science.** Solving $\hat{y} = \beta_0 + \beta_1 x$ for the x that achieves a target $\hat{y}$ (inverse prediction).
- **Physical application.** Solve $F = \frac{9}{5}C + 32$ for C: $C = \frac{5}{9}(F - 32)$; $F = 50 \Rightarrow C = 10$.

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A1.11 Absolute Value Equations & Inequalities

$|x|$ is the distance from x to 0, so it is never negative. Absolute-value equations and inequalities split into two ordinary cases.

Definition	$\lvert x\rvert = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$	Leaves non-negatives alone and flips the sign of negatives.
Equation	$\lvert x-a\rvert = r \iff x = a \pm r \ (r \geq 0)$	Two points at distance r from a; no solution if $r < 0$.
"Less than"	$\lvert x-a\rvert < r \iff a-r < x < a+r$	Within r of a: one connected interval (*and*).
"Greater than"	$\lvert x-a\rvert > r \iff x < a-r$ or $x > a+r$	Farther than r from a: two rays (*or*).
Triangle inequality	$\lvert a+b\rvert \leq \lvert a\rvert + \lvert b\rvert$	The size of a sum never exceeds the sum of sizes.

STATED-EQUATION *The absolute value of x equals x when x is at least zero, and minus x when x is negative. The absolute value of x minus a less than r means a minus r less than x less than a plus r. Greater than r means x less than a minus r, or x greater than a plus r. The absolute value of x minus a equals r means x equals a plus or minus r, for r at least zero. The absolute value of a plus b is at most the absolute value of a plus the absolute value of b.*

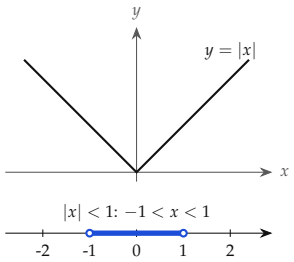
MEMORY AID "Less than" is a sandwich (and); "greater than" goes two ways (or).

METHOD

1. Isolate the absolute value: $|\cdots| = r$, $< r$ or $> r$.
2. If $r < 0$: an equation or "$<$" has no solution; "$>$" holds for all x.
3. Otherwise split into the two cases above and solve each linear piece A1.20. Example: $|x-3| \leq 2 \Rightarrow 1 \leq x \leq 5$.

TERMS

$|x|$ – distance from x to 0; always ≥ 0
a – center of the interval
r – radius or tolerance ($r > 0$)
Flip rule – "less than" $\rightarrow$ *and*; "greater than" $\rightarrow$ *or*
b – any real number (triangle inequality)



CONTEXT

- **Origin.** Notation $|x|$ introduced by Weierstrass (1841).
- **Use.** Tolerances, error bounds, distances on a number line.
- **Data science.** Mean absolute error and L1 (Lasso) regularization use $|\cdot|$ ML.8 §15 Machine Learning; its corner at 0 drives weights exactly to zero.
- **Physical application.** A bolt specified as 10 mm $\pm$ 0.05 mm must satisfy $|x-10| \leq 0.05$.

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A1.12 Systems of Linear Equations

Find the values that satisfy several linear equations at once; geometrically, the point(s) where the lines cross.

2 × 2 system	$\begin{cases} a_1x + b_1y = c_1 \\ a_2x + b_2y = c_2 \end{cases}$	Two lines: one, none, or infinitely many common points.
Cramer's rule (2 × 2)	$x = \dfrac{c_1b_2 - b_1c_2}{a_1b_2 - b_1a_2},\quad y = \dfrac{a_1c_2 - c_1a_2}{a_1b_2 - b_1a_2}$	Direct formula when the determinant $a_1b_2 - b_1a_2 \neq 0$ A2.25 §04 Algebra II.
Solution count test	$a_1b_2 - b_1a_2 \neq 0 \iff$ unique solution	Zero determinant: parallel (none) or identical (infinite) lines.

STATED-EQUATION *A system of two linear equations: a one x plus b one y equals c one, and a two x plus b two y equals c two. The solution is unique exactly when a one b two minus b one a two is not zero. Cramer's rule: x equals c one b two minus b one c two, over a one b two minus b one a two; y equals a one c two minus c one a two, over the same denominator.*

METHOD

1. **Substitution**: solve one equation for a variable, substitute into the other.
2. **Elimination**: scale equations so one variable has opposite coefficients, then add.
3. Back-substitute to get the second variable; check in both equations. Example: $x + y = 5$, $x - y = 1$; adding gives $2x = 6$, $x = 3$, $y = 2$.

DECISION POINT Substitution is quickest when a variable already has coefficient 1 or -1; elimination is cleaner when coefficients are larger; matrices LA.17 §09 Linear Algebra scale to 3+ variables. All give the same answer.

TERMS

a_i, b_i – coefficients; c_i constants
One solution – lines intersect (different slopes A1.6)
No solution – parallel lines A1.8 (inconsistent)
Infinite – same line (equations are multiples; dependent)
Determinant – $a_1b_2 - b_1a_2$ A2.25 §04 Algebra II

CONTEXT

- **Origin.** Elimination appears in the Chinese *Nine Chapters* (c. 200 BCE); formalized by Gauss.
- **Use.** Mixture, pricing, and break-even problems.
- **Data science.** Least-squares regression solves the normal equations $X^TX\beta = X^Ty$, a (large) linear system.
- **Physical application.** Mixing a 10% and a 30% saline solution to get 2 L of 25% solution is a two-equation system in the two volumes.

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A1.13 Polynomial Products & Special Products

Multiplying polynomials is repeated use of the distributive law; a few products appear so often they are worth memorizing.

Square of a sum	$(a+b)^2 = a^2 + 2ab + b^2$	Do not forget the middle term $2ab$.
Square of a difference	$(a-b)^2 = a^2 - 2ab + b^2$	Same as above with the middle sign negative.
Difference of squares	$(a+b)(a-b) = a^2 - b^2$	The middle terms cancel.
FOIL	$(a+b)(c+d) = ac + ad + bc + bd$	Every term in the first factor times every term in the second.
Cube of a sum	$(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$	Coefficients 1, 3, 3, 1 come from Pascal's triangle A2.23 §04 Algebra II.

STATED-EQUATION *The quantity a plus b, squared, equals a squared plus two a b plus b squared. a minus b, squared, equals a squared minus two a b plus b squared. a plus b times a minus b equals a squared minus b squared. a plus b times c plus d equals a c plus a d plus b c plus b d. a plus b, cubed, equals a cubed plus three a squared b plus three a b squared plus b cubed.*

MEMORY AID FOIL: First, Outer, Inner, Last.

TERMS

Polynomial – sum of terms $c\,x^k$ with whole-number k
Degree – largest exponent present
Like terms – same variable and power; combine them
Distributive law – the engine of every expansion A1.1
a, b, c, d – any real numbers or expressions
FOIL – First, Outer, Inner, Last: the four products of two binomials

CONTEXT

- **Origin.** Al-Khwarizmi's geometric "completing the square" (c. 820).
- **Use.** Expanding, simplifying, and setting up factoring A1.14.
- **Data science.** $\|x-y\|^2 = \|x\|^2 - 2\,x\cdot y + \|y\|^2$ lets libraries compute all pairwise distances with one matrix product (k-NN, k-means).
- **Physical application.** A square patio of side $x+1$ m has area $x^2 + 2x + 1$: the big square, two strips, and a corner square. Check with $x = 3$: $4^2 = 16 = 9 + 6 + 1$.

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A1.14 Factoring

Factoring rewrites a sum as a product, the reverse of expanding. A product equal to zero lets you solve by setting each factor to zero.

Greatest common factor	$ab + ac = a(b + c)$	Always pull out the GCF first.
Trinomial ($a = 1$)	$x^2 + (p + q)x + pq = (x + p)(x + q)$	Find p, q that multiply to the constant and add to the middle coefficient.
Difference of squares	$a^2 - b^2 = (a - b)(a + b)$	Two perfect squares separated by a minus.
Sum of cubes	$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$	Signs: Same, Opposite, Always Positive.
Difference of cubes	$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$	Same SOAP sign pattern.
AC method	$ax^2 + bx + c$: find $pq = ac,\ p + q = b$	Split bx into $px + qx$, then factor by grouping.

STATED-EQUATION *a b plus a c equals a times b plus c. x squared plus p plus q x plus p q equals x plus p times x plus q. a squared minus b squared equals a minus b times a plus b. a cubed plus b cubed equals a plus b times a squared minus a b plus b squared. a cubed minus b cubed equals a minus b times a squared plus a b plus b squared. AC method: for a x squared plus b x plus c, find p and q with p q equal to a c and p plus q equal to b.*

MEMORY AID Sum/difference of cubes signs: SOAP — Same, Opposite, Always Positive.

METHOD

1. Take out the GCF.
2. Count terms: 2 terms → difference of squares or sum/difference of cubes; 3 terms → trinomial (or AC method); 4 terms → grouping.
3. Factor each factor again until nothing more factors; check by expanding. Example: $x^2 + 5x + 6 = (x + 2)(x + 3)$ since $2 \cdot 3 = 6$, $2 + 3 = 5$.

DECISION POINT If no integer pair p, q works, the quadratic may still have real roots that are not whole numbers: switch to the quadratic formula A1.15 rather than searching longer. A sum of squares $a^2 + b^2$ does not factor over the reals.

TERMS

GCF – greatest common factor, pulled out first
p, q – numbers whose sum is the middle coefficient and product the constant
Zero product – if $AB = 0$ then $A = 0$ or $B = 0$ A1.1
Grouping – pair terms, factor each pair, then factor the common binomial
Reverse of – special products A1.13
a, b, c – coefficients of $ax^2 + bx + c$ (in the identities, any expressions)
AC method – split bx into $px + qx$ with $pq = ac$, then factor by grouping

CONTEXT

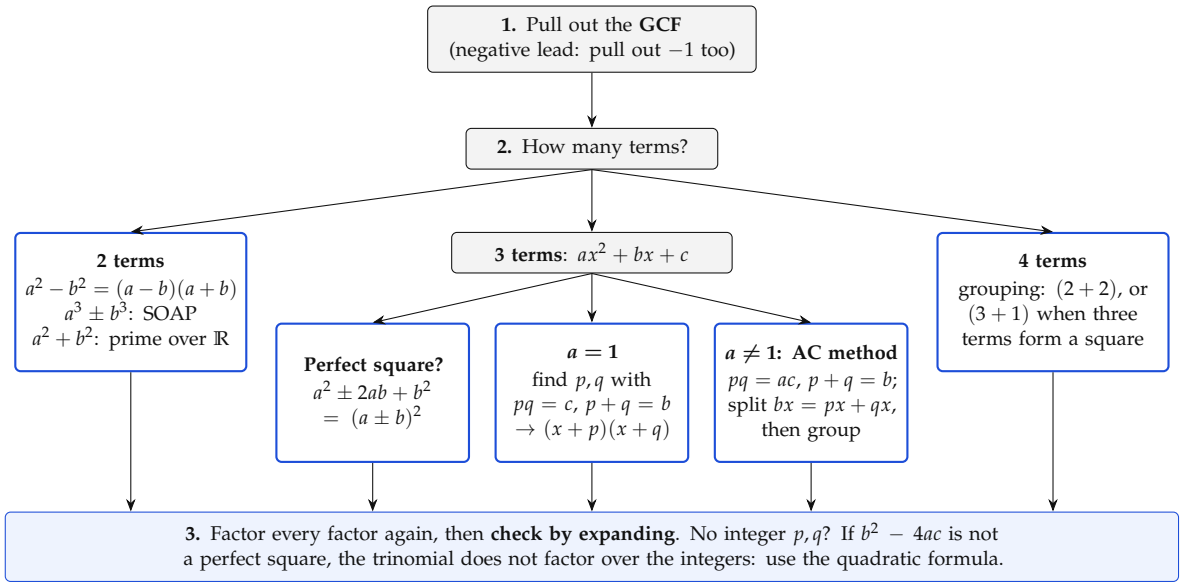
- **Origin.** The inverse of expansion; systematized in Renaissance algebra.
- **Use.** Solving polynomial equations and simplifying rational expressions A2.10 §04 Algebra II.
- **Data science.** Factoring out common terms cuts computation: Horner's rule evaluates degree-n polynomials with only n multiplications.
- **Physical application.** Factoring $h(t) = -4.9t^2 + 19.6t = -4.9t(t - 4)$ shows a ball thrown up at 19.6 m/s lands after 4 s.

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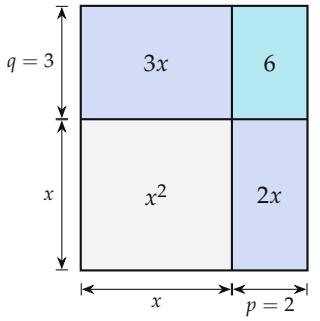
Factoring | *Key Topic A1.K2*

Factoring rewrites a sum as a product, the reverse of expanding A1.13. A product is easy to read: it is zero exactly when one factor is zero, so the factors hand you the roots, and common factors cancel A1.K1. Work through the same short checklist every time; the only real search is for the pair p, q.

Decision tree



Finding p and q: sum and product



corner $= pq = 6$
the constant c

two strips $= 2x + 3x$
$p + q = 5$, the middle b

diagonals match:
$x^2 \cdot 6 = 2x \cdot 3x = 6x^2$
(the AC check)

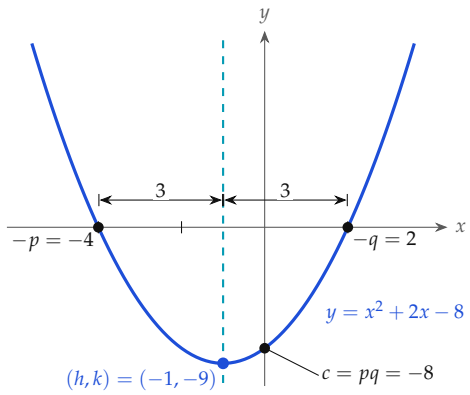
Expanding $(x+p)(x+q) = x^2 + (p+q)x + pq$ shows the target: two numbers whose **product is** c and whose **sum is** b. List the factor pairs of c and pick the pair with the right sum.

Signs first. $c > 0$: same signs, both the sign of b. $c < 0$: opposite signs; the larger size takes the sign of b.

Ex. $x^2 - x - 12$: pairs of -12 summing to $-1 \to 3, -4$, so $(x+3)(x-4)$.

When $a \neq 1$ the target is product ac, sum b: $6x^2 + 7x + 2$ needs $pq = 12$, $p + q = 7 \to 3, 4$; split and group: $6x^2 + 3x + 4x + 2 = 3x(2x+1) + 2(2x+1) = (3x+2)(2x+1)$.

p and q on the graph and in vertex form



In $y = (x+p)(x+q)$ the **roots are** $-p$ **and** $-q$. The axis of symmetry sits halfway between them, so the vertex form $y = (x-h)^2 + k$ A1.16 is tied to p, q by

$$h = -\frac{p+q}{2}, \qquad k = -\left(\frac{p-q}{2}\right)^2,$$

equivalently

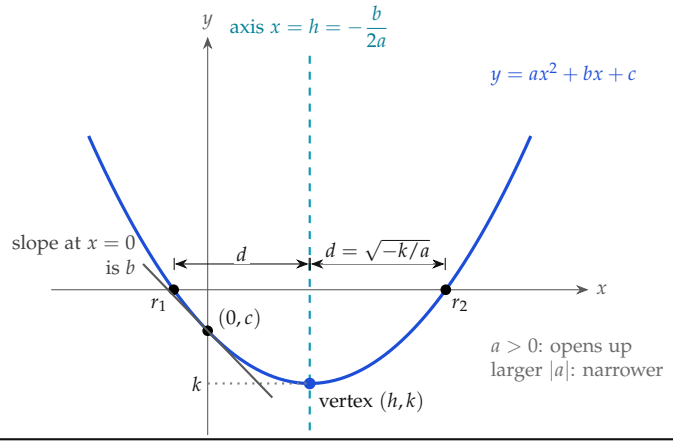
$$p + q = b = -2h, \qquad pq = c = h^2 + k.$$

Each root is $\sqrt{-k}$ from h: here $(x+1)^2 - 9$ gives roots -1 ± 3, i.e. 2 and -4, so $p, q = -2, 4$. If $-k$ is not a perfect square, no integer pair exists.

For any a the same facts read $r_1 + r_2 = -\frac{b}{a}$, $r_1 r_2 = \frac{c}{a}$ (Vieta's formulas).

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Every letter of a quadratic



Form	Letters it shows
Standard	$ax^2 + bx + c$: a, b, c
Vertex	$a(x-h)^2 + k$: a, h, k
Factored	$a(x-r_1)(x-r_2)$: a, r_1, r_2

$$h = -\frac{b}{2a}, \quad k = c - \frac{b^2}{4a}$$

$$r_{1,2} = h \pm \sqrt{-k/a} = \frac{-b \pm \sqrt{\Delta}}{2a}$$

$$b = -2ah = -a(r_1 + r_2), \quad c = ah^2 + k = ar_1r_2$$

$$\Delta = b^2 - 4ac = -4ak$$ A1.15

Letter	Meaning and what it controls	Letter	Meaning and what it controls
a	leading coefficient: up if $a > 0$, down if $a < 0$; larger $\lvert a \rvert$ is narrower	h	x of the vertex; the axis of symmetry is $x = h$
b	linear coefficient: the slope where the graph crosses the y-axis; with a it sets h	k	y of the vertex: the minimum ($a > 0$) or maximum ($a < 0$) value
c	constant: the y-intercept $(0, c)$	r_1, r_2	roots (zeros, x-intercepts), where $y = 0$
p, q	numbers in $(x+p)(x+q)$ when $a = 1$: $p = -r_1$, $q = -r_2$	Δ	discriminant: > 0 two roots, $= 0$ one, < 0 none real
d	half the distance between the roots: $d = \frac{\sqrt{\Delta}}{2\lvert a \rvert}$	x	the input; y the output A1.9

DECISION POINT **Factor, complete the square, or use the formula?** Try factoring when the numbers are small integers and $\Delta = b^2 - 4ac$ is a perfect square: it is fastest and gives the roots directly. If you need the vertex, complete the square A1.K3. If Δ is not a perfect square, or the search for p, q takes more than a few tries, use the quadratic formula A1.15.

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A1.15 Quadratic Formula & Discriminant

Every quadratic equation can be solved with one formula; the discriminant tells you in advance how many real solutions there are.

Quadratic formula	$ax^2 + bx + c = 0 \Rightarrow x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$	Plug in a, b, c; the $\pm$ gives the two roots.
Discriminant	$\Delta = b^2 - 4ac$	$\Delta > 0$: 2 real roots; $\Delta = 0$: 1 repeated root; $\Delta < 0$: 2 complex roots A2.6 §04 Algebra II.
Vieta's formulas	$x_1 + x_2 = -\dfrac{b}{a}, \quad x_1 x_2 = \dfrac{c}{a}$	Sum and product of the roots, without solving.

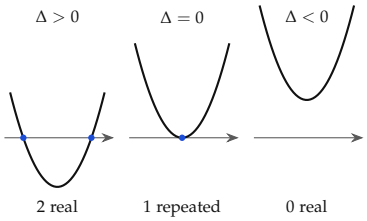
STATED-EQUATION *a x squared plus b x plus c equals zero implies x equals negative b plus or minus the square root of b squared minus four a c, all over two a. The discriminant, delta, equals b squared minus four a c. Vieta: x one plus x two equals negative b over a, and x one times x two equals c over a.*

MEMORY AID Sung to "Pop Goes the Weasel": "x equals negative b, plus or minus the square root, of b squared minus four a c, all over two a."

DECISION POINT Which method for a quadratic? Easy integer factoring A1.14 → factor. No x term ($ax^2 = c$) → take square roots. Need the vertex too → complete the square A1.16. Otherwise → quadratic formula, which always works.

TERMS

a, b, c – coefficients, $a \neq 0$
Δ – discriminant (the radicand)
Roots – solutions; x-intercepts of the parabola
Derived by – completing the square A1.16
x_1, x_2 – the two roots



CONTEXT

- **Origin.** Babylonian tablets (c. 2000 BCE); general rule by Brahmagupta (628 CE).
- **Use.** Projectile motion, area, break-even points.
- **Data science.** Quadratic discriminant analysis draws curved class boundaries; finding where two Gaussian class densities cross means solving a quadratic.
- **Physical application.** $x^2 - 5x + 6 = 0$: $\Delta = 25 - 24 = 1$, $x = \frac{5 \pm 1}{2} = 3$ or 2.

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A1.16 Vertex Form & Completing the Square

Rewriting a quadratic around its turning point (the vertex) shows its maximum or minimum directly.

Vertex form	$y = a(x-h)^2 + k$	Vertex at (h,k); opens up if $a>0$, down if $a<0$.
Vertex from standard form	$h = -\dfrac{b}{2a}, \quad k = f(h)$	Find h by formula, then evaluate the function there.
Completing the square	$x^2 + bx = \left(x + \frac{b}{2}\right)^2 - \frac{b^2}{4}$	Add and subtract $(\frac{b}{2})^2$ to create a perfect square.

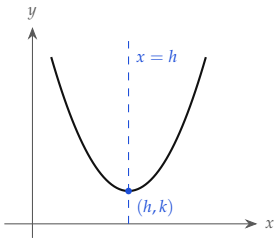
STATED-EQUATION *y equals a times the quantity x minus h squared, plus k. The vertex x-coordinate h equals negative b over two a, and k equals f of h. Completing the square: x squared plus b x equals the quantity x plus b over two, squared, minus b squared over four.*

METHOD

1. Factor a out of the x^2 and x terms.
2. Inside, add and subtract $(\frac{b}{2})^2$ (using the inner b).
3. Write the perfect square and collect the constants. Example: $x^2+6x+5 = (x+3)^2 - 9 + 5 = (x+3)^2 - 4$, vertex $(-3,-4)$.

TERMS

(h,k) – vertex: minimum if $a>0$, maximum if $a<0$
$x=h$ – axis of symmetry
a – stretch and direction, same a as in A1.15
$f(h)$ – the function evaluated at h A1.9
b – linear coefficient A1.15



CONTEXT

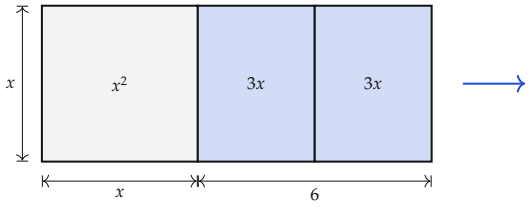
- **Origin.** Completing the square, Al-Khwarizmi (c. 820).
- **Use.** Finding a maximum height, minimum cost, or best price without calculus.
- **Data science.** Mean squared error in one parameter is a parabola; its vertex is the sample mean. Extrema in general: C1.23 §07 Calculus I.
- **Physical application.** The peak height of a thrown ball is the vertex of its parabolic path.

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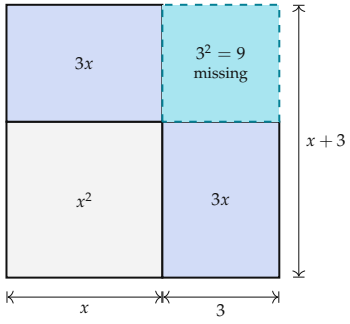
Completing the Square | *Key Topic A1.K3*

Completing the square rewrites x^2+bx as a perfect square minus a correction: $x^2+bx=\left(x+\frac{b}{2}\right)^2-\left(\frac{b}{2}\right)^2$. Geometrically it literally completes a square of area; on a graph it reveals the vertex, because $(x-h)^2+k$ is just $y=x^2$ moved. It is the move behind vertex form A1.16, the quadratic formula A1.15, circle equations G.19 §05 Geometry and many integrals.

The picture: it really is a square



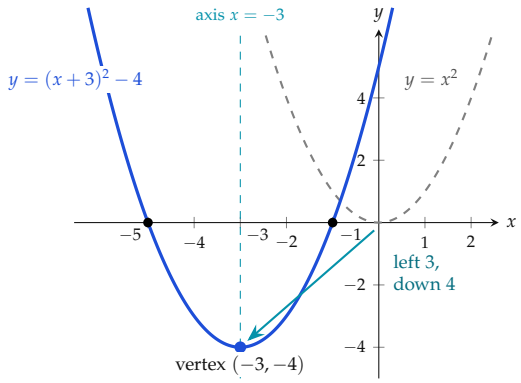
1. x^2+6x: split the $6x$ strip in half



2. add the corner: $x^2+6x+9=(x+3)^2$
so $x^2+6x=(x+3)^2-9$

Split the bx strip into two halves of width $\frac{b}{2}$ and lay them along two sides of the x-by-x square. The shape is a big square of side $x+\frac{b}{2}$ with one corner of area $\left(\frac{b}{2}\right)^2$ missing. Adding the corner completes the square; subtracting it again keeps the expression unchanged.

The graph: it reveals the vertex



$y=x^2+6x+5$ hides its shape. Completing the square,

$$y=(x^2+6x+9)-9+5=(x+3)^2-4,$$

shows it is the parent $y=x^2$ moved **left 3** and **down 4** A2.K1 §04 Algebra II: vertex $(-3,-4)$, axis $x=-3$, minimum value -4.

Setting $y=0$ solves it in one line: $(x+3)^2=4\Rightarrow x+3=\pm2\Rightarrow x=-1$ or $x=-5$, the two roots on the graph.

Since $(x+3)^2\geq 0$, the smallest y can be is -4, reached at $x=-3$: no calculus needed for the minimum C1.23 §07 Calculus I.

Method

When $a=1$: x^2+bx+c

1. Halve b and square it: $\left(\frac{b}{2}\right)^2$.
2. Add and subtract it: $x^2+bx+\left(\frac{b}{2}\right)^2-\left(\frac{b}{2}\right)^2+c$.
3. Fold the first three terms: $\left(x+\frac{b}{2}\right)^2+c-\frac{b^2}{4}$.

$x^2-8x+10=(x-4)^2-16+10=(x-4)^2-6$.

When $a\neq 1$: ax^2+bx+c

1. Factor a from the x-terms only: $a\left(x^2+\frac{b}{a}x\right)+c$.
2. Complete the square inside the bracket.
3. Multiply the correction by a when moving it out.

$2x^2+8x+3=2(x^2+4x)+3=2\left[(x+2)^2-4\right]+3=2(x+2)^2-5$.

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Where the quadratic formula comes from

$$\begin{aligned} ax^2+bx+c=0 &\Rightarrow x^2+\tfrac{b}{a}x=-\tfrac{c}{a} \\ &\Rightarrow \left(x+\tfrac{b}{2a}\right)^2=\tfrac{b^2}{4a^2}-\tfrac{c}{a}=\tfrac{b^2-4ac}{4a^2} \\ &\Rightarrow x+\tfrac{b}{2a}=\pm\tfrac{\sqrt{b^2-4ac}}{2a} \Rightarrow x=\tfrac{-b\pm\sqrt{b^2-4ac}}{2a}. \end{aligned}$$

The discriminant b^2-4ac A1.15 is the right-hand side of the completed square: negative means a square would have to equal a negative number, so there are no real roots.

Uses and common mistakes

Where it shows up

Vertex and max/min of any quadratic A1.16.

Center and radius of a circle: $x^2-4x+y^2+6y=3 \Rightarrow (x-2)^2+(y+3)^2=16$ G.19 §05 Geometry.

Integrals: $\int \frac{dx}{x^2+2x+5} = \int \frac{dx}{(x+1)^2+4}$ becomes an arctangent C2.3 §08 Calculus II.

Statistics: the Gaussian exponent is a completed square in $x-\mu$ S.20 §13 Statistics.

Common mistakes

Forgetting to subtract the added $\left(\frac{b}{2}\right)^2$: $x^2+6x \neq (x+3)^2$.

Not factoring a **first**: $2x^2+8x \neq (\sqrt{2}x+2)^2$ in general; use $2(x+2)^2-8$.

Sign of h: $(x+3)^2$ has vertex at $x=-3$, not $+3$.

Correction outside the bracket must be multiplied by a.

DECISION POINT **Complete the square or use the formula?** If you need the *vertex*, the *minimum or maximum*, a circle's center, or an integral in standard form, complete the square: the rewritten form is the answer. If you only need the roots and the numbers are messy, the quadratic formula is faster and less error-prone; it is the completed square done once in general.

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A1.17 Quadratic & Polynomial Inequalities

A polynomial can only change sign at its roots, so the roots cut the number line into intervals that are entirely positive or entirely negative.

Sign rule	$p(x) = (x - r_1)(x - r_2)\cdots$	Between consecutive roots the sign is constant; test one point per interval.
Upward parabola	$a > 0: \quad ax^2 + bx + c < 0 \iff r_1 < x < r_2$	Negative between the roots, positive outside.

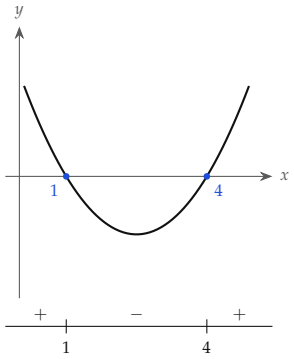
STATED-EQUATION *A polynomial changes sign only at its roots. For an upward parabola, the expression is negative strictly between its two roots.*

METHOD

1. Move everything to one side: $p(x) > 0$ (or $<, \leq, \geq$).
2. Find the roots of $p(x) = 0$ A1.15.
3. Mark them on a number line; test one value in each interval (sign chart).
4. Keep the intervals with the required sign; include roots for $\leq, \geq$. Example: $x^2 - 5x + 4 < 0 \Rightarrow (x-1)(x-4) < 0 \Rightarrow 1 < x < 4$.

TERMS

$p(x)$ – polynomial A1.13
r_i – roots of p
Sign chart – number line marked with $+$ / $-$ per interval
Interval notation – see A1.21



CONTEXT

- **Origin.** Sign analysis is implicit in Descartes' rule of signs (1637).
- **Use.** Feasible ranges, profit intervals, safe operating ranges.
- **Data science.** Finding where a quadratic loss is below a threshold; where one class density exceeds another.
- **Physical application.** A ball at $h(t) = -5t^2 + 20t$ is above 15 m when $-5t^2 + 20t > 15$, i.e. $1 < t < 3$ s.

A1.18 Direct & Inverse Variation

Variation describes proportional relationships: as one quantity grows the other grows in proportion (direct) or shrinks in proportion (inverse).

Direct	$y = kx$	Doubling x doubles y; the graph is a line through the origin.
Inverse	$y = \dfrac{k}{x}$	Doubling x halves y; the product $xy = k$ is constant.
Joint	$y = kxz$	y varies directly with each of several variables.

STATED-EQUATION *Direct variation: y equals k x. Inverse variation: y equals k over x. Joint variation: y equals k x z.*

METHOD

1. Write the variation equation with unknown k.
2. Substitute one known pair to find k.
3. Use k to answer the question. Example: y varies inversely with x, $y = 4$ at $x = 2 \Rightarrow k = 8$, so $y = 1$ at $x = 8$.

TERMS

k – constant of variation, $k \neq 0$
Direct – a line through the origin A1.7
Inverse – a hyperbola (reciprocal graph F6 §18 Graph Gallery)
z – a second input variable

CONTEXT

- **Origin.** Proportion theory in Euclid's *Elements*, Book V.
- **Use.** Ohm's law, unit pricing, speed and travel time.
- **Data science.** Ideal parallel speed-up: runtime $\approx k/n$ for n workers. The standard error of a mean varies inversely with $\sqrt{n}$ S.24 §13 Statistics.
- **Physical application.** Boyle's law $P = k/V$ (inverse); Hooke's law $F = kx$ (direct).

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A1.19 Ratios, Proportions & Percent Change

A ratio compares two quantities; a proportion states that two ratios are equal; percent change measures relative growth or decline.

Percent change	$\dfrac{\text{new} - \text{old}}{\text{old}} \times 100\%$	Always divide by the starting value.
Proportion	$\dfrac{a}{b} = \dfrac{c}{d} \iff ad = bc$	Cross-multiply to solve for a missing value.
Percent of	$p\%$ of $x = \dfrac{p}{100}\,x$	"Of" means multiply: 20% of 5 is 1.

STATED-EQUATION *Percent change equals new minus old, over old, times one hundred percent. a over b equals c over d if and only if a d equals b c. p percent of x equals p over one hundred times x.*

MEMORY AID Percent change: "difference over original."

TERMS

new, old – final and reference values (old $\neq 0$)
$a{:}b$ – ratio; $\frac{a}{b} = \frac{c}{d}$ is a proportion
Cross-multiplication – $ad = bc$
Asymmetry – $+p\%$ then $-p\%$ does not return to the start: $10 \to 11 \to 9.9$
$p,\ x$ – percentage and the base quantity

CONTEXT

- **Origin.** Latin *per centum*; Roman taxes were levied in hundredths.
- **Use.** Discounts, interest, scaling recipes and maps.
- **Data science.** "Lift" in an A/B test is a percent change in conversion rate; relative error and month-over-month growth metrics use the same formula.
- **Physical application.** Scaling a recipe from 4 servings to 6: $\frac{2\text{ cups}}{4} = \frac{x}{6}$, so $x = 3$ cups.

A1.20 Linear Inequalities

Inequalities are solved like equations, except that multiplying or dividing by a negative number reverses the direction of the sign.

Add / subtract	$a < b \implies a + c < b + c$	Adding the same number keeps the direction.
Multiply by positive	$a < b,\ c > 0 \implies ac < bc$	Direction unchanged.
Multiply by negative	$a < b,\ c < 0 \implies ac > bc$	Direction flips: $-2x < 6 \Rightarrow x > -3$.

STATED-EQUATION *If a is less than b, then a plus c is less than b plus c. If a is less than b and c is positive, a c is less than b c. If c is negative, the inequality flips: a c is greater than b c.*

MEMORY AID "Multiply or divide by a negative, flip the sign."

TERMS

a, b, c – real numbers
Flip rule – multiplying or dividing by a negative reverses the sign
Compound – $p < x < q$ means both $x > p$ and $x < q$
Open / closed dot – $<, >$ exclude the endpoint; $\leq, \geq$ include it
Related – absolute value inequalities A1.11

CONTEXT

- **Origin.** Thomas Harriot introduced the symbols $<$ and $>$ (1631).
- **Use.** Budgets, capacity limits, feasible ranges.
- **Data science.** Linear programming 0.3 §14 Optimization and SVMs are defined by linear inequality constraints, e.g. the margin condition $y_i(w \cdot x_i + b) \geq 1$.
- **Physical application.** An elevator rated for 1000 kg with a 75 kg operator carries passengers of 80 kg each: $75 + 80n \leq 1000$, so $n \leq 11$.

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A1.21 Sets, Set Operations & Interval Notation

A set is a collection of distinct elements. Set operations combine collections; interval notation describes sets of real numbers compactly.

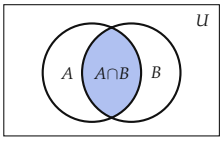
Union / intersection	$A \cup B$ (or), $A \cap B$ (and)	$\{a,b\} \cup \{b,c\} = \{a,b,c\}$; $\{a,b\} \cap \{b,c\} = \{b\}$.
Complement / difference	A^c (not), $A \setminus B$	Elements outside A; elements of A not in B.
Inclusion–exclusion	$\lvert A \cup B \rvert = \lvert A \rvert + \lvert B \rvert - \lvert A \cap B \rvert$	Subtract the overlap so it is not counted twice.
De Morgan's laws	$(A \cup B)^c = A^c \cap B^c$, $(A \cap B)^c = A^c \cup B^c$	Complementing swaps $\cup$ and $\cap$.
Interval notation	$[a,b) = \{x : a \le x < b\}$	Bracket = endpoint included; parenthesis = excluded; ∞ always gets a parenthesis.

STATED-EQUATION *A union B means elements in A or B. A intersect B means elements in both. A complement means elements not in A. A minus B means elements in A but not in B. The size of A union B equals the size of A plus the size of B minus the size of A intersect B. De Morgan's laws: the complement of a union is the intersection of the complements, and vice versa. The interval bracket a comma b parenthesis is the set of x with a less than or equal to x, less than b.*

MEMORY AID Square bracket = "included" (closed); parenthesis = "not included" (open). De Morgan: "break the bar, change the sign."

TERMS

$\in$, $\subseteq$ – "is an element of", "is a subset of"
$\varnothing$ – empty set
$|A|$ – cardinality (number of elements)
U – universal set (everything under discussion)
Number sets – $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$ (see Math Signs M.78 §01 Math Signs)
Inclusion–exclusion – the same idea as $P(A \cup B)$ S.7 §13 Statistics



$\{a,b\} \cap \{b,c\} = \{b\}$

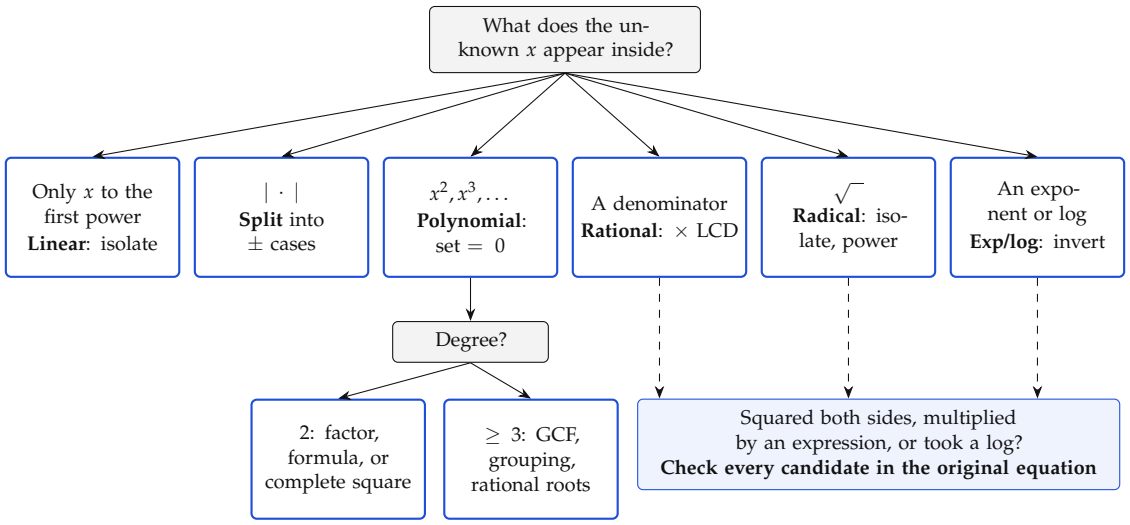
CONTEXT

- **Origin.** Georg Cantor (1874); De Morgan (1847); Venn diagrams by John Venn (1880).
- **Use.** The language of all mathematics: domains, solution sets, events.
- **Data science.** SQL `UNION`, `INTERSECT`, `EXCEPT` and joins are set operations; Jaccard similarity $|A \cap B|/|A \cup B|$ compares sets of items.
- **Physical application.** $A = \{1,2,3\}$, $B = \{3,4\}$: $|A \cup B| = 3 + 2 - 1 = 4$.

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Solving Equations: A Decision Map | *Key Topic A1.K4*

Most algebra problems reduce to "get the unknown alone". The type of equation decides the tool. Identify the type first, solve, then always check: some methods (squaring both sides, multiplying by an expression) can create *extraneous* solutions that do not satisfy the original equation.



The tools, in one line each

Linear A1.10: collect x-terms, divide by the coefficient. $2x + 3 = 11 \Rightarrow x = 4$.

Absolute value A1.11: isolate $|\cdot|$, split into $+$ and $-$ cases. $|x - 2| = 3 \Rightarrow x = 5$ or $x = -1$.

Quadratic A1.15: factor if easy, otherwise use the formula. $x^2 - 4 = 0 \Rightarrow x = \pm 2$.

Polynomial, degree ≥ 3 A2.8 §04 Algebra II: GCF, grouping, or rational-root test, then divide down to a quadratic.

Rational A2.10 §04 Algebra II: multiply by the LCD, solve, discard values that make a denominator 0. $\frac{1}{x} = \frac{2}{x+1} \Rightarrow x + 1 = 2x \Rightarrow x = 1$.

Radical: isolate the root, raise both sides to the root's index, *check*. $\sqrt{x+2} = x \Rightarrow x + 2 = x^2 \Rightarrow x = 2$ (and $x = -1$ is extraneous).

Exponential A2.16 §04 Algebra II: take logs of both sides. $2^x = 8 \Rightarrow x = 3$.

Logarithmic A2.14 §04 Algebra II: rewrite as an exponential. $\log_2 x = 3 \Rightarrow x = 8$.

Systems A1.12: substitution or elimination; matrices for 3+ variables LA.17 §09 Linear Algebra.

Inequalities A1.20, A1.17: solve like equations; flip on negatives; use a sign chart for nonlinear ones.

Decision points

DECISION POINT **When several methods apply**, prefer the one with the fewest steps you can do without error; they all give the same solutions. **When a step is irreversible** (squaring, multiplying by something that might be 0, taking even roots), plan to check every candidate in the original equation. **When nothing works algebraically**, a numerical method such as Newton's method C1.35 §07 Calculus I or graphing will approximate the root.

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04
Algebra II

Functions as objects: composing, inverting, and transforming them; polynomial, rational, radical, exponential and logarithmic models; sequences, series and proof by induction; and the matrix language that underlies all of modern data work.

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Functions and Their Graphs

A2.1 Functions: Composition & Inverses

Functions can be combined like numbers, chained one after another (composition), and, when they are one-to-one, undone (inverse). The graph of an inverse is the mirror image of the original in the line $y = x$.

Function arithmetic	$(f \pm g)(x) = f(x) \pm g(x), \quad (fg)(x) = f(x)\,g(x)$	Combine outputs point by point; the domain is the overlap of the two domains (and $g \neq 0$ for $\frac{f}{g}$).
Composition	$(f \circ g)(x) = f\big(g(x)\big)$	Apply g first, then f. Usually $f \circ g \neq g \circ f$.
Inverse identity	$f^{-1}\big(f(x)\big) = f\big(f^{-1}(x)\big) = x$	The inverse undoes f. Note $f^{-1}(x) \neq \frac{1}{f(x)}$.
Horizontal line test	$f(a) = f(b) \Rightarrow a = b$	f is one-to-one (has an inverse) iff no horizontal line meets its graph twice; otherwise restrict the domain.
Inverse graph reflection	(a, b) on $f \iff (b, a)$ on f^{-1}	Swapping coordinates reflects the graph in $y = x$; domain and range trade places.
Inverse of a composition	$(f \circ g)^{-1} = g^{-1} \circ f^{-1}$	Undo in reverse order, like taking off shoes before socks.

STATED-EQUATION *f composed with g of x equals f of g of x. f inverse of f of x equals f of f inverse of x, which equals x. A point a, b on f corresponds to the point b, a on f inverse. The inverse of f composed with g is g inverse composed with f inverse. f plus or minus g, of x, equals f of x plus or minus g of x, and f g of x equals f of x times g of x. Horizontal line test: f of a equals f of b implies a equals b, so f is one-to-one.*

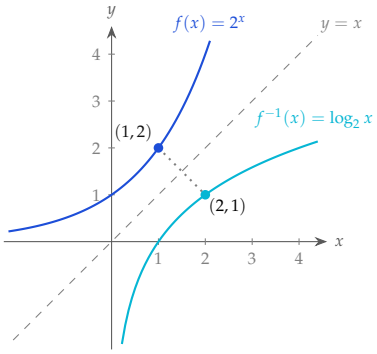
MEMORY AID Composition works inside-out: apply g first, then f. Inverses: "socks and shoes" — undo the last step first.

METHOD

1. Write $y = f(x)$ and check that f is one-to-one (restrict the domain if needed).
2. Swap x and y, then solve for y; that y is $f^{-1}(x)$.
3. Its domain is the range of f. Check $f\big(f^{-1}(x)\big) = x$. Example: $y = 2x + 3 \Rightarrow x = 2y + 3 \Rightarrow f^{-1}(x) = \frac{x-3}{2}$; check $2 \cdot \frac{x-3}{2} + 3 = x$.

TERMS

Domain, range – allowed inputs, produced outputs A1.9 §03 Algebra I
$f \circ g$ – "f of g": apply g first, then f; needs $g(x)$ in the domain of f
f^{-1} – inverse: swap x and y, solve for y
One-to-one – different inputs give different outputs; passes the horizontal line test
$y = x$ – the mirror line between f and f^{-1}
a, b – coordinates of a point on the graph
$f \pm g$, fg – functions built pointwise from f and g; domain is where both are defined



CONTEXT

- **Origin.** Leibniz coined "function" (1673); Euler introduced $f(x)$ notation (1734).
- **Use.** Chaining processes, unit conversions, undoing operations.
- **Data science.** A trained pipeline is a composition (scaler, then encoder, then model), and a deep network is a composition of layers. Predicting on a log target requires the inverse, $e^{\hat{y}}$ A2.14.
- **Physical application.** Converting Celsius to Fahrenheit, $F = \frac{9}{5}C + 32$, and its inverse $C = \frac{5}{9}(F - 32)$ undo each other.

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A2.2 Parent Functions & Function Families

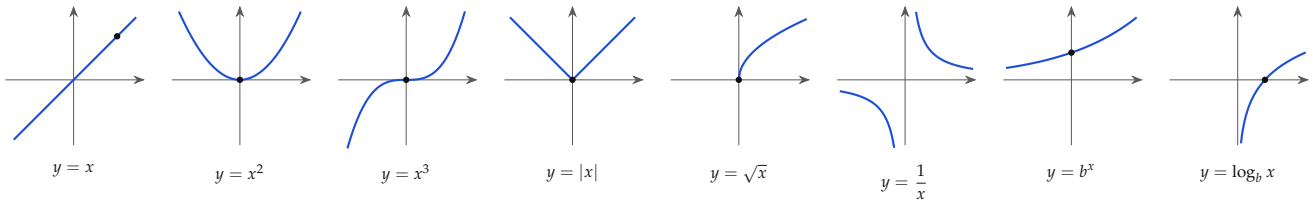
A function family is every graph obtainable from one simple "parent" by shifting, stretching and reflecting A2.5. Knowing eight parents by shape, key point, domain and range lets you sketch most functions on sight.

Linear (identity)	$y = x$	Domain and range $\mathbb{R}$; a line through $(0,0)$ and $(1,1)$.
Quadratic	$y = x^2$	U-shape, vertex $(0,0)$, range $[0,\infty)$; even A2.3.
Cubic	$y = x^3$	S-shape through the origin, range $\mathbb{R}$; odd; flat at 0.
Absolute value	$y = \lvert x \rvert$	V-shape, vertex $(0,0)$, range $[0,\infty)$ A1.11 §03 Algebra I.
Square root	$y = \sqrt{x}$	Starts at $(0,0)$, domain $[0,\infty)$; half of a sideways parabola.
Reciprocal	$y = \dfrac{1}{x}$	Two branches; asymptotes $x = 0$ and $y = 0$; domain $x \neq 0$.
Exponential	$y = b^x \quad (b > 1)$	Through $(0,1)$; range $(0,\infty)$; asymptote $y = 0$ A2.15.
Logarithmic	$y = \log_b x \quad (b > 1)$	Through $(1,0)$; domain $(0,\infty)$; asymptote $x = 0$ A2.14.

STATED-EQUATION *The eight parents: y equals x; y equals x squared; y equals x cubed; y equals the absolute value of x; y equals the square root of x; y equals one over x; y equals b to the x; y equals log base b of x.*

MEMORY AID One key point each: x, x^2, x^3, $|x|$, $\sqrt{x}$ pass through $(0,0)$ and $(1,1)$; $\frac{1}{x}$ through $(1,1)$; b^x through $(0,1)$; $\log_b x$ through $(1,0)$.

DECISION POINT To name the family of a formula, strip away the constants: the non-constant operation applied to x names the parent. $-3\sqrt{2x-4}+1$ is a square-root graph, $5 \cdot 2^{x+1}$ is exponential, $\frac{2}{x-3}+1$ is reciprocal. Then read the moves from the constants A2.5.



TERMS

Parent – simplest member of a family; all others are transformations of it
b – base, $b > 0$, $b \neq 1$
Asymptote – a line the graph approaches but never reaches
Vertex – turning point of a U- or V-shaped graph
$\mathbb{R}$ – all real numbers M.78 §01 Math Signs
Gallery – large plots of every parent: F.1 §18 Graph Gallery–F.9 §18 Graph Gallery

CONTEXT

- **Origin.** Studying curves by families comes from Descartes' analytic geometry (1637); "parent function" is a modern teaching term.
- **Use.** Recognizing a graph's formula, and a formula's graph, at a glance.
- **Data science.** Choosing a model family is the first modeling decision: linear, polynomial, log or exponential trends; a residual plot shaped like a parent suggests the missing transformation.
- **Physical application.** Free fall is quadratic in time; light intensity falls off like a reciprocal ($\frac{1}{d^2}$); population is exponential until resources run out.

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A2.3 Even & Odd Functions

Symmetry of a graph can be tested algebraically by replacing x with $-x$. Even functions mirror across the y-axis; odd functions look the same after a half-turn about the origin.

Even function	$f(-x) = f(x)$	Mirror image in the y-axis: x^2, $\lvert x\rvert$, $\cos x$.
Odd function	$f(-x) = -f(x)$	Half-turn symmetry about the origin: x^3, $\frac{1}{x}$, $\sin x$; if 0 is in the domain, $f(0) = 0$.
Polynomial rule	even powers only $\Rightarrow$ even; odd powers only $\Rightarrow$ odd	Hence the names. A constant counts as an even power (x^0).
Products	$\mathrm{E}\cdot\mathrm{E} = \mathrm{O}\cdot\mathrm{O} = \mathrm{E}, \quad \mathrm{E}\cdot\mathrm{O} = \mathrm{O}$	Combine like signs: even $\leftrightarrow +1$, odd $\leftrightarrow -1$.
Even–odd decomposition	$f(x) = \underbrace{\frac{f(x)+f(-x)}{2}}_{\text{even}} + \underbrace{\frac{f(x)-f(-x)}{2}}_{\text{odd}}$	Every function on a symmetric domain splits uniquely, e.g. $e^x = \cosh x + \sinh x$.
Symmetric integrals	$\int_{-a}^{a} f_{\text{odd}}\,dx = 0, \quad \int_{-a}^{a} f_{\text{even}}\,dx = 2\int_{0}^{a} f_{\text{even}}\,dx$	Odd areas cancel; even areas double C1.30 §07 Calculus I.

STATED-EQUATION *f is even if f of minus x equals f of x, and odd if f of minus x equals minus f of x. Every function equals its even part, f of x plus f of minus x over two, plus its odd part, f of x minus f of minus x over two. A polynomial with only even powers is even, and one with only odd powers is odd. Even times even and odd times odd are even; even times odd is odd. The integral from minus a to a of an odd function is zero, and of an even function is twice the integral from zero to a.*

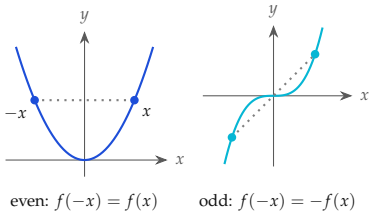
MEMORY AID Even $\leftrightarrow$ even powers $\leftrightarrow$ mirror in the y-axis. Odd $\leftrightarrow$ odd powers $\leftrightarrow$ spin 180° about the origin.

METHOD

1. Compute $f(-x)$ and simplify.
2. Compare: equals $f(x) \Rightarrow$ even; equals $-f(x) \Rightarrow$ odd; neither $\Rightarrow$ neither (most functions). Example: $f(x) = x^3 - x$, $f(-x) = -x^3 + x = -f(x)$, so odd.

TERMS

$f(-x)$ – f evaluated at the opposite input
E, O – even, odd function
$\cosh x$, $\sinh x$ – $\frac{e^x+e^{-x}}{2}$ (even) and $\frac{e^x-e^{-x}}{2}$ (odd)
a – half-width of the symmetric interval $[-a, a]$
Glossary – KT.4 §19 Glossary
$\int_{-a}^{a} \cdots\, dx$ – definite integral: signed area over $[-a, a]$ C1.30 §07 Calculus I



even: $f(-x) = f(x)$ odd: $f(-x) = -f(x)$

CONTEXT

- **Origin.** Euler classified functions as even and odd in his *Introductio* (1748).
- **Use.** Halving work in graphing and integration; Fourier series of even signals contain only cosines EE.26 §16 Electrical Eng..
- **Data science.** Squared and absolute error are even functions of the residual, so over- and under-predicting by the same amount cost the same; tanh is odd, so it is zero-centered F.15 §18 Graph Gallery.
- **Physical application.** A symmetric (even) charge distribution produces no net dipole moment; an odd one has zero average.

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A2.4 Piecewise & Step Functions

A piecewise function uses different formulas on different parts of its domain. Step functions are piecewise constant: they jump instead of varying smoothly.

Piecewise definition	$f(x) = \begin{cases} f_1(x), & x < c \\ f_2(x), & x \geq c \end{cases}$	First find which interval x lies in; each x uses exactly one piece.
Floor	$\lfloor x \rfloor = \text{greatest integer} \leq x$	Rounds down: $\lfloor 2.7 \rfloor = 2$, $\lfloor -2.3 \rfloor = -3$.
Ceiling	$\lceil x \rceil = \text{least integer} \geq x$	Rounds up: $\lceil 2.1 \rceil = 3$, $\lceil -2.9 \rceil = -2$.
Sign function	$\operatorname{sgn} x = \begin{cases} -1, & x < 0 \\ 0, & x = 0 \\ 1, & x > 0 \end{cases}$	Keeps only the sign; $\lvert x \rvert = x \operatorname{sgn} x$.
Heaviside step	$H(x) = \begin{cases} 0, & x < 0 \\ 1, & x \geq 0 \end{cases}$	Switches on at 0; $H(x-c)$ switches on at c.
ReLU	$\operatorname{ReLU}(x) = \max(0, x)$	Piecewise linear: 0 for negatives, x otherwise F.16 §18 Graph Gallery.
Continuity at a break	$\lim_{x \to c^-} f(x) = \lim_{x \to c^+} f(x) = f(c)$	The pieces join without a jump only if they agree at the boundary C1.7 §07 Calculus I.

STATED-EQUATION *f of x equals f one of x when x is less than c, and f two of x when x is at least c. The floor of x is the greatest integer less than or equal to x; the ceiling is the least integer greater than or equal to x. ReLU of x equals the maximum of zero and x. The sign of x is minus one, zero, or one as x is negative, zero, or positive. The Heaviside step H of x is zero for x less than zero and one for x at least zero. f is continuous at the break c when the left limit, the right limit, and f of c all agree.*

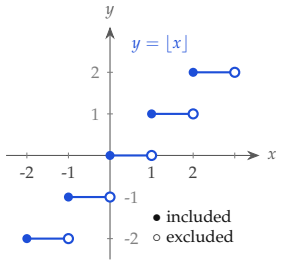
MEMORY AID Floor is the floor of the room (down); ceiling is up. Closed dot • = endpoint included; open dot ○ = excluded.

METHOD

1. Find the interval that contains x (watch $<$ versus $\leq$ at the boundaries).
2. Evaluate that piece's formula only.
3. To graph, draw each piece over its own interval, with a closed dot where the endpoint is included and an open dot where it is not. Example: $f(x) = x^2$ for $x < 1$, $3 - x$ for $x \geq 1$: $f(-2) = 4$, $f(1) = 2$.

TERMS

c – breakpoint between pieces
f_1, f_2 – the formulas used on each interval
$\lfloor \cdot \rfloor$, $\lceil \cdot \rceil$ – floor and ceiling (integer part, rounded down / up)
sgn, H – sign function; Heaviside (unit) step
c^-, c^+ – approaching c from the left / right C1.1 §07 Calculus I
Integers – $\mathbb{Z} = \{\ldots, -1, 0, 1, \ldots\}$ M.78 §01 Math Signs
ReLU, max – rectified linear unit; $\max(0, x)$ is the larger of 0 and x
lim – limit: the value $f(x)$ approaches C1.1 §07 Calculus I



CONTEXT

- **Origin.** Oliver Heaviside used the unit step for switched circuits (1890s); Kenneth Iverson introduced the $\lfloor\ \rfloor$, $\lceil\ \rceil$ notation (1962).
- **Use.** Tax brackets, shipping and parking rates, rounding rules.
- **Data science.** ReLU is the default neural-network activation ML.22 §15 Machine Learning; binning a feature is $\lfloor x/w \rfloor$; a decision tree predicts a piecewise-constant function ML.12 §15 Machine Learning.
- **Physical application.** Postage priced per started 100 g is a ceiling function; closing a switch at $t = c$ applies $V\,H(t-c)$ to a circuit.

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A2.5 Transformations of Functions

Every member of a family is the parent graph moved, stretched or flipped. Changes inside $f(\cdot)$ act on x (horizontally, and "backwards"); changes outside act on y (vertically, as written). Full visual catalog: A2.K1.

General form	$y = a\,f\big(b\,(x-h)\big)+k$	One formula holds all four moves: a, k act vertically, b, h horizontally.
Vertical shift	$y = f(x)+k$	Up k (down if $k<0$).
Horizontal shift	$y = f(x-h)$	Right h — opposite to the sign you see: $f(x+2)$ moves left 2.
Vertical stretch / reflection	$y = a\,f(x)$	$\lvert a\rvert>1$ stretches, $\lvert a\rvert<1$ compresses toward the x-axis; $a<0$ also reflects in the x-axis.
Horizontal scale / reflection	$y = f(bx)$	$\lvert b\rvert>1$ compresses by $\frac{1}{\lvert b\rvert}$ toward the y-axis; $b<0$ also reflects in the y-axis.
Point mapping	$(x,y)\mapsto\left(\frac{x}{b}+h,\ a\,y+k\right)$	Move the parent's key points: divide x by b then add h; multiply y by a then add k.

STATED-EQUATION *y equals a times f of b times x minus h, plus k. A point x, y on the parent moves to x over b plus h, a y plus k. y equals f of x plus k shifts up by k; y equals f of x minus h shifts right by h; y equals a f of x stretches vertically by a; y equals f of b x scales horizontally by one over b.*

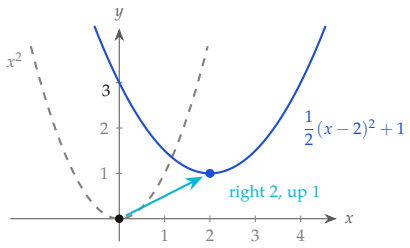
MEMORY AID Inside the parentheses acts on x and does the opposite (minus h shifts right); outside acts on y and does what it says.

METHOD

1. Identify the parent and three or four key points A2.2. Factor b out: $f(2x-4)=f\big(2(x-2)\big)$.
2. Horizontal: divide each x by b, then add h. Vertical: multiply each y by a, then add k.
3. Plot the new points, move asymptotes the same way, connect. Example: $y=\frac{1}{2}(x-2)^2+1$: vertex $(0,0)\to(2,1)$, $(2,4)\to(4,3)$.

TERMS

f – parent function
h – horizontal shift (right if $h>0$)
k – vertical shift (up if $k>0$)
a – vertical stretch by $\lvert a\rvert$; reflect over x-axis if $a<0$
b – horizontal compression by $1/\lvert b\rvert$; reflect over y-axis if $b<0$
$\mapsto$ – "moves to"



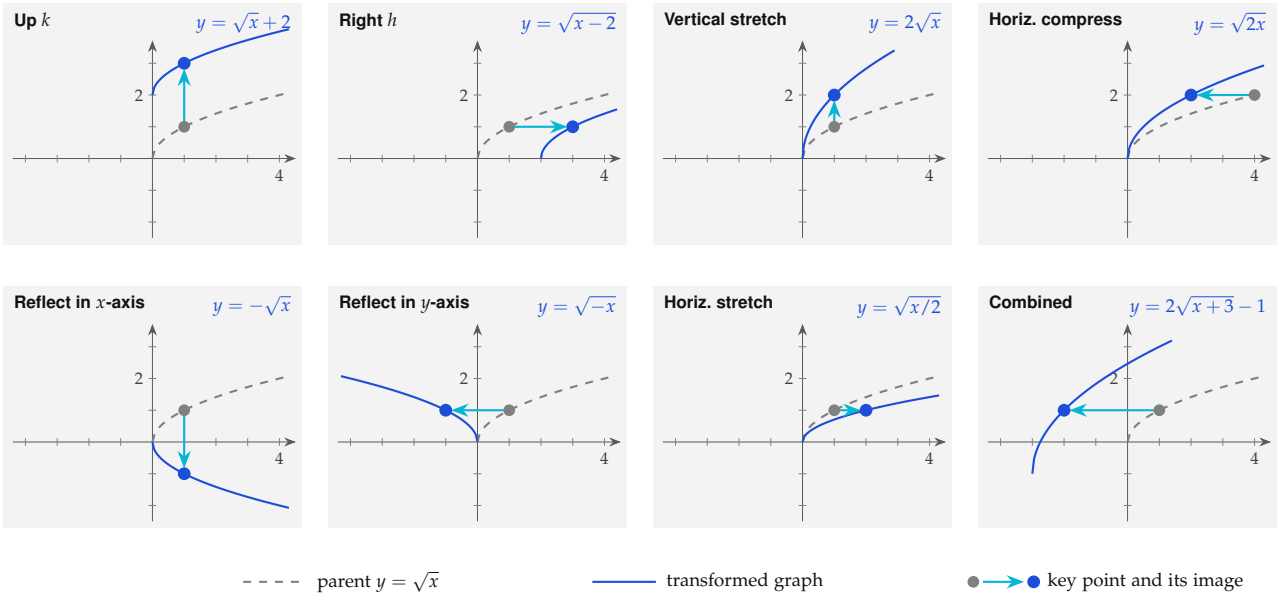
CONTEXT

- **Origin.** Systematized with graphing of function families in the 19th century.
- **Use.** Sketching any member of a family from one parent graph.
- **Data science.** Standardization $\frac{x-\mu}{\sigma}$ S.4 §13 Statistics is a shift and a scale; location–scale distribution families (normal, logistic) are one parent density transformed.
- **Physical application.** Shifting a sound wave in time by h delays it; multiplying by a makes it louder.

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Function Transformations | *Key Topic A2.K1*

Every transformation is one of four moves — shift, stretch, reflect, or a combination — applied either to the inputs (horizontal) or to the outputs (vertical). Below, each move is applied to the same parent, $y = \sqrt{x}$ (dashed), so the effect is easy to compare. Track a key point: the parent's $(1,1)$ goes to the blue dot.

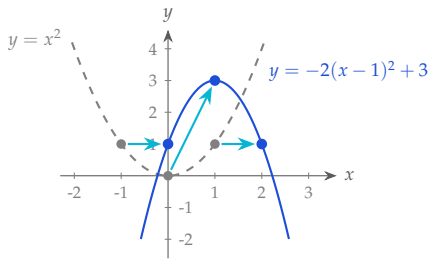


Rule table

Move	Equation	Point (x,y) becomes	Effect, and example on $f(x) = x^2$
Shift up / down	$f(x)+k$	$(x,\ y+k)$	up k if $k>0$, down if $k<0$. x^2+3: vertex $(0,3)$
Shift right / left	$f(x-h)$	$(x+h,\ y)$	right h if $h>0$; $f(x+h)$ is *left*. $(x-3)^2$: vertex $(3,0)$
Vertical stretch	$a\,f(x)$, $\lvert a\rvert>1$	$(x,\ ay)$	away from the x-axis; x-intercepts fixed. $3x^2$ is narrower
Vertical compress	$a\,f(x)$, $0<\lvert a\rvert<1$	$(x,\ ay)$	toward the x-axis. $\frac{1}{3}x^2$ is wider
Horizontal compress	$f(bx)$, $\lvert b\rvert>1$	$\left(\frac{x}{b},\ y\right)$	toward the y-axis by factor $\frac{1}{b}$; y-intercept fixed
Horizontal stretch	$f(bx)$, $0<\lvert b\rvert<1$	$\left(\frac{x}{b},\ y\right)$	away from the y-axis. $\sqrt{x/2}$ reaches $(2,1)$
Reflect in x-axis	$-f(x)$	$(x,\ -y)$	upside down. $-x^2$ opens down
Reflect in y-axis	$f(-x)$	$(-x,\ y)$	left–right mirror; no change for even f A2.3
Reflect in $y=x$	$f^{-1}(x)$	$(y,\ x)$	the inverse function A2.1
Absolute value (outside)	$\lvert f(x)\rvert$	$(x,\ \lvert y\rvert)$	parts below the x-axis flip up
Absolute value (inside)	$f(\lvert x\rvert)$	—	right half kept and mirrored onto the left

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Combining moves: $y = a\,f\big(b(x-h)\big)+k$



Order. Work *inside-out* for x and *outside-in* for y: horizontally, scale by $\frac{1}{b}$ then shift by h; vertically, scale by a (and reflect if $a<0$) then shift by k. The two directions are independent, so do them in either order.

Worked example. Sketch $y=-2(x-1)^2+3$ from $y=x^2$: $a=-2$, $b=1$, $h=1$, $k=3$, so $(x,y)\mapsto(x+1,\,-2y+3)$.

parent point	$(-1,1)$	$(0,0)$	$(1,1)$
$x+1$	0	1	2
$-2y+3$	1	3	1
new point	$(0,1)$	$(1,3)$	$(2,1)$

Vertex $(1,3)$, opens down, twice as narrow.

Common mistakes

Wrong direction. $f(x+2)$ moves *left* 2, not right: the input must be 2 smaller to give the same output.

Forgetting to factor b. $f(2x-6)=f\big(2(x-3)\big)$ shifts right 3, not 6.

Shift before stretch (vertical). $2f(x)+1$ is "double, then add 1": $(1,1)\to(1,3)$. Adding first would give $(1,4)$, which is $2\big(f(x)+1\big)$.

Mixing up the reflections. A minus *outside* flips upside down (x-axis); a minus *inside* flips left–right (y-axis).

Moving only the curve. Asymptotes, vertices and intercepts move too: $\frac{1}{x-2}+1$ has asymptotes $x=2$ and $y=1$.

Domain changes. $\sqrt{x-2}$ has domain $x\geq 2$; $\log(x+3)$ has domain $x>-3$.

DECISION POINT **Which reading to use?** For a quick sketch, track 2–4 key points with the point-mapping rule $(x,y)\mapsto(\frac{x}{b}+h,\;ay+k)$. To *find* the equation from a graph, read the shift from where the key point (vertex, start point, center, or asymptote crossing) went, then use one more point to solve for a. A horizontal stretch and a vertical stretch can look identical for some parents ($\sqrt{4x}=2\sqrt{x}$, $(2x)^2=4x^2$); either equation is correct.

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Complex Numbers, Polynomials & Rational Functions

A2.6 Complex Numbers

Complex numbers extend the reals with i, a number whose square is -1, so that every polynomial has roots. Each $z = a + bi$ is a point (a, b) in the complex plane.

Imaginary unit	$i^2 = -1, \qquad \sqrt{-n} = i\sqrt{n} \ \ (n > 0)$	Powers of i cycle with period 4: $i, -1, -i, 1$.
Standard form	$z = a + bi$	Real part a, imaginary part b; two complex numbers are equal iff both parts are.
Add / subtract	$(a + bi) \pm (c + di) = (a \pm c) + (b \pm d)\, i$	Combine real and imaginary parts separately, like vectors.
Multiply	$(a + bi)(c + di) = (ac - bd) + (ad + bc)\, i$	FOIL A1.13 §03 Algebra I, then replace i^2 with -1.
Conjugate & modulus	$\bar{z} = a - bi, \quad \lvert z\rvert = \sqrt{a^2 + b^2}, \quad z\bar{z} = \lvert z\rvert^2$	The conjugate mirrors z in the real axis; $z\bar{z}$ is real and ≥ 0.
Divide	$\dfrac{z}{w} = \dfrac{z\,\bar{w}}{w\,\bar{w}}$	Multiply top and bottom by the conjugate of the denominator to make it real.
Polar form	$z = r(\cos\theta + i\sin\theta) = re^{i\theta}$	$r = \lvert z\rvert$, $\theta = \arg z$; multiplying multiplies moduli and adds angles T.23 §06 Trigonometry.
Conjugate root theorem	$p(z) = 0 \Rightarrow p(\bar{z}) = 0$	For real coefficients, non-real roots come in conjugate pairs.

STATED-EQUATION *i squared equals minus one. z equals a plus b i; its conjugate, z bar, equals a minus b i; its modulus equals the square root of a squared plus b squared. a plus b i times c plus d i equals a c minus b d, plus a d plus b c times i. z over w equals z w bar over w w bar. In polar form, z equals r e to the i theta. The square root of minus n equals i root n for positive n. a plus b i, plus or minus c plus d i, equals a plus or minus c, plus b plus or minus d times i. z times z bar equals the modulus squared. In trigonometric form, z equals r times cos theta plus i sin theta. If a real polynomial p has root z, it also has root z bar.*

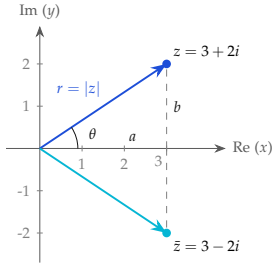
MEMORY AID Powers of i cycle $i, -1, -i, 1$: divide the exponent by 4 and use the remainder ($i^7 = i^3 = -i$).

METHOD

1. To divide, multiply numerator and denominator by the conjugate of the denominator.
2. Expand both, replacing i^2 by -1; the denominator becomes $c^2 + d^2$.
3. Split into $a + bi$ form. Example: $\frac{2+i}{1-i} = \frac{(2+i)(1+i)}{(1-i)(1+i)} = \frac{1+3i}{2} = \frac{1}{2} + \frac{3}{2}i$.

TERMS

a – real part, $\mathrm{Re}(z)$; b: imaginary part, $\mathrm{Im}(z)$
Complex plane – the xy-plane reused for numbers: the horizontal *real axis* (Re) carries a, the vertical *imaginary axis* (Im) carries b; $z = a + bi$ sits at (a, b), e.g. $3 + 2i$ at $(3, 2)$
c, d – real and imaginary parts of the second number $w = c + di$
$\bar{z}$ – complex conjugate
$|z|$, r – modulus: distance from 0 G.7 §05 Geometry
θ – argument: angle from the positive real axis T.19 §06 Trigonometry
p – polynomial with real coefficients; $\mathbb{C}$ the complex numbers M.78 §01 Math Signs
i – imaginary unit, $i^2 = -1$
z, w – complex numbers; $n > 0$ a real number
$e^{i\theta}$ – shorthand for $\cos\theta + i\sin\theta$ (Euler's formula) T.22 §06 Trigonometry



Complex plane: horizontal axis = real part $a = \mathrm{Re}(z)$, vertical axis = imaginary part $b = \mathrm{Im}(z)$. $z = a + bi$ is the point (a, b).

CONTEXT

- **Origin.** Cardano (1545) and Bombelli (1572), while solving cubics; Euler named i.
- **Use.** Roots of polynomials with $\Delta < 0$ A1.15 §03 Algebra I; AC circuits.
- **Data science.** The Fast Fourier Transform decomposes a time series into complex exponentials to expose seasonality; non-symmetric matrices can have complex eigenvalues.
- **Physical application.** AC circuits use complex impedance $Z = R + iX$; the modulus $|Z|$ sets the current amplitude.

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A2.7 Polynomial Long & Synthetic Division

Dividing one polynomial by another works like long division of numbers and produces a quotient and a remainder. Synthetic division is the shortcut for divisors of the form $x - c$.

Division algorithm	$P(x) = D(x)\,Q(x) + R(x), \quad \deg R < \deg D$	Every division ends with a remainder of lower degree than the divisor.
As a fraction	$\dfrac{P(x)}{D(x)} = Q(x) + \dfrac{R(x)}{D(x)}$	Splits an improper rational function; Q gives its slant asymptote A2.10.
Synthetic division	divide by $x - c$: bring down, $\times c$, add	Works on the coefficient list only; the last number is the remainder $P(c)$ A2.8.

STATED-EQUATION *P of x equals D of x times Q of x plus R of x, where the degree of R is less than the degree of D. P over D equals Q plus R over D. Synthetic division by x minus c: bring down the first coefficient, multiply by c, add to the next coefficient, and repeat.*

MEMORY AID Long-division loop: Divide, Multiply, Subtract, Bring down (“Does McDonald’s Sell Burgers?”).

METHOD

1. Write both polynomials in descending powers, inserting 0 for every missing power.
2. Divide the leading terms to get the next quotient term; multiply the divisor by it and subtract.
3. Bring down the next term and repeat until the remainder’s degree is less than the divisor’s.
4. Check: $DQ + R = P$. Example: $(x^2 + 3x + 5) \div (x + 1) = x + 2$ remainder 3, since $(x + 1)(x + 2) + 3 = x^2 + 3x + 5$.

DECISION POINT Divisor of the form $x - c$? Use synthetic division with c (for $x + 2$ use $c = -2$, not 2). Divisor of degree $\geq 2 \rightarrow$ long division. Only the remainder needed $\rightarrow$ just evaluate $P(c)$. Divisor $ax - r \rightarrow$ synthetic with $c = \frac{r}{a}$, then divide the quotient by a.

TERMS

P – dividend; D: divisor
Q – quotient; R: remainder
deg – degree: highest power present A1.13 §03 Algebra I
c – root of the linear divisor $x - c$
Placeholder 0 – coefficient written for a missing power, e.g. $x^3 + 1 \rightarrow 1, 0, 0, 1$
Synthetic division – shortcut for dividing by $x - c$ using only the coefficients

1	1	-6	11	-6
	1	-5	6	
	1	-5	6	$\boxed{0}$

$c = 1$; coefficients on top
bring down, multiply by c, add
$Q(x) = x^2 - 5x + 6$, remainder 0

CONTEXT

- **Origin.** Synthetic division is Ruffini’s rule (1809), rediscovered in Horner’s method (1819).
- **Use.** Factoring higher-degree polynomials once one root is known; slant asymptotes; partial fractions C2.4 §08 Calculus II.
- **Data science.** Horner’s scheme (the same arithmetic) evaluates a degree-n polynomial with n multiply–adds; CRC checksums are remainders of polynomial division over bits.
- **Physical application.** Control engineers divide transfer-function polynomials before inverting Laplace transforms to find a system’s response.

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A2.8 Remainder, Factor & Rational Root Theorems

These theorems link a polynomial's values, factors and roots, and reduce the search for exact roots of a higher-degree polynomial to a short list of candidates.

Remainder theorem	$P(x) = (x-c)\,Q(x) + P(c)$	The remainder on dividing by $x-c$ is simply $P(c)$.
Factor theorem	$P(c) = 0 \iff (x-c)$ is a factor	Every root gives a linear factor, and every linear factor gives a root.
Rational root theorem	$x = \pm\dfrac{p}{q},\ \ p \mid a_0,\ \ q \mid a_n$	The only possible rational roots; test them by substitution or synthetic division.
Fundamental theorem of algebra	$P(x) = a_n(x-r_1)(x-r_2)\cdots(x-r_n)$	A degree-n polynomial has exactly n complex roots, counted with multiplicity A2.6.
Descartes' rule of signs	# positive roots $= V,\ V-2,\ V-4, \ldots$	V = sign changes in the coefficients of $P(x)$; use $P(-x)$ for negative roots.
Vieta's formulas	$\sum r_i = -\dfrac{a_{n-1}}{a_n}, \qquad \prod r_i = (-1)^n\dfrac{a_0}{a_n}$	Sum and product of all roots straight from the coefficients (quadratic case: A1.15 §03 Algebra I).

STATED-EQUATION *P of x equals x minus c times Q of x, plus P of c. P of c equals zero if and only if x minus c is a factor. Possible rational roots are plus or minus p over q, where p divides a naught and q divides a n. A degree n polynomial factors into a n times n linear factors over the complex numbers. By Descartes' rule, the number of positive roots equals V, or V minus an even number. Vieta: the sum of the roots equals minus a n minus one over a n, and their product equals minus one to the n times a naught over a n.*

MEMORY AID Rational roots: "factors of the *last* over factors of the *first*."

METHOD

1. List the candidates $\pm\frac{p}{q}$.
2. Test them (start with ±1) by synthetic division A2.7 until the remainder is 0.
3. Divide out the root; repeat on the quotient until it is a quadratic.
4. Solve the quadratic A1.15 §03 Algebra I. Example: $x^3 - 6x^2 + 11x - 6$: candidates $\pm1, \pm2, \pm3, \pm6$; $P(1) = 0$, quotient $x^2 - 5x + 6 = (x-2)(x-3)$; roots $1, 2, 3$.

DECISION POINT Which candidate first? $P(1)$ is just the sum of the coefficients and $P(-1)$ the alternating sum, so test ±1 first. Use Descartes' rule and a rough graph to rule out signs and sizes. If no candidate works, the real roots are irrational: switch to Newton's method C1.35 §07 Calculus I or a calculator.

TERMS

$P(x)$ – polynomial $a_nx^n + \cdots + a_1x + a_0$
$Q(x)$ – quotient after dividing by $x-c$
a_n, a_0 – leading coefficient and constant term
$p \mid a_0$ – p divides a_0 evenly
r_i – the n roots; repeats counted by multiplicity A2.9
V – number of sign changes in the coefficient list
$q \mid a_n$ – q divides the leading coefficient; a_{n-1}: next coefficient
$\sum$, $\prod$ – sum and product over all roots A2.21

CONTEXT

- **Origin.** Descartes (1637) for the factor theorem and the rule of signs; Gauss proved the Fundamental Theorem of Algebra (1799).
- **Use.** Finding exact roots of higher-degree polynomials.
- **Data science.** Eigenvalues are roots of the characteristic polynomial $\det(A - \lambda I) = 0$ A2.25, which drive PCA and stability analysis.
- **Physical application.** Checking whether a design polynomial for a beam's deflection vanishes at a support point is a remainder-theorem evaluation.

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A2.9 Polynomial Graphs: End Behavior & Multiplicity

Far from the origin a polynomial behaves like its leading term; near each root, the root's multiplicity decides whether the graph crosses or bounces. Together they give a fast, accurate sketch.

Standard form	$P(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_0$	Degree n, leading coefficient $a_n \neq 0$.
End behavior	$P(x) \approx a_n x^n$ as $\lvert x\rvert \to \infty$	Even n: both ends point the same way (up if $a_n > 0$). Odd n: opposite ends (rises right if $a_n > 0$).
Multiplicity	$(x-r)^m$ a factor of P	Odd m: the graph crosses at r (flattening if $m \geq 3$). Even m: it touches and turns back.
Turning points	# turns $\leq n-1$	At most $n-1$ local maxima and minima C1.23 §07 Calculus I.
Real zeros	# x-intercepts $\leq n$	At most n real zeros; exactly n complex roots A2.8.
y-intercept	$P(0) = a_0$	The constant term.

STATED-EQUATION *P of x equals a n x to the n plus lower terms. As x grows large in size, P behaves like a n x to the n. A factor x minus r to the m crosses the axis if m is odd and bounces if m is even. There are at most n minus one turning points. There are at most n x-intercepts, and the y-intercept is P of zero, which equals a naught.*

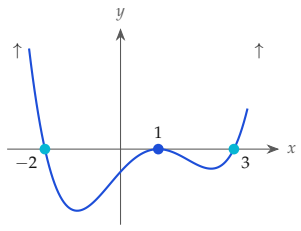
MEMORY AID The leading term rules the ends; multiplicity rules the middle: odd crosses, even bounces.

METHOD

1. Factor P completely A2.8.
2. Mark each zero with its multiplicity, and the y-intercept $P(0)$.
3. Read the end behavior from $a_n x^n$.
4. Trace from the left end, crossing or bouncing at each zero. Example: $(x+2)(x-1)^2(x-3)$: degree 4, $a_n = 1$ so both ends up; crosses at -2 and 3, bounces at 1; $P(0) = -6$.

TERMS

n – degree; a_n: leading coefficient
a_0 – constant term (y-intercept)
r – a real zero (root); m: its multiplicity (exponent of $x - r$)
$\approx$ – "behaves like" for large $\lvert x\rvert$ M.15 §01 Math Signs
Turning point – where the graph changes from rising to falling



cross (odd m) bounce (even m)

CONTEXT

- **Origin.** Newton classified cubic curves in his *Enumeratio* (1704).
- **Use.** Sketching and sanity-checking polynomial models.
- **Data science.** High-degree polynomial regression wiggles between points and explodes beyond the data (Runge's phenomenon), so polynomial fits extrapolate badly; splines use low-degree pieces instead.
- **Physical application.** Cubic splines in CAD software and font outlines are polynomial pieces joined smoothly.

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A2.10 Rational Functions & Asymptotes

A rational function is a ratio of polynomials. Its graph is shaped by asymptotes (lines it approaches) and holes (single missing points where a common factor cancels).

Rational function	$f(x) = \dfrac{P(x)}{Q(x)}$	Domain: every x with $Q(x) \neq 0$ A1.9 §03 Algebra I.
Vertical asymptote	$x = c$ where $Q(c) = 0,\ P(c) \neq 0$	The output blows up to $\pm\infty$ near c (check after canceling).
Hole	$(x - c)$ cancels from P and Q	A removable gap: open circle at c, height from the simplified f.
HA, bottom-heavy	$\deg P < \deg Q \ \Rightarrow\ y = 0$	The denominator wins: $f \to 0$.
HA, equal degrees	$\deg P = \deg Q \ \Rightarrow\ y = \dfrac{a_n}{b_n}$	Ratio of the leading coefficients.
Slant (oblique) asymptote	$\deg P = \deg Q + 1 \ \Rightarrow\ y =$ quotient of $P \div Q$	Divide A2.7 and drop the remainder.
Intercepts	x-int: $P(x) = 0$; $\quad y$-int: $f(0) = \dfrac{P(0)}{Q(0)}$	Zeros of the (simplified) numerator; value at 0.

STATED-EQUATION *f of x equals P of x over Q of x. A vertical asymptote at x equals c where Q of c is zero and P of c is not. If the degree of P is less than the degree of Q, the horizontal asymptote is y equals zero; if equal, y equals the ratio of leading coefficients; if one more, a slant asymptote. In that case the slant asymptote is y equals the quotient of P divided by Q. A factor x minus c that cancels from P and Q gives a hole. x-intercepts solve P of x equals zero, and the y-intercept is f of zero, which equals P of zero over Q of zero.*

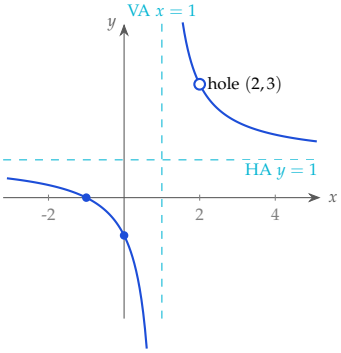
MEMORY AID Degrees: "Bigger on Bottom, $y = 0$; Exactly Even, divide the leads; Bigger on Top by one, slant."

METHOD

1. Factor P and Q; cancel common factors (each gives a hole).
2. Zeros of the remaining denominator $\to$ vertical asymptotes; of the remaining numerator $\to$ x-intercepts.
3. Compare degrees for the horizontal or slant asymptote.
4. Test the sign in each interval A1.17 §03 Algebra I and sketch. Example: $\frac{x+1}{x-1}$: VA $x = 1$, HA $y = 1$, intercepts $(-1, 0)$ and $(0, -1)$.

TERMS

P, Q – numerator and denominator polynomials
VA – vertical asymptote at a zero of Q that does not cancel
Hole – a factor common to P and Q cancels
HA – horizontal asymptote: end behavior (limits at infinity C1.1 §07 Calculus I)
a_n, b_n – leading coefficients of P and Q
deg – degree A2.9
c – a zero of Q



CONTEXT

- **Origin.** Ratios of polynomials studied since the 17th century; asymptote from Greek *asymptotos*, "not falling together".
- **Use.** Average cost, concentration, rates.
- **Data science.** Learning curves and saturation (Michaelis–Menten) models $\frac{ax}{b+x}$ level off at an asymptote; average cost per prediction $\frac{C_0+cn}{n} \to c$.
- **Physical application.** The average cost of producing x items, $\frac{500+2x}{x}$, levels off at the horizontal asymptote \$2 per item.

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A2.11 Rational Equations & Inequalities

Equations with the unknown in a denominator are cleared by multiplying by the LCD. That step can create candidates that make an original denominator zero, so every candidate must be checked; inequalities need a sign chart instead.

Clear denominators	$\frac{P}{Q} = \frac{R}{S} \;\Rightarrow\; PS = QR \quad (Q, S \neq 0)$	Cross-multiply (or multiply every term by the LCD) to get a polynomial equation.
Excluded values	$Q(x) = 0 \;\Rightarrow\; x \notin \text{domain}$	List them first; a candidate equal to one is extraneous.
Rational inequality	$\frac{P(x)}{Q(x)} > 0$	Never multiply by Q (its sign is unknown); sign chart on the zeros of P and Q A1.17 §03 Algebra I.
Work-rate equation	$\frac{1}{t_1} + \frac{1}{t_2} = \frac{1}{T}$	Rates add: workers taking t_1 and t_2 alone finish together in T.
Partial-fraction split	$\frac{1}{(x-a)(x-b)} = \frac{1}{a-b}\left(\frac{1}{x-a} - \frac{1}{x-b}\right)$	Undoes adding fractions; the basis of integration by partial fractions C2.4 §08 Calculus II.

STATED-EQUATION *P over Q equals R over S implies P S equals Q R, provided Q and S are not zero. One over t one plus one over t two equals one over T. Values making Q of x zero are excluded from the domain. P of x over Q of x greater than zero is solved with a sign chart of the zeros of P and Q. One over x minus a times x minus b equals one over a minus b, times the quantity one over x minus a minus one over x minus b.*

MEMORY AID Multiply by the LCD, then check every candidate against the "forbidden list".

METHOD

1. Factor every denominator and list the excluded values.
2. Multiply every term by the LCD A1.2 §03 Algebra I; solve the resulting polynomial equation.
3. Discard candidates on the excluded list. Example: $\frac{x}{x-1} = \frac{1}{x-1} + 2 \Rightarrow x = 1 + 2(x-1) \Rightarrow x = 1$, which is excluded: no solution.

DECISION POINT Equation ($=$): multiplying by the LCD is safe as long as you check. Inequality ($<, >$): move everything to one side, combine into one fraction, and use a sign chart; the zeros of Q are always open endpoints. Two painters taking 3 h and 6 h: $\frac{1}{3} + \frac{1}{6} = \frac{1}{T} \Rightarrow T = 2$ h.

TERMS

P, Q, R, S – polynomials; Q, S denominators
LCD – least common denominator A1.2 §03 Algebra I
Extraneous – a candidate that fails the original equation
t_1, t_2, T – times alone and together
a, b – distinct roots of the denominator ($a \neq b$)
$\notin$ – "is not an element of" A1.21 §03 Algebra I
Partial fractions – splitting a fraction into simpler fractions C2.4 §08 Calculus II

CONTEXT

- **Origin.** Cistern-filling (work-rate) puzzles appear in Greek, Chinese and medieval European texts.
- **Use.** Work, mixture, average-speed and lens problems.
- **Data science.** The F1 score $\frac{2PR}{P+R}$ is a rational expression ML.17 §15 Machine Learning; solving it for the recall needed to reach a target F1 is a rational equation.
- **Physical application.** Resistors in parallel, $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$, and the thin-lens equation $\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$ have the work-rate form.

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A2.12 Radical Equations & Extraneous Roots

To solve an equation with the unknown under a root, isolate the radical and raise both sides to the root's index. Squaring is not reversible, so it can add extraneous solutions that must be discarded.

Power both sides	$\sqrt[n]{u} = v \;\Rightarrow\; u = v^n$	Undo an n-th root with the n-th power A1.4 §03 Algebra I.
Why extra roots appear	$a = b \Rightarrow a^2 = b^2$, but $a^2 = b^2 \not\Rightarrow a = b$	Squaring also admits $a = -b$, which produces extraneous candidates.
Sign condition	$\sqrt{u} = v$ requires $v \geq 0$	A principal square root is never negative: candidates with $v < 0$ fail.
Two radicals	$\sqrt{u} + \sqrt{w} = c$	Isolate one radical, square; isolate the other, square again.
Rational exponent	$x^{m/n} = k \;\Rightarrow\; x = k^{n/m}$	Raise both sides to the reciprocal power; if m is even, use $\pm$.

STATED-EQUATION *The n-th root of u equals v implies u equals v to the n. Squaring is one-way: a equals b implies a squared equals b squared, but not the reverse. The square root of u equals v requires v at least zero. With two radicals, the square root of u plus the square root of w equals c, isolate one radical and square twice. x to the m over n equals k implies x equals k to the n over m.*

MEMORY AID Isolate, power, solve, *check*.

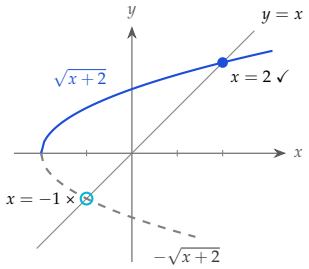
METHOD

1. Isolate the radical on one side.
2. Raise both sides to the index n; solve the resulting equation.
3. Substitute every candidate into the *original* equation; discard failures. Example: $\sqrt{x+2} = x \Rightarrow x+2 = x^2 \Rightarrow (x-2)(x+1) = 0$; $x = 2$ works, $x = -1$ gives $1 \neq -1$, extraneous.

DECISION POINT Odd index (cube root): raising to the power is reversible, so no extraneous roots appear. Even index: always check. With two radicals, isolate the more complicated one first so the second squaring has fewer terms.

TERMS

n – index of the root; u: radicand
v – the other side of the equation
Principal root – the non-negative even root, $\sqrt{9} = 3$ only
Extraneous root – solves the powered equation but not the original
w, c – second radicand and constant
m, k – exponent numerator and constant



CONTEXT

- **Origin.** Medieval Islamic algebraists such as al-Karaji (c. 1000) developed arithmetic with surds (irrational roots).
- **Use.** Distance, speed and geometry constraints G.12 §05 Geometry.
- **Data science.** Sample size for a target margin of error: $\frac{\sigma}{\sqrt{n}} = E \Rightarrow n = (\sigma/E)^2$ S.24 §13 Statistics.
- **Physical application.** A pendulum's period $T = 2\pi\sqrt{L/g}$ solved for length: $L = g\left(\frac{T}{2\pi}\right)^2$; a 2 s period needs $L \approx 0.99$ m.

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Exponential & Logarithmic Functions

A2.13 Exponential Growth & Decay

Exponential models change by a constant *factor* per unit of time, not a constant amount. Compound interest, unchecked populations and radioactive decay all follow this pattern.

Discrete growth	$y = y_0(1+r)^t$	Grows by the fraction r each period; $r<0$ decays (constant percent change A1.19 §03 Algebra I).
Compound interest	$A = P\left(1+\frac{r}{n}\right)^{nt}$	Annual rate r compounded n times a year for t years.
Continuous compounding	$A = Pe^{rt}$	The limit of the previous formula as $n \to \infty$.
Natural growth / decay	$y = y_0\, e^{kt}$	$k>0$ growth, $k<0$ decay; rate of change proportional to the amount C2.28 §08 Calculus II.
Half-life / doubling time	$t_{1/2} = \dfrac{\ln 2}{\lvert k\rvert}$	Time for the amount to halve (decay) or double (growth).
Half-life form	$y = y_0\left(\frac{1}{2}\right)^{t/t_{1/2}}$	Count how many half-lives have elapsed.

STATED-EQUATION *y equals y naught times one plus r to the t. A equals P times the quantity one plus r over n, raised to the n t. A equals P e to the r t. y equals y naught e to the k t. The half-life equals natural log of two over the absolute value of k. y equals y naught times one half to the t over t one half.*

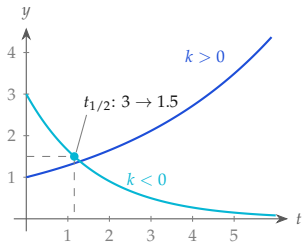
MEMORY AID "Pert": continuous compounding is $A = Pe^{rt}$. Rule of 72: doubling time $\approx 72/(\text{percent rate})$ FN.3 §17 Finance.

METHOD

1. y_0 is the amount at $t = 0$.
2. From one more data point (t_1, y_1): $k = \frac{\ln(y_1/y_0)}{t_1}$ A2.16.
3. Use $y_0 e^{kt}$ to predict. Example: 100 bacteria become 400 in 2 h: $k = \frac{\ln 4}{2} = \ln 2 \approx 0.69/\text{h}$, so they double every hour.

TERMS

P – principal (initial amount); A: amount after time t
r – rate per period as a decimal; n: compounding periods per year
y_0 – initial amount; t: elapsed time
e – $\approx 2.71828 = \lim(1+\frac{1}{n})^n$ C1.6 §07 Calculus I
k – growth ($k>0$) or decay ($k<0$) constant
$t_{1/2}$ – half-life (or doubling time for growth) A2.16



CONTEXT

- **Origin.** Jacob Bernoulli discovered e studying continuous compound interest (1683).
- **Use.** Interest, population growth, radioactive decay, cooling.
- **Data science.** Learning-rate decay, exponentially weighted moving averages, early epidemic curves, and user-retention decay all follow e^{kt}.
- **Physical application.** Carbon-14 has a half-life of about 5730 years, so $k = \ln 2/5730$ dates ancient wood.

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A2.14 Logarithms: Definition & Properties

A logarithm is an exponent: $\log_b x$ is the power to which b must be raised to give x. Logs turn multiplication into addition and powers into multiplication. Overview: A2.K2.

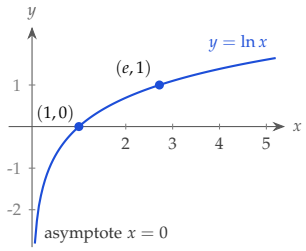
Definition	$\log_b x = y \iff b^y = x$	Converts between log form and exponential form.
Product rule	$\log_b(xy) = \log_b x + \log_b y$	The log of a product is the sum of the logs.
Quotient rule	$\log_b \dfrac{x}{y} = \log_b x - \log_b y$	The log of a quotient is the difference.
Power rule	$\log_b x^p = p \log_b x$	Brings an exponent down in front: the key to solving for exponents.
Special values	$\log_b 1 = 0, \quad \log_b b = 1$	Because $b^0 = 1$ and $b^1 = b$.
Inverse identities	$b^{\log_b x} = x, \quad \log_b(b^x) = x$	Log and exponential undo each other A2.1.

STATED-EQUATION *Log base b of x equals y if and only if b to the y equals x. Log of x y equals log x plus log y. Log of x over y equals log x minus log y. Log of x to the p equals p log x. Log base b of one equals zero, and log base b of b equals one. b to the log base b of x equals x, and log base b of b to the x equals x.*

MEMORY AID "A log is an exponent": $\log_b x$ asks "b to what power gives x?"

TERMS

b – base, $b > 0$, $b \neq 1$
x, y – arguments, $x, y > 0$ (in the definition, y is the exponent)
p – any real exponent
ln – natural log, base e A2.13; log alone often means base 10
Inverse of – the exponential, b^x A2.1
Pitfall – $\log(x+y) \neq \log x + \log y$; $\frac{\log x}{\log y} \neq \log \frac{x}{y}$



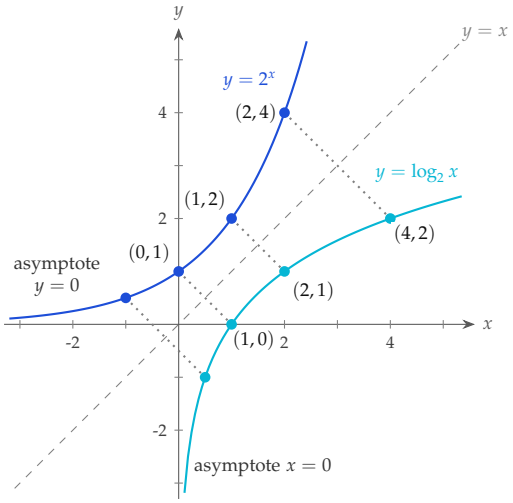
CONTEXT

- **Origin.** John Napier (1614) to speed astronomical multiplication; Henry Briggs produced base-10 tables.
- **Use.** pH, decibels, earthquake magnitudes, solving for exponents.
- **Data science.** Log-likelihood and log-loss (cross-entropy); log-transforming right-skewed features (income, counts); entropy in bits uses $\log_2$.
- **Physical application.** The decibel and Richter scales are logarithmic: each $+10$ dB is 10 times the sound intensity.

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Exponents & Logarithms | *Key Topic A2.K2*

Exponentials and logarithms are the same relationship read in two directions: $b^y = x$ says "raise b to y", $\log_b x = y$ asks "which power y?". Every exponent rule A1.3 §03 Algebra I therefore has a matching log rule, and the two graphs are mirror images in the line $y = x$ A2.1.



The two graphs ($b > 1$)

	$y = b^x$	$y = \log_b x$
domain	$\mathbb{R}$	$(0, \infty)$
range	$(0, \infty)$	$\mathbb{R}$
asymptote	$y = 0$	$x = 0$
key points	$(0,1)$, $(1,b)$	$(1,0)$, $(b,1)$
shape	increasing, ever steeper	increasing, ever flatter

For $0 < b < 1$ both are decreasing: $\left(\frac{1}{2}\right)^x$ decays and $\log_{1/2} x$ falls.

Translate between the forms

$$\log_b x = y \iff b^y = x$$

$\log_2 8 = 3$ because $2^3 = 8$. $\log_{10} 0.01 = -2$ because $10^{-2} = 0.01$. $\ln 1 = 0$ because $e^0 = 1$. $\log_9 3 = \frac{1}{2}$ because $9^{1/2} = 3$.

The rules side by side

Rule	Exponent form	Logarithm form	Example
Product	$b^m \cdot b^n = b^{m+n}$	$\log_b(xy) = \log_b x + \log_b y$	$\log_2(4 \cdot 8) = 2 + 3 = 5$
Quotient	$\frac{b^m}{b^n} = b^{m-n}$	$\log_b \frac{x}{y} = \log_b x - \log_b y$	$\log \frac{1000}{10} = 3 - 1 = 2$
Power	$(b^m)^n = b^{mn}$	$\log_b(x^p) = p \log_b x$	$\log_3 9^5 = 5 \cdot 2 = 10$
Zero / one	$b^0 = 1$, $b^1 = b$	$\log_b 1 = 0$, $\log_b b = 1$	$\ln e = 1$
Reciprocal	$b^{-n} = \frac{1}{b^n}$	$\log_b \frac{1}{x} = -\log_b x$	$\log_2 \frac{1}{8} = -3$
Inverse	$b^{\log_b x} = x$	$\log_b(b^x) = x$	$e^{\ln 5} = 5$
Change of base	$b^x = e^{x \ln b}$	$\log_b x = \frac{\ln x}{\ln b}$	$\log_2 7 = \frac{\ln 7}{\ln 2} \approx 2.81$

Solving: which move?

Unknown in an exponent A2.16. Isolate the power, take logs, bring the exponent down.
$5 \cdot 3^x = 45 \Rightarrow 3^x = 9 \Rightarrow x = \frac{\ln 9}{\ln 3} = 2$.

Same base available. Equate exponents.
$4^x = 8 \Rightarrow 2^{2x} = 2^3 \Rightarrow x = \frac{3}{2}$.

Unknown inside a log A2.17. Condense to one log, rewrite in exponential form, then check that every argument is positive.
$\log_2 x + \log_2(x-2) = 3 \Rightarrow x(x-2) = 8 \Rightarrow x = 4$ ($x = -2$ rejected).

Exponential of quadratic type. Substitute $u = b^x > 0$.
$4^x - 3 \cdot 2^x + 2 = 0 \Rightarrow u^2 - 3u + 2 = 0 \Rightarrow u = 1, 2 \Rightarrow x = 0, 1$.

Common mistakes. $\log(x+y) \neq \log x + \log y$; $\frac{\log x}{\log y} \neq \log \frac{x}{y}$; $(\log x)^2 \neq \log x^2$; log of a negative number or 0 is undefined; $e^{a+b} = e^a e^b$, *not* $e^a + e^b$.

Mental anchors. $\log_{10} 2 \approx 0.301$, $\ln 2 \approx 0.693$, $\ln 10 \approx 2.303$, $e \approx 2.718$, $2^{10} = 1024 \approx 10^3$.

DECISION POINT **Which base?** Any base works for solving: ln is standard in science and calculus (since $\frac{d}{dx} e^x = e^x$ C1.14 §07 Calculus I), $\log_{10}$ for orders of magnitude and decibels, $\log_2$ for bits, halvings and doublings. **Exponential or log model?** If equal steps in x multiply y by a constant factor, use $y = ab^x$ (a straight line on a log-y axis); if multiplying x adds a constant to y, use $y = a + c \ln x$ (a straight line on a log-x axis).

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A2.15 Graphs of Exponential & Logarithmic Functions

$y = b^x$ and $y = \log_b x$ are inverse functions, so their graphs are reflections in $y = x$. The base decides growth or decay; transformations A2.5 move the asymptote and the key point.

Exponential function	$f(x) = b^x$	Domain $\mathbb{R}$, range $(0, \infty)$, through $(0,1)$ and $(1,b)$; asymptote $y = 0$.
Growth vs. decay	$b > 1$ increasing; $\quad 0 < b < 1$ decreasing	$\left(\frac{1}{b}\right)^x = b^{-x}$ is the mirror image of b^x in the y-axis.
Logarithmic function	$g(x) = \log_b x$	Domain $(0, \infty)$, range $\mathbb{R}$, through $(1,0)$ and $(b,1)$; asymptote $x = 0$.
Base-e form	$b^x = e^{x \ln b}$	Every exponential is e^{kx} with $k = \ln b$; a larger base means a steeper curve.
Transformed exponential	$y = a\, b^{\,x-h} + k$	Asymptote moves to $y = k$; key point $(0,1)$ moves to $(h,\ a+k)$.
Transformed logarithm	$y = a \log_b(x - h) + k$	Asymptote moves to $x = h$; domain $x > h$; key point $(h+1,\ k)$.
Growth ranking	$\log x \ll x^p \ll b^x \quad (x \to \infty,\ p > 0,\ b > 1)$	Logs grow slower than any power; exponentials faster than any power O.10 §14 Optimization.

STATED-EQUATION *b to the x has domain all reals, range positive reals, passes through zero, one, and has asymptote y equals zero. Log base b of x has domain positive reals, passes through one, zero, and has asymptote x equals zero. b to the x equals e to the x ln b. b greater than one gives growth, and b between zero and one gives decay. y equals a b to the x minus h, plus k, and y equals a log base b of x minus h, plus k, are stretched and shifted copies. As x goes to infinity, log x grows much more slowly than x to the p, which grows much more slowly than b to the x.*

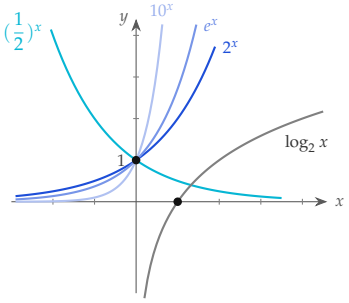
MEMORY AID Exponential: point $(0,1)$ and a floor at $y = 0$. Log: point $(1,0)$ and a wall at $x = 0$.

METHOD

1. Draw the asymptote ($y = k$ for exponentials, $x = h$ for logs).
2. Plot the moved key points: for $a\,b^{x-h} + k$ use $(h, a+k)$ and $(h+1, ab+k)$.
3. Draw the shape (growth or decay; flipped if $a < 0$). Example: $y = 2^{x-1} + 3$: asymptote $y = 3$, points $(1,4)$ and $(2,5)$.

TERMS

b – base, $b > 0$, $b \neq 1$
a, h, k – stretch and shifts as in A2.5
e, ln – natural base and natural log A2.13
p – any positive power
$\ll$ – "grows much more slowly than"
f, g – the exponential and logarithmic functions (inverses A2.1)



CONTEXT

- **Origin.** Euler (1748) first treated the exponential and the logarithm as inverse functions.
- **Use.** Reading growth and decay curves; choosing linear, semi-log or log–log axes.
- **Data science.** On a log-scaled y-axis an exponential becomes a straight line (semi-log plot); on log–log axes a power law becomes straight, which is how scaling laws for model size are read.
- **Physical application.** A capacitor discharges as $V_0 e^{-t/RC}$ EE.11 §16 Electrical Eng.: an exponential decay toward the asymptote $V = 0$.

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A2.16 Change of Base & Exponential Equations

Any logarithm can be computed from ln or $\log_{10}$, and any equation with the unknown in an exponent is solved by taking logs of both sides.

Change of base	$\log_b x = \dfrac{\ln x}{\ln b} = \dfrac{\log_{10} x}{\log_{10} b}$	Evaluate any base with a calculator's ln key.
Exponential equation	$b^x = c \implies x = \dfrac{\ln c}{\ln b}$	Take ln of both sides; the power rule brings x down.
Same base	$b^u = b^v \iff u = v$	If both sides are powers of one base, equate exponents.
Doubling time	$t = \dfrac{\ln 2}{\ln(1+r)}$	Periods needed to double at rate r per period.
Quadratic type	$a\,b^{2x} + c\,b^x + d = 0$	Substitute $u = b^x$, solve the quadratic A1.15 §03 Algebra I, keep only $u > 0$.

STATED-EQUATION *Log base b of x equals natural log of x over natural log of b, which equals log base ten of x over log base ten of b. If b to the x equals c, then x equals natural log of c over natural log of b. b to the u equals b to the v if and only if u equals v. The doubling time at rate r per period is t equals ln two over ln of one plus r. a b to the two x plus c b to the x plus d equals zero is a quadratic in u equals b to the x.*

MEMORY AID To get the exponent down, take a log.

METHOD

1. Isolate the exponential term.
2. Take ln of both sides and apply the power rule A2.14.
3. Solve the resulting linear equation. Example: $3 \cdot 2^x = 24 \Rightarrow 2^x = 8 \Rightarrow x \ln 2 = \ln 8 \Rightarrow x = 3$.

DECISION POINT Can both sides be written as powers of one base? Equate exponents (exact). Otherwise take ln. Different bases on both sides: take logs and collect the x terms: $2^x = 3^{x-1} \Rightarrow x \ln 2 = (x-1)\ln 3 \Rightarrow x = \frac{\ln 3}{\ln 3 - \ln 2} \approx 2.71$.

TERMS

b – original base; ln is any convenient base A2.14
c – target value, $c > 0$
u, v – exponent expressions; in the quadratic type, $u = b^x$
r – growth rate per period (decimal)
a, d – constant coefficients of the quadratic-type equation
Strategy – isolate the exponential, take logs of both sides
t – number of periods (doubling time)

CONTEXT

- **Origin.** A direct consequence of the power rule in A2.14.
- **Use.** Solving for time in growth problems; evaluating any base on a calculator.
- **Data science.** Converting nats to bits (divide by $\ln 2$); the depth of a balanced tree or binary search on n items is $\log_2 n$.
- **Physical application.** Solving $2^t = 1000$ for bacterial doublings: $t = \ln 1000 / \ln 2 \approx 9.97$ doublings.

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A2.17 Logarithmic Equations

Equations with the unknown inside a logarithm are solved by combining the logs into one and rewriting in exponential form. Logs need positive arguments, so every candidate must be checked in the original equation.

Exponentiate	$\log_b u = c \iff u = b^c$	One log equal to a number: rewrite in exponential form.
One-to-one property	$\log_b u = \log_b v \iff u = v \quad (u, v > 0)$	Equal logs in the same base have equal arguments.
Domain check	$u > 0$ for every $\log_b u$ in the original	Condensing logs can admit candidates that make an original argument ≤ 0.
Log inequality	$\log_b u < \log_b v \iff u < v \quad (b > 1)$	Logs keep the order for $b > 1$ and reverse it for $0 < b < 1$.

STATED-EQUATION *Log base b of u equals c if and only if u equals b to the c. Log base b of u equals log base b of v if and only if u equals v. Every argument in the original equation must be positive. For b greater than one, log base b of u is less than log base b of v if and only if u is less than v.*

MEMORY AID Condense, exponentiate, check that every argument is positive.

METHOD

1. Use the log rules A2.14 to condense each side to a single log.
2. Exponentiate (or use the one-to-one property) and solve.
3. Reject candidates that make any original argument ≤ 0. Example: $\log_2 x + \log_2(x-2) = 3 \Rightarrow x(x-2) = 2^3 \Rightarrow (x-4)(x+2) = 0$; keep $x = 4$, reject $x = -2$.

DECISION POINT Logs on both sides and nothing else → one-to-one property. Logs mixed with constants → condense all logs on one side, then exponentiate. Different bases → change of base first A2.16.

TERMS

b – base, $b > 0$, $b \neq 1$
u, v – arguments (expressions in x), required > 0
c – a constant
Extraneous – a candidate making an original argument ≤ 0

CONTEXT

- **Origin.** Briggs's base-10 tables (1624) made solving for unknown exponents and arguments routine.
- **Use.** Inverting log scales: pH, decibels, magnitudes.
- **Data science.** A model trained on $\log y$ predicts $\widehat{\log y} = 2.3$; the prediction in original units is $e^{2.3} \approx 10$ (the inverse of the log transform).
- **Physical application.** pH $= -\log_{10}[\mathrm{H}^+]$, so pH 3 means $[\mathrm{H}^+] = 10^{-3}$ mol/L; a 60 dB sound has intensity 10^6 times the reference.

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Sequences, Series & Induction

A2.18 Arithmetic Sequences & Series

An arithmetic sequence adds the same amount d at every step, so its terms lie on a line. Its sum is the number of terms times the average of the first and last.

nth term	$a_n = a_1 + (n-1)\,d$	Start at a_1 and take $n-1$ steps of size d.
Recursive form	$a_n = a_{n-1} + d$	Each term is the previous one plus d.
Sum (first and last)	$S_n = \dfrac{n\,(a_1 + a_n)}{2}$	Number of terms times the average of the ends.
Sum (no last term)	$S_n = \dfrac{n}{2}\left[\,2a_1 + (n-1)\,d\,\right]$	Same sum with a_n substituted.
Gauss sum	$1 + 2 + \cdots + n = \dfrac{n(n+1)}{2}$	The special case $a_1 = d = 1$.

STATED-EQUATION *a sub n equals a one plus n minus one times d. The sum S n equals n times a one plus a n, over two, which equals n over two times the quantity two a one plus n minus one d. Recursively, a n equals a n minus one plus d. One plus two plus up to n equals n times n plus one, over two.*

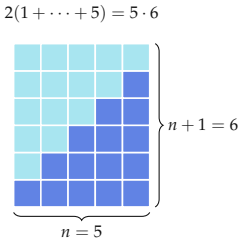
MEMORY AID Arithmetic sum: "average of first and last, times the number of terms" (Gauss's trick).

METHOD

1. Find d (difference of consecutive terms) and the number of terms $n = \frac{a_n - a_1}{d} + 1$.
2. Apply $S_n = \frac{n(a_1+a_n)}{2}$. Example: $5 + 8 + \cdots + 50$: $d = 3$, $n = 16$, $S = \frac{16 \cdot 55}{2} = 440$.

TERMS

a_1 – first term; a_n: nth term
d – common difference
n – number of terms (a positive integer)
S_n – sum of the first n terms (in Σ form A2.21)
Graph – terms lie on a line of slope d A1.6 §03 Algebra I



CONTEXT

- **Origin.** The (possibly apocryphal) story of young Gauss summing 1 to 100 by pairing terms.
- **Use.** Linear schedules, stacked seating, salary steps.
- **Data science.** $1 + 2 + \cdots + n = \frac{n(n+1)}{2}$ is why comparing every pair of n records costs $O(n^2)$; linear learning-rate warm-up is arithmetic.
- **Physical application.** Theater seating with 20 seats in row 1 and 2 more per row: row n has $20 + 2(n-1)$ seats.

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A2.19 Geometric Sequences & Series

A geometric sequence multiplies by the same ratio r at every step: discrete exponential growth. Its infinite sum is finite exactly when $|r| < 1$.

nth term	$a_n = a_1 r^{n-1}$	Start at a_1 and multiply by r, $n-1$ times.
Recursive form	$a_n = r\, a_{n-1}$	Each term is r times the previous one.
Finite sum	$S_n = \dfrac{a_1(1-r^n)}{1-r} \quad (r \neq 1)$	Subtract rS_n from S_n and almost every term cancels.
Infinite sum	$S_\infty = \displaystyle\sum_{n=1}^{\infty} a_1 r^{n-1} = \frac{a_1}{1-r}, \quad \lvert r\rvert < 1$	Diverges if $\lvert r\rvert \geq 1$ C2.16 §08 Calculus II.
Repeating decimal	$0.\overline{27} = \dfrac{0.27}{1-0.01} = \dfrac{27}{99} = \dfrac{3}{11}$	A repeating block is a geometric series with $r = 10^{-k}$.

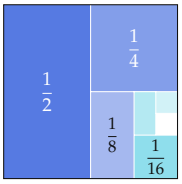
STATED-EQUATION *a sub n equals a one times r to the n minus one. S n equals a one times one minus r to the n, over one minus r. The infinite sum equals a one over one minus r, when the absolute value of r is less than one. Recursively, a n equals r times a n minus one. Zero point two seven repeating equals zero point two seven over one minus zero point zero one, which is twenty-seven over ninety-nine, or three over eleven.*

MEMORY AID Infinite geometric sum: "first term over one minus the ratio."

DECISION POINT Arithmetic or geometric? Constant *differences* → arithmetic A2.18; constant *ratios* → geometric. Neither: check for squares, factorials, or a recursion such as Fibonacci $a_n = a_{n-1} + a_{n-2}$.

TERMS

a_1 – first term
r – common ratio (each term divided by the previous)
n – number of terms
S_∞ – infinite sum; converges only if $|r| < 1$ C2.16 §08 Calculus II
Relation – exponential growth in discrete steps A2.13
$0.\overline{27}$ – repeating decimal: the digits under the bar repeat forever



$$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots = \frac{1/2}{1-1/2} = 1$$

CONTEXT

- **Origin.** Zeno's paradoxes; sum formula in Euclid's *Elements*, Book IX.
- **Use.** Loans, annuities, bouncing balls, present value.
- **Data science.** Discounted return in reinforcement learning $\sum_t \gamma^t r_t$; with constant reward r it equals $\frac{r}{1-\gamma}$. EMA weights form a geometric series summing to 1.
- **Physical application.** A ball rebounding to 60% of its previous height travels a finite total distance, a geometric series.

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A2.20 Arithmetic, Geometric & Harmonic Means

Three ways to average positive numbers, each matched to a kind of quantity: amounts that add (arithmetic), factors that multiply (geometric), and rates that combine through reciprocals (harmonic).

Arithmetic mean	$\mathrm{AM} = \dfrac{x_1 + x_2 + \cdots + x_n}{n}$	For quantities that add; the middle term of an arithmetic sequence A2.18.
Geometric mean	$\mathrm{GM} = \sqrt[n]{x_1 x_2 \cdots x_n}$	For growth factors and ratios; the middle term of a geometric sequence.
Harmonic mean	$\mathrm{HM} = \dfrac{n}{\frac{1}{x_1} + \frac{1}{x_2} + \cdots + \frac{1}{x_n}}$	For rates over equal amounts (average speed over equal distances).
AM–GM–HM inequality	$\mathrm{AM} \geq \mathrm{GM} \geq \mathrm{HM}$	Equality exactly when all the values are equal.
Two numbers	$\dfrac{a+b}{2} \geq \sqrt{ab} \geq \dfrac{2ab}{a+b}$	Quick bounds; $\mathrm{GM}^2 = \mathrm{AM} \cdot \mathrm{HM}$ for two numbers.
Weighted mean	$\bar{x}_w = \dfrac{\sum w_i x_i}{\sum w_i}$	Each value counts in proportion to its weight.

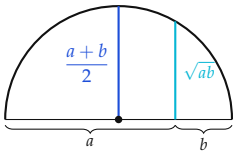
STATED-EQUATION *The arithmetic mean is the sum over n. The geometric mean is the n-th root of the product. The harmonic mean is n over the sum of reciprocals. The arithmetic mean is at least the geometric mean, which is at least the harmonic mean. For two numbers, a plus b over two is at least root a b, which is at least two a b over a plus b. The weighted mean x bar w equals the sum of w i x i over the sum of w i.*

MEMORY AID Add → arithmetic; multiply → geometric; "per" rates → harmonic.

DECISION POINT Which mean? Ask how the quantities combine. Totals that add (heights, costs) → AM. Successive growth factors ($\times 1.10$, then $\times 0.90$) → GM: $\sqrt{1.1 \cdot 0.9} \approx 0.995$, a net loss. Rates over equal distances (km/h out and back) → HM: 60 and 40 km/h average $\frac{2 \cdot 60 \cdot 40}{100} = 48$, not 50.

TERMS

x_i – the n positive values
a, b – two positive numbers
w_i – non-negative weights
$\sqrt[n]{\ }$ – n-th root A1.4 §03 Algebra I
n – number of values
$\bar{x}_w$ – weighted mean
$\sum$ – summation A2.21



CONTEXT

- **Origin.** The Pythagorean school (Archytas, c. 400 BCE) defined all three means from musical ratios.
- **Use.** Averaging speeds, prices, growth rates, ratios.
- **Data science.** Average return over years is a GM (CAGR FN.10 §17 Finance); the F1 score is the HM of precision and recall ML.17 §15 Machine Learning; the sample mean S.1 §13 Statistics is the AM.
- **Physical application.** Two resistors in parallel have $R = \frac{R_1 R_2}{R_1 + R_2}$, half their harmonic mean.

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A2.21 Sigma Notation & Summation Formulas

Σ notation writes a long sum compactly by an index running between limits. A handful of closed forms and the linearity rules evaluate most sums without adding term by term.

Constant sum	$\sum_{i=1}^{n} c = cn$	Adding c to itself n times.
Sum of integers	$\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$	Gauss's pairing A2.18.
Sum of squares	$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$	Proved by induction A2.22.
Sum of cubes	$\sum_{i=1}^{n} i^3 = \left[\frac{n(n+1)}{2}\right]^2$	The square of the sum of integers.
Linearity	$\sum_{i=1}^{n}(c\,a_i + b_i) = c\sum_{i=1}^{n} a_i + \sum_{i=1}^{n} b_i$	Sums split over $+$ and constants factor out (but $\sum a_i b_i \neq \sum a_i \sum b_i$).
Telescoping sum	$\sum_{i=1}^{n}(a_{i+1} - a_i) = a_{n+1} - a_1$	Consecutive terms cancel, leaving only the ends.

STATED-EQUATION *The sum from i equals one to n of c equals c n. The sum of i equals n times n plus one over two. The sum of i squared equals n times n plus one times two n plus one, over six. The sum of i cubed equals the square of n times n plus one over two. Sums split and constants factor out. The sum of c a i plus b i equals c times the sum of a i plus the sum of b i. The telescoping sum of a i plus one minus a i equals a n plus one minus a one.*

MEMORY AID Sum of cubes = (sum of integers)^2: $1 + 8 + 27 = 36 = 6^2$.

METHOD

1. Split the sum by linearity and pull out constants.
2. Replace each piece by its closed form. Example: $\sum_{i=1}^{4}(2i+1) = 2 \cdot \frac{4\cdot 5}{2} + 4 = 24$ (check: $3 + 5 + 7 + 9 = 24$).

TERMS

i – index of summation; n: upper limit
c – constant (does not depend on i)
a_i, b_i – terms of two sequences
Linearity – sums split and constants factor out
Series types – arithmetic A2.18, geometric A2.19; symbol M.53 §01 Math Signs
Telescoping – consecutive terms cancel, leaving only the ends

CONTEXT

- **Origin.** Euler introduced Σ for sums (1755).
- **Use.** Compactly expressing totals and averages.
- **Data science.** Every mean, loss, and gradient is a sum; linearity of sums becomes linearity of expectation S.13 §13 Statistics; integrals are limits of sums C1.30 §07 Calculus I.
- **Physical application.** Stacking cannonballs in a square pyramid of n layers uses $\sum i^2$ balls.

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A2.22 Mathematical Induction

A proof technique for statements about every integer $n \geq n_0$: prove the first case, then prove that each case implies the next. Like a line of dominoes, all cases then follow.

Base case	$P(n_0)$ is true	Check the first value directly.
Inductive step	$P(k) \Rightarrow P(k+1)$ for all $k \geq n_0$	Assume the statement for k (the inductive hypothesis) and derive it for $k+1$.
Conclusion	$P(n)$ is true for all $n \geq n_0$	Base case + step reach every integer.
Strong induction	$P(n_0), \ldots, P(k) \Rightarrow P(k+1)$	Assume all earlier cases; natural for recursions such as Fibonacci.

STATED-EQUATION *If P of n naught is true, and P of k implies P of k plus one for every k at least n naught, then P of n is true for every n at least n naught. Strong induction assumes P of n naught through P of k to prove P of k plus one.*

MEMORY AID Dominoes: the first one falls, and each falling domino knocks over the next.

METHOD

1. State $P(n)$ precisely and verify the base case n_0.
2. Assume $P(k)$; write what $P(k+1)$ claims.
3. Use the assumption to reach $P(k+1)$. Example: $1 + \cdots + n = \frac{n(n+1)}{2}$. Base $n = 1$: $1 = 1$. Step: $\frac{k(k+1)}{2} + (k+1) = \frac{(k+1)(k+2)}{2}$, which is the formula for $k+1$.

TERMS

$P(n)$ – the statement about the integer n
n_0 – first integer covered (often 0 or 1)
k – an arbitrary integer $\geq n_0$
Inductive hypothesis – the assumption that $P(k)$ holds
$\Rightarrow$ – "implies" M.95 §01 Math Signs
Strong induction – the step may use all earlier cases, not just $P(k)$

CONTEXT

- **Origin.** Used by Maurolico (1575) and Pascal (1654); named "mathematical induction" by De Morgan (1838).
- **Use.** Proving summation formulas A2.21, divisibility facts, and inequalities such as $2^n > n$.
- **Data science.** Correctness proofs of recursive algorithms and loop invariants are inductions; dynamic programming builds each answer from smaller solved cases O.21 §14 Optimization.
- **Physical application.** A row of dominoes, or a ladder you can climb if you can reach the first rung and step from any rung to the next.

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A2.23 Binomial Theorem

The binomial theorem expands any power $(a+b)^n$ directly; the coefficients are the binomial coefficients $\binom{n}{k}$, the entries of Pascal's triangle.

Binomial theorem	$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k} b^k$	Powers of a fall from n to 0 as powers of b rise from 0 to n.
Binomial coefficient	$\binom{n}{k} = \dfrac{n!}{k!\,(n-k)!}$	The number of ways to choose k items from n S.8 §13 Statistics.
General term	$T_{k+1} = \binom{n}{k} a^{n-k} b^k$	Find one specific term without expanding everything.
Pascal's rule	$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$	Each entry of Pascal's triangle is the sum of the two above it.
Symmetry & row sum	$\binom{n}{k} = \binom{n}{n-k}, \quad \sum_{k=0}^{n} \binom{n}{k} = 2^n$	Rows read the same both ways; set $a = b = 1$ for the sum.
Linear approximation	$(1+x)^n \approx 1 + nx \quad (\lvert x\rvert \ll 1)$	Keep the first two terms when x is small.

STATED-EQUATION *a plus b to the n equals the sum from k equals zero to n of n choose k, a to the n minus k, b to the k. n choose k equals n factorial over k factorial times n minus k factorial. The general term T k plus one equals n choose k, a to the n minus k, b to the k. Pascal's rule: n choose k equals n minus one choose k minus one plus n minus one choose k. n choose k equals n choose n minus k, and the row sum is two to the n. For small x, one plus x to the n is approximately one plus n x.*

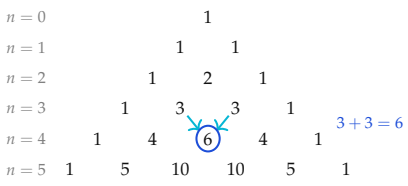
MEMORY AID Coefficients come from Pascal's triangle: 1; 1 1; 1 2 1; 1 3 3 1; …

METHOD

1. Match the power of the variable you want to a^{n-k} or b^k to find k.
2. Compute $\binom{n}{k} a^{n-k} b^k$ with the numbers substituted. Example: the x^2 coefficient of $(x+2)^5$ needs $n-k=2$, so $k=3$: $\binom{5}{3} 2^3 = 10 \cdot 8 = 80$.

TERMS

a, b – the two terms; n: non-negative integer power
k – term index, $0 \le k \le n$
$\binom{n}{k}$ – binomial coefficient, "n choose k" S.8 §13 Statistics
$n!$ – factorial $n(n-1)\cdots 1$, with $0! = 1$ M.48 §01 Math Signs
T_{k+1} – the $(k+1)$th term of the expansion
Pascal's triangle – each entry is the sum of the two above it
x, $|x| \ll 1$ – a number much smaller than 1 in size



CONTEXT

- **Origin.** Known to al-Karaji and Yang Hui; Pascal's *Traité* (1654); Newton extended it to any exponent.
- **Use.** Expanding powers, probability, approximations.
- **Data science.** Each term $\binom{n}{k} p^k q^{n-k}$ of $(p+q)^n$ is a binomial probability S.16 §13 Statistics, the model for conversions out of n trials.
- **Physical application.** Engineers approximate $(1+x)^n \approx 1+nx$ for small x, e.g., thermal expansion of a cube: $(1+\alpha\Delta T)^3 \approx 1 + 3\alpha\Delta T$.

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Matrices

A2.24 Matrix Operations

A matrix is a rectangular array of numbers. Matrices add entry by entry, but multiply "row times column", which composes the linear maps they represent. Full treatment: LA.K2 §09 Linear Algebra.

Matrix product	$(AB)_{ij} = \sum_{k=1}^{n} a_{ik}\, b_{kj}$	Entry (i, j) is row i of A dotted with column j of B.
Size rule	$(m \times n)(n \times p) = (m \times p)$	Inner sizes must match; outer sizes give the result.
Add & scale	$(A + B)_{ij} = a_{ij} + b_{ij}, \quad (cA)_{ij} = c\, a_{ij}$	Entry by entry; addition needs equal sizes.
2×2 **product**	$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} e & f \\ g & h \end{bmatrix} = \begin{bmatrix} ae + bg & af + bh \\ ce + dg & cf + dh \end{bmatrix}$	Each entry is a row of the first times a column of the second.
Transpose	$(A^{\mathsf{T}})_{ij} = a_{ji}, \qquad (AB)^{\mathsf{T}} = B^{\mathsf{T}} A^{\mathsf{T}}$	Rows become columns; transposing a product reverses the order.
Identity & order	$AI = IA = A, \qquad AB \neq BA$ in general	I acts like 1, but the order of a product matters.

STATED-EQUATION *The i j entry of A B equals the sum over k of a i k times b k j. For two-by-two matrices: first row is a e plus b g, a f plus b h; second row is c e plus d g, c f plus d h. A B transpose equals B transpose A transpose. An m by n matrix times an n by p matrix is m by p. Matrices add entry by entry, and c A multiplies every entry by c. The i j entry of A transpose is a j i. A I equals I A equals A, but A B generally does not equal B A.*

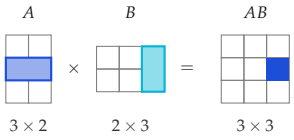
MEMORY AID "Row times column": slide across the row of A and down the column of B.

METHOD

1. Check the sizes: inner dimensions equal.
2. For each entry, multiply row i of A by column j of B term by term and add. Example: $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}$, since $1 \cdot 5 + 2 \cdot 7 = 19$.

TERMS

$m \times n$ – m rows, n columns
a_{ij} – entry in row i, column j; k: running index
$a, \ldots, h$ – entries of the two 2×2 matrices; c also a scalar in cA
A^{T} – transpose: rows become columns
I – identity matrix: 1s on the diagonal, 0s elsewhere
Non-commutative – $AB \neq BA$ in general
p – number of columns of B; b_{ij}: entries of B



row 2 of A · column 3 of $B = (AB)_{23}$

CONTEXT

- **Origin.** Arthur Cayley's matrix algebra (1858); Sylvester coined "matrix".
- **Use.** Solving systems A1.12 §03 Algebra I, geometric transformations G.24 §05 Geometry.
- **Data science.** A dataset is an $n \times p$ matrix; a dense neural layer is $XW + b$; embedding lookups and attention are matrix products.
- **Physical application.** Rotating and then scaling a robot arm's coordinates is a product of two transformation matrices.

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A2.25 Determinants, Inverses & Cramer's Rule

The determinant is a single number that measures how a square matrix scales area (or volume). A matrix is invertible exactly when its determinant is nonzero, and then small systems can be solved by formula.

2×2 **determinant**	$\det\begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$	Main diagonal product minus the off-diagonal product.
2×2 **inverse**	$A^{-1} = \dfrac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$	Exists only if $\det A \neq 0$; check $AA^{-1} = I$.
Cramer's rule	$x = \dfrac{\det A_x}{\det A}, \qquad y = \dfrac{\det A_y}{\det A}$	Solves $A\mathbf{x} = \mathbf{b}$ when $\det A \neq 0$ A1.12 §03 Algebra I.
3×3 **cofactor expansion**	$\det\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = a(ei-fh) - b(di-fg) + c(dh-eg)$	Expand along the top row with signs $+,-,+$.
Product rule	$\det(AB) = \det A \,\det B, \qquad \det(A^{-1}) = \dfrac{1}{\det A}$	Scale factors multiply.
Matrix solution	$A\mathbf{x} = \mathbf{b} \;\Rightarrow\; \mathbf{x} = A^{-1}\mathbf{b}$	Multiply both sides on the left by A^{-1}.

STATED-EQUATION *For A with entries a b c d, the determinant equals a d minus b c. A inverse equals one over a d minus b c times the matrix d, minus b, minus c, a. Cramer's rule: x equals det A x over det A, y equals det A y over det A. The three-by-three determinant equals a times e i minus f h, minus b times d i minus f g, plus c times d h minus e g. The determinant of A B equals det A times det B, and the determinant of A inverse equals one over det A. A x equals b gives x equals A inverse b.*

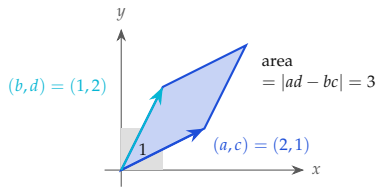
MEMORY AID Two-by-two inverse: "swap the diagonal, negate the off-diagonal, divide by the determinant."

METHOD

1. Compute $\det A$; if it is 0, there is no unique solution.
2. Form A_x (A_y) by replacing the x (y) column with the constants $\mathbf{b}$.
3. Divide determinants. Example: $2x + y = 5$, $x + y = 3$: $\det A = 1$, $x = \det\begin{bmatrix} 5 & 1 \\ 3 & 1 \end{bmatrix} = 2$, $y = \det\begin{bmatrix} 2 & 5 \\ 1 & 3 \end{bmatrix} = 1$.

TERMS

$\det A$ – determinant: signed area scale factor
$a, \ldots, i$ – matrix entries (here i is an entry, not $\sqrt{-1}$)
Singular – $\det A = 0$: no inverse, system has no unique solution A1.12 §03 Algebra I
A_x, A_y – A with the x (or y) column replaced by the constants
$\mathbf{x}, \mathbf{b}$ – unknown vector and constant vector
$AA^{-1} = I$ – identity matrix A2.24
A^{-1} – inverse matrix; exists only when $\det A \neq 0$



CONTEXT

- **Origin.** Leibniz (1693) and Gabriel Cramer (1750).
- **Use.** Solving small systems, testing invertibility, computing areas.
- **Data science.** A near-zero $\det(X^{\mathsf{T}}X)$ signals multicollinearity; $\det\Sigma$ appears in the multivariate normal density; normalizing flows use Jacobian determinants.
- **Physical application.** The determinant is the area scale factor: a transformation with $\det = 2$ doubles the area of any shape drawn on it.

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Conic Sections

A2.26 Conic Sections

The circle, parabola, ellipse and hyperbola are the curves cut from a double cone by a plane. Each has a standard equation centered at (h, k), and its foci determine its shape.

Circle	$(x-h)^2 + (y-k)^2 = r^2$	All points at distance r from the center G.19 §05 Geometry.
Parabola	$(x-h)^2 = 4p\,(y-k)$	Points equidistant from the focus and the directrix; opens up if $p > 0$.
Ellipse	$\dfrac{(x-h)^2}{a^2} + \dfrac{(y-k)^2}{b^2} = 1, \quad c^2 = a^2 - b^2$	Sum of distances to the two foci is constant ($2a$, with $a > b$).
Hyperbola	$\dfrac{(x-h)^2}{a^2} - \dfrac{(y-k)^2}{b^2} = 1, \quad c^2 = a^2 + b^2$	Difference of distances to the foci is constant ($2a$).
Hyperbola asymptotes	$y - k = \pm\dfrac{b}{a}\,(x-h)$	The diagonals of the $2a \times 2b$ box through the center.
Eccentricity	$e = \dfrac{c}{a}$	0: circle; < 1: ellipse; $= 1$: parabola; > 1: hyperbola.
General-form test	$Ax^2 + Cy^2 + Dx + Ey + F = 0$	$A = C$: circle; $AC > 0$: ellipse; $AC < 0$: hyperbola; $AC = 0$: parabola.

STATED-EQUATION *Circle: x minus h squared plus y minus k squared equals r squared. Parabola: x minus h squared equals four p times y minus k. Ellipse: x minus h squared over a squared plus y minus k squared over b squared equals one, with c squared equals a squared minus b squared. Hyperbola: the same with a minus sign, and c squared equals a squared plus b squared. Hyperbola asymptotes: y minus k equals plus or minus b over a times x minus h. Eccentricity e equals c over a. In the general form A x squared plus C y squared plus D x plus E y plus F equals zero: A equals C gives a circle, A C positive an ellipse, A C negative a hyperbola, and A C zero a parabola.*

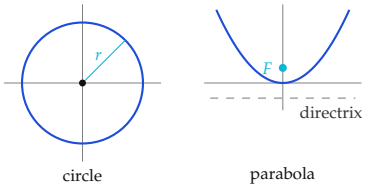
MEMORY AID Ellipse subtracts ($c^2 = a^2 - b^2$), hyperbola adds ($c^2 = a^2 + b^2$).

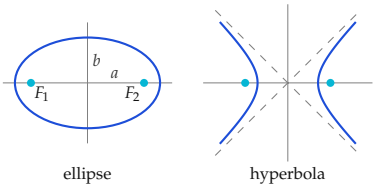
METHOD

1. Group the x terms and the y terms; move the constant to the right.
2. Complete the square in each A1.16 §03 Algebra I (factor out A or C first).
3. Divide to get 1 (or r^2) on the right and read off the center and axes. Example: $x^2 + y^2 - 4x + 6y - 12 = 0 \Rightarrow (x-2)^2 + (y+3)^2 = 25$: center $(2, -3)$, $r = 5$.

TERMS

(h, k) – center (or vertex of the parabola)
r – radius; a, b: semi-axes
c – center-to-focus distance
p – vertex-to-focus distance of a parabola
e – eccentricity (here not Euler's number)
$A, \ldots, F$ – coefficients of the general form (no xy term)





CONTEXT

- **Origin.** Apollonius of Perga (c. 200 BCE) named them as slices of a cone; Kepler showed planetary orbits are ellipses.
- **Use.** Orbits, reflectors, antennas, lenses.
- **Data science.** Contours of a bivariate normal density are ellipses (constant Mahalanobis distance); the L2 constraint in ridge regression is a circle, L1 a diamond.
- **Physical application.** Satellite dishes are parabolic so incoming signals reflect to the focus; planets orbit in ellipses with the Sun at one focus.

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05
Geometry

Shape, size, and space: points and lines, angles, triangles, polygons, circles, solids, and the transformations that move them. Geometry supplies the intuition behind distance, similarity, and dimensionality in data.

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G.1 Points, Lines, Rays, Segments & Planes

The undefined building blocks of Euclidean geometry. A point is a location, a line extends forever in both directions, and a plane is a flat surface extending forever; segments and rays are pieces of a line.

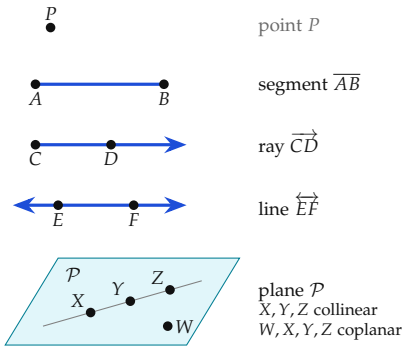
Segment	$\overline{AB}$ (length AB)	The part of a line between endpoints A and B; AB without the bar is its length.
Ray	$\overrightarrow{AB}$	Starts at endpoint A and runs through B forever; the first letter is always the endpoint.
Line	$\overleftrightarrow{AB}$	Extends forever both ways; two distinct points determine exactly one line.
Segment addition postulate	$AB + BC = AC$	Holds exactly when B lies on $\overline{AC}$, between A and C.
Collinearity (coordinates)	$m_{AB} = m_{BC}$	Three points are collinear when the slopes A1.6 §03 Algebra I agree (or all x are equal).
Plane postulate	A, B, C noncollinear $\Rightarrow$ exactly one plane	Three points not on one line determine a unique plane.
Intersections	$\ell_1 \cap \ell_2 =$ a point, $\varnothing$, or the whole line	Two lines meet once, never (parallel or skew), or coincide; two planes meet in a line or not at all.

STATED-EQUATION *Segment A B, written with a bar, has length A B. Ray A B starts at A and passes through B. Line A B extends both ways. If B is between A and C, then A B plus B C equals A C. Three points are collinear when the slopes A B and B C are equal. Three noncollinear points determine exactly one plane. Two lines in a plane meet in one point, in no point, or along the whole line.*

MEMORY AID Bar = stops at both ends (segment); one arrow = one way (ray); two arrows = both ways (line).

TERMS

Point – a location with no size; named by a capital letter
Line, plane – undefined terms: straight and flat, infinite
Collinear – lying on one line; *coplanar*: lying in one plane
Endpoint – where a segment or ray stops
Opposite rays – two rays from one point forming a line
Skew lines – not coplanar: never meet, not parallel
m_{AB} – slope of $\overleftrightarrow{AB}$ A1.6 §03 Algebra I
ℓ_1, ℓ_2 – lines; $\varnothing$ empty set A1.21 §03 Algebra I
$\cap$ – intersection: the points common to both A1.21 §03 Algebra I



CONTEXT

- **Origin.** Euclid, *Elements* Book I: "A point is that which has no part." Hilbert axiomatized the undefined terms (1899).
- **Use.** Stating any geometric figure precisely; CAD and drafting.
- **Data science.** A plane generalizes to a *hyperplane* $\mathbf{w} \cdot \mathbf{x} + b = 0$ in n dimensions, the decision boundary of logistic regression and SVMs A1.7 §03 Algebra I.
- **Physical application.** A three-legged stool never wobbles: three noncollinear feet always lie in one plane; four legs need not.

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G.2 Angles: Measure & Types

An angle is two rays (its sides) sharing an endpoint (its vertex). Its measure is the amount of turn from one side to the other, in degrees or radians; the size names the type.

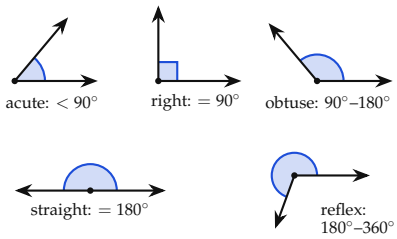
Acute angle	$0^\circ < \theta < 90^\circ$	Smaller than a right angle.
Right angle	$\theta = 90^\circ = \frac{\pi}{2}$	Marked with a small square; its sides are perpendicular ($\perp$).
Obtuse angle	$90^\circ < \theta < 180^\circ$	Larger than a right angle, smaller than a straight one.
Straight angle	$\theta = 180^\circ = \pi$	Two opposite rays: the angle is a line.
Reflex angle	$180^\circ < \theta < 360^\circ$	The "outside" of an angle: 360° minus the ordinary angle.
Full turn	$360^\circ = 2\pi$ rad	One revolution; convert with $\frac{\pi}{180^\circ}$ T.2 §06 Trigonometry.
Angle addition postulate	$m\angle ABC = m\angle ABD + m\angle DBC$	If D is inside $\angle ABC$, the two parts add to the whole.
Angle bisector	$m\angle ABD = m\angle DBC = \frac{1}{2}\, m\angle ABC$	Ray $\overrightarrow{BD}$ splits the angle into two equal halves.

STATED-EQUATION *An acute angle is between zero and ninety degrees, a right angle is ninety degrees, an obtuse angle is between ninety and one hundred eighty, a straight angle is one hundred eighty, and a reflex angle is between one hundred eighty and three hundred sixty. A full turn is three hundred sixty degrees, two pi radians. The measure of angle A B C equals angle A B D plus angle D B C. A bisector B D splits angle A B C into two equal angles, each one half of angle A B C.*

MEMORY AID *A-cute* angles are small; obtuse angles are wide; a straight angle is a straight line.

TERMS

Vertex – the common endpoint of the two rays
$\angle ABC$ – angle with vertex B (the middle letter)
$m\angle ABC$ – its measure; θ an angle measure
Degree, radian – $180^\circ = \pi$ rad T.2 §06 Trigonometry
$\perp$ – perpendicular: meeting at 90°
D – a point in the interior of the angle
Bisector – ray $\overrightarrow{BD}$ that divides an angle into two equal angles



CONTEXT

- **Origin.** Babylonian base-60 astronomy gave the 360° circle (c. 1500 BCE).
- **Use.** Navigation bearings, construction, machining.
- **Data science.** In cosine similarity T.21 §06 Trigonometry an acute angle between vectors means positive similarity, a right angle means none, an obtuse angle means opposite tendencies.
- **Physical application.** Clock hands at 3:00 form a right angle, at 6:00 a straight angle; at 1:00 the hands make 30° (acute) and 330° (reflex).

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G.3 Angle Relationships

Named pairs of angles formed by intersecting lines, and the equal or supplementary pairs created when a transversal crosses two parallel lines.

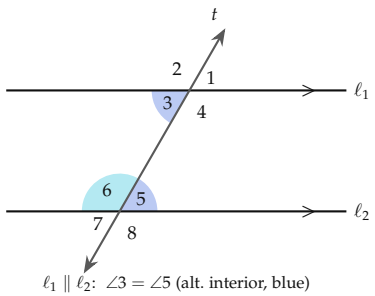
Complementary	$\alpha + \beta = 90^\circ$	Two angles that together make a right angle.
Supplementary	$\alpha + \beta = 180^\circ$	Two angles that together make a straight angle.
Linear pair	$\angle 1 + \angle 2 = 180^\circ$	Adjacent angles on a line are always supplementary.
Vertical angles	$\angle 1 = \angle 3$	Opposite angles at any intersection are equal.
Corresponding angles	$\angle 1 = \angle 5$	Same position at each crossing; equal exactly when the lines are parallel.
Alternate interior angles	$\angle 3 = \angle 5$	Between the parallels, on opposite sides of the transversal: equal.
Alternate exterior angles	$\angle 1 = \angle 7$	Outside the parallels, on opposite sides: equal.
Same-side interior angles	$\angle 4 + \angle 5 = 180^\circ$	Between the parallels, same side: supplementary.

STATED-EQUATION *Complementary angles: alpha plus beta equals ninety degrees. Supplementary: alpha plus beta equals one hundred eighty degrees. A linear pair sums to one hundred eighty degrees. Vertical angles are equal. With parallel lines cut by a transversal, corresponding angles are equal, alternate interior angles are equal, alternate exterior angles are equal, and same-side interior angles are supplementary.*

MEMORY AID Complementary makes a Corner (90°); Supplementary makes a Straight line (180°).

TERMS

α, β – angle measures
Vertical angles – opposite angles at an intersection
Transversal t – a line crossing two parallel lines $\ell_1 \parallel \ell_2$ A1.8 §03 Algebra I
Corresponding – same position at each intersection; equal
Alternate interior – opposite sides, between the parallels; equal
Same-side interior – supplementary (sum 180°)
Converse – any of these equalities forces $\ell_1 \parallel \ell_2$
Numbering – as in the figure: 1–4 at the upper crossing, 5–8 at the lower, counter-clockwise from upper right



$\ell_1 \parallel \ell_2$: $\angle 3 = \angle 5$ (alt. interior, blue)
$\angle 1 = \angle 5$ (corresp.), $\angle 5 + \angle 6 = 180^\circ$ (linear pair)

CONTEXT

- **Origin.** Euclid's *Elements*, Book I (c. 300 BCE).
- **Use.** Surveying, construction, proofs.
- **Data science.** The angle between two feature vectors underlies cosine similarity T.21 §06 Trigonometry: angle 0° means identical direction, 90° means unrelated.
- **Physical application.** A ramp at 20° to the ground meets a vertical wall at 70° (complementary angles).

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G.4 Classifying Triangles

Triangles are named by their sides (equilateral, isosceles, scalene) and by their largest angle (acute, right, obtuse). The name tells you which theorems apply.

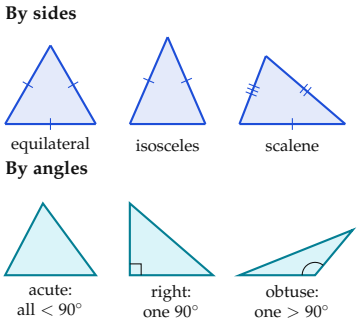
Equilateral	$a = b = c, \quad A = B = C = 60^\circ$	All sides and all angles equal.
Isosceles (base angles theorem)	$a = b \iff A = B$	Equal sides face equal angles, and conversely.
Scalene	$a,\ b,\ c$ all different	No equal sides, hence no equal angles.
By largest angle	$\max(A, B, C) < 90^\circ,\ = 90^\circ,\ > 90^\circ$	Acute, right, obtuse; test from the sides with G.7.
Side–angle order	$a > b \iff A > B$	The longest side is opposite the largest angle.
Equilateral area	$\text{Area} = \dfrac{\sqrt{3}}{4}\, s^2$	Half the base times the height $\frac{\sqrt{3}}{2}s$ G.8.

STATED-EQUATION *Equilateral: a equals b equals c and every angle is sixty degrees. Isosceles: a equals b if and only if angle A equals angle B. Scalene: all sides different. A triangle is acute, right or obtuse when its largest angle is less than, equal to, or greater than ninety degrees. The larger side is opposite the larger angle. An equilateral triangle of side s has area root three over four s squared.*

MEMORY AID *Isosceles* is Greek for "equal legs"; *scalene* for "uneven".

TERMS

a, b, c – side lengths, a opposite angle A, etc.
A, B, C – interior angles
Tick marks – equal number of ticks = equal sides
Legs, base – the equal sides and the third side of an isosceles triangle
s – side of the equilateral triangle
Inclusive – every equilateral triangle is also isosceles



CONTEXT

- **Origin.** Definitions 20–21 of Euclid's *Elements*, Book I.
- **Use.** Choosing the right formula: special right triangles, trigonometry T.1 §06 Trigonometry.
- **Data science.** Triangle meshes in graphics and finite-element models need near-equilateral elements; thin obtuse "slivers" make interpolation and solvers numerically unstable.
- **Physical application.** An isosceles roof truss spreads a ridge load equally to both walls.

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G.5 Triangle Angle Sum & Triangle Inequality

The interior angles of every (flat, Euclidean) triangle add to $180°$, and no side can be as long as the other two combined.

Angle sum	$A + B + C = 180°$	Any two angles determine the third.
Exterior angle theorem	$\angle_{\text{ext}} = \angle_{\text{remote}_1} + \angle_{\text{remote}_2}$	An exterior angle equals the sum of the two interior angles not next to it.
Triangle inequality	$\lvert a - b \rvert < c < a + b$	Each side is shorter than the sum, and longer than the difference, of the other two.

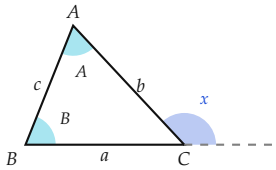
STATED-EQUATION *A plus B plus C equals one hundred eighty degrees. An exterior angle equals the sum of the two remote interior angles. The absolute value of a minus b is less than c, which is less than a plus b.*

METHOD

1. Sort the three lengths: $a \le b \le c$.
2. They form a triangle exactly when $a + b > c$ (the other two inequalities then hold automatically). Example: $3, 4, 8$: $3 + 4 = 7 < 8$, no triangle; $3, 4, 5$: $7 > 5$, a triangle.

TERMS

A, B, C – interior angles
a, b, c – side lengths (a opposite A, etc.)
Exterior angle – formed by extending one side
Remote interior – the two interior angles not adjacent to it
Triangle inequality – any side is shorter than the other two combined A1.11 §03 Algebra I



exterior $x = A + B$
$\lvert a - b \rvert < c < a + b$

CONTEXT

- **Origin.** Euclid, *Elements* I.20 and I.32.
- **Use.** Checking whether three lengths can form a triangle; finding missing angles.
- **Data science.** The triangle inequality $d(x, z) \le d(x, y) + d(y, z)$ defines a metric; ball trees and other nearest-neighbor indexes use it to prune whole regions of a search.
- **Physical application.** Truss bridges use triangles because three fixed side lengths fix the angles: the shape cannot deform.

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G.6 Triangle Congruence

Two triangles are congruent ($\cong$) when one can be moved onto the other by a rigid motion G.25. Five shortcuts prove it from three matching parts instead of all six.

SSS (side–side–side)	$AB = DE,\ BC = EF,\ CA = FD$	Three pairs of equal sides force $\triangle ABC \cong \triangle DEF$.
SAS (side–angle–side)	$AB = DE,\ \angle B = \angle E,\ BC = EF$	Two sides and the angle *between* them.
ASA (angle–side–angle)	$\angle A = \angle D,\ AB = DE,\ \angle B = \angle E$	Two angles and the side *between* them.
AAS (angle–angle–side)	$\angle A = \angle D,\ \angle B = \angle E,\ BC = EF$	Two angles and a non-included side; the third angle follows from G.5.
HL (hypotenuse–leg)	$\angle C = \angle F = 90°,\ AB = DE,\ BC = EF$	Right triangles only: by G.7 the other legs match too.
CPCTC	$\triangle ABC \cong \triangle DEF \Rightarrow AB = DE,\ \angle A = \angle D,\ \ldots$	Once congruent, every pair of corresponding parts is equal.

STATED-EQUATION *S S S, S A S, A S A, A A S and, for right triangles, H L each prove two triangles congruent. Corresponding parts of congruent triangles are congruent. S S S: all three pairs of sides are equal. S A S: two sides and the included angle. A S A: two angles and the included side. A A S: two angles and a non-included side. H L: equal hypotenuses and one equal pair of legs in right triangles.*

MEMORY AID The valid tests: SSS, SAS, ASA, AAS, HL. SSA ("the donkey rule") and AAA are *not* tests.

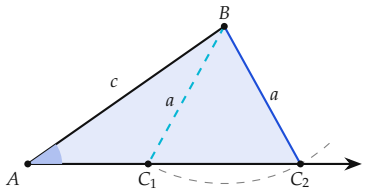
METHOD

1. Mark the given equal parts on both triangles (ticks for sides, arcs for angles).
2. Add free facts: a shared side, vertical angles G.3, angles from parallel lines.
3. Read the three marked parts in order around the triangle and match SSS, SAS, ASA, AAS or HL; write the vertices in corresponding order.
4. Conclude $\cong$, then use CPCTC. Example: a diagonal $\overline{AC}$ of parallelogram $ABCD$: $AB = CD$, $BC = DA$, $AC = CA \Rightarrow$ SSS.

DECISION POINT **Why SSA fails.** Fix $\angle A$, the adjacent side $c = AB$ and the opposite side $a = BC$: a circle of radius a about B can cut the base ray twice (figure), giving two different triangles. This is the ambiguous case of the law of sines T.16 §06 Trigonometry. SSA does work when $a \geq c$ (only one intersection on the ray) and when the angle is right, which is exactly HL. AAA fixes shape but not size: it proves similarity G.10, not congruence.

TERMS

$\cong$ – congruent: same size and shape
Corresponding parts – matched by the vertex order: $A \leftrightarrow D$, $B \leftrightarrow E$, $C \leftrightarrow F$
Included angle / side – the one between the two named parts
Hypotenuse, leg – see G.7
Reflexive property – a shared side equals itself
$\triangle ABC$ – triangle with vertices A, B, C
CPCTC – Corresponding Parts of Congruent Triangles are Congruent



same $\angle A$, c, $a \Rightarrow$ two triangles

CONTEXT

- **Origin.** Euclid, *Elements* I.4 (SAS), I.8 (SSS), I.26 (ASA, AAS).
- **Use.** Geometric proofs; interchangeable parts in manufacturing.
- **Data science.** Rigid motions preserve every distance, so distance-based methods (k-NN, k-means G.12) give identical results on a rotated or shifted copy of the data.
- **Physical application.** Triangle bracing makes frames rigid: by SSS, three fixed bar lengths allow only one shape.

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G.7 Pythagorean Theorem

In a right triangle the square on the hypotenuse equals the sum of the squares on the legs; comparing c^2 with $a^2 + b^2$ classifies any triangle.

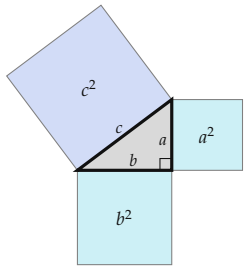
Pythagorean theorem	$a^2 + b^2 = c^2$	Right triangles only; c is the side opposite the right angle.
Missing side	$c = \sqrt{a^2 + b^2}, \quad a = \sqrt{c^2 - b^2}$	Add squares for the hypotenuse, subtract for a leg.
Converse	$c^2 = a^2 + b^2 \Rightarrow$ right angle opposite c	Use it to test for a right angle from three lengths.
Acute / obtuse test	$c^2 < a^2 + b^2 \Rightarrow$ acute, $\quad c^2 > a^2 + b^2 \Rightarrow$ obtuse	Apply to the longest side c; a special case of the law of cosines T.17 §06 Trigonometry.
Euclid's formula for triples	$(m^2 - n^2,\ 2mn,\ m^2 + n^2)$	Integers $m > n > 0$ generate every primitive triple: $m = 2, n = 1$ gives $3, 4, 5$.

STATED-EQUATION *a squared plus b squared equals c squared. If c squared is less than a squared plus b squared, the triangle is acute; if greater, obtuse; if equal, right. Pythagorean triples are m squared minus n squared, two m n, and m squared plus n squared. A missing hypotenuse is c equals the square root of a squared plus b squared; a missing leg is a equals the square root of c squared minus b squared.*

MEMORY AID Common triples: 3–4–5, 5–12–13, 8–15–17, 7–24–25.

TERMS

a, b – legs of a right triangle
c – hypotenuse (side opposite the right angle; the longest side)
Converse – classifies any triangle by its longest side c
Triples – integer solutions: $(3, 4, 5)$, $(5, 12, 13)$, $(8, 15, 17)$
m, n – positive integers, $m > n$; primitive if coprime and not both odd



$3^2 + 4^2 = 9 + 16 = 25 = 5^2$

CONTEXT

- **Origin.** Named for Pythagoras (6th c. BCE); Babylonian tablet Plimpton 322 lists triples a millennium earlier.
- **Use.** Distances, diagonals, construction squareness checks.
- **Data science.** Euclidean (L2) distance $\sqrt{\sum (x_i - y_i)^2}$ G.12 is the theorem in n dimensions, the default metric in k-NN and k-means.
- **Physical application.** A 3–4–5 rope loop lets builders lay out a perfect right-angle corner without a square.

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G.8 Special Right Triangles

Half a square and half an equilateral triangle give two right triangles whose side ratios are exact, so their lengths need no calculator.

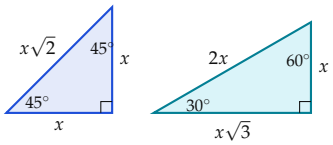
45–45–90 triangle	$x : x : x\sqrt{2}$	Isosceles right triangle: hypotenuse is leg $\times\sqrt{2}$.
30–60–90 triangle	$x : x\sqrt{3} : 2x$	Short leg (opposite 30°) is half the hypotenuse.
Square diagonal	$d = s\sqrt{2}$	The diagonal splits a square into two 45–45–90 triangles.
Equilateral height	$h = \frac{\sqrt{3}}{2}\,s$	The altitude splits an equilateral triangle into two 30–60–90 triangles.

STATED-EQUATION *A forty-five forty-five ninety triangle has sides x, x, x root two. A thirty sixty ninety triangle has sides x, x root three, two x. A square's diagonal is s root two; an equilateral triangle's height is root three over two times s.*

MEMORY AID 30–60–90: "1, root 3, 2" — the short leg is half the hypotenuse.

TERMS

x – length of the shortest side (or each leg)
$x\sqrt{2}$ – hypotenuse of the isosceles right triangle
$x\sqrt{3}$ – side opposite 60°
$2x$ – hypotenuse of the 30–60–90
s, d, h – side, diagonal, height; radicals A1.4 §03 Algebra I



CONTEXT

- **Origin.** Half of a square and half of an equilateral triangle, via G.7.
- **Use.** Exact trigonometric values on the unit circle T.4 §06 Trigonometry without a calculator.
- **Data science.** A 45° rotation G.24 of standardized features maps (x, y) to sums and differences scaled by $\frac{\sqrt{2}}{2}$, common in rotated-coordinate tricks.
- **Physical application.** A square tile's diagonal is side$\times\sqrt{2}$; an equilateral roof truss has height $\frac{\sqrt{3}}{2}$ of its side.

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G.9 Similar Triangles & Scale Factor

Similar figures have the same shape: equal angles and proportional lengths. One scale factor k controls lengths; areas scale by k^2 and volumes by k^3.

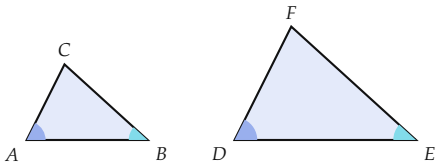
Similarity ratio	$\triangle ABC \sim \triangle DEF \implies \dfrac{AB}{DE} = \dfrac{BC}{EF} = \dfrac{AC}{DF} = k$	Corresponding sides (matched by vertex order) share one ratio.
Perimeter ratio	$\dfrac{P_1}{P_2} = k$	Every length scales by k: perimeters, altitudes, medians.
Area ratio	$\dfrac{\text{Area}_1}{\text{Area}_2} = k^2$	Area has two length dimensions.
Volume ratio	$\dfrac{\text{Vol}_1}{\text{Vol}_2} = k^3$	Volume has three length dimensions (similar solids).

STATED-EQUATION *If triangle A B C is similar to triangle D E F, then A B over D E equals B C over E F equals A C over D F equals k. The perimeter ratio is k, the area ratio is k squared, and the volume ratio is k cubed.*

MEMORY AID Length k, area k^2, volume k^3.

TERMS

$\sim$ – "is similar to": same shape, possibly different size
k – scale factor between corresponding lengths
P_1, P_2 – perimeters of the two figures
Tests – AA, SAS, SSS similarity G.10
Proportions – solved by cross-multiplication A1.19 §03 Algebra I



$\triangle ABC \sim \triangle DEF$

$\dfrac{DE}{AB} = \dfrac{EF}{BC} = \dfrac{DF}{AC} = 1.5$

areas $\times\, 1.5^2 = 2.25$

CONTEXT

- **Origin.** Thales measured the height of the Great Pyramid from its shadow (c. 600 BCE).
- **Use.** Maps, models, indirect measurement.
- **Data science.** Scaling laws: doubling image resolution multiplies pixel count (and convolution compute) by $k^2 = 4$; doubling a 3-D grid multiplies voxels by 8.
- **Physical application.** Doubling every dimension of a cup multiplies its volume (and mass of drink) by 8 but its surface by only 4.

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G.10 Similarity Criteria & Proportional Segments

Three shortcuts prove two triangles similar; similarity then yields the classic proportional-segment theorems.

AA similarity	$\angle A = \angle D,\ \angle B = \angle E \Rightarrow \triangle ABC \sim \triangle DEF$	Two equal angles suffice (the third then matches G.5).
SSS similarity	$\dfrac{AB}{DE} = \dfrac{BC}{EF} = \dfrac{CA}{FD}$	All three side ratios equal.
SAS similarity	$\dfrac{AB}{DE} = \dfrac{BC}{EF},\ \angle B = \angle E$	Two side ratios equal and the included angles equal.
Side-splitter theorem	$DE \parallel BC \iff \dfrac{AD}{DB} = \dfrac{AE}{EC}$	A line parallel to one side cuts the other two proportionally.
Angle bisector theorem	$\dfrac{BD}{DC} = \dfrac{AB}{AC}$	The bisector of $\angle A$ divides $\overline{BC}$ in the ratio of the adjacent sides.
Geometric mean (altitude)	$h^2 = pq$	The altitude to the hypotenuse is the geometric mean of the two pieces it cuts.

STATED-EQUATION *If two angles of one triangle equal two angles of another, the triangles are similar. If all three side ratios are equal, or two side ratios are equal and the included angles are equal, they are similar. A line parallel to B C cuts A B and A C proportionally. The angle bisector from A divides B C in the ratio A B to A C. The altitude to the hypotenuse squared equals p times q.*

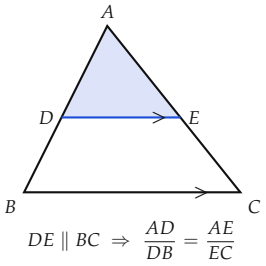
METHOD

1. Find two pairs of equal angles: a shared angle, vertical angles G.3, angles from parallel lines.
2. Write the similarity with vertices in corresponding order.
3. Set up one proportion and cross-multiply A1.19 §03 Algebra I. Example: a 2 m pole casts a 3 m shadow while a tree casts 12 m; $\frac{h}{2} = \frac{12}{3}$, so $h = 8$ m.

DECISION POINT **Which criterion?** Angles visible (parallel lines, a shared vertex, right angles) → AA, by far the most common. Only lengths known → SSS∼: sort each triangle's sides and compare smallest/smallest, middle/middle, largest/largest. Two sides and the angle between → SAS∼. As with congruence, SSA does not prove similarity.

TERMS

$\sim$ – similar G.9
D, E – points on sides $\overline{AB}$, $\overline{AC}$ (side-splitter); D on $\overline{BC}$ (bisector)
h – altitude from the right angle to the hypotenuse
p, q – the two parts of the hypotenuse
Geometric mean – $\sqrt{pq}$ of p and q
D (bisector theorem) – where the bisector of $\angle A$ meets $\overline{BC}$
$\parallel$ – parallel A1.8 §03 Algebra I



$DE \parallel BC \Rightarrow \dfrac{AD}{DB} = \dfrac{AE}{EC}$

CONTEXT

- **Origin.** Thales' intercept theorem; Euclid, *Elements* Book VI.
- **Use.** Indirect measurement, maps, perspective drawing.
- **Data science.** Scale-invariant comparisons: cosine similarity T.21 §06 Trigonometry ignores vector length and compares direction only, just as similarity ignores size and keeps shape.
- **Physical application.** A pinhole camera: image height over object height equals image distance over object distance, by AA similarity.

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G.11 Triangle Centers

Every triangle has four classical centers, each the common point of three special lines: medians, perpendicular bisectors, angle bisectors, or altitudes.

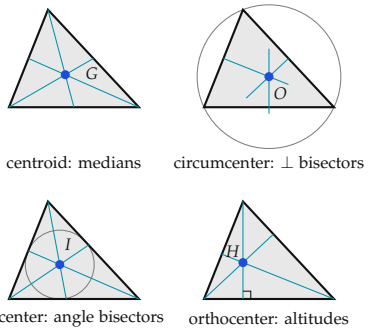
Centroid	$G = \left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}\right)$	Medians meet here, $\frac{2}{3}$ of the way from each vertex; the balance point.
Circumcenter	$OA = OB = OC = R$	Perpendicular bisectors meet here: center of the circle through all three vertices.
Incenter	$r = \dfrac{\text{Area}}{s}$	Angle bisectors meet here: center of the circle tangent to all three sides.
Orthocenter	$AH \perp BC,\ \ BH \perp CA,\ \ CH \perp AB$	Altitudes (lines through a vertex, perpendicular to the opposite side) meet here.
Euler line	$G = \frac{1}{3}H + \frac{2}{3}O$	H, G, O are collinear and $HG = 2\,GO$.
Circumradius	$R = \dfrac{abc}{4 \cdot \text{Area}}$	Radius of the circumscribed circle from the sides and area.

STATED-EQUATION *The centroid G is the average of the three vertices; it lies two thirds of the way along each median. The circumcenter O is equidistant from the vertices, at distance R. The incenter is equidistant from the sides, at distance r equals area over s. The altitudes meet at the orthocenter H. H, G and O lie on the Euler line, with G equal to one third H plus two thirds O. R equals a b c over four times the area.*

MEMORY AID Medians → Centroid, Perpendicular bisectors → Circumcenter, Angle bisectors → Incenter, Altitudes → Orthocenter: "MC–PC–AI–AO".

TERMS

Median – vertex to midpoint of the opposite side G.12
Perpendicular bisector – $\perp$ to a side through its midpoint
Angle bisector – splits an angle in half G.2
Altitude – from a vertex, $\perp$ to the opposite side
(x_i, y_i) – vertex coordinates; a, b, c sides
R, r – circumradius, inradius; s semi-perimeter G.16
Where – G, I always inside; O, H inside (acute), on (right), outside (obtuse)



centroid: medians | circumcenter: $\perp$ bisectors
incenter: angle bisectors | orthocenter: altitudes

CONTEXT

- **Origin.** Euler (1765) proved that H, G, O are collinear.
- **Use.** Balancing plates; siting a facility equidistant from three towns (circumcenter).
- **Data science.** The centroid is the mean of the points, exactly the k-means cluster center; Delaunay triangulation (meshes, interpolation) is defined by empty circumcircles.
- **Physical application.** A triangular metal plate balances on a pin placed at its centroid.

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G.12 Distance & Midpoint Formulas

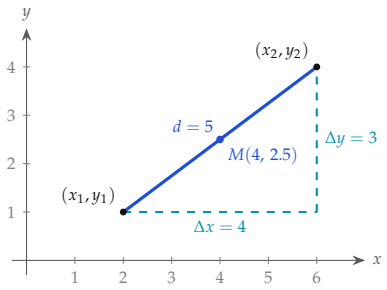
The Pythagorean theorem in coordinates: the straight-line distance between two points, the point halfway between them, and their generalizations.

Distance formula	$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$	Hypotenuse of the right triangle with legs Δx, Δy G.7.
Midpoint formula	$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$	Average the coordinates.
n-D Euclidean distance	$d(\mathbf{p}, \mathbf{q}) = \sqrt{\sum_{i=1}^{n} (p_i - q_i)^2}$	Same formula with one squared difference per coordinate.
Section formula	$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$	The point dividing the segment in the ratio $m : n$ from (x_1, y_1); $m = n$ gives the midpoint.
Manhattan distance	$d_1 = \lvert x_2 - x_1 \rvert + \lvert y_2 - y_1 \rvert$	Grid (L1) distance: walking along city blocks.

STATED-EQUATION *d equals the square root of x two minus x one squared plus y two minus y one squared. The midpoint M equals x one plus x two over two, y one plus y two over two. In n dimensions, d equals the square root of the sum of p i minus q i squared. The point dividing the segment in ratio m to n is the weighted average with weights n and m. The Manhattan distance d one equals the absolute value of x two minus x one plus the absolute value of y two minus y one.*

TERMS

$(x_1, y_1), (x_2, y_2)$ – two points
d – straight-line distance, from G.7
M – midpoint: average of coordinates
$\mathbf{p}, \mathbf{q}$ – points in n dimensions (feature vectors)
$m : n$ – ratio in which P divides the segment
$|\cdot|$ – absolute value A1.11 §03 Algebra I
P – the point dividing the segment
d_1 – Manhattan (taxicab) distance: along the grid
$\sum$ – summation over the n coordinates A2.21 §04 Algebra II



CONTEXT

- **Origin.** Descartes' coordinates applied to the Pythagorean theorem.
- **Use.** Navigation, mapping, geometry proofs.
- **Data science.** k-NN and k-means measure similarity by Euclidean distance; the k-means centroid generalizes the midpoint (the mean of the points in a cluster). L1 distance is more robust to outliers.
- **Physical application.** The straight-line distance between two GPS points on a local map is the distance formula.

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G.13 Polygon Angle Sums & Diagonals

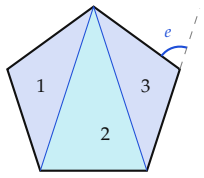
Any n-sided polygon splits into $n-2$ triangles from one vertex, which fixes its interior angle sum; the exterior angles always total one full turn.

Interior angle sum	$\Sigma_{\text{int}} = (n-2)\cdot 180^\circ$	$n-2$ triangles of 180° each G.5.
Each interior angle (regular)	$\dfrac{(n-2)\cdot 180^\circ}{n}$	Divide the sum equally among n equal angles.
Exterior angle sum	$\Sigma_{\text{ext}} = 360^\circ$	Walking around any convex polygon turns you once.
Each exterior angle (regular)	$e = \dfrac{360^\circ}{n}$	Reverse it to find n: $e = 40^\circ \Rightarrow n = 9$.
Number of diagonals	$\dfrac{n(n-3)}{2}$	Each vertex joins $n-3$ others; divide by 2 for double counting.

STATED-EQUATION *The interior angle sum equals n minus two times one hundred eighty degrees. Each interior angle of a regular polygon equals n minus two times one eighty, over n. The exterior angles sum to three hundred sixty degrees; each exterior angle of a regular polygon is three sixty over n. The number of diagonals equals n times n minus three, over two.*

TERMS

n – number of sides (and vertices)
Regular – all sides and angles equal
Derivation – split into $n-2$ triangles from one vertex G.5
Exterior angle e – supplement of the interior angle G.3
Convex – every interior angle $< 180^\circ$
Σ_{int}, Σ_{ext} – sums of the interior and exterior angles
Diagonal – segment joining two non-adjacent vertices



$n = 5$: $(5-2)\cdot 180^\circ = 540^\circ$

CONTEXT

- **Origin.** Triangulation arguments in Greek geometry.
- **Use.** Tiling, architecture, mesh design.
- **Data science.** The diagonal count is $\binom{n}{2} - n$ S.8 §13 Statistics: all pairs minus the edges. The same counting sizes a complete graph of n nodes in network analysis.
- **Physical application.** Honeycomb cells are regular hexagons with 120° angles that tile the plane with no gaps.

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G.14 Quadrilateral Properties

Special quadrilaterals form a family tree: each shape inherits every property of the shapes above it (a square is a rectangle and a rhombus, both parallelograms).

Parallelogram	$AB \parallel CD,\ AB = CD,\ \angle A = \angle C$	Opposite sides parallel and equal, opposite angles equal, consecutive angles supplementary, diagonals bisect each other.
Rectangle	$\angle A = 90^\circ,\ \ AC = BD$	A parallelogram with four right angles; its diagonals are equal.
Rhombus	$AB = BC = CD = DA,\ \ AC \perp BD$	A parallelogram with four equal sides; diagonals perpendicular and bisect the angles.
Square	rectangle $\cap$ rhombus	All of the above: equal sides, right angles, equal perpendicular diagonals.
Trapezoid midsegment	$m = \frac{1}{2}(b_1 + b_2)$	One pair of parallel sides (bases); the midsegment is parallel to them. Isosceles: equal legs, base angles and diagonals.
Kite	$AB = AD,\ \ CB = CD,\ \ AC \perp BD$	Two pairs of equal adjacent sides; the diagonals are perpendicular and one bisects the other.
Angle sum	$\angle A + \angle B + \angle C + \angle D = 360^\circ$	Two triangles G.13.
Area from diagonals	$\text{Area} = \frac{1}{2}\, d_1 d_2$	Rhombus, kite, square: any quadrilateral with perpendicular diagonals.

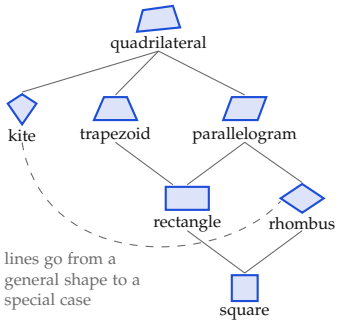
STATED-EQUATION *A parallelogram has opposite sides parallel and equal and opposite angles equal. A rectangle adds right angles and equal diagonals; a rhombus adds equal sides and perpendicular diagonals; a square is both. A trapezoid's midsegment is half the sum of the bases. A kite has two pairs of equal adjacent sides and perpendicular diagonals. The angles of any quadrilateral sum to three hundred sixty degrees. With perpendicular diagonals, area is one half d one d two.*

MEMORY AID Every square is a rectangle and a rhombus, but not the other way round.

DECISION POINT **Proving a parallelogram**: any one of — both pairs of opposite sides parallel; both pairs equal; one pair both parallel and equal; diagonals bisect each other; both pairs of opposite angles equal. **Then specialize**: equal diagonals ⇒ rectangle; perpendicular diagonals ⇒ rhombus; both ⇒ square. In coordinates use slopes, distances and midpoints G.20.

TERMS

$ABCD$ – vertices in order around the figure
$AC,\ BD$ – diagonals; d_1, d_2 their lengths
b_1, b_2 – parallel sides (bases) of a trapezoid; m midsegment
$\parallel,\ \perp$ – parallel, perpendicular A1.8 §03 Algebra I
Legs – the non-parallel sides of a trapezoid
$\cap$ – "and": a square is in both families A1.21 §03 Algebra I



CONTEXT

- **Origin.** Euclid, *Elements* I.34 (properties of parallelograms).
- **Use.** Framing, tiling, layout and drafting.
- **Data science.** Object detectors predict axis-aligned rectangles (bounding boxes); intersection-over-union is computed from rectangle overlaps.
- **Physical application.** Carpenters check that a frame is square by measuring both diagonals: equal diagonals mean a rectangle.

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Shapes at a Glance | *Key Topic G.K1*

Nearly every area formula is "base × height" in disguise, and nearly every volume formula is "base area × height", divided by 3 when the solid comes to a point. Perimeter is a length (units), area is square units, volume is cubic units. Heights are always *perpendicular* to the base; slant heights ℓ appear only in surface areas.

Plane figures: perimeter and area

	Figure	Perimeter	Area	Notes
	Square	$4s$	s^2	diagonal $s\sqrt{2}$ G.8
	Rectangle	$2(l+w)$	lw	diagonal $\sqrt{l^2+w^2}$
	Parallelogram	$2(a+b)$	bh	h is perpendicular, *not* the slanted side a
	Triangle	$a+b+c$	$\frac{1}{2}bh$	also $\frac{1}{2}ab\sin C$ T.16 §06 Trigonometry and Heron G.16
	Trapezoid	sum of sides	$\frac{1}{2}(b_1+b_2)h$	average base × height
	Rhombus, kite	$4s$ (rhombus)	$\frac{1}{2}d_1d_2$	diagonals perpendicular G.14
	Regular n-gon	$P=ns$	$\frac{1}{2}aP$	a = apothem (center to side midpoint)
	Circle	$2\pi r=\pi d$	πr^2	$d=2r$ G.17
	Sector	$2r+r\theta$	$\frac{1}{2}r^2\theta$	θ in radians; fraction $\frac{\theta^\circ}{360^\circ}$ of the circle

Common mistakes and worked examples

Common mistakes

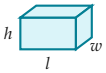
Using the slanted side as the height of a parallelogram, triangle or cone.
Using the diameter where the radius belongs (πd^2 is four times too big).
Degrees in $\frac{1}{2}r^2\theta$ or $r\theta$: these need radians T.2 §06 Trigonometry.
Counting both bases of an open container, or the glued faces of a composite solid.
Mixing units: convert first; $1\,\text{m}^2 = 10\,000\,\text{cm}^2$, $1\,\text{m}^3 = 1000\,\text{L}$.

Worked mini-examples

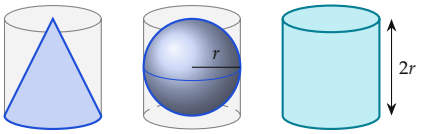
Composite area. Gable wall: rectangle 6×4 plus a triangle of base 6, height 2: $24+\frac{1}{2}(6)(2)=30$.
Sector. $r=4$, $\theta=\frac{\pi}{4}$: arc $=4\cdot\frac{\pi}{4}=\pi$; area $=\frac{1}{2}(16)\frac{\pi}{4}=2\pi$.
Cone. $r=3$, $h=4$: $\ell=5$; $V=\frac{1}{3}\pi(9)(4)=12\pi$; $SA=9\pi+15\pi=24\pi$.
Hollow solid. Pipe, outer radius 3, inner 2, length 10: $V=\pi(9-4)(10)=50\pi$.

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Solids: volume and surface area

	Solid	Volume	Surface area	Notes
	Cube	s^3	$6s^2$	space diagonal $s\sqrt{3}$
	Rectangular prism	lwh	$2(lw + lh + wh)$	space diagonal $\sqrt{l^2 + w^2 + h^2}$ G.21
	Any prism	Bh	$2B + Ph$	B base area, P base perimeter
	Cylinder	$\pi r^2 h$	$2\pi r^2 + 2\pi rh$	lateral $2\pi rh$ unrolls to a rectangle
	Pyramid	$\frac{1}{3}Bh$	$B + \frac{1}{2}P\ell$	ℓ = slant height of a face (regular pyramid)
	Cone	$\frac{1}{3}\pi r^2 h$	$\pi r^2 + \pi r\ell$	$\ell = \sqrt{r^2 + h^2}$ G.22
	Sphere	$\frac{4}{3}\pi r^3$	$4\pi r^2$	four great-circle areas
	Hemisphere	$\frac{2}{3}\pi r^3$	$3\pi r^2$	curved $2\pi r^2$ plus flat disk πr^2

Archimedes' cone : sphere : cylinder



cone: $\frac{2}{3}\pi r^3$ sphere: $\frac{4}{3}\pi r^3$ cylinder: $2\pi r^3$

1 : 2 : 3

Fit a sphere of radius r snugly inside a cylinder of radius r and height $2r$, and a cone with the same base and height. Their volumes are $\frac{2}{3}\pi r^3$, $\frac{4}{3}\pi r^3$ and $2\pi r^3$: exactly 1 : 2 : 3.

So *cone + sphere = cylinder*, the cone is $\frac{1}{3}$ of the cylinder G.22, and the sphere is $\frac{2}{3}$ of it. The sphere's area $4\pi r^2$ also equals the cylinder's lateral area $2\pi r \cdot 2r$: the basis of equal-area map projections.

Patterns that save memorizing

Families

Prism / cylinder: $V = Bh$ (stack copies of the base).
Pyramid / cone: $V = \frac{1}{3}Bh$ (a point-topped solid holds one third of its prism).
Sphere: $\frac{2}{3}$ of the circumscribed cylinder, in volume *and* in surface area (Archimedes).
Derivative link: $\frac{d}{dr}(\pi r^2) = 2\pi r$ and $\frac{d}{dr}(\frac{4}{3}\pi r^3) = 4\pi r^2$: the boundary is the rate of growth C1.9 §07 Calculus I.
Scaling by k: lengths $\times k$, areas $\times k^2$, volumes $\times k^3$ G.9.

DECISION POINT **Which height?** Volume and area use the perpendicular height h; lateral and surface areas of pyramids and cones use the slant height ℓ measured along a face. Convert between them with the Pythagorean theorem G.7. **Composite figures**: split into the shapes above, add the pieces, and subtract holes.

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G.15 Area of Plane Figures

Area measures the flat space inside a figure in square units. Most formulas reduce to base $\times$ perpendicular height, halved for triangles.

Rectangle	$A = lw$	Length times width; a square is s^2.
Parallelogram	$A = bh$	Slide a triangle across to make a rectangle; h is perpendicular to b.
Triangle	$A = \frac{1}{2}\,bh$	Half of a parallelogram; more triangle formulas in G.16.
Trapezoid	$A = \frac{1}{2}\,(b_1 + b_2)\,h$	Average of the parallel sides times the height.
Rhombus / kite	$A = \frac{1}{2}\,d_1 d_2$	Half the product of the perpendicular diagonals.
Regular polygon	$A = \frac{1}{2}\,aP$	n triangles of height a (the apothem) around the center.

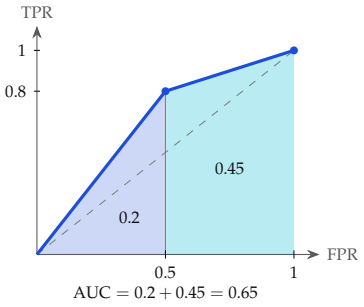
STATED-EQUATION *Rectangle: l w. Parallelogram: b h. Triangle: one half b h. Trapezoid: one half times b one plus b two, times h. Rhombus or kite: one half d one d two. Regular polygon: one half the apothem times the perimeter.*

METHOD

1. Split a composite figure into rectangles, triangles, trapezoids and circles.
2. Compute each area with its own base and perpendicular height.
3. Add the pieces and subtract holes. Example: a gable wall, a 6×4 rectangle plus a triangle of base 6 and height 2: $24 + 6 = 30$.

TERMS

b, h – base and perpendicular height
l, w – length and width
b_1, b_2 – parallel sides of a trapezoid
d_1, d_2 – diagonals G.14
a, P – apothem (center to side midpoint), perimeter
Heron, SAS area – G.16; $\frac{1}{2}ab\sin C$ T.16 §06 Trigonometry



CONTEXT

- **Origin.** Egyptian Rhind papyrus (c. 1650 BCE); Heron of Alexandria (c. 60 CE).
- **Use.** Land, materials, floor plans.
- **Data science.** ROC AUC is usually computed with the trapezoid rule, summing trapezoids between successive (FPR, TPR) points, a preview of numeric integration C1.30 §07 Calculus I.
- **Physical application.** Estimating the paint needed for a gable wall: rectangle plus triangle area.

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G.16 Triangle Area: Heron's Formula & Alternatives

A triangle's area can be computed from whatever is known: base and height, two sides and their angle, all three sides (Heron), its inscribed or circumscribed circle, or its vertex coordinates.

Base–height	$\text{Area} = \frac{1}{2}\,bh$	Needs the perpendicular height to the chosen base.
SAS (trigonometric) area	$\text{Area} = \frac{1}{2}\,ab\sin C$	Two sides and the included angle T.16 §06 Trigonometry.
Heron's formula	$\text{Area} = \sqrt{s(s-a)(s-b)(s-c)},\quad s = \dfrac{a+b+c}{2}$	Three sides only; no angles or heights needed.
Inradius form	$\text{Area} = rs$	Three triangles of height r from the incenter G.11.
Circumradius form	$\text{Area} = \dfrac{abc}{4R}$	From the circumradius R and the three sides.
Coordinate (shoelace) formula	$\text{Area} = \frac{1}{2}\,\lvert x_1(y_2-y_3) + x_2(y_3-y_1) + x_3(y_1-y_2)\rvert$	Half the absolute determinant of the vertices A2.25 §04 Algebra II; sign gives orientation.

STATED-EQUATION *Area equals one half b h, or one half a b sine C. Heron's formula: the square root of s times s minus a times s minus b times s minus c, where s equals a plus b plus c over two. Area equals r s, or a b c over four R. From coordinates, area is one half the absolute value of x one times y two minus y three, plus x two times y three minus y one, plus x three times y one minus y two.*

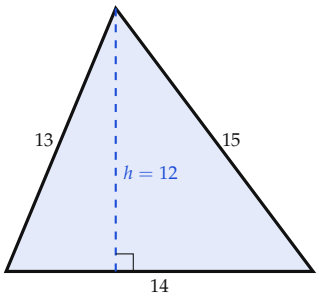
METHOD

1. Compute the semi-perimeter $s = \frac{a+b+c}{2}$.
2. Compute $s-a$, $s-b$, $s-c$; all must be positive (otherwise no triangle G.5).
3. Multiply the four numbers and take the square root. Example: $3, 4, 5$: $s = 6$, $\sqrt{6\cdot 3\cdot 2\cdot 1} = 6$.

DECISION POINT **Which formula?** Base and height known → $\frac{1}{2}bh$. Two sides and the angle between → $\frac{1}{2}ab\sin C$. Three sides → Heron. Vertex coordinates → shoelace (it extends to any simple polygon). Inradius or circumradius given → rs or $\frac{abc}{4R}$.

TERMS

a, b, c – side lengths; C the angle between a and b
s – semi-perimeter (half the perimeter)
b, h – base and perpendicular height
r, R – inradius and circumradius G.11
(x_i, y_i) – vertex coordinates



$s = 21,\ A = \sqrt{21\cdot 8\cdot 7\cdot 6} = 84$
$= \frac{1}{2}(14)(12)$

CONTEXT

- **Origin.** Heron of Alexandria, *Metrica* (c. 60 CE); likely known to Archimedes.
- **Use.** Surveying a plot from its side lengths alone.
- **Data science.** The shoelace formula computes polygon areas in GIS (geofences, land parcels) and mesh areas in graphics; its sign tests clockwise vs counterclockwise order.
- **Physical application.** A triangular plot with sides 13, 14 and 15 m covers $\sqrt{21\cdot 8\cdot 7\cdot 6} = 84\,\text{m}^2$.

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G.17 Circles: Circumference, Area, Arcs & Sectors

Every circle is a scaled copy of every other, so all its measurements are π times powers of the radius; arcs and sectors take the fraction $\frac{\theta}{2\pi}$ of the whole.

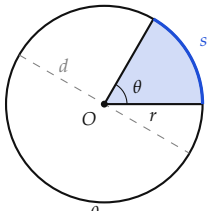
Circumference	$C = 2\pi r = \pi d$	Distance around; π is defined as C/d.
Area	$A = \pi r^2$	Square the radius, not the diameter.
Arc length	$s = r\theta = \dfrac{\theta^\circ}{360^\circ} \cdot 2\pi r$	The simple form $r\theta$ needs θ in radians T.2 §06 Trigonometry.
Sector area	$A_{\text{sector}} = \frac{1}{2} r^2\theta = \dfrac{\theta^\circ}{360^\circ} \cdot \pi r^2$	A "pizza slice": the same fraction of the disk.
Segment area	$A_{\text{seg}} = \frac{1}{2} r^2(\theta - \sin\theta)$	Region between a chord and its arc: sector minus triangle.
Chord length	$c = 2r \sin\dfrac{\theta}{2}$	Straight-line distance across the arc.

STATED-EQUATION *C equals two pi r, which equals pi d. A equals pi r squared. Arc length s equals r theta, or theta degrees over three sixty times two pi r. Sector area equals one half r squared theta, or theta over three sixty times pi r squared. A segment has area one half r squared times theta minus sine theta. A chord has length two r sine theta over two.*

MEMORY AID "Cherry pie is delicious" ($C = \pi d$); "apple pies are too" ($A = \pi r^2$).

TERMS

r, d – radius and diameter ($d = 2r$)
π – ≈ 3.14159, ratio of circumference to diameter
θ – central angle in radians T.2 §06 Trigonometry (or degrees, θ°)
s – arc length; c chord length
sin – sine T.1 §06 Trigonometry



sector $= \dfrac{\theta}{2\pi}$ of the disk

CONTEXT

- **Origin.** Archimedes bounded π with 96-sided polygons (c. 250 BCE).
- **Use.** Wheels, pipes, rotation, circular tracks.
- **Data science.** Pie and donut charts encode values as sector angles; people misjudge angles, one reason bar charts are preferred for precise comparisons.
- **Physical application.** A pizza slice with a 45° angle is one-eighth of the pizza's area.

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G.18 Circle Theorems

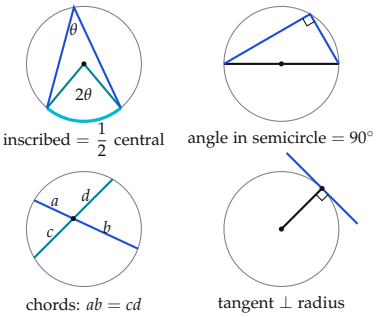
Angles and lengths in a circle are tied to the arcs they cut off and to the center, which makes many unknown angles and segments solvable at sight.

Inscribed angle theorem	$\angle_{\text{inscribed}} = \frac{1}{2}\angle_{\text{central}}$	Same arc: the angle at the circle is half the angle at the center.
Thales' theorem	angle in a semicircle $= 90^\circ$	An inscribed angle on a diameter is right.
Tangent–radius theorem	tangent $\perp$ radius	At the point of tangency.
Intersecting chords	$ab = cd$	Products of the pieces of two crossing chords are equal.
Power of a point (outside)	$PT^2 = PA \cdot PB$	Tangent length squared equals external part times whole secant.
Two tangents	$PT_1 = PT_2$	Tangent segments from one outside point are equal.
Cyclic quadrilateral	$\angle A + \angle C = 180^\circ$	Opposite angles of a quadrilateral inscribed in a circle are supplementary.

STATED-EQUATION *An inscribed angle equals one half the central angle on the same arc. An angle inscribed in a semicircle is ninety degrees. A tangent is perpendicular to the radius. Intersecting chords: a b equals c d. From an outside point P, P T squared equals P A times P B, and the two tangent segments are equal. Opposite angles of a cyclic quadrilateral sum to one hundred eighty degrees.*

TERMS

Inscribed angle – vertex on the circle, subtending an arc
Central angle – vertex at the center, same arc
Thales' theorem – inscribed angle on a diameter is right
a, b, c, d – segments of two chords crossing inside the circle
P – outside point; T, T_1, T_2 tangency points
A, B – where a secant from P meets the circle ($PA < PB$)
Cyclic – all four vertices on one circle



CONTEXT

- **Origin.** Euclid, *Elements* Book III; Thales (c. 600 BCE).
- **Use.** Constructions, optics, locating centers.
- **Data science.** "Tangent $\perp$ radius" generalizes to gradients: the gradient is perpendicular to the level curve of a loss, so gradient descent crosses contours at right angles C1.26 §07 Calculus I.
- **Physical application.** A carpenter finds the center of a round table by drawing two chords and their perpendicular bisectors.

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G.19 Equation of a Circle

A circle is every point at distance r from a center (h,k); writing that distance with the distance formula G.12 gives its equation.

Standard form	$(x-h)^2+(y-k)^2=r^2$	Center (h,k) and radius r read off directly; watch the signs.
Centered at the origin	$x^2+y^2=r^2$	The case $h=k=0$; $r=1$ is the unit circle T.4 §06 Trigonometry.
General form	$x^2+y^2+Dx+Ey+F=0$	Center $\left(-\frac{D}{2},-\frac{E}{2}\right)$, $r^2=\frac{D^2+E^2}{4}-F$ (must be >0).
Point position test	$(x_0-h)^2+(y_0-k)^2 \lessgtr r^2$	Less: inside; equal: on; greater: outside.
Tangent line at a point	$(x_0-h)(x-h)+(y_0-k)(y-k)=r^2$	Line through (x_0,y_0) on the circle, perpendicular to the radius G.18.
Parametric form	$x=h+r\cos t,\ \ y=k+r\sin t$	Traces the circle once as t runs over $[0,2\pi)$.

STATED-EQUATION *The circle with center h, k and radius r is x minus h squared plus y minus k squared equals r squared. Centered at the origin: x squared plus y squared equals r squared. General form: x squared plus y squared plus D x plus E y plus F equals zero. A point is inside, on, or outside as its squared distance is less than, equal to, or greater than r squared. The tangent line at the point x zero, y zero is x zero minus h times x minus h, plus y zero minus k times y minus k, equals r squared. Parametrically, x equals h plus r cos t and y equals k plus r sin t.*

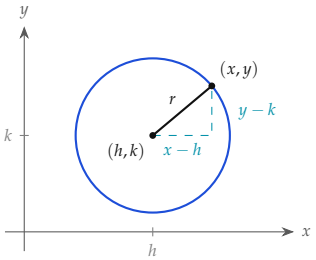
MEMORY AID The signs flip: $(x-2)^2+(y+3)^2=16$ has center $(2,-3)$.

METHOD

1. Group the x-terms and the y-terms; move F to the right.
2. Complete the square in x and in y A1.16 §03 Algebra I, adding the same amounts to the right.
3. Read the center and radius. Example: $x^2+y^2-4x+6y-3=0 \Rightarrow (x-2)^2+(y+3)^2=16$: center $(2,-3)$, $r=4$.

TERMS

(h,k) – center; $r>0$ radius
(x,y) – any point on the circle
D, E, F – constants of the general form
(x_0,y_0) – a given point
t – parameter (angle in radians) T.2 §06 Trigonometry
Conic – the circle is a special ellipse A2.26 §04 Algebra II
$\lessgtr$ – less than, equal to, or greater than: inside, on, outside
cos, sin – cosine and sine T.1 §06 Trigonometry



CONTEXT

- **Origin.** Descartes, *La Géométrie* (1637).
- **Use.** GPS trilateration, design, collision detection.
- **Data science.** Radius queries ("all points within r") are discs: DBSCAN ε-neighborhoods, k-NN balls. Ridge regression's constraint $\|\mathbf{w}\|^2 \le t$ is a circle (ball) ML.8 §15 Machine Learning.
- **Physical application.** A cell tower at $(2,1)$ with a 5 km range covers $(x-2)^2+(y-1)^2 \le 25$.

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G.20 Coordinate Geometry Proofs

Place a figure on axes with general (letter) coordinates, then prove its properties with algebra: slopes for parallel and perpendicular, distances for lengths, midpoints for bisection.

Parallel sides	$m_1 = m_2$	Equal slopes A1.8 §03 Algebra I.
Perpendicular sides	$m_1 m_2 = -1$	Negative-reciprocal slopes (or one horizontal, one vertical).
Equal lengths	$d_1 = d_2$	Distance formula G.12; compare d^2 to avoid roots.
Bisection	$M_{AC} = M_{BD}$	Two segments bisect each other when their midpoints coincide.
Collinear points	$m_{AB} = m_{BC}$	Equal slopes through a shared point G.1.
Right-angle (dot product) test	$(x_2 - x_1)(x_3 - x_1) + (y_2 - y_1)(y_3 - y_1) = 0$	Right angle at (x_1, y_1); works for vertical sides too T.21 §06 Trigonometry.

STATED-EQUATION *Parallel: m one equals m two. Perpendicular: m one times m two equals minus one. Equal lengths: equal distances. Segments bisect each other when their midpoints coincide. Points are collinear when slopes A B and B C are equal. A right angle at point one when the dot product of the two side vectors is zero.*

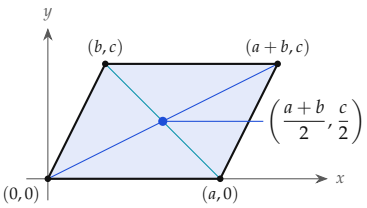
METHOD

1. Place the figure conveniently: a vertex at the origin, a side along the x-axis, letters such as $(a, 0)$, (b, c) for generality.
2. Translate the claim into slope, distance or midpoint statements.
3. Compute symbolically and simplify.
4. Conclude in words. Example (figure): for the parallelogram $(0,0), (a,0), (a+b,c), (b,c)$ both diagonals have midpoint $\left(\frac{a+b}{2}, \frac{c}{2}\right)$, so they bisect each other.

DECISION POINT **Choosing coordinates**: use as few letters as possible and exploit symmetry (an isosceles triangle as $(-a,0), (a,0), (0,b)$; a right angle at the origin along the axes). Specific numbers give an example, not a proof. **Choosing the test**: parallel or perpendicular → slopes; congruent or isosceles → distances; bisects → midpoints.

TERMS

m – slope A1.6 §03 Algebra I
d – distance G.12
M_{AC} – midpoint of $\overline{AC}$
a, b, c – arbitrary constants (general position)
(x_i, y_i) – vertex coordinates



CONTEXT

- **Origin.** Descartes and Fermat merged algebra and geometry (17th c.).
- **Use.** Proving properties of polygons; computer-checked geometry.
- **Data science.** Computational geometry (convex hulls, segment-intersection tests in GIS) uses the same orientation tests, written with cross products to avoid dividing by zero.
- **Physical application.** Checking from corner coordinates that a surveyed lot is a rectangle: perpendicular sides and equal diagonals.

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G.21 Prisms & Cylinders

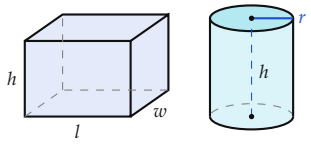
A prism or cylinder has two congruent parallel bases joined straight across, so its volume is base area times height.

Prism / cylinder volume	$V = Bh$	Base area times the perpendicular height between the bases.
Rectangular prism volume	$V = lwh$	Box: $B = lw$.
Rectangular prism surface area	$SA = 2(lw + lh + wh)$	Three pairs of equal rectangular faces.
Cylinder volume	$V = \pi r^2 h$	Circular base $B = \pi r^2$ G.17.
Cylinder surface area	$SA = 2\pi r^2 + 2\pi rh$	Two disks plus the label, a $2\pi r \times h$ rectangle.
Cube	$V = s^3, \quad SA = 6s^2$	All edges equal s.
Space diagonal	$d = \sqrt{l^2 + w^2 + h^2}$	Pythagoras twice G.7.

STATED-EQUATION *Volume equals base area times height. Rectangular prism: V equals l w h, surface area equals two times l w plus l h plus w h. Cylinder: V equals pi r squared h, surface area equals two pi r squared plus two pi r h. Cube: s cubed and six s squared. The space diagonal is the square root of l squared plus w squared plus h squared.*

TERMS

B – area of the base G.15
h – height between the parallel bases
l, w – length and width; s cube edge
r – radius of the cylinder's base
SA – surface area: total area of all faces G.23



CONTEXT

- **Origin.** Cavalieri's principle (1635): equal cross-sections at every height give equal volumes.
- **Use.** Containers, tanks, packaging.
- **Data science.** Tensor memory is a "box volume": a batch of shape $32 \times 224 \times 224 \times 3$ in float32 needs $32 \cdot 224 \cdot 224 \cdot 3 \cdot 4$ bytes.
- **Physical application.** A soda can with $r = 3.3$ cm and $h = 12.2$ cm holds $\pi(3.3)^2(12.2) \approx 417$ cm^3, about 355 mL of drink plus headspace.

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G.22 Pyramids, Cones & Spheres

Solids that taper to a point hold exactly one third of the prism or cylinder with the same base and height; the sphere is fixed entirely by its radius.

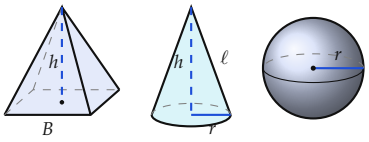
Pyramid volume	$V = \frac{1}{3}Bh$	One third of the prism on the same base.
Cone volume	$V = \frac{1}{3}\pi r^2 h$	One third of the matching cylinder.
Cone surface area	$SA = \pi r^2 + \pi r\ell$	Base disk plus lateral area $\pi r\ell$.
Sphere volume	$V = \frac{4}{3}\pi r^3$	Two thirds of the circumscribed cylinder.
Sphere surface area	$SA = 4\pi r^2$	Four great-circle areas.
Slant height	$\ell = \sqrt{r^2 + h^2}$	From the right triangle inside the cone G.7.
Frustum volume	$V = \frac{1}{3}\pi h\,(R^2 + Rr + r^2)$	A cone with its tip cut off: radii R and r.

STATED-EQUATION *Pyramid: V equals one third B h. Cone: V equals one third pi r squared h, surface area equals pi r squared plus pi r l, with slant height l equal to the square root of r squared plus h squared. Sphere: V equals four thirds pi r cubed, surface area equals four pi r squared. Frustum: one third pi h times R squared plus R r plus r squared.*

MEMORY AID "A cone is one-third of the cylinder" with the same base and height.

TERMS

B, h – base area and height G.21
ℓ – slant height, $\ell = \sqrt{r^2 + h^2}$ G.7
$\frac{1}{3}$ – a point-topped solid holds one third of its prism
r – radius; R larger radius of a frustum
Derivation – by integration, volumes of revolution C2.8 §08 Calculus II



CONTEXT

- **Origin.** Archimedes proved the sphere is $\frac{2}{3}$ of its circumscribed cylinder and asked for it on his tombstone.
- **Use.** Storage, physics, astronomy.
- **Data science.** Curse of dimensionality: in n dimensions the fraction of a ball's volume within radius $1 - \varepsilon$ is $(1-\varepsilon)^n$, so almost all volume sits near the surface.
- **Physical application.** Volume of a cup: a tapered cup is a frustum, $V = \frac{1}{3}\pi h\,(R^2 + Rr + r^2)$; a cup 10 cm tall with rim radius 4 cm and base radius 3 cm holds $\frac{1}{3}\pi(10)(16 + 12 + 9) \approx 387$ mL.

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G.23 Surface Area of Solids

Surface area is the total area of every face of a solid. Unfold the solid into a flat net and add the pieces; lateral area leaves out the bases.

Prism	$SA = 2B + Ph$	Two bases plus the lateral rectangles (base perimeter $\times$ height).
Cylinder lateral area	$L = 2\pi rh$	Unrolls to a $2\pi r$ by h rectangle.
Regular pyramid	$SA = B + \frac{1}{2}P\ell$	Base plus triangular faces of slant height ℓ.
Cone lateral area	$L = \pi r\ell$	Unrolls to a sector of radius ℓ and arc $2\pi r$.
Sphere	$SA = 4\pi r^2$	Equals $\frac{dV}{dr}$ of $\frac{4}{3}\pi r^3$ C1.9 §07 Calculus I.
Hemisphere (closed)	$SA = 3\pi r^2$	Curved half $2\pi r^2$ plus the flat disk πr^2.
Euler's polyhedron formula	$V - E + F = 2$	Vertices minus edges plus faces of any convex polyhedron; checks a net.

STATED-EQUATION *A prism's surface area is two B plus P h. A cylinder's lateral area is two pi r h; a cone's is pi r l. A regular pyramid has surface area B plus one half P l. A sphere has four pi r squared; a closed hemisphere three pi r squared. For a convex polyhedron, V minus E plus F equals two.*

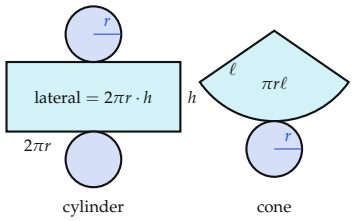
METHOD

1. Sketch the net: list every face that is actually there.
2. Find each face's area G.15.
3. Add them, leaving out open or glued faces. Example: a closed $2 \times 3 \times 4$ box: $2(6 + 8 + 12) = 52$.

DECISION POINT **Lateral or total?** Open containers (cup, pipe, tent without a floor) omit their bases; composite solids omit faces glued together. ℓ **or** h**?** Surface formulas use the slant height ℓ along a face, volume uses the perpendicular height h; convert with G.7.

TERMS

B, P – base area and base perimeter
h – perpendicular height; ℓ slant height
L – lateral area (sides only)
r – radius
V, E, F – numbers of vertices, edges, faces
Net – a flat pattern that folds into the solid



CONTEXT

- **Origin.** Archimedes, *On the Sphere and Cylinder*; Euler's formula (1750).
- **Use.** Paint, packaging material, heat loss.
- **Data science.** Renderers compute a mesh's area as a sum of triangle areas G.16 to sample points uniformly on a 3-D surface (point clouds, lighting).
- **Physical application.** Cells stay small: nutrient intake grows with surface (r^2) but demand with volume (r^3).

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G.24 Coordinate Transformations

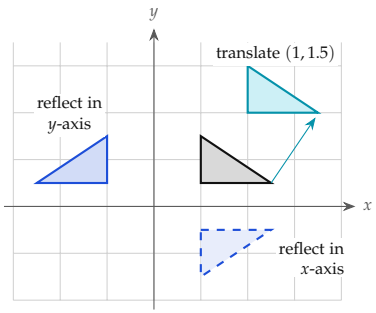
A transformation moves every point of the plane by a rule. Translations, reflections and rotations are rigid (they keep size and shape); dilations keep shape only.

Translation	$(x,y) \mapsto (x+a,\ y+b)$	Slide every point by the same vector (a,b).
Reflection in the x-axis	$(x,y) \mapsto (x,-y)$	Flip top to bottom.
Reflection in the y-axis	$(x,y) \mapsto (-x,y)$	Flip left to right.
Reflection in the line $y=x$	$(x,y) \mapsto (y,x)$	Swap the coordinates (this is how inverse functions are graphed).
Reflection in the line $y=-x$	$(x,y) \mapsto (-y,-x)$	Swap and negate both.
Rotation matrix	$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$	Rotation by θ counterclockwise about the origin.
Dilation	$(x,y) \mapsto (kx,\ ky)$	Scale by k from the origin: similar image G.9.
Dilation about a center	$(x,y) \mapsto \big(p+k(x-p),\ q+k(y-q)\big)$	Scale distances from the center (p,q) by k.

STATED-EQUATION *Translation: x, y maps to x plus a, y plus b. Reflection in the x-axis negates y; in the y-axis negates x. Reflection in the line y equals x swaps the coordinates; in y equals minus x swaps and negates them. Rotation by theta counterclockwise: x prime, y prime equals the matrix cosine theta, minus sine theta; sine theta, cosine theta, times x, y. Dilation: k x, k y; about a center p, q, distances from the center are multiplied by k.*

TERMS

a, b – horizontal and vertical shift
θ – rotation angle T.1 §06 Trigonometry
Matrix form – matrix–vector product A2.24 §04 Algebra II; linear maps LA.9 §09 Linear Algebra
Reflections – x-axis: $(x,-y)$; y-axis: $(-x,y)$
k – scale factor G.9; (p,q) center of dilation
(x',y') – image of (x,y) (prime = after)
Function graphs – same moves on $y=f(x)$ A2.5 §04 Algebra II
$\mapsto$ – "maps to": point $\to$ image



CONTEXT

- **Origin.** Felix Klein's Erlangen program (1872) defined geometries by the transformations they preserve.
- **Use.** Computer graphics, robotics, symmetry.
- **Data science.** Image augmentation applies flips, rotations, and shifts; PCA rotates the axes to align with variance; point-cloud registration finds the best rigid rotation.
- **Physical application.** Video-game and robotics engines rotate objects with exactly this matrix every frame.

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G.25 Rotations, Rigid Motions & Compositions

Quarter-turn rotations about the origin are coordinate swaps with sign changes. Rigid motions preserve distance, and composing reflections produces translations and rotations.

Rotation $90°$ counterclockwise	$(x, y) \mapsto (-y,\ x)$	Swap, then negate the new first coordinate.
Rotation $180°$	$(x, y) \mapsto (-x,\ -y)$	Negate both; also called point reflection through the origin.
Rotation $270°$ counterclockwise	$(x, y) \mapsto (y,\ -x)$	The same as $90°$ clockwise.
Rotation about a point	$\begin{bmatrix} x' \\ y' \end{bmatrix} = R_\theta \begin{bmatrix} x-a \\ y-b \end{bmatrix} + \begin{bmatrix} a \\ b \end{bmatrix}$	Translate the center (a, b) to the origin, rotate G.24, translate back.
Two reflections, parallel mirrors	translation by $2d$	Perpendicular to the mirrors, twice their distance apart.
Two reflections, crossing mirrors	rotation by 2φ	About the crossing point, twice the angle between the mirrors.
Isometry	$d(P', Q') = d(P, Q)$	Rigid motions preserve every distance (and angle): images are congruent G.6.
Similarity transformation	$d(P', Q') = k\, d(P, Q)$	An isometry followed by a dilation: images are similar G.9.

STATED-EQUATION *Rotating ninety degrees counterclockwise sends x, y to minus y, x; one hundred eighty degrees to minus x, minus y; two hundred seventy degrees to y, minus x. To rotate about a, b, subtract a, b, rotate, and add a, b back. Two reflections in parallel mirrors make a translation by twice their distance; in crossing mirrors, a rotation by twice their angle. Isometries keep distances; similarity transformations multiply them by k.*

MEMORY AID $90°$ counterclockwise: "swap and negate the first". Two mirrors double: $2d$ or 2φ.

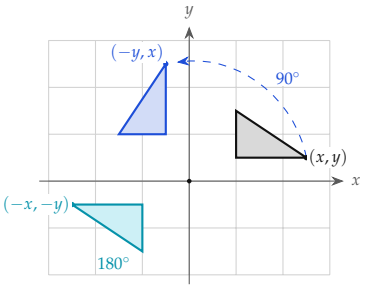
METHOD

1. Write the rule for each transformation in the sequence.
2. Apply the first rule to every vertex, then the next rule to the results (for $T_2 \circ T_1$, T_1 acts first).
3. Example: reflect $(2, 1)$ in the y-axis $\to (-2, 1)$, then rotate $90° \to (-1, -2)$.

DECISION POINT **Order matters**: rotating then translating differs from translating then rotating, just as matrix products do not commute A2.24 §04 Algebra II. **Identifying an unknown rigid motion**: if the vertex order stays counterclockwise (orientation kept) it is a translation or a rotation (a rotation has a fixed point); if orientation flips it is a reflection (a fixed line) or a glide reflection.

TERMS

R_θ – rotation matrix G.24
(a, b) – center of rotation
d – distance between parallel mirrors; φ angle between crossing mirrors
P', Q' – images of points P, Q
Isometry – distance-preserving map (rigid motion)
Glide reflection – reflection followed by a translation along the mirror
$T_2 \circ T_1$ – composition: T_1 first A2.1 §04 Algebra II
$d(P, Q)$ – distance from P to Q G.12
k – scale factor G.9



CONTEXT

- **Origin.** Classification of plane isometries (Chasles, 1831); the 17 wallpaper groups (Fedorov, 1891).
- **Use.** Animation, robotics, crystallography, pattern design.
- **Data science.** Rotations by multiples of $90°$ are exact on pixel grids (no interpolation), so they are cheap, lossless augmentations; equivariant CNNs build these symmetries into the network.
- **Physical application.** A kaleidoscope's two mirrors at $60°$ generate rotations by $120°$, giving its repeating pattern.

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06
Trigonometry

Angles become functions: ratios in right triangles extend to the unit circle, to periodic waves, to vectors and complex numbers, and (through e^x) to the hyperbolic functions. This is the mathematics of rotation, cycles, and similarity.

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Angles, Ratios & the Unit Circle

T.1 Right-Triangle Trigonometric Ratios

In a right triangle, each acute angle fixes the ratios between the sides, no matter how large the triangle is. The six trigonometric functions name those ratios.

Sine (SOH)	$\sin\theta = \dfrac{\text{opp}}{\text{hyp}}$	Side across from θ over the longest side; always between 0 and 1 for an acute angle.
Cosine (CAH)	$\cos\theta = \dfrac{\text{adj}}{\text{hyp}}$	Side next to θ (not the hypotenuse) over the hypotenuse.
Tangent (TOA)	$\tan\theta = \dfrac{\text{opp}}{\text{adj}} = \dfrac{\sin\theta}{\cos\theta}$	Ratio of the two legs: the slope of the hypotenuse A1.6 §03 Algebra I.
Cosecant	$\csc\theta = \dfrac{1}{\sin\theta} = \dfrac{\text{hyp}}{\text{opp}}$	Reciprocal of sine: flip SOH.
Secant	$\sec\theta = \dfrac{1}{\cos\theta} = \dfrac{\text{hyp}}{\text{adj}}$	Reciprocal of cosine: flip CAH.
Cotangent	$\cot\theta = \dfrac{1}{\tan\theta} = \dfrac{\text{adj}}{\text{opp}}$	Reciprocal of tangent: flip TOA.

STATED-EQUATION *Sine theta equals opposite over hypotenuse. Cosine theta equals adjacent over hypotenuse. Tangent theta equals opposite over adjacent, which equals sine over cosine. Cosecant is one over sine, secant is one over cosine, cotangent is one over tangent. Equivalently, cosecant equals hypotenuse over opposite, secant equals hypotenuse over adjacent, and cotangent equals adjacent over opposite.*

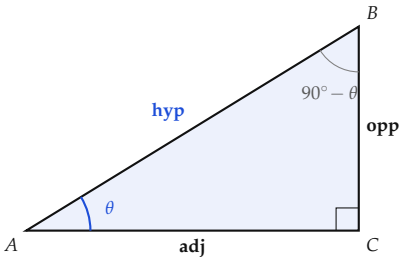
MEMORY AID SOH-CAH-TOA: Sine = Opposite/Hypotenuse, Cosine = Adjacent/Hypotenuse, Tangent = Opposite/Adjacent. Each "co-" reciprocal pairs with a non-"co" function: csc↔sin, sec↔cos.

METHOD

1. Label the sides relative to the angle you know or want: opp, adj, hyp.
2. Pick the ratio that contains the known side and the unknown side (one angle + one side known), or the two known sides (angle unknown).
3. Solve the equation; for an unknown angle use the inverse function T.14. Example: elevation 30° at distance 10 m gives height $10\tan 30^\circ = \frac{10}{\sqrt{3}} \approx 5.77$ m.

TERMS

θ – an acute angle of a right triangle
opp, adj – legs opposite and adjacent to θ
hyp – hypotenuse, opposite the right angle G.7 §05 Geometry
csc, sec, cot – the reciprocal ratios (cosecant, secant, cotangent)
Angle of elevation / depression – angle measured up / down from the horizontal
Mnemonic – SOH–CAH–TOA



CONTEXT

- **Origin.** Hipparchus' chord tables (c. 140 BCE); the sine (*jya*) from Aryabhata (499 CE).
- **Use.** Heights, slopes, navigation, engineering.
- **Data science.** Resolving a vector into components, $x = r\cos\theta$, $y = r\sin\theta$ T.19; sine and cosine features encode cyclical variables T.4.
- **Physical application.** Surveyors find a building's height from a distance and an angle of elevation: height $= d\tan\theta$.

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T.2 Radians, Degrees & Angular Speed

A radian measures an angle by the arc it cuts off on a circle, in units of the radius. Radians make arc length, area and speed formulas simple, and calculus requires them.

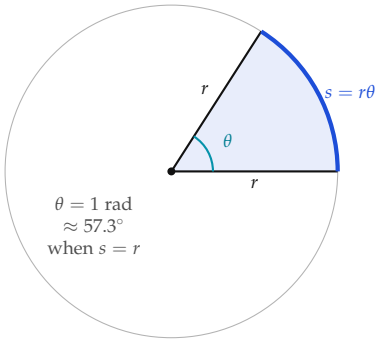
Conversion constant	$\pi \text{ rad} = 180^\circ$	One half-turn; a full turn is 2π rad $= 360^\circ$.
Degrees → radians	$\theta_{\text{rad}} = \theta^\circ \cdot \dfrac{\pi}{180^\circ}$	$150^\circ = \frac{5\pi}{6}$.
Radians → degrees	$\theta^\circ = \theta_{\text{rad}} \cdot \dfrac{180^\circ}{\pi}$	1 rad $\approx 57.3^\circ$.
Arc length	$s = r\theta$	Length of the arc cut off by θ (radians only) G.17 §05 Geometry.
Sector area	$A = \frac{1}{2}r^2\theta$	Fraction $\frac{\theta}{2\pi}$ of the disk area πr^2.
Angular speed	$\omega = \dfrac{\theta}{t} = 2\pi f$	Radians turned per unit time; f = revolutions per unit time.
Linear speed	$v = r\omega$	Speed of a point at radius r: farther out moves faster.

STATED-EQUATION *Pi radians equals one hundred eighty degrees. Theta in radians equals theta in degrees times pi over one eighty. Arc length s equals r theta. Sector area equals one half r squared theta. Angular speed omega equals theta over t, which equals two pi f. Linear speed v equals r omega. Theta in degrees equals theta in radians times one eighty over pi.*

MEMORY AID Degrees to radians: multiply by $\frac{\pi}{180}$ ("pi on top to go to radians").

TERMS

Radian – angle whose arc equals the radius
θ – angle; in radians for s, A, ω
r – radius; s arc length; A sector area
ω – angular speed (radians per unit time)
f – frequency (revolutions per unit time)
v – linear speed of a point at radius r
t – elapsed time



CONTEXT

- **Origin.** Roger Cotes (1714); the word "radian" is due to James Thomson (1873).
- **Use.** Calculus requires radians: $\frac{d}{dx}\sin x = \cos x$ only in radians C1.14 §07 Calculus I.
- **Data science.** Cyclical feature engineering: hour h maps to the angle $\frac{2\pi h}{24}$ before taking sine and cosine.
- **Physical application.** A bicycle wheel of radius 0.35 m spinning at 10 rad/s moves the bike at $v = r\omega = 3.5$ m/s.

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T.3 Standard Position, Coterminal & Reference Angles

An angle in standard position starts on the positive x-axis and turns to its terminal side. Angles with the same terminal side share all trig values; the reference angle reduces any angle to an acute one.

Standard position	$\theta > 0$: ↺, $\theta < 0$: ↻	Vertex at the origin, initial side on the $+x$-axis; positive turns counter-clockwise.
Coterminal angles	$\theta + 2\pi k \;\; (\theta + 360^\circ k),\ k \in \mathbb{Z}$	Add or subtract full turns: 150°, 510° and -210° share a terminal side.
Reference angle by quadrant	$\theta' = \begin{cases} \theta, & \text{QI} \\ \pi - \theta, & \text{QII} \end{cases} \quad \theta' = \begin{cases} \theta - \pi, & \text{QIII} \\ 2\pi - \theta, & \text{QIV} \end{cases} \quad (0 \le \theta < 2\pi)$	Acute angle between the terminal side and the *x-axis* (never the y-axis).
Reference-angle rule	$f(\theta) = \pm f(\theta')$	Same size as at θ'; the sign comes from the quadrant (ASTC) T.4.

STATED-EQUATION *An angle in standard position starts on the positive x-axis; positive angles turn counter-clockwise. Theta plus two pi k is coterminal with theta for every integer k. The reference angle is theta, pi minus theta, theta minus pi, or two pi minus theta in quadrants one to four. Any trig function of theta equals plus or minus the same function of the reference angle.*

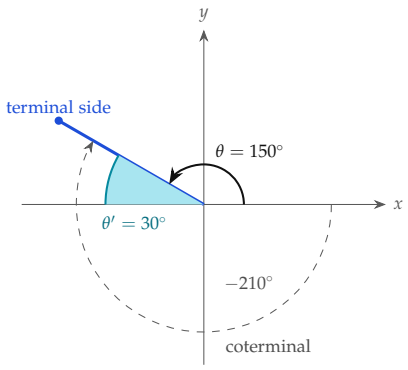
MEMORY AID Reference angle = "how far to the nearest x-axis".

METHOD

1. Add or subtract 2π until $0 \le \theta < 2\pi$ (coterminal angle).
2. Read the quadrant and compute θ' with the table above.
3. Evaluate at θ', then attach the quadrant sign. Example: $\sin\frac{4\pi}{3}$: QIII, $\theta' = \frac{\pi}{3}$, sine negative, so $-\frac{\sqrt{3}}{2}$.

TERMS

Initial / terminal side – the ray the angle starts from / ends on
k – any integer ($k \in \mathbb{Z}$, M.78 §01 Math Signs)
θ' – reference angle, $0 \le \theta' \le \frac{\pi}{2}$
Quadrantal angle – terminal side on an axis: $0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$
f – any of the six trig functions



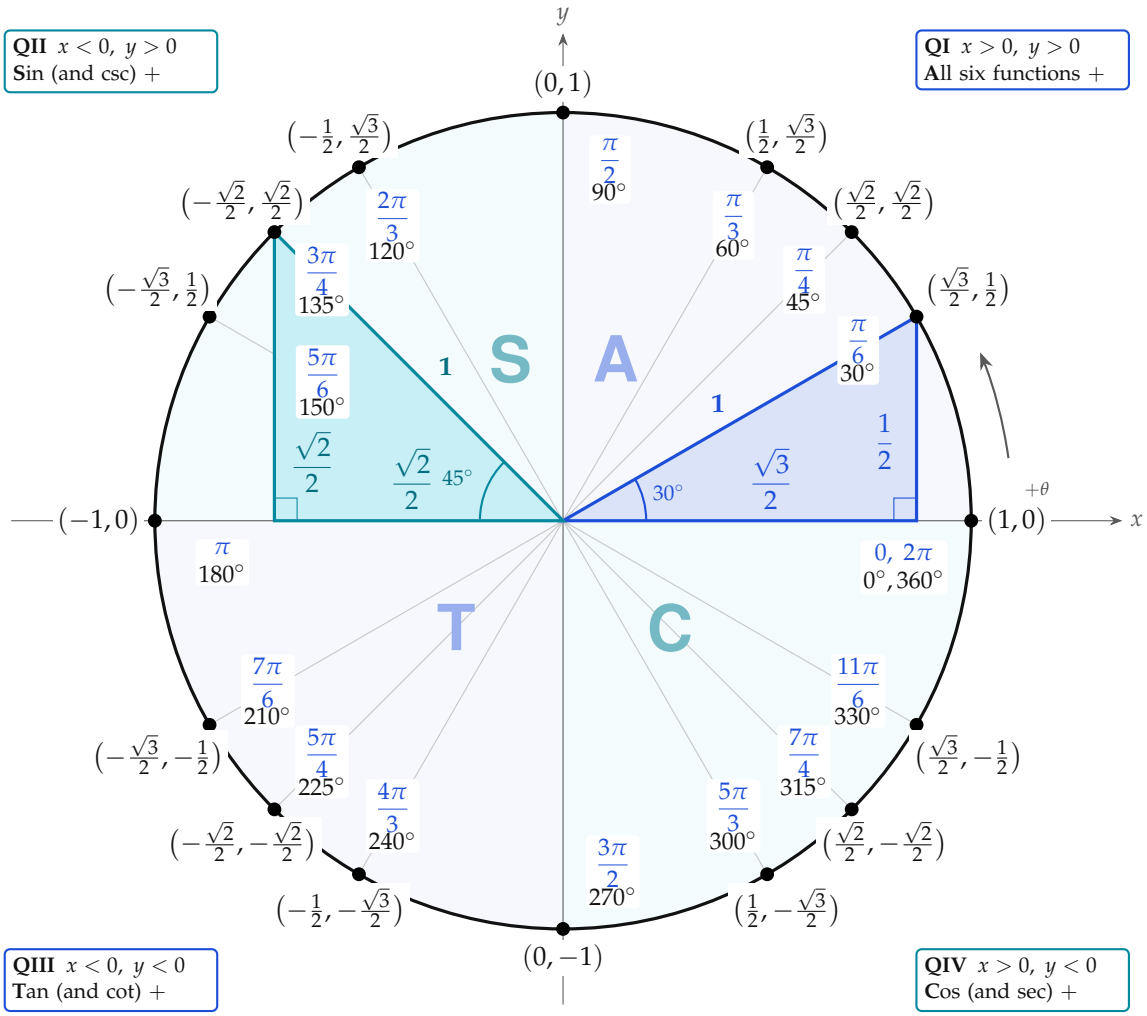
CONTEXT

- **Origin.** Angles beyond 180° entered astronomy with rotating-sphere models; the modern convention follows Euler.
- **Use.** Evaluating trig functions of any angle by hand; normalizing headings and bearings.
- **Data science.** Wrapping angles to $[0, 2\pi)$ or $(-\pi, \pi]$ (`np.mod`, `atan2`) before computing differences between headings, phases or wind directions.
- **Physical application.** A wheel turned 1000° points the same way as one turned 280° ($1000 - 2 \cdot 360$).

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The Unit Circle | *Key Topic T.K1*

The unit circle (radius 1, centered at the origin) turns trigonometry from triangles into functions of *any* angle: the point reached by rotating θ from $(1,0)$ is exactly $(\cos\theta, \sin\theta)$ T.4. Only three numbers, $\frac{1}{2}$, $\frac{\sqrt{2}}{2}$, $\frac{\sqrt{3}}{2}$, appear in the coordinates of all sixteen standard angles; the quadrant decides only the signs.



Reading the circle

x is cosine, y is sine. The point at angle θ is $(\cos\theta, \sin\theta)$; coordinates are always in that order (alphabetical: c before s).

Tangent is the slope of the terminal ray: $\tan\theta = \frac{y}{x} = \frac{\sin\theta}{\cos\theta}$; undefined when $x = 0$ ($\frac{\pi}{2}, \frac{3\pi}{2}$).

Denominators 6, 4, 3 are the 30°, 45°, 60° families: every $\frac{k\pi}{6}$ point has coordinates $\frac{\sqrt{3}}{2}$ and $\frac{1}{2}$ in some order; every $\frac{k\pi}{4}$ point has $\frac{\sqrt{2}}{2}$ twice.

Small angle, small sine. Near the x-axis ($\frac{\pi}{6}$) the y-value is the small one ($\frac{1}{2}$); near the y-axis ($\frac{\pi}{3}$) it is the big one ($\frac{\sqrt{3}}{2}$).

Symmetry. Reflecting across the y-axis ($\theta \to \pi - \theta$) flips the sign of x; across the x-axis ($\theta \to -\theta$) flips the sign of y; through the origin ($\theta \to \theta + \pi$) flips both.

Every point satisfies $x^2 + y^2 = 1$, i.e. $\cos^2\theta + \sin^2\theta = 1$ T.6. Use it to check a coordinate pair: $(\frac{1}{2})^2 + (\frac{\sqrt{3}}{2})^2 = 1$.

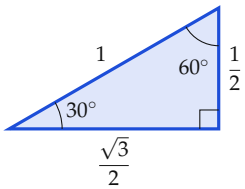
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Exact values: first quadrant and the axes

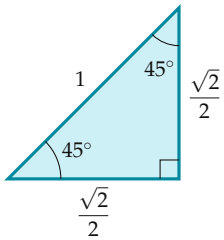
θ (rad)	θ (deg)	$\sin\theta$	$\cos\theta$	$\tan\theta$	$\csc\theta$	$\sec\theta$	$\cot\theta$
0	$0°$	0	1	0	undef.	1	undef.
$\frac{\pi}{6}$	$30°$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{3}$	2	$\frac{2\sqrt{3}}{3}$	$\sqrt{3}$
$\frac{\pi}{4}$	$45°$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1	$\sqrt{2}$	$\sqrt{2}$	1
$\frac{\pi}{3}$	$60°$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$	$\frac{2\sqrt{3}}{3}$	2	$\frac{\sqrt{3}}{3}$
$\frac{\pi}{2}$	$90°$	1	0	undef.	1	undef.	0
π	$180°$	0	-1	0	undef.	-1	undef.
$\frac{3\pi}{2}$	$270°$	-1	0	undef.	-1	undef.	0

Sine column: $\frac{\sqrt{0}}{2}, \frac{\sqrt{1}}{2}, \frac{\sqrt{2}}{2}, \frac{\sqrt{3}}{2}, \frac{\sqrt{4}}{2}$; cosine is the same list reversed. $\frac{\sqrt{3}}{3} = \frac{1}{\sqrt{3}}$ and $\frac{2\sqrt{3}}{3} = \frac{2}{\sqrt{3}}$ (rationalized A1.4 §03 Algebra I). "Undef." means a zero denominator.

Where the numbers come from: the special triangles G.8 §05 Geometry



sides $1 : \sqrt{3} : 2$ (scaled by $\frac{1}{2}$)



sides $1 : 1 : \sqrt{2}$ (scaled by $\frac{\sqrt{2}}{2}$)

Scale each special triangle so its hypotenuse is 1 (the radius). Its legs are then the coordinates of the point: the $30°$ triangle gives $(\frac{\sqrt{3}}{2}, \frac{1}{2})$, the $60°$ triangle (same triangle, turned) gives $(\frac{1}{2}, \frac{\sqrt{3}}{2})$, the $45°$ triangle gives $(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2})$. In other quadrants the same triangle is reflected (see the shaded triangle at $\frac{3\pi}{4}$): same lengths, new signs.

Any angle in four steps: the reference-angle method T.3

1. **Reduce**: add/subtract 2π until $0 \le \theta < 2\pi$.
2. **Locate** the quadrant; find the reference angle θ': QI θ, QII $\pi - \theta$, QIII $\theta - \pi$, QIV $2\pi - \theta$.
3. **Evaluate** the function at θ' from the table.
4. **Sign** it with ASTC (All, Sin, Tan, Cos are positive in QI–QIV).

Worked examples

$\cos\frac{11\pi}{3}$: $\frac{11\pi}{3} - 2\pi = \frac{5\pi}{3}$ (QIV), $\theta' = \frac{\pi}{3}$, $\cos\frac{\pi}{3} = \frac{1}{2}$, cosine + in QIV $\Rightarrow \frac{1}{2}$.

$\tan 210°$: QIII, $\theta' = 30°$, $\tan 30° = \frac{\sqrt{3}}{3}$, tangent + in QIII $\Rightarrow \frac{\sqrt{3}}{3}$.

$\sin(-\frac{3\pi}{4})$: $+2\pi \to \frac{5\pi}{4}$ (QIII), $\theta' = \frac{\pi}{4} \Rightarrow -\frac{\sqrt{2}}{2}$.

Common mistakes

Swapping sine and cosine: $\sin\frac{\pi}{6} = \frac{1}{2}$, not $\frac{\sqrt{3}}{2}$. Small angle $\Rightarrow$ small y.

Reference angle to the y-axis: always measure to the nearest x-axis.

Dropping the sign: the table gives magnitudes; ASTC supplies the sign.

Calculator in the wrong mode: $\sin 30 = -0.988$ in radians; $\sin 30° = 0.5$.

$\tan\frac{\pi}{2}$ **is not ∞ or 0**: it is undefined (vertical asymptote T.5).

$\sin^{-1} x \neq \frac{1}{\sin x}$: the inverse is arcsin T.14; the reciprocal is csc.

DECISION POINT **Exact or decimal?** If the angle is a multiple of $\frac{\pi}{6}$ or $\frac{\pi}{4}$ (or $30°$, $45°$), use the circle and give an exact answer. Otherwise there is no simple exact value: use a calculator in the right mode, or an identity T.8 to rewrite it in terms of standard angles (e.g. $15° = 45° - 30°$).

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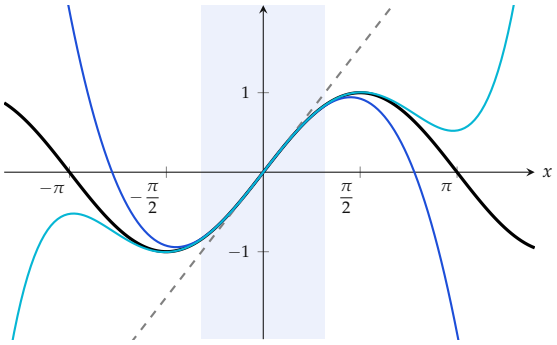
Taylor approximations: why polynomials can draw the wave

Near $\theta = 0$ the point $(\cos\theta, \sin\theta)$ can be computed with nothing but $+, -, \times, \div$. The Maclaurin series C2.22 §08 Calculus II of sine and cosine are

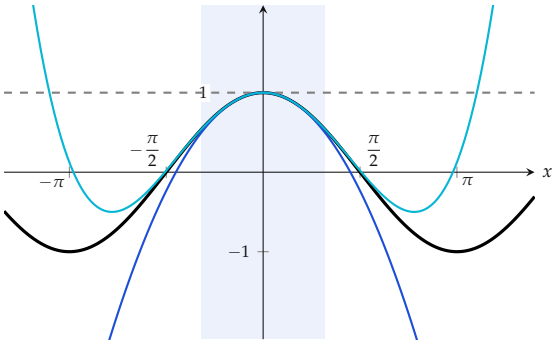
$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots \qquad \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots \qquad (x \text{ in radians})$$

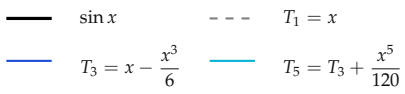
and cutting them off after a few terms gives the **Taylor polynomials** T_n plotted below (shaded: $|x| < 1$, where T_3 and T_2 are already within about 0.01 of the wave).

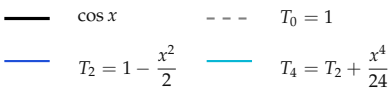
Sine: odd powers only



Cosine: even powers only







Where the coefficients come from. The n-th coefficient is $f^{(n)}(0)/n!$. Differentiating cycles $\sin \to \cos \to -\sin \to -\cos \to \sin$, so at 0 the derivatives of sin are $0, 1, 0, -1, 0, 1, \ldots$ and those of cos are $1, 0, -1, 0, 1, \ldots$: half the terms vanish and the survivors alternate in sign.

Why odd vs. even. sin is odd ($\sin(-x) = -\sin x$), so only odd powers survive; cos is even, so only even powers survive. Each graph keeps its symmetry at every order.

Why order 2 and 3 already look like the wave. $T_1 = x$ only matches the *slope* of sin at 0 (the small-angle rule $\sin\theta \approx \theta$). $T_3 = x - \frac{x^3}{6}$ adds a cubic that bends the line back down: it rises, levels off near $x \approx \sqrt{2}$ and turns, exactly the S-shaped crest of the sine wave. Likewise $T_2 = 1 - \frac{x^2}{2}$ is the parabola that fits the *top* of the cosine hump (same value, slope 0 and curvature -1 at 0). Each further term corrects the previous one in the opposite direction, so the polynomial follows the wave for one more half-turn: T_5 hugs sin to within 0.03 out to $x = \pm 2$.

How good is it? Alternating series error is at most the first omitted term: $|\sin x - T_3(x)| \le \frac{|x|^5}{120}$. At $x = 0.5$: $\sin 0.5 = 0.479426$, $T_3 = 0.479167$ (error $0.00026 \le 0.00026$); $\cos 0.5 = 0.877583$, $T_2 = 0.875$, $T_4 = 0.877604$.

On the unit circle. For a small angle the point is $(\cos\theta, \sin\theta) \approx (1 - \frac{\theta^2}{2},\ \theta - \frac{\theta^3}{6})$: the arc length θ is almost the height, and the circle drops below $x = 1$ by about $\frac{\theta^2}{2}$.

Common mistake. Degrees break the series: $\sin 30 \neq 30 - \frac{30^3}{6}$. Convert to radians first ($30^\circ = \frac{\pi}{6} \approx 0.5236$, $T_3 = 0.49967$).

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T.4 Unit Circle & Exact Values

Defining the trig functions as coordinates on the circle $x^2 + y^2 = 1$ extends them from acute angles to every real angle. The full picture is the key topic T.K1.

Unit-circle definition	$P(\theta) = (\cos\theta,\ \sin\theta)$	Rotate θ from $(1,0)$: the x-coordinate is cosine, the y-coordinate is sine.
Tangent as slope	$\tan\theta = \dfrac{y}{x} = \dfrac{\sin\theta}{\cos\theta}$	Slope of the terminal ray; also the height where it meets the line $x = 1$.
Circle of radius r	$\sin\theta = \dfrac{y}{r},\ \cos\theta = \dfrac{x}{r},\ \tan\theta = \dfrac{y}{x}$	For any point (x,y) on the terminal side, $r = \sqrt{x^2+y^2}$.
Exact values (first quadrant)		Every other standard angle reuses these numbers via its reference angle T.3.

θ	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
$\sin\theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos\theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0

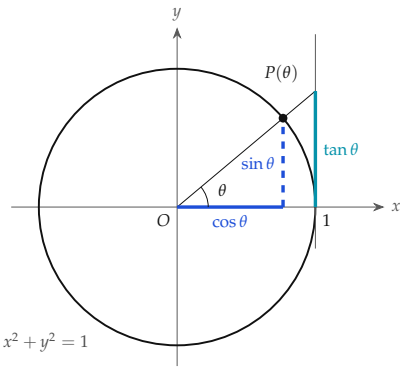
Quadrant signs (ASTC)	QI: all QII: sin QIII: tan QIV: cos	The functions listed (and their reciprocals) are positive there; the others are negative.

STATED-EQUATION *The point at angle theta on the unit circle is cosine theta, sine theta. Tangent theta is y over x. On a circle of radius r, sine is y over r and cosine is x over r. At zero, pi over six, pi over four, pi over three, pi over two, sine is zero, one half, root two over two, root three over two, one; cosine is the same list in reverse. Signs: all functions are positive in quadrant one, sine in quadrant two, tangent in quadrant three, and cosine in quadrant four.*

MEMORY AID Sine values are $\frac{\sqrt{0}}{2}, \frac{\sqrt{1}}{2}, \frac{\sqrt{2}}{2}, \frac{\sqrt{3}}{2}, \frac{\sqrt{4}}{2}$; cosine is the reverse. Signs by quadrant: "All Students Take Calculus" (All, Sin, Tan, Cos positive).

TERMS

Unit circle – radius 1 centered at the origin, $x^2 + y^2 = 1$ A2.26 §04 Algebra II
$P(\theta)$ – point reached by rotating θ from $(1,0)$
(x,y), r – any point on the terminal side and its distance from O G.12 §05 Geometry
Reference angle – acute angle to the x-axis T.3
Signs – QI all $+$, QII sin, QIII tan, QIV cos
Source of values – special triangles G.8 §05 Geometry



CONTEXT

- **Origin.** Euler's circle definitions (18th c.) extended trigonometry to every angle.
- **Use.** Evaluating trig functions of any angle exactly.
- **Data science.** Encoding a cyclical feature as $(\cos\theta, \sin\theta)$ places 23:00 next to 00:00 and December next to January, which a raw integer cannot do.
- **Physical application.** The height of a Ferris-wheel seat is $r\sin\theta$ above the axle as the wheel turns.

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T.5 Graphs of the Six Trigonometric Functions

Following $P(\theta)$ around the circle and plotting its coordinates against θ gives periodic graphs. Sine and cosine are smooth waves; the other four have vertical asymptotes wherever their denominator is zero.

Periodicity	$f(x+P)=f(x)$ for all x	The smallest such $P>0$ is the period: 2π for sin, cos, csc, sec; π for tan, cot.
Sine and cosine	$-1\le\sin x,\ \cos x\le 1$	Continuous waves with zeros at $k\pi$ (sin) and $\frac{\pi}{2}+k\pi$ (cos).
Cosine is shifted sine	$\cos x=\sin\left(x+\frac{\pi}{2}\right)$	Same curve moved $\frac{\pi}{2}$ left: every sinusoid is one shape T.12.
Tangent asymptotes	$x=\frac{\pi}{2}+k\pi$	Where $\cos x=0$; between them tan rises from $-\infty$ to ∞.
Cotangent asymptotes	$x=k\pi$	Where $\sin x=0$; cot falls between them.
Secant and cosecant range	$\lvert\sec x\rvert\ge 1,\quad \lvert\csc x\rvert\ge 1$	Reciprocals of waves bounded by 1, so they never take values in $(-1,1)$.

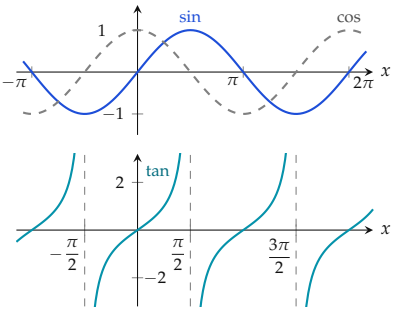
STATED-EQUATION *f of x plus P equals f of x for every x; P is the period. Sine and cosine stay between minus one and one. Cosine x equals sine of x plus pi over two. Tangent has vertical asymptotes at pi over two plus k pi; cotangent at k pi. Secant and cosecant are at least one in absolute value.*

MEMORY AID Asymptotes sit where the *denominator* is zero: tan, sec $\to\cos x=0$; cot, csc $\to\sin x=0$.

Function	Domain	Range	Period	Symmetry
$\sin x$	$\mathbb{R}$	$[-1,1]$	2π	odd
$\cos x$	$\mathbb{R}$	$[-1,1]$	2π	even
$\tan x$	$x\neq\frac{\pi}{2}+k\pi$	$\mathbb{R}$	π	odd
$\csc x$	$x\neq k\pi$	$(-\infty,-1]\cup[1,\infty)$	2π	odd
$\sec x$	$x\neq\frac{\pi}{2}+k\pi$	$(-\infty,-1]\cup[1,\infty)$	2π	even
$\cot x$	$x\neq k\pi$	$\mathbb{R}$	π	odd

TERMS

P – period (length of one cycle)
k – any integer
Asymptote – vertical line the graph approaches but never meets A2.10 §04 Algebra II
Odd / even – $f(-x)=-f(x)$ / $f(-x)=f(x)$ T.6
Domain, range – allowed inputs / outputs A1.9 §03 Algebra I
sec, csc – reciprocals of cos and sin T.1



CONTEXT

- **Origin.** Graphs of sine appear in Roberval's work on the cycloid (1630s).
- **Use.** Recognizing periodic behavior; sketching before solving equations T.15.
- **Data science.** Sinusoidal positional encodings in transformers use sin and cos at many periods so each position gets a unique, smoothly varying code.
- **Physical application.** Gallery tiles F.10 §18 Graph Gallery, F.11 §18 Graph Gallery, F.12 §18 Graph Gallery show the sine, cosine and tangent graphs on their own.

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Identities

T.6 Pythagorean & Fundamental Identities

Identities hold for every angle. The Pythagorean identity is the unit-circle equation $x^2 + y^2 = 1$; the symmetry identities describe how the circle reflects.

Pythagorean identity	$\sin^2\theta + \cos^2\theta = 1$	The point $(\cos\theta, \sin\theta)$ lies on $x^2 + y^2 = 1$ G.7 §05 Geometry.
Tangent–secant form	$1 + \tan^2\theta = \sec^2\theta$	Divide the first identity by $\cos^2\theta$.
Cotangent–cosecant form	$1 + \cot^2\theta = \csc^2\theta$	Divide the first identity by $\sin^2\theta$.
Even / odd (negative angle)	$\sin(-\theta) = -\sin\theta,\ \cos(-\theta) = \cos\theta,\ \tan(-\theta) = -\tan\theta$	Reflecting across the x-axis flips y only; cosine is the only even one of the three.
Periodicity	$\sin(\theta + 2\pi) = \sin\theta, \quad \tan(\theta + \pi) = \tan\theta$	A full turn returns to the same point; tan repeats after a half turn.
Supplement / half-turn	$\sin(\pi - \theta) = \sin\theta,\ \cos(\pi - \theta) = -\cos\theta,\ \sin(\theta + \pi) = -\sin\theta$	Reflection across the y-axis; a half turn negates both coordinates.

STATED-EQUATION *Sine squared theta plus cosine squared theta equals one. One plus tangent squared equals secant squared. One plus cotangent squared equals cosecant squared. Sine of minus theta equals minus sine theta; cosine of minus theta equals cosine theta; tangent of minus theta equals minus tangent theta. Adding two pi leaves sine unchanged; adding pi leaves tangent unchanged. Sine of pi minus theta equals sine theta, and cosine of pi minus theta equals minus cosine theta.*

MEMORY AID Divide $\sin^2 + \cos^2 = 1$ by $\cos^2$ or $\sin^2$ to get the other two. Cosine is the only even one.

METHOD

1. To find the other functions from one value, use $\sin^2 + \cos^2 = 1$ and pick the sign from the quadrant.
2. Example: $\sin\theta = \frac{3}{5}$ in QII $\Rightarrow \cos\theta = -\sqrt{1 - \frac{9}{25}} = -\frac{4}{5}$, $\tan\theta = -\frac{3}{4}$.

TERMS

$\sin^2\theta$ – means $(\sin\theta)^2$, not $\sin(\sin\theta)$
Identity – an equation true for every allowed θ A1.10 §03 Algebra I
Derivation – Pythagorean theorem G.7 §05 Geometry on the unit circle T.4
Odd / even – sine and tangent are odd; cosine is even
All identities – see T.K2

CONTEXT

- **Origin.** Follows from Pythagoras once trig is defined on the circle.
- **Use.** Simplifying expressions, verifying identities, trig integrals C2.2 §08 Calculus II.
- **Data science.** Because $\sin^2 + \cos^2 = 1$, every cyclical encoding $(\cos\theta, \sin\theta)$ has norm 1: no hour is artificially "larger" than another to a distance-based model.
- **Physical application.** In a pendulum or spring, $\sin^2 + \cos^2 = 1$ is why kinetic plus potential energy stays constant.

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T.7 Reciprocal, Quotient & Cofunction Identities

Every trig function can be written with sine and cosine alone; and because the two acute angles of a right triangle add to $\frac{\pi}{2}$, each function of θ equals its "co"-function of the complement.

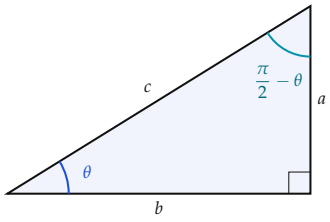
Reciprocal identities	$\csc\theta = \dfrac{1}{\sin\theta},\quad \sec\theta = \dfrac{1}{\cos\theta},\quad \cot\theta = \dfrac{1}{\tan\theta}$	Valid where the denominator is nonzero; each pair multiplies to 1.
Quotient identities	$\tan\theta = \dfrac{\sin\theta}{\cos\theta},\qquad \cot\theta = \dfrac{\cos\theta}{\sin\theta}$	Rewriting in sin and cos is the first move in most simplifications.
Cofunction: sine / cosine	$\sin\left(\frac{\pi}{2} - \theta\right) = \cos\theta,\quad \cos\left(\frac{\pi}{2} - \theta\right) = \sin\theta$	The side opposite θ is adjacent to its complement.
Cofunction: tangent / cotangent	$\tan\left(\frac{\pi}{2} - \theta\right) = \cot\theta$	Legs swap roles, so the ratio inverts.
Cofunction: secant / cosecant	$\sec\left(\frac{\pi}{2} - \theta\right) = \csc\theta$	Reciprocal of the sine/cosine cofunction rule.

STATED-EQUATION *Cosecant is one over sine, secant is one over cosine, cotangent is one over tangent. Tangent is sine over cosine, cotangent is cosine over sine. Sine of pi over two minus theta equals cosine theta, and cosine of pi over two minus theta equals sine theta. Tangent of the complement equals cotangent; secant of the complement equals cosecant.*

MEMORY AID "Co" means "complement's": cosine is the sine of the complementary angle ($\sin 30^\circ = \cos 60^\circ$).

TERMS

θ – any angle (acute in the triangle picture)
Complement – $\frac{\pi}{2} - \theta$ ($90^\circ - \theta$) G.3 §05 Geometry
Cofunction pairs – sin/cos, tan/cot, sec/csc
a, b, c – legs and hypotenuse in the figure
General form – special case of the difference formulas T.8



$\sin\theta = \frac{a}{c} = \cos(\frac{\pi}{2} - \theta)\quad \tan\theta = \frac{a}{b} = \cot(\frac{\pi}{2} - \theta)$

CONTEXT

- **Origin.** The name *co-sinus* ("sine of the complement") is due to Edmund Gunter (1620).
- **Use.** Rewriting expressions in one or two functions; reading tables backwards.
- **Data science.** Libraries implement only a few kernels (sin, cos, atan) and derive the rest through these identities; $\cot x$ is computed as $1/\tan x$.
- **Physical application.** Sun elevation θ and zenith angle $90^\circ - \theta$: solar irradiance on flat ground uses $\sin(\text{elevation}) = \cos(\text{zenith})$.

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T.8 Sum & Difference Formulas

The trig function of a sum is *not* the sum of the functions. These formulas expand $f(\alpha \pm \beta)$ exactly, and every multiple-angle identity follows from them.

Sine of a sum / difference	$\sin(\alpha \pm \beta) = \sin\alpha\cos\beta \pm \cos\alpha\sin\beta$	Mixed products, same sign as in the argument.
Cosine of a sum / difference	$\cos(\alpha \pm \beta) = \cos\alpha\cos\beta \mp \sin\alpha\sin\beta$	Like products, *opposite* sign.
Tangent of a sum / difference	$\tan(\alpha \pm \beta) = \dfrac{\tan\alpha \pm \tan\beta}{1 \mp \tan\alpha\tan\beta}$	Divide the sine formula by the cosine formula; undefined when the denominator is 0.

STATED-EQUATION *Sine of alpha plus or minus beta equals sine alpha cosine beta plus or minus cosine alpha sine beta. Cosine of alpha plus or minus beta equals cosine alpha cosine beta minus or plus sine alpha sine beta. Tangent of alpha plus or minus beta equals tangent alpha plus or minus tangent beta, over one minus or plus tangent alpha tangent beta.*

MEMORY AID Sine keeps the sign and mixes (sin cos + cos sin); cosine keeps the pairs and flips the sign.

METHOD

1. Write the angle as a sum or difference of standard angles T.K1.
2. Expand with the formula and substitute exact values.
3. Simplify. Example: $\sin 75^\circ = \sin(45^\circ + 30^\circ) = \frac{\sqrt{2}}{2}\cdot\frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{2}\cdot\frac{1}{2} = \frac{\sqrt{6}+\sqrt{2}}{4}$.

TERMS

α, β – any two angles
$\pm, \mp$ – use the top signs together, or the bottom signs together
Pitfall – $\sin(\alpha+\beta) \neq \sin\alpha + \sin\beta$
Special cases – double angles T.9 ($\alpha = \beta$), cofunctions T.7 ($\alpha = \frac{\pi}{2}$)
Matrix view – rotation by α then β is rotation by $\alpha+\beta$ T.20

CONTEXT

- **Origin.** Ptolemy's theorem on chords in the *Almagest* (c. 150 CE).
- **Use.** Exact values such as $\sin 75^\circ$; simplifying wave sums; deriving derivatives of sin and cos C1.14 §07 Calculus I.
- **Data science.** Rotary positional embeddings (RoPE) in transformers rotate queries and keys; angle addition makes their dot product depend only on relative position $m - n$.
- **Physical application.** Two sound waves that differ slightly in frequency combine into audible "beats" (sum-to-product identities T.11).

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T.9 Double-Angle & Half-Angle Formulas

Setting $\alpha = \beta$ in the sum formulas gives functions of 2θ; solving the cosine version backwards gives functions of $\frac{\theta}{2}$.

Sine double angle	$\sin 2\theta = 2\sin\theta\cos\theta$	$\sin(\theta+\theta)$ from T.8; not $2\sin\theta$.
Tangent double angle	$\tan 2\theta = \dfrac{2\tan\theta}{1-\tan^2\theta}$	Undefined when $\tan\theta = \pm 1$.
Cosine double angle (three forms)	$\cos 2\theta = \cos^2\theta - \sin^2\theta = 2\cos^2\theta - 1 = 1 - 2\sin^2\theta$	Swap forms with $\sin^2 + \cos^2 = 1$ to keep only the function you want.
Sine half angle	$\sin\dfrac{\theta}{2} = \pm\sqrt{\dfrac{1-\cos\theta}{2}}$	Sign from the quadrant of $\frac{\theta}{2}$, not of θ.
Cosine half angle	$\cos\dfrac{\theta}{2} = \pm\sqrt{\dfrac{1+\cos\theta}{2}}$	Same sign rule; "+" goes with cosine.
Tangent half angle	$\tan\dfrac{\theta}{2} = \dfrac{1-\cos\theta}{\sin\theta} = \dfrac{\sin\theta}{1+\cos\theta}$	No $\pm$ needed: the sign comes out automatically.
Triple angle	$\sin 3\theta = 3\sin\theta - 4\sin^3\theta,\quad \cos 3\theta = 4\cos^3\theta - 3\cos\theta$	From $\sin(2\theta+\theta)$; links $\cos 3\theta$ to a cubic in $\cos\theta$.

STATED-EQUATION *Sine two theta equals two sine theta cosine theta. Tangent two theta equals two tangent theta over one minus tangent squared. Cosine two theta equals cosine squared minus sine squared, which equals two cosine squared minus one, which equals one minus two sine squared. Sine of theta over two equals plus or minus the square root of one minus cosine theta over two; cosine of theta over two equals plus or minus the square root of one plus cosine theta over two. Tangent of theta over two equals one minus cosine theta over sine theta. Sine three theta equals three sine theta minus four sine cubed theta. It also equals sine theta over one plus cosine theta. Cosine three theta equals four cosine cubed theta minus three cosine theta.*

MEMORY AID Half-angle: "sine is minus, cosine is plus" under the root.

DECISION POINT Which form of $\cos 2\theta$? Choose the one that leaves a single function: if the rest of the expression is in $\sin\theta$, use $1 - 2\sin^2\theta$; if in $\cos\theta$, use $2\cos^2\theta - 1$. To *lower* a power, read the forms backwards (power reduction T.10).

TERMS

Double angle – $\alpha = \beta = \theta$ in T.8
$\pm$ – sign chosen by the quadrant of $\frac{\theta}{2}$ T.4
Power reduction – $\sin^2\theta = \frac{1-\cos 2\theta}{2}$ T.10
Pitfall – $\sin 2\theta \neq 2\sin\theta$

CONTEXT

- **Origin.** Special cases of Ptolemy's sum formulas.
- **Use.** Integrating $\sin^2 x$ C2.2 §08 Calculus II; exact values like $\sin 22.5°$; the substitution $t = \tan\frac{x}{2}$ in integrals.
- **Data science.** Power reduction explains why squaring a seasonal signal produces a component at *twice* the frequency, visible as a harmonic in a periodogram.
- **Physical application.** A projectile's range $R = \frac{v^2\sin 2\theta}{g}$ is largest at $\theta = 45°$ because $\sin 2\theta$ peaks there.

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T.10 Power-Reduction Formulas

Power reduction trades a squared (or higher) trig function for first powers of cosines of multiple angles. It is the standard way to integrate, average or analyze the frequency content of $\sin^n$ and $\cos^n$.

Sine squared	$\sin^2\theta = \dfrac{1-\cos 2\theta}{2}$	Solve $\cos 2\theta = 1 - 2\sin^2\theta$ for $\sin^2\theta$.
Cosine squared	$\cos^2\theta = \dfrac{1+\cos 2\theta}{2}$	Solve $\cos 2\theta = 2\cos^2\theta - 1$ for $\cos^2\theta$.
Tangent squared	$\tan^2\theta = \dfrac{1-\cos 2\theta}{1+\cos 2\theta}$	Ratio of the two above.
Product of squares	$\sin^2\theta\cos^2\theta = \dfrac{1-\cos 4\theta}{8}$	Equals $\frac{1}{4}\sin^2 2\theta$, then reduce again.
Cosine to the fourth	$\cos^4\theta = \dfrac{3+4\cos 2\theta+\cos 4\theta}{8}$	Square the $\cos^2$ formula and reduce the new $\cos^2 2\theta$ (for $\sin^4$, use $-4\cos 2\theta$).
Sine cubed	$\sin^3\theta = \dfrac{3\sin\theta-\sin 3\theta}{4}$	Rearranged triple-angle formula T.9.
Cosine cubed	$\cos^3\theta = \dfrac{3\cos\theta+\cos 3\theta}{4}$	Same idea for cosine.

STATED-EQUATION *Sine squared theta equals one minus cosine two theta, over two. Cosine squared theta equals one plus cosine two theta, over two. Tangent squared equals one minus cosine two theta over one plus cosine two theta. Sine squared cosine squared equals one minus cosine four theta, over eight. Cosine to the fourth equals three plus four cosine two theta plus cosine four theta, over eight. Sine cubed equals three sine theta minus sine three theta, over four; cosine cubed equals three cosine theta plus cosine three theta, over four.*

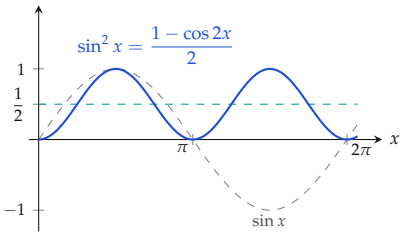
MEMORY AID "Squared becomes half, plus or minus half of the double": $\sin^2, \cos^2 = \frac{1}{2} \mp \frac{1}{2}\cos 2\theta$.

METHOD

1. Replace each even power $\sin^2$ or $\cos^2$ with its half-angle-free form above.
2. Expand; any new $\cos^2(2\theta)$ is reduced again to $\cos 4\theta$.
3. Stop when only first powers remain. Example: $\sin^2\frac{\pi}{8} = \frac{1-\cos\frac{\pi}{4}}{2} = \frac{2-\sqrt{2}}{4}$.

TERMS

θ – any angle
$\sin^n\theta$ – $(\sin\theta)^n$
Average value – $\sin^2$ and $\cos^2$ average $\frac{1}{2}$ over a period C2.12 §08 Calculus II
Related – half-angle formulas T.9 are the square roots of the first two



CONTEXT

- **Origin.** Rearrangements of the double-angle formulas known to Viète (16th c.).
- **Use.** Integrals $\int \sin^2 x\,dx = \frac{x}{2} - \frac{\sin 2x}{4} + C$ C2.2 §08 Calculus II.
- **Data science.** The RMS of a sine wave is $\frac{1}{\sqrt{2}}$ of its peak because $\sin^2$ averages $\frac{1}{2}$ EE.13 §16 Electrical Eng.; squaring a periodic feature doubles its frequency.
- **Physical application.** Average AC power $P = \frac{V_m^2}{2R}$ comes from averaging $\sin^2(\omega t)$.

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T.11 Product-to-Sum & Sum-to-Product

Products of sines and cosines can be rewritten as sums of single sinusoids, and sums as products. Products are easier to factor; sums are easier to integrate and to read as frequencies.

Sine times cosine	$\sin A\cos B = \frac{1}{2}\left[\sin(A+B)+\sin(A-B)\right]$	Add the two sine sum/difference formulas T.8.
Cosine times cosine	$\cos A\cos B = \frac{1}{2}\left[\cos(A-B)+\cos(A+B)\right]$	Add the two cosine formulas.
Sine times sine	$\sin A\sin B = \frac{1}{2}\left[\cos(A-B)-\cos(A+B)\right]$	Subtract them; note the order.
Sum of sines	$\sin A+\sin B = 2\sin\frac{A+B}{2}\cos\frac{A-B}{2}$	Average angle times half-difference angle.
Difference of sines	$\sin A-\sin B = 2\cos\frac{A+B}{2}\sin\frac{A-B}{2}$	Sine and cosine swap places.
Sum of cosines	$\cos A+\cos B = 2\cos\frac{A+B}{2}\cos\frac{A-B}{2}$	Both factors are cosines.
Difference of cosines	$\cos A-\cos B = -2\sin\frac{A+B}{2}\sin\frac{A-B}{2}$	The only one with a minus sign in front.

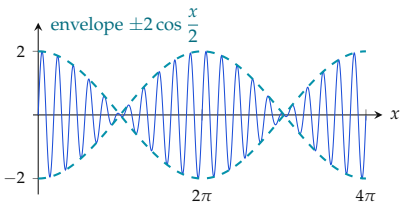
STATED-EQUATION *Sine A cosine B equals one half of sine of A plus B plus sine of A minus B. Cosine A cosine B equals one half of cosine of A minus B plus cosine of A plus B. Sine A sine B equals one half of cosine of A minus B minus cosine of A plus B. Sine A plus sine B equals two sine of the half-sum times cosine of the half-difference. Sine A minus sine B equals two cosine of the half-sum times sine of the half-difference. Cosine A plus cosine B equals two cosine of the half-sum times cosine of the half-difference. Cosine A minus cosine B equals minus two sine of the half-sum times sine of the half-difference.*

MEMORY AID Sum-to-product: "half-sum, half-difference". Product-to-sum: "average the sum and difference angles".

DECISION POINT Which direction? To *integrate*, take a Fourier view, or simplify a product, go product → sum. To *solve* an equation ($\sin 3x+\sin x=0$) or find an envelope, go sum → product and use the zero-product rule A1.1 §03 Algebra I. Example: $\sin 75^\circ+\sin 15^\circ = 2\sin 45^\circ\cos 30^\circ = \frac{\sqrt{6}}{2}$.

TERMS

A, B – any two angles
$\frac{A+B}{2}$ – half-sum (average) angle
$\frac{A-B}{2}$ – half-difference angle
Beat frequency – $|f_1-f_2|$, the rate of the slow envelope



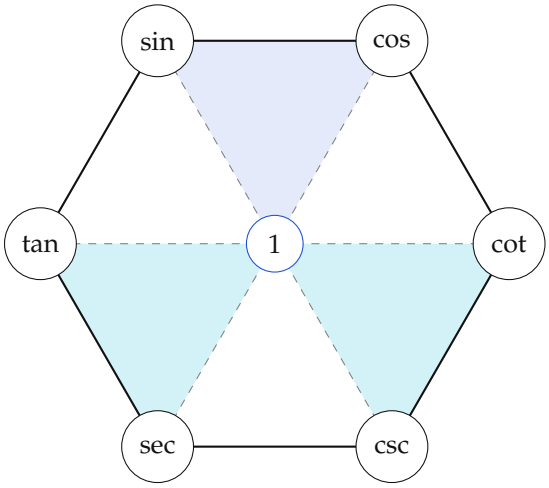
CONTEXT

- **Origin.** *Prosthaphaeresis* (16th c.): astronomers such as Tycho Brahe multiplied large numbers with these identities before logarithms existed.
- **Use.** Integrating $\sin mx\cos nx$ C2.2 §08 Calculus II; orthogonality of Fourier modes EE.26 §16 Electrical Eng..
- **Data science.** Multiplying two seasonal signals creates components at the sum and difference frequencies (mixing, aliasing in feature products).
- **Physical application.** Two guitar strings slightly out of tune produce beats: $\sin(10x)+\sin(11x) = 2\sin(10.5x)\cos(0.5x)$.

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Trig Identities at a Glance | *Key Topic T.K2*

Every identity in this chapter flows from two facts: the point $(\cos\theta, \sin\theta)$ lies on the unit circle (Pythagorean identities) and rotations add (sum formulas). The hexagon packs the first family into one picture; the panels list every formula you are expected to know, with the concept that explains it.



Every identity, by family

Reading the hexagon

Diagonals are reciprocals: $\sin \cdot \csc = \cos \cdot \sec = \tan \cdot \cot = 1$.

Each function is the product of its two neighbors: $\sin = \cos \cdot \tan$, $\cos = \sin \cdot \cot$, $\tan = \sin \cdot \sec$, $\cot = \cos \cdot \csc$.

Going around, each is the next over the one after: $\tan = \frac{\sin}{\cos}$, $\cot = \frac{\cos}{\csc} \cdot \csc^2 \ldots$ in practice: $\tan = \frac{\sin}{\cos}$, $\cot = \frac{\cos}{\sin}$.

Shaded triangles (Pythagorean): the squares of the two top corners add to the square of the bottom corner: $\sin^2 + \cos^2 = 1$, $\tan^2 + 1 = \sec^2$, $1 + \cot^2 = \csc^2$.

Reciprocal & quotient T.7

$\csc\theta = \frac{1}{\sin\theta}$, $\sec\theta = \frac{1}{\cos\theta}$, $\cot\theta = \frac{1}{\tan\theta}$

$\tan\theta = \frac{\sin\theta}{\cos\theta}$, $\quad \cot\theta = \frac{\cos\theta}{\sin\theta}$

Sum & difference T.8

$$\sin(\alpha \pm \beta) = \sin\alpha\cos\beta \pm \cos\alpha\sin\beta$$
$$\cos(\alpha \pm \beta) = \cos\alpha\cos\beta \mp \sin\alpha\sin\beta$$
$$\tan(\alpha \pm \beta) = \frac{\tan\alpha \pm \tan\beta}{1 \mp \tan\alpha\tan\beta}$$

Power reduction T.10

$\sin^2\theta = \frac{1-\cos 2\theta}{2}$, $\cos^2\theta = \frac{1+\cos 2\theta}{2}$

$\tan^2\theta = \frac{1-\cos 2\theta}{1+\cos 2\theta}$

Pythagorean T.6

$$\sin^2\theta + \cos^2\theta = 1$$
$$1 + \tan^2\theta = \sec^2\theta$$
$$1 + \cot^2\theta = \csc^2\theta$$

Double angle T.9

$$\sin 2\theta = 2\sin\theta\cos\theta$$
$$\cos 2\theta = \cos^2\theta - \sin^2\theta = 2\cos^2\theta - 1 = 1 - 2\sin^2\theta$$
$$\tan 2\theta = \frac{2\tan\theta}{1 - \tan^2\theta}$$

Product to sum T.11

$$\sin A\cos B = \tfrac{1}{2}[\sin(A+B) + \sin(A-B)]$$
$$\cos A\cos B = \tfrac{1}{2}[\cos(A-B) + \cos(A+B)]$$
$$\sin A\sin B = \tfrac{1}{2}[\cos(A-B) - \cos(A+B)]$$

Even / odd & periodic T.6

$\cos(-\theta) = \cos\theta$ (even)

$\sin(-\theta) = -\sin\theta$, $\tan(-\theta) = -\tan\theta$

$\sin, \cos$: period 2π; $\tan, \cot$: period π

Half angle T.9

$\sin\frac{\theta}{2} = \pm\sqrt{\frac{1-\cos\theta}{2}}$, $\cos\frac{\theta}{2} = \pm\sqrt{\frac{1+\cos\theta}{2}}$

$\tan\frac{\theta}{2} = \frac{1-\cos\theta}{\sin\theta} = \frac{\sin\theta}{1+\cos\theta}$

Sum to product T.11

$$\sin A \pm \sin B = 2\sin\tfrac{A\pm B}{2}\cos\tfrac{A\mp B}{2}$$
$$\cos A + \cos B = 2\cos\tfrac{A+B}{2}\cos\tfrac{A-B}{2}$$
$$\cos A - \cos B = -2\sin\tfrac{A+B}{2}\sin\tfrac{A-B}{2}$$

Cofunction T.7

$\sin(\frac{\pi}{2} - \theta) = \cos\theta$, $\cos(\frac{\pi}{2} - \theta) = \sin\theta$

$\tan(\frac{\pi}{2} - \theta) = \cot\theta$, $\sec(\frac{\pi}{2} - \theta) = \csc\theta$

Harmonic addition T.13

$a\sin x + b\cos x = R\sin(x + \varphi)$

$R = \sqrt{a^2 + b^2}$, $\varphi = \operatorname{atan2}(b, a)$

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Proving (verifying) an identity

1. Work on **one side only** (usually the more complicated one); never "do the same thing to both sides" as if solving.
2. Rewrite everything in sin and cos.
3. Combine fractions over a common denominator A1.2 §03 Algebra I; factor A1.14 §03 Algebra I.
4. Substitute a Pythagorean identity wherever $\sin^2$ and $\cos^2$ meet; multiply by a conjugate $(1 \pm \sin\theta)$ to create one.
5. Stop when the expression matches the other side. Example: $\frac{1}{1-\sin\theta} + \frac{1}{1+\sin\theta} = \frac{2}{1-\sin^2\theta} = \frac{2}{\cos^2\theta} = 2\sec^2\theta$.

DECISION POINT **Which identity next?** Look at what differs between the two sides. *Different functions* → reciprocal/quotient identities. *Squares* → Pythagorean, or power reduction if a 2θ appears on the other side. *Different angles* (θ vs. 2θ, $\frac{\theta}{2}$) → double/half angle. *A sum of sinusoids vs. a product* → sum-to-product. $a\sin + b\cos$ → harmonic addition. If stuck, test a value such as $\theta = \frac{\pi}{6}$ numerically: if the two sides differ, it is not an identity.

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Waves, Inverses & Equations

T.12 Graphs of Sinusoids

Every sine or cosine wave is the basic curve stretched, squeezed and shifted. Four numbers describe it: amplitude, period (or frequency), phase shift and midline.

General sinusoid	$y = A\sin\big(B(x-C)\big)+D$	The same four parameters work with cos; cos just starts a cycle at a maximum.
Amplitude	$\text{amplitude} = \lvert A\rvert$	Half the peak-to-trough height; $A<0$ also flips the wave.
Period	$\text{period} = \dfrac{2\pi}{\lvert B\rvert}$	Length of one cycle; larger $\lvert B\rvert$ squeezes the wave.
Phase shift	$\text{phase shift} = C \quad \left(= \frac{\varphi}{B} \text{ for } \sin(Bx-\varphi)\right)$	Horizontal shift: right if $C>0$ A2.5 §04 Algebra II; factor B out first.
Midline, max, min	$y = D, \quad y_{\max} = D+\lvert A\rvert, \quad y_{\min} = D-\lvert A\rvert$	Vertical shift: the wave oscillates about $y=D$.
Frequency	$f = \dfrac{\lvert B\rvert}{2\pi} = \dfrac{1}{\text{period}}$	Cycles per unit of x; $\lvert B\rvert = 2\pi f$ is the angular frequency T.2.

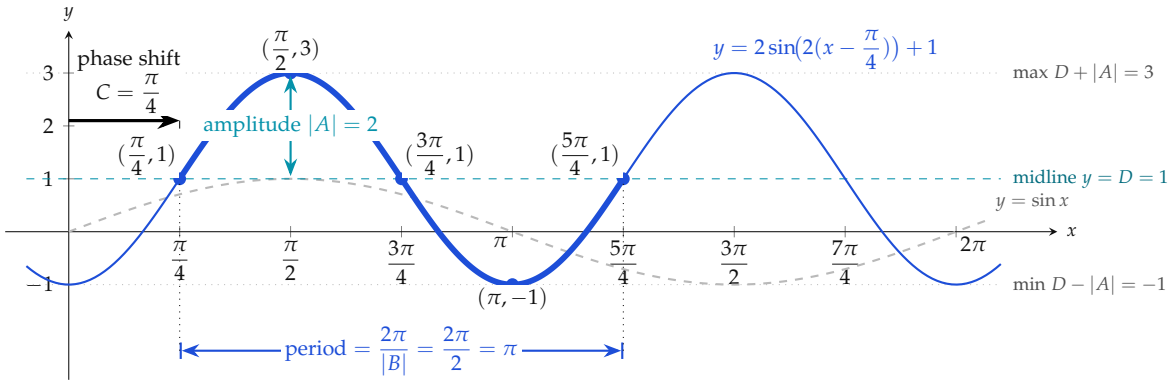
STATED-EQUATION *y equals A sine of B times x minus C, plus D. The amplitude is the absolute value of A. The period is two pi over the absolute value of B. The phase shift is C, which equals phi over B when the argument is B x minus phi. The midline is y equals D, with maximum D plus the amplitude and minimum D minus the amplitude. The frequency is the absolute value of B over two pi, one over the period.*

MEMORY AID "A-B-C-D": Amplitude, B for the period, C for the shift ("Copy the sign backwards": $x-C$ moves right), D for the Datum (midline).

METHOD

1. Write the function in the form $A\sin(B(x-C))+D$ (factor out B).
2. Read off $\lvert A\rvert$, period $\frac{2\pi}{\lvert B\rvert}$, shift C, midline D.
3. Mark the start of a cycle at $x=C$ and split one period into four equal steps.
4. Plot the five key points: midline, max, midline, min, midline (for sin, $A>0$), then repeat. Example (below): $y = 2\sin(2(x-\frac{\pi}{4}))+1$ has $A=2$, period π, shift $\frac{\pi}{4}$, midline $y=1$; key points at $x = \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}, \pi, \frac{5\pi}{4}$.

DECISION POINT Shift trap: in $\sin(2x-\frac{\pi}{2})$ the shift is *not* $\frac{\pi}{2}$; factor first, $\sin(2(x-\frac{\pi}{4}))$, so the shift is $\frac{\pi}{4}$. Sine or cosine model? Use cos if your data start at a peak (no shift needed), sin if they start on the midline going up.



TERMS

A – amplitude (with sign: reflection if negative)
B – angular frequency; period $\frac{2\pi}{\lvert B\rvert}$
C – phase (horizontal) shift; $\varphi = BC$ is the phase angle
D – vertical shift / midline A2.5 §04 Algebra II
f – frequency (cycles per unit x)

CONTEXT

- **Origin.** Joseph Fourier (1822): any periodic signal is a sum of sinusoids.
- **Use.** Tides, sound, alternating current, circadian rhythms.
- **Data science.** Seasonality in time-series models (e.g. Fourier terms in Prophet) uses sin and cos with periods of 7 or 365.25 days.
- **Physical application.** Household AC voltage is $170\sin(2\pi\cdot 60\,t)$ volts in the US: amplitude 170 V, period $\frac{1}{60}$ s EE.13 §16 Electrical Eng..

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T.13 Harmonic Addition: Combining Sine & Cosine

A sine and a cosine of the same frequency always add up to a single sinusoid of that frequency. The new amplitude and phase come from the right triangle with legs a and b.

Amplitude–phase form	$a\sin x + b\cos x = R\sin(x+\varphi)$	One wave instead of two; works for any a, b (same frequency only).
Combined amplitude	$R = \sqrt{a^2+b^2}$	Hypotenuse of the triangle with legs a and b G.7 §05 Geometry.
Phase angle	$\cos\varphi = \dfrac{a}{R},\ \ \sin\varphi = \dfrac{b}{R} \ \Rightarrow\ \varphi = \operatorname{atan2}(b,a)$	Use both signs to get the right quadrant T.14.
Cosine form	$a\sin x + b\cos x = R\cos(x-\delta),\ \ \delta = \operatorname{atan2}(a,b)$	Same R; phase measured from a cosine peak.
Maximum value	$\max_x (a\sin x + b\cos x) = \sqrt{a^2+b^2}$	Read directly from R; no calculus needed.

STATED-EQUATION *a sine x plus b cosine x equals R sine of x plus phi, where R is the square root of a squared plus b squared, cosine phi equals a over R and sine phi equals b over R. Equivalently it equals R cosine of x minus delta. Its maximum value is R. So phi equals a tan two of b, a, while delta equals a tan two of a, b.*

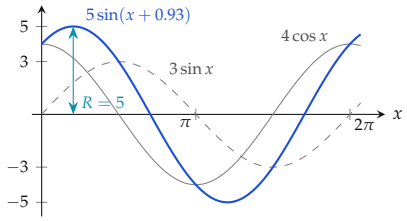
MEMORY AID "Legs a and b, hypotenuse R, angle φ": draw the triangle.

METHOD

1. Compute $R = \sqrt{a^2+b^2}$.
2. Find φ from $\cos\varphi = \frac{a}{R}$, $\sin\varphi = \frac{b}{R}$ (check the quadrant).
3. Write $R\sin(x+\varphi)$. Example: $3\sin x + 4\cos x = 5\sin(x+0.927)$, since $R = 5$, $\varphi = \arctan\frac{4}{3} \approx 0.927$; maximum value 5.

TERMS

a, b – coefficients of $\sin x$ and $\cos x$
R – amplitude of the combined wave
φ, δ – phase angles (radians)
$\operatorname{atan2}(y,x)$ – angle of the point (x,y), all four quadrants T.14
Derivation – expand $R\sin(x+\varphi)$ with T.8
$\max_x$ – the largest value over all x



CONTEXT

- **Origin.** Phasor addition in 19th-century AC theory (Steinmetz, 1893).
- **Use.** Solving $a\sin x + b\cos x = c$ T.15; adding AC signals EE.14 §16 Electrical Eng..
- **Data science.** Regression with $\beta_1\sin\frac{2\pi t}{P} + \beta_2\cos\frac{2\pi t}{P}$ fits a seasonal wave of *unknown* phase with a *linear* model; amplitude $= \sqrt{\beta_1^2+\beta_2^2}$.
- **Physical application.** Two speakers playing the same tone with a phase offset produce one tone whose loudness depends on the offset.

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T.14 Inverse Trigonometric Functions

Trig functions repeat, so they are not one-to-one. Restricting each to one branch (the principal values) gives inverse functions that return a single angle for a given ratio.

Arcsine	$y = \arcsin x \iff \sin y = x, \quad y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$	Domain $[-1, 1]$; returns an angle in QI or QIV.
Arccosine	$y = \arccos x \in [0, \pi], \quad x \in [-1, 1]$	Returns an angle in QI or QII.
Arctangent	$y = \arctan x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right), \quad x \in \mathbb{R}$	Accepts any real number; horizontal asymptotes $y = \pm\frac{\pi}{2}$.
Cancellation	$\sin(\arcsin x) = x; \quad \arcsin(\sin\theta) = \theta$ only if $\lvert\theta\rvert \le \frac{\pi}{2}$	Inverse-after-function returns the *principal* angle: $\arcsin(\sin \pi) = 0$.
Complementary sum	$\arcsin x + \arccos x = \frac{\pi}{2}$	Cofunction identity read backwards T.7.
Negative inputs	$\arcsin(-x) = -\arcsin x, \quad \arccos(-x) = \pi - \arccos x$	arcsin and arctan are odd; arccos is not.
Triangle composition	$\cos(\arcsin x) = \sqrt{1 - x^2}$	Draw a triangle with $\sin\theta = x$ and read the other side.
Two-argument arctangent	$\theta = \operatorname{atan2}(y, x) \in (-\pi, \pi]$	Angle of the point (x, y); uses both signs, so it gets the quadrant right.

STATED-EQUATION *y equals arcsine x if and only if sine y equals x, with y between minus pi over two and pi over two. Arccosine x lies between zero and pi. Arctangent x lies strictly between minus pi over two and pi over two. Sine of arcsine x equals x, but arcsine of sine theta equals theta only for theta between minus pi over two and pi over two. Arcsine x plus arccosine x equals pi over two. Cosine of arcsine x equals the square root of one minus x squared. Arcsine and arccosine accept x between minus one and one; arctangent accepts any real x. Arcsine of minus x equals minus arcsine x, and arccosine of minus x equals pi minus arccosine x. Theta equals a tan two of y, x gives the angle of the point x, y, in the interval from minus pi, exclusive, to pi, inclusive.*

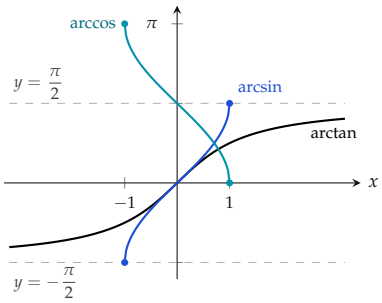
MEMORY AID Ranges: arcsin and arctan live on the *right* half of the circle (QI, QIV); arccos on the *top* half (QI, QII).

METHOD

1. Let $\theta =$ the inner inverse function, so one ratio of θ is known.
2. Draw a right triangle with that ratio; find the third side by Pythagoras G.7 §05 Geometry.
3. Read the requested ratio; fix its sign from the range of the inverse. Example: $\sin(\arctan\frac{3}{4})$: opp 3, adj 4, hyp 5, so $\frac{3}{5}$.

TERMS

$\sin^{-1} x$ – same as $\arcsin x$; *not* $\frac{1}{\sin x}$
Principal value – the one output in the restricted range
Restricted domain – makes the function one-to-one A2.1 §04 Algebra II
$\operatorname{atan2}(y, x)$ – full-circle angle using signs of both coordinates
Derivatives – $\frac{d}{dx}\arctan x = \frac{1}{1+x^2}$ C1.14 §07 Calculus I
$[a, b]$, $(a, b]$ – interval notation: bracket includes, parenthesis excludes A1.21 §03 Algebra I



CONTEXT

- **Origin.** The $\sin^{-1}$ notation is due to John Herschel (1813).
- **Use.** Recovering angles from ratios or measurements.
- **Data science.** arccos(cosine similarity) gives angular distance, a true metric; atan2 recovers a phase or hour from a $(\cos, \sin)$ encoding.
- **Physical application.** The angle of a ramp that rises 1 m over 12 m is $\arctan(1/12) \approx 4.8°$ (the ADA limit).

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T.15 Solving Trigonometric Equations

A trig equation usually has infinitely many solutions because the functions repeat. Find the solutions in one period, then add whole periods to get the general solution.

Sine equation	$\sin x = a \Rightarrow x = \arcsin a + 2\pi n$ or $x = \pi - \arcsin a + 2\pi n$	Two solutions per period (mirror images across $x = \frac{\pi}{2}$); none if $\lvert a\rvert > 1$.
Cosine equation	$\cos x = a \Rightarrow x = \pm\arccos a + 2\pi n$	Two solutions per period, symmetric about 0; none if $\lvert a\rvert > 1$.
Tangent equation	$\tan x = a \Rightarrow x = \arctan a + \pi n$	One solution per period π; every real a works.
Multiple angle	$\sin(kx) = a$: solve for kx, then divide by k	Divide the $2\pi n$ too: k times as many solutions in $[0, 2\pi)$.
Factored form	$f(x)\,g(x) = 0 \Rightarrow f(x) = 0$ or $g(x) = 0$	Zero-product rule A1.1 §03 Algebra I; never divide by a factor that could be 0.

STATED-EQUATION *If sine x equals a, then x equals arcsine a plus two pi n, or pi minus arcsine a plus two pi n. If cosine x equals a, x equals plus or minus arccosine a plus two pi n. If tangent x equals a, x equals arctangent a plus pi n. For sine of k x, solve for k x first and then divide everything by k. A product equal to zero gives one equation per factor.*

MEMORY AID "Find them in one lap, then go around again": $+2\pi n$ (or $+\pi n$ for tan).

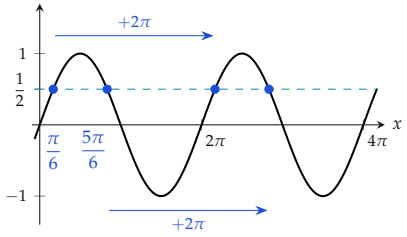
METHOD

1. Isolate a single trig function of a single angle (use identities if needed; see below).
2. If it is quadratic in that function, factor or use the quadratic formula A1.15 §03 Algebra I.
3. Find the reference angle and all solutions in $[0, 2\pi)$ with the unit circle T.K1.
4. Add $2\pi n$ (or πn), or list the solutions in the requested interval; check each one.
5. Example: $2\sin^2 x - \sin x - 1 = 0 \Rightarrow (2\sin x + 1)(\sin x - 1) = 0 \Rightarrow \sin x = -\frac{1}{2}$ or $1 \Rightarrow x = \frac{7\pi}{6}, \frac{11\pi}{6}, \frac{\pi}{2}$ $(+2\pi n)$.

DECISION POINT Which identity? *Both* sin *and* cos: if squared, use $\sin^2 + \cos^2 = 1$ to keep one function; if $a\sin x = b\cos x$, divide by $\cos x$ to get $\tan x = \frac{b}{a}$ (first check $\cos x = 0$ is not a solution); if $a\sin x + b\cos x = c$, use harmonic addition T.13. *Different angles* (x and $2x$): use double-angle T.9 to match them. *Sums of sines*: sum-to-product, then factor T.11. Avoid squaring both sides; if you must, check for extraneous roots.

TERMS

a – a constant; $\lvert a\rvert \le 1$ for sin and cos
n – any integer
k – coefficient of x inside the function
General solution – all solutions, written with n
Principal value – $\arcsin a$, $\arccos a$, $\arctan a$ T.14
f, g – trig expressions that are factors of the equation



CONTEXT

- **Origin.** Systematic general solutions appear in Euler's *Introductio* (1748).
- **Use.** Finding when a wave reaches a level: tides, AC voltage thresholds, angles of launch.
- **Data science.** Recovering times of peak or crossing from a fitted seasonal model; solving for the phase in a fitted $R\sin(\omega t + \varphi)$.
- **Physical application.** A tide $h(t) = 2\sin\frac{\pi t}{6} + 5$ is at 6 m when $\sin\frac{\pi t}{6} = \frac{1}{2}$, i.e. $t = 1, 5$ (h), then every 12 h.

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Solving Triangles

T.16 Law of Sines & Triangle Area

In any triangle (not only right ones), each side is proportional to the sine of the opposite angle. The same sine also gives the area from two sides and the included angle.

Law of sines	$\dfrac{a}{\sin A} = \dfrac{b}{\sin B} = \dfrac{c}{\sin C}$	Pair each side with its opposite angle; needs one complete pair.
Extended law of sines	$\dfrac{a}{\sin A} = 2R$	The common ratio is the circumcircle's diameter.
Area (SAS)	$K = \frac{1}{2}ab\sin C$	Two sides and the angle *between* them; $b\sin C$ is the height G.15 §05 Geometry.
Ambiguous-case height	$h = b\sin A$	For SSA, compare the side a opposite A with h and b to count triangles.

STATED-EQUATION *a over sine A equals b over sine B equals c over sine C, which equals two R. The area equals one half a b sine C. In the ambiguous case the height is b sine A.*

MEMORY AID "Side over sine of the angle across from it" is the same for all three pairs.

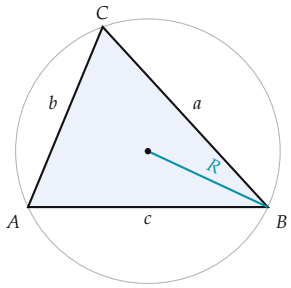
METHOD

1. Use the angle sum $A + B + C = \pi$ G.5 §05 Geometry to find any missing angle.
2. Use a known side–angle pair to get the common ratio.
3. Multiply by sin of each remaining angle. Example: $A = 30°$, $B = 45°$, $a = 4$: $b = \frac{4\sin 45°}{\sin 30°} = 4\sqrt{2} \approx 5.66$.

DECISION POINT SSA (two sides and a non-included angle A) may give 0, 1 or 2 triangles. With A acute and $h = b\sin A$: $a < h$ none; $a = h$ one (right angle); $h < a < b$ two (B and $180° - B$); $a \geq b$ one. With A obtuse: one triangle if $a > b$, none otherwise.

TERMS

a, b, c – sides opposite angles A, B, C G.5 §05 Geometry
R – radius of the circumscribed circle
K – area of the triangle
h – height from C to side c in the SSA test
Use when – you know AAS, ASA, or SSA (ambiguous case)



CONTEXT

- **Origin.** Stated in full by Nasir al-Din al-Tusi (13th c.).
- **Use.** Surveying and navigation by triangulation.
- **Data science.** Geospatial localization: a device's position is triangulated from bearings to known towers or beacons.
- **Physical application.** Triangulation: two observers a known distance apart measure angles to a ship to locate it.

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T.17 Law of Cosines

The Pythagorean theorem for any triangle: a correction term $-2ab\cos C$ accounts for the angle not being $90°$. It finds the third side from SAS, or any angle from SSS.

Law of cosines	$c^2 = a^2 + b^2 - 2ab\cos C$	Side opposite the known included angle.
Angle form	$\cos C = \dfrac{a^2 + b^2 - c^2}{2ab}$	Any angle from three sides; arccos returns the correct obtuse angle too.
Cyclic forms	$a^2 = b^2 + c^2 - 2bc\cos A, \quad b^2 = a^2 + c^2 - 2ac\cos B$	Same formula with the letters rotated.
Acute / obtuse test	$c^2 < a^2 + b^2 \iff C < 90°$	$=$: right angle; $>$: obtuse.

STATED-EQUATION *c squared equals a squared plus b squared minus two a b cosine C. Cosine C equals a squared plus b squared minus c squared, over two a b. The same holds for every side with its opposite angle. Angle C is acute exactly when c squared is less than a squared plus b squared.*

MEMORY AID It is the Pythagorean theorem with a correction: when $C = 90°$, $\cos C = 0$.

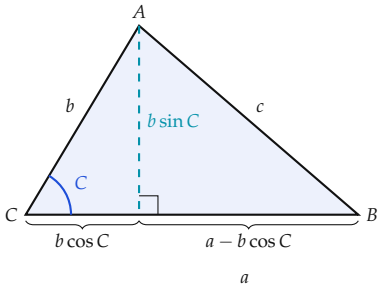
METHOD

1. SAS: compute the third side with the first formula. Example: $a = 3$, $b = 5$, $C = 60°$: $c^2 = 9 + 25 - 15 = 19$, $c \approx 4.36$.
2. SSS: compute the largest angle first with the angle form. Example: sides $3, 5, 7$: $\cos C = \frac{9+25-49}{30} = -\frac{1}{2}$, $C = 120°$.
3. Finish with the law of sines T.16 or the angle sum.

DECISION POINT Which law? **SAS** or **SSS** $\to$ law of cosines. **AAS** or **ASA** $\to$ law of sines. **SSA** $\to$ law of sines with the ambiguous-case check T.16. Right triangle $\to$ SOH-CAH-TOA T.1. After using cosines, find the *largest* angle with cosines (or only use the law of sines for the smaller angles), since arcsin cannot tell B from $180° - B$.

TERMS

C – angle between sides a and b (opposite c)
a, b, c – side lengths
Use when – you know SAS or SSS
$C = 90°$ – reduces to the Pythagorean theorem G.7 §05 Geometry
Vector form – $\|\mathbf{a} - \mathbf{b}\|^2 = \|\mathbf{a}\|^2 + \|\mathbf{b}\|^2 - 2\,\mathbf{a}\cdot\mathbf{b}$ T.21
A, B – angles opposite sides a, b



CONTEXT

- **Origin.** Geometric form in Euclid II.12–13; trigonometric form by al-Kashi (15th c.).
- **Use.** Distances across obstacles, navigation, robotics.
- **Data science.** $\|\mathbf{a} - \mathbf{b}\|^2 = \|\mathbf{a}\|^2 + \|\mathbf{b}\|^2 - 2\|\mathbf{a}\|\|\mathbf{b}\|\cos\theta$: for unit embeddings, Euclidean distance and cosine similarity T.21 carry the same ranking, $d^2 = 2 - 2\cos\theta$.
- **Physical application.** The distance between two ships leaving port on different headings is found by the law of cosines.

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T.18 Triangle Area, Radii & Law of Tangents

Once a triangle is solved, its area and its inscribed and circumscribed circles follow from a few formulas. The law of tangents and Mollweide's formula are classical alternatives and checks.

Area (SAS)	$K = \frac{1}{2}ab\sin C$	Two sides and the included angle T.16.
Area (ASA / AAS)	$K = \dfrac{a^2 \sin B \sin C}{2\sin A}$	One side and all angles; combine SAS area with the law of sines.
Heron's formula (SSS)	$K = \sqrt{s(s-a)(s-b)(s-c)},\quad s = \frac{a+b+c}{2}$	Three sides only, no angles G.15 §05 Geometry.
Circumradius	$R = \dfrac{abc}{4K}$	Radius of the circle through all three vertices.
Inradius	$r = \dfrac{K}{s}$	Radius of the circle tangent to all three sides.
Law of tangents	$\dfrac{a-b}{a+b} = \dfrac{\tan\frac{A-B}{2}}{\tan\frac{A+B}{2}}$	Optional SAS route avoiding square roots (pre-calculator).
Mollweide's check	$\dfrac{a+b}{c} = \dfrac{\cos\frac{A-B}{2}}{\sin\frac{C}{2}}$	Uses all six parts: a quick test that a solved triangle is consistent.

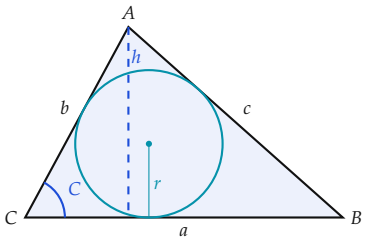
STATED-EQUATION *The area equals one half a b sine C, or a squared sine B sine C over two sine A, or the square root of s times s minus a times s minus b times s minus c, with s the semi-perimeter. The circumradius equals a b c over four K; the inradius equals K over s. The law of tangents: a minus b over a plus b equals tangent of half of A minus B over tangent of half of A plus B. Mollweide: a plus b over c equals cosine of half of A minus B over sine of half C.*

MEMORY AID Check with the 3–4–5 triangle: $s = 6$, $K = 6$, $R = \frac{60}{24} = 2.5$ (half the hypotenuse), $r = 1$.

DECISION POINT Which area formula? Two sides and the angle between them → SAS. Three sides → Heron. A side and angles → ASA form. Base and height known → $\frac{1}{2}bh$ G.15 §05 Geometry. Heron loses precision for needle-thin triangles; prefer SAS if an angle is available.

TERMS

K – area
s – semi-perimeter $\frac{a+b+c}{2}$
R, r – circumradius, inradius
a, b, c; A, B, C – sides and opposite angles G.5 §05 Geometry
Inscribed circle center – intersection of angle bisectors



$h = b\sin C, \quad K = \frac{1}{2}ah = rs$

CONTEXT

- **Origin.** Heron of Alexandria (c. 60 CE); law of tangents by Viète (1580); Mollweide (1808).
- **Use.** Land area from boundary measurements; mesh quality in engineering.
- **Data science.** Triangle meshes in graphics and finite-element models are rated by $\frac{r}{R}$; slivers with tiny $\frac{r}{R}$ cause numerical error.
- **Physical application.** A triangular plot with sides 30, 40, 50 m has area $\sqrt{60\cdot 30\cdot 20\cdot 10} = 600\,\text{m}^2$.

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Polar Form, Vectors & Complex Numbers

T.19 Polar Coordinates

Polar coordinates locate a point by its distance from the origin and its direction angle. They are natural whenever a problem has rotational symmetry.

Polar to Cartesian	$x = r\cos\theta, \qquad y = r\sin\theta$	Scale the unit-circle point $(\cos\theta, \sin\theta)$ by r T.4.
Radius	$r = \sqrt{x^2 + y^2}$	Distance from the origin G.12 §05 Geometry.
Angle	$\tan\theta = \dfrac{y}{x}, \qquad \theta = \operatorname{atan2}(y, x)$	$\arctan\frac{y}{x}$ alone is wrong in QII and QIII: add π there.
Non-uniqueness	$(r, \theta) = (r, \theta + 2\pi k) = (-r, \theta + \pi)$	One point has infinitely many polar names.
Basic polar curves	$r = a$ (circle), $\quad \theta = \alpha$ (line), $\quad r = 2a\cos\theta$	The last is the circle of radius a centered at $(a, 0)$.

STATED-EQUATION *x equals r cosine theta; y equals r sine theta. r equals the square root of x squared plus y squared. Tangent theta equals y over x; use a tan two of y, x to get the right quadrant. r, theta is the same point as r, theta plus two pi k, and as minus r, theta plus pi. Basic polar curves: r equals a is a circle of radius a about the origin, theta equals alpha is a line through the origin, and r equals two a cosine theta is a circle through the origin.*

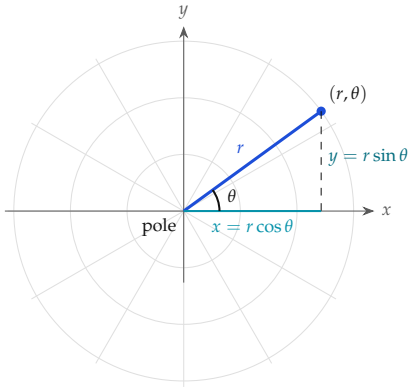
MEMORY AID Polar is "how far, which way"; Cartesian is "how far across, how far up".

METHOD

1. Compute $r = \sqrt{x^2 + y^2}$.
2. Compute $\arctan\frac{y}{x}$ and correct it by quadrant (or use atan2).
3. Example: $(-1, \sqrt{3})$: $r = 2$; $\arctan(-\sqrt{3}) = -\frac{\pi}{3}$, but the point is in QII, so $\theta = -\frac{\pi}{3} + \pi = \frac{2\pi}{3}$.

TERMS

r – distance from the origin (the pole) G.12 §05 Geometry
θ – angle from the positive x-axis T.2
Quadrant – choose θ from the signs of x, y T.14
a, α – constants in the curve equations
Polar area and calculus – see C2.26 §08 Calculus II; 3-D versions C3.21 §10 Calculus III
k – any integer



CONTEXT

- **Origin.** Used by Newton and Jacob Bernoulli (1691).
- **Use.** Circular motion, radar, antenna patterns, spirals.
- **Data science.** The Box–Muller transform generates normal samples from a random radius and angle; radar charts plot feature profiles in polar form.
- **Physical application.** Radar reports a plane's position as a distance and a bearing, which are polar coordinates.

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T.20 Vectors: Components, Magnitude & Direction

A plane vector can be described by its components (v_x, v_y) or by its length and direction angle. Trigonometry converts between the two; vectors are added in component form.

Components from polar	$\mathbf{v} = \langle \lVert\mathbf{v}\rVert \cos\theta,\ \lVert\mathbf{v}\rVert \sin\theta \rangle$	Resolve a force or velocity into horizontal and vertical parts.
Magnitude	$\lVert\mathbf{v}\rVert = \sqrt{v_x^2 + v_y^2}$	Length of the arrow (Pythagoras).
Direction angle	$\theta = \operatorname{atan2}(v_y, v_x)$	Angle from the $+x$-axis; check the quadrant T.19.
Unit vector	$\hat{\mathbf{u}} = \dfrac{\mathbf{v}}{\lVert\mathbf{v}\rVert} = \langle \cos\theta, \sin\theta \rangle$	Pure direction, length 1: a point on the unit circle.
Resultant (sum)	$\mathbf{u} + \mathbf{v} = \langle u_x + v_x,\ u_y + v_y \rangle$	Add components, never magnitudes (unless parallel).
Rotation by angle α	$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$	Turns any vector counter-clockwise by α without changing its length LA.9 §09 Linear Algebra.

STATED-EQUATION *A vector of length norm v at angle theta has components norm v cosine theta and norm v sine theta. Its magnitude is the square root of v x squared plus v y squared, and its direction is a tan two of v y, v x. The unit vector is v over its norm, cosine theta, sine theta. Vectors add component by component. Rotating by alpha sends x, y to x cosine alpha minus y sine alpha, x sine alpha plus y cosine alpha.*

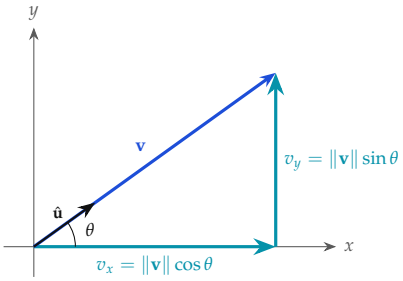
MEMORY AID "Cos goes with x, sin goes with y" (as on the unit circle).

METHOD

1. Convert each vector to components.
2. Add the components.
3. Convert back to magnitude and direction. Example: 3 N east plus 4 N north $= \langle 3, 4 \rangle$: magnitude 5 N, direction arctan $\frac{4}{3} \approx 53.1°$.

TERMS

$\mathbf{v}$ – vector with components v_x, v_y LA.K1 §09 Linear Algebra
$\lVert\mathbf{v}\rVert$ – magnitude (length)
θ – direction angle from the $+x$-axis
$\hat{\mathbf{u}}$ – unit vector in the direction of $\mathbf{v}$
α – rotation angle; the matrix is orthogonal LA.16 §09 Linear Algebra
$\mathbf{u}$ – a second vector with components u_x, u_y
$\langle \cdot, \cdot \rangle$ – component (coordinate) form of a vector
(x', y') – image of (x, y) after the rotation
atan2 – two-argument arctangent T.14



CONTEXT

- **Origin.** Stevin's parallelogram of forces (1586); component notation from Gibbs (1880s).
- **Use.** Forces, velocities, wind and current corrections in navigation.
- **Data science.** Data augmentation rotates images and point clouds with the rotation matrix; PCA rotates axes to the directions of greatest variance ML.38 §15 Machine Learning.
- **Physical application.** A plane flying 200 km/h north in a 50 km/h east wind has ground velocity $\langle 50, 200 \rangle$: about 206 km/h, 14° east of north.

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T.21 Vectors, Dot Product & Cosine Similarity

The dot product multiplies matching components and adds them. Geometrically it is length times length times the cosine of the angle between, so it measures how much two vectors point the same way.

Norm (length)	$\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}$	Pythagoras in n dimensions; $\|\mathbf{v}\|^2 = \mathbf{v} \cdot \mathbf{v}$.
Dot product	$\mathbf{u} \cdot \mathbf{v} = \sum_{i=1}^{n} u_i v_i = \|\mathbf{u}\| \, \|\mathbf{v}\| \cos\theta$	Algebraic and geometric definitions agree (law of cosines T.17).
Cosine similarity	$\cos\theta = \dfrac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \, \|\mathbf{v}\|}$	Between -1 and 1; ignores length, compares direction only.
Orthogonality test	$\mathbf{u} \cdot \mathbf{v} = 0 \iff \theta = \frac{\pi}{2}$	Perpendicular vectors (for nonzero $\mathbf{u}, \mathbf{v}$) A1.8 §03 Algebra I.
Scalar projection	$\text{comp}_{\mathbf{v}} \, \mathbf{u} = \dfrac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{v}\|} = \|\mathbf{u}\| \cos\theta$	Signed length of the shadow of $\mathbf{u}$ on $\mathbf{v}$.
Vector projection	$\text{proj}_{\mathbf{v}} \, \mathbf{u} = \dfrac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{v}\|^2} \mathbf{v}$	The shadow as a vector along $\mathbf{v}$.

STATED-EQUATION *The norm of v equals the square root of v one squared plus v two squared through v n squared. u dot v equals the sum of u i v i, which equals the norm of u times the norm of v times cosine theta. Cosine theta equals u dot v over the norms. u dot v equals zero exactly when the vectors are perpendicular. The scalar projection of u on v is u dot v over the norm of v. The projection of u onto v equals u dot v over the norm of v squared, times v.*

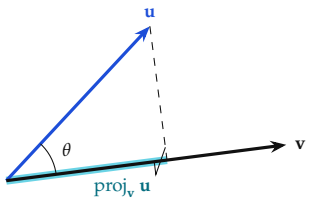
MEMORY AID Positive dot product: acute angle (same-ish direction); zero: perpendicular; negative: obtuse.

METHOD

1. Compute $\mathbf{u} \cdot \mathbf{v}$ and both norms.
2. Divide to get $\cos\theta$; take arccos for the angle. Example: $\mathbf{u} = (1, 2)$, $\mathbf{v} = (3, 1)$: $\mathbf{u} \cdot \mathbf{v} = 5$, $\cos\theta = \frac{5}{\sqrt{5}\sqrt{10}} = \frac{1}{\sqrt{2}}$, $\theta = 45°$.

TERMS

$\|\mathbf{v}\|$ – magnitude (L2 norm), from G.12 §05 Geometry
$\mathbf{u} \cdot \mathbf{v}$ – dot (inner) product, a scalar LA.K1 §09 Linear Algebra
u_i, v_i; n – components; dimension
θ – angle between the vectors T.17
Orthogonal – $\mathbf{u} \cdot \mathbf{v} = 0$, i.e. $\theta = 90°$ A1.8 §03 Algebra I
proj, comp – component of $\mathbf{u}$ along $\mathbf{v}$ (vector / scalar)



CONTEXT

- **Origin.** Vector analysis of Gibbs and Heaviside (1880s), from Hamilton's quaternions.
- **Use.** Work $W = \mathbf{F} \cdot \mathbf{d}$; components of forces.
- **Data science.** Cosine similarity ranks embeddings and TF-IDF documents; every linear model scores $\mathbf{w} \cdot \mathbf{x}$; projection is one-feature regression through the origin.
- **Physical application.** Work done by a force is a dot product: $W = \mathbf{F} \cdot \mathbf{d}$, pulling a sled at an angle.

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T.22 Polar (Trigonometric) Form of Complex Numbers

Plot $z = a + bi$ as the point (a, b) A2.6 §04 Algebra II; its polar coordinates give $z = r(\cos\theta + i\sin\theta)$. In this form multiplication is a rotation plus a scaling.

Trigonometric form	$z = r(\cos\theta + i\sin\theta) = r\,\mathrm{cis}\,\theta$	Modulus r, argument θ; "cis" abbreviates $\cos + i\sin$.
Modulus and argument	$r = \lvert z\rvert = \sqrt{a^2+b^2}, \qquad \theta = \arg z = \mathrm{atan2}(b,a)$	Polar conversion of the point (a,b) T.19.
Back to rectangular	$a = r\cos\theta, \qquad b = r\sin\theta$	Evaluate cos and sin, then distribute r.
Multiplication	$z_1 z_2 = r_1 r_2\,\mathrm{cis}(\theta_1 + \theta_2)$	Multiply moduli, add angles (proved by the sum formulas T.8).
Division	$\dfrac{z_1}{z_2} = \dfrac{r_1}{r_2}\,\mathrm{cis}(\theta_1 - \theta_2)$	Divide moduli, subtract angles ($z_2 \neq 0$).
Conjugate	$\bar{z} = r\,\mathrm{cis}(-\theta)$	Reflection across the real axis; $z\bar{z} = r^2$.

STATED-EQUATION *z equals r times cosine theta plus i sine theta, abbreviated r cis theta. The modulus r equals the square root of a squared plus b squared and the argument theta is a tan two of b, a. a equals r cosine theta and b equals r sine theta. To multiply, multiply the moduli and add the angles; to divide, divide the moduli and subtract the angles. The conjugate is r cis of minus theta.*

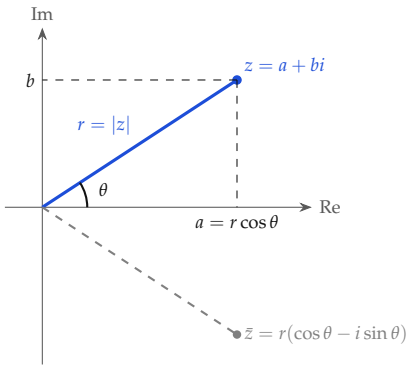
MEMORY AID "Multiply lengths, add angles."

METHOD

1. Convert each factor to polar form.
2. Multiply/divide the moduli and add/subtract the arguments.
3. Convert back if needed. Example: $(2\,\mathrm{cis}\,30^\circ)(3\,\mathrm{cis}\,60^\circ) = 6\,\mathrm{cis}\,90^\circ = 6i$.

TERMS

$z = a + bi$ – complex number; $a = \mathrm{Re}\,z$, $b = \mathrm{Im}\,z$ A2.6 §04 Algebra II
i – imaginary unit, $i^2 = -1$
$r = \lvert z\rvert$ – modulus (distance from 0)
$\theta = \arg z$ – argument, defined up to $2\pi k$
$\bar{z}$ – complex conjugate
Exponential form – $re^{i\theta}$ T.23
$\mathrm{cis}\,\theta$ – shorthand for $\cos\theta + i\sin\theta$
z_1, z_2 – two complex numbers with moduli r_1, r_2 and arguments θ_1, θ_2
atan2 – quadrant-correct arctangent T.14



CONTEXT

- **Origin.** Argand (1806) and Gauss (1831) popularized the complex plane.
- **Use.** Powers and roots of complex numbers T.23; AC circuit analysis with phasors EE.14 §16 Electrical Eng..
- **Data science.** Spectral analysis stores each frequency as a complex number: $\lvert z\rvert$ is its amplitude and $\arg z$ its phase EE.26 §16 Electrical Eng..
- **Physical application.** Impedance $Z = R + jX$ in polar form, $\lvert Z\rvert\angle\theta$, gives the voltage gain and phase lag of a circuit.

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T.23 Euler's Formula & De Moivre's Theorem

Euler's formula identifies $\operatorname{cis}\theta$ with the exponential $e^{i\theta}$, so the rules of exponents A1.3 §03 Algebra I become the rules of rotation: powers multiply angles and roots divide them.

Euler's formula	$e^{i\theta} = \cos\theta + i\sin\theta$	A point on the unit circle written as an exponential.
Exponential polar form	$z = re^{i\theta} = r(\cos\theta + i\sin\theta)$	Same as the trig form T.22, with exponent rules built in.
De Moivre's theorem	$z^n = r^n e^{in\theta} = r^n(\cos n\theta + i\sin n\theta)$	Raise the modulus to n, multiply the angle by n.
nth roots	$z^{1/n} = r^{1/n}\, e^{i(\theta+2\pi k)/n}, \qquad k = 0, 1, \ldots, n-1$	Exactly n roots, equally spaced by $\frac{2\pi}{n}$ on a circle of radius $r^{1/n}$.
Cosine and sine from exponentials	$\cos\theta = \dfrac{e^{i\theta}+e^{-i\theta}}{2}, \quad \sin\theta = \dfrac{e^{i\theta}-e^{-i\theta}}{2i}$	Add or subtract $e^{\pm i\theta}$; compare the hyperbolic functions T.24.
Euler's identity	$e^{i\pi} + 1 = 0$	$\theta = \pi$: links e, i, π, 1 and 0.

STATED-EQUATION *e to the i theta equals cosine theta plus i sine theta. z equals r e to the i theta. z to the n equals r to the n e to the i n theta. The n-th roots of z are r to the one over n times e to the i theta plus two pi k over n, for k from zero to n minus one. Cosine theta equals e to the i theta plus e to the minus i theta over two; sine theta equals their difference over two i. e to the i pi plus one equals zero.*

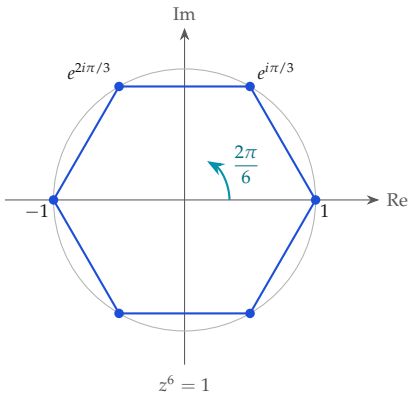
MEMORY AID Powers: "multiply the angle". Roots: "divide the angle, then add $\frac{2\pi}{n}$ steps".

METHOD

1. Write z in polar form $re^{i\theta}$.
2. Apply De Moivre (powers) or the root formula for each k.
3. Convert back. Example: $(1+i)^8 = (\sqrt{2}e^{i\pi/4})^8 = 16e^{2\pi i} = 16$; cube roots of 8: $2,\ 2e^{2\pi i/3},\ 2e^{4\pi i/3}$.

TERMS

r – modulus $|z|$ A2.6 §04 Algebra II
θ – argument (angle) of z T.19
i – imaginary unit, $i^2 = -1$
n, k – power / root index; root counter
Multiplication – multiply moduli, add angles (a rotation)
Proof idea – compare Taylor series C2.22 §08 Calculus II



CONTEXT

- **Origin.** Leonhard Euler (1748); Abraham de Moivre (1707).
- **Use.** Powers and roots of complex numbers; deriving multiple-angle identities; circuit and signal analysis.
- **Data science.** The discrete Fourier transform $X_k = \sum_n x_n e^{-2\pi ikn/N}$ powers spectral analysis of time series; RoPE treats embedding pairs as complex numbers and rotates them.
- **Physical application.** Electrical engineers write AC signals as $Ve^{i\omega t}$ so that phase shifts become simple multiplications.

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Hyperbolic Functions

T.24 Hyperbolic Functions

The hyperbolic functions are the even and odd parts of e^x. They satisfy identities that mirror the trig identities (with some signs flipped) and parametrize the hyperbola $x^2 - y^2 = 1$ the way cos, sin parametrize the circle.

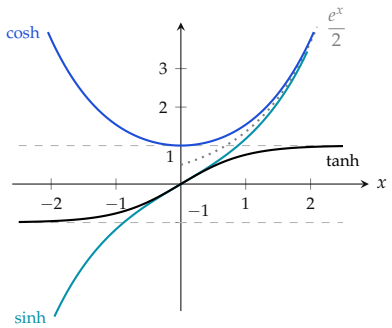
Hyperbolic sine	$\sinh x = \dfrac{e^x - e^{-x}}{2}$	Odd part of e^x; passes through 0, grows like $\frac{e^x}{2}$.
Hyperbolic cosine	$\cosh x = \dfrac{e^x + e^{-x}}{2}$	Even part of e^x; minimum 1 at $x = 0$ (the catenary shape).
Hyperbolic tangent	$\tanh x = \dfrac{\sinh x}{\cosh x} = \dfrac{e^x - e^{-x}}{e^x + e^{-x}}$	S-shaped, range $(-1, 1)$.
Fundamental identity	$\cosh^2 x - \sinh^2 x = 1$	Minus sign: $(\cosh t, \sinh t)$ lies on the hyperbola $x^2 - y^2 = 1$.
Exponential split	$e^x = \cosh x + \sinh x$	Compare $e^{i\theta} = \cos\theta + i\sin\theta$ T.23.
Addition and double angle	$\sinh(x+y) = \sinh x\cosh y + \cosh x \sinh y$, $\cosh 2x = \cosh^2 x + \sinh^2 x$	Same pattern as trig, but cosh formulas carry $+$ (Osborn's rule).
Derivatives	$\frac{d}{dx}\sinh x = \cosh x$, $\frac{d}{dx}\cosh x = \sinh x$	No minus sign, unlike cos C1.14 §07 Calculus I; $\frac{d}{dx}\tanh x = 1 - \tanh^2 x$.
Inverse functions	$\operatorname{arsinh} x = \ln\left(x + \sqrt{x^2+1}\right)$, $\operatorname{artanh} x = \frac{1}{2}\ln\frac{1+x}{1-x}$	Solve $y = \sinh x$ as a quadratic in e^x A2.14 §04 Algebra II.
Link to trigonometry	$\cos(ix) = \cosh x$, $\sin(ix) = i\sinh x$	Trig functions of imaginary angles are hyperbolic.

STATED-EQUATION *Hyperbolic sine of x equals e to the x minus e to the minus x, over two. Hyperbolic cosine equals e to the x plus e to the minus x, over two. Hyperbolic tangent equals sinh over cosh. Cosh squared minus sinh squared equals one. e to the x equals cosh x plus sinh x. Sinh of x plus y equals sinh x cosh y plus cosh x sinh y; cosh two x equals cosh squared plus sinh squared. The derivative of sinh is cosh and of cosh is sinh. Arsinh x equals the natural log of x plus the square root of x squared plus one; artanh x equals one half the log of one plus x over one minus x. Cosine of i x equals cosh x, and sine of i x equals i sinh x.*

MEMORY AID Osborn's rule: take a trig identity, change cos → cosh, sin → sinh, and flip the sign of every product of two sines ($\sin^2 \to -\sinh^2$).

TERMS

x – real number (not an angle)
e^x – natural exponential A2.13 §04 Algebra II
arsinh, artanh – inverse hyperbolic functions (also $\sinh^{-1}$, $\tanh^{-1}$)
Catenary – curve $y = a\cosh\frac{x}{a}$ of a hanging chain
Osborn's rule – converts trig identities to hyperbolic ones
Graph – gallery tile F.15 §18 Graph Gallery



CONTEXT

- **Origin.** Vincenzo Riccati (1757) and Johann Lambert (1768).
- **Use.** Hanging cables, special relativity (rapidity), integrals with $\sqrt{x^2 \pm a^2}$ C2.2 §08 Calculus II.
- **Data science.** tanh is a neural-network activation ML.22 §15 Machine Learning, $\tanh x = 2\sigma(2x) - 1$ with the logistic σ; Fisher's $z = \operatorname{artanh} r$ makes correlation coefficients approximately normal S.38 §13 Statistics.
- **Physical application.** Power lines and the Gateway Arch follow cosh curves; a falling body with drag has velocity $v_t \tanh\frac{gt}{v_t}$ P.22 §12 Physics.

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07
Calculus I

Limits, derivatives, and the first integrals. Calculus I is the mathematics of change and accumulation: the limit makes "instantaneous" precise, the derivative measures rates of change, and the integral adds rates back up into totals. The derivative is the engine inside every gradient-trained model.

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C1.1 Limits & Limit Laws

$\lim_{x\to a} f(x) = L$ means $f(x)$ can be made as close to L as we like by taking x close enough to (but not equal to) a. The limit laws split the limit of a combination into the limits of its pieces.

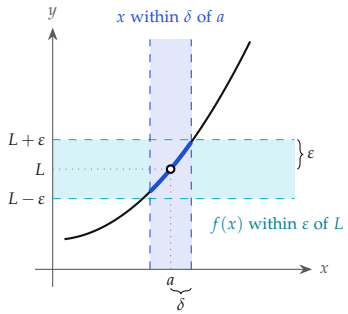
Limit	$\lim_{x\to a} f(x) = L$	What f approaches near a; the value $f(a)$ itself is irrelevant and may not even exist.
Sum & constant multiple laws	$\lim_{x\to a}\big[c\,f \pm g\big] = c\lim_{x\to a} f \pm \lim_{x\to a} g$	Limits split across sums and pull out constants, provided each piece's limit exists (and is finite).
Product law	$\lim_{x\to a}\big[f\,g\big] = \lim_{x\to a} f \cdot \lim_{x\to a} g$	The limit of a product is the product of the limits.
Quotient law	$\lim_{x\to a}\dfrac{f}{g} = \dfrac{\lim f}{\lim g} \quad (\lim g \neq 0)$	Fails when $\lim g = 0$: then classify the form C1.4.
Power & root laws	$\lim_{x\to a}[f]^n = \Big[\lim_{x\to a} f\Big]^n, \quad \lim_{x\to a}\sqrt[n]{f} = \sqrt[n]{\lim_{x\to a} f}$	For even roots the inner limit must be ≥ 0.
Direct substitution	$\lim_{x\to a} f(x) = f(a)$ if f is continuous at a	Polynomials, and rational, root, exp, log and trig functions inside their domains C1.7: just plug in.
Existence (two-sided)	$\lim_{x\to a} f(x)$ exists $\iff \lim_{x\to a^-} f(x) = \lim_{x\to a^+} f(x)$	Both one-sided limits C1.2 must exist and agree.
ε–δ definition	$\lim_{x\to a} f(x) = L \iff \forall \varepsilon > 0\ \exists \delta > 0 : \quad 0 < \lvert x-a\rvert < \delta \implies \lvert f(x) - L\rvert < \varepsilon$	Any output tolerance ε can be met by some input tolerance δ; the "$0 <$" excludes $x = a$ itself.

STATED-EQUATION *The limit as x approaches a of f of x equals L. The limit of c f plus or minus g is c times the limit of f plus or minus the limit of g; the limit of a product is the product of the limits; the limit of a quotient is the quotient of the limits, provided the denominator's limit is not zero; powers and roots pass through the limit. If f is continuous at a, the limit is f of a. The limit exists if and only if the left-hand and right-hand limits are equal. Formally: for every epsilon greater than zero there is a delta greater than zero such that zero less than the distance from x to a less than delta implies the distance from f of x to L is less than epsilon.*

MEMORY AID A limit asks where f is *heading*, not where it *arrives*.

TERMS

a – point approached (need not be in the domain)
L – limiting value (a finite real number)
a^-, a^+ – from the left, from the right
c, n – a constant; a positive integer
ε, δ – output and input tolerances (small positive numbers)
$\forall, \exists$ – for all, there exists M.89 §01 Math Signs, M.90 §01 Math Signs
$|x-a|$ – distance from x to a M.43 §01 Math Signs
At infinity – leading terms dominate C1.3, A2.10 §04 Algebra II
Procedure – see the key topic C1.K1
f, g – functions whose limits at a are combined



CONTEXT

- **Origin.** Cauchy (1821); rigorous ε–δ form by Weierstrass (1860s).
- **Use.** Foundation of derivatives C1.9 and integrals C1.30.
- **Data science.** Consistency of an estimator means $\hat{\theta}_n \to \theta$ as $n \to \infty$; "training converged" is a claim about a limit of the loss sequence.
- **Physical application.** Instantaneous speed is a limit of average speeds over shorter and shorter time intervals.

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C1.2 One-Sided & Infinite Limits

A one-sided limit lets x approach a from only one side. An infinite limit records that f grows without bound near a, which signals a vertical asymptote.

Left-hand limit	$\lim_{x\to a^-} f(x) = L$	Only $x < a$ counts. Needed at jumps, interval endpoints and piecewise boundaries.
Right-hand limit	$\lim_{x\to a^+} f(x) = L$	Only $x > a$ counts.
Infinite limit	$\lim_{x\to a} f(x) = +\infty$	f exceeds every bound near a. The limit does *not* exist as a number; "∞" says *how* it fails.
Vertical asymptote	$x = a$ if $\lim_{x\to a^\pm} f(x) = \pm\infty$	Any one infinite one-sided limit suffices. Typically a denominator $\to 0$ while the numerator does not A2.10 §04 Algebra II.
Nonzero over zero	$\lim_{x\to a^\pm} \frac{c}{x-a} = \pm\infty \quad (c>0)$	$\frac{c}{\text{tiny}}$ is huge; the sign comes from the signs of c and of the tiny denominator.
Standard examples	$\lim_{x\to 0^\pm} \frac{1}{x} = \pm\infty, \quad \lim_{x\to 0} \frac{1}{x^2} = +\infty, \quad \lim_{x\to 0^+} \ln x = -\infty$	$\frac{1}{x}$ has different one-sided limits; $\frac{1}{x^2}$ goes to $+\infty$ from both sides.

STATED-EQUATION *The limit as x approaches a from the left of f of x equals L; from the right, likewise. The limit of f equals infinity when f grows without bound. The line x equals a is a vertical asymptote when a one-sided limit is plus or minus infinity. c over x minus a goes to plus infinity from the right and minus infinity from the left when c is positive. One over x goes to plus or minus infinity as x approaches zero from the right or the left, one over x squared goes to plus infinity from both sides, and natural log of x goes to minus infinity as x approaches zero from the right.*

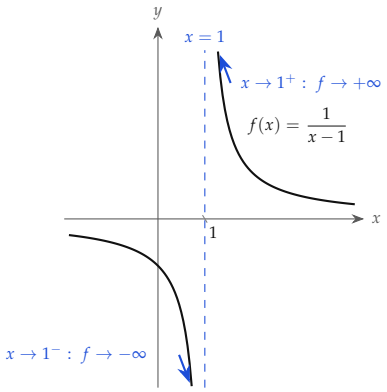
MEMORY AID "Nonzero over zero" is never indeterminate: it is always $\pm\infty$ (or does not exist). Only the *sign* needs work.

METHOD

1. Substitute $x = a$. A nonzero number over 0 means an infinite one-sided limit.
2. Find the sign of the denominator just left and just right of a (test $a \pm 0.01$).
3. Combine with the sign of the numerator; if the two sides disagree the two-sided limit does not exist. Example: $\lim_{x\to 2^-} \frac{3}{x-2}$: $x-2$ is a small negative number, so the limit is $-\infty$; from the right it is $+\infty$.

TERMS

a^- / a^+ – approach through $x < a$ / $x > a$
∞ – unbounded growth; not a real number
c – a nonzero constant
DNE – does not exist
ln – natural logarithm A2.14 §04 Algebra II
Piecewise function – different formulas on different intervals; check each side
L – a finite limiting value



CONTEXT

- **Origin.** Formalized with the rest of 19th-century analysis (Cauchy, Weierstrass).
- **Use.** Piecewise models, step functions, asymptotes of rational functions.
- **Data science.** Log-loss $-\ln p \to +\infty$ as $p \to 0^+$: a confident wrong prediction costs without bound, so libraries clip p to $[\epsilon, 1-\epsilon]$ ML.2 §15 Machine Learning.
- **Physical application.** Coulomb's force $F = kq_1q_2/r^2 \to \infty$ as $r \to 0^+$: the point-charge model breaks down at zero separation.

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C1.3 Limits at Infinity & Dominant Terms

As $x \to \pm\infty$ the fastest-growing term dominates everything else. For a rational function, compare the degrees of top and bottom. A finite limit at infinity is a horizontal asymptote.

Reciprocal power	$\lim_{x\to\pm\infty} \frac{c}{x^p} = 0 \quad (p > 0)$	The basic fact every other limit at infinity is reduced to.
Divide by the highest power	$\frac{3x^2+1}{5x^2-x} = \frac{3+1/x^2}{5-1/x} \xrightarrow[x\to\infty]{} \frac{3}{5}$	Divide every term by the highest power of x in the denominator, then apply the rule above.
Horizontal asymptote	$y = L$ if $\lim_{x\to\infty} f(x) = L$ or $\lim_{x\to-\infty} f(x) = L$	The graph levels off at height L; it may still cross the line for finite x.
Degree rule (rational functions)	$\lim_{x\to\pm\infty} \frac{a_n x^n + \cdots}{b_m x^m + \cdots} = \begin{cases} 0, & n < m \\ \frac{a_n}{b_m}, & n = m \\ \pm\infty, & n > m \end{cases}$	Bottom heavier: 0. Equal degrees: ratio of leading coefficients. Top heavier: infinite, with the sign of the leading terms.
Growth hierarchy	$\ln n \ll n^p \ll a^n \ll n! \ll n^n \quad (p > 0,\ a > 1)$	Each eventually beats the one before: $\frac{n^{10}}{2^n} \to 0$, $\frac{\ln n}{\sqrt{n}} \to 0$.
Standard limits at infinity	$e^{-x} \to 0, \quad \arctan x \to \pm\frac{\pi}{2}, \quad \frac{\ln x}{x} \to 0$	Exponential decay vanishes; arctan levels off; logs lose to every power.
Roots at $-\infty$	$\sqrt{x^2} = \lvert x\rvert = -x\ (x<0) \ \Rightarrow\ \lim_{x\to-\infty} \frac{\sqrt{x^2+1}}{x} = -1$	When pulling x out of a square root for $x \to -\infty$, it comes out as $-x$: the classic sign trap.

STATED-EQUATION *The limit of c over x to the p, as x goes to plus or minus infinity, is zero for positive p. Divide every term by the highest power in the denominator. If f approaches L at infinity, y equals L is a horizontal asymptote. For a rational function: degree of the top less than the bottom gives zero, equal degrees give the ratio of leading coefficients, a larger top degree gives infinity. Log n grows slower than any power, which grows slower than any exponential, then factorial, then n to the n. As x goes to infinity, e to the minus x goes to zero, arctangent of x goes to plus or minus pi over two, and log x over x goes to zero. For negative x the square root of x squared equals minus x, so the square root of x squared plus one, over x, goes to minus one as x goes to minus infinity.*

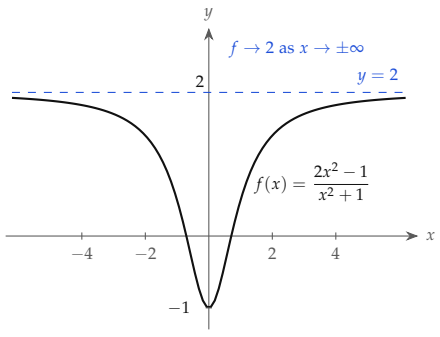
MEMORY AID BOBO BOTN EATS DC: *B*igger *O*n *B*ottom → *0*; *B*igger *O*n *T*op → *N*one (infinite); *E*xponents *A*re *T*he *S*ame → *D*ivide Coefficients.

METHOD

1. Identify the dominant (highest-power or fastest-growing) term in the numerator and in the denominator.
2. Divide every term by the highest power of x in the denominator (as $|x|$ under even roots).
3. Send every $\frac{c}{x^p}$ to 0 and read off the limit. Example: $\lim_{x\to\infty} \frac{2x+3}{x-1} = \lim \frac{2+3/x}{1-1/x} = \frac{2}{1} = 2$.

TERMS

n, m – degrees of numerator and denominator
a_n, b_m – leading coefficients
p, a – a positive power; an exponential base > 1
$f \ll g$ – f grows much slower: $\frac{f}{g} \to 0$
$n!$ – factorial M.48 §01 Math Signs
Oblique asymptote – if $n = m + 1$, the quotient line from long division A2.10 §04 Algebra II
c, L – a constant; a finite limit (asymptote height)
arctan – inverse tangent T.14 §06 Trigonometry
$|x|$ – absolute value; $\sqrt{x^2} = |x|$ M.43 §01 Math Signs



CONTEXT

- **Origin.** "Asymptote" (Greek: "not falling together") is Apollonius' word for the hyperbola's limiting lines.
- **Use.** Long-run behavior of models, end behavior of graphs, steady states.
- **Data science.** Big-O O.10 §14 Optimization is a statement about limits at infinity: an $O(n \log n)$ sort beats $O(n^2)$ because $\frac{n \log n}{n^2} \to 0$.
- **Physical application.** A falling object with drag approaches terminal velocity: $v(t) = v_T(1 - e^{-kt}) \to v_T$ P.22 §12 Physics.

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C1.4 Indeterminate Forms

An indeterminate form is a limit whose pieces' limits do not determine the answer: a $\frac{0}{0}$ limit can turn out to be any number, or $\pm\infty$. Each of the seven forms must be rewritten before it can be evaluated.

Quotient forms	$\frac{0}{0}$, $\frac{\infty}{\infty}$	Simplify algebraically C1.K1 or apply L'Hôpital's rule C1.28 directly.
Product form	$0 \cdot \infty \longrightarrow \frac{f}{1/g}$ or $\frac{g}{1/f}$	Move one factor into the denominator to get $\frac{0}{0}$ or $\frac{\infty}{\infty}$.
Difference form	$\infty - \infty \longrightarrow$ one fraction	Combine over a common denominator, or multiply by the conjugate, to get a quotient form.
Power forms	1^∞, 0^0, ∞^0	The base and the exponent pull in opposite directions; take logarithms first.
Exponential trick	$\lim f^g = \exp\left(\lim g \ln f\right)$	$g \ln f$ is a $0 \cdot \infty$ form; evaluate it, then exponentiate.
1^∞ shortcut	$f \to 1,\ g \to \infty:\quad \lim f^g = e^{\lim g\,(f-1)}$	Gives $\left(1+\frac{r}{n}\right)^n \to e^r$ at once C1.6.
Not indeterminate	$\frac{c}{0}\ (c \neq 0) \to \pm\infty$, $\frac{0}{\infty} \to 0$, $\infty + \infty \to \infty$, $0^{+\infty} \to 0$	These have definite outcomes. Do not apply L'Hôpital's rule to them.

STATED-EQUATION *Zero over zero and infinity over infinity are the quotient forms. Zero times infinity is rewritten as f over one over g. Infinity minus infinity is combined into one fraction. One to the infinity, zero to the zero and infinity to the zero are power forms: the limit of f to the g equals e to the limit of g times log f. For one to the infinity, the limit equals e to the limit of g times f minus one. Not indeterminate: c over zero, with c nonzero, is plus or minus infinity; zero over infinity is zero; infinity plus infinity is infinity; and zero to the plus infinity is zero.*

MEMORY AID Seven indeterminate forms: $\frac{0}{0}$, $\frac{\infty}{\infty}$, $0 \cdot \infty$, $\infty - \infty$, 1^∞, 0^0, ∞^0.

METHOD

1. Substitute to identify the form.
2. Quotient form: simplify algebraically or use L'Hôpital.
3. $0 \cdot \infty$ or $\infty - \infty$: rewrite as a single quotient first.
4. Power form: set $y = f^g$, compute $\lim \ln y = \lim g \ln f$, then $\lim y = e^{\lim \ln y}$. Example: $\lim_{x\to 0^+} x^x$: $\ln y = x \ln x = \frac{\ln x}{1/x} \to 0$, so $x^x \to e^0 = 1$.

Why "indeterminate"? As $x \to 0$, every one of these is a $\frac{0}{0}$ form, yet $\frac{x}{x} \to 1$, $\frac{x^2}{x} \to 0$, $\frac{x}{x^2} \to \infty$ (from the right), $\frac{5x}{x} \to 5$. The form alone cannot tell you the answer.

TERMS

f, g – functions whose limits produce the form
∞ – unbounded growth C1.2
$\exp(u)$ – e^u A2.13 §04 Algebra II; $\ln$ natural log A2.14 §04 Algebra II
c – a nonzero constant
Conjugate – $\sqrt{a} - \sqrt{b}$ for $\sqrt{a} + \sqrt{b}$ A1.4 §03 Algebra I

CONTEXT

- **Origin.** Euler treated vanishing fractions such as $\frac{0}{0}$ in his *Institutiones calculi differentialis* (1755).
- **Use.** Recognizing when a limit needs work before it can be evaluated.
- **Data science.** Entropy uses $0 \log 0 = 0$, justified by the $0 \cdot \infty$ limit $x \ln x \to 0$ as $x \to 0^+$ ML.3 §15 Machine Learning.
- **Physical application.** Compound interest $\left(1+\frac{r}{n}\right)^{nt}$ is a 1^∞ form whose limit is continuous compounding e^{rt} A2.13 §04 Algebra II.

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C1.5 Squeeze Theorem

If f is trapped between two functions that approach the same limit, f is forced to approach it too. The tool of choice when f oscillates or cannot be simplified.

Squeeze theorem	$g \le f \le h$ near a, $\lim_{x\to a} g = \lim_{x\to a} h = L \implies \lim_{x\to a} f = L$	The inequalities need only hold near a (not at a); works for $x \to \pm\infty$ too.
Bounded times vanishing	$\lvert f(x)\rvert \le M$, $\lim g = 0 \implies \lim f\,g = 0$	A bounded wiggle times something going to 0 goes to 0.
Absolute-value form	$\lvert f(x) - L\rvert \le g(x) \to 0 \implies f(x) \to L$	Bound the error by something that vanishes.
Classic oscillation	$-\lvert x\rvert \le x\sin\frac{1}{x} \le \lvert x\rvert \Rightarrow \lim_{x\to 0} x\sin\frac{1}{x} = 0$	$\sin\frac{1}{x}$ has no limit at 0, but multiplying by x crushes it.
Sine limit (proof)	$\cos x \le \dfrac{\sin x}{x} \le 1 \Rightarrow \lim_{x\to 0} \dfrac{\sin x}{x} = 1$	Comparing areas in the unit circle traps $\frac{\sin x}{x}$ C1.6.

STATED-EQUATION *If g is less than or equal to f, which is less than or equal to h near a, and g and h both approach L, then f approaches L. A bounded function times a function going to zero goes to zero. If the distance from f to L is at most g, and g goes to zero, f goes to L. Since x sine of one over x lies between minus absolute x and absolute x, its limit at zero is zero. Since sine x over x lies between cosine x and one, its limit at zero is one.*

MEMORY AID Two police officers walking a suspect to the same cell: wherever they end up, so does the suspect.

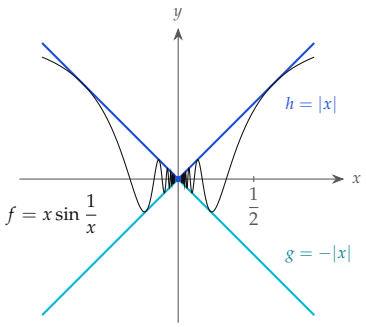
METHOD

1. Find a bound for the troublesome factor ($-1 \le \sin \le 1$, $-1 \le \cos \le 1$, $0 \le \text{anything}^2$).
2. Multiply the inequality through by the rest of the expression (reverse it if that factor is negative).
3. Check that both bounds have the *same* limit. Example: $\lim_{x\to\infty} \frac{\sin x}{x}$: $-\frac{1}{x} \le \frac{\sin x}{x} \le \frac{1}{x}$, both $\to 0$, so the limit is 0.

DECISION POINT Choosing the bounds is a judgment call: start from the known bounded factor and keep the rest exact. Bounds that are too loose (e.g. ≤ 1 when the limit is 0) prove nothing; if the two bounds have different limits, tighten one or try another method.

TERMS

g, h – lower and upper bounding functions
L – the common limit
M – a fixed bound: $|f| \le M$
Near a – on some open interval around a, except possibly at a
$|x|$ – absolute value M.43 §01 Math Signs



CONTEXT

- **Origin.** Archimedes trapped π between inscribed and circumscribed polygons; the modern form is part of 19th-century analysis.
- **Use.** Oscillating limits; proving the special limits C1.6.
- **Data science.** Convergence proofs squeeze the error: for gradient descent on a smooth convex loss, $0 \le L(\theta_k) - L^* \le \frac{C}{k}$, so the gap is forced to 0 C1.26.
- **Physical application.** Archimedes' polygon bounds gave $3\frac{10}{71} < \pi < 3\frac{1}{7}$.

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C1.6 Special Limits & the Number e

A handful of $\frac{0}{0}$ and 1^∞ limits that algebra cannot simplify. They are the reason $\sin x$, $\cos x$ and e^x have the derivatives they do, and they define e.

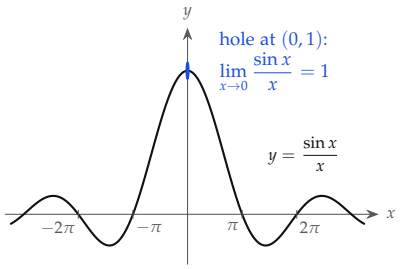
Sine limit	$\lim_{x\to 0}\dfrac{\sin x}{x}=1$	x in radians; proved by squeezing C1.5. Gives $\sin x \approx x$ for small x.
Scaled sine	$\lim_{x\to 0}\dfrac{\sin kx}{x}=k$	Rewrite as $k\cdot\frac{\sin kx}{kx}$ to match the sine limit.
Cosine limits	$\lim_{x\to 0}\dfrac{1-\cos x}{x}=0,\qquad \lim_{x\to 0}\dfrac{1-\cos x}{x^2}=\dfrac{1}{2}$	Multiply by $1+\cos x$ and use $\sin^2 x = 1-\cos^2 x$ T.6 §06 Trigonometry.
Exponential limit	$\lim_{x\to 0}\dfrac{e^x-1}{x}=1$	The slope of e^x at 0 is exactly 1: what makes e the natural base.
Logarithm limit	$\lim_{x\to 0}\dfrac{\ln(1+x)}{x}=1$	So $\ln(1+x)\approx x$ for small x.
Definition of e	$\lim_{n\to\infty}\left(1+\dfrac{1}{n}\right)^n=e$	Compounding 100% interest ever more often approaches $e\approx 2.71828$.
Continuous growth	$\lim_{n\to\infty}\left(1+\dfrac{r}{n}\right)^n=e^r$	A 1^∞ form C1.4; the basis of continuous compounding A2.13 §04 Algebra II.

STATED-EQUATION *The limit as x goes to zero of sine x over x equals one; of sine k x over x equals k. Of one minus cosine x over x equals zero; over x squared, one half. Of e to the x minus one over x equals one. Of natural log of one plus x, over x, equals one. The limit as n goes to infinity of one plus one over n, to the n, equals e; of one plus r over n, to the n, equals e to the r.*

MEMORY AID Small x (radians): $\sin x \approx x,\ 1-\cos x \approx \frac{x^2}{2},\ e^x \approx 1+x,\ \ln(1+x)\approx x$.

TERMS

x – in radians T.2 §06 Trigonometry
e – Euler's number ≈ 2.71828 A2.13 §04 Algebra II
k, r – constants (scale, rate)
n – positive integer, $n\to\infty$
Squeeze theorem – proves the sine limit by trapping it between $\cos x$ and 1
Consequence – $(1-\frac{1}{n})^n \to e^{-1}\approx 0.368$



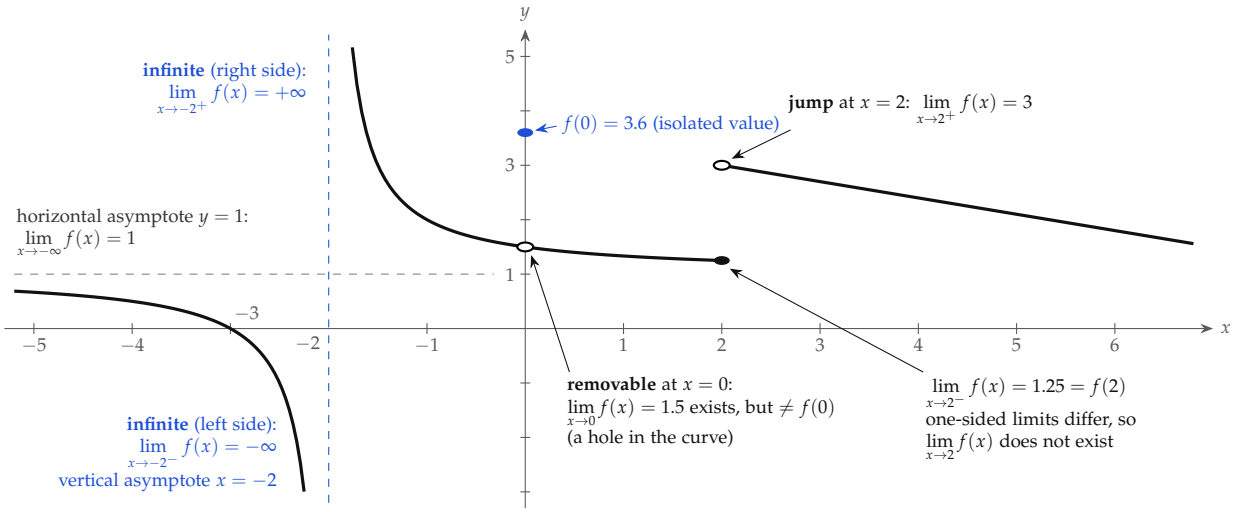
CONTEXT

- **Origin.** Jacob Bernoulli (1683) for e; the sine limit from the geometry of the unit circle.
- **Use.** Derivatives of $\sin x$ and e^x C1.14; continuous compounding.
- **Data science.** Bootstrap: a given row is left out of a resample with probability $(1-\frac{1}{n})^n \approx e^{-1}$, so about 36.8% of rows are "out-of-bag" in random forests ML.13 §15 Machine Learning.
- **Physical application.** The small-angle approximation $\sin\theta \approx \theta$ (radians) simplifies the pendulum equation P.15 §12 Physics.

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Limits: Reading, Evaluating & Defining | *Key Topic C1.K1*

A limit describes where $f(x)$ is heading as x approaches a point, regardless of what happens *at* the point. Read limits off a graph by tracing the curve toward $x = a$ from each side; evaluate them algebraically with the six-step procedure below. Step 1 settles most limits; the real skill is the algebra of step 3, where you must *choose* a move.



Reading limits from a graph

Two-sided limit exists when the curve approaches the same height from both sides; a hole or a moved point at $x = a$ does not matter (removable, $x = 0$ above).

One-sided limits C1.2: trace from the left ($x \to a^-$) and from the right ($x \to a^+$) separately. At a jump ($x = 2$) they differ, so the two-sided limit does not exist.

Infinite limits: near a vertical asymptote ($x = -2$) the curve shoots to $\pm\infty$; write $\lim = \pm\infty$ for each side.

Limits at infinity C1.3: the height the curve levels off to as $x \to \pm\infty$ is a horizontal asymptote ($y = 1$).

The evaluation procedure

1 Direct substitution

If f is continuous at a C1.7, the limit is $f(a)$. Always try this first. $\lim_{x\to 2}(x^2+3x) = 4+6 = 10$.

2 Classify the result

A number: done. $\frac{c}{0}$ with $c \neq 0$: infinite or DNE, go to step 6. $\frac{0}{0}$, $\frac{\infty}{\infty}$: indeterminate, continue. $0\cdot\infty$, $\infty-\infty$, 1^∞, 0^0, ∞^0: rewrite as a quotient first C1.4.

3 Algebra: remove the zero

Factor & cancel: $\lim\limits_{x\to 3}\dfrac{x^2-9}{x-3} = \lim\limits_{x\to 3}(x+3) = 6.$

Rationalize with the conjugate A1.4 §03 Algebra I: $\lim\limits_{x\to 4}\dfrac{\sqrt{x}-2}{x-4} = \lim\limits_{x\to 4}\dfrac{1}{\sqrt{x}+2} = \dfrac{1}{4}.$

Common denominator: $\lim\limits_{x\to 0}\dfrac{\frac{1}{x+2}-\frac{1}{2}}{x} = \lim\limits_{x\to 0}\dfrac{-1}{2(x+2)} = -\dfrac{1}{4}.$

Divide by the highest power ($x \to \infty$): $\lim\limits_{x\to\infty}\dfrac{3x^2+1}{x^2+5} = \lim\dfrac{3+1/x^2}{1+5/x^2} = 3.$

4 Special limits & squeeze

Rescale to match a known form C1.6: $\lim\limits_{x\to 0}\dfrac{\sin 3x}{x} = 3\cdot\lim\limits_{x\to 0}\dfrac{\sin 3x}{3x} = 3.$

Bounded oscillation C1.5: $-x^2 \le x^2\cos\frac{1}{x} \le x^2$, so $\lim_{x\to 0} x^2\cos\frac{1}{x} = 0$.

5 L'Hôpital's rule

Only for $\frac{0}{0}$ or $\frac{\infty}{\infty}$ C1.28: differentiate top and bottom separately.

$\lim\limits_{x\to 0}\dfrac{e^{2x}-1}{x} = \lim\limits_{x\to 0}\dfrac{2e^{2x}}{1} = 2.$

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6 One-sided check

At a $\frac{c}{0}$, a piecewise boundary, or an absolute value, compute both sides.
$\lim_{x\to 1^{\pm}} \frac{2}{x-1} = \pm\infty$ and $\lim_{x\to 0^{\pm}} \frac{|x|}{x} = \pm 1$: the two-sided limits do not exist.

Sanity check

Evaluate f at $a \pm 0.01$ and $a \pm 0.001$ (or graph it). For $\frac{x^2-9}{x-3}$: $f(2.99) = 5.99$, $f(3.01) = 6.01$, consistent with 6.

Decision point: which algebraic move?

DECISION POINT Step 3 is not deterministic: several moves may work, and the first one you try may not. Ask **what makes the denominator vanish**, and aim to cancel exactly that. **A polynomial that is 0 at** a has the factor $(x-a)$ A2.8 §04 Algebra II, in the top as well whenever the form is $\frac{0}{0}$: factor both and cancel. **A square root in a sum or difference** $(\sqrt{x}-2)$: multiply by the conjugate, so $(\sqrt{x})^2$ appears. **Fractions inside a fraction**: combine over a common denominator. $x \to \pm\infty$: divide by the highest power of the denominator. **Trig,** e^x **or** ln **near 0**: rescale to match a special limit. If two attempts stall, use L'Hôpital (when the derivatives are easy) or a Taylor expansion C2.22 §08 Calculus II, and confirm numerically.

Limits at infinity: the rules

Rational functions

Compare degrees n (top), m (bottom):
$n < m$: $\to 0$
$n = m$: $\to \frac{a_n}{b_m}$ (leading coefficients)
$n > m$: $\to \pm\infty$
$\lim_{x\to\infty} \frac{4x^3+x}{2x^3-7} = 2.$

Growth hierarchy

$\ln x \ll x^p \ll a^x \ll x! \ll x^x$
The faster-growing term wins:
$\lim_{x\to\infty} \frac{x^5}{e^x} = 0$, $\lim_{x\to\infty} \frac{e^x}{x^5} = \infty$.
Exponentials beat powers, powers beat logs.

Traps

$\sqrt{x^2} = |x|$, so for $x \to -\infty$ pull out $-x$:
$\lim_{x\to-\infty} \frac{\sqrt{x^2+1}}{x} = -1.$
$\infty - \infty$: use the conjugate:
$\sqrt{x^2+x} - x = \frac{x}{\sqrt{x^2+x}+x} \to \frac{1}{2}$.

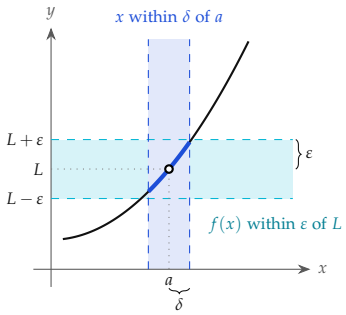
The ε–δ definition, explained

$$\lim_{x\to a} f(x) = L \iff \forall \varepsilon > 0\ \exists \delta > 0 :\ 0 < |x-a| < \delta \implies |f(x)-L| < \varepsilon.$$

Think of it as a challenge game. A skeptic names an output tolerance ε (the cyan band around L). You must answer with an input tolerance δ (the blue window around a) such that *every* x in the window, except possibly a itself, lands inside the band. If you can always answer, however small ε is, the limit is L. The value $f(a)$ never enters: the condition $0 < |x-a|$ excludes it.

Example. $\lim_{x\to 2}(3x+1) = 7$: $|(3x+1)-7| = 3|x-2| < \varepsilon$ whenever $|x-2| < \frac{\varepsilon}{3}$, so choose $\delta = \frac{\varepsilon}{3}$. For $\varepsilon = 0.3$, $\delta = 0.1$ works: $x \in (1.9, 2.1) \Rightarrow 3x+1 \in (6.7, 7.3)$.

Common mistakes. $\frac{0}{0}$ is not 0 or 1; $\frac{c}{0}$ is not indeterminate; "$f(a)$ undefined" does not mean the limit fails; L'Hôpital is not the quotient rule; $\frac{\sin x}{x} \to 1$ only in radians.



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C1.7 Continuity & the Intermediate Value Theorem

f is continuous at a when its limit there equals its value, so the graph has no break at a. A function continuous on a closed interval takes every value between its endpoint values (IVT).

Continuity at a point	f continuous at $a \iff \lim_{x\to a} f(x) = f(a)$	Three conditions: $f(a)$ is defined, the limit exists C1.1, and the two are equal.
Continuity on an interval	$f \in C[a,b]:$ continuous on (a,b), $\lim_{x\to a^+} f = f(a)$, $\lim_{x\to b^-} f = f(b)$	At the endpoints only the inside one-sided limit C1.2 is required.
Combinations	f, g continuous at $a \implies f \pm g$, fg, $\frac{f}{g}$ $(g(a)\neq 0)$ continuous at a	Follows from the limit laws; polynomials, rational, root, trig, exp and log functions are continuous on their domains.
Composition	g cont. at a, f cont. at $g(a) \implies \lim_{x\to a} f(g(x)) = f\left(\lim_{x\to a} g(x)\right)$	A continuous outer function lets the limit pass inside: $\lim e^{\sin x} = e^{\lim \sin x}$.
Intermediate value theorem	$f \in C[a,b]$, N between $f(a), f(b) \implies \exists\, c \in (a,b):\ f(c) = N$	Guarantees a solution exists but not where, nor how many; continuity on the *closed* interval is essential.
Root location (Bolzano)	$f \in C[a,b]$, $f(a)\,f(b) < 0 \implies \exists\, c \in (a,b):\ f(c) = 0$	The IVT with $N = 0$: a sign change of a continuous function traps a root.

STATED-EQUATION *f is continuous at a if and only if the limit as x approaches a of f of x equals f of a. Sums, products, quotients and compositions of continuous functions are continuous, and the limit of f of g of x is f of the limit of g. Intermediate value theorem: if f is continuous on a to b, and N is between f of a and f of b, then there is a c in a to b with f of c equal to N. If f of a and f of b have opposite signs, f has a root in between. f is continuous on the closed interval a to b when it is continuous on the open interval and its one-sided limits at a and at b equal f of a and f of b.*

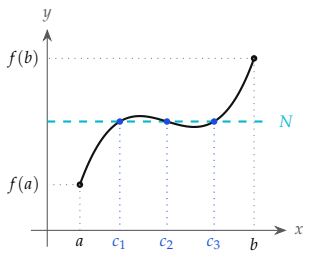
MEMORY AID Continuous = you can draw the graph without lifting your pencil.

METHOD

1. Find an interval $[a,b]$ on which f is continuous and $f(a)$, $f(b)$ have opposite signs.
2. Bisect: evaluate f at the midpoint $m = \frac{a+b}{2}$ and keep the half on which the sign still changes.
3. Repeat until the interval is shorter than the tolerance; each step halves the error. Example: $f(x) = x^2 - 2$ on $[1,2]$: $f(1.5) = 0.25 > 0$, keep $[1, 1.5]$; $f(1.25) < 0$, keep $[1.25, 1.5]$, closing in on $\sqrt{2} \approx 1.414$.

TERMS

$f(a)$ – value at a; must be defined
$C[a,b]$ – the functions continuous on $[a,b]$
N – any target value between $f(a)$ and $f(b)$
c – a point where the target is hit (may not be unique)
$\exists$ – there exists M.90 §01 Math Signs
Discontinuities – removable, jump, infinite, oscillating C1.8
a^+, b^- – approach from inside the interval C1.2



CONTEXT

- **Origin.** Bernard Bolzano (1817); Cauchy's definition of continuity (1821).
- **Use.** Guaranteeing that equations have solutions; justifying direct substitution in limits.
- **Data science.** The bisection method (and many threshold searches, such as finding where precision equals recall) relies on the IVT; decision-tree predictions are step functions, discontinuous, hence gradient-free ML.12 §15 Machine Learning.
- **Physical application.** If it was 10°C at 6 a.m. and 20°C at noon, the temperature was exactly 15°C at some moment in between.

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C1.8 Types of Discontinuity

A discontinuity is classified by how the limit fails: it exists but disagrees with $f(a)$ (removable), the one-sided limits differ (jump), one is infinite (infinite), or there is no limit at all (oscillating).

Removable	$\lim_{x\to a} f(x) = L$ exists, but $f(a) \neq L$ or $f(a)$ undefined	A hole: redefine $f(a) = L$ to repair it. Typical of a canceled factor: $\frac{x^2-1}{x-1}$ at $x = 1$.
Jump	$\lim_{x\to a^-} f(x) \neq \lim_{x\to a^+} f(x)$ (both finite)	The graph steps; jump size = right limit − left limit. Step and floor functions A2.5 §04 Algebra II.
Infinite	$\lim_{x\to a^\pm} f(x) = \pm\infty$	A vertical asymptote C1.2: a non-canceling zero of a denominator, $\ln x$ at 0, $\tan x$ at $\frac{\pi}{2}$.
Oscillating (essential)	$\lim_{x\to 0} \sin\frac{1}{x}$ does not exist	The function wiggles between values forever, so no one-sided limit exists.
Continuous extension	$\tilde{f}(x) = \begin{cases} f(x), & x \neq a \\ \lim_{x\to a} f(x), & x = a \end{cases}$	Fills a removable hole: $\frac{\sin x}{x}$ extended by 1 at 0 is the continuous sinc C1.6.
Piecewise boundary check	$\lim_{x\to a^-} f_{\text{left}} = \lim_{x\to a^+} f_{\text{right}} = f(a)$	A piecewise function is continuous at a seam only if both formulas meet the value there.

STATED-EQUATION *A removable discontinuity is a hole: the limit exists but differs from f of a, or f of a is undefined. A jump discontinuity has different finite one-sided limits. An infinite discontinuity has a one-sided limit equal to plus or minus infinity. An oscillating discontinuity has no limit at all, like sine of one over x at zero. A removable discontinuity is repaired by defining f of a as the limit. At a piecewise seam, f is continuous only if the left formula's limit, the right formula's limit and f of a all agree.*

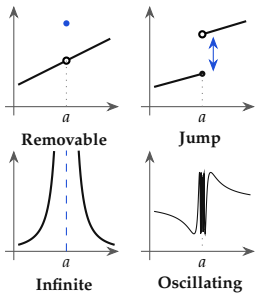
MEMORY AID Hole, step, wall, wiggle: only the hole can be patched.

METHOD

1. Find candidate points: zeros of denominators, piecewise seams, domain edges of logs and roots.
2. Compute the left and right limits at each candidate.
3. Classify: equal and finite (removable if $\neq f(a)$), finite but unequal (jump), infinite (infinite), neither (oscillating). Example: $f(x) = \frac{x^2-4}{x-2}$ has $\lim_{x\to 2} f = 4$ but $f(2)$ undefined, so a removable hole at $(2, 4)$.

TERMS

a – the point being tested
L – the common (finite) limit
a^-, a^+ – left and right approach C1.2
$\tilde{f}$ – the repaired (extended) function
$f_{\text{left}}, f_{\text{right}}$ – the formulas on either side of a seam
$\operatorname{sinc} x$ – $\frac{\sin x}{x}$, with value 1 at 0



CONTEXT

- **Origin.** The classification is due to Baire (1899); Dirichlet's function (1829), discontinuous everywhere, showed how wild functions can be.
- **Use.** Piecewise models, step responses, domain checks before integrating C2.6 §08 Calculus II.
- **Data science.** Thresholding a classifier score is a jump discontinuity, which is why training uses a smooth surrogate loss ML.2 §15 Machine Learning instead of the 0–1 loss.
- **Physical application.** Closing a switch makes a circuit's current jump; an RC capacitor voltage cannot jump, so it is continuous EE.11 §16 Electrical Eng..

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C1.9 Definition of the Derivative

The derivative $f'(x)$ is the limit of secant slopes as the second point slides onto the first: the slope of the tangent line, the instantaneous rate of change of f.

Derivative (limit definition)	$f'(x) = \dfrac{dy}{dx} = \lim_{h\to 0} \dfrac{f(x+h)-f(x)}{h}$	The difference quotient is a secant slope; the limit makes it a tangent slope.
Derivative at a point	$f'(a) = \lim_{x\to a} \dfrac{f(x)-f(a)}{x-a}$	The same limit with $x = a+h$; convenient for checking a single point or a piecewise seam.
Difference quotient	$\dfrac{\Delta y}{\Delta x} = \dfrac{f(x+h)-f(x)}{h}$	Average rate of change over $[x, x+h]$, the slope of the secant line A1.6 §03 Algebra I.
Notations	$f'(x) = \dfrac{dy}{dx} = \dfrac{d}{dx}f(x) = Df(x) = \dot{y}$	Lagrange, Leibniz, operator, and Newton's dot (time derivatives in physics).
Forward difference	$f'(x) \approx \dfrac{f(x+h)-f(x)}{h}$, error $O(h)$	The definition with a small fixed h; halving h only halves the error.
Central difference	$f'(x) \approx \dfrac{f(x+h)-f(x-h)}{2h}$, error $O(h^2)$	Symmetric, so the first error terms cancel: far more accurate for the same h.

STATED-EQUATION *f prime of x, or d y d x, equals the limit as h goes to zero of f of x plus h minus f of x, over h. Equivalently, f prime of a is the limit as x approaches a of f of x minus f of a over x minus a. The forward difference approximates f prime with error of order h; the central difference, f of x plus h minus f of x minus h, over two h, has error of order h squared. The difference quotient, delta y over delta x, equals f of x plus h minus f of x, over h. The derivative is also written d d x of f, D f, or y dot.*

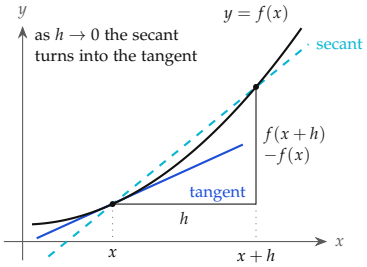
MEMORY AID "Rise over run, with the run shrinking to zero."

METHOD

1. Write the difference quotient $\frac{f(x+h)-f(x)}{h}$.
2. Expand $f(x+h)$ and simplify the numerator until a factor h appears (expand, rationalize, or combine fractions C1.K1).
3. Cancel h and let $h \to 0$. Example: $f(x) = x^2$: $\frac{(x+h)^2-x^2}{h} = \frac{2xh+h^2}{h} = 2x+h \to 2x$, so $f'(3) = 6$.

TERMS

h – small change in x ($h \neq 0$, $h \to 0$)
$\Delta x, \Delta y$ – change in input, change in output
$f'(x)$ – slope of the tangent: instantaneous rate of change
$\frac{dy}{dx}$ – Leibniz notation; f' is Lagrange's
$O(h)$ – error shrinks proportionally to h O.10 §14 Optimization
Differentiable – the limit exists; implies continuous C1.10
D, $\dot{y}$ – operator notation; Newton's dot for a time derivative



CONTEXT

- **Origin.** Fermat's tangents (1630s); Newton's "fluxions" (1666) and Leibniz's publication (1684).
- **Use.** Velocity, marginal cost, any instantaneous rate.
- **Data science.** Gradient checking: compare backprop gradients ML.22 §15 Machine Learning with a central-difference estimate to catch implementation bugs.
- **Physical application.** A car's speedometer shows the derivative of position with respect to time.

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C1.10 Differentiability vs. Continuity

A function is differentiable at a when the derivative limit exists there. Differentiable implies continuous, but not conversely: corners, cusps and vertical tangents are continuous yet have no derivative.

Differentiable implies continuous	$f'(a)$ exists $\implies \lim_{x\to a} f(x) = f(a)$	Proof: $f(x) - f(a) = \frac{f(x)-f(a)}{x-a}(x-a) \to f'(a)\cdot 0 = 0$.
Converse fails	f continuous at $a \not\Rightarrow f'(a)$ exists	Continuity is necessary, not sufficient. A discontinuity always kills the derivative.
One-sided derivatives	$f'(a)$ exists $\iff f'_-(a) = f'_+(a)$	$f'_\pm(a) = \lim_{h\to 0^\pm} \frac{f(a+h)-f(a)}{h}$; check both at piecewise seams and absolute values.
Corner	$f(x) = \lvert x\rvert$: $\quad f'_-(0) = -1 \neq 1 = f'_+(0)$	Two different finite one-sided slopes; ReLU $\max(0,x)$ has a corner at 0.
Cusp	$f(x) = x^{2/3}$: $\quad f'(x) = \frac{2}{3}x^{-1/3} \to \pm\infty$ as $x \to 0^\pm$	The one-sided slopes blow up with opposite signs; the graph comes to a point.
Vertical tangent	$f(x) = x^{1/3}$: $\quad f'(x) = \frac{1}{3}x^{-2/3} \to +\infty$ as $x \to 0$	The tangent line exists but is vertical, so its slope is undefined.

STATED-EQUATION *If f is differentiable at a, then f is continuous at a, but a continuous function need not be differentiable. The derivative exists exactly when the left and right derivatives agree. The absolute value has a corner at zero, x to the two-thirds has a cusp, and the cube root has a vertical tangent. At the corner of absolute x the left derivative is minus one and the right derivative is one; at the cusp of x to the two-thirds the slopes go to plus and minus infinity; at the vertical tangent of x to the one-third the slope goes to plus infinity.*

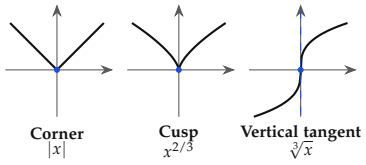
MEMORY AID Smooth $\Rightarrow$ connected, but connected $\not\Rightarrow$ smooth: a crease or a spike is still continuous.

METHOD

1. Check continuity at a; if it fails, f is not differentiable there.
2. Compute the one-sided derivatives (for a piecewise f, differentiate each formula and evaluate at the seam).
3. Differentiable iff both are finite and equal. Example: $f(x) = x^2$ for $x \le 1$, $2x - 1$ for $x > 1$: continuous ($1 = 1$), slopes $2\cdot 1 = 2$ and 2 agree, so $f'(1) = 2$.

TERMS

$f'_-(a), f'_+(a)$ – left and right derivatives
$|x|$ – absolute value M.43 §01 Math Signs
$x^{2/3}$, $x^{1/3}$ – rational powers A1.4 §03 Algebra I
ReLU – $\max(0, x)$, the standard network activation
Smooth – derivatives of all orders exist



CONTEXT

- **Origin.** Weierstrass (1872) built a function continuous everywhere and differentiable nowhere.
- **Use.** Knowing where derivative rules apply; piecewise models.
- **Data science.** ReLU and the lasso penalty $|w|$ ML.8 §15 Machine Learning are not differentiable at 0; frameworks use a subgradient (any slope between the one-sided ones, usually 0).
- **Physical application.** Brownian motion paths are continuous but nowhere differentiable: a particle in water has a position at every instant but no well-defined velocity.

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C1.11 Power, Constant & Sum Rules

Differentiation is linear: it splits across sums and passes through constant factors. With the power rule this differentiates every polynomial, root and reciprocal power on sight.

Constant rule	$\frac{d}{dx}\,c = 0$	A constant does not change, so its rate of change is 0.
Power rule	$\frac{d}{dx}\,x^n = n\,x^{n-1}$	Valid for every real n: $\frac{d}{dx}\sqrt{x} = \frac{1}{2\sqrt{x}}$, $\frac{d}{dx}\frac{1}{x} = -\frac{1}{x^2}$.
Constant multiple rule	$\frac{d}{dx}\big[c\,f(x)\big] = c\,f'(x)$	Constants factor out.
Sum & difference rule	$\frac{d}{dx}\big[f(x) \pm g(x)\big] = f'(x) \pm g'(x)$	Differentiate term by term.
Polynomial derivative	$\frac{d}{dx}\sum_{k=0}^{n} a_k x^k = \sum_{k=1}^{n} k\,a_k x^{k-1}$	Each term separately; the degree drops by one.
Linear function	$\frac{d}{dx}(mx+b) = m$	The tangent to a line is the line itself A1.7 §03 Algebra I.

STATED-EQUATION *The derivative of a constant is zero. The derivative of x to the n is n x to the n minus one. The derivative of c f is c f prime. The derivative of f plus or minus g is f prime plus or minus g prime. A polynomial is differentiated term by term, and the derivative of m x plus b is m.*

MEMORY AID Power rule: "bring the power down, subtract one."

METHOD

1. Rewrite roots and reciprocals as powers: $\sqrt[3]{x} = x^{1/3}$, $\frac{4}{x^2} = 4x^{-2}$ A1.4 §03 Algebra I.
2. Apply the power rule term by term, keeping constant factors.
3. Rewrite with positive exponents if wanted. Example: $\frac{d}{dx}\left(3x^4 - \frac{2}{x} + 5\right) = 12x^3 + 2x^{-2} = 12x^3 + \frac{2}{x^2}$.

TERMS

c – constant
n – any real exponent A1.3 §03 Algebra I; e.g. $\sqrt{x} = x^{1/2}$ A1.4 §03 Algebra I
a_k – polynomial coefficients; Σ summation M.53 §01 Math Signs
m, b – slope and intercept
Linearity – derivatives split across sums and pull out constants
f, g – differentiable functions of x

CONTEXT

- **Origin.** Derived from the limit definition C1.9 with the binomial theorem A2.23 §04 Algebra II; Leibniz (1684).
- **Use.** Differentiating any polynomial on sight.
- **Data science.** The gradient of squared error with respect to a prediction is $\frac{d}{d\hat{y}}(y-\hat{y})^2 = -2(y-\hat{y})$: the residual, scaled ML.2 §15 Machine Learning.
- **Physical application.** From $h(t) = -4.9t^2 + 20t$, velocity is $h'(t) = -9.8t + 20$, zero at the top of the throw P.2 §12 Physics.

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C1.12 Product & Quotient Rules

The derivative of a product is not the product of derivatives: each factor takes a turn being differentiated. The quotient rule follows from the product and chain rules.

Product rule	$\frac{d}{dx}[f\,g] = f'g + f\,g'$	Differentiate one factor at a time and add.
Quotient rule	$\frac{d}{dx}\left[\frac{f}{g}\right] = \frac{f'g - f\,g'}{g^2}$	Order matters in the numerator (minus sign); valid where $g \neq 0$.
Triple product	$(fgh)' = f'gh + fg'h + fgh'$	Every factor gets differentiated once, the others are left alone.
Reciprocal rule	$\frac{d}{dx}\left[\frac{1}{g}\right] = -\frac{g'}{g^2}$	The quotient rule with $f = 1$; quicker for $\frac{1}{\text{something}}$.
Sigmoid derivative	$\sigma(z) = \frac{1}{1+e^{-z}} \implies \sigma'(z) = \sigma(z)\big(1-\sigma(z)\big)$	From the reciprocal rule; the derivative is reused from the forward pass at no extra cost.

STATED-EQUATION *The derivative of f g equals f prime g plus f g prime. The derivative of f over g equals f prime g minus f g prime, over g squared. For three factors, differentiate each in turn and add. The derivative of one over g is minus g prime over g squared. The sigmoid's derivative is sigma times one minus sigma.*

MEMORY AID Quotient: "low d-high minus high d-low, over the square of what's below."

METHOD

1. Name the pieces: f = top (or first factor), g = bottom (or second factor).
2. Compute f' and g' separately.
3. Assemble and simplify. Example: $\frac{d}{dx}\frac{x}{x+1} = \frac{1\cdot(x+1)-x\cdot1}{(x+1)^2} = \frac{1}{(x+1)^2}$.

DECISION POINT Quotient rule or not? If the denominator is a single power, rewrite and use the power rule ($\frac{3}{x^4} = 3x^{-4}$); if the bottom is a constant, just factor it out. Otherwise, a quotient rule; or take logs when there are many factors C1.18.

TERMS

f, g, h – differentiable functions of x
$g \neq 0$ – required for the quotient
$\sigma(z)$ – logistic sigmoid ML.9 §15 Machine Learning
e – Euler's number A2.13 §04 Algebra II
Mnemonic – "low d-high minus high d-low, over low squared"

CONTEXT

- **Origin.** Leibniz (1684).
- **Use.** Derivatives of products and ratios of functions.
- **Data science.** The quotient rule gives the sigmoid identity $\sigma' = \sigma(1-\sigma)$ used in logistic regression, and sensitivities of ratio metrics such as CTR.
- **Physical application.** The rate of change of power $P = IV$ when both current and voltage change uses the product rule EE.1 §16 Electrical Eng..

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C1.13 Chain Rule

The derivative of a composition is the outer derivative, evaluated at the inner function, times the inner derivative. Rates multiply along a chain of dependencies.

Chain rule (Lagrange)	$\frac{d}{dx} f\big(g(x)\big) = f'\big(g(x)\big) \cdot g'(x)$	Outer derivative at the inside, times the inside's derivative.
Chain rule (Leibniz)	$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$	With $y = f(u)$, $u = g(x)$; the du's "cancel" as a mnemonic, not as algebra.
Longer chains	$\frac{d}{dx} f\Big(g\big(h(x)\big)\Big) = f'\big(g(h)\big)\, g'(h)\, h'(x)$	One factor per link, from the outside in.
General power rule	$\frac{d}{dx}\big[g(x)\big]^n = n\big[g(x)\big]^{n-1} g'(x)$	$\frac{d}{dx}(x^2+1)^5 = 5(x^2+1)^4 \cdot 2x$.
Exponential of a function	$\frac{d}{dx} e^{g(x)} = e^{g(x)}\, g'(x)$	The exponential is copied, then multiplied by the inner derivative.
Log of a function	$\frac{d}{dx} \ln g(x) = \frac{g'(x)}{g(x)}$	Derivative over function; the basis of logarithmic differentiation C1.18.

STATED-EQUATION *The derivative of f of g of x equals f prime of g of x times g prime of x. d y d x equals d y d u times d u d x. For longer chains, multiply one derivative per link. The derivative of g to the n is n g to the n minus one times g prime; of e to the g, e to the g times g prime; of natural log of g, g prime over g.*

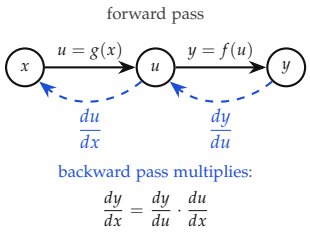
MEMORY AID "Derivative of the outside (inside left alone) times derivative of the inside."

METHOD

1. Identify the outermost operation and call everything inside it u.
2. Differentiate the outer function with respect to u, leaving u untouched.
3. Multiply by u', repeating the process if u is itself a composition. Example: $\frac{d}{dx}\sin(3x^2) = \cos(3x^2) \cdot 6x$.

TERMS

g – inner function; $u = g(x)$
f – outer function; $y = f(u)$
$f \circ g$ – composition A2.1 §04 Algebra II
n – any real exponent
ln – natural log A2.14 §04 Algebra II
Several variables – multivariable chain rule C3.13 §10 Calculus III



CONTEXT

- **Origin.** Leibniz (1676).
- **Use.** Differentiating nested functions: powers of polynomials, $e^{g(x)}$, $\ln g(x)$.
- **Data science.** Backpropagation is the chain rule run backward over a network's computation graph, multiplying local derivatives from the loss to each weight ML.22 §15 Machine Learning.
- **Physical application.** If a balloon's radius grows with time, its volume's rate is $\frac{dV}{dt} = \frac{dV}{dr}\frac{dr}{dt}$ C1.20.

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C1.14 Derivatives of Trig, Exponential & Log Functions

The transcendental derivatives, all following from the special limits C1.6. Sine and cosine cycle into each other; e^x is its own derivative; $\ln x$ differentiates to the algebraic $\frac{1}{x}$.

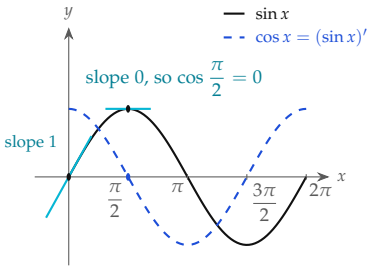
Sine & cosine	$\frac{d}{dx}\sin x = \cos x, \quad \frac{d}{dx}\cos x = -\sin x$	x in radians; four derivatives return to the start.
Tangent & cotangent	$\frac{d}{dx}\tan x = \sec^2 x, \quad \frac{d}{dx}\cot x = -\csc^2 x$	Quotient rule on $\frac{\sin x}{\cos x}$ with $\sin^2+\cos^2=1$ T.6 §06 Trigonometry.
Secant & cosecant	$\frac{d}{dx}\sec x = \sec x\tan x, \quad \frac{d}{dx}\csc x = -\csc x\cot x$	Reciprocal rule on $\frac{1}{\cos x}$, $\frac{1}{\sin x}$.
Natural exponential	$\frac{d}{dx}e^x = e^x$	The only function (up to a constant factor) equal to its own derivative.
General exponential	$\frac{d}{dx}a^x = a^x \ln a$	Write $a^x = e^{x\ln a}$ and use the chain rule; $a>0$.
Natural logarithm	$\frac{d}{dx}\ln x = \frac{1}{x} \quad (x>0)$	The inverse of e^x C1.15.
Log of absolute value	$\frac{d}{dx}\ln\lvert x\rvert = \frac{1}{x} \quad (x\neq 0)$	Extends to negative x; why $\int \frac{dx}{x} = \ln\lvert x\rvert + C$ C1.29.
General logarithm	$\frac{d}{dx}\log_a x = \frac{1}{x\ln a}$	Change of base A2.16 §04 Algebra II: $\log_a x = \frac{\ln x}{\ln a}$.

STATED-EQUATION *The derivative of sine is cosine; of cosine, minus sine; of tangent, secant squared; of cotangent, minus cosecant squared; of secant, secant tangent; of cosecant, minus cosecant cotangent. The derivative of e to the x is e to the x; of a to the x, a to the x natural log a; of natural log x, and of natural log of absolute x, one over x; of log base a of x, one over x natural log a.*

MEMORY AID Derivatives of "co" functions (cos, cot, csc) get a minus sign. The derivatives of tan, sec use sec; those of cot, csc use csc.

TERMS

x – in radians T.2 §06 Trigonometry; follows from C1.6
sec, csc, cot – $\frac{1}{\cos}$, $\frac{1}{\sin}$, $\frac{\cos}{\sin}$ T.1 §06 Trigonometry
e^x – its own derivative A2.13 §04 Algebra II
a – base, $a>0$, $a\neq 1$
$\ln x$ – natural log A2.14 §04 Algebra II, $x>0$
With inner functions – combine with the chain rule C1.13
Full table – C1.K2



CONTEXT

- **Origin.** Euler's *Introductio* (1748) and *Institutiones* (1755).
- **Use.** Growth models, oscillations, and log-transformed quantities.
- **Data science.** The log-loss gradient: $\frac{d}{dp}(-\ln p) = -\frac{1}{p}$, huge when a confident prediction is wrong; with the sigmoid it simplifies to $\hat{p} - y$ ML.9 §15 Machine Learning.
- **Physical application.** Radioactive decay $N = N_0 e^{-kt}$ has rate $-kN$: the derivative of an exponential is proportional to itself.

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C1.15 Inverse Functions & Inverse Trig Derivatives

The graph of f^{-1} is the mirror image of f in $y = x$, so slopes become reciprocals. This one rule produces the derivatives of $\ln x$ and of every inverse trig function.

Inverse function derivative	$\left(f^{-1}\right)'(b) = \dfrac{1}{f'(a)}, \quad b = f(a)$	Reciprocal slope at the mirrored point; requires $f'(a) \neq 0$.
Leibniz form	$\dfrac{dx}{dy} = \dfrac{1}{dy/dx}$	Swapping the roles of x and y flips the rate.
Arcsine	$\dfrac{d}{dx}\arcsin x = \dfrac{1}{\sqrt{1-x^2}} \quad (\lvert x\rvert < 1)$	From $\sin y = x$: $\cos y\, y' = 1$, $\cos y = \sqrt{1-x^2}$.
Arccosine	$\dfrac{d}{dx}\arccos x = -\dfrac{1}{\sqrt{1-x^2}}$	Since $\arccos x = \frac{\pi}{2} - \arcsin x$.
Arctangent	$\dfrac{d}{dx}\arctan x = \dfrac{1}{1+x^2}$	Defined for all x; integrating it back gives arctan C1.29.
Arcsecant	$\dfrac{d}{dx}\operatorname{arcsec} x = \dfrac{1}{\lvert x\rvert\sqrt{x^2-1}} \quad (\lvert x\rvert > 1)$	The "co" versions (arccot, arccsc) are the negatives of arctan, arcsec.

STATED-EQUATION *The derivative of the inverse function at b equals one over f prime at a, where b equals f of a. d x d y equals one over d y d x. The derivative of arcsine x is one over the square root of one minus x squared; of arccosine, the negative of that; of arctangent, one over one plus x squared; of arcsecant, one over absolute x times the square root of x squared minus one.*

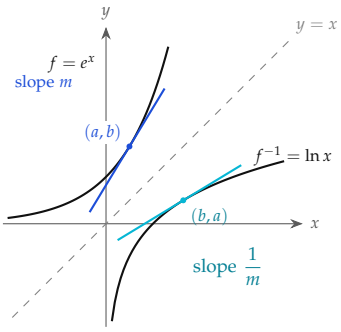
MEMORY AID Inverse trig derivatives are algebraic: no trig left. "co" inverses get a minus sign.

METHOD

1. Write $y = f^{-1}(x)$ as $f(y) = x$.
2. Differentiate implicitly C1.17: $f'(y)\, y' = 1$, so $y' = \frac{1}{f'(y)}$.
3. Rewrite $f'(y)$ in terms of x (a reference triangle helps). Example: $y = \ln x \iff e^y = x$: $e^y y' = 1$, so $y' = \frac{1}{e^y} = \frac{1}{x}$.

TERMS

f^{-1} – inverse function A2.1 §04 Algebra II; f must be one-to-one
(a, b), (b, a) – mirror points on f and f^{-1}
arcsin, arccos, arctan – inverse trig T.14 §06 Trigonometry, also $\sin^{-1}$ etc.
arcsec – inverse secant
$|x|$ – absolute value M.43 §01 Math Signs



CONTEXT

- **Origin.** Inverse trig derivatives appear in Leibniz's and Johann Bernoulli's work (1690s).
- **Use.** $\ln x$, inverse trig and root functions; integrals producing arctan, arcsin.
- **Data science.** Normalizing flows and the change-of-variables formula C1.34 need the derivative of an inverse transform; the probit link uses $(\Phi^{-1})'$ S.20 §13 Statistics.
- **Physical application.** A tracking camera's angle $\theta = \arctan\frac{h}{x}$ changes at $\frac{d\theta}{dt}$ given by the arctangent derivative.

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C1.16 Hyperbolic Functions & Their Derivatives

The even and odd parts of e^x. They obey identities like trig's (with sign changes) and differentiate without any minus signs on sinh and cosh.

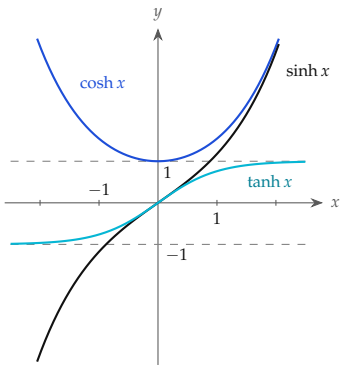
Hyperbolic sine & cosine	$\sinh x = \dfrac{e^x - e^{-x}}{2}, \qquad \cosh x = \dfrac{e^x + e^{-x}}{2}$	Odd and even parts of e^x: $e^x = \cosh x + \sinh x$.
Hyperbolic tangent	$\tanh x = \dfrac{\sinh x}{\cosh x} = \dfrac{e^x - e^{-x}}{e^x + e^{-x}}$	S-shaped, range $(-1, 1)$; $\tanh x = 2\sigma(2x) - 1$.
Fundamental identity	$\cosh^2 x - \sinh^2 x = 1$	$(\cosh t, \sinh t)$ traces the hyperbola $x^2 - y^2 = 1$ A2.26 §04 Algebra II, as $(\cos, \sin)$ traces a circle.
Derivatives of sinh & cosh	$\dfrac{d}{dx}\sinh x = \cosh x, \qquad \dfrac{d}{dx}\cosh x = \sinh x$	No minus sign, unlike cos.
Derivative of tanh	$\dfrac{d}{dx}\tanh x = \operatorname{sech}^2 x = 1 - \tanh^2 x$	Reuses the output, like the sigmoid C1.12.
Inverse hyperbolic derivatives	$\dfrac{d}{dx}\operatorname{arsinh} x = \dfrac{1}{\sqrt{x^2+1}}, \qquad \dfrac{d}{dx}\operatorname{artanh} x = \dfrac{1}{1-x^2}$	$\operatorname{arsinh} x = \ln\left(x + \sqrt{x^2+1}\right)$; artanh for $\lvert x\rvert < 1$.

STATED-EQUATION *Hyperbolic sine is e to the x minus e to the minus x, over two; hyperbolic cosine is e to the x plus e to the minus x, over two; hyperbolic tangent is their ratio. Cosh squared minus sinh squared equals one. The derivative of sinh is cosh, of cosh is sinh, of tanh is sech squared, which equals one minus tanh squared. The derivative of inverse sinh is one over the square root of x squared plus one; of inverse tanh, one over one minus x squared.*

MEMORY AID Hyperbolic = trig with the signs "fixed": $\cosh' = +\sinh$, and the identity has a minus: $\cosh^2 - \sinh^2 = 1$.

TERMS

sinh, cosh, tanh – hyperbolic sine, cosine, tangent ("sinch", "cosh", "tanch")
sech x – $\frac{1}{\cosh x}$
arsinh, artanh – inverse hyperbolic functions
$e^{\pm x}$ – exponentials A2.13 §04 Algebra II
σ – logistic sigmoid ML.9 §15 Machine Learning



CONTEXT

- **Origin.** Vincenzo Riccati (1757) and Johann Lambert (1768).
- **Use.** Catenaries, special relativity (rapidity), hyperbolic geometry.
- **Data science.** tanh is a classic activation (zero-centered, gradient $1 - \tanh^2 \le 1$ vanishes for large $|x|$) and gates LSTM cells ML.22 §15 Machine Learning.
- **Physical application.** A hanging cable takes the catenary shape $y = a\cosh\frac{x}{a}$; the Gateway Arch is an inverted catenary.

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Derivative Rules & Table | *Key Topic C1.K2*

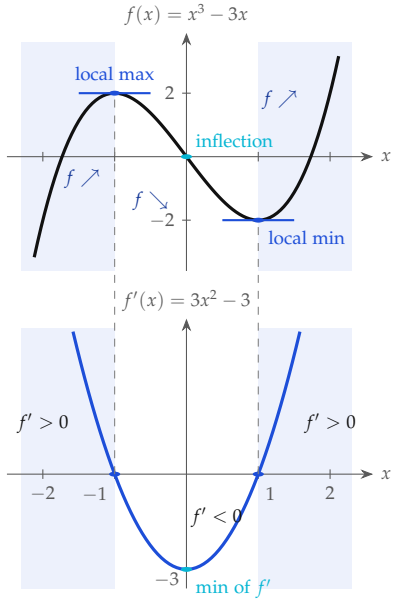
Every derivative you will meet is built from a short list of basic derivatives combined by five rules. Differentiation is mechanical: read the expression from the *outside in*, apply the rule for the outermost operation, and repeat on each piece. The derivative f' is itself a function whose sign and zeros describe the shape of f.

The five combination rules

Rule	Formula	Example
Linearity C1.11	$(af+bg)' = af' + bg'$	$(3x^2 - 5x)' = 6x - 5$
Power C1.11	$(x^n)' = nx^{n-1}$	$(\sqrt{x})' = \frac{1}{2\sqrt{x}}$
Product C1.12	$(fg)' = f'g + fg'$	$(x^2 e^x)' = (2x + x^2)e^x$
Quotient C1.12	$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$	$\left(\frac{\sin x}{x}\right)' = \frac{x\cos x - \sin x}{x^2}$
Chain C1.13	$\big(f(g)\big)' = f'(g)\,g'$	$\big(e^{x^2}\big)' = 2x\,e^{x^2}$

Reading f' from f (right)

$f' > 0$ where f rises, $f' < 0$ where it falls (shaded), and $f' = 0$ at the horizontal tangents: the local max and min of f sit exactly above the zeros of f' C1.23. The steepest descent of f (its inflection point) is where f' is smallest. Going the other way, f is an antiderivative of f' C1.29.



Basic derivatives (with the chain-rule version, $u = g(x)$)

$f(x)$	$f'(x)$	$\frac{d}{dx}f(u)$	$f(x)$	$f'(x)$	$\frac{d}{dx}f(u)$
c	0	—	$\sin x$	$\cos x$	$\cos u \cdot u'$
x^n	nx^{n-1}	$nu^{n-1}u'$	$\cos x$	$-\sin x$	$-\sin u \cdot u'$
e^x	e^x	$e^u\,u'$	$\tan x$	$\sec^2 x$	$\sec^2 u \cdot u'$
a^x	$a^x \ln a$	$a^u \ln a \cdot u'$	$\cot x$	$-\csc^2 x$	$-\csc^2 u \cdot u'$
$\ln\lvert x\rvert$	$\frac{1}{x}$	$\frac{u'}{u}$	$\sec x$	$\sec x \tan x$	$\sec u \tan u \cdot u'$
$\log_a x$	$\frac{1}{x \ln a}$	$\frac{u'}{u \ln a}$	$\csc x$	$-\csc x \cot x$	$-\csc u \cot u \cdot u'$
$\arcsin x$	$\frac{1}{\sqrt{1-x^2}}$	$\frac{u'}{\sqrt{1-u^2}}$	$\sinh x$	$\cosh x$	$\cosh u \cdot u'$
$\arccos x$	$-\frac{1}{\sqrt{1-x^2}}$	$-\frac{u'}{\sqrt{1-u^2}}$	$\cosh x$	$\sinh x$	$\sinh u \cdot u'$
$\arctan x$	$\frac{1}{1+x^2}$	$\frac{u'}{1+u^2}$	$\tanh x$	$\operatorname{sech}^2 x$	$\operatorname{sech}^2 u \cdot u'$
$\operatorname{arcsec} x$	$\frac{1}{\lvert x\rvert\sqrt{x^2-1}}$	$\frac{u'}{\lvert u\rvert\sqrt{u^2-1}}$	$\sqrt{x}$	$\frac{1}{2\sqrt{x}}$	$\frac{u'}{2\sqrt{u}}$

Decision point: which rule first?

DECISION POINT Ask **what is the last operation** you would perform to evaluate the expression at a number: that operation is the outermost layer, and its rule goes first. $x^2 \sin x$: last step is multiplying, so **product**. $\sin(x^2)$: last step is taking the sine, so **chain**. $\frac{x^2}{\sin x}$: dividing, so **quotient** (or product with $(\sin x)^{-1}$). **Simplify before differentiating**: $\frac{x^3+x}{x} = x^2 + 1$ needs no quotient rule; $\ln \frac{x^2}{\sqrt{x+1}} = 2\ln x - \frac{1}{2}\ln(x+1)$ needs no chain of quotients. **Many factors or a variable exponent** (x^x, $\frac{(x+1)^3\sqrt{x}}{e^x}$): take logs first C1.18. y **not isolated**: differentiate implicitly C1.17.

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Worked examples

Chain inside product

$\frac{d}{dx}\left[x^3 e^{2x}\right] = 3x^2 e^{2x} + x^3 \cdot 2e^{2x} = x^2 e^{2x}(3+2x)$.

Quotient with chain

$\frac{d}{dx}\frac{x}{\sqrt{x^2+1}} = \frac{\sqrt{x^2+1} - x\cdot\frac{x}{\sqrt{x^2+1}}}{x^2+1} = \frac{1}{(x^2+1)^{3/2}}$.

Nested chain

$\frac{d}{dx}\ln\big(\cos(4x)\big) = \frac{1}{\cos 4x}\cdot(-\sin 4x)\cdot 4 = -4\tan 4x$.

Inverse trig with chain

$\frac{d}{dx}\arctan(3x) = \frac{3}{1+9x^2}$; at $x=0$ the slope is 3.

Common mistakes

$(fg)' \neq f'g'$ and $\left(\frac{f}{g}\right)' \neq \frac{f'}{g'}$: $(x\cdot x)' = 2x$, not 1.

Forgotten inner derivative: $(\sin 3x)' = 3\cos 3x$, not $\cos 3x$; $(e^{-x})' = -e^{-x}$.

Power rule on an exponential: $(2^x)' = 2^x \ln 2$, not $x2^{x-1}$. The power rule needs a *constant* exponent; the exponential rule needs a *constant* base.

Sign slips: $(\cos x)' = -\sin x$; the quotient rule's numerator is $f'g - fg'$ ("low d-high" first); $(\frac{1}{x})' = -\frac{1}{x^2}$.

Degrees: the trig derivatives hold only in radians; in degrees $(\sin x^\circ)' = \frac{\pi}{180}\cos x^\circ$.

Constants of e, π, $\ln 2$: their derivatives are 0. $(e^2)' = 0$, not $2e$.

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C1.17 Implicit Differentiation

When y is tied to x by an equation rather than a formula, differentiate both sides with respect to x, treating y as an unknown function $y(x)$, and solve for y'.

Power of y	$\dfrac{d}{dx}\left[y^n\right] = n\,y^{n-1}\dfrac{dy}{dx}$	The chain rule: every y-term picks up a factor y'.
Product with y	$\dfrac{d}{dx}\left[x\,y\right] = y + x\dfrac{dy}{dx}$	Mixed terms need the product rule C1.12.
Function of y	$\dfrac{d}{dx}\,g(y) = g'(y)\,\dfrac{dy}{dx}$	$\frac{d}{dx}\sin y = \cos y\,y'$, $\frac{d}{dx}e^y = e^y\,y'$.
Implicit slope formula	$F(x,y) = 0 \implies \dfrac{dy}{dx} = -\dfrac{F_x}{F_y}$	Shortcut with partial derivatives C3.9 §10 Calculus III; valid where $F_y \neq 0$.
Circle example	$x^2 + y^2 = r^2 \implies \dfrac{dy}{dx} = -\dfrac{x}{y}$	Perpendicular to the radius, as geometry demands G.18 §05 Geometry.
Second derivative	$\dfrac{d^2y}{dx^2} = \dfrac{d}{dx}\left(-\dfrac{x}{y}\right) = -\dfrac{y - x\,y'}{y^2} = -\dfrac{r^2}{y^3}$	Differentiate y' again, then substitute y' and the original equation.

STATED-EQUATION *The derivative of y to the n with respect to x is n y to the n minus one, times d y d x. The derivative of x y is y plus x d y d x; of g of y, g prime of y times d y d x. If F of x and y equals zero, d y d x equals minus F sub x over F sub y. On the circle, d y d x equals minus x over y. Differentiating again, the second derivative on the circle is minus r squared over y cubed.*

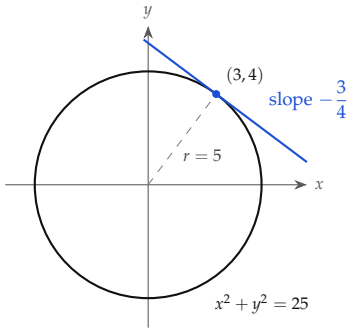
MEMORY AID "Every y gets a y'."

METHOD

1. Differentiate both sides with respect to x, attaching $\frac{dy}{dx}$ to every y-term (chain rule C1.13).
2. Collect all $\frac{dy}{dx}$ terms on one side, everything else on the other.
3. Factor out $\frac{dy}{dx}$ and divide. Example: $x^2 + y^2 = 25$: $2x + 2yy' = 0$, so $y' = -\frac{x}{y}$; at $(3,4)$ the slope is $-\frac{3}{4}$.

TERMS

$y(x)$ – y defined implicitly by an equation in x, y
y', $\frac{dy}{dx}$ – the unknown slope to solve for
$F(x,y)$ – the equation written as $F = 0$
F_x, F_y – partial derivatives C3.9 §10 Calculus III
r – radius
Result – usually depends on both x and y
$\frac{d^2y}{dx^2}$ – second derivative C1.19



CONTEXT

- **Origin.** Leibniz and the Bernoulli brothers (1690s); the implicit function theorem (Cauchy, Dini 1870s).
- **Use.** Slopes of circles and other curves that are not functions; related rates C1.20.
- **Data science.** The implicit function theorem lets frameworks differentiate through the solution of an optimization or equilibrium (implicit layers, hyperparameter gradients).
- **Physical application.** The slope of a circular track $x^2 + y^2 = r^2$ at any point is $-x/y$, found implicitly.

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C1.18 Logarithmic Differentiation

Take ln of both sides before differentiating. Logs turn products into sums, quotients into differences and powers into factors, so messy products and variable exponents become easy.

Logarithmic derivative	$\frac{d}{dx}\ln f(x) = \frac{f'(x)}{f(x)}$	The relative (percentage) rate of change of f.
Recover the derivative	$f'(x) = f(x)\cdot\frac{d}{dx}\ln f(x)$	Multiply back by the original function.
Variable base and exponent	$\frac{d}{dx}x^x = x^x(\ln x + 1)$	Neither the power rule nor the exponential rule applies; $\ln y = x\ln x$.
General f^g	$\frac{d}{dx}f^g = f^g\left(g'\ln f + \frac{g f'}{f}\right)$	From $\ln y = g\ln f$ and the product rule; needs $f > 0$.
Products of many factors	$\frac{(f_1 f_2\cdots f_n)'}{f_1 f_2\cdots f_n} = \frac{f_1'}{f_1} + \frac{f_2'}{f_2} + \cdots + \frac{f_n'}{f_n}$	Relative rates add: the log-derivative of a product is the sum of the log-derivatives.

STATED-EQUATION *The derivative of the natural log of f is f prime over f. So f prime equals f times the derivative of log f. The derivative of x to the x is x to the x times the quantity log x plus one. For f to the g, the derivative is f to the g times g prime log f plus g f prime over f. The log-derivative of a product is the sum of the log-derivatives of its factors.*

MEMORY AID "Log it, differentiate, multiply back."

METHOD

1. Write $y = f(x)$ and take ln of both sides (use $\ln|y|$ if f can be negative).
2. Expand with the log rules A2.14 §04 Algebra II: $\ln(ab) = \ln a + \ln b$, $\ln a^k = k\ln a$.
3. Differentiate implicitly, $\frac{y'}{y} = \cdots$, then multiply by y. Example: $y = \frac{(x+1)^2}{\sqrt{x}}$: $\ln y = 2\ln(x+1) - \frac{1}{2}\ln x$, $\frac{y'}{y} = \frac{2}{x+1} - \frac{1}{2x}$, so $y'(1) = \frac{4}{1}\left(1 - \frac{1}{2}\right) = 2$.

DECISION POINT Use it when the function has a **variable in the exponent and the base** ($x^{\sin x}$), or is a product/quotient of **three or more** factors. For a^x or x^n alone, the table is faster C1.K2.

TERMS

$\frac{f'}{f}$ – logarithmic derivative (relative rate)
f, g – differentiable, $f > 0$ for f^g
$f_1, \ldots, f_n$ – factors of a product
ln – natural log A2.14 §04 Algebra II
Implicit step – $\frac{d}{dx}\ln y = \frac{y'}{y}$ C1.17

CONTEXT

- **Origin.** Johann Bernoulli (1697) used it to differentiate x^x.
- **Use.** Products, quotients and powers with variable exponents.
- **Data science.** Log-likelihoods turn a product of densities into a sum, so the gradient of $\ln\prod p_i$ is $\sum \frac{p_i'}{p_i}$ S.26 §13 Statistics; elasticity in economics is a log-derivative.
- **Physical application.** Growth rates compound: if population and income per person grow 1% and 2% a year, total income grows about 3%.

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C1.19 Higher-Order Derivatives

Differentiating repeatedly gives f'', f''', …, $f^{(n)}$. The second derivative measures how the slope changes: acceleration, concavity, curvature.

Second derivative	$f''(x) = \dfrac{d^2y}{dx^2} = \dfrac{d}{dx}\left(\dfrac{dy}{dx}\right)$	The rate of change of the rate of change; its sign gives concavity C1.23.
nth derivative	$f^{(n)}(x) = \dfrac{d^n y}{dx^n}$	Parentheses distinguish the order n from a power.
Motion	$s(t), \quad v = s'(t), \quad a = v' = s''(t), \quad j = s'''(t)$	Position, velocity, acceleration, jerk.
Powers	$\dfrac{d^n}{dx^n}x^m = \dfrac{m!}{(m-n)!}x^{m-n}\ (n \le m), \qquad \dfrac{d^{m+1}}{dx^{m+1}}x^m = 0$	A degree-m polynomial vanishes after $m+1$ derivatives.
Exponential & trig patterns	$\dfrac{d^n}{dx^n}e^{kx} = k^n e^{kx}, \qquad \dfrac{d^n}{dx^n}\sin x = \sin\left(x + \dfrac{n\pi}{2}\right)$	Exponentials scale by k each time; sine cycles with period 4.
General Leibniz rule	$(fg)^{(n)} = \sum_{k=0}^{n} \binom{n}{k} f^{(k)} g^{(n-k)}$	The product rule iterated; binomial coefficients as in A2.23 §04 Algebra II.

STATED-EQUATION *The second derivative is the derivative of the derivative, written f double prime or d squared y d x squared. The n-th derivative is written f with n in parentheses. Velocity is the derivative of position, acceleration the second derivative, jerk the third. The n-th derivative of e to the k x is k to the n e to the k x, and sine cycles every four derivatives. The n-th derivative of f g is the sum of n choose k times f to the k times g to the n minus k. The n-th derivative of x to the m is m factorial over m minus n factorial, times x to the m minus n, and the m plus first derivative is zero. The n-th derivative of sine x is sine of x plus n pi over two.*

MEMORY AID Sine's derivatives: sin → cos → − sin → − cos → sin, a four-step cycle.

TERMS

f'', f''' – second, third derivatives
$f^{(n)}$ – nth derivative ($f^{(0)} = f$)
s, v, a, j – position, velocity, acceleration, jerk P.2 §12 Physics
$m!$ – factorial M.48 §01 Math Signs
$\binom{n}{k}$ – binomial coefficient A2.23 §04 Algebra II
k – a constant rate
m, n – a power; the order of the derivative

CONTEXT

- **Origin.** Leibniz's d^2y/dx^2 notation (1690s); Newton's dots $\ddot{x}$ for time derivatives.
- **Use.** Concavity tests, Taylor polynomials C2.22 §08 Calculus II, differential equations C2.28 §08 Calculus II.
- **Data science.** The second derivative (Hessian C3.16 §10 Calculus III) measures curvature of the loss: Newton's method O.13 §14 Optimization divides by it, and its largest value bounds the safe learning rate C1.26.
- **Physical application.** Ride designers limit jerk s''' so that elevators and roller coasters start and stop smoothly.

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C1.20 Related Rates

When several quantities change with time and are linked by an equation, differentiating the equation with respect to t links their rates. Know some rates, solve for another.

Time chain rule	$\dfrac{d}{dt}f\big(x(t)\big) = f'(x)\,\dfrac{dx}{dt}$	Every quantity is a function of t, so each term picks up its own rate.
Expanding circle	$A = \pi r^2 \;\Rightarrow\; \dfrac{dA}{dt} = 2\pi r\dfrac{dr}{dt}$	The area grows faster the larger the circle.
Expanding sphere	$V = \frac{4}{3}\pi r^3 \;\Rightarrow\; \dfrac{dV}{dt} = 4\pi r^2\dfrac{dr}{dt}$	Rate of volume = surface area × rate of radius G.22 §05 Geometry.
Sliding ladder	$x^2 + y^2 = \ell^2 \;\Rightarrow\; x\dfrac{dx}{dt} + y\dfrac{dy}{dt} = 0$	Pythagoras G.7 §05 Geometry with constant length ℓ; the top falls as the foot slides out.
Filling cone	$V = \frac{1}{3}\pi r^2 h,\ \dfrac{r}{h} = \dfrac{R}{H} \;\Rightarrow\; V = \dfrac{\pi R^2}{3H^2}h^3$	Similar triangles G.9 §05 Geometry eliminate r before differentiating.

STATED-EQUATION *The derivative with respect to t of f of x of t equals f prime of x times d x d t. For a circle, d A d t equals two pi r d r d t; for a sphere, d V d t equals four pi r squared d r d t. For a ladder, x d x d t plus y d y d t equals zero. For a cone, use similar triangles to write the volume in terms of h alone. The cone holds V equals one third pi r squared h, and with r over h equal to R over H, V equals pi R squared over three H squared, times h cubed.*

MEMORY AID "Differentiate first, substitute numbers after."

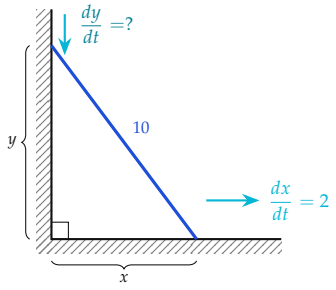
METHOD

1. Draw a picture; name the changing quantities and list the known and wanted rates (with units and signs).
2. Write an equation relating the quantities (geometry, Pythagoras, similar triangles); eliminate any variable whose rate you do not know.
3. Differentiate implicitly with respect to t C1.17.
4. Only now substitute the values at the instant in question and solve. Example: a 10 ft ladder, foot at $x = 6$ sliding out at 2 ft/s: $y = 8$, $6\cdot 2 + 8\,y' = 0$, so $y' = -1.5$ ft/s.

DECISION POINT Choosing the equation is the hard part: pick the one that contains *exactly* the quantities whose rates are given or wanted. If an extra variable appears with no known rate, use a fixed relation (similar triangles, constant length) to eliminate it *before* differentiating.

TERMS

t – time; every changing quantity is a function of t
$\frac{dr}{dt}$ – rate of change of r (negative if decreasing)
A, V – area, volume G.17 §05 Geometry
ℓ – fixed ladder length
R, H – fixed dimensions of the cone tank
Order – differentiate first, substitute numbers after
r, h – changing radius and height (water level)



CONTEXT

- **Origin.** Newton's fluxions were rates with respect to time.
- **Use.** Ladders, tanks, shadows, expanding circles.
- **Data science.** Metrics that are products change by the product rule C1.12: if revenue $R = pN$, then $\frac{dR}{dt} = N\frac{dp}{dt} + p\frac{dN}{dt}$, splitting growth into price and volume effects.
- **Physical application.** As oil spreads in a circle, the area grows faster the larger the slick, because $\frac{dA}{dt} = 2\pi r\frac{dr}{dt}$.

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C1.21 Tangent Lines & Linear Approximation

Zoomed in close enough, a differentiable curve looks like its tangent line. That line, the linearization, is the best straight-line approximation of f near a.

Tangent line	$y - f(a) = f'(a)\,(x - a)$	Point–slope form A1.7 §03 Algebra I with slope $f'(a)$.
Normal line	$y - f(a) = -\dfrac{1}{f'(a)}\,(x - a)$	Perpendicular to the tangent A1.8 §03 Algebra I; vertical if $f'(a) = 0$.
Linearization	$f(x) \approx L(x) = f(a) + f'(a)\,(x - a)$	Accurate for x near a; the first-order Taylor polynomial C2.22 §08 Calculus II.
Approximation error	$f(x) - L(x) \approx \frac{1}{2} f''(a)\,(x - a)^2$	The error shrinks quadratically; L overestimates where f is concave down.
Standard linearizations	$\sqrt{1+x} \approx 1 + \frac{x}{2}, \quad (1+x)^k \approx 1 + kx, \quad \sin x \approx x$	Near $x = 0$; the workhorse approximations of physics and finance.

STATED-EQUATION *The tangent line is y minus f of a equals f prime of a times x minus a. The normal line has slope minus one over f prime of a. Near a, f of x is approximately L of x, which equals f of a plus f prime of a times x minus a. The error is about one half f double prime of a times x minus a squared. Near zero, the square root of one plus x is about one plus x over two, one plus x to the k is about one plus k x, and sine x is about x.*

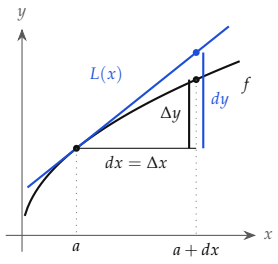
MEMORY AID "Near the point, the curve is the tangent line."

METHOD

1. Choose a near x where $f(a)$ and $f'(a)$ are easy.
2. Compute $f(a)$ and $f'(a)$ and write $L(x)$.
3. Evaluate L at the target. Example: $\sqrt{9.2}$ with $f = \sqrt{x}$, $a = 9$: $f'(9) = \frac{1}{6}$, so $\sqrt{9.2} \approx 3 + \frac{0.2}{6} \approx 3.0333$ (true: 3.0332).

TERMS

a – point of tangency (base point)
$L(x)$ – linearization: point–slope line A1.7 §03 Algebra I
$f'(a)$ – tangent slope C1.9
$f''(a)$ – curvature controls the error C1.19
k – any real exponent
Differentials – $dy = f'\,dx$ C1.22



CONTEXT

- **Origin.** Newton and Leibniz; the first-order Taylor polynomial.
- **Use.** Quick estimates and error propagation.
- **Data science.** Gradient descent relies on $L(\theta - \eta\nabla L) \approx L(\theta) - \eta\|\nabla L\|^2$; the delta method uses linearization to get standard errors of transformed estimates S.24 §13 Statistics.
- **Physical application.** Engineers estimate that a 1% error in a sphere's radius gives about a 3% error in its volume ($dV/V = 3\,dr/r$).

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C1.22 Differentials & Error Propagation

The differential $dy = f'(x)\,dx$ is the change along the tangent line. It approximates the true change Δy and turns a measurement error in x into an estimated error in y.

Differential	$dy = f'(x)\,dx$	dx is an independent increment; dy is the tangent-line rise over it.
Change estimate	$\Delta y = f(x+dx) - f(x) \approx dy$	Actual change versus tangent change; the gap is the linearization error.
Relative error	$\dfrac{dy}{y} = \dfrac{f'(x)}{f(x)}\,dx$	Percentage error $= 100\,\frac{dy}{y}\%$.
Power-law errors	$y = kx^n \implies \dfrac{dy}{y} = n\,\dfrac{dx}{x}$	Relative errors scale by the exponent: a 1% error in r is a 3% error in r^3.
Differential rules	$d(u+v) = du + dv, \quad d(uv) = u\,dv + v\,du$	Every derivative rule has a differential version.

STATED-EQUATION *The differential d y equals f prime of x d x. The actual change delta y is approximately d y. The relative error d y over y equals f prime over f times d x. For y equal to k x to the n, the relative error in y is n times the relative error in x. The differential of u plus v is d u plus d v, and the differential of u v is u d v plus v d u.*

MEMORY AID dy rides the tangent; Δy rides the curve.

METHOD

1. Write $y = f(x)$ and compute $f'(x)$.
2. Set $dx =$ the measurement error (or the change) and compute $dy = f'(x)\,dx$.
3. Divide by y for the relative error. Example: a cube of side $x = 10\,\text{cm}$ measured to $\pm 0.1\,\text{cm}$: $V = x^3$, $dV = 3x^2 dx = 3 \cdot 100 \cdot 0.1 = 30\,\text{cm}^3$, a relative error of $\frac{30}{1000} = 3\%$.

TERMS

dx – independent increment (error) in x
dy – change along the tangent line
Δy – actual change in y
$\frac{dy}{y}$ – relative error; $\times 100$ for percent
k, n – constant coefficient and exponent
u, v – differentiable functions

CONTEXT

- **Origin.** Leibniz treated dx, dy as infinitesimals (1684); made rigorous as linear maps in the 20th century.
- **Use.** Measurement uncertainty, tolerances, sensitivity analysis.
- **Data science.** The delta method: $\text{Var}(g(\hat{\theta})) \approx g'(\theta)^2\,\text{Var}(\hat{\theta})$, the standard error of a transformed estimate S.24 §13 Statistics.
- **Physical application.** A 2% error in measuring a pendulum's length gives about a 1% error in its period $T = 2\pi\sqrt{L/g}$ ($n = \frac{1}{2}$) P.15 §12 Physics.

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C1.23 Critical Points, Extrema & Concavity

The sign of f' says where f rises or falls; the sign of f'' says which way it bends. Extrema can occur only at critical points, where the slope is zero or undefined.

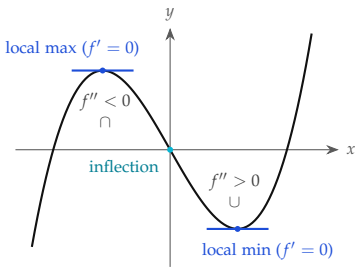
Critical point	c critical $\iff f'(c)=0$ or $f'(c)$ undefined $(c \in \operatorname{dom} f)$	The only interior candidates for local extrema.
Fermat's theorem	f has a local extremum at c, $f'(c)$ exists $\implies f'(c)=0$	Necessary, not sufficient: x^3 has $f'(0)=0$ but no extremum.
Increasing / decreasing test	$f'>0$ on $I \Rightarrow f \nearrow$; $\quad f'<0$ on $I \Rightarrow f \searrow$	Follows from the mean value theorem C1.27.
Concavity test	$f''>0 \Rightarrow$ concave up $(\cup)$; $\quad f''<0 \Rightarrow$ concave down $(\cap)$	Concave up: slopes increase, graph above its tangents.
Inflection point	f'' changes sign at c	Concavity flips; $f''(c)=0$ alone is not enough (x^4 at 0).
Second-derivative test	$f'(c)=0:\ f''(c)>0 \Rightarrow$ local min, $\ f''(c)<0 \Rightarrow$ local max	Inconclusive if $f''(c)=0$; then use the first-derivative test C1.25.

STATED-EQUATION *A critical point is where f prime of c equals zero or is undefined. Fermat: at a local extremum where f is differentiable, f prime is zero. f is increasing where f prime is positive and decreasing where it is negative. f is concave up where f double prime is positive, concave down where it is negative, and an inflection point is where f double prime changes sign. If f prime of c is zero and f double prime of c is positive, it is a local minimum; if negative, a local maximum.*

MEMORY AID Concave up holds water like a cup (a min); concave down spills like a frown (a max).

TERMS

c – critical number (in the domain of f)
I – an open interval
$\nearrow, \searrow$ – increasing, decreasing
Local extremum – max or min compared with nearby points
Convex function – $f'' \geq 0$ everywhere 0.2 §14 Optimization
Saddle point – $f'=0$ without an extremum



CONTEXT

- **Origin.** Fermat's method of adequality for maxima and minima (1638).
- **Use.** Finding peaks, troughs, and changes in trend.
- **Data science.** Convex losses (MSE, log-loss for linear models) have no bad local minima; deep networks are non-convex and full of saddle points where $f'=0$ but no extremum exists.
- **Physical application.** The highest point of a thrown ball is where its velocity (first derivative) is zero and acceleration is negative.

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C1.24 Extreme Value Theorem & Closed-Interval Method

A continuous function on a closed, bounded interval always attains an absolute maximum and minimum, and they occur either at critical points or at the endpoints.

Extreme value theorem	$f \in C[a,b] \implies \exists\, c,d \in [a,b] :\ f(c) \le f(x) \le f(d)\ \ \forall x \in [a,b]$	Both hypotheses matter: on $(0,1)$, $\frac{1}{x}$ has no max; a jump can skip the max.
Candidate set	abs. extrema $\in \{$critical points in $(a,b)\} \cup \{a,b\}$	Fermat's theorem C1.23 rules out all other interior points.
Absolute vs. local	$f(d) = \max_{[a,b]} f$ (global), $\quad f(c) \ge f(x)$ near c (local)	An absolute extremum may sit at an endpoint, where it is not a local one.

STATED-EQUATION *If f is continuous on the closed interval a to b, it attains an absolute minimum and an absolute maximum there. These occur at critical points inside the interval or at the endpoints. Compare the values at all candidates. An absolute maximum is the largest value on the whole interval; a local maximum is largest only compared with nearby points.*

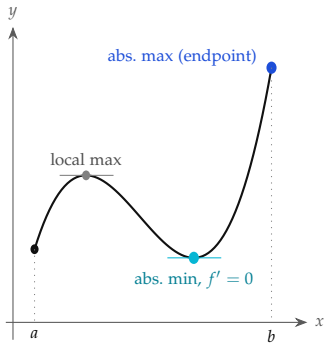
MEMORY AID "Closed and continuous: the max and min must show up." Check the ends!

METHOD

1. Check that f is continuous on the closed interval $[a,b]$.
2. Find the critical points of f in (a,b).
3. Evaluate f at each critical point and at a and b.
4. The largest value is the absolute max, the smallest the absolute min. Example: $f(x) = x^3 - 3x$ on $[0,3]$: $f' = 3x^2 - 3 = 0$ at $x = 1$; $f(0) = 0$, $f(1) = -2$, $f(3) = 18$: min -2, max 18.

TERMS

$[a,b]$ – closed, bounded interval
$C[a,b]$ – continuous on $[a,b]$ C1.7
c, d – locations of the absolute min and max
$\forall$ – for all M.89 §01 Math Signs
Absolute (global) – extreme over the whole interval
Several variables – closed bounded regions C3.16 §10 Calculus III



CONTEXT

- **Origin.** Bolzano (1830s) and Weierstrass (1860); a consequence of the completeness of $\mathbb{R}$.
- **Use.** Guaranteeing that an optimization problem has a solution before searching for it.
- **Data science.** Constraining parameters to a bounded box (weight clipping, bounded hyperparameter searches) guarantees an optimum exists O.1 §14 Optimization.
- **Physical application.** The hottest moment of a day on a continuous temperature record is either a turning point or the start or end of the recording.

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C1.25 Curve Sketching: First- & Second-Derivative Tests

The derivatives turn a formula into a shape. Sign charts of f' and f'' give the intervals of increase and concavity; the derivative tests classify each critical point.

First-derivative test	$f'\colon + \to -$ at $c \Rightarrow$ local max; $\quad - \to + \Rightarrow$ local min	No sign change: no extremum. Always conclusive, even where f'' fails.
Second-derivative test	$f'(c)=0,\ f''(c)>0 \Rightarrow$ min; $\ f''(c)<0 \Rightarrow$ max; $\ f''(c)=0 \Rightarrow$?	Quick, but silent when $f''(c)=0$: x^4 (min), $-x^4$ (max), x^3 (neither).
Sign chart	sign of f' (or f'') on each interval between its zeros	Test one point per interval; the sign cannot change without crossing a zero or discontinuity C1.7.
Symmetry	$f(-x)=f(x)$ (even), $\quad f(-x)=-f(x)$ (odd)	Sketch $x \geq 0$ and reflect A2.5 §04 Algebra II.
Asymptotes	$\lim_{x\to a} f = \pm\infty$ (vertical), $\quad \lim_{x\to\pm\infty} f = L$ (horizontal)	Frame the graph C1.2, C1.3.

STATED-EQUATION *First-derivative test: if f prime changes from positive to negative at c, f has a local maximum; from negative to positive, a local minimum; no change, no extremum. Second-derivative test: if f prime of c is zero and f double prime of c is positive, a minimum; negative, a maximum; zero, inconclusive. Use sign charts of f prime and f double prime, symmetry and asymptotes to sketch. f is even when f of minus x equals f of x, and odd when f of minus x equals minus f of x. A vertical asymptote is where f goes to plus or minus infinity as x approaches a; a horizontal asymptote is where f approaches L as x goes to plus or minus infinity.*

MEMORY AID f': up or down. f'': cup or cap.

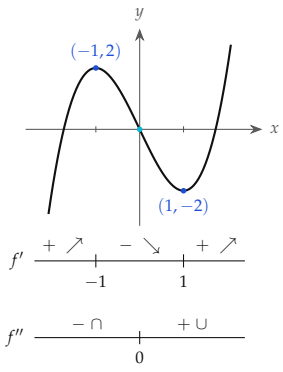
METHOD

1. Domain, intercepts, symmetry; vertical and horizontal asymptotes.
2. f': critical points and a sign chart (increasing/decreasing, local extrema by the first-derivative test).
3. f'': zeros and a sign chart (concavity, inflection points).
4. Plot the key points and join them with the right rise and bend. Example: $f = x^3 - 3x$: $f' = 3(x-1)(x+1)$ gives max $(-1,2)$, min $(1,-2)$; $f'' = 6x$ gives inflection $(0,0)$.

DECISION POINT Which test? Use the **second-derivative test** when f'' is easy and nonzero at c. Use the **first-derivative test** when $f''(c)=0$, when f' is undefined at c (corners, cusps), or when you need the sign chart anyway for the sketch.

TERMS

c – critical point C1.23
$+ \to - $ – sign of f' just left / just right of c
? – inconclusive
Even / odd – symmetric about the y-axis / the origin
L – horizontal asymptote height
a – location of a vertical asymptote C1.2



CONTEXT

- **Origin.** Systematized by Maria Agnesi's *Instituzioni analitiche* (1748), the first calculus textbook by a woman.
- **Use.** Graphing by hand; understanding a model's qualitative behavior.
- **Data science.** Reading learning curves: an inflection in validation loss marks where improvement starts slowing; the second derivative of a cumulative curve finds the elbow in k-means ML.35 §15 Machine Learning.
- **Physical application.** A logistic population curve has its inflection, the fastest growth, at half the carrying capacity C2.28 §08 Calculus II.

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C1.26 Optimization & Gradient Descent

To optimize, express the objective in one variable and set its derivative to zero. When that equation cannot be solved, gradient descent walks downhill in small steps against the slope.

Stationarity condition	$f'(x^*) = 0$	Solve for candidates; verify with C1.23 and the endpoints C1.24.
Single critical point	only one critical point on I, local min $\implies$ absolute min on I	Saves comparing values on an open interval.
Gradient descent (1-D)	$x_{k+1} = x_k - \eta\, f'(x_k)$	Step against the slope; the steps shrink as $f' \to 0$ near the minimum.
Gradient descent (vector)	$\boldsymbol{\theta}_{k+1} = \boldsymbol{\theta}_k - \eta\, \nabla L(\boldsymbol{\theta}_k)$	Same update with every partial derivative at once C3.12 §10 Calculus III.
Step-size condition	$0 < \eta < \dfrac{2}{M}, \quad M = \max \lvert f'' \rvert$	Guarantees descent on a smooth convex f; larger η overshoots and diverges.

STATED-EQUATION *Solve f prime of x star equals zero, then verify with the second derivative and endpoints. If there is only one critical point on an interval and it is a local minimum, it is the absolute minimum. Gradient descent: x k plus one equals x k minus eta f prime of x k; theta k plus one equals theta k minus eta times the gradient of L at theta k. The step size must be less than two over the largest curvature.*

MEMORY AID "Downhill is minus the gradient."

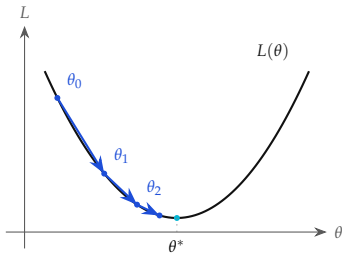
METHOD

1. Name the objective and the constraint; draw a picture.
2. Use the constraint to write the objective as a function of one variable, with its domain.
3. Solve $f'(x) = 0$; check critical points and endpoints (or use the second-derivative test).
4. Answer the question asked, with units. Example: a rectangle with perimeter 20 and maximum area: $A = x(10 - x)$, $A' = 10 - 2x = 0$ gives $x = 5$: a 5×5 square, area 25.

DECISION POINT **Which variable?** Choose the one that makes the objective simplest after substituting the constraint. **Closed or open domain?** Closed: compare all candidates C1.24; open: use the single-critical-point or second-derivative argument. **No closed-form solution of $f' = 0$?** Use gradient descent or Newton's method C1.35.

TERMS

Objective f – quantity to maximize or minimize
Constraint – condition used to reduce to one variable
x^* – optimal point
η – learning rate (step size)
∇L – gradient: vector of partial derivatives C3.12 §10 Calculus III
M – bound on curvature $|f''|$ (Lipschitz constant of f')



CONTEXT

- **Origin.** Augustin-Louis Cauchy proposed gradient descent in 1847.
- **Use.** Maximizing area or profit, minimizing cost or material.
- **Data science.** Training nearly every ML model: stochastic gradient descent, Adam, and friends are refinements of this update rule ML.27 §15 Machine Learning.
- **Physical application.** A can holding a fixed volume uses the least metal when its height equals its diameter.

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C1.27 Mean Value Theorem

Somewhere between a and b, the instantaneous rate of change equals the average rate of change: some tangent is parallel to the secant.

Mean value theorem	f continuous on $[a,b]$, differentiable on $(a,b) \implies \exists\, c \in (a,b):\ f'(c) = \dfrac{f(b)-f(a)}{b-a}$	Tangent slope at c equals secant slope.
Rolle's theorem	$f(a) = f(b) \implies \exists\, c \in (a,b):\ f'(c) = 0$	The special case of a level secant: what goes up and comes back has a flat point.
Zero derivative	$f' \equiv 0$ on $I \implies f$ constant on I	Hence $f' = g'$ means $f = g + C$: why antiderivatives differ by a constant C1.29.
Lipschitz bound	$\lvert f(x) - f(y)\rvert \le M\lvert x - y\rvert, \quad M = \sup\lvert f'\rvert$	A bounded slope bounds how far f can move.
Cauchy's mean value theorem	$\dfrac{f(b)-f(a)}{g(b)-g(a)} = \dfrac{f'(c)}{g'(c)}$	Two functions at the same c; the basis of L'Hôpital's rule C1.28.

STATED-EQUATION *If f is continuous on a to b and differentiable on the open interval, then there is a c in between with f prime of c equal to f of b minus f of a, over b minus a. Rolle: if f of a equals f of b, then f prime of c equals zero. A zero derivative means a constant function. The change in f is at most the largest slope times the change in x. Cauchy's version applies to a ratio of two functions. Cauchy: f of b minus f of a, over g of b minus g of a, equals f prime of c over g prime of c.*

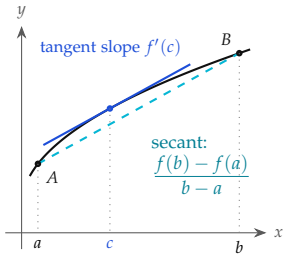
MEMORY AID "At some moment, your speedometer showed your average speed."

METHOD

1. Check the hypotheses: continuous on $[a,b]$, differentiable on (a,b).
2. Compute the average rate $\frac{f(b)-f(a)}{b-a}$.
3. Solve $f'(c) =$ that value for $c \in (a,b)$. Example: $f(x) = x^2$ on $[1,3]$: average rate $\frac{9-1}{2} = 4$, $2c = 4$, $c = 2$.

TERMS

Right side – average rate of change (secant slope A1.6 §03 Algebra I)
$f'(c)$ – instantaneous rate (tangent slope C1.9)
c – an interior point (existence only, not location)
$f' \equiv 0$ – zero at every point of I
M – Lipschitz constant; sup least upper bound
g – second function with $g' \neq 0$ on (a,b)



CONTEXT

- **Origin.** Michel Rolle (1691); general form by Lagrange (1797); Cauchy's form (1823).
- **Use.** Speed-trap logic; proving error bounds and the derivative tests.
- **Data science.** Lipschitz bounds from the MVT set safe step sizes (gradient descent converges for $\eta < 2/M$ on smooth convex losses C1.26) and certify model robustness to input perturbations.
- **Physical application.** Average-speed cameras rely on the MVT: if your average speed exceeded the limit, your instantaneous speed did too at some moment.

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C1.28 L'Hôpital's Rule

For a $\frac{0}{0}$ or $\frac{\infty}{\infty}$ limit, the ratio of the functions has the same limit as the ratio of their derivatives. Other indeterminate forms are rewritten as quotients first.

L'Hôpital's rule	$$\lim_{x\to a} f = \lim_{x\to a} g = 0 \text{ or } \pm\infty \implies \lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)}$$	Provided the right side exists (or is $\pm\infty$); a may be $\pm\infty$.
Repeated application	$\lim \frac{f}{g} = \lim \frac{f'}{g'} = \lim \frac{f''}{g''} = \cdots$	Reapply only while the form is still $\frac{0}{0}$ or $\frac{\infty}{\infty}$: $\lim_{x\to 0} \frac{1-\cos x}{x^2} = \lim \frac{\sin x}{2x} = \frac{1}{2}$.
Product form	$0 \cdot \infty: \ \lim f\,g = \lim \frac{f}{1/g}$	Put the factor that is easier to differentiate on top C1.4.
Power forms	$1^\infty, 0^0, \infty^0: \ \lim f^g = e^{\lim g \ln f}$	Apply the rule to $g \ln f$ written as a quotient.
Growth comparisons	$\lim_{x\to\infty} \frac{\ln x}{x^p} = 0, \qquad \lim_{x\to\infty} \frac{x^n}{e^x} = 0$	Proves the growth hierarchy C1.3.

STATED-EQUATION *If the limits of f and g as x approaches a are both zero or both infinite, then the limit of f over g equals the limit of f prime over g prime, provided that limit exists. Apply again while the form remains indeterminate. Rewrite zero times infinity as a quotient, and power forms with logarithms. Log x over any power, and any power over e to the x, go to zero.*

MEMORY AID Differentiate top and bottom separately, not the quotient rule.

METHOD

1. Confirm the form is $\frac{0}{0}$ or $\frac{\infty}{\infty}$ (rewrite other forms first).
2. Differentiate numerator and denominator separately.
3. Evaluate the new limit; repeat if still indeterminate. Example: $\lim_{x\to\infty} \frac{x^2}{e^x} = \lim \frac{2x}{e^x} = \lim \frac{2}{e^x} = 0$.

DECISION POINT L'Hôpital or algebra? If factoring, a conjugate, or a special limit C1.6 cancels the zero in one step, do that. Prefer L'Hôpital when the derivatives are simpler than the functions (ln, inverse trig, exponentials). Stop if derivatives grow messier or cycle ($\frac{e^x+e^{-x}}{e^x-e^{-x}}$ as $x \to \infty$): divide by e^x instead.

TERMS

Indeterminate – $\frac{0}{0}$, $\frac{\infty}{\infty}$; rewrite $0 \cdot \infty$, $\infty - \infty$, 1^∞ first
f', g' – differentiate separately, *not* the quotient rule C1.12
a – finite or $\pm\infty$ C1.1
p, n – positive powers
Proof – Cauchy's mean value theorem C1.27

CONTEXT

- **Origin.** Johann Bernoulli taught it to the Marquis de L'Hôpital, who published it (1696).
- **Use.** Evaluating indeterminate limits.
- **Data science.** Comparing growth rates: $\frac{\ln n}{n} \to 0$ ranks $O(\log n)$ below $O(n)$ O.10 §14 Optimization; $x \ln x \to 0$ justifies $0 \log 0 = 0$ in entropy ML.3 §15 Machine Learning.
- **Physical application.** Evaluating $\lim_{\theta\to 0} \frac{\sin\theta}{\theta}$ in small-angle optics or the damped-oscillator limit uses L'Hôpital's rule.

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C1.29 Antiderivatives & Basic Integrals

An antiderivative of f is a function F with $F' = f$. All antiderivatives differ by a constant, so the indefinite integral is written with $+C$. Every basic integral is a derivative read backward.

Antiderivative	$\int f(x)\,dx = F(x) + C \iff F'(x) = f(x)$	The whole family; C is fixed only by extra information.
Power rule	$\int x^n\,dx = \dfrac{x^{n+1}}{n+1} + C \quad (n \neq -1)$	Raise the power, divide by the new power.
Reciprocal	$\int \dfrac{1}{x}\,dx = \ln\lvert x\rvert + C$	The missing $n = -1$ case; the absolute value covers $x < 0$.
Exponentials	$\int e^x\,dx = e^x + C, \qquad \int a^x\,dx = \dfrac{a^x}{\ln a} + C$	Reverse of $(a^x)' = a^x \ln a$.
Sine & cosine	$\int \sin x\,dx = -\cos x + C, \qquad \int \cos x\,dx = \sin x + C$	Signs are the reverse of differentiation.
Secant squared & arctangent	$\int \sec^2 x\,dx = \tan x + C, \qquad \int \dfrac{dx}{1+x^2} = \arctan x + C$	Reverse of the tan and arctan derivatives.
Linearity	$\int [af(x) + bg(x)]\,dx = a\int f\,dx + b\int g\,dx$	Integrate term by term. There is no product or quotient rule.
Initial value problem	$F' = f,\ F(x_0) = y_0 \implies C = y_0 - F_0(x_0)$	One known value fixes the constant, where F_0 is any antiderivative.

STATED-EQUATION *The integral of f of x d x equals F of x plus C exactly when F prime equals f. The integral of x to the n is x to the n plus one over n plus one, plus C, for n not equal to minus one. The integral of one over x is natural log of the absolute value of x plus C. The integral of e to the x is e to the x; of a to the x, a to the x over natural log a; of sine, minus cosine; of cosine, sine; of secant squared, tangent; of one over one plus x squared, arctangent; each plus C. Integrals split over sums and constants. An initial value fixes C. Given F prime equals f and F of x naught equals y naught, the constant is C equals y naught minus F naught of x naught.*

MEMORY AID "Add one to the power, divide by the new power"; don't forget $+C$.

METHOD

1. Rewrite the integrand as a sum of table terms (expand, split fractions, roots as powers).
2. Integrate term by term with the table C1.K3; add one $+C$.
3. Check by differentiating; if an initial value is given, solve for C. Example: $F' = 6x^2$, $F(1) = 5$: $F = 2x^3 + C$, $2 + C = 5$, so $F = 2x^3 + 3$.

TERMS

F – antiderivative of f: $F' = f$
C – constant of integration (MVT C1.27)
$\int$ – Leibniz's elongated S for *summa*
dx – variable of integration
a, b – constants; $a > 0$ as an exponential base
(x_0, y_0) – a known point (initial condition)
Check – differentiate the answer C1.14
F_0 – any one antiderivative (the one with $C = 0$)

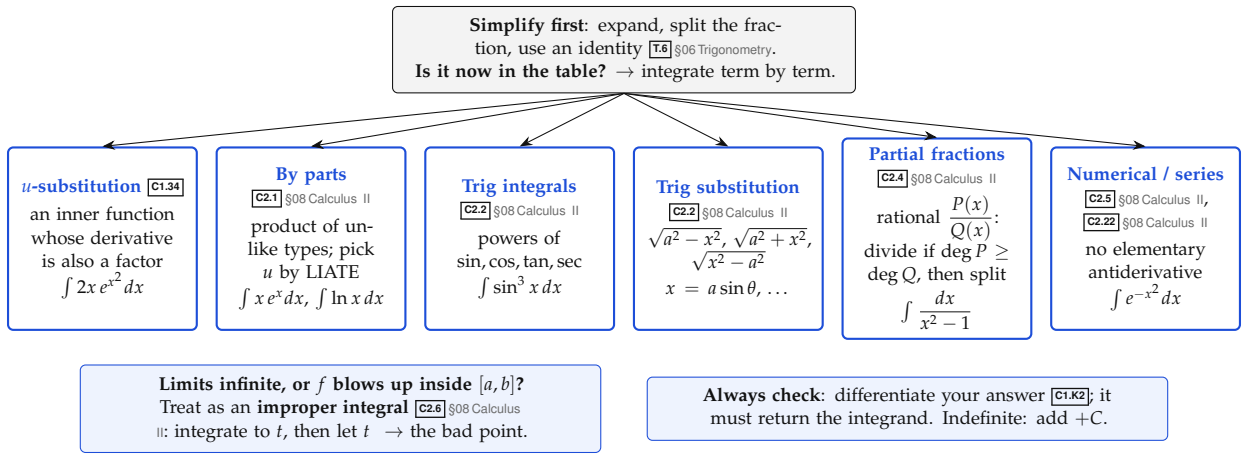
CONTEXT

- **Origin.** Leibniz introduced $\int$ (1675); Newton's inverse method of fluxions.
- **Use.** Recovering position from velocity, total from rate.
- **Data science.** A CDF is the antiderivative of its PDF, $F'(x) = f(x)$ S.20 §13 Statistics; cumulative metrics are running integrals of rates.
- **Physical application.** Integrating acceleration gives velocity, and integrating velocity gives position P.2 §12 Physics.

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Antiderivative Table | *Key Topic C1.K3*

Every entry below is a derivative from the derivative table C1.K2 read backward, so each can be checked by differentiating. Integration is not mechanical: first simplify the integrand into table form, then use linearity, and reach for u-substitution C1.34 (or the Calculus II techniques) only when no rewrite works. The decision map shows the order in which to try methods.



The table (add $+C$ to every entry)

$\int f(x)\,dx$	$F(x)$	Note	$\int f(x)\,dx$	$F(x)$
$\int k\,dx$	kx	constant	$\int \sin x\,dx$	$-\cos x$
$\int x^n\,dx$	$\frac{x^{n+1}}{n+1}$	$n \neq -1$	$\int \cos x\,dx$	$\sin x$
$\int \frac{1}{x}\,dx$	$\ln\lvert x\rvert$	$x \neq 0$	$\int \sec^2 x\,dx$	$\tan x$
$\int e^x\,dx$	e^x		$\int \csc^2 x\,dx$	$-\cot x$
$\int e^{kx}\,dx$	$\frac{1}{k}e^{kx}$	$k \neq 0$	$\int \sec x\tan x\,dx$	$\sec x$
$\int a^x\,dx$	$\frac{a^x}{\ln a}$	$a > 0$, $a \neq 1$	$\int \csc x\cot x\,dx$	$-\csc x$
$\int \ln x\,dx$	$x\ln x - x$	by parts C2.1 §08 Calculus II	$\int \tan x\,dx$	$-\ln\lvert\cos x\rvert = \ln\lvert\sec x\rvert$
$\int \frac{dx}{a^2+x^2}$	$\frac{1}{a}\arctan\frac{x}{a}$	$a > 0$	$\int \cot x\,dx$	$\ln\lvert\sin x\rvert$
$\int \frac{dx}{\sqrt{a^2-x^2}}$	$\arcsin\frac{x}{a}$	$\lvert x\rvert < a$	$\int \sec x\,dx$	$\ln\lvert\sec x+\tan x\rvert$
$\int \frac{dx}{x\sqrt{x^2-a^2}}$	$\frac{1}{a}\operatorname{arcsec}\frac{\lvert x\rvert}{a}$	$\lvert x\rvert > a$	$\int \csc x\,dx$	$-\ln\lvert\csc x+\cot x\rvert$
$\int \frac{dx}{a^2-x^2}$	$\frac{1}{2a}\ln\lvert\frac{a+x}{a-x}\rvert$	partial fractions C2.4 §08 Calculus II	$\int \sinh x\,dx$, $\int \cosh x\,dx$	$\cosh x$, $\sinh x$

Rules that extend the table

Linearity
$\int (af+bg)\,dx = a\int f\,dx + b\int g\,dx$.
$\int (4x^3 - \frac{2}{x})\,dx = x^4 - 2\ln\lvert x\rvert + C$.
No product rule: $\int fg \neq \int f \int g$.

Linear inside
$\int f(ax+b)\,dx = \frac{1}{a}F(ax+b) + C$.
$\int \cos(3x+1)\,dx = \frac{1}{3}\sin(3x+1) + C$.
$\int (2x-5)^4 dx = \frac{(2x-5)^5}{10} + C$.

Reverse chain rule
$\int f'(g)\,g'\,dx = f(g) + C$ C1.34.
$\int \frac{g'(x)}{g(x)}dx = \ln\lvert g(x)\rvert + C$:
$\int \frac{2x}{x^2+1}dx = \ln(x^2+1) + C$.

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Decision point: rewrite before you integrate

DECISION POINT The table only matches exact shapes, so the real choice is **which rewrite** turns the integrand into table entries. **Expand** products and powers: $\int (x+1)^2 dx = \int (x^2+2x+1)\,dx$. **Split** a fraction with a single-term denominator: $\int \frac{x^2+1}{x} dx = \int (x + \frac{1}{x})\,dx$. **Roots and reciprocals as powers**: $\int \frac{dx}{\sqrt{x}} = \int x^{-1/2} dx = 2\sqrt{x} + C$. **Trig identities** T.6 §06 Trigonometry: $\int \tan^2 x\,dx = \int (\sec^2 x - 1)\,dx = \tan x - x + C$. If no rewrite produces table forms, look for an inner function whose derivative is present (u-substitution); otherwise follow the map above into Calculus II C2.K1 §08 Calculus II.

Worked examples

Rewrite, then table

$\int \frac{x^3 - 2\sqrt{x}}{x}\,dx = \int (x^2 - 2x^{-1/2})\,dx = \frac{x^3}{3} - 4\sqrt{x} + C.$

Definite integral via FTC C1.32

$\int_0^{\pi/2} (\sin x + 2x)\,dx = \left[-\cos x + x^2\right]_0^{\pi/2} = \frac{\pi^2}{4} + 1 \approx 3.47.$

Initial value problem C1.29

$f'(x) = e^{2x} + \cos x$, $f(0) = 3$: $f = \frac{1}{2}e^{2x} + \sin x + C$, $\frac{1}{2} + C = 3$, so $C = \frac{5}{2}$.

Inverse-trig form

$\int_0^3 \frac{dx}{9+x^2} = \left[\frac{1}{3}\arctan\frac{x}{3}\right]_0^3 = \frac{1}{3}\cdot\frac{\pi}{4} = \frac{\pi}{12}.$

Common mistakes

The $n = -1$ trap: $\int x^{-1} dx = \ln|x|$, not $\frac{x^0}{0}$.

Missing $+C$ on indefinite integrals; missing absolute value in $\ln|x|$.

Sign of trig integrals: $\int \sin x\,dx = -\cos x$ (differentiation has $(\cos x)' = -\sin x$, so the sign moves).

No quotient rule: $\int \frac{1}{x^2+1} dx = \arctan x$, but $\int \frac{1}{x^2-1} dx$ needs partial fractions, and $\int \frac{x}{x^2+1} dx = \frac{1}{2}\ln(x^2+1)$.

Inner constants: $\int e^{3x} dx = \frac{1}{3}e^{3x}$, not e^{3x} or $3e^{3x}$.

Always check by differentiating: $\frac{d}{dx}\left(\frac{1}{3}e^{3x}\right) = e^{3x}$ ✓.

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C1.30 Riemann Sums & the Definite Integral

The definite integral is the limit of sums of thin rectangles: the signed area between f and the x-axis over $[a,b]$. Area above the axis counts positive, below negative.

Definite integral	$\displaystyle\int_a^b f(x)\,dx = \lim_{n\to\infty}\sum_{i=1}^{n} f(x_i^*)\,\Delta x$	Exists for every continuous (even piecewise continuous) f on $[a,b]$.
Width and grid	$\Delta x = \dfrac{b-a}{n}, \qquad x_i = a + i\,\Delta x$	n equal subintervals $[x_{i-1}, x_i]$.
Sample points	$x_i^* = x_{i-1}$ (left), $\quad x_i$ (right), $\quad \frac{x_{i-1}+x_i}{2}$ (midpoint)	All give the same limit; for finite n the midpoint is usually best.
Reversed limits	$\displaystyle\int_a^b f\,dx = -\int_b^a f\,dx$	Integrating right to left flips the sign.
Summation formulas	$\displaystyle\sum_{i=1}^{n} i = \frac{n(n+1)}{2}, \qquad \sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$	Needed to evaluate a Riemann-sum limit by hand A2.21 §04 Algebra II.
Over/underestimates	$f\nearrow:\ L_n \le \displaystyle\int_a^b f \le R_n$	For increasing f, left sums undershoot and right sums overshoot.

STATED-EQUATION *The integral from a to b of f of x d x equals the limit as n goes to infinity of the sum from i equals one to n of f of x i star times delta x, where delta x equals b minus a over n and x i equals a plus i delta x. The sample point can be the left endpoint, right endpoint or midpoint. Swapping the limits changes the sign. The sum of i is n times n plus one over two; the sum of i squared is n times n plus one times two n plus one, over six. For an increasing f, the left sum underestimates and the right sum overestimates the integral.*

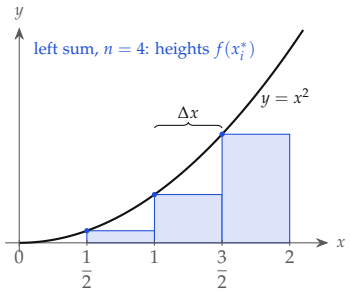
MEMORY AID "Height times width, added up, with the width shrinking to zero."

METHOD

1. Compute $\Delta x = \frac{b-a}{n}$ and the sample points x_i^*.
2. Write the sum $\sum f(x_i^*)\Delta x$; simplify with the summation formulas.
3. Take $n \to \infty$ (or stop at a finite n for an estimate). Example: $\int_0^2 x^2 dx$, right sum: $\Delta x = \frac{2}{n}$, $\sum\left(\frac{2i}{n}\right)^2\frac{2}{n} = \frac{8}{n^3}\cdot\frac{n(n+1)(2n+1)}{6} \to \frac{8}{3}$.

TERMS

Δx – width of each subinterval
x_i^* – sample point in the ith subinterval
n – number of subintervals
Σ – summation A2.21 §04 Algebra II
L_n, R_n – left and right sums
Integrand – f; a, b the lower and upper limits
Trapezoid rule – average of L_n and R_n C2.5 §08 Calculus II
x_i – grid points $a + i\,\Delta x$



CONTEXT

- **Origin.** Archimedes' method of exhaustion; formalized by Riemann (1854).
- **Use.** Area, distance, total accumulation.
- **Data science.** Expected values of continuous variables are integrals S.13 §13 Statistics; Monte Carlo integration averages random samples when dimensions are high O.27 §14 Optimization.
- **Physical application.** Estimating the distance a car traveled from speed readings every second is a Riemann sum.

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C1.31 Properties of Definite Integrals

The algebra of definite integrals: they are linear, add over adjacent intervals, respect inequalities, and simplify under symmetry. Each property is the corresponding fact about sums of rectangles.

Zero width	$\displaystyle\int_a^a f(x)\,dx = 0$	No interval, no area.
Linearity	$\displaystyle\int_a^b \big[cf \pm g\big]\,dx = c\int_a^b f\,dx \pm \int_a^b g\,dx$	As for sums A2.21 §04 Algebra II.
Additivity	$\displaystyle\int_a^b f\,dx + \int_b^c f\,dx = \int_a^c f\,dx$	Holds for any order of a, b, c; used to split at sign changes or piecewise seams.
Comparison	$f \le g$ on $[a,b] \implies \displaystyle\int_a^b f\,dx \le \int_a^b g\,dx$	In particular $f \ge 0 \Rightarrow \int_a^b f \ge 0$.
Bounds	$m \le f \le M \implies m(b-a) \le \displaystyle\int_a^b f\,dx \le M(b-a)$	Rectangles below and above the graph; quick sanity checks.
Absolute value	$\left\lvert\displaystyle\int_a^b f\,dx\right\rvert \le \int_a^b \lvert f\rvert\,dx$	Cancellation can only shrink the integral.
Even & odd symmetry	$\displaystyle\int_{-a}^a f\,dx = 2\int_0^a f\,dx$ (even), $\displaystyle\int_{-a}^a f\,dx = 0$ (odd)	$\int_{-2}^2 x^3\,dx = 0$ with no computation.
Mean value theorem for integrals	$\exists\, c \in [a,b] : \displaystyle\int_a^b f\,dx = f(c)\,(b-a)$	For continuous f; $f(c)$ is the average value C2.12 §08 Calculus II.

STATED-EQUATION *The integral from a to a is zero. Integrals are linear and add over adjacent intervals: a to b plus b to c equals a to c. If f is at most g, its integral is at most g's. If f is between m and M, the integral is between m times b minus a and M times b minus a. The absolute value of an integral is at most the integral of the absolute value. Over a symmetric interval, an even function gives twice the integral from zero to a, and an odd function gives zero. Some value f of c times b minus a equals the integral.*

MEMORY AID Odd over symmetric limits: zero. Even: double the half.

TERMS

c – a constant (linearity) or an intermediate point
m, M – minimum and maximum of f on $[a,b]$
Even / odd – $f(-x) = f(x)$ / $f(-x) = -f(x)$ A2.5 §04 Algebra II
$|f|$ – absolute value M.43 §01 Math Signs
Average value – $\frac{1}{b-a}\int_a^b f\,dx$ C2.12 §08 Calculus II

CONTEXT

- **Origin.** Properties established with the Riemann integral (1854); the integral mean value theorem goes back to Cauchy (1823).
- **Use.** Simplifying and bounding integrals before computing them.
- **Data science.** Linearity of expectation $E[aX + bY] = aE[X] + bE[Y]$ S.13 §13 Statistics is linearity of the integral; symmetric densities have zero odd moments, so a standard normal has mean and skewness 0.
- **Physical application.** The average temperature over a day is $\frac{1}{24}\int_0^{24} T(t)\,dt$, a temperature actually reached at some hour.

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C1.32 Fundamental Theorem of Calculus

Differentiation and integration are inverse operations. Part 1: the accumulated area under f grows at rate f. Part 2: a definite integral is the change in any antiderivative.

Part 1 (accumulation)	$\dfrac{d}{dx}\displaystyle\int_a^x f(t)\,dt = f(x)$	The area function $A(x)$ is an antiderivative of a continuous f.
Part 1 with chain rule	$\dfrac{d}{dx}\displaystyle\int_a^{g(x)} f(t)\,dt = f\big(g(x)\big)\,g'(x)$	Variable upper limit: substitute it, multiply by its derivative.
Variable limits on both ends	$\dfrac{d}{dx}\displaystyle\int_{h(x)}^{g(x)} f(t)\,dt = f\big(g(x)\big)g'(x) - f\big(h(x)\big)h'(x)$	Split at a constant using additivity C1.31.
Part 2 (evaluation)	$\displaystyle\int_a^b f(x)\,dx = F(b) - F(a) = \Big[F(x)\Big]_a^b$	Any antiderivative works; the $+C$ cancels.
Every continuous f has an antiderivative	$F(x) = \displaystyle\int_a^x f(t)\,dt$	Even if it has no elementary formula: $\int_0^x e^{-t^2}dt$ defines the error function.

STATED-EQUATION *Part one: the derivative of the integral from a to x of f of t d t equals f of x. With a variable upper limit g of x, it equals f of g of x times g prime of x; with two variable limits, subtract the same term for the lower limit. Part two: the integral from a to b of f of x d x equals F of b minus F of a. Every continuous function has an antiderivative, its own accumulation function.*

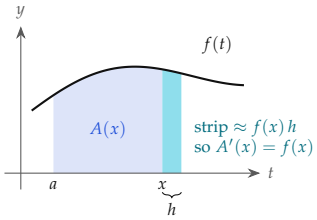
MEMORY AID "Integrate then differentiate: you get back where you started."

METHOD

1. Find an antiderivative F of the integrand (table C1.K3, or a substitution C1.34).
2. Evaluate F at the upper limit, then at the lower limit.
3. Subtract: upper minus lower. Example: $\int_1^3 x^2\,dx = \left[\frac{x^3}{3}\right]_1^3 = 9 - \frac{1}{3} = \frac{26}{3}$.

TERMS

F – any antiderivative of f C1.29
t – dummy variable of integration
$A(x)$ – $\int_a^x f(t)\,dt$, accumulated area
g, h – differentiable limit functions
$[F]_a^b$ – shorthand for $F(b) - F(a)$
Continuity – f continuous on $[a, b]$ C1.7



CONTEXT

- **Origin.** Isaac Barrow, Newton, and Leibniz (1660s–1680s).
- **Use.** Evaluating definite integrals exactly.
- **Data science.** Probabilities from a CDF: $P(a \le X \le b) = F(b) - F(a)$ S.20 §13 Statistics; the density is the derivative of the CDF.
- **Physical application.** Total water that flowed into a tank equals the integral of the inflow rate, found from any antiderivative.

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C1.33 Net Change Theorem

Integrating a rate of change over an interval gives the net change of the quantity. Signed areas cancel; to get a total amount (such as distance) integrate the absolute value.

Net change theorem	$\displaystyle\int_a^b F'(x)\,dx = F(b) - F(a)$	FTC Part 2 read as: total of the rate = net change of the amount.
Displacement	$\displaystyle s(t_2) - s(t_1) = \int_{t_1}^{t_2} v(t)\,dt$	Forward and backward motion cancel.
Total distance	$\displaystyle \text{distance} = \int_{t_1}^{t_2} \lvert v(t)\rvert\,dt$	Split at the zeros of v and add the absolute areas.
Amount from a rate	$\displaystyle Q(b) = Q(a) + \int_a^b Q'(t)\,dt$	Starting amount plus accumulated change: water in a tank, population, revenue.
Velocity from acceleration	$\displaystyle v(t) = v(0) + \int_0^t a(\tau)\,d\tau$	The same idea one derivative up.

STATED-EQUATION *The integral from a to b of F prime equals F of b minus F of a: the integral of a rate is the net change. Displacement is the integral of velocity; total distance is the integral of the absolute value of velocity. An amount at time b equals the amount at time a plus the integral of its rate. Velocity equals the initial velocity plus the integral of acceleration.*

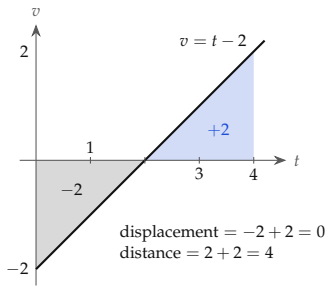
MEMORY AID Displacement keeps the signs; distance takes the absolute value.

METHOD

1. Identify the rate function and the interval.
2. For net change, integrate the rate directly.
3. For a total amount, find where the rate changes sign, split there, and add the absolute values. Example: $v = t - 2$ on $[0, 4]$: displacement $\int_0^4 (t-2)\,dt = 0$; distance $= 2 + 2 = 4$.

TERMS

F' – rate of change of F
s, v, a – position, velocity, acceleration P.2 §12 Physics
$|v|$ – speed
Q, Q' – an amount and its rate
τ – dummy time variable



CONTEXT

- **Origin.** A restatement of the FTC C1.32; used in physics since Newton's *Principia* (1687).
- **Use.** Physics, economics, any rate-to-total conversion.
- **Data science.** Cumulative metrics: integrating a daily-signups rate curve gives total signups; the area under a hazard-rate curve gives the cumulative hazard in survival analysis.
- **Physical application.** A car's odometer integrates speed (distance); its trip displacement is the integral of velocity.

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C1.34 Integration by u-Substitution

The chain rule run backward. If the integrand is a function of an inner expression times the inner expression's derivative, rename the inner expression u and integrate in u.

Substitution rule	$\int f(g(x))\,g'(x)\,dx = \int f(u)\,du$	The derivative of the inside must be present (up to a constant factor).
Differential	$u = g(x), \qquad du = g'(x)\,dx$	Replace *all* of the x's, including dx.
Definite integrals	$\int_a^b f(g(x))g'(x)\,dx = \int_{g(a)}^{g(b)} f(u)\,du$	Change the limits to u-values; no need to return to x.
Log pattern	$\int \frac{g'(x)}{g(x)}\,dx = \ln\lvert g(x)\rvert + C$	Numerator is the derivative of the denominator: $\int \tan x\,dx = -\ln\lvert\cos x\rvert + C$.
Linear substitution	$\int f(ax+b)\,dx = \frac{1}{a}F(ax+b) + C$	$u = ax+b$, $du = a\,dx$: divide by the inner slope.

STATED-EQUATION *The integral of f of g of x times g prime of x d x equals the integral of f of u d u, with u equals g of x and d u equals g prime of x d x. For definite integrals, change the limits to g of a and g of b. The integral of g prime over g is the natural log of the absolute value of g. The integral of f of a x plus b is one over a times F of a x plus b.*

MEMORY AID Look for an "inside function" whose derivative is also present.

METHOD

1. Choose u (see the decision point) and compute $du = g'(x)\,dx$.
2. Rewrite the whole integrand in u and du, adjusting a constant factor if needed; no x may remain.
3. Integrate in u.
4. Substitute back $u = g(x)$ (indefinite) or use the new limits (definite). Example: $\int_0^2 x(x^2+1)^3dx$: $u = x^2+1$, $du = 2x\,dx$, limits $1 \to 5$: $\frac{1}{2}\int_1^5 u^3du = \frac{625-1}{8} = 78$.

DECISION POINT Choosing u: take the **inner** part of a composition (inside a power, root, exponent, trig or log) whose **derivative also appears** as a factor, up to a constant. Good: $u = x^2+1$ in $\int x\sqrt{x^2+1}\,dx$; $u = \ln x$ in $\int \frac{\ln x}{x}dx$; $u = \sin x$ in $\int \sin^4 x\cos x\,dx$. If a leftover x remains, try solving $u = g(x)$ for x ($\int x\sqrt{x-1}\,dx$, $x = u+1$). If no choice leaves only u, switch technique: parts C2.1 §08 Calculus II for products of unlike types, trig substitution C2.2 §08 Calculus II, partial fractions C2.4 §08 Calculus II.

TERMS

u – inner function, chosen so du appears in the integrand
du – differential of u C1.22
$g(a), g(b)$ – new limits in u
F – antiderivative of f
a, b – constants, $a \neq 0$ (linear case)
Reverse of – the chain rule C1.13

CONTEXT

- **Origin.** The chain rule read backward (Leibniz).
- **Use.** Integrals of composite functions.
- **Data science.** Change of variables for densities: if $Y = g(X)$, then $f_Y(y) = f_X(g^{-1}(y))\left|\frac{d}{dy}g^{-1}(y)\right|$, which turns a normal into a log-normal S.20 §13 Statistics.
- **Physical application.** Computing the work to stretch a nonlinear spring often requires a substitution P.6 §12 Physics.

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C1.35 Newton's Method for Roots

Replace f by its tangent line at the current guess and jump to where the tangent crosses zero. Near a simple root the number of correct digits roughly doubles each step.

Newton's iteration	$x_{n+1} = x_n - \dfrac{f(x_n)}{f'(x_n)}$	The x-intercept of the tangent line C1.21 at x_n.
Stopping rule	$\lvert x_{n+1} - x_n\rvert < \text{tolerance}$ or $\lvert f(x_n)\rvert < \text{tolerance}$	Stop when the steps or the residual are tiny.
Square-root iteration	$x_{n+1} = \dfrac{1}{2}\left(x_n + \dfrac{a}{x_n}\right) \to \sqrt{a}$	Newton on $f(x) = x^2 - a$: the Babylonian method.
Quadratic convergence	$e_{n+1} \approx \dfrac{f''(r)}{2f'(r)}\, e_n^2$	Error squared each step near a simple root ($f'(r) \neq 0$).
Optimization version	$x_{n+1} = x_n - \dfrac{f'(x_n)}{f''(x_n)}$	Newton applied to f' finds critical points O.13 §14 Optimization.

STATED-EQUATION *x n plus one equals x n minus f of x n over f prime of x n. Stop when successive estimates differ by less than the tolerance. For a square root of a, average x n and a over x n. Near a simple root, the new error is about f double prime over two f prime times the old error squared. To optimize, apply the method to f prime.*

MEMORY AID "Follow the tangent line down to the axis, then repeat."

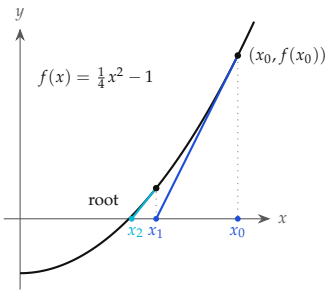
METHOD

1. Pick a starting guess x_0 near the root (a graph or a sign change helps C1.7).
2. Compute $f(x_n)$ and $f'(x_n)$; update $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$.
3. Repeat until the stopping rule is met. Example: $\sqrt{2}$ with $f = x^2 - 2$, $x_0 = 1$: $x_1 = 1.5$, $x_2 = 1.41\overline{6}$, $x_3 = 1.414216$ (true 1.414214).

DECISION POINT Newton or bisection? Newton is much faster when f' is available and the guess is close; bisection is slower but cannot fail once a sign change is bracketed. When Newton jumps away or oscillates (flat spots, $f'(x_n) \approx 0$), take a few bisection steps first, then switch.

TERMS

x_n – current estimate of the root
r – the true root; $e_n = x_n - r$ the error
a – number whose square root is sought
Convergence – quadratic near a simple root: correct digits roughly double each step
Failure – $f'(x_n) \approx 0$, bad starting point, or oscillation
Optimization version – apply to f': O.13 §14 Optimization



CONTEXT

- **Origin.** Newton (1669), Raphson (1690); Simpson gave the calculus form (1740).
- **Use.** Calculators computing square roots, solving equations with no closed form.
- **Data science.** Solving for an internal rate of return or implied volatility FN.8 §17 Finance; the core of Newton/IRLS fitting in logistic regression ML.9 §15 Machine Learning.
- **Physical application.** Engineers solve the Colebrook equation for pipe friction P.21 §12 Physics with Newton's method because it has no algebraic solution.

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08
Calculus II

Integration techniques, applications of the integral, infinite series, parametric and polar curves, and first differential equations. Calculus II is where approximation (Taylor series) and accumulation meet the models used every day in statistics and machine learning.

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Integration Techniques

C2.1 Integration by Parts

Integration by parts trades the integral of a product for a (hopefully simpler) integral by moving the derivative from one factor to the other. It is the product rule C1.12 §07 Calculus I read backwards.

Integration by parts	$\int u\,dv = uv - \int v\,du$	Differentiate u, integrate dv; worthwhile when $\int v\,du$ is simpler than the original.
Definite form	$\int_a^b u\,dv = \Big[uv\Big]_a^b - \int_a^b v\,du$	Evaluate the boundary term uv at both limits; no back-substitution needed.
Tabular method	$\int p\,g\,dx = p\,G_1 - p'G_2 + p''G_3 - \cdots$	Differentiate the polynomial p down to 0, integrate g repeatedly (G_k), alternate signs along the diagonals.
Cyclic (solve-for-I) case	$I = \int e^x \sin x\,dx = \frac{1}{2}e^x(\sin x - \cos x) + C$	After two rounds the original integral I reappears: move it to the left side and solve for I.
Lone log / inverse trig	$\int \ln x\,dx = x\ln x - x + C$	Take $u = \ln x$, $dv = dx$ (the "hidden 1"); the same trick integrates $\tan^{-1} x$.
Reduction formula	$\int x^n e^x\,dx = x^n e^x - n\int x^{n-1}e^x\,dx$	Each application lowers the power by one until it reaches $\int e^x\,dx$.

STATED-EQUATION *The integral of u d v equals u v minus the integral of v d u. For definite integrals: u v evaluated from a to b, minus the integral from a to b of v d u. Tabular method: differentiate the polynomial p down to zero while integrating g repeatedly, giving p G one minus p prime G two plus p double prime G three, and so on, with alternating signs. The integral of e to the x sine x equals one half e to the x times sine x minus cosine x, plus C. The integral of natural log x is x natural log x minus x, plus C. The integral of x to the n e to the x equals x to the n e to the x minus n times the integral of x to the n minus one e to the x.*

MEMORY AID LIATE picks u: Log, Inverse trig, Algebraic, Trig, Exponential. "Ultraviolet voodoo": $uv - \int v\,du$.

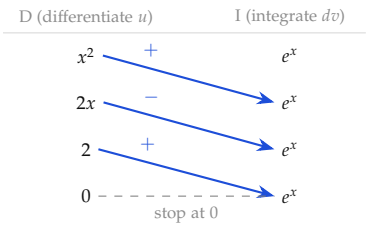
METHOD

1. Choose u by LIATE; dv is everything else, including dx. Compute $du = u'\,dx$ and $v = \int dv$.
2. Write $uv - \int v\,du$; repeat (or go tabular) if the new integral is still a product.
3. Example: $\int xe^x\,dx$: $u = x$, $dv = e^x\,dx \Rightarrow xe^x - \int e^x\,dx = e^x(x-1) + C$.

DECISION POINT Which factor is u? Pick the one that gets *simpler* when differentiated (logs and inverse trig turn algebraic; polynomials drop a degree) and let dv be something you can integrate. LIATE is a heuristic: if $\int v\,du$ looks worse, swap the roles. Two "eternal" factors (e^x, sin/cos) mean the cyclic case. For $f(g(x))\,g'(x)$, u-substitution C1.34 §07 Calculus I beats parts.

TERMS

u – part to differentiate; chosen by LIATE
dv – part to integrate C1.29 §07 Calculus I
du, v – $du = u'\,dx$, $v = \int dv$
Reverse of – the product rule C1.12 §07 Calculus I
Tabular method – repeated parts for a polynomial p; G_k is the kth antiderivative of g; C constant
p, G_k – a polynomial; the kth repeated antiderivative of g
I – the unknown integral, solved for algebraically
n – a positive integer (reduction formula)



$$\int x^2e^x\,dx = x^2e^x - 2xe^x + 2e^x + C$$

CONTEXT

- **Origin.** Brook Taylor (1715), from integrating the product rule.
- **Use.** Integrals of products such as xe^x, $x\sin x$, $\ln x$.
- **Data science.** Moments of distributions: $E[X] = \int_0^\infty x\,\lambda e^{-\lambda x}\,dx = \frac{1}{\lambda}$ for the exponential S.19 §13 Statistics, computed by parts.
- **Physical application.** The work done by a force that varies like $F = xe^{-x}$ along a path is integrated by parts.

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C2.2 Trigonometric Integrals

Integrals of powers and products of sin, cos, tan and sec are reduced, with a trig identity, either to a u-substitution or to lower powers.

Power reduction for $\sin^2$	$\sin^2 x = \dfrac{1-\cos 2x}{2}$	Use for even powers of sine: halves the power, doubles the angle T.9 §06 Trigonometry.
Power reduction for $\cos^2$	$\cos^2 x = \dfrac{1+\cos 2x}{2}$	Use for even powers of cosine; $\int_0^{2\pi} \cos^2 x\,dx = \pi$.
Odd power	$\displaystyle\int \sin^3 x\,dx = \int (1-\cos^2 x)\sin x\,dx$	Save one factor, convert the rest with $\sin^2+\cos^2=1$ T.6 §06 Trigonometry, then $u=\cos x$.
Tangent integral	$\displaystyle\int \tan x\,dx = \ln\lvert\sec x\rvert + C$	Write $\tan x = \frac{\sin x}{\cos x}$ and let $u=\cos x$.
Secant integral	$\displaystyle\int \sec x\,dx = \ln\lvert\sec x+\tan x\rvert + C$	For $\tan^m x \sec^n x$ save $\sec^2 x$ ($u=\tan x$) or $\sec x\tan x$ ($u=\sec x$).
Arctangent form	$\displaystyle\int \frac{dx}{a^2+x^2} = \frac{1}{a}\tan^{-1}\frac{x}{a} + C$	The integral behind every irreducible quadratic denominator C2.4.
Arcsine form	$\displaystyle\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\frac{x}{a} + C$	Valid for $\lvert x\rvert < a$; inverse trig as in T.14 §06 Trigonometry.
Product to sum	$\sin A\cos B = \frac{1}{2}\big[\sin(A-B)+\sin(A+B)\big]$	For $\sin mx\cos nx$ with different frequencies; gives the orthogonality behind Fourier series EE.26 §16 Electrical Eng..

STATED-EQUATION *Sine squared x equals one minus cosine two x, over two; cosine squared x equals one plus cosine two x, over two. The integral of d x over a squared plus x squared equals one over a arctangent of x over a, plus C. The integral of d x over the square root of a squared minus x squared equals arcsine of x over a, plus C. The integral of tangent is the log of the absolute secant; the integral of secant is the log of the absolute value of secant plus tangent. For an odd power of sine, save one sine and convert the rest with sine squared equals one minus cosine squared. Sine A cosine B equals one half of sine of A minus B plus sine of A plus B.*

MEMORY AID Odd power: "save one, convert the rest." Even powers: "reduce the power."

METHOD

1. For $\int \sin^m x\cos^n x\,dx$: if n is odd, save one $\cos x$, convert the other cosines to $1-\sin^2 x$, let $u=\sin x$.
2. If m is odd, save one $\sin x$, convert to $1-\cos^2 x$, let $u=\cos x$.
3. If both are even, apply power reduction (repeatedly) and integrate the cosines of multiple angles.
4. Example: $\int \sin^2 x\,dx = \int \frac{1-\cos 2x}{2}\,dx = \frac{x}{2} - \frac{\sin 2x}{4} + C$.

TERMS

m, n – non-negative integer powers
Power reduction – from double-angle identities T.9 §06 Trigonometry
Pythagorean identity – $\sin^2 x+\cos^2 x = 1$, $1+\tan^2 x = \sec^2 x$ T.6 §06 Trigonometry
u-sub – substitution after saving one factor C1.34 §07 Calculus I
$\tan^{-1}, \sin^{-1}$ – inverse trig T.14 §06 Trigonometry
a – positive constant
A, B – angles in the product-to-sum identity T.11 §06 Trigonometry

CONTEXT

- **Origin.** Developed by Euler and contemporaries in the 1700s.
- **Use.** Areas of circles and ellipses, arc lengths, physics of oscillation.
- **Data science.** The Cauchy distribution integrates to 1 because $\int_{-\infty}^{\infty} \frac{dx}{\pi(1+x^2)} = \frac{1}{\pi}\left[\tan^{-1} x\right]_{-\infty}^{\infty} = 1$; its heavy tails make the mean undefined.
- **Physical application.** The average power of an AC signal uses $\overline{\sin^2} = \frac{1}{2}$, which is why $V_{\text{rms}} = V_{\text{peak}}/\sqrt{2}$ EE.13 §16 Electrical Eng..

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C2.3 Trigonometric Substitution

A square root of a sum or difference of squares is removed by substituting a trig function, so that a Pythagorean identity turns the radicand into a single perfect square.

Sine substitution	$\sqrt{a^2-x^2}: \quad x=a\sin\theta \;\Rightarrow\; \sqrt{a^2-x^2}=a\cos\theta$	$dx=a\cos\theta\,d\theta$, $-\frac{\pi}{2}\le\theta\le\frac{\pi}{2}$; uses $1-\sin^2=\cos^2$.
Tangent substitution	$\sqrt{a^2+x^2}: \quad x=a\tan\theta \;\Rightarrow\; \sqrt{a^2+x^2}=a\sec\theta$	$dx=a\sec^2\theta\,d\theta$; uses $1+\tan^2=\sec^2$.
Secant substitution	$\sqrt{x^2-a^2}: \quad x=a\sec\theta \;\Rightarrow\; \sqrt{x^2-a^2}=a\tan\theta$	$dx=a\sec\theta\tan\theta\,d\theta$; uses $\sec^2-1=\tan^2$.
Back-substitution	$\sin\theta=\dfrac{x}{a} \;\Rightarrow\; \cos\theta=\dfrac{\sqrt{a^2-x^2}}{a}$	Read every other ratio off the reference triangle T.1 §06 Trigonometry to return to x.
Complete the square first	$\sqrt{x^2+2x+5}=\sqrt{(x+1)^2+4}$	Turns a general quadratic into one of the three forms with $u=x+1$, $a=2$ A1.16 §03 Algebra I.

STATED-EQUATION *For root a squared minus x squared use x equals a sine theta; root a squared plus x squared, x equals a tangent theta; root x squared minus a squared, x equals a secant theta. Convert back to x with the reference triangle. If sine theta equals x over a, then cosine theta equals the square root of a squared minus x squared, over a. Complete the square first: the square root of x squared plus two x plus five equals the square root of x plus one squared plus four.*

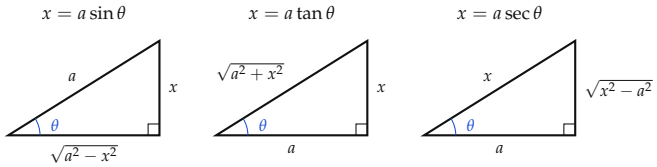
MEMORY AID Match the root to the identity: $a^2-x^2\to\sin$, $a^2+x^2\to\tan$, $x^2-a^2\to\sec$.

METHOD

1. Identify the form (complete the square if there is a linear term).
2. Substitute x and dx; the identity collapses the root to a single trig function.
3. Integrate in θ (trig integrals C2.2), then convert back with the triangle (or change the limits).
4. Example: $\int_0^1\sqrt{1-x^2}\,dx$, $x=\sin\theta$: $\int_0^{\pi/2}\cos^2\theta\,d\theta=\frac{\pi}{4}$, a quarter of the unit circle.

DECISION POINT Look for a shortcut first: if x multiplies the root ($\int x\sqrt{a^2-x^2}\,dx$), the plain substitution $u=a^2-x^2$ is faster. Use trig substitution only when a root (or a power such as $(a^2+x^2)^{3/2}$) survives that check. The sign between the squares decides the function, as in the table and triangles below.

Radicand	Substitute	Root becomes
a^2-x^2	$x=a\sin\theta$	$a\cos\theta$
a^2+x^2	$x=a\tan\theta$	$a\sec\theta$
x^2-a^2	$x=a\sec\theta$	$a\tan\theta$



TERMS

a – positive constant (the length that is squared)
θ – new variable; the angle of the reference triangle
Pythagorean identities – remove the root T.6 §06 Trigonometry
Reference triangle – SOH-CAH-TOA picture used to convert back T.1 §06 Trigonometry
$\sin^{-1},\tan^{-1},\sec^{-1}$ – inverse trig, to write θ in terms of x T.14 §06 Trigonometry

CONTEXT

- **Origin.** Developed by Euler and contemporaries in the 1700s.
- **Use.** Areas of circles and ellipses, arc lengths, electric fields of line charges.
- **Data science.** The arcsine and arctangent forms produce the closed-form CDFs of the arcsine and Cauchy distributions.
- **Physical application.** The area of an ellipse, $\frac{4b}{a}\int_0^a\sqrt{a^2-x^2}\,dx=\pi ab$, needs a sine substitution.

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C2.4 Partial Fractions

A proper rational function splits into a sum of simple fractions, one per factor of its denominator, and each simple fraction integrates to a logarithm, a power, or an arctangent.

Distinct linear factors	$\dfrac{px+q}{(x-a)(x-b)} = \dfrac{A}{x-a} + \dfrac{B}{x-b}$	One constant over each different linear factor.
Repeated linear factor	$\dfrac{r(x)}{(x-a)^2} = \dfrac{A}{x-a} + \dfrac{B}{(x-a)^2}$	One term for every power up to the multiplicity.
Irreducible quadratic factor	$\dfrac{r(x)}{x^2+c} = \dfrac{Ax+B}{x^2+c} \quad (c>0)$	Linear numerator over a quadratic with no real roots.
Log integral	$\displaystyle\int \frac{dx}{x-a} = \ln\lvert x-a\rvert + C$	Every distinct linear piece integrates to a log.
Power integral	$\displaystyle\int \frac{dx}{(x-a)^k} = \frac{-1}{(k-1)(x-a)^{k-1}} + C \quad (k \geq 2)$	Repeated pieces integrate by the power rule.
Heaviside cover-up	$A = \left.\dfrac{px+q}{x-b}\right\rvert_{x=a}$	For a simple linear factor: cover up $(x-a)$ and substitute $x=a$.

STATED-EQUATION *p x plus q over x minus a times x minus b equals A over x minus a plus B over x minus b. A repeated factor x minus a squared gives A over x minus a plus B over x minus a squared. An irreducible quadratic factor gets a numerator A x plus B. The integral of d x over x minus a equals natural log of the absolute value of x minus a, plus C. The integral of d x over x minus a to the k equals minus one over k minus one times x minus a to the k minus one, plus C, for k at least two. Cover-up: A equals p x plus q over x minus b, evaluated at x equals a.*

MEMORY AID "One fraction per factor; each numerator one degree below its denominator piece."

METHOD

1. If $\deg(\text{numerator}) \geq \deg(\text{denominator})$, long-divide first A2.8 §04 Algebra II.
2. Factor the denominator completely A1.14 §03 Algebra I and write the template, case by case.
3. Clear denominators; find constants by substituting roots (cover-up) and by matching coefficients.
4. Integrate each piece. Example: $\frac{1}{x(x+1)} = \frac{1}{x} - \frac{1}{x+1}$, so $\int = \ln\left\lvert\frac{x}{x+1}\right\rvert + C$.

DECISION POINT Which template? Read each factor of the denominator: a linear factor once $\to \frac{A}{x-a}$; a linear factor k times $\to k$ terms $\frac{A_1}{x-a} + \cdots + \frac{A_k}{(x-a)^k}$; a quadratic with negative discriminant A1.15 §03 Algebra I $\to \frac{Ax+B}{\text{quadratic}}$ (one per power if repeated). Substituting roots is fastest for linear factors; match coefficients for the rest.

TERMS

Requirement – degree of numerator $<$ degree of denominator (proper fraction)
Factors – from factoring the denominator A1.14 §03 Algebra I
A, B – constants found by clearing denominators and plugging in roots
$r(x)$ – numerator polynomial; k multiplicity
Irreducible quadratic – numerator $Ax+B$; integrates to ln and $\tan^{-1}$ C2.2
p, q – constants in a linear numerator
a, b; k – roots of the linear factors; power of a repeated factor ($k \geq 2$)
c – positive constant, so x^2+c has no real roots

CONTEXT

- **Origin.** Johann Bernoulli and Leibniz (1702).
- **Use.** Integrating rational functions; inverse Laplace transforms C2.33 in engineering.
- **Data science.** Solving the logistic equation C2.28: $\int \frac{dy}{y(1-y)} = \ln\left\lvert\frac{y}{1-y}\right\rvert + C$, the *logit*, which is exactly the link function of logistic regression ML.9 §15 Machine Learning.
- **Physical application.** Solving for the concentration in a two-step chemical reaction leads to partial-fraction integrals.

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C2.5 Numerical Integration: Trapezoid & Simpson's Rules

When no antiderivative is available, or f is only known at sample points, replace f on each small piece by a line or a parabola and integrate that exactly.

Step size	$h = \dfrac{b-a}{n}, \qquad f_i = f(a+ih)$	Split $[a,b]$ into n equal pieces; f_i are the sampled values.
Midpoint rule	$M_n = h\big[f(\bar{x}_1) + f(\bar{x}_2) + \cdots + f(\bar{x}_n)\big]$	Rectangles at the midpoints; error about half the trapezoid error, with the opposite sign.
Trapezoid rule	$T_n = \dfrac{h}{2}\Big[f_0 + 2f_1 + 2f_2 + \cdots + 2f_{n-1} + f_n\Big]$	Straight lines between neighboring points; ends weighted 1, interior 2.
Simpson's rule	$S_n = \dfrac{h}{3}\Big[f_0 + 4f_1 + 2f_2 + 4f_3 + \cdots + 4f_{n-1} + f_n\Big] \quad (n \text{ even})$	Parabolas through each triple of points; exact for cubics.
Error bounds	$\lvert E_T\rvert \le \dfrac{K_2(b-a)^3}{12n^2}, \qquad \lvert E_S\rvert \le \dfrac{K_4(b-a)^5}{180n^4}$	Doubling n cuts the trapezoid error by 4 and Simpson's by 16.
Simpson as a blend	$S_{2n} = \dfrac{2M_n + T_n}{3}$	Weighted average that cancels the leading errors of M_n and T_n.

STATED-EQUATION *h equals b minus a over n. The trapezoid rule: T n equals h over two times the quantity f naught plus two f one plus two f two, and so on, plus two f n minus one plus f n. Simpson's rule: S n equals h over three times f naught plus four f one plus two f two plus four f three, alternating, plus four f n minus one plus f n, for even n. The trapezoid error shrinks like one over n squared, Simpson's like one over n to the fourth. The midpoint rule: M n equals h times the sum of f at the midpoints of the subintervals. The trapezoid error is at most K two times b minus a cubed, over twelve n squared; Simpson's error is at most K four times b minus a to the fifth, over one hundred eighty n to the fourth. Simpson's S two n equals two M n plus T n, over three.*

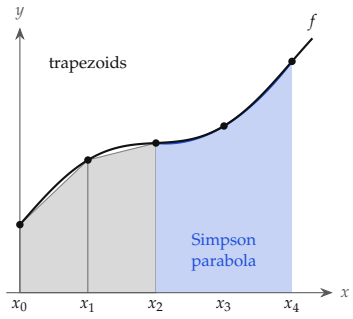
MEMORY AID Simpson's weights: 1, 4, 2, 4, 2, …, 4, 1. Trapezoid: 1, 2, 2, …, 2, 1.

METHOD

1. Choose n (even for Simpson) and compute $h = \frac{b-a}{n}$.
2. Tabulate $f_0, \ldots, f_n$ and apply the weights.
3. Example: $\int_0^2 x^2\,dx$ with $n = 2$, $h = 1$, $f = 0, 1, 4$: $T_2 = \frac{1}{2}(0+2+4) = 3$; $S_2 = \frac{1}{3}(0+4+4) = \frac{8}{3}$ (exact).

TERMS

f_i – $f(a+ih)$, values at equally spaced points; $\bar{x}_i$ midpoints
Trapezoid – straight lines between points (compare G.15 §05 Geometry, C1.30 §07 Calculus I); error $O(h^2)$
Simpson – parabolas through triples of points A1.16 §03 Algebra I; error $O(h^4)$
K_2, K_4 – bounds on $|f''|$ and $|f^{(4)}|$ on $[a,b]$
h – spacing; n number of subintervals
$\bar{x}_i$ – midpoint of the ith subinterval
M_n, T_n, S_n – midpoint, trapezoid and Simpson estimates with n subintervals
E_T, E_S – errors of the trapezoid and Simpson estimates



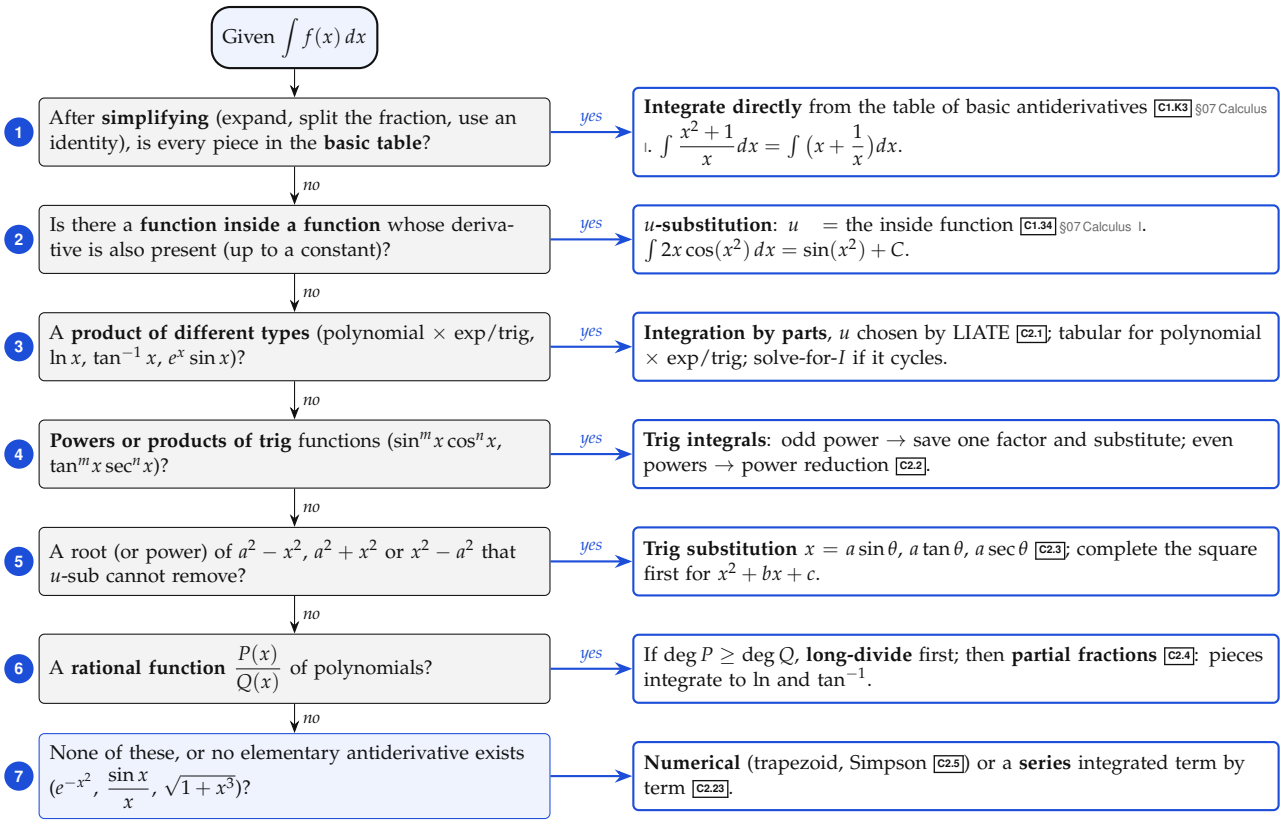
CONTEXT

- **Origin.** Trapezoid rule known to Babylonian astronomers; Simpson's rule by Thomas Simpson (1743), earlier by Kepler.
- **Use.** Integrating tabulated or measured data.
- **Data science.** `numpy.trapz` computes ROC AUC and cumulative metrics from sampled curves ML.18 §15 Machine Learning; Monte Carlo O.27 §14 Optimization takes over in high dimensions.
- **Physical application.** Estimating the total distance of a run from speed readings every 10 seconds.

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Integration Techniques: Which Method? | *Key Topic C2.K1*

Differentiation is mechanical; integration is pattern recognition. Work down the list: each question is cheaper to check than the next, and most integrals are settled by the first three. Simplify first (expand, split fractions, use identities), because a small algebraic rewrite often turns a "hard" integral into a table entry.



Recognize the pattern

Integrand looks like	Method	Mini-example
x^n, e^x, $\sin x$, $\frac{1}{x}$, $\frac{1}{1+x^2}$	Table C1.K3 §07 Calculus I	$\int(3x^2+e^x)\,dx = x^3+e^x+C$
$f(g(x))\,g'(x)$	u-substitution C1.34 §07 Calculus I	$\int \frac{2x}{x^2+1}\,dx = \ln(x^2+1)+C \quad (u=x^2+1)$
$x^n e^x$, $x\sin x$, $\ln x$, $\tan^{-1}x$	By parts C2.1	$\int x\cos x\,dx = x\sin x+\cos x+C$
$\sin^m x\cos^n x$, $\tan^m x\sec^n x$	Trig integrals C2.2	$\int \sin^3 x\,dx = -\cos x+\frac{\cos^3 x}{3}+C$
$\sqrt{a^2-x^2}$, $\sqrt{a^2+x^2}$, $\sqrt{x^2-a^2}$	Trig substitution C2.3	$\int \frac{dx}{\sqrt{4-x^2}} = \sin^{-1}\frac{x}{2}+C$
$\frac{P(x)}{Q(x)}$, Q factors	Partial fractions C2.4	$\int \frac{dx}{x^2-1} = \frac{1}{2}\ln\left\lvert\frac{x-1}{x+1}\right\rvert+C$
e^{-x^2}, $\frac{\sin x}{x}$, data points	Numerical / series C2.5	$\int_0^1 e^{-x^2}dx \approx 0.7468$ (Simpson, $n=4$: 0.7469)

Decision points

DECISION POINT **Parts vs. substitution:** if one factor is (a constant times) the derivative of something *inside* the other factor, substitute; if the factors are unrelated types, use parts. **Trig sub vs. u-sub:** $\int x\sqrt{1-x^2}\,dx$ is a u-sub ($u=1-x^2$); $\int \sqrt{1-x^2}\,dx$ needs $x=\sin\theta$. **Partial fractions vs. arctangent:** a denominator with negative discriminant does not factor; complete the square and use $\tan^{-1}$. **Definite integrals:** change the limits with the substitution and never return to x.

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Worked examples, one per method

1. u-substitution C1.34 §07 Calculus I

$\int xe^{x^2}\,dx$: the inside is x^2 and its derivative $2x$ is present. $u = x^2$, $du = 2x\,dx$: $\frac{1}{2}\int e^u\,du = \frac{1}{2}e^{x^2} + C$.

2. Integration by parts C2.1

$\int x\ln x\,dx$: LIATE puts the log first, $u = \ln x$, $dv = x\,dx$, $v = \frac{x^2}{2}$: $\frac{x^2}{2}\ln x - \int \frac{x}{2}\,dx = \frac{x^2}{2}\ln x - \frac{x^2}{4} + C$.

3. Trig integral C2.2

$\int \sin^2 x\cos^3 x\,dx$: cosine has the odd power, so save one $\cos x$: $\int \sin^2 x(1-\sin^2 x)\cos x\,dx$, $u = \sin x$: $\frac{\sin^3 x}{3} - \frac{\sin^5 x}{5} + C$.

4. Trig substitution C2.3

$\int \frac{dx}{(4+x^2)^{3/2}}$: $x = 2\tan\theta$, $dx = 2\sec^2\theta\,d\theta$, $(4+x^2)^{3/2} = 8\sec^3\theta$: $\frac{1}{4}\int \cos\theta\,d\theta = \frac{1}{4}\sin\theta = \frac{x}{4\sqrt{4+x^2}} + C$.

5. Partial fractions C2.4

$\int \frac{x+5}{(x-1)(x+2)}dx$: cover-up gives $A = \frac{1+5}{1+2} = 2$, $B = \frac{-2+5}{-2-1} = -1$, so the integral is $2\ln|x-1| - \ln|x+2| + C$.

6. Definite integral: change the limits

$\int_0^1 x(1+x^2)^3 dx$, $u = 1+x^2$ runs from 1 to 2: $\frac{1}{2}\int_1^2 u^3\,du = \frac{1}{2}\cdot\frac{16-1}{4} = \frac{15}{8}$. No return to x needed.

Common mistakes

Lost differentials. Forgetting dx when substituting ($du = 2x\,dx$, not $du = 2x$), or substituting u but leaving some x behind.

Illegal shortcuts. Integrating a quotient as a quotient of integrals; writing $\int \frac{dx}{x^2+1} = \ln(x^2+1)$ (that needs $2x$ on top); dropping the absolute value in $\ln|x|$.

Skipped division. Splitting $\frac{x^3}{x^2-1}$ into partial fractions without long-dividing first.

Always check by differentiating the answer: it must give back the integrand. A definite integral across a vertical asymptote is improper C2.6: do not just plug in the limits.

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C2.6 Improper Integrals

An integral over an infinite interval, or of a function that blows up, is defined as a limit of ordinary integrals. It converges when that limit is finite.

Type 1: infinite interval	$\displaystyle\int_a^\infty f(x)\,dx = \lim_{b\to\infty}\int_a^b f(x)\,dx$	Integrate to a finite b, then let $b\to\infty$.
Type 2: unbounded integrand	$\displaystyle\int_a^b f(x)\,dx = \lim_{t\to a^+}\int_t^b f(x)\,dx$	For f with a vertical asymptote at a; at an interior point split there.
Doubly infinite	$\displaystyle\int_{-\infty}^\infty f\,dx = \int_{-\infty}^c f\,dx + \int_c^\infty f\,dx$	Each half must converge separately (not $\lim_{R\to\infty}\int_{-R}^R$).
p-test at infinity	$\displaystyle\int_1^\infty \frac{dx}{x^p} = \begin{cases}\dfrac{1}{p-1}, & p>1\\ \text{diverges}, & p\le 1\end{cases}$	The benchmark for tails: decay faster than $\frac{1}{x}$ to converge.
p-test at zero	$\displaystyle\int_0^1 \frac{dx}{x^p}$ converges $\iff p<1$	The reverse condition near a singularity at 0.
Exponential tail	$\displaystyle\int_0^\infty e^{-ax}\,dx = \frac{1}{a}\quad (a>0)$	Exponential decay beats every power.
Comparison test	$0\le f\le g:\ \int g<\infty \Rightarrow \int f<\infty$	Smaller than convergent converges; bigger than divergent diverges.
Limit comparison	$\displaystyle\lim\frac{f(x)}{g(x)} = c\in(0,\infty)\Rightarrow$ same behavior	Compare with the dominant term; no inequality needed.

STATED-EQUATION *The integral from a to infinity of f of x d x equals the limit as b goes to infinity of the integral from a to b. If f is unbounded at a, take the limit as t approaches a from the right. The integral from one to infinity of d x over x to the p equals one over p minus one if p is greater than one, and diverges if p is at most one. An integral over the whole line splits at any c into two improper integrals, and both must converge. The integral from zero to one of d x over x to the p converges if and only if p is less than one. The integral from zero to infinity of e to the minus a x equals one over a, for positive a. If zero is at most f, which is at most g, and the integral of g converges, so does the integral of f. If f over g tends to a positive finite limit c, the two integrals converge or diverge together.*

MEMORY AID p-test: "p bigger than one converges" (at infinity); near zero it is the opposite.

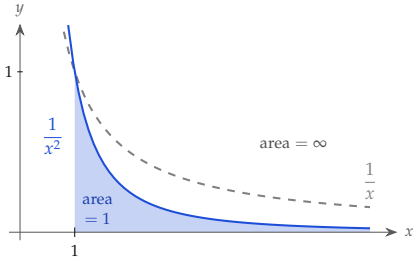
METHOD

1. Locate every problem point (an infinite limit, or where f is unbounded); split the integral so each piece has one.
2. Replace the problem point by a variable, integrate, take the limit C1.1 §07 Calculus I.
3. Example: $\int_1^\infty \frac{dx}{x^2} = \lim_{b\to\infty}\left(1-\frac{1}{b}\right) = 1$.

DECISION POINT Converge or diverge without integrating? Look at the integrand *near the problem point* only and keep its dominant term: $\frac{x+3}{x^3+1}\sim\frac{1}{x^2}$ at ∞ (converges); $\frac{1}{\sqrt{x}+x^2}\sim\frac{1}{\sqrt{x}}$ at 0 (converges). Then compare with a p-integral or e^{-x}.

TERMS

Type 1 – infinite interval of integration
Type 2 – integrand has a vertical asymptote A2.10 §04 Algebra II
Converges – the limit C1.1 §07 Calculus I exists and is finite
p-test – benchmark for comparison; mirrors p-series C2.17
c – any convenient split point
b, t – moving limit of integration, sent to the bad point
f, g – nonnegative integrands being compared
a – lower limit; in e^{-ax}, a positive decay rate



CONTEXT

- **Origin.** Formalized by Cauchy (1823).
- **Use.** Total accumulated quantities over unbounded time or space.
- **Data science.** Every density must satisfy $\int_{-\infty}^\infty f(x)\,dx = 1$ S.20 §13 Statistics; a Pareto tail $\propto x^{-(\alpha+1)}$ has a finite mean only when $\alpha>1$, which is why heavy-tailed revenue data breaks averages.
- **Physical application.** The escape energy from Earth's gravity is $\int_R^\infty \frac{GMm}{r^2}\,dr = \frac{GMm}{R}$, a finite improper integral.

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Applications of the Integral

C2.7 Area Between Curves

The area between two curves is the integral of the gap between them: slice the region into thin strips, take each strip's length times its width, and add.

Vertical slices	$A = \int_a^b \big[f(x) - g(x)\big]\,dx \quad (f \geq g \text{ on } [a,b])$	Top curve minus bottom curve, integrated in x.
Horizontal slices	$A = \int_c^d \big[x_{\text{right}}(y) - x_{\text{left}}(y)\big]\,dy$	Right curve minus left curve, integrated in y.
Crossing curves	$A = \int_a^b \big\lvert f(x) - g(x)\big\rvert\,dx$	Split at every intersection so each piece has a fixed top curve.

STATED-EQUATION *The area between curves equals the integral from a to b of f of x minus g of x, d x, with f above g. With horizontal slices, integrate x right minus x left with respect to y. If the curves cross, integrate the absolute value of the difference, splitting at the crossings.*

MEMORY AID "Top minus bottom" (or "right minus left").

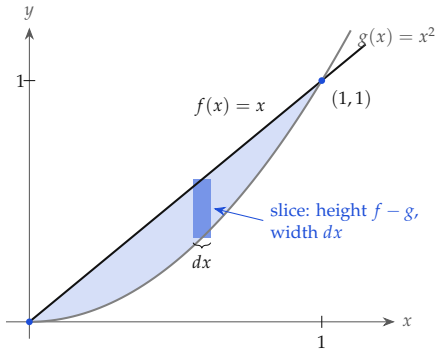
METHOD

1. Sketch the region and solve $f(x) = g(x)$ for the intersection points.
2. Decide which curve is on top on each interval (test one point).
3. Integrate top minus bottom on each interval and add. Example: $y = x$ and $y = x^2$ meet at $0, 1$: $\int_0^1 (x - x^2)\,dx = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}$.

DECISION POINT Slice in x or in y? Choose the direction in which the region has a single top and a single bottom (or right and left) boundary. A region bounded by $x = y^2$ and $y = x - 2$ needs two dx-integrals but only one dy-integral, $\int_{-1}^{2} (y + 2 - y^2)\,dy = \frac{9}{2}$.

TERMS

$f(x)$ – upper curve; $g(x)$ lower curve
a, b – intersection points: solve $f(x) = g(x)$
$x_{\text{right}}, x_{\text{left}}$ – boundaries written as functions of y
Built on – the definite integral C1.32 §07 Calculus I



CONTEXT

- **Origin.** Archimedes' method of exhaustion; integral form from Leibniz.
- **Use.** Consumer surplus, net displacement, profit regions.
- **Data science.** Model lift: the Gini coefficient is twice the area between the ROC curve and the diagonal, Gini $= 2 \cdot \text{AUC} - 1$ ML.18 §15 Machine Learning.
- **Physical application.** The area of a lens or a river cross-section between two bank profiles is an area between curves.

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C2.8 Volumes: Slicing, Disks & Washers

Volume is the integral of cross-sectional area. When a region is spun about an axis, slices perpendicular to the axis are disks, or washers if there is a hole.

Slicing	$V = \int_a^b A(x)\,dx$	Any solid with known cross-sectional area $A(x)$ (Cavalieri's principle).
Disk method	$V = \pi \int_a^b \big[R(x)\big]^2\,dx$	Each slice is a disk of radius $R(x)$, area πR^2.
Washer method	$V = \pi \int_a^b \Big(\big[R(x)\big]^2 - \big[r(x)\big]^2\Big)\,dx$	Outer disk minus inner hole; square *each* radius first.
Other axis	axis $y = k$: $\quad R(x) = \lvert f(x) - k\rvert$	Radii are always distances from the axis of rotation.

STATED-EQUATION *Slicing: V equals the integral of the cross-sectional area A of x d x. Disk method: V equals pi times the integral of R of x squared d x. Washer method: V equals pi times the integral of R squared minus r squared d x. For the axis y equals k, the radius is the absolute value of f of x minus k.*

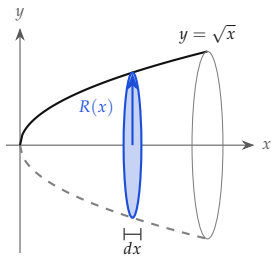
MEMORY AID Washer = "big disk minus small disk" (not "disk of the difference").

METHOD

1. Sketch the region and the axis; draw one slice perpendicular to the axis.
2. Measure the outer radius R (and inner radius r) from the axis to the curves.
3. Integrate $\pi(R^2 - r^2)$ along the axis. Example: $y = \sqrt{x}$, $0 \le x \le 4$, about the x-axis: $\pi \int_0^4 x\,dx = 8\pi$.

TERMS

$R(x), r(x)$ – outer and inner radius measured from the axis
πR^2 – area of a circular slice G.17 §05 Geometry
$A(x)$ – area of the cross-section at x (squares, triangles, …) G.21 §05 Geometry
k – position of a horizontal axis $y = k$
Alternative – cylindrical shells C2.9



axis of rotation: x-axis

CONTEXT

- **Origin.** Cavalieri's principle (1635); Kepler measured wine-barrel volumes this way.
- **Use.** Tanks, machine parts, and any solid with rotational symmetry.
- **Data science.** The same slice-and-sum idea computes probabilities over 2-D regions by integrating a joint density one strip at a time C3.19 §10 Calculus III.
- **Physical application.** Volume of a cup: rotate its side profile $y = R(x)$ about the axis and integrate $\pi R(x)^2$; a cup whose radius grows linearly from 3 to 4 cm over 10 cm holds $\pi \int_0^{10} (3 + 0.1x)^2 dx \approx 387$ mL.

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C2.9 Shell Method

Slice *parallel* to the axis of rotation instead: each thin strip sweeps out a cylindrical shell whose volume is circumference × height × thickness.

Shell method (y-axis)	$V = 2\pi \int_a^b x\, h(x)\, dx$	Vertical strip at distance x from the y-axis, height $h(x)$.
Shell, general axis	$V = 2\pi \int_a^b \rho(x)\, h(x)\, dx$	ρ is the strip's distance to the axis, e.g. $\rho = \lvert x - c\rvert$ for the axis $x = c$.
Shell (x-axis)	$V = 2\pi \int_c^d y\, w(y)\, dy$	Horizontal strips of width $w(y)$ rotated about the x-axis.

STATED-EQUATION *Shell method: V equals two pi times the integral of x times h of x d x. In general, two pi times the integral of radius times height times thickness. About the x-axis, V equals two pi times the integral from c to d of y times w of y, d y.*

MEMORY AID Shell = "circumference × height × thickness": $2\pi\rho \cdot h \cdot dx$.

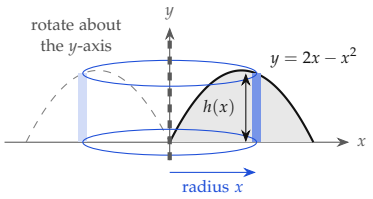
METHOD

1. Draw a strip parallel to the axis; find its distance ρ to the axis and its height h.
2. Integrate $2\pi\rho h$ across the region (perpendicular to the axis).
3. Example: $y = 2x - x^2$, $0 \le x \le 2$, about the y-axis: $2\pi \int_0^2 x(2x - x^2)\, dx = 2\pi\left(\frac{16}{3} - 4\right) = \frac{8\pi}{3}$.

DECISION POINT Disks/washers or shells? Disks slice *perpendicular* to the axis and integrate *along* it; shells slice *parallel* and integrate *across*. Pick the one whose radius and height are each a single formula over the whole interval, in the variable the curve is already written in: $y = f(x)$ spun about the y-axis → shells in x (no need to invert f). If the top or bottom boundary changes partway, the other method often avoids splitting.

TERMS

x, ρ – distance from the strip to the axis (shell radius)
$h(x)$ – height of the strip (top minus bottom curve)
$2\pi x$ – circumference of the shell G.17 §05 Geometry
dx – thickness; $w(y)$ width of a horizontal strip
Alternative – disks and washers C2.6
y, $w(y)$; c, d – radius and width of a horizontal shell; its y-limits



CONTEXT

- **Origin.** Equivalent to slicing by Fubini's theorem C3.19 §10 Calculus III; both methods always give the same volume.
- **Use.** Solids with holes, vases, parts turned on a lathe.
- **Data science.** The same "choose the slicing direction that makes the integrand simplest" decision is the choice of integration order in a double integral of a joint density.
- **Physical application.** A ring made by spinning the disk $(x-3)^2 + y^2 \le 1$ about the y-axis has volume $2\pi \cdot 3 \cdot \pi = 6\pi^2$ (Pappus, C2.14).

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C2.10 Arc Length

The length of a curve is the sum of infinitely many tiny hypotenuses $ds = \sqrt{dx^2 + dy^2}$, written in whatever variable describes the curve.

Arc-length element	$ds = \sqrt{dx^2 + dy^2}$	Pythagoras on an infinitesimal piece of the curve G.7 §05 Geometry.
Graph $y = f(x)$	$L = \int_a^b \sqrt{1 + [f'(x)]^2}\, dx$	Factor dx out of ds; needs f' continuous.
Graph $x = g(y)$	$L = \int_c^d \sqrt{1 + [g'(y)]^2}\, dy$	Same idea with y as the variable.
Parametric curve	$L = \int_\alpha^\beta \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$	Integral of speed: distance traveled along $(x(t), y(t))$ C2.25.
Polar curve	$L = \int_\alpha^\beta \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\, d\theta$	For $r = f(\theta)$ T.19 §06 Trigonometry, C2.26.

STATED-EQUATION *L equals the integral from a to b of the square root of one plus f prime of x squared, d x. For a parametric curve, L equals the integral of the square root of d x d t squared plus d y d t squared, d t. For a polar curve, the square root of r squared plus d r d theta squared, d theta. The arc-length element d s equals the square root of d x squared plus d y squared. For x equals g of y, L equals the integral from c to d of the square root of one plus g prime of y squared, d y.*

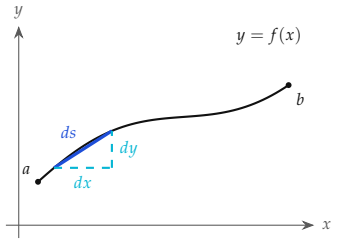
MEMORY AID "Pythagoras, then integrate": $ds = \sqrt{dx^2 + dy^2}$, factor out whichever differential you integrate in.

METHOD

1. Differentiate, square, add 1 (or add the squared derivatives).
2. Simplify under the root (perfect squares are common in textbook problems), then integrate.
3. Example: $y = \frac{2}{3}x^{3/2}$ on $[0, 3]$: $f' = \sqrt{x}$, $L = \int_0^3 \sqrt{1 + x}\, dx = \frac{2}{3}(8 - 1) = \frac{14}{3}$.

TERMS

$f'(x)$ – derivative C1.9 §07 Calculus I
Derivation – sum of tiny hypotenuses G.7 §05 Geometry, G.12 §05 Geometry
Parametric – $x(t), y(t)$ C2.25
Polar – $r = f(\theta)$ T.19 §06 Trigonometry
α, β – parameter limits
ds, L – arc-length element; total arc length
$g(y)$; c, d – curve written as a function of y; its y-limits



$ds = \sqrt{dx^2 + dy^2} = \sqrt{1 + (f')^2}\, dx$

CONTEXT

- **Origin.** First rectifications of curves by Neile and van Heuraet (1659).
- **Use.** Cable and road lengths, distance traveled along a path.
- **Data science.** Distance along a GPS trace, or the total path length of an optimizer's trajectory, is a discrete version of this integral.
- **Physical application.** The length of a sagging power cable (a catenary) between two towers is found with the arc-length integral.

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C2.11 Surface Area of Revolution

Spinning a curve about an axis sweeps a surface. Each small piece of curve of length ds traces a thin band of area $2\pi\rho\,ds$, where ρ is its distance to the axis.

About the x-axis	$S = 2\pi \int_a^b f(x)\sqrt{1+\left[f'(x)\right]^2}\,dx$	Radius $f(x)$ times arc-length element ($f \geq 0$).
About the y-axis	$S = 2\pi \int_a^b x\sqrt{1+\left[f'(x)\right]^2}\,dx$	Radius is now the distance x to the y-axis.
Parametric curve	$S = 2\pi \int_\alpha^\beta y(t)\sqrt{x'(t)^2+y'(t)^2}\,dt$	About the x-axis, for $y(t) \geq 0$.
General form	$S = \int 2\pi\rho\,ds$	Circumference of the band $\times$ slant length; never use dx in place of ds.

STATED-EQUATION *Surface area about the x-axis equals two pi times the integral of f of x times the square root of one plus f prime squared, d x. In general, S equals the integral of two pi rho d s. About the y-axis, S equals two pi times the integral of x times the square root of one plus f prime squared, d x. For a parametric curve, S equals two pi times the integral of y of t times the square root of x prime squared plus y prime squared, d t.*

MEMORY AID "Circumference $\times$ slant length": $2\pi\rho\,ds$.

METHOD

1. Find the radius ρ (distance to the axis) and ds C2.10.
2. Integrate $2\pi\rho\,ds$ over the curve.
3. Example: $f(x) = \sqrt{4-x^2}$ on $[-2,2]$: $f\sqrt{1+f'^2} = 2$, so $S = 2\pi\int_{-2}^{2} 2\,dx = 16\pi = 4\pi(2)^2$, the sphere.

TERMS

ρ – distance from the curve to the axis
ds – arc-length element C2.10
$f(x)$ – generating curve, $f \geq 0$
Band – frustum of a cone, lateral area $2\pi\rho\,\ell$ G.22 §05 Geometry
$x(t), y(t)$; α, β – parametric curve; parameter limits C2.25

CONTEXT

- **Origin.** Archimedes found the sphere's area $4\pi r^2$ (c. 225 BCE).
- **Use.** Paint, sheet metal, heat-loss areas of rotationally symmetric parts.
- **Data science.** Surface integrals V.11 §11 Vector Calculus generalize $\int \rho\,ds$; the surface area of a high-dimensional sphere explains why random vectors concentrate near its "equator".
- **Physical application.** Material needed for a parabolic satellite dish $y = x^2$, $0 \leq x \leq 1$, spun about the y-axis: $2\pi\int_0^1 x\sqrt{1+4x^2}\,dx \approx 5.33$.

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C2.12 Average Value of a Function

The average of a function over an interval is its total accumulation divided by the length of the interval: the height of the rectangle with the same area.

Average value	$f_{\text{avg}} = \dfrac{1}{b-a}\displaystyle\int_a^b f(x)\,dx$	Continuous version of "sum divided by count".
Mean Value Theorem for Integrals	$\exists\, c \in [a,b] :\ f(c) = f_{\text{avg}}$	A continuous f actually attains its average somewhere.
Root mean square	$f_{\text{rms}} = \sqrt{\dfrac{1}{b-a}\displaystyle\int_a^b f(x)^2\,dx}$	Average size regardless of sign; used for AC signals EE.13 §16 Electrical Eng..
Weighted average	$\bar{f}_w = \dfrac{\int_a^b f(x)\,w(x)\,dx}{\int_a^b w(x)\,dx}$	Weights $w \geq 0$; with a density as weight this is $E[f(X)]$.

STATED-EQUATION *f average equals one over b minus a times the integral from a to b of f of x d x. Mean value theorem for integrals: there is a c in the interval with f of c equal to the average. The root mean square is the square root of the average of f squared. The weighted average is the integral of f times w divided by the integral of w.*

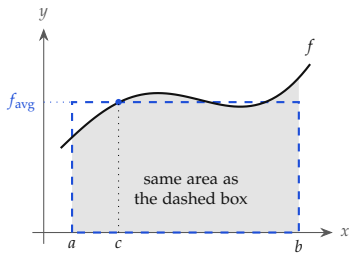
MEMORY AID "Average = total over length."

METHOD

1. Integrate f over $[a,b]$; divide by $b-a$.
2. Example: x^2 on $[0,3]$: $\frac{1}{3}\int_0^3 x^2\,dx = \frac{1}{3}\cdot 9 = 3$, attained at $c = \sqrt{3}$.

TERMS

$b-a$ – length of the interval
$\int_a^b f$ – total accumulation C1.32 §07 Calculus I
Discrete analog – the arithmetic mean S.1 §13 Statistics
MVT – requires f continuous C1.7 §07 Calculus I; compare C1.27 §07 Calculus I
$w(x)$ – non-negative weight function
$f_{\text{avg}}, f_{\text{rms}}$ – average and root-mean-square values
c – a point where f equals its average



CONTEXT

- **Origin.** A consequence of the Fundamental Theorem of Calculus.
- **Use.** Average temperature, average velocity, average power.
- **Data science.** If $X \sim \text{Uniform}(a,b)$ then $E[f(X)] = \frac{1}{b-a}\int_a^b f(x)\,dx$ S.19 §13 Statistics; Monte Carlo integration O.27 §14 Optimization estimates it by averaging f at random points.
- **Physical application.** The average temperature over a day is the integral of the temperature curve divided by 24 hours.

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C2.13 Work, Pumping & Hydrostatic Force

When a force, a distance, or a pressure varies, chop the problem into pieces on which it is constant, multiply, and integrate.

Work by a variable force	$W = \int_a^b F(x)\,dx$	Force times distance, added along the path P.6 §12 Physics.
Spring (Hooke's law)	$W = \int_0^d kx\,dx = \frac{1}{2}kd^2$	Stretching a spring a distance d from rest.
Pumping a liquid	$W = \int_a^b \rho g\, A(y)\, D(y)\,dy$	Weight of each thin slab times the distance D it is lifted.
Hydrostatic pressure	$P = \rho g h$	Pressure at depth h acts equally in all directions P.19 §12 Physics.
Hydrostatic force	$F = \int_a^b \rho g\, h(y)\, w(y)\,dy$	Pressure at each depth times the strip's area $w\,dy$.

STATED-EQUATION *Work equals the integral of force over distance; a spring needs one half k d squared. Pumping work is the integral of rho g times the slab area times the lift distance. Hydrostatic pressure is rho g h, and the force on a plate is the integral of rho g h times the strip width.*

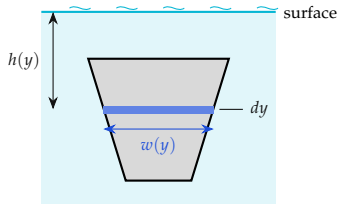
MEMORY AID "Slab, weight, distance" for pumping; "depth times width" for walls.

METHOD

1. Choose a coordinate (often depth y) and draw one thin slab or strip.
2. Write its force (weight $\rho g A\,dy$, or pressure $\times$ area) and the distance it moves (or its depth).
3. Integrate over all slabs. Example: a spring with $k = 10\,\text{N/m}$ stretched $2\,\text{m}$: $W = \frac{1}{2}\cdot 10\cdot 2^2 = 20\,\text{J}$.

TERMS

$F(x)$ – force at position x; k spring constant
ρ – density (water $1000\,\text{kg/m}^3$); $g = 9.8\,\text{m/s}^2$
$A(y)$ – cross-sectional area of the slab at height y
$D(y)$ – distance that slab is lifted
$h(y)$, $w(y)$ – depth and width of a strip on the plate
k, d – spring constant; stretch distance P.5 §12 Physics
g, P – gravitational acceleration; pressure



strip force $\approx \rho g\, h\, w\, dy$

CONTEXT

- **Origin.** Pascal's hydrostatic law (1653); Hooke's law (1678).
- **Use.** Dams, tanks, springs, lifting cables and chains.
- **Data science.** Total cost $= \int (\text{marginal cost})\,dq$ is the same "rate times small amount" integral; so is energy used by a GPU job, $\int P(t)\,dt$.
- **Physical application.** The force of water on a dam face grows with depth, which is why dams are thicker at the bottom.

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C2.14 Moments & Center of Mass

The center of mass is the balance point: the mass-weighted average position. For a uniform flat plate it is the geometric centroid.

Discrete masses	$\bar{x} = \dfrac{\sum m_i x_i}{\sum m_i}$	Moment about the origin divided by total mass.
Rod with density $\rho(x)$	$\bar{x} = \dfrac{\int_a^b x\,\rho(x)\,dx}{\int_a^b \rho(x)\,dx}$	The continuous version: replace sums by integrals.
Centroid $\bar{x}$ of a region	$\bar{x} = \dfrac{1}{A}\displaystyle\int_a^b x\big[f(x)-g(x)\big]\,dx$	Strip at position x with area $(f-g)\,dx$; A is the region's area C2.7.
Centroid $\bar{y}$ of a region	$\bar{y} = \dfrac{1}{A}\displaystyle\int_a^b \tfrac{1}{2}\Big(\big[f(x)\big]^2-\big[g(x)\big]^2\Big)\,dx$	Each strip's own center is at height $\frac{f+g}{2}$.
Theorem of Pappus	$V = 2\pi\bar{\rho}\,A$	Volume of revolution = area × distance traveled by the centroid.

STATED-EQUATION *x bar equals the sum of m i x i over the sum of m i; for a rod, the integral of x rho over the integral of rho. For a region, x bar equals one over A times the integral of x times f minus g; y bar equals one over A times the integral of one half f squared minus g squared. Pappus: volume equals two pi times the centroid distance times the area.*

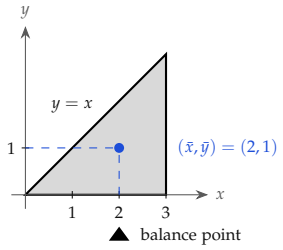
MEMORY AID "Moment over mass."

METHOD

1. Compute the area (or mass) A.
2. Compute the moments $\int x\,(\cdots)$ and $\int \frac{1}{2}(f^2-g^2)$; divide each by A.
3. Example: triangle under $y=x$, $0\le x\le 3$: $A=\frac{9}{2}$, $\bar{x}=\frac{2}{9}\int_0^3 x^2dx=2$, $\bar{y}=\frac{2}{9}\int_0^3 \frac{x^2}{2}dx=1$.

TERMS

m_i, x_i – point masses and positions
$\rho(x)$ – linear density (mass per length)
f, g – top and bottom curves; A area
$\bar{\rho}$ – distance from the centroid to the axis of rotation
Moment – mass × distance from a reference point
$\bar{x}, \bar{y}$ – coordinates of the center of mass (centroid)
A – area of the region



CONTEXT

- **Origin.** Archimedes, *On the Equilibrium of Planes*; Pappus of Alexandria (c. 320 CE).
- **Use.** Balancing structures, vehicles and aircraft; designing beams.
- **Data science.** The mean $E[X] = \int x\,f(x)\,dx$ is the center of mass of the density S.13 §13 Statistics; k-means centroids ML.35 §15 Machine Learning are discrete centers of mass.
- **Physical application.** A seesaw balances when $m_1x_1 = m_2x_2$: a 30 kg child 2 m out balances a 60 kg adult 1 m out.

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Sequences & Series

C2.15 Sequences

A sequence is an infinite ordered list $a_1, a_2, a_3, \ldots$ (a function on the positive integers). It converges when its terms settle down to a single finite value.

Sequence	$\{a_n\}_{n=1}^{\infty} = a_1, a_2, a_3, \ldots$	Given by a formula $a_n = f(n)$ or by a recurrence $a_{n+1} = g(a_n)$.
Limit of a sequence	$\lim_{n\to\infty} a_n = L$	For every tolerance $\varepsilon > 0$, all terms from some N on lie within ε of L.
Geometric limit	$\lim_{n\to\infty} r^n = 0 \ \ (\lvert r\rvert < 1)$	Diverges for $\lvert r\rvert > 1$ or $r = -1$; equals 1 for $r = 1$.
Monotone convergence	bounded and monotonic $\implies$ convergent	Proves convergence without knowing the limit.
Squeeze theorem	$b_n \le a_n \le c_n,\ \ b_n, c_n \to L \ \Rightarrow\ a_n \to L$	Trap a messy sequence between two simple ones.
Function link	$f(x) \to L$ as $x \to \infty \ \Rightarrow\ f(n) \to L$	Lets you use L'Hôpital's rule C1.28 §07 Calculus I on sequences.
Growth order	$\ln n \ll n^p \ll b^n \ll n! \ll n^n$	For large n each beats the one before it ($p > 0$, $b > 1$).

STATED-EQUATION *A sequence a n converges to L if the limit as n goes to infinity of a n equals L. r to the n goes to zero when the absolute value of r is less than one. A bounded, monotonic sequence converges. If b n is at most a n is at most c n and both outer sequences go to L, so does a n. The sequence a n is the ordered list a one, a two, a three, and so on. If f of x approaches L as x goes to infinity, then f of n approaches L. Log n grows slower than any power, which grows slower than b to the n, then n factorial, then n to the n.*

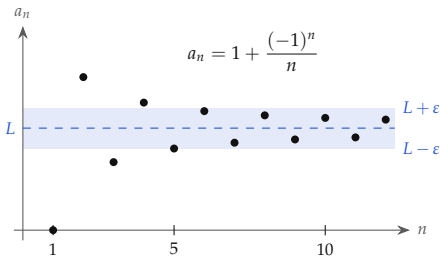
MEMORY AID Growth order: logs $\ll$ powers $\ll$ exponentials $\ll$ factorials.

METHOD

1. Divide by the dominant term, or rewrite $a_n = f(n)$ and take $\lim_{x\to\infty} f(x)$ (L'Hôpital if $\frac{\infty}{\infty}$).
2. For oscillating signs, show $|a_n| \to 0$ (then $a_n \to 0$) or squeeze.
3. Example: $\frac{n}{2n+1} = \frac{1}{2+1/n} \to \frac{1}{2}$; $\frac{\ln n}{n} \to 0$ by L'Hôpital.

TERMS

a_n – nth term; a function on the positive integers
Converges – terms approach a finite limit C1.1 §07 Calculus I
ε, N – tolerance and the index after which terms stay within it
Monotonic – always increasing or always decreasing
Examples – arithmetic A2.18 §04 Algebra II, geometric A2.19 §04 Algebra II
L, r – the limit; a common ratio
b_n, c_n – lower and upper bounding sequences
$\ll$; p, b – grows much slower (ratio $\to 0$); a positive power, a base > 1



CONTEXT

- **Origin.** Made rigorous by Cauchy and Bolzano (early 1800s).
- **Use.** Iterative algorithms, recurrences, compound interest.
- **Data science.** Learning-rate schedules $\eta_n = \frac{\eta_0}{1+kn}$ and estimator sequences $\hat{\theta}_n$ are sequences; "converges" is a statement about their limit.
- **Physical application.** The amount of a drug in the bloodstream just after each daily dose forms a sequence that converges to a steady level.

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C2.16 Series, Partial Sums & the Divergence Test

An infinite series is the limit of its partial sums. It has a value only if that limit exists; terms going to zero is necessary but not sufficient.

Partial sum	$S_N = \sum_{n=1}^{N} a_n$	The running total after N terms.
Sum of a series	$\sum_{n=1}^{\infty} a_n = \lim_{N\to\infty} S_N$	Converges if the limit is finite; otherwise diverges.
Divergence (nth-term) test	$\lim_{n\to\infty} a_n \neq 0 \;\Rightarrow\; \text{diverges}$	Can only prove divergence; $a_n \to 0$ proves nothing.
Geometric series	$\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r} \quad (\lvert r\rvert < 1)$	Diverges for $\lvert r\rvert \geq 1$; a is the first term.
Telescoping series	$\sum_{n=1}^{\infty} (b_n - b_{n+1}) = b_1 - \lim_{n\to\infty} b_n$	Inner terms cancel: $\sum \frac{1}{n(n+1)} = \sum \left(\frac{1}{n} - \frac{1}{n+1}\right) = 1$.
Harmonic series	$\sum_{n=1}^{\infty} \frac{1}{n} = \infty$	Terms go to 0, yet the sum diverges (slowly, like $\ln N$).
Linearity	$\sum(ca_n + b_n) = c\sum a_n + \sum b_n$	Valid when both series on the right converge.

STATED-EQUATION *S N equals the sum of the first N terms; the series equals the limit of S N as N goes to infinity. If the limit of a n is not zero, the series diverges. The geometric series, sum of a r to the n from zero, equals a over one minus r when the absolute value of r is less than one. The harmonic series diverges. A telescoping series, the sum of b n minus b n plus one, equals b one minus the limit of b n. Series split over sums and pull out constants.*

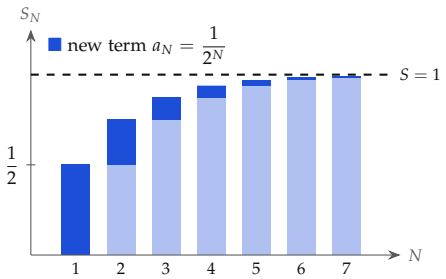
MEMORY AID "If the terms don't go to zero, the series won't go."

METHOD

1. Check $\lim a_n$ first; nonzero $\Rightarrow$ diverges.
2. Recognize a geometric (constant ratio $r = \frac{a_{n+1}}{a_n}$) or telescoping (partial fractions C2.4) pattern.
3. Example: $\sum_{n=0}^{\infty} 3\left(\frac{1}{2}\right)^n$ has $a = 3$, $r = \frac{1}{2}$: sum $\frac{3}{1-1/2} = 6$.

TERMS

S_N – Nth partial sum A2.21 §04 Algebra II
a, r – first term and common ratio A2.19 §04 Algebra II
b_n – the sequence whose consecutive differences form the terms
c – constant multiple
Choosing a test – see C2.K2
a_n, N – the nth term; the number of terms summed



CONTEXT

- **Origin.** Nicole Oresme proved the harmonic series diverges (~1350); Euler found $\sum \frac{1}{n^2} = \frac{\pi^2}{6}$ (1734).
- **Use.** Deciding whether an infinite sum has a finite value.
- **Data science.** Robbins–Monro conditions for SGD: step sizes need $\sum \eta_n = \infty$ yet $\sum \eta_n^2 < \infty$; $\eta_n = \frac{1}{n}$ works because of the harmonic and $p = 2$ series ML.27 §15 Machine Learning.
- **Physical application.** Zeno's paradox resolved: walking half the remaining distance forever covers $\frac{1}{2} + \frac{1}{4} + \cdots = 1$ total.

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C2.17 Integral Test & p-Series

A series of positive, decreasing terms $a_n = f(n)$ behaves like the improper integral of f: rectangles of height $f(n)$ sit just above or just below the curve.

Integral test	$\sum_{n=1}^{\infty} f(n)$ and $\int_1^{\infty} f(x)\,dx$ converge together	Requires f positive, continuous and decreasing (eventually).
p-series	$\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges $\iff p > 1$	From the p-integral C2.6; $p = 1$ is the harmonic series.
Remainder estimate	$\int_{N+1}^{\infty} f\,dx \le \sum_{n=N+1}^{\infty} a_n \le \int_N^{\infty} f\,dx$	Bounds the error of stopping at S_N.

STATED-EQUATION *If f is positive, continuous and decreasing, the sum of f of n and the integral from one to infinity of f of x converge or diverge together. The p-series, sum of one over n to the p, converges if and only if p is greater than one. The tail after N terms lies between the integrals from N plus one and from N to infinity.*

MEMORY AID p-series: "p bigger than one converges" (same rule as the p-integral).

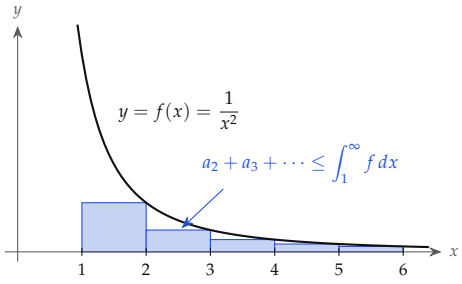
METHOD

1. Check $f(x) > 0$, continuous, decreasing (e.g. $f'(x) < 0$) for $x \ge$ some N.
2. Evaluate the improper integral C2.6; its fate is the series' fate.
3. Example: $\sum \frac{1}{n \ln n}$: $\int_2^{\infty} \frac{dx}{x \ln x} = \lim \ln(\ln b) = \infty$, so it diverges.

DECISION POINT When is the integral test worth it? When f has an easy antiderivative (often via u-sub) and no simpler comparison exists, as for $\frac{1}{n \ln n}$ or ne^{-n^2}. The series sum is *not* the integral's value: $\sum \frac{1}{n^2} = \frac{\pi^2}{6} \approx 1.64$ while $\int_1^{\infty} \frac{dx}{x^2} = 1$.

TERMS

f – positive, continuous, decreasing function with $f(n) = a_n$
p – exponent of the p-series
N – number of terms kept; the tail is the remainder
Improper integral – C2.6
a_n – the series terms, $a_n = f(n)$



CONTEXT

- **Origin.** Integral test due to Maclaurin (1742) and Cauchy; Euler evaluated $\sum \frac{1}{n^2} = \frac{\pi^2}{6}$ (the Basel problem, 1734).
- **Use.** The benchmark series for all comparison arguments.
- **Data science.** Zipf's law: word frequencies $\propto \frac{1}{n^s}$ with $s \approx 1$, so vocabulary "mass" keeps growing with corpus size, as the harmonic series does.
- **Physical application.** The total length of an infinitely stacked overhang of cards (harmonic stacking) is unbounded.

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C2.18 Comparison & Limit Comparison Tests

A positive series is judged by comparing it with a series whose behavior is already known, usually a p-series or a geometric series built from its dominant terms.

Comparison (convergence)	$0 \le a_n \le b_n,\ \sum b_n < \infty \ \Rightarrow\ \sum a_n < \infty$	Smaller than a convergent series converges.
Comparison (divergence)	$0 \le b_n \le a_n,\ \sum b_n = \infty \ \Rightarrow\ \sum a_n = \infty$	Bigger than a divergent series diverges.
Limit comparison	$\lim_{n\to\infty} \frac{a_n}{b_n} = c,\ 0 < c < \infty \ \Rightarrow$ same behavior	No inequality needed; both converge or both diverge.
Dominant-term rule	$\frac{3n+1}{n^3+2} \sim \frac{3n}{n^3} = \frac{3}{n^2}$	Keep the highest powers top and bottom to pick b_n.

STATED-EQUATION *If zero is at most a n, which is at most b n, and the sum of b n converges, then the sum of a n converges. If a n is at least b n and the sum of b n diverges, so does the sum of a n. If the limit of a n over b n is a positive finite number, both series behave the same. For limit comparison keep only the dominant terms: three n plus one over n cubed plus two behaves like three over n squared.*

MEMORY AID "Smaller than convergent converges; bigger than divergent diverges."

METHOD

1. Find b_n from the dominant terms of a_n.
2. Compute $\lim \frac{a_n}{b_n}$ (or prove an inequality).
3. Conclude from the known series. Example: $\sum \frac{3n+1}{n^3+2}$, $b_n = \frac{1}{n^2}$: $\lim \frac{a_n}{b_n} = 3$, so it converges.

DECISION POINT Direct or limit comparison? Direct comparison needs the inequality in the useful direction ("smaller than convergent" or "bigger than divergent"). If it points the wrong way, as with $\frac{1}{n^2-1}$ versus $\frac{1}{n^2}$, switch to limit comparison. A useless conclusion (smaller than a divergent series) means you picked the wrong test, not that the series diverges.

TERMS

a_n – terms of the series under test (non-negative)
b_n – comparison terms: p-series C2.17 or geometric C2.16
c – limit of the ratio, must be positive and finite
$\sim$ – "behaves like" for large n

CONTEXT

- **Origin.** Comparison arguments go back to Gauss and Cauchy (early 1800s).
- **Use.** Most textbook series are settled by a comparison with $\sum \frac{1}{n^p}$.
- **Data science.** Bounding a sum of losses or errors by a simpler series is the core of convergence proofs for iterative algorithms.
- **Physical application.** Asymptotic analysis in physics keeps only the dominant terms for the same reason.

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C2.19 Ratio, Root & Alternating Series Tests

Ratio and root tests compare a series with a geometric series by measuring how fast its terms shrink. The alternating series test handles terms of alternating sign.

Ratio test	$L = \lim_{n\to\infty} \left\lvert \frac{a_{n+1}}{a_n} \right\rvert$	Best with factorials and exponentials.
Root test	$L = \lim_{n\to\infty} \sqrt[n]{\lvert a_n \rvert}$	Best when a_n has the form $(\cdot)^n$.
Conclusion (both tests)	$\begin{cases} L<1 & \text{converges absolutely} \\ L>1 & \text{diverges} \\ L=1 & \text{inconclusive} \end{cases}$	$L=1$ for every p-series: use another test.
Alternating series test	$\sum(-1)^n b_n$ converges if $b_{n+1} \le b_n$ and $b_n \to 0$	Leibniz test; $b_n > 0$.
Alternating error bound	$\lvert S - S_N \rvert \le b_{N+1}$	The first omitted term bounds the error, and the sum lies between consecutive partial sums.

STATED-EQUATION *L equals the limit of the absolute value of a n plus one over a n, or the limit of the n-th root of the absolute value of a n. If L is less than one the series converges absolutely; greater than one, it diverges; equal to one, the test is inconclusive. An alternating series converges if its terms decrease to zero, and the error after N terms is at most b N plus one.*

MEMORY AID Ratio for factorials, root for nth powers; alternating: "decreasing to zero is enough."

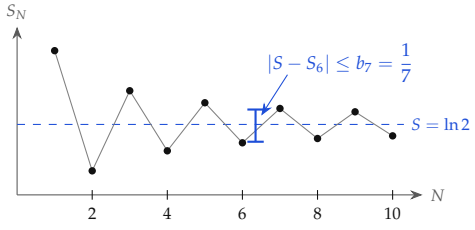
METHOD

1. Form $\left\lvert \frac{a_{n+1}}{a_n} \right\rvert$, cancel factorials ($\frac{(n+1)!}{n!} = n+1$) and powers, take the limit.
2. Compare L with 1.
3. Example: $\sum \frac{2^n}{n!}$: ratio $= \frac{2}{n+1} \to 0 < 1$, so it converges (to $e^2 - 1$ from $n = 1$).

DECISION POINT Ratio vs. root: factorials or products of many factors → ratio; the whole term raised to the nth power, $\left(\frac{n}{2n+1}\right)^n$ → root. If the answer is $L = 1$ (all rational functions of n give 1), go back to comparison C2.18 or the integral test.

TERMS

L – limiting ratio or root
Absolute – $\sum |a_n|$ converges C2.20
Conditional – $\sum a_n$ converges but $\sum |a_n|$ does not
b_n – positive size of the alternating terms
b_{N+1} – first omitted term bounds the error



CONTEXT

- **Origin.** Ratio test by d'Alembert (1768); alternating test by Leibniz (1705).
- **Use.** Radius of convergence of power series C2.21; error bounds.
- **Data science.** Factorials beat exponentials, so $\sum \frac{\lambda^k}{k!}$ converges and Poisson probabilities sum to 1 S.18 §13 Statistics; alternating bounds give guaranteed accuracy for truncated approximations.
- **Physical application.** Truncating the series for $\sin x$ in a calculator: the alternating-series bound certifies the error.

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C2.20 Absolute vs. Conditional Convergence

A series converges absolutely if the series of absolute values converges; it converges conditionally if it converges only thanks to cancellation between positive and negative terms.

Absolute convergence	$\sum \lvert a_n \rvert < \infty$	The strongest kind: any positive-term test may be applied to $\lvert a_n \rvert$.
Absolute implies convergent	$\sum \lvert a_n \rvert < \infty \;\Rightarrow\; \sum a_n$ converges	Signs can only help; the converse is false.
Conditional convergence	$\sum a_n$ converges, $\sum \lvert a_n \rvert = \infty$	E.g. $1 - \frac{1}{2} + \frac{1}{3} - \cdots = \ln 2$, while $\sum \frac{1}{n} = \infty$.
Riemann rearrangement	conditional $\Rightarrow$ rearranging can give any sum	Absolutely convergent sums are order-independent; conditional ones are not.

STATED-EQUATION *A series converges absolutely if the sum of the absolute values converges, and then it converges. It converges conditionally if it converges but the sum of absolute values diverges. Rearranging a conditionally convergent series can change its sum to any value.*

MEMORY AID "Absolute is robust, conditional is fragile."

METHOD

1. Test $\sum \lvert a_n \rvert$ with a positive-term test (comparison, p-series, ratio, integral). Converges $\Rightarrow$ absolutely convergent.
2. If not, test $\sum a_n$ itself (usually the alternating series test C2.19). Converges $\Rightarrow$ conditionally convergent; otherwise divergent.
3. Example: $\sum \frac{(-1)^n}{\sqrt{n}}$: $\sum \frac{1}{\sqrt{n}}$ diverges ($p = \frac{1}{2}$), but $\frac{1}{\sqrt{n}} \downarrow 0$, so it converges conditionally.

TERMS

$\lvert a_n \rvert$ – absolute value of the nth term M.43 §01 Math Signs
Rearrangement – the same terms summed in a different order
Ratio/root tests – give absolute convergence directly when $L < 1$ C2.19
Cancellation – positive and negative terms offsetting each other

CONTEXT

- **Origin.** Dirichlet (1837) noticed rearrangements change sums; Riemann proved the rearrangement theorem (1854).
- **Use.** Deciding whether a sum is stable under reordering and approximation.
- **Data science.** An expected value $E[X] = \sum x_i p_i$ is only defined when $\sum \lvert x_i \rvert p_i < \infty$; parallel (reordered) floating-point sums are reliable only for well-conditioned, absolutely convergent sums.
- **Physical application.** Lattice sums for crystal energies (Madelung constant) are conditionally convergent, so physicists must specify the summation order.

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C2.21 Power Series & Radius of Convergence

A power series is an "infinite polynomial" in $(x-a)$. It converges on an interval centered at a, and inside that interval it can be differentiated and integrated term by term.

Power series	$\displaystyle\sum_{n=0}^{\infty} c_n (x-a)^n$	Coefficients c_n, center a.
Radius (ratio form)	$\displaystyle R = \lim_{n\to\infty} \left\lvert \frac{c_n}{c_{n+1}} \right\rvert$	From the ratio test C2.19; R may be 0, finite, or ∞.
Radius (root form)	$\displaystyle \frac{1}{R} = \limsup_{n\to\infty} \sqrt[n]{\lvert c_n \rvert}$	Cauchy–Hadamard; always works, even when the ratio limit does not exist.
Interval of convergence	$\lvert x-a\rvert < R$ converges, $\lvert x-a\rvert > R$ diverges	Endpoints $x = a \pm R$ must be tested separately.
Term-by-term derivative	$\displaystyle f'(x) = \sum_{n=1}^{\infty} n\, c_n (x-a)^{n-1}$	Same radius R C1.11 §07 Calculus I.
Term-by-term integral	$\displaystyle \int f\,dx = C + \sum_{n=0}^{\infty} \frac{c_n (x-a)^{n+1}}{n+1}$	Same radius R C1.29 §07 Calculus I.
Geometric source series	$\displaystyle \frac{1}{1-x} = \sum_{n=0}^{\infty} x^n \quad (\lvert x\rvert < 1)$	The parent of many other series A2.19 §04 Algebra II.

STATED-EQUATION *A power series is the sum of c n times x minus a to the n. The radius R equals the limit of the absolute value of c n over c n plus one; the series converges for the absolute value of x minus a less than R. Inside, it may be differentiated and integrated term by term. One over one minus x equals the sum of x to the n for the absolute value of x less than one. Equivalently, one over R is the lim sup of the n-th root of the absolute value of c n. The derivative is the sum of n c n times x minus a to the n minus one; the integral is C plus the sum of c n times x minus a to the n plus one, over n plus one.*

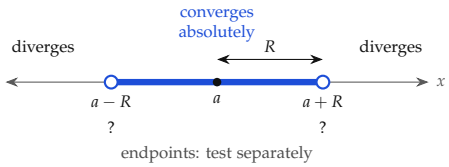
MEMORY AID "Ratio test on the whole term, solve $|x-a|<R$, then check both ends."

METHOD

1. Apply the ratio test to $|c_n(x-a)^n|$; require the limit <1 and solve for $|x-a|<R$.
2. Plug each endpoint into the series and test it (often a p-series or alternating series).
3. Example: $\sum \frac{x^n}{n}$: $R=1$; $x=1$ gives the harmonic series (diverges), $x=-1$ converges, so the interval is $[-1,1)$.

TERMS

c_n – coefficients; a the center
R – radius of convergence
lim sup – largest limit point of the sequence
Endpoints – $x = a \pm R$; may converge or diverge
Geometric source – $\frac{1}{1-x}$ is a geometric series A2.19 §04 Algebra II
a; C – center of the series; constant of integration



CONTEXT

- **Origin.** Newton's power series (1665); convergence theory by Abel and Cauchy.
- **Use.** Representing functions as "infinite polynomials."
- **Data science.** Probability generating functions $G(s) = \sum_k P(X=k)\,s^k$ encode a distribution; $G'(1) = E[X]$ S.13 §13 Statistics.
- **Physical application.** Physicists expand potentials in power series to approximate forces near equilibrium.

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C2.22 Taylor & Maclaurin Series

A smooth function can be rebuilt near a point from its derivatives there: the Taylor series matches the value, slope, curvature and every higher derivative at the center.

Taylor series

Coefficient $c_n = \frac{f^{(n)}(a)}{n!}$.

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots$$

Taylor polynomial	$P_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k$	Degree 1 is the tangent line C1.21 §07 Calculus I; degree 2 adds curvature.
Maclaurin series	$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n$	A Taylor series centered at $a = 0$.
Exponential	$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \cdots$	All real x; every derivative of e^x at 0 is 1.
Sine	$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \cdots$	All real x; odd powers only.
Cosine	$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \cdots$	All real x; even powers only.

STATED-EQUATION *f of x equals the sum of f n-th derivative at a over n factorial, times x minus a to the n: f of a plus f prime of a times x minus a plus f double prime of a over two factorial times x minus a squared, and so on. e to the x equals the sum of x to the n over n factorial; sine x equals the sum of minus one to the n x to the two n plus one over two n plus one factorial; cosine x uses the even powers. The Taylor polynomial P n keeps the terms up to degree n. A Maclaurin series is a Taylor series centered at zero.*

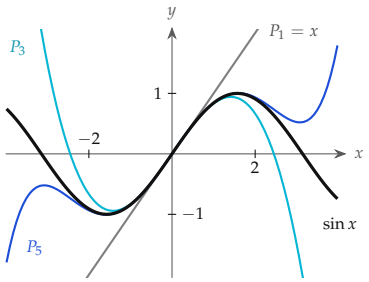
MEMORY AID sin uses odd powers (sine is odd), cos uses even powers (cosine is even), e^x uses all.

METHOD

1. Compute $f(a), f'(a), f''(a), \ldots$ until a pattern appears.
2. Divide the nth derivative by $n!$ and multiply by $(x-a)^n$.
3. Example: $\cos x$ at 0: derivatives $1, 0, -1, 0, 1, \ldots$, so $\cos x = 1 - \frac{x^2}{2} + \frac{x^4}{24} - \cdots$.

TERMS

$f^{(n)}(a)$ – nth derivative at the center C1.11 §07 Calculus I
$n!$ – factorial M.48 §01 Math Signs, S.8 §13 Statistics
a – center of the expansion
Error – see the Lagrange remainder C2.24
Degree 1 – the linearization C1.21 §07 Calculus I
$P_n(x)$ – Taylor polynomial of degree n



CONTEXT

- **Origin.** Brook Taylor (1715); Colin Maclaurin (1742); Gregory and Newton earlier.
- **Use.** How calculators compute sin, e^x, ln; physics approximations.
- **Data science.** Newton's method and XGBoost use a 2nd-order Taylor expansion of the loss (gradient and Hessian); the delta method uses a 1st-order expansion to get standard errors of transformed estimates S.24 §13 Statistics.
- **Physical application.** The pendulum period formula $T \approx 2\pi\sqrt{L/g}$ comes from the first Taylor term $\sin\theta \approx \theta$.

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C2.23 More Maclaurin Series & Building New Ones

A handful of standard series, each with its interval of convergence, generate almost every other series by substitution, differentiation, integration and multiplication.

Geometric	$\dfrac{1}{1-x} = \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + \cdots$	$\lvert x\rvert < 1$; the parent series.
Natural log	$\ln(1+x) = \sum_{n=1}^{\infty} \dfrac{(-1)^{n+1}x^n}{n} = x - \dfrac{x^2}{2} + \cdots$	$-1 < x \le 1$; integrate $\frac{1}{1+x}$.
Arctangent	$\tan^{-1} x = \sum_{n=0}^{\infty} \dfrac{(-1)^n x^{2n+1}}{2n+1} = x - \dfrac{x^3}{3} + \cdots$	$\lvert x\rvert \le 1$; integrate $\frac{1}{1+x^2}$; $x=1$ gives $\frac{\pi}{4}$.
Binomial series	$(1+x)^k = \sum_{n=0}^{\infty} \binom{k}{n} x^n = 1 + kx + \dfrac{k(k-1)}{2!}x^2 + \cdots$	$\lvert x\rvert < 1$ for any real k; finite when k is a whole number A2.23 §04 Algebra II.
Generalized binomial coefficient	$\binom{k}{n} = \dfrac{k(k-1)\cdots(k-n+1)}{n!}$	Defined for any real k.
Build by substitution	$e^{-x^2} = \sum_{n=0}^{\infty} \dfrac{(-1)^n x^{2n}}{n!}$	Replace x by $-x^2$ in e^x; works for any known series.

STATED-EQUATION *One over one minus x equals the sum of x to the n for x between minus one and one. Natural log of one plus x equals x minus x squared over two plus x cubed over three, and so on. Arctangent x equals x minus x cubed over three plus x to the fifth over five, and so on. One plus x to the k equals the sum of k choose n x to the n. k choose n equals k times k minus one, down to k minus n plus one, over n factorial. Substituting minus x squared into the series for e to the x gives e to the minus x squared equals the sum of minus one to the n x to the two n, over n factorial.*

MEMORY AID $\ln(1+x)$ and $\tan^{-1} x$ are "integrated geometric series": divide by the new power.

METHOD

1. Start from a known series (see the table in C2.K2).
2. Substitute, multiply by a power of x, differentiate or integrate term by term; adjust the interval.
3. Example: $\int_0^1 e^{-x^2}dx = 1 - \frac{1}{3} + \frac{1}{10} - \frac{1}{42} + \cdots \approx 0.743$ (true value 0.7468).

DECISION POINT Do not differentiate f twenty times to get a series. Ask which standard series it resembles: $\frac{1}{1+x^2}$ is geometric with $-x^2$; $\frac{x}{(1-x)^2}$ is $x\frac{d}{dx}\frac{1}{1-x}$; $\sqrt{1+x}$ is binomial with $k=\frac{1}{2}$. Substitution can shrink the interval: $\frac{1}{1-2x}$ needs $\lvert x\rvert < \frac{1}{2}$.

TERMS

k – any real exponent
$\binom{k}{n}$ – generalized binomial coefficient A2.23 §04 Algebra II
Interval – set of x where the series converges C2.21
Term by term – differentiate or integrate inside the interval C2.21

CONTEXT

- **Origin.** Newton's binomial series (1665); Gregory and Leibniz's arctangent series (1670s), found earlier by Madhava of Kerala (c. 1400).
- **Use.** Approximating integrals with no elementary antiderivative.
- **Data science.** Numerical libraries provide `log1p(x)` and `expm1(x)`, evaluated from these series for small x, to avoid catastrophic cancellation in log-likelihoods and softplus.
- **Physical application.** Relativity: $(1 - v^2/c^2)^{-1/2} \approx 1 + \frac{v^2}{2c^2}$ turns $mc^2\gamma$ into $mc^2 + \frac{1}{2}mv^2$.

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C2.24 Taylor's Theorem & the Lagrange Remainder

Taylor's theorem says exactly how far a Taylor polynomial is from the function: the error looks like the next term, with the derivative evaluated at some unknown point between a and x.

Taylor's theorem	$f(x) = P_n(x) + R_n(x)$	Function = degree-n Taylor polynomial + remainder.
Lagrange remainder	$R_n(x) = \dfrac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}$	For some c between a and x: "the next term, evaluated at c".
Remainder bound	$\lvert R_n(x)\rvert \le \dfrac{M\lvert x-a\rvert^{n+1}}{(n+1)!}$	M bounds $\lvert f^{(n+1)}\rvert$ between a and x.
Big-O form	$f(x) = P_n(x) + O\big((x-a)^{n+1}\big)$	Near a, the error shrinks like the next power.
Series equals f	$R_n(x) \to 0 \;\Rightarrow\; f(x) = \sum \dfrac{f^{(n)}(a)}{n!}(x-a)^n$	The remainder test for when a Taylor series really represents f.

STATED-EQUATION *f of x equals the Taylor polynomial P n of x plus the remainder R n of x. The Lagrange remainder equals the n plus first derivative at some c, over n plus one factorial, times x minus a to the n plus one. Its size is at most M times the absolute value of x minus a to the n plus one, over n plus one factorial. So f of x equals P n of x plus big O of x minus a to the n plus one. If R n goes to zero, f equals the sum of its Taylor series.*

MEMORY AID "The error is the first term you left out, with the derivative taken at some c."

METHOD

1. Find $M = \max\lvert f^{(n+1)}\rvert$ on the interval between a and x.
2. Plug into the bound; increase n until the bound is below the tolerance.
3. Example: $e^{0.1} \approx 1 + 0.1 + \frac{0.01}{2} = 1.105$; with $M = e^{0.1} < 2$, $\lvert R_2\rvert \le \frac{2(0.1)^3}{6} \approx 3.3 \times 10^{-4}$.

DECISION POINT Which error estimate? If the series is alternating with decreasing terms, the first omitted term is the bound C2.19 (simpler and sharper). Otherwise use Lagrange: for sin and cos, $M = 1$; for e^x, $M = e^{\max}$ on the interval.

TERMS

P_n – Taylor polynomial of degree n C2.22
R_n – remainder (error)
c – unknown point between a and x
M – bound on $\lvert f^{(n+1)}\rvert$ between a and x
$O(\cdot)$ – order of magnitude O.10 §14 Optimization

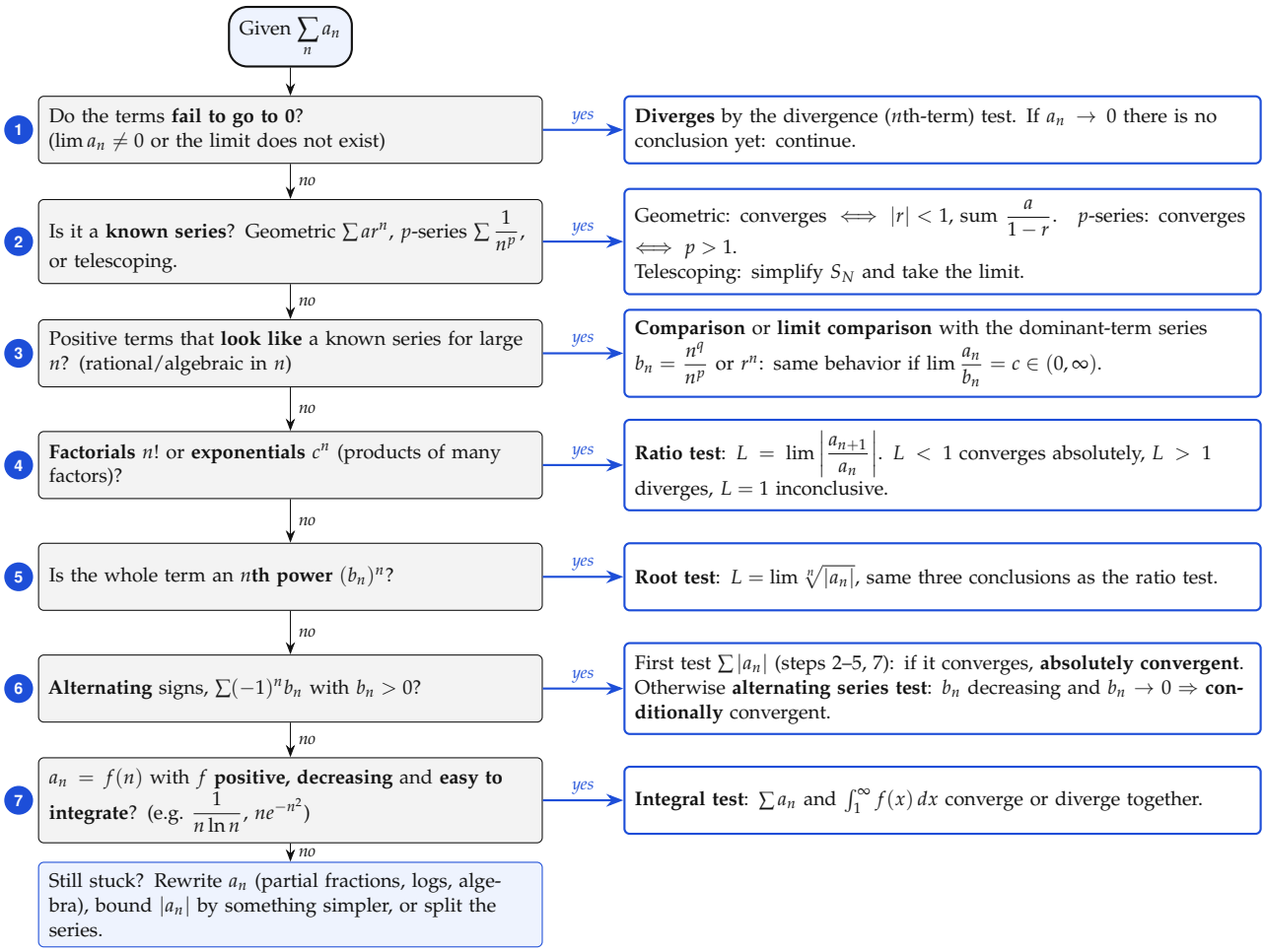
CONTEXT

- **Origin.** Joseph-Louis Lagrange (1797); a generalization of the Mean Value Theorem C1.27 §07 Calculus I (the case $n = 0$).
- **Use.** Guaranteeing accuracy of approximations; choosing how many terms to keep.
- **Data science.** Error analysis of gradient steps: a second-order Taylor bound $L(\theta + \Delta) \le L(\theta) + \nabla L \cdot \Delta + \frac{M}{2}\lVert\Delta\rVert^2$ is why step sizes below $\frac{1}{M}$ are safe.
- **Physical application.** Small-angle approximation: $\sin\theta \approx \theta$ has error at most $\frac{\theta^3}{6}$, about 1.7×10^{-4} at 0.1 rad.

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Series Convergence: Which Test? | *Key Topic C2.K2*

Every convergence question is answered by the same short checklist. Start with the cheapest test (the divergence test), recognize the standard series, and only then reach for the heavier tests. Read the *form* of a_n: it tells you which test will work. The second page collects every test with its conditions, and the standard Maclaurin series with their intervals.



Decision points

DECISION POINT **Ratio test returned $L = 1$?** That always happens for rational and algebraic terms ($\frac{n^2+1}{n^4}$, $\frac{1}{\sqrt{n}}$): it is a signal to use comparison with a p-series, not a verdict. **Comparison went the wrong way?** Switch to limit comparison. **Signs mixed but not alternating** ($\frac{\sin n}{n^2}$)? Test $\sum |a_n|$: here $\leq \sum \frac{1}{n^2}$, so it converges absolutely. **Power series?** Use the ratio test on the whole term to get R, then test the endpoints with the other tests.

Mini-examples. $\sum \frac{n}{n+1}$: terms $\to 1$, diverges (step 1). $\sum \frac{5}{3^n}$: geometric, $r = \frac{1}{3}$, sum $\frac{15}{2}$ (step 2). $\sum \frac{1}{n^2+n}$: $\sim \frac{1}{n^2}$, converges (step 3). $\sum \frac{n^3}{2^n}$: ratio $\to \frac{1}{2}$, converges (step 4). $\sum \left(\frac{n}{3n+1}\right)^n$: root $\to \frac{1}{3}$, converges (step 5). $\sum \frac{(-1)^n}{n}$: conditionally convergent (step 6). $\sum \frac{1}{n(\ln n)^2}$: integral test, converges (step 7).

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The tests at a glance

Test	Applies when / condition	Conclusion	Use it for
Divergence C2.16	any series	$\lim a_n \neq 0 \Rightarrow$ diverges	first check, always
Geometric C2.16	$\sum ar^n$	conv. iff $\lvert r\rvert < 1$; sum $\frac{a}{1-r}$	constant ratio
p-series C2.17	$\sum \frac{1}{n^p}$	conv. iff $p > 1$	comparison benchmark
Telescoping C2.16	$\sum(b_n - b_{n+1})$	sum $= b_1 - \lim b_n$	partial-fraction terms
Integral C2.17	$a_n = f(n)$, $f > 0$, continuous, decreasing	same as $\int_1^\infty f\,dx$	$\frac{1}{n\ln n}$, ne^{-n^2}
Direct comparison C2.18	$0 \le a_n \le b_n$ (or $\ge$)	conv. below conv.; div. above div.	clean inequalities
Limit comparison C2.18	$a_n, b_n > 0$, $\frac{a_n}{b_n} \to c \in (0,\infty)$	same behavior	rational / algebraic terms
Ratio C2.19	$L = \lim \left\lvert \frac{a_{n+1}}{a_n} \right\rvert$	$L < 1$ abs. conv.; $L > 1$ div.; $L = 1$?	$n!$, c^n, power series
Root C2.19	$L = \lim \sqrt[n]{\lvert a_n\rvert}$	same as ratio	$(b_n)^n$
Alternating C2.19	$\sum(-1)^n b_n$, $b_n \downarrow 0$	converges; $\lvert S - S_N\rvert \le b_{N+1}$	sign-alternating terms
Absolute C2.20	$\sum \lvert a_n\rvert < \infty$	$\sum a_n$ converges (absolutely)	mixed signs

Standard Maclaurin series C2.22, C2.23

Function	Series	First terms	Converges for
$\frac{1}{1-x}$	$\sum_{n=0}^{\infty} x^n$	$1 + x + x^2 + x^3 + \cdots$	$-1 < x < 1$
e^x	$\sum_{n=0}^{\infty} \frac{x^n}{n!}$	$1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$	all x
$\sin x$	$\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$	$x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$	all x
$\cos x$	$\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$	$1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots$	all x
$\ln(1+x)$	$\sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^n}{n}$	$x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots$	$-1 < x \le 1$
$\tan^{-1} x$	$\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1}$	$x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots$	$-1 \le x \le 1$
$(1+x)^k$	$\sum_{n=0}^{\infty} \binom{k}{n} x^n$	$1 + kx + \frac{k(k-1)}{2!}x^2 + \cdots$	$-1 < x < 1$

Common mistakes

"Terms go to 0, so it converges." False: $\sum \frac{1}{n}$ diverges. The divergence test only ever proves divergence.

Confusing sequence and series. $a_n = \frac{1}{n} \to 0$ (the sequence converges) while $\sum \frac{1}{n} = \infty$.

Forgetting the endpoints of a power series, or the absolute value in the ratio test.

Positive-term tests on signed series. Comparison and integral tests need $a_n \ge 0$: apply them to $\lvert a_n\rvert$.

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Parametric & Polar Calculus

C2.25 Parametric Equations

A parametric curve gives x and y separately as functions of a third variable t, so it can loop, cross itself and carry a direction. Slopes, tangents, areas and lengths all come from the chain rule applied to $x(t)$ and $y(t)$.

Parametric curve	$x = f(t), \quad y = g(t), \quad a \le t \le b$	t traces the curve in a direction; one curve has many parametrizations.
Eliminating the parameter	$x = r\cos t,\ y = r\sin t \ \Rightarrow\ x^2 + y^2 = r^2$	Solve for t, or use an identity such as $\sin^2 t + \cos^2 t = 1$ T.6 §06 Trigonometry, to get a Cartesian equation.
Parametric slope	$\dfrac{dy}{dx} = \dfrac{dy/dt}{dx/dt} \quad \left(\dfrac{dx}{dt} \neq 0\right)$	Chain rule C1.13 §07 Calculus I: $\frac{dy}{dt} = \frac{dy}{dx}\frac{dx}{dt}$, solved for $\frac{dy}{dx}$.
Second derivative	$\dfrac{d^2y}{dx^2} = \dfrac{\frac{d}{dt}\left(\frac{dy}{dx}\right)}{dx/dt}$	Differentiate the slope in t, then divide by $\frac{dx}{dt}$ again; *not* $\frac{y''(t)}{x''(t)}$.
Horizontal / vertical tangents	$\frac{dy}{dt} = 0$ (horiz.), $\quad \frac{dx}{dt} = 0$ (vert.)	If both vanish at once the test fails: take a limit of the slope.
Area under a parametric curve	$A = \displaystyle\int_\alpha^\beta y(t)\,x'(t)\,dt$	Substitute $x = x(t)$ in $\int y\,dx$; order the limits so that x runs left to right.
Speed & arc length	$L = \displaystyle\int_\alpha^\beta \sqrt{x'(t)^2 + y'(t)^2}\,dt$	The integrand is the speed; integrating it gives distance traveled C2.10.
Line segment	$x = x_0 + t\,\Delta x, \quad y = y_0 + t\,\Delta y, \quad 0 \le t \le 1$	From (x_0, y_0) to (x_1, y_1), with $\Delta x = x_1 - x_0$, $\Delta y = y_1 - y_0$.

STATED-EQUATION *x equals f of t, y equals g of t. $dy\,dx$ equals $dy\,dt$ over $dx\,dt$, when $dx\,dt$ is not zero. The second derivative $d^2y\,dx^2$ equals the derivative with respect to t of $dy\,dx$, divided by $dx\,dt$. The area under the curve is the integral of y times $x'\,dt$, and the arc length is the integral of the speed, the square root of x' squared plus y' squared. For x equals r cosine t and y equals r sine t, eliminating t gives $x^2 + y^2 = r^2$. Tangents are horizontal where $dy\,dt$ is zero and vertical where $dx\,dt$ is zero. The segment from x_0, y_0 moves by t times Δx and t times Δy, for t from zero to one.*

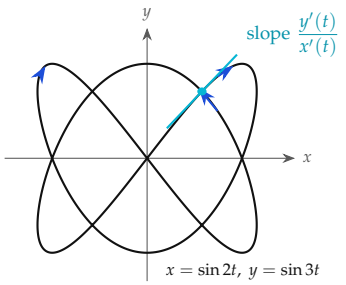
MEMORY AID "Everything over dx/dt": the slope is y'/x', and the second derivative is (slope)′ over x' once more.

METHOD

1. Compute $x'(t)$ and $y'(t)$; slope $m = y'(t_0)/x'(t_0)$.
2. Find the point $(x(t_0), y(t_0))$.
3. Write the tangent line $y - y_0 = m(x - x_0)$ A1.7 §03 Algebra I. Example: $x = t^2$, $y = t^3$ at $t = 1$: $m = \frac{3t^2}{2t} = \frac{3}{2}$, point $(1, 1)$, so $y - 1 = \frac{3}{2}(x - 1)$.

TERMS

t – parameter (often time); gives the curve a direction
$x'(t), y'(t)$ – derivatives of the coordinate functions C1.9 §07 Calculus I
α, β – parameter limits; a, b likewise
$(x_0, y_0), (x_1, y_1)$ – endpoints of a segment
Vector form – $\mathbf{r}(t) = (x(t), y(t))$, velocity $\mathbf{r}'(t)$ C3.5 §10 Calculus III
f, g; a, b – coordinate functions of t; parameter limits
r – circle radius
$\Delta x, \Delta y$ – $x_1 - x_0$ and $y_1 - y_0$



CONTEXT

- **Origin.** Used by Euler and in the study of projectile and planetary motion.
- **Use.** Trajectories, animation, robot paths, Bézier curves in fonts and CAD.
- **Data science.** A ROC curve is parametric in the threshold τ: $(\mathrm{FPR}(\tau), \mathrm{TPR}(\tau))$; precision–recall curves work the same way ML.18 §15 Machine Learning.
- **Physical application.** A projectile's path $x = v_0 t\cos\theta$, $y = v_0 t\sin\theta - \frac{1}{2}gt^2$ is parametric in time P.3 §12 Physics.

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C2.26 Polar Area & Calculus

A polar curve $r = f(\theta)$ is swept out by a rotating ray. Area is a sum of thin circular sectors; slopes and lengths come from treating $x = r\cos\theta$, $y = r\sin\theta$ as a parametric curve in θ.

Polar to Cartesian	$x = r\cos\theta, \quad y = r\sin\theta, \quad r^2 = x^2 + y^2$	Converts a polar curve T.19 §06 Trigonometry into a parametric curve in θ C2.25.
Polar area	$A = \dfrac{1}{2}\displaystyle\int_\alpha^\beta r^2\,d\theta$	Adds sectors of area $\frac{1}{2}r^2\,d\theta$ G.17 §05 Geometry; the angle limits must trace the region exactly once.
Area between polar curves	$A = \dfrac{1}{2}\displaystyle\int_\alpha^\beta \left(r_{\text{out}}^2 - r_{\text{in}}^2\right) d\theta$	Subtract the squares, not the radii: $(r_o - r_i)^2 \neq r_o^2 - r_i^2$.
Polar slope	$\dfrac{dy}{dx} = \dfrac{\frac{dr}{d\theta}\sin\theta + r\cos\theta}{\frac{dr}{d\theta}\cos\theta - r\sin\theta}$	The parametric slope $\frac{dy/d\theta}{dx/d\theta}$ with the product rule applied to $r\sin\theta$ and $r\cos\theta$.
Polar arc length	$L = \displaystyle\int_\alpha^\beta \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\,d\theta$	From $x'(\theta)^2 + y'(\theta)^2 = r^2 + r'^2$ C2.10.
Tangent at the pole	$r(\theta_0) = 0,\ r'(\theta_0) \neq 0 \ \Rightarrow\ \text{tangent line } \theta = \theta_0$	Where the curve passes through the origin, it leaves along the ray θ_0.

STATED-EQUATION *x equals r cosine theta and y equals r sine theta. The area equals one half the integral from alpha to beta of r squared d theta; between two curves, one half the integral of r outer squared minus r inner squared. The slope d y d x equals d r d theta sine theta plus r cosine theta, over d r d theta cosine theta minus r sine theta. Arc length is the integral of the square root of r squared plus d r d theta squared. If r is zero and r prime is not zero at theta naught, the curve passes through the pole tangent to the line theta equals theta naught.*

MEMORY AID Polar area is a sum of thin pizza slices: $\frac{1}{2}r^2\,d\theta$.

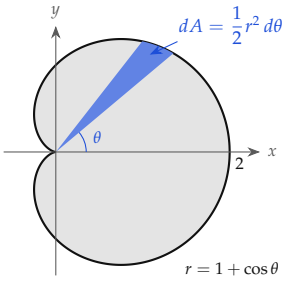
METHOD

1. Sketch the curve; find the angles where $r = 0$ (petal ends) or where two curves meet ($r_1 = r_2$; check the pole separately).
2. Use symmetry: integrate over one piece and multiply.
3. Integrate $\frac{1}{2}r^2$ (power-reduce $\cos^2$ T.10 §06 Trigonometry). Example: cardioid $r = 1 + \cos\theta$: $A = \frac{1}{2}\int_0^{2\pi}(1 + 2\cos\theta + \cos^2\theta)\,d\theta = \frac{1}{2}(2\pi + 0 + \pi) = \frac{3\pi}{2}$.

DECISION POINT Choosing the limits is the whole difficulty. **One petal** of $r = \cos n\theta$: between consecutive zeros of r. **Inside one curve, outside another**: from one intersection angle to the next. **Intersections**: solving $r_1 = r_2$ misses points where the curves reach the pole at *different* angles, so always sketch.

TERMS

$r = f(\theta)$ – polar curve T.19 §06 Trigonometry
$\frac{1}{2}r^2\,d\theta$ – area of a thin sector G.17 §05 Geometry
α, β – angle limits; one full loop or one petal
$r_{\text{out}}, r_{\text{in}}$ – outer and inner boundary curves
$\frac{dr}{d\theta}$ – rate of change of the radius
Slope – treat $x = r\cos\theta$, $y = r\sin\theta$ as parametric C2.25
θ_0 – an angle at which the curve passes through the pole



CONTEXT

- **Origin.** Area by sectors goes back to Archimedes' spiral; calculus form from Newton and the Bernoullis.
- **Use.** Antenna radiation patterns, orbits, rose and cardioid curves.
- **Data science.** Radar (spider) charts encode metrics as polar radii; their enclosed area depends on axis order, a known pitfall for model-comparison dashboards.
- **Physical application.** The area swept by a planet's radius line (Kepler's second law P.9 §12 Physics) is a polar area integral.

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Differential Equations

C2.27 Differential Equations: Terms, Slope Fields & Equilibria

A differential equation relates an unknown function to its derivatives; its solutions are functions. A slope field shows the solutions graphically, and equilibria show the long-run behavior without solving anything.

Order	$F\left(x, y, y', \ldots, y^{(n)}\right) = 0$	The order n is the highest derivative; the general solution has n arbitrary constants.
Initial value problem	$y' = f(x,y), \qquad y(x_0) = y_0$	The initial condition selects one particular solution from the family.
Slope field	segment of slope $f(x,y)$ at each (x,y)	Solution curves are tangent to the segments everywhere.
Equilibrium solution	$y' = f(y), \quad f(y^*) = 0 \;\Rightarrow\; y \equiv y^*$	Constant solutions of an autonomous equation: set the right side to zero.
Stability test	$f'(y^*) < 0$ stable, $\qquad f'(y^*) > 0$ unstable	Nearby solutions approach a stable equilibrium and move away from an unstable one.
Existence & uniqueness	f, $\frac{\partial f}{\partial y}$ continuous near $(x_0, y_0) \Rightarrow$ unique solution	Picard's theorem: solution curves then never cross.

STATED-EQUATION *A differential equation of order n involves derivatives up to the n-th. An initial value problem is y prime equals f of x and y, with y at x naught equal to y naught. An equilibrium of y prime equals f of y is a value y star with f of y star equal to zero; it is stable when f prime of y star is negative. A slope field draws a short segment of slope f of x and y at each point. The equilibrium is unstable when f prime of y star is positive. If f and its partial derivative with respect to y are continuous near x naught, y naught, the initial value problem has exactly one solution nearby.*

MEMORY AID Phase line: arrows point up where $f > 0$, down where $f < 0$; a stable equilibrium has arrows pointing *into* it.

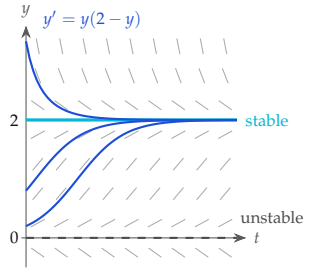
METHOD

1. Solve $f(y) = 0$ for the equilibria.
2. Test the sign of f on each interval between them; draw arrows up $(+)$ or down $(-)$ on a phase line.
3. Classify. Example: $y' = y(2-y)$: equilibria $0, 2$; $f > 0$ on $(0,2)$ and $f < 0$ above 2, so $y = 2$ is stable and $y = 0$ unstable.

DECISION POINT **Which solution method?** Write the equation as $y' = \ldots$ and classify. Right side factors as $g(x)h(y) \to$ separable C2.28. Linear in y and y' ($y' + P(x)y = Q(x)$) $\to$ integrating factor C2.30. Second order, constant coefficients $\to$ characteristic equation C2.31, or Laplace for an IVP with switched or impulsive input C2.33. None of these $\to$ qualitative analysis or a numerical method C2.29.

TERMS

$y^{(n)}$ – nth derivative of the unknown $y(x)$ C1.11 §07 Calculus I
$f(x,y)$ – right side: the slope at (x,y)
x_0, y_0 – initial point (initial condition)
Autonomous – $y' = f(y)$: no explicit x or t
y^* – equilibrium (steady state)
ODE / PDE – one / several independent variables V.13 §11 Vector Calculus
F – the equation written as $F = 0$
$\frac{\partial f}{\partial y}$ – partial derivative of f in y C3.9 §10 Calculus III



CONTEXT

- **Origin.** Newton's *fluxions* (1671); Leibniz's notation; slope fields and phase lines from Poincaré's qualitative theory (1880s).
- **Use.** Every law of change: motion, circuits, populations, chemistry.
- **Data science.** Neural ODEs, continuous-time RNNs and diffusion models describe a state by $\frac{d\mathbf{x}}{dt} = f_\theta(\mathbf{x}, t)$ and integrate it numerically.
- **Physical application.** A falling object with drag, $v' = g - kv$, has the stable equilibrium $v^* = g/k$: terminal velocity P.22 §12 Physics.

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C2.28 Differential Equations: Separable, Growth & Logistic

When the rate splits into an x-part times a y-part, move each variable to its own side and integrate. Exponential growth, logistic growth and Newton's cooling are the classic separable models.

Separation of variables	$\dfrac{dy}{dx} = g(x)\,h(y) \;\Rightarrow\; \displaystyle\int \frac{dy}{h(y)} = \int g(x)\,dx$	One constant C after integrating; solve for y if possible.
Exponential growth / decay	$\dfrac{dy}{dt} = ky \;\Rightarrow\; y = y_0 e^{kt}$	Rate proportional to amount: $k>0$ grows, $k<0$ decays A2.13 §04 Algebra II.
Doubling time / half-life	$T = \dfrac{\ln 2}{\lvert k\rvert}$	Time for the amount to double ($k>0$) or halve ($k<0$).
Logistic equation	$\dfrac{dP}{dt} = rP\left(1 - \dfrac{P}{K}\right)$	Nearly exponential when $P \ll K$; growth stops at the carrying capacity K.
Logistic solution	$P(t) = \dfrac{K}{1 + Ae^{-rt}}, \qquad A = \dfrac{K - P_0}{P_0}$	Solved by partial fractions C2.4; an S-curve from P_0 to K.
Fastest growth	$P = \dfrac{K}{2}, \qquad \left(\dfrac{dP}{dt}\right)_{\max} = \dfrac{rK}{4}$	The inflection point of the S-curve is at half capacity.
Newton's law of cooling	$\dfrac{dT}{dt} = -k(T - T_a) \;\Rightarrow\; T = T_a + (T_0 - T_a)e^{-kt}$	The difference from the surroundings decays exponentially P.25 §12 Physics.

STATED-EQUATION *If d y d x equals g of x times h of y, then the integral of d y over h of y equals the integral of g of x d x. d y d t equals k y gives y equals y naught e to the k t, with doubling time or half-life ln two over the absolute value of k. The logistic equation d P d t equals r P times one minus P over K gives P of t equals K over one plus A e to the minus r t, with A equals K minus P naught over P naught. Newton's law of cooling: d T d t equals minus k times T minus T a, so T equals T a plus T naught minus T a times e to the minus k t. Logistic growth is fastest at P equals K over two, where d P d t equals r K over four.*

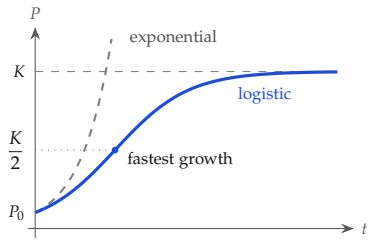
MEMORY AID Separable: "get all the y's with dy, all the x's with dx, then integrate both sides."

METHOD

1. Rewrite as $\frac{dy}{h(y)} = g(x)\,dx$; first note any y with $h(y)=0$ (these constant solutions are lost by dividing).
2. Integrate both sides C1.29 §07 Calculus I; write a single $+C$.
3. Solve for y and apply the initial condition. Example: $y' = xy$, $y(0)=2$: $\ln|y| = \frac{x^2}{2} + C$, so $y = 2e^{x^2/2}$.

TERMS

$g(x), h(y)$ – the x-factor and y-factor of the rate
k – growth ($k>0$) or decay ($k<0$) rate A2.13 §04 Algebra II
y_0, P_0, T_0 – initial values (initial conditions)
r – intrinsic growth rate; K carrying capacity
A – constant fixed by P_0
T_a – ambient temperature (Newton's law of cooling)
Lost solutions – equilibria $h(y)=0$ C2.27
P, K – population; carrying capacity
T – doubling time or half-life; in cooling, the temperature



CONTEXT

- **Origin.** Logistic model by Pierre-François Verhulst (1838); separation of variables by Leibniz and Bernoulli.
- **Use.** Population, radioactive decay, cooling, epidemics (SIR).
- **Data science.** The logistic solution is the sigmoid $\sigma(z) = \frac{1}{1+e^{-z}}$ of logistic regression ML.9 §15 Machine Learning and neural nets; adoption curves are fit with it; neural ODEs learn $\frac{dy}{dt}$ directly.
- **Physical application.** Newton's law of cooling predicts how fast coffee cools; logistic growth models bacteria in a dish reaching its capacity P.25 §12 Physics.

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C2.29 Euler's Method for Differential Equations

When an ODE cannot be solved exactly, walk along it: from the current point, step a distance h along the tangent slope the equation gives, and repeat. Higher-order methods sample the slope several times per step.

Initial value problem	$\dfrac{dy}{dt} = f(t,y), \qquad y(t_0) = y_0$	Any first-order IVP; systems work the same with vectors.
Euler step	$y_{n+1} = y_n + h\,f(t_n, y_n), \qquad t_{n+1} = t_n + h$	Follow the tangent line for one step C1.21 §07 Calculus I.
Global error	$\lvert y(t_N) - y_N \rvert = O(h)$	First-order: halving h roughly halves the final error (and doubles the work).
Improved Euler (Heun)	$y_{n+1} = y_n + \dfrac{h}{2}(k_1 + k_2)$ $k_1 = f(t_n, y_n), \quad k_2 = f(t_n + h,\, y_n + hk_1)$	Average the slopes at the start and at the Euler-predicted end; error $O(h^2)$.
Runge–Kutta 4 (RK4)	$y_{n+1} = y_n + \dfrac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4)$	Weighted average of four slope samples, Simpson-like weights C2.5; error $O(h^4)$.
RK4 stages	$k_1 = f(t_n, y_n), \qquad k_2 = f\left(t_n + \frac{h}{2},\, y_n + \frac{h}{2}k_1\right)$ $k_3 = f\left(t_n + \frac{h}{2},\, y_n + \frac{h}{2}k_2\right), \qquad k_4 = f(t_n + h,\, y_n + hk_3)$	Slopes at the start, twice at the midpoint, and at the end.

STATED-EQUATION *Given d y d t equals f of t and y, with y at t naught equal to y naught: y n plus one equals y n plus h times f of t n and y n, and t n plus one equals t n plus h. The global error is of order h. Improved Euler averages the start slope and the predicted end slope; Runge-Kutta four averages four slopes with weights one, two, two, one over six, with error of order h to the fourth. Heun: y n plus one equals y n plus h over two times k one plus k two, with k one the slope at the start and k two the slope at the Euler-predicted end. The RK4 stages are k one at the start, k two and k three at the midpoint using the previous stage, and k four at the end using k three.*

MEMORY AID "New value = old value + step × slope."

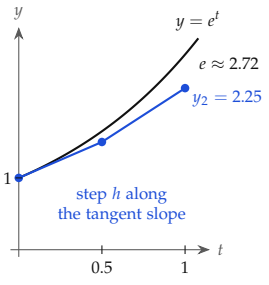
METHOD

1. Set t_0, y_0, step h, number of steps N.
2. Repeat: slope $m = f(t_n, y_n)$, then $y_{n+1} = y_n + hm$, $t_{n+1} = t_n + h$.
3. Example: $y' = y$, $y(0) = 1$, $h = 0.5$: $y_1 = 1 + 0.5(1) = 1.5$, $y_2 = 1.5 + 0.5(1.5) = 2.25$; the true $y(1) = e \approx 2.72$.

DECISION POINT **How small an h?** Run with h and $h/2$; if the answers agree to the digits you need, stop. Euler is for understanding; use RK4 or an adaptive solver (`solve_ivp`) in practice. If tiny steps are needed just to stay stable ("stiff" equations, very different time scales), switch to an implicit method.

TERMS

h – step size; N number of steps
t_n, y_n – the nth grid time and approximation
$f(t_n, y_n)$ – slope from the differential equation C2.27
$k_1, \ldots, k_4$ – RK4 slope samples
$O(h^p)$ – order p: halving h divides the error by about 2^p



CONTEXT

- **Origin.** Leonhard Euler, *Institutionum calculi integralis* (1768); Runge (1895) and Kutta (1901).
- **Use.** Simulating any ODE that has no closed-form solution.
- **Data science.** Epidemic (SIR) and population simulations; a residual network layer $\mathbf{x}_{k+1} = \mathbf{x}_k + f(\mathbf{x}_k)$ is one Euler step, the idea behind neural ODEs; gradient descent C1.26 §07 Calculus I is Euler's method on $\mathbf{x}' = -\nabla L$.
- **Physical application.** Game and physics engines advance position and velocity each frame with Euler-type steps: $x \mathrel{+}= v\,\Delta t$, $v \mathrel{+}= a\,\Delta t$.

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C2.30 First-Order Linear ODEs & Integrating Factors

A first-order linear equation has y and y' only to the first power. Multiplying by the right factor turns the left side into the derivative of a product, which integrates at once.

Standard form	$y' + P(x)\,y = Q(x)$	Divide by the coefficient of y' first; P and Q may be any continuous functions.
Integrating factor	$\mu(x) = e^{\int P(x)\,dx}$	No $+C$ needed here; any antiderivative of P works.
Product collapse	$\mu y' + \mu P y = (\mu y)' = \mu Q$	Works because $\mu' = \mu P$: the product rule C1.12 §07 Calculus I in reverse.
General solution	$y = \dfrac{1}{\mu(x)}\left[\int \mu(x) Q(x)\,dx + C\right]$	Keep the $+C$ inside the bracket: it becomes $\frac{C}{\mu}$, the transient.
Constant coefficients	$y' + ay = b \;\Rightarrow\; y = \dfrac{b}{a} + Ce^{-ax}$	Steady state $\frac{b}{a}$ plus a decaying transient (for $a > 0$).
Mixing problem	$\dfrac{dA}{dt} = r_{\text{in}} c_{\text{in}} - \dfrac{r_{\text{out}}}{V(t)} A$	Rate in minus rate out; linear in the amount A.
Bernoulli equation	$y' + Py = Qy^n, \quad v = y^{1-n}$ $\Rightarrow\; v' + (1-n)Pv = (1-n)Q$	A nonlinear equation that the substitution makes linear ($n \neq 0, 1$).

STATED-EQUATION *Standard form: y prime plus P of x times y equals Q of x. The integrating factor mu equals e to the integral of P. Multiplying gives the derivative of mu y equal to mu Q, so y equals one over mu times the quantity integral of mu Q plus C. For y prime plus a y equals b, y equals b over a plus C e to the minus a x. In a mixing problem, d A d t equals the inflow rate times the inflow concentration, minus the outflow rate over the volume, times A. A Bernoulli equation, y prime plus P y equals Q y to the n, becomes linear in v equals y to the one minus n.*

MEMORY AID "e to the integral of P", then the left side is $(\mu y)'$.

METHOD

1. Write the equation in standard form and read off $P(x)$.
2. Compute $\mu = e^{\int P\,dx}$ and multiply through: $(\mu y)' = \mu Q$.
3. Integrate both sides (with $+C$), divide by μ, apply the initial condition. Example: $y' + 2y = 4$, $y(0) = 5$: $\mu = e^{2x}$, $(e^{2x}y)' = 4e^{2x}$, $e^{2x}y = 2e^{2x} + C$, $y = 2 + 3e^{-2x}$.

DECISION POINT **Separable or linear?** $y' = ky$ or $y' = a - by$ are both: use whichever is quicker. $y' + y = x$ is linear but not separable. If y appears inside a nonlinear function (y^2, $\sin y$, e^y), it is not linear: try separation or Bernoulli's substitution, otherwise go numerical C2.29.

TERMS

$P(x), Q(x)$ – coefficient and forcing functions
$\mu(x)$ – integrating factor
C – arbitrary constant, fixed by $y(x_0)$
a, b – constants; b/a is the equilibrium C2.27
A, V – amount of solute and volume; r flow rates, c_{in} inflow concentration
n, v – Bernoulli exponent and new variable
$r_{\text{in}}, r_{\text{out}}$; c_{in} – inflow and outflow rates; inflow concentration

CONTEXT

- **Origin.** Leibniz and Johann Bernoulli (1690s); Euler named the integrating factor.
- **Use.** RC and RL circuits EE.11 §16 Electrical Eng., mixing tanks, drug dosing, falling with drag.
- **Data science.** An exponential moving average $s_k = s_{k-1} + \alpha(x_k - s_{k-1})$ is the Euler discretization of $s' + \frac{1}{\tau}s = \frac{1}{\tau}x(t)$; Adam's moment estimates ML.27 §15 Machine Learning are such filters.
- **Physical application.** Charging a capacitor: $RC\,v' + v = V_s$ gives $v = V_s(1 - e^{-t/RC})$.

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C2.31 Second-Order Linear ODEs with Constant Coefficients

Trying $y = e^{rx}$ turns $ay'' + by' + cy = 0$ into a quadratic in r. Its roots, real and distinct, repeated, or complex, give exactly three solution shapes: pure decay or growth, critical decay, or oscillation.

Homogeneous equation	$ay'' + by' + cy = 0$	Constant $a \neq 0$, b, c; two initial conditions $y(0)$, $y'(0)$ fix the solution.
Characteristic equation	$ar^2 + br + c = 0$	Substitute $y = e^{rx}$ and cancel e^{rx}; solve with the quadratic formula A1.15 §03 Algebra I.
Superposition	$y = c_1 y_1 + c_2 y_2$	Any combination of two independent solutions is again a solution, and gives all of them.
Distinct real roots ($\Delta > 0$)	$y = c_1 e^{r_1 x} + c_2 e^{r_2 x}$	Two exponentials; no oscillation (overdamped).
Repeated root ($\Delta = 0$)	$y = (c_1 + c_2 x)e^{rx}$	Multiply the second solution by x (critically damped).
Complex roots $\alpha \pm \beta i$ ($\Delta < 0$)	$y = e^{\alpha x}\left(c_1 \cos \beta x + c_2 \sin \beta x\right)$	Euler's formula T.23 §06 Trigonometry turns $e^{(\alpha \pm \beta i)x}$ into a sinusoid with envelope $e^{\alpha x}$.
Damped spring	$mx'' + cx' + kx = 0$	Mass, damping, stiffness; $c^2 - 4mk$ classifies the motion as over, critically or under damped.
Natural frequency	$\omega_0 = \sqrt{\dfrac{k}{m}}, \qquad \beta = \sqrt{\omega_0^2 - \left(\dfrac{c}{2m}\right)^2}$	Undamped angular frequency, and the slower damped frequency β P.15 §12 Physics.

STATED-EQUATION *a y double prime plus b y prime plus c y equals zero has characteristic equation a r squared plus b r plus c equals zero. Distinct real roots give c one e to the r one x plus c two e to the r two x; a repeated root gives c one plus c two x, times e to the r x; complex roots alpha plus or minus beta i give e to the alpha x times c one cosine beta x plus c two sine beta x. Superposition: c one y one plus c two y two is again a solution. A damped spring obeys m x double prime plus c x prime plus k x equals zero, with natural frequency omega naught equal to the square root of k over m, and damped frequency beta equal to the square root of omega naught squared minus c over two m, squared.*

MEMORY AID "Guess e^{rx}; the quadratic decides the shape." Repeated root: "multiply by x".

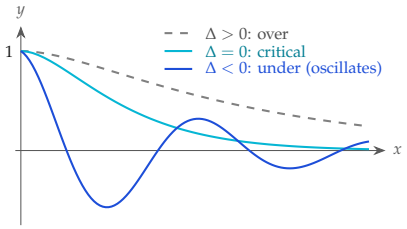
METHOD

1. Write the characteristic equation and find its roots.
2. Pick the solution form from the root case.
3. Use $y(0)$ and $y'(0)$ to find c_1, c_2. Example: $y'' - 3y' + 2y = 0$, $y(0) = 0$, $y'(0) = 1$: $r = 1, 2$; $c_1 + c_2 = 0$, $c_1 + 2c_2 = 1$, so $y = e^{2x} - e^x$.

DECISION POINT **Which form?** Read the discriminant $\Delta = b^2 - 4ac$ before solving: positive, zero, negative. In a mechanical or electrical system the sign of Δ is the physics: overdamped (sluggish return), critically damped (fastest return without overshoot, the design target for door closers and car suspensions), underdamped (rings).

TERMS

a, b, c – constant coefficients; $\Delta = b^2 - 4ac$
r, r_1, r_2 – roots of the characteristic equation
$\alpha \pm \beta i$ – complex roots A2.6 §04 Algebra II
c_1, c_2 – constants fixed by two initial conditions
m, c, k – mass, damping coefficient, spring constant P.5 §12 Physics
Independent – y_2 not a constant multiple of y_1
Δ – discriminant $b^2 - 4ac$ A1.15 §03 Algebra I
y_1, y_2 – two independent solutions
ω_0, β – natural and damped angular frequencies



CONTEXT

- **Origin.** Euler (1743) solved linear equations with constant coefficients by the exponential guess; d'Alembert treated repeated roots.
- **Use.** Springs, suspensions, pendulums, RLC circuits EE.12 §16 Electrical Eng., structural vibration.
- **Data science.** Gradient descent with momentum on a quadratic behaves like a damped oscillator: too little friction (momentum near 1) rings, too much crawls; the best momentum is near critical damping.
- **Physical application.** A car's shock absorbers are tuned near critical damping so the car settles after a bump without bouncing.

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C2.32 Nonhomogeneous Equations, Undetermined Coefficients & Resonance

With a forcing term $g(x)$, the solution is the homogeneous solution plus any one particular solution. For polynomial, exponential and sinusoidal forcing, guess a solution of the same shape and solve for its coefficients.

Nonhomogeneous equation	$ay'' + by' + cy = g(x)$	g is the forcing (input); the left side is the system C2.31.
Solution structure	$y = y_h + y_p$	General homogeneous solution plus one particular solution; apply initial conditions to the sum.
Polynomial forcing	$g = p_n(x) \;\to\; y_p = A_n x^n + \cdots + A_1 x + A_0$	p_n a degree-n polynomial; include every lower power, even if g lacks them.
Exponential forcing	$g = e^{kx} \;\to\; y_p = Ae^{kx}$	Substitute and match coefficients of e^{kx}.
Sinusoidal forcing	$g = \cos\omega x,\ \sin\omega x \;\to\; y_p = A\cos\omega x + B\sin\omega x$	Always include both sine and cosine.
Modification rule	$\text{trial} \subset y_h \;\Rightarrow\; \text{use } x\cdot\text{trial } (\text{or } x^2\cdot\text{trial})$	If the trial already solves the homogeneous equation it gives $0 = g$ and cannot match.
Resonance	$x'' + \omega_0^2 x = \frac{F_0}{m}\cos\omega_0 t$ $\Rightarrow\; x_p = \frac{F_0}{2m\omega_0}\, t\sin\omega_0 t$	Forcing at the natural frequency: amplitude grows linearly without bound (no damping).
Variation of parameters	Works for any continuous g (equation divided by a so the y'' coefficient is 1). $y_p = -y_1\int \frac{y_2 g}{W}\,dx + y_2\int \frac{y_1 g}{W}\,dx$ $W = y_1 y_2' - y_1' y_2$	

STATED-EQUATION *For a y double prime plus b y prime plus c y equals g of x, the general solution is y h plus y p. Guess y p with the same shape as g: a full polynomial, A e to the k x, or A cosine omega x plus B sine omega x, multiplying by x if the guess solves the homogeneous equation. Forcing an undamped oscillator at its natural frequency gives a response growing like t sine omega naught t. At resonance the particular solution is F naught over two m omega naught, times t sine omega naught t. Variation of parameters: y p equals minus y one times the integral of y two g over W, plus y two times the integral of y one g over W, where the Wronskian W equals y one y two prime minus y one prime y two.*

MEMORY AID "Guess what you see" (same family as g); if the guess is already in y_h, "bump it by x".

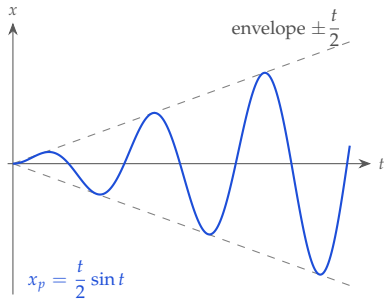
METHOD

1. Solve the homogeneous equation for y_h C2.31.
2. Choose the trial y_p from the form of g; apply the modification rule.
3. Substitute, match coefficients, add y_h, then apply initial conditions. Example: $y'' - y = 3e^{2x}$: $y_h = c_1e^x + c_2e^{-x}$; $y_p = Ae^{2x}$ gives $4A - A = 3$, $A = 1$; $y = c_1e^x + c_2e^{-x} + e^{2x}$.

DECISION POINT **Which method for y_p?** g a polynomial, exponential, sine/cosine, or a product of these $\to$ undetermined coefficients (fast). Anything else ($\tan x$, $\frac{1}{x}$, $\ln x$) $\to$ variation of parameters. An IVP with a switch-on, step or impulse input $\to$ Laplace transform C2.33.

TERMS

$g(x)$ – forcing term (input)
y_h, y_p – homogeneous and particular solutions
A, B, A_i – undetermined coefficients
ω, ω_0 – forcing and natural angular frequencies
F_0, m – forcing amplitude, mass
y_1, y_2; W – homogeneous solutions; Wronskian (a 2×2 determinant A2.25 §04 Algebra II)
$p_n(x)$; k – a degree-n polynomial; an exponential rate



CONTEXT

- **Origin.** Euler and Lagrange (variation of parameters, 1774).
- **Use.** Driven springs, AC circuits EE.16 §16 Electrical Eng., earthquakes and bridges.
- **Data science.** Linear systems respond to sinusoidal input at the same frequency, only scaled and shifted; this is why frequency-domain (Fourier) features and filters EE.17 §16 Electrical Eng. work.
- **Physical application.** Soldiers break step on bridges so their marching does not drive a structural mode at resonance.

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C2.33 Laplace Transform Basics

The Laplace transform turns functions of time into functions of s and derivatives into multiplication by s. A linear ODE with constant coefficients becomes algebra; the initial conditions are built in.

Definition	$\mathcal{L}\{f\}(s) = F(s) = \int_0^\infty e^{-st} f(t)\,dt$	An improper integral C2.6; exists for s large enough when f grows at most exponentially.
Linearity	$\mathcal{L}\{af + bg\} = aF + bG$	Transform term by term; the inverse $\mathcal{L}^{-1}$ is linear too.
Derivative rule	$\mathcal{L}\{f'\} = sF - f(0)$ $\mathcal{L}\{f''\} = s^2F - sf(0) - f'(0)$	Differentiation becomes multiplication by s; initial values enter automatically.
First shift (s-shift)	$\mathcal{L}\{e^{at}f(t)\} = F(s-a)$	An exponential factor shifts the transform; gives damped sinusoids.
Second shift (t-shift)	$\mathcal{L}\{u(t-c)f(t-c)\} = e^{-cs}F(s)$	A delay by c multiplies by e^{-cs}; models switching on at $t = c$.
Convolution	$\mathcal{L}\{f * g\} = F(s)\,G(s)$ $(f*g)(t) = \int_0^t f(\tau)g(t-\tau)\,d\tau$	Products in s are convolutions in t EE.22 §16 Electrical Eng..

STATED-EQUATION *The Laplace transform of f is F of s, the integral from zero to infinity of e to the minus s t times f of t, d t. It is linear. The transform of f prime is s F minus f of zero; of f double prime, s squared F minus s f of zero minus f prime of zero. Multiplying by e to the a t shifts F to F of s minus a; delaying by c multiplies by e to the minus c s; convolution becomes multiplication.*

MEMORY AID "Derivative in t → times s in s" (minus the initial values).

METHOD

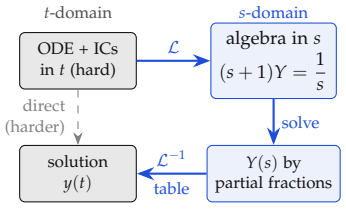
1. Transform both sides, using the derivative rule with the initial conditions.
2. Solve the algebraic equation for $Y(s)$.
3. Split $Y(s)$ by partial fractions C2.4 and invert each piece with the table. Example: $y' + y = 1$, $y(0) = 0$: $sY + Y = \frac{1}{s}$, $Y = \frac{1}{s(s+1)} = \frac{1}{s} - \frac{1}{s+1}$, so $y = 1 - e^{-t}$.

DECISION POINT **When to use Laplace?** Constant-coefficient linear IVPs with conditions at $t = 0$, especially with piecewise, step or impulse forcing, where undetermined coefficients C2.32 would need case-by-case work. For variable coefficients, or when no initial conditions are given, use the other methods.

$f(t)$	1	t^n	e^{at}	$\sin bt$	$\cos bt$	$e^{at}\sin bt$	$u(t-c)$	$\delta(t-c)$
$F(s)$	$\frac{1}{s}$	$\frac{n!}{s^{n+1}}$	$\frac{1}{s-a}$	$\frac{b}{s^2+b^2}$	$\frac{s}{s^2+b^2}$	$\frac{b}{(s-a)^2+b^2}$	$\frac{e^{-cs}}{s}$	e^{-cs}

TERMS

s – transform variable (a complex frequency)
F, G, Y – transforms of f, g, y
$u(t-c)$ – unit step: 0 before c, 1 after
$\delta(t-c)$ – unit impulse at $t = c$ (area 1)
a, b, c, n – constants; n a whole number, $n!$ factorial M.48 §01 Math Signs
$f * g$ – convolution
τ – dummy integration variable



CONTEXT

- **Origin.** Laplace (1782) in probability; Heaviside's operational calculus (1890s) for circuits.
- **Use.** Control systems, circuits and transfer functions EE.23 §16 Electrical Eng., mechanical vibration.
- **Data science.** The moment generating function $M_X(t) = E[e^{tX}]$ S.13 §13 Statistics is a Laplace transform with $s = -t$; sums of independent variables multiply their transforms (convolution theorem).
- **Physical application.** Engineers find a circuit's response to a switch closing at $t = 2$ s with one e^{-2s} factor instead of piecing solutions together.

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09
Linear Algebra

Vectors, bases, and the matrices that transform them. Linear algebra is the working language of data science: every dataset is a matrix, every model a transformation, and every embedding a vector in some basis.

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Vectors

LA.1 Vectors: Components, Magnitude & Unit Vectors

A vector is a quantity with a magnitude and a direction, recorded as an ordered list of components. Its length comes from the Pythagorean theorem; dividing a vector by its length leaves a unit vector that points the same way.

Vector between two points	$\overrightarrow{PQ} = Q - P = \langle q_1 - p_1,\ q_2 - p_2,\ q_3 - p_3 \rangle$	"Tip minus tail": $P = (1,2)$, $Q = (4,6)$ give $\overrightarrow{PQ} = \langle 3,4 \rangle$.
Component (basis) form	$\mathbf{v} = \langle v_1, v_2, v_3 \rangle = v_1\mathbf{i} + v_2\mathbf{j} + v_3\mathbf{k}$	Each component is the step taken along one axis.
Magnitude (Euclidean norm)	$\lVert\mathbf{v}\rVert = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}$	Pythagoras in n dimensions: $\lVert\langle 3,4\rangle\rVert = 5$; $\lVert\mathbf{v}\rVert = 0$ only for $\mathbf{v} = \mathbf{0}$.
Unit vector (normalizing)	$\hat{\mathbf{v}} = \dfrac{\mathbf{v}}{\lVert\mathbf{v}\rVert}$	Same direction, length 1. Undefined for the zero vector.
Magnitude–direction form (2-D)	$\mathbf{v} = \lVert\mathbf{v}\rVert\ \langle \cos\theta,\ \sin\theta \rangle$	Length times a unit direction; θ is measured from the $+x$ axis T.4 §06 Trigonometry.
Length of a multiple	$\lVert c\mathbf{v}\rVert = \lvert c\rvert\ \lVert\mathbf{v}\rVert$	Scaling by c multiplies the length by $\lvert c\rvert$; $c < 0$ also reverses the direction.
Distance between points	$d(P,Q) = \lVert Q - P\rVert$	The length of the vector joining them: the distance formula G.12 §05 Geometry.

STATED-EQUATION *The vector from P to Q equals Q minus P, componentwise. v equals v one i plus v two j plus v three k. The magnitude of v is the square root of the sum of the squares of its components. The unit vector v hat equals v divided by its magnitude. In two dimensions v equals its magnitude times cosine theta, sine theta. The length of c v is the absolute value of c times the length of v. The distance from P to Q is the length of Q minus P.*

MEMORY AID "Tip minus tail." A hat on a vector means "length 1".

METHOD

1. Find the components (subtract coordinates if the vector joins two points).
2. Square, add, take the square root to get $\lVert\mathbf{v}\rVert$.
3. Divide every component by $\lVert\mathbf{v}\rVert$. Example: $\mathbf{v} = \langle 6,8\rangle$: $\lVert\mathbf{v}\rVert = \sqrt{36+64} = 10$, $\hat{\mathbf{v}} = \langle 0.6, 0.8\rangle$.

TERMS

P, Q – points; p_i, q_i their coordinates
v_i – components of $\mathbf{v}$; n = number of components (dimension)
$\mathbf{i}, \mathbf{j}, \mathbf{k}$ – standard unit vectors along x, y, z (also $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ LA.7)
$\lVert\mathbf{v}\rVert$ – magnitude, length or 2-norm (also $\lvert\mathbf{v}\rvert$); other norms in LA.29
$\hat{\mathbf{v}}$ – "v-hat", the unit vector in the direction of $\mathbf{v}$
θ – direction angle; c a scalar (real number)
$\mathbf{0}$ – zero vector: magnitude 0, no direction
c, $d(P,Q)$ – a scalar; the distance between P and Q

CONTEXT

- **Origin.** Hamilton coined "vector" (1843, Latin *vehere*, "to carry"); bold-face notation from Gibbs and Heaviside.
- **Use.** Displacements, velocities, forces, any quantity with size and direction.
- **Data science.** Embeddings are normalized to unit length before cosine similarity T.21 §06 Trigonometry; the norm of a gradient measures how steep a loss is.
- **Physical application.** A hiker walks 3 km east then 4 km north: displacement $\langle 3,4\rangle$, distance from start 5 km.

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LA.2 Vectors & Linear Combinations

Vectors are added component by component and scaled by multiplying every component. A linear combination does both at once, and almost every construction in linear algebra is built from it.

Vector addition	$\mathbf{u}+\mathbf{v}=(u_1+v_1,\ \ldots,\ u_n+v_n)$	Add matching components; geometrically tip-to-tail, or the diagonal of the parallelogram.
Scalar multiplication	$c\,\mathbf{v}=(cv_1,\ \ldots,\ cv_n)$	Stretches by $\lvert c\rvert$; a negative scalar flips the direction.
Linear combination	$\mathbf{w}=c_1\mathbf{v}_1+c_2\mathbf{v}_2+\cdots+c_k\mathbf{v}_k$	Scale each vector by its weight, then add.
Subtraction	$\mathbf{u}-\mathbf{v}=\mathbf{u}+(-1)\mathbf{v}$	The arrow from the tip of $\mathbf{v}$ to the tip of $\mathbf{u}$.
Matrix form of a combination	$c_1\mathbf{v}_1+\cdots+c_k\mathbf{v}_k=V\mathbf{c}$	Put the $\mathbf{v}_i$ as the columns of V and the weights in $\mathbf{c}$ LA.10.
Algebra rules	$\mathbf{u}+\mathbf{v}=\mathbf{v}+\mathbf{u},\quad c(\mathbf{u}+\mathbf{v})=c\mathbf{u}+c\mathbf{v}$	Vectors obey the same commutative and distributive laws as numbers A1.1 §03 Algebra I.

STATED-EQUATION *u plus v is the vector of componentwise sums. c times v scales every component by c. w equals c one v one plus c two v two, and so on through c k v k: a linear combination of the vectors v one through v k. u minus v equals u plus minus one times v. A linear combination equals the matrix V, whose columns are the vectors, times the weight vector c. Addition is commutative and scalars distribute over sums.*

MEMORY AID Add tip-to-tail; scaling stretches (and a negative scalar flips).

METHOD

1. To test whether $\mathbf{w}$ is a combination of $\mathbf{v}_1,\ldots,\mathbf{v}_k$, write $c_1\mathbf{v}_1+\cdots+c_k\mathbf{v}_k=\mathbf{w}$ one component at a time.
2. Solve the resulting linear system for the weights A1.12 §03 Algebra I, LA.17; no solution means $\mathbf{w}$ is not in the span LA.7. Example: $\mathbf{w}=(5,1)$, $\mathbf{v}_1=(1,1)$, $\mathbf{v}_2=(1,-1)$: $c_1+c_2=5$, $c_1-c_2=1$, so $c_1=3$, $c_2=2$.

TERMS

$\mathbf{v}\in\mathbb{R}^n$ – an ordered list of n real numbers M.78 §01 Math Signs; an arrow from the origin LA.1
c, c_i – scalars (weights) of the combination
$\mathbf{0}$ – zero vector
Geometry – addition is the parallelogram rule (see LA.K1)
V, $\mathbf{c}$ – matrix with the $\mathbf{v}_i$ as columns; vector of weights A2.24 §04 Algebra II

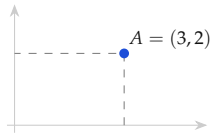
CONTEXT

- **Origin.** Grassmann (1844) and Hamilton; vectors in physics popularized by Gibbs and Heaviside (1880s).
- **Use.** Forces, velocities, displacements, colors (RGB), coordinates.
- **Data science.** A row of features is a vector; a linear model's prediction $\mathbf{w}\cdot\mathbf{x}$ is a weighted combination; word-embedding arithmetic (king − man + woman ≈ queen) is vector addition.
- **Physical application.** Two tugboats pulling a ship: the ship moves along the vector sum of their forces.

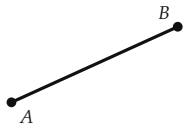
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Vectors: Points, Rays, Segments & Operations | *Key Topic LA.K1*

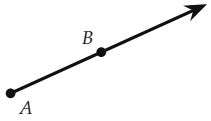
Geometry and linear algebra use similar pictures for different objects. A *point* is a location; a *segment*, *ray* and *line* are sets of points; a *vector* is a displacement, a length and a direction with no fixed starting place. Keep them apart, and the vector operations below (add, subtract, scale, measure, dot, cross, project) become pictures you can draw.



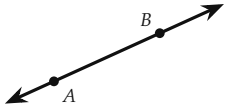
Point A
a location only:
no length, no direction



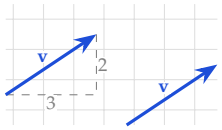
Segment $\overline{AB}$
two endpoints;
finite length AB



Ray $\overrightarrow{AB}$
starts at A, passes B,
goes on forever



Line $\overleftrightarrow{AB}$
no endpoints;
infinite both ways

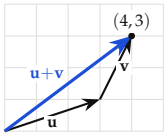


Vector $\mathbf{v} = \vec{v} = \langle 3,2 \rangle$
length + direction; any
parallel copy is the same $\mathbf{v}$

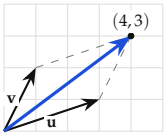
Notation. A point is written $A = (3,2)$; a vector is written in bold $\mathbf{v}$, with an arrow $\vec{v}$, or as $\langle 3,2 \rangle$ (angle brackets separate it from the point). The vector from A to B is $\overrightarrow{AB} = B - A$. In geometry, $\overrightarrow{AB}$ also names the *ray* from A through B, so read the context. Lengths: segment $AB = |\overline{AB}|$, vector $\|\mathbf{v}\|$. A point P and its *position vector* $\overrightarrow{OP}$ have the same numbers, but only the vector can be added and scaled.

The four basic operations

Addition: tip-to-tail and parallelogram



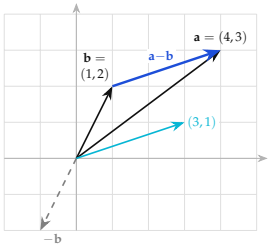
tip-to-tail



parallelogram

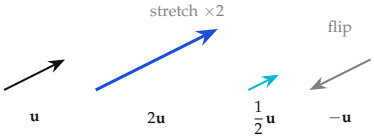
$\mathbf{u} + \mathbf{v} = (u_1+v_1,\ u_2+v_2)$.
Tail of $\mathbf{v}$ on the tip of $\mathbf{u}$; the sum runs from the first tail to the last tip.
Ex. $(3,1) + (1,2) = (4,3)$.

Subtraction: the vector from b to a



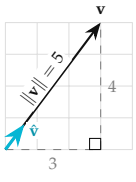
$\mathbf{a} - \mathbf{b} = \mathbf{a} + (-\mathbf{b})$: the arrow *from the tip of* $\mathbf{b}$ *to the tip of* $\mathbf{a}$ ("head minus tail").
Ex. $(4,3) - (1,2) = (3,1)$, drawn from the origin in cyan.

Scalar multiplication: stretch, shrink, flip



$c\mathbf{u} = (cu_1, cu_2)$: length $\times|c|$; $c < 0$ flips.
Ex. $\mathbf{u} = (2,1)$:
$2\mathbf{u} = (4,2)$, $\frac{1}{2}\mathbf{u} = (1, \frac{1}{2})$,
$-\mathbf{u} = (-2,-1)$.

Magnitude and unit vector

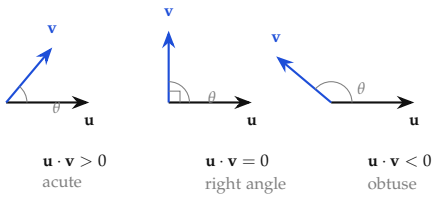


$\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2\ (+v_3^2)}$ (Pythagoras);
$\hat{\mathbf{v}} = \mathbf{v}/\|\mathbf{v}\|$ has length 1.
Ex. $\mathbf{v} = (3,4)$: $\|\mathbf{v}\| = 5$, $\hat{\mathbf{v}} = (\frac{3}{5}, \frac{4}{5})$; check $\frac{9}{25} + \frac{16}{25} = 1$.

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Products and projection

Dot product: a number that measures alignment LA.3

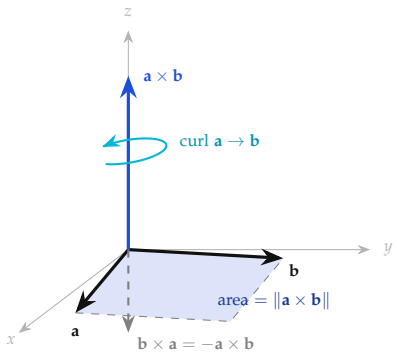


$\mathbf{u}\cdot\mathbf{v}>0$ acute | $\mathbf{u}\cdot\mathbf{v}=0$ right angle | $\mathbf{u}\cdot\mathbf{v}<0$ obtuse

Algebraic: $\mathbf{u}\cdot\mathbf{v}=u_1v_1+u_2v_2+u_3v_3$. **Geometric:** $\mathbf{u}\cdot\mathbf{v}=\|\mathbf{u}\|\|\mathbf{v}\|\cos\theta$. The sign gives the angle: positive ⇒ acute, zero ⇒ perpendicular, negative ⇒ obtuse. $\mathbf{v}\cdot\mathbf{v}=\|\mathbf{v}\|^2$.

Ex. $\mathbf{u}=(1,2)$, $\mathbf{v}=(3,1)$: $\mathbf{u}\cdot\mathbf{v}=3+2=5>0$; $\cos\theta=\frac{5}{\sqrt{5}\sqrt{10}}=\frac{1}{\sqrt{2}}$, so $\theta=45°$.

Cross product (3-D only): a vector perpendicular to both LA.5



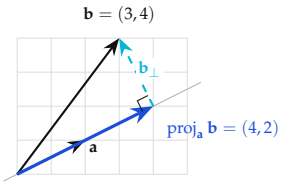
$$\mathbf{a}\times\mathbf{b}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\a_1&a_2&a_3\\b_1&b_2&b_3\end{vmatrix}=\langle a_2b_3-a_3b_2,\ a_3b_1-a_1b_3,\ a_1b_2-a_2b_1\rangle$$

Right-hand rule: fingers along $\mathbf{a}$, curl toward $\mathbf{b}$; the thumb points along $\mathbf{a}\times\mathbf{b}$. Swapping the order reverses it. **Length** $=\|\mathbf{a}\|\|\mathbf{b}\|\sin\theta=$ area of the parallelogram.

Ex. $(1,2,3)\times(4,5,6)=\langle 2\cdot6-3\cdot5,\ 3\cdot4-1\cdot6,\ 1\cdot5-2\cdot4\rangle=\langle-3,6,-3\rangle$. Check: $(1,2,3)\cdot\langle-3,6,-3\rangle=-3+12-9=0$ ✓.

Ex. $(3,1,0)\times(1,2,0)=\langle0,0,5\rangle$: points up $(+z)$, parallelogram area 5.

Projection: the shadow of b on the line of a LA.4



Scalar projection (signed length): $\text{comp}_{\mathbf{a}}\,\mathbf{b}=\dfrac{\mathbf{a}\cdot\mathbf{b}}{\|\mathbf{a}\|}$. **Vector projection:** $\text{proj}_{\mathbf{a}}\,\mathbf{b}=\dfrac{\mathbf{a}\cdot\mathbf{b}}{\mathbf{a}\cdot\mathbf{a}}\,\mathbf{a}$.

Ex. $\mathbf{a}=(2,1)$, $\mathbf{b}=(3,4)$: $\mathbf{a}\cdot\mathbf{b}=10$, $\mathbf{a}\cdot\mathbf{a}=5$, so $\text{proj}_{\mathbf{a}}\,\mathbf{b}=2(2,1)=(4,2)$ and $\text{comp}_{\mathbf{a}}\,\mathbf{b}=\frac{10}{\sqrt{5}}=2\sqrt{5}$. The remainder $\mathbf{b}_\perp=\mathbf{b}-\text{proj}_{\mathbf{a}}\,\mathbf{b}=(-1,2)$ is perpendicular to $\mathbf{a}$: $(2,1)\cdot(-1,2)=0$ ✓.

Common mistakes

Point ≠ vector. $(3,2)+(1,1)$ makes sense for vectors (displacements), not for two locations; $Q-P$ is a vector.

Wrong order in subtraction. $\overrightarrow{AB}=B-A$ (end minus start), not $A-B$.

Adding lengths. $\|\mathbf{u}+\mathbf{v}\|\le\|\mathbf{u}\|+\|\mathbf{v}\|$, with equality only when they point the same way: $\|(3,0)+(0,4)\|=5\neq7$.

Forgetting the square root in $\|\mathbf{v}\|$, or dividing by $\|\mathbf{a}\|$ instead of $\|\mathbf{a}\|^2$ in proj.

Dot gives a number, cross gives a vector. $\mathbf{a}\cdot\mathbf{b}$ is a scalar in any dimension; $\mathbf{a}\times\mathbf{b}$ is a vector and exists only in 3-D.

Middle sign of the cross product. The $\mathbf{j}$ term in the cofactor expansion carries a minus sign.

Order in the cross product. $\mathbf{b}\times\mathbf{a}=-\,\mathbf{a}\times\mathbf{b}$; the dot product is symmetric.

Normalizing 0. The zero vector has no direction, so $\hat{\mathbf{0}}$ is undefined.

DECISION POINT **Which product?** Need an *angle*, a test for perpendicularity, a projection or "how much of $\mathbf{b}$ goes along $\mathbf{a}$" (work, similarity) → **dot product**. Need a *direction perpendicular to two vectors* (a plane's normal, torque, rotation axis) or an *area / volume* → **cross product** (3-D only), or the scalar triple product for volume. Just combining displacements → addition and scaling.

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LA.3 Dot Product & the Angle Between Vectors

The dot product combines two vectors into a single number that measures how much they point the same way. It yields lengths, angles, perpendicularity and projections.

Algebraic dot product	$\mathbf{u}\cdot\mathbf{v}=\sum_{i=1}^{n}u_iv_i=u_1v_1+\cdots+u_nv_n$	Multiply matching components and add: $(1,2)\cdot(3,1)=5$.
Geometric dot product	$\mathbf{u}\cdot\mathbf{v}=\lVert\mathbf{u}\rVert\,\lVert\mathbf{v}\rVert\cos\theta$	Length times length times the cosine of the angle between them (law of cosines T.17 §06 Trigonometry).
Angle between vectors	$\theta=\arccos\dfrac{\mathbf{u}\cdot\mathbf{v}}{\lVert\mathbf{u}\rVert\,\lVert\mathbf{v}\rVert}$	Always $0\le\theta\le\pi$ T.14 §06 Trigonometry; for nonzero vectors only.
Sign test	$\mathbf{u}\cdot\mathbf{v}\ >0,\ =0,\ <0$	Acute, perpendicular (orthogonal), obtuse.
Length from the dot product	$\mathbf{v}\cdot\mathbf{v}=\lVert\mathbf{v}\rVert^2$	A vector dotted with itself is its squared length.
Algebraic properties	$\mathbf{u}\cdot\mathbf{v}=\mathbf{v}\cdot\mathbf{u},\quad\mathbf{u}\cdot(\mathbf{v}+\mathbf{w})=\mathbf{u}\cdot\mathbf{v}+\mathbf{u}\cdot\mathbf{w}$	Symmetric and distributive; scalars pull out: $(c\mathbf{u})\cdot\mathbf{v}=c(\mathbf{u}\cdot\mathbf{v})$.
Cauchy–Schwarz inequality	$\lvert\mathbf{u}\cdot\mathbf{v}\rvert\le\lVert\mathbf{u}\rVert\,\lVert\mathbf{v}\rVert$	Why $-1\le\cos\theta\le1$; equality exactly when the vectors are parallel.
Triangle inequality	$\lVert\mathbf{u}+\mathbf{v}\rVert\le\lVert\mathbf{u}\rVert+\lVert\mathbf{v}\rVert$	A side of a triangle is never longer than the other two together.
Cosine similarity	$\cos\theta=\hat{\mathbf{u}}\cdot\hat{\mathbf{v}}$	Dot product of the unit vectors: direction only, length ignored T.21 §06 Trigonometry.

STATED-EQUATION *u dot v equals the sum of u i v i. It also equals the length of u times the length of v times cosine theta, so theta equals the arc cosine of u dot v over the product of the lengths. A positive dot product means an acute angle, zero means perpendicular, negative means obtuse. v dot v equals the length of v squared. The dot product is symmetric and distributive. Cauchy-Schwarz: the absolute value of u dot v is at most the product of the lengths. The length of u plus v is at most the sum of the lengths. Cosine similarity is the dot product of the unit vectors.*

MEMORY AID "Dot" gives a *dot*-sized answer: a single number. Zero dot product = right angle.

METHOD

1. Compute $\mathbf{u}\cdot\mathbf{v}$ componentwise.
2. Compute $\lVert\mathbf{u}\rVert$ and $\lVert\mathbf{v}\rVert$ LA.1.
3. Divide and take arccos. Example: $\mathbf{u}=(1,2)$, $\mathbf{v}=(3,1)$: $\cos\theta=\frac{5}{\sqrt{5}\sqrt{10}}=\frac{1}{\sqrt{2}}$, $\theta=45°$.

TERMS

$\mathbf{u}\cdot\mathbf{v}$ – dot (inner, scalar) product; also $\mathbf{u}^\top\mathbf{v}$ LA.12
θ – angle between the vectors placed tail to tail
arccos – inverse cosine, output in $[0,\pi]$ T.14 §06 Trigonometry
$\sum$ – summation over components M.53 §01 Math Signs
Orthogonal – $\mathbf{u}\cdot\mathbf{v}=0$; the zero vector is orthogonal to everything
$\hat{\mathbf{u}}$ – unit vector LA.1

CONTEXT

- **Origin.** Grassmann's inner product (1844); the dot notation is Gibbs' (1881).
- **Use.** Angles, perpendicularity tests, work, components of forces.
- **Data science.** Cosine similarity ranks documents and embeddings; attention scores are dot products $\mathbf{q}\cdot\mathbf{k}$; every linear model scores $\mathbf{w}\cdot\mathbf{x}$.
- **Physical application.** Work is $W=\mathbf{F}\cdot\mathbf{d}$ P.6 §12 Physics: a force $\langle3,4\rangle$ N moving a cart $\langle2,0\rangle$ m does 6 J of work.

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LA.4 Scalar & Vector Projection

Projection splits a vector $\mathbf{b}$ into a part along $\mathbf{a}$ and a part perpendicular to $\mathbf{a}$. The parallel part is the "shadow" of $\mathbf{b}$ on the line through $\mathbf{a}$.

Scalar projection (component)	$\operatorname{comp}_{\mathbf{a}} \mathbf{b} = \dfrac{\mathbf{a}\cdot\mathbf{b}}{\lVert\mathbf{a}\rVert} = \lVert\mathbf{b}\rVert\cos\theta$	Signed length of the shadow; negative when the angle is obtuse.
Vector projection	$\operatorname{proj}_{\mathbf{a}} \mathbf{b} = \dfrac{\mathbf{a}\cdot\mathbf{b}}{\mathbf{a}\cdot\mathbf{a}}\,\mathbf{a}$	The shadow as a vector: a multiple of $\mathbf{a}$. Note $\mathbf{a}\cdot\mathbf{a} = \lVert\mathbf{a}\rVert^2$.
Orthogonal component	$\mathbf{b}_\perp = \mathbf{b} - \operatorname{proj}_{\mathbf{a}} \mathbf{b}$	What is left over; always $\mathbf{a}\cdot\mathbf{b}_\perp = 0$.
Unit-vector form	$\operatorname{proj}_{\mathbf{a}} \mathbf{b} = (\hat{\mathbf{a}}\cdot\mathbf{b})\,\hat{\mathbf{a}}$	Scalar projection times the unit direction.
Projection matrix onto a line	$P = \dfrac{\mathbf{a}\mathbf{a}^\top}{\mathbf{a}^\top\mathbf{a}}$	$P\mathbf{b} = \operatorname{proj}_{\mathbf{a}} \mathbf{b}$; rank 1, $P^2 = P$ [LA.16]; onto a subspace see [LA.20].

STATED-EQUATION *The scalar projection of b onto a equals a dot b over the length of a, which is the length of b times cosine theta. The vector projection of b onto a equals a dot b over a dot a, times a. The orthogonal component is b minus its projection. Equivalently the projection is a hat dot b, times a hat. The projection matrix onto the line of a is a a transpose over a transpose a.*

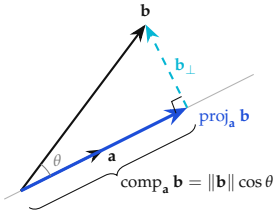
MEMORY AID "Dot over dot, times the direction": $\frac{\mathbf{a}\cdot\mathbf{b}}{\mathbf{a}\cdot\mathbf{a}}\,\mathbf{a}$; the vector you project *onto* appears three times.

METHOD

1. Compute $\mathbf{a}\cdot\mathbf{b}$ and $\mathbf{a}\cdot\mathbf{a}$.
2. Scale $\mathbf{a}$ by their ratio to get $\operatorname{proj}_{\mathbf{a}} \mathbf{b}$; subtract from $\mathbf{b}$ for $\mathbf{b}_\perp$.
3. Check $\mathbf{a}\cdot\mathbf{b}_\perp = 0$. Example: $\mathbf{a} = (2,1)$, $\mathbf{b} = (3,4)$: ratio $\frac{10}{5} = 2$, proj $= (4,2)$, $\mathbf{b}_\perp = (-1,2)$, $(2,1)\cdot(-1,2) = 0$.

TERMS

$\mathbf{a}$ – direction projected onto ($\mathbf{a} \neq \mathbf{0}$)
$\mathbf{b}$ – vector being projected
θ – angle between $\mathbf{a}$ and $\mathbf{b}$ [LA.3]
comp – scalar projection (a number)
proj – vector projection (a vector)
$\mathbf{a}\mathbf{a}^\top$ – outer product [LA.12]; $^\top$ transpose [LA.11]



CONTEXT

- **Origin.** Orthogonal decomposition goes back to Fourier's coefficients (1807); vector form by Gibbs.
- **Use.** Resolving forces into components, shadows, nearest point on a line.
- **Data science.** Regression through the origin on one feature has slope $\frac{\mathbf{x}\cdot\mathbf{y}}{\mathbf{x}\cdot\mathbf{x}}$: the projection coefficient of $\mathbf{y}$ onto $\mathbf{x}$.
- **Physical application.** On a ramp at angle α, gravity's component along the slope is $mg\sin\alpha$, a scalar projection.

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LA.5 Cross Product & Triple Products

In 3-D, the cross product $\mathbf{a} \times \mathbf{b}$ is a vector perpendicular to both $\mathbf{a}$ and $\mathbf{b}$, whose length is the area of the parallelogram they span and whose direction follows the right-hand rule.

Determinant formula	$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$	Expand along the top row LA.13; the $\mathbf{j}$ term gets a minus sign.
Component formula	$\mathbf{a} \times \mathbf{b} = \langle a_2 b_3 - a_3 b_2,\ a_3 b_1 - a_1 b_3,\ a_1 b_2 - a_2 b_1 \rangle$	Each component uses the *other* two indices in cyclic order (2-3, 3-1, 1-2).
Magnitude = area	$\|\mathbf{a} \times \mathbf{b}\| = \|\mathbf{a}\|\,\|\mathbf{b}\| \sin\theta$	Area of the parallelogram on $\mathbf{a}, \mathbf{b}$; half of it is the triangle's area.
Orthogonality (right-hand rule)	$(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{a} = (\mathbf{a} \times \mathbf{b}) \cdot \mathbf{b} = 0$	Perpendicular to both; fingers $\mathbf{a} \to \mathbf{b}$, thumb along $\mathbf{a} \times \mathbf{b}$.
Anticommutativity	$\mathbf{b} \times \mathbf{a} = -(\mathbf{a} \times \mathbf{b}), \quad \mathbf{a} \times \mathbf{a} = \mathbf{0}$	Order matters; parallel vectors give $\mathbf{0}$ ($\sin 0 = 0$).
Basis products	$\mathbf{i} \times \mathbf{j} = \mathbf{k}, \quad \mathbf{j} \times \mathbf{k} = \mathbf{i}, \quad \mathbf{k} \times \mathbf{i} = \mathbf{j}$	Cyclic order is positive; reversed order is negative.
Scalar triple product	$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \det \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix}$	Signed volume of the parallelepiped; 0 exactly when the three vectors are coplanar.
Vector triple product	$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = \mathbf{b}(\mathbf{a} \cdot \mathbf{c}) - \mathbf{c}(\mathbf{a} \cdot \mathbf{b})$	"BAC minus CAB"; note $\times$ is *not* associative.
Distributive law	$\mathbf{a} \times (\mathbf{b} + \mathbf{c}) = \mathbf{a} \times \mathbf{b} + \mathbf{a} \times \mathbf{c}$	Scalars pull out: $(c\mathbf{a}) \times \mathbf{b} = c(\mathbf{a} \times \mathbf{b})$.

STATED-EQUATION *a cross b is the determinant with i, j, k in the top row, the components of a in the second row and of b in the third. Its components are a two b three minus a three b two, a three b one minus a one b three, and a one b two minus a two b one. Its length is the length of a times the length of b times sine theta, the parallelogram area. It is perpendicular to both a and b. b cross a is minus a cross b, and a cross a is zero. i cross j is k, j cross k is i, k cross i is j. a dot b cross c is the determinant of the rows a, b, c: the signed volume. a cross b cross c equals b times a dot c minus c times a dot b. The cross product distributes over addition: a cross the sum of b and c equals a cross b plus a cross c.*

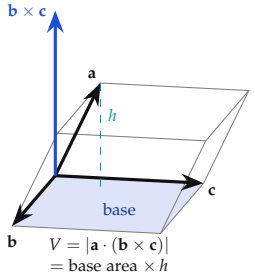
MEMORY AID Right hand: fingers along $\mathbf{a}$, curl to $\mathbf{b}$, thumb $= \mathbf{a} \times \mathbf{b}$. Triple product: "BAC − CAB".

METHOD

1. Write $\mathbf{i}, \mathbf{j}, \mathbf{k}$ over the rows $\mathbf{a}$ and $\mathbf{b}$.
2. Expand: $\mathbf{i}(a_2 b_3 - a_3 b_2) - \mathbf{j}(a_1 b_3 - a_3 b_1) + \mathbf{k}(a_1 b_2 - a_2 b_1)$.
3. Check by dotting the result with $\mathbf{a}$ and $\mathbf{b}$ (both must be 0). Example: $(1,2,3) \times (4,5,6) = \langle -3, 6, -3 \rangle$; area $= \sqrt{9 + 36 + 9} = 3\sqrt{6}$.

TERMS

$\mathbf{a}, \mathbf{b}, \mathbf{c}$ – vectors in $\mathbb{R}^3$ (the cross product is a 3-D operation)
$\mathbf{i}, \mathbf{j}, \mathbf{k}$ – standard unit vectors LA.1
$|\cdot|$ with rows – determinant LA.13, expanded formally
θ – angle between $\mathbf{a}$ and $\mathbf{b}$, $0 \le \theta \le \pi$
Parallelepiped – slanted box with edges $\mathbf{a}, \mathbf{b}, \mathbf{c}$
Coplanar – lying in one plane



CONTEXT

- **Origin.** Hamilton's quaternions (1843); the modern cross product from Gibbs and Heaviside (1880s).
- **Use.** Normals to planes, torque $\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}$, angular momentum, magnetic force $q\mathbf{v} \times \mathbf{B}$ EE.18 §16 Electrical Eng., curl V.5 §11 Vector Calculus.
- **Data science.** 3-D graphics and point-cloud processing compute surface normals with cross products for lighting and mesh orientation.
- **Physical application.** A 0.3 m wrench with a 50 N push at right angles gives torque $0.3 \cdot 50 \cdot \sin 90^\circ = 15\,\text{N}\cdot\text{m}$.

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Vector Spaces, Bases & Transformations

LA.6 Vector Spaces & Subspaces

A vector space is any set closed under addition and scalar multiplication. A subspace is a vector space inside another one; in $\mathbb{R}^n$ it is always a flat set through the origin (a point, line, plane, …).

Subspace test	$\mathbf{0} \in W; \quad \mathbf{u}, \mathbf{v} \in W \Rightarrow \mathbf{u} + \mathbf{v} \in W; \quad c\mathbf{u} \in W$	Three checks: contains zero, closed under addition, closed under scaling.
Span is a subspace	$\operatorname{span}\{\mathbf{v}_1, \ldots, \mathbf{v}_k\} \subseteq V$	The smallest subspace containing the $\mathbf{v}_i$ LA.7.
Homogeneous solutions	$\{\mathbf{x} : A\mathbf{x} = \mathbf{0}\}$	Always a subspace (the null space LA.19); $A\mathbf{x} = \mathbf{b} \neq \mathbf{0}$ gives a shifted copy, not a subspace.
Subspaces of $\mathbb{R}^3$	$\{\mathbf{0}\}$, lines through $\mathbf{0}$, planes through $\mathbf{0}$, $\mathbb{R}^3$	Dimensions 0, 1, 2, 3; nothing else.
Dimension of a sum	$\dim(W_1 + W_2) = \dim W_1 + \dim W_2 - \dim(W_1 \cap W_2)$	Inclusion–exclusion for dimensions A1.21 §03 Algebra I.
Orthogonal complement	$W^{\perp} = \{\mathbf{x} : \mathbf{x} \cdot \mathbf{w} = 0 \ \ \forall \mathbf{w} \in W\}$	Everything perpendicular to W; $\dim W + \dim W^{\perp} = n$.

STATED-EQUATION *W is a subspace if it contains the zero vector and is closed under addition and scalar multiplication. The span of any vectors is a subspace. The solutions of A x equals zero form a subspace. The subspaces of R three are the origin, lines and planes through the origin, and R three itself. The dimension of W one plus W two equals the sum of the dimensions minus the dimension of the intersection. W perp is the set of all x orthogonal to every w in W, and the dimensions of W and W perp add to n.*

MEMORY AID "Through the origin and flat." If $\mathbf{0}$ is missing, it is not a subspace.

METHOD

1. Check $\mathbf{0} \in W$ (fastest way to reject).
2. Take general $\mathbf{u}, \mathbf{v} \in W$ and check $\mathbf{u} + \mathbf{v}$ and $c\mathbf{u}$ still satisfy the defining condition. Example: $\{(x, y) : y = 2x\}$ passes; $\{(x, y) : y = 2x + 1\}$ fails since $(0, 0)$ is not on it.

TERMS

V – vector space (e.g. $\mathbb{R}^n$, polynomials, functions)
W, W_1, W_2 – subspaces; $W_1 + W_2$ all sums $\mathbf{w}_1 + \mathbf{w}_2$
$\in$, $\subseteq$, $\cap$ – element of, subset of, intersection A1.21 §03 Algebra I
$\forall$ – "for all" M.89 §01 Math Signs
dim – dimension LA.7; n = dimension of the whole space
$W^{\perp}$ – "W perp", orthogonal complement LA.3

CONTEXT

- **Origin.** Axiomatized by Peano (1888); abstract vector spaces by Banach and Weyl (1920s).
- **Use.** Solution sets, function spaces (Fourier series), polynomial spaces.
- **Data science.** A linear model's fitted values always lie in the column-space subspace of X; embeddings of related concepts cluster near low-dimensional subspaces.
- **Physical application.** Every force on a block sliding on a tabletop that does no vertical work lies in the horizontal plane, a 2-D subspace.

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LA.7 Span, Linear Independence, Basis & Dimension

The span of some vectors is everything their combinations can reach. They are independent if none is redundant. A basis is an independent spanning set, and its size is the dimension.

Span	$\operatorname{span}\{\mathbf{v}_1,\dots,\mathbf{v}_k\} = \{c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k\}$	All linear combinations LA.2.
Linear independence	$c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k = \mathbf{0} \Rightarrow c_1 = \cdots = c_k = 0$	Only the trivial combination gives $\mathbf{0}$; otherwise one vector is a combination of the others.
Dimension	$\dim V = \#$ vectors in any basis	Every basis of V has the same size; $\dim \mathbb{R}^n = n$.
Square determinant test	$\mathbf{v}_1,\dots,\mathbf{v}_n$ basis of $\mathbb{R}^n \iff \det[\mathbf{v}_1 \cdots \mathbf{v}_n] \neq 0$	Put the vectors as columns LA.13.
Counting rule	$k > n \Rightarrow$ dependent; $\quad k < n \Rightarrow$ cannot span $\mathbb{R}^n$	Too many vectors are redundant; too few miss directions.
Unique coordinates	$\mathbf{v} = c_1\mathbf{b}_1 + \cdots + c_n\mathbf{b}_n$ in exactly one way	A basis gives every vector one coordinate list LA.8.

STATED-EQUATION *The span of v one through v k is the set of all their linear combinations. The vectors are linearly independent if and only if the only combination equal to the zero vector is the one with every coefficient zero. The dimension of V is the number of vectors in any basis. n vectors form a basis of R n exactly when the determinant of the matrix with those columns is nonzero. More than n vectors in R n are dependent; fewer than n cannot span. In a basis every vector has exactly one set of coordinates.*

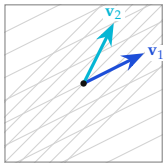
MEMORY AID A basis is a "minimal complete set": enough vectors to reach everything, none redundant. n vectors in $\mathbb{R}^n$ form a basis $\iff \det \neq 0$.

METHOD

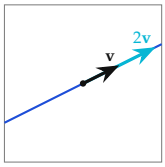
1. Place the vectors as the columns of a matrix.
2. Row reduce LA.17: a pivot in every column $\Rightarrow$ independent; a pivot in every row $\Rightarrow$ they span $\mathbb{R}^n$.
3. For n vectors in $\mathbb{R}^n$ a determinant check suffices. Example: $(1,2),(2,4)$: $\det = 4 - 4 = 0$, dependent (second $= 2\times$ first).

TERMS

Span – everything reachable by combinations LA.2
Dependent – some vector is a combination of the others (redundant)
Basis $\{\mathbf{b}_i\}$ – independent set that spans the space
Standard basis – $\mathbf{e}_1 = (1,0,\dots)$, $\mathbf{e}_2 = (0,1,\dots)$, …
Dimension – number of basis vectors; k = number of vectors given
Test – square matrix of the vectors has $\det \neq 0$ A2.25 §04 Algebra II



independent: span = plane



dependent: span = line

CONTEXT

- **Origin.** Formalized by Peano (1888) and Steinitz (1913).
- **Use.** Coordinates, degrees of freedom, solving systems.
- **Data science.** Multicollinearity is linear dependence among features; one-hot encoding with all categories plus an intercept is dependent (the "dummy-variable trap").
- **Physical application.** A point on a flat table needs 2 coordinates (dimension 2); a drone in the air needs 3.

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LA.8 Change of Basis

The same vector has different coordinates in different bases. A change-of-basis matrix converts coordinates, and $P^{-1}AP$ describes the same transformation in the new coordinates.

Change-of-basis matrix	$P = \begin{bmatrix} \mathbf{b}_1 & \mathbf{b}_2 & \cdots & \mathbf{b}_n \end{bmatrix}$	The new basis vectors, written in standard coordinates, as columns.
New → standard	$\mathbf{v} = P\,[\mathbf{v}]_B$	Multiply the new coordinates by P to get the ordinary coordinates.
Standard → new	$[\mathbf{v}]_B = P^{-1}\mathbf{v}$	Undo P; in practice solve $P[\mathbf{v}]_B = \mathbf{v}$.
Similar matrix	$[A]_B = P^{-1}AP$	The same transformation, seen in basis B.
Similarity invariants	det, tr, eigenvalues and rank of $P^{-1}AP$ = those of A	Basis-free properties do not change LA.24.

STATED-EQUATION *P is the matrix whose columns are the new basis vectors b one through b n. v equals P times the coordinates of v in basis B. The coordinates of v in B equal P inverse times v. The matrix of the transformation A in basis B equals P inverse A P. Similar matrices have the same determinant, trace, eigenvalues and rank.*

MEMORY AID "P takes you from the new basis to the old (standard) one; P^{-1} takes you back."

METHOD

1. Build P from the new basis vectors as columns.
2. Solve $P\mathbf{c} = \mathbf{v}$ for the new coordinates $\mathbf{c} = [\mathbf{v}]_B$. Example: $B = \{(1,1),(1,-1)\}$, $\mathbf{v} = (5,1)$: $c_1 + c_2 = 5$, $c_1 - c_2 = 1$, so $[\mathbf{v}]_B = (3,2)$.

TERMS

$B = \{\mathbf{b}_i\}$ – new basis LA.7
$[\mathbf{v}]_B$ – coordinates of $\mathbf{v}$ relative to B
P – change-of-basis matrix; invertible because its columns are independent LA.15
$P^{-1}AP$ – same transformation described in new coordinates (a similar matrix)
tr – trace, sum of the diagonal LA.11
Best basis – eigenvectors make A diagonal LA.25

CONTEXT

- **Origin.** Cayley and Sylvester's matrix theory (1850s).
- **Use.** Rotating coordinate frames, simplifying transformations.
- **Data science.** PCA is a change of basis to the principal axes ML.38 §15 Machine Learning; whitening, Fourier and wavelet transforms re-express data in a more useful basis.
- **Physical application.** Engineers describe forces on a ramp in tilted axes (along and perpendicular to the slope) instead of horizontal and vertical.

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LA.9 Linear Transformations & Their Matrices

A linear transformation preserves addition and scaling, so it maps lines to lines and the origin to itself. Every such map on $\mathbb{R}^n$ is multiplication by a matrix, whose columns are the images of the basis vectors.

Linearity	$T(a\mathbf{u}+b\mathbf{v}) = a\,T(\mathbf{u}) + b\,T(\mathbf{v})$	Preserves combinations; forces $T(\mathbf{0}) = \mathbf{0}$.
Matrix representation	$T(\mathbf{x}) = A\mathbf{x}$	Every linear map $\mathbb{R}^n \to \mathbb{R}^m$ is an $m \times n$ matrix.
Columns rule	$A = \begin{bmatrix} T(\mathbf{e}_1) & T(\mathbf{e}_2) & \cdots & T(\mathbf{e}_n) \end{bmatrix}$	Columns are where the basis vectors land.
Composition	$(S \circ T)(\mathbf{x}) = BA\,\mathbf{x}$	Do T (matrix A) first, then S (matrix B): the product reads right to left A2.1 §04 Algebra II.
Standard 2-D maps	$\underbrace{\begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix}}_{\text{scale}} \underbrace{\begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}}_{\text{shear}} \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}}_{\text{reflect in } x\text{-axis}}$	Rotations in LA.22.
Area scaling	$\text{area}(T(S)) = \lvert\det A\rvert \cdot \text{area}(S)$	Every region's area (volume) is multiplied by $\lvert\det A\rvert$ LA.13.
Kernel and image	$\ker T = \text{Null}\,A, \quad \text{im}\,T = \text{Col}\,A$	What is sent to $\mathbf{0}$, and everything that can be reached LA.19.

STATED-EQUATION *T of a u plus b v equals a T of u plus b T of v. T of x equals A x. The columns of A are T applied to the standard basis vectors e one through e n. Doing T then S corresponds to the product B A. Scaling, shear and reflection have simple two-by-two matrices. Areas are multiplied by the absolute value of det A. The kernel of T is the null space of A and the image is the column space.*

MEMORY AID "Columns are where the basis vectors land."

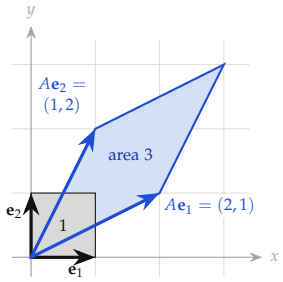
METHOD

1. Apply T to each standard basis vector $\mathbf{e}_1, \ldots, \mathbf{e}_n$.
2. Write the results as the columns of A, in order. Example: rotation by $90°$ sends $\mathbf{e}_1 \mapsto (0,1)$, $\mathbf{e}_2 \mapsto (-1,0)$, so $A = \left[\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right]$.

TERMS

Linear – preserves addition and scaling; so $T(\mathbf{0}) = \mathbf{0}$
$\mathbf{e}_i$ – standard basis LA.7
a, b – scalars; s_x, s_y scale factors; k shear factor
Examples – rotation G.24 §05 Geometry, reflection, scaling, projection, shear
Composition – matrix product, applied right to left LA.10
$\lvert\det A\rvert$ – factor by which areas/volumes scale A2.25 §04 Algebra II
s_x, s_y; k – scale factors; shear factor
S, B – a second transformation and its matrix
$\ker T$, $\text{im}\,T$ – kernel (inputs sent to $\mathbf{0}$); image (all outputs) LA.19

$A = \left[\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right]$, $\det A = 3$



CONTEXT

- **Origin.** Grassmann; matrices as transformations by Cayley (1858).
- **Use.** Computer graphics, robotics, signal processing.
- **Data science.** Each neural-network layer applies a linear map $W\mathbf{x}$ followed by a nonlinearity ML.22 §15 Machine Learning; image augmentation uses rotation, shear and scaling matrices.
- **Physical application.** Stretching a rubber sheet evenly in one direction is a linear transformation; every drawn square becomes a parallelogram.

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Matrices

LA.10 Matrix Multiplication & the Dimension Rule

The product AB is defined when A has as many columns as B has rows. Each entry is a row of A dotted with a column of B; equivalently, AB applies B first and then A.

Entry formula	$(AB)_{ij} = \sum_{k=1}^{n} a_{ik}\, b_{kj}$	Row i of A dotted with column j of B LA.3.
Dimension rule	$(m \times n)(n \times p) = m \times p$	Inner sizes must match; outer sizes give the shape of the product.
Matrix–vector, column view	$A\mathbf{x} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n$	A combination of the columns of A, weighted by the entries of $\mathbf{x}$ LA.2.
Matrix–vector, row view	$(A\mathbf{x})_i = \mathbf{r}_i \cdot \mathbf{x}$	Each output entry is one row of A dotted with $\mathbf{x}$.
Column-by-column	$AB = \begin{bmatrix} A\mathbf{b}_1 & A\mathbf{b}_2 & \cdots & A\mathbf{b}_p \end{bmatrix}$	Column j of AB is A times column j of B.
Identity	$I_m A = A I_n = A$	I has 1s on the diagonal and 0s elsewhere; it changes nothing.
Non-commutativity	$AB \neq BA$ in general	Order matters; BA may not even be defined.
Matrix powers	$A^k = \underbrace{AA\cdots A}_{k}, \quad A^0 = I$	Only for square A; not the entrywise powers.
Cost	$m \cdot n \cdot p$ multiplications	Two $n \times n$ matrices cost about n^3 O.10 §14 Optimization.

STATED-EQUATION *The i j entry of A B equals the sum over k of a i k times b k j: row i of A dotted with column j of B. An m by n matrix times an n by p matrix gives an m by p matrix. A x equals x one times column one plus x two times column two, and so on. Entry i of A x is row i of A dotted with x. Column j of A B is A times column j of B. The identity changes nothing. In general A B does not equal B A. A to the k is A multiplied by itself k times, and A to the zero is I. The cost is m n p multiplications.*

MEMORY AID "Row times column": slide across the row of A and down the column of B. "Inner must match, outer is the answer."

METHOD

1. Check the dimension rule: columns of A = rows of B.
2. Set up an $m \times p$ grid for the answer.
3. Fill entry (i, j) with row i of A dotted with column j of B. Example: $\left[\begin{smallmatrix}1&2\\3&4\end{smallmatrix}\right]\left[\begin{smallmatrix}5&6\\7&8\end{smallmatrix}\right] = \left[\begin{smallmatrix}19&22\\43&50\end{smallmatrix}\right]$ since $1\cdot 5 + 2\cdot 7 = 19$.

TERMS

a_{ik} – entry of A in row i, column k
$m \times n$ – m rows, n columns; p columns of B
$\mathbf{a}_j$, $\mathbf{b}_j$ – column j of A, of B
$\mathbf{r}_i$ – row i of A
I_n – $n \times n$ identity matrix
Full picture – see key topic LA.K2; basics in A2.24 §04 Algebra II

CONTEXT

- **Origin.** Arthur Cayley's matrix algebra (1858); Sylvester coined "matrix".
- **Use.** Composing transformations LA.9, solving systems, graph walks (A^k counts paths).
- **Data science.** A dense layer is $XW + \mathbf{b}$; a batch of n inputs is one $n \times d$ matrix times a $d \times h$ weight matrix; GPUs exist largely to do this fast.
- **Physical application.** Rotating and then scaling a robot arm's coordinates is a product of two transformation matrices.

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Matrix Multiplication | *Key Topic LA.K2*

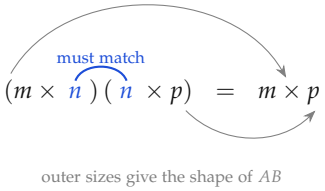
Matrix multiplication is not entry-by-entry. Each entry of AB is a *dot product* of a row of A with a column of B, so the rows of A must be as long as the columns of B. The picture below (Falk's scheme) places B above the answer and A to its left: every entry sits where its row and column meet.

$$B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}_{3\times 2}$$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}_{2\times 3} \quad \begin{bmatrix} 22 & 28 \\ 49 & 64 \end{bmatrix}_{2\times 2} = AB$$

$c_{12} = \underbrace{1\cdot 2}_{}+\underbrace{2\cdot 4}_{}+\underbrace{3\cdot 6}_{} = 28$ (row 1 of A · column 2 of B)

The dimension rule



Write the sizes side by side. If the two **inner** numbers agree, the product exists and its size is the two **outer** numbers.

- $(2\times 3)(3\times 2) = 2\times 2$, but $(3\times 2)(2\times 3) = 3\times 3$: same matrices, different sizes.
- $(2\times 2)(2\times 3) = 2\times 3$; the reverse $(2\times 3)(2\times 2)$ is **undefined** ($3 \neq 2$).
- $(1\times n)(n\times 1) = 1\times 1$: a dot product. $(n\times 1)(1\times n) = n\times n$: an outer product [LA.12].

Worked examples

2×2 times 2×2

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}\begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} = \begin{bmatrix} 1\cdot 5+2\cdot 7 & 1\cdot 6+2\cdot 8 \\ 3\cdot 5+4\cdot 7 & 3\cdot 6+4\cdot 8 \end{bmatrix} = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}$$

Reverse order: $\begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 23 & 34 \\ 31 & 46 \end{bmatrix} \neq AB.$

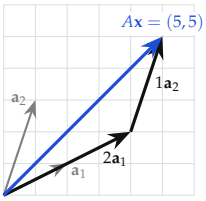
Non-commutative: $AB \neq BA$ in general, even when both exist.

2×3 times 3×2, and 2×2 times 2×3

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix} = \begin{bmatrix} 1{+}6{+}15 & 2{+}8{+}18 \\ 4{+}15{+}30 & 8{+}20{+}36 \end{bmatrix} = \begin{bmatrix} 22 & 28 \\ 49 & 64 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 & 2 \\ 3 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 7 & 2 & 4 \\ 3 & 1 & 1 \end{bmatrix} \quad (2\times 3 \text{ result})$$

Matrix × vector = combination of columns



$$\begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}\begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

Row view: $(2\cdot 2+1\cdot 1,\ 1\cdot 2+3\cdot 1) = (5,5)$.

Column view: $2\begin{bmatrix} 2 \\ 1 \end{bmatrix} + 1\begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \end{bmatrix}$.

Same answer; the column view shows $A\mathbf{x}$ lies in the column space [LA.19].

The identity and four views of AB

$I = \left[\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right]$: $IA = AI = A$, e.g. $\left[\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right]\left[\begin{smallmatrix} 1 & 2 \\ 3 & 4 \end{smallmatrix}\right] = \left[\begin{smallmatrix} 1 & 2 \\ 3 & 4 \end{smallmatrix}\right]$.

Entry: $(AB)_{ij} = \text{row}_i(A)\cdot\text{col}_j(B)$.
Columns: column j of AB is $A\,\mathbf{b}_j$.
Rows: row i of AB is $\text{row}_i(A)\,B$.
Outer products: $AB = \sum_k \mathbf{a}_k\mathbf{b}_k^\top$ (column k of A times row k of B).

Common mistakes

Multiplying entry by entry. That is the Hadamard product $A \odot B$ [LA.12], not AB.

Assuming $AB = BA$. Order matters; $(A+B)^2 = A^2+AB+BA+B^2$, not $A^2+2AB+B^2$.

Squaring entries for A^2. $A^2 = AA$; e.g. $\left[\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right]^2 = \left[\begin{smallmatrix} 1 & 2 \\ 0 & 1 \end{smallmatrix}\right]$.

Canceling. $AB = AC$ does not imply $B = C$ unless A is invertible; $AB = 0$ is possible with $A, B \neq 0$: $\left[\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right]\left[\begin{smallmatrix} 1 & -1 \\ -1 & 1 \end{smallmatrix}\right] = 0$.

Transposing a product. $(AB)^\top = B^\top A^\top$ (order reverses) [LA.11].

Skipping the size check. Write the sizes first; a mismatch means the product is undefined.

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DECISION POINT **Which view?** Computing numbers by hand → entry (row · column). Asking "what can $A\mathbf{x}$ reach?" (solvability, span, rank) → column view. Composing transformations → read right to left: $AB\mathbf{x}$ applies B first. Low-rank structure, SVD, attention → sum of outer products.

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LA.11 Matrix Algebra Rules & Identities

Matrices obey most of the rules of ordinary algebra, with two big exceptions: multiplication is not commutative, and transposes and inverses of products reverse the order. (Determinant rules are in LA.14.)

Identity & inverse	$AI = IA = A, \qquad AA^{-1} = A^{-1}A = I$	I is the "1" of matrices; A^{-1} exists only for square A with $\det A \neq 0$.
Associative & distributive	$A(BC) = (AB)C, \qquad A(B+C) = AB + AC$	Regroup freely, but keep the left-to-right order.
Not commutative	$AB \neq BA$ (in general)	Rotate-then-scale differs from scale-then-rotate LA.9.
Transpose	$(A^\top)_{ij} = a_{ji}, \qquad (A^\top)^\top = A$	Rows become columns; an $m \times n$ matrix becomes $n \times m$.
Transpose of sum & multiple	$(A+B)^\top = A^\top + B^\top, \qquad (cA)^\top = cA^\top$	Transpose is linear.
Transpose of a product	$(AB)^\top = B^\top A^\top$	Order reverses ("socks and shoes").
Inverse of a product	$(AB)^{-1} = B^{-1}A^{-1}$	Undo the last step first.
Inverse of transpose, multiple, inverse	$(A^\top)^{-1} = (A^{-1})^\top, \quad (cA)^{-1} = \frac{1}{c}A^{-1}, \quad (A^{-1})^{-1} = A$	Inverse and transpose commute; $c \neq 0$.
Trace rules	$\operatorname{tr}(A+B) = \operatorname{tr} A + \operatorname{tr} B, \qquad \operatorname{tr}(AB) = \operatorname{tr}(BA)$	Trace (sum of the diagonal) is linear and cyclic.
Gram matrix is symmetric	$(A^\top A)^\top = A^\top A$	$A^\top A$ and $AA^\top$ are always symmetric (and PSD LA.27).

STATED-EQUATION *A times the identity equals the identity times A equals A, and A times A inverse equals the identity. Multiplication is associative and distributive, but A B generally does not equal B A. The i j entry of A transpose is a j i, and the transpose of a transpose is A. The transpose of a sum is the sum of transposes, and the transpose of c A is c A transpose. The transpose of A B equals B transpose A transpose, and the inverse of A B equals B inverse A inverse. The inverse of A transpose equals the transpose of A inverse; the inverse of c A is one over c times A inverse; the inverse of A inverse is A. The trace is additive and the trace of A B equals the trace of B A. A transpose A is symmetric.*

MEMORY AID "Socks and shoes": to undo (or transpose) a product, reverse the order, $(AB)^{-1} = B^{-1}A^{-1}$.

TERMS

I – identity matrix: 1s on the diagonal, 0s elsewhere
$A^\top$ – transpose: rows become columns A2.24 §04 Algebra II
A^{-1} – inverse; exists iff $\det A \neq 0$ LA.15
a_{ij} – entry in row i, column j; c a nonzero scalar
$\operatorname{tr} A$ – trace: sum of diagonal entries $= \sum \lambda_i$ LA.23
Not commutative – order matters: rotate then scale $\neq$ scale then rotate LA.9

CONTEXT

- **Origin.** Arthur Cayley, "A Memoir on the Theory of Matrices" (1858).
- **Use.** Simplifying and rearranging every matrix equation.
- **Data science.** Deriving the OLS solution, writing neural-network layers as products, and the "trace trick" $\mathbf{x}^\top A\mathbf{x} = \operatorname{tr}(A\mathbf{x}\mathbf{x}^\top)$ used in expectations of quadratic forms.
- **Physical application.** In robotics, a chain of joint transformations $T = T_1T_2T_3$ must be multiplied in order; reversing it moves the arm somewhere else.

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LA.12 Inner, Outer, Hadamard, Kronecker & Block Products

Besides the ordinary matrix product there are several other ways to multiply vectors and matrices; knowing which one a formula (or a line of code) means avoids most shape bugs.

Inner product	$\mathbf{u}^\top \mathbf{v} = \sum_i u_i v_i$	$1 \times n$ times $n \times 1$: a scalar, the dot product LA.3.
Outer product	$\mathbf{u}\mathbf{v}^\top = [u_i v_j]$	$m \times 1$ times $1 \times n$: an $m \times n$ matrix of rank 1.
Hadamard (elementwise) product	$(A \odot B)_{ij} = a_{ij} b_{ij}$	Same-shape matrices, entry by entry.
Kronecker product	$A \otimes B = \begin{bmatrix} a_{11}B & \cdots & a_{1n}B \\ \vdots & \ddots & \vdots \\ a_{m1}B & \cdots & a_{mn}B \end{bmatrix}$	Each entry of A is replaced by that multiple of the whole block B.
Column view of Ax	$A\mathbf{x} = \sum_j x_j \, \mathbf{a}_j$	A combination of the columns of A LA.10.
Outer-product view of AB	$AB = \sum_k \mathbf{a}_k \mathbf{b}_k^\top$	Columns of A times rows of B, summed.
Block multiplication	$\begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} E \\ F \end{bmatrix} = \begin{bmatrix} AE + BF \\ CE + DF \end{bmatrix}$	Multiply blocks like entries when their sizes match (keep the order).

STATED-EQUATION *The inner product u transpose v is a scalar, the sum of u i v i. The outer product u v transpose is a matrix with entries u i v j. The Hadamard product multiplies entries elementwise. The Kronecker product replaces each entry a i j with the block a i j times B. A x is a combination of the columns of A weighted by the entries of x. A B is the sum of outer products of columns of A with rows of B. Block matrices multiply like matrices of numbers when the block sizes match.*

MEMORY AID Inner product "collapses" ($1 \times n$ times $n \times 1$ is 1×1); outer product "expands" ($n \times 1$ times $1 \times m$ is $n \times m$).

TERMS

Inner – dot product T.21 §06 Trigonometry
Outer – always rank 1 LA.19; SVD is a sum of these LA.28
Hadamard $\odot$ – same-shape elementwise product (`A*B` in NumPy)
Kronecker $\otimes$ – size $(mp) \times (nq)$ for $m \times n$ and $p \times q$
$\mathbf{a}_k$, $\mathbf{b}_k^\top$ – column k of A, row k of B
Block matrices – $A, \ldots, F$ here are sub-matrices of compatible sizes

CONTEXT

- **Origin.** Kronecker (1850s); Hadamard (1890s); the column and outer-product views popularized by Strang.
- **Use.** Organizing large computations, structured models.
- **Data science.** Attention scores $QK^\top$ and gating ($\sigma(\cdot) \odot \mathbf{h}$ in LSTMs) use these products; NumPy's `@` is matrix product while `*` is Hadamard, a common bug.
- **Physical application.** In quantum mechanics, the state of two combined systems is a Kronecker (tensor) product.

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LA.13 Determinants: Cofactor Expansion & Geometric Meaning

The determinant of a square matrix is one number that measures how the matrix scales area or volume (and whether it flips orientation). It is zero exactly when the matrix squashes space flat and has no inverse.

2×2 **determinant**	$\det\begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$	Main diagonal minus the other diagonal A2.25 §04 Algebra II.
Cofactor (Laplace) expansion	$\det A = \sum_{j=1}^{n} (-1)^{i+j} a_{ij} M_{ij}$	Along any row i (or column); each minor M_{ij} is one size smaller.
Checkerboard of signs	$(-1)^{i+j}:\ \begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$	The sign attached to each position in the expansion.
3×3 **rule of Sarrus**	Three "down-right" diagonals minus three "up-right" ones; only for 3×3.	
	$\det A = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33}$	
Triangular matrix	$\det A = a_{11}a_{22}\cdots a_{nn}$	Product of the diagonal; the fastest route is to row reduce to this form LA.14.
Geometric meaning	$\lvert\det A\rvert$ = area / volume scale factor	The unit square (cube) maps to a parallelogram (parallelepiped) of this size; $\det < 0$ means orientation flips.
Invertibility test	$\det A \neq 0 \iff A^{-1}$ exists	$\det A = 0$: columns dependent, space collapsed LA.7.

STATED-EQUATION *For a two-by-two matrix the determinant is a d minus b c. In general, expand along row i: the determinant is the sum over j of minus one to the i plus j, times a i j, times the minor M i j. The signs follow a checkerboard starting with plus. For three-by-three, Sarrus: add the three down-right diagonal products and subtract the three up-right ones. A triangular matrix's determinant is the product of its diagonal. The absolute value of the determinant is the factor by which area or volume is scaled, and a negative sign means orientation is reversed. A is invertible exactly when its determinant is nonzero.*

MEMORY AID Expand along the row or column with the *most zeros*. "$\det = 0 \iff$ flat $\iff$ no inverse."

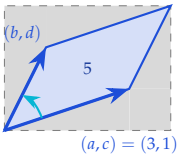
METHOD

1. Pick the row or column with the most zeros.
2. For each entry, cross out its row and column, take the determinant of what is left (the minor), and attach the checkerboard sign.
3. Sum entry $\times$ sign $\times$ minor. Example: $\begin{bmatrix} 2 & 0 & 1 \\ 1 & 3 & 2 \\ 1 & 1 & 2 \end{bmatrix}$ along row 1: $2(6-2) - 0 + 1(1-3) = 8 - 2 = 6$.

DECISION POINT **Which method?** 2×2: $ad - bc$. 3×3: Sarrus or a cofactor expansion along a row with zeros. 4×4 and larger: row reduce to triangular form (tracking swaps and scalings LA.14); cofactor expansion grows like $n!$ M.48 §01 Math Signs.

TERMS

a_{ij} – entry in row i, column j; n the size
M_{ij} – minor: determinant with row i and column j deleted
$C_{ij} = (-1)^{i+j} M_{ij}$ – cofactor
Orientation – $\det > 0$ keeps counterclockwise order of the columns; $\det < 0$ mirrors it
Singular – $\det A = 0$ (not invertible)



$\det\begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix} = 6 - 1 = 5$
$12 - 7 = 5$ (box minus gray)

CONTEXT

- **Origin.** Seki Takakazu (1683) and Leibniz (1693); Laplace's expansion (1772); Sarrus (1833).
- **Use.** Invertibility tests, areas and volumes, Cramer's rule, change of variables C3.19 §10 Calculus III.
- **Data science.** Normalizing flows track $\log\lvert\det J\rvert$ of each layer; the Gaussian density contains $\det\Sigma$; a near-zero $\det(X^\top X)$ flags multicollinearity.
- **Physical application.** A map drawn with a linear distortion of determinant 2 shows every field at twice its true area.

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LA.14 Determinant Properties & Row Operations

Determinants multiply, ignore transposes, and react predictably to each row operation, which makes row reduction the practical way to compute them.

Product rule	$\det(AB) = \det A \, \det B$	Scale factors multiply when maps are composed.
Transpose	$\det\left(A^{\top}\right) = \det A$	So every row rule is also a column rule.
Inverse	$\det\left(A^{-1}\right) = \dfrac{1}{\det A}$	Undoing a scaling divides by it.
Scalar multiple	$\det(cA) = c^n \det A$	All n rows are scaled; not $c \det A$.
Row swap	$R_i \leftrightarrow R_j : \ \det \to -\det$	Each swap flips the sign.
Row scaling	$R_i \to cR_i : \ \det \to c \det$	Scaling one row scales the determinant once.
Row replacement	$R_i \to R_i + kR_j : \ \det$ unchanged	The workhorse of elimination costs nothing LA.17.
Zero or repeated row	zero row or two equal rows $\Rightarrow \det A = 0$	Dependent rows mean a collapsed (flat) image.
Cramer's rule ($n \times n$)	$x_i = \dfrac{\det A_i}{\det A}$	A_i: replace column i by $\mathbf{b}$; elegant but slow for large n A2.25 §04 Algebra II.

STATED-EQUATION *The determinant of A B is det A times det B. The determinant of A transpose equals det A. The determinant of A inverse is one over det A. The determinant of c A is c to the n times det A. Swapping two rows changes the sign; multiplying a row by c multiplies the determinant by c; adding a multiple of one row to another leaves it unchanged. A zero row or two equal rows give determinant zero. Cramer's rule: x i equals det A i over det A.*

MEMORY AID "Swap flips, scale scales, replace keeps."

METHOD

1. Row reduce to upper-triangular form using replacements (free), swaps (each flips the sign) and scalings (divide them back out).
2. Multiply the diagonal and apply the sign. Example: $\left[\begin{smallmatrix}1&2&3\\2&5&3\\1&0&8\end{smallmatrix}\right]$: $R_2 - 2R_1$, $R_3 - R_1$, then $R_3 + 2R_2$ give diagonal $1, 1, -1$, so $\det = -1$.

TERMS

A, B – square $n \times n$ matrices; c a scalar
R_i – row i; k any scalar
A_i – A with column i replaced by the right-hand side $\mathbf{b}$ of $A\mathbf{x} = \mathbf{b}$
x_i – i-th unknown
Triangular – determinant is the diagonal product LA.13
c, k; n – scalars; the size of the $n \times n$ matrix

CONTEXT

- **Origin.** Cauchy proved the product rule (1812); Cramer's rule (1750).
- **Use.** Computing determinants efficiently, proving invertibility.
- **Data science.** log det of a covariance matrix is computed from its Cholesky factor LA.18 as $2\sum \log l_{ii}$, never by expansion.
- **Physical application.** Two successive lens magnifications of areas, $\times 2$ then $\times 3$, give $\times 6$: the product rule.

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LA.15 Matrix Inverse by Gauss–Jordan

The inverse A^{-1} undoes A. It exists exactly for square matrices with nonzero determinant, and it is computed by row reducing A side by side with the identity.

Definition	$AA^{-1} = A^{-1}A = I$	Only square matrices can have inverses; at most one.
Gauss–Jordan method	$\left[\,A \mid I\,\right] \xrightarrow{\text{row ops}} \left[\,I \mid A^{-1}\,\right]$	Reduce the left block to I; the right block becomes A^{-1}.
2×2 **inverse**	$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \dfrac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$	Swap the diagonal, negate the off-diagonal, divide by det A2.25 §04 Algebra II.
Adjugate formula	$A^{-1} = \dfrac{1}{\det A}\operatorname{adj} A, \qquad \operatorname{adj} A = C^{\top}$	Transposed cofactor matrix LA.13; useful in theory and for 3×3.
Solving with the inverse	$A\mathbf{x} = \mathbf{b} \Rightarrow \mathbf{x} = A^{-1}\mathbf{b}$	Clean on paper; numerically, solve by elimination instead LA.18.
Invertible matrix theorem		All the same fact, for square A.
	A^{-1} exists $\iff$ $\det A \neq 0$ $\iff$ $\operatorname{rank} A = n$ $\iff$ $\operatorname{Null} A = \{\mathbf{0}\}$ $\iff$ columns form a basis $\iff$ 0 is not an eigenvalue	

STATED-EQUATION *A times A inverse equals A inverse times A equals the identity. Row reduce the augmented matrix A given I until the left is I; the right is then A inverse. For two-by-two, swap a and d, negate b and c, and divide by a d minus b c. In general A inverse equals the adjugate over the determinant, where the adjugate is the transpose of the cofactor matrix. If A is invertible, A x equals b has the unique solution x equals A inverse b. A is invertible exactly when its determinant is nonzero, its rank is n, its null space is only zero, its columns are a basis, and zero is not an eigenvalue.*

MEMORY AID "Whatever turns A into I turns I into A^{-1}."

METHOD

1. Write $[A \mid I]$.
2. Row reduce LA.17: create pivots of 1 and zeros below, then zeros above.
3. If a row of zeros appears on the left, A is singular (stop).
4. Read A^{-1} on the right; check $AA^{-1} = I$. Example: $\left[\begin{smallmatrix} 1 & 2 & 1 & 0 \\ 3 & 7 & 0 & 1 \end{smallmatrix}\right] \xrightarrow{R_2-3R_1} \left[\begin{smallmatrix} 1 & 2 & 1 & 0 \\ 0 & 1 & -3 & 1 \end{smallmatrix}\right] \xrightarrow{R_1-2R_2} \left[\begin{smallmatrix} 1 & 0 & 7 & -2 \\ 0 & 1 & -3 & 1 \end{smallmatrix}\right]$.

DECISION POINT **Do you need A^{-1} at all?** To solve $A\mathbf{x} = \mathbf{b}$, eliminate or factor (LU, Cholesky, QR); forming A^{-1} costs more and loses accuracy. Compute an explicit inverse only for 2×2 work, closed-form derivations, or when the inverse itself is the object of interest (e.g. a precision matrix).

TERMS

I – identity matrix
$[A \mid I]$ – augmented matrix LA.17
C – cofactor matrix, $C_{ij} = (-1)^{i+j}M_{ij}$ LA.13
$\operatorname{adj} A$ – adjugate (classical adjoint)
n – size of A; rank LA.19; eigenvalue LA.23

CONTEXT

- **Origin.** Gauss–Jordan elimination: Jordan (1888) extended Gauss's method; Cayley defined the inverse matrix (1858).
- **Use.** Undoing transformations, closed-form solutions.
- **Data science.** The OLS formula $\hat{\beta} = (X^{\top}X)^{-1}X^{\top}\mathbf{y}$ is written with an inverse but computed by QR or Cholesky; the inverse covariance (precision) matrix encodes conditional independence in graphical models.
- **Physical application.** Converting a camera's pixel coordinates back to world coordinates applies the inverse of the projection-and-rotation matrix.

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LA.16 Special Matrices

Many matrices met in practice have structure (zeros in fixed places, symmetry, orthogonality) that makes them cheaper to store, faster to compute with, and easier to reason about.

Diagonal	$D = \operatorname{diag}(d_1, \ldots, d_n)$	Nonzeros only on the diagonal; $D^k = \operatorname{diag}(d_i^k)$, $D^{-1} = \operatorname{diag}(1/d_i)$.
Triangular	$\text{upper/lower; } \det = \prod a_{ii}$	Zeros below (or above) the diagonal; eigenvalues on the diagonal.
Symmetric & skew-symmetric	$A = A^\top, \qquad A^\top = -A$	Symmetric: real eigenvalues; skew: zero diagonal.
Orthogonal	$Q^\top Q = I$	Columns orthonormal; $Q^{-1} = Q^\top$ LA.22.
Idempotent	$P^2 = P$	Projections: applying twice changes nothing LA.20.
Nilpotent	$N^k = 0$ for some k	All eigenvalues are 0; e.g. $\left[\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right]^2 = 0$.
Permutation	$P\mathbf{x} = \text{reordered } \mathbf{x}$	The identity with rows reordered; orthogonal, $\det = \pm 1$.
Positive (semi)definite	$\mathbf{x}^\top A\mathbf{x} > 0 \ (\geq 0)$	Symmetric "bowl-shaped" matrices LA.27.

STATED-EQUATION *A diagonal matrix has nonzero entries only on the diagonal. A triangular matrix's determinant is the product of its diagonal entries. Symmetric means A equals A transpose; skew-symmetric means A transpose equals minus A. Orthogonal means Q transpose Q equals the identity. Idempotent means P squared equals P. Nilpotent means some power N to the k equals zero. A permutation matrix reorders rows. Positive definite means x transpose A x is positive for every nonzero x.*

MEMORY AID Orthogonal matrices are "rigid": they rotate or reflect without stretching, so their inverse is just the transpose.

TERMS

Diagonal – inverse is $1/d_{ii}$; powers are easy LA.25
Triangular – eigenvalues on the diagonal LA.23; solved by back substitution LA.17
Symmetric – real eigenvalues, orthogonal eigenvectors LA.25
Orthogonal – $\det = \pm 1$; preserves lengths and angles LA.20
Idempotent – projection matrices LA.20
Permutation – an identity with rows reordered; also orthogonal
Sparse – mostly zeros; stored and multiplied efficiently

CONTEXT

- **Origin.** Named and classified by Cayley, Sylvester, Hermite and Frobenius (19th century).
- **Use.** Exploiting structure makes computation faster and more stable.
- **Data science.** Covariance matrices are symmetric PSD LA.27; PCA loadings form an orthogonal matrix ML.38 §15 Machine Learning; the OLS hat matrix $H = X(X^\top X)^{-1}X^\top$ is idempotent; user–item matrices are sparse.
- **Physical application.** A rotation matrix describing a spacecraft's orientation is orthogonal: its inverse (to rotate back) is its transpose.

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Linear Systems

LA.17 Gaussian Elimination & Row Reduction

Any linear system is solved by applying three reversible row operations to its augmented matrix until it has a staircase (echelon) shape, then reading off or back-substituting the solution.

Augmented matrix	$A\mathbf{x} = \mathbf{b} \longleftrightarrow \left[\, A \mid \mathbf{b} \,\right]$	Coefficients with the right-hand side attached A1.12 §03 Algebra I.
Elementary row operations	$R_i \leftrightarrow R_j, \quad R_i \to cR_i \ (c \neq 0), \quad R_i \to R_i + cR_j$	Swap, scale, replace: none changes the solution set.
Row echelon form (REF)	zeros below each pivot; pivots step right	The staircase; solve by back substitution.
Reduced row echelon form (RREF)	pivots $= 1$, zeros above and below	Unique for each matrix; the solution can be read directly.
Number of solutions	unique / none / infinitely many	Pivot in every column; a row $[0 \cdots 0 \mid c \neq 0]$; a free variable.
Cost	$\approx \frac{2}{3}n^3$ operations	For n equations; back substitution adds only $\approx n^2$.

STATED-EQUATION *Row reduce the augmented matrix A given b to row echelon form. The three row operations: swap two rows; multiply a row by a nonzero c; add c times one row to another. In echelon form each pivot has zeros below it; in reduced form each pivot is one with zeros above and below. A pivot in every column gives one solution, a contradictory row gives none, and a free variable gives infinitely many. Elimination costs about two thirds n cubed operations.*

MEMORY AID Solutions: a pivot in every column → unique; a row $[0 \ \cdots \ 0 \mid c \neq 0]$ → none; a free variable → infinitely many.

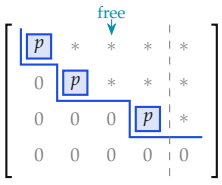
METHOD

1. Write $[A \mid \mathbf{b}]$.
2. Pick the leftmost column with a nonzero entry; swap a nonzero pivot to the top.
3. Subtract multiples of the pivot row to make zeros below the pivot.
4. Cover that row and repeat on the rows below until echelon form.
5. Back substitute (or continue to RREF); free variables become parameters. Example: $x+y+z = 6$, $2x+y-z = 1$, $x+2y+z = 8$: $R_2 - 2R_1$, $R_3 - R_1$ give $-y - 3z = -11$, $y = 2$, so $z = 3$, $x = 1$.

DECISION POINT **Which pivot?** By hand, choose a pivot equal to ± 1 (swap rows if needed) to avoid fractions. On a computer, choose the entry of largest absolute value (partial pivoting) to limit round-off. Stop at REF if you only need rank or solvability; go to RREF to write a general solution with free variables.

TERMS

Augmented matrix – coefficients with the right-hand side attached A1.12 §03 Algebra I
R_i – row i; c a scalar
Pivot – first nonzero entry of a row after elimination
Back substitution – solve from the bottom row up
Free variable – column without a pivot
Cost – in big-O terms $O(n^3)$ O.10 §14 Optimization; LU factorization LA.18



$p \neq 0$ pivots; $*$ any value;
column 3 has no pivot ⇒ free variable

CONTEXT

- **Origin.** Chinese text *Nine Chapters* (∼200 BCE); named after Gauss (1810).
- **Use.** Solving any linear system, computing inverses and ranks.
- **Data science.** Linear-regression solvers, circuit and network analysis EE.7 §16 Electrical Eng., and constraint systems behind LP O.3 §14 Optimization.
- **Physical application.** Balancing a chemical equation such as $a\,\mathrm{CH_4} + b\,\mathrm{O_2} \to c\,\mathrm{CO_2} + d\,\mathrm{H_2O}$ is a linear system in a, b, c, d.

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LA.18 LU & Cholesky Decompositions

Elimination can be recorded as a factorization: $A = LU$ with L lower- and U upper-triangular. Factor once, then each new right-hand side is solved by two cheap triangular solves. Symmetric positive definite matrices have the faster, stable Cholesky factor $A = LL^\top$.

LU factorization	$A = LU$	U is the echelon form; L is unit lower-triangular holding the multipliers l_{ij}.
With pivoting	$PA = LU$	Row swaps are collected in a permutation matrix P; always possible for square A.
Solving with LU	$L\mathbf{y} = \mathbf{b}$ (forward), $\quad U\mathbf{x} = \mathbf{y}$ (back)	Two triangular solves, $\approx n^2$ each.
Cholesky factorization	$A = LL^\top$	For symmetric positive definite A LA.27; L lower-triangular with positive diagonal.
Cholesky entries	$l_{jj} = \sqrt{a_{jj} - \sum_{k<j} l_{jk}^2}, \qquad l_{ij} = \frac{1}{l_{jj}}\left(a_{ij} - \sum_{k<j} l_{ik} l_{jk}\right) \quad (i > j)$	Column by column; a negative under the root means A is not PD.
Determinant via LU	$\det A = \pm \prod_i u_{ii}$	Sign $-$ for an odd number of row swaps LA.14.
Cost	$LU \approx \frac{2}{3}n^3, \quad$ Cholesky $\approx \frac{1}{3}n^3$	Cholesky does half the work by symmetry.

STATED-EQUATION *A equals L times U, with L unit lower triangular storing the elimination multipliers and U the upper triangular echelon form. With row swaps, P A equals L U. To solve A x equals b, solve L y equals b forward, then U x equals y backward. A symmetric positive definite A factors as L L transpose, with l j j equal to the square root of a j j minus the sum of squares of earlier entries in row j, and l i j equal to a i j minus the sum of l i k l j k, divided by l j j. The determinant is plus or minus the product of the pivots. LU costs about two thirds n cubed operations and Cholesky about one third n cubed.*

MEMORY AID LU = “elimination with a memory”. Cholesky = “square root of a matrix”.

METHOD

1. Eliminate below each pivot; store each multiplier $l_{ij} = a_{ij}/u_{jj}$ in L.
2. What remains is U. Example: $\left[\begin{smallmatrix}2&1\\6&5\end{smallmatrix}\right]$: $l_{21} = 3$, so $L = \left[\begin{smallmatrix}1&0\\3&1\end{smallmatrix}\right]$, $U = \left[\begin{smallmatrix}2&1\\0&2\end{smallmatrix}\right]$.
3. Cholesky example: $\left[\begin{smallmatrix}4&2\\2&5\end{smallmatrix}\right]$: $l_{11} = 2$, $l_{21} = \frac{2}{2} = 1$, $l_{22} = \sqrt{5-1} = 2$, so $L = \left[\begin{smallmatrix}2&0\\1&2\end{smallmatrix}\right]$.

DECISION POINT **Which factorization?** Symmetric positive definite (covariance, $X^\top X + \lambda I$, kernels) $\to$ Cholesky. General square $\to$ LU with partial pivoting. Rectangular or ill-conditioned least squares $\to$ QR LA.21, or SVD LA.28 if rank-deficient. Many right-hand sides with one A $\to$ factor once, reuse.

TERMS

L – lower-triangular (unit diagonal in LU)
U – upper-triangular (the echelon form LA.17)
P – permutation matrix LA.16
l_{ij}, u_{ij}, a_{ij} – entries of L, U, A
$\mathbf{y}$ – intermediate vector
SPD – symmetric positive definite LA.27

CONTEXT

- **Origin.** Turing (1948) described LU; Cholesky (c. 1910, published 1924) for geodesy.
- **Use.** Every linear solver in scientific computing; circuit simulation; finite elements.
- **Data science.** Sampling a multivariate normal: $\mathbf{x} = \boldsymbol{\mu} + L\mathbf{z}$ with $\Sigma = LL^\top$ and $\mathbf{z}$ standard normal; Gaussian processes factor their kernel matrix with Cholesky.
- **Physical application.** A bridge model with thousands of load cases factors its stiffness matrix once and back-substitutes for each load.

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LA.19 Rank, Column Space, Null Space & Rank–Nullity

The column space holds every output $A\mathbf{x}$ can produce; the null space holds every input sent to zero. The rank counts independent columns, and rank plus nullity always equals the number of columns.

Column space	$\operatorname{Col} A = \operatorname{span}\{\text{columns}\} = \{A\mathbf{x}\}$	All reachable outputs; $A\mathbf{x} = \mathbf{b}$ is solvable $\iff \mathbf{b} \in \operatorname{Col} A$.
Null space	$\operatorname{Null} A = \{\mathbf{x} : A\mathbf{x} = \mathbf{0}\}$	All inputs crushed to zero; adds freedom to every solution.
Rank	$\operatorname{rank} A = \dim \operatorname{Col} A = \dim \operatorname{Row} A$	Number of pivots; row rank always equals column rank.
Rank–nullity theorem	$\operatorname{rank} A + \dim \operatorname{Null} A = n$	Pivot columns plus free columns = all n columns.
Four fundamental subspaces	$\operatorname{Row} A \perp \operatorname{Null} A, \qquad \operatorname{Col} A \perp \operatorname{Null} A^\top$	Orthogonal complements LA.6, in $\mathbb{R}^n$ and $\mathbb{R}^m$.
General solution	$\mathbf{x} = \mathbf{x}_p + \mathbf{x}_n$	One particular solution plus anything in the null space.

STATED-EQUATION *The column space of A is the span of its columns, the set of all A x. The null space of A is the set of x with A x equal to zero. The rank of A is the dimension of the column space, which equals the dimension of the row space. Rank plus nullity equals n, the number of columns. The row space is orthogonal to the null space, and the column space is orthogonal to the null space of A transpose. Every solution is a particular solution plus a null-space vector.*

MEMORY AID "Rank = pivots; nullity = free variables; together = columns."

METHOD

1. Row reduce A to RREF LA.17; pivot columns of the *original* A form a basis of $\operatorname{Col} A$.
2. Write pivot variables in terms of free variables; each free variable gives one null-space basis vector. Example: $A = \left[\begin{smallmatrix} 1 & 2 \\ 2 & 4 \end{smallmatrix}\right] \to x_1 + 2x_2 = 0$, $\operatorname{Null} A = \operatorname{span}\{(-2, 1)\}$; rank 1+ nullity 1 = 2.

TERMS

Rank – number of pivots LA.17; number of independent columns (or rows)
Row A – span of the rows $= \operatorname{Col} A^\top$
Nullity – $\dim \operatorname{Null} A$ = number of free variables
Full rank – rank $= \min(m, n)$; square full-rank $\iff$ invertible LA.15
m, n – rows (equations), columns (unknowns)
$\mathbf{x}_p$, $\mathbf{x}_n$ – particular solution; null-space vector

CONTEXT

- **Origin.** Rank introduced by Frobenius (1879); rank–nullity is the fundamental theorem of linear algebra (Strang's name).
- **Use.** Counting independent equations and degrees of freedom.
- **Data science.** A rank-deficient design matrix means coefficients are not identifiable; low-rank structure is exploited by recommender systems and LoRA fine-tuning of LLMs.
- **Physical application.** A camera projecting 3-D space onto a 2-D image has a null space: every point along a line of sight maps to the same pixel.

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Orthogonality & Least Squares

LA.20 Orthogonality, Projections & Least Squares

When $A\mathbf{x} = \mathbf{b}$ has no exact solution, least squares chooses $\hat{\mathbf{x}}$ so that $A\hat{\mathbf{x}}$ is the point of the column space closest to $\mathbf{b}$: the orthogonal projection, whose error is perpendicular to every column.

Orthogonality test	$\mathbf{u} \perp \mathbf{v} \iff \mathbf{u} \cdot \mathbf{v} = 0$	Perpendicular vectors have zero dot product LA.3.
Projection onto a line	$\operatorname{proj}_{\mathbf{a}} \mathbf{b} = \dfrac{\mathbf{a} \cdot \mathbf{b}}{\mathbf{a} \cdot \mathbf{a}} \mathbf{a}$	One-dimensional case LA.4.
Least-squares problem	$\hat{\mathbf{x}} = \arg\min_{\mathbf{x}} \lVert A\mathbf{x} - \mathbf{b} \rVert^2$	Minimize the sum of squared residuals.
Normal equations	$A^\top A\,\hat{\mathbf{x}} = A^\top \mathbf{b}$	Set the residual orthogonal to the columns: $A^\top(\mathbf{b} - A\hat{\mathbf{x}}) = \mathbf{0}$.
Projection onto Col A	$\hat{\mathbf{b}} = A\left(A^\top A\right)^{-1} A^\top \mathbf{b}$	The fitted values; needs independent columns.
Projection matrix	$P = A(A^\top A)^{-1} A^\top, \quad P^2 = P = P^\top$	Idempotent and symmetric; $I - P$ projects onto the orthogonal complement.
Orthonormal columns	$Q^\top Q = I$	Mutually perpendicular unit columns.
Projection with orthonormal basis	$\hat{\mathbf{b}} = QQ^\top \mathbf{b} = \sum_i (\mathbf{q}_i \cdot \mathbf{b})\,\mathbf{q}_i$	No inverse needed: add the projections onto each $\mathbf{q}_i$.

STATED-EQUATION *u is perpendicular to v if and only if u dot v equals zero. The projection of b onto a equals a dot b over a dot a, times a. Least squares minimizes the squared length of A x minus b. The normal equations: A transpose A x hat equals A transpose b. The projection of b onto the column space is A times A transpose A inverse times A transpose b. The projection matrix P satisfies P squared equals P equals P transpose. An orthonormal matrix Q satisfies Q transpose Q equals the identity, and then the projection is Q Q transpose b.*

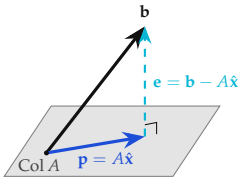
MEMORY AID Least squares: the error $\mathbf{b} - A\hat{\mathbf{x}}$ is perpendicular to every column of A.

METHOD

1. Write the model as $A\mathbf{x} \approx \mathbf{b}$ (one row per data point).
2. Form $A^\top A$ and $A^\top \mathbf{b}$.
3. Solve the normal equations. Example: fit $y = c + mx$ to $(0,2), (1,1), (2,6)$: $A^\top A = \left[\begin{smallmatrix} 3 & 3 \\ 3 & 5 \end{smallmatrix}\right]$, $A^\top \mathbf{b} = (9, 13)$, so $c = 1$, $m = 2$; residuals $1, -2, 1$ sum to 0.

TERMS

Orthogonal – dot product zero T.21 §06 Trigonometry
Orthonormal – orthogonal unit vectors; $Q^{-1} = Q^\top$ LA.22
$\hat{\mathbf{x}}$ – least-squares solution; arg min = the minimizing argument
Normal equations – least-squares solution S.40 §13 Statistics
Residual – $\mathbf{b} - A\hat{\mathbf{x}}$, orthogonal to Col A LA.19
$\mathbf{q}_i$ – columns of Q (Gram–Schmidt LA.21)



CONTEXT

- **Origin.** Least squares by Legendre (1805) and Gauss (1809); projections formalized by Hilbert and von Neumann.
- **Use.** Fitting data, shadows, signal denoising.
- **Data science.** OLS regression is the projection of $\mathbf{y}$ onto the column space of X; orthogonal features make coefficients independent of one another.
- **Physical application.** The shadow of a pole on flat ground at noon is its orthogonal projection onto the ground.

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LA.21 Gram–Schmidt & QR Factorization

Gram–Schmidt turns independent vectors into an orthonormal basis of the same span by subtracting from each vector its projections onto the earlier ones. Recorded as a factorization, it is $A = QR$.

First vector	$\mathbf{u}_1 = \mathbf{v}_1$	Start with the first vector as is.
Subtract the shadows	$\mathbf{u}_k = \mathbf{v}_k - \sum_{j=1}^{k-1} \frac{\mathbf{u}_j \cdot \mathbf{v}_k}{\mathbf{u}_j \cdot \mathbf{u}_j}\,\mathbf{u}_j$	Remove every component along the earlier directions LA.4.
Normalize	$\mathbf{q}_k = \frac{\mathbf{u}_k}{\lVert \mathbf{u}_k \rVert}$	Scale to length 1.
QR factorization	$A = QR$	Q has orthonormal columns; R is upper triangular.
Entries of R	$r_{ij} = \mathbf{q}_i \cdot \mathbf{a}_j \ (i \le j)$	How much of column j lies along $\mathbf{q}_i$.
Least squares via QR	$R\,\hat{\mathbf{x}} = Q^\top \mathbf{b}$	Back substitution; avoids forming $A^\top A$ LA.20.

STATED-EQUATION *u one equals v one. Each next u k equals v k minus its projections onto all the previous u j. Normalize: q k equals u k over its norm. The result factors A as Q times R, with Q orthonormal and R upper triangular, where r i j equals q i dot a j. Least squares then solves R x hat equals Q transpose b.*

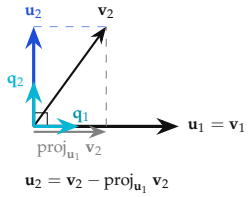
MEMORY AID "Subtract the shadows, then normalize."

METHOD

1. Set $\mathbf{u}_1 = \mathbf{v}_1$.
2. For each next $\mathbf{v}_k$, subtract its projection onto every earlier $\mathbf{u}_j$.
3. Normalize all $\mathbf{u}_k$. Example: $\mathbf{v}_1 = (1,1)$, $\mathbf{v}_2 = (2,0)$: $\mathbf{u}_2 = (2,0) - \frac{2}{2}(1,1) = (1,-1)$; $\mathbf{q}_1 = \frac{1}{\sqrt{2}}(1,1)$, $\mathbf{q}_2 = \frac{1}{\sqrt{2}}(1,-1)$.

TERMS

$\mathbf{v}_k$ – original independent vectors LA.7
$\mathbf{u}_k$ – orthogonal (not yet unit) vectors
$\mathbf{q}_k$ – orthonormal basis for the same span
Q – orthonormal columns; R upper triangular
$\mathbf{a}_j$ – column j of A (the $\mathbf{v}_j$)
Use – solve least squares stably: $R\hat{\mathbf{x}} = Q^\top \mathbf{b}$



CONTEXT

- **Origin.** Gram (1883) and Schmidt (1907); Laplace used it earlier.
- **Use.** Building orthonormal bases, numerically stable solving.
- **Data science.** `numpy.linalg.lstsq` and statsmodels use QR or SVD instead of forming $X^\top X$, which would square the condition number LA.29.
- **Physical application.** Setting up perpendicular axes on a tilted surveying site from two measured, non-perpendicular directions.

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LA.22 Orthogonal Matrices, Rotations & Reflections

An orthogonal matrix has orthonormal columns. It moves space rigidly (rotating or reflecting it) so lengths and angles are preserved and its inverse is simply its transpose.

Orthogonal matrix	$Q^\top Q = QQ^\top = I, \qquad Q^{-1} = Q^\top$	Square, with orthonormal columns (and rows).
Preserves geometry	$\|Q\mathbf{x}\| = \|\mathbf{x}\|, \qquad (Q\mathbf{x}) \cdot (Q\mathbf{y}) = \mathbf{x} \cdot \mathbf{y}$	Lengths, distances and angles are unchanged.
Determinant	$\det Q = \pm 1$	$+1$: rotation; -1: includes a reflection.
2-D rotation	$R(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$	Counterclockwise by θ G.24 §05 Geometry, T.8 §06 Trigonometry.
Composing rotations	$R(\alpha)R(\beta) = R(\alpha+\beta), \quad R(\theta)^{-1} = R(-\theta) = R(\theta)^\top$	Angles add; rotating back is the transpose.
2-D reflection	$F(\theta) = \begin{bmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{bmatrix}$	Reflects across the line through $\mathbf{0}$ at angle θ; $F^2 = I$.
3-D rotation about z	$R_z(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$	Rotates the xy-plane and leaves the z-axis fixed.
Householder reflection	$H = I - 2\hat{\mathbf{u}}\hat{\mathbf{u}}^\top$	Mirror across the plane perpendicular to $\hat{\mathbf{u}}$; the building block of stable QR.

STATED-EQUATION *Q transpose Q equals Q Q transpose equals the identity, so Q inverse is Q transpose. Q preserves lengths and dot products. Its determinant is plus or minus one. The two-dimensional rotation by theta has columns cosine theta, sine theta and minus sine theta, cosine theta. Rotation by alpha then beta is rotation by alpha plus beta, and the inverse rotation is the transpose. Reflection across the line at angle theta has entries cosine two theta, sine two theta, sine two theta, minus cosine two theta. Rotation about the z axis leaves z fixed. The Householder reflection is I minus two u hat u hat transpose.*

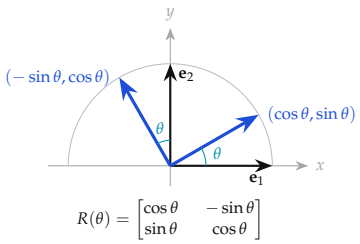
MEMORY AID Rotation matrix: "first column is where $\mathbf{e}_1$ lands: $(\cos\theta, \sin\theta)$."

METHOD

1. Write $R(\theta)$ with the angle (counterclockwise positive).
2. Multiply by the vector. Example: rotate $(2,1)$ by $90°$: $R = \left[\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right]$, giving $(-1,2)$; length $\sqrt{5}$ is unchanged.

TERMS

Q – orthogonal matrix (orthonormal columns LA.20)
θ, α, β – angles, counterclockwise positive T.2 §06 Trigonometry
R, F – rotation, reflection matrices
$\hat{\mathbf{u}}$ – unit normal of the mirror plane LA.1
I – identity matrix



CONTEXT

- **Origin.** Euler's rotation theorem (1775); Householder (1958).
- **Use.** Graphics, robotics, attitude of aircraft and spacecraft.
- **Data science.** PCA rotates data onto orthogonal axes ML.38 §15 Machine Learning; random rotations are used in data augmentation; orthogonal weight initialization keeps signal norms stable in deep networks.
- **Physical application.** A satellite's attitude is a 3×3 rotation matrix; un-rotating a camera image applies the transpose.

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Eigenvalues & Decompositions

LA.23 Eigenvalues & Eigenvectors

An eigenvector of A is a nonzero vector whose direction A does not change; A only stretches it by the eigenvalue λ. Eigenvalues are the roots of the characteristic polynomial.

Eigen-equation	$A\mathbf{v} = \lambda\mathbf{v} \quad (\mathbf{v} \neq \mathbf{0})$	A acts on $\mathbf{v}$ like the number λ.
Characteristic equation	$\det(A - \lambda I) = 0$	$A - \lambda I$ must be singular LA.13; a degree-n polynomial in λ.
Eigenvectors	$(A - \lambda I)\mathbf{v} = \mathbf{0}$	For each λ, the null space of $A - \lambda I$ LA.19.
Trace = sum	$\operatorname{tr} A = \sum_i \lambda_i$	Sum of the diagonal equals the sum of the eigenvalues.
Determinant = product	$\det A = \prod_i \lambda_i$	So $\det A = 0 \iff$ some $\lambda = 0$ M.54 §01 Math Signs.
2×2 **shortcut**	$\lambda^2 - (\operatorname{tr} A)\lambda + \det A = 0$	Quadratic in λ A1.15 §03 Algebra I.

STATED-EQUATION *A v equals lambda v, for a nonzero vector v. The eigenvalues solve the characteristic equation: the determinant of A minus lambda I equals zero. The eigenvectors for lambda solve A minus lambda I times v equals zero. The trace of A equals the sum of the eigenvalues, and the determinant equals their product. For two-by-two: lambda squared minus trace times lambda plus determinant equals zero.*

MEMORY AID "Eigen" is German for "own": an eigenvector keeps its own direction; λ is the stretch factor.

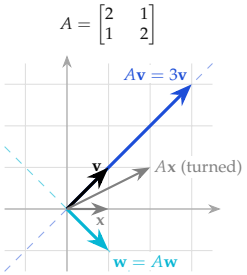
METHOD

1. Form $\det(A - \lambda I)$ and solve for λ (for 2×2, use trace and determinant).
2. For each λ, row reduce $A - \lambda I$ and solve $(A - \lambda I)\mathbf{v} = \mathbf{0}$.
3. Check $A\mathbf{v} = \lambda\mathbf{v}$. Example: $A = \left[\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right]$: $\lambda^2 - 4\lambda + 3 = 0$, $\lambda = 3, 1$; $(A - 3I)\mathbf{v} = \mathbf{0}$ gives $\mathbf{v} = (1, 1)$, and $\lambda = 1$ gives $(1, -1)$.

DECISION POINT **Shortcuts first.** Triangular or diagonal $\to$ read the diagonal. Every row sums to $s \to \lambda = s$ with $\mathbf{v} = (1, \ldots, 1)$. Singular $\to$ $\lambda = 0$ is one eigenvalue. Symmetric $\to$ all λ real. $2 \times 2 \to$ trace/determinant quadratic. Larger matrices in practice $\to$ numerical methods (QR algorithm, power iteration LA.24).

TERMS

λ – eigenvalue: how much $\mathbf{v}$ is stretched (negative flips it)
$\mathbf{v}$ – eigenvector; solve $(A - \lambda I)\mathbf{v} = \mathbf{0}$ LA.19
Characteristic polynomial – $\det(A - \lambda I)$ A2.25 §04 Algebra II
$\operatorname{tr} A$ – trace, the sum of the diagonal entries
Triangular A – eigenvalues are the diagonal entries
Complex λ – indicate rotation A2.6 §04 Algebra II



CONTEXT

- **Origin.** Euler and Lagrange (rotating bodies); Cauchy (1829); "eigen" by Hilbert (1904).
- **Use.** Vibration modes, stability, quantum mechanics, Markov chains.
- **Data science.** PCA ML.38 §15 Machine Learning, Google's PageRank (the dominant eigenvector of the link matrix), spectral clustering, and stability of RNN training.
- **Physical application.** A bridge's natural vibration frequencies are eigenvalues; soldiers break step on bridges so they don't excite one.

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LA.24 Eigenvalue Properties & Power Iteration

Eigenvalues transform simply under powers, inverses and shifts, are shared by similar matrices, and the largest one can be found by repeatedly multiplying by A.

Powers	$A^k\mathbf{v} = \lambda^k\mathbf{v}$	Same eigenvectors; eigenvalues raised to the power k.
Inverse	$A^{-1}\mathbf{v} = \frac{1}{\lambda}\mathbf{v}$	Invertible $\iff$ no eigenvalue is 0.
Shift	$(A + cI)\mathbf{v} = (\lambda + c)\mathbf{v}$	Adding cI shifts every eigenvalue by c (ridge regression adds λI).
Shared spectrum	$A,\ A^\top,\ P^{-1}AP$ have the same eigenvalues	Same characteristic polynomial LA.8.
Multiplicities	$1 \le \text{geometric} \le \text{algebraic}$	Distinct eigenvalues give independent eigenvectors; repeats may not LA.25.
Cayley–Hamilton	$p(A) = 0$	A satisfies its own characteristic polynomial $p(\lambda) = \det(A - \lambda I)$.
Power iteration	$\mathbf{x}_{k+1} = \dfrac{A\mathbf{x}_k}{\lVert A\mathbf{x}_k\rVert}$	Converges to the eigenvector of the largest $\lvert\lambda\rvert$, at rate $\lvert\lambda_2/\lambda_1\rvert$.
Rayleigh quotient	$R(\mathbf{x}) = \dfrac{\mathbf{x}^\top A\mathbf{x}}{\mathbf{x}^\top\mathbf{x}},\quad \lambda_{\min} \le R \le \lambda_{\max}$	For symmetric A; equals λ at an eigenvector.

STATED-EQUATION *A to the k times v equals lambda to the k times v. A inverse times v equals one over lambda times v. A plus c I times v equals lambda plus c times v. A, A transpose, and P inverse A P have the same eigenvalues. Geometric multiplicity is between one and the algebraic multiplicity. A satisfies its own characteristic polynomial. Power iteration repeatedly multiplies by A and normalizes. The Rayleigh quotient x transpose A x over x transpose x lies between the smallest and largest eigenvalues.*

MEMORY AID "Whatever you do to A, do to λ": $A^k \to \lambda^k$, $A^{-1} \to 1/\lambda$, $A + cI \to \lambda + c$.

METHOD

1. Start with any $\mathbf{x}_0$ (not orthogonal to the top eigenvector).
2. Repeat $\mathbf{x} \leftarrow A\mathbf{x}/\lVert A\mathbf{x}\rVert$; estimate $\lambda \approx R(\mathbf{x})$. Example: $A = \left[\begin{smallmatrix}2&1\\1&2\end{smallmatrix}\right]$, $\mathbf{x}_0 = (1,0)$: $(2,1)$, $(5,4)$, $(14,13)$, $\ldots \to (1,1)$ direction, $\lambda \to 3$.

TERMS

k – a power; c a scalar shift
Algebraic multiplicity – times λ repeats as a root of p
Geometric multiplicity – $\dim \text{Null}(A - \lambda I)$ LA.19
$p(\lambda)$ – characteristic polynomial LA.23
λ_1, λ_2 – largest and second-largest (in size) eigenvalues
$R(\mathbf{x})$ – Rayleigh quotient; quadratic form LA.27

CONTEXT

- **Origin.** Cayley (1858) and Hamilton; power iteration by von Mises (1929); Rayleigh (1877).
- **Use.** Stability analysis, numerical eigen-solvers, vibration estimates.
- **Data science.** PageRank is power iteration on the web's link matrix; ridge regression shifts the eigenvalues of $X^\top X$ by λ, improving conditioning ML.8 §15 Machine Learning.
- **Physical application.** Repeatedly pushing a swing at random times excites mostly its dominant mode, like power iteration.

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LA.25 Diagonalization & Matrix Powers

If A has n independent eigenvectors, switching to the eigenvector basis turns A into a diagonal matrix of eigenvalues, so powers and functions of A reduce to powers and functions of numbers.

Diagonalization	$A = PDP^{-1}$	Columns of P are eigenvectors; D holds the matching eigenvalues.
Eigenvalue matrix	$D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$	In the eigenbasis A just stretches each axis.
Matrix powers	$A^k = PD^kP^{-1}$	Only the diagonal is raised to the power.
Spectral theorem	$A = A^\top \Rightarrow A = Q\Lambda Q^\top$	Real symmetric: real eigenvalues, orthonormal eigenvectors.
When possible	n independent eigenvectors	Guaranteed if the n eigenvalues are distinct or A is symmetric.
Matrix exponential	$e^{At} = Pe^{Dt}P^{-1}$	Solves $\mathbf{x}' = A\mathbf{x}$: $\mathbf{x}(t) = e^{At}\mathbf{x}(0)$ C2.21 §08 Calculus II.

STATED-EQUATION *A equals P D P inverse, where the columns of P are eigenvectors and D is the diagonal matrix of eigenvalues. A to the k equals P D to the k P inverse. If A is symmetric, A equals Q Lambda Q transpose with Q orthonormal: the spectral theorem. A is diagonalizable when it has n independent eigenvectors. The matrix exponential e to the A t equals P e to the D t P inverse.*

MEMORY AID In the eigenbasis, a matrix just stretches each axis; powers become powers of numbers.

METHOD

1. Find eigenvalues and eigenvectors LA.23.
2. Put eigenvectors as columns of P and eigenvalues (same order) in D.
3. Compute $A^k = PD^kP^{-1}$. Example: $A = \left[\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right]$, $P = \left[\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right]$, $D = \operatorname{diag}(3, 1)$: $A^k = \frac{1}{2}\left[\begin{smallmatrix} 3^k+1 & 3^k-1 \\ 3^k-1 & 3^k+1 \end{smallmatrix}\right]$.

TERMS

P – eigenvector matrix LA.23; a change of basis LA.8
D, Λ – diagonal eigenvalue matrices
Diagonalizable – n independent eigenvectors
Spectral theorem – real symmetric matrices have real λ and orthonormal eigenvectors LA.20
e^{At} – matrix exponential (power series of At)
Steady state – Markov chain: eigenvector with $\lambda = 1$ LA.26

CONTEXT

- **Origin.** Jacobi (1846) for symmetric matrices; Jordan (1870) in general.
- **Use.** Solving linear recurrences and systems of ODEs, long-run behavior.
- **Data science.** PageRank and Markov-chain steady states; the long-run share of customers in each state of a churn model.
- **Physical application.** Coupled pendulums or masses on springs decouple into independent "normal modes" in the eigenbasis.

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LA.26 Markov Chains & Stochastic Matrices

A Markov chain moves between states with fixed transition probabilities. Its transition matrix multiplies the current distribution to give the next one, and the long-run distribution is an eigenvector for eigenvalue 1.

Transition matrix	$P_{ij} = \Pr(\text{next} = j \mid \text{now} = i), \quad \sum_j P_{ij} = 1$	Row-stochastic: nonnegative rows summing to 1 S.9 §13 Statistics.
Distribution update	$\boldsymbol{\pi}_{k+1} = \boldsymbol{\pi}_k P, \qquad \boldsymbol{\pi}_k = \boldsymbol{\pi}_0 P^k$	Row vector times P; $(P^k)_{ij}$ is the k-step probability.
Stationary distribution	$\boldsymbol{\pi} P = \boldsymbol{\pi}, \quad \sum_i \pi_i = 1$	A left eigenvector of P for $\lambda = 1$ LA.23.
Eigenvalue bound	$\lambda = 1$ always; $\quad \lvert\lambda\rvert \le 1$	Probabilities can neither grow nor vanish overall.
Convergence	P^k has all entries $> 0 \Rightarrow \boldsymbol{\pi}_k \to \boldsymbol{\pi}$	A regular chain forgets its start; speed set by $\lvert\lambda_2\rvert$.
Two-state chain	$P = \begin{bmatrix} 1-a & a \\ b & 1-b \end{bmatrix} \Rightarrow \boldsymbol{\pi} = \left(\frac{b}{a+b}, \frac{a}{a+b}\right)$	Long-run share is proportional to the flow *into* each state.

STATED-EQUATION *P i j is the probability of moving to state j given the chain is in state i, and each row sums to one. The next distribution is the current row vector times P, so pi k equals pi zero times P to the k. A stationary distribution satisfies pi P equals pi with entries summing to one. One is always an eigenvalue and no eigenvalue exceeds one in size. If some power of P is all positive, the chain converges to pi from any start. For two states with switching probabilities a and b, pi equals b over a plus b, a over a plus b.*

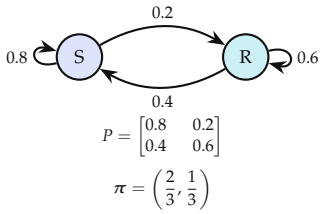
MEMORY AID "Rows sum to one; the steady state is the eigenvector for 1."

METHOD

1. Write $\boldsymbol{\pi} P = \boldsymbol{\pi}$ as equations, i.e. $(P^\top - I)\boldsymbol{\pi}^\top = \mathbf{0}$.
2. Replace one (redundant) equation by $\sum \pi_i = 1$ and solve. Example (figure): $a = 0.2$, $b = 0.4$: $\boldsymbol{\pi} = (\frac{0.4}{0.6}, \frac{0.2}{0.6}) = (\frac{2}{3}, \frac{1}{3})$; two days in three are sunny.

TERMS

P – transition (stochastic) matrix; i, j states
$\Pr(\cdot \mid \cdot)$ – conditional probability S.9 §13 Statistics
$\boldsymbol{\pi}_k$ – row vector of state probabilities at step k
$\boldsymbol{\pi}$ – stationary (steady-state) distribution
a, b – probabilities of leaving state 1, state 2
Regular – some power of P is strictly positive



$$P = \begin{bmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{bmatrix}$$

$$\boldsymbol{\pi} = \left(\frac{2}{3}, \frac{1}{3}\right)$$

CONTEXT

- **Origin.** Andrey Markov (1906); Perron–Frobenius theorem (1907–12) on positive matrices.
- **Use.** Queues, genetics, weather and credit-rating models, board games.
- **Data science.** PageRank, MCMC sampling 0.29 §14 Optimization, n-gram text models and customer-churn models are Markov chains.
- **Physical application.** A weather chain with $a = 0.2$, $b = 0.4$ (figure) spends $\frac{2}{3}$ of days sunny in the long run.

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LA.27 Symmetric, Positive Definite Matrices & Quadratic Forms

A quadratic form $\mathbf{x}^\top A\mathbf{x}$ is the matrix version of ax^2. A symmetric matrix is positive definite when this form is positive for every nonzero $\mathbf{x}$: its graph is a bowl with a unique minimum.

Quadratic form	$q(\mathbf{x}) = \mathbf{x}^\top A\mathbf{x} = \sum_{i,j} a_{ij}x_i x_j$	A degree-2 polynomial in the x_i; use symmetric A.
Positive definite	$A \succ 0 \iff \mathbf{x}^\top A\mathbf{x} > 0 \ \ \forall \mathbf{x} \neq \mathbf{0}$	The definition: a bowl (strictly convex).
Eigenvalue test	$A \succ 0 \iff$ all $\lambda_i > 0$	From $A = Q\Lambda Q^\top$ LA.25.
Sylvester's criterion	$A \succ 0 \iff$ all leading principal minors > 0	Check the top-left $1\times 1, 2\times 2, \ldots$ determinants.
2×2 **test**	$\begin{bmatrix} a & b \\ b & c \end{bmatrix} \succ 0 \iff a > 0,\ ac - b^2 > 0$	Sylvester's criterion for two variables.
Semidefinite & Gram matrices	$A \succeq 0 \iff \lambda_i \geq 0; \qquad X^\top X \succeq 0$	$\mathbf{x}^\top X^\top X\mathbf{x} = \|X\mathbf{x}\|^2 \geq 0$.

STATED-EQUATION *The quadratic form q of x equals x transpose A x, the sum over i and j of a i j x i x j. A is positive definite if and only if x transpose A x is positive for every nonzero x, if and only if all eigenvalues are positive, if and only if all leading principal minors are positive. For two by two: a positive and a c minus b squared positive. Positive semidefinite means all eigenvalues are nonnegative, and X transpose X is always positive semidefinite.*

MEMORY AID Positive definite = a bowl (unique minimum); indefinite = a saddle C3.16 §10 Calculus III.

METHOD

1. Make A symmetric (replace A by $\frac{1}{2}(A + A^\top)$; the form is unchanged).
2. Compute the leading principal minors (or eigenvalues).
3. All > 0: PD; all $\lambda \geq 0$: PSD; mixed signs: indefinite. Example: $\left[\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right]$: $2 > 0$, $4 - 1 = 3 > 0$, so PD.

TERMS

$A = A^\top$ – symmetric; a_{ij} entries; $\mathbf{x}$ any vector
$\succ 0$, $\succeq 0$ – positive definite, positive semidefinite (all $\lambda \geq 0$)
Leading principal minors – determinants of top-left $1\times 1, 2\times 2, \ldots$ blocks (Sylvester's criterion)
Indefinite – eigenvalues of both signs
Examples – covariance matrices S.38 §13 Statistics and $X^\top X$ are PSD

CONTEXT

- **Origin.** Sylvester (1852); quadratic forms studied by Lagrange and Gauss.
- **Use.** Energy functions, stability, second-derivative tests.
- **Data science.** Convexity of losses O.2 §14 Optimization, valid kernels and covariance matrices must be PSD; Cholesky factorization of a PD matrix samples correlated Gaussians LA.18.
- **Physical application.** The stiffness matrix of a stable structure is positive definite: every deformation stores positive energy.

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LA.28 Singular Value Decomposition (SVD)

Every matrix, square or not, factors as a rotation, a stretch along perpendicular axes, and another rotation. The stretch factors (singular values) rank the directions by importance.

SVD	$A = U\Sigma V^\top$	U ($m \times m$) and V ($n \times n$) orthogonal; Σ diagonal ($m \times n$).
Singular values	$\sigma_i = \sqrt{\lambda_i(A^\top A)}$	Square roots of the eigenvalues of $A^\top A$; $\sigma_1 \geq \sigma_2 \geq \cdots \geq 0$.
Singular vectors	$A\mathbf{v}_i = \sigma_i \mathbf{u}_i, \qquad A^\top A = V\Sigma^\top \Sigma V^\top$	$\mathbf{v}_i$: eigenvectors of $A^\top A$; $\mathbf{u}_i$: eigenvectors of $AA^\top$.
Best rank-k approximation	$A_k = \sum_{i=1}^{k} \sigma_i \, \mathbf{u}_i \mathbf{v}_i^\top$	Keep the k largest terms (Eckart–Young).
Spectral norm	$\|A\|_2 = \sigma_1$	The largest stretch of any unit vector LA.29.
Rank	$\operatorname{rank} A = \#\{\sigma_i > 0\}$	Numerically: count the singular values above a tolerance.

STATED-EQUATION *A equals U Sigma V transpose. The singular values sigma i are the square roots of the eigenvalues of A transpose A. A times v i equals sigma i times u i. The best rank-k approximation keeps the k largest terms, sigma i u i v i transpose. The spectral norm of A equals the largest singular value. The rank equals the number of nonzero singular values. A transpose A equals V times Sigma transpose Sigma times V transpose, so the columns of V are the eigenvectors of A transpose A.*

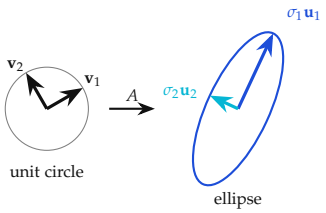
MEMORY AID Every matrix is "rotate, stretch, rotate" ($V^\top$, Σ, U).

METHOD

1. Compute $A^\top A$ and its eigenvalues λ_i and orthonormal eigenvectors $\mathbf{v}_i$.
2. Set $\sigma_i = \sqrt{\lambda_i}$ (largest first) and $\mathbf{u}_i = A\mathbf{v}_i/\sigma_i$. Example: $A = \left[\begin{smallmatrix} 3 & 0 \\ 4 & 5 \end{smallmatrix}\right]$: $A^\top A = \left[\begin{smallmatrix} 25 & 20 \\ 20 & 25 \end{smallmatrix}\right]$, $\lambda = 45, 5$, so $\sigma_1 = 3\sqrt{5}$, $\sigma_2 = \sqrt{5}$ (check: $\sigma_1\sigma_2 = 15 = |\det A|$).

TERMS

U, V – orthonormal matrices of left/right singular vectors $\mathbf{u}_i, \mathbf{v}_i$ LA.20
Σ – diagonal, $\sigma_1 \geq \sigma_2 \geq \cdots \geq 0$
Works for – any $m \times n$ matrix (unlike eigen-decomposition)
Eckart–Young – A_k minimizes $\|A - B\|$ over rank-k matrices B
k – number of terms kept; rank LA.19
σ_i; $\mathbf{u}_i, \mathbf{v}_i$ – singular values (largest first); left and right singular vectors
$\|A\|_2$ – spectral norm LA.29



CONTEXT

- **Origin.** Beltrami and Jordan (1870s); Eckart–Young (1936); Golub–Kahan algorithm (1965).
- **Use.** Compression, pseudo-inverses, noise reduction.
- **Data science.** PCA via SVD ML.38 §15 Machine Learning, latent semantic analysis, matrix-factorization recommenders (the Netflix Prize), and image compression.
- **Physical application.** A photo stored with only its top 50 singular values looks almost the same at a fraction of the size.

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LA.29 Norms, Trace & Condition Number

Norms measure the size of vectors and matrices in different ways; the condition number measures how much a linear system amplifies small errors in its data.

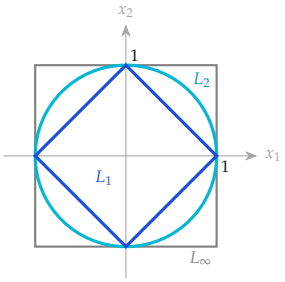
L_1 **norm**	$\|\mathbf{x}\|_1 = \sum_i \|x_i\|$	Sum of absolute values (Manhattan distance) M.43 §01 Math Signs.
L_2 **norm**	$\|\mathbf{x}\|_2 = \sqrt{\sum_i x_i^2}$	Ordinary length LA.1.
L_∞ **norm**	$\|\mathbf{x}\|_\infty = \max_i \|x_i\|$	Largest single component.
L_p **norm**	$\|\mathbf{x}\|_p = \left(\sum_i \|x_i\|^p\right)^{1/p}$	$p \geq 1$; includes the three above as $p = 1, 2, \infty$.
Frobenius norm	$\|A\|_F = \sqrt{\sum_{i,j} a_{ij}^2} = \sqrt{\operatorname{tr}\left(A^\top A\right)}$	Treat the matrix as one long vector; also $\sqrt{\sum \sigma_i^2}$.
Induced (operator) norm	$\|A\|_2 = \max_{\mathbf{x}\neq\mathbf{0}} \dfrac{\|A\mathbf{x}\|_2}{\|\mathbf{x}\|_2} = \sigma_{\max}$	The largest stretch LA.28.
Condition number	$\kappa(A) = \dfrac{\sigma_{\max}}{\sigma_{\min}} = \|A\|\,\|A^{-1}\|$	Relative error in $\mathbf{x}$ can be up to κ times the relative error in $\mathbf{b}$.
Trace	$\operatorname{tr} A = \sum_i a_{ii} = \sum_i \lambda_i$	Sum of the diagonal LA.23.

STATED-EQUATION *The L one norm is the sum of absolute values. The L two norm is the square root of the sum of squares. The infinity norm is the largest absolute entry. The L p norm is the p-th root of the sum of p-th powers. The Frobenius norm is the square root of the sum of squared entries, which equals the square root of the trace of A transpose A. The induced two-norm is the largest ratio of output length to input length, the largest singular value. The condition number kappa equals the largest singular value over the smallest. The trace is the sum of the diagonal, which equals the sum of the eigenvalues.*

MEMORY AID Large κ = "wobbly": tiny changes in the data cause big changes in the answer.

TERMS

Norm – length: positive, scales, triangle inequality LA.3
L_1 vs L_2 – lasso vs ridge penalties ML.8 §15 Machine Learning
p – norm order, $1 \leq p \leq \infty$
σ – singular values LA.28
κ – lose about $\log_{10} \kappa$ digits of accuracy when solving $A\mathbf{x} = \mathbf{b}$



CONTEXT

- **Origin.** Minkowski (1896) for L_p norms; condition number by Turing (1948).
- **Use.** Measuring error size, numerical stability.
- **Data science.** Multicollinear features make $X^\top X$ ill-conditioned (huge κ), so coefficients swing wildly; ridge regression lowers κ.
- **Physical application.** City-block (Manhattan) distance is the L_1 norm; a crow's flight is the L_2 norm.

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Matrix Calculus & Advanced Tools

LA.30 Matrix Calculus Identities

Gradients with respect to vectors and matrices follow the same patterns as one-variable derivatives, as long as shapes and transposes are tracked. These few identities derive most closed-form estimators.

Linear function	$\nabla_{\mathbf{x}}\left(\mathbf{a}^{\top}\mathbf{x}\right) = \mathbf{a}$	Like $\frac{d}{dx}(ax) = a$.
Quadratic form	$\nabla_{\mathbf{x}}\left(\mathbf{x}^{\top}A\mathbf{x}\right) = (A + A^{\top})\mathbf{x}$	$= 2A\mathbf{x}$ when A is symmetric; like $\frac{d}{dx}(ax^2) = 2ax$.
Squared norm	$\nabla_{\mathbf{x}}\|\mathbf{x}\|^2 = 2\mathbf{x}$	The special case $A = I$.
Least-squares gradient	$\nabla_{\beta}\,\|\mathbf{y} - X\beta\|^2 = -2X^{\top}(\mathbf{y} - X\beta)$	Set to zero: the normal equations LA.20.
Trace derivative	$\dfrac{\partial \operatorname{tr}(AB)}{\partial A} = B^{\top}$	Gradient with respect to a whole matrix has the shape of A.
Log-determinant derivative	$\dfrac{\partial \ln\det X}{\partial X} = X^{-\top}$	$X^{-\top} = (X^{-1})^{\top}$; appears in Gaussian likelihoods.

STATED-EQUATION *The gradient of a transpose x is a. The gradient of the quadratic form x transpose A x is A plus A transpose, times x, which is two A x when A is symmetric. The gradient of the squared norm of x is two x. The gradient of the squared residual norm, y minus X beta, is minus two X transpose times y minus X beta. The derivative of the trace of A B with respect to A is B transpose. The derivative of the log determinant of X is X inverse transpose.*

MEMORY AID Matrix calculus mirrors scalar calculus: $\frac{d}{dx}(ax) = a$, $\frac{d}{dx}(ax^2) = 2ax$; just watch transposes and shapes.

TERMS

$\nabla_{\mathbf{x}}$ – gradient with respect to a vector C3.12 §10 Calculus III
Shapes – gradient has the same shape as the variable
$\mathbf{a}$, A, B – constant vector, matrices; X data matrix; β coefficients
Quadratic form – LA.27
Residual – $\mathbf{y} - X\beta$ S.40 §13 Statistics
Log-determinant – appears in Gaussian log-likelihoods S.26 §13 Statistics

CONTEXT

- **Origin.** Developed for statistics and econometrics (Dwyer 1948; Magnus and Neudecker 1988).
- **Use.** Deriving estimators and gradients in closed form.
- **Data science.** Deriving OLS and ridge solutions ML.8 §15 Machine Learning, Gaussian MLE for covariance matrices, and backpropagation formulas for linear layers ML.22 §15 Machine Learning.
- **Physical application.** The stiffness-weighted energy $\frac{1}{2}\mathbf{x}^{\top}K\mathbf{x}$ of a structure has gradient $K\mathbf{x}$, the restoring force vector.

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LA.31 Pseudoinverse & Inverse-Update Identities

The pseudoinverse generalizes the inverse to any matrix and returns the least-squares, minimum-norm solution. Sherman–Morrison and Woodbury update a known inverse after a low-rank change without starting over.

Pseudoinverse (full column rank)	$A^+ = (A^\top A)^{-1} A^\top$	The least-squares solver as a matrix: $\hat{\mathbf{x}} = A^+\mathbf{b}$ LA.20.
Pseudoinverse via SVD	$A^+ = V\Sigma^+ U^\top$	Invert the nonzero singular values; works for any matrix LA.28.
Sherman–Morrison formula	$(A + \mathbf{u}\mathbf{v}^\top)^{-1} = A^{-1} - \dfrac{A^{-1}\mathbf{u}\mathbf{v}^\top A^{-1}}{1 + \mathbf{v}^\top A^{-1}\mathbf{u}}$	Rank-1 update of an inverse; requires $1 + \mathbf{v}^\top A^{-1}\mathbf{u} \neq 0$.
Woodbury identity	$(A + UCV)^{-1} = A^{-1} - A^{-1}U\left(C^{-1} + VA^{-1}U\right)^{-1}VA^{-1}$	Rank-k update: invert a small $k \times k$ matrix instead of $n \times n$.

STATED-EQUATION *For a matrix with full column rank, the Moore-Penrose pseudoinverse A plus equals A transpose A inverse, times A transpose. In general, A plus equals V Sigma plus U transpose, inverting the nonzero singular values. The Sherman-Morrison formula: the inverse of A plus u v transpose equals A inverse minus A inverse u v transpose A inverse, over one plus v transpose A inverse u. The Woodbury identity generalizes it to a low-rank update U C V. Explicitly, the inverse of A plus U C V equals A inverse minus A inverse U, times the inverse of C inverse plus V A inverse U, times V A inverse.*

MEMORY AID A rank-1 change to a matrix needs only a rank-1 correction to its inverse: $O(n^2)$ instead of $O(n^3)$.

TERMS

A^+ – pseudoinverse: gives the least-squares (minimum-norm) solution $\mathbf{x} = A^+\mathbf{b}$ LA.20
Σ^+ – replace each nonzero σ_i by $1/\sigma_i$ and transpose LA.28
$\mathbf{u}\mathbf{v}^\top$ – rank-1 update LA.12
U, C, V – $n \times k$, $k \times k$, $k \times n$ update factors (Woodbury)
Cost – O.10 §14 Optimization

CONTEXT

- **Origin.** Moore (1920), Penrose (1955); Sherman–Morrison (1949–50); Woodbury (1950).
- **Use.** Solving non-square and singular systems; fast updates.
- **Data science.** `numpy.linalg.pinv` and `lstsq`; recursive least squares and online regression update inverses one data point at a time; Gaussian-process and Kalman-filter computations use Woodbury.
- **Physical application.** GPS receivers solve an overdetermined system of satellite ranges with the pseudoinverse (least squares).

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10
Calculus III

Calculus in three dimensions and beyond: vectors, lines and surfaces in space; curves and motion; partial derivatives, gradients and tangent planes; optimization with and without constraints; and double and triple integrals. Gradients drive every optimizer, and Jacobians and Hessians appear throughout machine learning.

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Space, Vectors & Surfaces

C3.1 3-D Coordinates, Distance & Spheres

A point in space is an ordered triple (x, y, z) measured along three mutually perpendicular axes. Distance is the Pythagorean theorem applied twice; a sphere is every point at a fixed distance from a center.

Distance formula	$d = \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2 + (z_2-z_1)^2}$	Pythagoras across the floor diagonal, then up the height G.12 §05 Geometry.
Midpoint	$M = \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}, \frac{z_1+z_2}{2}\right)$	Average each coordinate separately.
Sphere (standard form)	$(x-h)^2 + (y-k)^2 + (z-l)^2 = r^2$	All points at distance r from the center (h, k, l).
Sphere (general form)	$x^2 + y^2 + z^2 + Gx + Hy + Iz + J = 0$	Complete the square in x, y and z A1.16 §03 Algebra I to read off center and radius; the result may be a point or empty.
Coordinate planes	$z = 0$ (xy), $\quad x = 0$ (yz), $\quad y = 0$ (xz)	In 3-D a single equation is a surface: $x = 3$ is a plane, not a line.

STATED-EQUATION *The distance between two points is the square root of the sum of the squared differences in x, y and z. The midpoint averages each coordinate. A sphere with center h, k, l and radius r is x minus h squared plus y minus k squared plus z minus l squared equals r squared. In general form a sphere is x squared plus y squared plus z squared plus G x plus H y plus I z plus J equals zero; complete the squares to find the center and radius. The coordinate planes are z equals zero, x equals zero and y equals zero.*

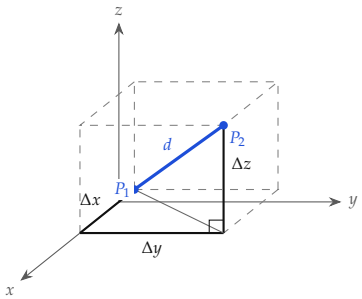
MEMORY AID In 3-D: one equation → a surface; two equations → a curve; three → points.

METHOD

1. Group the x, y and z terms; move the constant to the right.
2. Complete the square in each variable, adding the same amounts to the right side.
3. Read the center and r^2. Example: $x^2 + y^2 + z^2 - 2x + 4z = 4 \Rightarrow (x-1)^2 + y^2 + (z+2)^2 = 9$: center $(1, 0, -2)$, $r = 3$.

TERMS

(x, y, z) – coordinates of a point; (x_i, y_i, z_i) for P_i
(h, k, l), r – center and radius of a sphere
G, H, I, J – constants of the expanded (general) form
Right-handed axes – curl the fingers of the right hand from $+x$ to $+y$; the thumb points along $+z$
Octant – one of the 8 regions cut by the coordinate planes
Trace – intersection of a surface with a plane
d, M – distance; midpoint



CONTEXT

- **Origin.** Clairaut's study of curves in space (1731) and Euler's analytic solid geometry (1748).
- **Use.** Computer graphics, CAD, robotics, navigation.
- **Data science.** Euclidean distance in $\mathbb{R}^n$, the same formula with n terms, drives k-NN and k-means ML.11 §15 Machine Learning.
- **Physical application.** GPS: each satellite signal places the receiver on a sphere; intersecting several spheres pins down the position.

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C3.2 Vectors in Space: Dot & Cross Products

A vector in $\mathbb{R}^3$ has a length and a direction. The dot product measures alignment (a scalar); the cross product builds a vector perpendicular to both factors whose length is the area of the parallelogram they span. Vector basics: T.21 §06 Trigonometry, LA.K1 §09 Linear Algebra.

Magnitude & unit vector	$\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2 + v_3^2}, \qquad \hat{\mathbf{v}} = \dfrac{\mathbf{v}}{\|\mathbf{v}\|}$	Dividing by the length keeps the direction and makes the length 1.
Dot product	$\mathbf{u} \cdot \mathbf{v} = u_1v_1 + u_2v_2 + u_3v_3 = \|\mathbf{u}\|\|\mathbf{v}\| \cos\theta$	Zero exactly when $\mathbf{u} \perp \mathbf{v}$; positive when they point the same general way.
Vector projection	$\text{proj}_{\mathbf{v}}\,\mathbf{u} = \dfrac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{v}\|^2}\,\mathbf{v}$	The shadow of $\mathbf{u}$ on the line of $\mathbf{v}$; its signed length is $\text{comp}_{\mathbf{v}}\,\mathbf{u} = \frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{v}\|}$.
Cross product	Only in 3-D. Expand along the first row; the result is perpendicular to both $\mathbf{u}$ and $\mathbf{v}$. $\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} = (u_2v_3 - u_3v_2,\ u_3v_1 - u_1v_3,\ u_1v_2 - u_2v_1)$	
Length of the cross product	$\|\mathbf{u} \times \mathbf{v}\| = \|\mathbf{u}\|\|\mathbf{v}\| \sin\theta$	Area of the parallelogram spanned by $\mathbf{u}, \mathbf{v}$; zero exactly when they are parallel.
Cross-product algebra	$\mathbf{v} \times \mathbf{u} = -\mathbf{u} \times \mathbf{v}, \qquad \mathbf{i} \times \mathbf{j} = \mathbf{k}$	Anticommutative and *not* associative; direction by the right-hand rule.
Scalar triple product	$\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w}) = \det \begin{bmatrix} \mathbf{u} \\ \mathbf{v} \\ \mathbf{w} \end{bmatrix}$	Signed volume of the parallelepiped; zero exactly when the three vectors are coplanar A2.25 §04 Algebra II.
Triangle area	$A = \frac{1}{2}\|\overrightarrow{PQ} \times \overrightarrow{PR}\|$	Half the parallelogram on two edge vectors.

STATED-EQUATION *The length of v is the square root of the sum of its squared components. u dot v equals u one v one plus u two v two plus u three v three, which equals the lengths times cosine theta. The projection of u on v is u dot v over the squared length of v, times v. u cross v is the determinant with i, j, k on top; its length is the product of the lengths times sine theta. v cross u is minus u cross v. The triple product u dot v cross w is the determinant of the three vectors: the volume of the parallelepiped. In components, u cross v equals u two v three minus u three v two, u three v one minus u one v three, u one v two minus u two v one. i cross j equals k. The area of triangle P Q R is one half the length of P Q cross P R.*

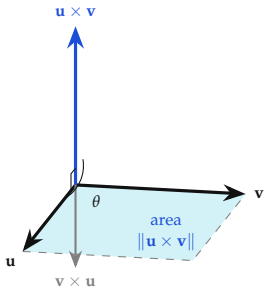
MEMORY AID Cycle $\mathbf{i} \to \mathbf{j} \to \mathbf{k} \to \mathbf{i}$: forward gives $+$ ($\mathbf{j} \times \mathbf{k} = \mathbf{i}$), backward gives $-$ ($\mathbf{k} \times \mathbf{j} = -\mathbf{i}$). Dot $\to$ cosine (alignment); cross $\to$ sine (area).

METHOD

1. Write $\mathbf{u}$ and $\mathbf{v}$ as rows 2 and 3 under $\mathbf{i}, \mathbf{j}, \mathbf{k}$.
2. Each component is a 2×2 determinant: cover the column, cross-multiply, and *subtract* for the middle ($\mathbf{j}$) term.
3. Check: the answer dotted with $\mathbf{u}$ and with $\mathbf{v}$ must be 0. Example: $(1,2,0) \times (0,1,3) = (6,-3,1)$; $(6,-3,1) \cdot (1,2,0) = 0$ ✓.

TERMS

$\mathbf{u}, \mathbf{v}, \mathbf{w}$ – vectors in $\mathbb{R}^3$ with components u_i, v_i, w_i
θ – angle between the vectors, $0 \le \theta \le \pi$
$\mathbf{i}, \mathbf{j}, \mathbf{k}$ – unit vectors along x, y, z
$\overrightarrow{PQ}$ – vector from P to Q: $Q - P$
$|\cdot|$ (vmatrix) – determinant A2.25 §04 Algebra II
Right-hand rule – fingers sweep $\mathbf{u} \to \mathbf{v}$, thumb gives $\mathbf{u} \times \mathbf{v}$
$\hat{\mathbf{v}}$ – unit vector in the direction of $\mathbf{v}$
$\text{proj}_{\mathbf{v}}\,\mathbf{u}$ – vector projection of $\mathbf{u}$ onto $\mathbf{v}$ LA.4 §09 Linear Algebra
P, Q, R; A – vertices of a triangle; its area



CONTEXT

- **Origin.** Hamilton's quaternions (1843); Gibbs and Heaviside's vector algebra (1880s).
- **Use.** Normals to planes, areas, volumes, torque, orientation tests.
- **Data science.** Cosine similarity of embeddings is the normalized dot product T.21 §06 Trigonometry; cross products give surface normals for lighting 3-D meshes and point clouds.
- **Physical application.** Torque $\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}$; the magnetic force on a moving charge is $q\,\mathbf{v} \times \mathbf{B}$.

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C3.3 Lines & Planes in Space

A line is fixed by a point and a direction vector; a plane by a point and a normal (perpendicular) vector. Distances and angles between them reduce to dot and cross products C3.2.

Line (vector / parametric)	$\mathbf{r}(t) = \mathbf{r}_0 + t\mathbf{v}: \quad x = x_0 + at,\ y = y_0 + bt,\ z = z_0 + ct$	Start at P_0 and move t steps of $\mathbf{v} = (a, b, c)$.
Line (symmetric)	$\dfrac{x - x_0}{a} = \dfrac{y - y_0}{b} = \dfrac{z - z_0}{c}$	Eliminate t; if, say, $a = 0$, write $x = x_0$ separately.
Plane (point–normal)	$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$	Every vector in the plane is perpendicular to $\mathbf{n} = (a, b, c)$: $\mathbf{n} \cdot (\mathbf{x} - \mathbf{x}_0) = 0$.
Plane (general)	$ax + by + cz = d$	Read the normal straight off the coefficients.
Plane through 3 points	$\mathbf{n} = \overrightarrow{PQ} \times \overrightarrow{PR}$	Two edge vectors lie in the plane; their cross product is normal to it.
Distance point to plane	$D = \dfrac{\lvert ax_1 + by_1 + cz_1 - d\rvert}{\sqrt{a^2 + b^2 + c^2}}$	Project onto the unit normal; the 3-D twin of the point–line distance A1.8 §03 Algebra I.
Angle between planes	$\cos\theta = \dfrac{\lvert \mathbf{n}_1 \cdot \mathbf{n}_2\rvert}{\lVert\mathbf{n}_1\rVert\lVert\mathbf{n}_2\rVert}$	The angle between the normals; parallel planes have parallel normals.
Distance point to line	$D = \dfrac{\lVert \overrightarrow{P_0P_1} \times \mathbf{v}\rVert}{\lVert\mathbf{v}\rVert}$	Parallelogram area divided by its base.
Distance between skew lines	$D = \dfrac{\lvert(\mathbf{r}_2 - \mathbf{r}_1) \cdot (\mathbf{v}_1 \times \mathbf{v}_2)\rvert}{\lVert\mathbf{v}_1 \times \mathbf{v}_2\rVert}$	Project the gap between the lines onto their common perpendicular.

STATED-EQUATION *A line is r of t equals r zero plus t v. Symmetric form: x minus x zero over a equals y minus y zero over b equals z minus z zero over c. A plane through x zero with normal a, b, c is a times x minus x zero plus b times y minus y zero plus c times z minus z zero equals zero. The distance from a point to the plane a x plus b y plus c z equals d is the absolute value of a x one plus b y one plus c z one minus d, over the length of the normal. In general form the plane is a x plus b y plus c z equals d. A plane through P, Q and R has normal P Q cross P R. The angle between two planes satisfies cosine theta equals the absolute value of n one dot n two, over the product of their lengths. The distance from P one to a line is the length of P zero P one cross v, over the length of v. The distance between skew lines is the absolute value of r two minus r one dot v one cross v two, over the length of v one cross v two.*

MEMORY AID Line = point + *direction*; plane = point + *normal*.

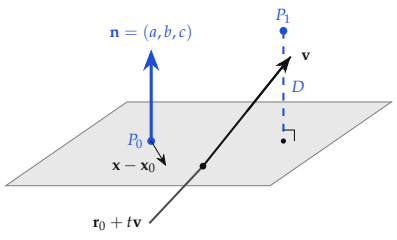
METHOD

1. Two edge vectors from three points: $\overrightarrow{PQ}$, $\overrightarrow{PR}$.
2. Normal $\mathbf{n} = \overrightarrow{PQ} \times \overrightarrow{PR}$; plug $\mathbf{n}$ and P into point–normal form.
3. Check with the other two points. Example: $P(1,0,0)$, $Q(0,2,0)$, $R(0,0,3)$: $\mathbf{n} = (-1,2,0) \times (-1,0,3) = (6,3,2)$, plane $6x + 3y + 2z = 6$.

DECISION POINT What fixes the direction or normal? Two points on a line $\to \mathbf{v} = \overrightarrow{PQ}$. A line perpendicular to a plane $\to \mathbf{v} = \mathbf{n}$. The line where two planes meet $\to \mathbf{v} = \mathbf{n}_1 \times \mathbf{n}_2$ (find a point by setting $z = 0$). A plane containing two directions $\to \mathbf{n} = \mathbf{v}_1 \times \mathbf{v}_2$. Two lines that are neither parallel nor intersecting are *skew*.

TERMS

$P_0 = (x_0, y_0, z_0)$ – known point; $\mathbf{r}_0$ its position vector
$\mathbf{v} = (a, b, c)$ – direction vector of a line
$\mathbf{n} = (a, b, c)$ – normal vector of a plane
t – parameter (any real number)
$P_1 = (x_1, y_1, z_1)$ – outside point
$\mathbf{r}_i, \mathbf{v}_i$ – point and direction of line i
d; D – constant in $ax + by + cz = d$; a distance
P, Q, R – three non-collinear points
θ; $\mathbf{n}_1, \mathbf{n}_2$ – angle between planes; their normal vectors



CONTEXT

- **Origin.** Analytic solid geometry of Lagrange and Monge (1770s–1790s).
- **Use.** Ray tracing, robotics, surveying, collision detection.
- **Data science.** A linear classifier's decision boundary is the hyperplane $\mathbf{w} \cdot \mathbf{x} + b = 0$; the point–plane distance is the SVM margin ML.15 §15 Machine Learning.
- **Physical application.** Ray tracing: a light ray (line) meets a mirror (plane) where the parametric line satisfies the plane equation.

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C3.4 Cylinders & Quadric Surfaces

Second-degree equations in x, y, z give six basic surfaces, plus cylinders when a variable is missing. Identify a surface from its traces: the conic sections A2.26 §04 Algebra II cut by planes parallel to the coordinate planes.

Ellipsoid	$\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}+\dfrac{z^2}{c^2}=1$	All traces are ellipses; a sphere when $a=b=c$.
Elliptic paraboloid	$z=\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}$	A bowl: horizontal traces are ellipses, vertical traces parabolas.
Hyperbolic paraboloid	$z=\dfrac{x^2}{a^2}-\dfrac{y^2}{b^2}$	A saddle: opposite signs; horizontal traces are hyperbolas.
Hyperboloid of one sheet	$\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}-\dfrac{z^2}{c^2}=1$	One minus sign: a connected waisted tube around the z-axis.
Hyperboloid of two sheets	$\dfrac{z^2}{c^2}-\dfrac{x^2}{a^2}-\dfrac{y^2}{b^2}=1$	Two minus signs: two separate bowls; empty for $\lvert z\rvert < c$.
Elliptic cone	$\dfrac{z^2}{c^2}=\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}$	Right side 0 in the "$=1$" form: a double cone through the origin.
Cylinder	$f(x,y)=0 \quad (z \text{ absent})$	The missing variable is free: the curve is extruded along that axis ($x^2+y^2=1$ is a tube).

STATED-EQUATION *Ellipsoid: x squared over a squared plus y squared over b squared plus z squared over c squared equals one. Elliptic paraboloid: z equals x squared over a squared plus y squared over b squared; with a minus sign it is a hyperbolic paraboloid. Hyperboloid of one sheet has one minus sign, of two sheets two minus signs; the cone has zero on the other side. An equation missing a variable is a cylinder.*

MEMORY AID In "$=1$" form count the minus signs: $0 \to$ ellipsoid, $1 \to$ one sheet, $2 \to$ two sheets; "$=0$" $\to$ cone. One variable to the first power $\to$ paraboloid (same signs: bowl; opposite: saddle).

METHOD

1. Complete squares and rearrange into a standard form (move the center to the origin).
2. Missing variable $\to$ cylinder; a linear variable $\to$ paraboloid; all squared $\to$ count signs.
3. Confirm with traces $z=k$, $y=k$, $x=k$. Example: $x^2+y^2-z^2=1$: trace $z=0$ is a circle of radius 1, $z=2$ a circle of radius $\sqrt{5} \Rightarrow$ hyperboloid of one sheet.



Ellipsoid $\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}+\dfrac{z^2}{c^2}=1$ — **Elliptic paraboloid** $z=\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}$ — **Hyperbolic paraboloid** $z=\dfrac{x^2}{a^2}-\dfrac{y^2}{b^2}$ — **Hyperboloid, 1 sheet** $\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}-\dfrac{z^2}{c^2}=1$ — **Hyperboloid, 2 sheets** $\dfrac{z^2}{c^2}-\dfrac{x^2}{a^2}-\dfrac{y^2}{b^2}=1$ — **Elliptic cone** $\dfrac{z^2}{c^2}=\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}$ — **Cylinder** $x^2+y^2=a^2$

TERMS

a, b, c – positive semi-axis lengths along x, y, z
Trace – cross-section by a plane $x=k$, $y=k$ or $z=k$
Axis – the variable with the odd sign (hyperboloids, cone) or to the first power (paraboloids)
Conics – ellipse, parabola, hyperbola A2.26 §04 Algebra II

CONTEXT

- **Origin.** Euler classified the quadric surfaces (1748).
- **Use.** Antennas, lenses, architecture, surface fitting.
- **Data science.** Level sets of a Gaussian density (constant Mahalanobis distance) are ellipsoids LA.27 §09 Linear Algebra; a quadratic loss with an indefinite Hessian is a saddle C3.16.
- **Physical application.** Satellite dishes are paraboloids (signals focus at one point); cooling towers are hyperboloids of one sheet, built from straight beams.

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Curves & Motion in Space

C3.5 Vector-Valued Functions & Motion

A vector-valued function $\mathbf{r}(t)$ traces a curve in space as t varies. Limits, derivatives and integrals act component by component; $\mathbf{r}'$ is the velocity (tangent) and $\mathbf{r}''$ the acceleration.

Vector function	$\mathbf{r}(t) = \big(x(t),\ y(t),\ z(t)\big)$	Three parametric equations C2.25 §08 Calculus II bundled as one position vector.
Derivative	$\mathbf{r}'(t) = \lim_{h\to 0} \dfrac{\mathbf{r}(t+h) - \mathbf{r}(t)}{h} = \big(x',\ y',\ z'\big)$	Differentiate each component; $\mathbf{r}'(t)$ is tangent to the curve.
Integral	$\displaystyle\int_a^b \mathbf{r}(t)\,dt = \Big(\int_a^b x\,dt,\ \int_a^b y\,dt,\ \int_a^b z\,dt\Big)$	Integrate each component; an indefinite integral adds a constant *vector* $\mathbf{C}$.
Velocity, speed, acceleration	$\mathbf{v} = \mathbf{r}', \qquad \text{speed} = \lVert\mathbf{r}'\rVert, \qquad \mathbf{a} = \mathbf{r}''$	Speed is a scalar; acceleration can be nonzero at constant speed (turning).
Product rules	$(\mathbf{u}\cdot\mathbf{w})' = \mathbf{u}'\cdot\mathbf{w} + \mathbf{u}\cdot\mathbf{w}', \quad (\mathbf{u}\times\mathbf{w})' = \mathbf{u}'\times\mathbf{w} + \mathbf{u}\times\mathbf{w}'$	As in one variable C1.12 §07 Calculus I; keep the order in cross products.
Constant length	$\lVert\mathbf{r}(t)\rVert = c \ \Rightarrow\ \mathbf{r}\cdot\mathbf{r}' = 0$	Differentiate $\mathbf{r}\cdot\mathbf{r} = c^2$: on a circle or sphere the velocity is perpendicular to the position.
Projectile motion	$\mathbf{r}(t) = \mathbf{r}_0 + \big(v_0\cos\alpha\ t,\ \ v_0\sin\alpha\ t - \tfrac{1}{2}gt^2\big)$	Integrate $\mathbf{a} = (0,-g)$ twice using the initial velocity and position P.2 §12 Physics.
Tangent line	$\mathbf{L}(s) = \mathbf{r}(t_0) + s\,\mathbf{r}'(t_0)$	Point on the curve plus multiples of the velocity there.

STATED-EQUATION *r of t equals x of t, y of t, z of t. Its derivative is the limit of the difference quotient, computed component by component; integrals also go component by component. Velocity is r prime, speed is the length of r prime, acceleration is r double prime. If r has constant length, r dot r prime equals zero. Derivatives of dot and cross products follow the product rule, keeping the order in the cross product. A projectile launched at speed v naught and angle alpha has position r naught plus v naught cosine alpha t, v naught sine alpha t minus one half g t squared. The tangent line at t naught is r of t naught plus s times r prime of t naught.*

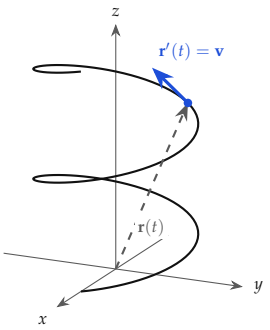
MEMORY AID Everything happens one component at a time.

METHOD

1. Integrate $\mathbf{a}(t)$ to get $\mathbf{v}(t) + \mathbf{C}_1$; fix $\mathbf{C}_1$ with $\mathbf{v}(0)$.
2. Integrate $\mathbf{v}(t)$ to get $\mathbf{r}(t) + \mathbf{C}_2$; fix $\mathbf{C}_2$ with $\mathbf{r}(0)$.
3. Answer the question (landing time, range, maximum height). Example: $\mathbf{a} = (0,-10)$, $\mathbf{v}(0) = (3,4)$, $\mathbf{r}(0) = \mathbf{0}$: $\mathbf{r} = (3t,\ 4t - 5t^2)$; lands at $t = 0.8$, $x = 2.4$.

TERMS

t – parameter (often time); t_0 a fixed value
$x(t), y(t), z(t)$ – component functions
$\mathbf{v}$, $\mathbf{a}$ – velocity and acceleration vectors
v_0, α – launch speed and angle; g gravitational acceleration
$\mathbf{u}, \mathbf{w}$ – differentiable vector functions
Helix – $\mathbf{r} = (\cos t, \sin t, ct)$: the standard space curve
$\mathbf{r}_0$, g – initial position; gravitational acceleration
c; s – a constant length; parameter along the tangent line



CONTEXT

- **Origin.** Newton's *Principia* (1687) treated curves as paths of moving bodies; vector notation from Gibbs (1880s).
- **Use.** Trajectories of projectiles, planets, robots and drones.
- **Data science.** Training traces a path $\boldsymbol{\theta}(t)$ through parameter space; neural ODEs model a hidden state as $\mathbf{h}'(t) = f(\mathbf{h}, t)$.
- **Physical application.** A thrown ball: $\mathbf{a} = (0,-g)$ integrates to a parabolic path; planetary orbits obey $\mathbf{a} = -GM\mathbf{r}/\lVert\mathbf{r}\rVert^3$ P.8 §12 Physics.

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C3.6 Arc Length, Curvature & the TNB Frame

Arc length measures distance along a curve; curvature measures how fast the direction turns per unit of length. The moving frame $\mathbf{T}, \mathbf{N}, \mathbf{B}$ splits motion into "along the path" and "turning".

Arc length	$L = \int_a^b \|\mathbf{r}'(t)\|\, dt$	Integrate the speed C2.10 §08 Calculus II.
Arc-length function	$s(t) = \int_a^t \|\mathbf{r}'(u)\|\, du, \qquad \frac{ds}{dt} = \|\mathbf{r}'(t)\|$	Distance traveled since $t = a$; reparametrizing by s gives unit speed.
Unit tangent	$\mathbf{T}(t) = \dfrac{\mathbf{r}'(t)}{\|\mathbf{r}'(t)\|}$	Direction of motion only.
Curvature (definition)	$\kappa = \left\|\dfrac{d\mathbf{T}}{ds}\right\| = \dfrac{\|\mathbf{T}'(t)\|}{\|\mathbf{r}'(t)\|}$	Turning rate per unit length; a circle of radius R has $\kappa = 1/R$.
Curvature (cross-product formula)	$\kappa = \dfrac{\|\mathbf{r}' \times \mathbf{r}''\|}{\|\mathbf{r}'\|^3}$	Usually the fastest route: no need to find $\mathbf{T}$ first.
Curvature of a graph	$\kappa = \dfrac{\|f''(x)\|}{\left(1 + f'(x)^2\right)^{3/2}}$	For $y = f(x)$; at a flat point ($f' = 0$) it is just $\|f''\|$.
Normal & binormal	$\mathbf{N} = \dfrac{\mathbf{T}'}{\|\mathbf{T}'\|}, \qquad \mathbf{B} = \mathbf{T} \times \mathbf{N}$	$\mathbf{N}$ points toward the inside of the turn; $\mathbf{B}$ completes a right-handed frame.
Acceleration components	$\mathbf{a} = a_T\mathbf{T} + a_N\mathbf{N}, \quad a_T = \dfrac{d}{dt}\|\mathbf{v}\|, \quad a_N = \kappa\|\mathbf{v}\|^2$	Speeding up along $\mathbf{T}$; turning along $\mathbf{N}$.
Frenet–Serret formulas	$\dfrac{d\mathbf{T}}{ds} = \kappa\mathbf{N}, \quad \dfrac{d\mathbf{N}}{ds} = -\kappa\mathbf{T} + \tau\mathbf{B}, \quad \dfrac{d\mathbf{B}}{ds} = -\tau\mathbf{N}$	Torsion τ measures twisting out of the plane of $\mathbf{T}, \mathbf{N}$.

STATED-EQUATION *Arc length is the integral of the speed. The unit tangent T is r prime over its length. Curvature kappa is the length of d T d s, equivalently the length of r prime cross r double prime over the cube of the speed. N is T prime over its length, and B equals T cross N. Acceleration splits into a T component, the rate of change of speed, and an N component, kappa times speed squared. The arc-length function s of t is the integral of the speed from a to t, so d s d t equals the speed. For a graph y equals f of x, kappa equals the absolute value of f double prime, over one plus f prime squared to the three halves. Frenet-Serret: d T d s equals kappa N, d N d s equals minus kappa T plus tau B, and d B d s equals minus tau N.*

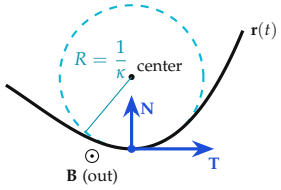
MEMORY AID Tight turn = big curvature: $\kappa = 1/R$, the reciprocal of the radius of the best-fitting (osculating) circle.

METHOD

1. Compute $\mathbf{r}'$ and $\mathbf{r}''$ at the point.
2. Cross them, take the length, divide by $\|\mathbf{r}'\|^3$.
3. Sanity check against $\kappa = 1/R$. Example: $\mathbf{r} = (t, t^2, 0)$ at $t = 0$: $\mathbf{r}' = (1,0,0)$, $\mathbf{r}'' = (0,2,0)$, $\mathbf{r}' \times \mathbf{r}'' = (0,0,2)$, $\kappa = 2$ (radius $\frac{1}{2}$).

TERMS

s – arc length; L total length
$\mathbf{T}, \mathbf{N}, \mathbf{B}$ – unit tangent, principal normal, binormal
κ – curvature; $R = 1/\kappa$ radius of curvature
τ – torsion (0 for a plane curve)
a_T, a_N – tangential and normal acceleration
Osculating circle – circle of radius $1/\kappa$ that best fits the curve
L; u – total arc length; dummy integration variable



CONTEXT

- **Origin.** Curvature of plane curves: Huygens and Newton (17th c.); the frame by Frenet (1847) and Serret (1851).
- **Use.** Road, rail and roller-coaster design; machining paths.
- **Data science.** Smoothness penalties on curvature for motion planning, splines and handwriting or trajectory features.
- **Physical application.** Centripetal acceleration $a_N = \kappa v^2 = v^2/R$ P8 §12 Physics limits the safe speed on a highway curve.

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Functions of Several Variables

C3.7 Functions of Several Variables & Level Curves

$f(x,y)$ assigns a number to each point of a region in the plane; its graph is a surface $z = f(x,y)$. Level curves $f = c$ slice the surface at fixed heights and draw it flat as a contour map.

Function of two variables	$z = f(x,y), \quad (x,y) \in D \subseteq \mathbb{R}^2$	The domain D follows the one-variable rules: no division by 0, no even roots or logs of bad inputs A1.9 §03 Algebra I.
Graph	$\{(x,y,z) : z = f(x,y),\ (x,y) \in D\}$	A surface lying over the domain.
Level curve	$f(x,y) = c$	Curve in the xy-plane at height c; with equally spaced c, crowded curves mean a steep surface.
Vertical traces	$z = f(x,b) \quad \text{and} \quad z = f(a,y)$	Slices by the planes $y = b$ and $x = a$; their slopes are the partial derivatives C3.9.
Level surface	$F(x,y,z) = c$	A function of three variables has a 4-D graph, so draw its level surfaces instead: $x^2 + y^2 + z^2 = c$ are spheres.
Function of n variables	$f : \mathbb{R}^n \to \mathbb{R}, \quad \mathbf{x} = (x_1, \ldots, x_n) \mapsto f(\mathbf{x})$	The setting of every loss function in machine learning.

STATED-EQUATION *z equals f of x and y, defined on a domain D in the plane. The graph is the set of points x, y, f of x, y. A level curve is the set where f of x, y equals a constant c. A level surface is where F of x, y, z equals c. Vertical traces fix y equals b or x equals a. A function of n variables sends a point x one through x n to a number f of x.*

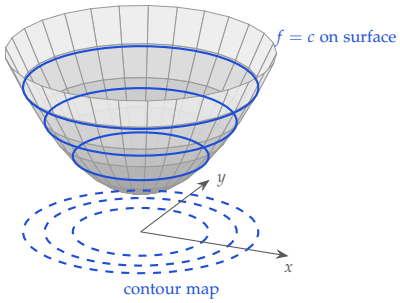
MEMORY AID Graph = landscape; level curves = the hiking map of that landscape.

METHOD

1. Set $f(x,y) = c$ for several equally spaced values of c.
2. Identify each curve (lines, circles, ellipses, hyperbolas …) and label it with c.
3. Read steepness from spacing. Example: $f = x^2 + y^2$, $c = 1, 4, 9$: circles of radius $1, 2, 3$ — evenly spaced radii, but in c they crowd as you move out.

TERMS

D – domain (allowed inputs)
c – a constant height (contour value)
a, b – fixed values of x and y
$F(x,y,z)$ – function of three variables
$\mathbb{R}^n$ – n-dimensional space M.78 §01 Math Signs
Contour map – a family of labeled level curves
$\mathbf{x}$ – point $(x_1, \ldots, x_n)$ in $\mathbb{R}^n$



CONTEXT

- **Origin.** Contour lines: Halley's chart of magnetic declination (1701) and Ducarla's topographic contours (1771).
- **Use.** Topographic maps, weather maps, engineering design spaces.
- **Data science.** Loss landscapes are shown as contour plots; a classifier's decision boundary is the level set $p(\mathbf{x}) = 0.5$.
- **Physical application.** Isobars (equal pressure) and isotherms (equal temperature) on a weather map are level curves.

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C3.8 Limits & Continuity in Several Variables

$\lim_{(x,y)\to(a,b)} f = L$ means f approaches L along *every* possible approach: infinitely many directions and curved paths. One disagreeing path kills the limit; agreement along many paths proves nothing.

ε–δ definition	$0 < \|(x,y)-(a,b)\| < \delta \;\Rightarrow\; \|f(x,y)-L\| < \varepsilon$	For every tolerance ε some disk of radius δ works, from all directions at once C1.1 §07 Calculus I.
Two-path test	$f \xrightarrow{C_1} L_1,\;\; f \xrightarrow{C_2} L_2,\;\; L_1 \neq L_2 \;\Rightarrow\;$ no limit	The standard way to prove a limit does not exist.
Polar substitution	$x = r\cos\theta,\; y = r\sin\theta:\;\; \|f-L\| \le g(r) \to 0$	If a bound free of θ goes to 0 as $r \to 0^+$, the limit is L T.19 §06 Trigonometry.
Squeeze theorem	$\|f(x,y)-L\| \le g(x,y),\;\; g \to 0 \;\Rightarrow\; f \to L$	Typical bounds: $\|x\| \le \sqrt{x^2+y^2}$, $\frac{x^2}{x^2+y^2} \le 1$.
Continuity	$\lim_{(x,y)\to(a,b)} f(x,y) = f(a,b)$	Polynomials, rational functions (off zeros of the denominator) and their compositions are continuous C1.7 §07 Calculus I.
Direction-dependent limit	$f = \dfrac{xy}{x^2+y^2}$ on $y = mx:\;\; f = \dfrac{m}{1+m^2}$	The value depends on the slope m, so the limit at $(0,0)$ does not exist.
Lines are not enough	$f = \dfrac{x^2y}{x^4+y^2}:\;\; y = mx \to 0,\quad y = x^2 \to \frac{1}{2}$	Every straight line gives 0, yet the parabola gives $\frac{1}{2}$: no limit.

STATED-EQUATION *The limit of f as x, y approaches a, b is L if for every epsilon there is a delta so that being within delta of a, b, but not equal to it, puts f within epsilon of L. If two paths give different limits, the limit does not exist. If the absolute value of f minus L is bounded by something that goes to zero, f goes to L. f is continuous at a, b when its limit there equals f of a, b. In polar form, if the absolute value of f minus L is bounded by a function of r alone that goes to zero, the limit is L. For x y over x squared plus y squared, the value along y equals m x is m over one plus m squared, which depends on m, so there is no limit. For x squared y over x to the fourth plus y squared, every line gives zero but the parabola y equals x squared gives one half.*

MEMORY AID Paths can only *disprove* a limit; to *prove* one, squeeze.

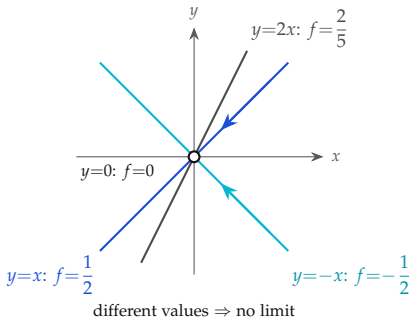
METHOD

1. Try direct substitution; if f is continuous there, you are done.
2. For $0/0$: test the axes and lines $y = mx$. A value depending on m means no limit.
3. Test curves $y = x^k$ chosen to balance the degrees of numerator and denominator.
4. If all paths agree, prove it with polar or squeeze. Example: $\left|\frac{x^2y}{x^2+y^2}\right| \le |y| \to 0$, so the limit at $(0,0)$ is 0.

DECISION POINT Which paths? Pick the path that makes the denominator's terms the same degree: in $\frac{x^2y}{x^4+y^2}$, x^4 and y^2 balance on $y = x^2$. If you suspect the limit exists, stop hunting paths and bound $|f-L|$; in polar form, a numerator of higher total degree than the denominator usually forces $|f| \le Cr^k \to 0$.

TERMS

(a,b) – the point approached; L the limit value
ε, δ – output tolerance and input radius
$\|\cdot\|$ – distance in the plane G.12 §05 Geometry
C_1, C_2 – two paths ending at (a,b)
r, θ – polar coordinates centered at (a,b)
m – slope of a test line; g a bounding function
L; g – candidate limit; a bounding function that goes to 0



different values ⇒ no limit

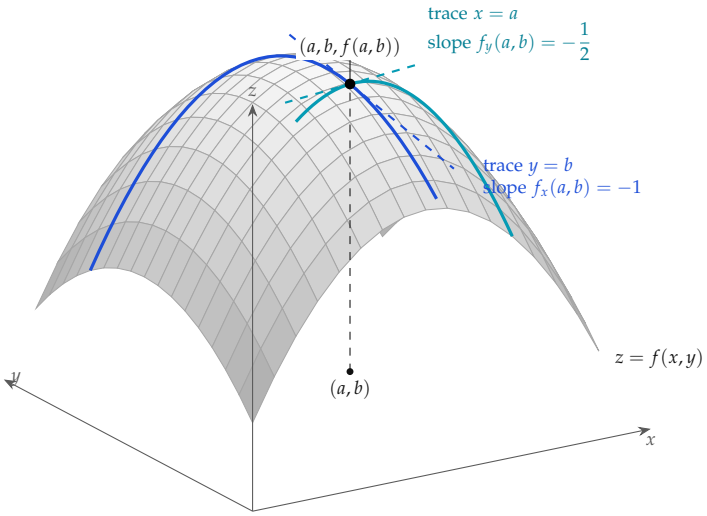
CONTEXT

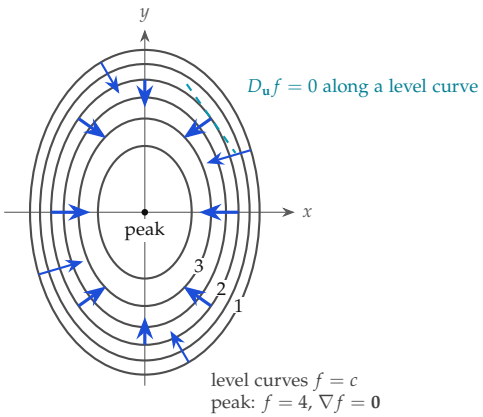
- **Origin.** Cauchy and Weierstrass made limits rigorous (1820s–1860s); the Genocchi–Peano text (1884) gave the classic counterexamples.
- **Use.** Deciding continuity and differentiability of multivariable models.
- **Data science.** Probing a model one feature at a time (partial dependence) can miss behavior that only appears when features change *together*: the multivariable-limit trap in practice.
- **Physical application.** Where two plates held at different temperatures meet at a corner, the temperature approached depends on the direction you come from.

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Partial Derivatives & the Gradient | *Key Topic C3.K1*

A function of several variables has a slope in *every* direction. Partial derivatives measure the two simplest ones (along the axes) by freezing all but one variable; the gradient collects them into a vector that points uphill and from which every other directional slope follows. Tangent planes, the chain rule and the Hessian are all built from these pieces. Surface used below: $f(x,y) = 4 - \frac{x^2}{2} - \frac{y^2}{4}$ at $(a,b) = (1,1)$.





Slices (left). Freeze $y = b$: the plane $y = b$ cuts the surface in a curve $z = f(x,b)$; its slope at $x = a$ is $f_x(a,b)$. Freeze $x = a$ instead for $f_y(a,b)$. Here $f_x = -x = -1$ and $f_y = -\frac{y}{2} = -\frac{1}{2}$: the surface falls in both directions.

Contour map (right). The gradient $\nabla f = (f_x, f_y) = (-x, -\frac{y}{2})$ is drawn at points of the level curves: always *perpendicular* to them, pointing *uphill*, longest where curves crowd. Along a level curve the slope is 0.

Computing partials, step by step

Example: $f(x,y) = x^2y + 3y$

1. **Pick the variable.** For f_x, x is the variable; y is a constant (think "$y = 5$").
2. **Differentiate term by term.** $x^2y \to 2x \cdot y$ (constant multiple), $3y \to 0$ (no x at all). So $f_x = 2xy$.
3. **Swap roles** for f_y: $x^2y \to x^2 \cdot 1$, $3y \to 3$. So $f_y = x^2 + 3$.
4. **Evaluate.** At $(1,2)$: $f = 8$, $f_x = 4$, $f_y = 4$, $\nabla f(1,2) = (4,4)$.
5. **Check** by a difference quotient: $\frac{f(1.01,2) - f(1,2)}{0.01} = \frac{8.0402 - 8}{0.01} \approx 4.02$ ✓.

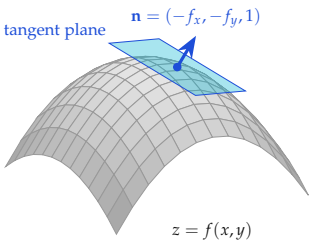
Notation (all mean the same thing)

	Leibniz	Subscript	Operator
x-partial	$\frac{\partial f}{\partial x}$	f_x	$\partial_x f,\ D_1 f$
y-partial	$\frac{\partial f}{\partial y}$	f_y	$\partial_y f,\ D_2 f$
second, x twice	$\frac{\partial^2 f}{\partial x^2}$	f_{xx}	$\partial_x^2 f$
mixed: x then y	$\frac{\partial^2 f}{\partial y\,\partial x}$	f_{xy}	$\partial_y \partial_x f$
gradient	$\left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right)$	(f_x, f_y)	∇f

Subscripts read left to right in the order taken; Leibniz reads right to left. Equal anyway when smooth (Clairaut C3.10).

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From partials to everything else



Tangent plane C3.11. The two tangent lines span a plane:

$$z = f(a,b) + f_x(a,b)(x-a) + f_y(a,b)(y-b).$$

For $x^2y + 3y$ at $(1,2)$: $z = 8 + 4(x-1) + 4(y-2)$, so $f(1.1, 2.1) \approx 8.8$ (true value 8.841).

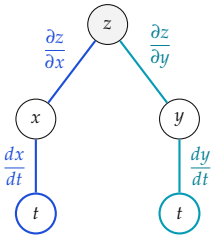
Directional derivative C3.12. Slope in a unit direction $\mathbf{u}$: $D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u}$. With $\nabla f = (4,4)$ and $\mathbf{u} = (\frac{3}{5}, \frac{4}{5})$: $D_{\mathbf{u}}f = 5.6$. The maximum, $\|\nabla f\| = 4\sqrt{2} \approx 5.66$, is attained along ∇f itself.

Second partials & the Hessian C3.16. $f_{xx} = 2y$, $f_{xy} = f_{yx} = 2x$, $f_{yy} = 0$, so

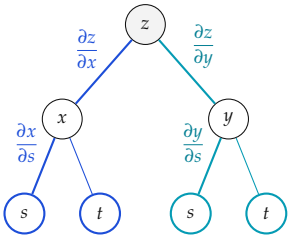
$$H(1,2) = \begin{bmatrix} 4 & 2 \\ 2 & 0 \end{bmatrix}, \qquad D = \det H = -4 < 0.$$

H holds the curvature in every direction ($\mathbf{u}^{\mathsf{T}} H \mathbf{u}$); at a critical point $D < 0$ would mean a saddle.

The chain rule as a tree



$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}$$



$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial s}$$

Draw the output on top, intermediate variables in the middle, and the independent variables at the bottom C3.13. **Multiply along each branch** from z down to the variable you differentiate by, then **add the branches**. Example: $z = x^2y$, $x = 2t$, $y = t + 1$, at $t = 1$ ($x = 2$, $y = 2$):

$$\tfrac{dz}{dt} = 2xy \cdot 2 + x^2 \cdot 1 = 16 + 4 = 20.$$

Check: $z = 4t^3 + 4t^2$, $z'(1) = 12 + 8 = 20$ ✓. Backpropagation ML.22 §15 Machine Learning is this rule run on a huge tree.

Common mistakes

Forgetting a frozen factor. $\partial_x(x^2y) = 2xy$, not $2x$: the frozen y stays as a coefficient.

Differentiating a constant term. $\partial_x(3y) = 0$, not 3: it has no x.

Non-unit direction. $D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u}$ needs $\|\mathbf{u}\| = 1$; normalize first.

Gradient lives in the domain. ∇f is a 2-D vector in the xy-plane (perpendicular to level curves), not a vector on the 3-D surface.

Partials exist $\neq$ differentiable. Both partials can exist while f is not even continuous C3.8; continuous partials guarantee differentiability.

Dropping chain-rule branches. Every path from output to input contributes a term; missing one is the most common chain-rule error.

DECISION POINT **Which derivative do you need?** Rate along an axis → a partial f_x, f_y. Rate in an arbitrary direction → the directional derivative $\nabla f \cdot \mathbf{u}$. Rate along a moving path $\mathbf{x}(t)$ → the chain rule $\nabla f \cdot \mathbf{x}'(t)$. Best direction to move (optimization) → the gradient, $\pm\nabla f$. Curvature, or classifying a critical point → the Hessian. Several outputs at once → the Jacobian C3.15.

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Partial Derivatives & Differentiability

C3.9 Partial Derivatives

The partial derivative $\partial f/\partial x$ is the ordinary derivative of f with respect to x while every other variable is held fixed; geometrically, the slope of the surface in the x-direction. Full visual treatment: C3.K1.

Partial derivative in x	$\dfrac{\partial f}{\partial x} = f_x = \lim_{h\to 0} \dfrac{f(x+h,y)-f(x,y)}{h}$	Only x is nudged; y stays frozen C1.9 §07 Calculus I.
Partial derivative in y	$\dfrac{\partial f}{\partial y} = f_y = \lim_{h\to 0} \dfrac{f(x,y+h)-f(x,y)}{h}$	Same idea with the roles swapped.
Geometric meaning	$f_x(a,b) = \dfrac{d}{dx} f(x,b)\Big\vert_{x=a}$	Slope of the trace curve cut by the plane $y=b$.
Notation	$\dfrac{\partial f}{\partial x} = f_x = \partial_x f = D_1 f$	All the same; ∂ ("partial") signals that other variables are held constant.
Mixed partial notation	$\dfrac{\partial^2 f}{\partial y\,\partial x} = (f_x)_y = f_{xy}$	Differentiate by x first, then by y; order rarely matters C3.10.
Numerical partial (central difference)	$f_x \approx \dfrac{f(x+h,y)-f(x-h,y)}{2h}$	Error $O(h^2)$; used to check analytic or autodiff gradients.

STATED-EQUATION *The partial derivative of f with respect to x, f sub x, is the limit as h goes to zero of f of x plus h, y minus f of x, y, over h, holding y fixed. f sub y is the same with y nudged. f x at a, b is the slope of the curve f of x, b at x equals a. The mixed partial f x y means differentiate by x, then by y. Other notations are del sub x f and D one f. Numerically, f x is approximately f of x plus h, y minus f of x minus h, y, over two h.*

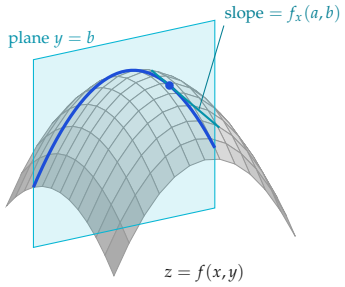
MEMORY AID "Freeze the other variables, then differentiate as usual."

METHOD

1. Choose the variable to differentiate; treat every other variable as a constant number.
2. Apply the one-variable rules C1.K2 §07 Calculus I.
3. Evaluate at the point. Example: $f = x^3 + 5xy^2$: $f_x = 3x^2 + 5y^2$, $f_y = 10xy$; at $(1,2)$: $f_x = 23$, $f_y = 20$.

TERMS

$f(x,y)$ – function of several variables (a surface)
∂ – "partial": other variables held constant
f_x – slope of the surface in the x-direction C1.9 §07 Calculus I
h – small step in one variable
(a,b) – point of evaluation
Rules – all single-variable rules apply C1.11 §07 Calculus I⊢C1.14 §07 Calculus I
$\partial_x f$, $D_1 f$ – other notations for f_x
f_{xy} – mixed partial: first x, then y C3.10



CONTEXT

- **Origin.** Euler, Clairaut and d'Alembert (1740s); the ∂ symbol by Legendre and Jacobi.
- **Use.** Rates of change of quantities depending on several inputs.
- **Data science.** Each partial derivative of the loss with respect to a weight is one entry of the gradient that backpropagation computes ML.22 §15 Machine Learning.
- **Physical application.** The ideal-gas pressure $P = nRT/V$ has $\partial P/\partial T = nR/V$: how pressure rises with temperature at fixed volume P.29 §12 Physics.

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C3.10 Higher-Order Partials & Clairaut's Theorem

Differentiating a partial derivative again gives second (and higher) partials. Under mild smoothness the order of differentiation does not matter, so only how many times each variable is used counts.

Pure second partials	$f_{xx} = \dfrac{\partial^2 f}{\partial x^2}, \qquad f_{yy} = \dfrac{\partial^2 f}{\partial y^2}$	Concavity of the surface along each axis C1.23 §07 Calculus I.
Mixed second partial	$f_{xy} = \dfrac{\partial}{\partial y}\left(\dfrac{\partial f}{\partial x}\right)$	How the x-slope changes as you move in y: the "twist" of the surface.
Clairaut's theorem	$f_{xy} = f_{yx}$ if both are continuous near (a, b)	The order of differentiation can be swapped; the Hessian is symmetric C3.16.
Higher orders	$f_{xxy} = f_{xyx} = f_{yxx}$	With continuous partials, only the number of x's and y's matters.
Laplace's equation	$f_{xx} + f_{yy} = 0$	A typical PDE check using second partials; solutions are *harmonic* V.13 §11 Vector Calculus.
Clairaut counterexample	$f = \dfrac{xy(x^2 - y^2)}{x^2 + y^2}$: $f_{xy}(0,0) = -1 \neq 1 = f_{yx}(0,0)$	Without continuity of the mixed partials the order can matter.

STATED-EQUATION *f x x is the second partial derivative in x; f y y in y. f x y is the partial with respect to y of f x. Clairaut's theorem: f x y equals f y x when they are continuous. Higher mixed partials depend only on how many times each variable is used. Laplace's equation: f x x plus f y y equals zero. Without continuity it can fail: for x y times x squared minus y squared, over x squared plus y squared, f x y at the origin is minus one while f y x is one.*

MEMORY AID Smooth functions forget the order: "x then y" = "y then x".

METHOD

1. Compute the first partials.
2. Differentiate each again with respect to each variable.
3. Use Clairaut as a check. Example: $f = x^2y^3$: $f_x = 2xy^3$, $f_{xy} = 6xy^2$; $f_y = 3x^2y^2$, $f_{yx} = 6xy^2$ ✓; at $(1,1)$ both are 6.

TERMS

f_{xx}, f_{yy} – pure second partials
f_{xy}, f_{yx} – mixed partials (order in the subscript, left to right)
Continuous partials – the hypothesis of Clairaut (also called Schwarz's theorem)
Harmonic – a solution of $f_{xx} + f_{yy} = 0$
Exactness test – $P_y = Q_x$ for conservative fields V.4 §11 Vector Calculus

CONTEXT

- **Origin.** Clairaut (1740) and Euler; first rigorous proof by Schwarz (1873).
- **Use.** PDEs, exact differentials, thermodynamics, symmetric Hessians.
- **Data science.** Hessian symmetry halves the work and storage of second-order methods; autodiff computes Hessian–vector products using it O.13 §14 Optimization.
- **Physical application.** Maxwell's relations in thermodynamics, such as $\left(\frac{\partial T}{\partial V}\right)_S = -\left(\frac{\partial P}{\partial S}\right)_V$, are equal mixed partials of an energy function.

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C3.11 Tangent Planes, Linear Approximation & Differentials

Near a point where f is differentiable, the surface looks like a plane. That tangent plane gives the best linear approximation, and its increments are the total differential.

Tangent plane	$z = f(a,b) + f_x(a,b)\,(x-a) + f_y(a,b)\,(y-b)$	Contains both trace tangent lines at $(a,b,f(a,b))$.
Linearization	$L(x,y) = f(a,b) + f_x\,\Delta x + f_y\,\Delta y \approx f(x,y)$	The tangent plane used as an approximation near (a,b) C1.21 §07 Calculus I.
Total differential	$dz = f_x\,dx + f_y\,dy$	Change in height along the tangent plane; $\Delta z \approx dz$ for small steps.
Differentiability	$\lim_{(h,k)\to(0,0)} \dfrac{f(a+h,b+k) - L(a+h,b+k)}{\sqrt{h^2+k^2}} = 0$	The error shrinks faster than the step. Continuous partials $\Rightarrow$ differentiable.
Normal vector to a graph	$\mathbf{n} = \langle f_x,\ f_y,\ -1 \rangle$	Perpendicular to the tangent plane; flip the sign for an upward normal.
Error propagation	$\lvert\Delta f\rvert \lesssim \lvert f_x\rvert\,\lvert\Delta x\rvert + \lvert f_y\rvert\,\lvert\Delta y\rvert$	Worst-case output error from small measurement errors.

STATED-EQUATION *The tangent plane at a, b is z equals f of a, b plus f x times x minus a plus f y times y minus b. The linearization L approximates f near a, b. The total differential d z equals f x d x plus f y d y. f is differentiable when the approximation error divided by the step length goes to zero. The normal vector to the graph is f x, f y, minus one. The error in f is at most about the absolute value of f x times the error in x, plus the absolute value of f y times the error in y.*

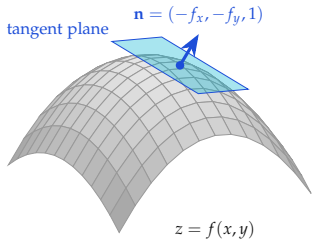
MEMORY AID Zoom in far enough on a smooth surface and it is flat.

METHOD

1. Evaluate f, f_x, f_y at a nearby convenient point (a,b).
2. Plug the small changes $\Delta x, \Delta y$ into L.
3. Example: $f = x^2y$ near $(2,3)$: $f = 12$, $f_x = 2xy = 12$, $f_y = x^2 = 4$; $f(2.1, 2.9) \approx 12 + 12(0.1) + 4(-0.1) = 12.8$ (true 12.789).

TERMS

(a,b) – base point; $\Delta x = x - a$, $\Delta y = y - b$
L – linearization (tangent-plane function)
dx, dy, dz – differentials (small changes)
h, k – steps in x and y
$\lesssim$ – "approximately at most"
Partials – C3.9; tangent planes to level surfaces: C3.12
$\mathbf{n}$ – normal vector to the surface
$\Delta x, \Delta y$ – changes (or measurement errors) in x and y



CONTEXT

- **Origin.** Lagrange and Cauchy; the modern (Fréchet) definition of differentiability (1911).
- **Use.** Approximation, sensitivity analysis, error propagation in measurement.
- **Data science.** Gradient descent minimizes the local linear model; the delta method propagates standard errors through a function of estimates S.24 §13 Statistics; LIME explains predictions with a local linear surrogate.
- **Physical application.** A cylinder $V = \pi r^2 h$ with $r = 3$, $h = 10$ and errors 0.1 in each: $dV = 2\pi rh\,dr + \pi r^2 dh = 6\pi + 0.9\pi \approx 21.7$.

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C3.12 Gradient & Directional Derivative

The gradient ∇f collects all first partials into one vector. Every directional slope is its dot product with a unit direction, so the gradient points straight uphill and is perpendicular to level sets.

Gradient	$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}\right)$	A vector in the domain, one entry per input.
Directional derivative	$D_{\mathbf{u}} f = \nabla f \cdot \mathbf{u} = \lVert\nabla f\rVert \cos\theta \quad (\lVert\mathbf{u}\rVert = 1)$	Slope when moving in direction $\mathbf{u}$; normalize $\mathbf{u}$ first.
Steepest ascent	$\max_{\mathbf{u}} D_{\mathbf{u}} f = \lVert\nabla f\rVert$ at $\mathbf{u} = \frac{\nabla f}{\lVert\nabla f\rVert}$	Steepest descent is $-\nabla f$; the slope is 0 across it ($\theta = 90°$).
Orthogonal to level sets	$\nabla f \perp \{f = c\}$	Along a level curve f does not change, so $\nabla f \cdot \mathbf{T} = 0$.
Tangent plane to a level surface	$\nabla F(P_0) \cdot (\mathbf{x} - \mathbf{x}_0) = 0$	The gradient of $F(x, y, z)$ is the normal of the surface $F = c$.
Normal line	$\mathbf{r}(t) = \mathbf{x}_0 + t\,\nabla F(P_0)$	Line through P_0 perpendicular to the level surface.
Gradient rules	$\nabla(fg) = f\nabla g + g\nabla f, \qquad \nabla(\mathbf{a}\cdot\mathbf{x}) = \mathbf{a}, \qquad \nabla\lVert\mathbf{x}\rVert^2 = 2\mathbf{x}$	Product rule and two identities used constantly in least squares LA.30 §09 Linear Algebra.

STATED-EQUATION *The gradient of f, del f, is the vector of partial derivatives. The directional derivative in the unit direction u equals the gradient dot u, which is the length of the gradient times cosine theta. The largest directional derivative equals the length of the gradient, in the gradient's direction. The gradient is perpendicular to the level curves; for a level surface F equals c, the gradient is the normal of the tangent plane. The tangent plane to F equals c at P naught is the gradient of F at P naught dot x minus x naught equals zero, and the normal line is x naught plus t times that gradient. The gradient of f g is f grad g plus g grad f; the gradient of a dot x is a; the gradient of the squared length of x is two x.*

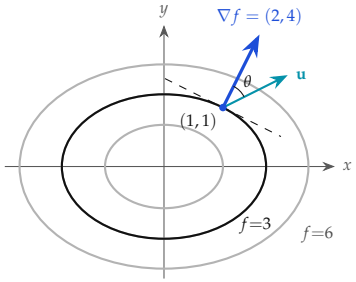
MEMORY AID The gradient points straight uphill; go the opposite way to descend C1.26 §07 Calculus I.

METHOD

1. Compute ∇f at the point.
2. Turn the given direction into a unit vector $\mathbf{u} = \mathbf{v}/\lVert\mathbf{v}\rVert$.
3. Dot them. Example: $f = x^2 + 2y^2$ at $(1,1)$ toward $(3,4)$: $\nabla f = (2,4)$, $\mathbf{u} = (\frac{3}{5}, \frac{4}{5})$, $D_{\mathbf{u}} f = \frac{6+16}{5} = 4.4$; maximum $\lVert\nabla f\rVert = \sqrt{20} \approx 4.47$.

TERMS

∇ – "del" or "nabla" operator
$\mathbf{u}$ – unit direction vector T.21 §06 Trigonometry
$D_{\mathbf{u}} f$ – rate of change moving in direction $\mathbf{u}$
θ – angle between ∇f and $\mathbf{u}$
$\lVert\nabla f\rVert$ – steepest slope
Level set – $f = c$ (contour line or surface); P_0 a point on it
F; $P_0 = \mathbf{x}_0$ – function defining a level surface; a point on it
$\mathbf{a}$; t – a constant vector; parameter along the normal line



CONTEXT

- **Origin.** Hamilton introduced ∇ (1837); gradient named by Maxwell's era (1870s).
- **Use.** Steepest ascent, contour maps, normal vectors.
- **Data science.** Gradient descent on a loss surface ML.27 §15 Machine Learning; saliency maps use $\nabla_{\mathbf{x}}$ of the output to show which pixels matter.
- **Physical application.** Heat flows opposite the temperature gradient P.24 §12 Physics; water runs down the negative gradient of terrain height.

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C3.13 Multivariable Chain Rule

When the inputs of f themselves depend on other variables, the rate of change of f is a sum over every dependency path of the products of derivatives along that path.

One parameter	$\dfrac{dz}{dt} = \dfrac{\partial z}{\partial x}\dfrac{dx}{dt} + \dfrac{\partial z}{\partial y}\dfrac{dy}{dt}$	$z = f(x,y)$ with $x = x(t)$, $y = y(t)$.
Two parameters	$\dfrac{\partial z}{\partial s} = \dfrac{\partial z}{\partial x}\dfrac{\partial x}{\partial s} + \dfrac{\partial z}{\partial y}\dfrac{\partial y}{\partial s}$	Partials because x, y also depend on t, which is held fixed.
Vector form	$\dfrac{d}{dt} f\big(\mathbf{x}(t)\big) = \nabla f \cdot \mathbf{x}'(t)$	Gradient dot velocity C3.12: the rate seen by a moving observer.
General form	$\dfrac{\partial u}{\partial t_j} = \displaystyle\sum_{i=1}^{n} \dfrac{\partial u}{\partial x_i}\dfrac{\partial x_i}{\partial t_j}$	One term per intermediate variable.
Matrix form	$J_{\mathbf{g}\circ\mathbf{f}}(\mathbf{x}) = J_{\mathbf{g}}\big(\mathbf{f}(\mathbf{x})\big)\, J_{\mathbf{f}}(\mathbf{x})$	Jacobians multiply C3.15; the sum above is one entry of this product.

STATED-EQUATION *d z d t equals partial z partial x times d x d t, plus partial z partial y times d y d t. Similarly for partial z partial s. In vector form, the derivative of f of x of t equals the gradient of f dot x prime of t. In general, Jacobians multiply: the Jacobian of g composed with f is the Jacobian of g times the Jacobian of f. In general, partial u partial t j is the sum over i of partial u partial x i times partial x i partial t j.*

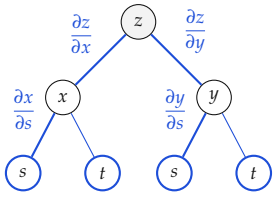
MEMORY AID Sum over every path from the input to the output: multiply along each path, add the paths.

METHOD

1. Draw the tree: output, intermediate variables, independent variables.
2. For each branch from the output down to the target variable, multiply the derivatives on its edges.
3. Add the branches; substitute values. Example: $z = x^2y$, $x = 2t$, $y = t + 1$: $\frac{dz}{dt} = 2xy \cdot 2 + x^2 \cdot 1$; at $t = 1$ ($x = y = 2$): $16 + 4 = 20$.

TERMS

Paths – each intermediate variable contributes one term
s, t; t_j – independent variables
x_i – intermediate variables; u the output
Single-variable case – C1.13 §07 Calculus I
Vector form – gradient dot velocity C3.12
Computation graph – backpropagation applies this rule node by node ML.22 §15 Machine Learning
u – the dependent variable
J – Jacobian matrix C3.15



CONTEXT

- **Origin.** Leibniz and Euler; systematized by Jacobi.
- **Use.** Related rates with several variables C1.20 §07 Calculus I, thermodynamics.
- **Data science.** Backpropagation is the multivariable chain rule on a computation graph; automatic differentiation (PyTorch, JAX) implements it.
- **Physical application.** Temperature felt by a moving car: $\frac{dT}{dt} = \frac{\partial T}{\partial x}\frac{dx}{dt} + \frac{\partial T}{\partial t}$, change along the road plus change over time.

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C3.14 Implicit Differentiation with Partials

When a curve or surface is given by an equation $F = 0$ rather than solved for one variable, the chain rule gives the slopes directly from partial derivatives of F C1.17 §07 Calculus I.

Implicit curve	$F(x,y)=0 \;\Rightarrow\; \dfrac{dy}{dx} = -\dfrac{F_x}{F_y} \quad (F_y \neq 0)$	Differentiate $F(x,y(x))=0$: $F_x + F_y\,y' = 0$.
Implicit surface	$F(x,y,z)=0 \;\Rightarrow\; \dfrac{\partial z}{\partial x} = -\dfrac{F_x}{F_z},\ \dfrac{\partial z}{\partial y} = -\dfrac{F_y}{F_z}$	Same idea with $z = z(x,y)$; requires $F_z \neq 0$.
Implicit function theorem	$F(a,b)=0,\ F_y(a,b)\neq 0 \;\Rightarrow\; y = g(x)$ near a	Guarantees the equation can be solved locally for y, with a differentiable g.
Cyclic (triple product) rule	$\left(\dfrac{\partial x}{\partial y}\right)_z\left(\dfrac{\partial y}{\partial z}\right)_x\left(\dfrac{\partial z}{\partial x}\right)_y = -1$	For three variables linked by one equation; the product is -1, not $+1$.

STATED-EQUATION *If F of x, y equals zero, d y d x equals minus F x over F y. If F of x, y, z equals zero, partial z partial x equals minus F x over F z, and partial z partial y equals minus F y over F z. If F y is not zero at a point, the equation defines y as a differentiable function of x nearby. Cyclic rule: partial x partial y at constant z, times partial y partial z at constant x, times partial z partial x at constant y, equals minus one.*

MEMORY AID "Minus the ratio, with the dependent variable's partial on the bottom."

METHOD

1. Move everything to one side: $F = 0$.
2. Compute the needed partials of F, treating all variables as independent.
3. Form $-F_x/F_z$ (or $-F_x/F_y$) and evaluate. Example: sphere $x^2+y^2+z^2=9$ at $(1,2,2)$: $\frac{\partial z}{\partial x} = -\frac{2x}{2z} = -\frac{1}{2}$, $\frac{\partial z}{\partial y} = -\frac{y}{z} = -1$.

TERMS

F – function whose zero set is the curve or surface
F_x, F_y, F_z – its partial derivatives C3.9
g – the (local) explicit solution $y = g(x)$
$(\cdot)_z$ – partial with z held fixed
Derivation – chain rule C3.13

CONTEXT

- **Origin.** Implicit functions: Cauchy (1831) and Dini (1878).
- **Use.** Curves and surfaces defined by constraints, thermodynamics.
- **Data science.** Implicit differentiation lets deep equilibrium models and differentiable optimization layers backpropagate through a solution $z^* = f(z^*, x)$ without unrolling the solver.
- **Physical application.** For an ideal gas, $\left(\frac{\partial P}{\partial V}\right)_T\left(\frac{\partial V}{\partial T}\right)_P\left(\frac{\partial T}{\partial P}\right)_V = -1$ P.29 §12 Physics.

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C3.15 The Jacobian

The Jacobian is the matrix of all first partial derivatives of a vector-valued function. It is the best linear approximation of the map near a point, and its determinant is the local area (volume) scale factor.

Jacobian matrix	$J = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \cdots & \frac{\partial f_m}{\partial x_n} \end{bmatrix}$	Row i is the gradient of output f_i; $m \times n$.
Linear approximation	$\mathbf{f}(\mathbf{x}+\mathbf{h}) \approx \mathbf{f}(\mathbf{x}) + J\mathbf{h}$	The vector version of the tangent plane C3.11.
Jacobian determinant	$\frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} x_u & x_v \\ y_u & y_v \end{vmatrix} = x_u y_v - x_v y_u$	For a square map; the sign records orientation.
Area scaling	$dx\,dy = \left\lvert \frac{\partial(x,y)}{\partial(u,v)} \right\rvert du\,dv$	Why change of variables needs $\lvert\det J\rvert$ C3.22.
Chain rule	$J_{\mathbf{g}\circ\mathbf{f}} = J_{\mathbf{g}}\,J_{\mathbf{f}}$	Composition multiplies Jacobians C3.13.
Inverse function theorem	$\det J_{\mathbf{f}}(\mathbf{a}) \neq 0 \;\Rightarrow\; J_{\mathbf{f}^{-1}} = (J_{\mathbf{f}})^{-1}$	A map is locally invertible where its Jacobian is invertible.

STATED-EQUATION *The Jacobian J is the matrix of all first partial derivatives: row i, column j is partial f i partial x j. f of x plus h is approximately f of x plus J h. The Jacobian determinant of x, y with respect to u, v is x u y v minus x v y u. Under a change of variables, d x d y equals the absolute value of the Jacobian determinant times d u d v. The Jacobian of a composition is the product of the Jacobians. If the Jacobian determinant is nonzero at a, f is locally invertible and the Jacobian of the inverse is the inverse of J.*

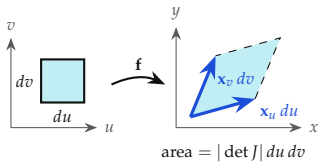
MEMORY AID Rows are outputs, columns are inputs; $\lvert\det J\rvert$ is the local area-stretch factor.

METHOD

1. List outputs as rows, inputs as columns.
2. Fill entry (i,j) with $\partial f_i/\partial x_j$.
3. For a square map take the determinant. Example: polar $x = r\cos\theta$, $y = r\sin\theta$: $J = \begin{bmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{bmatrix}$, $\det J = r$; at $r = 2$ areas double.

TERMS

$\mathbf{f} : \mathbb{R}^n \to \mathbb{R}^m$ – vector-valued function with outputs $f_1, \ldots, f_m$
J – $m \times n$; best linear approximation C1.21 §07 Calculus I, LA.9 §09 Linear Algebra
$\det J$ – area/volume scale factor A2.25 §04 Algebra II
x_u etc. – partial derivatives $\partial x/\partial u$
Gradient – Jacobian of a scalar function is ∇f^{T} C3.12
Chain rule – $J_{\mathbf{g}\circ\mathbf{f}} = J_{\mathbf{g}}J_{\mathbf{f}}$ C3.13
$\mathbf{h}$; $\mathbf{a}$ – a small displacement; a base point
u, v – new variables in a change of coordinates



CONTEXT

- **Origin.** Carl Jacobi (1841).
- **Use.** Coordinate changes, robotics kinematics, nonlinear systems.
- **Data science.** Normalizing flows use $\log\lvert\det J\rvert$ to transform densities; backprop multiplies Jacobians layer by layer.
- **Physical application.** A robot arm's Jacobian converts joint rotation speeds into the speed of its hand.

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Optimization

C3.16 Hessian & the Second-Derivative Test

At a critical point the gradient vanishes, so the second partials decide the shape: bowl (minimum), dome (maximum) or saddle. The Hessian matrix collects them.

Critical point	$\nabla f(a,b) = \mathbf{0} \quad (\text{or } \nabla f \text{ undefined})$	The only interior candidates for local extrema: the tangent plane is horizontal.
Hessian matrix	$H = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix}$	Symmetric when the mixed partials are continuous C3.10.
Discriminant	$D = \det H = f_{xx}f_{yy} - f_{xy}^2$	Product of the Hessian's eigenvalues LA.23 §09 Linear Algebra.
Second-derivative test	$\begin{cases} D>0,\ f_{xx}>0 & \text{local min} \\ D>0,\ f_{xx}<0 & \text{local max} \\ D<0 & \text{saddle point} \\ D=0 & \text{inconclusive} \end{cases}$	Evaluate at each critical point.
Second-order Taylor	$f(\mathbf{a}+\mathbf{h}) \approx f(\mathbf{a}) + \nabla f \cdot \mathbf{h} + \frac{1}{2}\,\mathbf{h}^{\mathsf{T}} H\,\mathbf{h}$	The quadratic model behind the test and Newton's method O.13 §14 Optimization.
Eigenvalue test (n variables)	all $\lambda_i > 0 \Rightarrow$ min; all $\lambda_i < 0 \Rightarrow$ max; mixed signs $\Rightarrow$ saddle	The 2-variable test is the special case LA.27 §09 Linear Algebra.

STATED-EQUATION *A critical point is where the gradient is zero. The Hessian H is the matrix of second partial derivatives. D, its determinant, equals f x x times f y y minus f x y squared. If D is positive and f x x is positive, it is a local minimum; if D is positive and f x x negative, a local maximum; if D is negative, a saddle point; if D is zero, the test is inconclusive. Near a point, f is approximately f plus gradient dot h plus one half h transpose H h. With n variables: all Hessian eigenvalues positive gives a minimum, all negative a maximum, and mixed signs a saddle.*

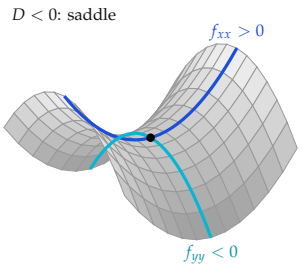
MEMORY AID Hessian positive definite → bowl (min); negative definite → dome (max); mixed → saddle LA.27 §09 Linear Algebra.

METHOD

1. Solve $f_x = 0$, $f_y = 0$ simultaneously for the critical points.
2. Compute f_{xx}, f_{yy}, f_{xy} and D at each.
3. Classify. Example: $f = x^2 + xy + y^2 - 3x$: $2x + y = 3$, $x + 2y = 0 \Rightarrow (2,-1)$; $D = 2\cdot 2 - 1^2 = 3 > 0$, $f_{xx} = 2 > 0$: local min $f(2,-1) = -3$.

TERMS

Critical point – $\nabla f = \mathbf{0}$ C3.12
H – symmetric (Clairaut C3.10); curvature in every direction
D – product of the eigenvalues of H LA.23 §09 Linear Algebra
λ_i – eigenvalues of H; $\mathbf{h}$ a small step
Saddle – up in one direction, down in another
One variable – reduces to the f'' test C1.23 §07 Calculus I
$\mathbf{a}, \mathbf{h}$ – base point; small displacement



CONTEXT

- **Origin.** Otto Hesse (1840s); second-derivative test from Lagrange.
- **Use.** Classifying extrema of surfaces; stability.
- **Data science.** Newton's method and second-order optimizers use H O.13 §14 Optimization; high-dimensional loss surfaces are full of saddle points rather than local minima.
- **Physical application.** A mountain pass is a saddle: the lowest point along the ridge but the highest point along the valley path.

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C3.17 Absolute Extrema on Closed, Bounded Regions

A continuous function on a closed, bounded region always attains a largest and a smallest value. They occur at interior critical points or somewhere on the boundary, so check both and compare.

Extreme value theorem	f continuous on closed, bounded $D \;\Rightarrow\; \max_D f,\ \min_D f$ exist	Both conditions matter: an open or unbounded region may have no extremum.
Candidate list	$\{\nabla f = \mathbf{0} \text{ in } D\} \cup \{\text{critical points on } \partial D\} \cup \{\text{corners}\}$	Evaluate f at every candidate; the largest and smallest values win.
Boundary reduction	$\partial D:\ \mathbf{r}(t),\ \ g(t) = f(\mathbf{r}(t)),\ \ g'(t) = 0$	Each boundary piece becomes a one-variable closed-interval problem C1.23 §07 Calculus I.

STATED-EQUATION *A continuous function on a closed, bounded region has an absolute maximum and minimum. The candidates are the interior critical points, the critical points of f restricted to the boundary, and the corners of the boundary. On the boundary, parametrize it by r of t, set g of t equal to f of r of t, and solve g prime of t equals zero.*

MEMORY AID "Inside, edges, corners — then compare."

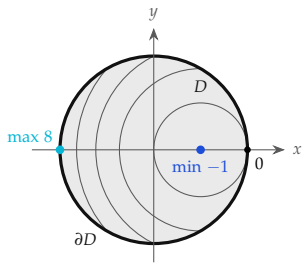
METHOD

1. Find interior critical points; keep those inside D.
2. Parametrize each boundary piece and find its one-variable critical points and endpoints (corners).
3. Evaluate f at all candidates. Example: $f = x^2 + y^2 - 2x$ on $x^2 + y^2 \le 4$: interior $(1,0)$, $f = -1$; boundary $f = 4 - 2x$, $x \in [-2,2]$: values 8 and 0. Max 8 at $(-2,0)$, min -1 at $(1,0)$.

DECISION POINT How to handle the boundary? Straight edges: substitute the edge equation (e.g. $y = 0$) and use the endpoints as corners. Circles and ellipses: parametrize ($x = 2\cos t$, $y = 2\sin t$) or substitute $x^2 + y^2 = 4$ directly as above. A smooth boundary $g = c$ that is hard to parametrize: use Lagrange multipliers C3.18.

TERMS

D – region; *closed* (contains its boundary) and *bounded* (fits in a disk)
∂D – boundary of D
$\mathbf{r}(t)$ – parametrization of a boundary piece
$g(t)$ – f restricted to that piece
Corner – endpoint of a boundary piece



CONTEXT

- **Origin.** Weierstrass' extreme value theorem (1860s).
- **Use.** Resource allocation with capacity limits; design within tolerances.
- **Data science.** Box-constrained training clips parameters to a closed box (projected gradient descent): the optimum may sit on the boundary, where $\nabla f \neq \mathbf{0}$.
- **Physical application.** The hottest and coldest points of a heated metal plate lie at interior critical points or on its edge.

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C3.18 Lagrange Multipliers

To optimize f subject to a constraint $g = c$, look for points where a level curve of f just touches the constraint curve: there the two gradients are parallel. The general theory is in O.17 §14 Optimization.

Tangency condition	$\nabla f = \lambda \nabla g, \qquad g(x,y) = c$	3 equations in x, y, λ; every constrained extremum (with $\nabla g \neq \mathbf{0}$) satisfies them.
Lagrangian	$\mathcal{L}(x,y,\lambda) = f(x,y) - \lambda\big(g(x,y) - c\big), \quad \nabla\mathcal{L} = \mathbf{0}$	Packages the same equations as one unconstrained critical-point problem.
Two constraints	$\nabla f = \lambda \nabla g + \mu \nabla h, \qquad g = c_1,\ h = c_2$	One multiplier per constraint; ∇f lies in the span of the constraint normals.
Meaning of λ	$\lambda = \dfrac{df^*}{dc}$	Shadow price: how much the optimal value f^* improves per unit relaxation of the constraint.

STATED-EQUATION *At a constrained extremum, the gradient of f equals lambda times the gradient of g, together with g equals c. Equivalently, the gradient of the Lagrangian f minus lambda times g minus c equals zero. With two constraints, the gradient of f equals lambda grad g plus mu grad h. Lambda is the rate of change of the optimal value with respect to c.*

MEMORY AID At the optimum the level curve of f *kisses* the constraint: parallel gradients.

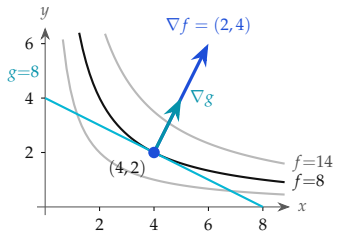
METHOD

1. Write $\nabla f = \lambda \nabla g$ component by component, plus $g = c$.
2. Eliminate λ (divide or compare equations) to relate x and y.
3. Substitute into $g = c$; evaluate f at every solution and compare. Example: max xy on $x + 2y = 8$: $y = \lambda$, $x = 2\lambda \Rightarrow x = 2y$, $4y = 8$: $(4,2)$, $f = 8$, $\lambda = 2$.

DECISION POINT Lagrange or substitution? If the constraint is easy to solve for one variable (a line), substitute and use one-variable calculus. If it is a closed curve or surface (circle, sphere, ellipse), or symmetric in the variables, use multipliers: they avoid square roots and missed branches. Multipliers only find candidates; compare values (and check endpoints of open curves). Inequality constraints need KKT O.18 §14 Optimization.

TERMS

f – objective; $g = c$ (and $h = c_2$) constraints
λ, μ – Lagrange multipliers
$\mathcal{L}$ – the Lagrangian
f^* – optimal value, as a function of c
Requirement – $\nabla g \neq \mathbf{0}$ at the optimum (constraint qualification)
g, h; c, c_1, c_2 – constraint functions; their constant levels



CONTEXT

- **Origin.** Lagrange, *Mécanique analytique* (1788).
- **Use.** Economics (utility under a budget), engineering design, physics with constraints.
- **Data science.** PCA maximizes $\mathbf{w}^\mathsf{T}\Sigma\mathbf{w}$ subject to $\|\mathbf{w}\| = 1$, giving $\Sigma\mathbf{w} = \lambda\mathbf{w}$ ML.38 §15 Machine Learning; maximum-entropy distributions and the SVM dual come from Lagrangians.
- **Physical application.** In mechanics the multiplier is the constraint force, e.g. the tension in a pendulum rod.

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Multiple Integrals

C3.19 Double & Triple Integrals

A double integral adds up $f\,dA$ over a region: the volume under a surface when $f \geq 0$. It is computed as two ordinary integrals in a row (an iterated integral), with inner limits that may depend on the outer variable.

Definition (Riemann sum)	$\displaystyle\iint_R f\,dA = \lim_{\|P\|\to 0} \sum_{i,j} f(x_i^*, y_j^*)\,\Delta A$	Columns of height f over tiny rectangles C1.30 §07 Calculus I.
Iterated integral (rectangle)	$\displaystyle\iint_R f\,dA = \int_a^b \int_c^d f(x,y)\,dy\,dx$	Inner integral in y with x frozen; order swappable C3.20.
Type I region	$\displaystyle\iint_D f\,dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x,y)\,dy\,dx$	Vertical strips: bottom curve to top curve.
Type II region	$\displaystyle\iint_D f\,dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x,y)\,dx\,dy$	Horizontal strips: left curve to right curve.
Area & volume	$\displaystyle\iint_D 1\,dA = \text{Area}(D), \qquad \iiint_E 1\,dV = \text{Vol}(E)$	Integrating 1 measures the region itself.
Triple integral	$\displaystyle\iiint_E f\,dV = \iint_D \Big[\int_{u_1(x,y)}^{u_2(x,y)} f\,dz\Big] dA$	Integrate z from the bottom surface to the top, then over the shadow D.
Average value	$\displaystyle f_{\text{avg}} = \frac{1}{\text{Area}(D)} \iint_D f\,dA$	Total over the region divided by its size C2.12 §08 Calculus II.
Separable integrand	$\displaystyle\iint_{[a,b]\times[c,d]} g(x)h(y)\,dA = \int_a^b g\,dx \cdot \int_c^d h\,dy$	On a rectangle a product factors into two single integrals.

STATED-EQUATION *The double integral of f over R is the limit of Riemann sums of f times delta A. Over a rectangle it equals the iterated integral from a to b of the integral from c to d of f d y d x. For a type one region, y runs from g one of x to g two of x; for type two, x runs from h one of y to h two of y. The double integral of one gives the area. Triple integrals work the same way over a solid E. The triple integral of one gives the volume, and the triple integral of f is the integral over D of the inner integral of f d z from u one to u two. The average value is the integral of f divided by the area. A separable integrand g of x times h of y over a rectangle integrates to the product of the two single integrals.*

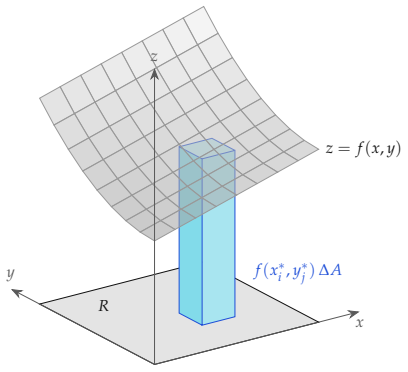
MEMORY AID Integrate the inside first (its limits may depend on the outer variable), treating the outer variable as a constant.

METHOD

1. Sketch the region; choose the order (vertical or horizontal strips).
2. Inner integral: outer variable held constant, limits from boundary curves.
3. Outer integral: constant limits only. Example: $\int_0^2 \int_0^1 (x+2y)\,dy\,dx = \int_0^2 \left[xy+y^2\right]_0^1 dx = \int_0^2 (x+1)\,dx = 4$.

TERMS

dA – area element $dx\,dy$; ΔA its finite version
R, D, E – rectangle, plane region, solid
$\|P\|$ – size of the largest sub-rectangle
g_i, h_i, u_i – boundary curves and surfaces
Iterated integral – two single integrals C1.32 §07 Calculus I; bounded regions C2.7 §08 Calculus II
dV – volume element
x_i^*, y_j^*; ΔA – sample points; area of a small rectangle
f_{avg} – average value of f on D



CONTEXT

- **Origin.** Euler (1769); Fubini (1907).
- **Use.** Mass, center of mass, total charge, volume under a surface C3.23.
- **Data science.** Joint probability: $P((X,Y) \in R) = \iint_R f(x,y)\,dA$; marginals integrate out one variable S.13 §13 Statistics.
- **Physical application.** Total rainfall over a region is the double integral of rainfall depth per area.

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C3.20 Fubini's Theorem & Changing the Order of Integration

For well-behaved integrands the order of integration does not change the answer, but it can change the difficulty enormously. Reversing the order means re-describing the same region with strips in the other direction.

Fubini's theorem	$\displaystyle\int_a^b\int_c^d f\,dy\,dx = \int_c^d\int_a^b f\,dx\,dy$	Valid for continuous f (or whenever $\iint \lvert f\rvert\,dA < \infty$).
Tonelli's theorem	$f \geq 0 \;\Rightarrow$ order can always be swapped	Both orders give the same value, possibly $+\infty$.
Reversing the order	$\displaystyle\int_0^1\int_x^1 f\,dy\,dx = \int_0^1\int_0^y f\,dx\,dy$	Same triangle $0 \leq x \leq y \leq 1$ seen with vertical, then horizontal strips.

STATED-EQUATION *Fubini's theorem: for continuous f the iterated integrals in either order are equal. Tonelli: for non-negative f the order can always be swapped. Reversing the order on the triangle where x is between zero and y and y is at most one: the integral from zero to one of the integral from x to one, d y d x, equals the integral from zero to one of the integral from zero to y, d x d y.*

MEMORY AID Same region, new strips: the limits change, the region does not.

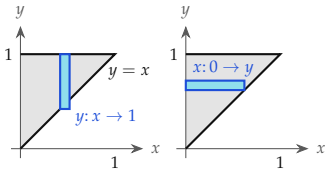
METHOD

1. Sketch the region from the given limits (inner limits are curves, outer limits are lines).
2. Read the new outer variable's full constant range from the sketch.
3. For a fixed value of it, read the new inner limits from the boundary curves.
4. Integrate. Example: $\int_0^1\int_x^1 e^{y^2}\,dy\,dx$ has no elementary inner antiderivative; swap: $\int_0^1\int_0^y e^{y^2}\,dx\,dy = \int_0^1 y e^{y^2}\,dy = \frac{e-1}{2}$.

DECISION POINT Which order? First choose the order whose *inner* integral you can actually do (e^{y^2}, $\frac{\sin x}{x}$, $\sqrt{1+x^3}$ need the other variable first). Then prefer the order that describes the region with *one* integral: if a boundary changes formula partway along an axis, strips in that direction need the region split into pieces.

TERMS

f – integrand; continuous or absolutely integrable
Vertical strip – x fixed, y from bottom to top curve ($dy\,dx$)
Horizontal strip – y fixed, x from left to right curve ($dx\,dy$)
Type I / II – regions described by vertical / horizontal strips C3.19



$dy\,dx$: vertical strip $dx\,dy$: horizontal strip

CONTEXT

- **Origin.** Guido Fubini (1907); Leonida Tonelli (1909).
- **Use.** Evaluating otherwise impossible iterated integrals; swapping sums and integrals.
- **Data science.** Justifies swapping expectations and integrals or sums, e.g. $\mathbb{E}\left[\sum_i X_i\right] = \sum_i \mathbb{E}[X_i]$ and computing marginal densities in either order S.13 §13 Statistics.
- **Physical application.** The mass of a triangular plate is the same whether you add up vertical or horizontal strips.

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C3.21 Polar, Cylindrical & Spherical Coordinates

Regions with circular, cylindrical or spherical symmetry become rectangles in the matching coordinates. Each system brings its own area or volume element: the Jacobian of the conversion.

Polar conversion	$x = r\cos\theta, \quad y = r\sin\theta, \quad r^2 = x^2 + y^2$	Coordinates of T.19 §06 Trigonometry; $r \geq 0$.
Polar double integral	$\iint_D f\,dA = \int_\alpha^\beta \int_{r_1(\theta)}^{r_2(\theta)} f(r\cos\theta, r\sin\theta)\, r\,dr\,d\theta$	$dA = r\,dr\,d\theta$: polar boxes get wider farther out.
Cylindrical coordinates	$(r,\theta,z): \quad dV = r\,dz\,dr\,d\theta$	Polar in the xy-plane plus height z.
Spherical conversion	$x = \rho\sin\phi\cos\theta, \quad y = \rho\sin\phi\sin\theta, \quad z = \rho\cos\phi$	ρ distance from origin, ϕ angle from $+z$, θ azimuth.
Spherical volume element	$dV = \rho^2\sin\phi\,d\rho\,d\phi\,d\theta$	The Jacobian of the spherical map C3.15.
Gaussian integral	$\int_{-\infty}^{\infty} e^{-x^2}dx = \sqrt{\pi}$	Square it and switch to polar: $\iint e^{-(x^2+y^2)}dA = \pi$.

STATED-EQUATION *In polar coordinates, x equals r cosine theta and y equals r sine theta, and d A equals r d r d theta. In cylindrical, d V equals r d z d r d theta. In spherical, x equals rho sine phi cosine theta, y equals rho sine phi sine theta, z equals rho cosine phi, and d V equals rho squared sine phi d rho d phi d theta. The Gaussian integral of e to the minus x squared over the whole real line equals the square root of pi. Also, r squared equals x squared plus y squared.*

MEMORY AID "Don't forget the extra r": it is the Jacobian C3.15.

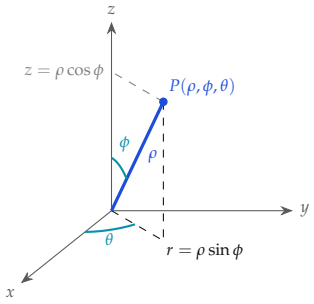
METHOD

1. Describe the region by ranges of r, θ (or ρ, ϕ, θ).
2. Substitute x, y (and z) in the integrand and append the extra factor r (or $\rho^2\sin\phi$).
3. Integrate. Example: $\iint_{x^2+y^2\leq 4}(x^2+y^2)\,dA = \int_0^{2\pi}\int_0^2 r^2 \cdot r\,dr\,d\theta = 2\pi\cdot 4 = 8\pi$.

DECISION POINT Which coordinates? Disks, annuli, sectors or x^2+y^2 in the integrand → polar. Solids with an axis of symmetry (cylinders, paraboloids $z = x^2+y^2$) → cylindrical. Balls, spherical shells, cones $\phi = \text{const}$, or $x^2+y^2+z^2$ → spherical. Boxes and polygons → stay Cartesian.

TERMS

r, θ – polar radius and angle T.19 §06 Trigonometry; $0 \leq \theta < 2\pi$
z – height (cylindrical)
ρ, ϕ, θ – distance from origin, angle from $+z$ axis ($0 \leq \phi \leq \pi$), azimuth
α, β; r_1, r_2 – angle limits; inner and outer radius curves
Extra factors – Jacobian determinants C3.15
Gaussian integral – square it and switch to polar: $\iint e^{-(x^2+y^2)}dA = \pi$



CONTEXT

- **Origin.** Polar integration by Euler; the Gaussian integral by Laplace (1774) and Gauss.
- **Use.** Circular, cylindrical and spherical symmetry: pipes, planets, antennas.
- **Data science.** The Gaussian integral is why the normal density has the constant $\frac{1}{\sqrt{2\pi}}$ S.20 §13 Statistics.
- **Physical application.** The mass of the atmosphere, a spherical shell, is computed in spherical coordinates.

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C3.22 Change of Variables in Multiple Integrals

Substituting $x = x(u,v)$, $y = y(u,v)$ turns an awkward region into a simple one; the area element picks up the absolute Jacobian determinant. Polar, cylindrical and spherical coordinates are special cases C3.21.

Change of variables (2-D)	$\iint_R f(x,y)\,dx\,dy = \iint_S f\big(x(u,v),\,y(u,v)\big)\left\lvert\dfrac{\partial(x,y)}{\partial(u,v)}\right\rvert du\,dv$	S in the uv-plane maps one-to-one onto R.
Change of variables (3-D)	$\iiint_E f\,dV = \iiint_S f\left\lvert\dfrac{\partial(x,y,z)}{\partial(u,v,w)}\right\rvert du\,dv\,dw$	Same rule with a 3×3 determinant C3.15.
Inverse Jacobian	$\dfrac{\partial(x,y)}{\partial(u,v)} = \left(\dfrac{\partial(u,v)}{\partial(x,y)}\right)^{-1}$	Use when u,v are given in terms of x,y; avoids solving for x,y.
Linear map	$\begin{pmatrix}x\\y\end{pmatrix} = A\begin{pmatrix}u\\v\end{pmatrix} \Rightarrow \lvert\det J\rvert = \lvert\det A\rvert$	Constant scale factor; parallelograms map to rectangles LA.9 §09 Linear Algebra.
Density transformation	$f_{\mathbf{Y}}(\mathbf{y}) = f_{\mathbf{X}}(\mathbf{x}(\mathbf{y}))\left\lvert\det\dfrac{\partial\mathbf{x}}{\partial\mathbf{y}}\right\rvert$	Probability is conserved, so densities rescale by the Jacobian S.13 §13 Statistics.

STATED-EQUATION *The double integral of f over R d x d y equals the double integral over S of f of x of u v, y of u v, times the absolute value of the Jacobian determinant of x, y with respect to u, v, d u d v. In three dimensions use the three by three Jacobian. The Jacobian of x, y with respect to u, v is the reciprocal of the Jacobian of u, v with respect to x, y. For a linear map, x, y equals A times u, v, the absolute Jacobian determinant is the absolute value of det A. A density transforms as f Y of y equals f X of x of y, times the absolute value of the determinant of partial x partial y.*

MEMORY AID New variables, new area element: always multiply by $|\det J|$.

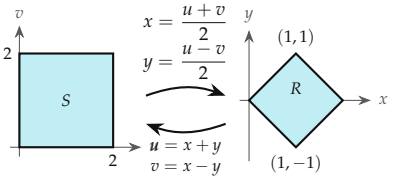
METHOD

1. Choose u,v from the boundary equations (or the integrand).
2. Find the new region S and $\left\lvert\frac{\partial(x,y)}{\partial(u,v)}\right\rvert$ (directly, or as $1/\left\lvert\frac{\partial(u,v)}{\partial(x,y)}\right\rvert$).
3. Rewrite and integrate. Example: R bounded by $x+y=0,2$ and $x-y=0,2$; $u=x+y$, $v=x-y$: $\left\lvert\frac{\partial(u,v)}{\partial(x,y)}\right\rvert = 2$, so $\iint_R (x+y)\,dA = \int_0^2\int_0^2 u\cdot\frac{1}{2}\,du\,dv = 2$.

DECISION POINT Which substitution? Make the boundary curves become $u=$ const and $v=$ const (lines $x+y=c$ suggest $u=x+y$; hyperbolas $xy=c$ suggest $u=xy$; ellipses suggest $x=au$, $y=bv$), or make the integrand simple. Check the map is one-to-one on the region.

TERMS

u,v,w – new variables; S the region in uv-space
$\frac{\partial(x,y)}{\partial(u,v)}$ – Jacobian determinant C3.15
A – constant 2×2 matrix
$f_{\mathbf{X}}, f_{\mathbf{Y}}$ – probability densities of $\mathbf{X}$ and $\mathbf{Y}$
R, S – a region in the xy-plane and its image in the uv-plane



CONTEXT

- **Origin.** Euler (1769) in two variables; Jacobi (1841) in general.
- **Use.** Elliptical and skewed regions, polar/spherical integration, physics in natural coordinates.
- **Data science.** Normalizing flows and the reparameterization trick rely on the density-transformation rule; it also gives the density of $\mathbf{Y} = A\mathbf{X} + \mathbf{b}$.
- **Physical application.** The area of an ellipse: $x = au$, $y = bv$ maps the unit disk to it with $|\det J| = ab$, so $A = \pi ab$.

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C3.23 Mass, Moments & Center of Mass

Integrating a density over a region gives total mass; weighting by position gives moments, and dividing moments by mass gives the balance point. The same integrals compute means of joint distributions.

Mass	$m = \iint_D \rho(x,y)\,dA \qquad \left(\text{solid: } m = \iiint_E \rho\,dV\right)$	Density times area, added up.
First moments	$M_x = \iint_D y\,\rho\,dA, \qquad M_y = \iint_D x\,\rho\,dA$	Moment *about the x-axis* uses the distance y from it.
Center of mass	$\bar{x} = \dfrac{M_y}{m}, \qquad \bar{y} = \dfrac{M_x}{m}$	The balance point; for constant ρ it is the centroid.
Moments of inertia	$I_x = \iint_D y^2\rho\,dA, \quad I_y = \iint_D x^2\rho\,dA, \quad I_0 = I_x + I_y$	Resistance to rotation about an axis (or the origin).
Centroid	$\bar{x} = \dfrac{1}{A}\iint_D x\,dA, \qquad \bar{y} = \dfrac{1}{A}\iint_D y\,dA$	Uniform density; the average position C2.12 §08 Calculus II.
Probability analog	$P\big((X,Y)\in D\big) = \iint_D f\,dA, \qquad \mathbb{E}[X] = \iint x\,f\,dA$	A density has mass 1; its center of mass is the mean S.13 §13 Statistics.

STATED-EQUATION *Mass is the double integral of the density rho. The moment about the x-axis is the integral of y rho, about the y-axis the integral of x rho. The center of mass is x bar equals M y over m, y bar equals M x over m. Moments of inertia integrate squared distance times density. For a probability density, the integral over D is a probability and the center of mass is the expected value. The polar moment I zero equals I x plus I y. The centroid is the center of mass with constant density: x bar equals one over A times the integral of x d A, and y bar likewise.*

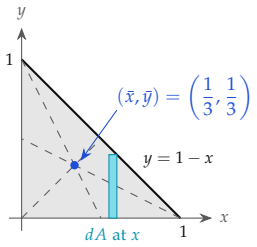
MEMORY AID Moment about an axis uses the *other* coordinate: M_x has y, M_y has x.

METHOD

1. Set up $m = \iint \rho\,dA$ with limits from a sketch.
2. Compute M_y and M_x over the same limits.
3. Divide. Example: triangle $(0,0),(1,0),(0,1)$, $\rho = 1$: $m = \frac{1}{2}$, $M_y = \int_0^1\int_0^{1-x} x\,dy\,dx = \frac{1}{6}$, $\bar{x} = \frac{1}{3}$; by symmetry $\bar{y} = \frac{1}{3}$.

TERMS

ρ – density (mass per area or volume)
m – mass; A area
M_x, M_y – first moments about the x- and y-axes
$(\bar{x},\bar{y})$ – center of mass
I_x, I_y, I_0 – moments of inertia
f – joint probability density S.13 §13 Statistics
A – area of D
$\mathbb{E}[X]$ – expected value S.12 §13 Statistics



CONTEXT

- **Origin.** Archimedes found centroids by balancing (3rd c. BCE); integral formulas from Euler.
- **Use.** Statics, balance and stability, structural design.
- **Data science.** The mean of a joint distribution is its center of mass; the variance is the moment of inertia about the mean divided by the mass.
- **Physical application.** A figure skater pulling in her arms lowers I, so her spin speeds up (conservation of angular momentum).

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C3.24 Surface Area

Over each small patch dA of the domain, the surface is nearly its tangent plane, whose area is dA stretched by the tilt. Adding up the tilted patches gives surface area.

Area of a graph	$A = \iint_D \sqrt{1 + f_x^2 + f_y^2}\, dA$	$\sqrt{1+f_x^2+f_y^2} = \|(f_x, f_y, -1)\|$, the length of the normal C3.11.
Area of a parametric surface	$A = \iint_D \|\mathbf{r}_u \times \mathbf{r}_v\|\, du\, dv$	General formula; parametrized surfaces in V.8 §11 Vector Calculus.
Sphere check	$A = \int_0^{2\pi}\int_0^{\pi} R^2 \sin\phi\, d\phi\, d\theta = 4\pi R^2$	Spherical parametrization with $\|\mathbf{r}_\phi \times \mathbf{r}_\theta\| = R^2 \sin\phi$ G.22 §05 Geometry.

STATED-EQUATION *The area of the graph z equals f of x, y over D is the double integral of the square root of one plus f x squared plus f y squared. For a parametric surface, integrate the length of r u cross r v. A sphere of radius R has area four pi R squared.*

MEMORY AID Steeper surface, bigger stretch factor; flat ($f_x = f_y = 0$) gives back the plain area.

METHOD

1. Compute f_x and f_y (or $\mathbf{r}_u \times \mathbf{r}_v$).
2. Form the stretch factor and integrate over the domain (polar if the domain is a disk).
3. Example: the plane $z = 1 + 2x + 2y$ over $[0,1]^2$: $\sqrt{1+4+4} = 3$, so $A = 3$.

TERMS

D – domain region under the surface
f_x, f_y – partial derivatives C3.9
$\mathbf{r}(u,v)$ – parametrization; $\mathbf{r}_u, \mathbf{r}_v$ its partials
R – sphere radius; ϕ, θ spherical angles C3.21
$\mathbf{r}_u, \mathbf{r}_v$ – partial derivatives of $\mathbf{r}(u,v)$
ϕ, θ – spherical angles C3.21

CONTEXT

- **Origin.** Surface area integrals: Euler and Lagrange (18th c.); the general definition by Lebesgue and Fréchet (early 20th c.).
- **Use.** Material and paint estimates, heat loss, membranes.
- **Data science.** 3-D vision and mesh processing weight per-face quantities by face area; the same stretch factor appears when integrating over a manifold.
- **Physical application.** The paraboloid dish $z = x^2 + y^2$, $r \le 1$, has area $\int_0^{2\pi}\int_0^1 \sqrt{1+4r^2}\, r\, dr\, d\theta = \frac{\pi}{6}(5\sqrt{5} - 1) \approx 5.33$.

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11
Vector Calculus

The calculus of fields: integrating along curves and across surfaces, measuring how a field spreads out (divergence) and spins (curl), and the great theorems that trade an integral over a boundary for an integral of a derivative inside. This is the mathematics of fluid flow, electromagnetism and every conservation law.

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Vector Fields & Line Integrals

V.1 Vector Fields & Line Integrals

A vector field attaches an arrow to every point of a region. A line integral adds something up along a curve: either a scalar density against arc length, or the component of a field along the direction of travel.

Vector field	$\mathbf{F}(x,y,z) = \langle P,\ Q,\ R \rangle$	Each component is a scalar function of position; in 2-D, $\mathbf{F} = \langle P, Q \rangle$.
Gradient field	$\mathbf{F} = \nabla f$	Arrows perpendicular to the level sets of f, pointing uphill C3.12 §10 Calculus III.
Flow line	$\mathbf{r}'(t) = \mathbf{F}(\mathbf{r}(t))$	A curve whose velocity is the field at every point (a streamline).
Scalar line integral	$\displaystyle\int_C f\,ds = \int_a^b f(\mathbf{r}(t))\,\lVert\mathbf{r}'(t)\rVert\,dt$	Weights f by arc length; $f = 1$ gives the length of C C3.6 §10 Calculus III.
Vector line integral	$\displaystyle\int_C \mathbf{F}\cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t))\cdot\mathbf{r}'(t)\,dt$	Adds up the component of $\mathbf{F}$ along the direction of motion.
Component form	$\displaystyle\int_C \mathbf{F}\cdot d\mathbf{r} = \int_C P\,dx + Q\,dy + R\,dz$	Substitute $dx = x'(t)\,dt$, etc.; the same integral written in differentials.
Orientation	$\displaystyle\int_{-C}\mathbf{F}\cdot d\mathbf{r} = -\int_C \mathbf{F}\cdot d\mathbf{r}, \qquad \int_{-C} f\,ds = \int_C f\,ds$	Reversing the path flips the sign of a vector line integral but not of a scalar one.
Additivity	$\displaystyle\int_{C_1\cup C_2} = \int_{C_1} + \int_{C_2}$	Split a piecewise-smooth path into smooth pieces.

STATED-EQUATION *A vector field F assigns the vector P, Q, R to each point. The scalar line integral of f with respect to arc length equals the integral of f at r of t times the speed, the norm of r prime of t. The line integral of F along C equals the integral from a to b of F at r of t, dotted with r prime of t, d t, which is also the integral of P d x plus Q d y plus R d z. Reversing the path changes the sign of the vector line integral only. A gradient field is F equal to the gradient of f, and a flow line is a curve whose velocity r prime of t equals F at r of t. Reversing the path leaves the scalar line integral unchanged, and the integral over C one joined with C two is the integral over C one plus the integral over C two.*

MEMORY AID $d\mathbf{r} = \mathbf{r}'(t)\,dt$ (direction matters); $ds = \lVert\mathbf{r}'(t)\rVert\,dt$ (only length matters).

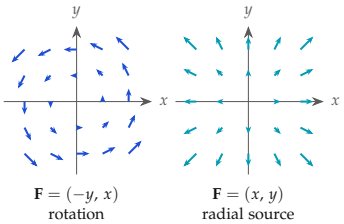
METHOD

1. Parametrize C as $\mathbf{r}(t)$, $a \le t \le b$, in the required direction.
2. Compute $\mathbf{r}'(t)$ and substitute $\mathbf{r}(t)$ into $\mathbf{F}$ (or f).
3. Dot (or multiply by the speed) and integrate in t. Example: $\mathbf{F} = \langle y, x\rangle$ along $\mathbf{r} = \langle t, t^2\rangle$, $0 \le t \le 1$: $\mathbf{F}\cdot\mathbf{r}' = \langle t^2, t\rangle \cdot \langle 1, 2t\rangle = 3t^2$, integral $= 1$.

DECISION POINT **Which parametrization?** Segment from A to B: $\mathbf{r} = (1-t)A + tB$. Circle of radius a: $\langle a\cos t, a\sin t\rangle$, counterclockwise. Graph $y = g(x)$: $\langle x, g(x)\rangle$. The value does not depend on the parametrization, only on the direction. Before parametrizing at all, check whether $\mathbf{F}$ is conservative V.4: then only the endpoints matter V.3.

TERMS

$\mathbf{F}$; P, Q, R – vector field and its component functions
C, $-C$ – oriented curve and the same curve reversed
$\mathbf{r}(t)$, $a \le t \le b$ – parametrized path C2.25 §08 Calculus II, C3.5 §10 Calculus III
$\mathbf{r}'(t)$ – velocity; $\lVert\mathbf{r}'(t)\rVert$ speed
$\mathbf{F}\cdot d\mathbf{r}$ – work along a tiny step P.6 §12 Physics, dot product T.21 §06 Trigonometry
ds – arc length element C2.10 §08 Calculus II
f – scalar function (density) on C
∇f – gradient of f C3.12 §10 Calculus III
C_1, C_2 – pieces of a piecewise-smooth path



$\mathbf{F} = \langle -y, x\rangle$ rotation $\qquad$ $\mathbf{F} = \langle x, y\rangle$ radial source

CONTEXT

- **Origin.** Developed by Green, Gauss, and Maxwell for electromagnetism (1820s–1870s).
- **Use.** Work, circulation, fluid flow along a curve; mass of a wire.
- **Data science.** Gradient fields visualize optimization (loss landscapes as arrows); score-based generative models learn the vector field $\nabla \log p(\mathbf{x})$ and follow its flow lines.
- **Physical application.** The work to push a cart along a winding track against a varying wind is a line integral.

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V.2 Work, Circulation & Flux Across a Curve

Split a field at each point of a curve into the part along the curve and the part across it. The tangential part gives work (and, around a loop, circulation); the normal part gives the flux of a 2-D flow across the curve.

Work	$W = \int_C \mathbf{F}\cdot\mathbf{T}\,ds = \int_C \mathbf{F}\cdot d\mathbf{r}$	Only the tangential component of a force does work P.6 §12 Physics.
Circulation	$\Gamma = \oint_C \mathbf{F}\cdot\mathbf{T}\,ds$	Work around a closed loop: the net swirl of the field along C.
Unit tangent & normal (2-D)	$\mathbf{T} = \dfrac{(x',y')}{\lVert\mathbf{r}'\rVert}, \qquad \mathbf{n} = \dfrac{(y',-x')}{\lVert\mathbf{r}'\rVert}$	$\mathbf{n}$ is $\mathbf{T}$ turned clockwise: outward for a counterclockwise loop.
Flux across a plane curve	$\int_C \mathbf{F}\cdot\mathbf{n}\,ds = \int_C P\,dy - Q\,dx$	Net rate at which a 2-D flow crosses C (left to right of the direction of travel).
Mass of a wire	$m = \int_C \rho\,ds$	A scalar line integral of the linear density ρ.
Center of mass of a wire	$\bar{x} = \dfrac{1}{m}\int_C x\,\rho\,ds, \qquad \bar{y} = \dfrac{1}{m}\int_C y\,\rho\,ds$	Density-weighted average position along the curve C3.23 §10 Calculus III.

STATED-EQUATION *Work equals the integral of F dot T d s, which is the integral of F dot d r. Circulation is the same integral around a closed loop. The flux across a plane curve equals the integral of F dot n d s, which is the integral of P d y minus Q d x. The mass of a wire is the integral of the density with respect to arc length. The unit tangent T is x prime, y prime over the speed, and the normal n is y prime, minus x prime over the speed. The center of mass of a wire has x bar equal to one over m times the integral of x rho d s, and likewise for y bar.*

MEMORY AID Work goes *along* (**T**), flux goes *across* (**n**). Flux: "$P\,dy - Q\,dx$" (swap and negate).

METHOD

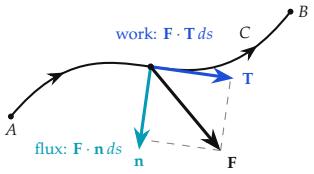
1. Parametrize C counterclockwise (for a loop).
2. Circulation: integrate $P\,dx + Q\,dy$. Flux: integrate $P\,dy - Q\,dx$.
3. Example: $\mathbf{F} = (x, y)$ on the unit circle $(\cos t, \sin t)$: circulation $\int(-\cos t\sin t + \sin t\cos t)\,dt = 0$; flux $\int(\cos^2 t + \sin^2 t)\,dt = 2\pi$. A pure source: outflow, no swirl.

DECISION POINT **Work or flux?** A force acting on a moving object, or a field "pushing along" a wire or path → tangential (work, circulation). A fluid, heat or charge crossing a boundary or fence → normal (flux). For a closed curve, either one can be converted to a double integral by Green's theorem V.7.

Integral	Integrand per ds	Reverse C?	Meaning
$\int_C f\,ds$	scalar f	unchanged	mass, length, average value
$\int_C \mathbf{F}\cdot\mathbf{T}\,ds$	component *along* C	sign flips	work
$\oint_C \mathbf{F}\cdot\mathbf{T}\,ds$	along a closed loop	sign flips	circulation
$\oint_C \mathbf{F}\cdot\mathbf{n}\,ds$	component *across* C	sign flips	flux (outflow) in 2-D

TERMS

$\mathbf{T}, \mathbf{n}$ – unit tangent and unit normal of C
Γ – circulation; $\oint$ marks a closed curve
x', y' – derivatives of the parametrization $\mathbf{r}(t) = (x(t), y(t))$
ρ – linear density (mass per length); m mass
$(\bar{x}, \bar{y})$ – center of mass
Positive orientation – counterclockwise: region on the left
W; m – work done by $\mathbf{F}$; mass of the wire



CONTEXT

- **Origin.** Kelvin's circulation theorem (1869); Kutta and Joukowski's lift theorem (1902–1906).
- **Use.** Aerodynamic lift, groundwater flow across a boundary, work done by forces.
- **Data science.** Wind or optical-flow data on a grid: the discrete flux through a region's boundary estimates net convergence or divergence, a feature used in weather nowcasting.
- **Physical application.** Lift on a wing per unit span is $L = \rho_{\text{air}} V\Gamma$ (Kutta–Joukowski): circulation of the airflow around the wing.

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V.3 Fundamental Theorem for Line Integrals

If a field is a gradient, its line integral depends only on where the path starts and ends: evaluate the potential at the end minus the start, exactly like the FTC C1.32 §07 Calculus I.

FT for line integrals	$\displaystyle\int_C \nabla f \cdot d\mathbf{r} = f\big(\mathbf{r}(b)\big) - f\big(\mathbf{r}(a)\big)$	Chain rule: $\frac{d}{dt} f(\mathbf{r}(t)) = \nabla f \cdot \mathbf{r}'(t)$ C3.13 §10 Calculus III, then the FTC.
Path independence	$\displaystyle\int_{C_1} \nabla f \cdot d\mathbf{r} = \int_{C_2} \nabla f \cdot d\mathbf{r}$	Any two paths with the same start and end give the same value.
Closed loops	$\displaystyle\oint_C \nabla f \cdot d\mathbf{r} = 0$	Start = end, so the difference of potentials is 0.
Equivalent conditions	$\mathbf{F} = \nabla f \iff$ path independent $\iff \displaystyle\oint_C \mathbf{F} \cdot d\mathbf{r} = 0$ for all closed C	On an open connected region the three statements are the same.
Conservation of energy	$\mathbf{F} = -\nabla U \;\Rightarrow\; K_A + U_A = K_B + U_B$	Work $= U_A - U_B = \Delta K$: kinetic plus potential energy is constant P.6 §12 Physics.

STATED-EQUATION *The line integral of the gradient of f along C equals f at the end point minus f at the start point. So the integral is the same along every path with the same endpoints, and zero around every closed loop. For a force F equal to minus the gradient of U, kinetic plus potential energy is conserved. Equivalently, F is a gradient field exactly when its line integrals are path independent, exactly when its integral around every closed loop is zero.*

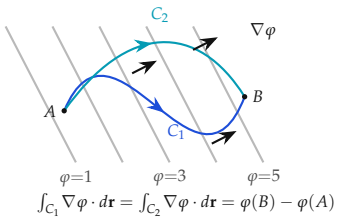
MEMORY AID The Fundamental Theorem of Calculus for line integrals: only the endpoints matter.

METHOD

1. Recognize (or test for) a gradient field and find a potential f V.4.
2. Evaluate f at the endpoints; ignore the path.
3. Example: $\mathbf{F} = (y, x) = \nabla(xy)$ from $(0,0)$ to $(1,2)$ along any curve: $f(1,2) - f(0,0) = 2$.

TERMS

f – potential function, $\nabla f = \mathbf{F}$ C3.12 §10 Calculus III
$\mathbf{r}(a), \mathbf{r}(b)$ – start and end points A, B of C
C_1, C_2 – two paths with the same endpoints
Connected – any two points can be joined by a path in the region
U; K – potential energy; kinetic energy $\frac{1}{2}mv^2$



CONTEXT

- **Origin.** Potential theory of Lagrange (1773), Laplace and Green (1828).
- **Use.** Gravity, electrostatics, energy conservation.
- **Data science.** Integrated-gradients attribution integrates ∇f along a straight path from a baseline $\mathbf{x}'$ to an input $\mathbf{x}$; by this theorem the attributions sum exactly to $f(\mathbf{x}) - f(\mathbf{x}')$.
- **Physical application.** The voltage between two points of an electrostatic field, $V_A - V_B = \int_A^B \mathbf{E} \cdot d\mathbf{r}$, is the same along any wire route.

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V.4 Conservative Fields & Potentials

A conservative field is the gradient of a scalar potential. Equal cross-partials (zero curl) detect one on a region without holes, and the potential is recovered by integrating one component at a time.

Conservative field	$\mathbf{F} = \nabla \varphi$	A field with a scalar potential behind it.
Path independence	$\int_C \mathbf{F} \cdot d\mathbf{r} = \varphi(B) - \varphi(A), \qquad \oint_C \mathbf{F} \cdot d\mathbf{r} = 0$	The fundamental theorem for line integrals V.3.
2-D test	$\dfrac{\partial P}{\partial y} = \dfrac{\partial Q}{\partial x}$	Necessary always; sufficient on a simply connected region.
3-D test	$\nabla \times \mathbf{F} = \mathbf{0}$	All three cross-partial pairs agree: $R_y = Q_z$, $P_z = R_x$, $Q_x = P_y$ V.5.
Why the test works	$P_y = \varphi_{xy} = \varphi_{yx} = Q_x$	Mixed partials of φ are equal (Clairaut C3.10 §10 Calculus III).
The hole counterexample	$\mathbf{F} = \left(\dfrac{-y}{x^2+y^2}, \dfrac{x}{x^2+y^2}\right)$ $P_y = Q_x, \quad \oint_{x^2+y^2=1} \mathbf{F} \cdot d\mathbf{r} = 2\pi$	Passes the test but is not conservative around the origin, where it is undefined.

STATED-EQUATION *F is conservative if F equals the gradient of a potential phi. Then the line integral from A to B equals phi of B minus phi of A, independent of path, and every closed-loop integral is zero. Test in two dimensions: partial P partial y equals partial Q partial x; in three dimensions: the curl of F is zero, on a region with no holes. The test works because P y equals phi x y, which equals phi y x, which equals Q x. The field minus y over x squared plus y squared, x over x squared plus y squared, passes the test yet circulates two pi around the unit circle, because of the hole at the origin.*

MEMORY AID The Fundamental Theorem of Calculus for line integrals: only the endpoints matter C1.32 §07 Calculus I.

METHOD

1. Check the test ($P_y = Q_x$, or curl $= \mathbf{0}$).
2. Integrate P in x: $\varphi = \int P\,dx + g(y)$, with an unknown $g(y)$ as the "constant".
3. Differentiate in y, set equal to Q, solve for $g'(y)$, integrate; check $\nabla\varphi = \mathbf{F}$. Example: $\mathbf{F} = (2xy,\ x^2 + 3y^2)$: $\varphi = x^2y + g(y)$, $x^2 + g' = x^2 + 3y^2$, $g = y^3$, so $\varphi = x^2y + y^3$.

DECISION POINT **Is it conservative?** Test the cross-partials first (cheap). Unequal → not conservative: integrate directly, or use Green or Stokes for a closed curve. Equal on a region without holes → find φ and use the endpoints. Equal, but the field blows up somewhere inside the loop → the test is not enough: compute one loop around the bad point.

TERMS

φ – potential function (negative of potential energy in physics)
P, Q, R – components of $\mathbf{F}$; $P_y = \partial P/\partial y$ C3.9 §10 Calculus III
Path independence – same answer for every route
$\oint$ – integral around a closed loop
Simply connected – every loop can shrink to a point inside the region (no holes)
$g(y)$ – integration "constant" that may depend on the other variable
A, B – start and end points of C
$\nabla \times \mathbf{F}$ – curl of $\mathbf{F}$ V.5

CONTEXT

- **Origin.** Potential theory by Lagrange (1773), Laplace and Green (1828).
- **Use.** Gravity, electrostatics, energy conservation.
- **Data science.** Only conservative fields have a scalar "loss" behind them; a field with circulation (like GAN training dynamics) can cycle instead of converging.
- **Physical application.** Lifting a box to a shelf takes the same gravitational work whether you use the stairs or the elevator.

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Divergence, Curl & Green's Theorem

V.5 Divergence & Curl

The del operator gives two derivatives of a vector field: the divergence, a scalar measuring net outflow per unit volume, and the curl, a vector whose direction is the local axis of rotation and whose length is twice the spin rate.

Del operator	$\nabla = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right)$	A vector of derivative operators; "dot" and "cross" it with $\mathbf{F}$.
Divergence	$\nabla \cdot \mathbf{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}$	Positive at a source, negative at a sink, zero for incompressible flow.
Curl	Expand like a cross product LA.5 §09 Linear Algebra; each component is the spin about that axis. $\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial_x & \partial_y & \partial_z \\ P & Q & R \end{vmatrix} = \left(R_y - Q_z,\ P_z - R_x,\ Q_x - P_y\right)$	
2-D (scalar) curl	$(\nabla \times \mathbf{F}) \cdot \mathbf{k} = Q_x - P_y$	For $\mathbf{F} = (P, Q, 0)$ only the $\mathbf{k}$-component survives: the integrand of Green's theorem V.7.
Divergence as flux density	$\nabla \cdot \mathbf{F}(p) = \lim_{V \to 0} \frac{1}{V} \iint_{\partial E} \mathbf{F} \cdot \mathbf{n}\, dS$	Outflow through a tiny closed surface, per unit volume V.11.
Curl as circulation density	$(\nabla \times \mathbf{F}) \cdot \mathbf{n} = \lim_{A \to 0} \frac{1}{A} \oint_C \mathbf{F} \cdot d\mathbf{r}$	Circulation around a tiny loop with normal $\mathbf{n}$, per unit area V.12.
Incompressible / irrotational	$\nabla \cdot \mathbf{F} = 0 \quad / \quad \nabla \times \mathbf{F} = \mathbf{0}$	No sources or sinks / no local spin (conservative on simply connected regions V.4).

STATED-EQUATION *The divergence of F, del dot F, equals partial P partial x plus partial Q partial y plus partial R partial z. The curl of F, del cross F, is the determinant with i, j, k in the first row, the partial derivative operators in the second, and P, Q, R in the third: R y minus Q z, P z minus R x, Q x minus P y. Divergence is flux per unit volume; the curl's component along n is circulation per unit area. Del is the vector of partial derivative operators with respect to x, y, and z. In two dimensions the scalar curl is Q x minus P y. A field with zero divergence is incompressible, and a field with zero curl is irrotational.*

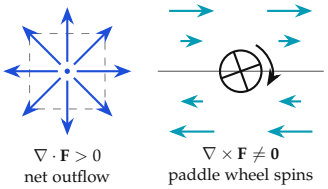
MEMORY AID Divergence = "spreading out" (source > 0, sink < 0); curl = "spinning" (right-hand rule).

METHOD

1. Write P, Q, R and list the partials you need.
2. Divergence: add the "matching" partials $P_x + Q_y + R_z$.
3. Curl: expand the determinant. Example: $\mathbf{F} = (xy, yz, zx)$: $\nabla \cdot \mathbf{F} = y + z + x$; $\nabla \times \mathbf{F} = (0 - y,\ 0 - z,\ 0 - x) = (-y, -z, -x)$.

TERMS

$\nabla\cdot$ – divergence: net outflow per volume (a scalar)
$\nabla\times$ – curl: rotation axis and strength (a vector)
$\mathbf{i}, \mathbf{j}, \mathbf{k}$ – unit vectors along x, y, z
R_y etc. – partial derivatives of the components C3.9 §10 Calculus III
V, A – volume of a tiny region, area of a tiny loop
Incompressible – $\nabla \cdot \mathbf{v} = 0$ (continuity P.20 §12 Physics)
Rigid rotation – $\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}$ has $\nabla \times \mathbf{v} = 2\boldsymbol{\omega}$
∇ – del: the vector of partial-derivative operators
∂E, C; $\mathbf{n}$ – small closed surface or loop shrinking to the point p; unit normal



CONTEXT

- **Origin.** Maxwell (1873) and Gibbs/Heaviside's vector notation (1880s).
- **Use.** Fluid dynamics, electromagnetism (Maxwell's equations), weather.
- **Data science.** Physics-informed neural networks enforce $\nabla \cdot \mathbf{v} = 0$ for fluids; optical-flow fields in video are analyzed with divergence (zoom) and curl (rotation).
- **Physical application.** Water draining from a bathtub swirls (curl) while flowing into the drain, a sink with negative divergence.

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V.6 Vector Identities: Product Rules, Curl of Gradient & Divergence of Curl

Gradient, divergence and curl obey product rules like ordinary derivatives, and two compositions always vanish. Those zeros explain exactly when potentials exist.

Curl of a gradient	$\nabla \times (\nabla f) = \mathbf{0}$	Gradient fields never swirl: equal mixed partials C3.10 §10 Calculus III.
Divergence of a curl	$\nabla \cdot (\nabla \times \mathbf{F}) = 0$	Curl fields have no sources: $\nabla \cdot \mathbf{B} = 0$ lets one write $\mathbf{B} = \nabla \times \mathbf{A}$.
Divergence of a gradient	$\nabla \cdot (\nabla f) = \nabla^2 f$	The Laplacian V.13.
Gradient of a product	$\nabla(fg) = f\,\nabla g + g\,\nabla f$	The ordinary product rule, component by component.
Divergence of $f\mathbf{F}$	$\nabla \cdot (f\mathbf{F}) = f\,\nabla \cdot \mathbf{F} + \nabla f \cdot \mathbf{F}$	Scalar times field: differentiate each factor in turn.
Curl of $f\mathbf{F}$	$\nabla \times (f\mathbf{F}) = f\,\nabla \times \mathbf{F} + \nabla f \times \mathbf{F}$	Same pattern with a cross product (keep the order).
Divergence of a cross product	$\nabla \cdot (\mathbf{F} \times \mathbf{G}) = \mathbf{G} \cdot (\nabla \times \mathbf{F}) - \mathbf{F} \cdot (\nabla \times \mathbf{G})$	Like a scalar triple product LA.5 §09 Linear Algebra; the minus sign comes from swapping.
Curl of a curl	$\nabla \times (\nabla \times \mathbf{F}) = \nabla(\nabla \cdot \mathbf{F}) - \nabla^2 \mathbf{F}$	"BAC–CAB" for operators; ∇^2 acts on each component.

STATED-EQUATION *The curl of a gradient is zero; the divergence of a curl is zero; the divergence of a gradient is the Laplacian. Product rules: the gradient of f g is f grad g plus g grad f; the divergence of f F is f div F plus grad f dot F; the curl of f F is f curl F plus grad f cross F. The divergence of F cross G is G dot curl F minus F dot curl G. The curl of the curl is the gradient of the divergence minus the Laplacian.*

MEMORY AID grad $\to$ curl $\to$ div: two steps in a row give zero ("the boundary of a boundary is empty").

METHOD

1. Write both sides in components (or use index notation).
2. Expand with the ordinary product rule and equal mixed partials.
3. Compare. Example: $\mathbf{F} = (xy, yz, zx)$ has $\nabla \times \mathbf{F} = (-y, -z, -x)$ V.5, and $\nabla \cdot (-y, -z, -x) = 0 + 0 + 0 = 0$ ✓.

$$f \xrightarrow{\nabla} \mathbf{F} \xrightarrow{\nabla\times} \mathbf{G} \xrightarrow{\nabla\cdot} h \qquad \text{any two consecutive arrows give } 0$$

TERMS

f, g – scalar functions (twice differentiable)
$\mathbf{F}, \mathbf{G}$ – vector fields; $\mathbf{A}$ vector potential, $\mathbf{B}$ magnetic field
∇^2 – Laplacian V.13
h – a scalar field
Converses – on a simply connected region, curl $\mathbf{F} = \mathbf{0} \Rightarrow \mathbf{F} = \nabla f$ V.4; on $\mathbb{R}^3$, div $\mathbf{G} = 0 \Rightarrow \mathbf{G} = \nabla \times \mathbf{A}$

CONTEXT

- **Origin.** Gibbs and Heaviside's vector analysis (1880s); the de Rham complex (1931) is the general form of "two steps give zero".
- **Use.** Electromagnetism (vector potential), fluid mechanics (vorticity equation).
- **Data science.** The Helmholtz–Hodge decomposition $\mathbf{F} = \nabla\varphi + \nabla \times \mathbf{A}$ splits a flow, or pairwise-comparison data in HodgeRank, into a rankable gradient part and a cyclic part.
- **Physical application.** Taking the curl of Faraday's law and using the curl-of-curl identity gives the wave equation for light, with speed $c = 1/\sqrt{\mu_0 \varepsilon_0}$.

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V.7 Green's Theorem

For a closed curve in the plane, the line integral around the boundary equals a double integral over the enclosed region: total circulation equals total 2-D curl, and total outward flux equals total divergence.

Circulation (tangential) form	$\oint_C P\,dx + Q\,dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA$	C counterclockwise, $\mathbf{F}$ smooth on all of D.
Flux (normal) form	$\oint_C \mathbf{F}\cdot\mathbf{n}\,ds = \oint_C P\,dy - Q\,dx = \iint_D \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y}\right) dA$	Outward flux equals total divergence: the 2-D divergence theorem.
Area formulas	$\text{Area}(D) = \frac{1}{2}\oint_C (x\,dy - y\,dx) = \oint_C x\,dy = -\oint_C y\,dx$	Choose P, Q with $Q_x - P_y = 1$.
Region with holes	$\partial D = C_{\text{outer}}\ (\circlearrowleft) \ \cup\ C_{\text{inner}}\ (\circlearrowright)$	Walk with D on your left: outer boundary counterclockwise, holes clockwise.
Shoelace formula	$A = \frac{1}{2}\left\lvert \sum_{i=1}^{n} (x_i y_{i+1} - x_{i+1} y_i) \right\rvert$	The area formula on a polygon (vertices in order, $x_{n+1} = x_1$).

STATED-EQUATION *The counterclockwise line integral around C of P d x plus Q d y equals the double integral over the enclosed region D of partial Q partial x minus partial P partial y. The outward flux, the integral of P d y minus Q d x, equals the double integral of the divergence. The area of D equals one half the loop integral of x d y minus y d x. The area also equals the loop integral of x d y, or minus the loop integral of y d x. For a region with holes, the boundary is the outer curve counterclockwise together with the inner curves clockwise. The shoelace formula: a polygon's area is one half the absolute value of the sum of x i times y i plus one, minus x i plus one times y i.*

MEMORY AID Circulation around the boundary = total spin inside. Counterclockwise is positive.

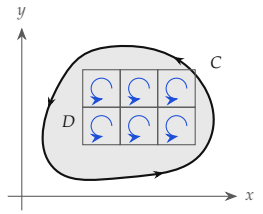
METHOD

1. Check: C closed, piecewise smooth, counterclockwise; $\mathbf{F}$ smooth on D (no singular points inside).
2. Compute $Q_x - P_y$ (or $P_x + Q_y$ for flux).
3. Integrate over D, in polar coordinates if D is round C3.21 §10 Calculus III. Example: $\oint(-y\,dx + x\,dy)$ around the circle of radius 2: $Q_x - P_y = 2$, so $\iint_D 2\,dA = 2 \cdot 4\pi = 8\pi$.

DECISION POINT **Green or direct?** Use Green when C is closed and made of several pieces, or when $Q_x - P_y$ is simpler than the parametrized integrand (often a constant). Curve not closed → close it with a segment, apply Green, subtract the segment's integral. Field singular inside D → cut out a small circle around the singularity.

TERMS

C – closed curve, counterclockwise
D – region enclosed by C
$Q_x - P_y$ – 2-D curl V.5; $P_x + Q_y$ 2-D divergence
$\mathbf{n}$ – outward unit normal V.2
Area formula – choose $P = -\frac{y}{2}$, $Q = \frac{x}{2}$
(x_i, y_i) – polygon vertices in counterclockwise order
Relation – 2-D case of Stokes V.12
∂D – boundary of D, positively oriented (region on the left)
n – number of polygon vertices; indices wrap, $x_{n+1} = x_1$



interior edges cancel; only C remains

CONTEXT

- **Origin.** George Green (1828), a self-taught miller.
- **Use.** Computing circulation and areas from boundaries.
- **Data science.** The shoelace formula for polygon area (GIS, computational geometry) is Green's theorem applied to straight edges.
- **Physical application.** A planimeter, a mechanical tool, measures the area of a map region just by tracing its border.

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Surfaces & Surface Integrals

V.8 Parametrized Surfaces & Surface Area

A surface is described by two parameters, $\mathbf{r}(u,v)$. Its partial derivatives are tangent vectors; their cross product is a normal vector whose length is the factor by which a small $du\,dv$ rectangle is stretched into surface area.

Parametric surface	$\mathbf{r}(u,v) = \big(x(u,v),\, y(u,v),\, z(u,v)\big), \quad (u,v) \in D$	Two parameters for a 2-D surface, as one parameter traces a curve C2.25 §08 Calculus II.
Tangent vectors	$\mathbf{r}_u = \dfrac{\partial \mathbf{r}}{\partial u}, \qquad \mathbf{r}_v = \dfrac{\partial \mathbf{r}}{\partial v}$	Velocities along the u- and v-grid curves on the surface.
Normal vector	$\mathbf{N} = \mathbf{r}_u \times \mathbf{r}_v$	Perpendicular to the surface LA.5 §09 Linear Algebra; smooth where $\mathbf{N} \neq \mathbf{0}$.
Tangent plane	$\mathbf{N} \cdot \big(\mathbf{x} - \mathbf{r}(u_0,v_0)\big) = 0$	Plane through the point with normal $\mathbf{N}$ C3.3 §10 Calculus III.
Surface area element	$dS = \lVert \mathbf{r}_u \times \mathbf{r}_v \rVert \, du\, dv$	Area of the small parallelogram spanned by $\mathbf{r}_u\,du$ and $\mathbf{r}_v\,dv$.
Surface area	$A(S) = \displaystyle\iint_D \lVert \mathbf{r}_u \times \mathbf{r}_v \rVert \, dA$	Add up the stretched patches C3.24 §10 Calculus III.
Graph $z = g(x,y)$	$\mathbf{r}_x \times \mathbf{r}_y = (-g_x, -g_y, 1), \quad dS = \sqrt{1 + g_x^2 + g_y^2}\, dA$	Parametrize by x, y themselves; upward normal.

STATED-EQUATION *A parametric surface is r of u and v equals x, y, z as functions of u and v. Its tangent vectors are r u and r v, and r u cross r v is a normal vector. The surface area element d S is the length of r u cross r v times d u d v, and the area is the double integral of that length over the parameter domain. The tangent plane at r of u naught, v naught is N dot the quantity x minus that point, equal to zero. For a graph z equals g of x and y, r x cross r y is minus g x, minus g y, one, and d S is the square root of one plus g x squared plus g y squared, d A.*

MEMORY AID "Cross the tangents, take the length": $\lVert \mathbf{r}_u \times \mathbf{r}_v \rVert$ is the area stretch.

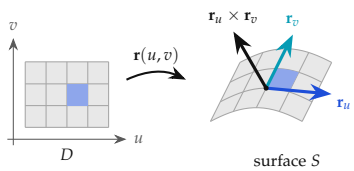
METHOD

1. Choose a parametrization and its domain D (table above).
2. Compute $\mathbf{r}_u$, $\mathbf{r}_v$, their cross product and its length.
3. Integrate over D. Example: cylinder of radius 2, height 3: $\lVert \mathbf{r}_\theta \times \mathbf{r}_z \rVert = 2$, so $A = \int_0^{2\pi}\int_0^3 2\,dz\,d\theta = 12\pi$.

Surface	Parametrization $\mathbf{r}$	dS
Sphere, radius a	$(a\sin\phi\cos\theta,\ a\sin\phi\sin\theta,\ a\cos\phi)$	$a^2\sin\phi\,d\phi\,d\theta$
Cylinder $x^2+y^2=a^2$	$(a\cos\theta,\ a\sin\theta,\ z)$	$a\,d\theta\,dz$
Cone $z=\sqrt{x^2+y^2}$	$(r\cos\theta,\ r\sin\theta,\ r)$	$\sqrt{2}\,r\,dr\,d\theta$
Plane through $\mathbf{r}_0$ along $\mathbf{a},\mathbf{b}$	$\mathbf{r}_0 + u\mathbf{a} + v\mathbf{b}$	$\lVert \mathbf{a}\times\mathbf{b} \rVert\, du\,dv$

TERMS

u, v – parameters; D their domain in the uv-plane
$\mathbf{r}_u, \mathbf{r}_v$ – partial derivatives (tangent vectors) C3.9 §10 Calculus III
$\mathbf{N}$ – normal vector (not unit); $\mathbf{x}$ a point of the plane
dS, dA – surface area element; area element in the parameter plane
ϕ, θ – spherical angles C3.21 §10 Calculus III
g_x, g_y – partials of the height function
$\mathbf{x}$; (u_0, v_0) – point on the tangent plane; parameter point of tangency
$A(S)$ – surface area of S



CONTEXT

- **Origin.** Gauss, *Disquisitiones generales circa superficies curvas* (1827).
- **Use.** CAD and graphics surfaces (Bézier and NURBS patches), meshes, cartography.
- **Data science.** Manifold learning assumes data lie near a low-dimensional parametrized surface; normalizing flows track the same kind of stretch factor (a Jacobian determinant C3.15 §10 Calculus III) to move densities.
- **Physical application.** Sheet metal for a cylindrical tank of radius 2 m and height 3 m: lateral area $12\pi \approx 37.7\ \text{m}^2$.

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V.9 Surface Integrals of Scalar Functions

Integrate a density over a curved surface by weighting each patch with its true area dS: the surface version of $\int_C f\,ds$. Mass, charge, averages and centroids of shells all come out this way.

Surface integral	$\displaystyle\iint_S f\,dS = \iint_D f(\mathbf{r}(u,v))\,\|\mathbf{r}_u \times \mathbf{r}_v\|\,dA$	Substitute the parametrization and multiply by the area stretch V.8.
Over a graph $z = g(x,y)$	$\displaystyle\iint_S f\,dS = \iint_D f(x,y,g(x,y))\sqrt{1+g_x^2+g_y^2}\,dA$	D is the shadow of S in the xy-plane.
Over a sphere	$\displaystyle\iint_S f\,dS = \int_0^{2\pi}\!\!\int_0^{\pi} f\,a^2\sin\phi\,d\phi\,d\theta$	Radius a, spherical angles C3.21 §10 Calculus III.
Mass of a shell	$\displaystyle m = \iint_S \rho\,dS$	ρ is mass per unit area; $f = 1$ gives the area.
Center of mass of a shell	$\displaystyle \bar{x} = \frac{1}{m}\iint_S x\,\rho\,dS \quad (\text{same for } \bar{y},\bar{z})$	Density-weighted average position C3.23 §10 Calculus III.
Average over a surface	$\displaystyle \bar{f} = \frac{1}{A(S)}\iint_S f\,dS$	Total divided by area, as for functions of one variable C2.12 §08 Calculus II.

STATED-EQUATION *The surface integral of f over S equals the double integral over the parameter domain of f at r of u and v times the length of r u cross r v. Over a graph z equals g of x and y, the stretch factor is the square root of one plus g x squared plus g y squared. The mass of a shell is the surface integral of its density. Over a sphere of radius a, d S is a squared sine phi d phi d theta. The center of mass of a shell has x bar equal to one over m times the surface integral of x rho, and likewise for y bar and z bar. The average of f over S is its surface integral divided by the area of S.*

MEMORY AID dS is *surface* area, never just dA: always include the stretch factor.

METHOD

1. Parametrize S and compute dS.
2. Write f in the parameters; integrate over D.
3. Example: $\iint_S z\,dS$ on the upper unit hemisphere: $z = \cos\phi$, $dS = \sin\phi\,d\phi\,d\theta$: $\int_0^{2\pi}\int_0^{\pi/2}\cos\phi\sin\phi\,d\phi\,d\theta = 2\pi\cdot\frac{1}{2} = \pi$. Average height $= \frac{\pi}{2\pi} = \frac{1}{2}$.

DECISION POINT **Which description of S?** A graph $z = g(x,y)$ over a simple shadow → use x, y. Spheres, cylinders and cones → the standard parametrizations V.8. A closed surface made of pieces (a can: side, top, bottom) → integrate each piece separately and add.

TERMS

f – scalar function on S
S; D – surface; its parameter domain (or shadow)
dS – surface area element V.8
ρ – surface density (mass per area); m mass
$(\bar{x},\bar{y},\bar{z})$ – center of mass; $A(S)$ surface area
a – sphere radius
$\mathbf{r}_u \times \mathbf{r}_v$; g_x, g_y – normal vector V.8; partial derivatives of g
ϕ, θ – spherical angles C3.22 §10 Calculus III
m; $\bar{f}$; $A(S)$ – mass; average value of f; surface area

CONTEXT

- **Origin.** Gauss and Green (1820s), for potentials of charged surfaces.
- **Use.** Mass and charge of thin shells, heat on surfaces, surface averages.
- **Data science.** Directional data and normalized embeddings live on a sphere; the von Mises–Fisher distribution is normalized by a surface integral over S^{n-1}.
- **Physical application.** The total charge on a curved metal plate is $\iint_S \sigma\,dS$ for surface charge density σ.

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V.10 Surface Integrals of Vector Fields (Flux)

The flux of $\mathbf{F}$ through an oriented surface is the rate at which the field flows across it: integrate the normal component $\mathbf{F}\cdot\mathbf{n}$ over the surface. The orientation (choice of $\mathbf{n}$) sets the sign.

Unit normal (orientation)	$\mathbf{n} = \dfrac{\mathbf{r}_u \times \mathbf{r}_v}{\lVert \mathbf{r}_u \times \mathbf{r}_v \rVert}$	An orientation is a continuous choice of $\mathbf{n}$; closed surfaces use outward.
Flux integral	$\displaystyle\iint_S \mathbf{F}\cdot d\mathbf{S} = \iint_S \mathbf{F}\cdot\mathbf{n}\, dS$	Only the normal component carries flow through the surface.
Parametric formula	$\displaystyle\iint_S \mathbf{F}\cdot d\mathbf{S} = \iint_D \mathbf{F}\big(\mathbf{r}(u,v)\big)\cdot\big(\mathbf{r}_u\times\mathbf{r}_v\big)\, dA$	The lengths cancel; flip the sign if $\mathbf{r}_u\times\mathbf{r}_v$ points the wrong way.
Graph $z=g(x,y)$, upward	$\displaystyle\iint_S \mathbf{F}\cdot d\mathbf{S} = \iint_D \big(-Pg_x - Qg_y + R\big)\, dA$	Uses $\mathbf{r}_x\times\mathbf{r}_y = (-g_x,-g_y,1)$.
Sphere, outward	$\mathbf{n} = \dfrac{\mathbf{r}}{a}, \qquad dS = a^2\sin\phi\, d\phi\, d\theta$	On a sphere centered at the origin the outward normal is the position over the radius.
Mass flow rate	$\dot m = \displaystyle\iint_S \rho\,\mathbf{v}\cdot\mathbf{n}\, dS$	Fluid of density ρ and velocity $\mathbf{v}$: kilograms per second through S.

STATED-EQUATION *The flux of F through S is the surface integral of F dot the unit normal n, d S. With a parametrization, it is the double integral of F at r of u and v dotted with r u cross r v. For an upward graph it is the double integral of minus P g x minus Q g y plus R. The unit normal n is r u cross r v divided by its length. On a sphere of radius a, the outward normal is r over a and d S is a squared sine phi d phi d theta. The mass flow rate through S is the surface integral of rho v dot n.*

MEMORY AID "Only what goes *through* counts": $\mathbf{F}\cdot\mathbf{n}$. No $\lVert\cdot\rVert$ in the parametric formula.

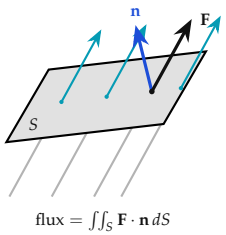
METHOD

1. Parametrize S; compute $\mathbf{r}_u\times\mathbf{r}_v$ and check it points in the required direction (test one point).
2. Dot $\mathbf{F}(\mathbf{r}(u,v))$ with it and integrate over D.
3. Example: $\mathbf{F}=(0,0,z)$ out through the upper unit hemisphere: $\mathbf{F}\cdot\mathbf{n} = z^2 = \cos^2\phi$, so flux $= \int_0^{2\pi}\int_0^{\pi/2}\cos^2\phi\sin\phi\, d\phi\, d\theta = \frac{2\pi}{3}$.

DECISION POINT **Shortcut available?** S closed $\to$ the divergence theorem V.11. $\mathbf{F}$ is a curl, $\nabla\times\mathbf{G}\to$ Stokes V.12, or swap to an easier surface with the same boundary. Otherwise compute directly, choosing the orientation from the wording ("upward", "outward", or induced by a boundary curve).

TERMS

$\mathbf{n}$ – unit normal giving the orientation
$d\mathbf{S}$ – vector area element $\mathbf{n}\,dS = (\mathbf{r}_u\times\mathbf{r}_v)\,dA$
P, Q, R – components of $\mathbf{F}$; g_x, g_y partials of the graph
$\mathbf{r}$, a – position vector, sphere radius
ρ, $\mathbf{v}$ – fluid density and velocity; $\dot m$ mass flow rate
One-sided surface – a Möbius strip has no orientation, so no flux
g_x, g_y – partial derivatives of the graph function g
ϕ, θ – spherical angles C3.22 §10 Calculus III
$\dot m$ – mass flow rate through S



CONTEXT

- **Origin.** Gauss's electric flux (1813, 1839); Maxwell's *Treatise* (1873).
- **Use.** Flow through membranes and pipes, heat flux, electric and magnetic flux.
- **Data science.** In diffusion and Fokker–Planck models, the flux of the probability current through a boundary is the rate at which probability mass leaves the region.
- **Physical application.** The magnetic flux $\Phi_B = \int\!\!\int_S \mathbf{B}\cdot\mathbf{n}\,dS$ through a coil sets the induced voltage in a generator EE.18 §16 Electrical Eng..

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The Big Theorems

V.11 Flux & the Divergence Theorem

The net outward flux of a field through a closed surface equals the total divergence (source strength) inside: what flows out through the skin is what is produced within.

Flux	$\text{flux} = \iint_S \mathbf{F}\cdot\mathbf{n}\,dS$	The surface integral of the normal component V.10.
Divergence theorem	$\iint_{S=\partial E} \mathbf{F}\cdot\mathbf{n}\,dS = \iiint_E \nabla\cdot\mathbf{F}\,dV$	S closed with outward $\mathbf{n}$; $\mathbf{F}$ smooth throughout the solid E.
Gauss's law	$\iint_{\partial E} \mathbf{E}\cdot\mathbf{n}\,dS = \frac{Q_{\text{enc}}}{\varepsilon_0}$	Electric flux out of any closed surface depends only on the enclosed charge.
Continuity equation	$\frac{\partial\rho}{\partial t} + \nabla\cdot(\rho\mathbf{v}) = 0$	Conservation of mass in local form: apply the theorem to every small volume P.20 §12 Physics.
Solid with a cavity	$\partial E = S_{\text{outer}} \cup S_{\text{inner}}$, $\mathbf{n}$ pointing out of E	On the inner surface "outward from E" points into the cavity.
Volume from the boundary	$V = \frac{1}{3}\iint_{\partial E} \mathbf{r}\cdot\mathbf{n}\,dS$	Since $\nabla\cdot\mathbf{r} = 3$; the 3-D analog of Green's area formula.

STATED-EQUATION *The flux of F through a surface S is the surface integral of F dot the unit normal n, d S. The divergence theorem: the outward flux through a closed surface equals the triple integral of the divergence of F over the enclosed solid E. Gauss's law: the electric flux out equals the enclosed charge over epsilon naught. Continuity: the rate of change of density plus the divergence of rho v is zero. For a solid with a cavity, the boundary is the outer and inner surfaces, with n pointing out of E. The volume of E is one third of the flux of the position vector r out through its boundary.*

MEMORY AID What flows out through the skin = what is produced inside.

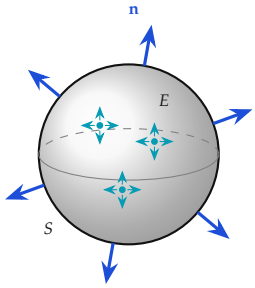
METHOD

1. Check that S is closed, oriented outward, and $\mathbf{F}$ is smooth inside.
2. Compute $\nabla\cdot\mathbf{F}$.
3. Set up the triple integral (spherical or cylindrical if round C3.21 §10 Calculus III). Example: $\mathbf{F} = (x,y,z)$ out of the unit sphere: $\nabla\cdot\mathbf{F} = 3$, flux $= 3\cdot\frac{4}{3}\pi = 4\pi$.

DECISION POINT **Use it when** S is closed and $\nabla\cdot\mathbf{F}$ is simpler than $\mathbf{F}\cdot\mathbf{n}$ (constant or zero). **Open surface?** Close it with a simple cap (often a flat disk), apply the theorem, subtract the cap's flux. **Singular point inside** (a point charge, $\mathbf{F} = \mathbf{r}/\|\mathbf{r}\|^3$)? The theorem fails there: surround the point with a small sphere; here the flux is 4π for every surface enclosing the origin.

TERMS

$S = \partial E$ – closed surface bounding the solid E
$\mathbf{n}$ – outward unit normal
$\mathbf{F}\cdot\mathbf{n}$ – component crossing the surface T.21 §06 Trigonometry
$\nabla\cdot\mathbf{F}$ – divergence V.5; dV volume element C3.19 §10 Calculus III
$\mathbf{E}$, Q_{enc}, ε_0 – electric field, enclosed charge, permittivity of free space
ρ, $\mathbf{v}$ – density and velocity of a flow
Also called – Gauss's theorem or Ostrogradsky's theorem
E; V, dV – solid region; its volume and volume element
$\mathbf{r}$ – position vector (x,y,z)



CONTEXT

- **Origin.** Lagrange (1762), Gauss (1813), Ostrogradsky (1826).
- **Use.** Conservation laws: mass, charge, heat.
- **Data science.** Conservation constraints in physics-informed models and flux balances in hydrology and climate data assimilation; finite-volume solvers apply the theorem cell by cell.
- **Physical application.** Gauss's law: the electric flux through any closed surface equals the enclosed charge divided by ε_0.

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V.12 Stokes' Theorem

The circulation of a field around a closed space curve equals the flux of its curl through any surface the curve bounds. Green's theorem is the flat case.

Stokes' theorem	$\oint_C \mathbf{F}\cdot d\mathbf{r} = \iint_S (\nabla\times\mathbf{F})\cdot\mathbf{n}\,dS$	$C=\partial S$, oriented by the right-hand rule with $\mathbf{n}$.
Green as a special case	$S\subset xy$-plane, $\mathbf{n}=\mathbf{k}$: $(\nabla\times\mathbf{F})\cdot\mathbf{k} = Q_x - P_y$	A flat surface gives Green's theorem V.7.
Surface independence	$\iint_{S_1}(\nabla\times\mathbf{F})\cdot d\mathbf{S} = \iint_{S_2}(\nabla\times\mathbf{F})\cdot d\mathbf{S}$	Any two surfaces with the same oriented boundary give the same flux of a curl.
Closed surface	$\iint_{\partial E}(\nabla\times\mathbf{F})\cdot\mathbf{n}\,dS = 0$	No boundary curve, so zero; agrees with $\nabla\cdot(\nabla\times\mathbf{F})=0$ V.6.
Faraday's law	$\oint_C \mathbf{E}\cdot d\mathbf{r} = -\frac{d}{dt}\iint_S \mathbf{B}\cdot\mathbf{n}\,dS$	A changing magnetic flux induces a circulating electric field (a voltage).
Ampère's law	$\oint_C \mathbf{B}\cdot d\mathbf{r} = \mu_0 I_{\text{enc}}$	Magnetic circulation around a loop measures the current through it.

STATED-EQUATION *The circulation of F around the boundary curve C equals the flux of the curl of F through any surface S bounded by C. Green's theorem is the special case of a flat surface in the plane. Faraday's law: the circulation of E equals minus the rate of change of magnetic flux. Ampere's law: the circulation of B equals mu naught times the enclosed current. So the curl flux is the same through any two surfaces S one and S two with the same boundary, and the curl flux out of any closed surface is zero.*

MEMORY AID Right-hand rule: curl your fingers along C, your thumb gives $\mathbf{n}$.

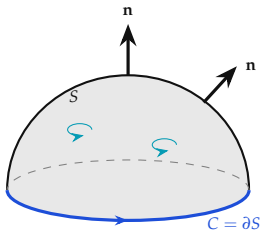
METHOD

1. Identify the boundary C and orient C and $\mathbf{n}$ consistently (right-hand rule).
2. Choose the easier side: compute $\oint_C$ directly, or compute $\nabla\times\mathbf{F}$ and pick the simplest surface with boundary C (often a flat disk).
3. Integrate. Example: $\mathbf{F}=(-y,x,z)$, C the unit circle in $z=0$, counterclockwise from above: $\nabla\times\mathbf{F}=(0,0,2)$, disk with $\mathbf{n}=\mathbf{k}$: $\iint 2\,dA = 2\pi$. Directly: $\oint(-y\,dx + x\,dy) = 2\pi$ ✓.

DECISION POINT **Line or surface?** C awkward (an intersection of two surfaces) but $\nabla\times\mathbf{F}$ simple → surface side. C simple but the curl messy → line side. Asked for the flux of a curl through a complicated surface → replace it by the flat surface with the same boundary.

TERMS

$C=\partial S$ – boundary of S with matching orientation
$\nabla\times\mathbf{F}$ – curl V.5
S_1, S_2 – surfaces sharing the boundary C
$\mathbf{E}, \mathbf{B}$ – electric and magnetic fields EE.18 §16 Electrical Eng.
μ_0, I_{enc} – permeability of free space; current through S
Big picture – FTC C1.32 §07 Calculus I, Green, Stokes, divergence: "boundary integral = interior integral of a derivative" V.K1
∂E – closed surface bounding a solid E



CONTEXT

- **Origin.** Stated by Kelvin (1850), set as an exam question by Stokes (1854).
- **Use.** Electromagnetism (Faraday's and Ampère's laws), aerodynamics (lift from circulation).
- **Data science.** Discrete versions underpin graph Laplacians and Hodge decompositions used in ranking and flow data on networks.
- **Physical application.** A changing magnetic flux through a coil induces a voltage around the wire (Faraday), which is how generators work.

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The Big Theorems | *Key Topic V.K1*

Five theorems, one idea: *integrating a derivative over a region gives the original function integrated over the region's oriented boundary.* The FTC is the one-dimensional case; the fundamental theorem for line integrals, Green's theorem, Stokes' theorem and the divergence theorem say the same thing for curves, plane regions, surfaces and solids. Recognizing which one applies turns hard integrals into easy ones.

FTC C1.32 §07 Calculus I



$$\int_a^b f'(x)\,dx = f(b) - f(a)$$

interval → 2 endpoints

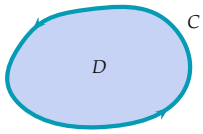
Line integrals V.3



$$\int_C \nabla f \cdot d\mathbf{r} = f(B) - f(A)$$

curve → 2 endpoints

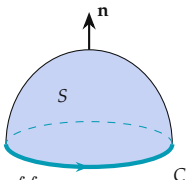
Green V.7



$$\iint_D (Q_x - P_y)\,dA = \oint_C P\,dx + Q\,dy$$

plane region → closed curve (ccw)

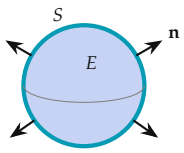
Stokes V.12



$$\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n}\,dS = \oint_C \mathbf{F} \cdot d\mathbf{r}$$

surface → closed curve (right hand)

Divergence V.11



$$\iiint_E \nabla \cdot \mathbf{F}\,dV = \iint_S \mathbf{F} \cdot \mathbf{n}\,dS$$

solid → closed surface (outward)

One pattern, five derivatives

$$\underbrace{\int_{\partial\Omega} (\text{field})}_{\text{boundary}} = \underbrace{\int_{\Omega} (\text{derivative of the field})}_{\text{inside}}$$

Why it works. Chop Ω into tiny cells. For one cell, the boundary integral is (derivative)×(cell size): that is the definition of f', curl or divergence as a density V.5. Neighboring cells share faces or edges that are traversed in opposite directions, so those contributions cancel; only the outer boundary survives (picture in V.7). The "derivative" is whichever one matches the dimension: f', ∇f, the 2-D curl $Q_x - P_y$, $\nabla \times \mathbf{F}$, $\nabla \cdot \mathbf{F}$.

The theorems at a glance

Theorem	Region Ω	Boundary $\partial\Omega$, orientation	Derivative	Statement	Typical use
FTC C1.32 §07 Calculus I	interval $[a,b]$	endpoints: b (+), a (−)	f'	$\int_a^b f'(x)\,dx = f(b) - f(a)$	evaluating integrals
Line integrals V.3	curve C from A to B	endpoints: B (+), A (−)	∇f	$\int_C \nabla f \cdot d\mathbf{r} = f(B) - f(A)$	work in a conservative field
Green V.7	plane region D	closed curve, counterclockwise	$Q_x - P_y$	$\iint_D (Q_x - P_y)\,dA = \oint_C P\,dx + Q\,dy$	circulation, flux and area in the plane
Stokes V.12	surface S	closed curve, right-hand rule	$\nabla \times \mathbf{F}$	$\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n}\,dS = \oint_C \mathbf{F} \cdot d\mathbf{r}$	circulation in space; swapping surfaces
Divergence V.11	solid E	closed surface, outward	$\nabla \cdot \mathbf{F}$	$\iiint_E \nabla \cdot \mathbf{F}\,dV = \iint_{\partial E} \mathbf{F} \cdot \mathbf{n}\,dS$	flux of closed surfaces; conservation laws

Worked checks: both sides agree

FTC. $\int_1^3 2x\,dx = 3^2 - 1^2 = 8$.

Line integral. $\mathbf{F} = (y,x) = \nabla(xy)$ from $(0,0)$ to $(1,2)$: $f(1,2) - f(0,0) = 2$, along any path.

Green. $\mathbf{F} = (-y,x)$, unit circle: $\oint = \int_0^{2\pi} (\sin^2 t + \cos^2 t)\,dt = 2\pi$; inside, $Q_x - P_y = 2$ and $\iint_D 2\,dA = 2\pi$.

Stokes. Same field with z-component 0, upper unit hemisphere: $\nabla \times \mathbf{F} = (0,0,2)$; its flux through the hemisphere equals its flux through the flat disk, $2 \cdot \pi = 2\pi$, which is the circulation around the equator.

Divergence. $\mathbf{F} = (x,y,z)$, unit sphere: on S, $\mathbf{F} \cdot \mathbf{n} = 1$, so flux = area $= 4\pi$; inside, $\nabla \cdot \mathbf{F} = 3$ and $3 \cdot \frac{4}{3}\pi = 4\pi$.

Consequences. $\nabla \times \nabla f = \mathbf{0}$ and $\nabla \cdot (\nabla \times \mathbf{F}) = 0$ V.6 are the same fact as "a closed curve or closed surface has no boundary": the circulation of a gradient and the flux of a curl through a closed surface are always 0.

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Decision point: which theorem?

DECISION POINT **Line integral along an open curve** → test for a gradient field V4; if conservative use endpoints (FT for line integrals); if not, parametrize, or close the curve and use Green/Stokes minus the added piece. **Closed curve in the plane** → Green (circulation form for $\oint \mathbf{F} \cdot d\mathbf{r}$, flux form for $\oint \mathbf{F} \cdot \mathbf{n}\, ds$). **Closed curve in space** → Stokes with the simplest surface it bounds. **Flux through a closed surface** → divergence theorem. **Flux through an open surface** → if $\mathbf{F}$ is a curl, Stokes; otherwise close it with a cap and use the divergence theorem, or compute directly. **Field undefined somewhere inside** → none applies directly: cut the bad point out with a small circle or sphere.

Orientation rules & common mistakes

Wrong orientation. Green: counterclockwise (region on your left; holes clockwise). Stokes: right-hand rule between C and $\mathbf{n}$. Divergence: outward normal. A reversed orientation flips the sign.

Using a theorem on an open curve or surface. Green and the divergence theorem need a *closed* boundary. Close it first, then subtract what you added.

Ignoring singularities. For $\mathbf{F} = \frac{\langle -y, x \rangle}{x^2+y^2}$, $Q_x - P_y = 0$ everywhere it is defined, yet $\oint = 2\pi$ around the origin: Green does not apply on a disk containing it.

Mixing the derivatives. Circulation pairs with curl, flux with divergence. In 2-D: $\oint P\,dx + Q\,dy \leftrightarrow Q_x - P_y$; $\oint P\,dy - Q\,dx \leftrightarrow P_x + Q_y$.

Forgetting the vector area. In Stokes and divergence the surface side is $\mathbf{G} \cdot \mathbf{n}\, dS = \mathbf{G} \cdot (\mathbf{r}_u \times \mathbf{r}_v)\, dA$, not $\mathbf{G}\, dA$.

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The Laplacian & Partial Differential Equations

V.13 The Laplacian & Classic PDEs

The Laplacian, the divergence of the gradient, measures how a function's value differs from the average of its neighbors. It is the spatial part of the heat, wave, Laplace and Poisson equations, the classic PDEs of physics.

Laplacian	$\nabla^2 f = \nabla \cdot \nabla f = \dfrac{\partial^2 f}{\partial x^2} + \dfrac{\partial^2 f}{\partial y^2} + \dfrac{\partial^2 f}{\partial z^2}$	Sum of the unmixed second partials C3.10 §10 Calculus III; also written Δf.
Heat (diffusion) equation	$u_t = \alpha \nabla^2 u$	Hot spots ($\nabla^2 u < 0$) cool and cold spots warm: smoothing over time P.31 §12 Physics.
Wave equation	$u_{tt} = c^2 \nabla^2 u$	Acceleration proportional to the Laplacian: disturbances travel at speed c P.17 §12 Physics.
Laplace's equation	$\nabla^2 u = 0$	Steady state of the heat equation; solutions are *harmonic*.
Poisson's equation	$\nabla^2 u = f$	Laplace with sources: $\nabla^2 V = -\rho/\varepsilon_0$ in electrostatics, $\nabla^2 \Phi = 4\pi G\rho$ in gravity.
Mean-value property	$u(p) = \dfrac{1}{4\pi r^2} \iint_{\text{sphere}(p,r)} u \, dS$	A harmonic function equals its average over every sphere about p (circle in 2-D); no interior max or min.
Discrete Laplacian	$\nabla^2 u \approx \dfrac{u_E + u_W + u_N + u_S - 4u_C}{h^2}$	Five-point stencil; in 1-D, $\frac{f(x+h)-2f(x)+f(x-h)}{h^2}$ C1.9 §07 Calculus I.
Laplacian in polar coordinates	$\nabla^2 u = u_{rr} + \dfrac{1}{r} u_r + \dfrac{1}{r^2} u_{\theta\theta}$	For round regions; radial solutions of $\nabla^2 u = 0$ are $a + b \ln r$ in 2-D C3.21 §10 Calculus III.

STATED-EQUATION *The Laplacian of f, del squared f, is the divergence of the gradient: the sum of the unmixed second partial derivatives. The heat equation: u t equals alpha times the Laplacian of u. The wave equation: u t t equals c squared times the Laplacian of u. Laplace's equation: the Laplacian of u equals zero; Poisson's equation: the Laplacian of u equals f. A harmonic function equals its average over any sphere. The discrete Laplacian is u east plus u west plus u north plus u south minus four u center, all over h squared. In polar coordinates the Laplacian is u r r plus one over r times u r, plus one over r squared times u theta theta.*

MEMORY AID The Laplacian compares a point to the average of its neighbors: positive means the point is lower than its surroundings.

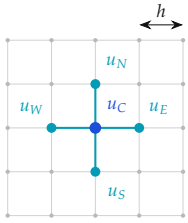
METHOD

1. Separation of variables: assume $u(x,t) = X(x)T(t)$ and substitute into $u_t = \alpha u_{xx}$.
2. Divide by αXT: $\frac{T'}{\alpha T} = \frac{X''}{X} = -\lambda$, a constant, giving two ODEs C2.31 §08 Calculus II, C2.28 §08 Calculus II.
3. Solve with the boundary conditions and superpose (Fourier series EE.26 §16 Electrical Eng.). Example: rod $0 \le x \le \pi$, ends at 0, $u(x,0) = \sin x$: $X = \sin x$, $\lambda = 1$, so $u = e^{-\alpha t} \sin x$.

DECISION POINT **Which PDE?** One time derivative → diffusion (smoothing, irreversible; needs u at $t = 0$). Two time derivatives → waves (propagation at speed c; needs u and u_t at $t = 0$). No time derivative → equilibrium (Laplace, or Poisson with sources; needs boundary values only).

TERMS

∇^2 – Laplacian, also written Δ
$u(x,y,z,t)$ – temperature, displacement or potential
α; c – diffusivity; wave speed
Harmonic – $\nabla^2 u = 0$: steady-state temperature, electrostatic potential
f, ρ, G – source term; charge or mass density; gravitational constant
h; $u_C, \ldots$ – grid spacing; center and neighbor values
X, T, λ – separated factors and separation constant
p; r, θ – center of the averaging sphere of radius r; polar coordinates T.20 §06 Trigonometry in the polar Laplacian



$$\nabla^2 u \approx \frac{u_E + u_W + u_N + u_S - 4u_C}{h^2}$$

CONTEXT

- **Origin.** Laplace (1780s) on gravitational potential; Fourier's heat equation (1822); d'Alembert's wave equation (1747).
- **Use.** Heat, waves, electrostatics, fluid flow.
- **Data science.** The graph Laplacian $L = D - A$ powers spectral clustering and graph neural networks; the Laplacian filter detects edges in images; diffusion models run a heat equation backwards.
- **Physical application.** The steady temperature across a metal plate with fixed edge temperatures solves Laplace's equation.

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12
Physics & Natural Phenomena

Units, motion, forces, energy, rotation, waves, fluids, heat, electricity, light and a first look at modern physics, plus the dimensionless numbers that tell you which physics dominates. Calculus turns these laws into predictions; the same equations govern weather, oceans, cooking, satellites and the cooling of electronics. Take $g = 9.81\,m/s^2$ throughout, and work in SI units.

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Units & Measurement

P.1 Units & Dimensional Analysis

Every physical quantity is a number times a unit. An equation must balance in dimensions as well as in numbers, and that requirement alone often fixes the form of a formula.

Unit of force (newton)	$[\text{force}] = \dfrac{\text{kg}\cdot\text{m}}{\text{s}^2} = \text{N}$	Mass times acceleration, from $F = ma$ P.4.
Unit of energy (joule)	$[\text{energy}] = \text{N}\cdot\text{m} = \text{J}$	Force times distance; in base units $\text{kg}\cdot\text{m}^2/\text{s}^2$ P.6.
Unit of power (watt)	$[\text{power}] = \dfrac{\text{J}}{\text{s}} = \text{W}$	Energy per unit time.
Dimensional homogeneity	$[\text{left side}] = [\text{right side}]$	Only like dimensions add or equate; arguments of e^x, log, sin must be dimensionless.
Power-law ansatz	$Q = C\,x_1^a x_2^b x_3^c$	Match the exponent of each base dimension (M, L, T) on both sides and solve for a, b, c.
Buckingham π theorem	$\#\text{ dimensionless groups} = n - k$	n variables built from k base dimensions collapse into $n - k$ independent π groups.

STATED-EQUATION *The unit of force is kilograms times meters per second squared, called a newton. A joule is a newton-meter, and a watt is a joule per second. Both sides of an equation must have the same dimensions. The Buckingham pi theorem: the number of dimensionless groups equals the number of variables, n, minus the number of base dimensions, k. A power-law guess writes Q as C times x one to the a, x two to the b, x three to the c, and matching dimensions on both sides fixes the exponents.*

MEMORY AID "Units first": if the units of both sides do not match, the equation is wrong.

METHOD

1. List every variable the answer could depend on.
2. Write each in base dimensions: mass M, length L, time T (add Θ for temperature).
3. Assume $Q = C\,x_1^a x_2^b x_3^c$ and equate the exponents of M, L and T on both sides.
4. Solve the linear system. Example: pendulum period $T = C\,L^a m^b g^c$; T: $1 = -2c$, L: $0 = a + c$, M: $0 = b$, so $T = C\sqrt{L/g}$ (C $= 2\pi$, P.16).

TERMS

SI base units – meter (m), kilogram (kg), second (s), kelvin (K), ampere (A), mole (mol), candela (cd); see P.K1

$[\,\cdot\,]$ – "dimensions of"; M mass, L length, T time

n – number of physical variables in the problem

k – number of independent base dimensions (e.g. M, L, T)

C – dimensionless constant (not found by this method)

Dimensionless group – a ratio with no units, like Re P.21 or Bi P.26

Conversions – multiply by 1 in a useful form, e.g. $\frac{1000\text{ m}}{1\text{ km}}$ A1.19 §03 Algebra I

Q; x_1, x_2, x_3; a, b, c – quantity sought; the variables it depends on; unknown exponents

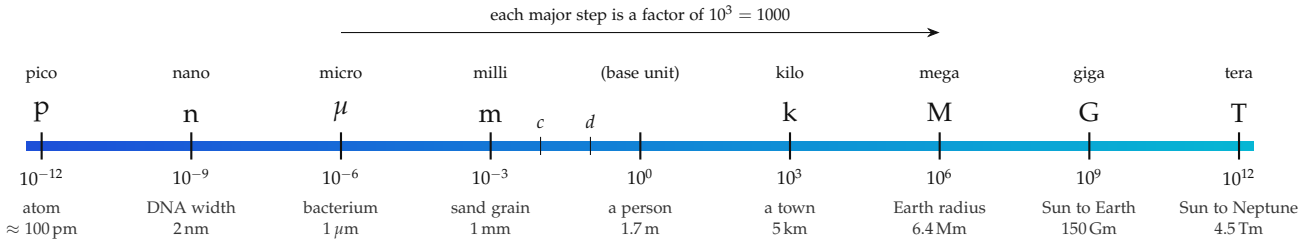
CONTEXT

- **Origin.** Fourier (1822) stated dimensional homogeneity; Buckingham (1914) the π theorem.
- **Use.** Checking equations, converting units, and building scale models (wind tunnels, ship tanks).
- **Data science.** Unit-consistent feature engineering (normalizing by physically meaningful scales), and dimensionless features that generalize across system sizes.
- **Physical application.** Engineers test a small model ship in a tank at the same Froude number as the full ship, so the waves match.

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Units & SI Prefixes | *Key Topic P.K1*

Every measurement is a number times a unit. The International System (SI) builds every unit from seven base units, and scales them with prefixes that are powers of ten. The single most effective habit in physics: **convert every input to base SI units (m, kg, s, K, A, mol) before substituting into a formula**; the answer then comes out in base SI units automatically, and a dimensional check P.1 catches most algebra slips.



The seven SI base units

Quantity	Unit	Symbol	Dim.
Length	meter	m	L
Mass	kilogram	kg	M
Time	second	s	T
Electric current	ampere	A	I
Temperature	kelvin	K	Θ
Amount of substance	mole	mol	N
Luminous intensity	candela	cd	J

Note: the base unit of mass is the **kilogram**, not the gram, so $1\ \text{g} = 10^{-3}$ kg.

SI prefixes, pico to tera

Prefix	Symbol	Factor	Example
tera	T	10^{12}	1 TB of storage
giga	G	10^{9}	2.4 GHz Wi-Fi
mega	M	10^{6}	5 MW wind turbine
kilo	k	10^{3}	1 km = 1000 m
centi	c	10^{-2}	1 cm = 0.01 m
milli	m	10^{-3}	250 mL of water
micro	μ	10^{-6}	10 μm hair detail
nano	n	10^{-9}	500 nm green light
pico	p	10^{-12}	10 pF capacitor

Less common: deci d 10^{-1}, deka da 10^{1}, hecto h 10^{2}; beyond: peta P 10^{15}, femto f 10^{-15}.

Derived units: every one is a product of base units

Quantity	Unit (symbol)	Defining formula	Other units	Base SI units	Where
Force	newton (N)	$F = ma$		$\text{kg}\cdot\text{m}\cdot\text{s}^{-2}$	P.4
Energy, work, heat	joule (J)	$W = Fd$	N·m	$\text{kg}\cdot\text{m}^2\cdot\text{s}^{-2}$	P.6
Power	watt (W)	$P = W/t$	J/s	$\text{kg}\cdot\text{m}^2\cdot\text{s}^{-3}$	P.6
Pressure, stress	pascal (Pa)	$P = F/A$	N/m^2	$\text{kg}\cdot\text{m}^{-1}\cdot\text{s}^{-2}$	P.19
Frequency	hertz (Hz)	$f = 1/T$	cycles/s	s^{-1}	P.15
Electric charge	coulomb (C)	$q = It$		A·s	P.32
Voltage	volt (V)	$V = P/I$	J/C, W/A	$\text{kg}\cdot\text{m}^2\cdot\text{s}^{-3}\cdot\text{A}^{-1}$	P.33
Resistance	ohm (Ω)	$R = V/I$	V/A	$\text{kg}\cdot\text{m}^2\cdot\text{s}^{-3}\cdot\text{A}^{-2}$	P.33
Capacitance	farad (F)	$C = q/V$	C/V	$\text{s}^4\cdot\text{A}^2\cdot\text{kg}^{-1}\cdot\text{m}^{-2}$	EE.10 §16 Electrical Eng.
Magnetic field	tesla (T)	$F = qvB$	N/(A·m)	$\text{kg}\cdot\text{s}^{-2}\cdot\text{A}^{-1}$	EE.18 §16 Electrical Eng.
Angle	radian (rad)	$\theta = s/r$	m/m	dimensionless	T.2 §06 Trigonometry

Converting units: the factor-label method

Method. (1) Write the quantity with its units. (2) Multiply by conversion factors written as fractions equal to 1, oriented so the unwanted unit cancels (top against bottom). (3) Raise a factor to a power when the unit is squared or cubed. (4) Cancel units, multiply numbers, and check that only the wanted unit is left.

Speed. $72\ \frac{\text{km}}{\text{h}} \cdot \frac{1000\ \text{m}}{1\ \text{km}} \cdot \frac{1\ \text{h}}{3600\ \text{s}} = 20\ \frac{\text{m}}{\text{s}}$. Shortcut: divide km/h by 3.6.

Area. $1\ \text{cm}^2 = 1\ (\text{cm})^2 \cdot \left(\frac{1\ \text{m}}{100\ \text{cm}}\right)^2 = 10^{-4}\ \text{m}^2$, and $1\ \text{km}^2 = 10^6\ \text{m}^2$.

Density. $1\ \frac{\text{g}}{\text{cm}^3} \cdot \frac{1\ \text{kg}}{1000\ \text{g}} \cdot \left(\frac{100\ \text{cm}}{1\ \text{m}}\right)^3 = 1000\ \frac{\text{kg}}{\text{m}^3}$ (water).

Energy. $1\ \text{kWh} = 1000\ \text{W} \cdot 3600\ \text{s} = 3.6 \times 10^6\ \text{J}$; $1\ \text{eV} = 1.602 \times 10^{-19}\ \text{J}$ P.37.

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Temperature. $T_K = T_C + 273.15$ (an offset, not a factor): convert absolute temperatures before using $PV = nRT$ or T^4 P.23. A temperature *difference* is the same size in K and °C.

Dimension check. Is $v = \sqrt{2gh}$ plausible? $[gh] = \frac{\text{m}}{\text{s}^2} \cdot \text{m} = \frac{\text{m}^2}{\text{s}^2}$, whose square root is m/s. ✓

DECISION POINT **Which units to use?** If a formula contains a constant with units (G, k, R, σ, h), work in base SI so the constant's units match. Pure ratios (efficiency, $\frac{V_{\text{sub}}}{V}$, decibels) can be computed in any consistent unit. **Squared and cubed units**: the prefix is raised too ($1\ \text{mm}^3 = 10^{-9}\ \text{m}^3$, not 10^{-3}). **Angles**: ω, $s = r\theta$ and calculus with sin need radians T.2 §06 Trigonometry.

Common mistakes

- Mixing m (milli) and M (mega): 1 mW is 10^9 times smaller than 1 MW.
- Using grams in $F = ma$ or $KE = \frac{1}{2}mv^2$: mass must be in kg.
- Celsius in $PV = nRT$ or T^4 laws: always kelvin.
- $1\ \text{L} = 10^{-3}\ \text{m}^3$, not $1\ \text{m}^3$; $1\ \text{mL} = 1\ \text{cm}^3$.
- Forgetting the 3.6 between km/h and m/s, or 60 between rpm and rev/s.
- Adding quantities with different units (5 m + 3 s is meaningless).

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Motion in One and Two Dimensions

P.2 Kinematics: Constant Acceleration

Kinematics describes motion without asking what causes it. With constant acceleration, four equations link displacement, initial and final velocity, acceleration and time; each one leaves out exactly one of these five quantities.

Velocity	$v = \dfrac{dx}{dt}$	Slope of the position–time graph C1.9 §07 Calculus I.
Acceleration	$a = \dfrac{dv}{dt} = \dfrac{d^2x}{dt^2}$	Slope of velocity; the second derivative of position.
Velocity–time	$v = v_0 + at$	Constant a adds the same velocity every second. Lacks Δx.
Position–time	$x = x_0 + v_0 t + \frac{1}{2}at^2$	Integrate $v = v_0 + at$ once more; lacks the final v.
Timeless equation	$v^2 = v_0^2 + 2a\,\Delta x$	Eliminate t between the two above; lacks t.
Average-velocity form	$\Delta x = \frac{1}{2}(v_0 + v)\,t$	Under constant a the average velocity is the midpoint; lacks a.

STATED-EQUATION *Velocity is d x d t; acceleration is d v d t, the second derivative of position. v equals v naught plus a t. x equals x naught plus v naught t plus one half a t squared. v squared equals v naught squared plus two a delta x. Delta x equals one half of v naught plus v, times t.*

MEMORY AID "Velocity is the slope of position; acceleration is the slope of velocity" (and area under v–t is distance).

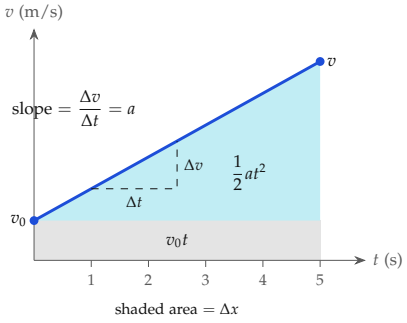
METHOD

1. Sketch the motion, choose a positive direction, and list the knowns with signs (e.g. braking: $a < 0$; free fall with up positive: $a = -g$).
2. Name the unknown you want, and the one quantity among $\Delta x, v_0, v, a, t$ that you neither know nor want.
3. Pick the equation that *lacks* that quantity (table in the decision point).
4. Solve, then check units and sign. Example: a car at 20 m/s brakes at $-5\,\text{m/s}^2$ to rest; t is not needed, so $0 = 20^2 + 2(-5)\Delta x \Rightarrow \Delta x = 40\,\text{m}$.

DECISION POINT Missing $\Delta x \to v = v_0 + at$. Missing $v \to$ position–time. Missing $t \to$ timeless. Missing $a \to$ average-velocity form. If the acceleration changes partway (speed up, then brake), split the trip into phases: the final v of one phase is v_0 of the next. If a varies continuously, integrate $a(t)$ instead C1.29 §07 Calculus I.

TERMS

x, x_0 – position and initial position (m)
v, v_0 – velocity and initial velocity (m/s); derivative C1.9 §07 Calculus I
a – acceleration (m/s^2); free fall $a = -g$
t – time (s)
Δx – displacement $x - x_0$ (m), signed
Derivation – integrate a twice C1.29 §07 Calculus I; the $\frac{1}{2}$ comes from $\int t\,dt$



CONTEXT

- **Origin.** Galileo (1638) with rolling balls on ramps; calculus form by Newton.
- **Use.** Braking distances, projectile motion P.3, elevator design.
- **Data science.** Sensor fusion and Kalman filters use these equations as the motion model to track vehicles and phones from noisy GPS and accelerometer data.
- **Physical application.** Stopping distance grows with v^2: doubling your speed quadruples the braking distance.

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P.3 Projectile Motion

Ignoring air resistance, a projectile moves at constant velocity horizontally and with constant acceleration $-g$ vertically. The two motions are independent and share only the time t; the path is a parabola.

Launch components	$v_{0x} = v_0 \cos\theta, \quad v_{0y} = v_0 \sin\theta$	Resolve the launch velocity with right-triangle trig T.1 §06 Trigonometry.
Horizontal motion	$x = v_{0x}\, t$	No horizontal force, so v_x never changes.
Vertical motion	$y = y_0 + v_{0y}t - \frac{1}{2}gt^2, \quad v_y = v_{0y} - gt$	Kinematics P.2 with $a = -g$.
Time of flight (level)	$t_f = \dfrac{2v_0 \sin\theta}{g}$	Set $y = 0$; twice the time to reach the top.
Maximum height	$H = \dfrac{v_0^2 \sin^2\theta}{2g}$	At the top $v_y = 0$; use the timeless equation vertically.
Range (level ground)	$R = \dfrac{v_0^2 \sin 2\theta}{g}$	Largest at $\theta = 45°$; angles θ and $90° - \theta$ give the same range.
Trajectory equation	$y = x\tan\theta - \dfrac{g\,x^2}{2v_0^2\cos^2\theta}$	Eliminate t: a downward parabola in x A1.16 §03 Algebra I.

STATED-EQUATION *The initial velocity components are v naught cos theta horizontally and v naught sin theta vertically. x equals v naught x times t. y equals y naught plus v naught y t minus one half g t squared. On level ground the time of flight is two v naught sin theta over g, the maximum height is v naught squared sin squared theta over two g, and the range is v naught squared sin two theta over g. The vertical velocity is v naught y minus g t. Eliminating time, the path is y equals x tan theta minus g x squared over two v naught squared cos squared theta.*

MEMORY AID "Horizontal: cruise. Vertical: free fall." The clock is the only thing the two directions share.

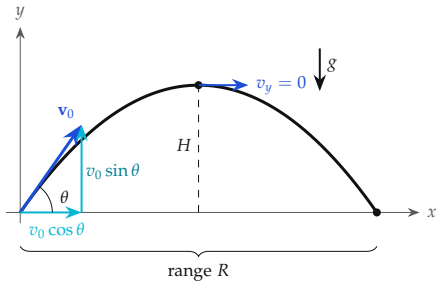
METHOD

1. Resolve v_0 into v_{0x} and v_{0y}.
2. Write $x(t)$ and $y(t)$ separately.
3. Use the vertical equation to find the time of the event: top ($v_y = 0$) or landing ($y = y_{\text{land}}$, a quadratic A1.15 §03 Algebra I; keep the positive root).
4. Substitute that t into $x(t)$. Example: $v_0 = 20\,\text{m/s}$ at $30°$: $v_{0y} = 10$, $t_f = \frac{20}{9.81} = 2.04\,\text{s}$, $R = 17.3 \cdot 2.04 = 35.3\,\text{m}$, $H = \frac{10^2}{2 \cdot 9.81} = 5.1\,\text{m}$.

DECISION POINT Launch and landing at the same height → use the shortcuts t_f, H, R. Landing higher or lower (cliff, basketball hoop) → the shortcuts are wrong; solve $y(t) = y_{\text{land}}$. Air drag matters (fast, light objects) → the path is no longer a parabola; integrate numerically with drag P.22.

TERMS

v_0 – launch speed (m/s); θ launch angle above horizontal
v_{0x}, v_{0y} – horizontal and vertical components (m/s)
g – 9.81 m/s^2, downward
y_0 – launch height (m)
t_f – time of flight (s)
H – maximum height above launch (m)
R – horizontal range (m)
θ – launch angle above horizontal
x, y; v_y – horizontal and vertical position; vertical velocity



CONTEXT

- **Origin.** Galileo (1638) proved the parabolic path by combining independent horizontal and vertical motions.
- **Use.** Sports, ballistics, fountains, firefighting hoses.
- **Data science.** Ball-tracking systems fit a parabola to camera detections by least squares S.40 §13 Statistics to predict where a ball will land; parametric curves C2.25 §08 Calculus II.
- **Physical application.** Long jumpers take off near 20°, not 45°: they cannot keep their run-up speed at steep angles, so the ideal-range formula is only the starting point.

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Forces & Newton's Laws

P.4 Newton's Laws & Friction

Forces change motion: the net force on an object equals its mass times its acceleration. Every force is an interaction between two bodies, and contact forces such as friction and the normal force respond to what the other forces are doing.

First law (inertia)	$\sum \mathbf{F} = \mathbf{0} \iff \mathbf{v} = \text{const}$	No net force means no change in velocity (rest is a special case).
Second law	$\sum \mathbf{F} = m\mathbf{a}$	Net force, not any single force, sets the acceleration; apply per component.
Third law	$\mathbf{F}_{AB} = -\mathbf{F}_{BA}$	Equal and opposite, acting on *different* bodies, so they never cancel on one diagram.
Weight	$W = mg$	Gravity's pull near Earth's surface, directed downward.
Kinetic friction	$f_k = \mu_k N$	Sliding surfaces; opposes the relative motion.
Static friction	$f_s \le \mu_s N$	Takes whatever value (up to the maximum) is needed to prevent slipping.
Incline components	$F_{\parallel} = mg\sin\theta, \quad N = mg\cos\theta$	Weight split along and into the slope (N is this only if nothing else pushes perpendicular).

STATED-EQUATION *First law: net force zero if and only if velocity is constant. Second law: the sum of the forces equals m a. Third law: the force of A on B equals minus the force of B on A. Weight equals m g. Kinetic friction equals mu k times the normal force N; static friction is at most mu s N. On an incline the weight has components m g sine theta along the slope and m g cosine theta into it.*

MEMORY AID "F equals m a" — every problem starts with a free-body diagram: draw all forces, then sum them.

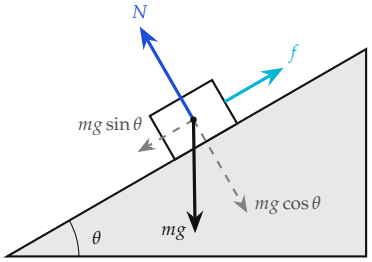
METHOD

1. Isolate one object and draw it as a box or dot.
2. Draw every force acting *on* it: weight, normal, tension, friction, spring, applied. Not the forces it exerts on others.
3. Choose axes with one along the (expected) acceleration; on a slope, along the slope.
4. Resolve each force into components and write $\sum F_x = ma_x$, $\sum F_y = ma_y$.
5. Solve. Example: block on a 30° slope, $\mu_k = 0.2$: $a = g(\sin\theta - \mu_k\cos\theta) = 9.81(0.5 - 0.173) = 3.2\,\text{m/s}^2$.

DECISION POINT **Static or kinetic?** Assume the object stays at rest and compute the friction needed. If that is $\le \mu_s N$, it stays put and f_s equals the needed value (not $\mu_s N$). Otherwise it slides and $f = \mu_k N$. **Several bodies** (rope, pulley): draw one diagram per body and link them by the shared acceleration and tension.

TERMS

$\sum \mathbf{F}$ – net (vector) force in newtons T.21 §06 Trigonometry
m – mass (kg); **a** acceleration (m/s^2) P.2
$W = mg$ – weight (N), the force of gravity near Earth
N – normal force (N) from a surface (equals mg on level ground)
μ_k, μ_s – kinetic and static friction coefficients (dimensionless), $\mu_k < \mu_s$
θ – incline angle; components via T.1 §06 Trigonometry
v, **a**; $\mathbf{F}_{AB}$ – velocity, acceleration; force exerted by A on B
f_k, f_s; $F_{\parallel}$ – kinetic and static friction forces; weight component along the slope



axes: x along the slope, y perpendicular

CONTEXT

- **Origin.** Isaac Newton, *Principia* (1687).
- **Use.** Designing structures, vehicles, and machines.
- **Data science.** Physics engines for simulation and robotics; physics-informed ML adds $F = ma$ as a constraint in the loss.
- **Physical application.** A heavy box and a light box on the same ramp slide identically when friction is proportional to weight, because m cancels.

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P.5 Hooke's Law & Springs

An ideal spring pulls or pushes back with a force proportional to how far it is stretched or compressed. The energy it stores grows with the square of that deformation.

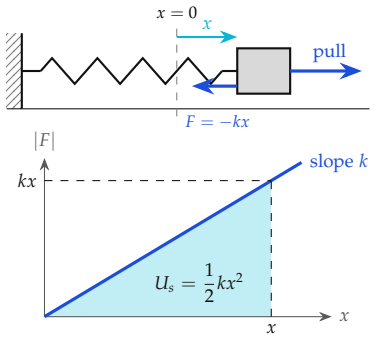
Hooke's law	$F = -kx$	Restoring force, opposite the displacement; valid up to the elastic limit.
Spring potential energy	$U_s = \frac{1}{2}kx^2$	Work to stretch it: the triangular area under the F–x line.
Springs in series	$\frac{1}{k_{eq}} = \frac{1}{k_1} + \frac{1}{k_2}$	Same force through each, stretches add: the chain is softer.
Springs in parallel	$k_{eq} = k_1 + k_2$	Same stretch, forces add: the pair is stiffer.
Stress–strain (Young's modulus)	$\frac{F}{A} = E\frac{\Delta L}{L_0}$	Hooke's law for a material; a rod acts as a spring with $k = \frac{EA}{L_0}$.

STATED-EQUATION *The spring force F equals minus k x. The stored energy is one half k x squared. For springs in series, one over k equivalent equals one over k one plus one over k two; in parallel, k equivalent equals k one plus k two. Stress, F over A, equals Young's modulus E times strain, delta L over L naught.*

MEMORY AID Hooke's anagram "ut tensio, sic vis": as the extension, so the force. Springs combine the *opposite* way to resistors P.33.

TERMS

F – spring force (N) on the attached object
k – spring constant, stiffness (N/m)
x – displacement from the natural length (m)
U_s – elastic potential energy (J) P.6
E – Young's modulus (Pa); steel $\approx$ 200 GPa, rubber $\approx$ 0.01 GPa
A, L_0, ΔL – cross-section (m^2), original length and extension (m)
k_1, k_2; k_{eq} – individual spring constants; equivalent spring constant



CONTEXT

- **Origin.** Robert Hooke published the law as an anagram (1676) and solved it in 1678; Thomas Young (1807).
- **Use.** Scales, vehicle suspensions, mattresses, building materials, oscillators P.15.
- **Data science.** L2 regularization $\frac{1}{2}\lambda\|w\|^2$ is a spring pulling weights toward zero ML.8 §15 Machine Learning; force-directed graph layouts model edges as springs.
- **Physical application.** A bathroom scale reads your weight by measuring how far a stiff spring compresses: $x = mg/k$.

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Energy & Momentum

P.6 Work, Energy & Power

Work is energy transferred by a force acting through a distance. Energy changes form (kinetic, potential, heat) but its total is conserved; power is the rate at which it moves.

Work	$W = Fd\cos\theta = \int \mathbf{F}\cdot d\mathbf{x}$	Only the force component along the motion does work (dot product T.21 §06 Trigonometry).
Kinetic energy	$KE = \frac{1}{2}mv^2$	Energy of motion; doubles with mass, quadruples with speed.
Gravitational potential energy	$PE = mgh$	Near Earth's surface; h measured from any chosen zero.
Work–energy theorem	$W_{\text{net}} = \Delta KE$	The net work by all forces equals the change in kinetic energy.
Conservation of mechanical energy	$KE_1 + PE_1 = KE_2 + PE_2$	Holds when only conservative forces (gravity, springs) do work.
Energy with losses	$KE_1 + PE_1 = KE_2 + PE_2 + E_{\text{loss}}$	Friction and drag turn mechanical energy into heat: $E_{\text{loss}} = f_k d$.
Power	$P = \dfrac{W}{t} = Fv$	Rate of doing work; Fv for a force pushing at speed v.
Efficiency	$\eta = \dfrac{P_{\text{out}}}{P_{\text{in}}}$	Fraction of input energy delivered usefully; always ≤ 1.

STATED-EQUATION *Work equals F d cosine theta, the integral of F dot d x. Kinetic energy equals one half m v squared. Potential energy equals m g h. Net work equals the change in kinetic energy. Kinetic plus potential energy before equals kinetic plus potential energy after, plus any energy lost. Power equals work over time, which equals F v. Efficiency is power out over power in.*

METHOD

1. Pick two states (start, end) and a zero level for PE.
2. Write every energy term in each state: $\frac{1}{2}mv^2$, mgh, spring $\frac{1}{2}kx^2$ P.5.
3. Add E_{loss} if friction acts, then solve. Example: a ball dropped from 5 m: $mgh = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{2 \cdot 9.81 \cdot 5} = 9.9\,\text{m/s}$.

DECISION POINT **Energy or forces?** If the question asks for a speed at a position (and not a time), energy is quicker and ignores the path's shape. If it asks for time or acceleration, use forces P.4 and kinematics P.2. Collisions: use momentum P.7, since kinetic energy is usually lost.

TERMS

W – work (J); θ angle between force and displacement
F, d – force (N), displacement (m)
KE – kinetic energy (J); m mass (kg), v speed (m/s)
PE – gravitational potential energy (J); h height (m)
P – power (W = J/s); t time (s)
η – efficiency (0 to 1)
Derivation – work–energy from $F = ma$ and the chain rule C1.13 §07 Calculus I; conservative forces: V.4 §11 Vector Calculus
m, v; g, h – mass, speed; gravitational acceleration, height
E_{loss}; $P_{\text{out}}, P_{\text{in}}$ – energy lost to friction, heat, sound; useful and supplied power

CONTEXT

- **Origin.** Leibniz's *vis viva* (1686); energy conservation by Joule, Mayer, and Helmholtz (1840s).
- **Use.** Roller coasters, hydroelectric dams, engine ratings.
- **Data science.** Energy-based models and Hamiltonian Monte Carlo O.29 §14 Optimization use "potential energy" $= -\log p(x)$ and "kinetic energy" to explore distributions.
- **Physical application.** A 60 W bulb left on for 10 hours uses 0.6 kWh; your electricity bill is an energy (work) bill.

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P.7 Momentum & Impulse

Momentum is mass times velocity. A force acting over time changes it (impulse), and when no external force acts, the total momentum of a system is conserved, which is what makes collisions solvable.

Momentum	$\mathbf{p} = m\mathbf{v}$	A vector: direction matters, so signs matter in 1-D.
Second law (momentum form)	$\sum \mathbf{F} = \frac{d\mathbf{p}}{dt}$	Newton's original statement; reduces to $m\mathbf{a}$ for constant mass.
Impulse–momentum theorem	$\mathbf{J} = \int \mathbf{F}\,dt = \mathbf{F}_{\text{avg}}\,\Delta t = \Delta\mathbf{p}$	Same Δp over a longer time means a smaller force.
Conservation of momentum	$m_1\mathbf{v}_1 + m_2\mathbf{v}_2 = m_1\mathbf{v}_1' + m_2\mathbf{v}_2'$	Total before equals total after when external forces are negligible.
Perfectly inelastic collision	$v' = \frac{m_1 v_1 + m_2 v_2}{m_1 + m_2}$	Objects stick together; the most kinetic energy is lost.
Elastic collision (1-D)	$v_1' - v_2' = -(v_1 - v_2)$	Kinetic energy also conserved: relative velocity reverses. Equal masses swap velocities.
Center of mass	$x_{\text{cm}} = \frac{\sum m_i x_i}{\sum m_i}$	The system moves as if all mass sat here, pushed by the net external force.

STATED-EQUATION *Momentum p equals m v. Net force equals d p d t. Impulse J equals the integral of F d t, the average force times delta t, which equals the change in momentum. m one v one plus m two v two equals m one v one prime plus m two v two prime. Objects that stick move at the mass-weighted average velocity. In an elastic collision the relative velocity reverses. The center of mass is the mass-weighted mean position.*

METHOD

1. Choose a positive direction; write each momentum with its sign.
2. Set total momentum before = total after.
3. If elastic, add the relative-velocity equation; if they stick, use one common v'.
4. Solve. Example: 2 kg at 3 m/s hits a 1 kg block at rest and sticks: $v' = \frac{6}{3} = 2$ m/s; KE drops from 9 J to 6 J.

DECISION POINT Momentum is conserved in every collision or explosion (internal forces dominate during the brief contact); kinetic energy only in elastic ones. Objects stick → perfectly inelastic. "Bounce with no loss" (billiard balls, gas molecules) → elastic. Anything in between needs a coefficient of restitution.

TERMS

p – momentum (kg·m/s), a vector
J – impulse (N·s); area under force–time curve C1.32 §07 Calculus I
v, **v**′ – velocities before and after (m/s)
Elastic – kinetic energy also conserved
Perfectly inelastic – objects stick: v' is a weighted mean S.1 §13 Statistics
x_{cm} – center of mass (m)
$\mathbf{F}_{\text{avg}}$, Δt – average force over the contact time
m_1, m_2; m_i, x_i – colliding masses; masses and positions of the parts

CONTEXT

- **Origin.** Descartes (1644) and Huygens; Newton's second law is originally $\mathbf{F} = \frac{d\mathbf{p}}{dt}$.
- **Use.** Airbags, crash testing, rocket propulsion.
- **Data science.** "Momentum" in optimizers is named by analogy: a velocity term accumulates past gradients ML.27 §15 Machine Learning.
- **Physical application.** Airbags and crumple zones lengthen Δt so the same Δp produces a much smaller force on your body.

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Circular Motion & Gravitation

P.8 Circular Motion & Gravitation

Moving in a circle requires a net inward (centripetal) force. Gravity is an inverse-square attraction between masses; for planets and satellites it supplies exactly that inward force.

Centripetal acceleration	$a_c = \dfrac{v^2}{r} = \omega^2 r$	Points to the center; present even at constant speed because the direction changes.
Centripetal force	$F_c = \dfrac{mv^2}{r}$	Not a new force: the net inward pull of tension, friction, gravity or normal force.
Speed and period	$v = \omega r = \dfrac{2\pi r}{T}$	One lap of circumference $2\pi r$ per period T G.17 §05 Geometry.
Newton's law of gravitation	$F = \dfrac{G\,m_1 m_2}{r^2}$	Attractive, along the line of centers; inverse square in distance.
Surface gravity	$g = \dfrac{GM}{R^2}$	Earth's mass and radius give $g \approx 9.81\ \text{m/s}^2$.
Orbital speed	$v_{\text{orbit}} = \sqrt{\dfrac{GM}{r}}$	Set gravity equal to mv^2/r; independent of the satellite's mass.

STATED-EQUATION *Centripetal acceleration equals v squared over r, which equals omega squared r. Centripetal force equals m v squared over r. Speed equals omega r, which is two pi r over the period. Gravitational force equals G m one m two over r squared. Little g equals G M over R squared. Orbital speed equals the square root of G M over r.*

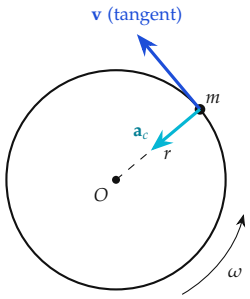
MEMORY AID Gravity is an inverse-square law: double the distance, one quarter the force.

METHOD

1. Draw the forces; identify which ones (or components) point toward the center.
2. Set $\sum F_{\text{inward}} = \frac{mv^2}{r}$ and solve. Example: flat curve, $r = 50\,\text{m}$, $\mu_s = 0.8$: friction $\mu_s mg = \frac{mv^2}{r} \Rightarrow v_{\max} = \sqrt{\mu_s g r} = 19.8\,\text{m/s}$.

TERMS

a_c – centripetal acceleration (m/s^2), pointing to the center
ω – angular speed (rad/s) T.2 §06 Trigonometry; T period (s)
r – radius of the circle / distance between centers (m)
G – $6.674 \times 10^{-11}\ \text{N·m}^2/\text{kg}^2$
M, R – Earth: 5.972×10^{24} kg, 6.371×10^{6} m
Orbit – gravity supplies the centripetal force: $\frac{GMm}{r^2} = \frac{mv^2}{r}$; more in P.9
v; T; m – speed; period of one revolution; moving mass



a_c points to the center

CONTEXT

- **Origin.** Huygens (1673) for circular motion; Newton's universal gravitation (1687).
- **Use.** Satellites, banked curves, centrifuges, planetary science.
- **Data science.** Inverse-square falloff inspires distance-decay weights in spatial models and gravity models of trade and migration flows.
- **Physical application.** A car on a curve relies on tire friction to supply mv^2/r; that is why icy curves are dangerous.

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P.9 Orbits, Escape Velocity & Kepler's Laws

Away from Earth's surface gravity weakens as $1/r^2$, so its potential energy is $-GMm/r$. That one formula gives escape speeds, orbital energies and Kepler's laws of planetary motion.

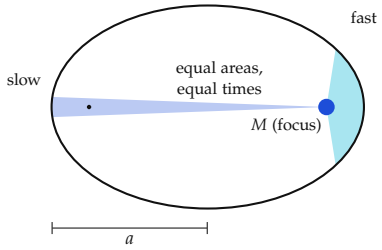
Gravitational potential energy	$U = -\dfrac{GMm}{r}$	Zero at infinite distance, negative when bound; mgh is its near-surface approximation.
Escape velocity	$v_{\text{esc}} = \sqrt{\dfrac{2GM}{r}} = \sqrt{2}\, v_{\text{orbit}}$	Kinetic energy just cancels U: total energy 0.
Field strength at distance r	$g(r) = \dfrac{GM}{r^2} = g\left(\dfrac{R}{r}\right)^2$	Weaker with altitude; ISS (400 km up) still feels 89% of g.
Kepler's first & second laws	ellipse, star at a focus; $\dfrac{dA}{dt} = \text{const}$	Equal areas in equal times: angular momentum is conserved P.13.
Kepler's third law	$T^2 = \dfrac{4\pi^2}{GM}\, a^3$	Period squared is proportional to the semi-major axis cubed.
Orbital energy	$E = KE + U = -\dfrac{GMm}{2a}$	Circular orbit: $KE = -\frac{1}{2}U$; lower orbits are faster but more tightly bound.

STATED-EQUATION *Gravitational potential energy equals minus G M m over r. Escape velocity equals the square root of two G M over r, root two times the orbital speed. The field at distance r is G M over r squared. Orbits are ellipses with the star at a focus and sweep equal areas in equal times. T squared equals four pi squared over G M, times a cubed. Orbital energy equals minus G M m over two a. Equivalently, the field is surface g times R over r, squared.*

MEMORY AID Kepler III: "square the period, cube the distance."

TERMS

U – gravitational potential energy (J)
r – distance from the planet's center (m), not altitude
a – semi-major axis of the ellipse (m) A2.26 §04 Algebra II; $= r$ for a circle
T – orbital period (s)
E – total orbital energy (J); $E < 0$ means bound
G, M, R, g – as in P.8
v_{esc}; v_{orbit} – escape speed; circular-orbit speed P.8
dA/dt – rate at which the line to the star sweeps out area



CONTEXT

- **Origin.** Kepler (1609, 1619) from Tycho Brahe's data; Newton derived all three laws from inverse-square gravity (1687).
- **Use.** Satellite and GPS orbits, launch planning, exoplanet detection.
- **Data science.** Gauss invented least squares S.40 §13 Statistics in 1801 to recover the orbit of the asteroid Ceres from a few noisy observations.
- **Physical application.** Escape velocity from Earth is 11.2 km/s; a geostationary satellite orbits at $r = 42{,}164$ km, where T equals one sidereal day.

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Rotation

P.10 Rotational Kinematics

Rotation about a fixed axis is described by an angle, an angular velocity and an angular acceleration. Every linear kinematics equation has a rotational twin, and a point at radius r moves with linear speed $r\omega$.

Arc length	$s = r\theta$	Requires θ in radians T.2 §06 Trigonometry.
Angular velocity & acceleration	$\omega = \dfrac{d\theta}{dt}, \quad \alpha = \dfrac{d\omega}{dt}$	Rates of change of angle and of angular velocity.
Linear–angular link	$v_t = r\omega, \quad a_t = r\alpha$	Points farther out move and speed up faster.
Angular velocity–time	$\omega = \omega_0 + \alpha t$	Twin of $v = v_0 + at$ P.2.
Angular position–time	$\theta = \theta_0 + \omega_0 t + \frac{1}{2}\alpha t^2$	Twin of $x = x_0 + v_0 t + \frac{1}{2}at^2$.
Rotational timeless equation	$\omega^2 = \omega_0^2 + 2\alpha\,\Delta\theta$	Twin of $v^2 = v_0^2 + 2a\Delta x$.
Frequency & rpm	$\omega = 2\pi f = \dfrac{2\pi \cdot \text{rpm}}{60}$	One revolution is 2π radians.

STATED-EQUATION *Arc length s equals r theta. Angular velocity omega is d theta d t, angular acceleration alpha is d omega d t. Tangential speed equals r omega and tangential acceleration r alpha. omega equals omega naught plus alpha t; theta equals theta naught plus omega naught t plus one half alpha t squared; omega squared equals omega naught squared plus two alpha delta theta. omega equals two pi f. In terms of revolutions per minute, omega equals two pi times rpm over sixty.*

MEMORY AID Swap $x \to \theta$, $v \to \omega$, $a \to \alpha$ and every kinematics equation still works.

METHOD

1. List the knowns among $\Delta\theta, \omega_0, \omega, \alpha, t$; convert rpm and degrees to rad/s and rad.
2. Pick the equation lacking the quantity you neither know nor need (as in P.2).
3. Solve; convert to linear quantities with r if asked. Example: a wheel starts from rest with $\alpha = 2\,\text{rad/s}^2$; after 3 s, $\omega = 6\,\text{rad/s}$ and $\Delta\theta = \frac{1}{2}\cdot 2 \cdot 9 = 9\,\text{rad}$.

Linear	x	v	a	m	$F = ma$	$p = mv$	$\frac{1}{2}mv^2$
Rotational	θ	ω	α	I P.12	$\tau = I\alpha$ P.13	$L = I\omega$	$\frac{1}{2}I\omega^2$ P.14

TERMS

θ – angular position (rad); s arc length (m); r radius (m)
ω – angular velocity (rad/s)
α – angular acceleration (rad/s^2)
v_t, a_t – tangential speed (m/s) and acceleration (m/s^2); the inward part is $a_c = \omega^2 r$ P.8
f – frequency (rev/s = Hz); rpm revolutions per minute
s, r; t – arc length, radius; time
rpm – revolutions per minute

CONTEXT

- **Origin.** Euler (1750s) formalized rigid-body rotation.
- **Use.** Motors, turbines, hard drives, gears, wheels.
- **Data science.** Gyroscopes in phones and drones measure ω; integrating it gives orientation, whose drift is corrected by sensor fusion.
- **Physical application.** A 7200 rpm hard disk spins at $\omega = 754\,\text{rad/s}$; its outer edge (4.5 cm) moves at 34 m/s.

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P.11 Torque & Static Equilibrium

Torque is the turning effect of a force: force times the perpendicular distance from the pivot to the force's line of action. An object is in static equilibrium when both the net force and the net torque are zero.

Torque (magnitude)	$\tau = rF\sin\phi = r_\perp F$	Only the force component perpendicular to $\mathbf{r}$ turns things.
Torque (vector)	$\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}$	Cross product: direction along the axis by the right-hand rule.
Law of the lever	$F_1 d_1 = F_2 d_2$	Balanced torques about the pivot: a long arm multiplies force.
Translational equilibrium	$\sum \mathbf{F} = \mathbf{0}$	No net force: $\sum F_x = \sum F_y = 0$.
Rotational equilibrium	$\sum \tau = 0$ about any point	No net torque; if it holds about one point in equilibrium, it holds about all.

STATED-EQUATION *Torque equals r F sine phi, which is the lever arm r perpendicular times F; as a vector, torque is r cross F. For a lever, F one d one equals F two d two. In static equilibrium the sum of the forces is zero and the sum of the torques about any point is zero.*

MEMORY AID Door handles sit far from the hinges: a long lever arm needs only a small force.

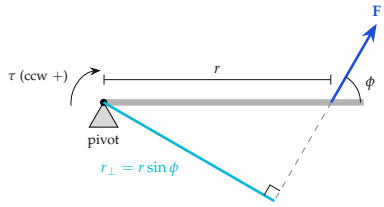
METHOD

1. Draw the extended body with each force at its actual point of application (weight at the center of mass).
2. Choose the pivot where an unknown force acts, so its torque vanishes.
3. Write $\sum \tau = 0$ (counter-clockwise positive), then $\sum F_x = 0$, $\sum F_y = 0$.
4. Solve. Example: a 30 kg child sits 2 m from a seesaw pivot; a 40 kg child balances at $d = \frac{30 \cdot 2}{40} = 1.5$ m.

DECISION POINT Pivot choice changes the arithmetic, not the answer. Put it where the most (or the most awkward) unknown forces act. Count signs carefully: a force pulling the body counter-clockwise about your pivot is positive.

TERMS

τ – torque (N·m)
r – distance from pivot to where the force acts (m)
ϕ – angle between $\mathbf{r}$ and $\mathbf{F}$
$r_\perp$ – lever (moment) arm: perpendicular distance to the line of action
d_1, d_2 – lever arms of the two forces (m)
Center of mass – where the weight acts P.7



CONTEXT

- **Origin.** Archimedes (~250 BCE): "give me a place to stand and I will move the Earth."
- **Use.** Bridges, cranes, wrenches, biomechanics of joints.
- **Data science.** The balance point of masses on a beam is their weighted mean S.1 §13 Statistics: the mean is the point where the "torques" of deviations cancel.
- **Physical application.** Your biceps attaches about 4 cm from the elbow but holds a load 35 cm away, so it must pull with roughly 9 times the load's weight.

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P.12 Moment of Inertia

The moment of inertia I is rotational mass: it measures how hard it is to change an object's spin. It depends on the mass and, quadratically, on how far that mass lies from the rotation axis.

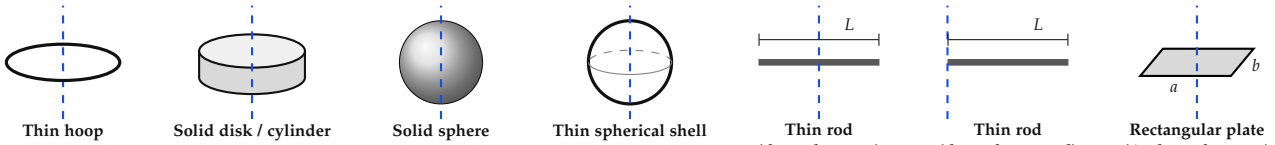
Point masses	$I = \sum m_i r_i^2$	Each mass weighted by its squared distance from the axis.
Continuous body	$I = \int r^2 \, dm$	Sum becomes an integral over the body C3.19 §10 Calculus III.
Parallel-axis theorem	$I = I_{\text{cm}} + Md^2$	Shift from a center-of-mass axis to a parallel axis a distance d away.
Perpendicular-axis theorem	$I_z = I_x + I_y$	Flat plates only: axes x, y in the plane, z perpendicular.
Radius of gyration	$I = MR_g^2$	The radius at which all the mass could sit to give the same I.

STATED-EQUATION *The moment of inertia equals the sum of m i r i squared, or the integral of r squared d m. Parallel-axis theorem: I equals I center of mass plus M d squared. For a flat plate, I z equals I x plus I y. The radius of gyration R g satisfies I equals M R g squared.*

MEMORY AID Mass far from the axis counts quadratically: a hoop (MR^2) is twice as hard to spin as a disk ($\frac{1}{2}MR^2$) of equal mass and radius.

METHOD

1. Find I_{cm} from the table (or integrate).
2. If the axis does not pass through the center of mass, add Md^2.
3. For composite bodies, add the I of each part about the same axis. Example: rod about its end: $\frac{1}{12}ML^2 + M\left(\frac{L}{2}\right)^2 = \frac{1}{3}ML^2$.



Thin hoop (central axis)	Solid disk / cylinder (central axis)	Solid sphere (diameter)	Thin spherical shell (diameter)	Thin rod (through center)	Thin rod (through one end)	Rectangular plate (⊥, through center)
MR^2	$\frac{1}{2}MR^2$	$\frac{2}{5}MR^2$	$\frac{2}{3}MR^2$	$\frac{1}{12}ML^2$	$\frac{1}{3}ML^2$	$\frac{1}{12}M(a^2+b^2)$

TERMS

I – moment of inertia (kg·m^2)
m_i, r_i – each mass (kg) and its distance from the axis (m)
M – total mass (kg); R radius, L length, a, b sides (m)
I_{cm} – moment about a parallel axis through the center of mass
d – distance between the two parallel axes (m)
R_g – radius of gyration (m)

CONTEXT

- **Origin.** Euler named and systematized the "moment of inertia" (1765); Huygens used it for compound pendulums (1673).
- **Use.** Flywheels, crankshafts, figure skating, vehicle and aircraft design.
- **Data science.** I is a second moment, like variance S.2 §13 Statistics: about the center of mass $I = M\cdot$(mean squared distance). The principal axes of the inertia tensor are the eigenvectors that PCA finds for data ML.38 §15 Machine Learning.
- **Physical application.** A tightrope walker's long pole has a large I, so any wobble develops slowly enough to correct.

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P.13 Rotational Dynamics & Angular Momentum

Net torque causes angular acceleration just as net force causes linear acceleration. Angular momentum $L = I\omega$ is conserved whenever no external torque acts, which is why spinning skaters speed up when they pull in their arms.

Newton's second law for rotation	$\sum \tau = I\alpha$	Torque about the axis divided by I gives the angular acceleration.
Angular momentum (rigid body)	$L = I\omega$	Rotational twin of $p = mv$.
Angular momentum (particle)	$\mathbf{L} = \mathbf{r} \times \mathbf{p}, \quad L = mvr_\perp$	A particle moving in a straight line still has L about a point off that line.
Torque–angular momentum	$\sum \boldsymbol{\tau} = \dfrac{d\mathbf{L}}{dt}$	General form of $\tau = I\alpha$; also explains gyroscope precession.
Conservation of angular momentum	$I_1\omega_1 = I_2\omega_2$	No external torque: shrinking I raises ω.
Rotational work & power	$W = \tau\,\Delta\theta, \quad P = \tau\omega$	Twins of $W = Fd$ and $P = Fv$; engine torque times rpm is power.

STATED-EQUATION *The net torque equals I alpha. Angular momentum L equals I omega; for a particle it is r cross p, with size m v r perpendicular. Net torque equals d L d t. Without external torque, I one omega one equals I two omega two. Rotational work equals torque times angle turned, and power equals torque times omega.*

MEMORY AID "Pull in, spin up": L fixed, so smaller I means larger ω.

METHOD

1. Draw forces with their lever arms; take torques about the rotation axis.
2. Write $\sum \tau = I\alpha$ for each rotating body and $\sum F = ma$ for each translating one.
3. Link them with the no-slip condition $a = r\alpha$ (rope on a pulley, rolling). Example: a $10\,\text{N·m}$ torque on a disk with $I = 2\,\text{kg·m}^2$ gives $\alpha = 5\,\text{rad/s}^2$.

DECISION POINT Is L conserved? Yes if the system is isolated, or if every external force passes through the axis (planet–sun gravity, a skater on smooth ice). Note kinetic energy is *not* conserved when a skater pulls in: $\frac{1}{2}I\omega^2 = \frac{L^2}{2I}$ rises, paid for by muscle work.

TERMS

τ – torque (N·m) P.11
I – moment of inertia (kg·m^2) P.12
α, ω – angular acceleration, velocity P.10
L – angular momentum ($\text{kg·m}^2/\text{s}$)
$\mathbf{r}$, $\mathbf{p}$ – position from the axis point (m), linear momentum P.7
$\Delta\theta$ – angle turned (rad); P power (W)

CONTEXT

- **Origin.** Euler's laws of rigid-body motion (1775); Kepler's second law P.9 is angular-momentum conservation.
- **Use.** Gyroscopes, reaction wheels on spacecraft, bicycles, diving and figure skating.
- **Data science.** Spacecraft attitude controllers and drone flight controllers solve $\tau = I\alpha$ in real time from gyroscope data.
- **Physical application.** A skater with $I = 4\,\text{kg·m}^2$ spinning at $2\,\text{rev/s}$ pulls in to $I = 1.6\,\text{kg·m}^2$ and spins at $\frac{4 \cdot 2}{1.6} = 5\,\text{rev/s}$.

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P.14 Rotational Kinetic Energy & Rolling

A spinning body stores kinetic energy $\frac{1}{2}I\omega^2$. A body rolling without slipping both translates and rotates, so part of its energy is tied up in spin, and it accelerates down a slope more slowly than a sliding block.

Rotational kinetic energy	$KE_{\text{rot}} = \frac{1}{2}I\omega^2$	Twin of $\frac{1}{2}mv^2$ with I for m and ω for v.
Rolling without slipping	$v_{\text{cm}} = R\omega, \quad a_{\text{cm}} = R\alpha$	The contact point is momentarily at rest; the top moves at $2v_{\text{cm}}$.
Total kinetic energy (rolling)	$KE = \frac{1}{2}Mv_{\text{cm}}^2 + \frac{1}{2}I_{\text{cm}}\omega^2$	Translation of the center plus spin about it.
Speed after descending h	$v = \sqrt{\dfrac{2gh}{1 + I_{\text{cm}}/(MR^2)}}$	Energy conservation; static friction does no work in pure rolling.
Acceleration down an incline	$a = \dfrac{g\sin\theta}{1 + I_{\text{cm}}/(MR^2)}$	Same factor slows the acceleration; a frictionless slider has $a = g\sin\theta$.

STATED-EQUATION *Rotational kinetic energy equals one half I omega squared. Rolling without slipping: v center of mass equals R omega, and a equals R alpha. Total kinetic energy equals one half M v squared plus one half I omega squared. After descending h, the speed is the square root of two g h over one plus I over M R squared. Down an incline the acceleration is g sine theta over the same factor.*

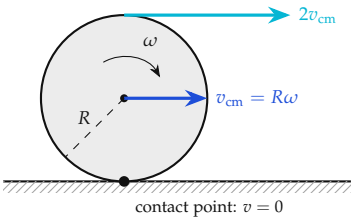
MEMORY AID Race down a ramp: sphere (0.4) beats disk (0.5) beats hoop (1), whatever their mass or size. The smaller I/MR^2, the faster.

METHOD

1. Look up $c = \frac{I_{\text{cm}}}{MR^2}$ from P.12: hoop 1, disk $\frac{1}{2}$, sphere $\frac{2}{5}$.
2. Write $Mgh = \frac{1}{2}Mv^2(1 + c)$ and solve. Example: a solid sphere rolling down a 1.4 m drop: $v = \sqrt{\frac{2 \cdot 9.81 \cdot 1.4}{1.4}} = 4.43\,\text{m/s}$ (sliding without friction would give 5.24 m/s).

TERMS

I_{cm} – moment of inertia about the center (kg·m^2)
ω – angular speed (rad/s); α angular acceleration
M, R – mass (kg) and rolling radius (m)
v_{cm}, a_{cm} – speed and acceleration of the center
h – height descended (m); θ incline angle
No slip – requires enough static friction P.4



CONTEXT

- **Origin.** Euler and d'Alembert (18th c.) on rigid-body energy.
- **Use.** Flywheel energy storage, wheels and bearings, ball sports.
- **Data science.** Flywheel and regenerative-braking models in EV energy simulations add $\frac{1}{2}I\omega^2$ terms alongside battery state.
- **Physical application.** A flywheel spinning at 20,000 rpm can store several kWh; buses in Switzerland once ran on them.

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Oscillations & Waves

P.15 Oscillations & Simple Harmonic Motion

When the restoring force is proportional to displacement, an object oscillates sinusoidally about equilibrium with a period that does not depend on amplitude. Energy sloshes back and forth between kinetic and potential.

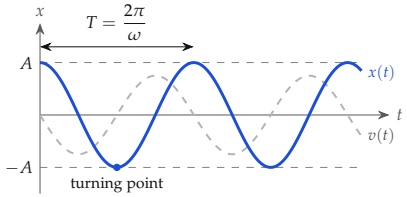
SHM equation	$m\ddot{x} = -kx \iff \ddot{x} = -\omega^2 x$	Any force obeying Hooke's law P.5 produces this ODE.
Solution	$x(t) = A\cos(\omega t + \phi)$	Sinusoid T.12 §06 Trigonometry; A and ϕ fixed by the starting position and velocity.
Angular frequency (spring–mass)	$\omega = \sqrt{\dfrac{k}{m}}$	Stiffer spring: faster; heavier mass: slower.
Period & frequency	$T = \dfrac{2\pi}{\omega} = 2\pi\sqrt{\dfrac{m}{k}}, \quad f = \dfrac{1}{T}$	Independent of amplitude: the hallmark of SHM.
Maximum speed & acceleration	$v_{\max} = A\omega, \quad a_{\max} = A\omega^2$	Speed peaks at equilibrium; acceleration peaks at the turning points.
Energy	$E = \frac{1}{2}kA^2 = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$	Constant total; all potential at $x = \pm A$, all kinetic at $x = 0$.
Damped oscillation	$x(t) = A\,e^{-\gamma t}\cos(\omega_d t + \phi), \qquad \gamma = \dfrac{b}{2m}, \quad \omega_d = \sqrt{\omega^2 - \gamma^2}$	Friction $-b\dot{x}$ shrinks the amplitude exponentially A2.13 §04 Algebra II.

STATED-EQUATION *m x double dot equals minus k x. x of t equals A cosine of omega t plus phi. Omega equals the square root of k over m; the period T equals two pi over omega, which is two pi root m over k, and f equals one over T. The maximum speed is A omega and the maximum acceleration A omega squared. The energy is one half k A squared. With damping, the amplitude decays as e to the minus gamma t. At any instant the energy also equals one half m v squared plus one half k x squared. The damping rate gamma is b over two m, and the damped angular frequency omega d is the square root of omega squared minus gamma squared.*

MEMORY AID SHM = "a spring and a cosine"; the period depends on the system (m, k), never on how hard you pluck it.

TERMS

A – amplitude (m); ϕ phase (rad) T.12 §06 Trigonometry
k – spring constant (N/m), from Hooke's law $F = -kx$
ω – angular frequency (rad/s) T.2 §06 Trigonometry
T, f – period (s) and frequency (Hz)
$\ddot{x}$ – second time derivative of x
b, γ – damping coefficient (kg/s) and decay rate (1/s)
Origin of the solution – $m\ddot{x} = -kx$ is solved by sinusoids C1.14 §07 Calculus I; driven systems resonate EE.16 §16 Electrical Eng.
See also – pendulum P.16, waves P.17
$x(t)$; ϕ; m – displacement from equilibrium; phase constant; mass
ω_d – damped angular frequency
$v_{\max}$, $a_{\max}$ – peak speed (at $x = 0$) and peak acceleration (at $x = \pm A$)



CONTEXT

- **Origin.** Hooke (1678); Galileo noticed pendulum isochronism (1602); Huygens built the pendulum clock (1656).
- **Use.** Clocks, suspensions, musical instruments, seismology.
- **Data science.** Fourier features and spectral analysis decompose signals into sinusoids; vibration data from machines feeds predictive-maintenance models.
- **Physical application.** A 1000 kg car on springs totaling $k = 40$ kN/m bounces at $f = \frac{1}{2\pi}\sqrt{40} \approx 1$ Hz; shock absorbers supply the damping.

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P.16 Pendulums

A pendulum swings because gravity's component along the arc pulls it back toward the bottom. For small angles that pull is proportional to displacement, so the motion is simple harmonic with a period set by length and g alone.

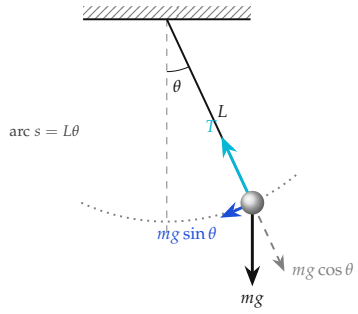
Restoring force	$F_t = -mg\sin\theta \approx -mg\,\theta$	Small-angle approximation $\sin\theta \approx \theta$ (Taylor C2.22 §08 Calculus II), valid to 1% up to about 14°.
Simple pendulum period	$T \approx 2\pi\sqrt{\dfrac{L}{g}}$	Independent of mass and (small) amplitude.
Physical pendulum period	$T = 2\pi\sqrt{\dfrac{I}{mgd}}$	Any rigid body swinging about a pivot d from its center of mass P.12.
Large-amplitude correction	$T \approx 2\pi\sqrt{\dfrac{L}{g}}\left(1+\dfrac{\theta_0^2}{16}\right)$	Big swings take slightly longer (θ_0 in radians).
Measuring g	$g = \dfrac{4\pi^2 L}{T^2}$	Time many swings, divide, and invert the period formula.

STATED-EQUATION *The restoring force along the arc is minus m g sine theta, about minus m g theta for small angles. The period of a simple pendulum is approximately two pi root L over g. A physical pendulum has period two pi root I over m g d. For a larger amplitude theta naught, multiply by one plus theta naught squared over sixteen. Rearranged, g equals four pi squared L over T squared.*

MEMORY AID "Period depends on length, not on mass": a 1 m pendulum ticks once per second.

TERMS

L – length from pivot to the bob's center (m)
θ, θ_0 – angle from vertical, and its maximum (rad)
I – moment of inertia about the pivot (kg·m^2)
d – pivot to center-of-mass distance (m)
m – mass (kg); g 9.81 m/s^2
T – period (s), one full back-and-forth



CONTEXT

- **Origin.** Galileo noticed pendulum isochronism (1602); Huygens built the pendulum clock (1656); Foucault's pendulum showed Earth's rotation (1851).
- **Use.** Clocks, seismometers, metronomes, gravimetry surveys.
- **Data science.** Estimating g from several (L, T) pairs is a regression of T^2 on L with slope $4\pi^2/g$ S.40 §13 Statistics.
- **Physical application.** A grandfather clock's 1 m pendulum swings with a period of about 2 s, one tick per second.

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P.17 Waves: Speed, Frequency & Wavelength

A wave carries energy without carrying matter along. Its speed is set by the medium; its frequency by the source; and the wavelength is whatever fits: $v = f\lambda$.

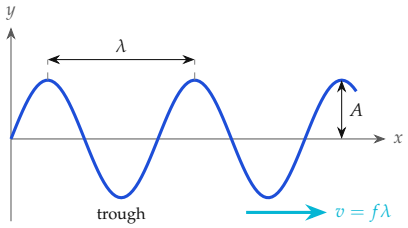
Wave speed	$v = f\lambda = \dfrac{\lambda}{T}$	One wavelength passes per period.
Traveling sinusoidal wave	$y(x,t) = A\sin(kx - \omega t)$	Moves in $+x$; use $kx + \omega t$ for $-x$.
Wave number & angular frequency	$k = \dfrac{2\pi}{\lambda}, \quad \omega = 2\pi f$	So $v = \frac{\omega}{k}$. This k is not the spring constant.
Wave speed on a string	$v = \sqrt{\dfrac{F_T}{\mu}}$	Tighter or lighter string: faster waves, higher pitch.
Standing waves (both ends fixed)	$\lambda_n = \dfrac{2L}{n}, \quad f_n = \dfrac{nv}{2L}$	Only whole numbers of half-wavelengths fit; harmonics $n = 1, 2, 3, \ldots$
Superposition & interference	$y = y_1 + y_2; \;\; \Delta r = m\lambda$ (constructive)	Waves add; half-integer path difference $(m + \frac{1}{2})\lambda$ cancels.
Electromagnetic waves	$c = f\lambda, \quad c = 3.00 \times 10^8$ m/s	Light, radio and X-rays differ only in f and λ.

STATED-EQUATION *Wave speed v equals frequency f times wavelength lambda. A traveling wave is y equals A sine of k x minus omega t, with k equal to two pi over lambda and omega equal to two pi f. On a string, v equals the square root of tension over linear density. Standing waves on a string of length L have wavelengths two L over n and frequencies n v over two L. Waves superpose; path differences of whole wavelengths interfere constructively. Light obeys c equals f lambda.*

MEMORY AID "The medium sets the speed, the source sets the frequency."

TERMS

v – wave speed (m/s); c speed of light
f – frequency (Hz); $T = 1/f$ period (s)
λ – wavelength (m); A amplitude
k – wave number (rad/m); ω angular frequency (rad/s)
F_T, μ – string tension (N) and mass per length (kg/m)
L, n – string length (m) and harmonic number
Δr – path difference (m); m integer
A; T; ω – amplitude; period; angular frequency
m (interference) – integer order $0, 1, 2, \ldots$
c – speed of light in vacuum



CONTEXT

- **Origin.** d'Alembert's wave equation (1747) V.13 §11 Vector Calculus; Young's double-slit interference (1801).
- **Use.** Music, radio, seismology, ultrasound imaging.
- **Data science.** Any signal is a superposition of sinusoids: the Fourier transform EE.26 §16 Electrical Eng. and sampling at twice the top frequency EE.27 §16 Electrical Eng..
- **Physical application.** 2.4 GHz Wi-Fi has $\lambda = \frac{3\times10^8}{2.4\times10^9} = 12.5$ cm, which is why antennas are a few centimeters long.

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P.18 Sound: Intensity, Decibels & the Doppler Effect

Sound is a pressure wave. Its intensity spreads over a growing sphere and is measured on a logarithmic decibel scale; motion of the source or listener shifts the frequency heard (Doppler effect).

Speed of sound in air	$v \approx 331 + 0.6\,T_C \ \text{m/s}$	343 m/s at 20°C; about 1480 m/s in water.
Intensity of a point source	$I = \dfrac{P}{4\pi r^2}$	Power spread over a sphere's area G.22 §05 Geometry: inverse-square law.
Sound intensity level	$\beta = 10 \log_{10} \dfrac{I}{I_0} \ \text{dB}$	Same decibel as in electronics EE.29 §16 Electrical Eng.; $I_0 = 10^{-12}$ W/m^2 is the hearing threshold.
Doppler effect	$f' = f\,\dfrac{v \pm v_o}{v \mp v_s}$ (upper signs: moving toward each other)	Approaching raises the pitch, receding lowers it.
Beat frequency	$f_{\text{beat}} = \lvert f_1 - f_2 \rvert$	Two close tones pulse in loudness; used to tune instruments.
Doppler for light (slow source)	$\dfrac{\Delta f}{f} \approx \dfrac{v_r}{c}$	Radar guns and redshift; valid for $v_r \ll c$.

STATED-EQUATION *The speed of sound is about three thirty-one plus zero point six times the Celsius temperature, in meters per second. Intensity equals power over four pi r squared. The level in decibels is ten log base ten of I over I naught. The Doppler-shifted frequency equals f times v plus or minus v observer, over v minus or plus v source. The beat frequency is the difference of the two frequencies. For light from a slowly moving source, the fractional frequency shift equals the radial velocity over c.*

MEMORY AID Doppler signs: pick the sign that makes f' *bigger* when the two approach. Decibels: $+10$ dB $= \times 10$ intensity, $+3$ dB $\approx \times 2$.

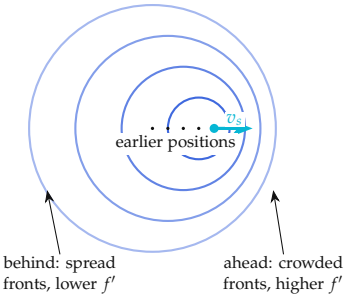
METHOD

1. Decibel arithmetic: convert with $I = I_0\,10^{\beta/10}$; add intensities, not decibels, for several sources A2.14 §04 Algebra II.
2. Doubling distance cuts I by 4, i.e. -6 dB. Example: 60 dB is $I = 10^{-12} \cdot 10^6 = 10^{-6}$ W/m^2; two such sources give 63 dB, not 120.

DECISION POINT Doppler signs: the numerator holds the observer's speed (plus if moving toward the source); the denominator holds the source's speed (minus if moving toward the observer). Check: approaching must raise f'. Example: a 700 Hz siren approaching at 30 m/s: $f' = 700 \cdot \frac{343}{313} = 767$ Hz; receding: 644 Hz.

TERMS

v – speed of sound (m/s); T_C air temperature (°C)
I – intensity (W/m^2); P source power (W); r distance (m)
β – sound level (dB); $I_0 = 10^{-12}$ W/m^2
f, f' – emitted and observed frequency (Hz)
v_o, v_s – observer and source speeds (m/s)
v_r – radial (line-of-sight) speed; c speed of light
T_C – air temperature in degrees Celsius
P, r – source power; distance from the source
I_0 – reference intensity 10^{-12} W/m^2 (threshold of hearing)
f_1, f_2; c – two nearby frequencies; speed of light



CONTEXT

- **Origin.** Christian Doppler (1842); Buys Ballot tested it with trumpeters on a train (1845); the bel honors Alexander Graham Bell.
- **Use.** Acoustics, noise regulation, medical ultrasound, weather radar.
- **Data science.** Audio models use log-scaled (decibel) spectrograms as input features; Doppler radar data feed precipitation nowcasting models.
- **Physical application.** Normal conversation is about 60 dB, a rock concert 110 dB: 10^5 times the intensity.

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Fluids

P.19 Pressure, Hydrostatics & Buoyancy

Pressure is force per area. In a fluid at rest it grows linearly with depth, and the pressure difference between the top and bottom of a submerged object produces an upward buoyant force.

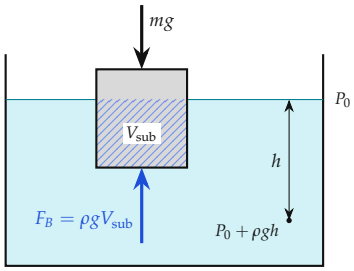
Pressure	$P = \dfrac{F}{A}$	Same force on a smaller area gives higher pressure (knives, stiletto heels).
Hydrostatic pressure	$P = P_0 + \rho g h$	Pressure rises by ρg per meter of depth; shape of the container is irrelevant.
Pascal's principle (hydraulics)	$\dfrac{F_1}{A_1} = \dfrac{F_2}{A_2}$	Pressure is transmitted unchanged; a big piston multiplies force.
Archimedes' principle	$F_B = \rho_{\text{fluid}}\, g\, V_{\text{displaced}}$	Upward force equals the weight of the fluid pushed aside.
Floating fraction	$\dfrac{V_{\text{sub}}}{V} = \dfrac{\rho_{\text{object}}}{\rho_{\text{fluid}}}$	Floats if less dense than the fluid; sinks if denser.

STATED-EQUATION *Pressure equals force over area. Pressure at depth h equals P naught plus rho g h. Pascal: F one over A one equals F two over A two. The buoyant force equals the fluid density times g times the displaced volume. For a floating object, the submerged fraction equals the object's density over the fluid's density.*

MEMORY AID Archimedes: "the fluid pushes up with the weight of the fluid you push aside."

TERMS

P – pressure (Pa $= \text{N/m}^2$); P_0 surface pressure (1 atm $= 101{,}325$ Pa)
F, A – force (N) and area (m^2)
ρ – density (kg/m^3); water 1000, seawater ≈ 1025
h – depth below the surface (m)
F_B – buoyant force (N)
$V_{\text{displaced}}$ – volume of fluid pushed aside (m^3) G.21 §05 Geometry, G.22 §05 Geometry
Gauge pressure – $P - P_0 = \rho g h$ (what tire gauges read)
P_0; g – pressure at the surface; gravitational acceleration
F_1, A_1; F_2, A_2 – force and piston area on each side
V_{sub}, V – submerged volume; total volume



CONTEXT

- **Origin.** Archimedes (∼250 BCE); Pascal (1653); Stevin (1586).
- **Use.** Dams, submarines, ship design, blood pressure, hydraulic lifts.
- **Data science.** Ocean-sensor data (pressure → depth) and weather-station barometric pressure are standard engineered features in climate models.
- **Physical application.** Only about 90% of an iceberg is underwater because ice is $917\,\text{kg/m}^3$ and seawater $1025\,\text{kg/m}^3$. At 10 m depth, water pressure doubles to about 2 atm.

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P.20 Continuity & Bernoulli's Equation

For a steady, incompressible flow, mass conservation fixes the speed in each part of a pipe, and energy conservation per unit volume (Bernoulli) then fixes the pressure: where the fluid speeds up, its pressure drops.

Volumetric flow rate	$Q = Av$	Volume per second through a cross-section.
Continuity equation	$A_1 v_1 = A_2 v_2$	Incompressible: what flows in must flow out, so narrow means fast.
Bernoulli's equation	$P_1 + \frac{1}{2}\rho v_1^2 + \rho g h_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho g h_2$	Pressure + kinetic + potential energy per volume is constant along a streamline.
Torricelli's law	$v_{\text{out}} = \sqrt{2gh}$	Outflow from a hole a depth h below the surface: free-fall speed.

STATED-EQUATION *Volumetric flow rate Q equals area times velocity, so A one v one equals A two v two. P one plus one half rho v one squared plus rho g h one equals P two plus one half rho v two squared plus rho g h two. Torricelli's law: the outflow speed equals the square root of two g h.*

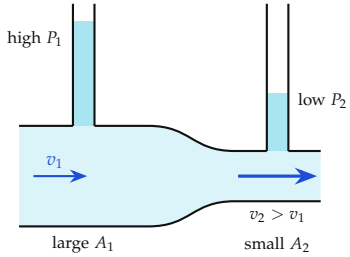
MEMORY AID "Fast fluid, low pressure" — squeeze the pipe, speed goes up, pressure goes down.

METHOD

1. Use continuity to get the unknown speed from the area ratio.
2. Apply Bernoulli between two points on the same streamline; cancel terms that are equal (same height, open to air $P = P_0$).
3. Solve for the pressure or speed. Example: a tank with a hole 5 m below the surface: $v = \sqrt{2 \cdot 9.81 \cdot 5} = 9.9\,\text{m/s}$.

TERMS

Q – volumetric flow rate (m^3/s)
A – cross-sectional area (m^2), πr^2 G.17 §05 Geometry
v – flow speed (m/s); P pressure (Pa); h height (m)
Continuity – conservation of mass for an incompressible fluid
Bernoulli – energy per volume P.6 along a streamline
Assumptions – steady, incompressible, inviscid (no friction P.21)
P; ρ; h – pressure; fluid density; height
v_{out} – speed of fluid leaving a hole a depth h below the surface



CONTEXT

- **Origin.** Daniel Bernoulli, *Hydrodynamica* (1738); Euler wrote the general equations (1757).
- **Use.** Venturi flow meters, carburetors, pitot tubes on aircraft, plumbing.
- **Data science.** Flow sensors (Venturi and pitot readings) are converted to velocity with Bernoulli before feeding pipeline anomaly-detection models.
- **Physical application.** Putting your thumb over a garden hose shrinks A, so the water speeds up and sprays farther.

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P.21 Viscosity & the Reynolds Number

Viscosity is a fluid's internal friction. The Reynolds number compares inertia to viscosity and predicts whether a flow is smooth (laminar) or chaotic (turbulent).

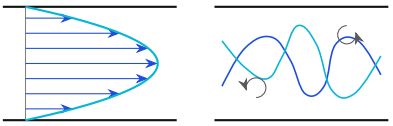
Newton's law of viscosity	$\tau = \mu \dfrac{du}{dy}$	Shear stress is proportional to how fast the velocity changes across the flow.
Reynolds number	$\mathrm{Re} = \dfrac{\rho v L}{\mu} = \dfrac{vL}{\nu} = \dfrac{\text{inertial forces}}{\text{viscous forces}}$	The key dimensionless group of fluid flow P.1.
Pipe flow regimes	$\mathrm{Re} < 2300$ laminar, $\mathrm{Re} > 4000$ turbulent	In between: transitional, sensitive to disturbances.
Poiseuille's law	$Q = \dfrac{\pi r^4 \Delta P}{8 \mu L}$	Laminar flow through a pipe; flow rises with the fourth power of radius.

STATED-EQUATION *Shear stress tau equals mu times d u d y. The Reynolds number, Re, equals rho v L over mu, which equals v L over nu: inertial forces over viscous forces. In a pipe, Reynolds below about twenty-three hundred is laminar and above about four thousand is turbulent. Poiseuille's law: flow rate Q equals pi r to the fourth delta P over eight mu L.*

MEMORY AID "Re is big, flow goes wild" — high Reynolds means turbulence; low means smooth, syrupy flow.

TERMS

μ – dynamic viscosity (Pa·s); water $\approx 10^{-3}$, honey ≈ 10
$\nu = \mu/\rho$ – kinematic viscosity (m^2/s)
τ – shear stress (Pa); du/dy velocity gradient ($1/s$) C1.9 §07 Calculus I
L – characteristic length (pipe diameter, wing chord) (m)
v – characteristic speed (m/s); ρ density
Q, r, ΔP – flow rate (m^3/s), pipe radius (m), pressure drop (Pa) over length L
r^4 – doubling pipe radius gives 16× the flow A1.3 §03 Algebra I



laminar, Re < 2300
parabolic profile

turbulent, Re > 4000
eddies, mixing

CONTEXT

- **Origin.** Stokes (1851); Osborne Reynolds' dye experiments (1883); Poiseuille (1840s) studying blood flow.
- **Use.** Pipeline design, aircraft, blood flow, microfluidics.
- **Data science.** Re is the classic example of a dimensionless feature: flows with the same Re look alike at any scale, so models trained on it generalize.
- **Physical application.** Stirring honey stays smooth ($\mathrm{Re} \ll 1$) while a river is turbulent (Re in the millions); water at 1 m/s in a 2 cm pipe has $\mathrm{Re} \approx 20{,}000$.

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P.22 Drag & Terminal Velocity

Moving through a fluid costs a drag force that grows with speed, roughly as v^2 for everyday objects. A falling body speeds up until drag balances its weight: terminal velocity.

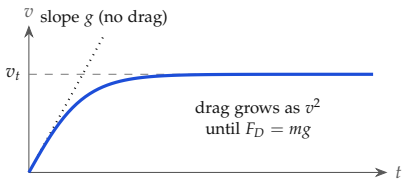
Quadratic drag	$F_D = \frac{1}{2}\rho\, C_D\, A\, v^2$	Dominant at high Re P.21: cars, balls, people.
Terminal velocity	$v_t = \sqrt{\dfrac{2mg}{\rho\, C_D A}}$	Set $F_D = mg$; heavier or more streamlined objects fall faster.
Stokes drag (Re $\ll 1$)	$F_D = 6\pi\mu r v$	Linear in v: dust, droplets, bacteria.
Power against drag	$P_{\text{drag}} = F_D v \propto v^3$	Going 10% faster costs about 33% more power.

STATED-EQUATION *Drag force equals one half rho C D A v squared. Terminal velocity equals the square root of two m g over rho C D A. For very small Reynolds number, Stokes drag equals six pi mu r v. The power to overcome drag, F D times v, is proportional to v cubed.*

MEMORY AID Drag goes as v^2, drag power as v^3.

TERMS

ρ – fluid density (air ≈ 1.2 kg/m^3)
C_D – drag coefficient (sphere ≈ 0.47, skydiver ≈ 1)
A – frontal area (m^2)
v_t – speed where drag equals weight, so $a = 0$ P.4
μ, r – viscosity (Pa·s) and sphere radius (m)
Stokes regime – viscous drag for tiny particles P.21
v^3 power – why fuel use rises quickly with highway speed
F_D; m; g – drag force; mass of the falling body; gravitational acceleration



CONTEXT

- **Origin.** Newton's drag law (1687); Rayleigh; Stokes (1851).
- **Use.** Vehicle efficiency, parachutes, sedimentation, sports ballistics.
- **Data science.** Fuel-economy and range models for EVs use a v^2 drag term as a physics-based feature alongside learned corrections.
- **Physical application.** Raindrops fall at only a few m/s instead of hundreds; an 80 kg skydiver ($C_D A \approx 0.7$ m^2) tops out near 43 m/s.

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Heat & Thermodynamics

P.23 Heat, Temperature & Phase Change

Temperature measures the average energy of molecular motion; heat is energy transferred because of a temperature difference. Added heat either raises the temperature or, at a phase change, breaks bonds at constant temperature.

Sensible heat	$Q = mc\,\Delta T$	Heat to change temperature without changing phase.
Latent heat	$Q = mL$	Heat to melt or boil at constant temperature.
Calorimetry balance	$\sum Q_i = 0$	In an insulated mix, heat lost by hot parts equals heat gained by cold parts.
Kelvin	$T_K = T_C + 273.15$	Absolute scale: use it in every formula with T itself (not ΔT).
Fahrenheit	$T_F = \frac{9}{5}T_C + 32$	Linear conversion A1.7 §03 Algebra I.

STATED-EQUATION *Heat Q equals m c delta T. Heat for a phase change equals m times L. In an insulated system, the heats sum to zero. Kelvin equals Celsius plus two seventy-three point one five. Fahrenheit equals nine fifths Celsius plus thirty-two.*

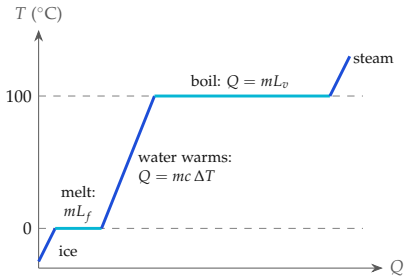
MEMORY AID Water: $c \approx 4186$ J/(kg·K); melting ice takes as much heat as warming the water by about 80°C.

METHOD

1. Write $Q = mc(T_f - T_i)$ for each body (negative for bodies that cool), plus $\pm mL$ for any phase change.
2. Set the sum to zero and solve for the final temperature T_f.
3. Check T_f lies between the starting temperatures. Example: 1 kg of water at 80°C mixed with 1 kg at 20°C ends at 50°C.

TERMS

Q – heat added (J)
m – mass (kg)
c – specific heat capacity (J/(kg·K)); water 4186
ΔT – temperature change (same size in K and °C)
L – latent heat (J/kg); ice melting 3.34×10^5, water boiling 2.26×10^6
Kelvin – absolute scale; 0 K is absolute zero



CONTEXT

- **Origin.** Joseph Black (1760s) distinguished heat from temperature and found latent heat; Kelvin scale (1848).
- **Use.** Cooking, HVAC, climate, industrial processes.
- **Data science.** Energy-demand forecasting uses heating and cooling degree-days, which are built on $Q = mc\Delta T$ reasoning.
- **Physical application.** Coastal cities have milder climates because water's large c makes oceans slow to heat and cool.

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P.24 Conduction: Fourier's Law & Thermal Resistance

Conduction moves heat through a solid from hot to cold at a rate proportional to the temperature gradient. Layers act like resistors in series, so heat flow is a temperature difference divided by a total resistance.

Fourier's law	$q = -k\dfrac{dT}{dx}$	Heat flux runs down the temperature gradient.
Heat flow through a slab	$\dot{Q} = \dfrac{kA\,\Delta T}{L}$	Steady state, uniform slab: thicker or less conductive walls pass less heat.
Conduction resistance	$R_{\text{cond}} = \dfrac{L}{kA}$	Thermal analog of electrical resistance.
Thermal circuit	$\dot{Q} = \dfrac{\Delta T}{R_{\text{total}}}, \quad R_{\text{series}} = R_1 + R_2 + \cdots$	Layers in series add; parallel paths combine like parallel resistors P.33.

STATED-EQUATION *Heat flux q equals minus k times d T d x. Heat flow Q dot equals k A delta T over L. Conduction resistance equals L over k A. Heat flow equals delta T over the total resistance, and resistances in series add.*

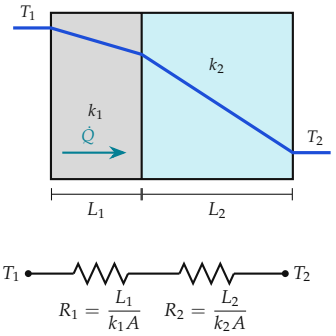
MEMORY AID Heat flows like current: ΔT is the "voltage," R the "resistor" P.33.

METHOD

1. Compute each layer's $R = \frac{L}{kA}$ (and convection films $\frac{1}{hA}$ P.25).
2. Add them, then $\dot{Q} = \frac{\Delta T}{R_{\text{total}}}$. Example: $10\,\text{m}^2$ of $0.1\,\text{m}$ fiberglass ($k = 0.04$) with a $20\,\text{K}$ difference: $R = 0.25\,\text{K/W}$, $\dot{Q} = 80\,\text{W}$.

TERMS

q – heat flux (W/m^2); minus sign: heat flows downhill in T
k – thermal conductivity (W/(m·K)); copper 400, glass 0.8, air 0.026
$\dot{Q}$ – heat flow rate (W)
A, L – area (m^2) and thickness (m)
R – thermal resistance (K/W)
dT/dx – temperature gradient (K/m) C1.9 §07 Calculus I



CONTEXT

- **Origin.** Joseph Fourier, *Théorie analytique de la chaleur* (1822), which also introduced Fourier series.
- **Use.** Insulation, building energy, CPU heat sinks, cookware.
- **Data science.** Building-energy models combine thermal-resistance networks with regression on smart-meter data to estimate insulation quality.
- **Physical application.** A metal spoon in hot soup gets hot fast; a wooden one does not, because k differs by a factor of about 100.

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P.25 Convection & Newton's Law of Cooling

Convection carries heat away by a moving fluid at a rate proportional to the surface–fluid temperature gap. A uniformly hot object therefore cools exponentially toward its surroundings with a time constant τ.

Convective heat flow	$\dot{Q} = hA\,(T_s - T_\infty)$	Rate proportional to the temperature gap; h captures the flow.
Newton's law of cooling	$T(t) = T_\infty + (T_0 - T_\infty)\,e^{-t/\tau}$	Exponential approach to the surroundings A2.13 §04 Algebra II.
Time constant	$\tau = \dfrac{\rho V c}{hA} = \dfrac{mc}{hA}$	Heat capacity divided by surface conductance: big, compact objects cool slowly.

STATED-EQUATION *Convective heat flow equals h A times the surface temperature minus the fluid temperature. T of t equals T infinity plus the quantity T naught minus T infinity, times e to the minus t over tau. The time constant tau equals rho V c over h A, which is m c over h A.*

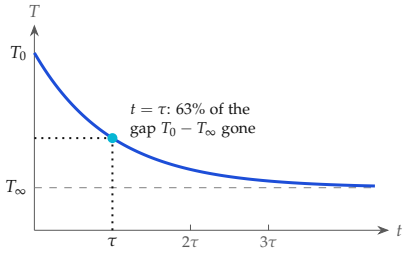
MEMORY AID "Hot things cool fastest when they are hottest": the rate is proportional to the temperature gap.

METHOD

1. Check Bi < 0.1 P.26 so the object has one temperature.
2. Compute $\tau = \frac{mc}{hA}$ (or fit it from two readings).
3. Evaluate $T(t)$, or solve for $t = \tau \ln \frac{T_0 - T_\infty}{T - T_\infty}$ A2.14 §04 Algebra II. Example: coffee at 90°C in a 20°C room with $\tau = 10$ min is $20 + 70e^{-1} = 46$°C after 10 min.

TERMS

h – convection coefficient (W/(m^2·K)): still air 2–25, moving air 25–250, water 50–20,000
T_s, T_∞ – surface and surrounding-fluid temperatures
T_0 – initial temperature; m, c, V, ρ as in P.23
Natural convection – fluid moves because warm fluid is less dense and rises P.27
Forced convection – a fan or pump drives the flow
τ – time constant (s); the solution of $\frac{dT}{dt} = -\frac{1}{\tau}(T - T_\infty)$ C2.28 §08 Calculus II
A; ρ, V, c; m – surface area; density, volume, specific heat; mass



CONTEXT

- **Origin.** Newton's law of cooling (1701); boundary-layer theory by Prandtl (1904).
- **Use.** Cooling electronics, HVAC, weather, forensic time-of-death estimates.
- **Data science.** Exponential-decay fits (with T_∞ and τ as parameters) are fit by nonlinear least squares O.13 §14 Optimization to sensor data for thermal digital twins.
- **Physical application.** Blowing on hot soup increases h (forced convection), so it cools faster.

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P.26 Biot Number & Lumped Capacitance

The Biot number compares how hard it is for heat to move inside a body with how easily it leaves the surface. When it is small, the body stays at one uniform temperature and the simple exponential cooling model applies.

Biot number	$\mathrm{Bi} = \dfrac{hL_c}{k_{\text{solid}}} = \dfrac{\text{internal conduction resistance}}{\text{surface convection resistance}}$	Small Bi: inside keeps up with the surface.
Characteristic length	$L_c = \dfrac{V}{A_s}$	Volume per unit surface area.
Lumped-model criterion	$\mathrm{Bi} < 0.1 \Rightarrow$ lumped model valid	Internal temperature differences are under about 5%.
Fourier number	$\mathrm{Fo} = \dfrac{\alpha t}{L_c^2}$	Dimensionless time: how far heat has diffused in.

STATED-EQUATION *The Biot number, B i, equals h L c over k of the solid: internal conduction resistance over surface convection resistance. The characteristic length L c equals volume over surface area. If Biot is less than zero point one, the lumped model is valid. The Fourier number, F o, equals alpha t over L c squared.*

MEMORY AID "Biot small, same temperature all" — below 0.1, the inside keeps up with the surface.

METHOD

1. Compute $L_c = V/A_s$ (sphere $r/3$).
2. Compute Bi with the *solid's* k.
3. $\mathrm{Bi} < 0.1$: use P.25; otherwise solve the heat equation P.31. Example: steel ball, $r = 1\,\text{cm}$, $k = 50$, $h = 100$: $\mathrm{Bi} = \frac{100 \cdot 0.0033}{50} = 0.007$, so lumped is fine.

TERMS

h – convection coefficient P.25
k_{solid} – conductivity of the object (not the fluid) P.24
L_c – characteristic length (m); sphere $r/3$, long cylinder $r/2$, plate $\frac{\text{thickness}}{2}$
V, A_s – volume (m^3) and surface area (m^2)
Lumped capacitance – treat the object as one temperature, giving the exponential in P.25
$\alpha = \frac{k}{\rho c}$ – thermal diffusivity (m^2/s); t time (s)
Fo – dimensionless time: how far heat has diffused P.31

CONTEXT

- **Origin.** Jean-Baptiste Biot (1804), a contemporary of Fourier.
- **Use.** Heat treating metals, food processing, battery thermal management.
- **Data science.** Choosing between a cheap lumped (one-parameter) model and an expensive spatial simulation is a model-complexity decision ML.4 §15 Machine Learning; Bi tells you which is justified.
- **Physical application.** A thick steak (large Bi) sears outside while the center stays rare; a thin slice cooks through evenly.

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P.27 Natural Convection: Nu, Pr, Gr & Ra

In natural convection, buoyancy of warm fluid drives the flow. Four dimensionless numbers organize the problem: Ra decides whether convection starts, and Nu reports how much it boosts heat transfer over pure conduction.

Nusselt number	$\mathrm{Nu} = \dfrac{hL}{k_{\text{fluid}}}$	Convective over conductive heat transfer; the usual unknown.
Prandtl number	$\mathrm{Pr} = \dfrac{\nu}{\alpha}$	Momentum diffusivity over heat diffusivity; a fluid property.
Grashof number	$\mathrm{Gr} = \dfrac{g\beta\,\Delta T\,L^3}{\nu^2}$	Buoyancy over viscous forces.
Rayleigh number	$\mathrm{Ra} = \mathrm{Gr}\cdot\mathrm{Pr} = \dfrac{g\beta\,\Delta T\,L^3}{\nu\,\alpha}$	Buoyancy over the combined damping of viscosity and heat diffusion.
Onset of convection	$\mathrm{Ra}_c \approx 1708$	A layer heated from below starts to roll into cells above this.
Empirical correlation	$\mathrm{Nu} = C\,\mathrm{Ra}^n$	Fit from experiments; $n \approx \frac{1}{4}$ laminar, $\frac{1}{3}$ turbulent.

STATED-EQUATION *The Nusselt number equals h L over the fluid's k. The Prandtl number equals nu over alpha. The Grashof number equals g beta delta T L cubed over nu squared. The Rayleigh number equals Grashof times Prandtl, which is g beta delta T L cubed over nu alpha. The critical Rayleigh number for a fluid layer heated from below is about seventeen hundred eight. Correlations take the form Nu equals C Ra to the n.*

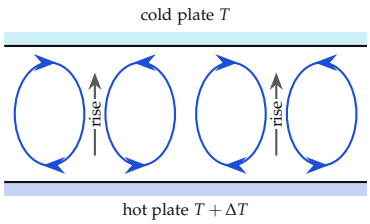
MEMORY AID Ra is the "Reynolds number of buoyancy": above ≈ 1708, a heated layer starts to roll into convection cells.

METHOD

1. Look up fluid properties ν, α, k, β at the mean temperature.
2. Compute Ra.
3. Pick the correlation for that geometry and Ra range; get Nu, then $h = \frac{\mathrm{Nu}\,k}{L}$.

TERMS

Nu – convective vs. conductive heat transfer (Nu = 1: pure conduction)
Pr – momentum vs. heat diffusivity (air ≈ 0.71, water ≈ 7)
Gr – buoyancy vs. viscous forces (natural-convection analog of Re^2 P.21)
β – thermal expansion coefficient (1/K); ideal gas $\beta = 1/T$ in kelvin P.29
ν, α – kinematic viscosity and thermal diffusivity (m^2/s)
L, ΔT – layer height or plate length (m); temperature difference (K)
Correlations – $\mathrm{Nu} = C\,\mathrm{Ra}^n$ fit on log–log axes A2.14 §04 Algebra II, S.40 §13 Statistics
h; k_{fluid} – convection coefficient; fluid thermal conductivity
Ra; Ra_c – Rayleigh number; its critical value for onset of convection



CONTEXT

- **Origin.** Nusselt (1915); Rayleigh (1916) analyzing Bénard's hexagonal convection cells (1900); Grashof, Prandtl.
- **Use.** Room heating, cooling of electronics, double-pane windows, mantle convection, weather.
- **Data science.** Empirical correlations $\mathrm{Nu} = C\,\mathrm{Ra}^n$ are power-law regressions; fitting them in log space is a classic linear-regression exercise.
- **Physical application.** Warm air rising from a radiator, thunderstorm updrafts, and the churning pattern in a pot of heated miso soup are all Rayleigh–Bénard convection.

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P.28 Thermal Radiation

Every object above absolute zero emits electromagnetic radiation. The power grows as the fourth power of absolute temperature, and the peak wavelength shifts shorter as the object gets hotter; no medium is needed.

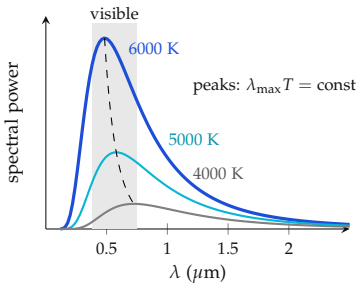
Stefan–Boltzmann law	$P = \varepsilon \sigma A T^4$	Total power radiated by a surface.
Net radiative exchange	$P_{\text{net}} = \varepsilon \sigma A \left(T^4 - T_{\text{surr}}^4 \right)$	Emission minus absorption from surroundings at T_{surr}.
Wien's displacement law	$\lambda_{\max} T = 2.898 \times 10^{-3}\ \text{m} \cdot \text{K}$	Hotter means bluer: peak wavelength is inversely proportional to T.
Stefan–Boltzmann constant	$\sigma = 5.670 \times 10^{-8}\ \dfrac{\text{W}}{\text{m}^2\text{K}^4}$	Universal constant.

STATED-EQUATION *Radiated power equals epsilon sigma A T to the fourth. Net power equals epsilon sigma A times T to the fourth minus T surroundings to the fourth. Wien's law: lambda max times T equals two point eight nine eight times ten to the minus three meter-kelvins. Sigma equals five point six seven times ten to the minus eight watts per square meter per kelvin to the fourth.*

MEMORY AID "Fourth power": double the absolute temperature, sixteen times the radiation. Always use kelvin.

TERMS

ε – emissivity, 0 to 1 (skin ≈ 0.97, polished metal ≈ 0.05)
σ – Stefan–Boltzmann constant
A – radiating surface area (m^2)
T – absolute temperature in kelvin P.23
$\lambda_{\max}$ – peak wavelength (m) (Wien's law; inverse variation A1.18 §03 Algebra I)
No medium needed – radiation crosses vacuum, unlike conduction and convection
T_{surr}; P – temperature of the surroundings; radiated power



CONTEXT

- **Origin.** Stefan (1879, experiment) and Boltzmann (1884, theory); Wien (1893); Planck (1900) launched quantum theory from this problem P.37.
- **Use.** Thermal cameras, solar panels, climate energy balance, furnaces.
- **Data science.** Satellite land-surface-temperature products are inverted from infrared radiance with these laws before being used as features in crop and climate models.
- **Physical application.** You feel the Sun's warmth across 150 million km of vacuum; the Sun (≈ 5778 K) peaks near 500 nm, green light. A resting person radiates a net ≈ 100 W indoors.

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P.29 Ideal Gas Law & Thermodynamics

An ideal gas ties pressure, volume and temperature together in one equation. The first law is energy conservation for heat and work; the second law limits how much heat can become work.

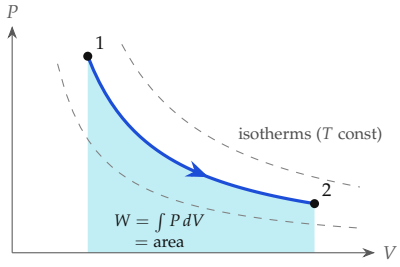
Ideal gas law	$PV = nRT$	Any two of P, V, T fix the third for a given amount of gas.
First law of thermodynamics	$\Delta U = Q - W$	Internal energy rises with heat in and falls with work done by the gas.
Work by a gas	$W = \int P\,dV$	Area under the path on a P–V diagram C1.30 §07 Calculus I.
Isothermal work	$W = nRT \ln \dfrac{V_2}{V_1}$	Ideal gas at constant T: $\Delta U = 0$, so $Q = W$.
Molecular kinetic energy	$\langle KE \rangle = \frac{3}{2} k_B T$	Temperature is average molecular kinetic energy.
Carnot efficiency	$\eta_{\text{Carnot}} = 1 - \dfrac{T_C}{T_H}$	The best any heat engine can do between two temperatures P.30.
Second law (entropy)	$\Delta S \geq 0$ (isolated system)	Entropy of an isolated system never decreases.

STATED-EQUATION *P V equals n R T. The change in internal energy equals heat added minus work done by the system. Work equals the integral of P d V; for an isothermal process it is n R T log of V two over V one. The average kinetic energy of a molecule is three halves k B T. Carnot efficiency equals one minus T cold over T hot. The entropy of an isolated system never decreases.*

MEMORY AID "You can't win, you can't break even, you can't get out of the game" (first, second, third laws).

TERMS

P, V – pressure (Pa) and volume (m^3)
n – moles; $R = 8.314$ J/(mol·K)
T – absolute temperature (K)
ΔU – change in internal energy (J); Q heat in (J); W work by the gas (J)
k_B – Boltzmann constant 1.381×10^{-23} J/K $= R/N_A$
η – maximum efficiency of any heat engine; T_H, T_C hot, cold reservoirs (K)
S – entropy (J/K) (second law); links to information entropy ML.3 §15 Machine Learning
Q, W – heat added to; work done by the gas
R – gas constant 8.314 J/(mol K)
V_1, V_2; T_C, T_H – initial and final volumes; cold and hot reservoir temperatures



CONTEXT

- **Origin.** Boyle (1662), Charles, Gay-Lussac; Clapeyron (1834); Carnot (1824); Clausius (1850s).
- **Use.** Engines, refrigerators, weather balloons, tire pressure.
- **Data science.** Boltzmann's $S = k \ln W$ inspired Shannon entropy and the "temperature" in softmax and simulated annealing ML.24 §15 Machine Learning, O.22 §14 Optimization.
- **Physical application.** Tire pressure drops on a cold morning: at fixed V, $P \propto T$. One mole of gas at 0°C and 1 atm fills 22.4 L.

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P.30 Entropy & the Second Law

Entropy measures how many microscopic arrangements are consistent with what we observe. Spontaneous processes increase total entropy, which is why heat flows from hot to cold and why no engine or refrigerator can beat the Carnot limits.

Clausius entropy change	$\Delta S = \int \frac{dQ_{\text{rev}}}{T} \quad \left(= \frac{Q}{T} \text{ at constant } T\right)$	Heat added reversibly, weighted by $1/T$.
Second law	$\Delta S_{\text{total}} \geq 0$	Equality only for reversible (idealized) processes.
Boltzmann entropy	$S = k_B \ln \Omega$	More microstates, more entropy: the statistical meaning.
Heat-engine efficiency	$\eta = \frac{W}{Q_H} = 1 - \frac{Q_C}{Q_H} \leq 1 - \frac{T_C}{T_H}$	Some heat must always be dumped to the cold side.
Refrigerator COP	$\text{COP}_R = \frac{Q_C}{W} \leq \frac{T_C}{T_H - T_C}$	Heat removed per unit work; can exceed 1.
Heat-pump COP	$\text{COP}_{HP} = \frac{Q_H}{W} \leq \frac{T_H}{T_H - T_C}$	Heat delivered per unit work; best when $T_H - T_C$ is small.

STATED-EQUATION *The entropy change equals the integral of d Q reversible over T, which is Q over T at constant temperature. The total entropy never decreases. Boltzmann: S equals k B log omega. Engine efficiency is work over heat in, one minus Q cold over Q hot, at most one minus T cold over T hot. A refrigerator's coefficient of performance is Q cold over W, at most T cold over T hot minus T cold; a heat pump's is Q hot over W, at most T hot over T hot minus T cold.*

MEMORY AID Heat flows downhill in T because that is the direction that raises total entropy: $\frac{Q}{T_C} > \frac{Q}{T_H}$.

TERMS

S – entropy (J/K)
Q_{rev} – heat added along a reversible path (J)
Ω – number of microstates; k_B Boltzmann constant P.29
Q_H, Q_C – heat exchanged with hot and cold reservoirs (J)
W – work (J); T_H, T_C reservoir temperatures (K)
COP – coefficient of performance (dimensionless)
Third law – $S \to$ minimum as $T \to 0$ K; absolute zero is unreachable
T; T_H, T_C – absolute temperature; hot and cold reservoir temperatures (kelvin)

CONTEXT

- **Origin.** Carnot (1824); Clausius coined "entropy" (1865); Boltzmann's $S = k \log W$ (1877) is engraved on his tombstone.
- **Use.** Power plants, refrigeration, heat pumps, chemistry (which reactions run).
- **Data science.** Shannon entropy $-\sum p \log p$ ML.3 §15 Machine Learning has the same form as Gibbs entropy; maximum-entropy models pick the least-committal distribution consistent with the data.
- **Physical application.** Melting 1 kg of ice at 273 K raises its entropy by $\frac{334{,}000}{273} = 1223$ J/K. A heat pump moving heat from 0°C outdoors to 20°C indoors could ideally deliver 14.7 J per joule of work; real ones achieve 3–4.

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P.31 Diffusion & the Heat Equation

Random molecular motion spreads concentration (or heat) from high to low. The flux follows the gradient, and the resulting partial differential equation makes a spike spread as a Gaussian whose width grows like $\sqrt{t}$.

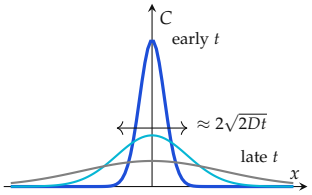
Fick's first law	$J = -D\dfrac{dC}{dx}$	Flux down the concentration gradient; twin of Fourier's law P.24.
Fick's second law (diffusion equation)	$\dfrac{\partial C}{\partial t} = D\dfrac{\partial^2 C}{\partial x^2}$	Concentration rises where the profile curves upward.
Heat equation	$\dfrac{\partial T}{\partial t} = \alpha\dfrac{\partial^2 T}{\partial x^2}$	The same equation for temperature V.13 §11 Vector Calculus.
Diffusion length	$L_{\text{diff}} \approx \sqrt{2Dt}$	Typical distance spread in time t.

STATED-EQUATION *Fick's law: the flux J equals minus D times d C d x. The rate of change of concentration, partial C partial t, equals D times the second partial of C with respect to x. The heat equation has the same form with alpha. The typical diffusion distance is about the square root of two D t.*

MEMORY AID Diffusion distance grows like $\sqrt{t}$: to go twice as far takes four times as long.

TERMS

J – flux (amount per area per time)
D – diffusion coefficient (m^2/s); small molecule in water $\sim 10^{-9}$, in air $\sim 10^{-5}$
C – concentration (e.g. mol/m^3)
∂ – partial derivative (other variables held fixed)
α – thermal diffusivity (m^2/s) P.26
$\sqrt{2Dt}$ – SD of a random walk S.2 §13 Statistics, S.24 §13 Statistics



CONTEXT

- **Origin.** Fick (1855), by analogy with Fourier's heat law P.24; Einstein linked diffusion to random walks (1905).
- **Use.** Drug delivery, pollution spread, semiconductor doping, cooking.
- **Data science.** Diffusion models for image generation add Gaussian noise step by step and learn to reverse it; the forward process is the heat equation.
- **Physical application.** Sugar sitting in unstirred tea spreads only about 2 mm in an hour ($D \approx 5 \times 10^{-10}\ m^2/s$): stirring (convection) is far faster than diffusion.

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Electricity

P.32 Coulomb's Law, Electric Field & Potential

Charges exert inverse-square forces on each other. The electric field is the force per unit charge a charge would feel at a point, and the electric potential (voltage) is the potential energy per unit charge.

Coulomb's law	$F = k\dfrac{\lvert q_1 q_2\rvert}{r^2}, \quad k = \dfrac{1}{4\pi\varepsilon_0}$	Like charges repel, opposites attract; same form as gravity P.8.
Electric field	$\mathbf{E} = \dfrac{\mathbf{F}}{q}$	Force per unit test charge; points the way a $+$ charge is pushed.
Field of a point charge	$E = \dfrac{k\,q}{r^2}$	Radially outward for $q > 0$, inward for $q < 0$.
Potential of a point charge	$V = \dfrac{k\,q}{r}$	Potential energy per unit charge, zero at infinity; a scalar, so just add.
Electric potential energy	$U = qV, \quad U_{12} = \dfrac{k\,q_1 q_2}{r}$	Energy to assemble the charges; $\Delta U = q\,\Delta V$.
Uniform field	$E = \dfrac{\Delta V}{d}$	Between parallel plates; units $\mathrm{V/m} = \mathrm{N/C}$.
Superposition	$\mathbf{E} = \sum_i \mathbf{E}_i$	Add the field vectors of all charges T.21 §06 Trigonometry.
Lorentz force	$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$	Magnetic part acts only on moving charges, at right angles to $\mathbf{v}$ EE.18 §16 Electrical Eng..

STATED-EQUATION *Coulomb's law: the force equals k times the product of the charges over r squared, with k equal to one over four pi epsilon naught. The electric field is force per unit charge; for a point charge it is k q over r squared. The potential is k q over r, and potential energy is q V. In a uniform field, E equals delta V over d. Fields from several charges add as vectors. The Lorentz force is q times E plus v cross B. The potential energy of two point charges is k q one q two over r.*

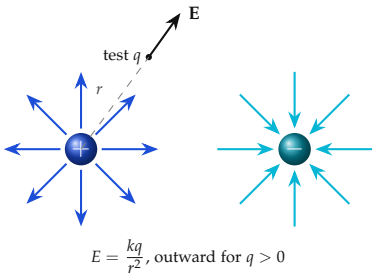
MEMORY AID Field is force per charge; potential is energy per charge. "Field $\sim 1/r^2$, potential $\sim 1/r$."

METHOD

1. For each charge, compute $E_i = \frac{k|q_i|}{r_i^2}$ at the point.
2. Give each a direction (away from $+$, toward $-$) and split into components.
3. Add components and recombine. Example: $q = +2\,\mu$C at 3 m: $E = \frac{8.99\times10^9\cdot2\times10^{-6}}{9} \approx 2000\,\mathrm{N/C}$, pointing away.

TERMS

q – charge (C); electron $-e$, $e = 1.602 \times 10^{-19}$ C EE.1 §16 Electrical Eng.
k – Coulomb constant 8.99×10^9 N·m^2/C^2
ε_0 – permittivity of free space 8.854×10^{-12} F/m
r – distance from the charge (m)
E – electric field (N/C = V/m)
V – electric potential (V = J/C); U potential energy (J)
d – plate separation (m); $\mathbf{B}$ magnetic field (T); $\mathbf{v}$ velocity
U, U_{12} – electric potential energy; that of a pair of charges
$\mathbf{F}$; $\mathbf{v}$, $\mathbf{B}$ – force on the charge; its velocity and the magnetic field



$E = \frac{kq}{r^2}$, outward for $q > 0$

CONTEXT

- **Origin.** Coulomb's torsion balance (1785); Faraday's field lines (1830s); Maxwell unified electricity, magnetism and light (1865).
- **Use.** Electronics, capacitors EE.10 §16 Electrical Eng., particle accelerators, printers, lightning protection.
- **Data science.** Force-directed graph layouts treat nodes as like charges that repel (Coulomb) while edges pull like springs P.5.
- **Physical application.** Rubbing a balloon on hair transfers electrons; the charged balloon then sticks to a wall by inducing opposite charge.

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P.33 Electric Circuits: Ohm's Law & Power

Voltage pushes current through resistance. Ohm's law relates the three, power is current times voltage, and resistors combine in series and parallel by simple rules.

Ohm's law	$V = IR$	Current is proportional to voltage for a resistor.
Electric power	$P = IV = I^2R = \dfrac{V^2}{R}$	Rate of energy conversion; pick the form with the knowns.
Resistivity	$R = \dfrac{\rho_e L}{A}$	Longer, thinner wire has more resistance.
Series resistors	$R_{\text{series}} = R_1 + R_2 + \cdots$	Same current through each; voltages add.
Parallel resistors	$\dfrac{1}{R_{\text{parallel}}} = \dfrac{1}{R_1} + \dfrac{1}{R_2} + \cdots$	Same voltage across each; currents add.
Electrical energy	$E = Pt$	Billed in kWh: 1 kWh $= 3.6 \times 10^6$ J.

STATED-EQUATION *Voltage equals current times resistance. Power equals I V, which equals I squared R, which equals V squared over R. Resistance equals resistivity times length over area. Series resistances add. One over the parallel resistance equals one over R one plus one over R two, and so on. Energy equals power times time.*

MEMORY AID "VIR" triangle: cover the one you want — $V = IR$, $I = V/R$, $R = V/I$.

TERMS

V – voltage (V), the electric "pressure" P.32
I – current (A), charge per second
R – resistance (Ω)
P – power (W); t time (s)
ρ_e – resistivity (Ω·m); copper 1.7×10^{-8}
L, A – wire length (m) and cross-section (m^2)
Parallel – total is less than the smallest resistor (harmonic-mean style ML.17 §15 Machine Learning)
Analogy – same math as heat flow P.24 and pipe flow P.21; full circuit analysis: Kirchhoff EE.4 §16 Electrical Eng.

CONTEXT

- **Origin.** Georg Ohm (1827); Kirchhoff's laws (1845).
- **Use.** Household wiring, electronics, batteries, power grids.
- **Data science.** Data-center power budgeting ($P = IV$ per rack) and GPU energy-per-training-run estimates.
- **Physical application.** A 1500 W kettle on 120 V draws 12.5 A, which is why kitchens use 15–20 A circuits.

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Optics

P.34 Reflection, Refraction & Total Internal Reflection

Light reflects at equal angles and bends (refracts) when it changes speed entering a new medium. Going from a slower to a faster medium beyond a critical angle, it cannot escape and reflects totally.

Law of reflection	$\theta_r = \theta_i$	Angles measured from the normal to the surface.
Index of refraction	$n = \dfrac{c}{v}$	How many times slower light travels in the medium; $n \geq 1$.
Snell's law	$n_1 \sin\theta_1 = n_2 \sin\theta_2$	Into a higher n: bends toward the normal.
Critical angle	$\sin\theta_c = \dfrac{n_2}{n_1} \quad (n_1 > n_2)$	Beyond θ_c all light reflects (total internal reflection).
Wavelength in a medium	$\lambda_n = \dfrac{\lambda_0}{n}$	Frequency stays fixed, so wavelength shrinks with speed.

STATED-EQUATION *The angle of reflection equals the angle of incidence. The index of refraction n equals c over v. Snell's law: n one sine theta one equals n two sine theta two. The critical angle satisfies sine theta c equals n two over n one, when n one exceeds n two. In a medium the wavelength is lambda naught over n.*

MEMORY AID "Slow down, bend toward the normal" (like a car whose wheels hit sand at an angle).

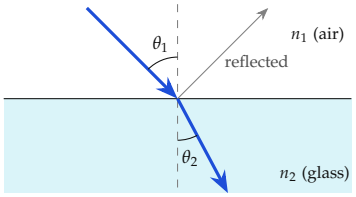
METHOD

1. Draw the normal; measure all angles from it.
2. Apply $n_1 \sin\theta_1 = n_2 \sin\theta_2$ and solve for θ_2 T.14 §06 Trigonometry.
3. If $\sin\theta_2 > 1$, there is no refracted ray: total internal reflection. Example: air to glass ($n = 1.5$) at 45°: $\sin\theta_2 = \frac{0.707}{1.5} = 0.471$, $\theta_2 = 28.1°$.

DECISION POINT Total internal reflection is possible only going from higher n to lower n (glass → air, water → air). For glass to air $\theta_c = \arcsin\frac{1}{1.5} = 41.8°$.

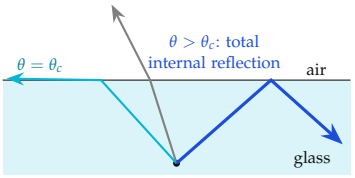
TERMS

θ_i, θ_r – angles of incidence and reflection (from the normal)
θ_1, θ_2 – angles in media 1 and 2
n – refractive index: air 1.000, water 1.33, glass ≈ 1.5, diamond 2.42
c – speed of light in vacuum 3.00×10^8 m/s; v speed in the medium
θ_c – critical angle
λ_0, λ_n – wavelength in vacuum and in the medium (m)
v – speed of light in the medium



CONTEXT

- **Origin.** Ibn Sahl (984) stated the sine law; Snell (1621) and Descartes (1637) rediscovered it; Fermat derived it from least time (1662).
- **Use.** Lenses, prisms, fiber optics, rainbows, mirages.
- **Data science.** Optical fibers carrying internet and data-center traffic guide light by total internal reflection.
- **Physical application.** A diamond ($\theta_c = 24.4°$) traps light by repeated total internal reflection, which makes it sparkle; a straw in water looks bent.



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P.35 Thin Lenses & Mirrors

A lens or curved mirror bends rays so that rays from an object point meet again at an image point. One equation locates the image, and the magnification tells its size and orientation.

Thin lens / mirror equation	$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$	Same equation for thin lenses and spherical mirrors (with sign rules).
Magnification	$m = \frac{h_i}{h_o} = -\frac{d_i}{d_o}$	$\lvert m\rvert > 1$ enlarged; $m < 0$ inverted.
Spherical mirror focal length	$f = \frac{R}{2}$	Half the radius of curvature.
Lensmaker's equation	$\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$	Focal length from glass index and surface curvatures.
Optical power	$P = \frac{1}{f}, \quad P_{\text{total}} = P_1 + P_2$	In diopters (1/m); thin lenses in contact add powers.

STATED-EQUATION *One over f equals one over d o plus one over d i. Magnification equals image height over object height, which equals minus d i over d o. A spherical mirror's focal length is half its radius. Lensmaker: one over f equals n minus one, times one over R one minus one over R two. Optical power is one over f, and powers of lenses in contact add.*

MEMORY AID Object inside $f \rightarrow$ magnifying glass: virtual, upright, bigger.

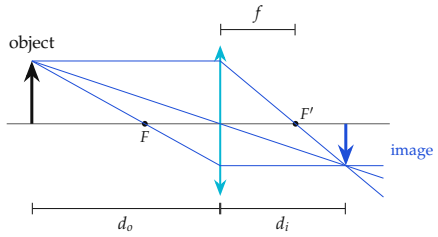
METHOD

1. Draw the three principal rays (parallel $\rightarrow$ through F'; through the center undeviated; through $F \rightarrow$ parallel) to sketch the image.
2. Apply the lens equation with signs; solve for d_i, then m.
3. Interpret the signs. Example: $f = 10\,\text{cm}$, $d_o = 30\,\text{cm}$: $\frac{1}{d_i} = \frac{1}{10} - \frac{1}{30}$, $d_i = 15\,\text{cm}$, $m = -0.5$: real, inverted, half size.

DECISION POINT Signs: $f > 0$ for converging lenses and concave mirrors, $f < 0$ for diverging lenses and convex mirrors. $d_i > 0$: real image (opposite side for a lens, same side for a mirror); $d_i < 0$: virtual. Diverging elements always give virtual, upright, smaller images.

TERMS

f – focal length (m)
d_o, d_i – object and image distances from the lens (m)
h_o, h_i – object and image heights; m magnification
R, R_1, R_2 – radii of curvature (m)
n – refractive index of the lens P.34
P – optical power (diopters, $\text{D} = 1/\text{m}$)
m – magnification (negative means inverted)



CONTEXT

- **Origin.** Kepler's *Dioptrice* (1611); lensmaker's equation from Huygens and Halley (17th c.).
- **Use.** Cameras, eyeglasses, microscopes, telescopes, projectors.
- **Data science.** The pinhole and thin-lens models underlie camera calibration in computer vision; focal length in pixels maps 3-D points to image coordinates.
- **Physical application.** Glasses of $-2.5\,\text{D}$ are diverging lenses with $f = -0.4\,\text{m}$, correcting nearsightedness.

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Modern Physics

P.36 Special Relativity Basics

The speed of light is the same for every observer. As a result, moving clocks run slow, moving lengths contract, and mass is a form of energy. The effects are governed by the Lorentz factor γ, which is 1 at everyday speeds.

Lorentz factor	$\gamma = \dfrac{1}{\sqrt{1 - v^2/c^2}}$	$\gamma \geq 1$; $\gamma = 1.25$ at $0.6c$, blowing up as $v \to c$.
Time dilation	$\Delta t = \gamma \, \Delta t_0$	A moving clock's ticks (Δt_0, proper time) look longer.
Length contraction	$L = \dfrac{L_0}{\gamma}$	Only along the direction of motion.
Mass–energy equivalence	$E_0 = mc^2$	Rest energy; tiny mass, enormous energy.
Relativistic energy & momentum	$E = \gamma mc^2, \quad p = \gamma mv$	Kinetic energy is $(\gamma - 1)mc^2$, reducing to $\frac{1}{2}mv^2$ at low speed.
Energy–momentum relation	$E^2 = (pc)^2 + (mc^2)^2$	Massless particles (photons) have $E = pc$.
Velocity addition	$u = \dfrac{u' + v}{1 + u'v/c^2}$	Speeds never add past c.

STATED-EQUATION *The Lorentz factor gamma equals one over the square root of one minus v squared over c squared. Moving time intervals dilate: delta t equals gamma delta t naught. Moving lengths contract: L equals L naught over gamma. Rest energy equals m c squared. Total energy equals gamma m c squared and momentum gamma m v. E squared equals p c squared plus m c squared, squared. Relativistic velocities add as u prime plus v over one plus u prime v over c squared.*

MEMORY AID "Moving clocks run slow, moving rulers shrink."

TERMS

v – relative speed (m/s); $c = 3.00 \times 10^8$ m/s
γ – Lorentz factor (dimensionless)
Δt_0, L_0 – proper time and proper length (measured at rest)
m – (rest) mass (kg); E energy (J); p momentum
u', u – object's speed in the moving frame and in the ground frame
c – speed of light
Δt, L; E, E_0, p – measured time and length; total and rest energy, momentum

CONTEXT

- **Origin.** Einstein (1905), building on Lorentz and Poincaré; confirmed with fast muons and atomic clocks on jets (1971).
- **Use.** GPS, particle accelerators, nuclear energy, astrophysics.
- **Data science.** GPS timestamps in location datasets are corrected for relativity: without it, satellite clocks would drift by 38 μs/day, about 10 km of position error per day.
- **Physical application.** Cosmic-ray muons live 2.2 μs at rest, yet reach the ground from 15 km up because at $0.998c$ their clocks run $\gamma \approx 16$ times slow. Converting 1 g of mass releases 9×10^{13} J.

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P.37 Photons & Matter Waves

Light comes in packets (photons) whose energy is set by frequency, and matter has a wavelength set by momentum. These two ideas, plus the uncertainty principle, are the entry point to quantum mechanics.

Photon energy	$E = hf = \dfrac{hc}{\lambda}$	Bluer light, more energetic photons.
Photon momentum	$p = \dfrac{h}{\lambda}$	Light carries momentum (radiation pressure, solar sails).
de Broglie wavelength	$\lambda = \dfrac{h}{mv}$	Particles diffract like waves when λ is comparable to the obstacle.
Photoelectric effect	$KE_{\max} = hf - \phi$	Electrons escape only if one photon carries more than the work function.
Heisenberg uncertainty	$\Delta x\,\Delta p \geq \dfrac{\hbar}{2}$	Position and momentum cannot both be sharp.

STATED-EQUATION *Photon energy equals h f, which equals h c over lambda. Photon momentum equals h over lambda. A particle's de Broglie wavelength equals h over m v. In the photoelectric effect, the maximum kinetic energy equals h f minus the work function. The uncertainties in position and momentum satisfy delta x delta p at least h bar over two.*

MEMORY AID $hc \approx 1240$ eV·nm: a 500 nm photon carries $\frac{1240}{500} \approx 2.5$ eV.

TERMS

h – Planck's constant 6.626×10^{-34} J·s; $\hbar = h/2\pi$
f, λ – frequency (Hz) and wavelength (m) P.17
ϕ – work function of the metal (eV); 1 eV $= 1.602 \times 10^{-19}$ J
m, v – particle mass (kg) and speed (m/s)
Δx, Δp – standard deviations of position and momentum S.2 §13 Statistics
c; p; $\hbar$ – speed of light; momentum; $h/2\pi$
$KE_{\max}$ – maximum kinetic energy of an ejected electron

CONTEXT

- **Origin.** Planck (1900) from blackbody radiation P.28; Einstein's photons (1905); de Broglie (1924); Heisenberg (1927).
- **Use.** Solar cells, LEDs, lasers, electron microscopes, medical imaging.
- **Data science.** Camera sensors count photons, so low-light images carry Poisson "shot" noise S.18 §13 Statistics, a key term in denoising models.
- **Physical application.** UV photons (≈ 4 eV) break chemical bonds in skin; visible photons ($\approx$ 2–3 eV) do not, which is why UV, not brightness, causes sunburn.

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13
Statistics & Probability

Describing data, quantifying uncertainty, and drawing conclusions from samples. This is the daily toolkit of data science: every A/B test, confidence band, and regression summary is built from the concepts below.

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Describing Data

S.1 Measures of Center

One number that says where the data sit. The mean uses every value and is pulled by outliers; the median and mode are robust alternatives.

Sample mean	$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$	Balance point of the data; the value that minimizes squared error.
Population mean	$\mu = \frac{1}{N}\sum_{i=1}^{N} x_i$	Same average over the whole population; usually unknown and estimated by $\bar{x}$.
Weighted mean	$\bar{x}_w = \frac{\sum_i w_i x_i}{\sum_i w_i}$	Values with larger weights count more; equals $\bar{x}$ when all weights are equal.
Median	$\tilde{x} = \begin{cases} x_{(\frac{n+1}{2})}, & n \text{ odd} \\ \frac{1}{2}\left(x_{(n/2)} + x_{(n/2+1)}\right), & n \text{ even} \end{cases}$	Middle of the sorted data; minimizes absolute error and ignores how extreme the extremes are.
Mode	$\text{mode} = \arg\max_x \ \text{freq}(x)$	Most frequent value; the only center for categorical data. Can be non-unique.
Trimmed mean	$\bar{x}_{\text{tr}} = \frac{1}{n-2g}\sum_{i=g+1}^{n-g} x_{(i)}$	Drop the g smallest and g largest values, average the rest: a compromise between mean and median.

STATED-EQUATION *x bar equals one over n times the sum of x i. Mu equals one over N times the sum of x i. The weighted mean equals the sum of w i x i over the sum of w i. The median is the middle value of the sorted data, or the average of the two middle values when n is even. The mode is the most frequent value. The trimmed mean averages the sorted values after dropping g from each end.*

MEMORY AID Mean = balance point; median = halfway point; mode = most popular.

DECISION POINT Which center? Roughly symmetric with no outliers → mean. Skewed or with outliers (incomes, house prices, latencies) → median. Categories → mode. Mean > median signals a long right tail S.5. Data 1, 2, 3, 4, 100: mean 22, median 3.

TERMS

$\bar{x}$, μ – sample mean, population mean
n, N – sample size, population size
$\sum$ – summation notation A2.21 §04 Algebra II
w_i – weights (importance, frequency, or inverse variance)
$x_{(k)}$ – kth value after sorting (order statistic)
$\tilde{x}$ – median; g number trimmed from each end (e.g. 10% of n)
$\text{freq}(x)$ – count of x; arg max returns the x that maximizes it
$\bar{x}_w$; $\bar{x}_{\text{tr}}$, g – weighted mean; trimmed mean and the number of values trimmed from each end

CONTEXT

- **Origin.** The arithmetic mean dates to antiquity; the median was popularized by Galton (1880s).
- **Use.** Summarizing a typical value.
- **Data science.** Median is robust to outliers (median house price, latency p50); the mean minimizes squared error and the median minimizes absolute error, which is why MSE and MAE losses predict different things ML.2 §15 Machine Learning.
- **Physical application.** The center of mass of objects on a seesaw is a weighted mean of their positions, weighted by mass.

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S.2 Variance & Standard Deviation

Spread measured as the average squared distance from the mean; its square root, the standard deviation, is in the data's own units.

Population variance	$\sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2$	Mean squared deviation from μ.
Sample variance	$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$	Divide by $n-1$ (Bessel's correction) so that $E[s^2] = \sigma^2$.
Standard deviation	$\sigma = \sqrt{\sigma^2}, \qquad s = \sqrt{s^2}$	Typical distance from the mean, in the original units.
Computational formula	$\sigma^2 = \frac{1}{N}\sum_{i=1}^{N}x_i^2 - \mu^2$	Mean of the squares minus the square of the mean; loses precision when values are large and close together.
Coefficient of variation	$CV = \frac{s}{\bar{x}}$	Unit-free spread relative to the mean; compares variability across scales (positive data only).
Range	$R = x_{\max} - x_{\min}$	Quickest spread measure; depends entirely on the two most extreme values.

STATED-EQUATION *Sigma squared equals one over N times the sum of x i minus mu, squared. s squared equals one over n minus one times the sum of x i minus x bar, squared. Sigma and s are the square roots. Also, sigma squared equals the mean of the squares minus the square of the mean. The coefficient of variation is s over x bar; the range is the maximum minus the minimum.*

MEMORY AID Divide by $n-1$ for a Sample ("s for sample, subtract one").

METHOD

1. Compute the mean.
2. Subtract it from every value and square the deviations.
3. Sum the squares (SS), divide by N (population) or $n-1$ (sample).
4. Take the square root for the SD. Example: 2, 4, 4, 4, 5, 5, 7, 9 has mean 5 and $SS = 32$; $\sigma^2 = 32/8 = 4$, $\sigma = 2$; $s^2 = 32/7 \approx 4.57$.

TERMS

σ^2, σ – population variance and standard deviation
s^2, s – sample variance and standard deviation
μ, $\bar{x}$ – means S.1
$n-1$ – Bessel's correction: makes s^2 unbiased (one degree of freedom S.21 used by $\bar{x}$) S.25
$x_{\max}, x_{\min}$ – largest and smallest values
Units – σ has the data's units; σ^2 has squared units
CV; R – coefficient of variation (unitless); range
x_i; N, n – data values; population and sample sizes

CONTEXT

- **Origin.** Standard deviation named by Karl Pearson (1894); variance by R. A. Fisher (1918).
- **Use.** Measuring spread, risk, and consistency.
- **Data science.** Feature scaling divides by s S.4; the bias–variance trade-off ML.4 §15 Machine Learning; volatility of a metric determines how long an A/B test must run S.29.
- **Physical application.** Machining tolerance: a process with SD 0.01 mm produces far more consistent parts than one with SD 0.05 mm.

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S.3 Quartiles, IQR & Outliers

Quartiles cut the sorted data into four equal-count parts. The middle half's width (IQR) is a robust spread, and Tukey's fences flag values far outside it.

Interquartile range	$\mathrm{IQR} = Q_3 - Q_1$	Spread of the middle 50%; unaffected by the tails.
Tukey's fences	$x < Q_1 - 1.5\,\mathrm{IQR}$ or $x > Q_3 + 1.5\,\mathrm{IQR}$	Points beyond the fences are flagged as outliers (3 IQR: extreme outliers).
Five-number summary	$\min,\ Q_1,\ \tilde{x},\ Q_3,\ \max$	Exactly what a box plot draws.
Percentile position	$L = \frac{p}{100}(n-1) + 1$	Position of the pth percentile in the sorted list; interpolate between neighbors (NumPy's default).
Percentile rank	$\mathrm{PR}(x) = \dfrac{\#\{x_i \le x\}}{n} \times 100\%$	Share of the data at or below x: the empirical CDF S.12.

STATED-EQUATION *The IQR equals Q three minus Q one. A value is an outlier if it is below Q one minus one point five IQR or above Q three plus one point five IQR. The five-number summary: minimum, Q one, median, Q three, maximum. The p-th percentile sits at position p over one hundred times n minus one, plus one. The percentile rank of x is the percent of data values at or below x.*

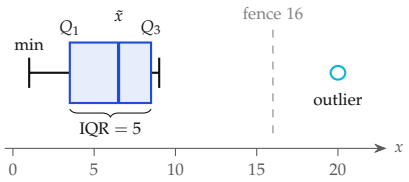
MEMORY AID The box holds the middle half; whiskers reach to 1.5 boxes.

METHOD

1. Sort the data and find the median $\tilde{x}$.
2. Q_1 is the median of the lower half, Q_3 of the upper half (conventions differ slightly between software).
3. Compute IQR and the fences; flag values beyond them. Example: 1, 3, 4, 6, 7, 8, 9, 20 gives $Q_1 = 3.5$, $\tilde{x} = 6.5$, $Q_3 = 8.5$, IQR $= 5$, upper fence $8.5 + 7.5 = 16$, so 20 is an outlier.

TERMS

Q_1, Q_3 – 25th and 75th percentiles: medians of the lower and upper halves S.1
IQR – spread of the middle 50%; robust to outliers
Fences – 1.5 IQR beyond the box (Tukey's rule)
Whiskers – extend to the most extreme non-outlier values
Percentile p – value below which p% of data fall; L its position
$\#\{\cdot\}$ – number of data values satisfying the condition
L; $\tilde{x}$ – position (1-based) in the sorted data; median S.1
$\mathrm{PR}(x)$; n – percentile rank of x; sample size



CONTEXT

- **Origin.** John Tukey introduced the box plot in *Exploratory Data Analysis* (1977).
- **Use.** Comparing distributions and spotting unusual values quickly.
- **Data science.** Outlier filtering in data cleaning, robust scaling (`RobustScaler` uses median and IQR), and latency SLOs defined by percentiles (p95, p99).
- **Physical application.** Quality engineers flag a batch whose tensile strength falls outside the 1.5 IQR fences.

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S.4 z-Scores & Standardization

A z-score re-expresses a value as "how many standard deviations from the mean", which puts different scales on a common footing.

Population z-score	$z = \dfrac{x - \mu}{\sigma}$	Positive above the mean, negative below; $\lvert z\rvert > 3$ is rare for bell-shaped data.
Sample z-score	$z = \dfrac{x - \bar{x}}{s}$	Same idea with estimated mean and SD.
Back-transform	$x = \mu + z\,\sigma$	Turns a standard score (or a normal quantile) back into data units.
Standardized data	$\bar{z} = 0, \qquad s_z = 1$	Subtracting the mean centers; dividing by the SD rescales. Shape is unchanged.
Min–max scaling	$x' = \dfrac{x - x_{\min}}{x_{\max} - x_{\min}}$	Maps data to $[0, 1]$ instead; sensitive to outliers.

STATED-EQUATION *z equals x minus mu over sigma; for a sample, x minus x bar over s. Conversely, x equals mu plus z sigma. Standardized data have mean zero and standard deviation one. Min-max scaling maps x to x minus the minimum, over the range.*

MEMORY AID "How many SDs from the mean?"

METHOD

1. Compute each value's z against its own group's mean and SD.
2. Compare the z's: larger means relatively more extreme. Example: SAT 1300 with $\mu = 1050$, $\sigma = 200$ gives $z = 1.25$; ACT 28 with $\mu = 21$, $\sigma = 5$ gives $z = 1.4$, so the ACT score is relatively higher.

TERMS

z – number of standard deviations from the mean
μ, σ – population mean S.1 and SD S.2
$\bar{x}, s$ – sample versions; $\bar{z}, s_z$ mean and SD of the z's
$x_{\min}, x_{\max}$ – smallest and largest values
With normal data – look up $\Phi(z)$ for probabilities S.20
x' – min–max scaled value in $[0, 1]$
$\bar{z}$, s_z – mean and standard deviation of the z-scores

CONTEXT

- **Origin.** Standardization grew out of Gauss's and Pearson's work on the normal curve.
- **Use.** Comparing values on different scales (SAT vs ACT).
- **Data science.** `StandardScaler` before regularized regression, k-NN, SVMs, PCA, and gradient descent ML.11 §15 Machine Learning; $\lvert z\rvert > 3$ is a quick anomaly flag.
- **Physical application.** Comparing a hot day in Phoenix and in Seattle: standardize each temperature by its own city's mean and SD.

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S.5 Shape: Skewness & Kurtosis

Beyond center and spread, the third and fourth standardized moments describe asymmetry (skewness) and tail heaviness (kurtosis).

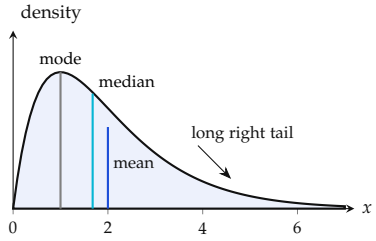
Sample skewness	$g_1 = \dfrac{\frac{1}{n}\sum(x_i-\bar{x})^3}{\left(\frac{1}{n}\sum(x_i-\bar{x})^2\right)^{3/2}}$	Average cubed z-score: >0 long right tail, <0 long left tail, ≈ 0 symmetric.
Excess kurtosis	$g_2 = \dfrac{\frac{1}{n}\sum(x_i-\bar{x})^4}{\left(\frac{1}{n}\sum(x_i-\bar{x})^2\right)^2} - 3$	Average fourth-power z-score minus the normal's 3: >0 heavy tails, more outliers.
Skew and the center	right skew: $\text{mode} < \tilde{x} < \bar{x}$	The tail drags the mean furthest; reversed for left skew.

STATED-EQUATION *Skewness g one is the average cubed deviation divided by the variance to the three-halves. Excess kurtosis g two is the average fourth-power deviation over the squared variance, minus three. In a right-skewed distribution, mode is less than median, which is less than mean.*

MEMORY AID Skew points to the tail: the tail is where the mean goes.

TERMS

g_1 – skewness (third standardized moment)
g_2 – excess kurtosis (fourth standardized moment -3); normal $= 0$
$\bar{x}, \tilde{x}$ – mean S.1, median
Moment – $E[(X-\mu)^k]$ S.13
Fix – log transform tames right skew A2.14 §04 Algebra II



CONTEXT

- **Origin.** Karl Pearson named skewness (1895) and kurtosis (1905).
- **Use.** Choosing summaries and checking normality before t-tests S.30.
- **Data science.** Revenue, latency and counts are right-skewed: log-transform features or report medians; heavy-tailed returns break normal-based risk models S.20.
- **Physical application.** Wind speeds and earthquake magnitudes have long right tails.

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Probability

S.6 Sample Spaces, Events & the Axioms

A probability model is a sample space of outcomes plus a rule P that assigns each event (a set of outcomes) a number obeying three axioms; every other rule follows.

Sample space and event	$S = \{\text{all outcomes}\}, \qquad A \subseteq S$	One die: $S = \{1, \dots, 6\}$; "even" is the event $A = \{2, 4, 6\}$ A1.21 §03 Algebra I.
Axiom 1: non-negativity	$P(A) \geq 0$	No event has negative probability.
Axiom 2: normalization	$P(S) = 1$	Something in the sample space must happen.
Axiom 3: additivity	$P\left(\bigcup_i A_i\right) = \sum_i P(A_i) \;\; (A_i \text{ disjoint})$	Probabilities of non-overlapping events add (also for countably many).
Classical probability	$P(A) = \dfrac{\lvert A\rvert}{\lvert S\rvert}$	Only when all outcomes are equally likely; count with S.8.
Frequentist probability	$P(A) = \lim_{n\to\infty} \dfrac{n_A}{n}$	Long-run relative frequency in repeated trials S.23.

STATED-EQUATION *The sample space S is the set of all outcomes; an event A is a subset of S. P of A is at least zero, P of S equals one, and the probability of a union of disjoint events is the sum of their probabilities. With equally likely outcomes, P of A is the size of A over the size of S; in the long run it is the fraction of trials in which A occurs.*

MEMORY AID Kolmogorov's three: never negative, total one, disjoint pieces add.

TERMS

S (or Ω) – sample space; outcome = element of S
Event A – a subset of S; $\subseteq$ "is a subset of"
$\bigcup$ – union of events ("or") A1.21 §03 Algebra I
$|A|$ – number of outcomes in A
n_A/n – relative frequency of A in n trials
Bayesian view – P as a degree of belief S.37

CONTEXT

- **Origin.** Andrey Kolmogorov's *Foundations of the Theory of Probability* (1933).
- **Use.** Building any probability model; checking that a model is valid.
- **Data science.** A classifier's softmax output must satisfy the axioms (non-negative, sums to 1) ML.24 §15 Machine Learning; simulations estimate probabilities as frequencies.
- **Physical application.** Quantum measurement outcomes are assigned probabilities that obey the same axioms.

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S.7 Basic Probability Rules

The everyday rules derived from the axioms: bounds, complements, and the addition rule for "or".

Bounds	$0 \le P(A) \le 1$	0 is impossible, 1 is certain.
Complement rule	$P(A^c) = 1 - P(A)$	Often easier to compute what does *not* happen.
Addition rule	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$	"Or": add, then subtract the double-counted overlap A1.21 §03 Algebra I.
Disjoint events	$P(A \cup B) = P(A) + P(B)$	Mutually exclusive: no overlap to subtract.
At least one	$P(\text{at least one}) = 1 - \prod_{i=1}^{k}(1 - p_i)$	One minus "none happen", for independent events S.9.
Three-event inclusion–exclusion	$P(A \cup B \cup C) = \sum P(A) - \sum P(A \cap B) + P(A \cap B \cap C)$	Add singles, subtract pairs, add back the triple.

STATED-EQUATION *The probability of A is between zero and one. The probability of not A equals one minus P of A. The probability of A or B equals P of A plus P of B minus P of A and B; for disjoint events, just P of A plus P of B. The probability of at least one event equals one minus the product of one minus p i. For three events, add the single probabilities, subtract the pairwise intersections, and add back the triple intersection.*

MEMORY AID "Or" adds (minus the overlap); "at least one" = 1− "none."

METHOD

1. Decide whether the question is "and", "or", "not", or "at least one".
2. For "at least one", compute the probability that none occur and subtract from 1.
3. Example: 4 independent smoke detectors, each failing with probability 0.1: $P(\text{at least one works}) = 1 - 0.1^4 = 0.9999$.

TERMS

A^c – complement: "not A"
$A \cup B$ – A or B (or both)
$A \cap B$ – A and B
Mutually exclusive – $A \cap B = \varnothing$, so $P(A \cap B) = 0$
p_i – probability of the ith event; $\prod$ product M.54 §01 Math Signs
Sample space – S, with $P(S) = 1$ S.6
k; C – number of events; a third event

CONTEXT

- **Origin.** Pascal and Fermat's letters on gambling (1654); axioms by Kolmogorov (1933).
- **Use.** Games of chance, reliability, risk.
- **Data science.** "At least one" via complements: the chance that at least one of many A/B metrics looks significant by luck S.35.
- **Physical application.** The chance that at least one of four independent smoke detectors works is $1 - P(\text{all fail})$.

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S.8 Counting: Permutations & Combinations

With equally likely outcomes, probability is counting. Two questions decide the formula: does order matter, and can items repeat?

Multiplication principle	m ways then n ways $= m \cdot n$ ways	Independent stages multiply.
Factorial	$n! = n(n-1)(n-2)\cdots 1, \quad 0! = 1$	Arrangements of n distinct objects in a row M.48 §01 Math Signs.
Permutations	$${}_nP_r = \frac{n!}{(n-r)!}$$	Ordered selections of r from n, no repeats.
Combinations	$$\binom{n}{r} = {}_nC_r = \frac{n!}{r!\,(n-r)!}$$	Unordered selections; divide out the $r!$ orderings.
With repetition, ordered	$$n^r$$	Each of r positions has n choices (PIN codes, strings).
With repetition, unordered	$$\binom{n+r-1}{r}$$	"Stars and bars": choosing r items from n types.
Arrangements with repeats	$$\frac{n!}{n_1!\,n_2!\cdots n_k!}$$	Orderings of n items with n_j identical copies of type j; MISSISSIPPI has $\frac{11!}{1!4!4!2!}$.

STATED-EQUATION *n factorial equals n times n minus one down to one; zero factorial equals one. n P r equals n factorial over n minus r factorial. n choose r equals n factorial over r factorial times n minus r factorial. The multiplication principle: m ways then n ways gives m times n. With repetition: n to the r ordered, n plus r minus one choose r unordered. Arrangements with repeats: n factorial over the product of the n j factorials.*

MEMORY AID Permutation: Position matters. Combination: Committee (order doesn't matter).

DECISION POINT Ask two questions. **Does order matter?** (president/treasurer vs a committee). **Can items repeat?** (digits of a PIN vs people). Ordered + no repeats → ${}_nP_r$; unordered + no repeats → $\binom{n}{r}$; ordered + repeats → n^r; unordered + repeats → $\binom{n+r-1}{r}$. Example: a 3-person committee from 5 is $\binom{5}{3} = 10$; a president, VP and treasurer is ${}_5P_3 = 60$.

TERMS

$n!$ – factorial: arrangements of n distinct objects
${}_nP_r$ – ordered selections of r from n (order matters)
$\binom{n}{r}$ – unordered selections ("n choose r"); binomial coefficient A2.23 §04 Algebra II
m, n – numbers of ways at each stage; r items chosen
n_j – copies of identical type j, $\sum n_j = n$
Relationship – ${}_nP_r = \binom{n}{r} r!$; a set of n items has 2^n subsets
r – number of items chosen or arranged

CONTEXT

- **Origin.** Combinatorics in Indian mathematics (Pingala, ∼200 BCE); Pascal's triangle (1654).
- **Use.** Probabilities of equally likely outcomes S.7; lotteries and card games.
- **Data science.** Size of search spaces: 2^p feature subsets, $\binom{p}{2}$ pairwise interactions, grid-search combinations of hyperparameters.
- **Physical application.** The number of distinct orderings of 5 lab samples in a queue is $5! = 120$.

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S.9 Conditional Probability & Independence

Conditioning on B restricts attention to outcomes where B happened. Events are independent when that restriction does not change the probability of the other.

Conditional probability	$P(A \mid B) = \dfrac{P(A \cap B)}{P(B)}$	Share of B's probability that also lies in A ($P(B) > 0$).
Multiplication rule	$P(A \cap B) = P(A \mid B)\,P(B)$	"And" for any two events; chain events in sequence.
Chain rule	$P(A \cap B \cap C) = P(A)\,P(B \mid A)\,P(C \mid A \cap B)$	Each factor conditions on everything before it.
Independence	$P(A \cap B) = P(A)\,P(B) \iff P(A \mid B) = P(A)$	Knowing B tells nothing about A.
Conditional independence	$P(A \cap B \mid C) = P(A \mid C)\,P(B \mid C)$	Independent once C is known; neither implies nor follows from plain independence.
Disjoint ⇒ dependent	$A \cap B = \varnothing,\ P(A), P(B) > 0 \Rightarrow P(A \mid B) = 0 \neq P(A)$	Mutually exclusive events are *never* independent: one happening rules the other out.

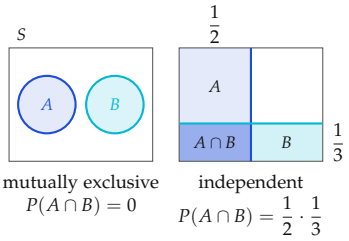
STATED-EQUATION *P of A given B equals P of A and B over P of B. P of A and B equals P of A given B times P of B. P of A and B and C equals P of A times P of B given A times P of C given A and B. A and B are independent if and only if P of A and B equals P of A times P of B, which is the same as P of A given B equals P of A. A and B are conditionally independent given C when P of A and B given C equals P of A given C times P of B given C. Disjoint events with positive probabilities are dependent, since knowing B makes P of A given B zero.*

MEMORY AID "Given" shrinks the sample space to B. Disjoint means "can't both happen"; independent means "doesn't matter if the other happened".

DECISION POINT Independent or mutually exclusive? Ask: if B happens, what is $P(A)$ now? If it drops to 0, they are mutually exclusive (and therefore dependent). If it stays the same, they are independent. Example: one die, A = "even", B = "≤ 2": $P(A \cap B) = \frac{1}{6} = \frac{1}{2} \cdot \frac{1}{3}$, so independent, though they overlap.

TERMS

$P(A \mid B)$ – probability of A given that B occurred
$P(B)$ – must be > 0; B becomes the new sample space
$\cap$ – "and" (intersection); $\varnothing$ empty set A1.21 §03 Algebra I
Independent – knowing B tells nothing about A
C – a third (conditioning) event
Not the same as – mutually exclusive S.7
$\varnothing$ – empty set: A and B cannot both happen



CONTEXT

- **Origin.** Formalized by de Moivre (1718) and Bayes (1763).
- **Use.** Sequential events, drawing without replacement, reliability.
- **Data science.** Naive Bayes assumes features are conditionally independent given the class ML.10 §15 Machine Learning; funnel conversion rates are conditional probabilities $P(\text{buy} \mid \text{click})$.
- **Physical application.** Reliability of components in series: if each part works independently with probability 0.99, three in series work with probability 0.99^3.

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S.10 Law of Total Probability

Split the sample space into non-overlapping cases, find the probability of B within each case, and add them up weighted by how likely each case is.

Partition	$A_i \cap A_j = \varnothing \ (i \neq j), \quad \bigcup_i A_i = S$	Cases that are mutually exclusive and cover every outcome.
Law of total probability	$P(B) = \sum_i P(B \mid A_i)\, P(A_i)$	Weighted average of the conditional probabilities of B.
Two-case form	$P(B) = P(B \mid A)P(A) + P(B \mid A^c)P(A^c)$	The most common partition: A or not A.

STATED-EQUATION *The events A i form a partition when they are pairwise disjoint and their union is S. Then P of B equals the sum over i of P of B given A i times P of A i; with two cases, P of B given A times P of A plus P of B given not A times P of not A.*

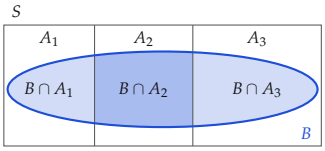
MEMORY AID "Divide and conquer": sum over cases of (chance of case) $\times$ (chance of B in that case).

METHOD

1. Choose a partition where each $P(B \mid A_i)$ is easy (machines, groups, disease status).
2. Multiply each conditional by its case probability; add.
3. Example: three machines make 50%, 30%, 20% of parts with defect rates 1%, 2%, 5%: $P(\text{defect}) = 0.005 + 0.006 + 0.010 = 0.021$.

TERMS

$A_1, A_2, \ldots$ – partition of S S.6
B – event of interest
$P(B \mid A_i)$ – conditional probability S.9
A^c – complement of A
Continuous version – $P(B) = \int P(B \mid X = x) f(x)\, dx$ S.14
S; $\varnothing$ – sample space S.6; empty set



CONTEXT

- **Origin.** Implicit in Bayes (1763) and Laplace (1774).
- **Use.** Mixtures of populations, multi-stage experiments.
- **Data science.** Overall conversion as a traffic-weighted mix of channel rates; the denominator of Bayes' theorem S.11; marginal likelihoods in mixture models ML.37 §15 Machine Learning.
- **Physical application.** Total failure rate of a fleet from per-model failure rates and fleet shares.

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S.11 Bayes' Theorem

Bayes' theorem reverses a conditional probability: from how likely the evidence is under each hypothesis to how likely each hypothesis is given the evidence.

Bayes' theorem	$P(A \mid B) = \dfrac{P(B \mid A)\,P(A)}{P(B)}$	Posterior = likelihood × prior / evidence.
Evidence (two cases)	$P(B) = P(B \mid A)P(A) + P(B \mid A^c)P(A^c)$	Denominator from the law of total probability S.10.
Bayes over a partition	$P(A_j \mid B) = \dfrac{P(B \mid A_j)P(A_j)}{\sum_i P(B \mid A_i)P(A_i)}$	Posterior for each of several hypotheses; they sum to 1.
Odds form	$\dfrac{P(A \mid B)}{P(A^c \mid B)} = \dfrac{P(B \mid A)}{P(B \mid A^c)} \cdot \dfrac{P(A)}{P(A^c)}$	Posterior odds = likelihood ratio × prior odds; easy to update repeatedly.
Proportional form	$P(A \mid B) \propto P(B \mid A)\,P(A)$	Compute numerators for every hypothesis, then normalize to sum 1.

STATED-EQUATION *P of A given B equals P of B given A times P of A, over P of B. P of B equals P of B given A P of A plus P of B given not A P of not A. Over a partition, the posterior of A j is its likelihood times prior over the sum of all such products. Posterior odds equal the likelihood ratio times prior odds. Equivalently, the posterior is proportional to the likelihood times the prior.*

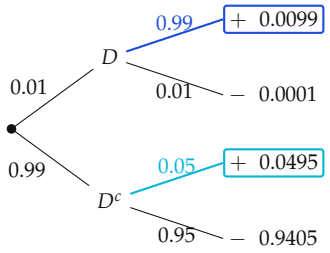
MEMORY AID Posterior $\propto$ likelihood $\times$ prior.

METHOD

1. Write the prior $P(A)$ and the likelihoods $P(B \mid A)$, $P(B \mid A^c)$.
2. Multiply to get the joint probabilities of each branch (tree diagram).
3. Add the branches consistent with the evidence to get $P(B)$.
4. Divide the branch of interest by $P(B)$. Example: prevalence 1%, sensitivity 99%, false-positive rate 5%: $P(D \mid +) = \frac{0.0099}{0.0099+0.0495} \approx 0.17$.

TERMS

$P(A)$ – prior: belief before seeing evidence
$P(B \mid A)$ – likelihood of the evidence if A is true
$P(B)$ – evidence (marginal), by total probability
$P(A \mid B)$ – posterior: updated belief S.9
A_i – partition of the sample space; $\propto$ "proportional to"
Base-rate fallacy – ignoring a small prior
A^c – complement, "not A"



$$P(D \mid +) = \frac{0.0099}{0.0099 + 0.0495} \approx 0.17$$

CONTEXT

- **Origin.** Rev. Thomas Bayes (published 1763); generalized by Laplace (1774).
- **Use.** Medical testing, forensic evidence, updating beliefs.
- **Data science.** Naive Bayes spam filters ML.10 §15 Machine Learning, Bayesian A/B testing S.37, and interpreting classifier precision when the positive class is rare ML.17 §15 Machine Learning.
- **Physical application.** Interpreting a positive result from a sensitive but imperfect sensor (e.g., a gas-leak detector) requires the base rate of real leaks.

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Random Variables

S.12 Random Variables, PMFs, PDFs & CDFs

A random variable turns outcomes into numbers. Its distribution is described by a mass function (discrete), a density (continuous), or, for both, a cumulative distribution function.

PMF (discrete)	$p(x) = P(X = x), \qquad \sum_x p(x) = 1$	Probability sits on separate points (counts, categories).
PDF (continuous)	$P(a \le X \le b) = \int_a^b f(x)\,dx, \quad \int_{-\infty}^{\infty} f = 1$	Probability is area; $P(X = x) = 0$ and f may exceed 1 C1.30 §07 Calculus I.
CDF	$F(x) = P(X \le x)$	Non-decreasing from 0 to 1; defined for every random variable.
Density from CDF	$f(x) = F'(x)$	The PDF is the slope of the CDF (fundamental theorem C1.32 §07 Calculus I).
Interval from CDF	$P(a < X \le b) = F(b) - F(a)$	Any interval probability is a difference of two CDF values.
Quantile function	$Q(p) = F^{-1}(p) = \min\{x : F(x) \ge p\}$	Value with a fraction p below it; median $= Q(0.5)$ S.3.

STATED-EQUATION *The PMF p of x is P of X equals x, summing to one. For a density, P of a less than or equal X less than or equal b is the integral of f from a to b, and the total area is one. The CDF F of x is P of X less than or equal x; f is the derivative of F; P of a less than X at most b is F of b minus F of a. The quantile Q of p is the inverse CDF.*

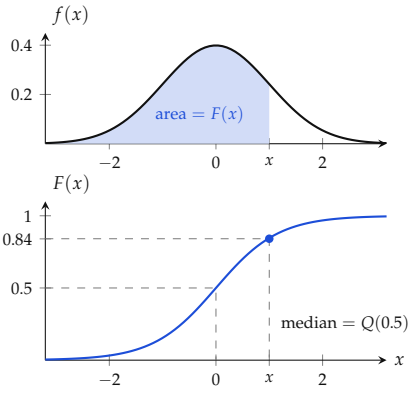
MEMORY AID PMF = point masses; PDF = area; CDF = "so far" total; quantile = CDF read backwards.

METHOD

1. Set $F(x) = p$.
2. Solve for x. Example: exponential $F(x) = 1 - e^{-\lambda x}$ S.19; $p = 0.5$ gives the median $x = \frac{\ln 2}{\lambda}$, so $\lambda = 2$ has median ≈ 0.35.

TERMS

X – random variable (capital); x a value it can take
$p(x)$ – probability mass function (PMF)
$f(x)$ – probability density function (PDF)
$F(x)$ – cumulative distribution function (CDF)
Q, F^{-1} – quantile (inverse CDF) A2.1 §04 Algebra II; $p \in (0, 1)$
Survival – $1 - F(x) = P(X > x)$
a, b; p – interval endpoints; probability level for a quantile



CONTEXT

- **Origin.** Distribution functions were formalized by Kolmogorov (1933); the quantile function by Galton and Pearson.
- **Use.** Describing any random quantity; computing interval probabilities.
- **Data science.** Inverse-transform sampling $X = F^{-1}(U)$ 0.26 §14 Optimization; quantile regression; percentile-based SLOs; the empirical CDF behind Q–Q plots and the KS test.
- **Physical application.** The CDF of component lifetimes gives the fraction failed by time t.

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S.13 Expected Value & Variance of a Random Variable

The expected value is the probability-weighted average of a random variable; the variance is the expected squared distance from it. A few algebra rules handle sums and rescaling.

Expected value (discrete)	$E[X] = \mu = \sum_x x\, p(x)$	Long-run average of many independent draws.
Expected value (continuous)	$E[X] = \int_{-\infty}^{\infty} x\, f(x)\, dx$	Same idea with the density; the balance point of f.
LOTUS	$E[g(X)] = \sum_x g(x)\, p(x)$	Average a function of X without finding its distribution (integral for continuous).
Variance	$\mathrm{Var}(X) = E\big[(X-\mu)^2\big] = E[X^2] - \big(E[X]\big)^2$	Second form is quicker by hand.
Linear transformation	$E[aX+b] = aE[X]+b, \quad \mathrm{Var}(aX+b) = a^2\,\mathrm{Var}(X)$	Shifts move the mean only; scaling by a scales the SD by $\lvert a\rvert$.
Linearity of expectation	$E[X+Y] = E[X] + E[Y]$	Always true, even for dependent variables.
Variance of a sum	$\mathrm{Var}(X+Y) = \mathrm{Var}\,X + \mathrm{Var}\,Y + 2\,\mathrm{Cov}(X,Y)$	The covariance term vanishes for independent variables S.38.
Product of independents	$E[XY] = E[X]\,E[Y]$	Only for independent (or uncorrelated) variables.

STATED-EQUATION *The expected value E of X equals the sum of x p of x, or the integral of x f of x d x. E of g of X equals the sum of g of x p of x. The variance equals the expected value of X minus mu squared, which equals E of X squared minus E of X, squared. E of a X plus b equals a E X plus b; the variance of a X plus b equals a squared times variance of X. Expectations always add. Var of X plus Y equals Var X plus Var Y plus two Cov X Y. For independent X and Y, E of X Y equals E of X times E of Y.*

MEMORY AID Adding a constant shifts the mean but not the spread; scaling by a scales variance by a^2.

METHOD

1. List values and probabilities; compute $E[X] = \sum x\, p(x)$ and $E[X^2] = \sum x^2 p(x)$.
2. $\mathrm{Var} = E[X^2] - (E[X])^2$. Example: a fair die has $E[X] = 3.5$, $E[X^2] = \frac{91}{6}$, $\mathrm{Var} = \frac{91}{6} - 12.25 = \frac{35}{12} \approx 2.92$.

TERMS

X, Y – random variables: numeric outcomes of a random process S.12
$p(x), f(x)$ – PMF and PDF; integral C1.32 §07 Calculus I
g – any function of X; LOTUS "law of the unconscious statistician"
a, b – constants
$\mathrm{Cov}(X,Y)$ – covariance S.38
μ – shorthand for $E[X]$
$E[\cdot]$, $\mathrm{Var}(\cdot)$ – expected value; variance

CONTEXT

- **Origin.** Christiaan Huygens (1657) on the value of a gamble.
- **Use.** Fair prices, insurance premiums, long-run averages.
- **Data science.** Expected revenue per user, expected loss (risk) minimized by ML training ML.1 §15 Machine Learning, and the variance of an ensemble average shrinking as models are added ML.13 §15 Machine Learning.
- **Physical application.** The expected energy of a gas molecule is an expected value over the Boltzmann distribution of speeds.

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S.14 Joint, Marginal & Conditional Distributions

A joint distribution describes two random variables together. Summing (or integrating) out one variable gives a marginal; dividing by a marginal gives a conditional.

Joint PMF	$p(x,y) = P(X = x,\ Y = y)$	All pairs' probabilities; sums to 1 (density $f(x,y)$ integrates to 1).
Marginal	$p_X(x) = \sum_y p(x,y), \qquad f_X(x) = \int f(x,y)\,dy$	"Sum out" the other variable: the row or column totals.
Conditional	$p(y \mid x) = \dfrac{p(x,y)}{p_X(x)}$	Distribution of Y within the slice $X = x$ S.9.
Independence of variables	$p(x,y) = p_X(x)\,p_Y(y)$ for all x, y	Joint table equals the product of its margins.
Conditional expectation	$E[Y \mid X = x] = \sum_y y\,p(y \mid x)$	Mean of Y in each slice; regression models estimate it S.40.
Law of total expectation	$E[Y] = E\big[E[Y \mid X]\big]$	Average the slice means, weighted by p_X.
Law of total variance	$\mathrm{Var}(Y) = E\big[\mathrm{Var}(Y \mid X)\big] + \mathrm{Var}\big(E[Y \mid X]\big)$	Total = within-group + between-group variance (the idea behind ANOVA S.33).

STATED-EQUATION *The joint PMF p of x y is P of X equals x and Y equals y. The marginal of X sums the joint over y. The conditional p of y given x is the joint over the marginal of x. X and Y are independent when the joint equals the product of the marginals. E of Y equals the expectation of E of Y given X. Var Y equals the expected conditional variance plus the variance of the conditional mean. The conditional expectation of Y given X equals x is the sum of y times p of y given x.*

MEMORY AID Joint = the whole table; marginal = the margins; conditional = one row, rescaled to sum to 1.

METHOD

1. Write the joint table and add row and column totals (marginals).
2. For $p(y \mid x)$ divide the row x by its total. Example: with the table below, $P(Y = 1 \mid X = 1) = \frac{0.4}{0.5} = 0.8$, while $P(Y = 1) = 0.6$, so X and Y are dependent.

	$Y = 0$	$Y = 1$	p_X
$X = 0$	0.3	0.2	0.5
$X = 1$	0.1	0.4	0.5
p_Y	0.4	0.6	1

TERMS

$p(x,y), f(x,y)$ – joint PMF, joint PDF
p_X, p_Y, f_X – marginal distributions
$p(y \mid x)$ – conditional distribution of Y given $X = x$
$E[Y \mid X]$ – conditional mean, itself a random variable (a function of X)
$\mathrm{Var}(Y \mid X)$ – variance within a slice; Var, E from S.13

CONTEXT

- **Origin.** Developed with multivariate statistics by Pearson and Fisher (early 1900s); measure-theoretic conditioning by Kolmogorov (1933).
- **Use.** Contingency tables, multi-sensor data, hierarchical models.
- **Data science.** Every supervised model approximates $E[Y \mid X]$ or $p(y \mid x)$; marginalizing latent variables in mixture models ML.37 §15 Machine Learning; pivot tables are joint and marginal tables.
- **Physical application.** Joint distribution of wind speed and direction at a turbine site.

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Discrete Distributions

S.15 Bernoulli, Geometric & Negative Binomial

Built from independent yes/no trials with success probability p: one trial (Bernoulli), the wait for the first success (geometric), and the wait for the rth success (negative binomial).

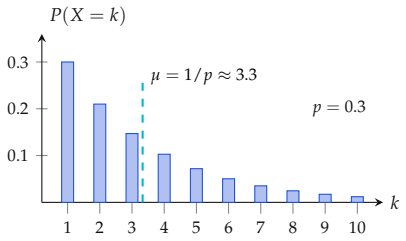
Bernoulli	$P(X=1)=p,\ P(X=0)=1-p;\quad \mu=p,\ \sigma^2=p(1-p)$	A single trial; variance is largest at $p=\frac{1}{2}$ D.1 §18 Graph Gallery.
Geometric PMF	$P(X=k)=(1-p)^{k-1}p,\quad k=1,2,\dots$	$k-1$ failures then the first success D.3 §18 Graph Gallery.
Geometric mean, variance	$\mu=\dfrac{1}{p},\qquad \sigma^2=\dfrac{1-p}{p^2}$	With $p=0.2$ expect 5 trials.
Geometric tail	$P(X>k)=(1-p)^k$	First k trials all fail; memoryless like the exponential S.19.
Negative binomial PMF	$P(X=k)=\dbinom{k-1}{r-1}p^r(1-p)^{k-r},\ k\geq r$	The rth success on trial k: $r-1$ successes among the first $k-1$.
Negative binomial mean, variance	$\mu=\dfrac{r}{p},\qquad \sigma^2=\dfrac{r(1-p)}{p^2}$	A sum of r independent geometric waits.

STATED-EQUATION *Bernoulli: X is one with probability p and zero otherwise, mean p, variance p times one minus p. Geometric: P of X equals k is one minus p to the k minus one, times p; mean one over p, variance one minus p over p squared; P of X greater than k is one minus p to the k. Negative binomial: k minus one choose r minus one, p to the r, one minus p to the k minus r; mean r over p. The negative binomial variance is r times one minus p, over p squared.*

MEMORY AID Geometric: "how long until the first win?" $\Rightarrow 1/p$ tries on average.

TERMS

p – success probability on each independent trial S.9
k – trial on which the (first or rth) success occurs
r – number of successes waited for
μ, σ^2 – mean and variance S.13
$\binom{k-1}{r-1}$ – combinations S.8
Convention – some texts count failures before success ($k=0,1,\dots$; mean $\frac{1-p}{p}$)



CONTEXT

- **Origin.** Bernoulli trials from Jacob Bernoulli's *Ars Conjectandi* (1713); the negative binomial from Pascal and Montmort (1700s).
- **Use.** Repeated attempts: sales calls, retries, games until a win.
- **Data science.** Bernoulli is the likelihood behind log-loss ML.9 §15 Machine Learning; geometric models retries and sessions until conversion; the negative binomial models over-dispersed counts (variance > mean) better than Poisson.
- **Physical application.** Number of photon emissions attempted before a single-photon detector clicks.

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S.16 Binomial Distribution

The number of successes in a fixed number n of independent trials, each with the same success probability p.

Binomial PMF	$P(X=k)=\binom{n}{k}p^k(1-p)^{n-k},\ \ k=0,\dots,n$	Ways to place k successes times the probability of any one such sequence.
Mean	$\mu=np$	Expected successes: 100 trials at 30% gives 30.
Variance	$\sigma^2=np(1-p)$	Sum of n Bernoulli variances S.15.
Sum of Bernoullis	$X=B_1+B_2+\cdots+B_n$	Why mean and variance are n times the Bernoulli ones S.13.
CDF	$P(X\le k)=\sum_{j=0}^{k}\binom{n}{j}p^j(1-p)^{n-j}$	Add the PMF up to k ("at most k").
Normal approximation	$X\approx N\big(np,\ np(1-p)\big)$	Good when $np\ge 10$ and $n(1-p)\ge 10$; use $k\pm 0.5$ (continuity correction) S.20.

STATED-EQUATION *P of X equals k equals n choose k, p to the k, times one minus p to the n minus k. The mean is n p, and the variance is n p times one minus p. X is a sum of n Bernoulli trials. The CDF adds the PMF from zero to k. For large n, X is approximately normal with mean n p and variance n p one minus p.*

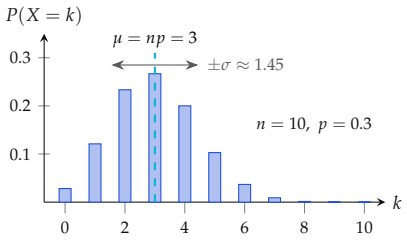
MEMORY AID BINS conditions: Binary outcomes, Independent trials, fixed Number, Same p.

METHOD

1. Check BINS; identify n, p and k.
2. For "at least" use the complement S.7: $P(X\ge 1)=1-(1-p)^n$.
3. Example: 4 fair coin flips, $P(X=2)=\binom{4}{2}\left(\frac{1}{2}\right)^4=\frac{6}{16}=0.375$.

TERMS

n – number of independent trials S.9
k – number of successes
p – probability of success on each trial
$\binom{n}{k}$ – ways to place k successes S.8
B_i – Bernoulli indicator of trial i
Large n, small p – approximately Poisson, $\lambda=np$ S.18
X; j – number of successes; summation index



CONTEXT

- **Origin.** Jacob Bernoulli, *Ars Conjectandi* (1713).
- **Use.** Quality control, polling, free-throw percentages.
- **Data science.** Conversions out of n visitors in an A/B test S.31; the Bernoulli likelihood is the basis of logistic regression's log-loss ML.9 §15 Machine Learning.
- **Physical application.** The number of defective chips in a batch of 100 with a 2% defect rate is binomial with $n=100$, $p=0.02$.

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S.17 Hypergeometric & Discrete Uniform

The hypergeometric counts successes when sampling *without* replacement from a finite population; the discrete uniform gives every one of m values equal probability.

Hypergeometric PMF	$P(X=k)=\dfrac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}$	Choose k of the K successes and $n-k$ of the failures, out of all samples of size n.
Hypergeometric mean	$\mu = n\,\dfrac{K}{N}$	Same as the binomial with $p=K/N$.
Hypergeometric variance	$\sigma^2 = n\dfrac{K}{N}\left(1-\dfrac{K}{N}\right)\dfrac{N-n}{N-1}$	Binomial variance times the finite-population correction $\frac{N-n}{N-1}<1$.
Discrete uniform PMF	$P(X=x)=\dfrac{1}{m},\quad x=a,\dots,b,\ m=b-a+1$	Fair die, random index, random digit.
Discrete uniform mean, variance	$\mu=\dfrac{a+b}{2},\qquad \sigma^2=\dfrac{m^2-1}{12}$	Die: $\mu=3.5$, $\sigma^2=\frac{35}{12}$.

STATED-EQUATION *Hypergeometric: P of X equals k is K choose k times N minus K choose n minus k, over N choose n; mean n K over N; variance n K over N times one minus K over N times N minus n over N minus one. Discrete uniform on a to b: each value has probability one over m, with m equal to b minus a plus one; mean a plus b over two; variance m squared minus one over twelve.*

MEMORY AID Hypergeometric = "draws without putting back".

DECISION POINT Binomial or hypergeometric? Sampling *with* replacement, or from a huge population → binomial. Sampling *without* replacement from a finite population (cards, defective items in a lot) → hypergeometric. If the sample is under about 5% of the population ($n \le 0.05N$) the binomial is an excellent approximation. Example: 5 cards from 52, $P(\text{exactly 2 aces}) = \frac{\binom{4}{2}\binom{48}{3}}{\binom{52}{5}} \approx 0.040$.

TERMS

N – population size; K successes in it
n – sample size; k successes in the sample
$\binom{\cdot}{\cdot}$ – combinations S.8
a, b – smallest and largest values; m how many values
$\frac{N-n}{N-1}$ – finite-population correction
K; k – successes in the population; successes in the sample
m – number of equally likely values

CONTEXT

- **Origin.** Hypergeometric probabilities appear in de Moivre (1711); the name is from the hypergeometric series.
- **Use.** Card games, lotteries, acceptance sampling of lots.
- **Data science.** Fisher's exact test for 2×2 tables is a hypergeometric probability S.32; gene-set enrichment tests; uniform random indices for shuffling and train/test splits.
- **Physical application.** Capture–recapture estimates of animal populations use the hypergeometric model.

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S.18 Poisson Distribution

The number of events in a fixed interval of time or space when events occur independently at a constant average rate.

Poisson PMF	$P(X=k)=\dfrac{\lambda^k e^{-\lambda}}{k!},\ \ k=0,1,2,\dots$	Probability of exactly k events when λ are expected.
Mean and variance	$\mu=\sigma^2=\lambda$	Equal mean and variance is the Poisson signature; variance $>$ mean means over-dispersion.
Rate scaling	$\lambda=r\,t$	Rate per unit times the interval length: 2 calls/min over 5 min gives $\lambda=10$.
Sum of Poissons	$X_1+X_2\sim\text{Pois}(\lambda_1+\lambda_2)$	Independent counts add their rates.
Binomial limit	$\binom{n}{k}p^k(1-p)^{n-k}\to\dfrac{\lambda^k e^{-\lambda}}{k!}$	Large n, small p, $\lambda=np$ fixed S.16.

STATED-EQUATION *P of X equals k equals lambda to the k times e to the minus lambda, over k factorial. The mean and variance both equal lambda. Lambda equals rate times time. Sums of independent Poissons are Poisson with the rates added. The binomial tends to the Poisson as n grows with n p fixed.*

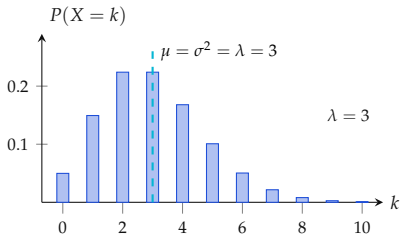
MEMORY AID Poisson: mean = variance = λ.

METHOD

1. Convert the rate to the interval asked about: $\lambda=rt$.
2. Plug into the PMF; use the complement for "at least one".
3. Example: 3 typos per page on average; $P(\text{no typos})=e^{-3}\approx 0.050$, $P(\text{at least one})\approx 0.95$.

TERMS

λ – average number of events per interval (rate $\times$ time)
k – number of events observed
r, t – rate per unit, interval length
e – Euler's number A2.13 §04 Algebra II; $k!$ factorial S.8
Sum to 1 – by the series for e^{λ} C2.22 §08 Calculus II
Waiting times – between events are exponential S.19
n, p – binomial trials and success probability, with $\lambda=np$
X_1, X_2; λ_1, λ_2 – independent Poisson counts; their rates



CONTEXT

- **Origin.** Siméon-Denis Poisson (1837); famously fit to Prussian cavalry deaths from horse kicks (Bortkiewicz, 1898).
- **Use.** Arrivals, defects, calls per hour, rare events.
- **Data science.** Modeling event counts (clicks, support tickets, server errors per minute); Poisson regression for count targets; alerting when counts are improbably high.
- **Physical application.** The number of radioactive decays detected by a Geiger counter per second is Poisson.

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Continuous Distributions

S.19 Continuous Distributions: Uniform & Exponential

The uniform spreads probability evenly over an interval; the exponential models the waiting time until the next event of a Poisson process.

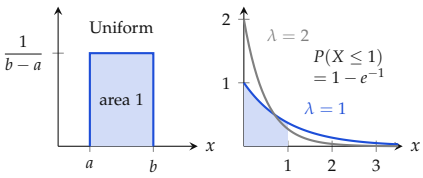
Uniform PDF	$f(x) = \dfrac{1}{b-a}, \quad a \le x \le b$	Flat density; probability is proportional to interval length.
Uniform CDF	$F(x) = \dfrac{x-a}{b-a}$	Fraction of the interval already passed.
Uniform mean, variance	$\mu = \dfrac{a+b}{2}, \qquad \sigma^2 = \dfrac{(b-a)^2}{12}$	Midpoint; the 12 comes from integrating x^2.
Exponential PDF	$f(x) = \lambda e^{-\lambda x}, \quad x \ge 0$	Short waits are most likely; decays at rate λ.
Exponential CDF	$F(x) = 1 - e^{-\lambda x}$	Probability the event has happened by time x.
Exponential mean, variance	$\mu = \dfrac{1}{\lambda}, \qquad \sigma^2 = \dfrac{1}{\lambda^2}$	Mean wait is one over the rate; SD equals the mean.
Memoryless property	$P(X > s+t \mid X > s) = P(X > t)$	Having waited s does not change the remaining wait.

STATED-EQUATION *Uniform on a to b: f of x equals one over b minus a; F of x is x minus a over b minus a; the mean is a plus b over two; the variance is b minus a squared over twelve. Exponential with rate lambda: f of x equals lambda e to the minus lambda x; F of x equals one minus e to the minus lambda x; the mean is one over lambda and the variance one over lambda squared. It is memoryless. Memoryless means the probability of lasting t more units, given it has already lasted s, equals the probability of lasting t from the start.*

MEMORY AID Exponential: mean $= 1/\lambda$; "memoryless" — the past doesn't matter.

TERMS

a, b – endpoints of the interval; every value equally likely
λ – rate (events per unit time), same as Poisson S.18
$F(x)$ – CDF $= \int_0^x f$ C1.32 §07 Calculus I, S.12
Memoryless – conditional probability S.9; $s, t \ge 0$ times
Mean – $E[X]$ S.13, computed by parts C2.1 §08 Calculus II
Sum of k exponentials – Gamma distribution D.9 §18 Graph Gallery
s, t – elapsed time; additional time



CONTEXT

- **Origin.** The exponential arises as the waiting time between Poisson events (Erlang, early 1900s).
- **Use.** Waiting times, lifetimes of components, random numbers.
- **Data science.** Uniform draws power random search and Monte Carlo O.27 §14 Optimization; exponential models time-to-churn, inter-arrival times in queueing, and survival analysis baselines.
- **Physical application.** The waiting time until the next radioactive decay is exponential; its memorylessness is why a half-life never "resets."

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S.20 Normal Distribution & the CDF

The bell curve, fully described by its mean and standard deviation. Probabilities come from the standard normal CDF Φ after converting to z-scores.

Normal PDF	$f(x) = \dfrac{1}{\sigma\sqrt{2\pi}}\, e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	Symmetric, peaked at μ, inflection points at $\mu \pm \sigma$.
Notation	$X \sim N(\mu, \sigma^2)$	Second parameter is the *variance* (software often takes the SD).
Normal CDF	$F(x) = P(X \le x) = \displaystyle\int_{-\infty}^{x} f(t)\, dt$	No closed form; tabulated as Φ.
Standardizing	$Z = \dfrac{X-\mu}{\sigma} \sim N(0,1)$	Every normal becomes the standard normal S.4.
Interval probability	$P(a \le X \le b) = \Phi(z_b) - \Phi(z_a)$	Convert both ends to z, subtract the CDF values.
Symmetry	$\Phi(-z) = 1 - \Phi(z)$	Left-tail areas from right-tail tables.
Empirical rule	$P(\lvert Z\rvert \le 1,2,3) \approx 0.68,\ 0.95,\ 0.997$	Quick mental probabilities.
Sum of independent normals	$X + Y \sim N(\mu_X + \mu_Y,\ \sigma_X^2 + \sigma_Y^2)$	Means add, variances add S.13.

STATED-EQUATION *f of x equals one over sigma root two pi, times e to the minus x minus mu squared over two sigma squared. X is normal with mean mu and variance sigma squared. The CDF F of x is the integral of f from minus infinity to x. Z equals X minus mu over sigma is standard normal. P of a less than or equal X less than or equal b equals Phi of z b minus Phi of z a. Phi of minus z equals one minus Phi of z. About sixty-eight, ninety-five, and ninety-nine point seven percent of values lie within one, two, and three standard deviations of the mean. A sum of independent normals is normal, with the means adding and the variances adding.*

MEMORY AID 68–95–99.7: within 1, 2, 3 SDs.

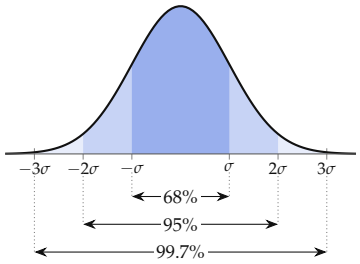
METHOD

1. Standardize both endpoints: $z = \frac{x-\mu}{\sigma}$.
2. Look up Φ (table below or software) and subtract.
3. Example: $X \sim N(100, 15^2)$; $P(85 \le X \le 130) = \Phi(2) - \Phi(-1) = 0.977 - 0.159 = 0.818$.

z	0	0.5	1	1.282	1.645	1.96	2.326	2.576
$\Phi(z)$	0.500	0.691	0.841	0.900	0.950	0.975	0.990	0.995

TERMS

μ, σ – mean and standard deviation S.2
PDF $f(x)$ – density: height of the curve; area = probability S.12
CDF $F(x)$ – accumulated area C1.32 §07 Calculus I; $f = F'$
$\Phi(z)$ – standard normal CDF, z from S.4; z_a, z_b standardized ends
e, π – constants A2.13 §04 Algebra II; total area $= 1$ C2.6 §08 Calculus II
X, Y; Z – normal random variables; standard normal
z_a, z_b – z-scores of a and b



CONTEXT

- **Origin.** De Moivre (1733) as a binomial limit; Gauss (1809) for measurement error.
- **Use.** Heights, measurement error, test scores, sums of many small effects.
- **Data science.** Gaussian noise in linear regression, Gaussian processes, weight initialization in neural nets, and the sampling distribution of almost every estimator S.24.
- **Physical application.** Measurement errors on a precision scale are approximately normal, which is why repeated weighings are averaged.

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S.21 Degrees of Freedom

Degrees of freedom (df) count how many values in a calculation are free to vary once its constraints are fixed. Each parameter you estimate from the same data uses one up, which is why many formulas divide by $n-1$ or $n-p$ instead of n.

General rule	$df = n - (\text{number of estimated parameters / constraints})$	Start with the number of observations; subtract one for every quantity estimated from them.
Sample variance	$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$	The deviations must sum to 0, so only $n-1$ of them are free (Bessel's correction S.2).
One-sample t	$df = n - 1$	One parameter ($\bar{x}$) is estimated before s.
Two-sample t (pooled)	$df = n_1 + n_2 - 2$	One mean estimated per group.
χ^2 goodness of fit	$df = k - 1 - m$	k categories, minus 1 because counts sum to n, minus m fitted parameters S.32.
χ^2 independence	$df = (r-1)(c-1)$	Row and column totals are fixed by the data.
Regression residuals	$df = n - p - 1$	p slopes plus one intercept estimated S.40.
ANOVA	$df_{\text{between}} = k - 1, \quad df_{\text{within}} = N - k$	k group means; N total observations S.33.

STATED-EQUATION *Degrees of freedom equal the number of observations minus the number of estimated parameters or constraints. The sample variance divides by n minus one. A one-sample t test has n minus one degrees of freedom, a pooled two-sample t test n one plus n two minus two. Chi-square goodness of fit has k minus one minus m; a test of independence has r minus one times c minus one. Regression residuals have n minus p minus one; ANOVA has k minus one between groups and N minus k within groups.*

MEMORY AID "One estimate, one freedom lost."

METHOD

1. Count the observations (or cells) n.
2. Count every constraint the data must satisfy and every parameter you estimated from the same data before computing the statistic.
3. Subtract. Example: 3 numbers with mean 5: pick 2 and 6 freely; the third must be 7. So $df = 3 - 1 = 2$.

DECISION POINT Which n to start from? Use the number of independent pieces of information entering the statistic: observations for t and variance, categories (cells) for χ^2, observations for regression. If unsure, ask "after fixing the totals and estimates, how many values could I still change at will?"

TERMS

df, ν – degrees of freedom (Greek nu GR.13 §02 Greek Alphabet)
n – number of observations; N total across groups
$\bar{x}$ – sample mean S.1; s^2 sample variance
k – number of categories (or groups); m fitted parameters
r, c – rows and columns of a contingency table
p – number of predictors in a regression
Effect – small df gives heavier-tailed t and χ^2 distributions; as $df \to \infty$, $t \to N(0,1)$
m – number of parameters estimated from the data
n_1, n_2; N – two sample sizes; total observations over all groups

CONTEXT

- **Origin.** Introduced by R. A. Fisher (1922) while correcting Pearson's chi-square test.
- **Use.** Choosing the right t, χ^2 or F table row; unbiased variance.
- **Data science.** Model complexity: a model with as many parameters as data points has 0 residual df and fits noise perfectly (overfitting ML.4 §15 Machine Learning); adjusted R^2 and AIC penalize spent df.
- **Physical application.** A rigid body in 3-D space has 6 mechanical degrees of freedom (3 translations, 3 rotations); same idea: independent ways to vary.

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S.22 Sampling Distributions: χ^2, t & F

Three distributions built from normal samples that power classical inference: χ^2 for variances, t for means with unknown σ, and F for ratios of variances.

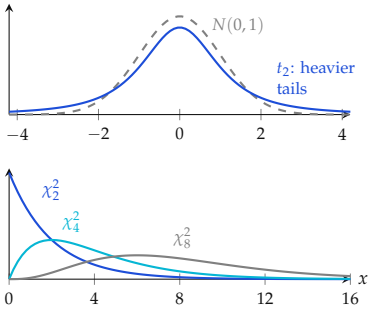
Chi-square	$Q = \sum_{i=1}^{k} Z_i^2 \sim \chi^2_k, \quad \mu = k,\ \sigma^2 = 2k$	Sum of k squared independent standard normals; right-skewed D.11 §18 Graph Gallery.
Sample variance	$\dfrac{(n-1)s^2}{\sigma^2} \sim \chi^2_{n-1}$	For normal data; gives confidence intervals for σ^2.
Student's t	$T = \dfrac{Z}{\sqrt{V/k}} \sim t_k$	Normal over the root of an independent χ^2_k/k; heavier tails D.7 §18 Graph Gallery.
t-statistic	$\dfrac{\bar{X}-\mu}{s/\sqrt{n}} \sim t_{n-1}$	Replacing σ by s costs extra uncertainty, hence the fatter tails S.30.
F distribution	$F = \dfrac{V_1/d_1}{V_2/d_2} \sim F_{d_1,d_2}$	Ratio of two scaled independent χ^2's: compares variances (ANOVA S.33).
Relationships	$t_k^2 = F_{1,k}, \qquad t_k \to N(0,1)\ (k \to \infty)$	By $k \approx 30$ the t is close to normal.

STATED-EQUATION *Chi-square with k degrees of freedom is the sum of k squared standard normals, mean k, variance two k. n minus one s squared over sigma squared is chi-square with n minus one degrees of freedom. Student t is Z over the square root of V over k. The t statistic x bar minus mu over s over root n has n minus one degrees of freedom. F is the ratio of two chi-squares each divided by its degrees of freedom. t squared is F with one and k degrees of freedom.*

MEMORY AID χ^2 = squares; t = mean with estimated SD; F = ratio of variances.

TERMS

Z, Z_i – independent standard normals S.20
k, d_1, d_2 – degrees of freedom (df): independent values free to vary S.21
V, V_1, V_2 – independent chi-square variables
s, σ – sample and population SD S.2
n – sample size; $\bar{X}$ sample mean
χ^2_k, t_k, F_{d_1,d_2} – chi-square, Student t, and F distributions with the given degrees of freedom S.21



CONTEXT

- **Origin.** Helmert (1876) and Pearson (1900) for χ^2; Gosset ("Student", 1908) for t; Fisher and Snedecor (1920s–1934) for F.
- **Use.** Critical values for tests and intervals on means and variances.
- **Data science.** Regression output: coefficient t-values, the overall F-statistic S.41, and χ^2 tests on contingency tables S.32.
- **Physical application.** Comparing the precision (variance) of two measuring instruments uses an F-test.

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Sampling & Estimation

S.23 Law of Large Numbers & Probability Inequalities

Averages of many independent draws settle down to the true mean. Markov's and Chebyshev's inequalities bound tail probabilities using only the mean and variance.

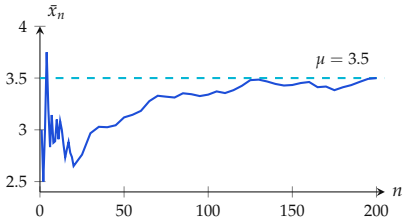
Mean of the sample mean	$E[\bar{X}] = \mu$	The sample mean is unbiased S.25.
Variance of the sample mean	$\mathrm{Var}(\bar{X}) = \dfrac{\sigma^2}{n}$	Averaging n independent values divides the variance by n S.13.
Weak law of large numbers	$P\big(\lvert\bar{X}_n - \mu\rvert > \varepsilon\big) \to 0 \quad (n \to \infty)$	For any tolerance ε, the average eventually stays close to μ.
Markov's inequality	$P(X \geq a) \leq \dfrac{E[X]}{a} \quad (X \geq 0)$	A non-negative variable rarely exceeds a large multiple of its mean.
Chebyshev's inequality	$P\big(\lvert X - \mu\rvert \geq k\sigma\big) \leq \dfrac{1}{k^2}$	Any distribution: at least 75% within 2σ, 89% within 3σ.
Monte Carlo estimate	$\dfrac{1}{n}\sum_{i=1}^{n} g(X_i) \to E[g(X)]$	The LLN justifies estimating expectations by simulation O.27 §14 Optimization.

STATED-EQUATION *The expected value of x bar is mu and its variance is sigma squared over n. The weak law: the probability that x bar n differs from mu by more than epsilon goes to zero. Markov: P of X at least a is at most E X over a for non-negative X. Chebyshev: P of being at least k sigma from mu is at most one over k squared. Averages of g of X i converge to E of g of X.*

MEMORY AID LLN: the average converges. CLT S.24: and here is the shape of its error.

TERMS

$\bar{X}_n$ – mean of n independent, identically distributed (i.i.d.) draws
μ, σ^2 – population mean and variance
ε – any small positive tolerance
$a > 0, k > 0$ – threshold, number of SDs
g – any function with finite expectation
Gambler's fallacy – the LLN does not "correct" short runs; it dilutes them



CONTEXT

- **Origin.** Jacob Bernoulli (1713) proved the first LLN; Chebyshev (1867), Markov (1880s); Kolmogorov's strong law (1930s).
- **Use.** Why frequencies estimate probabilities; insurance pooling.
- **Data science.** Monte Carlo and simulation, SGD averaging, why more data reduces variance; Chebyshev gives distribution-free anomaly bounds.
- **Physical application.** A casino's average profit per bet converges to the house edge over millions of bets.

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S.24 Central Limit Theorem & Standard Error

Whatever the population's shape, the mean of a large random sample is approximately normal, centered at μ with spread $\sigma/\sqrt{n}$ (the standard error).

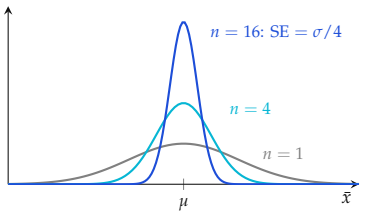
Central limit theorem	$\bar{X} \approx N\left(\mu, \dfrac{\sigma^2}{n}\right)$ $(n \text{ large})$	Holds for almost any population with finite variance; $n \geq 30$ is a rule of thumb.
Standardized mean	$z = \dfrac{\bar{x} - \mu}{\sigma/\sqrt{n}}$	Number of standard errors between the sample mean and μ.
SE of a mean	$SE_{\bar{x}} = \dfrac{\sigma}{\sqrt{n}} \approx \dfrac{s}{\sqrt{n}}$	Halving the error needs four times the data.
SE of a proportion	$SE_{\hat{p}} = \sqrt{\dfrac{p(1-p)}{n}}$	A proportion is a mean of 0/1 values S.15.
SE of a difference of means	$SE_{\bar{x}_1 - \bar{x}_2} = \sqrt{\dfrac{s_1^2}{n_1} + \dfrac{s_2^2}{n_2}}$	Variances (not SEs) of independent estimates add.
SE of a difference of proportions	$SE_{\hat{p}_1 - \hat{p}_2} = \sqrt{\dfrac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \dfrac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$	Basis of A/B-test intervals S.31.

STATED-EQUATION *X bar is approximately normal with mean mu and variance sigma squared over n, for large n. z equals x bar minus mu over sigma over root n. The standard error of x bar equals sigma over root n. The standard error of p hat equals the square root of p times one minus p over n. The SE of a difference adds the two variances under one square root.*

MEMORY AID "Root-n law": four times the data halves the error.

TERMS

$\bar{X}$ – sample mean, itself a random variable S.13
σ, s – population SD and its estimate S.2
n, n_1, n_2 – sample sizes; rule of thumb $n \geq 30$ (more if very skewed)
SE – standard deviation of a statistic's sampling distribution
$p, \hat{p}$ – true and sample proportions
Requires – independent observations; LLN S.23



CONTEXT

- **Origin.** De Moivre (1733), Laplace (1810); named by Pólya (1920).
- **Use.** Justifies normal-based inference for averages and proportions.
- **Data science.** Why A/B tests, confidence intervals S.27, and bootstrap estimates S.36 work; sample-size planning for experiments; SGD gradient noise shrinks like $1/\sqrt{\text{batch size}}$.
- **Physical application.** Averaging 100 noisy thermometer readings cuts the random error by a factor of $\sqrt{100} = 10$.

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S.25 Point Estimators: Bias, Variance & MSE

An estimator is a rule that turns data into a guess $\hat{\theta}$ for a parameter θ. Its quality is judged by bias, variance and their combination, the mean squared error.

Bias	$\text{Bias}(\hat{\theta}) = E[\hat{\theta}] - \theta$	Systematic error; unbiased if 0 ($\bar{x}$ for μ, s^2 for σ^2).
Mean squared error	$\text{MSE}(\hat{\theta}) = E\left[(\hat{\theta} - \theta)^2\right] = \text{Var}(\hat{\theta}) + \text{Bias}^2$	Total error: a little bias can pay for a big variance cut ML.4 §15 Machine Learning.
Standard error	$SE(\hat{\theta}) = \sqrt{\text{Var}(\hat{\theta})}$	Typical sampling error of the estimate S.24.
Consistency	$\hat{\theta}_n \to \theta \;\text{ as } n \to \infty$	More data drives the estimate to the truth (e.g. when bias and variance both $\to 0$).
Cramér–Rao bound	$\text{Var}(\hat{\theta}) \geq \dfrac{1}{n\, I(\theta)}$	No unbiased estimator beats this variance; efficient estimators reach it.
Fisher information	$I(\theta) = E\left[\left(\frac{\partial}{\partial\theta} \ln f(X \mid \theta)\right)^2\right]$	How sharply one observation pins down θ C3.K1 §10 Calculus III.

STATED-EQUATION *The bias of theta hat is its expected value minus theta. The MSE equals the variance plus the bias squared. The standard error is the square root of the variance. A consistent estimator converges to theta as n grows. Cramer-Rao: the variance of an unbiased estimator is at least one over n times the Fisher information. The Fisher information is the expected square of the derivative of the log density with respect to theta.*

MEMORY AID MSE = variance + bias^2: the dartboard picture (spread vs off-center).

TERMS

θ – true parameter (fixed, unknown)
$\hat{\theta}$ – estimator: a statistic, random because the data are
E, Var – expectation, variance S.13
n – sample size
$f(x \mid \theta)$ – model PMF/PDF S.26
Example – $\hat{\sigma}^2 = \frac{1}{n}\sum(x_i - \bar{x})^2$ is biased low by $\frac{\sigma^2}{n}$
$I(\theta)$ – Fisher information per observation
MSE; SE – mean squared error; standard error

CONTEXT

- **Origin.** Gauss and Laplace on least error; Fisher (1922) defined consistency, efficiency and information; Cramér and Rao (1945).
- **Use.** Comparing estimation methods; justifying $n - 1$ S.2.
- **Data science.** Shrinkage and regularization trade bias for variance ML.8 §15 Machine Learning; MSE is the default regression loss ML.19 §15 Machine Learning; reporting an estimate means reporting its SE.
- **Physical application.** Averaging repeated measurements reduces variance; calibrating an instrument removes bias.

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S.26 Maximum Likelihood Estimation (MLE)

Choose the parameter value under which the observed data would have been most probable. It is the standard way to fit a probability model.

Likelihood	$L(\theta) = \prod_{i=1}^{n} f(x_i \mid \theta)$	Probability (density) of the whole i.i.d. sample, viewed as a function of θ.
Log-likelihood	$\ell(\theta) = \ln L(\theta) = \sum_{i=1}^{n} \ln f(x_i \mid \theta)$	Same maximizer; sums are easier to differentiate and do not underflow.
MLE condition	$\hat{\theta}_{\text{MLE}} = \arg\max_{\theta} \ell(\theta) : \ \frac{d\ell}{d\theta} = 0$	Set the score to zero; check it is a maximum.
Known MLEs	$\hat{p} = \bar{x}, \quad \hat{\lambda}_{\text{Pois}} = \bar{x}, \quad \hat{\lambda}_{\text{Exp}} = \frac{1}{\bar{x}}, \quad \hat{\mu} = \bar{x}$	Bernoulli, Poisson, exponential and normal mean.
Normal variance MLE	$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^{n} (x_i - \bar{x})^2$	Divides by n: slightly biased, consistent S.25.
Asymptotic normality	$\hat{\theta} \approx N\left(\theta, \ \frac{1}{n\, I(\theta)}\right)$	For large n the MLE is unbiased and efficient; gives Wald intervals.

STATED-EQUATION *The likelihood L of theta is the product over the data of f of x i given theta. The log-likelihood, ell of theta, is the sum of the logs. The maximum likelihood estimate theta hat is the value that maximizes ell, found by setting d ell d theta equal to zero. For large n, theta hat is approximately normal with variance one over n I of theta. The MLEs of a Bernoulli p, a Poisson lambda, and a normal mean are all x bar, and the exponential rate's MLE is one over x bar. The normal variance MLE divides the sum of squared deviations by n, not n minus one.*

MEMORY AID "Which parameter makes the data I saw most probable?" Take logs to turn products into sums.

METHOD

1. Write the likelihood of the sample and take logs.
2. Differentiate with respect to θ C1.23 §07 Calculus I and set to 0.
3. Solve, then confirm a maximum (second derivative < 0).
4. Example: exponential data 1, 2, 3: $\ell = n \ln \lambda - \lambda \sum x_i = 3 \ln \lambda - 6\lambda$; $\frac{3}{\lambda} - 6 = 0 \Rightarrow \hat{\lambda} = 0.5 = 1/\bar{x}$.

TERMS

$f(x \mid \theta)$ – PMF or PDF with parameter θ S.16–S.20
Product – data assumed independent S.9; $\prod$ M.54 §01 Math Signs
Log – monotone, so same maximizer; sums are easier A2.14 §04 Algebra II
arg max – the θ achieving the maximum; solve analytically or numerically O.13 §14 Optimization
$I(\theta)$ – Fisher information S.25
$L(\theta)$, $\ell(\theta)$; $\hat{\theta}$ – likelihood and log-likelihood; the estimate
$\bar{x}$; n – sample mean; sample size

CONTEXT

- **Origin.** R. A. Fisher (1912–1922); earlier ideas from Gauss and Lagrange.
- **Use.** The standard way to fit parameters of probability models.
- **Data science.** Minimizing MSE is MLE under Gaussian noise; minimizing log-loss is MLE for Bernoulli labels ML.2 §15 Machine Learning; most `fit()` methods are MLE or regularized MLE (MAP S.37).
- **Physical application.** Estimating the decay rate of a radioactive sample from measured decay times.

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S.27 Confidence Intervals

A range of plausible values for a parameter: estimate ± critical value × standard error. The confidence level is the long-run share of such intervals that capture the truth.

z-interval for a mean	$\bar{x} \pm z^* \dfrac{\sigma}{\sqrt{n}}$	When σ is known (rare) or n is very large.
t-interval for a mean	$\bar{x} \pm t^*_{n-1} \dfrac{s}{\sqrt{n}}$	The usual case: σ estimated by s S.22.
Interval for a proportion	$\hat{p} \pm z^* \sqrt{\dfrac{\hat{p}(1-\hat{p})}{n}}$	Wald interval; needs $n\hat{p} \geq 10$ and $n(1-\hat{p}) \geq 10$ (else Wilson).
Difference of two means	$(\bar{x}_1 - \bar{x}_2) \pm t^* \sqrt{\dfrac{s_1^2}{n_1} + \dfrac{s_2^2}{n_2}}$	If it excludes 0, the two-sided test rejects at the same α.
Sample size for a mean	$n = \left(\dfrac{z^* \sigma}{E}\right)^2$	Smallest n for margin of error E; round up.
Sample size for a proportion	$n = \dfrac{(z^*)^2\, p(1-p)}{E^2}$	Use $p = 0.5$ when unknown (worst case): ±3% at 95% needs $n \approx 1068$.

STATED-EQUATION *x bar plus or minus z star sigma over root n; x bar plus or minus t star with n minus one degrees of freedom, times s over root n; p hat plus or minus z star times the square root of p hat times one minus p hat over n. For a difference of means, the difference plus or minus t star times the combined standard error. Required sample size n equals z star sigma over E, squared. For a proportion, n equals z star squared times p times one minus p, over E squared.*

MEMORY AID Estimate ± (critical value)×(standard error).

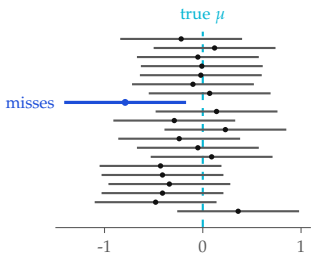
METHOD

1. Pick the confidence level; find the critical value (z^* from S.20, or t^* with $n-1$ df).
2. Compute the standard error S.24.
3. Margin E = critical value × SE; interval = estimate $\pm E$.
4. Interpret in context. Example: $n = 16$, $\bar{x} = 50$, $s = 8$: $SE = 2$, $t^*_{15} = 2.131$, 95% CI $= 50 \pm 4.26 = (45.7,\ 54.3)$.

DECISION POINT **z or t?** Proportions use z. Means use t whenever σ is estimated from the data (almost always); the difference vanishes for large n. **Meaning:** "95% confident" describes the method: 95% of intervals built this way cover μ. It is *not* a 95% probability that this particular fixed interval contains μ (that is a Bayesian credible interval S.37).

TERMS

z^* – critical value: 1.645 (90%), 1.96 (95%), 2.576 (99%)
t^*_{n-1} – critical value from the t distribution S.30
$\hat{p}$ – sample proportion; p planning guess
E – margin of error (half-width) = critical value × SE
s_1, s_2, n_1, n_2 – SDs and sizes of the two samples
$\bar{x}$; σ, s; n – sample mean; population and sample SD; sample size
p – planning guess for the proportion (0.5 is the safe choice)
df – degrees of freedom S.21



CONTEXT

- **Origin.** Jerzy Neyman (1937).
- **Use.** Reporting estimates with uncertainty; polling margins of error.
- **Data science.** Error bars on dashboards, uplift estimates from experiments, and sample-size calculators; always report the interval, not just the point estimate.
- **Physical application.** Reporting a lab measurement as 9.81 ± 0.02 m/s^2 is a confidence interval.

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Hypothesis Testing

S.28 Hypothesis Testing, p-Values & Errors

Assume no effect (H_0), measure how far the data are from it in standard errors, and ask how surprising that distance would be if H_0 were true.

Hypotheses	$H_0:\ \mu = \mu_0 \quad \text{vs} \quad H_a:\ \mu \neq \mu_0 \ (\text{or } >, <)$	H_0 is the default "no effect"; H_a is what you hope to show.
Test statistic (z)	$z = \dfrac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$	(Estimate − null value) / standard error S.24.
Two-sided p-value	$p = P(\lvert Z\rvert \geq \lvert z\rvert \mid H_0) = 2(1 - \Phi(\lvert z\rvert))$	Chance of a result at least this extreme, in either direction, if H_0 holds.
One-sided p-value	$p = P(Z \geq z \mid H_0) = 1 - \Phi(z)$	For $H_a: \mu > \mu_0$; use $\Phi(z)$ for $H_a: \mu < \mu_0$.
Decision rule	reject H_0 if $p < \alpha \iff \lvert z\rvert > z^*_{\alpha/2}$	p-value and critical-value approaches always agree.
Error rates	$\alpha = P(\text{reject } H_0 \mid H_0), \quad \beta = P(\text{keep } H_0 \mid H_a), \quad \text{power} = 1 - \beta$	Type I (false alarm), Type II (miss), detection rate S.29.

STATED-EQUATION *H naught: mu equals mu naught, versus H a: mu not equal to mu naught. z equals x bar minus mu naught over sigma over root n. The two-sided p-value is the probability of a statistic at least as extreme as observed, two times one minus Phi of the absolute z. Reject H naught if p is less than alpha. Alpha is P of rejecting H naught when it is true; beta is P of failing to reject when H a is true; power equals one minus beta. The one-sided p-value is one minus Phi of z. Equivalently, reject when the absolute z exceeds the critical value z star alpha over two.*

MEMORY AID "If p is low, the null must go." Type I = false alarm; Type II = missed detection.

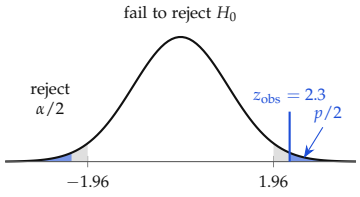
METHOD

1. State H_0, H_a and α (and one- or two-sided) *before* looking at the data.
2. Check conditions (random, independent, normal or large n).
3. Compute the test statistic and its p-value.
4. Compare p to α; conclude in context with an effect size and CI. Example: $\mu_0 = 500$, $\sigma = 10$, $n = 25$, $\bar{x} = 504$: $z = \frac{4}{2} = 2$, $p = 2(1 - 0.977) = 0.046 < 0.05$, reject H_0.

DECISION POINT **One- or two-tailed?** Two-tailed is the default: use it whenever a difference in either direction would matter or be reported. Use one-tailed only if the direction was fixed in advance for a substantive reason and an effect in the other direction would lead to the same action as no effect. Never switch after seeing the data: that doubles the real α.

TERMS

H_0, H_a – null (no effect) and alternative hypotheses
μ_0 – value of the parameter claimed by H_0
z, Φ – test statistic S.4, standard normal CDF S.20
p-value – $P(\text{data this extreme} \mid H_0)$, *not* $P(H_0 \mid \text{data})$ S.11
α – significance level; $z^*_{\alpha/2}$ its critical value
β, power – miss rate; detection rate (rises with n)
$z^*_{\alpha/2}$ – critical value cutting off $\alpha/2$ in each tail
Z; $\bar{x}, \sigma, n$ – standard normal; sample mean, SD, size



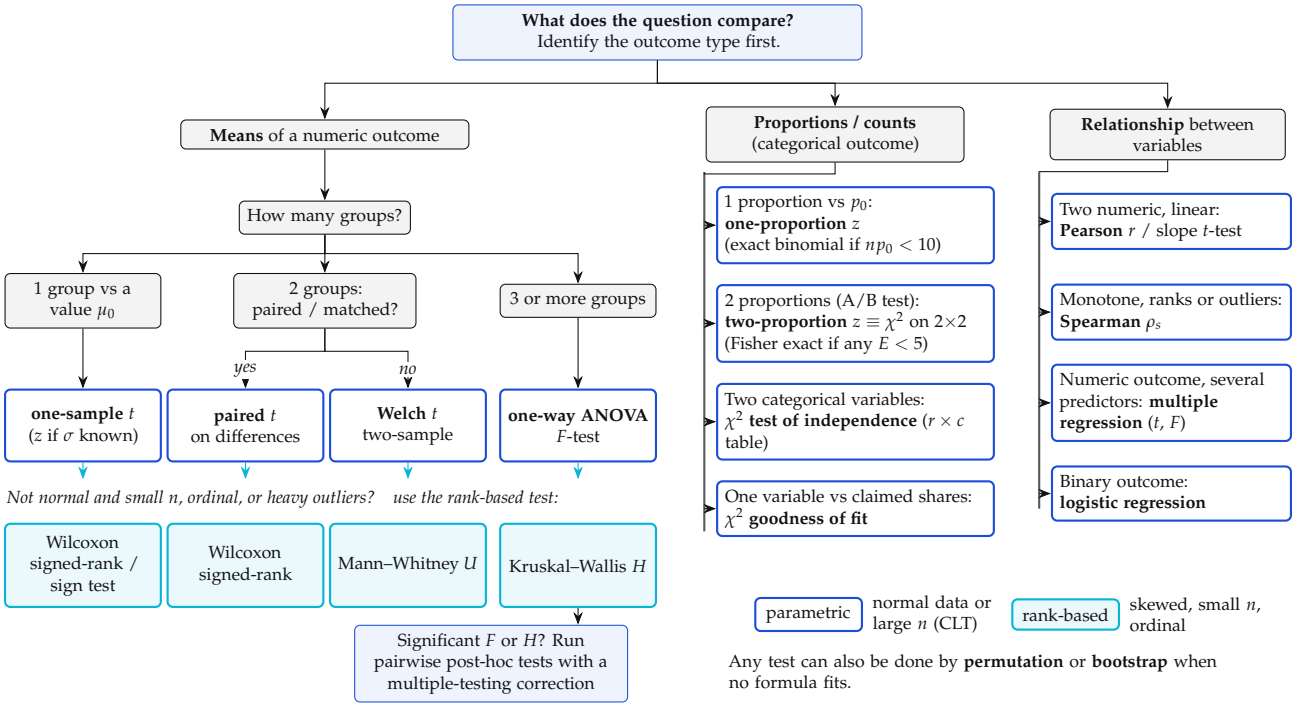
CONTEXT

- **Origin.** Fisher's significance tests (1925); Neyman–Pearson framework (1933).
- **Use.** Scientific claims, quality control, clinical trials.
- **Data science.** A/B testing; multiple comparisons inflate false positives, so use Bonferroni or FDR control S.35; statistical significance is not practical significance S.29.
- **Physical application.** A factory tests whether a machine's mean fill volume has drifted from 500 mL.

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Which Test? Which Distribution? | *Key Topic S.K1*

Most applied statistics questions are answered by recognizing the situation. For a test, the deciding facts are the **type of outcome** (numeric, categorical, relationship), the **number of groups**, whether observations are **paired**, and whether the data are **roughly normal or the sample large**. For a probability model, it is the **story** that generated the numbers: counting successes, counting events, or waiting.



Every test follows the same five steps

1. Hypotheses and α. Write H_0 (no effect, equality) and H_a; choose one- or two-sided and α (usually 0.05) *before* seeing data S.28.

2. Conditions. Random sampling or assignment; independence (sample $< 10\%$ of population); normal shape or large n S.24; expected counts ≥ 5 for χ^2.

3. Test statistic. Almost always $\dfrac{\text{estimate} - \text{null value}}{\text{standard error}}$ (z, t), or a ratio of variability (F, χ^2).

4. p-value. Area beyond the statistic under its null distribution ($N(0,1)$, t_{df}, χ^2_{df}, F_{d_1,d_2}; df S.21).

5. Conclusion. Reject H_0 if $p < \alpha$. State it in context, with an effect size and a confidence interval S.29; "fail to reject" is not "accept".

DECISION POINT **One- or two-tailed?** Two-tailed unless the direction was fixed in advance and the opposite direction is irrelevant. **z or t?** z for proportions or a known σ; t whenever σ is estimated by s S.30. **Paired or independent?** Paired if each value in one group has a natural partner (same subject before/after, same fold, twins); otherwise independent. **Parametric or rank-based?** Parametric if roughly normal or $n \gtrsim 30$ per group; rank-based (or bootstrap S.36) for small, skewed, ordinal or outlier-heavy data S.34. **Many tests?** Correct with Bonferroni or Benjamini–Hochberg S.35.

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Which distribution? Match the story

Situation (the story)	Distribution	Mean	Example
One yes/no trial	Bernoulli(p) S.15	p	one visitor converts
Successes in n independent trials, same p	Binomial(n, p) S.16	np	7 heads in 10 flips
Successes in n draws *without* replacement	Hypergeometric S.17	nK/N	aces in a 5-card hand
Trials until the first success	Geometric(p) S.15	$1/p$	calls until a sale
Trials until the rth success; over-dispersed counts	Negative binomial	r/p	games to win 3
Count of events in a fixed time/space at a constant rate	Poisson(λ) S.18	λ	emails per hour
Waiting time between Poisson events	Exponential(λ) S.19	$1/\lambda$	time to next request
Waiting time until the kth event	Gamma(k, λ) D.9 §18 Graph Gallery	k/λ	time to 3 failures
Every value in $[a, b]$ equally likely	Uniform(a, b) S.19	$\frac{a+b}{2}$	random arrival in a slot
Sum or average of many small independent effects	Normal(μ, σ^2) S.20	μ	heights, measurement error
Product of many positive factors (log is normal)	Log-normal D.12 §18 Graph Gallery	$e^{\mu+\sigma^2/2}$	incomes, file sizes
Mean standardized with an estimated SD	t_{n-1} S.22	0	t-tests, CIs
Sum of squared standard normals; variances; count tables	χ^2_k S.22	k	χ^2 tests
Ratio of two variance estimates	F_{d_1,d_2} S.22	≈ 1	ANOVA
A probability or proportion itself (prior/posterior)	Beta(α, β) S.37	$\frac{\alpha}{\alpha+\beta}$	conversion rate belief

Worked mini-examples

Same users, before vs after a redesign, time-on-page, $n = 40$: paired t on the 40 differences S.30.

Control vs treatment conversion, 50/500 vs 70/500: two-proportion z, $z \approx 1.95$, $p \approx 0.05$ S.31.

Order value in 3 regions, heavily skewed, $n = 12$ each: Kruskal–Wallis, not ANOVA S.34.

Device type × churned (yes/no): χ^2 test of independence on the 3×2 table S.32.

Server errors per minute, alerting on spikes: Poisson with λ from history S.18.

Does ad spend predict sales? Regression slope t-test; correlation alone does not show causation S.43.

Common mistakes

- ► Treating paired data as two independent samples (loses power).
- ► Using a pooled-variance t by default; Welch is safer S.30.
- ► Reading p as the probability H_0 is true S.11.
- ► Running many tests and reporting the one with $p < 0.05$ S.35.
- ► χ^2 with expected counts below 5: use Fisher's exact test.
- ► Equating "not significant" with "no effect": check power S.29.

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S.29 Errors, Power, Sample Size & Effect Size

A test can be wrong in two ways. Power is the chance of detecting a real effect of a given size; planning n to reach adequate power, and reporting effect sizes, separates real findings from noise.

Power of a z-test	$1-\beta=\Phi\left(\frac{\lvert\delta\rvert\sqrt{n}}{\sigma}-z^*_{\alpha/2}\right)$	Two-sided, ignoring the far tail: grows with effect size and $\sqrt{n}$, shrinks with noise.
Sample size (one sample)	$n=\left(\frac{(z^*_{\alpha/2}+z^*_{\beta})\,\sigma}{\delta}\right)^2$	n needed to detect a shift δ with power $1-\beta$; round up.
Sample size per group (two groups)	$n=\frac{2\,(z^*_{\alpha/2}+z^*_{\beta})^2\sigma^2}{\delta^2}$	For $\alpha=0.05$, power 0.8: $n\approx\frac{16\sigma^2}{\delta^2}$ (Lehr's rule).
Cohen's d	$d=\frac{\bar{x}_1-\bar{x}_2}{s_p}$	Difference in SD units: 0.2 small, 0.5 medium, 0.8 large.
Pooled SD	$s_p=\sqrt{\frac{(n_1-1)s_1^2+(n_2-1)s_2^2}{n_1+n_2-2}}$	Weighted average of the two group variances.

STATED-EQUATION *Power equals Phi of the absolute delta root n over sigma, minus z star. The required n equals z alpha over two plus z beta, times sigma over delta, all squared; for two groups, two times that per group. Cohen's d equals the difference in means over the pooled standard deviation. The pooled standard deviation is the square root of n one minus one times s one squared plus n two minus one times s two squared, over n one plus n two minus two.*

MEMORY AID Power rises with: bigger effect, bigger n, less noise, larger α. "16 over d^2" per group.

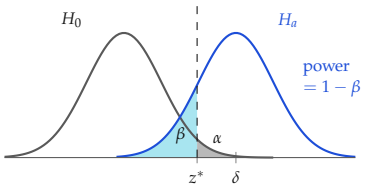
METHOD

1. Fix α (0.05) and target power (0.8, so $z^*_{\beta}=0.84$).
2. Choose the smallest effect δ worth detecting and estimate σ (pilot data).
3. Compute n and round up. Example: two groups, $\sigma=10$, $\delta=5$: $n=\frac{2(1.96+0.84)^2\cdot 100}{25}\approx 63$ per group.

	H_0 true (no effect)	H_0 false (real effect)
Reject H_0	**Type I error** (false positive), prob. α	Correct detection, prob. $1-\beta$ (power)
Fail to reject	Correct, prob. $1-\alpha$	**Type II error** (false negative), prob. β

TERMS

α, β – Type I and Type II error rates S.28
δ – true effect (difference in means) to detect
σ – outcome SD; n sample size (per group)
$z^*_{\alpha/2}, z^*_{\beta}$ – normal critical values (1.96; 0.84 for 80% power)
s_1, s_2, n_1, n_2 – group SDs and sizes; Φ normal CDF S.20
d; s_p – Cohen's effect size; pooled standard deviation
$\bar{x}_1, \bar{x}_2$; n – group means; sample size



CONTEXT

- **Origin.** Neyman and Pearson (1933); Jacob Cohen's *Statistical Power Analysis* (1969).
- **Use.** Planning studies; judging whether a null result means anything.
- **Data science.** A/B test duration and minimum detectable effect (MDE); with huge n trivial effects become "significant", so report d or lift with a CI.
- **Physical application.** Deciding how many prototypes to test to detect a 5% strength improvement.

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S.30 t-Tests

Compare means when σ is unknown: the statistic is a difference in means divided by its estimated standard error, referred to a t distribution.

One-sample t	$t = \dfrac{\bar{x} - \mu_0}{s/\sqrt{n}}, \quad df = n - 1$	One group against a fixed value.
Welch two-sample t	$t = \dfrac{\bar{x}_1 - \bar{x}_2}{\sqrt{\dfrac{s_1^2}{n_1} + \dfrac{s_2^2}{n_2}}}$	Two independent groups; no equal-variance assumption (the safe default).
Welch degrees of freedom	$df \approx \dfrac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}}$	Welch–Satterthwaite; lies between $\min(n_1, n_2) - 1$ and $n_1 + n_2 - 2$.
Pooled two-sample t	$t = \dfrac{\bar{x}_1 - \bar{x}_2}{s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}, \; df = n_1 + n_2 - 2$	Only if variances are truly equal; s_p from S.29.
Paired t	$t = \dfrac{\bar{d}}{s_d/\sqrt{n}}, \quad df = n - 1$	One-sample t on the within-pair differences.

STATED-EQUATION *One-sample: t equals x bar minus mu naught over s over root n, with n minus one degrees of freedom. Welch two-sample: t equals x bar one minus x bar two over the square root of s one squared over n one plus s two squared over n two, with Welch-Satterthwaite degrees of freedom. Pooled: the difference over s p times root of one over n one plus one over n two. Paired: t equals d bar over s d over root n.*

MEMORY AID Paired = one-sample test on differences.

METHOD

1. Pairs? Compute differences d_i. Otherwise compute both groups' $\bar{x}$, s.
2. Compute t and df; get p from t_{df} (two-sided: $2P(T \geq |t|)$).
3. Example (paired): differences 2, 4, 3, 5, 1: $\bar{d} = 3$, $s_d = 1.58$, $t = \frac{3}{1.58/\sqrt{5}} = 4.24$, $df = 4$, $p \approx 0.013$.

DECISION POINT **Paired or two-sample?** Same subjects measured twice, matched pairs, or models scored on the same folds → paired (removes between-subject noise). Different subjects → two-sample. **Welch or pooled?** Welch, unless there is a strong reason to believe equal variances; it costs almost nothing when they are equal. **t or nonparametric?** Small n with strong skew or outliers → Mann–Whitney / Wilcoxon S.34.

TERMS

s – sample SD S.2, replacing unknown σ
df – degrees of freedom S.21; $t \to N(0,1)$ as $df \to \infty$
t distribution – like the normal but with heavier tails S.22
$\bar{d}, s_d$ – mean and SD of paired differences; n number of pairs
s_p – pooled SD; μ_0 null value
Decision – compare p to α S.28
$\bar{x}$, μ_0, n – sample mean, null value, sample size
s_1, s_2, n_1, n_2 – group SDs and sizes

CONTEXT

- **Origin.** William Sealy Gosset, writing as "Student," at the Guinness brewery (1908); Welch (1947).
- **Use.** Comparing means with small samples and unknown σ.
- **Data science.** Comparing a continuous metric (revenue per user, session length) between control and treatment; paired tests for before/after on the same users or models on the same folds ML.5 §15 Machine Learning.
- **Physical application.** Comparing the mean strength of concrete from two suppliers with small samples uses a two-sample t-test.

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S.31 z-Tests for Proportions

Test a single proportion against a claim, or compare two proportions (the classic A/B conversion test), using the normal approximation to the binomial.

One-proportion z	$z = \dfrac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}}$	SE uses the *null* p_0; needs $np_0 \geq 10$ and $n(1-p_0) \geq 10$.
Pooled proportion	$\hat{p} = \dfrac{x_1 + x_2}{n_1 + n_2}$	Under $H_0 : p_1 = p_2$ both groups share one rate.
Two-proportion z	$z = \dfrac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$	Equivalent to a χ^2 test on the 2×2 table ($z^2 = \chi^2$) S.32.
Interval for $p_1 - p_2$	$(\hat{p}_1 - \hat{p}_2) \pm z^* \sqrt{\dfrac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \dfrac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$	Unpooled SE for the interval S.24.
Relative lift	$\text{lift} = \dfrac{\hat{p}_2 - \hat{p}_1}{\hat{p}_1}$	Percent change in conversion A1.19 §03 Algebra I.

STATED-EQUATION *One-proportion z equals p hat minus p naught over the square root of p naught one minus p naught over n. The pooled proportion is total successes over total trials. Two-proportion z equals p hat one minus p hat two over the square root of p hat one minus p hat times one over n one plus one over n two. The interval for the difference uses the unpooled standard error. The relative lift is p hat two minus p hat one, over p hat one.*

MEMORY AID Test with the null (pooled) SE; estimate with the unpooled SE.

METHOD

1. Compute $\hat{p}_1$, $\hat{p}_2$ and the pooled $\hat{p}$.
2. Compute z and its two-sided p-value.
3. Example: control 50/500, treatment 70/500: $\hat{p} = 0.12$, $SE = \sqrt{0.12 \cdot 0.88 \cdot \frac{2}{500}} = 0.0206$, $z = \frac{0.04}{0.0206} = 1.95$, $p \approx 0.051$: not quite significant at 0.05, lift $= 40\%$.

TERMS

$\hat{p}, \hat{p}_1, \hat{p}_2$ – sample proportions x/n
p_0 – proportion claimed by H_0
x_1, x_2 – successes; n_1, n_2 trials
z^* – critical value S.27
Small samples – exact binomial test S.16 or Fisher's exact test
n, n_1, n_2 – sample sizes
Lift – relative change of variant 2 over baseline 1

CONTEXT

- **Origin.** Normal approximation to the binomial (de Moivre–Laplace); routine since early 20th-century survey statistics.
- **Use.** Polls, defect rates, clinical response rates.
- **Data science.** The default A/B test on click-through and conversion; sample-ratio-mismatch checks use a one-proportion test.
- **Physical application.** Comparing failure rates of components from two production lines.

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S.32 Chi-Square Tests

Compare observed category counts with the counts expected under H_0; large squared discrepancies, scaled by the expected counts, are evidence against H_0.

Chi-square statistic	$\chi^2 = \sum \frac{(O-E)^2}{E}$	Sum over all cells; always ≥ 0; compared with χ^2_{df} S.22.
Goodness of fit	$E_i = n\,p_{0i}, \quad df = k-1$	One categorical variable vs claimed shares.
Expected count (two-way table)	$E = \frac{(\text{row total})(\text{column total})}{\text{grand total}}$	What independence predicts for each cell S.14.
Independence / homogeneity	$df = (r-1)(c-1)$	Same statistic for "are two variables related?" and "do groups share one distribution?"
Cramér's V	$V = \sqrt{\frac{\chi^2}{n\,(\min(r,c)-1)}}$	Effect size from 0 (none) to 1 (perfect association).

STATED-EQUATION *Chi squared equals the sum of observed minus expected, squared, over expected. Goodness of fit has expected counts n times p naught i and k minus one degrees of freedom. For independence, the expected count is row total times column total over grand total, with r minus one times c minus one degrees of freedom. Cramer's V is the square root of chi squared over n times min of r and c, minus one.*

MEMORY AID "Observed minus expected, squared, over expected."

METHOD

1. Compute expected counts under H_0; check all $E \geq 5$.
2. Sum $(O-E)^2/E$; find df and the p-value from χ^2_{df}.
3. Example: a die rolled 60 times gives 5, 8, 9, 8, 10, 20; $E = 10$ each, $\chi^2 = \frac{25+4+1+4+0+100}{10} = 13.4$, $df = 5$, $p \approx 0.02$: not fair.

TERMS

O, E – observed and expected counts (need $E \geq 5$)
k – number of categories; p_{0i} claimed shares
r, c – rows and columns of a contingency table
n – total count
Large χ^2 – data far from H_0; small p-value S.28
Independence – E uses $P(A \cap B) = P(A)P(B)$ S.9
E_i, p_{0i} – expected count and null share of category i
V – Cramér's V, 0 (none) to 1 (perfect association)
df – degrees of freedom S.21

CONTEXT

- **Origin.** Karl Pearson (1900).
- **Use.** Testing fairness of dice, genetics ratios, survey cross-tabs.
- **Data science.** Conversion-rate A/B tests as 2×2 tables, sample-ratio-mismatch checks, and χ^2 feature selection for categorical features.
- **Physical application.** Testing whether a Geiger counter's counts per interval follow a Poisson distribution is a goodness-of-fit test.

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S.33 One-Way ANOVA & the F-Test

Test whether $k \geq 3$ group means are all equal by comparing variability *between* group means with variability *within* groups.

Sum-of-squares split	$SS_T = SS_B + SS_W$	Total variation = between-group + within-group S.14.
Between groups	$SS_B = \sum_{j=1}^{k} n_j(\bar{y}_j - \bar{y})^2$	How far group means sit from the grand mean.
Within groups	$SS_W = \sum_{j=1}^{k}\sum_{i=1}^{n_j}(y_{ij} - \bar{y}_j)^2$	Noise around each group's own mean.
F-statistic	$F = \dfrac{MS_B}{MS_W} = \dfrac{SS_B/(k-1)}{SS_W/(N-k)}$	≈ 1 under H_0; large F means the groups differ; compare to $F_{k-1,N-k}$.
Effect size	$\eta^2 = \dfrac{SS_B}{SS_T}$	Share of total variation explained by group membership (like R^2).

STATED-EQUATION *The total sum of squares equals the between plus the within sums of squares. SS B is the sum of n j times group mean minus grand mean squared. SS W is the sum of squared deviations from each group mean. F equals MS B over MS W, which is SS B over k minus one, over SS W over N minus k. Eta squared is SS B over SS T.*

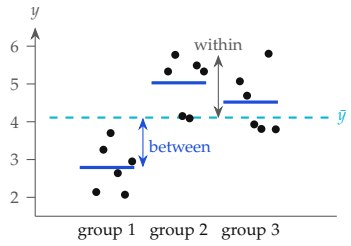
MEMORY AID F = signal (between) / noise (within).

METHOD

1. Compute group means $\bar{y}_j$ and the grand mean $\bar{y}$.
2. Compute SS_B, SS_W, their mean squares and F; get p from $F_{k-1,N-k}$.
3. If significant, run post-hoc pairwise tests (Tukey HSD) with correction S.35. Example: $k = 3$, $N = 18$, $SS_B = 10$, $SS_W = 15$: $F = \frac{10/2}{15/15} = 5$, $p \approx 0.02$.

TERMS

k – number of groups; N total observations
$n_j, \bar{y}_j$ – size and mean of group j; y_{ij} observation i in group j
$\bar{y}$ – grand mean
MS – mean square $= SS/df$
Assumptions – independence, normal residuals, equal variances (else Welch ANOVA or Kruskal–Wallis S.34)
$k = 2$ – $F = t^2$ of the pooled t-test S.30
SS_T, SS_B, SS_W – total, between-group, within-group sums of squares
y_{ij}; N – observation i in group j; total sample size
F; η^2 – ratio of mean squares S.22; share of variance explained
df – degrees of freedom S.21



CONTEXT

- **Origin.** R. A. Fisher, agricultural field trials at Rothamsted (1918–1925).
- **Use.** Comparing several treatments, fertilizers, teaching methods.
- **Data science.** Multi-arm experiments (A/B/C/D), comparing models across several datasets; the regression F-test is the same idea S.41.
- **Physical application.** Testing whether three furnace temperatures give different mean alloy hardness.

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S.34 Nonparametric (Rank) Tests

Rank-based tests replace values by their ranks, so they need no normality and resist outliers. They test whether one group tends to have larger values (a shift in location).

Mann–Whitney U	$U_1 = R_1 - \dfrac{n_1(n_1+1)}{2}$	Two independent groups; counts how often a group-1 value beats a group-2 value.
U normal approximation	$\mu_U = \dfrac{n_1 n_2}{2}, \quad \sigma_U^2 = \dfrac{n_1 n_2 (n_1+n_2+1)}{12}$	Standardize U for $n_1, n_2 \gtrsim 10$.
Wilcoxon signed-rank	$W^+ = \sum_{d_i>0} \operatorname{rank}(\lvert d_i \rvert)$	Paired data / one sample: rank $\lvert d_i \rvert$, add ranks of positive differences.
W^+ normal approximation	$\mu_W = \dfrac{n(n+1)}{4}, \quad \sigma_W^2 = \dfrac{n(n+1)(2n+1)}{24}$	For $n \gtrsim 20$ nonzero differences.
Kruskal–Wallis H	$H = \dfrac{12}{N(N+1)} \sum_{j=1}^{k} \dfrac{R_j^2}{n_j} - 3(N+1)$	Rank version of ANOVA; $H \approx \chi^2_{k-1}$ S.33.
Sign test	$S = \#\{d_i > 0\} \sim \operatorname{Bin}\left(n, \frac{1}{2}\right)$	Simplest paired test: only the signs, binomial under H_0 S.16.

STATED-EQUATION *Mann-Whitney U one equals the rank sum of group one minus n one times n one plus one over two; its mean is n one n two over two. Wilcoxon W plus is the sum of the ranks of the absolute differences that are positive, with mean n times n plus one over four. Kruskal-Wallis H equals twelve over N times N plus one, times the sum of R j squared over n j, minus three times N plus one. The sign test counts positive differences, binomial with one half. Their variances are n one n two times n one plus n two plus one, over twelve, and n times n plus one times two n plus one, over twenty-four.*

MEMORY AID Rank first, then test: Mann–Whitney (2 independent), Wilcoxon (paired), Kruskal–Wallis (3+).

METHOD

1. Pool and rank all values (average ranks for ties).
2. Compute the rank statistic and its p-value (exact tables for small n, normal approximation otherwise).
3. Example: group 1 = 1, 3, 5; group 2 = 2, 6, 7, 8: ranks of group 1 are 1, 3, 4, so $R_1 = 8$, $U_1 = 8 - 6 = 2$ of the 12 possible pairs.

TERMS

R_1, R_j – sum of ranks in group 1, group j
n_1, n_2, n_j – group sizes; N total size; k groups
d_i – paired differences (zeros dropped); n their number
rank – position in the sorted list (1 = smallest)
Spearman ρ_s – rank correlation S.38
U_1; W^+; H; S – Mann–Whitney, signed-rank, Kruskal–Wallis, and sign-test statistics
N; k; n – total size; number of groups; number of pairs

CONTEXT

- **Origin.** Wilcoxon (1945); Mann and Whitney (1947); Kruskal and Wallis (1952).
- **Use.** Ordinal scales (Likert), small or skewed samples.
- **Data science.** Comparing latency or revenue distributions with heavy tails; AUC equals $U/(n_1 n_2)$, the probability a random positive outranks a random negative ML.18 §15 Machine Learning.
- **Physical application.** Comparing pain ratings or ranked quality scores between two process settings.

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S.35 Multiple Testing: Bonferroni & Benjamini–Hochberg

Running many tests makes some false positives almost certain. Corrections control either the chance of any false positive (FWER) or the expected share of false discoveries (FDR).

Family-wise error rate	$\text{FWER} = P(\geq 1 \text{ false positive}) = 1 - (1-\alpha)^m$	For m independent true nulls; 20 tests at 0.05 gives 64%.
Bonferroni	reject H_i if $p_i \leq \dfrac{\alpha}{m}$	Guarantees FWER $\leq \alpha$ under any dependence; conservative.
Holm step-down	$p_{(i)} \leq \dfrac{\alpha}{m-i+1}$ for $i = 1, 2, \ldots$	Test sorted p-values in order, stop at the first failure; uniformly more powerful than Bonferroni.
False discovery rate	$\text{FDR} = E\left[\dfrac{V}{\max(R,1)}\right]$	Expected fraction of rejections that are false.
Benjamini–Hochberg	$k = \max\left\{i : \ p_{(i)} \leq \dfrac{i}{m}\, q\right\}$; reject $H_{(1)}, \ldots, H_{(k)}$	Controls FDR at q (independent or positively dependent tests); many more discoveries.

STATED-EQUATION *The family-wise error rate is one minus one minus alpha to the m. Bonferroni rejects when p i is at most alpha over m. Holm compares the i-th smallest p-value with alpha over m minus i plus one. The FDR is the expected false rejections over total rejections. Benjamini-Hochberg finds the largest i with p sub i at most i over m times q, and rejects the k smallest.*

MEMORY AID Bonferroni: "divide α by m". BH: "line through the origin with slope q/m".

METHOD

1. Sort the m p-values ascending: $p_{(1)} \leq \cdots \leq p_{(m)}$.
2. Compare $p_{(i)}$ with $\frac{i}{m}q$; find the largest i that passes.
3. Reject that one and all smaller. Example: $m = 5$, $q = 0.05$, $p = 0.004, 0.012, 0.027, 0.038, 0.20$ vs $0.01, 0.02, 0.03, 0.04, 0.05$: $k = 4$, reject 4. Bonferroni (0.01) rejects only 1.

DECISION POINT FWER or FDR? Use Bonferroni/Holm when even one false claim is costly (confirmatory trials, a few key metrics). Use BH when screening many hypotheses and a known share of false leads is acceptable (genes, features, many A/B metrics).

TERMS

m – number of tests; α target FWER
$p_{(i)}$ – ith smallest p-value; $H_{(i)}$ its hypothesis
V, R – false rejections, total rejections
q – target FDR (e.g. 0.05 or 0.10)
k – number of hypotheses rejected by BH
α; H_i, p_i – target error rate; hypothesis i and its p-value

CONTEXT

- **Origin.** Bonferroni inequality (1936); Holm (1979); Benjamini and Hochberg (1995).
- **Use.** Genomics, neuroimaging, multi-arm trials.
- **Data science.** Dashboards with dozens of A/B metrics, feature screening, "p-hacking" by trying many analyses; pre-register a primary metric S.28.
- **Physical application.** Testing many sensors for drift at once without flooding operators with false alarms.

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S.36 Bootstrap & Permutation Tests

Resampling replaces formulas with computation: the bootstrap resamples the data to approximate a statistic's sampling distribution; a permutation test shuffles labels to build the null distribution.

Bootstrap resample	$x_1^*, \dots, x_n^*$ drawn with replacement from $x_1, \dots, x_n$	Same size as the data; some points repeat, about 37% are left out.
Bootstrap standard error	$SE^* = \sqrt{\frac{1}{B-1}\sum_{b=1}^{B}\left(\hat{\theta}_b^* - \bar{\theta}^*\right)^2}$	SD of the statistic across B resamples (e.g. $B = 2000$).
Percentile interval	$\left[\hat{\theta}^*_{(\alpha/2)},\ \hat{\theta}^*_{(1-\alpha/2)}\right]$	Middle $1-\alpha$ of the bootstrap distribution S.27.
Permutation p-value	$p = \frac{1 + \#\{\lvert T_b^*\rvert \geq \lvert T_{\text{obs}}\rvert\}}{B+1}$	Shuffle group labels B times; how often is the shuffled statistic as extreme as observed?
Resampling probability	$P(x_i \notin \text{resample}) = \left(1 - \frac{1}{n}\right)^n \to e^{-1}$	Why ≈36.8% of points are "out-of-bag" C1.6 §07 Calculus I.

STATED-EQUATION *A bootstrap resample draws n values with replacement from the data. The bootstrap SE is the standard deviation of the B resampled statistics. The percentile interval takes the alpha over two and one minus alpha over two quantiles of those statistics. The permutation p-value is one plus the number of shuffled statistics at least as extreme, over B plus one. Each data point is left out of a given resample with probability one minus one over n, to the n, which approaches one over e.*

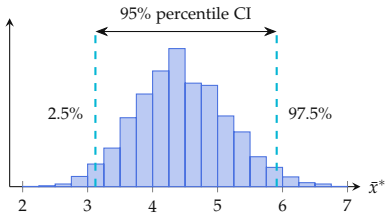
MEMORY AID Bootstrap: "the sample is my population". Permutation: "if labels don't matter, shuffle them".

METHOD

1. Compute the statistic $\hat{\theta}$ on the data (mean, median, AUC, …).
2. Draw B resamples with replacement; recompute $\hat{\theta}_b^*$ each time.
3. Use the SD of the $\hat{\theta}_b^*$ as the SE and their 2.5% and 97.5% quantiles as a 95% CI. Example: 30 order values with mean 4.45; 4000 resampled means give the 95% CI $(3.1,\ 5.9)$ (figure).

TERMS

x^* – resampled data; $\hat{\theta}_b^*$ statistic on resample b
B – number of resamples (1000–10 000)
$\bar{\theta}^*$ – mean of the bootstrap statistics
$\hat{\theta}^*_{(q)}$ – q-quantile of the bootstrap statistics S.12
T_{obs}, T_b^* – observed and shuffled test statistics
Limits – fails for extremes (max) and tiny n; assumes i.i.d. data
$\hat{\theta}_b^*$; α; n – statistic on resample b; one minus confidence level; sample size



CONTEXT

- **Origin.** Bradley Efron (1979); permutation tests from Fisher (1935) and Pitman (1937).
- **Use.** Inference for medians, ratios, and any statistic without a formula.
- **Data science.** CIs for model metrics (AUC, F1) on a test set; bagging and random forests are built on bootstrap samples ML.13 §15 Machine Learning; permutation importance of features.
- **Physical application.** Uncertainty of a fitted material constant from a small set of stress tests.

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S.37 Bayesian Inference & Conjugate Priors

Treat the unknown parameter as random: start with a prior, multiply by the likelihood of the data, and normalize to get the posterior, which carries all conclusions.

Posterior	$p(\theta \mid x) = \dfrac{p(x \mid \theta)\, p(\theta)}{\int p(x \mid \theta')\, p(\theta')\, d\theta'}$	Bayes' theorem S.11 for a parameter; the denominator is the evidence.
Proportional form	$p(\theta \mid x) \propto p(x \mid \theta)\, p(\theta)$	Posterior $\propto$ likelihood $\times$ prior; normalize at the end.
Beta prior	$p(\theta) = \dfrac{\theta^{\alpha-1}(1-\theta)^{\beta-1}}{B(\alpha,\beta)}, \quad 0 \le \theta \le 1$	Belief about a probability; mean $\frac{\alpha}{\alpha+\beta}$ D.10 §18 Graph Gallery.
Beta–Binomial update	$\mathrm{Beta}(\alpha,\beta) \xrightarrow{k \text{ of } n} \mathrm{Beta}(\alpha+k,\ \beta+n-k)$	Conjugate: add successes to α, failures to β.
Posterior mean	$E[\theta \mid x] = \dfrac{\alpha+k}{\alpha+\beta+n}$	Shrinks the sample rate k/n toward the prior mean; prior acts like $\alpha+\beta$ extra trials.
Normal–normal update	$\mu_{\text{post}} = \dfrac{\mu_0/\tau^2 + n\bar{x}/\sigma^2}{1/\tau^2 + n/\sigma^2}$	Precision-weighted average of prior mean and data mean (σ known).
Credible interval	$P(a \le \theta \le b \mid x) = 0.95$	A direct probability statement about θ, unlike a confidence interval S.27.
MAP estimate	$\hat{\theta}_{\text{MAP}} = \arg\max_{\theta}\ p(\theta \mid x)$	Posterior mode; MLE plus a log-prior penalty S.26.

STATED-EQUATION *The posterior p of theta given x equals the likelihood times the prior over the evidence. A Beta alpha beta prior with k successes in n trials gives a Beta alpha plus k, beta plus n minus k posterior, with mean alpha plus k over alpha plus beta plus n. For normal data the posterior mean is the precision-weighted average of the prior mean and x bar. A credible interval contains theta with posterior probability point nine five. Equivalently, the posterior is proportional to the likelihood times the prior. The beta prior density is theta to the alpha minus one times one minus theta to the beta minus one, over the beta function of alpha and beta. The MAP estimate is the theta that maximizes the posterior.*

MEMORY AID Prior + data $\to$ posterior; today's posterior is tomorrow's prior.

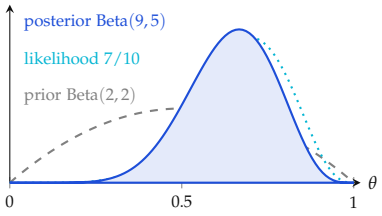
METHOD

1. Choose a prior (e.g. $\mathrm{Beta}(1,1)$ uniform, or one encoding past data).
2. Observe the data and update the parameters (conjugate) or sample the posterior (MCMC O.29 §14 Optimization).
3. Report the posterior mean and a 95% credible interval. Example: prior $\mathrm{Beta}(2,2)$, 7 successes in 10 trials $\to$ $\mathrm{Beta}(9,5)$, posterior mean $\frac{9}{14} \approx 0.64$ (vs MLE 0.70).

DECISION POINT Frequentist or Bayesian? Frequentist methods treat θ as fixed and talk about long-run error rates of procedures; Bayesian methods treat θ as uncertain and give direct probabilities like $P(\text{B beats A} \mid \text{data})$. Choose Bayesian when there is useful prior information, small data, or decisions need such probabilities; check sensitivity to the prior.

TERMS

θ – parameter; $p(\theta)$ prior; $p(\theta \mid x)$ posterior
$p(x \mid \theta)$ – likelihood S.26
α, β – Beta shape parameters (pseudo-successes, pseudo-failures)
$B(\alpha,\beta)$ – Beta function (normalizing constant)
k, n – successes, trials
μ_0, τ^2 – prior mean and variance; σ^2 data variance
$\bar{x}$, σ^2; μ_{post} – sample mean and known data variance; posterior mean
a, b; $\hat{\theta}_{\text{MAP}}$ – credible-interval endpoints; posterior mode



CONTEXT

- **Origin.** Bayes (1763) and Laplace (1774); revived with MCMC computing in the 1990s.
- **Use.** Updating beliefs with evidence; small-sample estimation.
- **Data science.** Bayesian A/B testing (Beta posteriors per arm), Thompson sampling for bandits, smoothing click-through rates of rare items; ridge and lasso are MAP estimates with normal and Laplace priors ML.8 §15 Machine Learning.
- **Physical application.** Updating an estimate of a component's failure probability as field data arrive.

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Relationships & Regression

S.38 Covariance & Correlation

Covariance measures whether two variables move together; correlation rescales it to a unit-free number between -1 and 1 that measures the strength of a *linear* relationship.

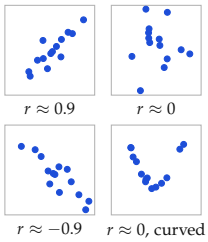
Sample covariance	$\mathrm{Cov}(X,Y) = \dfrac{1}{n-1}\sum_{i=1}^{n}(x_i-\bar{x})(y_i-\bar{y})$	Positive if above-average x go with above-average y; units are x-units $\times$ y-units.
Pearson correlation	$r = \dfrac{\mathrm{Cov}(X,Y)}{s_x\, s_y} = \dfrac{1}{n-1}\sum_{i=1}^{n} z_{x_i} z_{y_i}$	Average product of z-scores S.4.
Range of r	$-1 \le r \le 1$	± 1 only for a perfect line; $r = 0$ means no *linear* trend.
Population covariance	$\mathrm{Cov}(X,Y) = E\big[(X-\mu_X)(Y-\mu_Y)\big] = E[XY] - \mu_X\mu_Y$	Zero for independent variables (not conversely).
Spearman rank correlation	$\rho_s = 1 - \dfrac{6\sum d_i^2}{n(n^2-1)}$	Pearson on ranks: measures monotone association, robust to outliers.
Test for correlation	$t = \dfrac{r\sqrt{n-2}}{\sqrt{1-r^2}}, \quad df = n-2$	Tests $H_0 : \rho = 0$; equals the regression slope t-test S.40.

STATED-EQUATION *The covariance of X and Y equals one over n minus one times the sum of x i minus x bar, times y i minus y bar. r equals the covariance over s x s y, which equals one over n minus one times the sum of z x times z y. r is between minus one and one. Spearman's rho equals one minus six times the sum of d i squared over n times n squared minus one. To test rho equals zero, t equals r root n minus two over root one minus r squared. The population covariance is E of X minus mu X, times Y minus mu Y, which equals E of X Y minus mu X mu Y.*

MEMORY AID Correlation is not causation S.43.

TERMS

$\bar{x}, \bar{y}$ – means S.1; s_x, s_y SDs S.2
r, ρ – sample and population Pearson correlation
z_{x_i}, z_{y_i} – z-scores S.4
d_i – difference between the ranks of x_i and y_i
Geometry – r is the cosine similarity of centered vectors T.21 §06 Trigonometry
Caution – r measures *linear* association; $r \approx 0$ can hide a curve
s_x, s_y; n – sample standard deviations; sample size
μ_X, μ_Y – population means
df – degrees of freedom S.21



CONTEXT

- **Origin.** Galton (1888) and Pearson (1896); Spearman (1904).
- **Use.** Measuring how two variables move together.
- **Data science.** Correlation matrices and heatmaps in EDA, detecting multicollinearity S.41, covariance matrices in PCA and portfolio risk S.39.
- **Physical application.** Air temperature and air density are strongly negatively correlated at fixed pressure.

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S.39 Covariance Matrix & Multivariate Normal

For a vector of variables, all variances and covariances are collected in one symmetric matrix. It controls how linear combinations vary and defines the shape of the multivariate normal.

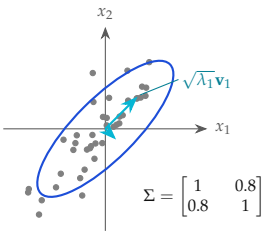
Covariance matrix	$\Sigma = E\big[(\mathbf{X}-\boldsymbol{\mu})(\mathbf{X}-\boldsymbol{\mu})^\top\big]$	Variances on the diagonal, covariances off it; symmetric, positive semidefinite LA.27 §09 Linear Algebra.
Sample covariance matrix	$S = \dfrac{1}{n-1} X_c^\top X_c$	X_c is the data matrix with each column centered.
Variance of a combination	$\mathrm{Var}(\mathbf{a}^\top \mathbf{X}) = \mathbf{a}^\top \Sigma\, \mathbf{a}$	Portfolio variance; generalizes $\mathrm{Var}(X+Y)$ S.13.
Linear transformation	$\mathrm{Cov}(A\mathbf{X}+\mathbf{b}) = A\,\Sigma A^\top$	Shifts do nothing; matrices act on both sides.
Correlation matrix	$R = D^{-1/2}\,\Sigma\, D^{-1/2}$	Divide each entry by both SDs; ones on the diagonal.
Multivariate normal density	$f(\mathbf{x}) = \dfrac{1}{(2\pi)^{p/2}\lvert\Sigma\rvert^{1/2}} \exp\!\Big(-\tfrac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x}-\boldsymbol{\mu})\Big)$	Elliptical contours aligned with the eigenvectors of Σ; the exponent is the squared Mahalanobis distance.

STATED-EQUATION *Sigma is the expected outer product of the centered random vector. The sample version is one over n minus one times X c transpose X c. The variance of a transpose X is a transpose Sigma a. The covariance of A X plus b is A Sigma A transpose. The correlation matrix is D to the minus one half, Sigma, D to the minus one half. The multivariate normal density is one over two pi to the p over two times the root determinant of Sigma, times e to the minus one half the Mahalanobis distance squared.*

MEMORY AID Diagonal = variances, off-diagonal = covariances; eigenvectors = ellipse axes.

TERMS

$\mathbf{X}, \boldsymbol{\mu}$ – random vector of p variables and its mean vector
Σ, S – population and sample covariance matrices
X_c – $n \times p$ centered data matrix A2.24 §04 Algebra II
$\mathbf{a}$, A, $\mathbf{b}$ – weight vector, matrix, shift vector
D – diagonal matrix of the variances
$|\Sigma|, \Sigma^{-1}$ – determinant and inverse A2.25 §04 Algebra II
p; n – number of variables (dimension); sample size



CONTEXT

- **Origin.** Pearson and Galton (bivariate normal, 1880s–1890s); Mahalanobis (1936).
- **Use.** Multivariate data, measurement error propagation.
- **Data science.** PCA diagonalizes S ML.38 §15 Machine Learning; Gaussian mixture components ML.37 §15 Machine Learning; Mahalanobis anomaly detection; mean–variance portfolio optimization.
- **Physical application.** Error ellipses for a GPS position estimate come from its 2×2 covariance matrix.

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S.40 Least-Squares Linear Regression & R^2

Fit the line that minimizes the sum of squared vertical errors (residuals). R^2 reports the share of the variation in y that the line explains.

Fitted line	$\hat{y} = b_0 + b_1 x$	Predicted mean of y at each x (an estimate of $E[Y \mid x]$ S.14).
Slope	$b_1 = \dfrac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2} = r\,\dfrac{s_y}{s_x}$	Covariance over variance of x.
Intercept	$b_0 = \bar{y} - b_1 \bar{x}$	Forces the line through $(\bar{x}, \bar{y})$.
Coefficient of determination	$R^2 = 1 - \dfrac{SS_{\text{res}}}{SS_{\text{tot}}} = 1 - \dfrac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2}$	Share of variance explained; equals r^2 in simple regression.
Normal equations	$\hat{\boldsymbol{\beta}} = (X^\top X)^{-1} X^\top \mathbf{y}$	Matrix form of least squares for any number of predictors LA.20 §09 Linear Algebra.
Residual standard error	$s_e = \sqrt{\dfrac{SS_{\text{res}}}{n-2}}$	Typical size of a residual; estimates the noise SD σ.
Slope t-test	$t = \dfrac{b_1}{SE(b_1)}, \quad SE(b_1) = \dfrac{s_e}{\sqrt{\sum(x_i - \bar{x})^2}}$	Tests $H_0 : \beta_1 = 0$ with $n-2$ df; CI $b_1 \pm t^* SE(b_1)$.

STATED-EQUATION *y hat equals b naught plus b one x. b one equals the sum of x i minus x bar times y i minus y bar, over the sum of x i minus x bar squared, which equals r times s y over s x. b naught equals y bar minus b one x bar. R squared equals one minus the residual sum of squares over the total sum of squares. In matrix form, beta hat equals X transpose X inverse, X transpose y. The slope t equals b one over its standard error. The residual standard error is the square root of the residual sum of squares over n minus two, and the slope's standard error is s e over the square root of the sum of x i minus x bar squared.*

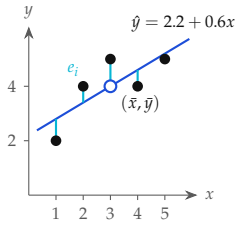
MEMORY AID The regression line always passes through $(\bar{x}, \bar{y})$.

METHOD

1. Compute $\bar{x}$, $\bar{y}$, $S_{xy} = \sum(x_i - \bar{x})(y_i - \bar{y})$ and $S_{xx} = \sum(x_i - \bar{x})^2$.
2. $b_1 = S_{xy}/S_{xx}$, then $b_0 = \bar{y} - b_1\bar{x}$.
3. Example: $(1,2), (2,4), (3,5), (4,4), (5,5)$: $\bar{x} = 3$, $\bar{y} = 4$, $S_{xy} = 6$, $S_{xx} = 10$, so $b_1 = 0.6$, $b_0 = 2.2$, $\hat{y} = 2.2 + 0.6x$.

TERMS

$\hat{y}$ – predicted value; $e_i = y_i - \hat{y}_i$ residual
b_1 – slope A1.6 §03 Algebra I: change in $\hat{y}$ per unit of x
b_0 – intercept; β_0, β_1 true coefficients
r – correlation S.38; s_x, s_y SDs
Least squares – minimizes $\sum e_i^2$ by setting derivatives to 0 C1.26 §07 Calculus I
X, $\mathbf{y}$ – design matrix (column of 1s plus x), response vector A2.24 §04 Algebra II, A2.25 §04 Algebra II
s_e; $SS_{\text{res}}, SS_{\text{tot}}$ – residual standard error; residual and total sums of squares
s_x, s_y; $\hat{\boldsymbol{\beta}}$ – SDs of x and y; coefficient vector
df – degrees of freedom S.21



CONTEXT

- **Origin.** Legendre (1805) and Gauss (1809); "regression" coined by Galton (1886).
- **Use.** Prediction and quantifying relationships.
- **Data science.** The baseline model for everything; interpretable coefficients for stakeholders; R^2 is the share of variance explained, and the same MSE loss drives neural-net regression ML.19 §15 Machine Learning.
- **Physical application.** Calibrating a spring: fit force versus stretch by least squares; the slope is the spring constant k.

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S.41 Multiple Regression & Adjusted R^2

Model a numeric outcome as a linear function of several predictors. Each coefficient is the effect of its predictor *holding the others fixed*.

Model	$\mathbf{y} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad \varepsilon_i \sim N(0, \sigma^2)$	$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p + \varepsilon$ for each row.
Coefficient covariance	$\mathrm{Cov}(\hat{\boldsymbol{\beta}}) = \hat{\sigma}^2 (X^\top X)^{-1}, \quad \hat{\sigma}^2 = \dfrac{SS_{\text{res}}}{n-p-1}$	$SE(\hat{\beta}_j)$ is the root of the jth diagonal entry.
Coefficient t-test	$t_j = \dfrac{\hat{\beta}_j}{SE(\hat{\beta}_j)}, \quad df = n-p-1$	Is predictor j useful given the others?
Adjusted R^2	$R^2_{\text{adj}} = 1 - (1-R^2)\dfrac{n-1}{n-p-1}$	Penalizes extra predictors; rises only if a new term helps more than chance.
Overall F-test	$F = \dfrac{R^2/p}{(1-R^2)/(n-p-1)}$	Tests whether any predictor matters ($F_{p,n-p-1}$) S.33.
Variance inflation factor	$\mathrm{VIF}_j = \dfrac{1}{1-R_j^2}$	Multicollinearity check; $>$ 5–10 means unstable, inflated SEs.

STATED-EQUATION *y equals X beta plus epsilon with normal errors. The covariance of beta hat is sigma hat squared times X transpose X inverse, with sigma hat squared equal to the residual sum of squares over n minus p minus one. Each t j is beta hat j over its standard error. Adjusted R squared equals one minus one minus R squared, times n minus one over n minus p minus one. The overall F equals R squared over p, over one minus R squared over n minus p minus one. VIF j equals one over one minus R j squared.*

MEMORY AID Assumptions LINE: Linear, Independent errors, Normal errors, Equal variance.

METHOD

1. Fit by least squares S.40; inspect residual plots for curvature and non-constant spread.
2. Check the overall F, then each t_j; check VIFs.
3. Compare models by R^2_{adj} or cross-validation. Example: $R^2 = 0.50$, $n = 21$, $p = 4$: $R^2_{\text{adj}} = 1 - 0.5 \cdot \frac{20}{16} = 0.375$.

TERMS

X – $n \times (p+1)$ design matrix (first column 1s); $\mathbf{y}$ response
$\boldsymbol{\beta}, \hat{\boldsymbol{\beta}}$ – true and fitted coefficients
p – number of predictors; n observations
$\boldsymbol{\varepsilon}, \sigma^2$ – errors and their variance
R_j^2 – R^2 from regressing x_j on the other predictors
Categorical x – enter as 0/1 dummy variables
$\hat{\sigma}^2$; SS_{res} – estimated error variance; residual sum of squares
R^2, R^2_{adj}; VIF_j – fit and adjusted fit S.40; variance inflation factor
df – degrees of freedom S.21

CONTEXT

- **Origin.** Gauss–Markov theorem (1821, 1900); matrix formulation by Fisher and Yule (early 1900s).
- **Use.** Controlling for several factors at once; forecasting.
- **Data science.** Baseline tabular model; coefficients are interpretable effects; regularize with ridge or lasso when p is large ML.8 §15 Machine Learning; adjusting for confounders S.43.
- **Physical application.** Predicting a building's energy use from floor area, insulation and outdoor temperature.

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S.42 Logistic Regression (Statistical View)

For a yes/no outcome, model the log-odds as linear in the predictors. It is the generalized linear model for Bernoulli data; the machine-learning view is in ML.9 §15 Machine Learning.

Logit link	$\ln \frac{p}{1-p} = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$	Log-odds are linear; p itself is not.
Sigmoid (inverse link)	$p = \sigma(z) = \frac{1}{1+e^{-z}},\ z = \mathbf{x}^\top \boldsymbol{\beta}$	Squashes any real z into a probability F.14 §18 Graph Gallery.
Odds ratio	$\mathrm{OR}_j = e^{\beta_j}$	Multiplicative change in the odds per one-unit increase in x_j.
Log-likelihood	$\ell(\boldsymbol{\beta}) = \sum_{i=1}^{n} \left[y_i \ln p_i + (1-y_i) \ln(1-p_i) \right]$	Bernoulli MLE S.26; negative of the log-loss ML.2 §15 Machine Learning; maximized numerically.
Wald test	$z_j = \frac{\hat{\beta}_j}{SE(\hat{\beta}_j)}$	Is predictor j associated with the outcome?

STATED-EQUATION *The log-odds, log of p over one minus p, equals beta naught plus beta one x one and so on. p equals the sigmoid of z, one over one plus e to the minus z. The odds ratio for x j is e to the beta j. The log-likelihood sums y i log p i plus one minus y i times log of one minus p i. The Wald z is beta hat j over its standard error.*

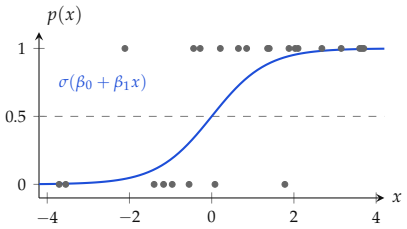
MEMORY AID Logistic: linear in the log-odds; exponentiate a coefficient to get an odds ratio.

METHOD

1. Fit by maximum likelihood (software).
2. Interpret each $\hat{\beta}_j$ as a change in log-odds, $e^{\hat{\beta}_j}$ as an odds ratio.
3. Example: $\hat{\beta}_1 = 0.69$ for "has coupon": $e^{0.69} \approx 2$, so the odds of buying double.

TERMS

p – $P(Y = 1 \mid \mathbf{x})$; odds $= \frac{p}{1-p}$
β_j – coefficients; $\mathbf{x}$ predictor vector
$\sigma(z)$ – logistic sigmoid; z linear predictor
$y_i \in \{0, 1\}$ – observed outcomes; p_i fitted probabilities
GLM – generalized linear model: Poisson regression uses a log link S.18
z, $\mathbf{x}$ – linear predictor; feature vector
OR_j; ℓ; SE – odds ratio; log-likelihood S.26; standard error



CONTEXT

- **Origin.** Cox (1958); GLM framework by Nelder and Wedderburn (1972).
- **Use.** Medical risk factors, credit default, any binary outcome.
- **Data science.** The default classifier baseline ML.9 §15 Machine Learning; calibrated probabilities; odds ratios explain feature effects to stakeholders.
- **Physical application.** Probability that a component fails as a function of operating temperature.

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S.43 Correlation vs Causation

An association between X and Y can arise from X causing Y, Y causing X, a common cause (confounder), or selection. Randomization or adjustment for confounders is needed to claim causation.

Average treatment effect	$\mathrm{ATE} = E\big[Y(1) - Y(0)\big]$	Difference between potential outcomes with and without treatment; only one is ever observed per unit.
Randomized experiment	$\mathrm{ATE} = E[Y \mid T = 1] - E[Y \mid T = 0]$	Random assignment makes groups comparable, so the simple difference is causal.
Adjustment for a confounder	$\mathrm{ATE} = \sum_z \big(E[Y \mid T{=}1, z] - E[Y \mid T{=}0, z]\big) P(z)$	Compare within strata of Z, then average (needs all confounders measured).
Simpson's paradox	$P(A \mid B, z) > P(A \mid B^c, z)\ \forall z \not\Rightarrow P(A \mid B) > P(A \mid B^c)$	A trend in every subgroup can reverse when groups are pooled.

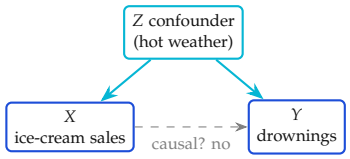
STATED-EQUATION *The average treatment effect is the expected difference of the potential outcomes Y of one and Y of zero. In a randomized experiment it equals the difference in group means. With a confounder Z, average the within-stratum differences weighted by P of z. Simpson's paradox: an association that holds in every subgroup can reverse in the combined data.*

MEMORY AID Correlation is not causation: look for the confounder, reverse causation, and selection.

DECISION POINT Can I claim causation? Randomized assignment → yes, for the population studied. Observational data → only with a design that removes confounding (adjustment, matching, difference-in-differences, instrumental variables, regression discontinuity) and stated assumptions. Otherwise report "associated with", not "causes".

TERMS

$Y(1), Y(0)$ – potential outcomes with and without treatment
T – treatment indicator $(0/1)$
Z – confounder: a common cause of T and Y
$E[\cdot \mid \cdot]$ – conditional expectation S.14
Reverse causation – Y drives X
Selection bias – the sample depends on the outcome (survivorship)



correlated, but only through Z

CONTEXT

- **Origin.** Fisher's randomized experiments (1935); Rubin's potential outcomes (1974); Pearl's causal graphs (1990s).
- **Use.** Policy evaluation, medicine, economics.
- **Data science.** A/B tests are randomized experiments; feature importance and correlation in observational logs are not causal; uplift modeling and causal inference libraries estimate the ATE.
- **Physical application.** Ice-cream sales and drownings rise together because both follow hot weather.

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14
Optimization & Simulation

Choosing the best decision subject to constraints, and estimating what is too hard to compute exactly. Linear, integer, and nonlinear programming decide schedules, routes, and portfolios; iterative algorithms such as gradient descent and Newton's method train models; Monte Carlo methods and discrete-event simulation turn randomness into answers.

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Formulating & Classifying Problems

O.1 Anatomy of an Optimization Problem

Every optimization problem has three parts: decision variables you control, an objective to minimize or maximize, and constraints that define which decisions are allowed (the feasible set).

Standard form	Every problem can be rewritten this way; solvers and theory assume it. $\min_{\mathbf{x}\in\mathbb{R}^n} f(\mathbf{x}) \quad \text{subject to} \quad g_i(\mathbf{x}) \le 0 \;\; (i=1,\dots,m), \qquad h_j(\mathbf{x}) = 0 \;\; (j=1,\dots,p)$	
Max–min equivalence	$\max_{\mathbf{x}} f(\mathbf{x}) = -\min_{\mathbf{x}} \big(-f(\mathbf{x})\big)$	Maximize profit = minimize −profit; the optimal $\mathbf{x}$ is the same.
Feasible set	$\mathcal{F} = \{\mathbf{x} : \; g_i(\mathbf{x}) \le 0,\; h_j(\mathbf{x}) = 0\}$	All decisions satisfying every constraint; empty means *infeasible*.
Global minimum	$f(\mathbf{x}^*) \le f(\mathbf{x}) \;\; \forall\, \mathbf{x} \in \mathcal{F}$	Best over the whole feasible set.
Local minimum	$f(\mathbf{x}^*) \le f(\mathbf{x}) \;\; \forall\, \mathbf{x} \in \mathcal{F},\; \|\mathbf{x} - \mathbf{x}^*\| < \varepsilon$	Best only in a neighborhood; most algorithms only guarantee this unless the problem is convex O.2.
Constraint rewrites	$g \ge 0 \iff -g \le 0; \quad h = 0 \iff h \le 0,\; -h \le 0$	Flip signs to reach standard form; an equality is two inequalities.

STATED-EQUATION *Minimize f of x over x in R n, subject to g i of x less than or equal to zero and h j of x equal to zero. Maximizing f is the same as minimizing minus f. The feasible set is all x satisfying every constraint. A global minimum beats every feasible point; a local minimum beats every feasible point nearby. A constraint g at least zero is the same as minus g at most zero, and an equality h equals zero is the pair h at most zero and minus h at most zero.*

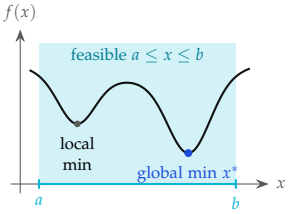
MEMORY AID Three parts: decisions, objective, constraints.

METHOD

1. Name the decision variables and their units ("x_1 = tons of product 1 per week").
2. Write the objective as a function of those variables; decide min or max.
3. Translate every limit (budget, capacity, demand, logic) into a constraint; add sign and integrality restrictions.
4. Classify the problem (linear? integer? convex? smooth?) to pick a solver, see the key topic O.K1. Example: fence a rectangle with 8 m of fencing for maximum area: max xy s.t. $2x + 2y = 8$; answer $x = y = 2$.

TERMS

$\mathbf{x}$ – decision variables (what you control), a vector in $\mathbb{R}^n$ M.78 §01 Math Signs
$f(\mathbf{x})$ – objective: cost, loss, or −profit
g_i, h_j – inequality and equality constraints
$\mathcal{F}$ – feasible region (compare linear inequalities A1.20 §03 Algebra I)
$\mathbf{x}^*$ – an optimal solution; $f(\mathbf{x}^*)$ the optimal value
ε – radius of the neighborhood in the local definition
Classes – LP: linear f, g, h O.3; MIP: some x_k integer O.6; NLP: nonlinear O.13
Convex problem – convex f and feasible set O.2: every local optimum is global
m, p – numbers of inequality and equality constraints



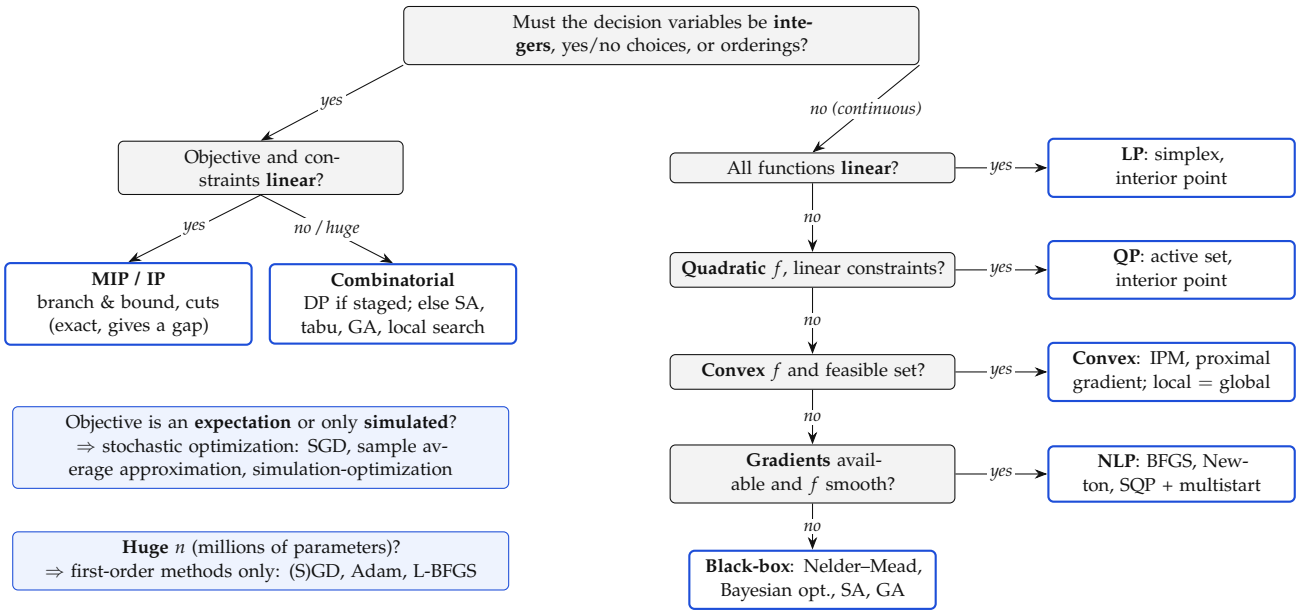
CONTEXT

- **Origin.** Calculus of variations (Euler, Lagrange, 1700s); operations research born in WWII logistics.
- **Use.** Scheduling, routing, pricing, portfolio allocation, engineering design.
- **Data science.** Training a model *is* an optimization problem: $\min_\theta$ loss + penalty ML.1 §15 Machine Learning; hyperparameter tuning is optimization over a black-box function.
- **Physical application.** Choosing the shape of a beam to minimize weight subject to a maximum allowed deflection is a constrained optimization problem.

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Optimization Problem Types at a Glance | *Key Topic O.K1*

The single most useful skill in optimization is recognizing *what kind* of problem you have. The structure of the objective and constraints (linear, quadratic, convex, smooth, discrete) decides which solvers apply, whether the answer is guaranteed to be globally optimal, and how large a problem is tractable. Classify first, then model in a form your solver accepts.



The main classes

Type	Form	Solver & guarantee	Example
LP 0.3	$\min \mathbf{c}^\top \mathbf{x}$ s.t. $A\mathbf{x} \le \mathbf{b},\ \mathbf{x} \ge \mathbf{0}$	Simplex, interior point; global optimum at a vertex; millions of variables	Production mix, diet, transport
IP / MIP 0.6	LP + some $x_j \in \mathbb{Z}$ or $\{0,1\}$	Branch & bound/cut; NP-hard but optimality gap reported	Scheduling, facility location
QP 0.5	$\min \frac{1}{2}\mathbf{x}^\top Q\mathbf{x} + \mathbf{c}^\top \mathbf{x}$ s.t. linear	Active set, IPM; global if $Q \succeq 0$	Markowitz portfolio, SVM, lasso
Convex	convex f, convex g_i, affine h_j 0.2	IPM, (proximal/projected) gradient; every local min is global	Logistic regression, SOCP/SDP
NLP 0.13	smooth nonlinear f, g, h	Newton, BFGS, SQP, KKT 0.18; local optimum only	Engineering design, MLE
Combinatorial	finite set: subsets, tours, matchings	Exact: DP 0.21, B&B; else heuristics 0.22	Knapsack 0.7, TSP 0.8
Network	flows on a graph	Dijkstra, max-flow; LP with integral solutions	Shortest path, max flow 0.9
Stochastic	min $E[F(\mathbf{x}, \xi)]$	SGD, sample average approx. 0.16	Training ML models

What information the algorithm uses

Zeroth order (values only)

Grid / random search, Nelder–Mead, simulated annealing 0.22, genetic algorithms 0.24, Bayesian optimization. Robust to noise and kinks; slow in high dimension.

First order (gradients)

Gradient descent 0.11, SGD 0.16, coordinate descent 0.15, projected/proximal gradient. Cheap steps, $O(n)$ memory; slowed by bad conditioning.

Second order (curvature)

Newton 0.13, Gauss–Newton, quasi-Newton BFGS / L-BFGS 0.14, interior point. Few iterations, quadratic or superlinear convergence; each step costs a linear solve.

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Decision points and common mistakes

DECISION POINT **Convex or not?** Check whether f is convex (Hessian $\succeq 0$ or built from convex pieces) and the feasible set is convex (convex $g_i \leq 0$, affine $h_j = 0$). If yes, any local solver finds the global optimum and KKT conditions 0.18 are sufficient. If not, use multistart, a global method, or accept a good local optimum. **Exact or heuristic?** If a MIP solver closes the gap in the time available, use it; heuristics give no optimality certificate. **Reformulate before you solve:** $|x| \leq t$ becomes $-t \leq x \leq t$; $\max_i a_i^\top x \leq t$ becomes m linear constraints; a product xy of binaries becomes linear with $z \leq x,\ z \leq y,\ z \geq x + y - 1$.

Mistake: rounding an LP to get integers. The rounded point can be infeasible or far from optimal; solve the MIP 0.6.

Mistake: assuming a local solver found the global optimum of a non-convex problem. Try several starting points.

Mistake: an unbounded model. A missing constraint (e.g. $x \geq 0$) lets the objective run to $\pm\infty$.

Mistake: badly scaled variables (dollars next to millions of dollars) slow gradient methods 0.11 and cause numerical trouble in every solver; rescale to similar magnitudes.

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O.2 Convex Sets & Convex Functions

A set is convex if it contains the segment between any two of its points; a function is convex if its graph lies below every chord. Convex problems have no spurious local minima, which makes them reliably solvable.

Convex set	$\lambda\mathbf{x}+(1-\lambda)\mathbf{y}\in C \quad \forall\,\mathbf{x},\mathbf{y}\in C,\ \lambda\in[0,1]$	No dents or holes: every segment stays inside.
Convex function (chord)	$f\big(\lambda\mathbf{x}+(1-\lambda)\mathbf{y}\big)\le\lambda f(\mathbf{x})+(1-\lambda)f(\mathbf{y})$	The graph lies on or below every chord; reverse the inequality for *concave*.
First-order condition	$f(\mathbf{y})\ge f(\mathbf{x})+\nabla f(\mathbf{x})^{\top}(\mathbf{y}-\mathbf{x})$	Every tangent plane is a global under-estimator; so $\nabla f=\mathbf{0}$ means global min.
Second-order condition	$f''(x)\ge 0 \quad\text{or}\quad \nabla^2 f(\mathbf{x})\succeq 0$	Concave up everywhere; Hessian positive semidefinite LA.27 §09 Linear Algebra.
Strong convexity	$\nabla^2 f(\mathbf{x})\succeq mI, \quad m>0$	Curvature at least m in every direction: unique minimizer, fast convergence O.11.
Local = global	convex f, convex $\mathcal{F}\ \Rightarrow$ local min = global min	The payoff of convexity: any descent method that stops has found the best answer.

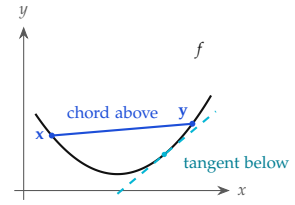
STATED-EQUATION *f of lambda x plus one minus lambda y is at most lambda f of x plus one minus lambda f of y, for lambda between zero and one. f of y is at least f of x plus the gradient at x dotted with y minus x. Equivalently, f double prime at least zero, or the Hessian is positive semidefinite. For convex problems a local minimum is a global minimum. A set C is convex when lambda x plus one minus lambda y stays in C for all x and y in C. f is strongly convex when its Hessian is at least m times the identity, with m positive.*

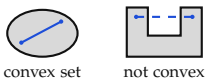
MEMORY AID Convex looks like a bowl: any chord lies above the curve, any tangent below it.

DECISION POINT How to show convexity: (1) one or two variables: compute f'' or the Hessian and check its eigenvalues or leading minors C3.16 §10 Calculus III; (2) larger expressions: build f from convex pieces with the closure rules (non-negative sums, max, composition with an affine map $A\mathbf{x}+\mathbf{b}$); (3) to *disprove*: find two points where the midpoint lies above the chord. Example: $f(x,y)=x^2+xy+y^2$ has Hessian $\begin{bmatrix}2&1\\1&2\end{bmatrix}$ with eigenvalues $1,3>0$: convex.

TERMS

C – a set; $\mathbf{x},\mathbf{y}$ any two of its points
λ – mixing weight; $\lambda\mathbf{x}+(1-\lambda)\mathbf{y}$ traces the segment
∇f – gradient C3.12 §10 Calculus III
$\nabla^2 f$ – Hessian, matrix of second partials C3.16 §10 Calculus III; $\succeq 0$: all eigenvalues ≥ 0 LA.23 §09 Linear Algebra
$f''\ge 0$ – concave up C1.23 §07 Calculus I
m – strong-convexity constant; I identity matrix
Closed under – sums, positive scaling, max, composition with affine maps
Jensen – $f(E[X])\le E[f(X)]$ for convex f S.13 §13 Statistics
$\succeq$; I – positive semidefinite ordering LA.27 §09 Linear Algebra; identity matrix
$\mathbf{x},\mathbf{y}$; $\mathcal{F}$ – any two points; feasible set O.1





CONTEXT

- **Origin.** Jensen's inequality (1906); modern convex optimization by Rockafellar, Boyd and Vandenberghe.
- **Use.** Guarantees that algorithms find the true optimum efficiently.
- **Data science.** MSE, log-loss, and hinge loss are convex in linear-model parameters, so linear/logistic regression and SVMs have one global optimum; deep networks are non-convex.
- **Physical application.** Potential energy of a stretched spring $\frac{1}{2}kx^2$ is convex: the equilibrium is the unique minimum.

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Linear, Quadratic & Integer Programming

O.3 Linear Programming (LP)

Optimize a linear objective over a region cut out by linear inequalities. The feasible region is a convex polygon (polyhedron), and an optimum, if one exists, sits at a corner.

Canonical form	$\max_{\mathbf{x}} \; z = \mathbf{c}^\top \mathbf{x} \;\text{ s.t. }\; A\mathbf{x} \le \mathbf{b}, \;\; \mathbf{x} \ge \mathbf{0}$	Linear objective, linear constraints, non-negative variables.
Standard (equality) form	$A\mathbf{x} + \mathbf{s} = \mathbf{b}, \quad \mathbf{x}, \mathbf{s} \ge \mathbf{0}$	Slack variables $\mathbf{s}$ turn inequalities into equations for the simplex method O.4.
Corner-point theorem	optimum (if any) is attained at a vertex	So only finitely many candidate points need checking.
Iso-profit line	$\mathbf{c}^\top \mathbf{x} = k$	Parallel lines; slide in the direction of $\mathbf{c}$ until the line last touches the region.

STATED-EQUATION *Maximize z equals c transpose x, subject to A x less than or equal to b and x greater than or equal to zero. Adding slacks gives A x plus s equals b. An optimum, if one exists, occurs at a vertex of the feasible region. The iso-profit lines c transpose x equal to k are parallel; slide them in the improving direction to the last feasible vertex.*

MEMORY AID "Check the corners."

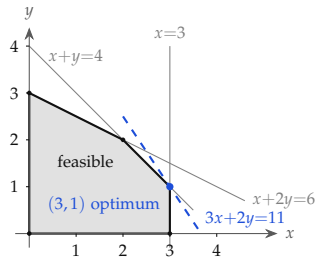
METHOD

1. Define variables, objective and constraints (formulation, O.1).
2. Graph each constraint as a line and shade the side that satisfies it; the overlap is the feasible region.
3. List the corner points (intersections of pairs of boundary lines that are feasible).
4. Evaluate z at each corner; the best is optimal. Example: $\max 3x + 2y$ s.t. $x + 2y \le 6$, $x + y \le 4$, $x \le 3$, $x, y \ge 0$. Corners $(0,0), (0,3), (2,2), (3,1), (3,0)$ give $z = 0, 6, 10, 11, 9$: optimum $(3,1)$, $z = 11$.

DECISION POINT Outcomes: unique optimum; *multiple optima* (iso-profit line parallel to an edge); *infeasible* (empty region: contradictory constraints); *unbounded* (region open in the direction of $\mathbf{c}$). For more than 2–3 variables use a solver: simplex is fast in practice and gives a vertex and shadow prices O.4; interior-point methods are faster on very large sparse LPs.

TERMS

$\mathbf{c}$ – objective coefficients (profit per unit)
A, $\mathbf{b}$ – constraint matrix and right-hand sides A2.24 §04 Algebra II
$\mathbf{s}$ – slack: unused amount of each resource
Feasible region – convex polyhedron formed by half-planes A1.20 §03 Algebra I
Vertex – corner where n independent constraints are tight
k – a fixed objective value (iso-profit level) A1.7 §03 Algebra I
Solvers – simplex O.4 and interior-point methods O.20
z; $\mathbf{x}$ – objective value; decision variables



CONTEXT

- **Origin.** Kantorovich (1939) and Dantzig (1947); Nobel Prize in economics 1975.
- **Use.** Production mix, diets, blending, transportation, staffing.
- **Data science.** Budget allocation across ad channels, supply-chain planning, L1 (least absolute deviation) regression and quantile regression are solvable as LPs.
- **Physical application.** Blending ores or fuels to meet chemical specifications at minimum cost is a classic LP.

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O.4 Simplex Method & LP Duality

The simplex method walks from vertex to adjacent vertex of the feasible region, improving the objective at every pivot. Every LP has a dual LP whose optimal value equals the primal's; the dual variables are the prices of the constraints.

Ratio test	$\text{leave: } \arg\min_i \left\{ \frac{b_i}{a_{ij}} : a_{ij} > 0 \right\}$	The first constraint to become tight as the entering variable x_j grows.
Optimality test	$\bar{c}_j = c_j - \mathbf{c}_B^\top B^{-1} A_j \le 0 \ \ \forall j$	(For max.) No variable can enter profitably: the current vertex is optimal.
Primal–dual pair	$\text{Primal: } \max \ \mathbf{c}^\top \mathbf{x} \ \text{ s.t. } A\mathbf{x} \le \mathbf{b},\ \mathbf{x} \ge \mathbf{0} \quad \Longleftrightarrow \quad \text{Dual: } \min \ \mathbf{b}^\top \mathbf{y} \ \text{ s.t. } A^\top \mathbf{y} \ge \mathbf{c},\ \mathbf{y} \ge \mathbf{0}$	Each primal constraint becomes a dual variable.
Weak duality	$\mathbf{c}^\top \mathbf{x} \le \mathbf{b}^\top \mathbf{y}$	Any dual-feasible $\mathbf{y}$ bounds every primal-feasible objective.
Strong duality	$\mathbf{c}^\top \mathbf{x}^* = \mathbf{b}^\top \mathbf{y}^*$	At the optimum the bounds meet: a certificate of optimality.
Complementary slackness	$y_i^* \left(b_i - (A\mathbf{x}^*)_i \right) = 0$	A resource with leftover slack has price zero.

STATED-EQUATION *The leaving row minimizes b i over a i j among positive a i j. The vertex is optimal when every reduced cost is at most zero. Primal: maximize c transpose x subject to A x at most b, x at least zero. Dual: minimize b transpose y subject to A transpose y at least c, y at least zero. Weak duality: c transpose x is at most b transpose y. Strong duality: at the optimum they are equal. Complementary slackness: y i star times the slack of constraint i equals zero.*

MEMORY AID Each primal constraint becomes a dual variable; max becomes min.

METHOD

1. Add slacks; start at the vertex $\mathbf{x} = \mathbf{0}$ (all slacks basic).
2. Entering variable: the one with the largest positive reduced cost $\bar{c}_j$ (most negative entry in the z-row of the tableau).
3. Leaving variable: ratio test on the entering column.
4. Pivot (row-reduce LA.17 §09 Linear Algebra) so the entering variable becomes basic; repeat until all $\bar{c}_j \le 0$.
5. Example (LP of O.3): x enters ($\bar{c} = 3$); ratios $6, 4, 3$: $x \le 3$ leaves; $x = 3$, $z = 9$. Then y enters ($\bar{c} = 2$); ratios $1.5, 1$: $y = 1$, $z = 11$; done. Dual: $\mathbf{y}^* = (0, 2, 1)$, $\mathbf{b}^\top \mathbf{y}^* = 4 \cdot 2 + 3 \cdot 1 = 11$.

TERMS

Basic variables – the m variables solved for at a vertex; B their columns, $\mathbf{c}_B$ their costs

$\bar{c}_j$ – reduced cost of x_j: objective gain per unit if it enters; A_j its column

Slack variable – $s_i = b_i - (A\mathbf{x})_i \ge 0$ turns $\le$ into $=$

$\mathbf{y}$ – dual variables $=$ shadow prices: value of one more unit of resource b_i

$A^\top$ – transpose A2.24 §04 Algebra II

Binding constraint – $s_i = 0$; only these can have $y_i > 0$

Degeneracy – ties in the ratio test; can cycle (Bland's rule prevents it)

a_{ij}, b_i – entries of A and of the right-hand side $\mathbf{b}$

B, $\mathbf{c}_B$, A_j – basis matrix, costs of basic variables, column j of A

$\mathbf{x}^*, \mathbf{y}^*$ – primal and dual optima

CONTEXT

- **Origin.** Dantzig's simplex (1947); duality theory by von Neumann (1947) and Gale–Kuhn–Tucker (1951).
- **Use.** Pricing scarce resources; certifying that a solution is optimal.
- **Data science.** Shadow prices tell a planner which capacity to expand; duality is the same idea behind the SVM dual and kernel trick ML.15 §15 Machine Learning and Lagrangian relaxation O.19.
- **Physical application.** Shadow prices tell a power-grid operator how much an extra megawatt of line capacity is worth.

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O.5 Quadratic Programming (QP)

Minimize a quadratic objective subject to linear constraints. When the quadratic is convex the problem is as reliable as an LP, but the optimum may lie on an edge or inside the region rather than at a vertex.

QP form	$\min_{\mathbf{x}} \ \frac{1}{2}\mathbf{x}^\top Q\mathbf{x} + \mathbf{c}^\top\mathbf{x} \quad \text{s.t.} \quad A\mathbf{x} \le \mathbf{b}, \quad E\mathbf{x} = \mathbf{d}$	Quadratic objective, linear constraints.
Convexity test	$Q \succeq 0$	Convex QP: global optimum, polynomial-time; Q indefinite makes it NP-hard.
Unconstrained solution	$\nabla f = Q\mathbf{x} + \mathbf{c} = \mathbf{0} \Rightarrow \mathbf{x}^* = -Q^{-1}\mathbf{c}$	One linear solve when $Q \succ 0$ and no constraint is active.
Equality-constrained KKT system	$\begin{bmatrix} Q & E^\top \\ E & 0 \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \boldsymbol{\nu} \end{bmatrix} = \begin{bmatrix} -\mathbf{c} \\ \mathbf{d} \end{bmatrix}$	Lagrange conditions O.17 for a QP are linear: solve one system.
Markowitz portfolio	$\min_{\mathbf{w}} \ \mathbf{w}^\top\Sigma\mathbf{w} \ \text{ s.t. } \boldsymbol{\mu}^\top\mathbf{w} \ge r, \ \mathbf{1}^\top\mathbf{w} = 1$	Least risk for a target return; $\Sigma \succeq 0$ so it is convex FN.12 §17 Finance.

STATED-EQUATION *Minimize one half x transpose Q x plus c transpose x, subject to A x at most b and E x equal to d. It is convex when Q is positive semidefinite. Without active constraints, x star equals minus Q inverse c. With equality constraints, solve the block system Q, E transpose; E, zero, times x, nu, equals minus c, d. Markowitz: minimize the portfolio variance w transpose Sigma w, subject to expected return mu transpose w at least r and weights summing to one.*

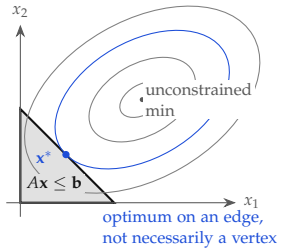
MEMORY AID LP with a bowl on top: the bottom of the bowl may sit on a face, not a corner.

METHOD

1. Check $Q \succeq 0$ (eigenvalues LA.27 §09 Linear Algebra); if so any KKT point is the global optimum.
2. Solve the unconstrained or equality-constrained system; if all inequalities hold, done.
3. Otherwise guess which inequalities are active, add them as equalities, re-solve, and check the multipliers' signs (active-set method, O.18).
4. Example: $\min x^2 + y^2$ s.t. $x + y = 2$: $2x + \nu = 0$, $2y + \nu = 0$, $x + y = 2$ give $x = y = 1$, $\nu = -2$, minimum value 2.

TERMS

Q – symmetric $n \times n$ matrix (the Hessian of the objective)
$\mathbf{c}$ – linear cost vector; $A, \mathbf{b}$ inequality data; $E, \mathbf{d}$ equality data
$\boldsymbol{\nu}$ – multipliers of the equality constraints O.17
$\mathbf{w}, \Sigma, \boldsymbol{\mu}, r$ – portfolio weights, return covariance S.38 §13 Statistics, mean returns, target return
$\mathbf{1}$ – vector of ones (weights sum to 1)
Solvers – active set, interior point, ADMM/OSQP
$A, \mathbf{b}$; $E, \mathbf{d}$ – inequality and equality constraint data
$\succeq$; $\mathbf{x}^*$ – positive semidefinite LA.27 §09 Linear Algebra; optimum



CONTEXT

- **Origin.** Frank and Wolfe (1956); Markowitz portfolio theory (1952, Nobel 1990).
- **Use.** Portfolio optimization, model-predictive control, least squares with bounds.
- **Data science.** The SVM training problem is a convex QP ML.15 §15 Machine Learning; lasso and ridge ML.8 §15 Machine Learning and constrained least squares are QPs.
- **Physical application.** Model-predictive control solves a small QP every few milliseconds to steer cars and drones.

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O.6 Integer & Mixed-Integer Programming (MIP)

An LP in which some variables must be integers or yes/no (binary). Integrality makes the model far more expressive (logic, fixed costs, choices) and far harder to solve.

MIP form	$\max\ \mathbf{c}^\top \mathbf{x}\ \text{ s.t. } A\mathbf{x} \le \mathbf{b},\ \ x_j \in \mathbb{Z}\ (j \in I)$	Some (MIP) or all (IP) variables integer; binaries $y_k \in \{0,1\}$ model decisions.
LP-relaxation bound	$z_{\text{MIP}} \le z_{\text{LP relax}}$	(For max.) Dropping integrality can only help, so the LP value is an upper bound.
Big-M switch	$x \le M\,y, \quad y \in \{0,1\}$	$y = 0$ forces $x = 0$; $y = 1$ allows x up to M (fixed costs, "only if open").
Branching	$x_j \le \lfloor v \rfloor\ \text{ or }\ x_j \ge \lceil v \rceil$	Split on a fractional LP value v; the two children exclude it.
Logical constraints	$y_1 + y_2 \le 1; \quad y_1 \le y_2; \quad \sum_k y_k \ge K$	At most one; "if 1 then 2"; at least K chosen.
Optimality gap	$\text{gap} = \dfrac{z_{\text{bound}} - z_{\text{incumbent}}}{\lvert z_{\text{incumbent}} \rvert}$	Solver progress: 0 proves optimality.

STATED-EQUATION *Maximize c transpose x subject to A x at most b, with x j integer for j in I. The MIP optimum is at most the LP-relaxation optimum. Big M: x at most M times y. Branch on x j at most the floor of v or at least the ceiling of v. The gap is the bound minus the incumbent, over the incumbent. Binary logic: y one plus y two at most one means at most one of the two; y one at most y two means one implies two; the sum of the y k at least K means at least K are chosen.*

MEMORY AID Relax, bound, branch, prune.

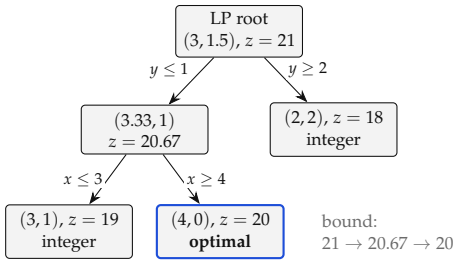
METHOD

1. Solve the LP relaxation. If its solution is integer, it is optimal.
2. Otherwise pick a fractional variable and branch into two subproblems.
3. Prune a node if it is infeasible, or its LP bound is no better than the best integer solution found (the incumbent).
4. Repeat until no open nodes remain. Example: $\max 5x + 4y$ s.t. $6x + 4y \le 24$, $x + 2y \le 6$: LP $(3, 1.5)$, $z = 21$; branching on y then x finds $(4, 0)$, $z = 20$; rounding the LP would give only $(3, 1)$, $z = 19$.

DECISION POINT Keep M as small as is valid (a huge M weakens the LP bound and causes numerical errors). Prefer tight formulations, since a stronger relaxation prunes more. If the gap will not close in time, accept the incumbent with its gap, or switch to a heuristic O.22.

TERMS

I – index set of integer variables (the rest are continuous)
Binary y – yes/no decisions: open a warehouse, pick an item
LP relaxation – drop integrality; gives an upper bound (for max) O.3
$\lfloor v \rfloor, \lceil v \rceil$ – floor and ceiling of v
Branch and bound – split on a fractional x_j and prune by bounds
Big-M – large constant that switches a constraint on or off
Incumbent – best integer solution so far; cutting planes tighten the bound
Hardness – NP-hard in general O.10; rounding the LP can be infeasible or far from optimal
M; v – constant large enough never to bind when $y = 1$; fractional LP value of x_j
z_{bound}, $z_{\text{incumbent}}$ – best proven bound; best feasible solution found
K; $\mathbb{Z}$ – minimum number of options chosen; the integers



CONTEXT

- **Origin.** Gomory cutting planes (1958); branch and bound by Land and Doig (1960).
- **Use.** Facility location, crew scheduling, production with setup costs.
- **Data science.** Optimal assignment of promotions under budget, shift scheduling, sparse (best-subset) regression, and decision optimization after an ML forecast ("predict, then optimize").
- **Physical application.** Deciding which power plants to switch on each hour (unit commitment) is a mixed-integer program.

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Combinatorial & Network Optimization

O.7 Knapsack Problem

Choose a subset of items, each with a value and a weight, to maximize total value without exceeding a capacity. It is the simplest NP-hard integer program and the classic example of dynamic programming.

0/1 knapsack model	$\max \sum_{i=1}^{n} v_i x_i \;\text{ s.t. }\; \sum_{i=1}^{n} w_i x_i \le W, \;\; x_i \in \{0,1\}$	One budget constraint, binary take/skip decisions O.6.
DP recurrence	$V(i,w) = \max\{V(i-1,w),\; v_i + V(i-1,w-w_i)\}$	Skip item i, or take it (if $w_i \le w$) and solve the rest with less capacity.
Base case	$V(0,w) = 0, \qquad V(i,w) = V(i-1,w) \;\text{ if } w_i > w$	No items, no value; an item too heavy must be skipped.
Fractional (greedy) bound	sort by $\frac{v_i}{w_i}$, fill, split the last item	Optimal when items are divisible; an upper bound for the 0/1 problem.
Complexity	$O(nW)$ table vs 2^n subsets	Pseudo-polynomial: fast when W is a small integer O.10.

STATED-EQUATION *Maximize the sum of v i x i subject to the sum of w i x i at most W, with each x i zero or one. The recurrence: V of i, w equals the maximum of V of i minus one, w, and v i plus V of i minus one, w minus w i. With zero items the value is zero. If item i does not fit, V of i, w equals V of i minus one, w. The fractional version sorts by value per weight and splits the last item, which gives an upper bound. The table costs order n W, versus two to the n subsets by brute force.*

MEMORY AID Each item: "skip it or take it" — keep the better.

METHOD

1. Make a table with rows $i = 0..n$ (items) and columns $w = 0..W$ (capacity); fill row 0 with zeros.
2. Fill row by row with the recurrence.
3. The answer is $V(n,W)$; trace back (where the "take" branch won) to recover the items.
4. Example: $(w,v) = (1,1),(3,4),(4,5)$, $W = 7$: $V(3,7) = \max\{V(2,7),\, 5 + V(2,3)\} = \max\{5,\, 5+4\} = 9$: take items 2 and 3.

DECISION POINT Small integer $W \to$ DP. Large or real-valued weights $\to$ MIP solver O.6 (the knapsack is easy in practice). Need a fast approximate answer $\to$ greedy by v_i/w_i (also take the single best item: this guarantees at least half the optimum).

TERMS

v_i, w_i – value and weight of item i
W – capacity (budget)
x_i – binary: take item i or not O.6
$V(i,w)$ – best value using items $1..i$ with capacity w (dynamic programming O.21)
Cost – $O(nW)$ table vs 2^n subsets S.8 §13 Statistics
Fractional version – greedy by v_i/w_i is optimal; gives the LP bound
n; w – number of items; remaining capacity

CONTEXT

- **Origin.** Studied by Dantzig (1957); named by Tobias Dantzig; one of Karp's 21 NP-complete problems (1972).
- **Use.** Capital budgeting, cargo loading, cutting stock.
- **Data science.** Selecting features or experiments under a compute budget, choosing which campaigns to fund for maximum expected lift, and packing jobs onto GPUs.
- **Physical application.** Packing a spacecraft's limited mass budget with the most valuable instruments is a knapsack problem.

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O.8 Traveling Salesman Problem (TSP)

Find the shortest closed tour that visits every city exactly once. The number of tours explodes factorially, so exact methods use integer programming with clever cuts, and large instances use heuristics.

Objective	$\min \sum_{i \neq j} c_{ij} x_{ij}, \quad x_{ij} \in \{0,1\}$	Total length of the chosen edges.
Degree constraints	$\sum_j x_{ij} = 1 \ \ \forall i, \qquad \sum_i x_{ij} = 1 \ \ \forall j$	Leave each city once and enter each city once.
Subtour elimination	$\sum_{i,j \in S} x_{ij} \leq \lvert S \rvert - 1 \quad \forall\, S \subset V,\ 2 \leq \lvert S \rvert \leq n-1$	Forbids closed loops on a proper subset of cities.
Number of tours	$\dfrac{(n-1)!}{2}$	Symmetric distances; 10 cities already give 181 440 tours.
2-opt move gain	$\Delta = c_{ac} + c_{bd} - c_{ab} - c_{cd}$	Replace edges ab, cd by ac, bd (reversing the segment between); accept if $\Delta < 0$.

STATED-EQUATION *Minimize the sum of c i j x i j subject to: each city is left once, each city is entered once, no subtours on any subset S, and x i j binary. There are n minus one factorial over two distinct tours. A 2-opt move changes the length by c a c plus c b d minus c a b minus c c d.*

MEMORY AID $(n-1)!/2$: fix the start, and a tour and its reverse are the same.

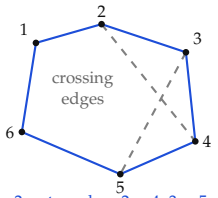
METHOD

1. **Nearest neighbor**: start anywhere; repeatedly go to the closest unvisited city; return home.
2. **2-opt**: for every pair of edges, compute Δ; if $\Delta < 0$, swap them (this removes crossings).
3. Repeat until no improving swap exists (a 2-opt local optimum); restart or anneal O.22 to escape. Example: 4 cities on a unit square: the crossing tour has length $2 + 2\sqrt{2} \approx 4.83$, the 2-opt move gives the perimeter 4.

DECISION POINT Up to a few thousand cities: exact branch-and-cut (Concorde, MIP O.6). Larger, or with time windows and vehicle capacities (vehicle routing): heuristics such as Lin–Kernighan, local search O.23, or genetic algorithms O.24. Christofides' algorithm guarantees $\leq 1.5\times$ optimal for metric distances.

TERMS

c_{ij} – distance or cost from city i to j G.12 §05 Geometry
x_{ij} – 1 if the tour goes $i \to j$ O.6
V, S – the set of all n cities; a subset of them
a, b, c, d – endpoints of the two edges in a 2-opt move
$(n-1)!/2$ – distinct symmetric tours S.8 §13 Statistics, M.48 §01 Math Signs
Heuristics – nearest neighbor, 2-opt, simulated annealing O.22
Δ; n – change in tour length; number of cities



2-opt: replace 2→4, 3→5 by 2→3, 4→5 (shorter)

CONTEXT

- **Origin.** Posed by Menger (1930s); Dantzig–Fulkerson–Johnson solved 49 cities (1954); Concorde now solves tens of thousands.
- **Use.** Delivery routes, drilling circuit boards, genome sequencing.
- **Data science.** Last-mile delivery and ride-pooling routing (vehicle routing problems), warehouse pick paths, and ordering items to minimize changeover.
- **Physical application.** Routing a laser or drill head over thousands of points on a circuit board is a TSP.

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O.9 Shortest Paths & Network Flow

Graphs model roads, pipes and networks. Shortest-path algorithms find the cheapest route from a source; flow algorithms push the most material through capacitated edges, limited by the tightest cut.

Bellman optimality	$d(s)=0,\quad d(v)=\min_{(u,v)\in E}\big[d(u)+w(u,v)\big]$	The best route to v comes through the best route to some predecessor u.
Edge relaxation	$d(v)\leftarrow\min\{d(v),\ d(u)+w(u,v)\}$	The update every shortest-path algorithm repeats.
Complexities	Dijkstra $O\big((V+E)\log V\big)$; Bellman–Ford $O(VE)$	Dijkstra needs $w\geq 0$; Bellman–Ford allows negative edges and detects negative cycles.
Maximum flow	$\max\sum_v f_{sv}\quad\text{s.t.}\quad 0\leq f_{uv}\leq c_{uv},\qquad \sum_u f_{uv}=\sum_w f_{vw}\ \ (v\neq s,t)$	Capacity limits and conservation at every interior node.
Max-flow min-cut	max flow $=$ min cut capacity	The bottleneck set of edges limits throughput (LP duality 0.4).

STATED-EQUATION *d of s equals zero, and d of v equals the minimum over edges u v of d of u plus w of u v. Relaxation replaces d of v by d of u plus w of u v when that is smaller. Max flow: maximize the flow out of s subject to capacities and flow conservation at each node. Max flow equals min cut. Dijkstra runs in order V plus E times log V time, and Bellman-Ford in order V times E.*

MEMORY AID A chain is only as strong as its weakest link: max flow = min cut.

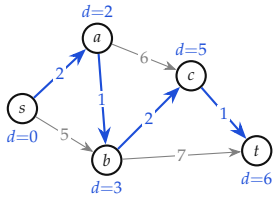
METHOD

1. Dijkstra: set $d(s)=0$, all others ∞; all nodes unsettled.
2. Settle the unsettled node u with the smallest $d(u)$; relax every edge leaving u.
3. Repeat until the target is settled; follow predecessor links back for the path. Example (figure): settle s (a=2, b=5), a (b=3, c=8), b (c=5, t=10), c (t=6): path $s\,a\,b\,c\,t$, length 6.

DECISION POINT Unweighted graph → breadth-first search. Non-negative weights → Dijkstra (A* with a good distance heuristic if one target). Negative weights → Bellman–Ford. All pairs on a dense graph → Floyd–Warshall $O(V^3)$. Capacities and throughput → max flow (Ford–Fulkerson / Edmonds–Karp).

TERMS

$d(v)$ – shortest distance from source s to node v
$w(u,v)$ – edge weight (length, time, cost)
V, E – numbers of nodes and edges
f_{uv}, c_{uv} – flow and capacity on edge $u\to v$; t the sink
Conservation – flow in = flow out at every non-terminal node
Cut – set of edges separating s from t; its capacity bounds the flow (duality 0.4)
s, t; u, v – source and sink; nodes



CONTEXT

- **Origin.** Dijkstra (1956), Bellman (1958); max-flow min-cut by Ford and Fulkerson (1956).
- **Use.** GPS navigation, network routing, pipeline and traffic capacity.
- **Data science.** Graph analytics (shortest paths for centrality), matching riders to drivers (bipartite flow), image segmentation by min cut, and the Bellman equation in RL 0.21.
- **Physical application.** Water-distribution and traffic networks: the maximum throughput equals the capacity of the tightest bottleneck cut.

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O.10 Algorithmic Complexity (Big-O)

Big-O describes how an algorithm's running time or memory grows with the input size, ignoring constant factors. It separates tractable (polynomial) problems from those that explode (exponential).

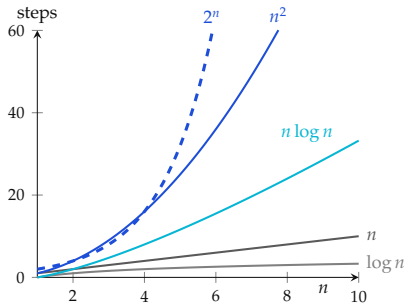
Big-O (upper bound)	$f(n) = O(g(n)) \iff \exists\, c > 0, n_0 :\ f(n) \le c\, g(n)\ \ \forall n \ge n_0$	Grows no faster than g, eventually.
Big-Ω and Big-Θ	$f = \Omega(g) \iff g = O(f); \quad f = \Theta(g) \iff$ both	Lower bound; tight bound (same growth rate).
Simplification rule	$3n^2 + 5n + 7 = O(n^2)$	Keep the dominant term, drop its constant.
Growth hierarchy	$O(1) < O(\log n) < O(n) < O(n \log n) < O(n^2) < O(n^3) < O(2^n) < O(n!)$	Polynomial on the left is "tractable"; exponential and factorial are not.
P and NP	P: solvable in poly time; NP: checkable in poly time	NP-hard problems (TSP, MIP) have no known polynomial algorithm.

STATED-EQUATION *f of n is big O of g of n if and only if there are constants c and n naught such that f of n is at most c times g of n for all n at least n naught. Omega is the reverse bound, Theta both. The common orders, from fastest to slowest: constant, logarithmic, linear, n log n, quadratic, cubic, exponential, factorial. Drop constants and lower-order terms: three n squared plus five n plus seven is big O of n squared. P is the class of problems solvable in polynomial time; NP is the class whose solutions can be checked in polynomial time.*

MEMORY AID Drop constants and lower-order terms: $3n^2 + 5n + 7 = O(n^2)$.

TERMS

n – input size
c, n_0 – constant factor and threshold in the definition
$O(\log n)$ – binary search: halve each step A2.14 §04 Algebra II
$O(n \log n)$ – efficient sorting, FFT
$O(n^2)$ – all pairs (e.g. naive k-NN, distance matrices)
$O(n^3)$ – dense matrix multiplication and inversion LA.17 §09 Linear Algebra
$O(2^n), O(n!)$ – brute-force subsets and tours O.7, O.8
Space – memory complexity is measured the same way
f, g – cost function and comparison function of n
Ω; Θ – asymptotic lower bound; tight bound



CONTEXT

- **Origin.** Bachmann (1894) and Landau (1909) for number theory; Knuth popularized it in computing (1976); P vs NP posed by Cook (1971).
- **Use.** Predicting how runtime scales with data size.
- **Data science.** Choosing algorithms for big data: exact k-NN is $O(n)$ per query, approximate nearest-neighbor indexes are sublinear; transformer attention is $O(n^2)$ in sequence length.
- **Physical application.** Simulating gravity between n stars by all pairs is $O(n^2)$; tree codes cut it to $O(n \log n)$, making galaxy simulations feasible.

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Unconstrained Continuous Optimization

O.11 Gradient Descent & Convergence

Repeatedly step opposite the gradient, the direction of steepest decrease. Its speed is governed by the step size and by the condition number: the ratio of the largest to the smallest curvature.

Gradient descent update	$\mathbf{x}_{k+1} = \mathbf{x}_k - \eta\,\nabla f(\mathbf{x}_k)$	Move downhill by a step proportional to the slope C1.26 §07 Calculus I.
Step-size rule	$0 < \eta < \dfrac{2}{L}, \qquad \eta = \dfrac{1}{L}$ safe	Too large overshoots and diverges; L is the largest curvature.
Convex rate	$f(\mathbf{x}_k) - f^* \le \dfrac{\lVert \mathbf{x}_0 - \mathbf{x}^* \rVert^2}{2\eta k}$	Sublinear $O(1/k)$ for convex L-smooth f with $\eta \le 1/L$.
Strongly convex rate	$f(\mathbf{x}_k) - f^* \le \left(1 - \dfrac{1}{\kappa}\right)^k \left(f(\mathbf{x}_0) - f^*\right)$	Linear (geometric) convergence with $\eta = 1/L$: about $\kappa \ln(1/\varepsilon)$ steps.
Condition number	$\kappa = \dfrac{L}{m} = \dfrac{\lambda_{\max}(\nabla^2 f)}{\lambda_{\min}(\nabla^2 f)}$	Large κ: long narrow valleys and zig-zagging LA.29 §09 Linear Algebra.
Momentum (heavy ball)	$\mathbf{x}_{k+1} = \mathbf{x}_k - \eta\nabla f(\mathbf{x}_k) + \beta(\mathbf{x}_k - \mathbf{x}_{k-1})$	Reuses the last step to damp zig-zags and roll through flat regions ML.27 §15 Machine Learning.
Nesterov acceleration	$O(1/k^2)$ convex; $\sim\sqrt{\kappa}\ln\frac{1}{\varepsilon}$ steps	Gradient at a look-ahead point; optimal for first-order methods.

STATED-EQUATION *x k plus one equals x k minus eta times the gradient of f at x k. The step size must be between zero and two over L. For convex f the error after k steps is at most the squared initial distance over two eta k. For strongly convex f it shrinks by a factor one minus one over kappa each step, where kappa, the condition number, is L over m. Momentum adds beta times the previous step. Nesterov acceleration improves the convex rate to one over k squared, and needs about root kappa times log of one over epsilon steps when f is strongly convex.*

MEMORY AID "Walk downhill; take smaller steps where it is steep." Bigger κ, narrower valley, slower descent.

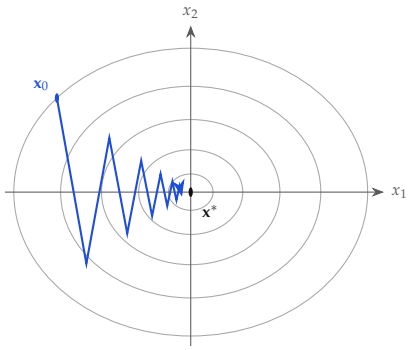
METHOD

1. Choose $\mathbf{x}_0$, a step size η (or a line search O.12) and a tolerance.
2. Compute $\nabla f(\mathbf{x}_k)$; stop if $\lVert\nabla f\rVert$ is below tolerance.
3. Update $\mathbf{x}_{k+1} = \mathbf{x}_k - \eta\nabla f(\mathbf{x}_k)$ and repeat.
4. Example: $f(x) = x^2$, $\eta = 0.25$, $x_0 = 4$: $x_1 = 4 - 0.25\cdot 8 = 2$, $x_2 = 1$, $x_3 = 0.5$; each step halves the error. With $\eta = 1.1 > 2/L = 1$ it diverges.

DECISION POINT Step size: start with $1/L$ if L is known, otherwise backtrack O.12 or try a log-spaced grid $(10^{-1}, 10^{-2}, \dots)$ and keep the largest that decreases f. Slow zig-zagging means large κ: standardize features S.4 §13 Statistics, precondition, add momentum, or switch to a quasi-Newton method O.14.

TERMS

η – step size (learning rate)
∇f – gradient C3.12 §10 Calculus III
L – smoothness constant: largest Hessian eigenvalue (Lipschitz constant of ∇f)
m – strong-convexity constant: smallest Hessian eigenvalue O.2
$\lambda_{\max}, \lambda_{\min}$ – extreme eigenvalues LA.23 §09 Linear Algebra
$f^*, \mathbf{x}^*$ – optimal value and minimizer
β – momentum coefficient, typically 0.9
ε – target accuracy
$\mathbf{x}_k$, k; $\mathbf{x}_0$ – iterate and iteration count; starting point
κ – condition number LA.29 §09 Linear Algebra



$\kappa = 8$: steep across, flat along

CONTEXT

- **Origin.** Cauchy (1847); Polyak's heavy ball (1964); Nesterov (1983).
- **Use.** Large-scale smooth optimization where Hessians are too expensive.
- **Data science.** The workhorse of model training: every neural network is fit by a gradient variant ML.22 §15 Machine Learning; feature scaling speeds it up by lowering κ.
- **Physical application.** A ball rolling downhill in a viscous fluid follows the negative gradient of the potential energy.

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0.12 Line Search: Armijo & Wolfe Conditions

Given a descent direction, a line search chooses how far to move. Instead of minimizing exactly, practical methods accept any step that decreases the function enough and is not too short.

Descent direction	$\nabla f(\mathbf{x})^\top \mathbf{p} < 0$	$\mathbf{p}$ points downhill (e.g. $\mathbf{p} = -\nabla f$ or the Newton direction).
Exact line search	$\alpha^* = \arg\min_{\alpha>0} \phi(\alpha),\ \ \phi(\alpha) = f(\mathbf{x}+\alpha\mathbf{p})$	A 1-D minimization; rarely worth its cost except for quadratics.
Armijo (sufficient decrease)	$\phi(\alpha) \le \phi(0) + c_1\,\alpha\,\nabla f^\top \mathbf{p}$	Actual decrease must be a fraction c_1 of what the slope predicts.
Curvature (Wolfe)	$\nabla f(\mathbf{x}+\alpha\mathbf{p})^\top \mathbf{p} \ge c_2\,\nabla f(\mathbf{x})^\top \mathbf{p}$	Rules out tiny steps: the slope must have flattened.
Backtracking	$\alpha \leftarrow \rho\,\alpha$ until Armijo holds	Start at $\alpha = 1$ and shrink by $\rho \in (0,1)$.

STATED-EQUATION *p is a descent direction when the gradient dotted with p is negative. Exact line search minimizes phi of alpha, which is f of x plus alpha p. Armijo: phi of alpha is at most phi of zero plus c one alpha times the directional derivative. Wolfe curvature: the new directional derivative is at least c two times the old one. Backtracking multiplies alpha by rho until Armijo holds.*

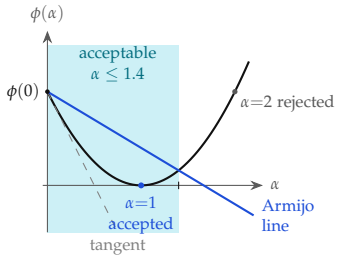
MEMORY AID Armijo: “not too long”. Wolfe curvature: “not too short”.

METHOD

1. Set $\alpha = 1$ (the natural step for Newton/quasi-Newton), $c_1 = 10^{-4}$, $\rho = 0.5$.
2. If the Armijo condition fails, set $\alpha \leftarrow \rho\alpha$ and test again.
3. Accept α and update $\mathbf{x} \leftarrow \mathbf{x} + \alpha\mathbf{p}$. Example: $f(x) = x^2$, $x = 1$, $\mathbf{p} = -2$: $\alpha = 1$ gives $f(-1) = 1 > 1 - 0.0004$, rejected; $\alpha = 0.5$ gives $f(0) = 0$, accepted.

TERMS

$\mathbf{p}$ – search direction
α – step length along $\mathbf{p}$
$\phi(\alpha)$ – the function restricted to the line
c_1, c_2 – constants $0 < c_1 < c_2 < 1$; typically 10^{-4} and 0.9
ρ – backtracking factor, e.g. 0.5
Strong Wolfe – uses $|\nabla f(\mathbf{x}+\alpha\mathbf{p})^\top \mathbf{p}| \le c_2 |\nabla f^\top \mathbf{p}|$



CONTEXT

- **Origin.** Armijo (1966); Wolfe (1969).
- **Use.** Globalizes Newton and quasi-Newton methods so they converge from far away.
- **Data science.** L-BFGS in SciPy and PyTorch uses a strong-Wolfe line search; backtracking replaces learning-rate tuning in full-batch fitting.
- **Physical application.** Like a hiker who steps forward, checks the altitude dropped enough, and otherwise steps back halfway.

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O.13 Nonlinear Programming & Newton's Method

For smooth nonlinear objectives, a minimum has zero gradient and positive curvature. Newton's method jumps to the minimum of the local quadratic (Taylor) model, converging extremely fast once close.

First-order condition	$\nabla f(\mathbf{x}^*) = \mathbf{0}$	Necessary at an interior local minimum (stationary point) C1.23 §07 Calculus I.
Second-order condition	$\nabla^2 f(\mathbf{x}^*) \succ 0$	Together with $\nabla f = \mathbf{0}$, sufficient for a strict local minimum C3.16 §10 Calculus III.
Newton (1-D)	$x_{k+1} = x_k - \dfrac{f'(x_k)}{f''(x_k)}$	Root-finding C1.35 §07 Calculus I applied to f'.
Newton step (n-D)	$\mathbf{x}_{k+1} = \mathbf{x}_k - \left[\nabla^2 f(\mathbf{x}_k)\right]^{-1} \nabla f(\mathbf{x}_k)$	Minimizes the 2nd-order Taylor model C2.22 §08 Calculus II; solve $H\mathbf{p} = -\nabla f$, never invert.
Quadratic convergence	$\lVert \mathbf{x}_{k+1} - \mathbf{x}^* \rVert \le C \lVert \mathbf{x}_k - \mathbf{x}^* \rVert^2$	Correct digits roughly double each step near the optimum.
Gauss–Newton	$\mathbf{p} = -(J^\top J)^{-1} J^\top \mathbf{r}$	For least squares $\frac{1}{2}\lVert \mathbf{r}(\mathbf{x}) \rVert^2$: drop second derivatives of $\mathbf{r}$ LA.20 §09 Linear Algebra.

STATED-EQUATION *First order: the gradient at x star equals zero. Second order: the Hessian at x star is positive definite. Newton's method: x k plus one equals x k minus f prime over f double prime; in several variables, x k plus one equals x k minus the Hessian inverse times the gradient. The error is squared each step. Gauss-Newton uses J transpose J in place of the Hessian.*

MEMORY AID Newton fits a parabola and jumps to its bottom.

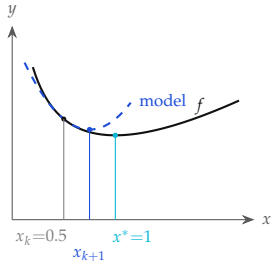
METHOD

1. Compute ∇f and $\nabla^2 f$ at $\mathbf{x}_k$.
2. Solve $\nabla^2 f(\mathbf{x}_k)\,\mathbf{p} = -\nabla f(\mathbf{x}_k)$ for the Newton direction $\mathbf{p}$.
3. Update $\mathbf{x}_{k+1} = \mathbf{x}_k + \alpha \mathbf{p}$ ($\alpha = 1$, or damped by a line search O.12).
4. Stop when $\lVert \nabla f \rVert$ is tiny. Example: $f(x) = x - \ln x$: $x_{k+1} = 2x_k - x_k^2$; from 0.5: 0.75, 0.9375, 0.996 (errors $0.25, 0.0625, 0.004$) $\to x^* = 1$.

DECISION POINT Newton when n is modest (Hessian solve $O(n^3)$) and second derivatives are available. If $\nabla^2 f$ is not positive definite the step may go uphill: add λI (Levenberg–Marquardt) or use a trust region. Least squares → Gauss–Newton / LM. Large n → quasi-Newton O.14 or gradient methods O.11.

TERMS

∇f – gradient; zero at a stationary point C3.12 §10 Calculus III
$\nabla^2 f$ – Hessian; positive definite ⇒ strict local min O.2
$\mathbf{p}$ – Newton direction; α step length
C – constant depending on f near $\mathbf{x}^*$
$\mathbf{r}$, J – residual vector and its Jacobian C3.15 §10 Calculus III
Convergence – quadratic near the optimum, but each step costs a linear solve
$\succ$; $\mathbf{x}^*$ – positive definite LA.27 §09 Linear Algebra; minimizer
f', f'' – first and second derivatives (1-D)



CONTEXT

- **Origin.** Newton (1669) and Raphson (1690) for roots; applied to optimization by Simpson (1740); Gauss–Newton (1809).
- **Use.** Engineering design, curve fitting, maximum-likelihood estimation S.26 §13 Statistics.
- **Data science.** Logistic regression is fit by Newton's method (IRLS); `scikit-learn` defaults to L-BFGS; XGBoost uses gradient and Hessian of the loss.
- **Physical application.** Finding the equilibrium shape of a hanging chain by minimizing potential energy is an NLP.

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O.14 Quasi-Newton Methods (BFGS, L-BFGS)

Quasi-Newton methods build an approximation of the inverse Hessian from successive gradient differences, getting most of Newton's speed using only gradients.

Search direction	$\mathbf{p}_k = -H_k \nabla f(\mathbf{x}_k)$	$H_k \approx [\nabla^2 f]^{-1}$; no Hessian is ever formed or inverted.
Step and gradient change	$\mathbf{s}_k = \mathbf{x}_{k+1} - \mathbf{x}_k, \quad \mathbf{y}_k = \nabla f_{k+1} - \nabla f_k$	The data used to learn curvature.
Secant condition	$H_{k+1}\mathbf{y}_k = \mathbf{s}_k$	The new approximation must reproduce the observed gradient change.
BFGS inverse update	$H_{k+1} = (I - \rho_k \mathbf{s}_k \mathbf{y}_k^\top) H_k (I - \rho_k \mathbf{y}_k \mathbf{s}_k^\top) + \rho_k \mathbf{s}_k \mathbf{s}_k^\top, \qquad \rho_k = \dfrac{1}{\mathbf{y}_k^\top \mathbf{s}_k}$	Rank-two correction; keeps H positive definite if $\mathbf{y}_k^\top \mathbf{s}_k > 0$.
Curvature condition	$\mathbf{y}_k^\top \mathbf{s}_k > 0$	Guaranteed by a Wolfe line search O.12.
L-BFGS cost	$O(mn)$ time and memory per step	Keeps only the last m (5–20) pairs $(\mathbf{s}, \mathbf{y})$ instead of an $n \times n$ matrix.

STATED-EQUATION *The direction p k equals minus H k times the gradient. s k is the step and y k the change in gradient. The secant condition: H k plus one times y k equals s k. BFGS: H k plus one equals I minus rho s y transpose, times H k, times I minus rho y s transpose, plus rho s s transpose, with rho equal to one over y transpose s. The update needs y transpose s positive. L-BFGS costs order m n per step.*

MEMORY AID "Learn the curvature from how the gradient changes."

METHOD

1. Start with $H_0 = I$ (or a scaled identity).
2. Direction $\mathbf{p}_k = -H_k \nabla f_k$; choose α_k by a Wolfe line search.
3. Step, then compute $\mathbf{s}_k, \mathbf{y}_k$ and apply the BFGS update.
4. Repeat until $\|\nabla f\|$ is small. Example: on $f(x) = x^2$ from $x_0 = 1$ with $H_0 = 1$ and exact line search, $x_1 = 0$ immediately; on an n-D quadratic with exact line search BFGS finishes in at most n steps.

DECISION POINT Which method? Huge n or noisy minibatch gradients → (S)GD, Adam O.16. Smooth, deterministic, medium-to-large n → L-BFGS (the default in many libraries). Small n with cheap exact Hessian → Newton O.13. Bound constraints → L-BFGS-B.

	Gradient descent	BFGS / L-BFGS	Newton
Uses	∇f	∇f (curvature learned)	∇f, $\nabla^2 f$
Cost per step	$O(n)$	$O(n^2)$ / $O(mn)$	$O(n^3)$
Memory	$O(n)$	$O(n^2)$ / $O(mn)$	$O(n^2)$
Convergence	linear (depends on κ)	superlinear	quadratic

TERMS

H_k – inverse-Hessian approximation (symmetric positive definite LA.27 §09 Linear Algebra)
$\mathbf{s}_k, \mathbf{y}_k$ – step and gradient difference
ρ_k – scalar $1/(\mathbf{y}_k^\top \mathbf{s}_k)$
m – number of stored pairs in L-BFGS
I – identity matrix; $\mathbf{s}\mathbf{y}^\top$ is an outer product LA.12 §09 Linear Algebra
Superlinear – error ratio → 0: faster than linear, slower than quadratic
n – number of variables

CONTEXT

- **Origin.** Davidon (1959); Broyden, Fletcher, Goldfarb and Shanno (1970); L-BFGS by Nocedal (1980).
- **Use.** General-purpose smooth unconstrained optimization.
- **Data science.** Default solver for logistic regression in `scikit-learn`, for Gaussian-process hyperparameters, and for `scipy.optimize.minimize`.
- **Physical application.** Fitting the parameters of a physics simulator (e.g. a climate or orbit model) to observations.

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O.15 Coordinate Descent

Optimize one variable (or block) at a time while holding the others fixed, cycling through the coordinates. Each subproblem is cheap, often solvable in closed form.

Exact coordinate update	$x_i \leftarrow \arg\min_t \; f(x_1, \ldots, x_{i-1}, t, x_{i+1}, \ldots, x_n)$	Minimize along one axis; repeat for $i = 1, \ldots, n$.
Coordinate gradient step	$x_i \leftarrow x_i - \dfrac{1}{L_i} \dfrac{\partial f}{\partial x_i}$	When the 1-D minimization has no closed form.
Soft-thresholding	$S_\lambda(z) = \operatorname{sign}(z) \max(\lvert z \rvert - \lambda, 0)$	Closed-form minimizer of $\frac{1}{2}(t-z)^2 + \lambda \lvert t \rvert$; sets small values exactly to 0.
Lasso coordinate update	$\beta_j \leftarrow \dfrac{S_\lambda(\mathbf{x}_j^\top \mathbf{r}_{(j)})}{\mathbf{x}_j^\top \mathbf{x}_j}$	Partial residual $\mathbf{r}_{(j)}$ excludes feature j ML.8 §15 Machine Learning.

STATED-EQUATION *Coordinate descent replaces x i by the minimizer of f along coordinate i with the others fixed. A gradient version steps x i by minus one over L i times the partial derivative. Soft-thresholding S lambda of z is the sign of z times the larger of absolute z minus lambda and zero. The lasso update sets beta j to S lambda of x j transpose r j, over x j transpose x j.*

MEMORY AID "One knob at a time."

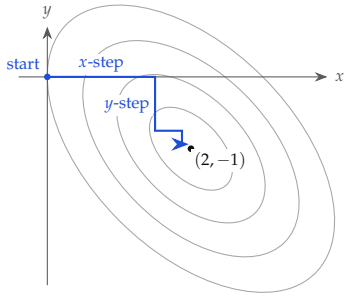
METHOD

1. Pick a starting point and an order of coordinates (cyclic or random).
2. For each coordinate, set $\partial f / \partial x_i = 0$ and solve for x_i with the others fixed.
3. Sweep until the change is below tolerance. Example: $f = x^2 + xy + y^2 - 3x$: $x = (3-y)/2$, $y = -x/2$; from $(0,0)$: $(1.5, 0)$, $(1.5, -0.75)$, $(1.875, -0.75)$, $(1.875, -0.94) \to (2, -1)$.

DECISION POINT Coordinate descent shines when coordinate updates are cheap and the non-smooth part is *separable* (lasso, elastic net, SVM duals). It can stall on non-smooth, non-separable terms (e.g. $\lvert x - y \rvert$), and is slow when variables are strongly coupled (tilted narrow valleys); then use full-gradient or quasi-Newton methods.

TERMS

x_i – the i-th coordinate; others held fixed
L_i – curvature bound along coordinate i
$\partial f / \partial x_i$ – partial derivative C3.9 §10 Calculus III
S_λ – soft-thresholding operator; $\lambda \geq 0$ penalty weight
$\beta_j, \mathbf{x}_j$ – coefficient and data column of feature j
$\mathbf{r}_{(j)}$ – residual $\mathbf{y} - \sum_{k \neq j} \mathbf{x}_k \beta_k$
Block CD – update groups of variables (e.g. alternating least squares)
λ; sign – lasso penalty weight ML.8 §15 Machine Learning; ± 1 by the sign of z
t; z – trial value of coordinate i; input to S_λ



CONTEXT

- **Origin.** Gauss–Seidel iteration (1820s); revived for statistics by Friedman, Hastie and Tibshirani (`glmnet`, 2010).
- **Use.** Large sparse problems with separable structure.
- **Data science.** `glmnet` and `scikit-learn`'s Lasso/ElasticNet use coordinate descent; SMO for SVMs and alternating least squares for recommender matrix factorization are block versions.
- **Physical application.** Tuning a radio by adjusting one dial at a time until no single dial improves the signal.

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O.16 Stochastic Optimization & SGD

When the objective is an expectation or an average over a huge dataset, use cheap, noisy but unbiased gradient estimates. Decreasing step sizes average the noise away.

Stochastic program	$\min_{\mathbf{x}} f(\mathbf{x}) = E_{\xi}[F(\mathbf{x},\xi)]$	Expected cost over random data or scenarios ξ; ERM is the empirical version ML.1 §15 Machine Learning.
SGD update	$\mathbf{x}_{k+1} = \mathbf{x}_k - \eta_k \mathbf{g}_k, \quad E[\mathbf{g}_k] = \nabla f(\mathbf{x}_k)$	Use the gradient of one example or a minibatch.
Minibatch variance	$\mathrm{Var}(\mathbf{g}_k) = \dfrac{\sigma^2}{B}$	Averaging B examples cuts the noise by $1/B$ S.24 §13 Statistics.
Robbins–Monro steps	$\sum_k \eta_k = \infty, \qquad \sum_k \eta_k^2 < \infty$	Steps large enough to get anywhere, small enough to quiet the noise, e.g. $\eta_k = \eta_0/k$.
Convergence rates	$O(1/\sqrt{k})$ convex; $O(1/k)$ strongly convex	Slower per step than full GD, but each step is N times cheaper.
Sample average approximation	$\min_{\mathbf{x}} \dfrac{1}{N}\sum_{i=1}^{N} F(\mathbf{x},\xi_i)$	Replace the expectation by a Monte Carlo average O.27, then solve deterministically.

STATED-EQUATION *Minimize the expected value over xi of F of x and xi. SGD: x k plus one equals x k minus eta k times g k, where g k is an unbiased estimate of the gradient. A minibatch of size B has variance sigma squared over B. Step sizes must sum to infinity while their squares sum to a finite number. Sample average approximation minimizes the average of F over N samples. SGD converges at rate one over root k for convex problems and one over k for strongly convex ones.*

MEMORY AID "Many cheap noisy steps beat a few expensive exact ones."

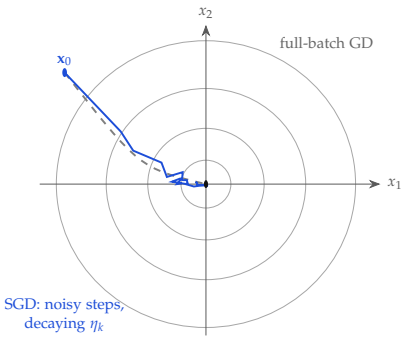
METHOD

1. Shuffle the data; split into minibatches of size B.
2. For each minibatch: compute $\mathbf{g}_k$ (average gradient over the batch), update $\mathbf{x}$.
3. After each epoch, decay η (step, cosine, or $1/k$ schedule) and monitor validation loss ML.5 §15 Machine Learning.
4. Example: estimate a mean μ by $\min_x E[(x-\xi)^2]/2$: SGD with $\eta_k = 1/k$ gives $x_k = x_{k-1} - \frac{1}{k}(x_{k-1} - \xi_k)$, exactly the running sample mean.

DECISION POINT Batch size: larger B gives smoother gradients and better hardware use, but diminishing returns; scale η roughly with B. Constant η converges only to a noise ball around $\mathbf{x}^*$; decay it (or average iterates) for precise answers. Adaptive methods (Adam, RMSProp ML.27 §15 Machine Learning) reduce tuning; variance-reduced methods (SVRG, SAGA) recover linear rates on finite sums.

TERMS

ξ – random data point or scenario; F its cost
$\mathbf{g}_k$ – stochastic gradient (unbiased estimate)
η_k – step size at iteration k (learning-rate schedule)
B – minibatch size; σ^2 per-example gradient variance S.2 §13 Statistics
N – number of samples or scenarios
$E[\cdot]$ – expected value S.13 §13 Statistics
Epoch – one pass through the training data
$F(\mathbf{x},\xi)$ – cost for one random scenario or data point ξ
σ^2; k – variance of a single-sample gradient; iteration count



CONTEXT

- **Origin.** Robbins and Monro (1951); Kiefer–Wolfowitz (1952); revived by large-scale ML (Bottou, 2010).
- **Use.** Optimization under uncertainty: inventory with random demand, energy with random prices.
- **Data science.** Every deep network is trained by minibatch SGD or a variant ML.27 §15 Machine Learning; online learning updates a model one event at a time.
- **Physical application.** Adaptive filters in noise-canceling headphones (LMS algorithm) are SGD on the squared error.

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Constrained Optimization & Duality

O.17 Lagrange Multipliers

To optimize subject to an equality constraint, look for points where the objective's gradient is parallel to the constraint's gradient: the level curve just touches the constraint. The multiplier measures how much the optimum improves per unit of relaxed constraint.

Lagrangian	$\mathcal{L}(\mathbf{x},\lambda) = f(\mathbf{x}) - \lambda\big(g(\mathbf{x}) - c\big)$	Folds the constraint into the objective with a price λ.
Tangency (stationarity)	$\nabla f(\mathbf{x}) = \lambda\,\nabla g(\mathbf{x})$	Gradients parallel: no feasible direction improves f.
Constraint	$g(\mathbf{x}) = c$	Equivalent to $\partial\mathcal{L}/\partial\lambda = 0$.
Sensitivity (shadow price)	$\lambda = \dfrac{\partial f^*}{\partial c}$	Change in the optimal value per unit increase in c O.4.
Several constraints	$\nabla f = \sum_{i=1}^{k} \lambda_i \nabla g_i, \quad g_i(\mathbf{x}) = c_i$	One multiplier per constraint (gradients must be independent).

STATED-EQUATION *The Lagrangian L of x and lambda equals f of x minus lambda times g of x minus c. At the optimum, the gradient of f equals lambda times the gradient of g, and g of x equals c. Lambda equals the partial derivative of the optimal value with respect to c. With several constraints, the gradient of f is a combination of the constraint gradients.*

MEMORY AID At the optimum the level curve kisses (is tangent to) the constraint.

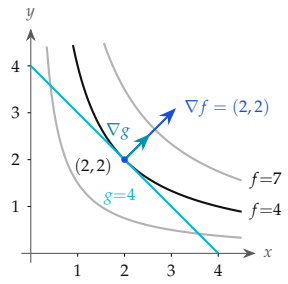
METHOD

1. Write $\mathcal{L} = f - \lambda(g - c)$.
2. Set every partial derivative to zero: $\nabla_{\mathbf{x}}\mathcal{L} = \mathbf{0}$ and $g = c$.
3. Solve the system (often divide equations to eliminate λ).
4. Evaluate f at every candidate (and at boundary/endpoints) to identify the max and min. Example: max xy s.t. $x + y = 4$: $y = \lambda$, $x = \lambda$, so $x = y = 2$, $f = 4$, $\lambda = 2$ (raising the sum to 5 adds about 2 to the product).

DECISION POINT Lagrange gives *candidates*, not a verdict: compare f values, check the second-order (bordered Hessian) condition, or argue from compactness. If the constraint is an inequality, use KKT O.18 instead. If you can solve the constraint for one variable, direct substitution may be quicker.

TERMS

f – objective; $g(\mathbf{x}) = c$ equality constraint
λ – multiplier: rate the optimum changes as c changes (a shadow price O.4)
f^* – optimal value as a function of c
Geometry – at the optimum, the level curve of f is tangent to the constraint, so gradients are parallel T.21 §06 Trigonometry
$\mathcal{L}$ – Lagrangian; set all partial derivatives to 0 C3.9 §10 Calculus III
Inequalities – generalize to KKT O.18
g, c; g_i, c_i, k – constraint function and level; several constraints and their number



CONTEXT

- **Origin.** Joseph-Louis Lagrange, *Mécanique analytique* (1788).
- **Use.** Constrained design, economics (utility under a budget), physics.
- **Data science.** Derives maximum-entropy distributions (the normal and softmax), PCA as $\max \mathbf{v}^\top S\mathbf{v}$ s.t. $\|\mathbf{v}\| = 1$ ML.38 §15 Machine Learning, and regularization as a constraint $\|\beta\| \le t$ ML.8 §15 Machine Learning.
- **Physical application.** Designing a cylindrical tank of fixed volume with minimum surface area uses a Lagrange multiplier.

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O.18 KKT Conditions

The Karush–Kuhn–Tucker conditions extend Lagrange multipliers to inequality constraints. Each inequality is either active (on its boundary, with a non-negative multiplier) or inactive (multiplier zero).

Stationarity	$\nabla f(\mathbf{x}^*) + \sum_i \mu_i \nabla g_i(\mathbf{x}^*) + \sum_j \nu_j \nabla h_j(\mathbf{x}^*) = \mathbf{0}$	For $\min f$ s.t. $g_i \le 0$, $h_j = 0$: no feasible descent direction.
Primal feasibility	$g_i(\mathbf{x}^*) \le 0, \quad h_j(\mathbf{x}^*) = 0$	The point satisfies the constraints.
Dual feasibility	$\mu_i \ge 0$	Inequality multipliers push *out* of the feasible region only.
Complementary slackness	$\mu_i\, g_i(\mathbf{x}^*) = 0$	Either the constraint is active ($g_i = 0$) or its multiplier is 0.
Sufficiency (convex case)	f, g_i convex, h_j affine $\Rightarrow$ KKT = global optimal	Necessity needs a constraint qualification (e.g. Slater O.19).

STATED-EQUATION *Minimize f subject to g i at most zero and h j equal to zero: the gradient of f plus the sum of mu i times the gradient of g i plus the sum of nu j times the gradient of h j equals zero; each g i is at most zero and each h j zero; each mu i is at least zero; and mu i times g i equals zero. For convex problems these conditions are sufficient for a global optimum.*

MEMORY AID Complementary slackness: either the constraint is tight or its multiplier is zero.

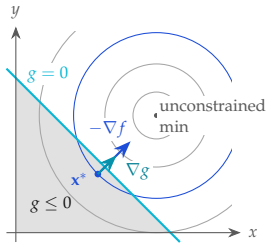
METHOD

1. Write the KKT conditions for the problem in standard form O.1.
2. Guess an active set (which $g_i = 0$); set $\mu_i = 0$ for the others.
3. Solve the resulting equations; accept if all $g_i \le 0$ and all $\mu_i \ge 0$, otherwise try another active set.
4. Example: $\min (x-3)^2$ s.t. $x \le 1$. Inactive: $x = 3$ violates $x \le 1$. Active: $x = 1$, $2(1-3) + \mu = 0 \Rightarrow \mu = 4 \ge 0$. Optimum $x^* = 1$.

DECISION POINT With m inequalities there are 2^m active sets; start from the unconstrained optimum (if feasible, done), then activate the violated constraints. A negative μ_i means that constraint wants to be released. Solvers automate this (active-set and interior-point methods O.20).

TERMS

Stationarity – Lagrange condition O.17 with inequality multipliers μ_i
μ_i, ν_j – multipliers for inequalities and equalities
Primal feasibility – $g_i \le 0$, $h_j = 0$
Dual feasibility – $\mu_i \ge 0$
Complementary slackness – either the constraint is active ($g_i = 0$) or $\mu_i = 0$ O.4
Active set – the constraints with $g_i(\mathbf{x}^*) = 0$
Sufficiency – for convex problems O.2, KKT points are global optima
f; g_i, h_j – objective; inequality and equality constraints O.1



CONTEXT

- **Origin.** Karush (1939 thesis) and Kuhn–Tucker (1951).
- **Use.** The optimality test built into every constrained NLP solver.
- **Data science.** In SVMs, the training points with $\mu_i > 0$ are exactly the *support vectors* ML.15 §15 Machine Learning; KKT also explains why lasso sets coefficients exactly to zero ML.8 §15 Machine Learning.
- **Physical application.** A ball resting against a wall: the wall pushes back (multiplier > 0) only when the ball touches it (constraint active).

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O.19 Lagrangian Duality

Every constrained problem has a dual problem built from its Lagrangian. The dual is always concave, its value always bounds the primal, and for convex problems the two values usually coincide.

Lagrangian	$\mathcal{L}(\mathbf{x},\boldsymbol{\mu},\boldsymbol{\nu}) = f(\mathbf{x}) + \sum_i \mu_i g_i(\mathbf{x}) + \sum_j \nu_j h_j(\mathbf{x})$	Constraints priced into the objective.
Dual function	$q(\boldsymbol{\mu},\boldsymbol{\nu}) = \inf_{\mathbf{x}} \mathcal{L}(\mathbf{x},\boldsymbol{\mu},\boldsymbol{\nu})$	Concave even if the primal is not (a minimum of affine functions).
Dual problem	$d^* = \max_{\boldsymbol{\mu}\geq\mathbf{0},\ \boldsymbol{\nu}} q(\boldsymbol{\mu},\boldsymbol{\nu})$	The best lower bound obtainable this way.
Weak duality	$d^* \leq p^*$	Always true; $p^* - d^* \geq 0$ is the *duality gap*.
Strong duality (Slater)	$\text{convex} + \exists\, \tilde{\mathbf{x}} :\ g_i(\tilde{\mathbf{x}}) < 0 \ \Rightarrow\ d^* = p^*$	A strictly feasible point closes the gap.
Saddle point	$\min_{\mathbf{x}} \max_{\boldsymbol{\mu}\geq 0} \mathcal{L} = \max_{\boldsymbol{\mu}\geq 0} \min_{\mathbf{x}} \mathcal{L}$	Under strong duality the order of min and max can be swapped.

STATED-EQUATION *The Lagrangian is f plus the sum of mu i g i plus the sum of nu j h j. The dual function q is the infimum over x of the Lagrangian. The dual problem maximizes q over mu at least zero. Weak duality: d star is at most p star. Strong duality: for a convex problem with a strictly feasible point, d star equals p star, and min-max equals max-min.*

MEMORY AID The dual is a "price the constraints, then let x respond" game; its best price never beats the true optimum.

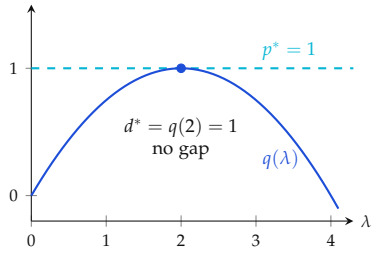
METHOD

1. Form $\mathcal{L}$ with $\mu_i \geq 0$ for each $g_i \leq 0$.
2. Minimize $\mathcal{L}$ over $\mathbf{x}$ (set $\nabla_{\mathbf{x}}\mathcal{L} = \mathbf{0}$) to get q.
3. Maximize q over $\boldsymbol{\mu} \geq \mathbf{0}$; recover $\mathbf{x}$ from step 2.
4. Example: $\min x^2$ s.t. $1 - x \leq 0$: $\mathcal{L} = x^2 + \mu(1-x)$, $x = \mu/2$, $q = \mu - \mu^2/4$, maximized at $\mu = 2$: $d^* = 1 = p^*$ at $x = 1$.

DECISION POINT Solve the dual when it is smaller (fewer constraints than variables), decomposes into independent pieces (one per machine or time period), or exposes structure (kernels in SVMs ML.15 §15 Machine Learning). For non-convex or integer problems the dual still gives a valid *bound* (Lagrangian relaxation), used to prune branch and bound O.6.

TERMS

$\boldsymbol{\mu}, \boldsymbol{\nu}$ – dual variables (multipliers) for $g_i \leq 0$, $h_j = 0$
inf – infimum (greatest lower bound; a minimum when attained)
q – dual function; d^* dual optimum; p^* primal optimum
$\tilde{\mathbf{x}}$ – a strictly feasible point (Slater's condition)
Duality gap – $p^* - d^*$; zero under strong duality
LP case – reduces to the LP dual O.4
p^*, d^* – optimal primal and dual values
f, g_i, h_j – objective and constraints O.1



CONTEXT

- **Origin.** Von Neumann's minimax theorem (1928); Slater (1950); Rockafellar's convex analysis (1970).
- **Use.** Bounds, decomposition of large problems, sensitivity analysis.
- **Data science.** The SVM dual and kernel trick ML.15 §15 Machine Learning; Wasserstein GANs use Kantorovich duality; adversarial training is a min-max saddle problem.
- **Physical application.** In economics, dual variables are market prices that make supply meet demand.

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O.20 Penalty, Barrier & Projection Methods

Turn a constrained problem into a sequence of unconstrained ones: penalties charge for leaving the feasible set, barriers forbid it by blowing up at the boundary, and projections snap each step back inside.

Quadratic penalty	$\min_{\mathbf{x}} \; f(\mathbf{x}) + \frac{\rho}{2}\left[\sum_i \max\left(0, g_i(\mathbf{x})\right)^2 + \sum_j h_j(\mathbf{x})^2\right]$	Iterates approach from *outside*; exact only as $\rho \to \infty$.
Log barrier	$\min_{\mathbf{x}} \; f(\mathbf{x}) - \mu \sum_i \ln\left(-g_i(\mathbf{x})\right)$	Iterates stay strictly inside; the basis of interior-point methods.
Central path gap	$f(\mathbf{x}^*(\mu)) - p^* \le m\,\mu$	Shrinking μ (e.g. $\times 0.1$) drives the barrier solution to the optimum.
Augmented Lagrangian	$\mathcal{L}_\rho = f + \boldsymbol{\nu}^\top \mathbf{h} + \frac{\rho}{2}\|\mathbf{h}\|^2, \qquad \boldsymbol{\nu} \leftarrow \boldsymbol{\nu} + \rho\,\mathbf{h}(\mathbf{x})$	Multiplier updates make it exact with finite ρ (basis of ADMM).
Projected gradient	$\mathbf{x}_{k+1} = \Pi_C\left(\mathbf{x}_k - \eta \nabla f(\mathbf{x}_k)\right)$	Gradient step, then the nearest feasible point; ideal for boxes, balls, simplices.

STATED-EQUATION *Quadratic penalty: minimize f plus rho over two times the squared constraint violations. Log barrier: minimize f minus mu times the sum of log of minus g i. The barrier solution is within m mu of the optimum. Augmented Lagrangian adds nu transpose h plus rho over two norm h squared, then updates nu by rho h. Projected gradient takes a gradient step and projects onto C.*

MEMORY AID Penalty: "a fine for trespassing". Barrier: "an electric fence". Projection: "step, then snap back".

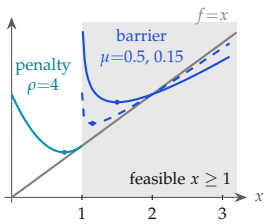
METHOD

1. Barrier method: start strictly feasible with μ_0 (e.g. 1).
2. Minimize $f - \mu \sum \ln(-g_i)$ by Newton's method O.13, warm-started from the previous solution.
3. Decrease $\mu \leftarrow \mu/10$; stop when $m\mu <$ tolerance.
4. Example: $\min x$ s.t. $x \ge 1$: barrier $x - \mu \ln(x-1)$ is minimized at $x = 1 + \mu$: $\mu = 1 \to 2$, $\mu = 0.1 \to 1.1$, $\mu = 0.01 \to 1.01 \to 1$.

DECISION POINT Penalty: simplest, works from infeasible starts, but large ρ makes the problem ill-conditioned. Barrier / interior point: high accuracy for LP, QP and convex problems; needs a strictly feasible start. Augmented Lagrangian / ADMM: equality constraints and distributed problems. Projection: when Π_C is cheap (box: clip; ball: rescale).

TERMS

ρ – penalty weight ($\rho \to \infty$ enforces feasibility)
μ – barrier parameter ($\mu \to 0$); m number of inequalities
$\mathbf{x}^*(\mu)$ – barrier minimizer; the curve it traces is the central path
$\boldsymbol{\nu}$ – multiplier estimates O.19
Π_C – Euclidean projection onto the convex set C LA.20 §09 Linear Algebra
ln – natural log A2.14 §04 Algebra II; $-\ln(-g) \to \infty$ as $g \to 0^-$
m; p^* – number of inequality constraints; optimal value
$\mathbf{h}$; C – vector of equality constraints; convex feasible set



CONTEXT

- **Origin.** Courant (1943) penalties; Fiacco and McCormick (1968) barriers; Karmarkar's interior-point LP algorithm (1984); Hestenes–Powell augmented Lagrangian (1969).
- **Use.** Engineering design, optimal control, large LPs.
- **Data science.** Regularization is a penalty form of a constraint ML.8 §15 Machine Learning; projected gradient enforces non-negativity in NMF; ADMM fits distributed and federated models.
- **Physical application.** Soft constraints in robot motion planning (penalize getting close to obstacles) are penalty and barrier terms.

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Dynamic Programming & Heuristics

O.21 Dynamic Programming & the Bellman Equation

Solve a problem by breaking it into overlapping subproblems, solving each once and storing the answer. For sequential decisions the value of a state equals the best immediate reward plus the value of where you land.

Bellman optimality equation	$V(s) = \max_a \left[r(s,a) + \gamma \sum_{s'} P(s' \mid s,a)\, V(s') \right]$	Best long-run value from state s under uncertainty.
Finite-horizon recursion	$V_t(s) = \max_a \left[r(s,a) + V_{t+1}(s') \right], \quad V_T \equiv 0$	Work backward from the last stage.
Requirements	optimal substructure + overlapping subproblems	Optimal answers contain optimal sub-answers, and subproblems repeat.
Q-function	$Q(s,a) = r(s,a) + \gamma \sum_{s'} P(s' \mid s,a) \max_{a'} Q(s',a')$	Value of taking a in s then acting optimally (Q-learning).
Value iteration	$V_{k+1} = \mathcal{T} V_k, \quad \lVert V_k - V^* \rVert \le \gamma^k \lVert V_0 - V^* \rVert$	The Bellman operator is a γ-contraction: converges geometrically.
Discounted constant reward	$\sum_{t=0}^{\infty} \gamma^t r = \frac{r}{1-\gamma}$	Geometric series A2.19 §04 Algebra II; why $\gamma < 1$ keeps values finite.

STATED-EQUATION *V of s equals the maximum over actions a of r of s, a plus gamma times the sum over s prime of P of s prime given s, a times V of s prime. With a finite horizon, V t equals the max of r plus V t plus one, with V T equal to zero. It requires optimal substructure and overlapping subproblems. Value iteration applies the Bellman operator repeatedly and converges like gamma to the k. The discounted sum of a constant reward r is r over one minus gamma. The Q-function Q of s, a equals r of s, a plus gamma times the expected value, over s prime, of the maximum over a prime of Q of s prime, a prime.*

MEMORY AID "Solve each subproblem once and remember it."

METHOD

1. Define the **state** (what you need to know to continue) and the **decision** at each step.
2. Write the **recurrence** for the optimal value in terms of smaller states.
3. Set the **base cases**; choose an evaluation order so dependencies are computed first (bottom-up table) or memoize a recursion (top-down).
4. Store choices to reconstruct the solution. Example: fewest coins for 6 with $\{1,3,4\}$: $C(n) = 1 + \min\{C(n-1), C(n-3), C(n-4)\}$, $C(0) = 0$: $C(1..6) = 1,2,1,1,2,2$: $6 = 3+3$ (greedy $4+1+1$ uses 3).

DECISION POINT Use DP when subproblems overlap (otherwise plain recursion / divide-and-conquer suffices) and the state space is small enough to tabulate; if it is huge, approximate V with a function (reinforcement learning). Greedy is correct only when a local choice can never be regretted (e.g. fractional knapsack O.7).

TERMS

$V(s)$ – value: best total reward from state s
a, $r(s,a)$ – action and its immediate reward
γ – discount factor, $0 \le \gamma < 1$ (geometric series A2.19 §04 Algebra II)
$P(s' \mid s,a)$ – transition probability S.9 §13 Statistics
t, T – stage and horizon
$\mathcal{T}$ – Bellman operator (the right-hand side as a map on V)
Memoization – store subproblem answers (knapsack table O.7, shortest paths O.9)
s, s'; V^* – current and next state; optimal value function
$Q(s,a)$ – value of taking action a in state s, then acting optimally

CONTEXT

- **Origin.** Richard Bellman (1950s) at RAND.
- **Use.** Inventory control, resource allocation, sequence alignment, speech decoding (Viterbi).
- **Data science.** Reinforcement learning (value iteration, Q-learning), edit distance for fuzzy matching, and the Viterbi algorithm in HMMs and sequence labeling.
- **Physical application.** Optimal control of a rocket's fuel burn stage by stage is solved with dynamic programming.

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O.22 Heuristics & Simulated Annealing

Heuristics find good (not provably optimal) solutions fast. Simulated annealing is local search that sometimes accepts worse moves, with a probability that shrinks as a "temperature" cools, so it can escape local minima.

Metropolis acceptance	$P(\text{accept}) = \begin{cases} 1, & \Delta \le 0 \\ e^{-\Delta/T}, & \Delta > 0 \end{cases}$	Always take improvements; take a worsening by Δ with probability $e^{-\Delta/T}$.
Cost change	$\Delta = f(\mathbf{x}_{\text{new}}) - f(\mathbf{x}_{\text{current}})$	For minimization; positive means the move is worse.
Geometric cooling	$T_k = T_0\,\alpha^k \quad (0 < \alpha < 1)$	The common practical schedule, $\alpha \approx 0.9$–0.99.
Logarithmic cooling	$T_k = \dfrac{c}{\ln(1+k)}$	Provably reaches the global optimum, but far too slowly for practice.

STATED-EQUATION *The acceptance probability is one if delta is at most zero, and e to the minus delta over T if delta is positive. Delta equals the new cost minus the current cost. The temperature cools as T k equals T naught alpha to the k, with alpha between zero and one; logarithmic cooling uses c over log of one plus k.*

MEMORY AID Hot $\rightarrow$ explore; cool $\rightarrow$ settle.

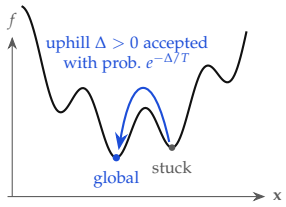
METHOD

1. Start from any solution $\mathbf{x}$ and a temperature T_0.
2. Propose a random neighbor (e.g. a 2-opt swap O.8); compute Δ.
3. Accept if $\Delta \le 0$, else accept if a uniform $U < e^{-\Delta/T}$ O.25.
4. After a fixed number of moves, cool $T \leftarrow \alpha T$; keep the best solution seen. Example: $\Delta = 2$, $T = 1$: accept with $e^{-2} \approx 0.14$; at $T = 10$: $e^{-0.2} \approx 0.82$.

DECISION POINT Set T_0 so roughly 80% of uphill moves are accepted at first; cool slowly enough that the cost keeps improving; stop when almost nothing is accepted. The neighborhood matters more than the schedule: small moves that keep solutions feasible work best.

TERMS

Δ – change in cost of a proposed move (minimization)
T – temperature: high T explores, low T exploits
T_0, α – initial temperature and cooling rate (exponential decay A2.13 §04 Algebra II)
c – constant at least the depth of the deepest local minimum
Greedy – best immediate choice; fast but can be suboptimal
Local search – improve by small moves (2-opt for TSP O.8), see O.23
Genetic algorithms – selection, crossover, mutation on a population O.24



CONTEXT

- **Origin.** Kirkpatrick, Gelatt, Vecchi (1983), inspired by the Metropolis algorithm O.29 and annealing of metals.
- **Use.** Near-optimal answers to NP-hard problems in reasonable time.
- **Data science.** Hyperparameter search, feature-subset search, and large routing/scheduling problems where exact MIP O.6 is too slow.
- **Physical application.** Annealing steel: slow cooling lets atoms settle into a low-energy crystal instead of a brittle, disordered one.

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O.23 Local Search, Greedy & Tabu Search

Local search improves a solution by small moves within a neighborhood until no move helps. Tabu search keeps moving even when every neighbor is worse, forbidding recently visited moves so it does not cycle back.

Neighborhood	$N(\mathbf{x}) = \{\text{solutions one move from } \mathbf{x}\}$	Moves: flip a bit, swap two jobs, 2-opt O.8.
Hill climbing (descent)	$\mathbf{x} \leftarrow \arg\min_{\mathbf{y} \in N(\mathbf{x})} f(\mathbf{y})$ if $f(\mathbf{y}) < f(\mathbf{x})$	Stops at a local optimum of the neighborhood.
Greedy construction	add the element with best immediate gain	Builds a starting solution (nearest neighbor, v/w ratio).
Tabu move	$\mathbf{x} \leftarrow \arg\min_{\mathbf{y} \in N(\mathbf{x}) \setminus \mathcal{T}} f(\mathbf{y})$	Take the best non-tabu neighbor, even if worse.
Aspiration criterion	allow tabu $\mathbf{y}$ if $f(\mathbf{y}) < f_{\text{best}}$	Never forbid a new overall best.
Iterated local search	$\mathbf{x} \leftarrow \text{LocalSearch}(\text{Perturb}(\mathbf{x}))$	Kick the solution, re-optimize, keep if better; multistart is the random-restart version.

STATED-EQUATION *The neighborhood of x is all solutions one move away. Hill climbing moves to the best neighbor if it improves f. Tabu search moves to the best neighbor not on the tabu list, accepting worse moves, unless a tabu move would beat the best so far. Iterated local search perturbs and re-optimizes. Greedy construction builds a solution by repeatedly adding the element with the best immediate gain.*

MEMORY AID Tabu: "don't undo what you just did."

METHOD

1. Build an initial solution (greedy or random); set an empty tabu list $\mathcal{T}$ of length (tenure) k.
2. Evaluate all (or a sample of) neighbors; move to the best one that is not tabu (or satisfies aspiration).
3. Add the reverse of that move to $\mathcal{T}$; drop the oldest entry when $|\mathcal{T}| > k$.
4. Track the best solution; stop after a budget or no improvement for many moves. Example: in a 0/1 problem, after flipping bit 3, flipping bit 3 back is tabu for the next $k = 5$ moves.

DECISION POINT Neighborhood design: small neighborhoods are fast but trap easily; large ones (e.g. 3-opt) find better optima per move at higher cost. Choose tabu tenure around 5–20 or $\sqrt{n}$; too short cycles, too long blocks good moves. If evaluation is expensive, use incremental (delta) evaluation of moves.

TERMS

$N(\mathbf{x})$ – neighborhood of $\mathbf{x}$
$\mathcal{T}$ – tabu list of recently made (reversed) moves; k its tenure
f_{best} – best objective value found so far
Local optimum – no neighbor is better (depends on the neighborhood) O.1
Perturb – a larger random change (kick)
Related – simulated annealing O.22, genetic algorithms O.24

CONTEXT

- **Origin.** Hill climbing is folklore; tabu search by Fred Glover (1986); iterated local search (Lourenço et al., 2003).
- **Use.** Scheduling, timetabling, vehicle routing, facility layout.
- **Data science.** Greedy forward/backward feature selection is hill climbing over feature subsets; k-means ML.35 §15 Machine Learning is a local search that converges to a local optimum.
- **Physical application.** Nurse and airline-crew rostering with many soft rules is commonly solved by tabu search.

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O.24 Genetic Algorithms & Population Methods

Evolve a population of candidate solutions: select the fitter ones, recombine them (crossover), and randomly perturb them (mutation). Population methods explore many regions at once and need only objective values.

Fitness-proportional selection	$p_i = \dfrac{F_i}{\sum_j F_j}$	Roulette wheel: fitter individuals are chosen more often ($F_i \geq 0$).
Tournament selection	pick k at random, keep the best	Simple, scale-free; larger k means stronger selection pressure.
One-point crossover	$(a_{1:c}, b_{c+1:L}),\ (b_{1:c}, a_{c+1:L})$	Children swap tails after a random cut point c.
Mutation	flip each bit w.p. $p_m \approx \dfrac{1}{L}$	Keeps diversity; about one change per child.
Elitism	copy the best e individuals unchanged	The best solution can never be lost.
Particle swarm velocity	$\mathbf{v} \leftarrow \omega\mathbf{v} + c_1 r_1(\mathbf{p} - \mathbf{x}) + c_2 r_2(\mathbf{g} - \mathbf{x}), \qquad \mathbf{x} \leftarrow \mathbf{x} + \mathbf{v}$	Each particle is pulled toward its own best $\mathbf{p}$ and the swarm's best $\mathbf{g}$.

STATED-EQUATION *Fitness-proportional selection picks individual i with probability F i over the sum of all fitnesses. Tournament selection keeps the best of k random individuals. One-point crossover swaps the tails of two parents after cut c. Mutation flips each bit with probability about one over L. Elitism copies the best unchanged. Particle swarm: v becomes omega v plus c one r one times p minus x plus c two r two times g minus x.*

MEMORY AID Survival of the fittest, mixing, and mutation: select, cross, mutate.

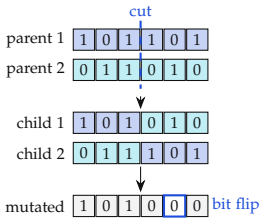
METHOD

1. Encode a solution (bit string, permutation, or real vector); create a random population of size P.
2. Evaluate fitness; select parents (tournament or roulette).
3. Apply crossover (probability ≈ 0.8) and mutation to make children; keep e elites.
4. Repeat for G generations; return the best. Example (figure): parents `101101`, `011010` cut after bit 3 give `101010` and `011101`.

DECISION POINT Use a GA or swarm when the objective is a black box (no gradients), noisy, multimodal, or mixed discrete; prefer gradient methods O.11 for smooth problems and MIP O.6 when the model is linear. Choose an encoding where crossover of two good parents tends to give good children (for permutations use order crossover, not bit crossover).

TERMS

F_i – fitness (higher is better; e.g. $-f$ shifted to be positive)
k – tournament size; P population size; G generations
L – chromosome length; c cut point; a, b parents
p_m – mutation rate per gene; e number of elites
ω, c_1, c_2 – inertia and attraction weights; $r_1, r_2 \sim U(0,1)$ S.19 §13 Statistics
$\mathbf{p}, \mathbf{g}$ – particle's best and swarm's best positions
a, b; c – parent strings; crossover cut point
e; p_i – number of elites; selection probability
$\mathbf{x}, \mathbf{v}$; r_1, r_2 – particle position and velocity; uniform random numbers



CONTEXT

- **Origin.** John Holland (1975); particle swarm by Kennedy and Eberhart (1995); evolution strategies (CMA-ES, Hansen 2001).
- **Use.** Design optimization, scheduling, antenna and circuit design.
- **Data science.** Neural architecture search, hyperparameter tuning (CMA-ES), symbolic regression, and evolving game-playing policies.
- **Physical application.** NASA's evolved X-band antenna (ST5 mission, 2006) was designed by a genetic algorithm.

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Simulation & Monte Carlo

O.25 Pseudo-Random Number Generation

Simulations run on deterministic sequences that only look random. A generator produces integers from a seed; dividing by the modulus gives uniforms on $[0,1)$, from which every other distribution is built O.26.

Linear congruential generator	$X_{n+1} = (aX_n + c) \bmod m$	Fast and simple; quality depends entirely on a, c, m.
Uniform output	$U_n = \dfrac{X_n}{m} \in [0,1)$	Rescale with $a' + (b'-a')U$ for any interval.
Full period (Hull–Dobell)	$\gcd(c,m) = 1;\ \ p \mid m \Rightarrow p \mid a-1;\ \ 4 \mid m \Rightarrow 4 \mid a-1$	Conditions for the period to equal m (every value appears once).
Box–Muller	$Z = \sqrt{-2\ln U_1}\,\cos(2\pi U_2)$	Two uniforms $\to$ an exact standard normal S.20 §13 Statistics (the sin gives a second).
Lattice structure	$(U_n, U_{n+1}, \dots)$ lie on few hyperplanes	LCG points fall on parallel lines; fails tests in high dimensions.

STATED-EQUATION *The LCG: X n plus one equals a X n plus c, mod m, and U n equals X n over m. The period is the full m when c and m are coprime, every prime factor of m divides a minus one, and four divides a minus one if four divides m. Box-Muller: Z equals the square root of minus two log U one, times cosine of two pi U two. Successive tuples of LCG outputs lie on only a few hyperplanes.*

MEMORY AID Same seed, same "random" numbers: reproducible by design.

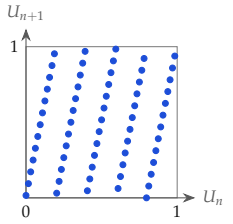
METHOD

1. Choose parameters satisfying Hull–Dobell and a seed X_0.
2. Iterate $X_{n+1} = (aX_n + c) \bmod m$; output $U_n = X_n/m$.
3. Example: $a = 5$, $c = 3$, $m = 16$, $X_0 = 7$: $X = 6, 1, 8, 11, 10, 5, \dots$ (period 16, since $\gcd(3,16) = 1$ and $4 \mid a-1 = 4$); $U_1 = 6/16 = 0.375$.

DECISION POINT Never write your own generator for real work: use a modern one (PCG64, Mersenne Twister, xoshiro; `numpy.random.default_rng`). Use independent streams (spawned seeds) for parallel simulations, and a cryptographic generator (`secrets`) for anything security-related. Always record the seed.

TERMS

X_n – integer state; X_0 the seed
a, c, m – multiplier, increment, modulus (mod: remainder after division)
gcd – greatest common divisor; $p \mid m$: prime p divides m
Period – number of steps before the sequence repeats
U_1, U_2 – independent $U(0,1)$ draws S.19 §13 Statistics; $Z \sim N(0,1)$
Quasi-random – low-discrepancy sequences (Sobol, Halton) for QMC O.27
U_n; Z – uniform output; standard normal
p – any prime factor of m



a=5, c=1, m=64: points on 5 lines

CONTEXT

- **Origin.** Lehmer's LCG (1949); von Neumann's middle-square method (1946); Mersenne Twister (Matsumoto–Nishimura, 1998).
- **Use.** Every stochastic simulation, randomized algorithm and game.
- **Data science.** Seeds make train/test splits, weight initialization and dropout reproducible; random-number streams drive bootstraps and MCMC O.29.
- **Physical application.** Monte Carlo neutron transport codes consume billions of random numbers per run.

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O.26 Random Sampling Methods

Turn uniform random numbers into draws from a target distribution: invert its CDF when possible, otherwise propose from an easy distribution and accept or reweight.

Inverse transform	$X = F^{-1}(U), \quad U \sim \text{Unif}(0,1)$	$P(X \le x) = P(U \le F(x)) = F(x)$: works for any distribution with an invertible CDF.
Exponential case	$X = -\dfrac{\ln(1-U)}{\lambda}$	Inverse of $F(x) = 1 - e^{-\lambda x}$ S.19 §13 Statistics.
Discrete inverse	$X = x_k$ where $F(x_{k-1}) < U \le F(x_k)$	Walk the cumulative probabilities until they pass U.
Rejection sampling	accept $x \sim g$ if $U \le \dfrac{f(x)}{M\,g(x)}$	Needs an envelope $f \le Mg$; acceptance rate $1/M$.
Importance sampling	$E_p\big[h(X)\big] = E_q\Big[h(X)\dfrac{p(X)}{q(X)}\Big]$	Sample from q, reweight by p/q (see variance reduction O.28).
Uniform on $[a,b]$	$X = a + (b-a)U$	The simplest inverse transform.

STATED-EQUATION *Inverse transform: X equals F inverse of U, with U uniform on zero to one. For the exponential, X equals minus log of one minus U over lambda. Rejection sampling: draw x from g and accept it if U is at most f of x over M g of x. Importance sampling: the expectation under p of h of X equals the expectation under q of h of X times p of X over q of X. For a discrete variable, X is the first value x k whose CDF reaches U. To sample uniformly on a to b, X equals a plus b minus a, times U.*

MEMORY AID "Read the CDF backwards": pick a height U, find the x where the CDF reaches it.

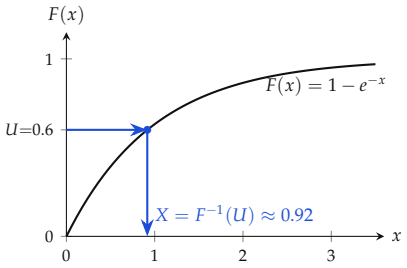
METHOD

1. Write the CDF $F(x)$ S.20 §13 Statistics.
2. Solve $u = F(x)$ for x to get $F^{-1}(u)$.
3. Plug in uniforms. Example: exponential with $\lambda = 2$ and $U = 0.5$: $X = -\ln 0.5/2 \approx 0.35$.

DECISION POINT Closed-form F^{-1} (exponential, uniform, Weibull, Cauchy, discrete) → inverse transform. Normal → Box–Muller O.25 or a library. Density known but CDF not invertible, low dimension → rejection. Density known only up to a constant, or high dimension → MCMC O.29.

TERMS

F, F^{-1} – CDF and its inverse (quantile function) A2.1 §04 Algebra II, S.20 §13 Statistics
λ – rate of the exponential distribution S.19 §13 Statistics
x_k – possible values of a discrete X, in increasing order
f, g, M – target density, easy proposal density, and a bound with $f \le Mg$
p, q, h – target and proposal densities; function whose mean is wanted
Weights – p/q (likelihood ratios)
U; a, b – uniform random number on $(0,1)$; interval endpoints
E_p, E_q – expectation when $X \sim p$ or $X \sim q$



CONTEXT

- **Origin.** Von Neumann described inverse transform and rejection sampling (1951).
- **Use.** Generating random variates for any simulation O.27.
- **Data science.** Synthetic data generation, off-policy evaluation of recommenders with importance weights, and correcting for sampling bias with inverse-propensity weights.
- **Physical application.** Simulating the random lifetimes of light bulbs from uniform random numbers uses the inverse CDF of the exponential.

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O.27 Monte Carlo Estimation & Integration

Estimate an expectation or integral by averaging a function over random samples. The error shrinks like $1/\sqrt{N}$ no matter how many dimensions the problem has.

Monte Carlo estimator	$\hat{\theta}_N = \frac{1}{N}\sum_{i=1}^{N} f(X_i) \longrightarrow E[f(X)]$	Law of large numbers; unbiased for every N.
Monte Carlo integration	$\int_a^b f(x)\,dx \approx \frac{b-a}{N}\sum_{i=1}^{N} f(U_i)$	Average height $\times$ length, $U_i \sim \text{Unif}(a,b)$ C2.12 §08 Calculus II.
Standard error	$SE = \frac{\hat{\sigma}}{\sqrt{N}}$	Central limit theorem S.24 §13 Statistics: $100\times$ the samples for one more digit.
Confidence interval	$\hat{\theta}_N \pm 1.96\,\frac{\hat{\sigma}}{\sqrt{N}}$	Approximate 95% interval S.27 §13 Statistics; always report it.
Sample size	$N = \left(\frac{z\,\hat{\sigma}}{\varepsilon}\right)^2$	Samples needed for half-width ε at confidence level z.
Estimating π	$\hat{\pi} = 4 \cdot \frac{\#\{U_1^2 + U_2^2 \le 1\}}{N}$	Fraction of random points in the quarter circle estimates $\pi/4$.

STATED-EQUATION *Theta hat N equals one over N times the sum of f of X i, which converges to the expected value of f of X. The integral from a to b of f of x d x is approximately b minus a over N times the sum of f of U i. The standard error equals sigma hat over root N, and the 95 percent interval is theta hat plus or minus 1.96 standard errors. To reach half-width epsilon, N equals z sigma hat over epsilon, squared. Pi is estimated as four times the fraction of random points U one, U two that fall inside the unit circle.*

MEMORY AID Error shrinks like $1/\sqrt{N}$: 100 times the samples for one more digit.

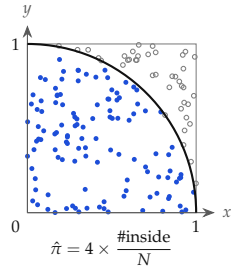
METHOD

1. Write the quantity as an expectation $E[f(X)]$ (or an integral as average $\times$ volume).
2. Draw N independent samples X_i O.26; compute $f(X_i)$.
3. Report the mean, $\hat{\sigma}$, and the interval $\hat{\theta} \pm 1.96\,\hat{\sigma}/\sqrt{N}$.
4. Example: $\int_0^1 x^2 dx$ with $U = 0.2, 0.5, 0.9$: $\frac{1}{3}(0.04 + 0.25 + 0.81) \approx 0.37$ (true $\frac{1}{3}$).

DECISION POINT Dimension 1–3 with a smooth integrand $\rightarrow$ deterministic quadrature (Simpson C2.5 §08 Calculus II) is far more accurate. High dimension, complex simulation, or random systems $\rightarrow$ Monte Carlo. To go faster than $1/\sqrt{N}$, use variance reduction O.28 or quasi-Monte Carlo (low-discrepancy points, error near $1/N$).

TERMS

X_i – independent random draws
$\hat{\theta}_N$ – sample-mean estimate (law of large numbers S.1 §13 Statistics)
$b - a$ – interval length; the formula is average value $\times$ length C2.12 §08 Calculus II
$\hat{\sigma}$ – sample SD of $f(X_i)$ S.2 §13 Statistics
z – normal critical value (1.96 for 95%); ε target half-width
Error rate – $\propto 1/\sqrt{N}$ regardless of dimension S.24 §13 Statistics
π estimate – $4\times$ fraction of random points inside the quarter circle
N; f – number of samples; function being averaged
U_i; ε; SE – uniform draws; target half-width; standard error S.24 §13 Statistics



CONTEXT

- **Origin.** Ulam, von Neumann, and Metropolis at Los Alamos (1946); named after the casino.
- **Use.** High-dimensional integrals, physics simulation, financial risk (VaR FN.30 §17 Finance).
- **Data science.** Simulating revenue under uncertainty, bootstrap confidence intervals, estimating expected loss, and Monte Carlo dropout for prediction uncertainty.
- **Physical application.** Physicists estimate radiation shielding by simulating millions of random particle paths.

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O.28 Variance Reduction Techniques

Get the same Monte Carlo accuracy with fewer samples by exploiting structure: negatively correlated pairs, a related quantity with a known mean, sampling where it matters, or splitting into strata.

Antithetic variates	$\hat{\theta} = \frac{1}{N}\sum_{i=1}^{N} \frac{f(U_i) + f(1-U_i)}{2}$	Pairs a draw with its mirror; helps when f is monotone (negative correlation).
Antithetic variance	$\mathrm{Var} = \frac{\sigma^2}{2N}(1+\rho)$	Versus $\sigma^2/2N$ for $2N$ independent draws: a gain whenever $\rho < 0$.
Control variate	$\hat{\theta}_c = \bar{Y} - c\,(\bar{X} - \mu_X)$	Correct $\bar{Y}$ using a variable X with known mean μ_X.
Optimal coefficient	$c^* = \frac{\mathrm{Cov}(Y,X)}{\mathrm{Var}(X)}, \quad \mathrm{Var} = (1-\rho_{XY}^2)\frac{\sigma_Y^2}{N}$	A regression slope S.40 §13 Statistics; $\rho = 0.9$ cuts variance by 81%.
Importance sampling	$\hat{\theta}_{IS} = \frac{1}{N}\sum_{i=1}^{N} h(X_i)\frac{p(X_i)}{q(X_i)}, \quad X_i \sim q$	Oversample the important region, reweight to stay unbiased.
Optimal proposal	$q^*(x) \propto \lvert h(x)\rvert\, p(x)$	Ideal q puts mass where $\lvert h\rvert p$ is large; heavy-tailed q keeps weights bounded.
Stratified sampling	$\hat{\theta}_{st} = \sum_s w_s \bar{Y}_s$	Sample within each stratum and weight by its probability w_s; removes between-strata variance.

STATED-EQUATION *Antithetic: average f of U and f of one minus U; the variance is sigma squared over two N times one plus rho. Control variate: Y bar minus c times X bar minus mu X, with c star equal to the covariance of Y and X over the variance of X, reducing variance by the factor one minus rho squared. Importance sampling averages h times p over q with samples from q; the best q is proportional to absolute h times p. Stratified: weight the stratum means by the stratum probabilities.*

MEMORY AID Same answer, less noise: mirror it, correct it, focus it, or split it.

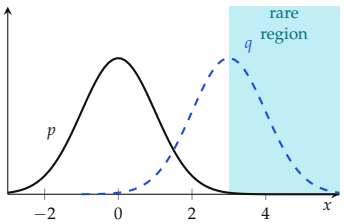
METHOD

1. Control variates: pick X correlated with Y whose mean μ_X is known exactly.
2. Run N paired simulations of (Y_i, X_i); estimate c^* from a pilot run.
3. Report $\bar{Y} - c^*(\bar{X} - \mu_X)$. Example: estimate $E[e^U]$ with control $X = U$ ($\mu_X = 0.5$): $\rho \approx 0.99$, variance drops about 60-fold.

DECISION POINT Monotone integrand → antithetic (nearly free). A simpler model with a known answer (e.g. geometric vs arithmetic Asian option, linearized system) → control variate. Rare events ($P \ll 10^{-3}$) → importance sampling, otherwise almost no samples hit the event. Known subgroups or input ranges → stratify. Comparing two systems → common random numbers (same U_i for both) to reduce the variance of the difference.

TERMS

U_i – uniform draws O.25; $1 - U_i$ is their antithetic mirror
ρ – correlation S.38 §13 Statistics (between the pair, or between X and Y)
σ^2 – variance of a single $f(U)$ S.2 §13 Statistics
μ_X – known mean of the control variable
p, q, h – target, proposal, and integrand (as in O.26)
$w_s, \bar{Y}_s$ – stratum probability and stratum sample mean
N; $\bar{X}, \bar{Y}$ – number of samples; sample means of control and target
c, c^*; ρ_{XY} – control coefficient and its optimum; correlation of X and Y



CONTEXT

- **Origin.** Hammersley and Handscomb, *Monte Carlo Methods* (1964); Kahn and Marshall importance sampling (1953).
- **Use.** Option pricing, reliability and rare-event estimation, particle transport.
- **Data science.** Off-policy evaluation and covariate-shift correction use importance weights; CUPED in A/B testing is a control variate using pre-experiment data.
- **Physical application.** Estimating the failure probability of a bridge or a power grid under rare extreme loads uses importance sampling.

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O.29 Markov Chain Monte Carlo (Metropolis–Hastings)

When a distribution is known only up to a constant, build a random walk whose long-run distribution is the target. After a burn-in period, the visited states are (correlated) samples from it.

Metropolis–Hastings acceptance	$\alpha(x \to x') = \min\left\{1,\ \dfrac{p(x')\,q(x \mid x')}{p(x)\,q(x' \mid x)}\right\}, \qquad \text{accept if } U \le \alpha$	The normalizing constant of p cancels in the ratio.
Symmetric proposal (Metropolis)	$\alpha = \min\left\{1,\ \dfrac{p(x')}{p(x)}\right\}$	For $q(x' \mid x) = q(x \mid x')$, e.g. a Gaussian random walk.
Detailed balance	$p(x)\,P(x \to x') = p(x')\,P(x' \to x)$	Guarantees p is the stationary distribution of the chain.
Gibbs sampling	$x_i \sim p(x_i \mid \mathbf{x}_{-i})$	Update one coordinate at a time from its full conditional; always accepted.
Effective sample size	$\mathrm{ESS} = \dfrac{N}{1 + 2\sum_{k \ge 1} \rho_k}$	Autocorrelated draws carry less information than independent ones.

STATED-EQUATION *The acceptance probability alpha from x to x prime equals the minimum of one and p of x prime q of x given x prime, over p of x q of x prime given x; accept if U is at most alpha. For a symmetric proposal, alpha equals the minimum of one and p of x prime over p of x. Detailed balance: p of x times the transition probability equals the reverse. Gibbs samples each coordinate from its conditional. The effective sample size is N over one plus twice the sum of autocorrelations.*

MEMORY AID Always accept uphill in probability; accept downhill sometimes, with probability equal to the ratio.

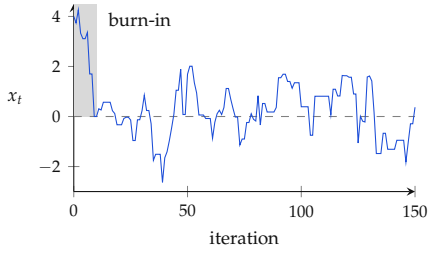
METHOD

1. Start at any x_0; choose a proposal q (e.g. $x' = x + \varepsilon$, $\varepsilon \sim N(0, s^2)$).
2. Propose x'; compute α.
3. Draw $U \sim \mathrm{Unif}(0,1)$; if $U \le \alpha$ move to x', else stay at x (record x again).
4. Repeat; discard burn-in; use the rest. Example: $p \propto e^{-x^2/2}$, $x = 0$, $x' = 1$: $\alpha = e^{-1/2} \approx 0.61$; with $U = 0.4$ accept.

DECISION POINT Tune the proposal scale s: too small means high acceptance but tiny moves; too large means rare acceptance. Aim for $\approx 44\%$ acceptance in 1-D and $\approx 23\%$ in high dimensions. Check convergence with trace plots, several chains, and $\hat{R} \approx 1$; prefer Hamiltonian MC (NUTS) for continuous high-dimensional posteriors, Gibbs when conditionals are standard.

TERMS

$p(x)$ – target density, needed only up to a constant (e.g. an unnormalized posterior S.11 §13 Statistics)
$q(x' \mid x)$ – proposal distribution (e.g. $x' = x + \varepsilon$, $\varepsilon \sim N(0, s^2)$ S.20 §13 Statistics)
α – acceptance probability; U uniform draw O.25
$\mathbf{x}_{-i}$ – all coordinates except x_i
ρ_k – lag-k autocorrelation of the chain; N number of draws
Markov chain – next state depends only on the current state S.9 §13 Statistics
Burn-in – early draws discarded before the chain reaches the target
Mixing – autocorrelation reduces the effective sample size
$P(x \to x')$; U; N – transition probability; uniform draw; chain length



CONTEXT

- **Origin.** Metropolis et al. (1953); generalized by Hastings (1970); Gibbs sampling by Geman and Geman (1984).
- **Use.** Statistical physics, Bayesian inference with intractable normalizing constants.
- **Data science.** Bayesian models in PyMC and Stan (Hamiltonian MC), posterior uncertainty for A/B tests and hierarchical models.
- **Physical application.** The original 1953 Metropolis algorithm simulated atoms in a liquid at a fixed temperature.

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O.30 Queueing Basics: Little's Law & M/M/1

Queueing theory predicts waiting lines from arrival and service rates. Little's law holds for almost any stable system; the M/M/1 queue (random arrivals, random service, one server) has simple closed forms that show delays exploding as utilization nears 1.

Little's law	$L = \lambda W$	Average number in system = arrival rate × average time in system; also $L_q = \lambda W_q$.
Utilization	$\rho = \dfrac{\lambda}{\mu} < 1$	Fraction of time the server is busy; $\rho \geq 1$ means the queue grows forever.
M/M/1 state probabilities	$P(N = n) = (1-\rho)\rho^n$	Geometric distribution of the number in system.
M/M/1 number in system	$L = \dfrac{\rho}{1-\rho}, \qquad L_q = \dfrac{\rho^2}{1-\rho}$	Blows up as $\rho \to 1$: 80% busy gives 4 in system, 90% gives 9.
M/M/1 time in system	$W = \dfrac{1}{\mu-\lambda}, \qquad W_q = \dfrac{\rho}{\mu-\lambda}$	Waiting in line plus one mean service time $1/\mu$.
Kendall notation	$A/S/c$	Arrival process / service distribution / servers; M = Markovian (Poisson/exponential), D = deterministic, G = general.

STATED-EQUATION *Little's law: L equals lambda W. Utilization rho equals lambda over mu and must be less than one. In an M/M/1 queue, the probability of n customers is one minus rho times rho to the n; L equals rho over one minus rho; L q equals rho squared over one minus rho; W equals one over mu minus lambda; and W q equals rho over mu minus lambda. Kendall notation A slash S slash c names the arrival process, the service-time distribution, and the number of servers.*

MEMORY AID "L equals lambda W": the line is as long as the rate times the wait. Near 100% busy, waits explode.

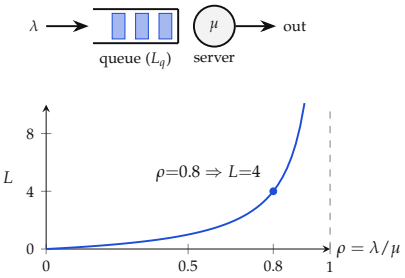
METHOD

1. Measure λ (arrivals per unit time) and μ (services per unit time, 1/mean service time).
2. Check stability $\rho = \lambda/\mu < 1$.
3. Apply the formulas, then Little's law to convert between L and W. Example: $\lambda = 4$/h, $\mu = 5$/h: $\rho = 0.8$, $L = 4$, $W = 1$ h, $W_q = 0.8$ h, $L_q = 3.2$.

DECISION POINT Closed forms need specific assumptions (Poisson arrivals, exponential service, FIFO). With non-exponential service (M/G/1) use the Pollaczek–Khinchine formula; with several servers use M/M/c (Erlang C). For networks of queues, priorities, time-varying rates, or finite buffers, use discrete-event simulation O.31. Little's law itself needs only stability.

TERMS

λ – arrival rate (Poisson arrivals S.18 §13 Statistics)
μ – service rate; service times exponential with mean $1/\mu$ S.19 §13 Statistics
ρ – utilization (traffic intensity)
L, L_q – mean number in system and in queue
W, W_q – mean time in system and waiting in queue
N – number in system (random); c number of servers
n; $A/S/c$ – number of customers; e.g. M/M/1 = Poisson arrivals, exponential service, 1 server



CONTEXT

- **Origin.** Erlang (1909) for telephone exchanges; Kendall (1953); Little's proof (1961).
- **Use.** Call centers, hospitals, manufacturing lines, traffic, computer networks.
- **Data science.** Capacity planning for model-serving endpoints and data pipelines: latency rises sharply above about 70–80% utilization; Little's law links throughput, latency and concurrency.
- **Physical application.** A checkout lane at 95% utilization averages 19 customers in the system; adding a bit of capacity cuts waits dramatically.

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O.31 Discrete-Event Simulation

Model a system as a state that changes only at discrete events (arrivals, service completions, failures). The clock jumps from event to event, so long stretches of time cost nothing to simulate.

Next-event time advance	$t \leftarrow \min\{\text{scheduled event times}\}$	Pop the earliest event from the event list (priority queue) and process it.
Arrival times	$A_n = A_{n-1} + T_n$	Cumulative sum of random interarrival times T_n O.26.
Lindley recursion	$W_{n+1} = \max\left(0,\ W_n + S_n - T_{n+1}\right)$	Waiting time of the next customer in a single-server FIFO queue.
Time-average estimator	$\bar{L} = \dfrac{1}{T_{\text{end}}} \displaystyle\int_0^{T_{\text{end}}} N(t)\, dt$	Area under the state curve divided by the run length.
Replication interval	$\bar{x} \pm t_{R-1} \dfrac{s}{\sqrt{R}}$	Independent replications give an honest confidence interval S.30 §13 Statistics.

STATED-EQUATION *The clock jumps to the minimum of the scheduled event times. Arrival n happens at arrival n minus one plus the interarrival time T n. The Lindley recursion: W n plus one equals the max of zero and W n plus S n minus T n plus one. The time-average number in system is the integral of N of t over the run length. Report x bar plus or minus t times s over root R across R replications.*

MEMORY AID "Jump from event to event; nothing happens in between."

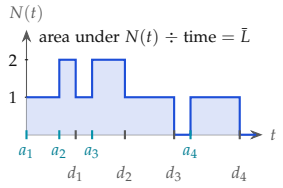
METHOD

1. Define the state (queue length, server busy?), the event types, and the statistics to collect.
2. Initialize the clock, state and event list (schedule the first arrival).
3. Loop: remove the earliest event, advance the clock, update the state and statistics, schedule future events (e.g. an arrival schedules the next arrival and, if the server is free, a departure).
4. Stop at T_{end}; repeat for R replications. Example (Lindley): service 3 min, arrivals every 2 min: $W_1 = 0$, $W_2 = \max(0, 0 + 3 - 2) = 1$, $W_3 = 2$: waits grow ($\rho = 1.5 > 1$).

DECISION POINT Terminating system (a shop open 9–5) → independent replications from the real start state. Steady-state behavior → delete a warm-up period, then use long runs with batch means or replications. Use enough replications that the interval is narrower than the differences you care about; compare designs with common random numbers O.28. Validate against a closed-form case (e.g. M/M/1 O.30) first.

TERMS

t – simulation clock; event list: future events ordered by time
A_n, T_n – arrival time and interarrival time of customer n
W_n, S_n – waiting time and service time of customer n
$N(t)$ – number in system at time t (a step function)
T_{end} – run length; R number of replications
$\bar{x}, s$ – mean and SD across replications; t_{R-1} the t critical value
Warm-up – initial transient discarded in steady-state studies
t_{R-1}; R – t critical value with $R-1$ degrees of freedom S.21 §13 Statistics; number of replications



CONTEXT

- **Origin.** Tocher's General Simulation Program (1960); GPSS (Gordon, 1961); SimPy and AnyLogic today.
- **Use.** Factories, hospitals, airports, supply chains, network protocols.
- **Data science.** Digital twins and what-if analysis of data pipelines or support queues; simulators generate training data for reinforcement-learning agents.
- **Physical application.** Airports simulate passenger flows through check-in and security before building new terminals.

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15
Machine Learning

Supervised learning (predicting a labeled target), unsupervised learning (finding structure without labels), and reinforcement learning (acting to earn reward). Every method below is assembled from earlier pieces: linear algebra, calculus, probability, and optimization. The chapter moves from the learning problem and classical models through evaluation, neural networks and modern architectures, to clustering, generative models and reinforcement learning.

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The Learning Problem

ML.1 The Learning Problem & Empirical Risk

A model f_θ is fit by choosing parameters that make the average loss on training data small, plus a penalty for complexity. What we really want is small loss on *new* data from the same distribution.

Regularized empirical risk	The template for almost every model: average training loss plus a complexity penalty. $$\hat{\theta} = \arg\min_\theta \underbrace{\frac{1}{n}\sum_{i=1}^{n} L\big(y_i,\ f_\theta(\mathbf{x}_i)\big)}_{\text{empirical risk } \hat{R}_n(\theta)} + \underbrace{\lambda\, R(\theta)}_{\text{regularization}}$$	
Expected (true) risk	$R(\theta) = E_{(\mathbf{x},y)\sim\mathcal{D}}\Big[L\big(y, f_\theta(\mathbf{x})\big)\Big]$	The goal: small loss on new data; estimated on held-out test data, never on training data.
Generalization gap	$R(\hat{\theta}) - \hat{R}_n(\hat{\theta})$	How much training error understates true error; a large gap is overfitting ML.4.
Likelihood as a loss	$L = -\log p_\theta(y \mid \mathbf{x}) \ \Rightarrow\ \text{ERM} = \text{MLE}$	With negative log-likelihood loss, ERM is maximum likelihood S.26 §13 Statistics: MSE ↔ Gaussian noise, log-loss ↔ Bernoulli.

STATED-EQUATION *Theta hat equals the arg min over theta of one over n times the sum of L of y i and f theta of x i, the empirical risk, plus lambda R of theta, the regularization. The goal is small expected loss on new data; the difference between true and training risk is the generalization gap. Using the negative log-likelihood as the loss makes empirical risk minimization the same as maximum likelihood.*

MEMORY AID Train to fit, validate to choose, test once to report.

METHOD

1. Split the data: train / validation / test (e.g. 70/15/15), or train/test plus cross-validation ML.5.
2. Fit a trivial baseline (mean, majority class) and a simple model (linear or logistic).
3. Fit candidate models on train; choose hyperparameters on validation or by CV.
4. Refit the chosen model and report its metric *once* on the untouched test set. Example: baseline MAE 10, linear 6, boosted trees 4 on validation ⇒ report the trees' test MAE.

DECISION POINT **What kind of problem is it?** Numeric target → regression; categorical target → classification; no target → clustering or dimension reduction; sequential decisions with reward → reinforcement learning. See the decision map ML.K1. **How to split?** Random for i.i.d. rows, by group (customer, patient) when rows are related, and by time (train on past, test on future) for forecasting.

TERMS

$\mathbf{x}_i$, y_i – features and label (target) of example i; n examples
f_θ – model with parameters θ
L – loss function ML.2
$\lambda R(\theta)$ – complexity penalty, strength λ ML.8
arg min – the minimizing θ, found by optimization O.1 §14 Optimization, ML.27
$\mathcal{D}$ – the (unknown) data distribution; E expectation S.13 §13 Statistics
$p_\theta(y \mid \mathbf{x})$ – model's conditional probability of y
Tasks – regression (numeric y), classification (categorical y), clustering and dimension reduction (no y)
$R(\theta)$ (risk); $\hat{R}_n$ – expected loss on new data; empirical (training) risk
θ, $\hat{\theta}$; n – model parameters and their fitted value; number of training examples
ERM; MLE – empirical risk minimization; maximum likelihood estimation S.26 §13 Statistics

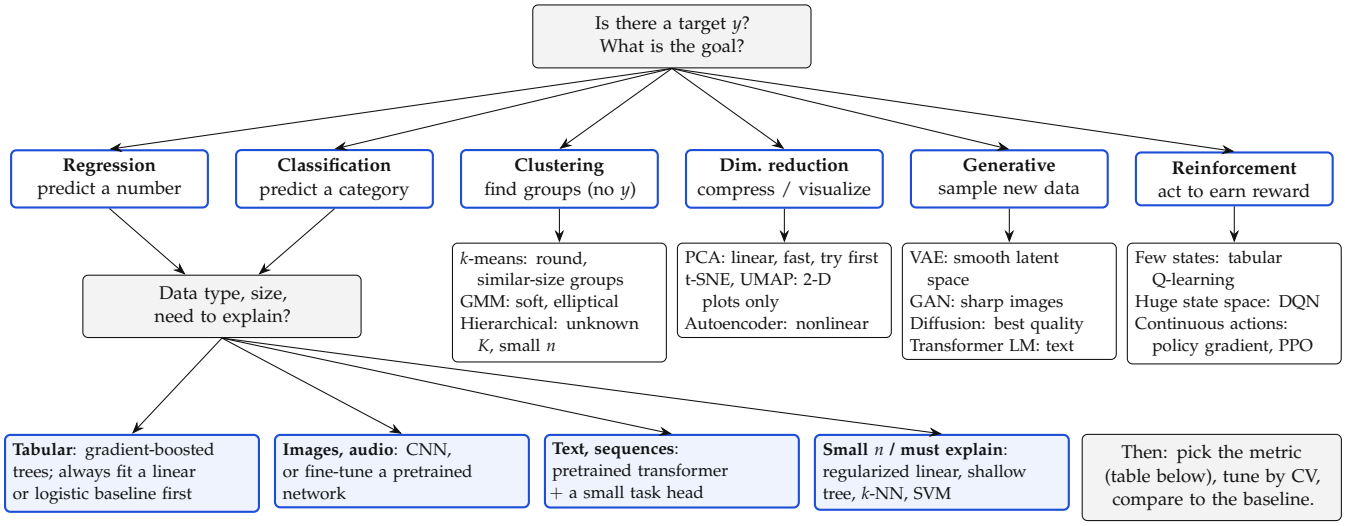
CONTEXT

- **Origin.** Statistical learning theory of Vapnik and Chervonenkis (1970s); "machine learning" coined by Arthur Samuel (1959).
- **Use.** Prediction from data instead of hand-written rules.
- **Data science.** The template for every model in this chapter; held-out test data estimates the true (expected) risk S.13 §13 Statistics.
- **Physical application.** Fitting a drag coefficient to wind-tunnel measurements is empirical-risk minimization with a physical model.

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Choosing a Model & Metric | *Key Topic ML.K1*

Two decisions come before any tuning: *which model family* fits the task and data, and *which metric* measures success. Start from the goal (is there a target? what type?), then the data type and size. Always fit a trivial baseline first, and judge every model by the metric that matches the real cost of its mistakes.



Which metric? (task → situation → metric)

Task	Situation	Metric	Why / catch
Regression	errors in units, outliers present	MAE ML.19	robust; "off by 3 on average"
	large misses are costly	RMSE	squares errors; $\geq$ MAE
	compare across targets or scales	R^2, MAPE	scale-free; MAPE fails at $y = 0$
Binary class.	balanced classes, ranking quality	accuracy, ROC AUC ML.18	threshold-free AUC $= P(s_+ > s_-)$
	rare positives (fraud, disease)	PR AUC, F_1, recall at fixed precision	accuracy and ROC flatter imbalanced data
	probabilities feed decisions	log-loss, Brier, ECE ML.21	rewards calibrated probabilities
	one balanced number	MCC ML.20	0 random, 1 perfect, uses all four cells
Multi-class	every class matters equally	macro-F_1	averages per-class F_1
	overall hit rate	accuracy (micro), top-k	dominated by big classes
Clustering	no labels / with labels	silhouette ML.36 / adjusted Rand	silhouette favors round clusters
Dim. reduction	compression quality	explained variance ML.38	keep ≈90% as a default
RL	policy quality	average return ML.42	report over many seeds

Data size rules of thumb

Small: $n < 10^3$

Regularized linear/logistic ML.8, k-NN, SVM ML.15, shallow trees. Use 10-fold or repeated CV; report a confidence interval.

Medium tabular: 10^3–10^7

Gradient-boosted trees ML.14 usually win; random forests are a strong, low-tuning alternative ML.13. 5-fold CV.

Large or unstructured

Images, audio, text: deep networks ML.29, ML.34; fine-tune a pretrained model when labels are few. A single held-out split suffices.

DECISION POINT **Interpretability required** (credit, medicine)? Prefer linear/logistic models or shallow trees, or explain a black box carefully. **Accuracy vs. cost of errors**: pick the threshold and metric from the cost of a false positive vs. a false negative, not by default 0.5. **Mistakes**: reporting accuracy on imbalanced data; tuning on the test set; fitting scalers or feature selection on all data (leakage); random splits on time series; comparing models on different splits.

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ML.2 Loss Functions

A loss scores a single prediction against the truth; the average loss is what training minimizes. The loss decides which errors matter and what the optimal prediction is.

Mean squared error	$\mathrm{MSE} = \frac{1}{n}\sum_i (y_i - \hat{y}_i)^2$	Penalizes large errors heavily; the best constant prediction is the mean S.1 §13 Statistics.
Mean absolute error	$\mathrm{MAE} = \frac{1}{n}\sum_i \lvert y_i - \hat{y}_i \rvert$	Robust to outliers A1.11 §03 Algebra I; the best constant prediction is the median.
Huber loss	$L_\delta(r) = \begin{cases} \frac{1}{2}r^2, & \lvert r \rvert \le \delta \\ \delta\left(\lvert r \rvert - \frac{1}{2}\delta\right), & \lvert r \rvert > \delta \end{cases}$	Quadratic near 0, linear in the tails: smooth like MSE, robust like MAE.
Log-loss (binary cross-entropy)	$-\left[y \ln p + (1-y)\ln(1-p)\right]$	Negative Bernoulli log-likelihood S.16 §13 Statistics; punishes confident wrong probabilities without bound.
Hinge loss	$\max\left(0,\, 1 - y\, f(\mathbf{x})\right)$	Zero once a point is on the correct side by margin 1; used by SVMs ML.15.
Quantile (pinball) loss	$L_\tau(r) = \max\left(\tau r,\, (\tau - 1)r\right)$	Minimized by the τ-quantile; $\tau = 0.5$ gives half the MAE.

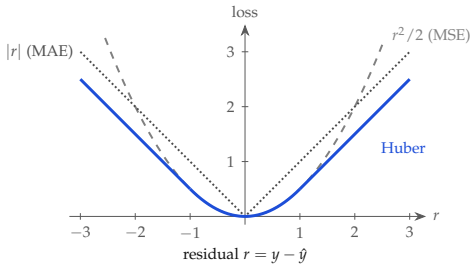
STATED-EQUATION *MSE equals one over n times the sum of y i minus y hat i, squared. MAE equals one over n times the sum of the absolute errors. Huber loss is one half r squared for small residuals and delta times the absolute residual minus one half delta otherwise. Log-loss equals minus the quantity y natural log p plus one minus y natural log of one minus p. Hinge loss equals the maximum of zero and one minus y f of x. Quantile loss is the maximum of tau r and tau minus one times r.*

MEMORY AID MSE → mean; MAE → median; pinball → quantile.

DECISION POINT **Which loss?** Match the business cost: MAE or Huber when outliers should not dominate; MSE when large misses are disproportionately bad; log-loss when calibrated probabilities are needed; quantile loss for prediction intervals and inventory buffers (e.g. $\tau = 0.9$).

TERMS

$\hat{y}$ – prediction; $r = y - \hat{y}$ residual S.40 §13 Statistics
δ – Huber threshold between quadratic and linear
p – predicted probability, $y \in \{0, 1\}$ (log-loss)
$y \in \{-1, +1\}$ – label coding for hinge loss; $f(\mathbf{x})$ raw score
$\tau \in (0, 1)$ – target quantile
Convexity – all six are convex in $\hat{y}$ O.2 §14 Optimization
r – residual $y - \hat{y}$
y_i, n; $f(\mathbf{x})$ – true values and count; raw classifier score



CONTEXT

- **Origin.** Least squares (Gauss, Legendre); Huber (1964); log-loss from maximum likelihood (Fisher, 1920s); quantile regression by Koenker and Bassett (1978).
- **Use.** Defines what "good" means for a model.
- **Data science.** Match the loss to the business cost: MAE for forecast error in units, log-loss for calibrated probabilities, quantile (pinball) loss for inventory buffers.
- **Physical application.** Calibrating a thermometer by minimizing squared error against a reference is least squares.

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ML.3 Entropy, Cross-Entropy & KL Divergence

Information theory measures uncertainty (entropy) and the mismatch between a true distribution and a model's (cross-entropy, KL). Classification losses and tree splits are built from these.

Entropy	$H(p) = -\sum_k p_k \log p_k$	Average surprise; 0 for a certain outcome, maximal ($\log K$) for uniform.
Cross-entropy	$H(p,q) = -\sum_k p_k \log q_k$	Average code length using model q when data follow p; the classification loss ML.24.
KL divergence	$D_{\mathrm{KL}}(p \,\|\, q) = \sum_k p_k \log \frac{p_k}{q_k} \geq 0$	Extra cost of using q instead of p; zero iff $p = q$; not symmetric.
Decomposition	$H(p,q) = H(p) + D_{\mathrm{KL}}(p \,\|\, q)$	Since $H(p)$ is fixed by the data, minimizing cross-entropy minimizes KL.
Mutual information	$I(X;Y) = H(Y) - H(Y \mid X)$	Uncertainty in Y removed by knowing X; information gain ML.12 is its sample version.

STATED-EQUATION *Entropy H of p equals minus the sum of p k log p k. Cross-entropy H of p, q equals minus the sum of p k log q k. The KL divergence of p from q equals the sum of p k log p k over q k, which is at least zero. Cross-entropy equals entropy plus KL divergence. Mutual information equals the entropy of Y minus the conditional entropy of Y given X.*

MEMORY AID KL is not symmetric: $D(p\|q) \neq D(q\|p)$.

TERMS

p – true distribution; q model's predicted distribution
k – outcome / class index, K classes
log – base 2 gives bits, base e gives nats A2.14 §04 Algebra II, A2.16 §04 Algebra II
$H(p)$ – uncertainty; maximal for a uniform distribution
Cross-entropy – average code length using q when data follow p
KL – extra cost of using q; zero iff $p = q$; not symmetric
$H(Y \mid X)$ – expected entropy of Y after observing X S.14 §13 Statistics
q – model (approximating) distribution
$I(X;Y)$ – mutual information between X and Y

CONTEXT

- **Origin.** Claude Shannon, "A Mathematical Theory of Communication" (1948); KL by Kullback and Leibler (1951).
- **Use.** Data compression, communication, measuring information.
- **Data science.** Cross-entropy is the classification loss ML.24; decision-tree splits maximize information gain ML.12; KL appears in VAEs ML.40, t-SNE ML.39, and drift monitoring.
- **Physical application.** Thermodynamic entropy $S = k_B \ln W$ P.29 §12 Physics is the physical ancestor of Shannon entropy.

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ML.4 Bias–Variance Trade-off

Expected prediction error splits into systematic error (bias), sensitivity to the particular training sample (variance), and noise no model can remove. Increasing complexity lowers bias but raises variance.

Bias–variance decomposition

Expectation over random training sets and noise, at a fixed $\mathbf{x}$, for squared loss.

$$E\left[\left(y-\hat{f}(\mathbf{x})\right)^2\right] = \underbrace{\left(E[\hat{f}(\mathbf{x})]-f(\mathbf{x})\right)^2}_{\text{Bias}^2} + \underbrace{\text{Var}\left(\hat{f}(\mathbf{x})\right)}_{\text{Variance}} + \underbrace{\sigma^2}_{\text{irreducible noise}}$$

Bias	$\text{Bias} = E[\hat{f}(\mathbf{x})] - f(\mathbf{x})$	Average miss over many training sets: too-simple models underfit.
Variance	$\text{Var}(\hat{f}) = E\left[\left(\hat{f}-E[\hat{f}]\right)^2\right]$	How much the fit jumps between training sets: too-flexible models overfit.

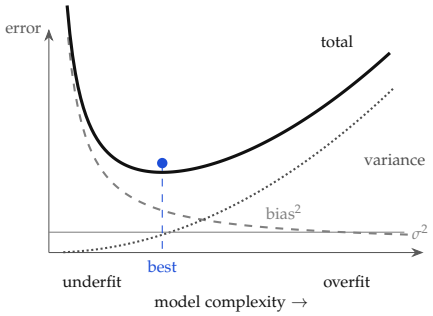
STATED-EQUATION *The expected squared error equals bias squared, the square of the expected prediction minus the true value, plus variance, the variance of the prediction, plus sigma squared, the irreducible noise.*

MEMORY AID Too simple → bias (underfit); too complex → variance (overfit).

DECISION POINT **Diagnose from learning curves.** High training *and* validation error ⇒ bias: add features, a more flexible model, less regularization. Low training but high validation error ⇒ variance: more data, regularization ML.8, bagging ML.13, early stopping ML.28, simpler model. Both low ⇒ done.

TERMS

f – true function; $\hat{f}$ model fit on a random training set
Bias – systematic error from an overly simple model (underfitting)
Variance – sensitivity to the particular training sample (overfitting) S.13 §13 Statistics
σ^2 – noise in y that no model can remove
Estimator view – same split as MSE = bias^2+variance S.25 §13 Statistics
Diagnosis – low train error but high validation error ⇒ variance
$\hat{f}$; $\mathbf{x}$, y – fitted model (random through the training set); input and noisy target



CONTEXT

- **Origin.** Formalized for neural networks by Geman, Bienenstock and Doursat (1992).
- **Use.** Choosing model complexity.
- **Data science.** Tune depth, k, or λ with validation data ML.5; bagging reduces variance ML.13; regularization trades a little bias for less variance ML.8.
- **Physical application.** A very flexible curve through noisy lab measurements (high variance) predicts a new trial worse than a simple physical law.

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ML.5 Cross-Validation

Rotate which part of the data is held out, train on the rest, and average the held-out errors. This estimates out-of-sample error and is how hyperparameters are chosen.

k-fold CV estimate	$\mathrm{CV}_{(k)} = \frac{1}{k}\sum_{j=1}^{k} \mathrm{Err}_j$	Average validation error over the k folds.
Fold sizes	train on $\frac{k-1}{k}n$, validate on $\frac{n}{k}$	Each row is validated exactly once.
Leave-one-out	LOOCV: $k = n$	Nearly unbiased but costly and high-variance; closed form for linear smoothers.
Standard error of CV	$SE_{CV} \approx \frac{s_{\mathrm{Err}}}{\sqrt{k}}$	Spread of the fold errors; used by the one-standard-error rule.

STATED-EQUATION *The k-fold CV error equals one over k times the sum of the fold errors. Each fold trains on k minus one over k of the rows and evaluates on n over k. Leave-one-out uses k equals n. The standard error is about s of the errors over root k.*

MEMORY AID Every row gets one turn as the test set.

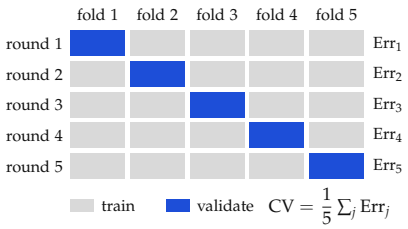
METHOD

1. Shuffle and split the rows into k folds (stratify by class; group or time-order if needed).
2. For $j = 1..k$: fit all preprocessing and the model on the other $k-1$ folds; compute Err_j on fold j.
3. Average to get $\mathrm{CV}_{(k)}$; repeat for each hyperparameter setting and pick the best (or the simplest within one SE).
4. Refit on all training data with the chosen setting. Example: fold errors $4, 6, 5, 5, 5 \Rightarrow \mathrm{CV}_{(5)} = 5$.

DECISION POINT **Which k?** 5 or 10 balances bias (small training folds) and cost; LOOCV only for tiny n. **Which scheme?** Stratified for classification; grouped when rows share an entity; forward-chaining for time series; nested CV when you need an unbiased score *and* tuning.

TERMS

k – number of folds (typically 5 or 10); n rows
Err_j – validation error on fold j (any loss ML.2)
s_{Err} – standard deviation of the k fold errors S.2 §13 Statistics
Mean – averaging reduces variance of the estimate S.1 §13 Statistics, S.24 §13 Statistics
Stratified – keeps class proportions in each fold
Time series – use forward-chaining splits, never future data
Leakage – fit scalers and feature selection inside each fold
n; $\mathrm{CV}_{(k)}$ – number of rows; k-fold cross-validation estimate
LOOCV – leave-one-out cross-validation



CONTEXT

- **Origin.** Stone (1974) and Geisser (1975).
- **Use.** Estimating out-of-sample performance without wasting data.
- **Data science.** Selecting λ, depth, or k ML.4; comparing models with paired tests on the same folds S.30 §13 Statistics.
- **Physical application.** Validating a heat-transfer correlation on held-out experiments, rather than the ones used to fit it.

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ML.6 Feature Scaling & Encoding

Put features on comparable scales and turn categories into numbers. Distance-based, regularized and gradient-trained models depend on scale; tree models do not.

Standardization	$z = \dfrac{x - \mu}{\sigma}$	Mean 0, SD 1 S.4 §13 Statistics; default for linear models, SVM, k-NN, PCA, neural nets.
Min–max scaling	$x' = \dfrac{x - x_{\min}}{x_{\max} - x_{\min}} \in [0,1]$	Fixed range (e.g. pixels); one outlier squashes everything else.
Robust scaling	$x' = \dfrac{x - \text{median}}{\text{IQR}}$	Uses the median and IQR S.3 §13 Statistics, so outliers barely move it.
Log transform	$x' = \log(1 + x)$	Tames right-skewed, positive data (income, counts) A2.14 §04 Algebra II.
One-hot encoding	$\text{color} \in \{r,g,b\} \mapsto (1,0,0),(0,1,0),(0,0,1)$	One binary column per category; drop one for unpenalized linear models.
Target (mean) encoding	$x'_c = \dfrac{n_c \bar{y}_c + m\,\bar{y}}{n_c + m}$	Smoothed mean target of category c; compute out-of-fold to avoid leakage.

STATED-EQUATION *Standardize with z equals x minus mu over sigma. Min-max scaling maps x to x minus the minimum over the range. Robust scaling subtracts the median and divides by the IQR. Log transform uses log of one plus x. One-hot encoding makes one binary column per category. Target encoding replaces a category by its smoothed mean target.*

MEMORY AID Fit the scaler on train, apply it to test.

DECISION POINT **Which scaling?** Roughly symmetric features → standardize; bounded/known range → min–max; heavy outliers → robust or log; trees and boosting → none needed. **Categories:** few levels → one-hot; many levels (zip codes) → target encoding, hashing, or learned embeddings ML.32.

TERMS

μ, σ – training-set mean and SD of the feature S.1 §13 Statistics, S.2 §13 Statistics
$x_{\min}, x_{\max}$ – training-set minimum and maximum
IQR – interquartile range $Q_3 - Q_1$
$n_c, \bar{y}_c$ – count and mean target in category c; $\bar{y}$ global mean
m – smoothing strength (pseudo-count)
Leakage – using test or future information during fitting
x'; x'_c – transformed feature; encoded value of category c
median; $\tilde{y}$ – middle value S.1 §13 Statistics; overall target mean

CONTEXT

- **Origin.** Standard scores from Pearson-era statistics; one-hot ("dummy") coding from regression analysis (Suits, 1957).
- **Use.** Preparing raw data for any model.
- **Data science.** Pipelines (fit transformers inside CV folds) prevent leakage ML.5; unscaled features make ridge/lasso penalties ML.8 and k-NN distances ML.11 meaningless.
- **Physical application.** Comparing sensor channels in volts and in kilopascals requires standardizing them first.

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Linear Models & Simple Classifiers

ML.7 Linear Regression: Closed Form & Gradient

Predict y as a weighted sum of features and choose the weights that minimize squared error, either exactly (normal equations) or iteratively (gradient descent).

Model	$\hat{\mathbf{y}} = X\boldsymbol{\beta}$	Column of 1s in X carries the intercept β_0.
Least-squares objective	$J(\boldsymbol{\beta}) = \frac{1}{n}\lVert \mathbf{y} - X\boldsymbol{\beta}\rVert_2^2$	Convex quadratic bowl O.2 §14 Optimization: one global minimum when X has full column rank.
Normal equations	$X^\top X\,\boldsymbol{\beta} = X^\top \mathbf{y}$	Residuals orthogonal to every column of X LA.20 §09 Linear Algebra.
Closed form (OLS)	$\hat{\boldsymbol{\beta}} = (X^\top X)^{-1} X^\top \mathbf{y}$	Exact solution; solve by QR/Cholesky, not an explicit inverse LA.18 §09 Linear Algebra; pseudoinverse if rank-deficient LA.31 §09 Linear Algebra.
Gradient	$\nabla_{\boldsymbol{\beta}} J = -\frac{2}{n} X^\top (\mathbf{y} - X\boldsymbol{\beta})$	From matrix calculus LA.30 §09 Linear Algebra; zero gradient gives the normal equations.
Gradient-descent step	$\boldsymbol{\beta} \leftarrow \boldsymbol{\beta} + \eta\,\frac{2}{n} X^\top (\mathbf{y} - X\boldsymbol{\beta})$	Move along the residual-weighted features C1.26 §07 Calculus I; scale features first ML.6.
Simple regression	$\hat{\beta}_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2},\ \ \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$	The one-feature special case S.40 §13 Statistics.

STATED-EQUATION *The prediction y hat equals X beta. The objective J equals one over n times the squared norm of y minus X beta. Normal equations: X transpose X beta equals X transpose y, so beta hat equals X transpose X inverse times X transpose y. The gradient equals minus two over n times X transpose times the residual vector; gradient descent adds eta times that residual direction. For one feature, beta one hat equals the sum of x i minus x bar times y i minus y bar, over the sum of x i minus x bar squared, and beta naught hat equals y bar minus beta one hat times x bar.*

MEMORY AID Normal equations: residuals are *normal* (perpendicular) to the features.

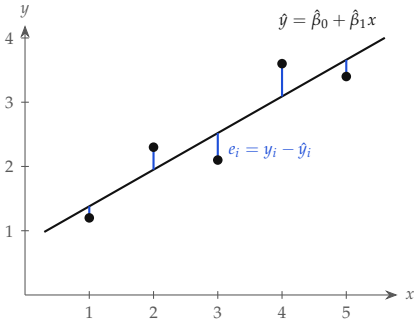
METHOD

1. Build X with a leading column of 1s.
2. Form $X^\top X$ and $X^\top \mathbf{y}$.
3. Solve the linear system $X^\top X\boldsymbol{\beta} = X^\top \mathbf{y}$ LA.17 §09 Linear Algebra. Example: points $(0,1),(1,3),(2,5)$: $\bar{x} = 1, \bar{y} = 3$, $\hat{\beta}_1 = \frac{(-1)(-2)+0+(1)(2)}{1+0+1} = 2$, $\hat{\beta}_0 = 3 - 2 = 1$, so $\hat{y} = 1 + 2x$.

DECISION POINT **Closed form or gradient descent?** Few features ($d \lesssim 10^4$) → closed form via QR or Cholesky ($O(nd^2)$). Huge d or streaming data → (stochastic) gradient descent ML.27. Correlated features or $d > n$ → add ridge ML.8.

TERMS

X – $n \times (d+1)$ design matrix; rows are examples
$\boldsymbol{\beta}$ – coefficients; $\hat{\boldsymbol{\beta}}$ fitted values
$\mathbf{y}, \hat{\mathbf{y}}$ – target vector and predictions
$\lVert \cdot \rVert_2$ – Euclidean norm LA.29 §09 Linear Algebra
η – learning rate
$\bar{x}, \bar{y}$ – sample means S.1 §13 Statistics
Assumptions – for inference: linearity, independent errors, constant variance S.41 §13 Statistics
J; n – least-squares cost; number of rows
$\nabla_{\boldsymbol{\beta}} J$; $\hat{\boldsymbol{\beta}}$ – gradient of the cost LA.30 §09 Linear Algebra; fitted coefficients



CONTEXT

- **Origin.** Legendre (1805) and Gauss (1809), least squares.
- **Use.** Trend lines, calibration curves, the baseline for every regression task.
- **Data science.** The first model to fit on any numeric target; coefficients give per-unit effects; the closed form is a single linear solve.
- **Physical application.** Fitting Hooke's law $F = kx$ to spring measurements.

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ML.8 Regularization: Ridge, Lasso & Elastic Net

Add a penalty on coefficient size to the loss. This trades a little bias for much less variance; the L1 penalty also sets some coefficients exactly to zero.

Ridge objective	$\min_{\boldsymbol{\beta}} \|\mathbf{y} - X\boldsymbol{\beta}\|_2^2 + \lambda\|\boldsymbol{\beta}\|_2^2$	Shrinks all coefficients smoothly toward 0.
Ridge closed form	$\hat{\boldsymbol{\beta}} = \left(X^\top X + \lambda I\right)^{-1} X^\top \mathbf{y}$	Adding λI makes the matrix invertible and well-conditioned LA.29 §09 Linear Algebra.
Lasso objective	$\min_{\boldsymbol{\beta}} \|\mathbf{y} - X\boldsymbol{\beta}\|_2^2 + \lambda\|\boldsymbol{\beta}\|_1$	Corners of the L1 ball give exact zeros: built-in feature selection.
Norms	$\|\boldsymbol{\beta}\|_1 = \sum_j \|\beta_j\|, \quad \|\boldsymbol{\beta}\|_2^2 = \sum_j \beta_j^2$	L1 sums absolute values; squared L2 sums squares.
Elastic net	$\lambda\left[\alpha\|\boldsymbol{\beta}\|_1 + (1-\alpha)\|\boldsymbol{\beta}\|_2^2\right]$	Mixes both: sparse like lasso, stable with correlated features like ridge.
Weight decay (gradient)	$\nabla\left(\lambda\|\boldsymbol{\beta}\|_2^2\right) = 2\lambda\boldsymbol{\beta}$	Each step shrinks weights by a constant fraction; L2 in neural nets.

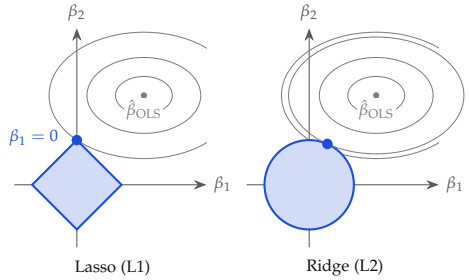
STATED-EQUATION *Ridge: minimize the squared norm of y minus X beta plus lambda times the squared L2 norm of beta; beta hat equals X transpose X plus lambda I, inverse, X transpose y. Lasso: the same with lambda times the L1 norm. The L1 norm is the sum of absolute values; the squared L2 norm is the sum of squares. Elastic net mixes them with weight alpha. The L2 penalty adds two lambda beta to the gradient.*

MEMORY AID Lasso has a Lasso that ropes coefficients to exactly zero; ridge just shrinks.

DECISION POINT **Ridge, lasso, or elastic net?** Many small effects or correlated features → ridge; you expect few relevant features or want selection → lasso; correlated groups and selection → elastic net. **How strong (λ)?** Search a log-spaced grid ($10^{-4} \ldots 10^{2}$) by CV ML.5; pick the minimum or the largest λ within one SE of it (simpler model).

TERMS

λ – penalty strength; $\lambda = 0$ gives OLS ML.7
$\|\cdot\|_2$ – Euclidean norm T.21 §06 Trigonometry; $\|\cdot\|_1$ sum of absolute values
I – identity; λI makes $X^\top X$ invertible LA.15 §09 Linear Algebra
$\alpha \in [0, 1]$ – L1 share in elastic net
Lasso – sets some coefficients exactly to 0 (feature selection) O.18 §14 Optimization
Scaling – standardize features first S.4 §13 Statistics; do not penalize β_0
$\|\cdot\|_1$; β_j – L1 norm (sum of absolute values); coefficient j
X, $\mathbf{y}$; $\hat{\boldsymbol{\beta}}$ – design matrix and targets ML.7; fitted coefficients



CONTEXT

- **Origin.** Ridge by Hoerl and Kennard (1970); lasso by Tibshirani (1996); elastic net by Zou and Hastie (2005).
- **Use.** Stable estimates with many correlated predictors.
- **Data science.** Preventing overfitting in wide data ML.4; weight decay in neural networks is L2; Bayesian view: ridge = Gaussian prior, lasso = Laplace prior S.11 §13 Statistics.
- **Physical application.** Tikhonov regularization (ridge) stabilizes ill-posed inverse problems like deblurring images or CT reconstruction.

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ML.9 Logistic Regression

A linear score passed through the sigmoid gives a probability for a binary outcome. Fitted by minimizing log-loss, it is linear in the log-odds.

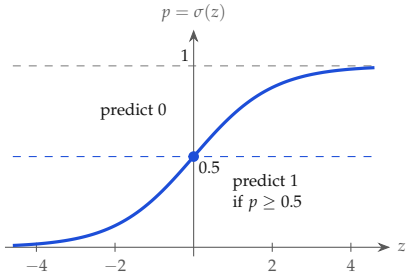
Model	$p = P(y = 1 \mid \mathbf{x}) = \sigma(z) = \dfrac{1}{1 + e^{-z}}, \quad z = \beta_0 + \boldsymbol{\beta}^\top \mathbf{x}$	The sigmoid squashes any score into $(0, 1)$.
Log-odds (logit)	$\ln \dfrac{p}{1 - p} = \beta_0 + \boldsymbol{\beta}^\top \mathbf{x}$	Linear in the features: each unit of x_j adds β_j to the log-odds.
Odds ratio	$\mathrm{OR}_j = e^{\beta_j}$	Multiplicative change in odds per unit of x_j A2.13 §04 Algebra II.
Log-loss gradient	$\nabla_\beta L = \dfrac{1}{n} X^\top (\mathbf{p} - \mathbf{y})$	Same shape as linear regression's: features times (prediction − truth).
Decision boundary	$p = 0.5 \iff \beta_0 + \boldsymbol{\beta}^\top \mathbf{x} = 0$	A hyperplane; other thresholds shift it parallel.

STATED-EQUATION *p, the probability y equals one given x, equals sigma of z, which is one over one plus e to the minus z, where z equals beta naught plus beta transpose x. The log-odds, natural log of p over one minus p, equals beta naught plus beta transpose x. The odds ratio for feature j is e to the beta j. The gradient of the log-loss is one over n X transpose times p minus y. The boundary p equals one half is the hyperplane z equals zero.*

MEMORY AID Logistic regression is linear in the log-odds.

TERMS

$\sigma(z)$ – sigmoid (logistic curve C2.28 §08 Calculus II); maps $\mathbb{R} \to (0, 1)$
$\frac{p}{1-p}$ – odds; its log is the logit C2.4 §08 Calculus II
β_j – change in log-odds per unit of x_j
e^{β_j} – multiplicative change in odds A2.13 §04 Algebra II
$\mathbf{p}, \mathbf{y}$ – vectors of predicted probabilities and 0/1 labels
Fitting – minimize log-loss ML.2 (convex O.2 §14 Optimization) by Newton/L-BFGS O.13 §14 Optimization
Boundary – $p = 0.5 \iff z = 0$: a hyperplane
Statistics view – S.42 §13 Statistics
z; $\beta_0, \boldsymbol{\beta}$ – linear score (logit); intercept and weights
X; n; p – feature matrix; number of rows; predicted probability



CONTEXT

- **Origin.** Berkson (1944); David Cox (1958).
- **Use.** Probability of a binary outcome: default, disease, purchase.
- **Data science.** The standard baseline classifier: churn, conversion, credit scoring; interpretable odds ratios; the output layer of binary neural networks.
- **Physical application.** The probability that a material fractures versus applied stress is often modeled with a logistic curve.

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ML.10 Naive Bayes Classifier

Apply Bayes' theorem with the "naive" assumption that features are independent given the class, so the likelihood factorizes into one term per feature.

Posterior	$P(y \mid \mathbf{x}) \propto P(y)\prod_{j=1}^{d} P(x_j \mid y)$	Bayes S.11 §13 Statistics plus conditional independence S.9 §13 Statistics.
Decision rule (logs)	$\hat{y} = \arg\max_y \left[\log P(y) + \sum_j \log P(x_j \mid y)\right]$	Sum of logs avoids underflow of a long product A2.14 §04 Algebra II.
Laplace smoothing	$\hat{P}(x_j \mid y) = \dfrac{n_{jy}+1}{n_y + V}$	Add one pseudo-count so an unseen word never zeroes the product.
Gaussian likelihood	$P(x_j \mid y) = \dfrac{1}{\sqrt{2\pi\sigma_{jy}^2}} e^{-(x_j-\mu_{jy})^2/2\sigma_{jy}^2}$	For continuous features: one normal per feature and class S.20 §13 Statistics.

STATED-EQUATION *P of y given x is proportional to P of y times the product over j of P of x j given y. Predict the y that maximizes log P of y plus the sum of log P of x j given y. With Laplace smoothing, P hat of x j given y equals n j y plus one over n y plus V. Gaussian naive Bayes uses a normal density for each feature and class.*

MEMORY AID Naive = pretend the features don't talk to each other.

METHOD

1. Estimate priors $P(y)$ from class frequencies.
2. Estimate each $P(x_j \mid y)$ (counts with smoothing, or a per-class mean and variance).
3. For a new $\mathbf{x}$, add log-prior and log-likelihoods per class and take the largest. Example: $P(\text{spam}) = 0.4$, $P(\text{"win"} \mid \text{spam}) = 0.5$, $P(\text{"win"} \mid \text{ham}) = 0.1$: scores 0.20 vs 0.06 $\Rightarrow$ spam.

TERMS

$P(y)$ – class prior S.11 §13 Statistics
$P(x_j \mid y)$ – likelihood of feature j given the class; d features
Naive assumption – features conditionally independent given y S.9 §13 Statistics
Logs – sum of logs avoids underflow A2.14 §04 Algebra II
Laplace smoothing – $+1$ counts; n_{jy} count of feature j in class y, n_y total count in y, V vocabulary size
μ_{jy}, σ_{jy}^2 – per-class mean and variance of feature j
Variants – multinomial (counts), Bernoulli, Gaussian
n_{jy}, n_y; V – count of feature value x_j in class y, count of class y; number of possible values
d; $\hat{y}$ – number of features; predicted class

CONTEXT

- **Origin.** Applied to text classification by Maron (1961); spam filtering popularized by Paul Graham (2002).
- **Use.** Fast, strong baseline for text and categorical data.
- **Data science.** Spam and sentiment classification, document tagging, and quick baselines on high-dimensional sparse features.
- **Physical application.** Diagnosing a machine fault from several independent sensor alarms combines their likelihoods naively.

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ML.11 k-Nearest Neighbors & Distance Metrics

Predict by looking up the k most similar training examples and voting (classification) or averaging (regression). Everything depends on the distance used.

Euclidean distance	$d_2(\mathbf{x},\mathbf{z}) = \sqrt{\sum_j (x_j - z_j)^2}$	Straight-line distance G.12 §05 Geometry; the default.
Manhattan distance	$d_1(\mathbf{x},\mathbf{z}) = \sum_j \lvert x_j - z_j \rvert$	Grid (taxicab) distance; less dominated by one large difference.
Minkowski distance	$d_p(\mathbf{x},\mathbf{z}) = \left(\sum_j \lvert x_j - z_j \rvert^p\right)^{1/p}$	$p=1$ Manhattan, $p=2$ Euclidean, $p\to\infty$ largest coordinate gap.
Cosine similarity	$\cos\theta = \dfrac{\mathbf{x}\cdot\mathbf{z}}{\lVert\mathbf{x}\rVert\,\lVert\mathbf{z}\rVert}$	Angle only, ignores length T.21 §06 Trigonometry; standard for text and embeddings.
k-NN prediction	$\hat{y} =$ majority (or mean) of $N_k(\mathbf{x})$	No training step: all work happens at prediction time.

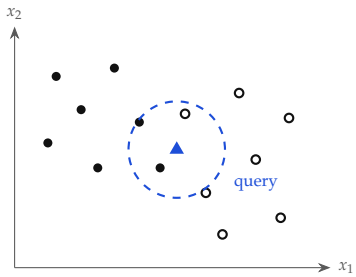
STATED-EQUATION *The Euclidean distance d two equals the square root of the sum of x j minus z j, squared. The Manhattan distance d one equals the sum of the absolute differences. Minkowski distance raises absolute differences to the p, sums, and takes the p-th root. Cosine similarity equals x dot z over the product of the norms. Predict the majority, or mean, of the k nearest neighbors.*

MEMORY AID "You are the company you keep."

DECISION POINT **Choosing k**: small $k \to$ jagged boundary, high variance; large $k \to$ smooth, high bias ML.4. Try odd k from 1 to $\sqrt{n}$ by CV. **Choosing the distance**: Euclidean on standardized numeric features; cosine for text/embeddings; Manhattan for high-dimensional or outlier-prone data. Beyond ~20 dimensions all distances look alike (curse of dimensionality): reduce dimension first ML.38.

TERMS

$\mathbf{x}, \mathbf{z}$ – two feature vectors; j feature index
d_2 – Euclidean distance G.12 §05 Geometry
d_1 – Manhattan (taxicab) distance
p – Minkowski order, $p \geq 1$
$\cos\theta$ – cosine similarity T.21 §06 Trigonometry, LA.3 §09 Linear Algebra
$N_k(\mathbf{x})$ – the k training points closest to $\mathbf{x}$
Scaling – standardize so no feature dominates S.4 §13 Statistics
k; $\hat{y}$ – number of neighbors; prediction



$k = 5$: 3 • vs 2 ○ ⇒ predict •

CONTEXT

- **Origin.** Fix and Hodges (1951); Cover and Hart (1967).
- **Use.** Simple, assumption-free classification and regression.
- **Data science.** Recommendation ("users like you"), vector search over embeddings with cosine similarity, and imputation of missing values from nearest rows.
- **Physical application.** Classifying a mineral sample by comparing its measured density and hardness to the nearest known samples.

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Trees, Ensembles & Kernel Methods

ML.12 Decision Trees: Gini, Entropy & Information Gain

A tree splits the data by one feature at a time, choosing each split to make the child nodes as pure as possible; leaves predict the majority class or mean.

Gini impurity	$\text{Gini} = 1 - \sum_k p_k^2$	Probability two random picks from the node disagree; 0 when pure.
Entropy	$H = -\sum_k p_k \log_2 p_k$	Uncertainty of the node's class mix ML.3; 1 bit for a 50/50 binary node.
Information gain	$IG = H(\text{parent}) - \sum_{c \in \text{children}} \frac{n_c}{n} H(c)$	Impurity removed by the split, children weighted by size.
Variance reduction (regression)	$\Delta = \text{Var}(\text{parent}) - \sum_c \frac{n_c}{n} \text{Var}(c)$	Regression trees split to lower the within-node variance S.2 §13 Statistics.
Cost-complexity pruning	$R_\alpha(T) = R(T) + \alpha \lvert T \rvert$	Penalize the number of leaves; choose α by CV.

STATED-EQUATION *Gini impurity equals one minus the sum of p k squared. Entropy H equals minus the sum of p k log base two of p k. Information gain equals the parent entropy minus the weighted sum of child entropies, n c over n times H of c. Regression trees use the reduction in variance. Pruning minimizes the tree's error plus alpha times its number of leaves.*

MEMORY AID Split to purify; prune to generalize.

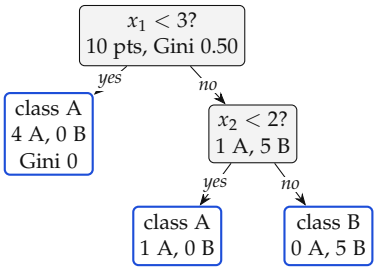
METHOD

1. At a node, for every feature and every candidate threshold, compute the weighted child impurity.
2. Take the split with the largest gain (greedy O.23 §14 Optimization); send rows left/right.
3. Recurse on each child until a stopping rule (max depth, min samples, pure node).
4. Prune back with R_α if needed. Example: parent 4 yes/4 no (Gini 0.5) splits into (4,0) and (0,4) (Gini 0): gain $0.5 - 0 = 0.5$.

DECISION POINT **How deep?** A single interpretable tree: depth 3–5 or pruning by CV. Trees in a random forest: grow deep (averaging removes variance). Trees in boosting: shallow (depth 3–8) ML.14. Gini vs. entropy rarely matters; Gini is faster.

TERMS

p_k – fraction of class k in a node
Gini – probability two random picks disagree; 0 for a pure node
H – entropy ML.3
IG – reduction in impurity; the weighted average uses $\frac{n_c}{n}$ S.1 §13 Statistics
n, n_c – rows in the parent and in child c
$R(T), |T|$ – training error and number of leaves of tree T; $\alpha \geq 0$
Greedy growth – choose the split with largest gain at each node O.22 §14 Optimization
α; T – complexity penalty per leaf; a candidate subtree
Δ – variance reduction of a split



CONTEXT

- **Origin.** CART by Breiman et al. (1984); ID3/C4.5 by Quinlan (1986, 1993).
- **Use.** Interpretable if–then rules; handles mixed feature types.
- **Data science.** The building block of random forests and gradient boosting ML.13; decision rules stakeholders can read.
- **Physical application.** A field guide's key for identifying trees ("leaves opposite? then…") is a decision tree.

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ML.13 Ensembles: Bagging, Random Forests & Boosting

Combine many models. Bagging averages independent fits to cut variance; boosting adds models sequentially, each correcting the last, to cut bias.

Bagging average	$\bar{f} = \frac{1}{B}\sum_{b=1}^{B} \hat{f}_b$	Average B models each trained on a bootstrap sample S.36 §13 Statistics.
Variance of an average	$\mathrm{Var}(\bar{f}) = \rho\,\sigma^2 + \frac{1-\rho}{B}\,\sigma^2$	More models remove the second term; only decorrelation ($\rho \downarrow$) shrinks the first.
Out-of-bag fraction	$\left(1 - \frac{1}{n}\right)^n \to e^{-1} \approx 0.368$	Each tree never sees ≈37% of rows: free validation C1.6 §07 Calculus I.
Boosting update	$F_m(\mathbf{x}) = F_{m-1}(\mathbf{x}) + \nu\, h_m(\mathbf{x})$	Add a small multiple of a new weak learner each round ML.14.
AdaBoost learner weight	$\alpha_m = \frac{1}{2}\ln\frac{1-\varepsilon_m}{\varepsilon_m}$	Accurate learners (small ε_m) get a bigger vote.
AdaBoost reweighting	$w_i \leftarrow w_i\, e^{-\alpha_m y_i h_m(\mathbf{x}_i)}$	Misclassified points gain weight, so the next learner focuses on them.

STATED-EQUATION *Bagging averages B models: f bar equals one over B times the sum of f hat b, with variance rho sigma squared plus one minus rho over B times sigma squared. A bootstrap sample leaves out about e to the minus one of the rows. Boosting: F m equals F m minus one plus nu h m. AdaBoost weight alpha m equals one half natural log of one minus epsilon m over epsilon m, and each point's weight is multiplied by e to the minus alpha m y i h m of x i.*

MEMORY AID Bagging fights variance; boosting fights bias.

TERMS

- B – number of models, each fit on a bootstrap sample (OOB fraction $\approx e^{-1}$ C1.6 §07 Calculus I)
- ρ – correlation between models S.38 §13 Statistics; random forests lower it by sampling $\approx \sqrt{d}$ features per split
- σ^2 – variance of a single model S.13 §13 Statistics
- h_m – weak learner fit to the current residuals (negative gradient) C1.26 §07 Calculus I
- ν – learning rate (shrinkage)
- α_m, ε_m – AdaBoost weight and weighted error of learner m
- w_i – weight of example i (renormalized to sum 1); $y_i, h_m \in \{-1, +1\}$
- $\hat{f}_b$; F_m; n – model fit on bootstrap sample b; boosted ensemble after m rounds; number of rows

CONTEXT

- **Origin.** Bagging and random forests by Breiman (1996, 2001); AdaBoost by Freund and Schapire (1997); gradient boosting by Friedman (2001).
- **Use.** State-of-the-art accuracy on tabular data.
- **Data science.** XGBoost, LightGBM, and CatBoost dominate tabular ML (churn, fraud, pricing); random forests give out-of-bag validation and feature importances for free.
- **Physical application.** Weather forecasters run ensembles of simulations with perturbed initial conditions and average them.

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ML.14 Gradient Boosting

Build an additive model one small tree at a time; each new tree is fit to the negative gradient of the loss at the current predictions, which for squared error is just the residuals. It is gradient descent in function space.

Additive model	$F_M(\mathbf{x}) = F_0 + \nu \sum_{m=1}^{M} h_m(\mathbf{x})$	Start from a constant, add M shrunken trees.
Initial model	$F_0 = \arg\min_c \sum_i L(y_i, c)$	The mean for MSE, the median for MAE, the log-odds of $\bar{y}$ for log-loss.
Pseudo-residuals	$r_{im} = -\left.\frac{\partial L(y_i, F)}{\partial F}\right\|_{F=F_{m-1}(\mathbf{x}_i)}$	The direction that most reduces each point's loss C1.26 §07 Calculus I.
Squared-error residuals	$L = \frac{1}{2}(y-F)^2 \Rightarrow r_{im} = y_i - F_{m-1}(\mathbf{x}_i)$	For MSE the gradient step is literally "fit the residuals".
Log-loss residuals	$r_{im} = y_i - p_{m-1}(\mathbf{x}_i)$	Label minus current predicted probability.
Second-order leaf weight	$w_j^* = -\frac{\sum_{i\in j} g_i}{\sum_{i\in j} h_i + \lambda}$	XGBoost: Newton step per leaf O.13 §14 Optimization, with gradients g_i, Hessians h_i, L2 penalty λ.

STATED-EQUATION *The boosted model F M equals F naught plus nu times the sum of M trees h m. F naught is the best constant. The pseudo-residual r i m is minus the derivative of the loss with respect to the prediction, evaluated at the previous model. For squared error it is y i minus the previous prediction; for log-loss, y i minus the predicted probability. XGBoost sets each leaf weight to minus the sum of gradients over the sum of Hessians plus lambda.*

MEMORY AID Each tree fixes what the previous trees got wrong.

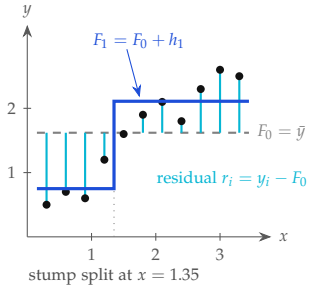
METHOD

1. Set F_0 to the best constant.
2. For $m = 1..M$: compute pseudo-residuals r_{im} at the current predictions.
3. Fit a shallow regression tree h_m to $(\mathbf{x}_i, r_{im})$; set leaf values (line search or Newton step).
4. Update $F_m = F_{m-1} + \nu h_m$; stop when validation loss stops improving. Example: $y = (2, 4, 9)$, MSE: $F_0 = 5$, residuals $(-3, -1, 4)$; a stump predicting -2 for the first two and 4 for the third with $\nu = 0.5$ gives $F_1 = (4, 4, 7)$.

DECISION POINT **Tuning:** smaller ν (0.01–0.1) needs more trees M but generalizes better; pick M by early stopping on a validation set. Tree depth 3–8 controls interaction order. Subsample rows and columns (≈0.8) to add randomness and cut variance.

TERMS

F_m – ensemble after m trees; M total trees
h_m – regression tree fit to pseudo-residuals
ν – learning rate (shrinkage), $0 < \nu \leq 1$
L – differentiable loss ML.2
$p_{m-1} - \sigma(F_{m-1})$, current probability ML.9
g_i, h_i – first and second derivatives of L at $F_{m-1}(\mathbf{x}_i)$
j – a leaf; λ L2 penalty on leaf weights
M; F_0; c – number of boosting rounds; initial constant model; candidate constant
r_{im} – pseudo-residual of point i at round m
w_j^*; λ – optimal weight of leaf j; L2 penalty on leaf weights



stump split at $x = 1.35$

CONTEXT

- **Origin.** Friedman, "Greedy Function Approximation" (2001); XGBoost by Chen and Guestrin (2016).
- **Use.** Best-in-class accuracy on structured/tabular data.
- **Data science.** Default winner for tabular competitions, credit risk, click-through and demand forecasting; native handling of missing values; SHAP values explain its predictions.
- **Physical application.** Iteratively correcting a numerical model with the residuals between measurement and simulation (residual correction) follows the same idea.

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ML.15 Support Vector Machines

Find the separating hyperplane with the widest margin between classes, allowing some violations (soft margin). Only the points on or inside the margin, the support vectors, determine it.

Soft-margin primal	$\min_{\mathbf{w},b,\boldsymbol{\xi}} \frac{1}{2}\|\mathbf{w}\|^2 + C\sum_i \xi_i \quad \text{s.t.} \quad y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1 - \xi_i, \;\; \xi_i \geq 0$	A quadratic program O.5 §14 Optimization: wide margin vs. training errors, traded by C.
Margin width	$\text{margin} = \frac{2}{\|\mathbf{w}\|}$	Distance between the two dashed hyperplanes $\mathbf{w}^\top \mathbf{x} + b = \pm 1$.
Hinge-loss form	$\min \frac{1}{2}\|\mathbf{w}\|^2 + C\sum_i \max\left(0, 1 - y_i f(\mathbf{x}_i)\right)$	Same problem, unconstrained: L2-regularized hinge loss ML.2.
Dual decision function	$f(\mathbf{x}) = \sum_i \alpha_i y_i K(\mathbf{x}_i, \mathbf{x}) + b$	From the Lagrangian dual O.19 §14 Optimization; $\alpha_i > 0$ only for support vectors.
RBF kernel	$K(\mathbf{x}, \mathbf{z}) = e^{-\gamma\|\mathbf{x}-\mathbf{z}\|^2}$	Similarity decaying with distance; gives smooth nonlinear boundaries ML.16.

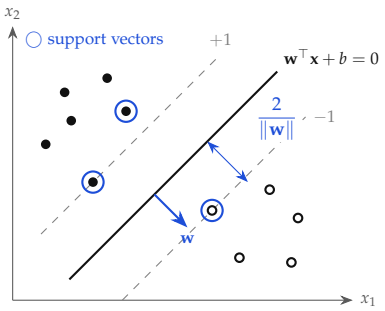
STATED-EQUATION *Minimize one half the squared norm of w plus C times the sum of the slacks xi i, subject to y i times w transpose x i plus b being at least one minus xi i, and xi i at least zero. The margin equals two over the norm of w. Equivalently minimize one half norm w squared plus C times the total hinge loss. The decision function is the sum of alpha i y i K of x i and x, plus b. The RBF kernel K of x, z equals e to the minus gamma times the squared distance between x and z.*

MEMORY AID Maximize the street width; only the points on the curb (support vectors) matter.

DECISION POINT **Choosing C and γ**: grid-search both on a log scale ($C, \gamma \in 10^{-3} \ldots 10^{3}$) with CV. Large $C \to$ few violations, narrow margin, overfit risk; large $\gamma \to$ wiggly boundary. Start with a linear kernel for wide sparse data (text), RBF otherwise; standardize features first.

TERMS

$\mathbf{w}, b$ – normal vector and offset of the hyperplane $\mathbf{w}^\top \mathbf{x} + b = 0$ T.21 §06 Trigonometry
$y_i \in \{-1, +1\}$ – class labels
ξ_i – slack; equivalent to hinge loss ML.2
C – trade-off between wide margin and training errors
α_i – dual variables (Lagrange multipliers)
Support vectors – points on or inside the margin ($\mu_i > 0$ O.18 §14 Optimization)
Kernel K – inner product in a feature space; RBF width $\gamma > 0$
γ; $\mathbf{x}_i$ – RBF width parameter; training points
$f(\mathbf{x})$ – decision score; predict sign f



CONTEXT

- **Origin.** Vapnik and Chervonenkis (1963); soft margin and kernels by Cortes and Vapnik (1995).
- **Use.** Robust classification in high dimensions.
- **Data science.** Text and image classification before deep learning; still strong for small, wide datasets; the kernel trick underlies Gaussian processes.
- **Physical application.** Classifying particle-detector events (signal versus background) was a classic SVM application in physics.

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ML.16 Kernels & the Kernel Trick

A kernel computes an inner product in a (possibly infinite-dimensional) feature space without ever building the features. Any algorithm written only in terms of inner products becomes nonlinear by swapping in a kernel.

Kernel definition	$K(\mathbf{x},\mathbf{z}) = \boldsymbol{\phi}(\mathbf{x})^\top \boldsymbol{\phi}(\mathbf{z})$	An inner product LA.3 §09 Linear Algebra after the feature map $\boldsymbol{\phi}$.
Polynomial kernel	$K(\mathbf{x},\mathbf{z}) = (\mathbf{x}^\top \mathbf{z} + c)^p$	All monomials up to degree p, at the cost of one dot product.
Worked identity	$(\mathbf{x}^\top \mathbf{z})^2 = \boldsymbol{\phi}(\mathbf{x})^\top \boldsymbol{\phi}(\mathbf{z}),\ \boldsymbol{\phi}(\mathbf{x}) = (x_1^2, \sqrt{2}x_1x_2, x_2^2)$	The trick in 2-D: a squared dot product equals a 3-D inner product.
Gram matrix (Mercer)	$K_{ij} = K(\mathbf{x}_i,\mathbf{x}_j), \quad K \succeq 0$	A valid kernel gives a symmetric positive semidefinite Gram matrix LA.27 §09 Linear Algebra.
Representer theorem	$f(\mathbf{x}) = \sum_{i=1}^{n} \alpha_i K(\mathbf{x}_i,\mathbf{x})$	The regularized optimum is a weighted sum of kernels at the training points.
Kernel ridge regression	$\boldsymbol{\alpha} = (K + \lambda I)^{-1}\mathbf{y}$	Ridge ML.8 in feature space; an $n \times n$ solve, so cost grows as n^3.

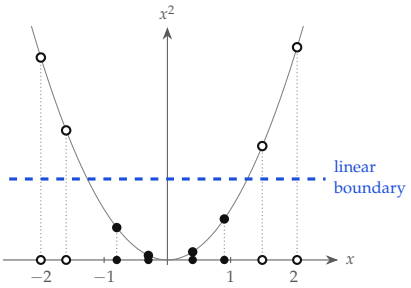
STATED-EQUATION *A kernel K of x and z equals phi of x transpose phi of z. The polynomial kernel is x transpose z plus c, to the p. For example x transpose z squared equals the inner product of the features x one squared, root two x one x two, x two squared. A valid kernel's Gram matrix is positive semidefinite. The representer theorem says f is a weighted sum of kernels at the training points; kernel ridge sets alpha to K plus lambda I inverse times y.*

MEMORY AID Never compute $\boldsymbol{\phi}$; only compute K.

DECISION POINT **Which kernel?** Linear for high-dimensional sparse data; RBF as the general default (tune γ); polynomial when interactions of known order matter; string/graph kernels for structured data. Beyond $n \approx 10^5$ the $n \times n$ Gram matrix is too big: use random features or a neural network.

TERMS

$\boldsymbol{\phi}$ – feature map into a higher-dimensional space
$c \geq 0,\ p$ – polynomial kernel offset and degree
K – kernel function, or the $n \times n$ Gram matrix
$\succeq 0$ – positive semidefinite: $\mathbf{v}^\top K\mathbf{v} \geq 0$ for all $\mathbf{v}$
α_i – weights on training points; λ ridge penalty
RBF – Gaussian kernel ML.15; infinite-dimensional $\boldsymbol{\phi}$
K_{ij}; λ, I – Gram matrix entry; ridge penalty and identity
$\mathbf{x}, \mathbf{z}$; $\mathbf{y}$ – input vectors; training targets



on the x-line no threshold separates them

CONTEXT

- **Origin.** Mercer's theorem (1909); kernel trick in machine learning by Aizerman et al. (1964) and Boser, Guyon and Vapnik (1992).
- **Use.** Nonlinear models with linear-algebra solutions.
- **Data science.** Kernel SVMs, kernel PCA, Gaussian processes (the kernel is the covariance function S.39 §13 Statistics) for Bayesian optimization of hyperparameters.
- **Physical application.** Gaussian-process regression (kriging) interpolates ore grades or temperatures between measurement sites.

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Evaluating Models

ML.17 Classification Metrics

Every binary prediction is a true/false positive/negative. Precision, recall and their relatives summarize the confusion matrix from different points of view.

Precision	$\text{Precision} = \dfrac{TP}{TP+FP}$	Of those flagged positive, the fraction that truly are.
Recall (TPR, sensitivity)	$\text{Recall} = \dfrac{TP}{TP+FN}$	Of the true positives, the fraction caught.
F_1 **score**	$F_1 = \dfrac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$	Harmonic mean: high only when both are high.
Accuracy	$\text{Accuracy} = \dfrac{TP+TN}{N}$	Fraction correct; misleading when one class dominates.
False positive rate	$\text{FPR} = \dfrac{FP}{FP+TN} = 1 - \text{Specificity}$	Fraction of negatives wrongly flagged; the ROC x-axis ML.18.
F_β **score**	$F_\beta = \dfrac{(1+\beta^2)\,PR}{\beta^2 P + R}$	$\beta > 1$ weights recall more, $\beta < 1$ weights precision more.

STATED-EQUATION *Precision equals T P over T P plus F P. Recall, the true positive rate, equals T P over T P plus F N. F one equals two times precision times recall over precision plus recall. Accuracy equals T P plus T N over N. The false positive rate equals F P over F P plus T N, one minus specificity. F beta weights recall beta squared times as much as precision.*

MEMORY AID Precision: "of those I flagged, how many were right?" Recall: "of the real ones, how many did I catch?"

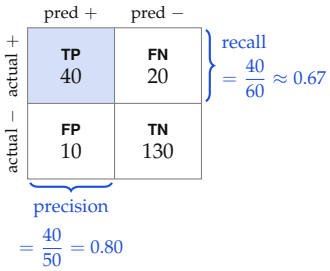
METHOD

1. Count TP, FP, FN, TN at the chosen threshold.
2. Divide by the relevant column (precision) or row (recall). Example (figure): $TP = 40, FN = 20, FP = 10, TN = 130$: precision $= \frac{40}{50} = 0.8$, recall $= \frac{40}{60} \approx 0.67$, $F_1 \approx 0.73$, accuracy $= \frac{170}{200} = 0.85$.

DECISION POINT **Which metric?** False alarms expensive (spam filter, ad targeting) → precision; misses expensive (cancer screening, fraud) → recall; need one number → F_1 or F_β; balanced classes and equal costs → accuracy. Multi-class: macro-average to weight classes equally, micro-average to weight rows equally. See ML.K1.

TERMS

TP, FP – true and false positives (FP = Type I error S.28 §13 Statistics)
TN, FN – true and false negatives (FN = Type II error)
N – total count; P, R precision and recall
Precision – $P(\text{actual+} \mid \text{predicted+})$ S.9 §13 Statistics
Recall – $P(\text{predicted+} \mid \text{actual+})$; sensitivity
Specificity – $\frac{TN}{TN+FP}$, true negative rate
F_1 – harmonic mean of precision and recall
ROC AUC – area under TPR vs FPR as the threshold moves G.15 §05 Geometry, C2.25 §08 Calculus II
β; P, R – weight on recall; precision and recall



CONTEXT

- **Origin.** Precision and recall from information retrieval (Kent et al., 1955); ROC from WWII radar.
- **Use.** Evaluating any classifier or diagnostic test.
- **Data science.** With rare positives (fraud, 1%), accuracy is misleading; choose thresholds by the cost of FP vs FN; report PR curves for imbalanced data.
- **Physical application.** An earthquake early-warning system trades recall (catching quakes) against false alarms.

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ML.18 ROC, AUC & Precision–Recall Curves

Sweep the decision threshold over all values and plot the resulting trade-off. The area under the curve summarizes ranking quality independent of any one threshold.

ROC curve	$\big(\mathrm{FPR}(\tau),\ \mathrm{TPR}(\tau)\big),\ \tau \in \mathbb{R}$	A parametric curve C2.25 §08 Calculus II from $(0,0)$ to $(1,1)$.
AUC	$\mathrm{AUC} = \int_0^1 \mathrm{TPR}\, d(\mathrm{FPR}) = P\big(s(\mathbf{x}_+) > s(\mathbf{x}_-)\big)$	Probability a random positive outranks a random negative.
PR curve	$\big(\mathrm{Recall}(\tau),\ \mathrm{Precision}(\tau)\big)$	Focuses on the positive class; sensitive to imbalance.
PR baseline	$\mathrm{precision}_{\mathrm{random}} = \pi_+$	A random classifier's precision is the prevalence; judge PR AUC against it.
Youden's J	$J = \max_\tau \big(\mathrm{TPR} - \mathrm{FPR}\big)$	Threshold farthest above the diagonal when costs are equal.

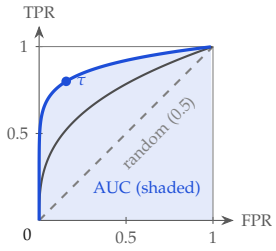
STATED-EQUATION *The ROC curve plots the false positive rate against the true positive rate as the threshold tau varies. The AUC, the area under it, equals the probability that a random positive gets a higher score than a random negative. The precision-recall curve plots precision against recall. The precision-recall baseline equals the prevalence of positives, pi plus. Youden's J picks the threshold maximizing TPR minus FPR.*

MEMORY AID Lower the threshold ⇒ flag more ⇒ recall up, precision (usually) down. Raise it ⇒ the reverse.

DECISION POINT **Choosing the threshold**: minimize expected cost $c_{FP} \cdot FP + c_{FN} \cdot FN$ on validation data; or fix the constraint that matters ("recall $\geq$ 95%") and maximize the other. Use Youden's J only when costs are equal. **ROC or PR?** PR when positives are rare.

TERMS

τ – decision threshold on the score $s(\mathbf{x})$
$\mathbf{x}_+, \mathbf{x}_-$ – a random positive and a random negative example
TPR, FPR – ML.17
AUC – threshold-free ranking quality; area C1.30 §07 Calculus I, parametric curve C2.25 §08 Calculus II
Diagonal – random guessing, AUC $= 0.5$
π_+ – fraction of positives; a random classifier's precision
Imbalanced data – prefer PR curves: ROC can look good while precision is poor
$s(\mathbf{x})$; J – classifier score; Youden's statistic



CONTEXT

- **Origin.** ROC analysis from WWII radar operators; adopted in medicine (1970s) and machine learning (1990s).
- **Use.** Comparing classifiers and choosing operating thresholds.
- **Data science.** Pick the threshold by business cost: fraud screening favors high recall, marketing lists favor high precision; report PR-AUC when positives are rare.
- **Physical application.** Setting the sensitivity of a smoke detector: more sensitive catches every fire but triggers on burnt toast.

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ML.19 Regression Metrics

Summaries of the residuals $y - \hat{y}$: in the units of y (RMSE, MAE), as percentages (MAPE), or as a share of variance explained (R^2).

RMSE	$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_i (y_i - \hat{y}_i)^2}$	Units of y; dominated by the largest errors.
MAE	$\mathrm{MAE} = \frac{1}{n}\sum_i \lvert y_i - \hat{y}_i \rvert$	Units of y; the typical miss; RMSE $\geq$ MAE always.
MAPE	$\mathrm{MAPE} = \frac{100\%}{n}\sum_i \left\lvert \frac{y_i - \hat{y}_i}{y_i} \right\rvert$	Scale-free percentage; undefined at $y = 0$, penalizes over-forecasts more.
Coefficient of determination	$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$	Share of variance explained vs. predicting the mean S.40 §13 Statistics; can be negative on test data.
Adjusted R^2	$R^2_{\mathrm{adj}} = 1 - (1 - R^2)\frac{n-1}{n-p-1}$	Penalizes predictors that do not pull their weight S.41 §13 Statistics.

STATED-EQUATION *RMSE is the square root of the mean squared error. MAE is the mean absolute error. MAPE is one hundred percent over n times the sum of the absolute errors divided by the actual values. R squared equals one minus the residual sum of squares over the total sum of squares. Adjusted R squared equals one minus the quantity one minus R squared, times n minus one over n minus p minus one.*

MEMORY AID RMSE punishes big misses; MAE treats all misses equally; both are in the units of y.

TERMS

$y, \hat{y}$ – actual and predicted values; $\bar{y}$ mean of y
RMSE – same units as y; RMSE $\geq$ MAE always
MAPE – scale-free percentage; undefined when $y = 0$
R^2 – share of variance explained S.40 §13 Statistics
p – number of predictors; adjusted R^2 penalizes adding useless ones
Losses – ML.2
n; $\bar{y}$ – number of predictions; mean of actual values

CONTEXT

- **Origin.** Least squares (Gauss); adjusted R^2 by Ezekiel (1930).
- **Use.** Evaluating forecasts and continuous predictions.
- **Data science.** Report RMSE or MAE in business units ("off by $12 on average"); compare to a naive baseline such as predicting the mean or last value.
- **Physical application.** Evaluating a weather model's temperature forecasts: an MAE of 1.5 °C is useful, 8 °C is not.

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ML.20 Reading Metrics: Bad, Good & Perfect

A metric only means something next to its baseline (what a trivial model scores) and its ceiling. The table gives the range, the "no skill" value and rough targets for common metrics.

Matthews correlation coefficient

$$\mathrm{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}}$$

Correlation between predicted and actual labels; robust to imbalance.

Brier score

$$\mathrm{Brier} = \frac{1}{n}\sum_i (p_i - y_i)^2$$

MSE of predicted probabilities; rewards calibration ML.21.

Baseline log-loss (balanced)

$$-\ln \tfrac{1}{2} = \ln 2 \approx 0.693$$

Score of always predicting $p = 0.5$; a useful model must beat it.

STATED-EQUATION *The Matthews correlation coefficient equals T P times T N minus F P times F N, over the square root of the product of the four sums. The Brier score is the mean squared difference between predicted probability and outcome. A balanced binary classifier that always predicts one half has log-loss natural log of two, about zero point six nine three.*

MEMORY AID A metric means nothing without its baseline: always ask "compared to what?"

Metric	Range	Bad / baseline	Good (rule of thumb)	Perfect
Accuracy	0 to 1	majority-class rate	well above the baseline	1
Precision	0 to 1	prevalence (random)	high when false alarms are costly	1 (no false alarms)
Recall	0 to 1	fraction flagged at random	high when misses are costly	1 (no misses)
F_1	0 to 1	near 0	> 0.7 common target	1
ROC AUC	0 to 1	0.5 (random); < 0.5 inverted	0.7–0.8 fair, 0.8–0.9 good, > 0.9 excellent	1
PR AUC	0 to 1	prevalence π_+	many times π_+	1
Log-loss	0 to ∞	0.693 (balanced, $p = 0.5$)	well below baseline	0
Brier	0 to 1	$\pi_+(1-\pi_+)$; 0.25 balanced	well below baseline	0
MCC	-1 to 1	0 (random); -1 inverted	> 0.5 strong	1
R^2	$-\infty$ to 1	0 (predict the mean); < 0 worse	0.5–0.9 depends on field	1
RMSE, MAE	0 to ∞	SD of y (mean baseline)	small vs. SD and business tolerance	0
Silhouette	-1 to 1	≤ 0 overlapping/misassigned	> 0.5 reasonable structure	1

TERMS

Baseline – simplest possible model (majority class, mean, last value)
MCC – balanced summary of the confusion matrix, robust to imbalance ML.17
p_i, y_i – predicted probability and 0/1 outcome
Brier – MSE of probabilities; rewards calibration ML.2
Calibration – predicted 70% should happen about 70% of the time
Context – "good" depends on the field: $R^2 = 0.3$ can be excellent in social science, poor in physics
n – number of predictions

CONTEXT

- **Origin.** MCC by Brian Matthews (1975); Brier score by Glenn Brier for weather forecasts (1950).
- **Use.** Judging whether a model is useful, not just whether a number is high.
- **Data science.** A 95%-accurate fraud model is worthless if 95% of transactions are legitimate; always report metrics next to the baseline and with confidence intervals S.27 §13 Statistics.
- **Physical application.** Weather services score rain forecasts with the Brier score against climatology (the historical rain rate) as the baseline.

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ML.21 Probability Calibration

A model is calibrated if, among cases where it predicts probability p, the event happens a fraction p of the time. Good ranking (AUC) does not imply good calibration; post-hoc maps fix it.

Perfect calibration	$P(y = 1 \mid \hat{p} = p) = p \quad \forall p$	Predicted confidence equals observed frequency.
Expected calibration error	$\mathrm{ECE} = \sum_{b=1}^{B} \frac{n_b}{n} \lvert \bar{y}_b - \bar{p}_b \rvert$	Bin predictions; average the gap between accuracy and confidence.
Platt scaling	$\hat{p} = \sigma(a\,s + b)$	Fit a 1-feature logistic regression ML.9 on the score s using held-out data.
Temperature scaling	$\hat{\mathbf{p}} = \mathrm{softmax}(\mathbf{z}/T)$	One parameter $T > 1$ softens an overconfident network ML.24; accuracy unchanged.
Isotonic regression	$\hat{p} = g(s),\ g$ non-decreasing	Flexible monotone step map; needs more calibration data.

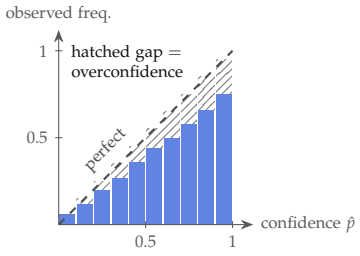
STATED-EQUATION *Perfect calibration: the probability y equals one given predicted p equals p, for all p. Expected calibration error is the weighted average over bins of the gap between observed frequency and mean confidence. Platt scaling passes the score through a fitted sigmoid a s plus b. Temperature scaling divides logits by T before the softmax. Isotonic regression fits a non-decreasing map from score to probability.*

MEMORY AID "70% should mean 70%."

DECISION POINT **Which calibrator?** Few calibration rows or sigmoid-shaped distortion (SVMs, boosted trees) → Platt; neural nets → temperature scaling; thousands of rows and odd distortion → isotonic. Always fit on data not used for training (a validation split or CV).

TERMS

$\hat{p}$ – predicted probability; $y \in \{0, 1\}$ outcome
B, n_b – number of bins and rows in bin b
$\bar{y}_b, \bar{p}_b$ – observed frequency and mean confidence in bin b
s – raw model score; a, b fitted constants
$\mathbf{z}, T$ – logits and temperature
Reliability diagram – plot of $\bar{y}_b$ vs. $\bar{p}_b$ (figure)
Brier score – combines calibration and sharpness ML.20
a, b; g – fitted Platt parameters; fitted monotone map
softmax; σ – normalized exponential ML.24; sigmoid



CONTEXT

- **Origin.** Brier (1950) and Murphy's decomposition in weather forecasting; Platt (1999); temperature scaling by Guo et al. (2017).
- **Use.** Making scores usable as probabilities.
- **Data science.** Expected-value decisions (pricing, risk, triage) need calibrated probabilities; deep nets are typically overconfident until temperature-scaled.
- **Physical application.** A 30% rain forecast should verify on about 30% of such days; forecasters check this with reliability diagrams.

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Neural Networks & Training

ML.22 Neural Networks & Backpropagation

A neural network stacks layers of (linear map → nonlinearity). Backpropagation applies the chain rule layer by layer, from the loss back to the input, to get every weight's gradient in one backward pass.

Layer (forward pass)	$\mathbf{z}^{(l)} = W^{(l)}\mathbf{a}^{(l-1)} + \mathbf{b}^{(l)}, \;\; \mathbf{a}^{(l)} = \phi(\mathbf{z}^{(l)})$	Affine map then elementwise activation; $\mathbf{a}^{(0)} = \mathbf{x}$.
ReLU	$\mathrm{ReLU}(z) = \max(0, z)$	The default hidden activation ML.23.
Sigmoid derivative	$\sigma'(z) = \sigma(z)\big(1 - \sigma(z)\big)$	At most $\frac{1}{4}$, which is why deep sigmoid nets suffer vanishing gradients.
Output error	$\delta^{(L)} = \nabla_{\mathbf{a}^{(L)}} L \odot \phi'(\mathbf{z}^{(L)})$	Starts the backward pass; equals $\hat{\mathbf{p}} - \mathbf{y}$ for softmax + cross-entropy ML.24.
Backpropagation	$\delta^{(l)} = \left(W^{(l+1)\top}\delta^{(l+1)}\right) \odot \phi'(\mathbf{z}^{(l)})$	Send the error back through the transposed weights, gated by the activation slope C1.13 §07 Calculus I.
Weight & bias gradients	$\dfrac{\partial L}{\partial W^{(l)}} = \delta^{(l)}\,\mathbf{a}^{(l-1)\top}, \;\; \dfrac{\partial L}{\partial \mathbf{b}^{(l)}} = \delta^{(l)}$	Outer product of the error and the layer's input LA.12 §09 Linear Algebra.

STATED-EQUATION *z l equals W l times a l minus one plus b l, and a l equals ϕ of z l. ReLU of z equals the maximum of zero and z. Sigma prime of z equals sigma of z times one minus sigma of z. The output error delta L is the gradient of the loss with respect to the output, elementwise times phi prime. Backpropagation: delta l equals W l plus one transpose times delta l plus one, elementwise times phi prime of z l. The gradient with respect to W l equals delta l times a l minus one transpose; the bias gradient is delta l.*

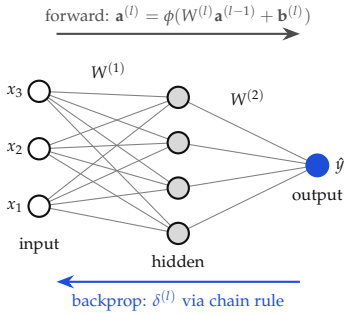
MEMORY AID Forward pass computes; backward pass blames (chain rule, layer by layer).

METHOD

1. Forward: compute and store every $\mathbf{z}^{(l)}, \mathbf{a}^{(l)}$ and the loss.
2. Output: compute $\delta^{(L)}$.
3. Backward: for $l = L-1, \ldots, 1$ compute $\delta^{(l)}$ from $\delta^{(l+1)}$.
4. Gradients: $\partial L/\partial W^{(l)} = \delta^{(l)}\mathbf{a}^{(l-1)\top}$; take an optimizer step ML.27. Example: one neuron $\hat{y} = wx$, $L = \frac{1}{2}(\hat{y} - y)^2$, $x = 2$, $w = 1$, $y = 5$: $\delta = \hat{y} - y = -3$, $\partial L/\partial w = \delta x = -6$, so w increases.

TERMS

$W^{(l)}, \mathbf{b}^{(l)}$ – weights and biases of layer l (matrix product LA.10 §09 Linear Algebra)
ϕ – activation (ReLU, sigmoid ML.9, tanh)
$\mathbf{a}^{(0)}$ – input $\mathbf{x}$; superscript (L) the output layer, L the loss
$\delta^{(l)}$ – error signal $\partial L/\partial \mathbf{z}^{(l)}$, passed backward by the chain rule C1.13 §07 Calculus I
$\odot$ – elementwise (Hadamard) product
Autodiff – frameworks build this backward pass automatically (reverse mode) C3.15 §10 Calculus III
Universal approximation – one wide hidden layer can approximate any continuous function
$\mathbf{z}^{(l)}$, $\mathbf{a}^{(l)}$; L – pre-activation and activation of layer l; loss (as a superscript, the last layer)
σ; ReLU – sigmoid ML.23; rectified linear unit



CONTEXT

- **Origin.** Perceptron by Rosenblatt (1958); backpropagation popularized by Rumelhart, Hinton and Williams (1986).
- **Use.** Learning features automatically from raw data.
- **Data science.** Deep learning for images (CNNs), sequences and language (transformers), and tabular embeddings; trained with SGD/Adam ML.27.
- **Physical application.** Neural networks now emulate expensive physics simulations (weather, fluid flow) thousands of times faster.

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ML.23 Activation Functions & Their Derivatives

The nonlinearity applied after each linear layer. Without it a deep network collapses to one linear map; its derivative controls how well gradients flow backward.

Sigmoid	$\sigma(z) = \dfrac{1}{1+e^{-z}}, \quad \sigma' = \sigma(1-\sigma)$	Output $(0,1)$; for binary outputs and gates; saturates, so gradients vanish.
Hyperbolic tangent	$\tanh z = \dfrac{e^{z}-e^{-z}}{e^{z}+e^{-z}}, \quad \tanh' = 1-\tanh^2$	Zero-centered, output $(-1,1)$; $\tanh z = 2\sigma(2z) - 1$ C1.16 §07 Calculus I.
ReLU	$\max(0,z), \quad \mathrm{ReLU}' = \begin{cases} 1, & z>0 \\ 0, & z<0 \end{cases}$	Cheap, no saturation for $z>0$; units stuck at $z<0$ can "die".
Leaky ReLU	$\max(\alpha z, z), \quad \alpha \approx 0.01$	Small negative slope keeps gradients alive.
GELU	$\mathrm{GELU}(z) = z\,\Phi(z) \approx \frac{z}{2}\left[1+\tanh\left(\sqrt{\frac{2}{\pi}}\left(z+0.0447z^3\right)\right)\right]$	Smooth ReLU weighted by the normal CDF S.20 §13 Statistics; default in transformers.
Softmax & Jacobian	$s_k = \dfrac{e^{z_k}}{\sum_j e^{z_j}}, \quad \dfrac{\partial s_k}{\partial z_j} = s_k(\delta_{kj} - s_j)$	Turns K scores into probabilities; output layer for multi-class ML.24.

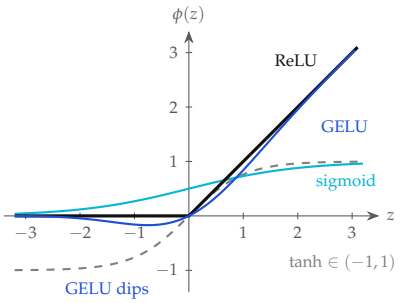
STATED-EQUATION *The sigmoid is one over one plus e to the minus z, with derivative sigma times one minus sigma. Tanh has derivative one minus tanh squared. ReLU is the max of zero and z, with derivative one for positive z and zero for negative z. Leaky ReLU is the max of alpha z and z. GELU is z times the standard normal CDF of z. Softmax s k is e to the z k over the sum of e to the z j, with derivative s k times delta k j minus s j. Tanh of z is e to the z minus e to the minus z, over e to the z plus e to the minus z.*

MEMORY AID Hidden layers: ReLU/GELU. Output: identity (regression), sigmoid (binary), softmax (multi-class).

DECISION POINT **Which activation?** Hidden layers of CNNs/MLPs: ReLU (or leaky ReLU if many units die); transformers: GELU (or SiLU $z\sigma(z)$); RNN states: tanh; gates: sigmoid. The output activation is set by the task and must match the loss: sigmoid + log-loss, softmax + cross-entropy, identity + MSE.

TERMS

z – pre-activation (input to ϕ); $\mathbf{z}$ vector of logits
Saturation – flat regions where $\phi' \approx 0$ stall learning
α – leaky slope for $z<0$
Φ – standard normal CDF S.20 §13 Statistics
δ_{kj} – Kronecker delta: 1 if $k=j$, else 0
K – number of classes; Jacobian C3.15 §10 Calculus III
σ; tanh – sigmoid; hyperbolic tangent C1.16 §07 Calculus I
s_k; z_k – softmax output k; logit k



CONTEXT

- **Origin.** Sigmoid units from early neural models (1940s–80s); ReLU popularized by Nair and Hinton (2010) and AlexNet (2012); GELU by Hendrycks and Gimpel (2016).
- **Use.** Giving networks nonlinear expressive power.
- **Data science.** Switching sigmoid → ReLU made deep nets trainable; GELU/SiLU power modern language models.
- **Physical application.** The sigmoid has the same shape as a neuron's firing-rate response and the Fermi–Dirac distribution.

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ML.24 Softmax & Cross-Entropy Loss

The standard multi-class output: softmax turns logits into probabilities and cross-entropy scores them against the true class. Together their gradient is simply prediction minus truth.

Softmax	$\hat{p}_k = \text{softmax}(\mathbf{z})_k = \frac{e^{z_k}}{\sum_{j=1}^{K} e^{z_j}}$	Positive outputs summing to 1; larger logits get exponentially more mass.
Cross-entropy loss	$L = -\sum_{k=1}^{K} y_k \ln \hat{p}_k = -\ln \hat{p}_{\text{true}}$	With one-hot $\mathbf{y}$, only the true class's log-probability counts ML.3.
Gradient	$\frac{\partial L}{\partial \mathbf{z}} = \hat{\mathbf{p}} - \mathbf{y}$	The softmax Jacobian and the log cancel neatly.
Log-sum-exp trick	$\ln \sum_j e^{z_j} = m + \ln \sum_j e^{z_j - m},\ m = \max_j z_j$	Subtracting the max prevents overflow without changing the result A2.13 §04 Algebra II.
Temperature	$\hat{p}_k \propto e^{z_k/T}$	$T < 1$ sharpens, $T > 1$ flattens the distribution.

STATED-EQUATION *p hat k equals softmax of z at k, which is e to the z k over the sum over j of e to the z j. The loss L equals minus the sum of y k natural log p hat k, which is minus the log-probability of the true class. The gradient with respect to z equals p hat minus y. Subtract the maximum logit before exponentiating. Temperature divides the logits by T.*

MEMORY AID Gradient is "prediction minus truth."

TERMS

$\mathbf{z}$ – logits (unnormalized scores) from the last layer
$\hat{p}_k$ – predicted probabilities; positive and sum to 1 S.7 §13 Statistics
$\mathbf{y}$ – one-hot true label; K classes
L – cross-entropy ML.3; with $K = 2$ it is log-loss ML.2
Stability – subtract $\max_j z_j$ before exponentiating A2.13 §04 Algebra II
Temperature T – $e^{z_k/T}$ sharpens ($T < 1$) or flattens ($T > 1$)
K; m; y_k – number of classes; largest logit; one-hot target entry

CONTEXT

- **Origin.** Boltzmann distribution in statistical physics; used in multinomial logit models (Luce, 1959).
- **Use.** Multi-class classification.
- **Data science.** The output layer of image and text classifiers and language models; temperature controls randomness when an LLM samples the next token.
- **Physical application.** Softmax is the Boltzmann distribution: states with energy E_k occur with probability $\propto e^{-E_k/k_B T}$.

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ML.25 Weight Initialization

Start weights random (to break symmetry) with a variance chosen so that activations and gradients neither explode nor vanish as they pass through many layers.

Variance propagation	$\mathrm{Var}(z) = n_{\text{in}}\,\mathrm{Var}(w)\,\mathrm{Var}(x)$	Each layer multiplies the signal variance by $n_{\text{in}}\mathrm{Var}(w)$ S.13 §13 Statistics; keep this ≈ 1.
Xavier / Glorot	$\mathrm{Var}(w) = \dfrac{2}{n_{\text{in}} + n_{\text{out}}}$	Balances forward and backward signals; for tanh and sigmoid.
He (Kaiming)	$\mathrm{Var}(w) = \dfrac{2}{n_{\text{in}}}$	Factor 2 compensates ReLU zeroing half its inputs.
Uniform version	$w \sim U\big[-a, a\big],\ \ a = \sqrt{3\,\mathrm{Var}(w)}$	A uniform on $[-a, a]$ has variance $a^2/3$ S.19 §13 Statistics; Xavier gives $a = \sqrt{6/(n_{\text{in}} + n_{\text{out}})}$.
Biases	$\mathbf{b} = \mathbf{0}$	Fine because random weights already break symmetry.

STATED-EQUATION *The variance of a pre-activation equals n in times the weight variance times the input variance. Xavier initialization sets the weight variance to two over n in plus n out; He initialization sets it to two over n in. A uniform initialization uses bound a equal to root three times the variance. Biases start at zero.*

MEMORY AID ReLU $\to$ He ($2/n_{\text{in}}$); tanh $\to$ Xavier ($2/(n_{\text{in}} + n_{\text{out}})$).

TERMS

$n_{\text{in}}, n_{\text{out}}$ – fan-in and fan-out: inputs to and outputs from a unit
w, x, z – a weight, an input, a pre-activation (independent, mean 0)
$U[-a, a]$ – uniform distribution
Symmetry – identical initial weights get identical gradients forever
Exploding/vanishing – signal scale grows/shrinks geometrically with depth A2.19 §04 Algebra II
$\mathrm{Var}(\cdot)$; $\mathbf{b}$ – variance S.13 §13 Statistics; bias vector

CONTEXT

- **Origin.** Glorot and Bengio (2010); He et al. (2015).
- **Use.** Making deep networks trainable from the first step.
- **Data science.** Framework defaults (PyTorch, Keras) implement these; residual connections and normalization ML.26 further relax the requirement.
- **Physical application.** Like setting amplifier gains in a long chain of repeaters so the signal neither clips nor fades.

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ML.26 Dropout, Batch & Layer Normalization

Dropout randomly silences units during training, a cheap regularizer. Normalization layers re-center and re-scale activations so training is faster and more stable.

Dropout mask	$m_j \sim \text{Bernoulli}(1-p)$	Each unit is kept with probability $1-p$ at training time S.15 §13 Statistics.
Inverted dropout	$\tilde{\mathbf{h}} = \dfrac{\mathbf{m} \odot \mathbf{h}}{1-p}$	Rescale during training so the expected activation is unchanged; do nothing at test time.
Batch statistics	$\mu_B = \dfrac{1}{m}\sum_{i=1}^{m} x_i, \;\; \sigma_B^2 = \dfrac{1}{m}\sum_{i=1}^{m}(x_i-\mu_B)^2$	Mean and variance of each feature over the mini-batch.
Batch norm	$\hat{x}_i = \dfrac{x_i-\mu_B}{\sqrt{\sigma_B^2+\epsilon}}, \;\; y_i = \gamma\hat{x}_i + \beta$	Standardize S.4 §13 Statistics, then learn a scale and shift; uses running averages at test time.
Layer norm	$\hat{\mathbf{x}} = \dfrac{\mathbf{x}-\mu}{\sqrt{\sigma^2+\epsilon}}, \;\; \mu, \sigma^2$ over features	Per example, across its features: batch-size independent; used in transformers.
RMSNorm	$\hat{\mathbf{x}} = \dfrac{\mathbf{x}}{\sqrt{\frac{1}{d}\sum_j x_j^2+\epsilon}} \odot \gamma$	Layer norm without mean-centering; cheaper, common in LLMs.

STATED-EQUATION *Dropout keeps each unit with probability one minus p and divides the kept activations by one minus p. Batch norm subtracts the mini-batch mean, divides by the square root of the batch variance plus epsilon, then multiplies by gamma and adds beta. Layer norm does the same across the features of one example. RMSNorm divides by the root-mean-square of the features.*

MEMORY AID Batch norm: across the batch; layer norm: across the features.

DECISION POINT **Batch norm or layer norm?** CNNs with reasonable batch sizes → batch norm; transformers, RNNs, small or variable batches → layer norm / RMSNorm. **Dropout rate?** 0.1 for large pretrained transformers, 0.2–0.5 for dense layers of small nets; skip it when data are plentiful.

TERMS

p – drop probability; $\mathbf{m}$ binary mask
$\mathbf{h}$ – a layer's activations
m – mini-batch size; x_i one feature for example i
ϵ – small constant (e.g. 10^{-5}) avoiding division by 0
γ, β – learned scale and shift (γ vector in RMSNorm)
d – number of features
Ensemble view – dropout trains many thinned networks that share weights ML.13
$\mathbf{m}$, m_j – dropout mask: 1 keeps the unit, 0 drops it
μ_B, σ_B^2; $\hat{x}_i$; y_i – mini-batch mean and variance; normalized value; layer output

CONTEXT

- **Origin.** Dropout by Srivastava, Hinton et al. (2014); batch norm by Ioffe and Szegedy (2015); layer norm by Ba, Kiros and Hinton (2016); RMSNorm by Zhang and Sennrich (2019).
- **Use.** Regularizing and stabilizing deep networks.
- **Data science.** Standard ingredients of ResNets (batch norm) and transformers (layer norm) ML.34; forgetting `model.eval()` leaves dropout and batch statistics in training mode.
- **Physical application.** Automatic gain control in radios normalizes signal level the way normalization layers rescale activations.

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ML.27 SGD, Momentum & Adam

Iterative optimizers that step against noisy mini-batch gradients. Momentum smooths the direction; Adam also adapts each parameter's step size.

SGD	$\theta \leftarrow \theta - \eta \nabla_\theta L_{\text{batch}}$	Unbiased but noisy gradient from a random mini-batch O.16 §14 Optimization.
Momentum	$\mathbf{v} \leftarrow \beta\mathbf{v} + \nabla L, \;\; \theta \leftarrow \theta - \eta\mathbf{v}$	Velocity accumulates consistent directions and damps oscillation.
Adam moments	$\mathbf{m} \leftarrow \beta_1\mathbf{m} + (1-\beta_1)\mathbf{g}, \;\; \mathbf{v} \leftarrow \beta_2\mathbf{v} + (1-\beta_2)\mathbf{g}^2$	Running means of the gradient and its elementwise square.
Adam bias correction	$\hat{\mathbf{m}} = \dfrac{\mathbf{m}}{1-\beta_1^t}, \;\; \hat{\mathbf{v}} = \dfrac{\mathbf{v}}{1-\beta_2^t}$	Undo the pull toward 0 from starting the averages at 0.
Adam update	$\theta \leftarrow \theta - \eta \dfrac{\hat{\mathbf{m}}}{\sqrt{\hat{\mathbf{v}}} + \epsilon}$	Per-parameter step $\approx \eta$ regardless of gradient scale.
AdamW (decoupled decay)	$\theta \leftarrow \theta - \eta\left(\dfrac{\hat{\mathbf{m}}}{\sqrt{\hat{\mathbf{v}}} + \epsilon} + \lambda\theta\right)$	Weight decay applied directly, not through $\mathbf{g}$; the modern default.

STATED-EQUATION *SGD: theta becomes theta minus eta times the gradient of the mini-batch loss. Momentum: v becomes beta v plus the gradient, and theta becomes theta minus eta v. Adam: m becomes beta one m plus one minus beta one times g; v becomes beta two v plus one minus beta two times g squared; m hat equals m over one minus beta one to the t; v hat equals v over one minus beta two to the t; theta becomes theta minus eta m hat over the square root of v hat plus epsilon. AdamW adds lambda theta to the step.*

MEMORY AID Adam = momentum (m) + per-parameter step sizes (v).

DECISION POINT **Which optimizer?** AdamW ($\eta \approx 10^{-3}$ to 10^{-4}) for transformers and most new problems; SGD with momentum 0.9 often generalizes slightly better for CNNs given a tuned schedule ML.28. **Batch size**: as large as memory allows, scaling η roughly linearly with it.

TERMS

η – learning rate C1.26 §07 Calculus I
θ – all model parameters; $\mathbf{g} = \nabla_\theta L_{\text{batch}}$
Mini-batch – random subset of rows; gradient is an unbiased Monte Carlo estimate O.27 §14 Optimization
β, β_1 – momentum decay (≈ 0.9); $\beta_2 \approx 0.999$
$\mathbf{m}, \mathbf{v}$ – running means of the gradient and its square (exponential moving averages A2.19 §04 Algebra II)
Bias correction – divides by $1-\beta^t$ since averages start at 0; t step count
ϵ, λ – stabilizer (10^{-8}) and weight-decay rate
Epoch – one pass through the data: $\lceil n/\text{batch} \rceil$ steps
$\mathbf{g}$; t – mini-batch gradient; step number
$\hat{\mathbf{m}}, \hat{\mathbf{v}}$; β_2 – bias-corrected moments; decay rate of the second moment

CONTEXT

- **Origin.** Robbins–Monro (1951); momentum by Polyak (1964); Adam by Kingma and Ba (2014); AdamW by Loshchilov and Hutter (2019).
- **Use.** Optimizing huge models where exact gradients are too expensive.
- **Data science.** Default optimizers for deep learning (Adam, AdamW); step-size schedules satisfy convergence conditions C2.16 §08 Calculus II.
- **Physical application.** A heavy ball rolling down a bumpy hill (momentum) coasts through small dips that would trap a massless particle.

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ML.28 Training Loop, Learning-Rate Schedules & Gradient Clipping

The learning rate is the most important hyperparameter, and it usually changes during training: warm up, then decay. Clipping caps rare huge gradients; early stopping ends training when validation loss stops improving.

Step decay	$\eta_t = \eta_0 \, \gamma^{\lfloor t/s \rfloor}$	Multiply by γ (e.g. 0.1) every s epochs.
Exponential decay	$\eta_t = \eta_0 \, e^{-kt}$	Smooth geometric decrease A2.13 §04 Algebra II.
Linear warmup	$\eta_t = \eta_{\max} \dfrac{t}{T_w}, \quad t \le T_w$	Ramp up from 0 so early, poorly-scaled Adam statistics do not blow up.
Cosine annealing	$\eta_t = \eta_{\min} + \frac{1}{2}(\eta_{\max} - \eta_{\min})\left(1 + \cos \dfrac{\pi t}{T}\right)$	Slow start, fast middle, gentle landing; the common default after warmup.
Gradient clipping (norm)	$\mathbf{g} \leftarrow \mathbf{g} \cdot \min\left(1, \dfrac{c}{\lVert \mathbf{g} \rVert}\right)$	Rescales only when $\lVert \mathbf{g} \rVert > c$, keeping the direction; essential for RNNs ML.30.
Early stopping	$t^* = \arg\min_t \; L_{\text{val}}(t)$	Keep the checkpoint with best validation loss; stop after a patience of P epochs.

STATED-EQUATION *Step decay multiplies the learning rate by gamma every s epochs. Exponential decay uses eta naught e to the minus k t. Linear warmup grows the rate in proportion to t up to T w. Cosine annealing goes from eta max to eta min along half a cosine. Gradient clipping rescales g by the minimum of one and c over the norm of g. Early stopping keeps the step with the lowest validation loss.*

MEMORY AID Warm up, cruise, cool down.

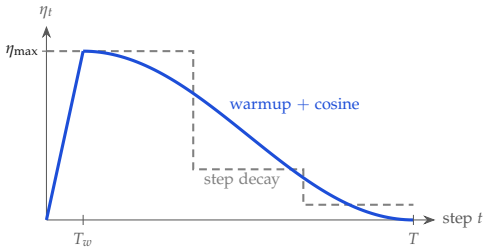
METHOD

1. For each epoch, shuffle the training data into mini-batches.
2. For each batch: forward pass, compute loss, backward pass ML.22.
3. Clip the gradient norm, take an optimizer step ML.27, update the learning rate from the schedule.
4. After each epoch, evaluate on validation data; save the best checkpoint; stop when no improvement for P epochs. Example: validation losses $0.9, 0.7, 0.6, 0.62, 0.65$ with $P = 2$: stop after epoch 5, restore epoch 3.

DECISION POINT **Picking η**: run an LR range test (increase η exponentially over a few hundred steps, plot loss) and choose about $\frac{1}{10}$ of the value where loss starts to rise. Loss explodes or is NaN $\rightarrow$ lower η, add warmup or clipping; loss plateaus early $\rightarrow$ raise η or check the data pipeline.

TERMS

η_t – learning rate at step/epoch t; $\eta_0, \eta_{\max}, \eta_{\min}$ initial, peak, final
γ, s – decay factor and step interval
k – exponential decay rate
T_w, T – warmup length and total schedule length
c – clipping threshold (e.g. 1.0); $\lVert \mathbf{g} \rVert$ global gradient norm LA.29 §09 Linear Algebra
L_{val} – validation loss; P patience
η_0; $\eta_{\max}, \eta_{\min}$ – initial rate; peak and floor rates
t; t^*; $\mathbf{g}$ – step or epoch; chosen stopping point; gradient



CONTEXT

- **Origin.** Step schedules from early CNN training; cosine annealing by Loshchilov and Hutter (2016); gradient clipping by Pascanu et al. (2013).
- **Use.** Fast, stable convergence of large models.
- **Data science.** Warmup + cosine decay + AdamW + clipping at 1.0 is the standard recipe for training transformers.
- **Physical application.** Simulated annealing cools a temperature schedule the same way O.22 §14 Optimization.

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Deep Architectures: Convolution, Recurrence & Attention

ML.29 Convolutional Networks: Convolution, Stride, Padding & Pooling

A convolutional layer slides a small learned kernel across the input, reusing the same weights at every position. This builds in locality and translation equivariance and needs far fewer parameters than a dense layer.

2-D convolution	$Y[i,j] = \sum_{c=1}^{C_{\text{in}}} \sum_{u=0}^{F-1} \sum_{v=0}^{F-1} K[c,u,v]\; X[c,\, iS+u,\, jS+v] + b$	Kernel dotted with each window (cross-correlation); one output channel per kernel.
Output size	$O = \left\lfloor \dfrac{W - F + 2P}{S} \right\rfloor + 1$	Applies to height and width separately.
"Same" padding	$P = \dfrac{F-1}{2} \quad (S = 1)$	Keeps the spatial size: $F = 3 \Rightarrow P = 1$.
Parameter count	$(F^2 C_{\text{in}} + 1)\, C_{\text{out}}$	Independent of image size, unlike a dense layer.
Max pooling	$Y[i,j] = \max_{(u,v) \in \text{window}} X[iS+u,\, jS+v]$	Downsamples (often 2×2, $S = 2$, halving each side); small shifts barely change it.
Receptive field (stacked)	$r_l = r_{l-1} + (F_l - 1) \prod_{k<l} S_k$	How much input one deep unit "sees"; grows with depth and stride.

STATED-EQUATION *Each output Y i j is the sum over input channels and kernel positions of the kernel times the matching input window, plus a bias. Output size O equals the floor of W minus F plus two P, over S, plus one. Same padding uses P equals F minus one over two. A layer has F squared C in plus one, times C out parameters. Max pooling takes the largest value in each window. The receptive field grows by F minus one times the product of earlier strides.*

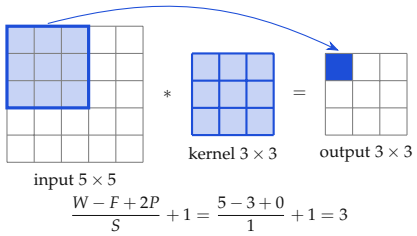
MEMORY AID "W minus F plus 2P, over S, plus 1."

METHOD

1. Write down input size W, kernel F, padding P, stride S.
2. Compute $O = \lfloor (W - F + 2P)/S \rfloor + 1$ for each layer in turn; pooling uses the same formula with $P = 0$.
3. Output channels = number of kernels. Example: $32 \times 32 \times 3$ image, 16 kernels 5×5, $P = 2$, $S = 1$: $O = \frac{32-5+4}{1} + 1 = 32$, output $32 \times 32 \times 16$, $(25 \cdot 3 + 1) \cdot 16 = 1216$ parameters; then 2×2 max pool $\to 16 \times 16 \times 16$.

TERMS

X, Y – input and output feature maps (channels × height × width)
K, b – learned kernel (filter) weights and bias
W, F – input width and kernel size
P, S – zero-padding on each side and stride (step)
$C_{\text{in}}, C_{\text{out}}$ – input and output channels
r_l – receptive field at layer l ($r_0 = 1$)
Convolution – true convolution flips the kernel C2.33 §08 Calculus II; deep learning uses cross-correlation
O; i, j; u, v – output width; output position; offsets within the kernel
F_l, S_k – kernel size and stride of layers l, k



CONTEXT

- **Origin.** Fukushima's Neocognitron (1980); LeCun's LeNet (1989); AlexNet (2012) launched the deep-learning era.
- **Use.** Images, audio spectrograms, any grid-structured signal.
- **Data science.** Image classification, detection and segmentation; 1-D convolutions for time series and audio; pretrained CNN backbones for transfer learning.
- **Physical application.** Edge detection with a Sobel kernel and image blurring are fixed convolutions; CNNs learn their kernels.

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ML.30 Recurrent Neural Networks

An RNN processes a sequence one step at a time, carrying a hidden state that summarizes everything seen so far. The same weights are reused at every step.

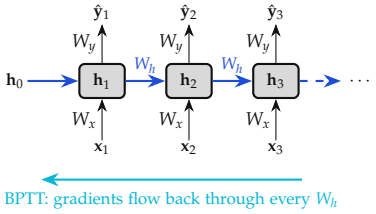
Hidden-state update	$\mathbf{h}_t = \tanh\left(W_h \mathbf{h}_{t-1} + W_x \mathbf{x}_t + \mathbf{b}\right)$	New memory from old memory and the current input.
Output	$\hat{\mathbf{y}}_t = \text{softmax}\left(W_y \mathbf{h}_t + \mathbf{c}\right)$	A prediction at every step (or only the last, for classification).
Sequence loss	$L = \sum_{t=1}^{T} L_t(\mathbf{y}_t, \hat{\mathbf{y}}_t)$	Sum of per-step losses; gradients flow back through time (BPTT).
BPTT gradient chain	$\frac{\partial \mathbf{h}_T}{\partial \mathbf{h}_k} = \prod_{t=k+1}^{T} \text{diag}\left(1 - \mathbf{h}_t^2\right) W_h$	A product of $T-k$ Jacobians C3.15 §10 Calculus III: shrinks or grows geometrically.
Vanishing / exploding test	$\|\partial \mathbf{h}_T / \partial \mathbf{h}_k\| \sim \rho(W_h)^{T-k}$	Spectral radius < 1 vanishes, > 1 explodes LA.23 §09 Linear Algebra; clipping fixes explosion, gates fix vanishing.

STATED-EQUATION *The hidden state h t equals tanh of W h times h t minus one plus W x times x t plus b. The output y hat t is softmax of W y h t plus c. The loss sums the per-step losses. The gradient of h T with respect to h k is a product of T minus k Jacobians, so it behaves like the spectral radius of W h to the power T minus k.*

MEMORY AID Same cell, every time step; the hidden state is the memory.

TERMS

$\mathbf{x}_t, \mathbf{h}_t$ – input and hidden state at time t; $\mathbf{h}_0 = \mathbf{0}$
W_h, W_x, W_y – recurrent, input and output weight matrices (shared across time)
$\mathbf{b}, \mathbf{c}$ – biases
T – sequence length
BPTT – backpropagation through time: backprop on the unrolled graph ML.22
$\rho(W_h)$ – spectral radius: largest $|\lambda|$ of W_h LA.24 §09 Linear Algebra
Fixes – gradient clipping ML.28; LSTM/GRU ML.31
$\hat{\mathbf{y}}_t, \mathbf{y}_t$; L_t – prediction and target at step t; per-step loss
$\text{diag}(1 - \mathbf{h}_t^2)$ – tanh derivative as a diagonal matrix



CONTEXT

- **Origin.** Elman networks (1990); BPTT by Werbos (1990); vanishing gradients analyzed by Hochreiter (1991) and Bengio et al. (1994).
- **Use.** Language, speech, time series, any ordered data.
- **Data science.** Forecasting, sequence labeling and streaming inference; largely replaced by transformers ML.34 for text, but still used on-device and in state-space models.
- **Physical application.** An RNN is a discrete-time nonlinear dynamical system $\mathbf{h}_t = f(\mathbf{h}_{t-1}, \mathbf{x}_t)$, like a state-space model of a controller.

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ML.31 LSTM & GRU Gates

An LSTM adds a cell state that is changed only by elementwise forgetting and adding, so gradients can flow across many steps. Sigmoid gates decide what to forget, write and output.

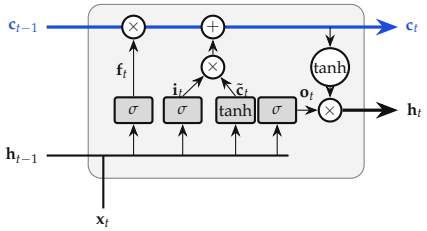
Forget gate	$\mathbf{f}_t = \sigma\big(W_f[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_f\big)$	1 keeps the old memory, 0 erases it.
Input gate & candidate	$\mathbf{i}_t = \sigma\big(W_i[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_i\big), \quad \tilde{\mathbf{c}}_t = \tanh\big(W_c[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_c\big)$	How much to write, and what to write.
Cell update	$\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{c}}_t$	Additive path: $\partial \mathbf{c}_t / \partial \mathbf{c}_{t-1} = \mathbf{f}_t$, no repeated matrix product.
Output gate & hidden state	$\mathbf{o}_t = \sigma\big(W_o[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_o\big), \quad \mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t)$	Exposes a filtered view of the memory.
GRU	$\mathbf{h}_t = (1 - \mathbf{z}_t) \odot \mathbf{h}_{t-1} + \mathbf{z}_t \odot \tilde{\mathbf{h}}_t$	Two gates (update $\mathbf{z}_t$, reset $\mathbf{r}_t$) and no separate cell; fewer parameters.

STATED-EQUATION *The forget gate f t is sigma of W f times the concatenation of h t minus one and x t, plus b f. The input gate i t is a sigmoid and the candidate c tilde t is a tanh of the same input. The cell state c t equals f t times c t minus one plus i t times c tilde t, elementwise. The output gate o t is a sigmoid and h t equals o t times tanh of c t. A GRU interpolates between the old state and a candidate with update gate z t.*

MEMORY AID Forget, write, output: the cell state is a conveyor belt.

TERMS

σ – sigmoid [ML.23]: gate values in $(0, 1)$
$[\mathbf{h}_{t-1}, \mathbf{x}_t]$ – concatenation of previous hidden state and input
$W_*, \mathbf{b}_*$ – weights and biases of each gate
$\mathbf{c}_t$ – cell state (long-term memory); $\mathbf{h}_t$ hidden state (output)
$\odot$ – elementwise product [LA.12] §09 Linear Algebra
$\mathbf{z}_t, \mathbf{r}_t, \tilde{\mathbf{h}}_t$ – GRU update gate, reset gate, candidate state
Forget bias – initialize $\mathbf{b}_f \approx 1$ so memory is kept early in training
$\mathbf{f}_t, \mathbf{i}_t, \mathbf{o}_t$; $\tilde{\mathbf{c}}_t$ – forget, input, output gates; candidate cell values



CONTEXT

- **Origin.** LSTM by Hochreiter and Schmidhuber (1997), forget gate by Gers et al. (2000); GRU by Cho et al. (2014).
- **Use.** Learning long-range dependencies in sequences.
- **Data science.** Speech recognition and machine translation (2014–2017), time-series forecasting, handwriting recognition.
- **Physical application.** The gated cell state behaves like a leaky integrator whose leak rate is set by the forget gate.

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ML.32 Embeddings & Cosine Similarity

An embedding maps a discrete item (word, token, product, user) to a dense vector learned so that similar items get nearby vectors. Similarity is measured by the angle between vectors.

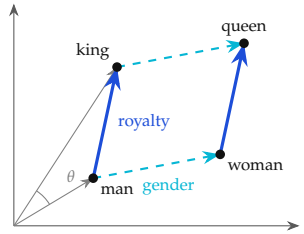
Embedding lookup	$\mathbf{e}_w = E^\top \mathbf{o}_w$	Row w of the table E: a one-hot times a matrix is a lookup.
Cosine similarity	$\cos(\mathbf{u}, \mathbf{v}) = \dfrac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \, \|\mathbf{v}\|}$	1 same direction, 0 unrelated, -1 opposite LA.3 §09 Linear Algebra; length ignored.
Cosine and Euclidean	$\|\hat{\mathbf{u}} - \hat{\mathbf{v}}\|^2 = 2 - 2\cos(\mathbf{u}, \mathbf{v})$	For unit vectors, nearest by cosine = nearest by Euclidean distance.
Skip-gram probability	$P(o \mid c) = \dfrac{\exp(\mathbf{u}_o^\top \mathbf{v}_c)}{\sum_{w=1}^{V} \exp(\mathbf{u}_w^\top \mathbf{v}_c)}$	Predict context word o from center word c; a softmax over the vocabulary ML.24.
Negative sampling	$\log \sigma(\mathbf{u}_o^\top \mathbf{v}_c) + \sum_{k=1}^{K} \log \sigma(-\mathbf{u}_{n_k}^\top \mathbf{v}_c)$	Cheap substitute: pull true pairs together, push K random pairs apart.
Analogy arithmetic	$\mathbf{e}_{\text{king}} - \mathbf{e}_{\text{man}} + \mathbf{e}_{\text{woman}} \approx \mathbf{e}_{\text{queen}}$	Directions encode relations (approximately).

STATED-EQUATION *The embedding of item w is E transpose times the one-hot vector, that is row w of E. Cosine similarity is u dot v over the product of their norms; for unit vectors the squared distance equals two minus two times the cosine. Skip-gram models the probability of context o given center c as a softmax of u o dot v c. Negative sampling maximizes log sigma of the true pair's score plus log sigma of minus each negative pair's score. Analogies become vector arithmetic: king minus man plus woman lands near queen.*

MEMORY AID Similar meaning → small angle.

TERMS

E – $V \times d$ embedding matrix; V vocabulary size, d dimension
$\mathbf{o}_w$ – one-hot vector for item w ML.6
$\mathbf{u}, \mathbf{v}$ – vectors; $\hat{\mathbf{u}} = \mathbf{u}/\|\mathbf{u}\|$ unit vector
$\mathbf{v}_c, \mathbf{u}_o$ – center-word and context-word vectors
n_k – randomly sampled negative words, K of them
σ – sigmoid ML.9
$\mathbf{e}_w$; $\hat{\mathbf{u}}, \hat{\mathbf{v}}$ – embedding of item w; unit-normalized vectors
o, c; V; K – context and center words; vocabulary size; number of negatives



CONTEXT

- **Origin.** Distributional hypothesis (Firth, 1957: "know a word by the company it keeps"); word2vec by Mikolov et al. (2013); GloVe (2014).
- **Use.** Representing categorical and textual data numerically.
- **Data science.** Semantic search and retrieval-augmented generation (vector databases, cosine top-k), recommendation (user/item embeddings), high-cardinality categorical features, clustering documents.
- **Physical application.** Molecules and materials are embedded as vectors so that similar compounds can be retrieved by cosine similarity.

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ML.33 Scaled Dot-Product Attention

Each position forms a query, compares it to every position's key, and takes a softmax-weighted average of their values. Every token can look directly at every other token in one step.

Scaled dot-product attention	Rows of the softmax are weights over keys; output row i is a weighted mix of value rows. $$\text{Attention}(Q,K,V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V$$	
Projections	$Q = XW_Q,\ \ K = XW_K,\ \ V = XW_V$	Queries, keys and values are learned linear maps of the same input (self-attention).
Why $\sqrt{d_k}$	$\text{Var}(\mathbf{q}^\top\mathbf{k}) = d_k$ for unit-variance entries	Scaling keeps scores $O(1)$ so the softmax does not saturate S.13 §13 Statistics.
Causal mask	$S_{ij} = -\infty$ for $j > i$	A token cannot attend to the future; used for text generation.

STATED-EQUATION *Attention of Q, K, V equals softmax of Q K transpose over the square root of d k, times V. Queries, keys and values are X times W Q, W K and W V. A dot product of d k unit-variance entries has variance d k, so dividing by root d k keeps scores of order one. A causal mask sets scores for future positions to minus infinity.*

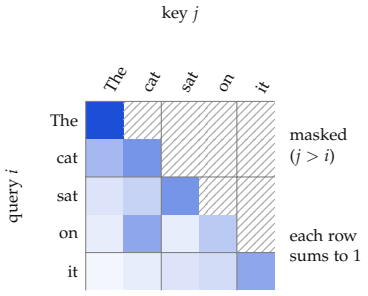
MEMORY AID Query asks, key answers "how relevant?", value is what gets passed along.

METHOD

1. Compute Q, K, V from the input X.
2. Scores $S = QK^\top/\sqrt{d_k}$ $(n \times n)$; apply the mask if causal.
3. Row-wise softmax gives weights A; output AV. Example: one query $\mathbf{q} = (1,0)$, keys $(1,0), (0,1)$, $d_k = 2$: scores $0.71, 0$; weights $\approx 0.67, 0.33$; output $= 0.67\mathbf{v}_1 + 0.33\mathbf{v}_2$.

TERMS

X – $n \times d$ input: one row per token
Q, K, V – query, key, value matrices; d_k key/query dimension
W_Q, W_K, W_V – learned projection matrices
softmax – row-wise ML.24
S, A – score and attention-weight matrices $(n \times n)$
Cost – $O(n^2 d)$ time and $O(n^2)$ memory in sequence length n O.10 §14 Optimization
d_k; $\mathbf{q}, \mathbf{k}$ – key dimension; a single query and key vector



CONTEXT

- **Origin.** Additive attention for translation by Bahdanau et al. (2014); scaled dot-product attention in "Attention Is All You Need" (Vaswani et al., 2017).
- **Use.** Relating every element of a set or sequence to every other.
- **Data science.** The core operation of transformers and LLMs; attention maps are inspected for interpretability; cross-attention connects text to images in diffusion models.
- **Physical application.** Like a content-addressable memory: a query retrieves stored values in proportion to how well it matches each key.

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ML.34 Transformers: Multi-Head Attention & Positional Encoding

A transformer stacks blocks of multi-head self-attention and a feed-forward network, each wrapped in a residual connection with layer normalization. Positional encodings tell it the order of tokens.

Attention head	$\text{head}_h = \text{Attention}\big(XW_Q^{(h)}, XW_K^{(h)}, XW_V^{(h)}\big)$	Each head attends in its own learned subspace of size d/H.
Multi-head attention	$\text{MHA}(X) = \big[\text{head}_1, \ldots, \text{head}_H\big] W_O$	Concatenate heads and mix them; different heads track different relations.
Feed-forward sublayer	$\text{FFN}(\mathbf{x}) = W_2\,\text{GELU}(W_1\mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2$	Applied to each token separately; hidden width usually $4d$.
Pre-LN block	$\mathbf{x} \leftarrow \mathbf{x} + \text{MHA}(\text{LN}(\mathbf{x})),\ \ \mathbf{x} \leftarrow \mathbf{x} + \text{FFN}(\text{LN}(\mathbf{x}))$	Residual paths keep gradients flowing through deep stacks ML.26.
Sinusoidal encoding (even)	$PE_{(p,2i)} = \sin\big(p/10000^{2i/d}\big)$	Each dimension pair is a sinusoid of a different wavelength T.K1 §06 Trigonometry.
Sinusoidal encoding (odd)	$PE_{(p,2i+1)} = \cos\big(p/10000^{2i/d}\big)$	Added to token embeddings; relative shifts are rotations, so offsets are easy to learn.

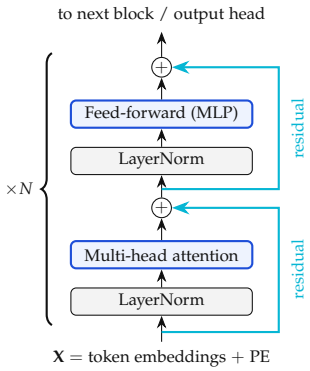
STATED-EQUATION *Each head is attention applied to X times its own query, key and value matrices. Multi-head attention concatenates the H heads and multiplies by W O. The feed-forward network is W two times GELU of W one x plus b one, plus b two. A pre-LN block adds multi-head attention of the layer-normed input, then adds the feed-forward network of the layer-normed result. Positional encodings use sine for even dimensions and cosine for odd dimensions of p over ten thousand to the two i over d.*

MEMORY AID Attention mixes across tokens; the FFN thinks within each token.

DECISION POINT **Encoder, decoder, or both?** Understanding tasks (classification, embeddings, retrieval) → encoder-only (BERT-style, bidirectional); generation → decoder-only with a causal mask (GPT-style); sequence-to-sequence (translation) → encoder–decoder. Long sequences → relative/rotary positions (RoPE) and efficient attention.

TERMS

H – number of heads; d model width
W_O – output projection mixing the heads
$W_1, W_2, \mathbf{b}_1, \mathbf{b}_2$ – FFN weights and biases
LN – layer normalization ML.26
p, i – token position and dimension-pair index
PE – positional encoding added to embeddings ML.32
RoPE – rotary embedding: rotates $\mathbf{q}, \mathbf{k}$ by position-dependent angles
head_h; MHA; FFN – output of head h; multi-head attention; feed-forward network
GELU; d – activation ML.23; model dimension



CONTEXT

- **Origin.** Vaswani et al. (2017); BERT (2018) and GPT (2018–) scaled it up.
- **Use.** Language, vision (ViT), audio, proteins, code.
- **Data science.** Large language models, machine translation, code completion, AlphaFold-style protein structure prediction; fine-tuning pretrained transformers is the default for text tasks.
- **Physical application.** Transformers trained on simulation data serve as fast surrogates for weather and fluid-dynamics models.

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Unsupervised Learning & Generative Models

ML.35 k-Means Clustering

Partition the data into K groups by repeatedly assigning each point to its nearest centroid and moving each centroid to the mean of its points.

Objective (inertia)	$J = \sum_{k=1}^{K} \sum_{\mathbf{x}_i \in C_k} \lVert \mathbf{x}_i - \boldsymbol{\mu}_k \rVert^2$	Within-cluster sum of squares; never increases under Lloyd's algorithm.
Assignment step	$c_i = \arg\min_k \lVert \mathbf{x}_i - \boldsymbol{\mu}_k \rVert^2$	Each point joins the nearest centroid (Voronoi cells).
Update step	$\boldsymbol{\mu}_k = \frac{1}{\lvert C_k \rvert} \sum_{\mathbf{x}_i \in C_k} \mathbf{x}_i$	The mean minimizes squared distance within each cluster S.1 §13 Statistics.
k-means++ seeding	$P(\mathbf{x}_i \text{ chosen}) = \frac{D(\mathbf{x}_i)^2}{\sum_j D(\mathbf{x}_j)^2}$	Spread initial centroids out; far points are more likely.

STATED-EQUATION *J equals the sum over clusters k of the sum over points in C k of the squared distance from x i to mu k. Assign each point to the nearest centroid; update each centroid to the mean of its points. k-means plus plus picks each new seed with probability proportional to its squared distance to the nearest existing seed.*

MEMORY AID Assign, update, repeat.

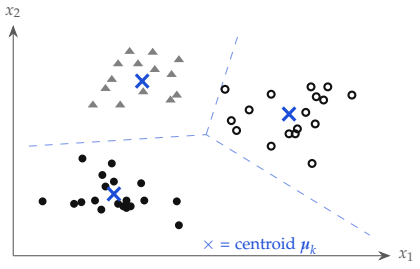
METHOD

1. Choose K; seed K centroids (k-means++).
2. Assign every point to its nearest centroid.
3. Move each centroid to the mean of its assigned points.
4. Repeat 2–3 until assignments stop changing; rerun from several seeds and keep the lowest J. Example: points $1, 2, 9, 10$, $K = 2$, seeds $1, 2$: assign $\{1\}, \{2, 9, 10\} \to \mu = 1, 7 \to$ assign $\{1, 2\}, \{9, 10\} \to \mu = 1.5, 9.5$, converged.

DECISION POINT **Choosing K**: plot J vs. K and look for the elbow; maximize average silhouette ML.36; or let the business decide ("5 segments"). **Is k-means right?** It assumes round, similar-size clusters; elongated or unequal clusters → GMM ML.37; arbitrary shapes or noise → density-based (DBSCAN) or hierarchical.

TERMS

K – number of clusters (chosen by elbow or silhouette ML.36)
C_k – set of points in cluster k; $|C_k|$ its size
$\boldsymbol{\mu}_k$ – centroid: the mean S.1 §13 Statistics
c_i – cluster label of point i
J – within-cluster sum of squares (inertia), distance ML.11
$D(\mathbf{x})$ – distance from $\mathbf{x}$ to the nearest chosen seed
Lloyd's algorithm – alternate assign and update; J never increases
Caveats – local optima (use k-means++ init), scale features S.4 §13 Statistics



CONTEXT

- **Origin.** Lloyd (1957, published 1982); MacQueen named it (1967); k-means++ by Arthur and Vassilvitskii (2007).
- **Use.** Grouping similar items; vector quantization.
- **Data science.** Customer segmentation, color quantization of images, and clustering embeddings to discover topics or intents.
- **Physical application.** Grouping weather stations with similar climate records into climate zones.

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ML.36 Hierarchical Clustering & Silhouette

Agglomerative clustering starts with every point alone and repeatedly merges the two closest clusters, producing a tree (dendrogram). The silhouette scores how well each point fits its cluster.

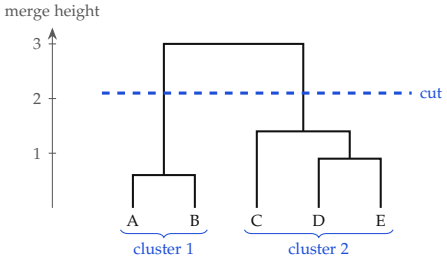
Silhouette	$s(i) = \dfrac{b(i) - a(i)}{\max\{a(i),\, b(i)\}} \in [-1, 1]$	Near 1 well placed, near 0 on a boundary, negative likely misassigned.
Single linkage	$d_{\text{single}} = \min_{\mathbf{x}\in A,\, \mathbf{z}\in B} d(\mathbf{x}, \mathbf{z})$	Closest pair: finds chains and odd shapes, but "chaining" risk.
Complete linkage	$d_{\text{complete}} = \max_{\mathbf{x}\in A,\, \mathbf{z}\in B} d(\mathbf{x}, \mathbf{z})$	Farthest pair: compact, similar-diameter clusters.
Average linkage	$d_{\text{avg}} = \dfrac{1}{\lvert A\rvert\lvert B\rvert} \sum_{\mathbf{x}\in A} \sum_{\mathbf{z}\in B} d(\mathbf{x}, \mathbf{z})$	Mean pairwise distance: a compromise.
Ward linkage	$\Delta J = \dfrac{\lvert A\rvert\lvert B\rvert}{\lvert A\rvert + \lvert B\rvert} \lVert \boldsymbol{\mu}_A - \boldsymbol{\mu}_B \rVert^2$	Merge the pair that least increases within-cluster variance (like k-means).

STATED-EQUATION *The silhouette s of i equals b of i minus a of i, over the maximum of a of i and b of i, between minus one and one. Single linkage uses the minimum pairwise distance, complete linkage the maximum, and average linkage the mean. Ward linkage merges the pair whose merge least increases the within-cluster sum of squares.*

MEMORY AID Bottom-up: merge the closest pair, record the height, repeat.

TERMS

$a(i)$ – mean distance from point i to its own cluster
$b(i)$ – mean distance to the nearest other cluster
$s(i)$ – near 1: well placed; near 0: on a boundary; < 0: likely misassigned
A, B – two clusters; $|A|$ size; $\boldsymbol{\mu}_A$ centroid
Agglomerative – start with singletons and repeatedly merge the closest pair
Linkage – rule for the distance between clusters ML.11
Dendrogram – tree of merges; cut it at a height to get clusters
$d(\mathbf{x}, \mathbf{z})$; $|A|$ – distance between points; cluster size
$\boldsymbol{\mu}_A, \boldsymbol{\mu}_B$; ΔJ – cluster centroids; increase in within-cluster sum of squares



CONTEXT

- **Origin.** Hierarchical methods from numerical taxonomy (Sokal and Sneath, 1963); Ward (1963); silhouette by Rousseeuw (1987).
- **Use.** Taxonomies, phylogenetic trees, choosing the number of clusters.
- **Data science.** Gene-expression heatmaps, grouping products into a hierarchy, and choosing K for k-means ML.35 by average silhouette.
- **Physical application.** Building a family tree of species from DNA distances (phylogenetics) uses hierarchical clustering.

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ML.37 Gaussian Mixtures & the EM Algorithm

Model the data as drawn from K Gaussians with unknown weights, means and variances. EM alternates between computing each point's soft membership (E-step) and refitting the Gaussians with those weights (M-step).

Mixture density	$p(x) = \sum_{k=1}^{K} \pi_k \mathcal{N}(x \mid \mu_k, \sigma_k^2)$	A weighted sum of normal bumps S.20 §13 Statistics.
Log-likelihood	$\ell = \sum_{i=1}^{n} \ln \sum_{k=1}^{K} \pi_k \mathcal{N}(x_i \mid \mu_k, \sigma_k^2)$	The log of a sum has no closed-form maximizer, hence EM S.26 §13 Statistics.
E-step (responsibilities)	$\gamma_{ik} = \dfrac{\pi_k \mathcal{N}(x_i \mid \mu_k, \sigma_k^2)}{\sum_j \pi_j \mathcal{N}(x_i \mid \mu_j, \sigma_j^2)}$	Posterior probability point i came from component k (Bayes S.11 §13 Statistics).
M-step: means	$\mu_k = \dfrac{\sum_i \gamma_{ik} x_i}{\sum_i \gamma_{ik}}$	Responsibility-weighted average.
M-step: variances	$\sigma_k^2 = \dfrac{\sum_i \gamma_{ik}(x_i - \mu_k)^2}{\sum_i \gamma_{ik}}$	Weighted variance; multivariate: weighted covariance matrix S.39 §13 Statistics.
M-step: weights	$\pi_k = \dfrac{N_k}{n}, \quad N_k = \sum_i \gamma_{ik}$	Share of the total responsibility.

STATED-EQUATION *p of x equals the sum over k of pi k times the normal density with mean mu k and variance sigma k squared. The log-likelihood is the sum over points of the log of that mixture. E-step: the responsibility gamma i k equals pi k N of x i given component k, over the sum over j of the same. M-step: mu k equals the sum of gamma i k x i over the sum of gamma i k; sigma k squared is the weighted variance; pi k equals N k over n.*

MEMORY AID E-step: guess memberships; M-step: refit the Gaussians.

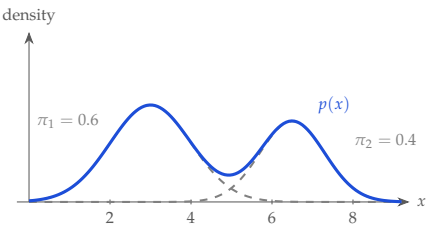
METHOD

1. Initialize π_k, μ_k, σ_k^2 (e.g. from k-means).
2. E-step: compute all γ_{ik}.
3. M-step: update μ_k, σ_k^2, π_k with the weighted formulas.
4. Repeat until ℓ stops increasing. Example: $x = 0$ with equal weights and variances, $\mu_1 = 0$, $\mu_2 = 2$, $\sigma = 1$: $\gamma_{01} = \frac{e^0}{e^0 + e^{-2}} \approx 0.88$.

DECISION POINT **Choosing** K: minimize BIC $= -2\ell + q \ln n$ (q free parameters) over K. **Covariance type**: spherical (like k-means) → diagonal → full, trading flexibility for parameters. Guard against a component collapsing onto one point (add a small variance floor).

TERMS

π_k – mixing weights, $\sum \pi_k = 1$
$\mathcal{N}$ – normal density S.20 §13 Statistics; μ_k, σ_k^2 component mean and variance
γ_{ik} – responsibility: posterior $P(\text{component } k \mid x_i)$ by Bayes S.11 §13 Statistics
N_k – effective count $\sum_i \gamma_{ik}$; M-step uses weighted means S.1 §13 Statistics
ℓ – log-likelihood; BIC penalized fit
EM – alternates E and M; never decreases the likelihood
Relation – soft version of k-means ML.35



CONTEXT

- **Origin.** Dempster, Laird and Rubin (1977); mixtures studied by Pearson (1894).
- **Use.** Density estimation and soft clustering; missing-data problems.
- **Data science.** Anomaly detection by low density, soft customer segments with membership probabilities, and speaker identification.
- **Physical application.** Separating overlapping peaks in a spectrometer reading is fitting a Gaussian mixture.

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ML.38 Principal Component Analysis (PCA)

Rotate the centered data onto orthogonal directions ordered by how much variance they capture, then keep the top few. These directions are eigenvectors of the covariance matrix.

Covariance matrix	$S = \frac{1}{n-1} X_c^\top X_c$	Centered data's feature-by-feature covariances S.39 §13 Statistics.
Eigen-decomposition	$S\,\mathbf{v}_k = \lambda_k \mathbf{v}_k, \;\; \lambda_1 \geq \lambda_2 \geq \cdots$	Directions $\mathbf{v}_k$ (orthonormal) and the variance λ_k along each LA.23 §09 Linear Algebra.
Explained variance ratio	$\frac{\lambda_k}{\sum_j \lambda_j}$	Fraction of total variance captured by component k.
Scores (projection)	$\mathbf{z} = V_k^\top (\mathbf{x} - \bar{\mathbf{x}})$	New coordinates in the top-k basis.
SVD route	$X_c = U\Sigma V^\top, \;\; \lambda_j = \frac{\sigma_j^2}{n-1}$	Same V, numerically better; how PCA is computed LA.28 §09 Linear Algebra.
Reconstruction	$\hat{\mathbf{x}} = \bar{\mathbf{x}} + V_k \mathbf{z}$	Best rank-k approximation in squared error (Eckart–Young).

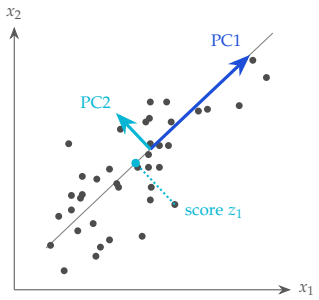
STATED-EQUATION *S equals one over n minus one times X c transpose X c. S v k equals lambda k v k, with eigenvalues in decreasing order. The explained variance ratio equals lambda k over the sum of all lambda j. The scores z equal V k transpose times x minus x bar. From the SVD of X c, lambda j equals sigma j squared over n minus one. Reconstruct with x bar plus V k z.*

MEMORY AID PC1 points along the longest direction of the data cloud.

DECISION POINT **How many components?** Keep enough for ≈90–95% cumulative explained variance; or look for the elbow in the scree plot; or pick k by downstream CV score. Always standardize first if features have different units ML.6; nonlinear structure → t-SNE/UMAP for plots ML.39 or an autoencoder ML.40.

TERMS

X_c – centered data matrix (subtract column means S.1 §13 Statistics)
S – covariance matrix S.38 §13 Statistics
$\mathbf{v}_k, \lambda_k$ – eigenvectors (directions) and eigenvalues (variances) LA.23 §09 Linear Algebra
PC1 – direction of maximum variance: $\max \mathbf{v}^\top S\mathbf{v}$, $\|\mathbf{v}\| = 1$ O.17 §14 Optimization
V_k – $d \times k$ matrix of the top k eigenvectors
$\mathbf{z}$ – coordinates after projecting onto the top k components (dot products T.21 §06 Trigonometry)
SVD – $X_c = U\Sigma V^\top$ gives the same V, with $\lambda = \sigma^2/(n-1)$
U, Σ; σ_j – SVD factors LA.28 §09 Linear Algebra; singular values
n; $\bar{\mathbf{x}}$; $\hat{\mathbf{x}}$ – number of rows; mean vector; reconstruction



CONTEXT

- **Origin.** Karl Pearson (1901); Harold Hotelling (1933).
- **Use.** Dimension reduction, visualization, and decorrelation.
- **Data science.** Compressing hundreds of correlated features, 2-D plots of embeddings, noise reduction, and removing multicollinearity before regression.
- **Physical application.** The principal axes of a rigid body's inertia tensor are its eigenvectors, the same math as PCA.

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ML.39 t-SNE & UMAP (Nonlinear Embedding for Visualization)

Place high-dimensional points in 2-D so that neighbors stay neighbors. Both methods match a neighborhood-probability structure in the original space to one in the low-dimensional map.

High-dim affinities	$p_{j\|i} = \dfrac{\exp(-\\|\mathbf{x}_i - \mathbf{x}_j\\|^2/2\sigma_i^2)}{\sum_{k\neq i}\exp(-\\|\mathbf{x}_i - \mathbf{x}_k\\|^2/2\sigma_i^2)}$	Gaussian neighbor probabilities; σ_i set so each point has a fixed perplexity.
Symmetrized	$p_{ij} = \dfrac{p_{j\|i} + p_{i\|j}}{2n}$	One joint distribution over pairs.
Low-dim affinities	$q_{ij} = \dfrac{(1 + \\|\mathbf{y}_i - \mathbf{y}_j\\|^2)^{-1}}{\sum_{k\neq l}(1 + \\|\mathbf{y}_k - \mathbf{y}_l\\|^2)^{-1}}$	Heavy-tailed Student-t: moderately far points can spread out (no crowding).
t-SNE objective	$C = D_{\mathrm{KL}}(P \,\\|\, Q) = \sum_{i\neq j} p_{ij} \log \dfrac{p_{ij}}{q_{ij}}$	Minimized by gradient descent on the map $\mathbf{y}$ ML.3.
UMAP objective	$\sum_{i\neq j}\left[w_{ij}\log\dfrac{w_{ij}}{v_{ij}} + (1-w_{ij})\log\dfrac{1-w_{ij}}{1-v_{ij}}\right]$	Fuzzy-set cross-entropy: also pushes non-neighbors apart; faster, keeps more global layout.

STATED-EQUATION *The conditional affinity p j given i is a Gaussian kernel of the distance, normalized over all k; sigma i is set by the perplexity. p i j averages the two conditionals over two n. In the map, q i j uses a Student-t kernel one over one plus the squared distance. t-SNE minimizes the KL divergence of P from Q. UMAP minimizes a binary cross-entropy between high- and low-dimensional edge weights w i j and v i j.*

MEMORY AID Distances between clusters in a t-SNE plot mean nothing.

DECISION POINT **Use for visualization, not features or clustering proof.** Run PCA to ≈50 dimensions first. Try several perplexities (5–50) or UMAP `n_neighbors`; trust structure that survives all of them. Cluster sizes and gaps are not interpretable; for a reusable, linear map use PCA ML.38.

TERMS

$\mathbf{x}_i, \mathbf{y}_i$ – original point and its 2-D (or 3-D) map position
σ_i – per-point Gaussian width
Perplexity – $2^{H(P_i)}$: effective number of neighbors ML.3
P, Q – affinity distributions in high and low dimension
w_{ij}, v_{ij} – UMAP edge weights in high and low dimension, in $[0, 1]$
KL – Kullback–Leibler divergence ML.3
n; C – number of points; cost

CONTEXT

- **Origin.** t-SNE by van der Maaten and Hinton (2008); UMAP by McInnes, Healy and Melville (2018).
- **Use.** Exploratory 2-D plots of high-dimensional data.
- **Data science.** Visualizing embeddings, single-cell RNA-seq clusters, and image features; spotting label noise and outliers.
- **Physical application.** Mapping high-dimensional sensor states of a plant onto a 2-D chart where normal operation and fault regimes form separate islands.

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ML.40 Autoencoders & Variational Autoencoders

An autoencoder compresses the input to a small code and reconstructs it. A VAE makes the code probabilistic and regularizes it toward a prior, so new data can be generated by sampling codes.

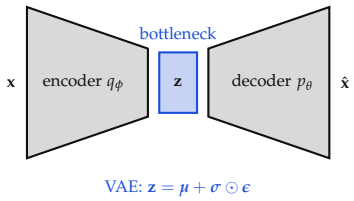
Autoencoder loss	$L = \left\|\mathbf{x} - g_\phi(f_\theta(\mathbf{x}))\right\|^2$	Encoder f_θ, decoder g_ϕ; a bottleneck forces a compact code.
Evidence lower bound (ELBO)	$\log p(\mathbf{x}) \;\geq\; \underbrace{E_{q_\phi(\mathbf{z}\mid\mathbf{x})}\left[\log p_\theta(\mathbf{x}\mid\mathbf{z})\right]}_{\text{reconstruction}} - \underbrace{D_{\mathrm{KL}}\left(q_\phi(\mathbf{z}\mid\mathbf{x}) \,\|\, p(\mathbf{z})\right)}_{\text{regularization}}$	The VAE maximizes this bound; the gap is $D_{\mathrm{KL}}(q \,\|\, p(\mathbf{z}\mid\mathbf{x})) \geq 0$.
Reparameterization	$\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\epsilon}, \;\; \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, I)$	Moves the randomness outside so gradients flow through $\boldsymbol{\mu}, \boldsymbol{\sigma}$.
Gaussian KL (closed form)	$D_{\mathrm{KL}} = \frac{1}{2}\sum_j \left(\mu_j^2 + \sigma_j^2 - \ln \sigma_j^2 - 1\right)$	KL from $\mathcal{N}(\boldsymbol{\mu}, \mathrm{diag}\,\sigma^2)$ to $\mathcal{N}(\mathbf{0}, I)$.

STATED-EQUATION *An autoencoder minimizes the squared error between x and the decoder of the encoder of x. The VAE maximizes the evidence lower bound: the expected log-likelihood of x given z under the encoder's distribution, minus the KL divergence from the encoder's distribution to the prior. Sample z as mu plus sigma times epsilon. For Gaussians the KL equals one half the sum of mu squared plus sigma squared minus log sigma squared minus one.*

MEMORY AID Reconstruct well, but keep the codes close to the prior.

TERMS

f_θ, g_ϕ – encoder and decoder networks (AE)
$\mathbf{z}$ – latent code, dimension $\ll$ input dimension
$q_\phi(\mathbf{z}\mid\mathbf{x})$ – VAE encoder: Gaussian with outputs $\boldsymbol{\mu}, \boldsymbol{\sigma}$
$p_\theta(\mathbf{x}\mid\mathbf{z})$ – VAE decoder likelihood
$p(\mathbf{z})$ – prior, usually $\mathcal{N}(\mathbf{0}, I)$ S.39 §13 Statistics
D_{KL} – KL divergence ML.3
$\boldsymbol{\epsilon}$ – standard normal noise; $\odot$ elementwise
$\boldsymbol{\mu}, \boldsymbol{\sigma}$; I – encoder's mean and SD outputs; identity covariance



CONTEXT

- **Origin.** Autoencoders from 1980s neural-network research (Rumelhart, Hinton); VAE by Kingma and Welling (2013).
- **Use.** Compression, denoising, representation learning, generation.
- **Data science.** Anomaly detection by reconstruction error, denoising, learned latent spaces for images and molecules; the VAE's latent space underlies latent-diffusion image generators.
- **Physical application.** Compressing high-resolution turbulence simulation snapshots into a small latent state for fast reduced-order modeling.

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ML.41 Generative Adversarial Networks

A generator turns noise into fake samples while a discriminator tries to tell fake from real. Training is a two-player minimax game whose equilibrium is a generator that matches the data distribution.

GAN minimax objective	$\min_G \max_D \; V(D,G) = E_{\mathbf{x}\sim p_{\text{data}}}\big[\log D(\mathbf{x})\big] + E_{\mathbf{z}\sim p(\mathbf{z})}\big[\log\big(1 - D(G(\mathbf{z}))\big)\big]$	D maximizes classification log-likelihood; G minimizes it.
Optimal discriminator	$D^*(\mathbf{x}) = \dfrac{p_{\text{data}}(\mathbf{x})}{p_{\text{data}}(\mathbf{x}) + p_g(\mathbf{x})}$	For a fixed generator; equals $\frac{1}{2}$ everywhere when $p_g = p_{\text{data}}$.
Value at the optimum	$V(D^*, G) = -\log 4 + 2\,\mathrm{JS}(p_{\text{data}} \,\|\, p_g)$	So training G minimizes the Jensen–Shannon divergence.
Non-saturating generator loss	$L_G = -E_{\mathbf{z}}\big[\log D(G(\mathbf{z}))\big]$	Strong gradients early, when D easily rejects fakes.
Jensen–Shannon divergence	$\mathrm{JS}(p\|q) = \frac{1}{2}D_{\mathrm{KL}}(p\|m) + \frac{1}{2}D_{\mathrm{KL}}(q\|m),\; m = \frac{p+q}{2}$	Symmetric, bounded version of KL ML.3.

STATED-EQUATION *The generator minimizes and the discriminator maximizes the expected log of D on real data plus the expected log of one minus D of G of z. For a fixed generator the best discriminator is p data over p data plus p g. At that optimum the value equals minus log four plus twice the Jensen-Shannon divergence. In practice the generator minimizes minus the log of D of G of z. The Jensen-Shannon divergence is the average of the KL divergences of p and of q from their midpoint m.*

MEMORY AID Counterfeiter vs. detective: both improve until fakes are indistinguishable.

DECISION POINT **GAN, VAE, or diffusion?** GANs: sharp samples, fast sampling, but unstable training and mode collapse. VAEs: stable, smooth latent space, blurrier samples ML.40. Diffusion models: best quality and coverage, slower sampling. Stabilize GANs with Wasserstein loss/gradient penalty, spectral normalization, and two learning rates.

TERMS

G – generator network: noise $\mathbf{z} \mapsto$ sample
D – discriminator: probability that its input is real
p_{data}, p_g – data distribution and generator's distribution
$p(\mathbf{z})$ – noise prior, e.g. $\mathcal{N}(\mathbf{0}, I)$
JS – Jensen–Shannon divergence; m the average distribution
Mode collapse – G produces only a few kinds of output
$V(D,G)$; D^* – minimax value; optimal discriminator
L_G; m – generator loss; mixture $(p+q)/2$
D_{KL} – KL divergence ML.3

CONTEXT

- **Origin.** Goodfellow et al. (2014); DCGAN (2015), StyleGAN (2019).
- **Use.** Realistic synthetic images and data.
- **Data science.** Image synthesis and super-resolution, synthetic tabular data for privacy, data augmentation; adversarial training ideas in domain adaptation.
- **Physical application.** GANs generate synthetic particle-detector showers far faster than full Monte Carlo simulation O.27 §14 Optimization.

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Reinforcement Learning

ML.42 Markov Decision Processes & the Bellman Equation

An agent repeatedly observes a state, takes an action, and receives a reward and a next state. Values (expected discounted future reward) satisfy recursive Bellman equations.

Discounted return	$G_t = \sum_{k=0}^{\infty} \gamma^k r_{t+k+1}$	Future rewards count less; finite for $\gamma < 1$ A2.19 §04 Algebra II.
State value	$V^\pi(s) = E_\pi\left[G_t \mid s_t = s\right]$	Expected return from s when following policy π.
Action value	$Q^\pi(s,a) = E_\pi\left[G_t \mid s_t = s, a_t = a\right]$	Expected return after taking a in s, then following π.
Bellman expectation equation	$V^\pi(s) = \sum_a \pi(a \mid s) \sum_{s'} P(s' \mid s,a)\left[R(s,a,s') + \gamma V^\pi(s')\right]$	Value = immediate reward + discounted value of where you land; linear in V^π.
Bellman optimality equation	$Q^*(s,a) = \sum_{s'} P(s' \mid s,a)\left[R(s,a,s') + \gamma \max_{a'} Q^*(s',a')\right]$	The best policy acts greedily on Q^*: $\pi^*(s) = \arg\max_a Q^*(s,a)$ O.21 §14 Optimization.

STATED-EQUATION *The return G t is the sum of gamma to the k times future rewards. V pi of s is the expected return from s under policy pi; Q pi of s, a is the expected return after taking a in s. Bellman expectation: V pi of s equals the policy-weighted, transition-weighted sum of reward plus gamma times V pi of the next state. Bellman optimality: Q star of s, a equals the expected reward plus gamma times the max over next actions of Q star.*

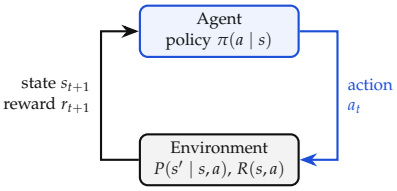
MEMORY AID Value now = reward now + $\gamma\times$ value next.

METHOD

1. Value iteration: start with $V(s) = 0$ for all s.
2. Sweep: $V(s) \leftarrow \max_a \sum_{s'} P(s' \mid s,a)[R + \gamma V(s')]$.
3. Repeat until changes are tiny (a contraction, so it converges); read off the greedy policy. Example: a single state with reward 1 every step and $\gamma = 0.9$: $V = 1 + 0.9V \Rightarrow V = 10$.

TERMS

s, a, s' – state, action, next state; t time step
r, $R(s,a,s')$ – reward and expected reward function
$\gamma \in [0,1)$ – discount factor
$\pi(a \mid s)$ – policy: probability of action a in state s
$P(s' \mid s,a)$ – transition probabilities (Markov property LA.26 §09 Linear Algebra)
V^*, Q^*, π^* – optimal value, action value and policy
G_t; t, k – discounted return; time step and look-ahead
V^π, Q^π; E_π – state and action values under π; expectation following π



CONTEXT

- **Origin.** Bellman's dynamic programming (1957); Howard's policy iteration (1960).
- **Use.** Sequential decisions under uncertainty: control, robotics, inventory, games.
- **Data science.** Recommendation and ad bidding as sequential decisions; RL from human feedback fine-tunes language models; the Bellman equation also prices customer lifetime value.
- **Physical application.** Optimal control of a robot arm or HVAC system is an MDP whose state is the physical configuration.

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ML.43 Q-Learning & Policy Gradients

When the transition model is unknown, learn from experience. Q-learning nudges action values toward a bootstrapped Bellman target; policy-gradient methods adjust a parameterized policy directly in the direction that increases expected return.

TD error	$\delta_t = r_{t+1} + \gamma \max_{a'} Q(s_{t+1}, a') - Q(s_t, a_t)$	Surprise: observed one-step target minus current estimate.
Q-learning update	$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha\, \delta_t$	Off-policy: learns Q^* while behaving exploratorily ML.42.
ε-greedy	$a_t = \begin{cases} \text{random action}, & \text{prob. } \varepsilon \\ \arg\max_a Q(s_t, a), & \text{prob. } 1-\varepsilon \end{cases}$	Balances exploring new actions and exploiting known good ones.
Policy gradient theorem	REINFORCE: raise log-probability of actions in proportion to how much better than baseline b they did. $\nabla_\theta J(\theta) = E_{\pi_\theta}\left[\sum_t \nabla_\theta \log \pi_\theta(a_t \mid s_t)\,(G_t - b(s_t))\right]$	
Advantage	$A(s,a) = Q(s,a) - V(s)$	How much better a is than average in s; the ideal baseline-corrected signal (actor–critic).

STATED-EQUATION *The TD error is the reward plus gamma times the best next Q, minus the current Q. Q-learning adds alpha times the TD error to Q of s t, a t. Epsilon-greedy takes a random action with probability epsilon, otherwise the best one. The policy gradient is the expected sum of the gradient of the log-policy times the return minus a baseline. The advantage is Q minus V.*

MEMORY AID Q-learning: "learn a guess from a guess." Policy gradient: "do more of what worked."

METHOD

1. Initialize $Q(s,a) = 0$; choose α, γ, ε.
2. In state s, pick a ε-greedily; observe r and s'.
3. Compute δ and update $Q(s,a)$; set $s \leftarrow s'$; decay ε over time.
4. Repeat over many episodes. Example: $Q(s,a) = 2$, $r = 1$, $\gamma = 0.9$, $\max_{a'} Q(s',a') = 5$, $\alpha = 0.5$: $\delta = 1 + 4.5 - 2 = 3.5$, new $Q = 3.75$.

DECISION POINT **Value-based or policy-based?** Small discrete action sets → Q-learning/DQN (replay buffer + target network for stability). Continuous or very large action spaces, or stochastic policies needed → policy gradient / actor–critic (PPO is the robust default). **Exploration**: start $\varepsilon \approx 1$ and decay to 0.01–0.1.

TERMS

$Q(s,a)$ – current action-value estimate
α – learning rate (step size)
γ – discount factor; r reward
ε – exploration probability
π_θ – policy with parameters θ; $J(\theta)$ expected return
$b(s)$ – baseline (often $V(s)$); reduces variance without bias O.28 §14 Optimization
G_t – sampled return ML.42; gradient is a Monte Carlo estimate O.27 §14 Optimization
δ_t; $A(s,a)$ – temporal-difference error; advantage
$J(\theta)$; $V(s)$ – expected return of the policy; state value ML.42

CONTEXT

- **Origin.** TD learning by Sutton (1988); Q-learning by Watkins (1989); REINFORCE by Williams (1992); DQN (2015); PPO (2017).
- **Use.** Learning control policies from trial and error.
- **Data science.** Game-playing agents (Atari, Go), robotics, recommendation and bidding policies, and RLHF (PPO) for aligning language models.
- **Physical application.** Learning to balance an inverted pendulum (cart-pole) from reward alone is the classic benchmark.

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16
Electrical Engineering

Circuits, signals, control, electronics and digital logic. Electrical engineering applies algebra, complex numbers, differential equations, linear algebra and Fourier analysis to the devices and grids that power and connect the world, including the hardware your models run on.

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Circuit Fundamentals

EE.1 Charge, Current, Voltage, Power & Energy

Charge is the conserved quantity that moves; current is its rate of flow; voltage is the energy each unit of charge gains or loses between two points; power is the rate at which that energy is delivered.

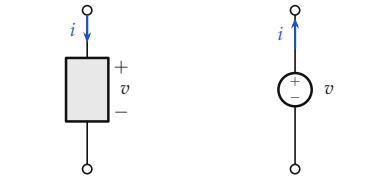
Current	$I = \dfrac{dQ}{dt}$	Rate of charge flow past a point; integrate to recover charge: $Q = \int I\,dt$.
Voltage	$V = \dfrac{dW}{dQ}$	Energy per coulomb between two points; always a *difference* (one point is the reference).
Electric power	$P = VI$	Rate of energy transfer; $P > 0$ means the element absorbs (passive sign convention).
Energy	$E = \displaystyle\int P\,dt = Pt \quad (\text{constant } P)$	Power accumulated over time.
Kilowatt-hour	$1\ \text{kWh} = 3.6 \times 10^6\ \text{J}$	The utility unit: 1000 W for 3600 s.
Charge quantization	$Q = ne, \quad e = 1.602 \times 10^{-19}\ \text{C}$	1 A is about 6.24×10^{18} electrons per second.

STATED-EQUATION *Current I equals the rate of flow of charge, d Q d t. Voltage V equals energy per unit charge, d W d Q. Power P equals V times I. Energy equals the integral of power over time, which is P t for constant power. One kilowatt-hour equals three point six million joules. A charge is a whole number of electron charges.*

MEMORY AID Water analogy: voltage = pressure, current = flow rate, resistance = pipe narrowness, power = pressure × flow.

TERMS

Q – charge, coulombs (C)
I – current, amperes (A = C/s); a derivative C1.9 §07 Calculus I
V – voltage (potential difference), volts (V = J/C)
W, E – energy, joules (J) or kWh P.6 §12 Physics
P – power, watts (W = J/s)
t – time (s); n number of electrons; e elementary charge
Passive sign convention – current enters the + terminal ⇒ $P = VI$ is power absorbed (figure)
n; e – whole number of charges; elementary charge



CONTEXT

- **Origin.** Volta's battery (1800); Ampère (1820s); the watt named for James Watt.
- **Use.** Sizing wiring and power supplies; reading utility bills.
- **Data science.** Estimating the energy and carbon cost of training runs: GPUs × watts × hours; data-center power budgeting.
- **Physical application.** A 1500 W space heater run for 3 hours uses 4.5 kWh, about $0.68 at $0.15/kWh.

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EE.2 Resistance, Ohm's Law & Resistivity

Resistance is the ratio of voltage to current in a conductor. It is set by the material (resistivity) and the geometry (length and cross-section), and the energy it absorbs becomes heat.

Ohm's law	$V = IR$	Linear voltage–current law of a resistor P.33 §12 Physics; the V–I graph is a line of slope R.
Conductance	$G = \dfrac{1}{R}$	Siemens (S); conductances add in parallel.
Joule heating	$P = I^2R = \dfrac{V^2}{R}$	Substitute Ohm's law into $P = VI$; all of it becomes heat.
Wire resistance	$R = \rho \dfrac{L}{A}$	Longer ⇒ more resistance, thicker ⇒ less.
Conductivity	$\sigma = \dfrac{1}{\rho}$	Material's ease of conduction (S/m).
Temperature coefficient	$R(T) \approx R_0\left[1 + \alpha\,(T - T_0)\right]$	Metals: $\alpha > 0$ (copper 0.0039/°C); thermistors can have $\alpha < 0$.

STATED-EQUATION *V equals I R. Conductance G equals one over R. The power dissipated is I squared R, which equals V squared over R. A wire's resistance equals resistivity rho times length over cross-sectional area. Conductivity sigma is one over rho. Resistance grows linearly with temperature: R of T is about R naught times one plus alpha times T minus T naught.*

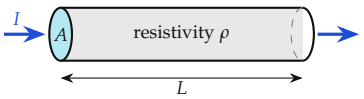
MEMORY AID "Long and thin, resistance wins"; "twice the current, four times the heat" (I^2R).

METHOD

1. Convert everything to SI: area in m^2 ($1\,\text{mm}^2 = 10^{-6}\,\text{m}^2$), length in m.
2. Look up ρ (copper 1.68×10^{-8}, aluminum $2.65 \times 10^{-8}\,\Omega\cdot\text{m}$).
3. Compute $R = \rho L/A$. Example: 10 m of $1\,\text{mm}^2$ copper: $R = 1.68 \times 10^{-8} \cdot 10/10^{-6} \approx 0.17\,\Omega$; at 10 A it wastes $I^2R = 17\,\text{W}$.

TERMS

V, I – voltage (V), current (A) EE.1
R – resistance, ohms ($\Omega = \text{V/A}$)
G – conductance, siemens ($\text{S} = 1/\Omega$)
ρ – resistivity ($\Omega\cdot\text{m}$), a material property
σ – conductivity (S/m)
L, A – conductor length (m), cross-sectional area (m^2) G.15 §05 Geometry
α – temperature coefficient ($1/°\text{C}$); T, T_0 temperature and reference (°C)
R_0; T, T_0 – resistance at the reference temperature; temperature and reference temperature



$R = \rho \dfrac{L}{A}$

$2L \Rightarrow 2R \qquad 2A \Rightarrow R/2$

CONTEXT

- **Origin.** Georg Ohm, *Die galvanische Kette* (1827); Joule's heating law (1841).
- **Use.** Wire-gauge selection, heaters, fuses, strain gauges and RTD temperature sensors.
- **Data science.** Estimating R from noisy (I, V) readings is a least-squares slope through the origin S.40 §13 Statistics.
- **Physical application.** A long, thin extension cord gets warm under a heater's load: its I^2R loss is heat.

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EE.3 Series & Parallel Combinations

Elements in series carry the same current; elements in parallel share the same voltage. Either group can be replaced by one equivalent element without changing the rest of the circuit.

Resistors in series	$R_{\text{eq}} = R_1 + R_2 + \cdots + R_n$	Same current, voltages add; R_{eq} exceeds the largest.
Resistors in parallel	$\dfrac{1}{R_{\text{eq}}} = \dfrac{1}{R_1} + \dfrac{1}{R_2} + \cdots + \dfrac{1}{R_n}$	Same voltage, currents add; R_{eq} is below the smallest.
Product over sum	$R_1 \parallel R_2 = \dfrac{R_1 R_2}{R_1 + R_2}$	Two-resistor shortcut; n equal resistors in parallel give R/n.
Inductors	$L_{\text{ser}} = L_1 + L_2, \quad \dfrac{1}{L_{\text{par}}} = \dfrac{1}{L_1} + \dfrac{1}{L_2}$	Combine like resistors (no magnetic coupling).
Capacitors	$C_{\text{par}} = C_1 + C_2, \quad \dfrac{1}{C_{\text{ser}}} = \dfrac{1}{C_1} + \dfrac{1}{C_2}$	Combine "upside down": parallel plates add area.
Impedances	$Z_{\text{ser}} = \sum_k Z_k, \quad \dfrac{1}{Z_{\text{par}}} = \sum_k \dfrac{1}{Z_k}$	The AC generalization for any mix of R, L, C EE.14.

STATED-EQUATION *Resistors in series add. In parallel, their reciprocals add, and two in parallel give the product over the sum. Inductors combine like resistors. Capacitors in parallel add, and in series their reciprocals add. Impedances combine exactly like resistors.*

MEMORY AID "R and L add in series; C adds in parallel."

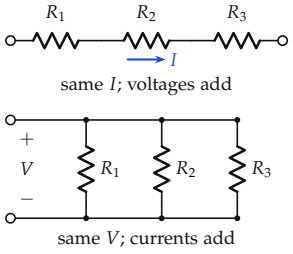
METHOD

1. Find two elements that share *both* nodes (parallel) or a node with nothing else attached (series).
2. Replace the pair by its equivalent and redraw.
3. Repeat until one element remains. Example: $6\,\Omega \parallel 3\,\Omega = \frac{18}{9} = 2\,\Omega$, then in series with $4\,\Omega$: $R_{\text{eq}} = 6\,\Omega$.

DECISION POINT Series or parallel? Ask "is the current forced to be the same?" (series) or "are both ends on the same two nodes?" (parallel). Some networks (a bridge) are neither: switch to nodal analysis EE.7 or a Thévenin equivalent EE.9.

TERMS

R – resistance (Ω); L inductance (H); C capacitance (F) EE.10
Z – impedance (Ω), complex EE.14
$\parallel$ – "in parallel with"
Equivalent – same V–I behavior at the terminals
Harmonic mean – $R_1 \parallel R_2$ is half the harmonic mean of R_1, R_2 S.1 §13 Statistics
$R_1, \ldots, R_n$; R_{eq} – individual resistors; equivalent resistance
L; C – inductance; capacitance EE.10



CONTEXT

- **Origin.** Follows directly from Kirchhoff's laws EE.4 (1845).
- **Use.** Simplifying networks, building non-standard values from standard parts.
- **Data science.** The F1 score is a harmonic mean, the same algebra as parallel resistance ML.17 §15 Machine Learning; parallel servers combine throughput like conductances.
- **Physical application.** Old holiday lights in series all go dark when one bulb fails; household outlets are wired in parallel so each gets the full 120 V.

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EE.4 Kirchhoff's Laws

Two conservation laws that hold in every lumped circuit: charge cannot pile up at a node (KCL), and the energy per charge around any closed loop returns to its starting value (KVL).

Kirchhoff's current law (KCL)	$\sum_k I_k = 0 \iff \sum I_{\text{in}} = \sum I_{\text{out}}$	At every node; take currents leaving as $+$ and entering as $-$ (or vice versa).
Kirchhoff's voltage law (KVL)	$\sum_{\text{loop}} V_k = 0$	Around any closed path: rises equal drops.
Single-loop current	$I = \dfrac{\sum V_{\text{src}}}{\sum R}$	KVL plus Ohm's law for one series loop.

STATED-EQUATION *Kirchhoff's current law: at any node the currents sum to zero, so current in equals current out. Kirchhoff's voltage law: around any closed loop the voltages sum to zero. In a single series loop, the current equals the total source voltage over the total resistance.*

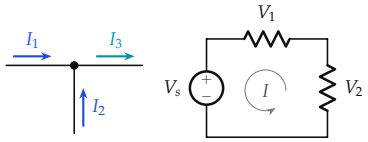
MEMORY AID KCL: "what flows in must flow out" (conservation of charge). KVL: "every round trip ends where it started" (conservation of energy).

METHOD

1. Assign a current direction to every branch (a wrong guess just comes out negative).
2. Mark each resistor's $+$ where the assumed current enters (passive sign convention EE.1).
3. Walk the loop: add $+V$ when you go from $-$ to $+$ (a rise), $-V$ for a drop; set the sum to 0.
4. Example: 12 V source with 2 Ω and 4 Ω in series: $12 - 2I - 4I = 0 \Rightarrow I = 2$ A.

TERMS

Node – junction where two or more elements meet
Loop – closed path through the circuit; a *mesh* has no loop inside it
I_k – branch currents (A); V_k element voltages (V)
Sign – rises and drops take opposite signs around a loop
Ohm's law – $V = IR$ for each resistor EE.2
Basis of – nodal and mesh analysis EE.7
V_k; V_{src} – voltage across element k; source voltages
R – resistances in the loop



KCL: $I_1 + I_2 = I_3$ KVL: $V_s - V_1 - V_2 = 0$

CONTEXT

- **Origin.** Gustav Kirchhoff (1845), as a student.
- **Use.** Analyzing any circuit.
- **Data science.** KCL is flow conservation, the same constraint as network flow O.9 §14 Optimization; the incidence matrix of a circuit graph encodes it.
- **Physical application.** The current into a house's breaker panel equals the sum of the currents to all running appliances.

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EE.5 Voltage & Current Dividers

Two series resistors split a voltage in proportion to their resistances; two parallel resistors split a current in inverse proportion. The fastest way to get one voltage or current without solving the whole circuit.

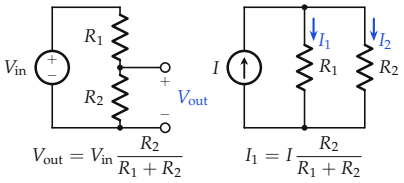
Voltage divider	$V_{\text{out}} = V_{\text{in}} \dfrac{R_2}{R_1 + R_2}$	V_{out} measured across R_2; only valid if no current is drawn from the midpoint.
General divider	$V_k = V \dfrac{R_k}{R_1 + \cdots + R_n}$	Each series resistor gets its share of the total.
Current divider (two)	$I_1 = I_{\text{total}} \dfrac{R_2}{R_1 + R_2}$	The *other* resistor is on top: more current takes the easier path.
General current divider	$I_k = I \dfrac{G_k}{G_1 + \cdots + G_n}$	With conductances $G = 1/R$ it looks like the voltage divider.
Loaded divider	$V_{\text{out}} = V_{\text{in}} \dfrac{R_2 \parallel R_L}{R_1 + R_2 \parallel R_L}$	A load in parallel with R_2 pulls V_{out} down.

STATED-EQUATION *The output voltage of a divider equals the input voltage times R two over R one plus R two. In a current divider with two parallel resistors, the current through R one equals the total current times R two over R one plus R two: the other resistor on top. With conductances, each branch gets its conductance's share of the current. A load resistor in parallel with R two lowers the output. In general, each of n series resistors gets the fraction R k over the total resistance of the voltage.*

MEMORY AID Voltage divider: "my resistor on top"; current divider: "the other resistor on top."

TERMS

R_1, R_2 – resistors (Ω); V_{out} is across R_2 EE.2
V_{in}, V_{out} – input and output voltages (V)
I, I_k – total and branch currents (A); G_k conductance (S)
R_L – load resistance; loading is negligible if $R_L \gg R_2$
$\parallel$ – parallel combination EE.3
Proportion – a weighted share A1.19 §03 Algebra I
V, V_k – total voltage; voltage across R_k
G_k – conductance $1/R_k$ EE.2
I_{total} – current entering the parallel pair



CONTEXT

- **Origin.** Follows from Ohm's and Kirchhoff's laws EE.4.
- **Use.** Sensor interfaces, level shifting, biasing, potentiometers.
- **Data science.** Reading analog sensors (thermistors, light sensors) into an ADC EE.27 for IoT data pipelines.
- **Physical application.** A volume knob (potentiometer) is an adjustable voltage divider: 9 V across 1 kΩ and 2 kΩ gives 6 V across the 2 kΩ.

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EE.6 Independent & Dependent Sources; Source Transformation

An independent source fixes a voltage or a current by itself. A dependent (controlled) source, drawn as a diamond, is set by a voltage or current elsewhere in the circuit; it is how transistors and amplifiers are modeled.

VCVS	$v = \mu\, v_x$	Voltage-controlled voltage source; μ is a dimensionless gain (op-amp model).
VCCS	$i = g_m\, v_x$	Voltage-controlled current source; g_m is a transconductance (MOSFET model).
CCVS	$v = r_m\, i_x$	Current-controlled voltage source; r_m is a transresistance.
CCCS	$i = \beta\, i_x$	Current-controlled current source; β is a current gain (BJT model).
Source transformation	V_s in series with $R \iff I_s = \dfrac{V_s}{R}$ in parallel with R	Same terminal behavior; use it to merge sources or simplify ladders.
Practical source	$v = V_s - i\,R_s$	Terminal voltage sags as load current rises (internal resistance).

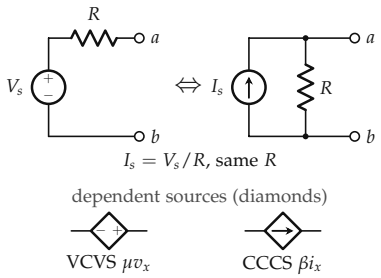
STATED-EQUATION *A voltage-controlled voltage source gives v equals mu times v x. A voltage-controlled current source gives i equals g m times v x. A current-controlled voltage source gives v equals r m times i x. A current-controlled current source gives i equals beta times i x. A voltage source V s in series with R is equivalent to a current source V s over R in parallel with R. A real source's terminal voltage is V s minus i R s.*

MEMORY AID Diamond = dependent ("it depends"). Never switch off a dependent source when finding R_{th} or using superposition.

DECISION POINT When to transform? When a voltage source sits in series with a resistor and you want to merge it with parallel branches (or the reverse). Do not transform away the resistor that carries the controlling variable v_x or i_x of a dependent source. An ideal source with no series (or parallel) resistor cannot be transformed.

TERMS

v_x, i_x – controlling voltage (V) or current (A) elsewhere in the circuit
μ, β – dimensionless voltage and current gains
g_m – transconductance (S = A/V) EE.33
r_m – transresistance (Ω = V/A)
V_s, I_s – source voltage (V), current (A); R_s internal resistance (Ω)
Symbols – independent: circle EE.41; dependent: diamond
v, i – output voltage and current of the source
R; R_s – source-transformation resistor; internal source resistance



CONTEXT

- **Origin.** Controlled sources arose with vacuum-tube and transistor small-signal models (1920s–1950s).
- **Use.** Modeling amplifiers and transistors; simplifying circuits before solving.
- **Data science.** A dependent source is a computational-graph edge: one node's value is a function of another's, exactly what backpropagation differentiates through ML.22 §15 Machine Learning.
- **Physical application.** A phone battery's voltage dips under heavy load because of its internal resistance R_s.

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Systematic Analysis & Network Theorems

EE.7 Nodal & Mesh Analysis

Systematic ways to write exactly enough independent equations for any linear circuit: KCL at each node in terms of node voltages (nodal), or KVL around each window in terms of mesh currents (mesh).

Nodal equation	$\sum_{\text{branches}} \frac{V_n - V_m}{R_{nm}} = I_{\text{src},n}$	KCL at node n: current leaving through each resistor equals current injected by sources.
Nodal matrix form	$G\,\mathbf{v} = \mathbf{i}$	G_{nn} = sum of conductances at n; $G_{nm} = -$(conductance between n and m).
Mesh equation	$\sum_{\text{loop}} R\,I = \sum V_{\text{source}} \;\Rightarrow\; R\,\mathbf{i} = \mathbf{v}$	KVL per mesh; a shared resistor carries the *difference* of the two mesh currents.

STATED-EQUATION *Nodal analysis: at each node, the sum over branches of the voltage difference over resistance equals the source current, giving the matrix equation G v equals i. Mesh analysis: around each loop, the sum of R times I equals the source voltages, giving R i equals v.*

MEMORY AID Nodal: "voltages unknown, KCL equations". Mesh: "currents unknown, KVL equations".

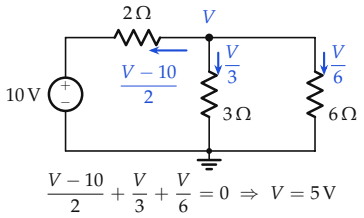
METHOD

1. Pick a reference node (ground, usually the one with most connections); label the other node voltages.
2. At each non-reference node write KCL with every current leaving: $\frac{V_n - V_m}{R}$; add injected source currents on the right.
3. A voltage source between two nodes forms a *supernode*: write one KCL around both plus $V_a - V_b = V_s$. (Mesh: a current source shared by two meshes forms a *supermesh*.)
4. Solve the linear system LA.17 §09 Linear Algebra. Example: node V fed from 10 V through 2 Ω, with 3 Ω and 6 Ω to ground: $\frac{V-10}{2} + \frac{V}{3} + \frac{V}{6} = 0 \Rightarrow 6V = 30$, $V = 5$ V.

DECISION POINT Nodal or mesh? Count unknowns: nodal needs (nodes $-1-$ voltage sources); mesh needs (meshes $-$ current sources). Choose the smaller. Current sources favor nodal, voltage sources favor mesh. Mesh works only for planar circuits; nodal always works (simulators use it).

TERMS

V_n – node voltage relative to ground (V)
G – conductance matrix (S), symmetric for resistor networks
$\mathbf{v}$, $\mathbf{i}$ – vectors of node voltages and injected currents
Mesh current – a loop current circulating in one window (A)
Solving – Gaussian elimination LA.17 §09 Linear Algebra, Cramer A2.25 §04 Algebra II
SPICE – circuit simulators assemble and solve these systems
R_{nm}; $I_{\text{src},n}$ – resistance between nodes n and m; source current into node n
R (matrix) – mesh resistance matrix



CONTEXT

- **Origin.** Formalized with matrix methods in the 20th century; SPICE simulator from Berkeley (1973).
- **Use.** Solving any size of linear circuit systematically.
- **Data science.** The graph Laplacian of a network is exactly a conductance matrix; "effective resistance" between nodes is used in graph ML and network centrality.
- **Physical application.** Power-grid operators solve huge nodal systems every few minutes to schedule generation.

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EE.8 Superposition Theorem

In a linear circuit with several independent sources, every voltage and current is the sum of the contributions of each source acting alone, with all other independent sources switched off.

Superposition	$x = \sum_k x_k, \quad x_k =$ response to source k alone	Holds for voltages and currents in any linear circuit.
Switching a source off	$V_s = 0 \to$ short, $\qquad I_s = 0 \to$ open	Replace, do not remove; dependent sources stay on.
Power is not superposable	$P = R\left(\sum_k i_k\right)^2 \neq \sum_k R\, i_k^2$	Add the currents first, then compute power.

STATED-EQUATION *The response x equals the sum of the responses to each source acting alone. A switched-off voltage source becomes a short circuit and a switched-off current source becomes an open circuit. Power is not superposable, because it depends on the square of the current.*

MEMORY AID "One at a time, then add"; "dead voltage source = wire, dead current source = gap."

METHOD

1. Keep one independent source; short the other voltage sources and open the other current sources.
2. Solve the simpler circuit for the wanted quantity (dividers often suffice EE.5).
3. Repeat for each source and add the results with their signs. Example (figure): 10 V alone gives $V_a = 10 \cdot \frac{2}{2+2} = 5$ V; 1 A alone sees $2 \parallel 2 = 1\,\Omega$, so $V_a = 1$ V; total $V_a = 6$ V.

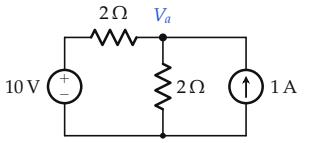
DECISION POINT Superposition pays off when sources are of different kinds or, essentially, at different frequencies (a DC bias plus an AC signal: each needs its own impedances EE.14). For one frequency and many sources, nodal analysis EE.7 is usually fewer steps.

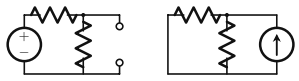
TERMS

Linear circuit – only R, L, C, linear dependent sources and independent sources
Linearity – additivity + homogeneity LA.9 §09 Linear Algebra
x, x_k – total response and response to source k (V or A)
V_s, I_s – independent source values (V, A)
P – power (W), quadratic in i EE.2
i_k; R – current due to source k; resistance dissipating the power

CONTEXT

- **Origin.** A consequence of linearity, used by Helmholtz (1853) and Thévenin.
- **Use.** Circuits with several sources, DC + AC biasing, noise analysis.
- **Data science.** Linear models decompose a prediction into per-feature contributions $\beta_j x_j$ for the same reason; for linear models SHAP values are exactly these terms.
- **Physical application.** An audio mixer's output is the sum of what each microphone alone would produce.





10 V alone (1 A open): 5 V
1 A alone (10 V shorted): 1 V

$V_a = 5 + 1 = 6$ V

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EE.9 Thévenin & Norton Equivalents; Maximum Power Transfer

Seen from two terminals, any linear circuit behaves exactly like one voltage source with one series resistor (Thévenin) or one current source with one parallel resistor (Norton). A load gets the most power when it matches that resistance.

Thévenin voltage	$V_{\text{th}} = V_{\text{oc}}$	The open-circuit voltage at the terminals.
Thévenin resistance	$R_{\text{th}} = \dfrac{V_{\text{oc}}}{I_{\text{sc}}}$	Or: switch sources off and combine resistors; or apply a test source.
Norton current	$I_N = I_{\text{sc}} = \dfrac{V_{\text{th}}}{R_{\text{th}}}$	Short-circuit current; Norton = source transformation of Thévenin EE.6.
Load voltage	$V_L = V_{\text{th}} \dfrac{R_L}{R_{\text{th}} + R_L}$	A divider once the equivalent is drawn EE.5.
Maximum power transfer	$R_L = R_{\text{th}} \;\Rightarrow\; P_{\max} = \dfrac{V_{\text{th}}^2}{4R_{\text{th}}}$	Found by setting $dP/dR_L = 0$ C1.26 §07 Calculus I.
Efficiency	$\eta = \dfrac{R_L}{R_{\text{th}} + R_L}$	Only 50% at maximum power; grids use $R_L \gg R_{\text{th}}$.

STATED-EQUATION *The Thevenin voltage is the open-circuit voltage. The Thevenin resistance equals the open-circuit voltage over the short-circuit current. The Norton current equals V t h over R t h. The load voltage is V t h times R L over R t h plus R L. Maximum power transfer: when the load resistance equals R t h, the load power is V t h squared over four R t h, at fifty percent efficiency.*

MEMORY AID Any linear circuit, seen from two terminals, is "one battery plus one resistor."

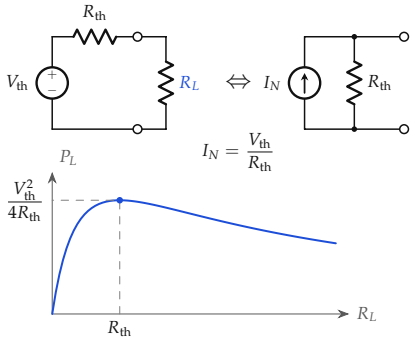
METHOD

1. Remove the load; label the terminals a, b.
2. Find $V_{\text{oc}} = V_{ab}$ with any method (divider, nodal, superposition).
3. Find R_{th}: independent sources only $\rightarrow$ switch them off and combine resistors; with dependent sources $\rightarrow$ apply a test source V_t and use $R_{\text{th}} = V_t / I_t$ (or $V_{\text{oc}} / I_{\text{sc}}$).
4. Redraw V_{th} in series with R_{th} and reconnect the load. Example: 12 V, 4 Ω series, 12 Ω shunt: $V_{\text{th}} = 12 \cdot \frac{12}{16} = 9\,\text{V}$, $R_{\text{th}} = 4 \parallel 12 = 3\,\Omega$, $P_{\max} = \frac{81}{12} = 6.75\,\text{W}$.

DECISION POINT Thévenin or Norton? They are interchangeable; pick the one that makes the next step a single divider: Thévenin when the load sits in series with other parts, Norton when it sits in parallel with other branches or the circuit is full of current sources. Use either when you must try many loads: solve the network once, then each load is one line.

TERMS

V_{th}, R_{th} – equivalent source (V) and series resistance (Ω)
V_{oc}, I_{sc} – open-circuit voltage (V), short-circuit current (A)
I_N – Norton current source (A), in parallel with R_{th}
R_L, V_L – load resistance (Ω) and voltage (V)
η – efficiency, fraction of power reaching the load



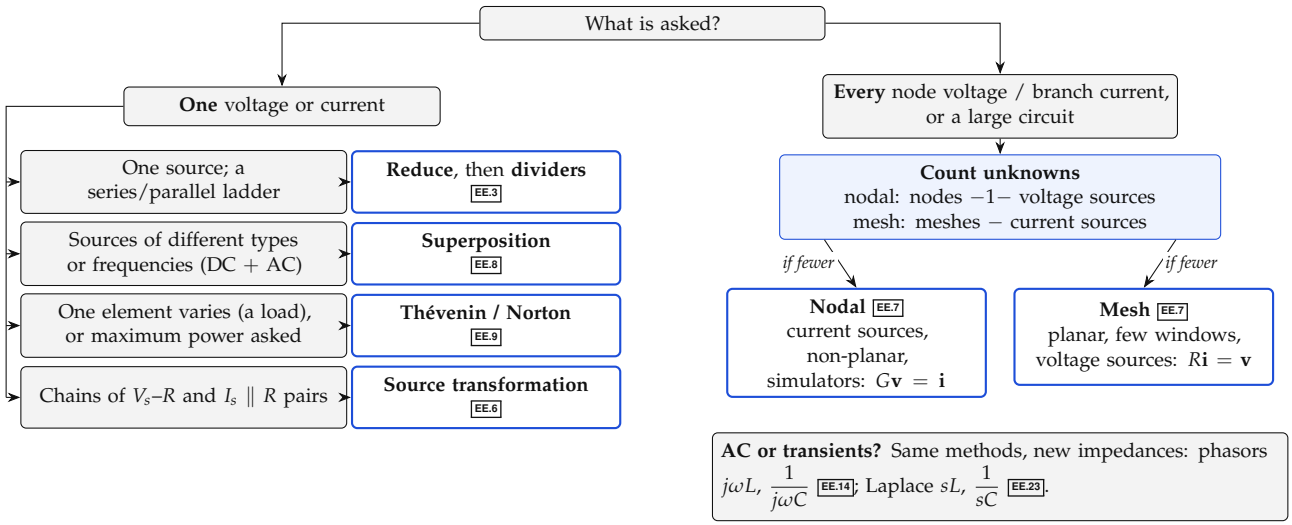
CONTEXT

- **Origin.** Léon Charles Thévenin (1883); Edward Norton (1926); Jacobi's maximum power theorem (1840).
- **Use.** Simplifying circuits, matching amplifiers and antennas.
- **Data science.** A "black-box" surrogate model: replace a complex subsystem by a simple equivalent that reproduces its input–output behavior.
- **Physical application.** Audio amplifiers are matched to speaker impedance (e.g. 8 Ω) to deliver maximum power.

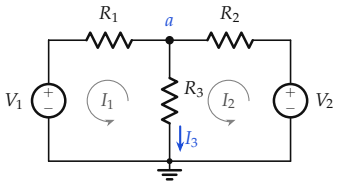
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Circuit Analysis Toolkit: Which Method When | *Key Topic EE.K1*

Every linear circuit can be solved by several methods, and they always agree. Speed and error rate depend on choosing well: count the unknowns, look at the kinds of sources, and ask whether you need one quantity or all of them. Below, one circuit is solved four ways, followed by the decision map and the most common mistakes.



One circuit, four methods



Find V_a and the current I_3 down through R_3. $V_1 = 10\,\text{V}$, $V_2 = 2\,\text{V}$, $R_1 = R_2 = R_3 = 2\,\Omega$.

1 Nodal EE.7 (1 unknown: fastest)

KCL at a, currents leaving: $\frac{V_a-10}{2} + \frac{V_a-2}{2} + \frac{V_a}{2} = 0 \Rightarrow 3V_a = 12$, so $V_a = 4\,\text{V}$ and $I_3 = V_a/R_3 = 2\,\text{A}$.

2 Mesh EE.7 (2 unknowns)

Clockwise I_1 (left), I_2 (right). Left: $10 - 2I_1 - 2(I_1 - I_2) = 0$. Right: $-2I_2 - 2 - 2(I_2 - I_1) = 0$. So $4I_1 - 2I_2 = 10$, $-2I_1 + 4I_2 = -2 \Rightarrow I_1 = 3\,\text{A}$, $I_2 = 1\,\text{A}$, $I_3 = I_1 - I_2 = 2\,\text{A}$.

3 Superposition EE.8 (2 easy dividers)

V_1 alone: $R_2 \parallel R_3 = 1\,\Omega$, $V_a' = 10 \cdot \frac{1}{2+1} = \frac{10}{3}$. V_2 alone: $V_a'' = 2 \cdot \frac{1}{3} = \frac{2}{3}$. Sum: $V_a = 4\,\text{V}$.

4 Thévenin seen by R_3 EE.9 (best if R_3 will change)

Remove R_3: $V_{\text{th}} = \frac{10+2}{2} = 6\,\text{V}$ (midpoint of equal resistors), $R_{\text{th}} = 2 \parallel 2 = 1\,\Omega$. Reconnect: $V_a = 6 \cdot \frac{2}{1+2} = 4\,\text{V}$.

Quick reference

Series/parallel reduction EE.3: one source, a ladder network; collapse, then expand back out with dividers.

Dividers EE.5: one voltage across a series string, or one branch current of a parallel pair.

Nodal EE.7: default general method; best with current sources and many parallel branches; the only choice for non-planar circuits.

Mesh EE.7: planar circuits with few windows and voltage sources in series with resistors.

Superposition EE.8: sources of different types or frequencies (DC bias + AC signal).

Source transformation EE.6: chains of V_s–R and $I_s \parallel R$ pairs that collapse into one.

Thévenin / Norton EE.9: one element varies (a load, a sensor), or maximum power transfer is asked.

Phasors / Laplace EE.14, EE.23: the same methods with impedances $j\omega L$, $\frac{1}{j\omega C}$ (steady AC) or sL, $\frac{1}{sC}$ (transients).

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Common mistakes

- Mixing sign conventions: decide "currents leaving" for every nodal equation and keep it.
- Switching off a *dependent* source in superposition or when finding R_{th}.
- Adding powers from superposition; add currents or voltages, then square.
- Using a divider when the midpoint is loaded EE.5.
- Forgetting that a voltage source between two non-ground nodes needs a supernode.
- Treating a capacitor as a short (it is open in DC steady state) or an inductor as open (it is a short).

DECISION POINT **Count before you solve:** nodal unknowns $=$ nodes $-1-$ voltage sources; mesh unknowns $=$ meshes $-$ current sources; pick the smaller. **Need just one quantity?** Try dividers, superposition or a Thévenin equivalent first. **Need everything, or the circuit is large?** Nodal in matrix form $G\mathbf{v} = \mathbf{i}$, solved numerically.

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Energy Storage & Transients

EE.10 Capacitors & Inductors

A capacitor stores energy in an electric field and resists sudden changes in voltage; an inductor stores energy in a magnetic field and resists sudden changes in current. Their laws involve derivatives, which is what makes circuits dynamic.

Capacitance	$Q = CV$	Charge stored per volt across the plates.
Capacitor law	$i = C\,\dfrac{dv}{dt}$	Current flows only while the voltage changes; DC steady state ⇒ open circuit.
Inductor law	$v = L\,\dfrac{di}{dt}$	Voltage appears only while the current changes; DC steady state ⇒ short circuit.
Stored energy	$E_C = \tfrac{1}{2}CV^2, \qquad E_L = \tfrac{1}{2}LI^2$	Compare $\tfrac{1}{2}mv^2$ and $\tfrac{1}{2}kx^2$ P.6 §12 Physics.
Combination	$C_{\parallel} = C_1 + C_2, \qquad \dfrac{1}{C_{\text{series}}} = \dfrac{1}{C_1} + \dfrac{1}{C_2}$	Capacitors combine like resistors "upside down"; inductors like resistors EE.3.
Parallel-plate capacitor	$C = \varepsilon_r \varepsilon_0\,\dfrac{A}{d}$	Bigger plates, smaller gap, better dielectric ⇒ more capacitance.
Continuity	$v_C(0^+) = v_C(0^-), \qquad i_L(0^+) = i_L(0^-)$	Capacitor voltage and inductor current cannot jump (that would need infinite power).

STATED-EQUATION *Charge equals capacitance times voltage. Capacitor current equals C times d v d t. Inductor voltage equals L times d i d t. Stored energy is one half C V squared for a capacitor and one half L I squared for an inductor. Capacitors in parallel add; in series, their reciprocals add. A parallel-plate capacitance equals epsilon r epsilon naught A over d. Capacitor voltage and inductor current are continuous.*

MEMORY AID Capacitors combine like resistors "upside down." A capacitor resists sudden voltage changes; an inductor resists sudden current changes.

TERMS

C – capacitance, farads (F = C/V); typically μF or pF
L – inductance, henries (H = V·s/A)
v, i – instantaneous voltage (V) and current (A); dv/dt, di/dt derivatives C1.9 §07 Calculus I
E_C, E_L – stored energy (J)
ε_0 – 8.85×10^{-12} F/m; ε_r relative permittivity (dielectric constant)
A, d – plate area (m^2), gap (m)
0^-, 0^+ – instants just before and after a switch at $t = 0$
Q; V – stored charge; capacitor voltage
ε_r – relative permittivity of the dielectric
v_C, i_L – capacitor voltage; inductor current

CONTEXT

- **Origin.** Leyden jar (1745); Faraday's induction (1831); Joseph Henry.
- **Use.** Energy storage, filtering, timing, power conversion.
- **Data science.** Supercapacitors and battery buffers smooth GPU power spikes in data centers; capacitive touchscreens sense changes in C.
- **Physical application.** A camera flash charges a capacitor slowly, then dumps its energy in a millisecond.

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EE.11 RC & RL Transients

A circuit with one capacitor or one inductor (first order) moves exponentially from its initial value to its new steady state after a switch, with a single time constant τ.

Capacitor charging	$v_C(t) = V_s\left(1 - e^{-t/\tau}\right)$	From 0 V toward the source voltage V_s.
Capacitor discharging	$v_C(t) = V_0\, e^{-t/\tau}$	From V_0 toward 0 through a resistor.
Time constants	$\tau_{RC} = RC, \qquad \tau_{RL} = \dfrac{L}{R}$	R is the Thévenin resistance seen by the C or L EE.9.
RL current rise	$i_L(t) = \dfrac{V_s}{R}\left(1 - e^{-t/\tau}\right)$	Inductor current builds toward V_s/R.
General first-order solution	$x(t) = x(\infty) + \left[x(0^+) - x(\infty)\right]e^{-t/\tau}$	Works for any voltage or current in a first-order circuit.
Governing ODE	$RC\,\dfrac{dv}{dt} + v = V_s$	KVL around the loop; a linear first-order ODE C2.28 §08 Calculus II.

STATED-EQUATION *While charging, the capacitor voltage equals V s times one minus e to the minus t over tau. While discharging, it equals V naught e to the minus t over tau. The time constant is R C for an RC circuit and L over R for an RL circuit. Inductor current rises as V s over R times one minus e to the minus t over tau. In general, x of t equals final value plus initial minus final, times e to the minus t over tau. The capacitor voltage obeys R C times d v d t plus v equals V s.*

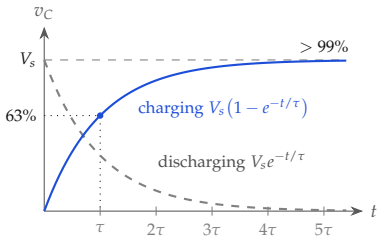
MEMORY AID After 1τ: 63% of the way; after 5τ: over 99% ("five tau and done").

METHOD

1. Before the switch (DC steady state, C open, L short): find $v_C(0^-)$ or $i_L(0^-)$; continuity gives $x(0^+)$ EE.10.
2. After the switch, long time: find $x(\infty)$ again with C open, L short.
3. Switch sources off, find R_{th} seen by the C or L; $\tau = R_{\text{th}}C$ or L/R_{th}.
4. Substitute into the general solution. Example: 10 V, 2 kΩ, 5 μF, uncharged: $\tau = 10\,\text{ms}$, $v_C(10\text{ ms}) = 10(1 - e^{-1}) \approx 6.3\,\text{V}$.

TERMS

τ – time constant (s)
V_s, V_0 – source voltage and initial capacitor voltage (V)
$x(0^+)$, $x(\infty)$ – value just after switching and final value
$e^{-t/\tau}$ – exponential decay A2.13 §04 Algebra II
Same math as – Newton's cooling P.25 §12 Physics
R, C, L; t – circuit elements; time since switching
v_C, i_L – capacitor voltage; inductor current



CONTEXT

- **Origin.** Solved with the differential equations of Kirchhoff and Heaviside (1880s).
- **Use.** Timers, debouncing buttons, sensor filtering, pulse shaping.
- **Data science.** An exponential moving average is a discrete RC low-pass filter: $s_t = \alpha x_t + (1 - \alpha)s_{t-1}$ EE.28.
- **Physical application.** A car's interior light fades out slowly after you close the door, thanks to an RC circuit.

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EE.12 Second-Order (RLC) Transients & Damping

A circuit with both an inductor and a capacitor stores energy in two places, so it obeys a second-order ODE. The damping ratio decides whether the response creeps, snaps, or rings on its way to steady state.

Standard form	$\ddot{x} + 2\alpha\,\dot{x} + \omega_0^2\,x = \omega_0^2\,x(\infty)$	Series RLC capacitor voltage: $LC\ddot{v} + RC\dot{v} + v = V_s$.
Characteristic roots	$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$	Roots of $s^2 + 2\alpha s + \omega_0^2 = 0$ A1.15 §03 Algebra I.
Damping parameters	$\alpha_{\text{ser}} = \dfrac{R}{2L},\quad \alpha_{\text{par}} = \dfrac{1}{2RC},\quad \omega_0 = \dfrac{1}{\sqrt{LC}},\quad \zeta = \dfrac{\alpha}{\omega_0}$	α sets the decay rate, ω_0 the natural frequency.
Overdamped ($\zeta > 1$)	$x_n = A_1 e^{s_1 t} + A_2 e^{s_2 t}$	Two real roots: slow, no overshoot.
Critically damped ($\zeta = 1$)	$x_n = (A_1 + A_2 t)\,e^{-\alpha t}$	Repeated root: fastest response with no overshoot.
Underdamped ($\zeta < 1$)	$x_n = e^{-\alpha t}\left(B_1 \cos \omega_d t + B_2 \sin \omega_d t\right)$	Complex roots A2.6 §04 Algebra II: decaying ringing at $\omega_d = \sqrt{\omega_0^2 - \alpha^2}$.
Overshoot & settling	$\%OS = 100\,e^{-\pi\zeta/\sqrt{1-\zeta^2}},\qquad t_s \approx \dfrac{4}{\alpha}$	Step-response specs; t_s is the 2% settling time.

STATED-EQUATION *The standard form is x double-dot plus two alpha x dot plus omega naught squared x equals omega naught squared times the final value. The characteristic roots are minus alpha plus or minus the square root of alpha squared minus omega naught squared. For a series RLC, alpha is R over two L; for a parallel RLC, one over two R C; omega naught is one over root L C; the damping ratio zeta is alpha over omega naught. Overdamped: two real exponentials. Critically damped: A one plus A two t, times e to the minus alpha t. Underdamped: a decaying sinusoid at omega d, the square root of omega naught squared minus alpha squared. The percent overshoot is one hundred times e to the minus pi zeta over root one minus zeta squared, and the settling time is about four over alpha.*

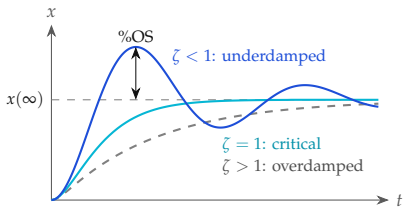
MEMORY AID $\zeta > 1$ "sluggish", $\zeta = 1$ "just right", $\zeta < 1$ "ringing". Series RLC: more $R \Rightarrow$ more damping; parallel RLC: more $R \Rightarrow$ less.

METHOD

1. Find $x(0^+)$ and $\dot{x}(0^+)$ from continuity: $\dot{v}_C = i_C/C$, $\dot{i}_L = v_L/L$ EE.10; find $x(\infty)$ from DC steady state.
2. Compute α, ω_0; compare them to choose the case.
3. Write $x(t) = x(\infty) + x_n(t)$ and fit the two constants to $x(0^+)$ and $\dot{x}(0^+)$. Example: series $R = 2\,\Omega$, $L = 1\,\text{H}$, $C = \frac{1}{5}\,\text{F}$: $\alpha = 1$, $\omega_0 = \sqrt{5}$, underdamped with $\omega_d = 2\,\text{rad/s}$.

TERMS

α – neper frequency (1/s); ω_0 undamped natural frequency (rad/s)
ζ – damping ratio (dimensionless)
ω_d – damped frequency (rad/s)
$s_{1,2}$ – natural frequencies (1/s), the poles EE.24
A_i, B_i – constants set by the two initial conditions
x_n – natural (homogeneous) response; same ODE as a mass–spring–damper P.15 §12 Physics
ω_0; $x(\infty)$ – undamped natural frequency; final value
$\dot{x}$, $\ddot{x}$ – first and second time derivatives
$\%OS$; t_s – percent overshoot; 2% settling time
R, L, C – resistance, inductance, capacitance



CONTEXT

- **Origin.** Kelvin's analysis of the Leyden-jar discharge (1853) showed it can oscillate.
- **Use.** Power-supply ringing, suspension and servo design, filter step responses.
- **Data science.** Gradient descent with momentum is a discretized damped oscillator: too little friction rings, too much crawls ML.27 §15 Machine Learning.
- **Physical application.** Car suspensions are tuned slightly underdamped ($\zeta \approx 0.3$–0.7) for comfort and control.

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AC Circuits & Power

EE.13 AC Signals & RMS Values

Alternating current is a sinusoid described by amplitude, frequency and phase. The RMS value is the DC level that would deliver the same average power to a resistor.

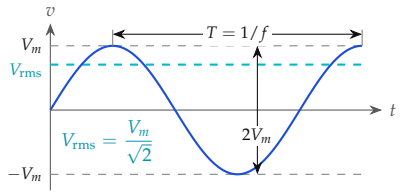
Sinusoid	$v(t) = V_m \sin(\omega t + \phi)$	Amplitude V_m, angular frequency ω, phase ϕ T.12 §06 Trigonometry.
Frequency and period	$\omega = 2\pi f, \qquad T = \dfrac{1}{f}$	f in cycles per second (Hz), ω in rad/s T.2 §06 Trigonometry.
RMS value	$V_{\text{rms}} = \sqrt{\dfrac{1}{T}\displaystyle\int_0^T v^2\,dt}$	Root of the mean of the square over one period C2.12 §08 Calculus II.
RMS of a sine	$V_{\text{rms}} = \dfrac{V_m}{\sqrt{2}} \approx 0.707\,V_m$	Because the average of $\sin^2$ is $\frac{1}{2}$ C2.2 §08 Calculus II.
Other waveforms	square: V_m, triangle: $\dfrac{V_m}{\sqrt{3}}$	RMS depends on the waveform shape.
Average power to a resistor	$P = \dfrac{V_{\text{rms}}^2}{R} = I_{\text{rms}}^2 R$	The DC formulas work with RMS values.

STATED-EQUATION *v of t equals V m sine of omega t plus phi. Omega equals two pi f, and the period equals one over f. The RMS voltage is the square root of the average of v squared over one period, which for a sine wave equals V m over root two. A square wave's RMS is V m and a triangle's is V m over root three. Average power to a resistor is V r m s squared over R.*

MEMORY AID RMS = Root of the Mean of the Square; for sine waves, peak $\approx 1.414\times$ RMS.

TERMS

V_m – peak (amplitude) (V); peak-to-peak $= 2V_m$
f – frequency (Hz); ω angular frequency (rad/s)
T – period (s); ϕ phase (rad or degrees)
RMS – effective value: equivalent DC for heating
Mains – US 120 V_{rms} at 60 Hz; Europe 230 V at 50 Hz
ω; ϕ – angular frequency; phase
I_{rms}; R – RMS current; resistance



CONTEXT

- **Origin.** AC systems championed by Tesla and Westinghouse (1880s "war of the currents").
- **Use.** Power grids, audio, communications.
- **Data science.** RMS is the same as the standard deviation of a zero-mean signal S.2 §13 Statistics; RMSE is RMS of errors ML.19 §15 Machine Learning.
- **Physical application.** A US outlet labeled 120 V actually swings to about ± 170 V peak.

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EE.14 Phasors & Impedance

At a single frequency, every voltage and current in a linear circuit is a sinusoid at that frequency; a phasor records only its amplitude and phase as a complex number, turning differential equations into complex algebra.

Phasor	$V_m\cos(\omega t+\phi) \;\leftrightarrow\; \tilde{V}=V_m\angle\phi = V_m e^{j\phi}$	Drop $e^{j\omega t}$, keep amplitude and phase (Euler T.23 §06 Trigonometry).
Impedances	$Z_R=R,\qquad Z_L=j\omega L,\qquad Z_C=\dfrac{1}{j\omega C}=-\dfrac{j}{\omega C}$	Inductors block high frequencies, capacitors block low ones.
Ohm's law for AC	$\tilde{V}=\tilde{I}\,Z$	All DC methods (dividers, nodal, Thévenin) work with complex Z.
Rectangular and polar	$Z=R+jX,\quad \lvert Z\rvert=\sqrt{R^2+X^2},\quad \theta=\tan^{-1}\dfrac{X}{R}$	$\lvert Z\rvert=\lvert\tilde{V}\rvert/\lvert\tilde{I}\rvert$; θ is how far v leads i T.19 §06 Trigonometry.
Admittance	$Y=\dfrac{1}{Z}=G+jB$	Admittances add in parallel.
Derivative rule	$\dfrac{d}{dt}\;\leftrightarrow\; j\omega$	Why $Z_L=j\omega L$ and $Z_C=1/(j\omega C)$.

STATED-EQUATION *The phasor of V m cosine of omega t plus phi is V m at angle phi, which is V m e to the j phi. The impedance of a resistor is R; of an inductor, j omega L; of a capacitor, one over j omega C, which equals minus j over omega C. Phasor voltage equals phasor current times impedance. Z equals R plus j X, with magnitude the square root of R squared plus X squared and angle the inverse tangent of X over R. Admittance Y is one over Z. Differentiation becomes multiplication by j omega.*

MEMORY AID ELI the ICE man: in an inductor (L), voltage (E) leads current (I); in a capacitor (C), current (I) leads voltage (E).

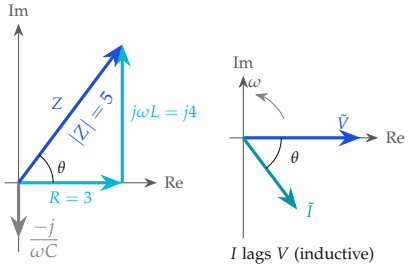
METHOD

1. Write every source as a cosine phasor ($\sin x=\cos(x-90^\circ)$).
2. Replace each element by its impedance at the source's ω.
3. Solve with any DC technique using complex arithmetic A2.6 §04 Algebra II.
4. Convert back: $\tilde{I}=I_m\angle\theta \Rightarrow i(t)=I_m\cos(\omega t+\theta)$. Example: $10\angle0^\circ$ V across $3\,\Omega$ in series with $\omega L=4\,\Omega$: $Z=3+j4=5\angle53.1^\circ$, $\tilde{I}=2\angle-53.1^\circ$ A, so $i=2\cos(\omega t-53.1^\circ)$ A.

DECISION POINT Phasors only describe sinusoidal steady state at one frequency. Several frequencies → phasors per frequency, then superposition EE.8. Switching transients or non-sinusoidal inputs → Laplace EE.23.

TERMS

j – imaginary unit ($j^2=-1$); engineers use j because i is current A2.6 §04 Algebra II
$\tilde{V}$, $\tilde{I}$ – phasors (V, A), complex numbers
Z – impedance (Ω); R resistance, X reactance: ωL or $-\frac{1}{\omega C}$
Y – admittance (S); G conductance, B susceptance
θ – impedance angle; $\angle$ polar angle T.19 §06 Trigonometry
R, X – resistance and reactance
G, B – conductance and susceptance
ω; V_m, ϕ – angular frequency; amplitude and phase



CONTEXT

- **Origin.** Charles Steinmetz (1893) introduced phasors, turning differential equations into algebra.
- **Use.** AC circuit analysis, power systems, RF design.
- **Data science.** Complex-valued features from FFTs EE.26 carry magnitude and phase exactly like phasors.
- **Physical application.** Motors are mostly inductive, so their current lags the voltage.

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EE.15 AC Power & Power Factor

With AC, only the part of the current in phase with the voltage does work (real power); the rest sloshes back and forth (reactive power). Their vector sum, apparent power, sizes the equipment.

Real power	$P = V_{\text{rms}} I_{\text{rms}} \cos\theta$	Average power that does work (W).
Reactive power	$Q = V_{\text{rms}} I_{\text{rms}} \sin\theta$	Energy exchanged with L and C each cycle (var); $Q > 0$ inductive.
Complex power	$S = \tilde{V}\tilde{I}^{*} = P + jQ, \qquad \lvert S\rvert^2 = P^2 + Q^2$	The power triangle (rms phasors); $\lvert S\rvert$ in VA.
Power factor	$\text{pf} = \cos\theta = \dfrac{P}{\lvert S\rvert}$	1 for a pure resistor; "lagging" when inductive.
Power-factor correction	$Q_C = P\,(\tan\theta_1 - \tan\theta_2), \qquad C = \dfrac{Q_C}{\omega V_{\text{rms}}^2}$	Shunt capacitors supply the reactive power locally.

STATED-EQUATION *Real power P equals V r m s times I r m s times cosine theta. Reactive power Q equals V r m s times I r m s times sine theta. Complex power S equals V times the conjugate of I, which is P plus j Q, and the magnitude of S squared equals P squared plus Q squared. The power factor equals cosine theta, which is P over the magnitude of S. Correcting from theta one to theta two needs Q C equals P times tan theta one minus tan theta two, from a capacitor C equals Q C over omega V r m s squared.*

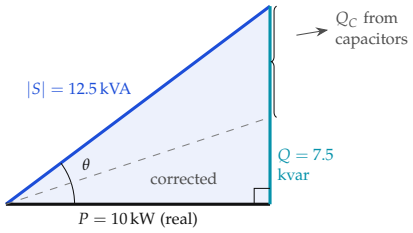
MEMORY AID The power triangle: P is the "beer" (useful), Q the "foam," $|S|$ the whole glass.

METHOD

1. From P and the old pf: $\theta_1 = \cos^{-1}\text{pf}_1$, $Q_1 = P\tan\theta_1$.
2. Target $Q_2 = P\tan\theta_2$; the capacitor supplies $Q_C = Q_1 - Q_2$.
3. Size $C = Q_C/(\omega V_{\text{rms}}^2)$. Example: 10 kW at pf 0.8 lagging: $Q_1 = 7.5$ kvar, $|S| = 12.5$ kVA; full correction needs 7.5 kvar of capacitors and $|S|$ drops to 10 kVA.

TERMS

P – real power, watts (W)
Q – reactive power, volt-ampere reactive (var)
$|S|$ – apparent power, volt-amperes (VA); sizes wires and transformers
θ – angle by which v leads i EE.14; $\tilde{I}^{*}$ complex conjugate
Right triangle – Pythagorean G.7 §05 Geometry, trig ratios T.1 §06 Trigonometry
C – correction capacitance (F); $\omega = 2\pi f$
S; $\tilde{I}^{*}$ – complex power; complex conjugate of the current phasor
Q_C; θ_1, θ_2 – reactive power the capacitor supplies; power-factor angles before and after
pf – power factor



$\text{pf} = \cos\theta = P/|S| = 0.8$

CONTEXT

- **Origin.** Steinmetz and early utility engineers (1890s–1900s).
- **Use.** Utility billing, motor sizing, grid efficiency.
- **Data science.** Industrial energy analytics flag low power factor from smart-meter data to cut penalties.
- **Physical application.** Factories are charged extra when their power factor is low, because the grid must carry the extra current.

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EE.16 RLC Circuits & Resonance

At one frequency the inductor's and capacitor's reactances cancel. A series RLC then has minimum impedance and maximum current; the quality factor Q measures how sharp that peak is.

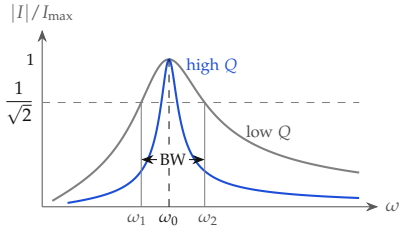
Resonant frequency	$\omega_0 = \dfrac{1}{\sqrt{LC}}, \qquad f_0 = \dfrac{1}{2\pi\sqrt{LC}}$	Where $\omega L = \frac{1}{\omega C}$.
Series impedance	$Z = R + j\left(\omega L - \dfrac{1}{\omega C}\right)$	Purely resistive ($Z = R$) at ω_0.
Quality factor (series)	$Q_s = \dfrac{1}{R}\sqrt{\dfrac{L}{C}} = \dfrac{\omega_0 L}{R}$	High Q: tall, narrow peak; $Q = \frac{1}{2\zeta}$ EE.12.
Quality factor (parallel)	$Q_p = R\sqrt{\dfrac{C}{L}}$	For parallel RLC, larger R means sharper resonance.
Bandwidth	$\mathrm{BW} = \omega_2 - \omega_1 = \dfrac{\omega_0}{Q}$	Width between the half-power (-3 dB) frequencies EE.17.
Half-power frequencies	$\omega_{1,2} \approx \omega_0 \mp \dfrac{\mathrm{BW}}{2} \quad (Q \gtrsim 5)$	Nearly symmetric about ω_0 for high Q.

STATED-EQUATION *The resonant angular frequency omega naught equals one over the square root of L C, so f naught equals one over two pi root L C. The series impedance is R plus j times omega L minus one over omega C. The quality factor of a series RLC equals one over R times the square root of L over C, which is omega naught L over R; for a parallel RLC it is R root C over L. The bandwidth equals omega naught over Q, centered on omega naught. The half-power frequencies sit about half a bandwidth below and above omega naught, when Q is at least about five.*

MEMORY AID At resonance the inductor and capacitor reactances cancel: $\omega L = \frac{1}{\omega C}$.

TERMS

ω_0 – resonant (natural) frequency (rad/s); compare $\sqrt{k/m}$ P.15 §12 Physics
Q – quality factor (dimensionless): sharpness of resonance
BW – bandwidth (rad/s) of the -3 dB band
ω_1, ω_2 – lower and upper half-power frequencies
Series resonance – impedance minimum; parallel resonance: maximum
Transient view – same ω_0 and damping EE.12
f_0; Z – resonant frequency in hertz; series impedance
R, L, C – circuit elements



CONTEXT

- **Origin.** Resonance studied by Hertz (1887) and Tesla; radio tuning circuits from Marconi's era.
- **Use.** Radio tuning, oscillators, wireless charging, filters.
- **Data science.** Damped oscillator models describe seasonal and business-cycle dynamics; the same second-order equation underlies momentum SGD ML.27 §15 Machine Learning.
- **Physical application.** Turning a radio dial changes C so f_0 matches the station's frequency: $L = 10\,\mu$H, $C = 253$ pF gives $f_0 \approx 3.2$ MHz.

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EE.17 Filters, Frequency Response & Decibels

A filter's transfer function $H(j\omega)$ gives, for each frequency, how much a sinusoid is scaled and shifted. A first-order RC filter passes one side of its cutoff frequency and rolls off the other at 20 dB per decade.

RC low-pass	$H(j\omega) = \dfrac{V_{\text{out}}}{V_{\text{in}}} = \dfrac{1}{1 + j\omega RC}$	Output across C: passes low frequencies.
Magnitude	$\lvert H\rvert = \dfrac{1}{\sqrt{1 + (f/f_c)^2}}$	Flat below f_c, falling as f_c/f above it.
Cutoff frequency	$f_c = \dfrac{1}{2\pi RC}$	Where $\lvert H\rvert = \frac{1}{\sqrt{2}}$: half power, -3 dB.
Gain in decibels	$\lvert H\rvert_{\text{dB}} = 20\log_{10}\lvert H\rvert$	Turns products of gains into sums A2.14 §04 Algebra II.
Phase	$\angle H = -\tan^{-1}\dfrac{f}{f_c}$	$-45°$ at f_c, approaching $-90°$.
RC high-pass	$H(j\omega) = \dfrac{j\omega RC}{1 + j\omega RC}$	Output across R: blocks DC, passes high frequencies.

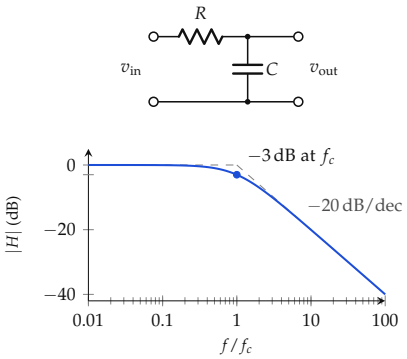
STATED-EQUATION *The transfer function of an RC low-pass filter, V out over V in, equals one over one plus j omega R C. Its magnitude equals one over the square root of one plus f over f c squared. The cutoff frequency f c equals one over two pi R C. In decibels, the gain is twenty log base ten of the magnitude. The phase is minus the inverse tangent of f over f c. The high-pass version is j omega R C over one plus j omega R C.*

MEMORY AID At the cutoff, $\lvert H\rvert = \frac{1}{\sqrt{2}}$, i.e. -3 dB (half power). Past it, -20 dB per decade for each RC stage.

DECISION POINT Which filter? Remove noise above the band of interest → low-pass (put f_c just above the highest wanted frequency). Remove drift or DC offset → high-pass. Keep one band → band-pass (RLC EE.16). Kill one interfering frequency (60 Hz hum) → notch. Need steeper roll-off → higher order (each order adds -20 dB/decade).

TERMS

$H(j\omega)$ – transfer function, a ratio of phasors EE.14
f_c – cutoff (corner) frequency (Hz)
R, C – resistance (Ω), capacitance (F); $RC = \tau$ (s) EE.11
dB – decibel, logarithmic ratio EE.29
Decade – a factor of 10 in frequency
Bode plot – $\lvert H\rvert_{\text{dB}}$ and $\angle H$ vs. $\log f$ EE.25
ω, f – angular frequency and frequency



CONTEXT

- **Origin.** Filter theory by Campbell and Wagner (1915); Bode plots (1930s).
- **Use.** Audio equalizers, anti-aliasing, noise removal, communications.
- **Data science.** Smoothing time-series data (moving averages are low-pass filters); convolution layers in CNNs learn filters ML.22 §15 Machine Learning.
- **Physical application.** A subwoofer's crossover is a low-pass filter that sends only bass to the big speaker.

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Electromagnetism & Power Systems

EE.18 Electric & Magnetic Fields

Charges create electric fields; moving charges (currents) create magnetic fields; a changing magnetic flux induces a voltage. These laws underlie capacitors, inductors, motors, generators and transformers.

Coulomb's law	$F = k\,\dfrac{q_1 q_2}{r^2}$	Force between point charges; like charges repel.
Electric field	$\mathbf{E} = \dfrac{\mathbf{F}}{q}, \qquad E = \dfrac{V}{d}$ (uniform)	Force per unit charge; between plates, volts per meter.
Field of a long wire	$B = \dfrac{\mu_0 I}{2\pi r}$	Circles the wire (right-hand rule); falls as $1/r$.
Solenoid	$B = \mu_0 n I$	Uniform field inside a long coil; basis of inductors.
Faraday's law	$\mathcal{E} = -N\,\dfrac{d\Phi_B}{dt}$	A changing flux induces an EMF; the minus sign is Lenz's law.
Lorentz force	$\mathbf{F} = q\big(\mathbf{E} + \mathbf{v} \times \mathbf{B}\big)$	On a wire: $\mathbf{F} = I\,\mathbf{L} \times \mathbf{B}$, the motor principle.

STATED-EQUATION *Coulomb's law: the force equals k q one q two over r squared. The electric field equals force per unit charge, or V over d between plates. The magnetic field around a long wire equals mu naught I over two pi r; inside a solenoid it is mu naught n I. Faraday's law: the induced E M F equals minus N times the rate of change of magnetic flux. The Lorentz force equals q times the quantity E plus v cross B.*

MEMORY AID Faraday: "change the flux, get a voltage"; the minus sign (Lenz) means the induced current opposes the change.

TERMS

$k = \frac{1}{4\pi\varepsilon_0} \approx 8.99 \times 10^9\ \mathrm{N{\cdot}m^2/C^2}$; inverse-square like gravity P.8 §12 Physics

q – charge (C); r distance (m); d plate gap (m)

$\mathbf{E}$ – electric field (V/m = N/C)

$\mathbf{B}$ – magnetic field, tesla (T); Earth $\approx 50\,\mu$T

$\mu_0 = 4\pi \times 10^{-7}\ \mathrm{T{\cdot}m/A}$; n turns per meter; N turns

Φ_B – magnetic flux $\iint \mathbf{B}\cdot\mathbf{n}\,dA$ (Wb) V.11 §11 Vector Calculus

$\mathcal{E}$ – induced EMF (V); $\mathbf{v}$ velocity; $\times$ cross product

Maxwell's equations – divergence and curl of $\mathbf{E}$, $\mathbf{B}$ V.5 §11 Vector Calculus, V.12 §11 Vector Calculus

F; r – force; distance

n; N – turns per meter; number of turns in the coil

V, d; I; $\mathbf{v}$ – plate voltage and gap; current; charge velocity

CONTEXT

- **Origin.** Coulomb (1785); Ørsted and Ampère (1820); Faraday (1831); Maxwell (1865); Lorentz (1895).
- **Use.** Motors, generators, MRI, wireless communication.
- **Data science.** Magnetometer and accelerometer fusion in phones; MRI image reconstruction with deep learning.
- **Physical application.** Induction cooktops heat pans with a rapidly changing magnetic field that induces currents in the metal.

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EE.19 Transformers & Three-Phase Power

A transformer uses Faraday's law to trade voltage for current at nearly constant power. High transmission voltage keeps the current, and therefore the I^2R loss, small; three-phase systems deliver that power most efficiently.

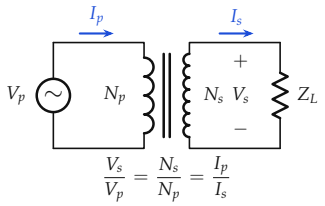
Turns ratio	$\dfrac{V_s}{V_p} = \dfrac{N_s}{N_p} = \dfrac{I_p}{I_s}$	Voltage scales with turns, current inversely.
Ideal power balance	$V_p I_p = V_s I_s$	Power in = power out (no losses).
Reflected impedance	$Z_{\text{in}} = \left(\dfrac{N_p}{N_s}\right)^2 Z_L$	A load seen through a transformer is scaled by the turns ratio squared.
Three-phase power	$P_{3\phi} = \sqrt{3}\, V_L I_L \cos\theta$	Line voltage, line current, power factor EE.20.
Line loss	$P_{\text{loss}} = I^2 R = \left(\dfrac{P}{V}\right)^2 R$	For fixed delivered power, loss falls as $1/V^2$.

STATED-EQUATION *For an ideal transformer, the secondary voltage over the primary voltage equals the turns ratio, N s over N p, which also equals the primary current over the secondary current, so V p I p equals V s I s. A load impedance appears at the primary multiplied by N p over N s squared. Three-phase power equals root three times the line voltage times the line current times the power factor. Line losses equal I squared R, which is P over V squared, times R.*

MEMORY AID "Step up voltage, step down current": power in = power out (ideal).

TERMS

N_p, N_s – turns on the primary and secondary coils
V_p, V_s, I_p, I_s – primary/secondary voltages (V) and currents (A)
Z_L – load impedance (Ω) EE.14
V_L, I_L – line voltage (V) and line current (A)
$\sqrt{3}$ – from phases 120° apart T.17 §06 Trigonometry
R – line resistance (Ω) EE.2
$P_{3\phi}$; θ – total three-phase power EE.20; power-factor angle
P_{loss}; P, V – power lost in the line; delivered power and transmission voltage
Z_{in} – impedance seen at the primary



CONTEXT

- **Origin.** Faraday's induction (1831); practical transformers by Gaulard, Stanley (1880s); three-phase by Tesla and Dolivo-Dobrovolsky.
- **Use.** Power transmission, chargers, industrial motors, impedance matching.
- **Data science.** Grid load forecasting and transformer failure prediction are major utility ML applications.
- **Physical application.** Raising transmission voltage 10× cuts line loss 100×; your phone charger steps 120 V down to 5 V.

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EE.20 Three-Phase Systems: Wye & Delta

Three equal sinusoids 120° apart. A balanced load draws constant total power and needs no return current, so three wires carry three times the single-phase power of two.

Balanced phase voltages	$\tilde{V}_a = V_p\angle 0^\circ,\ \ \tilde{V}_b = V_p\angle -120^\circ,\ \ \tilde{V}_c = V_p\angle 120^\circ$	They sum to zero (positive *abc* sequence).
Wye (Y) connection	$V_L = \sqrt{3}\,V_{\text{ph}}, \qquad I_L = I_{\text{ph}}$	Line voltage is between two phase tips; leads its phase voltage by 30°.
Delta (Δ) connection	$V_L = V_{\text{ph}}, \qquad I_L = \sqrt{3}\,I_{\text{ph}}$	Each load sits across two lines.
Total real power	$P = 3V_{\text{ph}}I_{\text{ph}}\cos\theta = \sqrt{3}\,V_L I_L \cos\theta$	Same formula for Y and Δ in line quantities.
Apparent and reactive	$\lvert S\rvert = \sqrt{3}\,V_L I_L, \qquad Q = \sqrt{3}\,V_L I_L \sin\theta$	The power triangle EE.15, three times over.
Y–Δ load conversion	$Z_Y = \dfrac{Z_\Delta}{3}$	Balanced loads: convert, then solve one phase.

STATED-EQUATION *Balanced phase voltages have equal magnitude V p at zero, minus one hundred twenty, and plus one hundred twenty degrees. In wye, line voltage is root three times phase voltage and line current equals phase current. In delta, line voltage equals phase voltage and line current is root three times phase current. Total power is three V phase I phase cosine theta, which equals root three V L I L cosine theta. A balanced delta load equals a wye load of one third the impedance. Apparent power is root three V L I L, and reactive power is root three V L I L sine theta.*

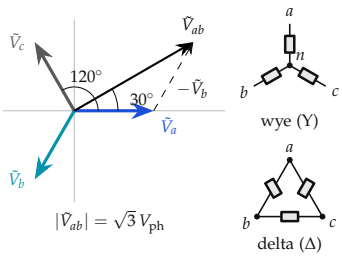
MEMORY AID "Wye: voltage gets the root three; Delta: current gets the root three."

METHOD

1. Convert any Δ parts to Y ($Z_Y = Z_\Delta/3$).
2. Solve one phase (line-to-neutral) as a single-phase circuit: $\tilde{I} = \tilde{V}_{\text{ph}}/Z_Y$.
3. Shift the result by $\mp 120^\circ$ for the other phases; total power is $3\times$ one phase. Example: 400 V line (Y), 10 A, pf 0.9: $V_{\text{ph}} = 231\,\text{V}$, $P = \sqrt{3}\cdot 400\cdot 10\cdot 0.9 \approx 6.2\,\text{kW}$.

TERMS

V_{ph}, I_{ph} – phase voltage (V) and current (A) of one winding or load
V_L, I_L – line-to-line voltage (V), line current (A)
V_p – phase amplitude used in the phasors
θ – load impedance angle; $\cos\theta$ power factor
Z_Y, Z_Δ – per-phase impedances (Ω) EE.14
Neutral – common point of a Y; carries zero current when balanced
P; $|S|$; Q – real, apparent, reactive power EE.15



CONTEXT

- **Origin.** Dolivo-Dobrovolsky and Tesla (1888–1891); the Lauffen–Frankfurt demonstration (1891).
- **Use.** Grids, industrial motors, data-center power distribution.
- **Data science.** Phase-imbalance detection in smart-meter and SCADA streams is a standard anomaly-detection task.
- **Physical application.** US commercial 208 V outlets are line-to-line on a 120 V wye: $120\sqrt{3} \approx 208$.

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EE.21 Batteries & Energy Storage

A battery stores chemical energy and delivers it at a roughly constant voltage. Capacity in amp-hours is charge; multiply by voltage to get energy.

Energy	$E\,(\text{Wh}) = V_{\text{nominal}} \times \text{capacity (Ah)}$	Watt-hours compare batteries fairly; mAh does not.
Runtime	$t_{\text{run}} \approx \dfrac{\text{capacity (Ah)}}{I\,(\text{A})}$	Ideal estimate; real runtime is shorter at high current.
C-rate	$\text{C-rate} = \dfrac{I}{\text{capacity}}$	1C empties the battery in one hour; 2C in 30 minutes.
Round-trip efficiency	$\eta_{\text{round trip}} = \dfrac{E_{\text{out}}}{E_{\text{in}}}$	Li-ion ≈ 90%; pumped hydro ≈ 75%.
Pack wiring	$V_{\text{pack}} = n_s V_{\text{cell}}, \qquad \text{Ah}_{\text{pack}} = n_p\,\text{Ah}_{\text{cell}}$	Series adds voltage, parallel adds capacity.
Terminal voltage	$V = V_{\text{oc}} - IR_{\text{int}}$	Internal resistance makes voltage sag under load EE.6.

STATED-EQUATION *Stored energy in watt-hours equals the nominal voltage times the capacity in amp-hours. Runtime is approximately capacity in amp-hours over current in amps. The C-rate equals current over capacity. Round-trip efficiency equals energy out over energy in. Cells in series add voltage and in parallel add capacity. Terminal voltage equals open-circuit voltage minus I times the internal resistance.*

MEMORY AID "mAh is charge, Wh is energy": multiply by voltage to compare batteries fairly.

TERMS

Ah, mAh – charge capacity (1 Ah = 3600 C) EE.1
Wh – energy capacity (1 Wh = 3600 J)
1C – current that empties the battery in one hour
n_s, n_p – cells in series, strings in parallel
R_{int} – internal resistance (Ω); V_{oc} open-circuit voltage
State of charge – fraction of capacity remaining
Real runtime – shorter at high C-rate and low temperature
I; E – discharge current; stored energy
V_{oc}; V_{cell} – open-circuit voltage; single-cell voltage
η; E_{in}, E_{out} – efficiency; energy charged and recovered

CONTEXT

- **Origin.** Volta's pile (1800); lead–acid (Planté, 1859); lithium-ion (Goodenough, Whittingham, Yoshino; Nobel 2019).
- **Use.** Phones, laptops, EVs, grid storage.
- **Data science.** Battery state-of-health and remaining-useful-life prediction from charge–discharge curves is a common time-series ML task.
- **Physical application.** A 75 kWh EV battery using 250 Wh per mile has about a 300-mile range.

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Signals & Systems

EE.22 LTI Systems, Impulse Response & Convolution

A linear time-invariant (LTI) system is completely described by its impulse response h: the output for any input is the convolution of the input with h. Transforms turn that convolution into multiplication.

Linearity	$ax_1 + bx_2 \;\mapsto\; ay_1 + by_2$	Scaled, summed inputs give scaled, summed outputs EE.8.
Time invariance	$x(t - t_0) \;\mapsto\; y(t - t_0)$	Delaying the input just delays the output.
Convolution (continuous)	$y(t) = (x * h)(t) = \int_{-\infty}^{\infty} x(\tau)\, h(t-\tau)\, d\tau$	Sum of shifted, scaled impulse responses.
Convolution (discrete)	$y[n] = \sum_k x[k]\, h[n-k]$	Flip h, slide, multiply, add.
Convolution theorem	$y = x * h \iff Y = X \cdot H$	Holds for Fourier, Laplace and Z transforms.
Sinusoidal steady state	$\cos \omega t \;\mapsto\; \lvert H(j\omega)\rvert \cos\big(\omega t + \angle H(j\omega)\big)$	Sinusoids are eigenfunctions: same frequency out EE.17.
BIBO stability	$\int_{-\infty}^{\infty} \lvert h(t)\rvert\, dt < \infty, \qquad \sum_n \lvert h[n]\rvert < \infty$	Every bounded input gives a bounded output.

STATED-EQUATION *A linear system maps a x one plus b x two to a y one plus b y two. A time-invariant system maps x of t minus t naught to y of t minus t naught. The output is the convolution of x with h: the integral of x of tau times h of t minus tau, or in discrete time the sum of x of k times h of n minus k. Convolution in time is multiplication of transforms. A sinusoid comes out at the same frequency, scaled by the magnitude of H and shifted by its angle. The system is BIBO stable if the impulse response is absolutely integrable or summable.*

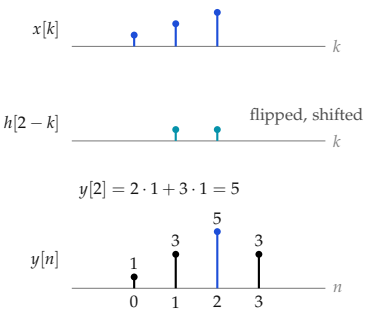
MEMORY AID "Flip and slide": convolution flips h and slides it across x.

METHOD

1. Flip $h[k]$ to $h[-k]$.
2. Shift by n to get $h[n-k]$.
3. Multiply by $x[k]$ where they overlap and add: that is $y[n]$; repeat for each n.
4. Example: $x = [1,2,3]$, $h = [1,1]$: $y = [1,\ 1{+}2,\ 2{+}3,\ 3] = [1,3,5,3]$ (length $3+2-1$).

TERMS

x, y – input and output signals; t time (s), n sample index
h – impulse response: output for input δ
δ – unit impulse (area 1 at $t=0$; $\delta[0]=1$)
$*$ – convolution; τ, k dummy variables
X, H, Y – transforms EE.26, EE.23, EE.28
BIBO – bounded-input bounded-output
a, b; t_0 – constants; time shift
τ; k, n – integration variable; sample indices
$H(j\omega)$ – frequency response EE.17



CONTEXT

- **Origin.** Duhamel's integral (1833); systems theory of the 1940s–60s (Wiener, Zadeh).
- **Use.** Filtering, echo and reverb, channel modeling, image blurring.
- **Data science.** CNN layers compute discrete convolutions (strictly cross-correlations) ML.22 §15 Machine Learning; a moving average is convolution with a box.
- **Physical application.** A concert hall's impulse response (a clap) lets software "convolve" any recording to sound as if played there.

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EE.23 Laplace Transform & Transfer Functions

The Laplace transform converts a linear ODE with initial conditions into an algebraic equation in the complex frequency s. Circuits become impedance networks in s, and a system is summarized by its transfer function $H(s)$.

Laplace transform	$F(s) = \int_{0^-}^{\infty} f(t)\, e^{-st}\, dt$	Defined for $\operatorname{Re} s$ large enough for convergence C2.6 §08 Calculus II.
Derivative and integral rules	$\mathcal{L}\{f'\} = sF(s) - f(0^-), \qquad \mathcal{L}\left\{\int_0^t f\right\} = \dfrac{F(s)}{s}$	Differentiation becomes multiplication by s; initial conditions enter automatically.
Core pairs	$u(t) \to \dfrac{1}{s}, \quad e^{-at} \to \dfrac{1}{s+a}, \quad \sin \omega t \to \dfrac{\omega}{s^2+\omega^2}$	More pairs in the table below.
s-domain impedances	$Z_R = R, \qquad Z_L = sL, \qquad Z_C = \dfrac{1}{sC}$	Phasor impedances with $j\omega \to s$ EE.14.
Transfer function	$H(s) = \dfrac{Y(s)}{X(s)}\Big\|_{\text{zero ICs}} = \mathcal{L}\{h(t)\}$	Output over input; the transform of the impulse response EE.22.
Final and initial value theorems	$f(\infty) = \lim_{s\to 0} sF(s), \qquad f(0^+) = \lim_{s\to\infty} sF(s)$	FVT only if $sF(s)$ has all poles in the left half-plane.

STATED-EQUATION *The Laplace transform F of s is the integral from zero to infinity of f of t times e to the minus s t. The transform of a derivative is s F of s minus f of zero; of an integral, F of s over s. The unit step transforms to one over s, e to the minus a t to one over s plus a, and sine omega t to omega over s squared plus omega squared. Impedances are R, s L and one over s C. The transfer function is Y of s over X of s with zero initial conditions. The final value is the limit of s F of s as s goes to zero. The initial value is the limit of s F of s as s goes to infinity.*

MEMORY AID "s is a derivative, $1/s$ is an integral."

METHOD

1. Transform the ODE (or redraw the circuit with sL, $\frac{1}{sC}$ and initial-condition sources).
2. Solve algebraically for $Y(s)$.
3. Split into partial fractions C2.4 §08 Calculus II.
4. Invert term by term with the table. Example: $Y = \frac{1}{s(s+2)} = \frac{1}{2}\left(\frac{1}{s} - \frac{1}{s+2}\right) \Rightarrow y(t) = \frac{1}{2}\left(1 - e^{-2t}\right)$.

$f(t)$	$F(s)$	$f(t)$	$F(s)$	$f(t)$	$F(s)$	$f(t)$	$F(s)$
$\delta(t)$	1	t	$\frac{1}{s^2}$	te^{-at}	$\frac{1}{(s+a)^2}$	$\cos \omega t$	$\frac{s}{s^2+\omega^2}$
$u(t)$	$\frac{1}{s}$	e^{-at}	$\frac{1}{s+a}$	$\sin \omega t$	$\frac{\omega}{s^2+\omega^2}$	$e^{-at}\cos \omega t$	$\frac{s+a}{(s+a)^2+\omega^2}$

TERMS

s – complex frequency $\sigma + j\omega$ (1/s)
$f(t)$, $F(s)$ – time function and its transform
$u(t)$ – unit step (0 before $t = 0$, 1 after); $\delta(t)$ unit impulse
a, ω – decay rate (1/s), angular frequency (rad/s)
$f(0^-)$ – initial condition just before $t = 0$
$H(s)$ – transfer function; poles and zeros EE.24
$\mathcal{L}\{\cdot\}$; $X(s), Y(s)$ – Laplace transform; input and output transforms
$h(t)$; Z – impulse response; impedance

CONTEXT

- **Origin.** Laplace (1780s, probability); Heaviside's operational calculus (1890s); made rigorous by Bromwich and Doetsch.
- **Use.** Circuit transients, control design, signal processing.
- **Data science.** The moment-generating function $E[e^{tX}]$ is a (two-sided) Laplace transform of a density S.13 §13 Statistics.
- **Physical application.** Engineers model a car's cruise control as $H(s) = \frac{K}{\tau s+1}$ and read off its time constant directly.

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EE.24 Poles, Zeros & Stability

Factoring a transfer function exposes its poles (where it blows up) and zeros (where it vanishes). Pole locations in the s-plane fix the shape of the natural response: decay rate, oscillation, and stability.

Pole–zero form	$H(s) = K\,\dfrac{(s-z_1)\cdots(s-z_m)}{(s-p_1)\cdots(s-p_n)}$	Zeros z_i: roots of the numerator; poles p_k: roots of the denominator.
Natural response	$y_n(t) = \sum_k A_k\, e^{p_k t}$	Each pole contributes one exponential mode.
Stability	$\mathrm{Re}(p_k) < 0 \;\text{ for all } k$	All poles in the open left half-plane ⇒ every mode decays.
Standard second-order	$H(s) = \dfrac{\omega_0^2}{s^2 + 2\zeta\omega_0 s + \omega_0^2}$	Poles $-\zeta\omega_0 \pm j\omega_0\sqrt{1-\zeta^2}$ EE.12.
DC gain	$H(0) = K\,\dfrac{\prod(-z_i)}{\prod(-p_k)}$	Steady-state output per unit step input.
Order and properness	$n = \deg(\text{den}) \;\geq\; m = \deg(\text{num})$	Physical systems are proper; n counts energy-storage elements.

STATED-EQUATION *H of s equals K times the product of s minus each zero, over the product of s minus each pole. The natural response is a sum of exponentials e to the p k t. The system is stable when every pole has negative real part. The standard second-order system has poles at minus zeta omega naught plus or minus j omega naught root of one minus zeta squared. The DC gain is H of zero. A proper system has at least as many poles as zeros.*

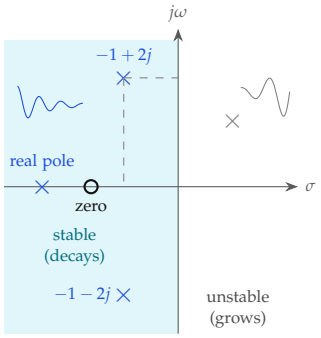
MEMORY AID "Left is safe, right is unstable, the $j\omega$ axis rings forever"; distance from the axis = decay rate, height = oscillation frequency.

METHOD

1. Write $H(s)$ as a ratio of polynomials in s.
2. Find the denominator's roots (quadratic formula A1.15 §03 Algebra I or numerically).
3. Read each pole: real part → decay rate, imaginary part → ringing frequency. Example: $s^2 + 2s + 5 = 0 \Rightarrow p = -1 \pm 2j$: stable, decays as e^{-t}, rings at 2 rad/s.

TERMS

z_i, p_k – zeros and poles (1/s), complex numbers A2.6 §04 Algebra II
K – gain constant
A_k – mode amplitudes, set by initial conditions
s-plane – real axis σ (decay), imaginary axis $j\omega$ (frequency)
State space – poles = eigenvalues of the system matrix LA.23 §09 Linear Algebra
Marginal – simple poles on the $j\omega$ axis: sustained oscillation
m, n – number of zeros and poles
ζ, ω_0 – damping ratio and natural frequency EE.12
$y_n(t)$; Re – natural response; real part



CONTEXT

- **Origin.** Pole–zero analysis grew from Heaviside, Nyquist and Bode at Bell Labs (1920s–40s).
- **Use.** Filter and controller design, oscillator design, stability checks.
- **Data science.** An AR(p) time-series model is stationary exactly when its characteristic roots lie inside the unit circle, the discrete version of this rule EE.28.
- **Physical application.** A bridge with lightly damped poles (close to the $j\omega$ axis) can resonate dangerously with wind or footsteps.

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EE.25 Bode Plots

A Bode plot draws gain in dB and phase in degrees against log frequency. Because logs turn products into sums, each pole and zero adds a simple straight-line piece, so a sketch takes minutes.

Bode (time-constant) form	$H(j\omega) = K\,\dfrac{\prod(1+j\omega/\omega_z)}{\prod(1+j\omega/\omega_p)}$	Each factor equals 1 at low frequency, so K is the DC gain.
Gain adds in dB	$\lvert H\rvert_{\text{dB}} = 20\log_{10}K + \sum 20\log_{10}\lvert\text{zero factors}\rvert - \sum 20\log_{10}\lvert\text{pole factors}\rvert$	Plot each factor and add.
Real pole at ω_p	-20 dB/dec above ω_p, -3 dB and -45° at ω_p	Phase falls from 0° to -90° between $0.1\,\omega_p$ and $10\,\omega_p$.
Real zero at ω_z	$+20$ dB/dec, $\quad 0^\circ \to +90^\circ$	Mirror image of a pole.
Pole at the origin	$\dfrac{1}{j\omega}$: -20 dB/dec through 0 dB at $\omega = 1$, -90°	An integrator; a zero at the origin is a differentiator.
Complex pole pair	-40 dB/dec above ω_0, $\quad$ peak $\approx \dfrac{1}{2\zeta}$ $(\zeta \ll 1)$	Phase drops 180°; sharp near ω_0 when ζ is small.

STATED-EQUATION *In Bode form, H equals K times the product of one plus j omega over omega z, over the product of one plus j omega over omega p. Gains add in decibels. Each real pole bends the gain down by twenty decibels per decade, is three decibels down at the corner, and contributes minus forty-five degrees there. A zero does the opposite. A pole at the origin slopes minus twenty decibels per decade through zero decibels at omega equals one. A complex pole pair slopes minus forty decibels per decade and peaks near one over two zeta.*

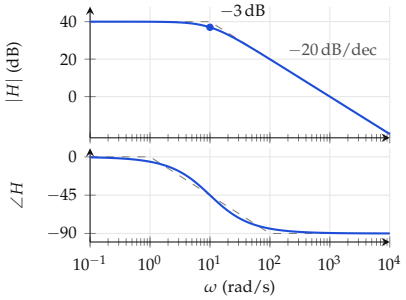
MEMORY AID "Pole: down 20 and back 90; zero: up 20 and forward 90." Phase changes over two decades centered on the corner.

METHOD

1. Rewrite H in Bode form; list the corner frequencies in increasing order.
2. Start at low frequency: level $20\log_{10}K$ (slope ± 20 per origin pole/zero).
3. At each corner change the slope by -20 (pole) or $+20$ (zero) dB/decade; round each corner by 3 dB.
4. Sketch phase the same way. Example: $H = \frac{100}{1+s/10}$: flat at 40 dB to 10 rad/s, then -20 dB/dec, crossing 0 dB at 1000 rad/s.

TERMS

ω_z, ω_p – zero and pole corner (break) frequencies (rad/s)
K – DC gain; $20\log_{10}K$ in dB EE.29
Decade – factor of 10 in ω; log scale A2.14 §04 Algebra II
ζ, ω_0 – damping ratio and natural frequency EE.12
Asymptote – straight-line approximation of each factor
First-order example – the RC filter EE.17
ω; j – angular frequency; imaginary unit



CONTEXT

- **Origin.** Hendrik Bode, Bell Labs (1938–1945), for feedback amplifier design.
- **Use.** Filter, amplifier and control-loop design; reading datasheets.
- **Data science.** Log–log plots reveal power laws as straight lines, the same trick used for scaling laws of model loss vs. compute.
- **Physical application.** Loudspeaker and headphone frequency-response charts are Bode magnitude plots.

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EE.26 Fourier Series, the DFT & FFT

Any periodic signal is a sum of harmonics of its fundamental frequency; any sampled signal of N points is a sum of N complex sinusoids. The FFT computes those amplitudes fast.

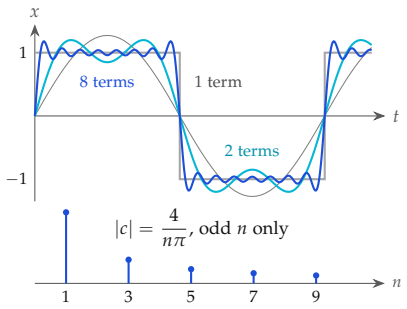
Complex Fourier series	$x(t) = \sum_{n=-\infty}^{\infty} c_n \, e^{jn\omega_0 t}$	Periodic signal = sum of harmonics $n\omega_0$.
Fourier coefficient	$c_n = \frac{1}{T}\int_0^T x(t)\, e^{-jn\omega_0 t}\, dt$	Projects x onto harmonic n (an inner product LA.20 §09 Linear Algebra).
Fourier transform	$X(f) = \int_{-\infty}^{\infty} x(t)\, e^{-j2\pi f t}\, dt$	Non-periodic signals: a continuous spectrum.
Discrete Fourier transform	$X_k = \sum_{n=0}^{N-1} x_n \, e^{-j2\pi kn/N}$	N samples $\to$ N frequency bins.
Frequency resolution	$\Delta f = \frac{f_s}{N}$	Bin spacing; record longer to resolve closer tones.
FFT speed-up	DFT $O(N^2)$ $\to$ FFT $O(N \log N)$	Divide and conquer on even and odd samples O.10 §14 Optimization.
Parseval's theorem	$\sum_n \lvert x_n\rvert^2 = \frac{1}{N}\sum_k \lvert X_k\rvert^2$	Energy is the same in time and frequency.

STATED-EQUATION *A periodic signal x of t equals the sum of coefficients c n times e to the j n omega naught t, where c n is one over T times the integral over a period of x times e to the minus j n omega naught t. The Fourier transform X of f is the integral of x of t e to the minus j two pi f t. The discrete Fourier transform: X k equals the sum from n equals zero to N minus one of x n times e to the minus j two pi k n over N. The frequency resolution equals f s over N. The FFT computes the DFT in order N log N instead of N squared operations. Parseval: energy is preserved. Parseval: the sum of x n squared equals one over N times the sum of X k squared.*

MEMORY AID Every signal is a sum of spinning arrows (sinusoids) T.23 §06 Trigonometry; the FFT finds how much of each.

TERMS

c_n – Fourier coefficient (amplitude and phase of harmonic n)
$\omega_0 = 2\pi/T$ – fundamental angular frequency (rad/s); T period (s)
$X(f)$ – spectrum (amplitude density per Hz)
X_k – frequency content at bin k, i.e. $f = kf_s/N$ (complex EE.14)
N – number of samples; f_s sampling rate EE.27
Square wave – odd harmonics with amplitudes $\frac{4}{n\pi}$ (figure)
T; f – period; frequency
f_s; Δf – sampling rate; spacing of DFT bins
x_n; n, k – samples; time and frequency indices



CONTEXT

- **Origin.** Fourier (1807, heat conduction P.24 §12 Physics); FFT by Cooley and Tukey (1965), anticipated by Gauss (1805).
- **Use.** Audio compression (MP3), spectrum analyzers, image compression (JPEG), vibration analysis.
- **Data science.** Spectral features for time series and audio (spectrograms feed CNNs); detecting seasonality; FFT-based fast convolution EE.22.
- **Physical application.** Noise-canceling headphones analyze and invert the frequency content of ambient noise in real time.

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EE.27 Sampling, Nyquist & Quantization

Converting an analog signal to digital samples it in time and rounds it in amplitude. Sampling too slowly folds high frequencies onto low ones (aliasing); each bit of resolution halves the rounding error.

Nyquist criterion	$f_s > 2f_{\max}$	Sample faster than twice the highest frequency present.
Alias frequency	$f_{\text{alias}} = \lvert f - k f_s \rvert$	Integer k nearest f/f_s: a tone above $f_s/2$ masquerades as a lower one.
ADC resolution (LSB)	$\Delta = \dfrac{V_{\text{ref}}}{2^N}$	Smallest voltage step an N-bit converter distinguishes.
ADC output code	$D = \left\lfloor \dfrac{V_{\text{in}}}{V_{\text{ref}}} 2^N \right\rfloor$	Integer from 0 to $2^N - 1$; DAC inverts it: $V_{\text{out}} = D\,\Delta$.
Quantization noise	$e_{\text{rms}} = \dfrac{\Delta}{\sqrt{12}}$	Rounding error is roughly uniform on $[-\frac{\Delta}{2}, \frac{\Delta}{2}]$ S.19 §13 Statistics.
SQNR	$\text{SQNR} \approx 6.02N + 1.76 \text{ dB}$	Full-scale sine; each extra bit buys about 6 dB.

STATED-EQUATION *The sampling rate f s must exceed twice the highest frequency, the Nyquist criterion. A tone above half the sampling rate appears at the alias frequency, the absolute value of f minus k f s. The resolution of an N-bit converter is V ref over two to the N. The output code is the floor of V in over V ref times two to the N. Quantization noise has RMS delta over root twelve. The signal-to-quantization-noise ratio is about six point oh two N plus one point seven six decibels.*

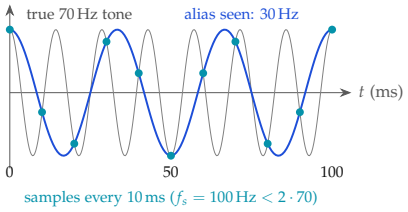
MEMORY AID "Sample at least twice as fast as the fastest wiggle"; each extra bit buys about 6 dB.

METHOD

1. LSB: $\Delta = V_{\text{ref}}/2^N$.
2. Code: divide V_{in} by Δ and round down.
3. Example: 10-bit ADC, $V_{\text{ref}} = 5\,\text{V}$: $\Delta = 4.88\,\text{mV}$; $V_{\text{in}} = 2\,\text{V}$ gives $D = \lfloor 409.6 \rfloor = 409$. Aliasing: a 70 Hz tone sampled at 100 Hz appears at $\lvert 70 - 100 \rvert = 30\,\text{Hz}$.

TERMS

f_s – sampling rate (samples/s, Hz); $f_s/2$ Nyquist frequency
$f_{\max}$ – highest frequency in the signal (Hz)
Aliasing – undersampled high frequencies masquerade as low ones (figure); prevent with an anti-alias low-pass EE.17
N – ADC bit depth; 2^N levels EE.37
V_{ref} – full-scale reference voltage (V); Δ LSB (V)
$\lfloor \cdot \rfloor$ – floor (round down); SQNR in dB EE.29
f; k – signal frequency; integer
Δ; D – LSB step size; output code
V_{in}; e_{rms}; SQNR – input voltage; RMS quantization error; signal-to-quantization-noise ratio



CONTEXT

- **Origin.** Nyquist (1928), Shannon (1949), Kotelnikov (1933).
- **Use.** Digital audio, sensors, imaging, telecommunications.
- **Data science.** Resampling time series and sensor streams; choosing logging frequency for IoT data; aliasing in downsampled images; 8-bit model quantization uses the same Δ.
- **Physical application.** CDs sample at 44.1 kHz because human hearing tops out near 20 kHz.

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EE.28 The Z-Transform & Digital Filters

The Z-transform is the discrete-time counterpart of the Laplace transform: a delay of one sample becomes multiplication by z^{-1}, so difference equations become ratios of polynomials in z.

Z-transform	$X(z) = \sum_{n=0}^{\infty} x[n]\, z^{-n}$	A power series in z^{-1} with the samples as coefficients.
Delay property	$x[n-k] \;\leftrightarrow\; z^{-k}X(z)$	z^{-1} is a one-sample delay (a register).
Core pairs	$\delta[n] \to 1, \qquad a^n u[n] \to \dfrac{z}{z-a} = \dfrac{1}{1-az^{-1}}$	Geometric series A2.19 §04 Algebra II; $a = 1$ gives the unit step.
Transfer function of a difference equation	$y[n] = \sum_k b_k x[n-k] - \sum_{k\geq 1} a_k y[n-k] \;\Rightarrow\; H(z) = \dfrac{\sum b_k z^{-k}}{1+\sum a_k z^{-k}}$	Feed-forward taps on top, feedback taps below.
Stability	$\lvert p_k\rvert < 1$ for all poles	Poles inside the unit circle (the image of the left half-plane, $z = e^{sT_s}$).
Frequency response	$H(e^{j\Omega}), \qquad \Omega = 2\pi\dfrac{f}{f_s}$	Evaluate on the unit circle; $\Omega = \pi$ is the Nyquist frequency.

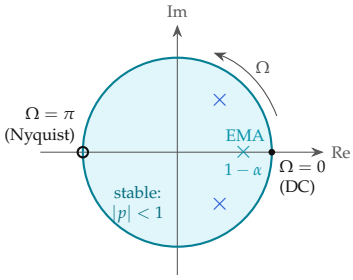
STATED-EQUATION *The Z-transform X of z is the sum of x of n times z to the minus n. A delay of k samples multiplies the transform by z to the minus k. The impulse transforms to one and a to the n times the step to z over z minus a. A difference equation with feed-forward coefficients b and feedback coefficients a has transfer function sum of b k z to the minus k over one plus sum of a k z to the minus k. The filter is stable when all poles are inside the unit circle. The frequency response is H evaluated at e to the j Omega, with Omega equal to two pi f over f s.*

MEMORY AID "s-plane left half $\leftrightarrow$ z-plane unit disk." z^{-1} = "one sample ago".

DECISION POINT FIR or IIR? FIR (no feedback, $a_k = 0$): always stable, can have exactly linear phase, but needs many taps for sharp cutoffs. IIR (feedback): sharp responses from few coefficients, but can be unstable and distorts phase. Default to FIR for offline data or when phase matters; IIR for cheap real-time smoothing (the EMA).

TERMS

$x[n]$, $y[n]$ – input and output sequences; n sample index
z – complex variable; z^{-1} unit delay
b_k, a_k – feed-forward and feedback coefficients
$u[n]$ – unit step sequence; $\delta[n]$ unit impulse
p_k – poles of $H(z)$ EE.24
Ω – digital frequency (rad/sample); f_s sampling rate EE.27
FIR / IIR – finite / infinite impulse response
$\delta[n]$; a – unit impulse; constant base
$X(z)$, $H(z)$ – transform of x; transfer function
f, f_s – frequency and sampling rate



CONTEXT

- **Origin.** Hurewicz (1947); named and developed by Ragazzini and Zadeh (1952).
- **Use.** Digital filters, audio effects, digital control loops.
- **Data science.** The EMA $y[n] = \alpha x[n] + (1-\alpha)y[n-1]$ has $H(z) = \frac{\alpha}{1-(1-\alpha)z^{-1}}$: one pole at $1-\alpha$, stable for $0 < \alpha < 2$; Adam's moment estimates are such filters ML.27 §15 Machine Learning.
- **Physical application.** A 3-tap moving average $y[n] = \frac{1}{3}(x[n] + x[n-1] + x[n-2])$ is an FIR filter that smooths sensor jitter.

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EE.29 Decibels, SNR & Channel Capacity

Decibels express ratios on a log scale so that gains in a chain simply add. The signal-to-noise ratio limits how many bits per second a channel can carry.

Decibel (power)	$\mathrm{dB} = 10\log_{10}\dfrac{P_2}{P_1}$	Logarithmic power ratio.
Decibel (voltage)	$\mathrm{dB} = 20\log_{10}\dfrac{V_2}{V_1}$	Factor 20 because $P \propto V^2$ EE.2.
Absolute levels	$\mathrm{dBm} = 10\log_{10}\dfrac{P}{1\ \mathrm{mW}}$	dB relative to a fixed reference (dBV: 1 V; dB SPL: 20 μPa).
Signal-to-noise ratio	$\mathrm{SNR} = \dfrac{P_{\text{signal}}}{P_{\text{noise}}}, \quad \mathrm{SNR}_{\mathrm{dB}} = 10\log_{10}\mathrm{SNR}$	Usually quoted in dB.
Cascaded gains	$G_{\text{total,dB}} = G_{1,\text{dB}} + G_{2,\text{dB}} + \cdots$	Multiplying ratios = adding dB A2.14 §04 Algebra II.
Shannon–Hartley capacity	$C = B\log_2(1+\mathrm{SNR})$	Use the linear SNR, not dB.

STATED-EQUATION *A power ratio in decibels is ten log base ten of P two over P one, which equals twenty log base ten of the voltage ratio. d B m is decibels relative to one milliwatt. The signal-to-noise ratio is signal power over noise power. Gains in decibels add along a chain. Shannon's capacity: C equals the bandwidth B times log base two of one plus the SNR.*

MEMORY AID $+3\,\mathrm{dB} \approx 2\times$ power, $+10\,\mathrm{dB} = 10\times$ power, $+20\,\mathrm{dB} = 100\times$ power ($10\times$ voltage).

TERMS

dB – decibel, logarithmic ratio A2.14 §04 Algebra II
P_1, P_2 – powers (W); V_1, V_2 voltages (V)
dBm – dB relative to 1 mW (0 dBm = 1 mW)
C – maximum error-free data rate (bits/s)
B – bandwidth (Hz)
Entropy link – ML.3 §15 Machine Learning
P; V_1, V_2 – power; voltage amplitudes
SNR; G – signal-to-noise ratio; stage gain

CONTEXT

- **Origin.** The bel named for Alexander Graham Bell (1920s); Shannon–Hartley theorem (1948).
- **Use.** Audio levels, Wi-Fi and cellular link budgets, amplifier specs.
- **Data science.** SNR thinking applies to features and metrics: a small effect in a noisy metric needs a larger sample S.24 §13 Statistics.
- **Physical application.** Wi-Fi drops to a slower rate as you move away because SNR falls; a 20 MHz channel at 30 dB SNR caps near $20\mathrm{M}\cdot\log_2 1001 \approx 200$ Mbit/s.

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Feedback & Control

EE.30 Feedback & Stability

Negative feedback measures the output, compares it with the goal, and corrects. It trades raw gain for accuracy, robustness and speed, but too much loop gain with too much delay makes a system oscillate.

Closed-loop transfer function	$T(s) = \dfrac{Y}{R} = \dfrac{G(s)}{1 + G(s)H(s)}$	Negative feedback with forward path G and sensor path H.
Loop gain	$L(s) = G(s)H(s), \qquad \lvert L\rvert \gg 1 \Rightarrow T \approx \dfrac{1}{H}$	High loop gain makes the result depend only on the (precise) feedback path.
Sensitivity	$S(s) = \dfrac{1}{1 + L(s)}, \qquad E = S\,R \;\; (H = 1)$	Fraction of the reference that remains as error; also how plant changes leak through.
Steady-state error (step)	$e_{\text{ss}} = \dfrac{1}{1 + K_p}, \qquad K_p = \lim_{s \to 0} L(s)$	Final value theorem EE.23; an integrator in L makes $e_{\text{ss}} = 0$.
Characteristic equation	$1 + L(s) = 0$	Its roots are the closed-loop poles; all must be in the left half-plane EE.24.
Stability margins	$\text{GM} = -20\log_{10}\lvert L(j\omega_{180})\rvert, \qquad \text{PM} = 180^\circ + \angle L(j\omega_c)$	How much extra gain or delay the loop tolerates.

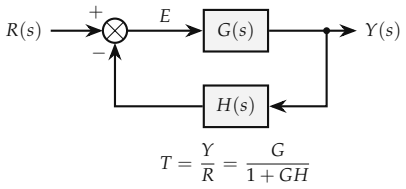
STATED-EQUATION *The closed-loop transfer function T equals G over one plus G H. The loop gain L is G times H; when it is large, T is about one over H. The sensitivity S is one over one plus L, and the error equals S times the reference. The steady-state step error is one over one plus K p, where K p is the low-frequency loop gain. Closed-loop poles are the roots of one plus L equals zero. The gain margin is minus twenty log of the loop gain where the phase is minus one eighty; the phase margin is one eighty plus the loop phase where the gain is one.*

MEMORY AID "Big loop gain, small error"; but "gain + lag = wobble." Aim for phase margin $\geq 45^\circ$.

DECISION POINT How to check stability? Low order with known numbers → solve $1 + L = 0$ directly. Polynomial with parameters → Routh–Hurwitz table. Designing or tuning a loop → Bode plot of L and read GM, PM EE.25. Open loop itself unstable, or delays → Nyquist criterion or simulation.

TERMS

R, Y, E – reference, output, error ($E = R - Y$ for $H = 1$)
$G(s)$ – forward path: controller × plant
$H(s)$ – feedback (sensor) path
L, T, S – loop, closed-loop and sensitivity functions; $T + S = 1$ when $H = 1$
K_p – position-error constant (DC loop gain)
ω_c, ω_{180} – gain-crossover ($\lvert L\rvert = 1$) and phase-crossover ($\angle L = -180^\circ$) frequencies



CONTEXT

- **Origin.** Harold Black's negative-feedback amplifier (1927); Nyquist's stability criterion (1932); Bode (1945).
- **Use.** Op-amps, voltage regulators, autopilots, thermostats, grid frequency control.
- **Data science.** Gradient descent is a feedback loop whose gain is the learning rate: too high a gain oscillates or diverges C1.26 §07 Calculus I; models retrained on data they influenced form feedback loops too.
- **Physical application.** A shower that reacts slowly to the tap invites overcorrection: delay plus high gain = oscillation between hot and cold.

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EE.31 PID Control

The PID controller acts on the error with three terms: the present (proportional), the accumulated past (integral) and the predicted future (derivative). It runs the vast majority of industrial control loops.

PID law	$u(t) = K_p e(t) + K_i \int_0^t e(\tau)\,d\tau + K_d \frac{de}{dt}$	Error $e = r - y$; output u drives the plant.
Transfer function	$C(s) = K_p + \frac{K_i}{s} + K_d s$	Integrator (pole at 0) and differentiator (zero) EE.23.
Standard (ISA) form	$C(s) = K_p\left(1 + \frac{1}{T_i s} + T_d s\right)$	$T_i = K_p/K_i$, $T_d = K_d/K_p$ in seconds.
Discrete PID	$u_k = K_p e_k + K_i T_s \sum_{j \le k} e_j + K_d \frac{e_k - e_{k-1}}{T_s}$	Sampled every T_s; running sum and difference.
Ziegler–Nichols tuning	$K_p = 0.6K_u, \quad T_i = \frac{T_u}{2}, \quad T_d = \frac{T_u}{8}$	From the gain K_u and period T_u of sustained oscillation under P-only control.

STATED-EQUATION *The PID control signal u equals K p times the error, plus K i times the integral of the error, plus K d times the derivative of the error. Its transfer function is K p plus K i over s plus K d s, or K p times one plus one over T i s plus T d s. The discrete version uses a running sum and a difference. Ziegler and Nichols: K p is zero point six K u, T i is half T u, and T d is one eighth T u.*

MEMORY AID P = present, I = past, D = future. "P reacts, I removes offset, D damps."

METHOD

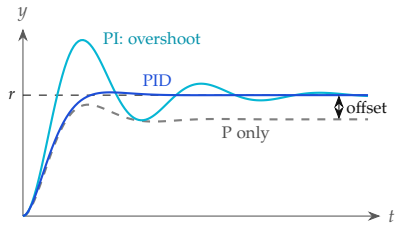
1. Set $K_i = K_d = 0$; raise K_p until the response is fast with modest overshoot.
2. Add K_i until the steady-state offset disappears in acceptable time.
3. Add K_d to cut overshoot; low-pass filter the derivative.
4. Example: P-only oscillates steadily at $K_u = 10$, $T_u = 2\,\text{s} \Rightarrow K_p = 6$, $T_i = 1\,\text{s}$, $T_d = 0.25\,\text{s}$ ($K_i = 6$, $K_d = 1.5$).

DECISION POINT Which terms? P alone if a small offset is acceptable. PI (most process loops: flow, temperature) to remove offset. PD for fast mechanical positioning where overshoot matters. Full PID when you need both. Derivative amplifies sensor noise; integral "winds up" when the actuator saturates (clamp it).

Increasing	Rise time	Overshoot	Settling time	Steady-state error
K_p	decreases	increases	small change	decreases
K_i	decreases	increases	increases	eliminated
K_d	small change	decreases	decreases	no effect

TERMS

e, r, y – error, setpoint (reference), measured output
u – control signal (actuator command)
K_p, K_i, K_d – proportional, integral (1/s), derivative (s) gains
T_i, T_d – integral and derivative times (s); T_s sample period (s)
K_u, T_u – ultimate gain and period
Windup – integral builds up while the output is saturated
T_s; e_k, u_k – sample period; error and control at step k
$C(s)$; s – controller transfer function; Laplace variable EE.23



CONTEXT

- **Origin.** Minorsky's ship-steering analysis (1922); Ziegler and Nichols (1942).
- **Use.** Thermostats, cruise control, drones, motor drives, chemical plants.
- **Data science.** The momentum and Adam updates resemble PI(D) control of the gradient ML.27 §15 Machine Learning; PID-style controllers also tune autoscalers and rate limiters.
- **Physical application.** Cruise control: P alone leaves you slightly slow on a hill; the I term removes that offset.

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Electronics

EE.32 Diodes, LEDs & Transistors

A diode conducts in one direction, with current rising exponentially with voltage; an LED emits light as it conducts. Transistors are controlled valves: a small base current or gate voltage controls a large current EE.33.

Shockley diode equation	$I = I_s\left(e^{V/(nV_T)} - 1\right)$	Exponential forward current; tiny $-I_s$ in reverse.
Thermal voltage	$V_T = \dfrac{kT}{q} \approx 25.9\ \text{mV at } 300\ \text{K}$	Sets how steep the exponential is.
LED series resistor	$R_{\text{LED}} = \dfrac{V_s - V_f}{I_{\text{LED}}}$	KVL: the resistor drops what the LED does not.
Constant-drop model	$V_D \approx 0.7\ \text{V (Si) when on,} \qquad I = 0 \text{ when off}$	Hand analysis: assume on/off, then check.
Half-wave rectifier	$V_{\text{out,pk}} \approx V_m - 0.7\ \text{V}$	A diode passes only the positive half-cycles.
Ripple with filter capacitor	$V_r \approx \dfrac{I_{\text{load}}}{fC}$	Capacitor tops up each cycle; full-wave: use $2f$.

STATED-EQUATION *Shockley's diode equation: I equals I s times e to the V over n V T, minus one. The thermal voltage V T equals k T over q, about twenty-five point nine millivolts at room temperature. An LED's series resistor equals the supply voltage minus the LED forward voltage, over the LED current. A conducting silicon diode drops about zero point seven volts. A half-wave rectifier's peak output is V m minus zero point seven. The ripple is about the load current over f C.*

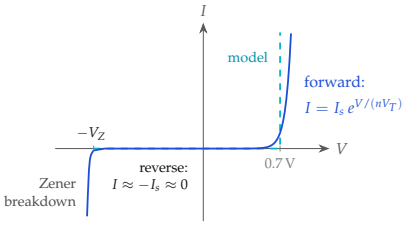
MEMORY AID Silicon diode "turns on" near 0.7 V; the current grows about 10× for every 60 mV more (with $n = 1$).

METHOD

1. Guess each diode's state (on: $V_D = 0.7\,\text{V}$; off: $I = 0$).
2. Solve the linear circuit.
3. Check: an "on" diode needs $I > 0$, an "off" diode needs $V_D < 0.7\,\text{V}$; if not, flip the guess. Example: 5 V supply, red LED $V_f = 2\,\text{V}$, 15 mA: $R = \frac{5-2}{0.015} = 200\,\Omega$.

TERMS

I_s – saturation current (A, tiny: $\sim 10^{-14}$)
n – ideality factor (1 to 2)
k, q, T – Boltzmann constant, electron charge, temperature (K)
V_f – LED forward voltage (red ≈ 2 V, blue ≈ 3 V)
V_m, f – AC peak (V), frequency (Hz); C filter capacitance (F)
Exponential – A2.13 §04 Algebra II; semilog plot is a straight line
Transistor – a current- or voltage-controlled switch/amplifier; billions form a GPU
V, I; V_T – diode voltage and current; thermal voltage
V_D; V_s – diode drop; supply voltage
R_{LED}, I_{LED} – series resistor; desired LED current
V_r; I_{load}; C – peak-to-peak ripple; load current; filter capacitance



CONTEXT

- **Origin.** Shockley (1949); the transistor invented at Bell Labs by Bardeen, Brattain and Shockley (1947).
- **Use.** Rectifiers, LEDs, solar cells, all digital electronics.
- **Data science.** Moore's law, the doubling of transistor counts roughly every two years, is an exponential growth trend fitted on a log scale A2.13 §04 Algebra II.
- **Physical application.** Solar panels are large diodes: light generates current across the junction.

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EE.33 BJT & MOSFET Operating Regions

A transistor's behavior depends on its bias region: off (cutoff), fully on (a switch), or in between where it acts as a controlled current source (an amplifier). Digital logic lives in the first two; amplifiers in the third.

BJT currents	$I_E = I_B + I_C, \qquad I_C = \beta I_B \ \ \text{(active)}$	Base current controls collector current; $\beta \approx 50$–300.
BJT regions	cutoff: $V_{BE} < 0.7$; active: $V_{CE} > 0.2$; saturation: $V_{CE} \approx 0.2\,\text{V}$	Active amplifies; saturation is a closed switch (npn, volts).
MOSFET cutoff	$V_{GS} < V_t \ \Rightarrow \ I_D = 0$	Below threshold the channel is off.
MOSFET triode (linear)	$I_D = k_n\left[(V_{GS} - V_t)V_{DS} - \tfrac{1}{2}V_{DS}^2\right], \ \ V_{DS} < V_{GS} - V_t$	Acts like a voltage-controlled resistor; digital "on".
MOSFET saturation	$I_D = \tfrac{1}{2}k_n(V_{GS} - V_t)^2, \ \ V_{DS} \geq V_{GS} - V_t$	Current set by V_{GS} alone: amplifier region.
Transconductance	$g_m^{\text{MOS}} = \dfrac{2I_D}{V_{GS} - V_t}, \qquad g_m^{\text{BJT}} = \dfrac{I_C}{V_T}$	Small-signal gain: $\Delta I_{\text{out}} = g_m \Delta V_{\text{in}}$ (a VCCS EE.6).

STATED-EQUATION *For a BJT, the emitter current is the base plus collector current, and in the active region the collector current is beta times the base current. Cutoff: V B E below zero point seven volts. Active: V C E above about zero point two volts. Saturation: V C E about zero point two volts. A MOSFET is off when V G S is below the threshold. In triode, I D equals k n times the quantity V G S minus V t times V D S minus half V D S squared. In saturation, I D equals one half k n times V G S minus V t squared. The transconductance is two I D over the overdrive for a MOSFET and I C over V T for a BJT.*

MEMORY AID The names clash: BJT "saturation" = switch fully on; MOSFET "saturation" = amplifier region. MOSFET switch on = triode.

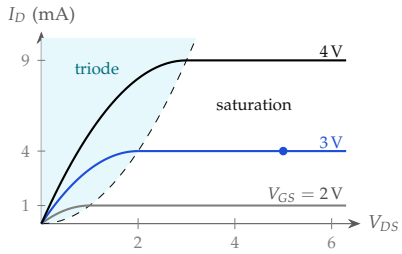
METHOD

1. Assume a region (e.g. MOSFET saturation).
2. Compute the current with that region's equation.
3. Check the region's condition; if it fails, switch region and recompute. Example: $k_n = 2\,\text{mA/V}^2$, $V_t = 1\,\text{V}$, $V_{GS} = 3\,\text{V}$, $V_{DS} = 5\,\text{V}$: overdrive $2\,\text{V} < 5\,\text{V} \Rightarrow$ saturation, $I_D = \frac{1}{2} \cdot 2 \cdot 2^2 = 4\,\text{mA}$.

DECISION POINT Switch or amplifier? For logic and power switching, drive between cutoff and fully on (BJT saturation, MOSFET triode) to minimize power loss. For linear amplification, bias in the middle of the BJT active / MOSFET saturation region so the signal swings without leaving it.

TERMS

I_B, I_C, I_E – base, collector, emitter currents (A)
β – BJT current gain (dimensionless)
V_{BE}, V_{CE} – base–emitter and collector–emitter voltages (V)
V_{GS}, V_{DS} – gate–source, drain–source voltages (V); I_D drain current
V_t – threshold voltage (V); $V_{GS} - V_t$ overdrive
k_n – MOSFET process/size parameter (A/V^2)
g_m – transconductance (S); $V_T \approx 26\,\text{mV}$ EE.32
I_D; V_T – drain current; thermal voltage EE.32



cutoff ($V_{GS} < V_t$): $I_D = 0$

CONTEXT

- **Origin.** BJT at Bell Labs (1947–48); MOSFET by Atalla and Kahng (1959); CMOS by Wanlass (1963).
- **Use.** Logic gates, power converters, audio and RF amplifiers.
- **Data science.** Every GPU multiply–accumulate is CMOS switching: MOSFETs toggling between cutoff and triode; energy per switch $\approx CV^2$ limits AI power budgets.
- **Physical application.** A microcontroller pin drives a MOSFET fully on to switch a 12 V LED strip it could never power directly.

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EE.34 Operational Amplifiers

An op-amp is a very-high-gain differential amplifier. Wrapped in negative feedback, its behavior is set almost entirely by the external resistors, captured by two "golden rules".

Open-loop gain	$V_{\text{out}} = A\,(V_+ - V_-)$	$A \sim 10^5$–10^6; output clips at the supply rails.
Ideal rules (negative feedback)	$V_+ = V_-, \qquad I_+ = I_- = 0$	Virtual short between the inputs; no input current.
Inverting amplifier	$\dfrac{V_{\text{out}}}{V_{\text{in}}} = -\dfrac{R_f}{R_{\text{in}}}$	Input to the $-$ terminal through R_{in}; $+$ grounded.
Non-inverting amplifier	$\dfrac{V_{\text{out}}}{V_{\text{in}}} = 1 + \dfrac{R_f}{R_g}$	Input to $+$; gain ≥ 1, very high input impedance.
Voltage follower	$V_{\text{out}} = V_{\text{in}}$	$R_f = 0$, $R_g = \infty$: gain 1, used as a buffer.
Gain–bandwidth product	$A_{\text{cl}} \cdot f_{-3\text{dB}} \approx \text{GBW}$	Higher closed-loop gain $\Rightarrow$ lower bandwidth.

STATED-EQUATION *The output equals the open-loop gain A times V plus minus V minus. For an ideal op-amp with negative feedback, the two input voltages are equal and no current flows into the inputs. The inverting amplifier's gain equals minus R f over R in. The non-inverting amplifier's gain equals one plus R f over R g. A follower has gain one. Closed-loop gain times bandwidth is about the gain-bandwidth product.*

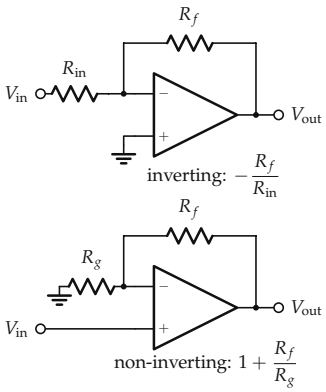
MEMORY AID The two golden rules: "the inputs are at the same voltage" and "no current enters the inputs."

METHOD

1. Confirm negative feedback (output connected back to the $-$ input).
2. Set $V_- = V_+$ (golden rule 1).
3. Write KCL at the $-$ node with zero current into the op-amp (golden rule 2); solve for V_{out}.
4. Check that V_{out} stays inside the rails. Example: inverting, $R_{\text{in}} = 1\,\text{k}\Omega$, $R_f = 10\,\text{k}\Omega$, $V_{\text{in}} = 0.2\,\text{V}$: $\frac{0-0.2}{1\text{k}} + \frac{0-V_{\text{out}}}{10\text{k}} = 0 \Rightarrow V_{\text{out}} = -2\,\text{V}$.

TERMS

V_+, V_- – non-inverting and inverting input voltages (V)
A – open-loop gain; A_{cl} closed-loop gain
R_f – feedback resistor; R_{in}, R_g input / ground resistors (Ω)
GBW – gain–bandwidth product (Hz)
Feedback theory – $T \approx 1/H$ for large loop gain EE.30
Limits – output clipped at the supply rails; finite slew rate
$V_{\text{in}}, V_{\text{out}}$; I_+, I_- – input and output voltages; input currents
R_{in}, R_g – input resistor; resistor from V_- to ground
A_{cl}; $f_{-3\text{dB}}$ – closed-loop gain; bandwidth



CONTEXT

- **Origin.** Harold Black's negative-feedback amplifier (1927); "operational" amplifiers in analog computers (1940s).
- **Use.** Sensor amplification, filters, analog computation.
- **Data science.** Analog in-memory AI chips compute neural-network weighted sums with op-amp style summing circuits EE.35.
- **Physical application.** A guitar pickup's tiny signal is amplified by op-amp circuits in the pedal board.

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EE.35 Op-Amp Configurations

Changing what sits in the input and feedback positions of an inverting op-amp turns it into an adder, a subtractor, an integrator or a differentiator. With no feedback at all it is a comparator.

Generalized inverting	$\dfrac{V_{\text{out}}}{V_{\text{in}}} = -\dfrac{Z_f}{Z_{\text{in}}}$	Use impedances EE.14: every configuration below follows.
Summing amplifier	$V_{\text{out}} = -R_f\left(\dfrac{V_1}{R_1} + \dfrac{V_2}{R_2} + \cdots\right)$	Weighted sum (a neuron's $\sum w_i x_i$); virtual ground isolates the inputs.
Difference amplifier	$V_{\text{out}} = \dfrac{R_2}{R_1}(V_2 - V_1)$	Matched resistor ratios; rejects common-mode noise.
Integrator	$V_{\text{out}} = -\dfrac{1}{RC}\int V_{\text{in}}\,dt, \qquad H(s) = -\dfrac{1}{sRC}$	C in the feedback path; add a large R across C to stop DC drift.
Differentiator	$V_{\text{out}} = -RC\dfrac{dV_{\text{in}}}{dt}, \qquad H(s) = -sRC$	C at the input; amplifies high-frequency noise.
Buffer (follower)	$V_{\text{out}} = V_{\text{in}}, \quad Z_{\text{in}} \to \infty,\ Z_{\text{out}} \to 0$	Isolates a weak source from a heavy load EE.5.
Comparator	$V_{\text{out}} = \begin{cases} +V_{\text{sat}}, & V_+ > V_- \\ -V_{\text{sat}}, & V_+ < V_- \end{cases}$	No feedback: a 1-bit ADC; add hysteresis against chatter.

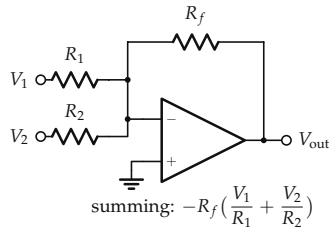
STATED-EQUATION *The generalized inverting gain is minus Z f over Z in. A summing amplifier outputs minus R f times the sum of each V i over R i. A difference amplifier outputs R two over R one times V two minus V one. An integrator outputs minus one over R C times the integral of the input; a differentiator outputs minus R C times its derivative. A buffer copies its input with infinite input and zero output impedance. A comparator saturates to plus or minus V sat according to which input is larger.*

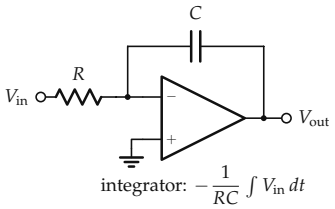
MEMORY AID "Capacitor in feedback integrates; capacitor at the input differentiates."

DECISION POINT Which configuration? Keep the sign or need high input impedance → non-inverting EE.34 or buffer. Mix or weight several signals → summing. Sensor bridge or noisy long wires → difference (instrumentation) amplifier. Accumulate or build a PI controller EE.31 → integrator. Prefer integrators to differentiators in practice: differentiate digitally after filtering.

TERMS

Z_f, Z_{in} – feedback and input impedances (Ω)
V_i, R_i – input voltages (V) and their resistors (Ω)
R, C – integrator/differentiator time constant RC (s)
Common mode – the average $\frac{V_1+V_2}{2}$, ideally ignored
V_{sat} – output saturation level, near the rails (V)
Virtual ground – $V_- \approx 0$ when V_+ is grounded
$H(s)$; s – transfer function; Laplace variable EE.23
Z_{out}; V_1, V_2, R_1, R_2 – output impedance; difference-amplifier inputs and resistors





CONTEXT

- **Origin.** Analog computers (1940s–60s) solved ODEs with integrators and summers; Philbrick's first commercial op-amp (1952).
- **Use.** Mixers, instrumentation, active filters, PID controllers, ADC front ends.
- **Data science.** The summing amplifier is literally an analog neuron: crossbar AI accelerators sum weighted currents the same way ML.22 §15 Machine Learning.
- **Physical application.** A strain-gauge scale uses a difference amplifier to read a tiny bridge voltage riding on a large common-mode level.

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Digital Systems

EE.36 Digital Logic, Binary & Boolean Algebra

Digital circuits represent information with two voltage levels, 0 and 1. Boolean algebra is the arithmetic of those values, and any truth table can be written as a sum of products and built from gates.

Binary value	$N = \sum_{i=0}^{n-1} b_i \, 2^i$	Each bit weighs a power of two; $1011_2 = 8 + 2 + 1 = 11$.
Number of values	n bits $\to 2^n$ values	8 bits $\to$ 256; also 2^n rows in a truth table.
Basic operations	$A \cdot B$ (AND), $A + B$ (OR), $\bar{A}$ (NOT)	AND is 1 only if both are 1; OR if either is.
De Morgan's laws	$\overline{A \cdot B} = \bar{A} + \bar{B}$, $\overline{A + B} = \bar{A} \cdot \bar{B}$	Break the bar, change the operator.
Key identities	$A + \bar{A} = 1$, $A \cdot \bar{A} = 0$, $A + AB = A$	Complement and absorption laws simplify expressions.
Sum of products	$f = \sum(\text{minterms where } f = 1)$	Read straight from a truth table; simplify with K-maps EE.38.

STATED-EQUATION *A binary number equals the sum of each bit b i times two to the i. n bits represent two to the n values. A dot B is AND, A plus B is OR, A bar is NOT. De Morgan's laws: NOT of A AND B equals NOT A OR NOT B, and NOT of A OR B equals NOT A AND NOT B. A or not A is one; A and not A is zero; A plus A B is A. Any function is the OR of its minterms.*

MEMORY AID De Morgan: "break the bar, change the operator" (same as for sets A1.21 §03 Algebra I).

TERMS

Bit – 0 or 1; byte = 8 bits
b_i – bit i (weight 2^i); n number of bits
A, B – Boolean variables
Gates – AND, OR, NOT, NAND, NOR, XOR EE.43
Truth table – all input combinations, 2^n rows S.8 §13 Statistics
Minterm – product of all variables (complemented or not) that is 1 for exactly one row
Hexadecimal, two's complement – EE.37
N; n – value; number of bits

CONTEXT

- **Origin.** George Boole (1854); Claude Shannon linked Boolean algebra to switching circuits (1937).
- **Use.** All digital computers, controllers, and memory.
- **Data science.** Bit masks, hashing, and quantized models (8-bit and 4-bit weights) reduce memory for large neural networks.
- **Physical application.** A two-switch hallway light (either switch toggles it) is an XOR circuit.

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EE.37 Number Systems & Two's Complement

Computers store integers in fixed-width binary. Hexadecimal is shorthand for groups of 4 bits; two's complement represents negative numbers so that the same adder handles signed and unsigned arithmetic.

Positional value (base b)	$N = \sum_i d_i\, b^i$	Binary $b = 2$, octal 8, decimal 10, hex 16 (digits 0–9, A–F).
Hex and octal grouping	1 hex digit $= 4$ bits, 1 octal digit $= 3$ bits	$1101\,0110_2 = \mathrm{D6}_{16} = 214$.
Unsigned range	0 to $2^n - 1$	8 bits: 0 to 255.
Two's complement value	$N = -b_{n-1}2^{n-1} + \sum_{i=0}^{n-2} b_i 2^i$	The top bit has negative weight; it is the sign bit.
Signed range	-2^{n-1} to $2^{n-1} - 1$	8 bits: -128 to 127 (one more negative).
Negation	$-x = \bar{x} + 1$	Invert every bit, then add 1.
Overflow rule	same-sign operands $\to$ opposite-sign result	Signed overflow; the carry-out bit is ignored.
IEEE 754 floating point	$x = (-1)^s \times 1.f \times 2^{e-\text{bias}}$	Binary scientific notation A1.5 §03 Algebra I; float32: bias 127, 23 fraction bits.

STATED-EQUATION *A number's value is the sum of each digit times the base to the power of its position. One hex digit is four bits and one octal digit is three. Unsigned n-bit numbers run from zero to two to the n minus one. In two's complement the top bit has weight minus two to the n minus one, so the range is minus two to the n minus one up to two to the n minus one minus one. To negate, invert all bits and add one. Overflow occurs when two same-sign numbers give a result of the opposite sign. A float is minus one to the s, times one point f, times two to the e minus bias.*

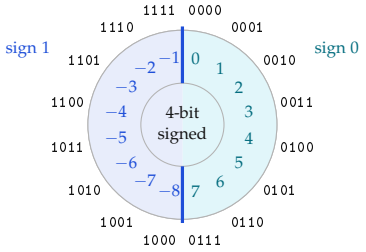
MEMORY AID "Flip and add one." The most negative number has no positive partner (-128 in 8 bits).

METHOD

1. Decimal $\to$ binary: divide by 2 repeatedly; the remainders, read bottom-up, are the bits.
2. Pad to n bits. For a negative number, write $|x|$, invert all bits, add 1.
3. Example: $13 = 1101_2 \to 00001101$; -13: invert 11110010, add 1: $11110011_2 = \mathrm{F3}_{16}$. Check: $-128 + 64 + 32 + 16 + 2 + 1 = -13$.

TERMS

b – base (radix); d_i digit in position i
n – word width in bits; b_{n-1} the sign bit
$\bar{x}$ – bitwise complement (one's complement)
s, f, e – sign bit, fraction (mantissa) bits, stored exponent
bias – exponent offset (127 for float32, 1023 for float64)
Overflow – result outside the representable range
d_i; b_i – digit in position i; bit i
x; N – a binary number; its value



overflow: $0111 + 1 = 1000$ ($7 \to -8$)

-128	64	32	16	8	4	2	1
1	1	1	1	0	0	1	1

$-128 + 64 + 32 + 16 + 2 + 1 = -13$

CONTEXT

- **Origin.** Leibniz's binary arithmetic (1703); two's complement in von Neumann's EDVAC report (1945); IEEE 754 (1985).
- **Use.** CPU arithmetic, memory addresses, color codes (#FF8800), network masks.
- **Data science.** Integer overflow in feature hashing and IDs; int8 quantization of model weights uses the -128 to 127 range; float16/bfloat16 trade precision for range.
- **Physical application.** A car odometer rolling over from 999999 to 000000 is unsigned overflow.

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EE.38 Karnaugh Maps & Logic Minimization

A Karnaugh map arranges a truth table so that neighboring cells differ in one variable. Circling rectangular groups of 1s visually applies $AB + A\bar{B} = A$, giving a minimal sum-of-products circuit.

Combining rule	$AB + A\bar{B} = A$	Two terms that differ in one variable merge and drop it.
Gray-code order	00, 01, 11, 10	Row and column labels change one bit at a time; edges wrap around.
Group size	2^k cells $\Rightarrow$ k variables eliminated	Groups must be rectangles of 1, 2, 4, 8 ... cells.
Minterm list	$f(A,B,C) = \sum m(\dots)$	Indices of the rows where $f = 1$ (read ABC as binary).
Don't cares	$X = 0$ or 1 as convenient	Inputs that never occur; include them only to enlarge groups.
Product of sums	group the 0s $\Rightarrow$ $f = \prod$(sum terms)	Dual method; useful when there are fewer 0s.

STATED-EQUATION *A B plus A B-bar equals A. Rows and columns follow Gray code: zero zero, zero one, one one, one zero, so neighbors differ in one bit and edges wrap. A group of two to the k cells eliminates k variables. A function is listed by its minterms. Don't-care cells may be treated as zero or one. Grouping the zeros gives a product of sums.*

MEMORY AID "Biggest boxes, fewest boxes, powers of two, wrap the edges."

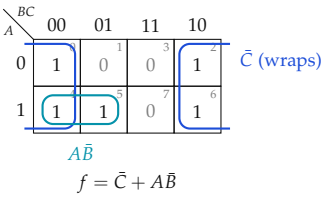
METHOD

1. Fill the map from the truth table (Gray-code order).
2. Circle the largest power-of-two rectangles of 1s (wrapping allowed) until every 1 is covered; overlaps are fine.
3. Each group gives one product of the variables that stay constant in it.
4. OR the products. Example (figure): $f = \sum m(0,2,4,5,6)$: the four cells with $C = 0$ give $\bar{C}$; cells 4, 5 give $A\bar{B}$; $f = \bar{C} + A\bar{B}$.

DECISION POINT Which groups? Start with 1s that can be covered only one way (essential prime implicants), then cover the rest with the fewest, largest groups. Several minimal answers can exist. Beyond 4–5 variables, use Quine–McCluskey or a tool (Espresso).

TERMS

A, B, C – input variables; $\bar{C}$ complement
m_i – minterm i (row i of the truth table) EE.36
Implicant – a group of 1s; prime: cannot be enlarged
Gray code – ordering where neighbors differ in one bit
SOP / POS – sum of products / product of sums
X – don't-care entry



CONTEXT

- **Origin.** Maurice Karnaugh, Bell Labs (1953), refining Veitch (1952).
- **Use.** Simplifying control logic, reducing gate count and delay.
- **Data science.** Rule-learning and decision-tree simplification merge conditions the same way; Boolean minimization underlies logic synthesis for ML accelerators.
- **Physical application.** A seven-segment display decoder is designed with one K-map per segment.

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EE.39 Flip-Flops, Registers & Counters

Sequential logic has memory. A flip-flop stores one bit and updates only on a clock edge; registers are banks of flip-flops, and counters are registers that step through a sequence of states.

D flip-flop	$Q^+ = D$	Copies D at the clock edge; the basic register bit.
JK flip-flop	$Q^+ = J\bar{Q} + \bar{K}Q$	J sets, K resets, $J = K = 1$ toggles.
T flip-flop	$Q^+ = T \oplus Q$	Toggles when $T = 1$; the counter building block.
SR latch	$Q^+ = S + \bar{R}Q \quad (SR = 0)$	Set/reset memory; $S = R = 1$ is forbidden.
Binary counter	n flip-flops $\rightarrow$ mod-2^n, $\quad f_{Q_i} = \dfrac{f_{\text{clk}}}{2^{i+1}}$	Each stage halves the frequency (figure).
Maximum clock frequency	$f_{\max} = \dfrac{1}{t_{cq} + t_{\text{logic}} + t_{\text{setup}}}$	The slowest register-to-register path sets the clock.

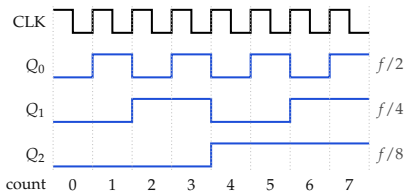
STATED-EQUATION *A D flip-flop's next state equals D. A JK flip-flop's next state is J Q-bar plus K-bar Q. A T flip-flop's next state is T XOR Q. An SR latch's next state is S plus R-bar Q, with S and R never both one. An n-bit counter counts modulo two to the n and stage i toggles at f clock over two to the i plus one. The maximum clock frequency is one over the clock-to-Q delay plus logic delay plus setup time.*

MEMORY AID "D for Data (copy), T for Toggle, JK: J jumps up, K knocks down."

DECISION POINT Latch or flip-flop? Latches are level-sensitive (transparent while enabled) and smaller; flip-flops are edge-triggered and make timing analysis simple. Synchronous designs use edge-triggered D flip-flops almost exclusively; derive JK/T behavior with logic in front of a D.

TERMS

Q, Q^+ – present state and next state (after the clock edge)
D, J, K, T, S, R – flip-flop inputs (data, set/reset, toggle)
$\oplus$ – exclusive OR EE.43
f_{clk} – clock frequency (Hz); Q_i counter bit i
t_{cq}, t_{logic}, t_{setup} – clock-to-output, combinational, setup delays (s)
Register – n flip-flops sharing a clock
n; Q_i – number of flip-flops; output of stage i



CONTEXT

- **Origin.** Eccles–Jordan trigger circuit (1918); edge-triggered flip-flops in 1960s TTL logic.
- **Use.** CPU registers, counters, timers, state machines, frequency dividers.
- **Data science.** Clock speed and pipeline depth set GPU throughput; a finite-state machine is the hardware analog of a Markov chain O.29 §14 Optimization.
- **Physical application.** A 32 768 Hz watch crystal divided by a 15-stage counter gives exactly 1 tick per second ($2^{15} = 32768$).

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Schematic Symbols

EE.40 Schematic Symbols: Passive Components & Switches

Passive components absorb or store energy but cannot amplify. Each symbol is a picture of what the part does: friction, a gap, a coil.

Resistor	$V = IR$	Zigzag (US) or rectangle (IEC) EE.2.
Capacitor	$i = C\,\frac{dv}{dt}$	Two parallel plates EE.10.
Inductor	$v = L\,\frac{di}{dt}$	A coil of wire EE.10.
Transformer	$\frac{V_s}{V_p} = \frac{N_s}{N_p}$	Two coupled coils on a core EE.19.
Fuse	opens when $I > I_{\text{rated}}$	One-time overcurrent protection.
Lamp	$P = \frac{V^2}{R}$	A resistive load.

STATED-EQUATION *A resistor obeys V equals I R. A capacitor's current equals C times d v d t. An inductor's voltage equals L times d i d t. A transformer's voltage ratio equals its turns ratio, N s over N p. A fuse opens when the current exceeds its rating. A lamp dissipates V squared over R.*

MEMORY AID Zigzag = resistor (friction); two parallel plates = capacitor (a gap that stores charge); coils = inductor (a wound wire).



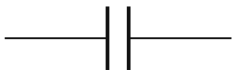
SY.1 Resistor

Zigzag (US) or rectangle (IEC); $V = IR$.



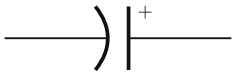
SY.2 Potentiometer

Resistor with an arrow wiper; volume knobs.



SY.3 Capacitor

Two parallel plates; blocks DC, passes AC.



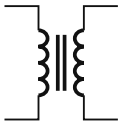
SY.4 Polarized Capacitor

Curved plate is negative; large values.



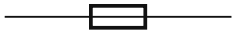
SY.5 Inductor

Coil; resists changes in current.



SY.6 Transformer

Coupled coils; steps AC voltage up/down.



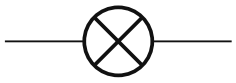
SY.7 Fuse

Overcurrent protection (one-time).



SY.8 Switch (SPST)

Opens or closes a single path.



SY.9 Lamp

Circle with an X; a resistive load.

TERMS

US vs. IEC – US zigzag resistor; IEC (Europe) uses a plain rectangle
Values – written beside the symbol (e.g. 4.7 kΩ, 10 μF, 1 mH)
V, I, R, C, L – volts, amperes, ohms, farads, henries EE.1
N_p, N_s – primary and secondary turns; I_{rated} fuse rating (A)
Passive – absorbs or stores energy; cannot amplify EE.10
Polarity – polarized parts must be installed the right way round
v, i; I_{rated} – instantaneous voltage and current; fuse rating

CONTEXT

- **Origin.** Standardized symbols from IEEE 315 (1975) and IEC 60617.
- **Use.** Reading and drawing circuit diagrams (schematics).
- **Data science.** Hardware and IoT data teams read schematics to know what a sensor channel measures and where noise enters.
- **Physical application.** A light-dimmer wall switch is a potentiometer (or its electronic equivalent) controlling a lamp.

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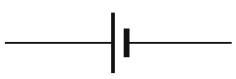
EE.41 Schematic Symbols: Sources, Meters & Ground

Sources supply energy; meters measure without (ideally) disturbing the circuit; ground is the 0 V reference every node voltage is measured from.

Ideal voltage source	$V = \text{const}$ for any I	Circle with $+/-$, or a battery EE.6.
Ideal current source	$I = \text{const}$ for any V	Circle with an arrow.
AC source	$v(t) = V_m \sin(\omega t + \phi)$	Circle with a sine EE.13.
Voltmeter	$R_{\text{in}} \to \infty$, in parallel	Measures the voltage across a part.
Ammeter	$R_{\text{in}} \to 0$, in series	Measures the current through a branch.
Ground	$V_{\text{gnd}} = 0$ V	Reference node for nodal analysis EE.7.

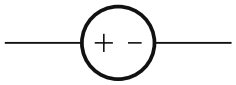
STATED-EQUATION *An ideal voltage source holds a constant voltage for any current. An ideal current source holds a constant current for any voltage. An AC source produces V m sine of omega t plus phi. A voltmeter has very high resistance and is connected in parallel. An ammeter has very low resistance and is connected in series. Ground is the zero-volt reference.*

MEMORY AID "Volts across, amps through": voltmeter across the part, ammeter in the path.



SY.10 Battery

Long plate is +; DC voltage.



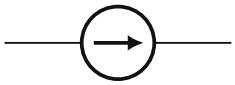
SY.11 DC Voltage Source

Circle with $+/-$; fixed voltage.



SY.12 AC Voltage Source

Circle with a sine wave; mains, signals.



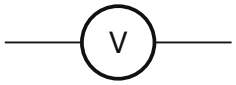
SY.13 Current Source

Circle with an arrow; fixed current.



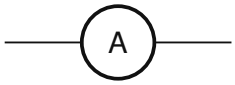
SY.14 Ground

Stacked lines; the 0 V reference.



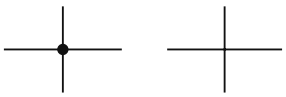
SY.15 Voltmeter

Circle with V; connect across.



SY.16 Ammeter

Circle with A; connect in series.



SY.17 Junction vs. Crossing

Dot = connected; no dot = crossing.

TERMS

Independent source – value set by itself EE.1
Dependent source – diamond shape; value controlled by another voltage or current EE.6
V_m, ω, ϕ – amplitude (V), angular frequency (rad/s), phase
R_{in} – meter's internal resistance (Ω)
Ground – reference node for nodal analysis EE.7
Node – all points joined by wire and dots share one voltage EE.4
Meters – ideal meters do not disturb the circuit

CONTEXT

- **Origin.** Symbols evolved from telegraph and power engineering drawings (late 1800s).
- **Use.** Wiring measurements and power supplies correctly.
- **Data science.** Understanding where a data-acquisition system measures voltage vs. current prevents misinterpreted sensor data.
- **Physical application.** Using a multimeter: measure battery voltage across its terminals; measure current by breaking the circuit and inserting the meter.

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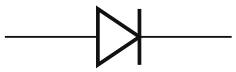
EE.42 Schematic Symbols: Semiconductors & Op-Amps

Semiconductor symbols show direction: the diode's arrow is the direction conventional current may flow, and a transistor's emitter arrow shows the polarity. Active devices need a power supply and can amplify.

Diode	$I = I_s\left(e^{V/(nV_T)} - 1\right)$	Conducts along the arrow (anode → cathode) EE.32.
Zener diode	$V \approx V_Z$ in reverse breakdown	A voltage reference.
BJT	$I_C \approx \beta I_B$	Base current controls collector current EE.33.
MOSFET	$I_D = f(V_{GS})$	Gate voltage controls the drain current.
Op-amp	$V_{\text{out}} = A\,(V_+ - V_-)$	Triangle with + and − inputs EE.34.

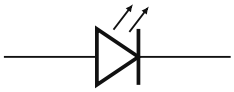
STATED-EQUATION *A diode's current equals I s times e to the V over n V T minus one, and it conducts in the direction of its arrow. A Zener diode holds about V Z in reverse breakdown. A bipolar transistor's collector current is about beta times the base current. In a MOSFET, the gate voltage controls the drain current. An op-amp's output equals its open-loop gain A times the difference between its plus and minus inputs.*

MEMORY AID The diode's arrow points the way conventional current flows (anode → cathode; the bar is the cathode).



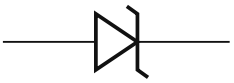
SY.18 Diode

Arrow into a bar; one-way current.



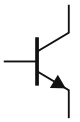
SY.19 LED

Diode with outgoing arrows (light).



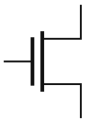
SY.20 Zener Diode

Bent bar; holds a fixed reverse voltage.



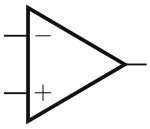
SY.21 NPN Transistor (BJT)

Base, collector, emitter (arrow out).



SY.22 MOSFET (n-channel)

Gate, drain, source; digital switch.



SY.23 Op-Amp

Triangle with + and − inputs.

TERMS

Anode / cathode – diode terminals; the bar marks the cathode EE.32
I_s, n, V_T – saturation current, ideality, thermal voltage EE.32
V_Z – Zener breakdown voltage (V)
β – current gain of a BJT (often 50–300); I_B, I_C (A)
Gate, drain, source – MOSFET terminals; V_{GS}, I_D
A – op-amp open-loop gain; rules EE.34
Active device – can amplify (needs a power supply)
I_C, I_B; I_D, V_{GS} – collector and base currents; drain current and gate–source voltage EE.33
V_+, V_- – op-amp inputs

CONTEXT

- **Origin.** Diode (Fleming valve 1904; semiconductor diodes 1940s); BJT (1947); MOSFET (Atalla and Kahng, 1959).
- **Use.** Rectification, switching, amplification, all digital logic.
- **Data science.** A GPU contains tens of billions of MOSFETs switching billions of times per second to run neural networks.
- **Physical application.** A phone charger rectifies AC to DC with a bridge of four diodes.

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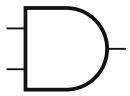
EE.43 Schematic Symbols: Logic Gates

Logic gates implement Boolean operations EE.36. The shape names the operation; a small circle (bubble) inverts.

AND	$Y = A \cdot B$	1 only if all inputs are 1.
OR	$Y = A + B$	1 if any input is 1.
NOT	$Y = \bar{A}$	Inverts.
NAND	$Y = \overline{A \cdot B}$	Universal: builds every other gate.
XOR	$Y = A \oplus B = A\bar{B} + \bar{A}B$	1 when the inputs differ; a 1-bit adder's sum.

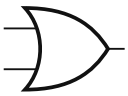
STATED-EQUATION *AND: Y equals A and B. OR: Y equals A or B. NOT: Y equals not A. NAND: Y equals not of A and B. XOR: Y equals A exclusive-or B, which is A and not B, or not A and B.*

MEMORY AID A small circle ("bubble") on an output means NOT. AND has a flat back, OR has a curved back, XOR adds an extra curved line.



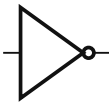
SY.24 AND Gate

Flat back, round front.



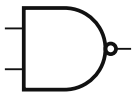
SY.25 OR Gate

Curved back, pointed front.



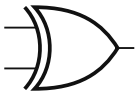
SY.26 NOT Gate (Inverter)

Triangle with an output bubble.



SY.27 NAND Gate

AND with a bubble.



SY.28 XOR Gate

OR with an extra back curve.

A	B	AND	OR	NAND	NOR	XOR	NOT A
0	0	0	0	1	1	0	1
0	1	0	1	1	0	1	1
1	0	0	1	1	0	1	0
1	1	1	1	0	0	0	0

TERMS

A, B – inputs; Y output (0 or 1)
Truth table – output for every input combination EE.36
Bubble – inversion (NOT)
$\oplus$ – exclusive OR
Universal gates – NAND (or NOR) alone can build any logic
Boolean algebra – De Morgan EE.36, A1.21 §03 Algebra I; minimization EE.38
Y – gate output

CONTEXT

- **Origin.** Boole (1854); Shannon (1937); standardized gate shapes in ANSI/IEEE 91.
- **Use.** Designing digital circuits, CPUs, and controllers.
- **Data science.** Boolean filters in SQL and pandas masks (`&`, `|`, `~`) are the software form of these gates.
- **Physical application.** A car's "door open AND key in ignition ⇒ chime" warning is an AND gate.

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17
Finance

The time value of money, valuation, risk and return, fixed income, and derivatives. Finance is applied exponentials, series, probability, and optimization: every loan, bond, portfolio, and option below is built from earlier concepts.

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Time Value of Money

FN.1 Simple vs. Compound Interest

Interest is the price of using money over time. Simple interest is paid on the principal only (linear growth); compound interest is also paid on interest already earned (exponential growth).

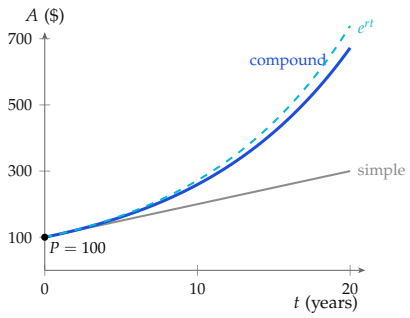
Simple interest	$I = Prt, \qquad A = P(1+rt)$	Interest on the original principal only; the balance grows by the same amount every year.
Compound interest	$A = P\left(1+\frac{r}{n}\right)^{nt}$	The rate $\frac{r}{n}$ is credited n times a year and each credit earns interest afterwards.
Continuous compounding	$A = Pe^{rt}$	The limit $n \to \infty$; the most a nominal rate r can grow to C1.6 §07 Calculus I.
Solving for time	$t = \dfrac{\ln(A/P)}{n\ln\left(1+\frac{r}{n}\right)}$	Take logs A2.16 §04 Algebra II. \$100 at 10% yearly doubles in $\frac{\ln 2}{\ln 1.1} \approx 7.3$ years.
Solving for the rate	$r = n\left[\left(\dfrac{A}{P}\right)^{1/(nt)} - 1\right]$	Take the nt-th root; continuous: $r = \frac{\ln(A/P)}{t}$.

STATED-EQUATION *Simple interest: I equals P r t, so A equals P times one plus r t. Compound interest: A equals P times one plus r over n, raised to the n t. Continuous compounding: A equals P e to the r t. Solving for time: t equals the log of A over P, over n times the log of one plus r over n. Solving for the rate: r equals n times the quantity A over P to the one over n t, minus one.*

MEMORY AID Compounding is "interest on interest"; "Pert" for continuous.

TERMS

P – principal (present amount)
A – accumulated amount (future value)
I – interest earned
r – annual nominal rate (decimal)
n – compounding periods per year
t – years
Growth – linear (simple) vs. exponential (compound) A1.7 §03 Algebra I, A2.13 §04 Algebra II
Limit – $n \to \infty$ gives e^{rt} C1.6 §07 Calculus I



CONTEXT

- **Origin.** Compound interest tables in Babylon (∼2000 BCE); Jacob Bernoulli discovered e studying continuous compounding (1683).
- **Use.** Savings, loans, credit cards, inflation.
- **Data science.** Customer lifetime value and growth projections compound monthly retention and revenue rates.
- **Physical application.** Bacterial and radioactive growth follow the same compounding law as money.

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FN.2 APR vs. Effective Annual Rate

A nominal rate (APR) is a quoting convention; the effective annual rate (EAR) is the growth actually achieved in a year after compounding. Convert every quote to a common basis before comparing or plugging into formulas.

Effective annual rate	$\mathrm{EAR} = \left(1 + \frac{\mathrm{APR}}{n}\right)^n - 1$	Compounds the periodic rate $\frac{\mathrm{APR}}{n}$ over one year; EAR $\geq$ APR when $n > 1$.
Continuous EAR	$\mathrm{EAR} = e^r - 1$	Effective rate of a continuously compounded nominal rate r.
Nominal from effective	$\mathrm{APR}_n = n\left[(1+\mathrm{EAR})^{1/n} - 1\right]$	The APR with frequency n that produces a given EAR.
Continuous equivalent	$r_c = \ln(1+\mathrm{EAR}) = n\ln\left(1 + \frac{\mathrm{APR}}{n}\right)$	The continuously compounded rate with the same annual growth.
Periodic rate conversion	$i_m = (1+i_n)^{n/m} - 1$	Converts a rate per $\frac{1}{n}$ year into a rate per $\frac{1}{m}$ year with the same growth.

STATED-EQUATION *The effective annual rate equals one plus the APR over n, raised to the n, minus one. With continuous compounding, the effective annual rate equals e to the r minus one. The APR with n periods that gives an effective rate is n times the quantity one plus EAR to the one over n, minus one. The equivalent continuous rate is the natural log of one plus EAR. A periodic rate converts by raising one plus the rate to the n over m and subtracting one.*

MEMORY AID APR is the "sticker" rate; EAR is what you actually pay or earn. EAR $\geq$ APR whenever $n > 1$.

METHOD

1. Convert the quoted rate to EAR.
2. Convert the EAR to the basis you need (another APR, a continuous rate, or a rate per payment period).
3. Example: 12% APR monthly $\Rightarrow$ EAR $= 1.01^{12} - 1 = 12.68\%$; the quarterly APR with the same EAR is $4[(1.1268)^{1/4} - 1] = 12.12\%$.

DECISION POINT Compare offers only on EAR. Inside a formula, use the rate *per payment period*: monthly payments need a monthly rate, which is $\frac{\mathrm{APR}}{12}$ if the APR compounds monthly but $(1+\mathrm{EAR})^{1/12} - 1$ if you were given an effective rate.

TERMS

APR – annual percentage rate (nominal)
EAR / APY – effective annual rate / annual percentage yield, after compounding
n, m – compounding frequencies per year (12 monthly, 365 daily)
r, r_c – continuously compounded rate FN.1
i_n – rate per $\frac{1}{n}$ year
Comparison – only compare offers using EAR
i_m – equivalent rate per period when there are m periods per year

CONTEXT

- **Origin.** US Truth in Lending Act (1968) required APR disclosure.
- **Use.** Comparing loans, credit cards and savings accounts.
- **Data science.** Normalizing rates quoted with different conventions before feeding them into pricing or credit-risk models.
- **Physical application.** Growth rates quoted per year vs. per month (e.g. population or reaction rates) must be converted the same way.

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FN.3 Doubling Time & the Rule of 72

How long a quantity growing at a constant rate takes to double, exactly by logarithms or mentally by the rule of 72.

Exact doubling time	$t_{\text{double}} = \dfrac{\ln 2}{\ln(1+r)}$	Solve $(1+r)^t = 2$ by logarithms A2.16 §04 Algebra II.
Rule of 72	$t_{\text{double}} \approx \dfrac{72}{r\ (\text{in }\%)}$	Mental shortcut, most accurate near 8%: $72/8 = 9$ years.
Continuous doubling	$t_{\text{double}} = \dfrac{\ln 2}{r} \approx \dfrac{69.3}{r\,(\%)}$	Exact for continuous growth e^{rt}; also the half-life formula for decay.
Tripling (rule of 114)	$t_{\text{triple}} = \dfrac{\ln 3}{\ln(1+r)} \approx \dfrac{114}{r\,(\%)}$	Same idea with $\ln 3 \approx 1.10$.

STATED-EQUATION *The exact doubling time equals the natural log of two over the natural log of one plus r. The rule of seventy-two: doubling time is approximately seventy-two divided by the rate in percent. With continuous growth the doubling time is ln two over r, about sixty-nine point three over the rate in percent. Tripling takes about one hundred fourteen over the rate in percent.*

MEMORY AID Rule of 72 (use 69.3 for continuous compounding, since $\ln 2 \approx 0.693$).

TERMS

r – growth rate per period (decimal; in % for the rules)
$\ln 2$ – from solving $(1+r)^t = 2$ A2.16 §04 Algebra II
Why 72 – close to 69.3 and divisible by 2, 3, 4, 6, 8, 9, 12
Units – t is in the same periods as r
t_{double}, t_{triple} – periods needed to double or triple

CONTEXT

- **Origin.** Luca Pacioli, *Summa de arithmetica* (1494).
- **Use.** Mental math for investments, inflation and debt.
- **Data science.** Quick sanity checks of growth metrics: 10% monthly user growth doubles in about 7 months.
- **Physical application.** Moore's law and bacterial growth doubling times use the same logarithmic calculation.

FN.4 Inflation & Real Returns

Inflation erodes purchasing power. Real (inflation-adjusted) rates measure growth in what money can buy; nominal rates measure growth in currency units.

Fisher equation	$(1+i) = (1+r_{\text{real}})(1+\pi)$	Nominal growth is real growth compounded with price growth.
Exact real rate	$r_{\text{real}} = \dfrac{1+i}{1+\pi} - 1$	Solve the Fisher equation for the real rate.
Approximate real rate	$r_{\text{real}} \approx i - \pi$	Good when rates are small; drops the cross term $r\pi$.
Real value (deflating)	$\text{real value} = \dfrac{\text{nominal}}{(1+\pi)^t}$	Expresses a future amount in today's purchasing power.
Inflation rate from CPI	$\pi_t = \dfrac{\text{CPI}_t}{\text{CPI}_{t-1}} - 1$	Percent change of a price index A1.19 §03 Algebra I.

STATED-EQUATION *The Fisher equation: one plus the nominal rate equals one plus the real rate times one plus inflation. So the real rate equals one plus i over one plus pi, minus one, approximately i minus pi. A future nominal amount's real value equals it divided by one plus pi to the t. Inflation is the percent change in the consumer price index.*

MEMORY AID "Real ≈ nominal minus inflation."

TERMS

i – nominal interest or return
π – inflation rate (here π is not 3.14!)
r_{real} – real (inflation-adjusted) rate
CPI – consumer price index used to measure π
t – number of years
Deflating – same as discounting FN.5

CONTEXT

- **Origin.** Irving Fisher, *The Theory of Interest* (1930).
- **Use.** Comparing salaries, returns and prices across years.
- **Data science.** Always deflate multi-year revenue and price series to real terms before modeling trends.
- **Physical application.** Comparing energy costs over decades requires inflation-adjusted prices.

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FN.5 Present Value & Discounting

Discounting is compounding run backwards: it converts a future amount into the equivalent amount today at a chosen rate of return.

Present value	$PV = \dfrac{FV}{(1+r)^t}$	What a future amount is worth today at discount rate r.
Future value	$FV = PV\,(1+r)^t$	The inverse operation: compound today's amount forward.
Discount factor	$DF_t = \dfrac{1}{(1+r)^t} = e^{-r_c t}$ (continuous)	Multiply a cash flow at t by DF_t to bring it to today.
PV of a cash-flow stream	$PV = \sum_{t=1}^{T} \dfrac{CF_t}{(1+r)^t}$	Values add: discount each flow separately and sum A2.21 §04 Algebra II.

STATED-EQUATION *Present value equals future value over one plus r, to the t. Future value equals present value times one plus r to the t. The discount factor is one over one plus r to the t, or e to the minus r t with continuous compounding. A stream of cash flows is worth the sum of each discounted flow.*

MEMORY AID "A dollar today is worth more than a dollar tomorrow."

DECISION POINT Choosing r: it is the return of the best alternative with the *same risk*. Certain flows → a risk-free (Treasury) rate; a typical project of the firm → WACC FN.17; equity cash flows → the cost of equity FN.14. Match nominal flows with nominal rates and real flows with real rates FN.4, and the rate's period with the cash-flow period FN.2.

TERMS

PV – value today
FV – value at time t
r – discount rate per period (opportunity cost of capital)
r_c – continuously compounded rate FN.2
CF_t – cash flow at time t
Inverse – discounting undoes compounding FN.1, A2.1 §04 Algebra II
DF_t; t, T – discount factor; period and final period

CONTEXT

- **Origin.** Fibonacci's *Liber Abaci* (1202) included present-value problems; formalized by Irving Fisher (1930).
- **Use.** Valuing any future payment: pensions, settlements, projects.
- **Data science.** Discount factors appear in reinforcement learning (γ O.21 §14 Optimization) and in customer-lifetime-value models.
- **Physical application.** Energy-efficiency upgrades (insulation, solar panels) are evaluated by discounting future energy savings.

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FN.6 Annuities & Perpetuities

An annuity is a level payment for a fixed number of periods; a perpetuity pays forever. Their closed forms are finite and infinite geometric series of discounted payments.

PV of an ordinary annuity	$PV = C\,\dfrac{1-(1+r)^{-n}}{r}$	Payments C at the end of each of n periods.
FV of an ordinary annuity	$FV = C\,\dfrac{(1+r)^{n}-1}{r}$	Value at the date of the last payment; $= PV(1+r)^{n}$.
Perpetuity	$PV = \dfrac{C}{r}$	Level payments forever; the annuity with $n \to \infty$.
Growing perpetuity	$PV = \dfrac{C_1}{r-g} \quad (r > g)$	Payments grow at rate g forever; C_1 is the *first* payment, one period from now.

STATED-EQUATION *The present value of an annuity equals C times one minus one plus r to the minus n, over r. The future value of an annuity equals C times one plus r to the n minus one, over r. A perpetuity is worth C over r. A growing perpetuity is worth C one over r minus g, when r exceeds g.*

MEMORY AID A perpetuity is an annuity with $n \to \infty$: the $(1+r)^{-n}$ term vanishes.

METHOD

1. Draw the timeline and confirm the first payment is one period from now (otherwise see FN.7).
2. Use the rate per payment period and the number of payments.
3. Apply the formula. Example: \$100 a year for 10 years at 5%: $PV = 100 \cdot \frac{1-1.05^{-10}}{0.05} = \$772.17$.

TERMS

C – payment per period (end of period)
C_1 – first payment of a growing stream
n – number of payments
r – rate per period
g – growth rate of payments
Derivation – geometric series A2.19 §04 Algebra II, C2.16 §08 Calculus II

CONTEXT

- **Origin.** Annuity valuation by Edmond Halley and de Moivre (1690s–1720s) for life insurance.
- **Use.** Retirement income, leases, pensions, loan payments.
- **Data science.** Subscription revenue with churn is a growing (or shrinking) perpetuity: CLV $\approx \frac{m}{r+\text{churn}}$.
- **Physical application.** The present value of a hydroelectric dam's steady output is valued like a perpetuity.

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FN.7 Annuities Due & Growing Annuities

Two common variations: payments at the *start* of each period (annuity due), and payments that grow at a constant rate for a finite term (growing annuity).

PV of an annuity due	$PV_{\text{due}} = C\,\dfrac{1-(1+r)^{-n}}{r}\,(1+r)$	Every payment arrives one period earlier, so multiply by $(1+r)$.
FV of an annuity due	$FV_{\text{due}} = C\,\dfrac{(1+r)^{n}-1}{r}\,(1+r)$	Measured one period after the last payment.
PV of a growing annuity	$PV = \dfrac{C_1}{r-g}\left[1-\left(\dfrac{1+g}{1+r}\right)^{n}\right]$	n payments growing at g; tends to $\frac{C_1}{r-g}$ as $n\to\infty$.
FV of a growing annuity	$FV = C_1\,\dfrac{(1+r)^{n}-(1+g)^{n}}{r-g}$	Value at the last payment date $= PV(1+r)^{n}$.
Equal-rate case	$g = r:\quad PV = \dfrac{nC_1}{1+r}$	Growth exactly offsets discounting; the general formula is $\frac{0}{0}$ here.

STATED-EQUATION *An annuity due is worth the ordinary annuity times one plus r, both in present and in future value. A growing annuity's present value is C one over r minus g, times one minus the quantity one plus g over one plus r, to the n. Its future value is C one times one plus r to the n minus one plus g to the n, over r minus g. If g equals r, the present value is n C one over one plus r.*

MEMORY AID "Due" = "do it now": payments at the start, one extra factor $(1+r)$.

DECISION POINT Ordinary or due? Ask *when the first payment happens*: today (rent, leases, insurance premiums, saving at the start of each month) → due; one period from now (loan payments, bond coupons) → ordinary. Growing? Use the growing forms when payments are indexed to salary, inflation or sales.

TERMS

C – level payment
C_1 – first payment of the growing stream
g – growth rate per period
r – discount rate per period
n – number of payments
Ordinary forms – FN.6

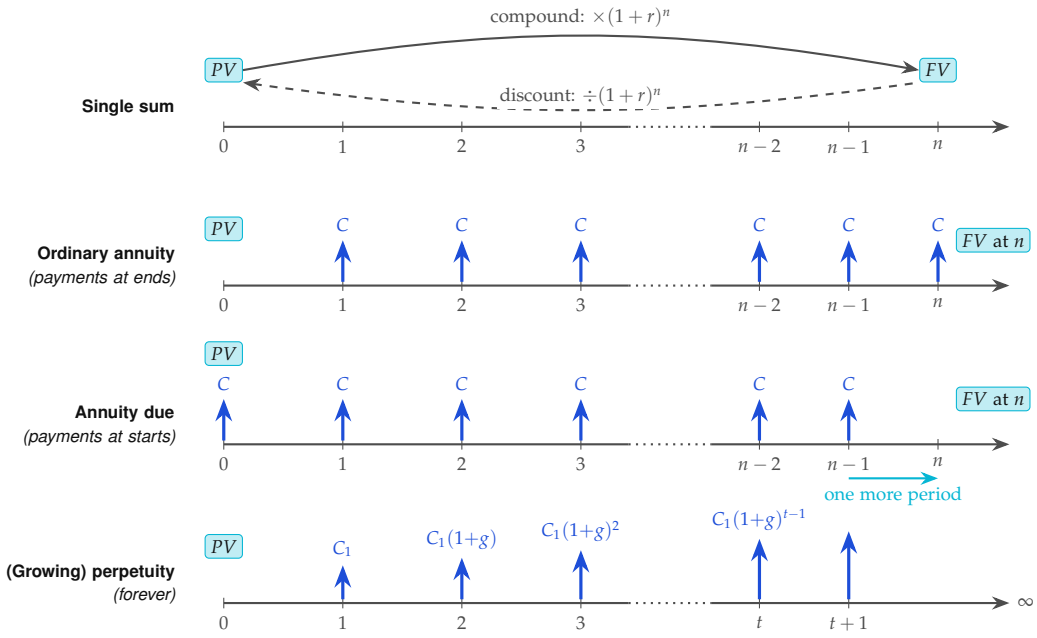
CONTEXT

- **Origin.** Actuarial science (18th–19th c.) tabulated annuity-due factors for life insurance.
- **Use.** Retirement saving with rising contributions, leases, salary-linked pensions.
- **Data science.** Forecasting the value of a subscription cohort whose revenue per user grows while the cohort shrinks.
- **Physical application.** Valuing a stream of energy savings that rise with electricity prices.

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Time Value of Money at a Glance | *Key Topic FN.K1*

Every valuation in this chapter is the same move: put each cash flow on a timeline, move it to one date with $(1+r)^{\pm t}$, and add. The closed forms below are shortcuts for common patterns of payments. Before using one, check three things: the *rate per period*, the *number of periods*, and *when the first payment occurs*.



All present- and future-value formulas

CASH FLOWS	PRESENT VALUE (AT 0)	FUTURE VALUE (AT n)	NOTES
Single sum	$\dfrac{FV}{(1+r)^n}$	$PV(1+r)^n$	FN.5; continuous: $e^{\mp rt}$
Ordinary annuity	$C\,\dfrac{1-(1+r)^{-n}}{r}$	$C\,\dfrac{(1+r)^n-1}{r}$	payments at $1,\dots,n$ FN.6
Annuity due	$C\,\dfrac{1-(1+r)^{-n}}{r}(1+r)$	$C\,\dfrac{(1+r)^n-1}{r}(1+r)$	payments at $0,\dots,n-1$ FN.7
Perpetuity	$\dfrac{C}{r}$	—	$n\to\infty$
Growing perpetuity	$\dfrac{C_1}{r-g}$	—	needs $r>g$; Gordon model FN.18
Growing annuity	$\dfrac{C_1}{r-g}\left[1-\left(\dfrac{1+g}{1+r}\right)^n\right]$	$C_1\dfrac{(1+r)^n-(1+g)^n}{r-g}$	$g=r$: $PV=\dfrac{nC_1}{1+r}$
Loan payment	$M=P\,\dfrac{i(1+i)^n}{(1+i)^n-1}$	—	solve $P=PV_{\text{annuity}}$ for C FN.9
Uneven stream	$\sum_t \dfrac{CF_t}{(1+r)^t}$	$\sum_t CF_t(1+r)^{n-t}$	NPV / IRR FN.8

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Worked mini-examples ($r = 5\%$)

Single sum. \$1000 in 10 years: $PV = 1000 \cdot 1.05^{-10} = \$613.91$.

Annuity. \$100 a year for 10 years: $PV = \$772.17$; $FV = 100 \cdot \frac{1.05^{10}-1}{0.05} = \$1257.79$.

Due. Same payments made at the start of each year: $772.17 \times 1.05 = \$810.78$.

Perpetuity. \$100 a year forever: $\frac{100}{0.05} = \$2000$; growing at 2%: $\frac{100}{0.03} = \$3333$.

Common mistakes

- Annual rate with monthly payments: use $i = \frac{\text{APR}}{12}$ and $n = 12\times$years FN.2.
- Growing perpetuity with this year's payment C_0 instead of next year's $C_1 = C_0(1+g)$.
- A perpetuity starting at year k: $\frac{C}{r}$ is its value at $k-1$; discount by $(1+r)^{-(k-1)}$.
- Mixing nominal cash flows with a real rate FN.4.
- Using $r \leq g$ in a growing perpetuity: the value is infinite, the model is wrong.

Decision points

DECISION POINT **Which formula?** Is it one amount → single sum. Level payments → annuity (ordinary if the first one is a period away, due if it is now). Forever → perpetuity. Growing at a constant rate → the growing forms. Anything irregular → discount each flow and add (a spreadsheet's NPV). **Which date?** You can move any value to any date by multiplying by $(1+r)^{\Delta t}$, so compute at the most convenient date and shift.

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FN.8 Net Present Value & Internal Rate of Return

NPV is the value a project adds today: discounted inflows minus discounted outflows. IRR is the discount rate at which NPV is exactly zero, i.e. the project's own rate of return.

Net present value	$NPV = \sum_{t=0}^{T} \frac{CF_t}{(1+r)^t}$	Discount every cash flow (investment negative) at the hurdle rate and add.
Internal rate of return	$\sum_{t=0}^{T} \frac{CF_t}{(1+\text{IRR})^t} = 0$	A root of a polynomial in $\frac{1}{1+\text{IRR}}$; solved numerically C1.35 §07 Calculus I.
Decision rule	accept if $NPV > 0$ ($\Leftrightarrow$ IRR $> r$ for normal projects)	Normal = one outflow then inflows; NPV falls as r rises.
Profitability index	$PI = \frac{PV(\text{future cash flows})}{\text{initial investment}}$	Value per dollar invested; $PI > 1 \Leftrightarrow NPV > 0$. Ranks projects under a budget limit.
Payback period	$\min\left\{T : \sum_{t=0}^{T} CF_t \geq 0\right\}$	Years to recover the outlay; ignores discounting and everything after payback.
Modified IRR	$\text{MIRR} = \left(\frac{FV(\text{inflows at } r_{\text{re}})}{PV(\text{outflows at } r_{\text{fin}})}\right)^{1/T} - 1$	One unique rate with explicit reinvestment; avoids multiple IRRs.

STATED-EQUATION *Net present value equals the sum over time of each cash flow C F t divided by one plus r to the t. The internal rate of return is the rate that makes that sum equal zero. Accept the project if the NPV is positive, which for normal projects means the IRR exceeds r. The profitability index is the present value of future cash flows over the initial investment. Payback is the first year the cumulative cash flow turns non-negative. Modified IRR compounds inflows forward, discounts outflows back, and takes the T-th root of their ratio, minus one.*

MEMORY AID NPV answers "how much value?"; IRR answers "what rate?" When they disagree, trust NPV.

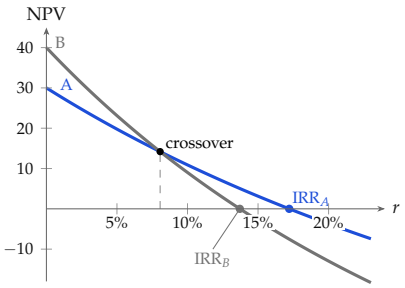
METHOD

1. Lay out the cash flows on a timeline ($CF_0 < 0$ is the investment).
2. NPV: multiply each CF_t by $(1+r)^{-t}$ and add.
3. IRR: find two rates with NPVs of opposite sign, then bisect or use Newton's method until NPV ≈ 0.
4. Example: $-100, +60, +60$ at $r = 10\%$: $NPV = -100 + 54.55 + 49.59 = 4.13 > 0$; IRR $= 13.1\% > 10\%$, accept.

DECISION POINT **NPV vs. IRR conflicts.** For *mutually exclusive* projects that differ in size or timing, IRR and NPV can rank them differently: below the *crossover rate* (where the NPV profiles cross, figure) the project with later or larger flows has the higher NPV even with a lower IRR. Choose the higher NPV, because NPV is measured in dollars of value created and IRR implicitly assumes reinvestment at the IRR. With cash flows that change sign more than once, there may be several IRRs or none: use NPV or MIRR.

TERMS

CF_t – cash flow at time t (negative = investment)
r – hurdle rate / cost of capital FN.17
T – final period
IRR – discount rate giving $NPV = 0$
$r_{\text{re}}, r_{\text{fin}}$ – reinvestment and financing rates
Pitfalls – multiple IRRs with sign-changing cash flows
Sum – finite series A2.21 §04 Algebra II
NPV; PI – net present value; profitability index
MIRR – modified internal rate of return



CONTEXT

- **Origin.** Irving Fisher (1907, 1930); widespread corporate use after Joel Dean (1951).
- **Use.** Capital budgeting: factories, software projects, acquisitions.
- **Data science.** Valuing a data-science project: upfront build cost vs. discounted future savings; payback of an ML platform investment.
- **Physical application.** Evaluating a solar installation: pay now, save on electricity for 25 years.

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FN.9 Loan Amortization & Mortgages

An amortizing loan is repaid by equal payments; each payment covers the interest due on the outstanding balance and the rest repays principal, so the balance falls to zero at the last payment.

Level payment	$M = P\,\dfrac{i\,(1+i)^n}{(1+i)^n - 1}$	The annuity payment whose present value equals the loan FN.6.
Monthly conventions	$i = \dfrac{\text{APR}}{12}, \quad n = 12 \times \text{years}$	Rate and count per payment period.
Interest / principal split	$I_k = i\,B_{k-1}, \qquad \text{Pr}_k = M - I_k$	Interest on last month's balance; the remainder repays principal.
Remaining balance	$B_k = P(1+i)^k - M\,\dfrac{(1+i)^k - 1}{i}$	Loan grown at i minus the payments grown at i; equals the PV of the remaining payments.
Total interest	$nM - P$	All payments minus the amount borrowed.

STATED-EQUATION *The monthly payment M equals the principal P times i times one plus i to the n, over one plus i to the n minus one, where i is the APR over twelve and n is twelve times the number of years. Each month, interest equals i times the previous balance, and the rest of the payment reduces principal. The balance after k payments is P grown for k periods minus the future value of k payments. Total interest is n M minus P.*

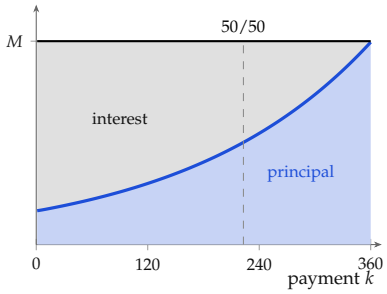
MEMORY AID Early payments are mostly interest; late payments are mostly principal.

METHOD

1. Compute i, n and the payment M.
2. Row k: interest $I_k = iB_{k-1}$; principal $\text{Pr}_k = M - I_k$; new balance $B_k = B_{k-1} - \text{Pr}_k$.
3. Repeat until $B_n = 0$ (rounding leaves cents). Example: \$1000 at $i = 1\%$, $n = 3$: $M = 340.02$; rows $(10.00, 330.02, 669.98)$, $(6.70, 333.32, 336.66)$, $(3.37, 336.66, 0)$.

TERMS

P – loan amount
i – rate per payment period
n – number of payments
M – level payment
I_k, Pr_k – interest and principal parts of payment k
B_k – balance after k payments
Amortization schedule – table of interest, principal, balance



CONTEXT

- **Origin.** Amortized mortgages became standard in the US after the 1930s (FHA).
- **Use.** Mortgages, car loans, student loans.
- **Data science.** Prepayment and default models (credit risk) simulate amortization schedules for millions of loans.
- **Physical application.** Financing a home solar or heat-pump system: the monthly payment vs. monthly energy savings.

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Returns & Risk

FN.10 Returns: Simple, Log & CAGR

A return measures relative change in value. Simple returns aggregate across assets, log returns aggregate across time, and geometric averages (CAGR) summarize multi-period growth honestly.

Simple return	$R_t = \dfrac{P_t - P_{t-1} + D_t}{P_{t-1}}$	Percent gain over one period, including any dividend D_t (total return).
Log return	$r_t = \ln \dfrac{P_t}{P_{t-1}} = \ln(1 + R_t)$	Continuously compounded return; $r \approx R$ for small moves.
Multi-period return	$1 + R_{0,T} = \prod_{t=1}^{T}(1 + R_t), \quad r_{0,T} = \sum_{t=1}^{T} r_t$	Simple returns compound (multiply); log returns add A2.14 §04 Algebra II.
CAGR	$\text{CAGR} = \left(\dfrac{V_T}{V_0}\right)^{1/T} - 1$	The constant yearly rate that turns V_0 into V_T in T years.
Geometric mean return	$\bar{R}_{\text{geo}} = \left[\prod(1 + R_t)\right]^{1/T} - 1 \le \bar{R}_{\text{arith}}$	The realized compound rate; never above the arithmetic mean S.1 §13 Statistics.
Volatility drag	$\bar{R}_{\text{geo}} \approx \bar{R}_{\text{arith}} - \frac{1}{2}\sigma^2$	Variance lowers compound growth.

STATED-EQUATION *The simple return R t equals price change plus dividend over the previous price. The log return r t equals the natural log of P t over P t minus one, which is the log of one plus R t. Over many periods, one plus the total return is the product of one plus each return, and log returns simply add. The compound annual growth rate equals ending value over starting value, to the one over T, minus one. The geometric mean return is at most the arithmetic mean, smaller by about half the variance.*

MEMORY AID Log returns add over time; simple returns add across assets. +50% then −50% loses 25%.

DECISION POINT **Which return measure?** Combining assets in a portfolio for one period → simple returns (they are weight-averaged FN.12). Aggregating over time, modeling, or assuming normality → log returns. Reporting realized multi-year performance → CAGR / geometric mean. Forecasting next period's expected return → arithmetic mean. A fund with deposits and withdrawals: time-weighted return to judge the manager, money-weighted (IRR FN.8) to judge the investor's experience.

TERMS

P_t – price at time t
D_t – dividend or coupon paid during period t
R_t, r_t – simple and log return
V_0, V_T – starting and ending values; T years
σ^2 – variance of returns FN.11
CAGR – constant rate giving the same growth FN.1
$R_{0,T}$, $r_{0,T}$; T – total simple and log return over T periods; number of periods
$\bar{R}_{\text{geo}}$, $\bar{R}_{\text{arith}}$ – geometric and arithmetic mean returns

CONTEXT

- **Origin.** Compounding math is ancient; log returns popularized by Osborne (1959) and Black–Scholes (1973).
- **Use.** Reporting investment performance honestly.
- **Data science.** Use log returns as model targets (roughly symmetric, additive); report CAGR for growth metrics like revenue or users.
- **Physical application.** Population growth over decades is summarized by its compound annual growth rate.

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FN.11 Volatility & Risk

Volatility is the standard deviation of returns. Under independent returns, variance grows linearly with the horizon, so volatility grows with its square root.

Sample volatility	$\sigma = \sqrt{\frac{1}{n-1}\sum_t (r_t - \bar{r})^2}$	Standard deviation of periodic (usually log) returns S.2 §13 Statistics.
Annualizing volatility	$\sigma_{\text{annual}} = \sigma_{\text{daily}}\sqrt{252}$	252 trading days; use $\sqrt{12}$ for monthly, $\sqrt{52}$ for weekly data.
Square-root-of-time	$\sigma_T = \sigma\sqrt{T}$	Variances of independent returns add S.13 §13 Statistics.
Annualizing the mean	$\mu_{\text{annual}} = 252\,\mu_{\text{daily}}$	Means scale linearly with time.
EWMA volatility	$\sigma_t^2 = \lambda\sigma_{t-1}^2 + (1-\lambda)r_{t-1}^2$	Exponentially weighted estimate that reacts to recent shocks (RiskMetrics: $\lambda = 0.94$).

STATED-EQUATION *Volatility sigma is the sample standard deviation of returns. Annual volatility equals daily volatility times the square root of two hundred fifty-two trading days. Over a horizon T, volatility scales with the square root of T, while the mean scales linearly with T. The exponentially weighted variance mixes yesterday's variance, weight lambda, with yesterday's squared return, weight one minus lambda.*

MEMORY AID "Risk grows with the square root of time; expected return grows with time."

TERMS

r_t – periodic (usually log) returns FN.10; $\bar{r}$ their mean
σ – standard deviation S.2 §13 Statistics
μ – mean return
T – horizon in periods
λ – decay factor, $0 < \lambda < 1$
252 – trading days per year (12 months, 52 weeks)
Assumption – i.i.d. returns; real markets show volatility clustering
n; $\bar{r}$ – number of returns; their mean
σ_t^2 – EWMA variance estimate for day t

CONTEXT

- **Origin.** Markowitz (1952) defined risk as variance; RiskMetrics EWMA (1994).
- **Use.** Measuring and comparing investment risk.
- **Data science.** GARCH and ML models forecast volatility; $\sqrt{T}$ scaling is the same as the standard error law S.24 §13 Statistics.
- **Physical application.** Random-walk diffusion distance also grows like $\sqrt{t}$ P.31 §12 Physics.

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Portfolio Theory & Asset Pricing

FN.12 Portfolio Return, Variance & Diversification

A portfolio's expected return is the weighted average of its assets' returns, but its risk is less than the weighted average risk whenever the assets are not perfectly correlated.

Expected portfolio return	$E[R_p] = \sum_i w_i E[R_i] = \mathbf{w}^\top \boldsymbol{\mu}$	Linearity of expectation S.13 §13 Statistics.
Portfolio variance	$\sigma_p^2 = \mathbf{w}^\top \Sigma\, \mathbf{w}$	A quadratic form in the weights LA.27 §09 Linear Algebra.
Two-asset variance	$\sigma_p^2 = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\,\rho\,\sigma_1\sigma_2$	The cross term shrinks as ρ falls; $\rho = -1$ allows zero risk.
Diversification limit	$\sigma_p^2 = \dfrac{\overline{\sigma^2}}{N} + \left(1 - \dfrac{1}{N}\right)\overline{\text{cov}}$	Equal weights over N assets: individual risk vanishes, average covariance remains.

STATED-EQUATION *Expected portfolio return equals the weighted sum of expected asset returns. Portfolio variance equals w transpose Sigma w. For two assets: w one squared sigma one squared plus w two squared sigma two squared plus two w one w two rho sigma one sigma two. With N equally weighted assets the variance is the average variance over N plus one minus one over N times the average covariance.*

MEMORY AID "Diversification is the only free lunch": with $\rho < 1$, portfolio risk is less than the weighted average risk.

TERMS

w_i – portfolio weights, $\sum w_i = 1$
$\boldsymbol{\mu}$ – vector of expected returns $E[R_i]$
Σ – covariance matrix S.38 §13 Statistics
σ_i – volatility of asset i FN.11
ρ – correlation between assets
$\overline{\sigma^2}$, $\overline{\text{cov}}$ – average variance, average covariance
Efficient frontier – FN.13
R_p; $E[R_i]$ – portfolio return; expected return of asset i
$\mathbf{w}$; N – weight vector; number of assets

CONTEXT

- **Origin.** Harry Markowitz, "Portfolio Selection" (1952); Nobel Prize 1990.
- **Use.** Asset allocation, fund construction, risk budgeting.
- **Data science.** Ensemble averaging of models reduces variance by the same formula when errors are not perfectly correlated ML.13 §15 Machine Learning.
- **Physical application.** Grid operators combine wind farms in different regions: uncorrelated wind makes total output steadier.

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FN.13 Markowitz Efficient Frontier & Minimum-Variance Portfolio

Mean–variance optimization finds, for each target return, the portfolio with the least variance. The upper edge of that curve is the efficient frontier; adding a risk-free asset turns the best risky portfolio into a straight line of optimal choices.

Mean–variance problem	$\min_{\mathbf{w}} \ \mathbf{w}^\top \Sigma \mathbf{w} \ \text{ s.t. } \mathbf{w}^\top \boldsymbol{\mu} = \mu_p, \ \mathbf{w}^\top \mathbf{1} = 1$	A quadratic program O.18 §14 Optimization, solved with Lagrange multipliers O.17 §14 Optimization.
Two-asset minimum variance	$w_1^* = \dfrac{\sigma_2^2 - \rho\sigma_1\sigma_2}{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2}$	Weight on asset 1 that minimizes two-asset variance FN.12.
Global minimum-variance portfolio	$\mathbf{w}_{\text{GMV}} = \dfrac{\Sigma^{-1}\mathbf{1}}{\mathbf{1}^\top \Sigma^{-1}\mathbf{1}}$	The leftmost point of the frontier; needs no expected returns.
Tangency portfolio	$\mathbf{w}_T = \dfrac{\Sigma^{-1}(\boldsymbol{\mu} - R_f\mathbf{1})}{\mathbf{1}^\top \Sigma^{-1}(\boldsymbol{\mu} - R_f\mathbf{1})}$	The risky portfolio with the highest Sharpe ratio FN.16.
Capital market line	$E[R_p] = R_f + \dfrac{E[R_T] - R_f}{\sigma_T}\,\sigma_p$	Mixes of cash and the tangency portfolio; every investor holds a point on it.

STATED-EQUATION *Minimize w transpose Sigma w subject to a target return and weights summing to one. With two assets, the minimum-variance weight on asset one is sigma two squared minus rho sigma one sigma two, over sigma one squared plus sigma two squared minus two rho sigma one sigma two. The global minimum-variance weights are Sigma inverse times the ones vector, normalized to sum to one. The tangency weights are Sigma inverse times the excess-return vector, normalized. The capital market line: expected return equals the risk-free rate plus the tangency Sharpe ratio times portfolio risk.*

MEMORY AID Two-fund separation: every efficient choice is "some cash + the tangency portfolio".

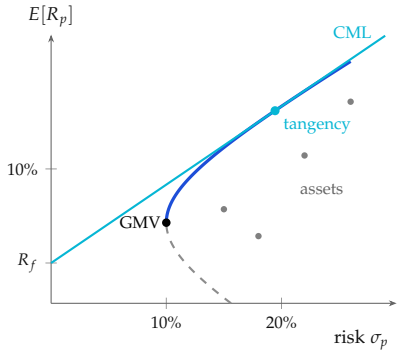
METHOD

1. Estimate $\boldsymbol{\mu}$ and Σ from returns (annualized FN.11).
2. Solve the quadratic program for a grid of target returns μ_p; plot (σ_p, μ_p).
3. Keep the upper branch above the GMV point; with a risk-free asset, draw the line from R_f tangent to it.
4. Example (two assets): $\sigma_1 = 20\%$, $\sigma_2 = 10\%$, $\rho = 0$: $w_1^* = \frac{0.01}{0.05} = 0.2$, $\sigma_p = \sqrt{0.2^2(0.04) + 0.8^2(0.01)} = 8.9\%$, below both.

DECISION POINT Estimated expected returns are very noisy, and the optimizer amplifies their errors (extreme long/short weights). Practical choices: the GMV portfolio (ignores $\boldsymbol{\mu}$), shrink Σ toward a simple target (Ledoit–Wolf), add no-short-sale or weight limits, or blend views with market-implied returns (Black–Litterman).

TERMS

$\mathbf{w}$ – portfolio weights; $\mathbf{1}$ vector of ones
$\boldsymbol{\mu}$, Σ – expected returns and covariance matrix FN.12
μ_p, σ_p – target return and risk of the portfolio
R_f – risk-free rate
$E[R_T]$, σ_T – tangency portfolio return and risk
Σ^{-1} – matrix inverse A2.25 §04 Algebra II
$\mathbf{1}$; w_1^* – vector of ones; minimum-variance weight on asset 1
$\mathbf{w}_{\text{GMV}}$, $\mathbf{w}_T$ – global minimum-variance and tangency portfolios
ρ; σ_1, σ_2 – correlation of the two assets; their volatilities



CONTEXT

- **Origin.** Markowitz (1952); Tobin's separation theorem (1958).
- **Use.** Strategic asset allocation, robo-advisers, risk parity variants.
- **Data science.** The same constrained quadratic program allocates budget across uncertain, correlated ad channels or model ensembles.
- **Physical application.** Choosing a mix of generation sources that minimizes variability of total power output.

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FN.14 CAPM & Beta

The capital asset pricing model says investors are paid only for systematic (market) risk, measured by beta. Expected return rises linearly with beta along the security market line.

CAPM (security market line)	$E[R_i] = R_f + \beta_i\big(E[R_m] - R_f\big)$	Risk-free rate plus beta times the market risk premium.
Beta	$\beta_i = \dfrac{\mathrm{Cov}(R_i, R_m)}{\mathrm{Var}(R_m)} = \rho_{im}\dfrac{\sigma_i}{\sigma_m}$	Sensitivity of the asset to market moves; a regression slope S.40 §13 Statistics.
Jensen's alpha	$\alpha_i = R_i - \big[R_f + \beta_i(R_m - R_f)\big]$	Return beyond what CAPM predicts for the risk taken.
Portfolio beta	$\beta_p = \sum_i w_i \beta_i$	Betas average linearly with the weights.
Risk decomposition	$\sigma_i^2 = \beta_i^2\sigma_m^2 + \sigma_\varepsilon^2$	Systematic part (priced) plus idiosyncratic part (diversifiable FN.12).

STATED-EQUATION *The capital asset pricing model: the expected return on asset i equals the risk-free rate plus beta i times the market risk premium. Beta equals the covariance of the asset with the market over the market's variance, which equals rho times sigma i over sigma m. Alpha is the actual return minus the CAPM-predicted return. A portfolio's beta is the weighted average beta. Total variance splits into beta squared times market variance plus idiosyncratic variance.*

MEMORY AID $\beta > 1$: amplifies the market; $\beta < 1$: dampens it; β is just a regression slope.

TERMS

R_f – risk-free rate
R_m – market return; $E[R_m] - R_f$ market risk premium
β – systematic risk; slope of R_i on R_m S.40 §13 Statistics
Cov, Var – S.38 §13 Statistics, S.2 §13 Statistics
ρ_{im} – correlation of asset and market
α – excess return not explained by the market (intercept)
σ_ε^2 – idiosyncratic (residual) variance
Estimation – FN.15
R_i; $E[\cdot]$ – return on asset i; expected value
σ_i, σ_m; β_p, w_i – asset and market volatility; portfolio beta and weights

CONTEXT

- **Origin.** Sharpe (1964), Lintner (1965), Mossin (1966), Treynor.
- **Use.** Cost of equity, performance attribution.
- **Data science.** Factor models (Fama–French) are multiple regressions of returns on factors; "alpha" is the intercept.
- **Physical application.** Like measuring how strongly a building sways (β) relative to ground motion.

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FN.15 Estimating Beta; Levered & Unlevered Beta

Beta is estimated by regressing the asset's excess returns on the market's. Because debt amplifies equity risk, betas are "unlevered" to compare businesses and "relevered" to a target capital structure.

Market model	$R_{i,t} - R_f = \alpha_i + \beta_i(R_{m,t} - R_f) + \varepsilon_t$	Ordinary least squares on paired excess returns S.40 §13 Statistics.
OLS beta	$\hat{\beta} = \frac{\sum_t (x_t - \bar{x})(y_t - \bar{y})}{\sum_t (x_t - \bar{x})^2} = \frac{s_{xy}}{s_x^2}$	Sample covariance over sample market variance.
Systematic share	$R^2 = \frac{\hat{\beta}^2 \sigma_m^2}{\sigma_i^2}$	Fraction of the asset's variance explained by the market.
Adjusted (Blume) beta	$\beta_{\text{adj}} = 0.67\,\hat{\beta} + 0.33$	Shrinks toward 1, since betas mean-revert.
Unlevering (Hamada)	$\beta_U = \frac{\beta_L}{1 + (1 - T_c)\,D/E}$	Removes the effect of debt: the risk of the business alone.
Relevering	$\beta_L = \beta_U \left[1 + (1 - T_c)\frac{D}{E}\right]$	Equity beta at a new debt-to-equity ratio.

STATED-EQUATION *Regress the asset's excess return on the market's excess return; the slope is beta and the intercept is alpha. The OLS beta is the sample covariance over the sample variance of the market. R squared is beta squared times market variance over the asset's variance. The adjusted beta is zero point six seven beta plus zero point three three. Unlevered beta is levered beta over one plus one minus the tax rate times debt over equity; relevering multiplies by the same factor.*

MEMORY AID Debt makes equity "louder": $\beta_L > \beta_U$ whenever $D > 0$.

METHOD

1. Collect 60 monthly (or 104 weekly) returns of the asset and a broad index; subtract R_f.
2. Regress $y = R_i - R_f$ on $x = R_m - R_f$; read $\hat{\beta}$, its standard error, and R^2.
3. Optionally adjust toward 1; for a private firm, average unlevered betas of comparable firms and relever.
4. Example: $s_{xy} = 0.0024$, $s_x^2 = 0.0016 \Rightarrow \hat{\beta} = 1.5$; at $D/E = 0.5$, $T_c = 25\%$: $\beta_U = \frac{1.5}{1.375} = 1.09$.

DECISION POINT Window and frequency trade off relevance against noise: longer windows give tighter estimates of a beta that may have changed. Daily data on thinly traded stocks bias beta toward 0 (prices lag the market). The index should match the investor's opportunity set.

TERMS

$R_{i,t}, R_{m,t}$ – asset and market returns in period t
x_t, y_t – market and asset excess returns; $\bar{x}, \bar{y}$ means
s_{xy}, s_x^2 – sample covariance and variance S.38 §13 Statistics
ε_t – residual
β_L, β_U – levered (equity) and unlevered (asset) beta
D/E – debt-to-equity ratio; T_c corporate tax rate
α_i; R_f – intercept (abnormal return); risk-free rate
R^2; σ_i, σ_m – share of variance explained S.40 §13 Statistics; asset and market volatility
β_{adj}; T_c – Blume-adjusted beta; corporate tax rate

CONTEXT

- **Origin.** Blume (1971); Hamada (1972), building on Modigliani–Miller.
- **Use.** Cost of equity for WACC FN.17, valuing private companies.
- **Data science.** A textbook use of linear regression with heteroscedasticity and outlier concerns; rolling-window betas are a time-varying feature.
- **Physical application.** Like calibrating a sensor's gain by regressing its readings on a reference instrument.

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FN.16 Sharpe Ratio & Risk-Adjusted Return

Risk-adjusted ratios divide a return by the risk taken to earn it, so strategies with different risk levels can be compared. They differ in which risk goes in the denominator.

Sharpe ratio	$S = \dfrac{E[R_p] - R_f}{\sigma_p}$	Excess return per unit of total volatility.
Sortino ratio	$\text{Sortino} = \dfrac{E[R_p] - R_f}{\sigma_{\text{down}}}$	Penalizes only downside volatility.
Downside deviation	$\sigma_{\text{down}} = \sqrt{\frac{1}{n}\sum_t \min(R_t - \tau, 0)^2}$	Root-mean-square shortfall below a target τ (often 0 or R_f).
Treynor ratio	$\text{Treynor} = \dfrac{E[R_p] - R_f}{\beta_p}$	Excess return per unit of systematic risk FN.14.
Information ratio	$IR = \dfrac{E[R_p - R_b]}{\sigma(R_p - R_b)}$	Active return over tracking error versus a benchmark.
Annualizing Sharpe	$S_{\text{annual}} = S_{\text{daily}}\sqrt{252}$	Mean scales by 252, volatility by $\sqrt{252}$.

STATED-EQUATION *The Sharpe ratio equals the expected portfolio return minus the risk-free rate, over the portfolio's volatility. The Sortino ratio uses only downside deviation, the root-mean-square of returns below a target. The Treynor ratio divides excess return by beta. The information ratio divides the average return over a benchmark by the volatility of that difference, the tracking error. An annual Sharpe ratio equals the daily Sharpe ratio times the square root of two hundred fifty-two.*

MEMORY AID Sharpe = "excess return per unit of risk." Rough guide: < 0.5 poor, ≈ 1 good, > 2 excellent (and suspicious).

DECISION POINT **Which ratio?** A stand-alone investment that is your whole portfolio → Sharpe (total risk matters). Skewed returns or when only losses hurt → Sortino. One sleeve inside a well-diversified portfolio → Treynor (only beta risk remains). An active manager judged against an index → information ratio. Always compare ratios computed over the same period and frequency.

TERMS

R_p, R_f, R_b – portfolio, risk-free (Treasury bills) and benchmark returns
Excess return – $R_p - R_f$
σ_p – volatility FN.11
τ – target return for downside deviation
β_p – portfolio beta FN.14
Tracking error – $\sigma(R_p - R_b)$
Similar to – a z-score / t-statistic of returns S.4 §13 Statistics, S.30 §13 Statistics
S; IR – Sharpe ratio; information ratio
σ_{down}; R_t, n – downside deviation; period returns and their count

CONTEXT

- **Origin.** William Sharpe (1966); Treynor (1965); Sortino (1980s); Nobel Prize to Sharpe 1990.
- **Use.** Comparing funds and strategies with different risk levels.
- **Data science.** Backtests of trading models report Sharpe; a high backtest Sharpe from many tried strategies is often overfitting ML.4 §15 Machine Learning.
- **Physical application.** Engineering analog: signal-to-noise ratio of returns EE.29 §16 Electrical Eng..

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Corporate Finance & Valuation

FN.17 Weighted Average Cost of Capital (WACC)

WACC is the blended return a firm must earn for all its investors: the costs of equity and (after-tax) debt weighted by their market values. It is the discount rate for the firm's typical project and its free cash flows.

WACC	$\text{WACC} = \frac{E}{V} r_E + \frac{D}{V} r_D (1 - T_c)$	Interest is tax-deductible, so debt's cost enters after tax.
Firm value	$V = E + D$	Market value of equity plus market value of debt.
Cost of equity (CAPM)	$r_E = R_f + \beta_L \big(E[R_m] - R_f\big)$	Uses the levered equity beta FN.14, FN.15.
Cost of debt	$r_D \approx$ YTM of the firm's bonds	The market rate on new borrowing FN.23; not the old coupon rate.

STATED-EQUATION *The weighted average cost of capital equals equity's share of value times the cost of equity, plus debt's share times the cost of debt times one minus the tax rate. Firm value is equity plus debt. The cost of equity comes from CAPM with the levered beta; the cost of debt is the current yield to maturity on the firm's debt.*

MEMORY AID "Pay each investor their price, in proportion to their share, and let the tax shield discount the debt."

METHOD

1. Find market values E (shares $\times$ price) and D; compute weights $\frac{E}{V}$, $\frac{D}{V}$.
2. Estimate r_E with CAPM and r_D from bond yields.
3. Combine. Example: $E = 60$, $D = 40$, $r_E = 10\%$, $r_D = 5\%$, $T_c = 25\%$: $0.6(10\%) + 0.4(5\%)(0.75) = 7.5\%$.

DECISION POINT Use market (not book) weights, ideally the *target* capital structure. Use the firm's WACC only for projects as risky as the firm; a riskier division or project needs its own beta (unlever a comparable's beta, relever at the project's financing).

TERMS

E, D – market values of equity and debt; $V = E + D$
r_E, r_D – costs of equity and debt
T_c – corporate tax rate
β_L – levered equity beta
R_f, $E[R_m]$ – risk-free rate, expected market return
Use – discount rate in DCF FN.19 and NPV FN.8
V – firm value $E + D$

CONTEXT

- **Origin.** Modigliani–Miller (1958, 1963) on capital structure and the debt tax shield.
- **Use.** Hurdle rates for capital budgeting, DCF valuation, regulated utility returns.
- **Data science.** Choosing the discount rate when valuing a data or ML product line inside a company.
- **Physical application.** Regulators set allowed returns of power utilities from a WACC calculation.

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FN.18 Stock Valuation & Multiples

A share is worth the present value of the cash it will pay. With constant dividend growth this collapses to the Gordon formula; multiples are shortcuts that compare a firm's price with peers' prices.

Gordon growth model	$P_0 = \dfrac{D_1}{r - g}$	Growing perpetuity of dividends FN.6; needs $g < r$.
Implied required return	$r = \dfrac{D_1}{P_0} + g$	Dividend yield plus growth: the return the current price implies.
Sustainable growth	$g = \text{ROE} \times b$	Growth funded by retained earnings; b = retention ratio FN.20.
Price–earnings ratio	$\dfrac{P}{E} = \dfrac{\text{price}}{\text{earnings per share}}$	Dollars paid per dollar of annual profit.
Enterprise value	$\text{EV} = \text{market cap} + \text{debt} - \text{cash}$	Value of the operating business to all investors; pairs with EBITDA FN.21.

STATED-EQUATION *The Gordon growth model: today's price equals next year's dividend over r minus g. Rearranged, the required return is the dividend yield plus growth. Sustainable growth equals return on equity times the retention ratio. The P E ratio equals price over earnings per share. Enterprise value equals market capitalization plus debt minus cash. A discounted cash flow valuation sums discounted free cash flows plus a discounted terminal value.*

MEMORY AID Price is the present value of all future cash; growth near the discount rate makes values explode.

TERMS

P_0 – price today
D_1 – next year's dividend
r – required return (e.g. from CAPM FN.14)
g – constant growth rate, $g < r$ FN.6
b – retention ratio, $1-$ payout ratio
EPS – net income per share
DCF – full discounted-cash-flow valuation with WACC and terminal value TV FN.19
ROE; P/E – return on equity; price–earnings ratio
EV – enterprise value

CONTEXT

- **Origin.** John Burr Williams (1938); Myron Gordon (1959).
- **Use.** Equity research, M&A, startup valuation.
- **Data science.** Sensitivity of valuations to r and g is a classic Monte Carlo exercise O.27 §14 Optimization.
- **Physical application.** Valuing a mine or oil field by discounting the cash from future extraction.

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FN.19 Discounted Cash Flow & Terminal Value

A DCF values a business as the present value of its forecast free cash flows for a few explicit years plus a terminal value for everything after, discounted at WACC.

Free cash flow to the firm	$FCF = \text{EBIT}(1 - T_c) + \text{D\&A} - \text{CapEx} - \Delta\text{NWC}$	Cash the operations generate for all investors after reinvestment.
DCF value	$V_0 = \sum_{t=1}^{T} \frac{FCF_t}{(1+\text{WACC})^t} + \frac{TV_T}{(1+\text{WACC})^T}$	Explicit forecast years plus the discounted terminal value.
Gordon terminal value	$TV_T = \frac{FCF_T(1+g)}{\text{WACC} - g}$	Growing perpetuity from year $T+1$ on FN.6.
Exit-multiple terminal value	$TV_T = \text{multiple} \times \text{EBITDA}_T$	Assumes the firm sells at peers' EV/EBITDA in year T.
Equity value per share	$\frac{V_0 - \text{debt} + \text{cash}}{\text{shares outstanding}}$	Enterprise value minus net debt, divided among shares.

STATED-EQUATION *Free cash flow equals after-tax operating profit plus depreciation and amortization, minus capital expenditure, minus the increase in net working capital. Value equals the sum of free cash flows discounted at WACC plus the terminal value discounted from year T. The Gordon terminal value is next year's cash flow over WACC minus g; the exit-multiple terminal value is a multiple times final-year EBITDA. Equity per share is enterprise value minus net debt, over shares.*

MEMORY AID In most DCFs the terminal value is 60–80% of the total: the answer lives in WACC and g.

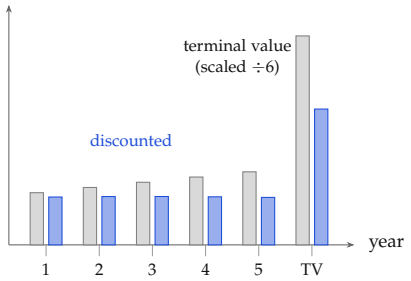
METHOD

1. Forecast revenue, margins, CapEx and working capital for 5–10 years; compute FCF_t.
2. Estimate WACC FN.17; discount each FCF_t.
3. Compute TV_T and discount it by $(1+\text{WACC})^{-T}$; add to get EV, then subtract net debt.
4. Example (figure): $FCF = 10, \ldots, 14$, WACC 9%, $g = 3\%$: PV of FCFs $= 46.0$, $TV_5 = 240.3$, PV of TV $= 156.2$, EV $= 202.2$.

DECISION POINT Gordon vs. exit multiple: Gordon is internally consistent but very sensitive to g; keep g at or below long-run nominal GDP growth. Exit multiples anchor to the market but import its mood. Compute both and check that the implied g and multiple agree; always show a WACC $\times$ g sensitivity table.

TERMS

EBIT – earnings before interest and taxes; T_c tax rate
D&A – depreciation and amortization (non-cash)
CapEx – capital expenditures
ΔNWC – increase in net working capital
T – last explicit forecast year; g perpetual growth
TV_T – terminal value at T
EBITDA – EBIT plus D&A
FCF_t; V_0 – free cash flow in year t; firm value today
WACC; g – discount rate FN.17; long-run growth rate



CONTEXT

- **Origin.** John Burr Williams, *The Theory of Investment Value* (1938); popularized by McKinsey's *Valuation* (1990).
- **Use.** Investment banking, private equity, impairment testing.
- **Data science.** Scenario and Monte Carlo DCFs propagate forecast uncertainty O.27 §14 Optimization; revenue forecasts often come from time-series models.
- **Physical application.** Valuing a wind farm from forecast energy output, prices and maintenance costs.

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FN.20 Financial Ratios & DuPont Analysis

Ratios turn financial statements into comparable numbers. DuPont analysis explains return on equity as profitability times efficiency times leverage.

Return on equity	$\text{ROE} = \dfrac{\text{net income}}{\text{equity}}$	Profit earned per dollar of shareholders' book capital.
DuPont decomposition	$\text{ROE} = \underbrace{\dfrac{\text{NI}}{\text{sales}}}_{\text{margin}} \cdot \underbrace{\dfrac{\text{sales}}{\text{assets}}}_{\text{turnover}} \cdot \underbrace{\dfrac{\text{assets}}{\text{equity}}}_{\text{leverage}}$	Sales and assets cancel; shows which driver moved ROE.
Current ratio	$\dfrac{\text{current assets}}{\text{current liabilities}}$	Short-term liquidity; > 1 covers bills due within a year.
Debt-to-equity	$\dfrac{D}{E} = \dfrac{\text{debt}}{\text{equity}}$	Financial leverage: how much is funded by lenders.

STATED-EQUATION *Return on equity equals net income over equity, which DuPont analysis splits into profit margin times asset turnover times financial leverage. The current ratio equals current assets over current liabilities. Debt-to-equity equals debt over equity.*

MEMORY AID Accounting identity: assets $=$ liabilities $+$ equity.

TERMS

NI – net income (profit after all expenses and taxes)
Equity – assets minus liabilities (book value)
Margin, turnover, leverage – three drivers of ROE
Current assets / liabilities – due or convertible to cash within a year
Liquidity – current ratio > 1 covers short-term bills
Ratios – A1.19 §03 Algebra I; more in FN.21

CONTEXT

- **Origin.** DuPont Corporation's analysts (1920s).
- **Use.** Credit analysis, equity research, management dashboards.
- **Data science.** Financial ratios are standard features for credit-scoring and bankruptcy-prediction models (e.g. Altman Z-score).
- **Physical application.** Energy intensity (energy per unit of output) is the engineering counterpart of asset turnover.

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FN.21 More Financial Statement Ratios

The working set of ratios analysts compute first, grouped by the question each answers: is the firm profitable, liquid, solvent, efficient, and how is it priced?

Gross / operating margin	$\frac{\text{revenue} - \text{COGS}}{\text{revenue}}, \quad \frac{\text{EBIT}}{\text{revenue}}$	Profitability before and after operating expenses.
Return on assets	$\text{ROA} = \frac{\text{net income}}{\text{total assets}}$	Profit per dollar of assets, independent of financing mix.
Quick ratio	$\frac{\text{current assets} - \text{inventory}}{\text{current liabilities}}$	Stricter liquidity test: inventory may be hard to sell fast.
Interest coverage	$\frac{\text{EBIT}}{\text{interest expense}}$	Times operating profit covers interest; low values signal distress.
Inventory turnover	$\frac{\text{COGS}}{\text{average inventory}}$	How many times inventory is sold and replaced per year.
Days sales outstanding	$\text{DSO} = \frac{\text{receivables}}{\text{revenue}} \times 365$	Average days to collect cash from customers.
EV/EBITDA	$\frac{\text{EV}}{\text{EBITDA}}$	Capital-structure-neutral valuation multiple FN.18.
Price-to-book	$\frac{P}{B} = \frac{\text{market cap}}{\text{book equity}}$	Market value relative to accounting net worth.
Dividend yield	$\frac{\text{annual dividend per share}}{\text{price}}$	Cash return from dividends alone.

STATED-EQUATION *Gross margin is revenue minus cost of goods sold, over revenue; operating margin is EBIT over revenue. Return on assets is net income over total assets. The quick ratio removes inventory from current assets. Interest coverage is EBIT over interest expense. Inventory turnover is cost of goods sold over average inventory. Days sales outstanding is receivables over revenue times three hundred sixty-five. EV over EBITDA and price to book are valuation multiples; dividend yield is dividend over price.*

MEMORY AID Five questions: profitable? liquid? solvent? efficient? cheap?

DECISION POINT A ratio means little alone: compare it with the same firm's history and with peers in the same industry (a grocer's margins and turnover look nothing like a software firm's). For a flow over a stock (income over assets, COGS over inventory), use the *average* of the opening and closing balance.

TERMS

COGS – cost of goods sold
EBIT, EBITDA – operating profit; plus depreciation & amortization
Receivables – money customers owe the firm
Book equity B – balance-sheet equity
EV – enterprise value FN.18
DuPont, ROE – FN.20
ROA; DSO – return on assets; days sales outstanding

CONTEXT

- **Origin.** Ratio analysis grew with bank credit analysis (late 19th c.); Altman's Z-score (1968) combined five ratios.
- **Use.** Credit decisions, equity screening, management KPIs.
- **Data science.** Ratios are scale-free features for models; watch for division by near-zero denominators and industry effects.
- **Physical application.** Plant "capacity utilization" and "mean time to repair" are operational analogs of turnover and DSO.

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Fixed Income

FN.22 Bond Pricing & Yield

A bond's price is the present value of its coupons and face value, discounted at its yield. Because yield sits in the denominator, prices and yields move in opposite directions.

Bond price	$P = \sum_{t=1}^{n} \frac{C}{(1+y)^t} + \frac{F}{(1+y)^n} = C\,\frac{1-(1+y)^{-n}}{y} + \frac{F}{(1+y)^n}$	Coupon annuity FN.6 plus discounted face value.
Zero-coupon bond	$P = \frac{F}{(1+y)^n}$	A single payment at maturity FN.5.
Premium / discount rule	$c > y \Rightarrow P > F, \quad c < y \Rightarrow P < F$	Coupon rate above the yield sells above par, and vice versa; $P \to F$ at maturity.
Dirty price	$P_{\text{dirty}} = P_{\text{clean}} + C \cdot \frac{\text{days since last coupon}}{\text{days in period}}$	Buyers pay the seller the coupon interest accrued so far.

STATED-EQUATION *A bond's price equals the sum of its discounted coupons C over one plus y to the t, plus the discounted face value F over one plus y to the n. That equals a coupon annuity plus the discounted face. A zero-coupon bond's price is F over one plus y to the n. When the coupon rate exceeds the yield the bond trades at a premium, and below the yield at a discount. The dirty price is the quoted clean price plus accrued interest.*

MEMORY AID Bond prices and yields move in opposite directions ("see-saw"). Coupon $>$ yield $\Rightarrow$ premium; coupon $<$ yield $\Rightarrow$ discount.

TERMS

C – coupon payment per period (= coupon rate $c \times F$ / periods per year)
F – face (par) value
y – yield to maturity per period: the IRR of the bond FN.8
n – periods to maturity
Clean / dirty – quoted price without / with accrued interest
Yield measures – FN.23
P; c – bond price; coupon rate C/F
P_{clean}, P_{dirty} – quoted price; price paid including accrued interest

CONTEXT

- **Origin.** Government bonds date to 12th-century Venice; modern yield math by Macaulay (1938).
- **Use.** Pricing government and corporate debt, interest-rate risk.
- **Data science.** Yield-curve modeling (Nelson–Siegel fits) and credit-spread prediction from company data.
- **Physical application.** Infrastructure projects (bridges, power plants) are often financed with long-term bonds.

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FN.23 Bond Yield Measures

Several "yields" describe a bond's return, each answering a different question. Yield to maturity is the full IRR; the others are approximations or variants for calls and compounding conventions.

Current yield	$\text{CY} = \dfrac{\text{annual coupon}}{P}$	Income return only; ignores the gain or loss as price pulls to par.
Yield to maturity	$P = \sum_{t=1}^{n} \dfrac{C}{(1+y)^t} + \dfrac{F}{(1+y)^n}$	The IRR that equates price and promised cash flows FN.22; solved numerically.
Approximate YTM	$y \approx \dfrac{C + (F-P)/n}{(F+P)/2}$	Coupon plus straight-line pull to par, over the average price.
Bond-equivalent & effective yield	$\text{BEY} = 2\,y_{\text{semi}}, \quad \text{EAY} = \left(1 + \frac{\text{BEY}}{2}\right)^2 - 1$	US bonds quote twice the semiannual yield; EAY compounds it FN.2.
Yield to call	$P = \sum_{t=1}^{n_c} \dfrac{C}{(1+y_c)^t} + \dfrac{\text{call price}}{(1+y_c)^{n_c}}$	YTM computed as if the issuer calls at the first call date.
Spot & forward rates	$s_n = \left(\dfrac{F}{P_n}\right)^{1/n} - 1, \quad (1+s_2)^2 = (1+s_1)(1+f_{1,2})$	Zero-coupon yields by maturity; forwards are the rates they imply between dates.

STATED-EQUATION *Current yield is the annual coupon over price. Yield to maturity is the rate that makes the discounted cash flows equal the price; approximately the coupon plus face minus price over n, all over the average of face and price. The bond-equivalent yield is twice the semiannual yield, and the effective annual yield compounds it. Yield to call replaces maturity with the call date and face with the call price. The n-year spot rate is face over the zero price to the one over n, minus one; forward rates link successive spot rates.*

MEMORY AID Current yield "now"; YTM "to the end"; YTC "if they call"; worst of them for callable bonds.

METHOD

1. List price P, face F, coupon C per period, periods n.
2. Get a starting guess from the approximation; refine by solving the price equation (bisection/Newton C1.35 §07 Calculus I).
3. Example: 10-year, $F = 1000$, $C = 50$, $P = 950$: $y \approx \frac{50+5}{975} = 5.64\%$ (exact 5.67%); current yield 5.26%.

DECISION POINT Current yield understates the return on a discount bond and overstates it on a premium bond. YTM assumes you hold to maturity and reinvest coupons at the YTM. For a callable bond quote *yield to worst*: the lowest of YTM and every yield to call.

TERMS

P – price; P_n price of an n-year zero
C, F – coupon per period, face value
y, y_c – yield to maturity, yield to call
n, n_c – periods to maturity, to the call date
s_n – spot (zero) rate for maturity n
$f_{1,2}$ – forward rate from year 1 to 2
CY; BEY; EAY – current yield; bond-equivalent yield; effective annual yield
P_n; s_1, s_2 – price of an n-year zero; 1- and 2-year spot rates

CONTEXT

- **Origin.** Yield conventions standardized by bond dealers in the 20th century; term structure theory by Fisher, Hicks and Lutz.
- **Use.** Comparing bonds of different coupons, maturities and call features.
- **Data science.** Bootstrapping a zero curve from coupon bonds is solving a triangular linear system; fitting it smoothly is regression.
- **Physical application.** Utilities' callable bonds are refinanced when rates fall, like replacing old equipment when efficiency gains pay off.

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FN.24 Duration & Convexity

Duration is the sensitivity of a bond's price to its yield (the slope of the price–yield curve); convexity is the curvature that corrects the linear estimate for larger rate moves.

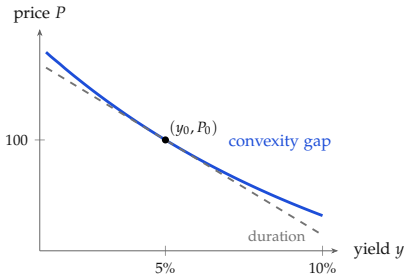
Macaulay duration	$D_{\text{Mac}} = \frac{1}{P}\sum_t t\,\frac{CF_t}{(1+y)^t}$	PV-weighted average time of the cash flows, in periods.
Modified duration	$D_{\text{mod}} = \frac{D_{\text{Mac}}}{1+y} = -\frac{1}{P}\frac{dP}{dy}$	Percent price change per unit change in yield.
Convexity	$\text{Cx} = \frac{1}{P}\frac{d^2P}{dy^2} = \frac{1}{P}\sum_t \frac{t(t+1)\,CF_t}{(1+y)^{t+2}}$	Curvature of the price–yield curve; always positive for plain bonds.
Price change approximation	$\frac{\Delta P}{P} \approx -D_{\text{mod}}\,\Delta y + \frac{1}{2}\,\text{Cx}\,(\Delta y)^2$	Second-order Taylor expansion C2.22 §08 Calculus II.
DV01	$\text{DV01} = D_{\text{mod}} \times P \times 0.0001$	Dollar value change for a one-basis-point move in yield.
Zero-coupon duration	$D_{\text{Mac}} = n$	All cash at maturity, so the average time is n.

STATED-EQUATION *Macaulay duration is the present-value-weighted average time of the cash flows. Modified duration equals Macaulay duration over one plus y, which equals minus one over P times d P d y. Convexity is one over P times the second derivative of price. The percent price change is approximately minus modified duration times the yield change, plus one half convexity times the yield change squared. DV01 is modified duration times price times one basis point. A zero's Macaulay duration equals its maturity.*

MEMORY AID Duration ≈ "the percent price drop for a 1-point rise in rates."

TERMS

D_{Mac} – weighted mean of payment times (years) S.1 §13 Statistics
D_{mod} – price sensitivity: a derivative C1.9 §07 Calculus I
Cx – convexity; second derivative term C2.22 §08 Calculus II
CF_t – cash flow at t; P price; y yield
Basis point – 0.01%
Longer maturity – higher duration, more rate risk
P; y; t – bond price; yield; time of each cash flow
Δy; DV01; n – yield change; dollar value of one basis point; maturity



CONTEXT

- **Origin.** Frederick Macaulay (1938); Redington's immunization (1952).
- **Use.** Managing interest-rate risk for banks, pensions and insurers.
- **Data science.** Sensitivity analysis: duration is a first-order Taylor sensitivity, the same idea as gradients and feature attributions.
- **Physical application.** Like linearizing a spring's force for small displacements, adding a curvature correction for larger ones.

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Derivatives

FN.25 Options: Payoffs & Put–Call Parity

An option gives the right, not the obligation, to buy (call) or sell (put) at a fixed strike price. Its payoff at expiry is a hinge; no-arbitrage ties call and put prices together.

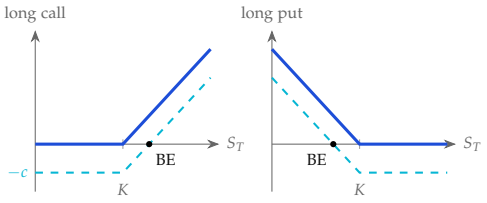
Call payoff	$\max(S_T - K, 0)$	Pays when the price ends above the strike.
Put payoff	$\max(K - S_T, 0)$	Pays when the price ends below the strike.
Profit	$\text{profit} = \text{payoff} - \text{premium}$	Subtract what you paid (ignoring the time value of the premium).
Put–call parity	$C - P = S_0 - Ke^{-rT}$	Call minus put replicates the stock minus a bond paying K (European options, no dividends).
Break-even price	$S_T = K + C$ (call), $S_T = K - P$ (put)	Price at which a long option's profit is zero.

STATED-EQUATION *A call's payoff is the maximum of S T minus K and zero. A put's payoff is the maximum of K minus S T and zero. Profit equals payoff minus premium. Put-call parity: call price minus put price equals the stock price minus the discounted strike, K e to the minus r T. A long call breaks even at K plus the premium, a long put at K minus the premium.*

MEMORY AID "Call up, put down": a call profits when price goes up, a put when it goes down. The call payoff is a ReLU ML.22 §15 Machine Learning!

TERMS

S_0, S_T – underlying price today and at expiry
K – strike price
C, P – call and put prices (premiums)
r, T – risk-free rate (continuous), time to expiry
European – exercise only at expiry
Moneyness – in / at / out of the money: positive / zero / no payoff if exercised now
Parity – no-arbitrage relation linking calls, puts, stock and cash FN.5



CONTEXT

- **Origin.** Options traded in 17th-century Amsterdam; parity formalized by Stoll (1969).
- **Use.** Hedging, speculation, employee stock options.
- **Data science.** The call payoff $\max(0, S - K)$ is exactly a ReLU; option-like payoffs appear in pricing models for insurance and cloud reserved capacity.
- **Physical application.** A home generator is like a put option on grid reliability: you pay a premium to be protected when power fails.

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FN.26 Binomial Option Pricing

Model the stock as moving up or down each step. An option can then be replicated by stock and cash, which gives its price as a discounted expectation under "risk-neutral" probabilities; with many steps it converges to Black–Scholes.

Up / down factors (CRR)	$u = e^{\sigma\sqrt{\Delta t}}, \qquad d = \dfrac{1}{u}$	Calibrates the tree to the volatility σ.
Risk-neutral probability	$p = \dfrac{e^{r\Delta t} - d}{u - d}$	Makes the stock earn the risk-free rate on average; needs $d < e^{r\Delta t} < u$.
One-step value	$V_0 = e^{-r\Delta t}\left[p\,V_u + (1-p)\,V_d\right]$	Discounted risk-neutral expectation of the next-step values.
Hedge ratio	$\Delta = \dfrac{V_u - V_d}{S_u - S_d}$	Shares per option in the replicating portfolio; $V_0 = \Delta S_0 + B$.
n-step European option	$V_0 = e^{-rT}\sum_{k=0}^{n}\binom{n}{k}p^k(1-p)^{n-k}\,\text{payoff}\left(S_0u^kd^{n-k}\right)$	A binomial expectation S.16 §13 Statistics of the payoff.
American early exercise	$V = \max\left(\text{exercise value},\ \text{continuation value}\right)$	Checked at every node while rolling back.

STATED-EQUATION *The up factor u is e to the sigma root delta t and the down factor d is one over u. The risk-neutral probability p is e to the r delta t minus d, over u minus d. The value one step back is the discounted p-weighted average of the up and down values. The hedge ratio delta is the change in option value over the change in stock price. With n steps the European price is the discounted binomial expectation of the payoff; for American options take the larger of exercising now and continuing.*

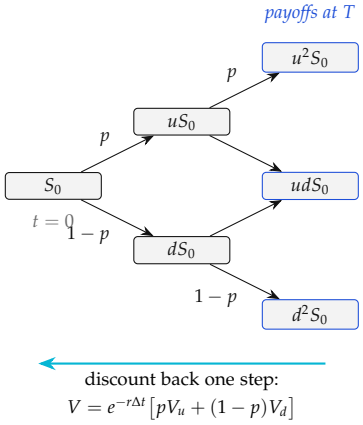
MEMORY AID "Replicate, don't predict": the real-world probability of an up move never enters the price.

METHOD

1. Build the price tree $S_0u^kd^j$ forward to expiry.
2. Fill the final nodes with the payoff.
3. Roll back with $V = e^{-r\Delta t}[pV_u + (1-p)V_d]$ (and the early-exercise check for American options).
4. Example: $S_0 = 100$, $u = 1.2$, $d = 0.8$, $r = 0$, $K = 100$ call: $p = \frac{1-0.8}{0.4} = 0.5$, $V_u = 20$, $V_d = 0$, so $C_0 = 10$ and $\Delta = \frac{20}{40} = 0.5$.

TERMS

u, d – up and down price factors
Δt – length of one step; $n = T/\Delta t$ steps
p – risk-neutral up probability
V_u, V_d – option values after an up / down move
S_u, S_d – stock prices uS_0, dS_0
B – cash (bond) position in the replication
σ, r, K – volatility, risk-free rate, strike FN.25
V_0; T – option value today; time to expiry
n; k – number of steps; number of up moves



CONTEXT

- **Origin.** Cox, Ross & Rubinstein (1979); Sharpe (1978).
- **Use.** Pricing American options, convertible bonds, real options in capital budgeting.
- **Data science.** Backward induction on the tree is dynamic programming O.21 §14 Optimization; the limit is a random walk becoming Brownian motion FN.27.
- **Physical application.** Real options: valuing the flexibility to delay or expand a power plant as prices evolve.

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FN.27 Geometric Brownian Motion

The standard model of a stock price: returns are normally distributed with constant drift and volatility, so the price itself is log-normal and never negative.

Stochastic differential equation	$dS = \mu S\,dt + \sigma S\,dW_t$	Proportional drift plus proportional random shocks.
Closed-form solution	$S_T = S_0 \exp\left[\left(\mu - \frac{\sigma^2}{2}\right)T + \sigma W_T\right]$	From Itô's lemma; the $-\frac{\sigma^2}{2}$ is the volatility drag FN.10.
Log-return distribution	$\ln\frac{S_T}{S_0} \sim N\left((\mu - \frac{\sigma^2}{2})T,\ \sigma^2 T\right)$	Log returns are normal, so S_T is log-normal.
Mean and median	$E[S_T] = S_0 e^{\mu T}, \quad \text{median}(S_T) = S_0 e^{(\mu-\sigma^2/2)T}$	Mean exceeds median: right-skewed outcomes.
Exact simulation step	$S_{t+\Delta t} = S_t \exp\left[\left(\mu - \frac{\sigma^2}{2}\right)\Delta t + \sigma\sqrt{\Delta t}\,Z\right], \ \ Z \sim N(0,1)$	Generates paths without discretization error O.27 §14 Optimization.

STATED-EQUATION *Geometric Brownian motion: d S equals mu S d t plus sigma S d W. The solution: S T equals S naught times e to the quantity mu minus sigma squared over two, times T, plus sigma W T. The log return over T is normal with mean mu minus sigma squared over two times T and variance sigma squared T. The expected value of S T equals S naught e to the mu T, while the median is S naught e to the mu minus sigma squared over two, times T. Paths are simulated by multiplying by the exponential of a drift step plus sigma root delta t times a standard normal draw.*

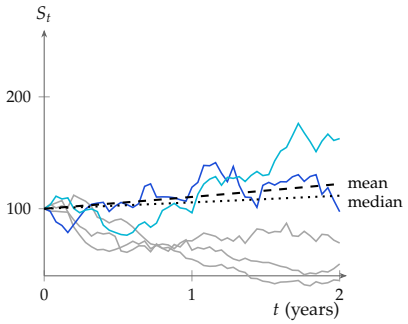
MEMORY AID Mean > median: a few huge winners pull the average up (log-normal skew).

METHOD

1. Choose S_0, μ, σ, horizon T and steps N; set $\Delta t = T/N$.
2. For each step draw $Z \sim N(0,1)$ and apply the exact step.
3. Repeat for many paths; average the quantity of interest. Example: $\mu = 10\%$, $\sigma = 30\%$, $T = 2$: mean $100e^{0.2} = 122$, median $100e^{0.11} = 112$.

TERMS

μ – drift (expected return)
σ – volatility FN.11
W_t – Brownian motion: $W_T \sim N(0,T)$ S.20 §13 Statistics
Z – standard normal draw
$\ln S_T$ – normally distributed, so S_T is log-normal
Simulation – Euler steps with random shocks C2.29 §08 Calculus II, O.27 §14 Optimization
S_t; S_0, S_T – stock price at time t; start and end prices
dS, dt; Δt – infinitesimal changes; simulation step



CONTEXT

- **Origin.** Bachelier (1900) modeled prices with Brownian motion; Samuelson (1965) proposed the geometric form; Itô calculus (1944).
- **Use.** Option pricing FN.28, risk simulation, retirement planning.
- **Data science.** Monte Carlo simulations of portfolios and revenue paths; stochastic differential equations underlie diffusion generative models.
- **Physical application.** Brownian motion was first observed as pollen jiggling in water (Brown, 1827) and explained by Einstein (1905).

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FN.28 Black–Scholes Option Pricing

Closed-form prices for European options when the stock follows geometric Brownian motion: the cost of the replicating portfolio of stock and cash, which does not depend on the stock's expected return.

Call price	$C = S_0\,N(d_1) - Ke^{-rT}N(d_2)$	Stock delta-weighted minus discounted strike times the exercise probability.
d_1 **and** d_2	$d_1 = \dfrac{\ln(S_0/K) + \left(r + \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}$	Standardized distance of the forward price from the strike.
Put price	$P = Ke^{-rT}N(-d_2) - S_0N(-d_1)$	Consistent with put–call parity FN.25.
Black–Scholes PDE	$\dfrac{\partial V}{\partial t} + \dfrac{1}{2}\sigma^2S^2\dfrac{\partial^2V}{\partial S^2} + rS\dfrac{\partial V}{\partial S} - rV = 0$	Every derivative on S satisfies it; the payoff is the boundary condition.
Implied volatility	$\sigma_{\text{imp}}:\ C_{\text{BS}}(\sigma_{\text{imp}}) = C_{\text{market}}$	Invert the formula numerically for σ C1.35 §07 Calculus I.

STATED-EQUATION *The Black-Scholes call price equals S naught times N of d one, minus K e to the minus r T times N of d two. d one equals the natural log of S naught over K, plus r plus sigma squared over two times T, all over sigma root T. d two equals d one minus sigma root T. The put price equals K e to the minus r T N of minus d two, minus S naught N of minus d one. Any derivative value V satisfies the Black-Scholes partial differential equation. Implied volatility is the sigma that makes the formula match the market price.*

MEMORY AID $N(d_2) \approx$ risk-neutral probability the call finishes in the money; $N(d_1)$ is the hedge ratio (delta).

METHOD

1. Gather S_0, K, r (continuous), σ (annual), T (years).
2. Compute d_1, then $d_2 = d_1 - \sigma\sqrt{T}$.
3. Look up $N(d_1)$, $N(d_2)$ (normal CDF table or software).
4. Plug in. Example: $S_0 = K = 100$, $r = 0$, $\sigma = 0.2$, $T = 1$: $d_1 = 0.1$, $d_2 = -0.1$, $C = 100(0.5398 - 0.4602) = \$7.97$.

TERMS

$N(\cdot)$ – standard normal CDF Φ S.20 §13 Statistics
S_0, K – stock price, strike
σ – volatility FN.11
r – risk-free rate (continuous)
T – time to expiry (years)
$V(S,t)$ – value of any derivative on S
Assumes – geometric Brownian motion FN.27, constant σ, no arbitrage
Origin – a heat equation P.31 §12 Physics in disguise
$C,\ P$; d_1, d_2 – call and put prices; standardized moneyness terms
σ_{imp} – implied volatility

CONTEXT

- **Origin.** Black, Scholes (1973) and Merton; Nobel Prize 1997.
- **Use.** Pricing and hedging options; implied volatility.
- **Data science.** Implied-volatility surfaces are fit and forecast with ML; neural networks are trained to approximate pricing functions quickly.
- **Physical application.** The Black–Scholes PDE transforms into the heat equation, the same one that governs diffusion of heat.

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FN.29 Option Greeks

The Greeks are partial derivatives of an option's value: how much it changes when the stock price, volatility, time or interest rate changes. Traders hedge by neutralizing them.

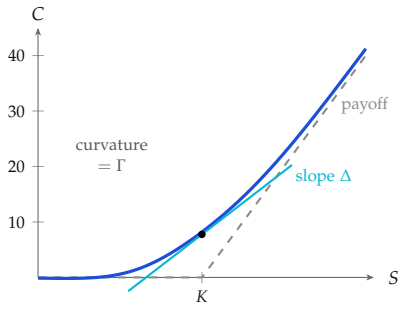
Delta	$\Delta = \dfrac{\partial V}{\partial S} = N(d_1)\ \text{(call)},\quad N(d_1) - 1\ \text{(put)}$	Shares needed to hedge one option; slope of value vs. price.
Gamma	$\Gamma = \dfrac{\partial^2 V}{\partial S^2} = \dfrac{\varphi(d_1)}{S\sigma\sqrt{T}}$	Rate of change of delta; largest near the strike close to expiry.
Vega	$\mathcal{V} = \dfrac{\partial V}{\partial \sigma} = S\,\varphi(d_1)\sqrt{T}$	Value gained per unit rise in volatility (same for calls and puts).
Theta (call)	$\Theta = \dfrac{\partial V}{\partial t} = -\dfrac{S\,\varphi(d_1)\,\sigma}{2\sqrt{T}} - rKe^{-rT}N(d_2)$	Time decay: value lost per year as expiry approaches (usually negative).
Rho (call)	$\rho = \dfrac{\partial V}{\partial r} = KTe^{-rT}N(d_2)$	Sensitivity to the interest rate; negative for puts.
Delta–gamma approximation	$\Delta V \approx \Delta \cdot \Delta S + \tfrac{1}{2}\Gamma(\Delta S)^2$	Taylor expansion in S C2.22 §08 Calculus II; the option analog of duration–convexity FN.24.

STATED-EQUATION *Delta is the first derivative of value with respect to the stock price: N of d one for a call, N of d one minus one for a put. Gamma is the second derivative, phi of d one over S sigma root T. Vega is S phi of d one root T. Call theta is minus S phi of d one sigma over two root T, minus r K e to the minus r T N of d two. Call rho is K T e to the minus r T N of d two. The change in value is about delta times the price change plus half gamma times its square.*

MEMORY AID Delta = slope, Gamma = curvature, Vega = volatility, Theta = time, Rho = rate.

TERMS

V – option value; S stock price
$\varphi(\cdot)$ – standard normal density S.20 §13 Statistics
$N(\cdot)$ – standard normal CDF
d_1, d_2 – from Black–Scholes FN.28
σ, r, T, K – volatility, rate, time to expiry, strike
Partial derivatives – C3.9 §10 Calculus III
Hedging – delta-neutral: $\Delta_{\text{book}} = 0$
$\Delta, \Gamma, \mathcal{V}, \Theta, \rho$ – delta, gamma, vega, theta, rho
S; ΔS, ΔV – stock price; small changes in price and value



CONTEXT

- **Origin.** Named by traders in the 1970s–80s ("vega" is not a Greek letter).
- **Use.** Risk reports, hedging and market-making in options.
- **Data science.** Greeks are gradients of a pricing model; automatic differentiation computes them for complex models ("AAD").
- **Physical application.** Like sensitivity coefficients of a control system with respect to its parameters.

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Risk Management & Position Sizing

FN.30 Value at Risk (VaR) & Expected Shortfall

VaR is a loss threshold that is exceeded only with small probability over a horizon; expected shortfall is the average loss in those bad cases, and so also sees how heavy the tail is.

Parametric VaR	$\mathrm{VaR}_\alpha = z_\alpha \sigma\sqrt{t}\, V$	Normal returns with zero mean; V is the portfolio value.
Expected shortfall	$\mathrm{ES}_\alpha = E\left[\text{loss} \mid \text{loss} > \mathrm{VaR}_\alpha\right]$	Average of the worst $1-\alpha$ of outcomes; always $\geq$ VaR.
Normal quantiles	$z_{0.95} = 1.645, \;\; z_{0.99} = 2.326$	One-sided critical values S.20 §13 Statistics.
Normal ES	$\mathrm{ES}_\alpha = \dfrac{\varphi(z_\alpha)}{1-\alpha}\sigma\sqrt{t}\, V$	At 95%: $2.063\,\sigma\sqrt{t}V$; at 99%: $2.665\,\sigma\sqrt{t}V$.
Historical VaR	$\mathrm{VaR}_\alpha = -\text{quantile}_{1-\alpha}(\text{past P\&L})$	No distributional assumption; uses the empirical percentile S.3 §13 Statistics.

STATED-EQUATION *Parametric value at risk at confidence alpha equals z alpha times sigma times the square root of t times the portfolio value V. Expected shortfall is the expected loss given that the loss exceeds VaR. z at ninety-five percent is one point six four five; at ninety-nine percent, two point three two six. For normal returns, expected shortfall is phi of z alpha over one minus alpha, times sigma root t times V. Historical VaR is minus the one-minus-alpha percentile of past profit and loss.*

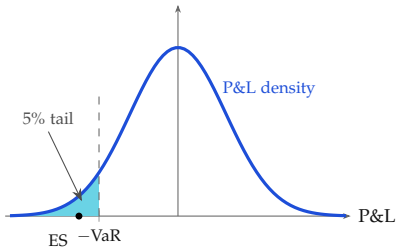
MEMORY AID VaR: "the loss you won't exceed α of the time"; ES: "how bad it gets when you do."

METHOD

1. Collect N past daily returns; revalue today's portfolio under each to get P&L scenarios.
2. Sort them from worst to best.
3. VaR is the loss at position $(1-\alpha)N$; ES is the average of the losses beyond it. Example: 500 days at 99%: VaR is the 5th-worst loss, ES the mean of the 5 worst.

TERMS

α – confidence level (95% or 99%)
z_α – normal quantile S.20 §13 Statistics; φ normal density
$\sigma\sqrt{t}$ – horizon volatility FN.11
V – portfolio value
Monte Carlo VaR – simulated losses O.27 §14 Optimization
Fat tails – normal VaR underestimates extreme losses
φ; t – standard normal density; horizon in the units of σ



CONTEXT

- **Origin.** J.P. Morgan's RiskMetrics (1994); Basel accords adopted VaR and later ES.
- **Use.** Bank capital requirements, risk limits.
- **Data science.** Quantile regression and extreme-value models estimate tail risk; the same idea sets SLOs on p99 latency.
- **Physical application.** Engineers design levees for the "100-year flood," a 1% annual exceedance level, like a 99% VaR.

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FN.31 Expected Value of Bets & the Kelly Criterion

A bet is worth taking only if its expected value is positive; the Kelly criterion then sizes it to maximize the long-run growth rate of wealth, balancing growth against ruin.

Expected value	$EV = p \cdot W - q \cdot L$	Probability-weighted win minus probability-weighted loss S.13 §13 Statistics.
Kelly fraction	$f^* = \frac{bp - q}{b} = p - \frac{q}{b}, \quad q = 1 - p$	Fraction of bankroll to stake on a bet paying b to 1.
Growth rate	$G(f) = p\ln(1 + bf) + q\ln(1 - f)$	Expected log-wealth growth per bet; f^* maximizes it.
Continuous Kelly	$f^* = \frac{\mu - r}{\sigma^2}$	Leverage on a risky asset with drift μ and volatility σ FN.27.

STATED-EQUATION *The expected value of a bet equals the probability of winning times the win, minus the probability of losing times the loss. The Kelly fraction f star equals b p minus q, over b, which equals p minus q over b, where q is one minus p. It maximizes the expected log of wealth, G of f equals p log one plus b f, plus q log one minus f. For an investment, the Kelly fraction is excess drift over variance.*

MEMORY AID Never bet on negative EV; even with an edge, bet a fraction (many use "half Kelly").

METHOD

1. Estimate p and the net odds b; check $EV > 0$.
2. Compute $f^* = p - \frac{q}{b}$; if $f^* \leq 0$ do not bet.
3. Scale down for estimation error. Example: $p = 0.6$, even odds $b = 1$: $f^* = 0.6 - 0.4 = 0.2$ (20% of bankroll; half Kelly 10%).

TERMS

p, q – win and loss probabilities
W, L – amount won, amount lost
b – net odds: profit per \$1 staked on a win
f^* – fraction of bankroll to wager
μ, r, σ – drift, risk-free rate, volatility
Log utility – maximizes long-run growth (geometric mean FN.10)
EV – expected value S.13 §13 Statistics

CONTEXT

- **Origin.** John Kelly at Bell Labs (1956), linked to Shannon's information theory ML.3 §15 Machine Learning.
- **Use.** Position sizing, betting, portfolio growth.
- **Data science.** Budget allocation to experiments or campaigns with uncertain returns; multi-armed bandits balance exploration and exploitation.
- **Physical application.** Choosing how much of a limited energy reserve to spend on a risky maneuver is the same growth-versus-ruin trade-off.

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18
Graph Gallery

Recognize functions and probability distributions by shape. Each graph shows only tick marks and key points; the tile below it names the curve and gives its formula. Every tile is also a flashcard: the front shows only the graph, the back gives the name and formula. Blue is the featured curve; cyan and gray curves are companions for comparison.

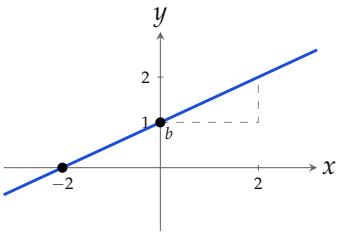
In this chapter

FUNCTIONS

PROBABILITY DISTRIBUTIONS

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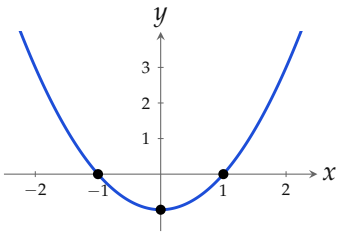
Functions



F.1 Linear

$y = mx + b$

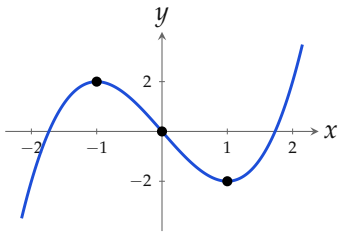
slope m, intercept b



F.2 Quadratic

$y = ax^2 + bx + c$

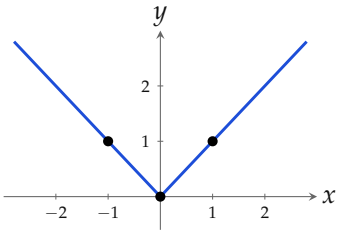
parabola; vertex at $x = -\dfrac{b}{2a}$



F.3 Cubic

$y = ax^3 + bx^2 + cx + d$

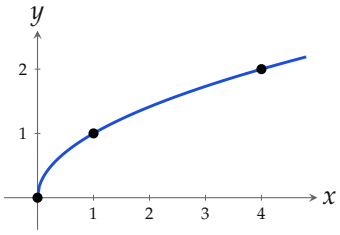
odd end behavior; up to 2 turns



F.4 Absolute Value

$y = |x|$

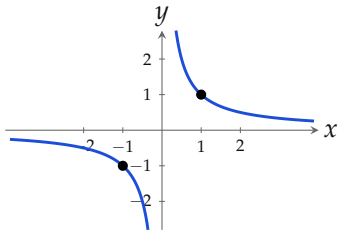
V shape, vertex at $(0, 0)$



F.5 Square Root

$y = \sqrt{x}$

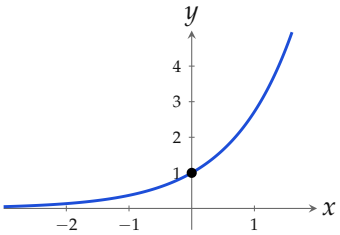
domain $x \geq 0$



F.6 Reciprocal

$y = \dfrac{1}{x}$

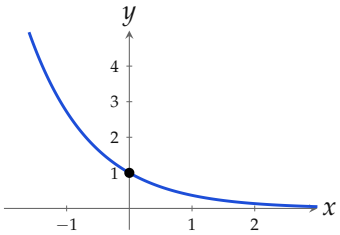
asymptotes $x = 0,\ y = 0$



F.7 Exponential Growth

$y = e^x$ $(b^x,\ b > 1)$

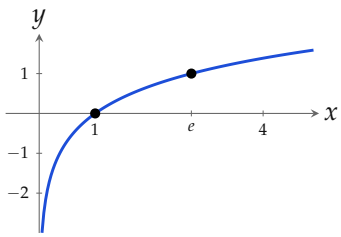
y-intercept 1; HA $y = 0$



F.8 Exponential Decay

$y = e^{-x}$ $(b^x,\ 0 < b < 1)$

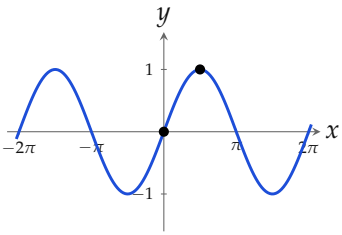
half-life $\dfrac{\ln 2}{k}$



F.9 Natural Logarithm

$y = \ln x$

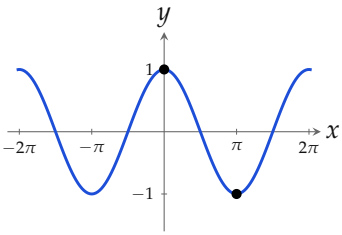
domain $x > 0$; VA $x = 0$; $\ln 1 = 0$



F.10 Sine

$y = \sin x$

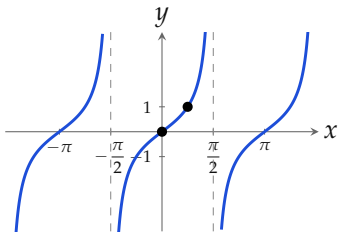
period 2π; $\sin 0 = 0$



F.11 Cosine

$y = \cos x$

period 2π; $\cos 0 = 1$

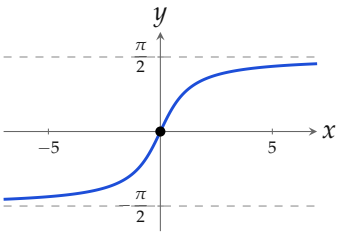


F.12 Tangent

$y = \tan x = \dfrac{\sin x}{\cos x}$

period π; VA at $x = \pm\dfrac{\pi}{2}$

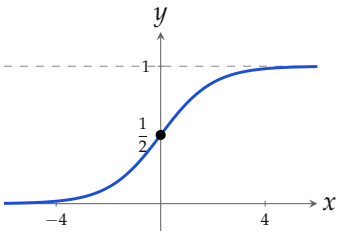
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F.13 Arctangent

$y = \arctan x$

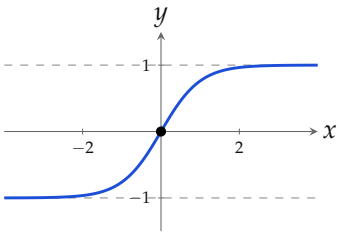
HA $y = \pm\dfrac{\pi}{2}$



F.14 Logistic Sigmoid

$\sigma(x) = \dfrac{1}{1+e^{-x}}$

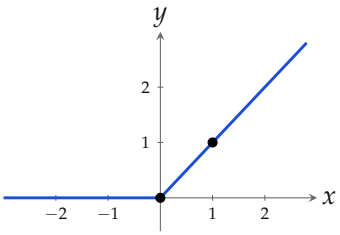
range $(0,1)$; $\sigma(0) = \dfrac{1}{2}$



F.15 Hyperbolic Tangent

$\tanh x = \dfrac{e^x - e^{-x}}{e^x + e^{-x}}$

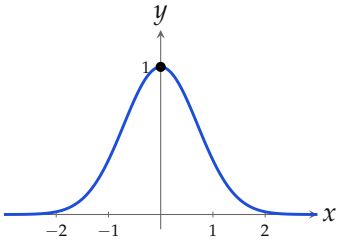
range $(-1,1)$; odd



F.16 ReLU

$\mathrm{ReLU}(x) = \max(0, x)$

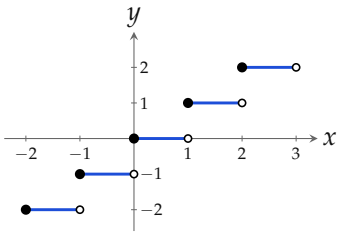
kink at 0



F.17 Gaussian (Bell Curve)

$y = e^{-x^2}$

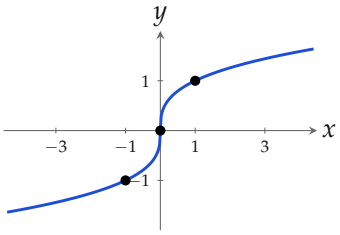
even; maximum 1 at $x = 0$



F.18 Floor (Step) Function

$y = \lfloor x \rfloor$

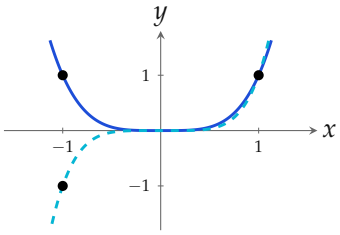
greatest integer $\leq x$



F.19 Cube Root

$y = \sqrt[3]{x} = x^{1/3}$

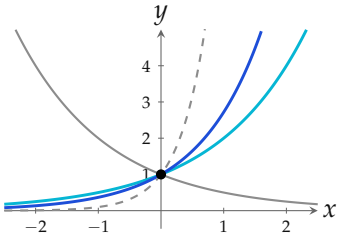
all reals; odd; vertical tangent at 0



F.20 Even vs. Odd Powers

$y = x^4$ vs. $y = x^5$

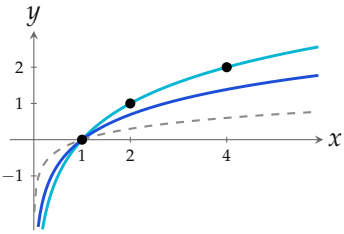
even n: ends both up; odd n: ends opposite



F.21 Exponential Family

$y = b^x \quad (b = \frac{1}{2},\ 2,\ e,\ 10)$

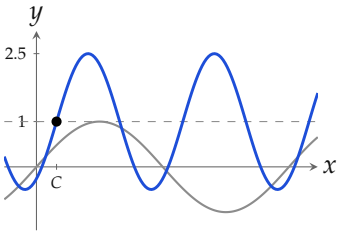
all pass $(0,1)$; larger b steeper; $b < 1$ decays



F.22 Logarithm Bases

$y = \log_b x = \dfrac{\ln x}{\ln b}$

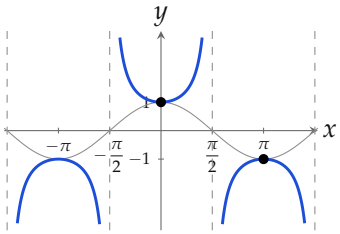
all pass $(1,0)$; $\log_2 > \ln > \log_{10}$ for $x > 1$



F.23 Sinusoid with Phase Shift

$y = A\sin\big(B(x - C)\big) + D$

amplitude $|A|$, period $\dfrac{2\pi}{|B|}$, shift C, midline D

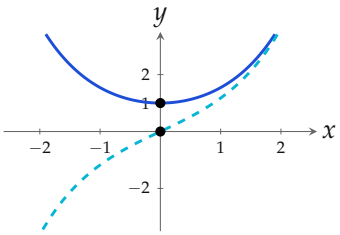


F.24 Secant

$y = \sec x = \dfrac{1}{\cos x}$

period 2π; $|y| \geq 1$; VA where $\cos x = 0$

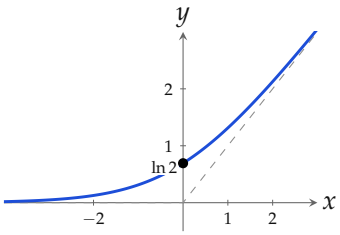
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F.25 Hyperbolic Sine & Cosine

$\cosh x = \dfrac{e^x + e^{-x}}{2}$, $\sinh x = \dfrac{e^x - e^{-x}}{2}$

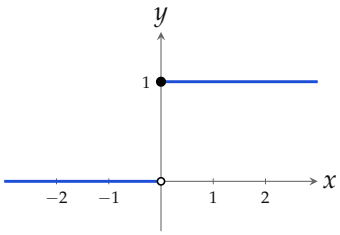
cosh even (catenary), sinh odd;
$\cosh^2 x - \sinh^2 x = 1$



F.26 Softplus

$\operatorname{softplus}(x) = \ln(1 + e^x)$

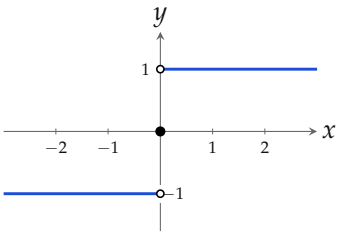
smooth ReLU; derivative is $\sigma(x)$



F.27 Heaviside Step

$$H(x) = \begin{cases} 0, & x < 0 \\ 1, & x \geq 0 \end{cases}$$

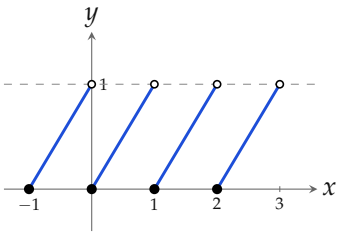
$H'(x) = \delta(x)$; $H(0)$ is a convention



F.28 Sign Function

$\operatorname{sgn} x = \dfrac{x}{|x|}$ $(x \neq 0)$, $\operatorname{sgn} 0 = 0$

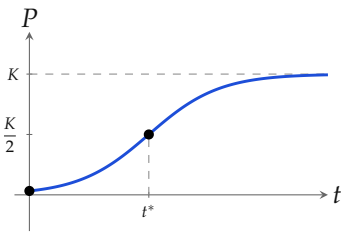
$\dfrac{d}{dx}|x| = \operatorname{sgn} x$ for $x \neq 0$



F.29 Sawtooth (Fractional Part)

$y = x - \lfloor x \rfloor$

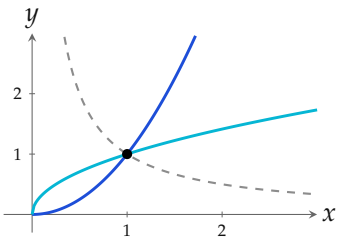
period 1; range $[0, 1)$; jumps at integers



F.30 Logistic Growth

$P(t) = \dfrac{K}{1 + Ae^{-rt}}$, $A = \dfrac{K - P_0}{P_0}$

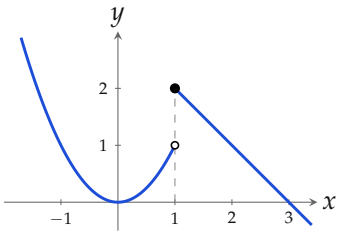
fastest growth at $P = \dfrac{K}{2}$; capacity K



F.31 Power Law

$y = Cx^k$

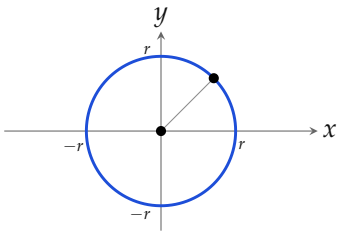
$k > 1$ convex, $0 < k < 1$ concave, $k < 0$ decays; line on log–log axes



F.32 Piecewise Function

$$f(x) = \begin{cases} x^2, & x < 1 \\ 3 - x, & x \geq 1 \end{cases}$$

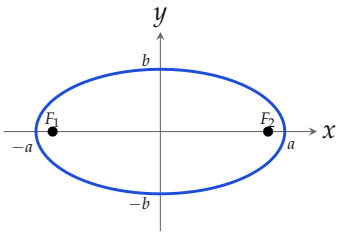
use the branch whose condition holds; dots show which end is included



F.33 Circle

$x^2 + y^2 = r^2$

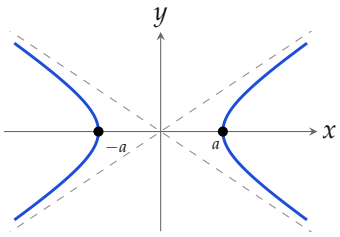
center (h, k): $(x - h)^2 + (y - k)^2 = r^2$; not a function



F.34 Ellipse

$\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$

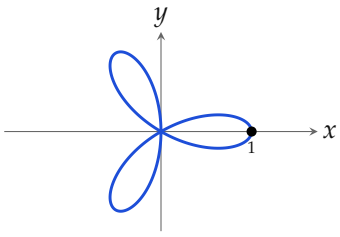
foci $(\pm c, 0)$, $c^2 = a^2 - b^2$; $d_1 + d_2 = 2a$



F.35 Hyperbola

$\dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = 1$

asymptotes $y = \pm \dfrac{b}{a}x$; $c^2 = a^2 + b^2$; $|d_1 - d_2| = 2a$



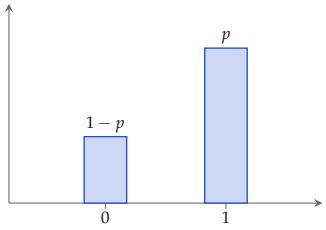
F.36 Polar Rose

$r = \cos(n\theta)$

n odd: n petals; n even: $2n$ petals; length 1

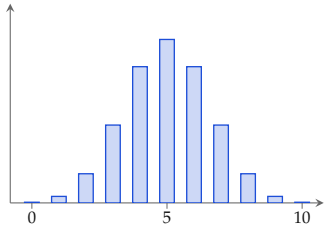
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Probability Distributions



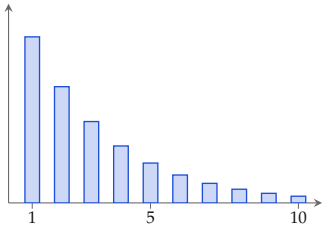
D.1 Bernoulli

$$P(X=1)=p,\;\; P(X=0)=1-p$$
$$\mu=p,\;\; \sigma^2=p(1-p)$$



D.2 Binomial

$$P(X=k)=\binom{n}{k}p^k(1-p)^{n-k}$$
$$\mu=np,\;\; \sigma^2=np(1-p)$$

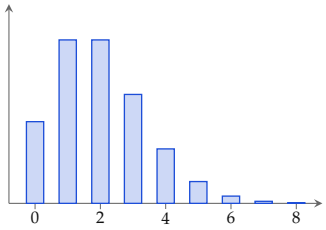


D.3 Geometric

$$P(X=k)=(1-p)^{k-1}p,\;\; k=1,2,\ldots$$

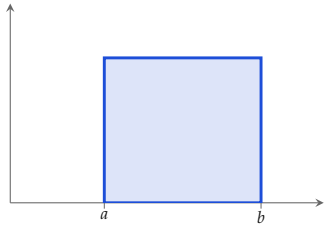
trials to first success;

$$\mu=\frac{1}{p},\ \sigma^2=\frac{1-p}{p^2}$$



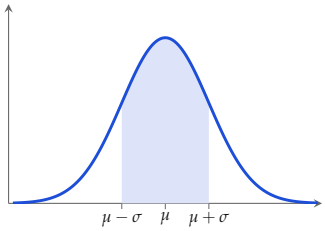
D.4 Poisson

$$P(X=k)=\frac{\lambda^k e^{-\lambda}}{k!}$$
$$\mu=\sigma^2=\lambda$$



D.5 Uniform (Continuous)

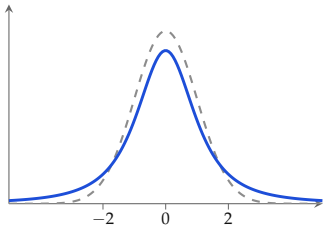
$$f(x)=\frac{1}{b-a},\;\; a\le x\le b$$
$$\mu=\frac{a+b}{2},\;\; \sigma^2=\frac{(b-a)^2}{12}$$



D.6 Normal

$$f(x)=\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

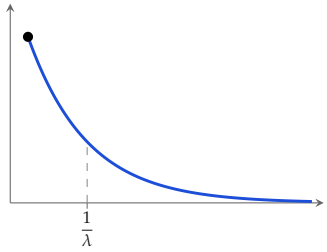
symmetric bell; 68–95–99.7



D.7 Student's t (vs. normal)

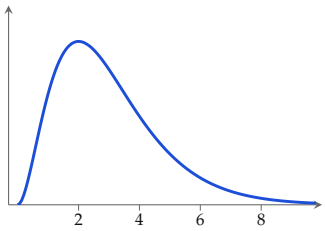
$$f(t)\propto\left(1+\frac{t^2}{\nu}\right)^{-\frac{\nu+1}{2}}$$

heavier tails; $t\to N(0,1)$ as $\nu\to\infty$



D.8 Exponential

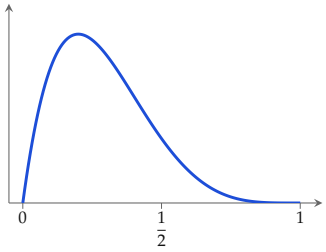
$$f(x)=\lambda e^{-\lambda x},\;\; x\ge 0$$
$$\mu=\frac{1}{\lambda};\ \text{memoryless}$$



D.9 Gamma

$$f(x)=\frac{x^{k-1}e^{-x/\theta}}{\Gamma(k)\,\theta^k}$$
$$\mu=k\theta,\;\; \sigma^2=k\theta^2$$

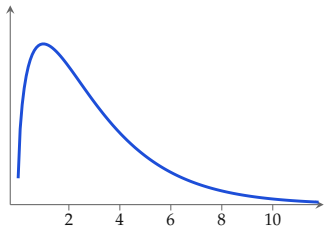
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D.10 Beta

$$f(x) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha,\beta)}, \quad 0 \le x \le 1$$

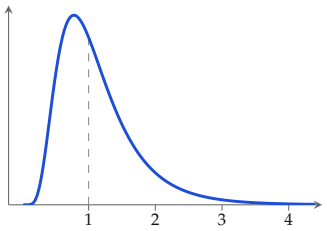
$$\mu = \frac{\alpha}{\alpha+\beta}$$



D.11 Chi-Square

$$\chi^2_k = \sum_{i=1}^{k} Z_i^2$$

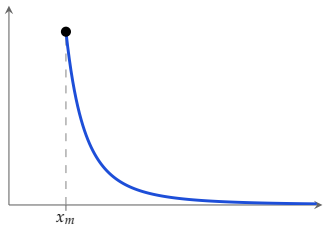
$\mu = k$, $\sigma^2 = 2k$; right-skewed



D.12 Log-Normal

$$\ln X \sim N(\mu, \sigma^2)$$

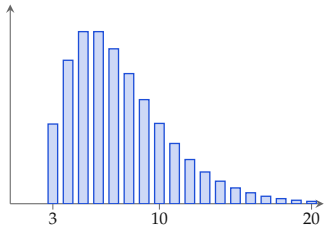
$$f(x) = \frac{1}{x\sigma\sqrt{2\pi}} e^{-\frac{(\ln x - \mu)^2}{2\sigma^2}}$$



D.13 Pareto (Power Law)

$$f(x) = \frac{\alpha\, x_m^{\alpha}}{x^{\alpha+1}}, \quad x \ge x_m$$

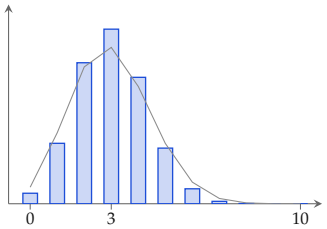
heavy tail; mean finite iff $\alpha > 1$



D.14 Negative Binomial

$$P(X = k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}$$

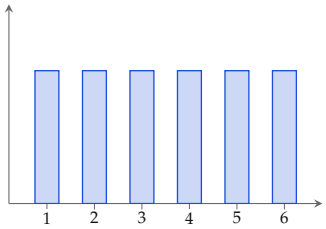
trials to r-th success;
$\mu = \frac{r}{p}$, $\sigma^2 = \frac{r(1-p)}{p^2}$



D.15 Hypergeometric

$$P(X = k) = \frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}$$

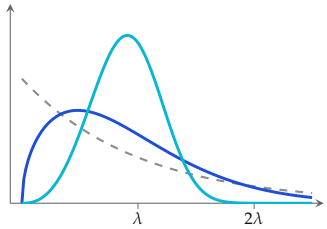
draws without replacement; $\mu = n\frac{K}{N}$



D.16 Discrete Uniform

$$P(X = k) = \frac{1}{n}, \quad k = 1, \ldots, n$$

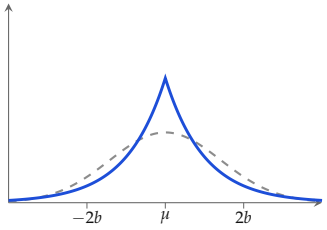
fair die; $\mu = \frac{n+1}{2}$, $\sigma^2 = \frac{n^2-1}{12}$



D.17 Weibull

$$f(x) = \frac{k}{\lambda}\left(\frac{x}{\lambda}\right)^{k-1} e^{-(x/\lambda)^k}$$

shape $k < 1, = 1, > 1$: failure rate falls, flat, rises

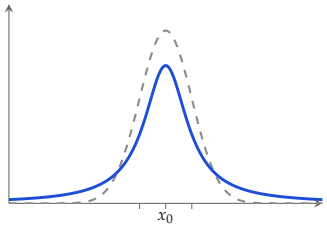


D.18 Laplace (vs. normal)

$$f(x) = \frac{1}{2b} e^{-|x-\mu|/b}$$

$\sigma^2 = 2b^2$; L1 loss / lasso prior

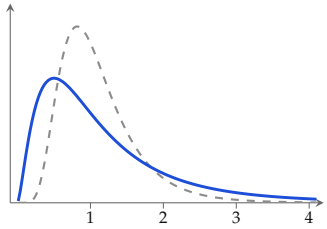
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D.19 Cauchy (vs. normal)

$$f(x) = \frac{1}{\pi\gamma\left[1 + \left(\frac{x - x_0}{\gamma}\right)^2\right]}$$

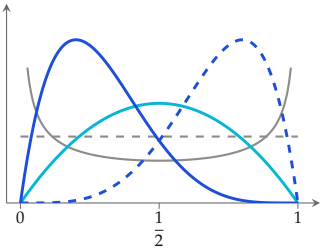
no mean or variance; ratio of two normals



D.20 F-Distribution

$$F = \frac{\chi^2_{d_1}/d_1}{\chi^2_{d_2}/d_2}$$

ANOVA, nested models; $\mu = \frac{d_2}{d_2 - 2}$



D.21 Beta Shapes

$$\text{mode} = \frac{\alpha - 1}{\alpha + \beta - 2} \quad (\alpha, \beta > 1)$$

$\alpha = \beta < 1$: U; $= 1$: flat; $\alpha < \beta$: right-skewed

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Glossary & Reference Cards

Machine-learning and statistics algorithms, acronyms, key terms, and electrical-engineering and finance vocabulary not covered elsewhere, with links to the concepts that treat them in depth. Every entry is also a flashcard: the front shows the name, the back gives the definition.

In this chapter

ALGORITHMS & WHERE THEY ARE USED

ACRONYMS

KEY TERMS

ELECTRICAL ENGINEERING TERMS

FINANCE TERMS

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Algorithms & Where They Are Used

AL.1 Linear Regression *Supervised, regression.* Fits $\hat{y} = X\beta$ by least squares. Used for: pricing, forecasting, effect sizes.

AL.2 Logistic Regression *Supervised, classification.* Sigmoid of a linear score. Used for: churn, credit risk, conversion.

AL.3 Ridge / Lasso *Supervised, regularized regression.* L2 / L1 penalties. Used for: many correlated features, feature selection.

AL.4 k-Nearest Neighbors *Supervised, classification or regression.* Vote of the k closest points. Used for: recommendations, simple baselines, imputation.

AL.5 Naive Bayes *Supervised, classification.* Bayes' rule with independent features. Used for: spam filtering, text tagging.

AL.6 Decision Tree *Supervised.* Splits data by if–then rules. Used for: interpretable rules, triage.

AL.7 Random Forest *Supervised.* Averages many decorrelated trees (bagging). Used for: robust tabular baselines, feature importance.

AL.8 Gradient Boosting (XGBoost, LightGBM) *Supervised.* Adds trees that fix previous errors. Used for: top accuracy on tabular data (fraud, pricing, ranking).

AL.9 Support Vector Machine *Supervised, classification.* Maximum-margin boundary, kernels. Used for: small high-dimensional data, text.

AL.10 Neural Network (MLP) *Supervised.* Layers of linear maps and nonlinearities. Used for: flexible function fitting, embeddings.

AL.11 CNN *Supervised (deep learning).* Convolutional filters share weights across space. Used for: images, medical scans, audio spectrograms.

AL.12 RNN / LSTM *Supervised (deep learning).* Recurrent state over a sequence. Used for: time series, speech (largely replaced by transformers).

AL.13 Transformer *Self-supervised / supervised.* Attention over all tokens. Used for: LLMs, translation, vision transformers.

AL.14 k-Means *Unsupervised, clustering.* Minimizes squared distance to K centroids. Used for: customer segmentation, quantization.

AL.15 DBSCAN *Unsupervised, clustering.* Groups dense regions and labels sparse points as noise; no K needed. Used for: geospatial hot spots, anomaly detection, irregular-shaped clusters.

AL.16 Hierarchical Clustering *Unsupervised.* Merges clusters into a dendrogram. Used for: taxonomies, gene expression.

AL.17 Gaussian Mixture Model *Unsupervised.* Soft clustering with EM. Used for: density estimation, soft segments.

AL.18 PCA *Unsupervised, dimension reduction.* Projects onto directions of maximum variance. Used for: compression, decorrelation, visualization.

AL.19 t-SNE / UMAP *Unsupervised, nonlinear embedding.* Preserves local neighborhoods in 2-D. Used for: visualizing embeddings and single-cell data (not for distances).

AL.20 Isolation Forest *Unsupervised, anomaly detection.* Anomalies are isolated in few random splits. Used for: fraud, sensor faults, intrusion detection.

AL.21 Autoencoder *Unsupervised (self-supervised).* Network compresses and reconstructs input. Used for: denoising, anomaly detection by reconstruction error.

AL.22 Apriori / Association Rules *Unsupervised.* Finds frequent itemsets with support, confidence, lift. Used for: market-basket analysis.

AL.23 Matrix Factorization *Unsupervised / collaborative filtering.* Low-rank user $\times$ item factors. Used for: recommender systems.

AL.24 LDA (Topic Model) *Unsupervised.* Latent Dirichlet Allocation: documents as mixtures of topics. Used for: topic discovery in text.

AL.25 ARIMA *Supervised time series.* AutoRegressive Integrated Moving Average. Used for: demand and sales forecasting.

AL.26 Q-Learning *Reinforcement learning.* Learns action values via the Bellman equation. Used for: games, control, recommendations over time.

AL.27 Spectral Clustering *Unsupervised.* Clusters using eigenvectors of a graph Laplacian. Used for: non-convex clusters, community detection.

AL.28 GAN *Unsupervised generative.* Generator vs. discriminator game. Used for: synthetic images and data augmentation.

AL.29 k-Medoids (PAM) *Unsupervised, clustering.* Like k-means ML.35 §15 Machine Learning but each center is an actual data point, so any distance works and outliers matter less. Used for: clustering with Manhattan, edit or Gower distance.

AL.30 HDBSCAN *Unsupervised, clustering.* Hierarchical DBSCAN: keeps the most stable clusters across all density levels, so it handles clusters of varying density; only a minimum cluster size is needed. Used for: embeddings, geospatial and log data.

AL.31 Mean Shift *Unsupervised, clustering.* Moves points uphill on a kernel density estimate until they reach a mode; the number of modes is the number of clusters. Used for: image segmentation, tracking.

AL.32 Elastic Net *Supervised, regularized regression.* Mixes L1 and L2 penalties ML.8 §15 Machine Learning: sparse like lasso but stable with correlated features. Used for: genomics, wide data ($p \gg n$).

AL.33 Generalized Linear Model (GLM) *Supervised.* A linear score passed through a link function with an exponential-family outcome (logistic, Poisson, gamma). Used for: insurance claims, count data, rates.

AL.34 Generalized Additive Model (GAM) *Supervised.* Sum of smooth one-feature curves (splines) inside a GLM: flexible yet interpretable. Used for: risk curves, dose–response, ecology.

AL.35 Quantile Regression *Supervised, regression.* Minimizes the pinball loss to predict a chosen quantile instead of the mean. Used

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for: prediction intervals, delivery-time and demand ranges.

AL.36 **Gaussian Process Regression** *Supervised, Bayesian.* A distribution over functions set by a kernel; gives predictions with uncertainty bands. Used for: small data, spatial kriging, Bayesian optimization.

AL.37 **Bayesian Optimization** *Optimization.* Fits a surrogate (often a GP) to an expensive objective and picks the next point by an acquisition function. Used for: hyperparameter tuning, experiment design.

AL.38 **Survival Analysis (Kaplan–Meier, Cox)** *Supervised, time-to-event.* Handles censored data; Kaplan–Meier estimates $S(t)$, Cox regression models hazard ratios. Used for: churn timing, clinical trials, failure analysis.

AL.39 **Exponential Smoothing (Holt–Winters)** *Time series.* Weighted averages that decay geometrically into the past, with optional trend and seasonal terms. Used for: inventory and short-horizon demand forecasts.

AL.40 **Prophet** *Time series.* Additive model: piecewise trend + Fourier seasonalities + holiday effects, fitted by regression. Used for: business forecasts with strong seasonality and missing data.

AL.41 **Hidden Markov Model / Viterbi** *Unsupervised / sequence.* Hidden states follow a Markov chain and emit observations; Viterbi finds the most likely state path by dynamic programming O.21 §14 Optimization. Used for: speech, gene finding, regime detection.

AL.42 **Kalman Filter** *State estimation.* Recursive predict–update for a linear system with Gaussian noise; optimal blend of model and measurement. Used for: GPS/IMU tracking, navigation, smoothing noisy sensors.

AL.43 **Word2Vec** *Self-supervised, embeddings.* A shallow network predicts a word from its context (CBOW) or the context from a word (skip-gram), producing word vectors where similar words are close. Used for: text features, similarity, analogies.

AL.44 **Contrastive Learning (SimCLR, CLIP)** *Self-supervised.* Pulls embeddings of matching pairs together and pushes non-matching pairs apart. Used for: image–text search, pre-training without labels.

AL.45 **Diffusion Model** *Generative.* Learns to reverse a gradual noising process, generating data by iterative denoising from pure noise. Used for: image, audio and video generation.

AL.46 **Graph Neural Network (GNN)** *Supervised / self-supervised.* Each node updates its vector by aggregating messages from its neighbors. Used for: molecules, fraud rings, recommendation graphs.

AL.47 **PageRank** *Graph ranking.* Stationary distribution of a random surfer on a link graph: the dominant eigenvector LA.23 §09 Linear Algebra of the transition matrix. Used for: web search, influence and citation ranking.

AL.48 **Approximate Nearest Neighbors (HNSW, FAISS)** *Search.* Index structures that find near neighbors in sublinear time with small recall loss. Used for: vector databases, semantic search, RAG retrieval.

AL.49 **LoRA & Fine-Tuning** *Transfer learning.* Fine-tuning updates a pretrained model on task data; LoRA freezes the weights W and learns a low-rank update $W + BA$, cutting trainable parameters by orders of magnitude. Used for: adapting LLMs and diffusion models cheaply.

AL.50 **RLHF / DPO** *Alignment.* Reinforcement Learning from Human Feedback trains a reward model on human preferences and optimizes the LLM against it (PPO); DPO optimizes the preferences directly. Used for: chat assistants.

AL.51 **Multi-Armed Bandit (Thompson Sampling, UCB)** *Online learning.* Balances exploring uncertain options and exploiting the best so far. Used for: adaptive A/B tests, ad and recommendation selection.

AL.52 **Bootstrap** *Resampling.* Resample the data with replacement many times and recompute a statistic to estimate its sampling distribution. Used for: confidence intervals when no formula exists S.27 §13 Statistics.

AL.53 **SMOTE** *Preprocessing.* Synthetic Minority Over-sampling: creates new minority-class points by interpolating between neighbors. Used for: imbalanced classification (apply to training folds only).

AL.54 **SHAP** *Interpretability.* SHapley Additive exPlanations: splits a prediction into per-feature contributions using game-theoretic Shapley values. Used for: explaining tree models and credit decisions.

Acronyms

AC.1 **CNN** Convolutional Neural Network: learns local filters (convolutions) shared across an image or signal.

AC.2 **RNN** Recurrent Neural Network: processes sequences with a hidden state carried forward.

AC.3 **LSTM** Long Short-Term Memory: an RNN with gates that preserve long-range information.

AC.4 **GRU** Gated Recurrent Unit: a simpler gated RNN.

AC.5 **GAN** Generative Adversarial Network: a generator and a discriminator trained against each other.

AC.6 **VAE** Variational Autoencoder: probabilistic autoencoder that learns a latent distribution.

AC.7 **LLM** Large Language Model: a very large transformer trained to predict text.

AC.8 **NLP** Natural Language Processing: computing with human language.

AC.9 **RAG** Retrieval-Augmented Generation: an LLM answers using

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documents retrieved by search.

AC.10 **MLP** Multilayer Perceptron: a fully connected feed-forward neural network.

AC.11 **SGD** Stochastic Gradient Descent: gradient steps on random mini-batches.

AC.12 **ReLU** Rectified Linear Unit: activation $\max(0, x)$.

AC.13 **ROC / AUC** Receiver Operating Characteristic curve / Area Under the Curve.

AC.14 **MSE / RMSE / MAE** Mean Squared Error / Root Mean Squared Error / Mean Absolute Error.

AC.15 **PCA** Principal Component Analysis.

AC.16 **SVD** Singular Value Decomposition: $A = U\Sigma V^\top$.

AC.17 **SVM** Support Vector Machine.

AC.18 **k-NN** k-Nearest Neighbors.

AC.19 **EDA** Exploratory Data Analysis: summarizing and plotting data before modeling.

AC.20 **ETL** Extract, Transform, Load: moving and reshaping data into a warehouse.

AC.21 **TF-IDF** Term Frequency–Inverse Document Frequency: weights words that are frequent in a document but rare overall.

AC.22 **BERT** Bidirectional Encoder Representations from Transformers: an encoder language model.

AC.23 **GPT** Generative Pre-trained Transformer: a decoder language model.

AC.24 **OLS** Ordinary Least Squares.

AC.25 **MLE / MAP** Maximum Likelihood Estimate / Maximum A Posteriori estimate (likelihood $\times$ prior).

AC.26 **MCMC** Markov Chain Monte Carlo.

AC.27 **KL** Kullback–Leibler divergence.

AC.28 **i.i.d.** Independent and Identically Distributed.

AC.29 **PDF / PMF / CDF** Probability Density Function / Probability Mass Function / Cumulative Distribution Function.

AC.30 **IQR** Interquartile Range, $Q_3 - Q_1$.

AC.31 **CI** Confidence Interval (also Continuous Integration in software).

AC.32 **RL** Reinforcement Learning: learning actions from rewards.

AC.33 **LP / MIP / NLP** Linear / Mixed-Integer / Nonlinear Programming (in optimization, NLP means nonlinear programming).

AC.34 **ODE / PDE** Ordinary / Partial Differential Equation.

AC.35 **GPU** Graphics Processing Unit: parallel hardware used to train neural networks.

AC.36 **MLOps** Machine Learning Operations: deploying, monitoring, and maintaining models.

AC.37 **GLM / GAM** Generalized Linear Model / Generalized Additive Model.

AC.38 **ANOVA** Analysis of Variance: tests whether several group means differ using an F statistic.

AC.39 **AIC / BIC** Akaike / Bayesian Information Criterion: fit minus a complexity penalty; lower is better.

AC.40 **KDE** Kernel Density Estimation: a smooth density built from a bump on each data point.

AC.41 **EM** Expectation–Maximization: alternates soft assignment and parameter updates ML.37 §15 Machine Learning.

AC.42 **HMM** Hidden Markov Model.

AC.43 **GP** Gaussian Process.

AC.44 **MDP** Markov Decision Process: states, actions, transition probabilities, rewards; the setting of RL.

AC.45 **TD** Temporal Difference learning: updates value estimates from the next estimate (bootstrapping).

AC.46 **PPO** Proximal Policy Optimization: a stable policy-gradient RL method with clipped updates.

AC.47 **RLHF** Reinforcement Learning from Human Feedback.

AC.48 **LoRA / PEFT** Low-Rank Adaptation / Parameter-Efficient Fine-Tuning.

AC.49 **MoE** Mixture of Experts: a router sends each token to a few expert sub-networks.

AC.50 **ViT** Vision Transformer: a transformer applied to image patches.

AC.51 **GNN** Graph Neural Network.

AC.52 **NER** Named Entity Recognition: tagging people, places, organizations in text.

AC.53 **ANN** Approximate Nearest Neighbor (search), or Artificial Neural Network; check context.

AC.54 **TPR / FPR** True Positive Rate (recall) / False Positive Rate: the axes of the ROC curve ML.18 §15 Machine Learning.

AC.55 **F1** Harmonic mean of precision and recall ML.17 §15 Machine Learning.

AC.56 **MAPE** Mean Absolute Percentage Error; undefined when actual values are 0.

AC.57 R^2 Coefficient of determination: share of variance explained S.40 §13 Statistics.

AC.58 **FDR** False Discovery Rate: expected share of false positives among discoveries (Benjamini–Hochberg).

AC.59 **VIF** Variance Inflation Factor, $1/(1 - R_j^2)$: flags multicollinearity (above 5–10).

AC.60 **RCT** Randomized Controlled Trial: random assignment makes groups comparable.

AC.61 **ATE** Average Treatment Effect: mean outcome difference if everyone vs. no one were treated.

AC.62 **DAG** Directed Acyclic Graph: used for causal diagrams, pipelines and Bayesian networks.

AC.63 **SQL** Structured Query Language: querying relational databases.

AC.64 **API** Application Programming Interface: a defined way for programs to call each other.

AC.65 **FLOPs** Floating-Point Operations: the unit of compute for train-

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ing and inference.

AC.66 TPU Tensor Processing Unit: Google's accelerator for matrix math.

AC.67 NPV / IRR Net Present Value / Internal Rate of Return FN.8 §17 Finance.

AC.68 APR / EAR Annual Percentage Rate / Effective Annual Rate FN.2 §17 Finance.

AC.69 CAGR Compound Annual Growth Rate FN.10 §17 Finance.

AC.70 VaR / ES Value at Risk / Expected Shortfall FN.30 §17 Finance.

AC.71 CAPM Capital Asset Pricing Model FN.14 §17 Finance.

AC.72 P/E Price-to-Earnings ratio: share price divided by earnings per share.

AC.73 ROE / ROA Return on Equity / Return on Assets FN.20 §17 Finance.

AC.74 KCL / KVL Kirchhoff's Current / Voltage Law EE.4 §16 Electrical Eng..

AC.75 RMS Root Mean Square: the DC-equivalent value of an AC signal EE.13 §16 Electrical Eng..

AC.76 SNR Signal-to-Noise Ratio, usually in dB EE.29 §16 Electrical Eng..

AC.77 DFT / FFT Discrete Fourier Transform / Fast Fourier Transform EE.26 §16 Electrical Eng..

Key Terms

KT.1 Domain & Range Domain: all allowed inputs of a function. Range: all outputs it produces.

KT.2 Monotonic Always increasing or always decreasing.

KT.3 One-to-One (Injective) Different inputs give different outputs; needed for an inverse.

KT.4 Even & Odd Function Even: $f(-x) = f(x)$ (symmetric about the y-axis). Odd: $f(-x) = -f(x)$ (symmetric about the origin).

KT.5 Asymptote A line a graph approaches but does not reach (vertical, horizontal, or slant).

KT.6 Identity vs. Equation An identity is true for all values; an equation is true only for its solutions.

KT.7 Theorem, Lemma, Corollary Theorem: major proven result. Lemma: helper result. Corollary: follows directly from a theorem.

KT.8 Necessary vs. Sufficient A is necessary for B if $B \Rightarrow A$; sufficient if $A \Rightarrow B$.

KT.9 Population vs. Sample Population: every unit of interest. Sample: the subset you observe.

KT.10 Parameter vs. Statistic Parameter describes a population (μ); statistic describes a sample ($\bar{x}$).

KT.11 Bias (Statistical) Systematic difference between an estimator's average and the truth.

KT.12 Confounder A variable that affects both the cause and the outcome, creating a spurious association.

KT.13 Stationarity A time series whose mean, variance and autocorrelation do not change over time.

KT.14 Heteroscedasticity Residual variance that changes with the predictors.

KT.15 Multicollinearity Predictors that are highly correlated, making coefficients unstable.

KT.16 Feature & Label Feature: an input variable. Label (target): the value to predict.

KT.17 Training, Validation, Test Sets Fit on training; tune on validation; report once on the untouched test set.

KT.18 Overfitting & Underfitting Overfitting: memorizes noise (great train, poor test). Underfitting: too simple for both.

KT.19 Hyperparameter A setting chosen before training (learning rate, depth, k), not learned from data.

KT.20 Epoch & Batch Epoch: one full pass over the training data. Batch: the subset used for one gradient step.

KT.21 Data Leakage Information from outside the training data (e.g. the future or the test set) sneaks into training.

KT.22 One-Hot Encoding Representing a category as a vector with a single 1.

KT.23 Normalization vs. Standardization Normalization rescales to $[0, 1]$; standardization gives mean 0, SD 1.

KT.24 Imputation Filling in missing values (mean, median, model-based).

KT.25 Embedding A dense vector representing an item (word, user, image) so that similar items are close.

KT.26 Tensor A multi-dimensional array (scalar, vector, matrix, and higher).

KT.27 Loss vs. Metric Loss: what training minimizes. Metric: what you report to judge usefulness.

KT.28 Class Imbalance One class is much rarer; use PR curves, resampling, or class weights.

KT.29 Calibration Predicted probabilities match observed frequencies.

KT.30 Ensemble Combining several models to improve accuracy or stability.

KT.31 Transfer Learning & Fine-Tuning Reusing a pre-trained model and adapting it to a new task.

KT.32 Attention Weighting inputs by relevance: $\text{softmax}\left(QK^\top/\sqrt{d}\right)V$.

KT.33 Supervised vs. Unsupervised Supervised learns from labeled examples; unsupervised finds structure without labels.

KT.34 Self-Supervised Learning Labels are created from the data itself (e.g. predict the next word).

KT.35 Latent Variable A hidden quantity inferred from data, such as a

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cluster or topic.

KT.36 **Convex vs. Non-Convex** Convex problems have one global minimum; non-convex ones can have many local minima.

KT.37 **Vectorization** Replacing explicit loops with whole-array operations (NumPy, GPU kernels); often 10–100× faster.

KT.38 **Broadcasting** Automatically stretching arrays of compatible shapes so element-wise operations work, e.g. subtracting a row of column means from every row of a matrix.

KT.39 **Sparsity** Most entries are zero (one-hot data, text counts, graphs); store and compute only the nonzeros. Also: a model with many zero weights (lasso).

KT.40 **Curse of Dimensionality** In high dimensions volume concentrates at the edges and distances become nearly equal, so data are sparse and nearest-neighbor methods degrade.

KT.41 **Regularization** Any constraint or penalty that reduces variance at the cost of a little bias: L1/L2 penalties ML.8 §15 Machine Learning, dropout, early stopping, data augmentation.

KT.42 **Early Stopping** Stop training when validation loss stops improving; a cheap regularizer.

KT.43 **Dropout** Randomly zeroing a fraction of activations during training so the network cannot rely on any single unit.

KT.44 **Batch / Layer Normalization** Rescaling activations to mean 0, variance 1 (over the batch or over features), then learning a scale and shift; stabilizes training.

KT.45 **Vanishing & Exploding Gradients** Gradients shrink or blow up as they are multiplied back through many layers; fixed by ReLU, residual connections, normalization, gradient clipping.

KT.46 **Learning Rate & Schedule** Step size of gradient descent ML.27 §15 Machine Learning; too big diverges, too small crawls. Schedules (warm-up, cosine decay) change it during training.

KT.47 **Logits & Temperature** Logits: raw scores before softmax ML.24 §15 Machine Learning. Temperature T divides logits: $T < 1$ sharpens, $T > 1$ flattens the distribution.

KT.48 **Tokenization** Splitting text into units (words, subwords such as BPE) that a model maps to IDs; costs and context limits are counted in tokens.

KT.49 **Context Window** The maximum number of tokens a language model can attend to at once.

KT.50 **Hallucination** A fluent but false output from a generative model; reduced by retrieval (RAG) and verification.

KT.51 **In-Context Learning** A large model performs a new task from instructions or examples in the prompt, without weight updates.

KT.52 **Inductive Bias** The assumptions a model builds in (locality in CNNs, linearity in regression) that let it generalize.

KT.53 **No Free Lunch** Averaged over all possible problems, no learning algorithm beats another; good performance comes from matching assumptions to the data.

KT.54 **Feature Engineering** Creating informative inputs: ratios, lags, interactions, aggregates, encodings.

KT.55 **Distribution Shift (Drift)** Production data differ from training data: covariate shift (inputs change) or concept drift (input–label relation changes). Monitor and retrain.

KT.56 **Target Leakage** A feature that is only known after the outcome (or is a proxy for it); gives great validation scores and a useless model. A form of leakage KT.21.

KT.57 **A/B Test** A randomized experiment comparing a control (A) and a variant (B) on a pre-chosen metric, with sample size set in advance from the desired power.

KT.58 **Statistical Power** $1 - \beta$: the probability a test detects a real effect of a given size S.28 §13 Statistics; rises with sample size and effect size.

KT.59 **Effect Size** The magnitude of a difference (e.g. Cohen's $d = \Delta\mu/\sigma$), as opposed to whether it is statistically significant.

KT.60 **Statistical vs. Practical Significance** With huge n, tiny useless effects become "significant"; always report effect size and a confidence interval.

KT.61 **p-Hacking** Trying many analyses, subsets or stopping points until $p < 0.05$; produces false discoveries. Prevented by preregistration and correction.

KT.62 **Multiple Comparisons** Running m tests at level α gives about $m\alpha$ false positives; correct with Bonferroni (α/m) or FDR control.

KT.63 **Simpson's Paradox** A trend in every subgroup reverses when the groups are combined, because group sizes differ; look for a lurking variable.

KT.64 **Selection & Survivorship Bias** The sample is not representative because of how it was chosen, e.g. studying only the funds or companies that survived.

KT.65 **Correlation vs. Causation** Association alone does not show cause; causal claims need randomization or explicit assumptions (confounders KT.12).

KT.66 **Heavy Tail** A distribution whose tail decays slower than exponential (Pareto, Cauchy); extremes dominate sums and averages converge slowly.

KT.67 **Outlier vs. Anomaly** Outlier: a point far from the rest (1.5 IQR rule S.3 §13 Statistics). Anomaly: an outlier that signals a different process (fraud, fault).

KT.68 **Markov Property** The future depends only on the present state, not on the path that led there.

KT.69 **Numerical Stability** Avoiding overflow and cancellation, e.g. the log-sum-exp trick $\log \sum e^{x_i} = m + \log \sum e^{x_i - m}$ with $m = \max x_i$.

KT.70 **Ill-Conditioned Problem** Tiny input changes cause large output changes (large condition number LA.29 §09 Linear Algebra); solve with QR/SVD or regularization.

KT.71 **Reproducibility** Getting the same result again: fixed random seeds, versioned data and code, recorded environment.

KT.72 **Tidy (Long) vs. Wide Data** Tidy: one row per observation, one column per variable. Wide: repeated measures spread across columns. Pivot to convert.

KT.73 **Interpolation vs. Extrapolation** Predicting inside the range of the training data vs. beyond it; extrapolation is far less reliable.

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KT.74 **Big-O Notation** How cost grows with input size n, ignoring constants: $O(n)$, $O(n \log n)$, $O(n^2)$ O.10 §14 Optimization.

Electrical Engineering Terms

ET.1 **AC vs. DC** Alternating current reverses direction periodically (grid power); direct current flows one way (batteries, electronics).

ET.2 **Ground** The reference node defined as 0 V; also a safety connection to earth.

ET.3 **Short Circuit** A near-zero-resistance path; current can become dangerously large.

ET.4 **Open Circuit** A break in the path; no current flows.

ET.5 **Series vs. Parallel** Series: same current through each element. Parallel: same voltage across each element.

ET.6 **Impedance** AC generalization of resistance, $Z = R + jX$, in ohms.

ET.7 **Reactance** The imaginary part of impedance from inductors ($+$) and capacitors ($-$).

ET.8 **Conductance** Reciprocal of resistance, $G = 1/R$, in siemens.

ET.9 **Bandwidth** The range of frequencies a system passes (Hz); in data links, the achievable data rate.

ET.10 **Gain** Output-to-input ratio of an amplifier, often in dB.

ET.11 **Feedback** Feeding part of the output back to the input; negative feedback stabilizes, positive feedback oscillates.

ET.12 **ADC / DAC** Analog-to-Digital / Digital-to-Analog Converter.

ET.13 **PWM** Pulse-Width Modulation: controlling average power by switching on and off at a varying duty cycle.

ET.14 **Semiconductor** A material (e.g. silicon) whose conductivity can be controlled by doping; basis of diodes and transistors.

ET.15 **MOSFET** Metal-Oxide-Semiconductor Field-Effect Transistor: the voltage-controlled switch in all modern chips.

ET.16 **Breaker & Fuse** Devices that interrupt current when it exceeds a safe level.

ET.17 **Duty Cycle** Fraction of each period that a signal is on.

ET.18 **Harmonics** Frequency components at integer multiples of the fundamental frequency.

ET.19 **Microcontroller** A small computer on one chip with CPU, memory and I/O, used in embedded devices.

ET.20 **Kilowatt-hour** Energy of 1 kW for 1 hour (3.6 MJ); the unit on electricity bills.

ET.21 **Capacitance** Charge stored per volt, $C = Q/V$, in farads; a capacitor blocks DC and passes high frequencies EE.10 §16 Electrical Eng..

ET.22 **Inductance** Flux linkage per ampere, L, in henries; an inductor resists changes in current and passes DC EE.10 §16 Electrical Eng..

ET.23 **Time Constant** $\tau = RC$ or L/R: time to reach 63% of the final change; about 5τ to settle EE.11 §16 Electrical Eng..

ET.24 **RMS Value** The DC value that delivers the same power; $V_{\text{peak}}/\sqrt{2}$ for a sine EE.13 §16 Electrical Eng..

ET.25 **Phasor** A complex number holding a sinusoid's amplitude and phase, turning AC circuit ODEs into algebra EE.14 §16 Electrical Eng..

ET.26 **Power Factor** $\cos\phi$ = real power / apparent power; low values waste line capacity EE.15 §16 Electrical Eng..

ET.27 **Resonance** Frequency where inductive and capacitive reactances cancel, $f_0 = \frac{1}{2\pi\sqrt{LC}}$ EE.16 §16 Electrical Eng..

ET.28 **Cutoff Frequency** The -3 dB point of a filter, where output power is half; $f_c = \frac{1}{2\pi RC}$ for an RC filter EE.17 §16 Electrical Eng..

ET.29 **Bode Plot** Gain (dB) and phase vs. log frequency; shows a system's frequency response at a glance.

ET.30 **Decibel (dB)** Logarithmic ratio: $10\log_{10}(P_2/P_1)$ for power, $20\log_{10}(V_2/V_1)$ for voltage; $+3$ dB $\approx$ double power.

ET.31 **Aliasing** Sampling below twice the highest frequency folds high frequencies onto low ones EE.27 §16 Electrical Eng.; prevent with an anti-aliasing low-pass filter.

ET.32 **Sampling Rate & Resolution** Samples per second (Hz) and bits per sample of an ADC; n bits give 2^n levels.

ET.33 **Pull-Up / Pull-Down Resistor** Holds a digital input at a defined high or low level when nothing drives it.

ET.34 **Debouncing** Filtering the rapid on–off chatter of a mechanical switch (RC filter or software delay).

ET.35 **Decoupling Capacitor** A capacitor placed next to a chip's supply pins to absorb fast current spikes and keep the voltage steady.

ET.36 **Impedance Matching** Making source and load impedances equal (or conjugate) to maximize power transfer EE.9 §16 Electrical Eng. or prevent reflections.

ET.37 **Transformer** Two coupled coils that step AC voltage up or down by the turns ratio, $V_2/V_1 = N_2/N_1$ EE.19 §16 Electrical Eng..

ET.38 **Three-Phase Power** Three sinusoids 120° apart; delivers constant total power with less conductor EE.19 §16 Electrical Eng..

ET.39 **Ripple** Residual AC variation on a DC supply after rectification and filtering.

ET.40 **Noise Floor** The level of background noise below which signals cannot be distinguished; sets the useful dynamic range.

ET.41 **Clock** A periodic square wave that synchronizes digital logic; its frequency limits how fast operations occur.

ET.42 **Integrated Circuit (IC) & PCB** IC: many components fabricated on one chip. PCB: printed circuit board with copper traces connecting components.

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Finance Terms

FT.1 Time Value of Money Money now is worth more than the same amount later, because it can earn a return.

FT.2 Liquidity How quickly an asset can be sold near its fair price.

FT.3 Arbitrage A riskless profit from price differences; its absence underlies pricing models.

FT.4 Hedge A position taken to reduce the risk of another position.

FT.5 Leverage Using borrowed money to magnify returns (and losses).

FT.6 Bid–Ask Spread Difference between the highest buy and lowest sell price; a trading cost.

FT.7 Market Capitalization Share price times shares outstanding.

FT.8 Dividend A cash payment from a company's profits to shareholders.

FT.9 EBITDA Earnings Before Interest, Taxes, Depreciation and Amortization; a proxy for operating cash profit.

FT.10 Free Cash Flow Operating cash flow minus capital expenditures.

FT.11 Balance Sheet Snapshot of assets, liabilities and equity at a point in time.

FT.12 Income Statement Revenue, expenses and profit over a period.

FT.13 Yield Curve Plot of bond yields vs. maturity; an inverted curve often precedes recessions.

FT.14 Basis Point One hundredth of a percent (0.01%).

FT.15 Bull vs. Bear Market Rising market (bull) vs. falling market, typically -20% or more (bear).

FT.16 Index Fund / ETF A fund that tracks a market index; ETFs trade like stocks.

FT.17 Short Selling Selling borrowed shares to profit if the price falls.

FT.18 Drawdown Decline from a peak to a trough in portfolio value.

FT.19 Implied Volatility The volatility that makes an option-pricing model match the market price.

FT.20 Credit Score / Default Credit score: estimate of a borrower's reliability. Default: failure to make required payments.

FT.21 Efficient Market Hypothesis Prices already reflect available information, so consistent excess returns are hard to earn.

FT.22 Backtesting Testing a strategy on historical data; prone to overfitting and look-ahead bias.

FT.23 Compounding Earning returns on previous returns, so growth is exponential FN.1 §17 Finance.

FT.24 Discount Rate The rate used to convert future cash flows to present value; reflects time value and risk FN.5 §17 Finance.

FT.25 Opportunity Cost The return given up on the best alternative use of the money or time.

FT.26 Nominal vs. Real Nominal: in current money. Real: adjusted for inflation, $1 + r_{\text{real}} = \frac{1+r_{\text{nom}}}{1+\pi}$ FN.4 §17 Finance.

FT.27 Risk Premium Extra expected return demanded for bearing risk, e.g. equity return minus the risk-free rate.

FT.28 Risk-Free Rate Return on a (nearly) default-free asset such as short-term government bills.

FT.29 Alpha & Beta Beta: sensitivity to market moves FN.14 §17 Finance. Alpha: return beyond what beta explains.

FT.30 Volatility Standard deviation of returns, usually annualized ($\sigma_{\text{annual}} = \sigma_{\text{daily}}\sqrt{252}$) FN.11 §17 Finance.

FT.31 Diversification Combining imperfectly correlated assets lowers portfolio risk without lowering expected return FN.12 §17 Finance.

FT.32 Systematic vs. Idiosyncratic Risk Market-wide risk that cannot be diversified away vs. company-specific risk that can.

FT.33 Coupon & Face Value Coupon: periodic interest a bond pays. Face (par) value: amount repaid at maturity.

FT.34 Yield to Maturity The single discount rate that makes a bond's cash flows equal its price FN.22 §17 Finance.

FT.35 Duration Weighted-average time of a bond's cash flows; approximate % price change per 1% yield change FN.24 §17 Finance.

FT.36 Credit Spread Yield of a risky bond minus the yield of a comparable government bond; compensation for default risk.

FT.37 Call & Put Option The right (not obligation) to buy (call) or sell (put) at the strike price by expiry FN.25 §17 Finance.

FT.38 Futures & Forwards Agreements to trade an asset at a set price on a future date; futures are standardized and exchange-traded.

FT.39 Swap Exchange of cash-flow streams, e.g. fixed for floating interest payments.

FT.40 Market vs. Limit Order Market: execute now at the best available price. Limit: execute only at the stated price or better.

FT.41 Order Book & Market Maker Order book: all resting buy and sell limit orders. Market maker: firm quoting both sides, earning the spread.

FT.42 Margin & Margin Call Margin: collateral posted for a leveraged position. Margin call: demand for more collateral after losses.

FT.43 Amortization Repaying a loan through scheduled payments that cover interest and principal FN.9 §17 Finance.

FT.44 Total Return Price change plus dividends or coupons, as a fraction of the starting value.

FT.45 Look-Ahead Bias Using information in a backtest that would not have been available at decision time.

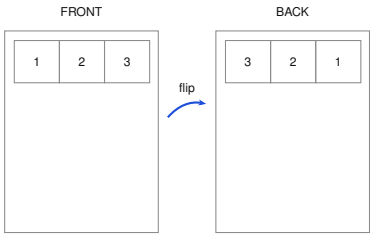
FT.46 Mean Reversion vs. Momentum Mean reversion: prices drift back toward an average. Momentum: recent winners keep winning for a while.

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Flashcards

Printing. Cards are grouped by chapter, and every chapter's cards start on a fresh sheet (an odd page), so you can print or remove one chapter's cards on their own. Each sheet is a pair of pages: a **front** (concept ID, name and the numbered names of the formulas to recall) followed by its **back** (the numbered formulas, key terms and memory aid). Busy concepts are split into Part 1, Part 2, ... cards.

- Print only the flashcard page range, double-sided.
- Scale: **100% / Actual size** (do not "fit to page").
- Duplex: **flip on long edge** (the 11-inch side).
- The back layout is mirrored column-by-column, and the grid is centered on the sheet, so every back aligns with its front.
- Cut along the dashed lines: 3 columns $\times$ 4 rows of $2.5'' \times 2.5''$ cards.



Columns reverse on the back so card 1's answer prints directly behind card 1.

Studying. Read the front, recall the formula and what each symbol means, flip to check. Every card's back also repeats the concept ID so you can jump to the full entry.

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M.1 MATH SIGNS & OPERATIONS



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M.2 MATH SIGNS & OPERATIONS

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M.3 MATH SIGNS & OPERATIONS



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M.4 MATH SIGNS & OPERATIONS

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M.5 MATH SIGNS & OPERATIONS

ab

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M.6 MATH SIGNS & OPERATIONS



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M.7 MATH SIGNS & OPERATIONS

a / b

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M.8 MATH SIGNS & OPERATIONS



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M.9 MATH SIGNS & OPERATIONS



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M.10 MATH SIGNS & OPERATIONS

mod

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M.11 MATH SIGNS & OPERATIONS

$a \mid b$

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M.12 MATH SIGNS & OPERATIONS

$\gcd$

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Times

Multiplication of numbers. Between vectors it is the cross product (see Vectors below).

EXAMPLE

$3 \times 4 = 12$

M.3

Minus

Subtraction; as a prefix, the negative (additive inverse) of a number.

EXAMPLE

$9 - 5 = 4; \quad -(-2) = 2$

M.2

Plus

Addition; as a prefix, marks a positive number.

EXAMPLE

$3 + 4 = 7$

M.1

Divided by

Division (obelus). Rare beyond arithmetic; replaced by a fraction bar or slash.

EXAMPLE

$8 \div 2 = 4$

M.6

Juxtaposition, "a b"

Writing symbols side by side means multiplication. It binds tighter than $\div$ in most texts, so add parentheses when in doubt.

EXAMPLE

$3x$ at $x = 2$ is 6; $\quad 2(3+1) = 8$

M.5

Dot, "times"

Multiplication where $\times$ could be confused with the letter x; between vectors, the dot product.

EXAMPLE

$2 \cdot 5 = 10$

M.4

Minus or plus

Used together with $\pm$: take the opposite sign (top sign with top sign).

EXAMPLE

$\cos(a \pm b) = \cos a \cos b \mp \sin a \sin b$

M.9

Plus or minus

Two values at once (both signs), or a tolerance around a central value.

EXAMPLE

$\pm\sqrt{9} = \pm 3; \quad 5 \pm 1$ means 4 to 6

M.8

Slash, "a over b"

In-line division. Everything to the right of / is *not* automatically in the denominator.

EXAMPLE

$6/3 = 2; \quad$ write $1/(2x)$, not $1/2x$

M.7

GCD (and lcm)

Greatest common divisor: largest integer dividing both. lcm: least common multiple. $\gcd \cdot \operatorname{lcm} = ab$.

EXAMPLE

$\gcd(6,9) = 3; \quad \operatorname{lcm}(4,6) = 12$

M.12

a divides b

b is an integer multiple of a; $\nmid$ means "does not divide".

EXAMPLE

$3 \mid 9; \quad 3 \nmid 10$

M.11

a mod n (modulo)

Remainder after dividing a by n (with $0 \le r < n$); clock arithmetic.

EXAMPLE

$10 \bmod 3 = 1; \quad 9 \bmod 3 = 0$

M.10

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M.14 MATH SIGNS & OPERATIONS



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M.15 MATH SIGNS & OPERATIONS



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M.17 MATH SIGNS & OPERATIONS



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M.18 MATH SIGNS & OPERATIONS



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M.19 MATH SIGNS & OPERATIONS



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M.20 MATH SIGNS & OPERATIONS



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M.21 MATH SIGNS & OPERATIONS



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M.22 MATH SIGNS & OPERATIONS



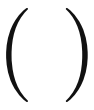
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Approximately equal

Close in value; the result of rounding or an approximation (linearization, series truncation C1.21 §07 Calculus I).

EXAMPLE

$\pi \approx 3.14; \quad \sqrt{2} \approx 1.41$

M.15

Not equal to

The two sides have different values.

EXAMPLE

$3 \neq 4$

M.14

Equals

Both sides have the same value.

EXAMPLE

$2 + 3 = 5$

M.13

Similar / equivalent / asymptotic

Geometry: same shape, any size G.9 §05 Geometry. Analysis: $f \sim g$ means $f/g \to 1$. Loosely: "on the order of". (In statistics $X \sim$ means "distributed as".)

EXAMPLE

Triangles $3, 4, 5 \sim 6, 8, 10$; $n^2 + n \sim n^2$

M.18

Congruent / isomorphic to

Geometry: same shape *and* size G.9 §05 Geometry. Algebra: same structure.

EXAMPLE

$\triangle ABC \cong \triangle DEF$ if all sides match: $3, 4, 5$ and $3, 4, 5$

M.17

Identical to; congruent mod n

True for *every* value of the variables (an identity), or logically equivalent; with $\pmod{n}$, same remainder.

EXAMPLE

$(x+1)^2 \equiv x^2 + 2x + 1; \quad 10 \equiv 1 \pmod{3}$

M.16

At most / at least

Less than or equal; greater than or equal. Equality allowed.

EXAMPLE

$x \leq 5$ includes 5; $\quad 4 \geq 4$

M.21

Less than / greater than

Strict order; the open end faces the larger number.

EXAMPLE

$3 < 7; \quad 9 > 2$

M.20

Proportional to

$y = kx$ for some constant k: doubling x doubles y A1.18 §03 Algebra I.

EXAMPLE

$y \propto x$ with $y = 3x$: $x = 2 \to y = 6$

M.19

Parentheses

Group: evaluate inside first. Also function arguments, ordered pairs, open intervals.

EXAMPLE

$2(3+4) = 14; \quad f(2); \quad$ point $(1, 2)$

M.24

Is defined as

Defines the left side by the right side (assignment, not a claim). Also written $\triangleq$ or $\overset{\text{def}}{=}$.

EXAMPLE

$f(x) := x^2$, so $f(3) = 9$

M.23

Much less / greater

Smaller (larger) by orders of magnitude, so the smaller term can often be neglected.

EXAMPLE

$x \ll 1 \Rightarrow x^2 \approx 0; \quad 1000 \gg 1$

M.22

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M.25 MATH SIGNS & OPERATIONS

$[\ ]$

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M.26 MATH SIGNS & OPERATIONS

$\{\ \}$

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M.28 MATH SIGNS & OPERATIONS

x^n

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M.29 MATH SIGNS & OPERATIONS

x^{-n}

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M.30 MATH SIGNS & OPERATIONS

$x^{1/n}$

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M.31 MATH SIGNS & OPERATIONS

$\sqrt{x}$

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M.32 MATH SIGNS & OPERATIONS

$\sqrt[n]{x}$

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M.33 MATH SIGNS & OPERATIONS

$\exp(x)$

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M.34 MATH SIGNS & OPERATIONS

$\log_b x$

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M.35 MATH SIGNS & OPERATIONS

$\ln x$

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M.36 MATH SIGNS & OPERATIONS

$\log x$

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Vinculum (overbar)

A bar groups like parentheses (fraction bar, root). Also marks repeating decimals, complex conjugates, complements and means.

EXAMPLE

$0.\overline{3} = \frac{1}{3}$; $\overline{3+2i} = 3-2i$

M.27

Braces

Outermost grouping; sets; piecewise cases.

EXAMPLE

$\{1,2,3\}$; $3\{2+[1+(1+1)]\} = 15$

M.26

Brackets

Outer grouping around parentheses; closed intervals; matrices; $\mathbb{E}[X]$.

EXAMPLE

$2[1+(3+1)] = 10$; $[1,3]$

M.25

Fractional exponent

The n-th root; $x^{m/n} = (\sqrt[n]{x})^m$ A1.4 §03 Algebra I.

EXAMPLE

$8^{1/3} = 2$; $4^{3/2} = 8$

M.30

x to the minus n

Reciprocal of the power: $1/x^n$. Not a negative number.

EXAMPLE

$2^{-3} = \frac{1}{8}$

M.29

x to the n (power)

n factors of x multiplied together A1.3 §03 Algebra I. x^2: "x squared"; x^3: "x cubed".

EXAMPLE

$2^3 = 2 \cdot 2 \cdot 2 = 8$

M.28

Exponential, "exp of x"

e^x, written as a function when the exponent is long A2.13 §04 Algebra II.

EXAMPLE

$\exp(0) = 1$; $\exp(\ln 5) = 5$

M.33

n-th root

The number whose n-th power is x; for even n, the non-negative one.

EXAMPLE

$\sqrt[3]{8} = 2$; $\sqrt[4]{16} = 2$

M.32

Square root

The *non-negative* number whose square is x (principal root).

EXAMPLE

$\sqrt{9} = 3$, not ± 3

M.31

Log of x (no base)

Base depends on the field: 10 in school and engineering, e in math and ML, 2 in computer science. Check the context.

EXAMPLE

$\log 1000 = 3$ (base 10); $\log_2 8 = 3$ (base 2)

M.36

Natural log, "L-N of x"

Logarithm base e; the inverse of e^x.

EXAMPLE

$\ln e = 1$; $\ln 1 = 0$

M.35

Log base b of x

The exponent b must be raised to in order to give x: $b^y = x$ A2.14 §04 Algebra II.

EXAMPLE

$\log_2 8 = 3$; $\log_{10} 100 = 2$

M.34

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M.37 MATH SIGNS & OPERATIONS

$1e{-}3$

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M.38 MATH SIGNS & OPERATIONS

π

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M.39 MATH SIGNS & OPERATIONS

e

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M.40 MATH SIGNS & OPERATIONS

i

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M.41 MATH SIGNS & OPERATIONS

φ

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M.42 MATH SIGNS & OPERATIONS

∞

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M.43 MATH SIGNS & OPERATIONS

$|x|$

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M.44 MATH SIGNS & OPERATIONS

$\lfloor x \rfloor$

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M.45 MATH SIGNS & OPERATIONS

$\lceil x \rceil$

NAME IT / GIVE AN EXAMPLE

M.46 MATH SIGNS & OPERATIONS

$\lfloor x \rceil$

NAME IT / GIVE AN EXAMPLE

M.47 MATH SIGNS & OPERATIONS

$\operatorname{sgn} x$

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M.48 MATH SIGNS & OPERATIONS

$n!$

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Euler's number

≈ 2.71828, the base of natural growth: $\lim_{n\to\infty}(1+\frac{1}{n})^n$ C1.6 §07 Calculus I.

EXAMPLE

$e^2 \approx 7.39; \quad (1+\frac{1}{10})^{10} \approx 2.59$

M.39

Pi

Circumference ÷ diameter of any circle, ≈ 3.14159 G.17 §05 Geometry.

EXAMPLE

Circle of radius 2: $C = 2\pi r = 4\pi \approx 12.57$

M.38

E-notation

Computer scientific notation: `aEk` means $a \times 10^k$ A1.5 §03 Algebra I.

EXAMPLE

`2.5e3` $= 2500$; `1e-3` $= 0.001$

M.37

Infinity

Unbounded growth; not a real number. Appears in limits, intervals and sums.

EXAMPLE

$\frac{1}{x} \to 0$ as $x \to \infty$; $[1, \infty)$

M.42

Golden ratio, "phi"

$\varphi = \frac{1+\sqrt{5}}{2} \approx 1.618$; satisfies $\varphi^2 = \varphi + 1$.

EXAMPLE

Fibonacci ratios $\frac{8}{5} = 1.6$, $\frac{13}{8} \approx 1.625 \to \varphi$

M.41

Imaginary unit

$i^2 = -1$ A2.6 §04 Algebra II. Engineers write j (since i is current).

EXAMPLE

$(1+i)(1-i) = 1 - i^2 = 2$

M.40

Ceiling of x

Least integer $\geq x$ (round up).

EXAMPLE

$\lceil 3.2 \rceil = 4; \quad \lceil -2.5 \rceil = -2$

M.45

Floor of x

Greatest integer $\leq x$ (round down) F.18 §18 Graph Gallery.

EXAMPLE

$\lfloor 3.7 \rfloor = 3; \quad \lfloor -2.5 \rfloor = -3$

M.44

Absolute value

Distance from x to 0, never negative A1.11 §03 Algebra I. Same bars mean size of a set, determinant, or modulus of a complex number.

EXAMPLE

$|-3| = 3; \quad |4| = 4; \quad |3+4i| = 5$

M.43

n factorial

Product $1 \cdot 2 \cdots n$; counts orderings of n items. By convention $0! = 1$ S.8 §13 Statistics.

EXAMPLE

$4! = 24; \quad 5! = 120$

M.48

Sign of x, "signum"

$+1$ if $x > 0$, 0 if $x = 0$, -1 if $x < 0$; $x = \operatorname{sgn}(x)\,|x|$.

EXAMPLE

$\operatorname{sgn}(-5) = -1; \quad \operatorname{sgn}(3) = 1$

M.47

Round (nearest integer)

Round to the closest integer. Ties: half up in school, half to even in most software.

EXAMPLE

$\lfloor 2.6 \rceil = 3$; `round(2.5)` $= 2$ in Python

M.46

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M.49 MATH SIGNS & OPERATIONS



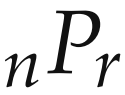
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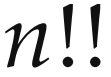
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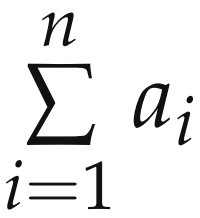
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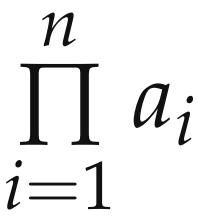
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M.54 MATH SIGNS & OPERATIONS



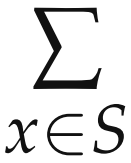
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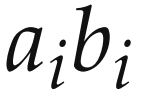
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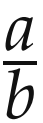
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M.57 MATH SIGNS & OPERATIONS



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M.58 MATH SIGNS & OPERATIONS



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M.59 MATH SIGNS & OPERATIONS



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M.60 MATH SIGNS & OPERATIONS



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n P r (permutations)

Ordered arrangements of r of n items: $\frac{n!}{(n-r)!}$ S.8 §13 Statistics.

EXAMPLE

$${}_5P_2 = 5 \cdot 4 = 20$$

M.51

n C r (combinations)

Same as $\binom{n}{r}$; also written $C(n,r)$ or ${}^nC_r$.

EXAMPLE

$${}_5C_2 = 10$$

M.50

n choose k (binomial coefficient)

Number of ways to choose k of n items, order ignored: $\frac{n!}{k!(n-k)!}$ S.8 §13 Statistics, A2.23 §04 Algebra II.

EXAMPLE

$$\binom{5}{2} = 10; \quad \binom{4}{2} = 6$$

M.49

Pi, “product from i = 1 to n”

Multiply the terms a_i for $i = 1, \ldots, n$; logs turn it into a sum.

EXAMPLE

$$\prod_{i=1}^{4} i = 24 = 4!$$

M.54

Sigma, “sum from i = 1 to n”

Add the terms a_i for every integer i from the lower limit to the upper limit A2.21 §04 Algebra II.

EXAMPLE

$$\sum_{i=1}^{4} i = 1+2+3+4 = 10$$

M.53

Double factorial

Product of every second integer down from n; not $(n!)!$.

EXAMPLE

$$7!! = 7 \cdot 5 \cdot 3 \cdot 1 = 105; \quad 6!! = 48$$

M.52

Einstein summation

A repeated index implies a sum (physics, `einsum`): $a_i b_i = \sum_i a_i b_i$.

EXAMPLE

$a = (1,2)$, $b = (3,4)$: $a_i b_i = 3 + 8 = 11$

M.57

Sum over a set

Add over every element of S (no counting index needed). $\sum_{i,j}$ sums over all pairs.

EXAMPLE

$$\sum_{x \in \{1,3,5\}} x = 9$$

M.56

a sub i

The i-th element of a sequence or vector; a_{ij} is row i, column j of a matrix.

EXAMPLE

$a = (5,8,2)$: $a_2 = 8$; $A = \left[\begin{smallmatrix} 1 & 2 \\ 3 & 4 \end{smallmatrix}\right]$: $a_{21} = 3$

M.55

Proportion, “as”

$a : b :: c : d$ reads “a is to b as c is to d”: two equal ratios; solve by cross-multiplying $ad = bc$ A1.19 §03 Algebra I.

EXAMPLE

$2 : 3 :: 4 : 6$ since $2 \cdot 6 = 3 \cdot 4$

M.60

Ratio, “a to b”

Compares two quantities; equals $\frac{a}{b}$. Odds $a : b$ mean probability $\frac{a}{a+b}$.

EXAMPLE

$6 : 4 = 3 : 2$; odds $1 : 3 \Rightarrow P = \frac{1}{4}$

M.59

Fraction, “a over b”

Numerator divided by denominator $b \neq 0$; the bar groups both parts A1.2 §03 Algebra I.

EXAMPLE

$$\frac{6+2}{4} = 2; \quad \frac{1}{2} + \frac{1}{3} = \frac{5}{6}$$

M.58

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M.61 MATH SIGNS & OPERATIONS

$\%$

NAME IT / GIVE AN EXAMPLE

M.62 MATH SIGNS & OPERATIONS

$\{a, b, c\}$

NAME IT / GIVE AN EXAMPLE

M.63 MATH SIGNS & OPERATIONS

$\{x \mid P\}$

NAME IT / GIVE AN EXAMPLE

M.64 MATH SIGNS & OPERATIONS

$\in$

NAME IT / GIVE AN EXAMPLE

M.65 MATH SIGNS & OPERATIONS

$\notin$

NAME IT / GIVE AN EXAMPLE

M.66 MATH SIGNS & OPERATIONS

$\subseteq$

NAME IT / GIVE AN EXAMPLE

M.67 MATH SIGNS & OPERATIONS

$\subset$

NAME IT / GIVE AN EXAMPLE

M.68 MATH SIGNS & OPERATIONS

$\supseteq$

NAME IT / GIVE AN EXAMPLE

M.69 MATH SIGNS & OPERATIONS

$\cup$

NAME IT / GIVE AN EXAMPLE

M.70 MATH SIGNS & OPERATIONS

$\cap$

NAME IT / GIVE AN EXAMPLE

M.71 MATH SIGNS & OPERATIONS

$\setminus$

NAME IT / GIVE AN EXAMPLE

M.72 MATH SIGNS & OPERATIONS

A^c

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Set-builder, “the set of x such that”

All x for which the condition P holds; also written $\{x : P\}$.

EXAMPLE

$$\{x \in \mathbb{Z} \mid 1 \le x \le 3\} = \{1, 2, 3\}$$

M.63

Set (roster)

An unordered collection of distinct elements; order and repetition do not matter.

EXAMPLE

$$\{a, b, c\} = \{c, a, b\} = \{a, a, b, c\}$$

M.62

Percent

Per hundred: $p\% = \frac{p}{100}$ A1.19 §03 Algebra I.

EXAMPLE

25% of 8 is 2; $\quad 3 \to 4$ is $+33.\overline{3}\%$

M.61

Is a subset of

Every element of the left set is in the right set (equality allowed).

EXAMPLE

$$\{a, b\} \subseteq \{a, b, c\}; \quad \{a, b\} \subseteq \{a, b\}$$

M.66

Is not an element of

Non-membership.

EXAMPLE

$$d \notin \{a, b, c\}$$

M.65

Is an element of

Membership of a single element.

EXAMPLE

$$a \in \{a, b, c\}$$

M.64

Union, “or”

Elements in either set (or both) A1.21 §03 Algebra I. $\bigcup_i A_i$: union of many sets.

EXAMPLE

$$\{a, b, c\} \cup \{d, e, f\} = \{a, b, c, d, e, f\}$$

M.69

Is a superset of

The left set contains the right ($\supset$: proper superset).

EXAMPLE

$$\{a, b, c\} \supseteq \{c\}$$

M.68

Proper subset

Subset but not equal ($\subsetneq$). Some authors use $\subset$ for $\subseteq$: check.

EXAMPLE

$$\{a\} \subset \{a, b, c\}$$

M.67

Complement

Everything in the universal set U not in A; also $\overline{A}$ or A'.

EXAMPLE

$U = \{a, \ldots, f\}$, $A = \{a, b, c\}$:
$A^c = \{d, e, f\}$

M.72

Set difference, “minus”

Elements of the first set that are not in the second.

EXAMPLE

$$\{a, b, c\} \setminus \{b\} = \{a, c\}$$

M.71

Intersection, “and”

Elements in both sets.

EXAMPLE

$$\{a, b, c\} \cap \{b, c, d\} = \{b, c\}$$

M.70

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M.73 MATH SIGNS & OPERATIONS

$A \triangle B$

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M.74 MATH SIGNS & OPERATIONS

$\varnothing$

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M.75 MATH SIGNS & OPERATIONS

$|A|$

NAME IT / GIVE AN EXAMPLE

M.76 MATH SIGNS & OPERATIONS

$A \times B$

NAME IT / GIVE AN EXAMPLE

M.77 MATH SIGNS & OPERATIONS

$\mathcal{P}(A)$

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M.78 MATH SIGNS & OPERATIONS

$\mathbb{N}\,\mathbb{Z}\,\mathbb{Q}\,\mathbb{R}\,\mathbb{C}$

NAME IT / GIVE AN EXAMPLE

M.79 MATH SIGNS & OPERATIONS

$\mathbb{R}^n$

NAME IT / GIVE AN EXAMPLE

M.80 MATH SIGNS & OPERATIONS

(a, b)

NAME IT / GIVE AN EXAMPLE

M.81 MATH SIGNS & OPERATIONS

$[a, b]$

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M.82 MATH SIGNS & OPERATIONS

$[a, b)$

NAME IT / GIVE AN EXAMPLE

M.83 MATH SIGNS & OPERATIONS

$\neg p$

NAME IT / GIVE AN EXAMPLE

M.84 MATH SIGNS & OPERATIONS

$p \wedge q$

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Cardinality, “size of A”

Number of elements; also $\#A$ or $n(A)$.

EXAMPLE

$|\{a,b,c\}| = 3; \quad |\varnothing| = 0$

M.75

Empty set

The set with no elements, $\{\}$; a subset of every set. Not the Greek ϕ.

EXAMPLE

$\{a,b,c\} \cap \{d,e,f\} = \varnothing$

M.74

Symmetric difference

In exactly one of the two sets: $(A \setminus B) \cup (B \setminus A)$; the set version of XOR.

EXAMPLE

$\{a,b,c\} \triangle \{b,c,d\} = \{a,d\}$

M.73

Number sets

Naturals $\{1,2,\dots\}$ (some include 0), integers, rationals $\frac{p}{q}$, reals, complex numbers: $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$. A superscript $^+$ keeps the positives.

EXAMPLE

$3 \in \mathbb{N}$; $-3 \in \mathbb{Z}$; $\frac{1}{2} \in \mathbb{Q}$; $\sqrt{2} \in \mathbb{R}$; $2+i \in \mathbb{C}$

M.78

Power set

The set of all subsets of A (also 2^A); it has $2^{|A|}$ elements.

EXAMPLE

$\mathcal{P}(\{a,b\}) = \{\varnothing, \{a\}, \{b\}, \{a,b\}\}$

M.77

Cartesian product

All ordered pairs (x,y) with $x \in A$, $y \in B$; $|A \times B| = |A|\,|B|$.

EXAMPLE

$\{a,b\} \times \{d,e\} = \{(a,d),(a,e),(b,d),(b,e)\}$

M.76

Closed interval

All reals from a to b, endpoints included.

EXAMPLE

$[2,5] = \{x \mid 2 \le x \le 5\}; \quad 2 \in [2,5]$

M.81

Open interval

All reals strictly between a and b; parentheses exclude the endpoints A1.21 §03 Algebra I. (Same notation as a point: read the context.)

EXAMPLE

$(2,5) = \{x \mid 2 < x < 5\}; \quad 2 \notin (2,5)$

M.80

R n, real n-space

All ordered lists of n real numbers: vectors with n components LA.2 §09 Linear Algebra.

EXAMPLE

$(1,2,3) \in \mathbb{R}^3$

M.79

p and q (conjunction)

True only when both are true; like $\cap$.

EXAMPLE

$(2 < 3) \wedge (4 < 5)$: true

M.84

Not p (negation)

Flips the truth value; also written $\sim p$, $\overline{p}$ or $!p$. See the truth tables M.K1 §01 Math Signs.

EXAMPLE

$\neg(3 > 5)$ is true

M.83

Half-open interval

Includes a, excludes b; ∞ always takes a parenthesis.

EXAMPLE

$[2,5) = \{x \mid 2 \le x < 5\}; \quad [1,\infty)$

M.82

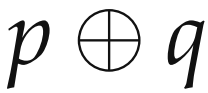
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M.85 MATH SIGNS & OPERATIONS



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M.86 MATH SIGNS & OPERATIONS



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M.87 MATH SIGNS & OPERATIONS



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M.88 MATH SIGNS & OPERATIONS



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M.89 MATH SIGNS & OPERATIONS



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M.90 MATH SIGNS & OPERATIONS



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M.91 MATH SIGNS & OPERATIONS



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M.92 MATH SIGNS & OPERATIONS



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M.94 MATH SIGNS & OPERATIONS



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M.96 MATH SIGNS & OPERATIONS



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Top, “true”

The always-true statement (tautology).

EXAMPLE

$p \vee \neg p \equiv \top$

M.87

Exclusive or (XOR)

True when exactly one is true; on bits, addition mod 2 EE.36 §16 Electrical Eng.. ($\oplus$ also denotes a direct sum.)

EXAMPLE

$1 \oplus 1 = 0$;
$5 \oplus 3 = 101_2 \oplus 011_2 = 110_2 = 6$

M.86

p or q (disjunction)

True when at least one is true (inclusive or); like $\cup$.

EXAMPLE

$(2 > 3) \vee (4 < 5)$: true

M.85

There exists

Existential quantifier: at least one element works. $\nexists$: there is none.

EXAMPLE

$\exists x \in \mathbb{N} :\ x + 2 = 5$ (namely $x = 3$)

M.90

For all

Universal quantifier: the statement holds for every element. Negation: $\neg \forall x\, P \equiv \exists x\, \neg P$.

EXAMPLE

$\forall x \in \mathbb{R},\ x^2 \geq 0$

M.89

Bottom, “false”

The always-false statement (contradiction). In geometry the same shape means perpendicular.

EXAMPLE

$p \wedge \neg p \equiv \bot$

M.88

Because

Introduces a reason.

EXAMPLE

6 is even $\because\ 6 = 2 \cdot 3$

M.93

Therefore

Introduces a conclusion.

EXAMPLE

$x + 2 = 5\ \therefore\ x = 3$

M.92

Exists exactly one

Existence *and* uniqueness.

EXAMPLE

$\exists!\, x :\ 2x = 6$ (only $x = 3$)

M.91

Is implied by, “if”

The reverse direction: $q \Leftarrow p$ means $p \Rightarrow q$.

EXAMPLE

$x^2 = 4 \Leftarrow x = 2$

M.96

Implies

Asserts that whenever the left side holds, so does the right (p is *sufficient* for q; q is *necessary* for p). The converse need not hold.

EXAMPLE

$x = 2 \Rightarrow x^2 = 4$, but $x^2 = 4 \not\Rightarrow x = 2$ ($x = -2$)

M.95

If p then q (conditional)

The logical connective: false only when p is true and q is false. Equivalent to $\neg p \vee q$.

EXAMPLE

$(n = 4) \to (n$ even): true for $n = 1, \ldots, 10$

M.94

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M.97 MATH SIGNS & OPERATIONS

$\leftarrow$

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M.98 MATH SIGNS & OPERATIONS

$p \leftrightarrow q$

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M.99 MATH SIGNS & OPERATIONS

$\Leftrightarrow$

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M.100 MATH SIGNS & OPERATIONS

iff

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M.101 MATH SIGNS & OPERATIONS

$q \rightarrow p$

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M.102 MATH SIGNS & OPERATIONS

$\neg p \rightarrow \neg q$

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M.103 MATH SIGNS & OPERATIONS

$\neg q \rightarrow \neg p$

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M.104 MATH SIGNS & OPERATIONS

$\vdash$

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M.105 MATH SIGNS & OPERATIONS

$\models$

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M.106 MATH SIGNS & OPERATIONS

$f : A \rightarrow B$

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M.107 MATH SIGNS & OPERATIONS

$\mapsto$

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M.108 MATH SIGNS & OPERATIONS

$f(x)$

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If and only if (iff)

Implication in both directions: each side is necessary and sufficient for the other; the statements are equivalent.

EXAMPLE

$2x = 6 \Leftrightarrow x = 3$

M.99

p if and only if q (biconditional)

The connective that is true when p and q have the same truth value; $(p \to q) \wedge (q \to p)$.

EXAMPLE

$(x + 1 = 3) \leftrightarrow (x = 2)$

M.98

Gets (assignment)

In algorithms: overwrite the variable on the left with the value on the right.

EXAMPLE

$x \leftarrow x + 1$ turns 3 into 4

M.97

Inverse

Negate both parts. Equivalent to the converse, not to the original.

EXAMPLE

"If $n \neq 4$ then n is odd": false ($n = 2$)

M.102

Converse

Swap hypothesis and conclusion. *Not* equivalent to $p \to q$.

EXAMPLE

"If n is even then $n = 4$": false ($n = 6$)

M.101

If and only if

Written abbreviation of $\Leftrightarrow$; used in definitions and theorems.

EXAMPLE

n is even iff $n \bmod 2 = 0$

M.100

Entails / satisfies

Semantic consequence: every interpretation making the left true makes the right true.

EXAMPLE

$p \wedge q \models p$

M.105

Proves, "turnstile"

Syntactic consequence: the right side can be derived from the left by proof rules.

EXAMPLE

$p,\ p \to q \vdash q$ (modus ponens)

M.104

Contrapositive

Swap *and* negate. Always equivalent to $p \to q$; proving it proves the original.

EXAMPLE

"If n is odd then $n \neq 4$": true

M.103

f of x

The output of f at input x; not multiplication.

EXAMPLE

$f(x) = x + 1$: $f(2) = 3$

M.108

Maps to

Where one element goes (element level, while $\to$ is set level).

EXAMPLE

$x \mapsto x^2$ sends $3 \mapsto 9$

M.107

f maps A to B

A function with domain A and codomain B A1.9 §03 Algebra I.

EXAMPLE

$f : \mathbb{R} \to \mathbb{R}$, $f(x) = 2x$

M.106

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M.109 MATH SIGNS & OPERATIONS

$g \circ f$

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M.110 MATH SIGNS & OPERATIONS

f^{-1}

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M.111 MATH SIGNS & OPERATIONS

$f'(x)$

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M.112 MATH SIGNS & OPERATIONS

$\max$

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M.113 MATH SIGNS & OPERATIONS

$\arg\max$

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M.114 MATH SIGNS & OPERATIONS

$\mathbf{1}_A$

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M.115 MATH SIGNS & OPERATIONS

$O(g)$

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M.116 MATH SIGNS & OPERATIONS

$f * g$

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M.117 MATH SIGNS & OPERATIONS

$\lim_{x \to a}$

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M.118 MATH SIGNS & OPERATIONS

$x \to a$

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M.119 MATH SIGNS & OPERATIONS

Δx

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M.120 MATH SIGNS & OPERATIONS

dx

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f prime of x

The derivative of f (Lagrange notation); f'': second derivative C1.9 §07 Calculus I.

EXAMPLE

$f(x) = x^2$: $f'(3) = 6$, $f''(3) = 2$

M.111

f inverse

The function that undoes f A2.1 §04 Algebra II. Not $\frac{1}{f}$ (that is $f(x)^{-1}$).

EXAMPLE

$f(x) = 2x$: $f^{-1}(6) = 3$

M.110

g composed with f, "g after f"

Apply f first, then g: $(g \circ f)(x) = g(f(x))$ A2.1 §04 Algebra II.

EXAMPLE

$f(x) = x + 1$, $g(x) = 2x$: $(g \circ f)(3) = 8$

M.109

Indicator function

Equals 1 when the condition or membership holds, 0 otherwise; $\mathbb{E}[\mathbf{1}_A] = P(A)$.

EXAMPLE

$\mathbf{1}_{x>3}(5) = 1; \quad \mathbf{1}_{x>3}(2) = 0$

M.114

Argmax (argmin)

The *input* at which the maximum (minimum) is reached, not the value itself. Training a model is an argmin ML.1 §15 Machine Learning.

EXAMPLE

$\arg\max_{x \in \{1,2,3\}} (4 - (x-2)^2) = 2$

M.113

Maximum (and min)

Largest (smallest) value of a list or of a function over a set.

EXAMPLE

$\max(3,7,5) = 7; \quad \min_x (x-2)^2 = 0$

M.112

Limit

The value $f(x)$ approaches as x gets close to a (from both sides; a^+, a^- for one side) C1.1 §07 Calculus I.

EXAMPLE

$\lim_{x \to 2}(x+3) = 5; \quad \lim_{x \to 0^+} \frac{1}{x} = \infty$

M.117

Convolution

Slide one function over the flipped other and sum the products; used in signals and CNNs EE.26 §16 Electrical Eng..

EXAMPLE

$(1,2) * (1,1) = (1,3,2)$

M.116

Big-O, "order g"

$f = O(g)$: f grows no faster than a constant times g O.10 §14 Optimization. Ω: at least; Θ: exactly.

EXAMPLE

$3n^2 + 5n = O(n^2)$

M.115

Differential, "d x"

An infinitesimal change in x; in an integral it names the variable of integration.

EXAMPLE

$dy = f'(x)\,dx$: $y = x^2$, $x = 3$, $dx = 0.1 \Rightarrow dy = 0.6$

M.120

Delta x, "change in x"

A finite change: final minus initial A1.6 §03 Algebra I.

EXAMPLE

x from 2 to 5: $\Delta x = 3$

M.119

x tends to a

x approaches a (in limits); in logic $\to$ is "implies", between sets it is "maps".

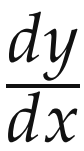
EXAMPLE

$\frac{1}{x} \to 0$ as $x \to \infty$

M.118

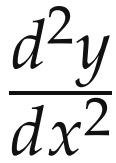
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M.121 MATH SIGNS & OPERATIONS



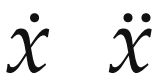
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M.122 MATH SIGNS & OPERATIONS



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M.123 MATH SIGNS & OPERATIONS



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M.124 MATH SIGNS & OPERATIONS



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M.125 MATH SIGNS & OPERATIONS



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M.126 MATH SIGNS & OPERATIONS



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M.127 MATH SIGNS & OPERATIONS



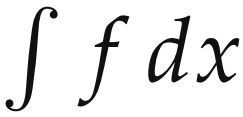
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M.128 MATH SIGNS & OPERATIONS



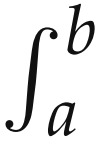
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M.129 MATH SIGNS & OPERATIONS



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M.130 MATH SIGNS & OPERATIONS



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M.131 MATH SIGNS & OPERATIONS



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M.132 MATH SIGNS & OPERATIONS



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x dot, x double-dot

Newton's notation for time derivatives (velocity, acceleration) P.2 §12 Physics.

EXAMPLE

$x = t^2$: $\dot{x} = 2t$, $\ddot{x} = 2$

M.123

Second derivative

Derivative of the derivative: concavity, acceleration C1.23 §07 Calculus I.

EXAMPLE

$y = x^3$: $\frac{d^2y}{dx^2} = 6x$; at $x = 1$: 6

M.122

d y d x (Leibniz)

Derivative of y with respect to x: instantaneous rate of change. $\frac{d}{dx}$ is the operator C1.9 §07 Calculus I.

EXAMPLE

$\frac{d}{dx}x^3 = 3x^2$; at $x = 2$: 12

M.121

Divergence

Net outflow per unit volume of a vector field V.5 §11 Vector Calculus.

EXAMPLE

$\mathbf{F} = (x, y)$: $\nabla \cdot \mathbf{F} = 1 + 1 = 2$

M.126

Gradient, "del f"

Vector of all partial derivatives; points uphill fastest C3.12 §10 Calculus III.

EXAMPLE

$f = x^2 + y^2$: $\nabla f = (2x, 2y)$; at $(1, 2)$: $(2, 4)$

M.125

Partial, "del" or "dee"

Partial derivative: differentiate in one variable, hold the others fixed C3.9 §10 Calculus III.

EXAMPLE

$f = x^2y$: $\frac{\partial f}{\partial x} = 2xy$; at $(1, 3)$: 6

M.124

Integral (indefinite)

Antiderivative: a function whose derivative is f; add $+C$ C1.29 §07 Calculus I.

EXAMPLE

$\int 2x\,dx = x^2 + C$

M.129

Laplacian (or Hessian)

Physics: $\nabla^2 f = \sum \partial^2 f / \partial x_i^2$, also Δf V.13 §11 Vector Calculus. Optimization texts use $\nabla^2 f$ for the Hessian matrix C3.16 §10 Calculus III.

EXAMPLE

$f = x^2 + y^2$: Laplacian 4; Hessian $\left[\begin{smallmatrix} 2 & 0 \\ 0 & 2 \end{smallmatrix}\right]$

M.128

Curl

Local rotation (circulation density) of a vector field V.5 §11 Vector Calculus.

EXAMPLE

$\mathbf{F} = (-y, x, 0)$: $\nabla \times \mathbf{F} = (0, 0, 2)$

M.127

Double integral

Integrate over a 2-D region (triple: $\iiint$) C3.19 §10 Calculus III.

EXAMPLE

$\int_0^2 \int_0^3 1\,dy\,dx = 6$ (area of a 2×3 rectangle)

M.132

Evaluated from a to b

$F(x)\big|_a^b = F(b) - F(a)$, the last step of the Fundamental Theorem C1.32 §07 Calculus I.

EXAMPLE

$x^2\big|_1^3 = 9 - 1 = 8$

M.131

Definite integral

Signed area under f between a and b C1.30 §07 Calculus I.

EXAMPLE

$\int_0^3 2x\,dx = 9$

M.130

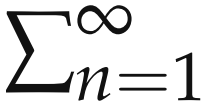
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M.133 MATH SIGNS & OPERATIONS



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M.136 MATH SIGNS & OPERATIONS



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M.137 MATH SIGNS & OPERATIONS



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M.138 MATH SIGNS & OPERATIONS



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M.139 MATH SIGNS & OPERATIONS



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M.140 MATH SIGNS & OPERATIONS



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M.141 MATH SIGNS & OPERATIONS



NAME IT / GIVE AN EXAMPLE

M.142 MATH SIGNS & OPERATIONS



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M.143 MATH SIGNS & OPERATIONS



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M.144 MATH SIGNS & OPERATIONS



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Vector v

A quantity with magnitude and direction; arrow (handwriting) or bold (print) LA.2 §09 Linear Algebra.

EXAMPLE

$\mathbf{v} = (3, 4)$

M.135

Infinite series

The limit of the partial sums, if it exists C2.16 §08 Calculus II.

EXAMPLE

$\sum_{n=1}^{\infty} \frac{1}{2^n} = \frac{1}{2} + \frac{1}{4} + \cdots = 1$

M.134

Contour (closed-loop) integral

Integral around a closed curve: circulation or flux V.7 §11 Vector Calculus.

EXAMPLE

$\oint_C dx = 0$ for any closed curve C

M.133

p-norm

$\left(\sum |v_i|^p\right)^{1/p}$; $p = 1$: taxicab, $p = 2$: Euclidean, $p = \infty$: largest entry ML.11 §15 Machine Learning.

EXAMPLE

$\mathbf{v} = (3, -4)$: $\|\mathbf{v}\|_1 = 7$, $\|\mathbf{v}\|_\infty = 4$

M.138

Norm, "length of v"

Euclidean length $\sqrt{\sum v_i^2}$ LA.29 §09 Linear Algebra.

EXAMPLE

$\|(3, 4)\| = 5$

M.137

u hat (unit vector)

A vector of length 1: $\hat{\mathbf{u}} = \mathbf{u}/\|\mathbf{u}\|$; $\hat{\imath}, \hat{\jmath}, \hat{k}$ are the axis directions.

EXAMPLE

$(3, 4) \to (0.6, 0.8)$

M.136

Inner product

General dot product; also $\langle 1, 2 \rangle$ for a vector in some texts.

EXAMPLE

$\langle (1, 2), (3, 4) \rangle = 11$

M.141

Cross product

A vector perpendicular to both (3-D only), length = parallelogram area.

EXAMPLE

$(1, 0, 0) \times (0, 1, 0) = (0, 0, 1)$

M.140

Dot product

$\sum u_i v_i$, a scalar; $= \|\mathbf{u}\| \|\mathbf{v}\| \cos\theta$ T.21 §06 Trigonometry.

EXAMPLE

$(1, 2) \cdot (3, 4) = 3 + 8 = 11$

M.139

A transpose

Rows become columns: $(A^\mathsf{T})_{ij} = a_{ji}$ LA.11 §09 Linear Algebra. Also A' or $A^\top$.

EXAMPLE

$\left[\begin{smallmatrix} 1 & 2 \\ 3 & 4 \end{smallmatrix}\right]^\mathsf{T} = \left[\begin{smallmatrix} 1 & 3 \\ 2 & 4 \end{smallmatrix}\right]$

M.144

Hadamard product

Multiply matching entries; same shape in and out LA.12 §09 Linear Algebra.

EXAMPLE

$(1, 2) \odot (3, 4) = (3, 8)$

M.143

Outer (tensor) product

Also the Kronecker product of matrices. $\mathbf{u} \otimes \mathbf{v} = \mathbf{u}\mathbf{v}^\mathsf{T}$: every product $u_i v_j$ LA.12 §09 Linear Algebra.

EXAMPLE

$(1, 2) \otimes (3, 4) = \left[\begin{smallmatrix} 3 & 4 \\ 6 & 8 \end{smallmatrix}\right]$

M.142

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M.145 MATH SIGNS & OPERATIONS

A^{-1}

NAME IT / GIVE AN EXAMPLE

M.146 MATH SIGNS & OPERATIONS

A^{+}

NAME IT / GIVE AN EXAMPLE

M.147 MATH SIGNS & OPERATIONS

$\det A$

NAME IT / GIVE AN EXAMPLE

M.148 MATH SIGNS & OPERATIONS

$\operatorname{tr} A$

NAME IT / GIVE AN EXAMPLE

M.149 MATH SIGNS & OPERATIONS

$I \quad \mathbf{0}$

NAME IT / GIVE AN EXAMPLE

M.150 MATH SIGNS & OPERATIONS

$\operatorname{rank} A$

NAME IT / GIVE AN EXAMPLE

M.151 MATH SIGNS & OPERATIONS

δ_{ij}

NAME IT / GIVE AN EXAMPLE

M.152 MATH SIGNS & OPERATIONS

$A \succ 0$

NAME IT / GIVE AN EXAMPLE

M.153 MATH SIGNS & OPERATIONS

$P(A)$

NAME IT / GIVE AN EXAMPLE

M.154 MATH SIGNS & OPERATIONS

$P(A \mid B)$

NAME IT / GIVE AN EXAMPLE

M.155 MATH SIGNS & OPERATIONS

$P(A, B)$

NAME IT / GIVE AN EXAMPLE

M.156 MATH SIGNS & OPERATIONS

$\mathbb{E}[X]$

NAME IT / GIVE AN EXAMPLE

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Determinant

Scalar giving the volume scaling of A; also $|A|$. Zero means singular A2.25 §04 Algebra II.

EXAMPLE

$$\det \left[\begin{smallmatrix} 1 & 2 \\ 3 & 4 \end{smallmatrix}\right] = 4 - 6 = -2$$

M.147

Pseudoinverse

Moore–Penrose inverse; gives least-squares solutions when A^{-1} does not exist LA.31 §09 Linear Algebra.

EXAMPLE

$$\left[\begin{smallmatrix} 2 & 0 \\ 0 & 0 \end{smallmatrix}\right]^{+} = \left[\begin{smallmatrix} 1/2 & 0 \\ 0 & 0 \end{smallmatrix}\right]$$

M.146

A inverse

The matrix with $AA^{-1} = I$; exists iff $\det A \neq 0$ A2.25 §04 Algebra II.

EXAMPLE

$$\left[\begin{smallmatrix} 2 & 0 \\ 0 & 4 \end{smallmatrix}\right]^{-1} = \left[\begin{smallmatrix} 1/2 & 0 \\ 0 & 1/4 \end{smallmatrix}\right]$$

M.145

Rank

Number of linearly independent rows (or columns) LA.19 §09 Linear Algebra.

EXAMPLE

$$\operatorname{rank} \left[\begin{smallmatrix} 1 & 2 \\ 2 & 4 \end{smallmatrix}\right] = 1$$

M.150

Identity / zero matrix

I: 1s on the diagonal, $AI = A$; $\mathbf{0}$: all zeros, $A + \mathbf{0} = A$ LA.16 §09 Linear Algebra.

EXAMPLE

$$I_2 = \left[\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right]$$

M.149

Trace

Sum of the diagonal entries; equals the sum of the eigenvalues LA.29 §09 Linear Algebra.

EXAMPLE

$$\operatorname{tr} \left[\begin{smallmatrix} 1 & 2 \\ 3 & 4 \end{smallmatrix}\right] = 5$$

M.148

Probability of A

A number in $[0, 1]$; $P(A^c) = 1 - P(A)$ S.7 §13 Statistics. Also $\Pr(A)$.

EXAMPLE

Fair die: $P(\text{even}) = \frac{3}{6} = \frac{1}{2}$

M.153

Positive definite

$\mathbf{x}^{\mathsf{T}} A\mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0}$ ($\succeq$: semidefinite) LA.27 §09 Linear Algebra.

EXAMPLE

$$\left[\begin{smallmatrix} 2 & 0 \\ 0 & 3 \end{smallmatrix}\right] \succ 0$$

M.152

Kronecker delta

1 if $i = j$, 0 otherwise; the entries of I.

EXAMPLE

$$\delta_{11} = 1; \quad \delta_{12} = 0$$

M.151

Expected value of X

Probability-weighted average of X S.13 §13 Statistics; also $\mathrm{E}(X)$, $\langle X \rangle$, μ.

EXAMPLE

Fair die: $\mathbb{E}[X] = 3.5$

M.156

Joint probability

Probability that both happen, $P(A \cap B)$.

EXAMPLE

Die: $P(\text{even}, > 3) = P(\{4, 6\}) = \frac{1}{3}$

M.155

Probability of A given B

Probability of A once B is known to have happened: $\frac{P(A \cap B)}{P(B)}$ S.9 §13 Statistics.

EXAMPLE

Die: $P(6 \mid \text{even}) = \frac{1}{3}$

M.154

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M.157 MATH SIGNS & OPERATIONS

$\mathrm{Var}(X)$

NAME IT / GIVE AN EXAMPLE

M.158 MATH SIGNS & OPERATIONS

μ

NAME IT / GIVE AN EXAMPLE

M.159 MATH SIGNS & OPERATIONS

$\bar{x}$

NAME IT / GIVE AN EXAMPLE

M.160 MATH SIGNS & OPERATIONS

$\sigma \quad s$

NAME IT / GIVE AN EXAMPLE

M.161 MATH SIGNS & OPERATIONS

$r \quad \rho$

NAME IT / GIVE AN EXAMPLE

M.162 MATH SIGNS & OPERATIONS

$X \sim$

NAME IT / GIVE AN EXAMPLE

M.163 MATH SIGNS & OPERATIONS

$\underset{\sim}{\text{iid}}$

NAME IT / GIVE AN EXAMPLE

M.164 MATH SIGNS & OPERATIONS

$\mathcal{N}(\mu, \sigma^2)$

NAME IT / GIVE AN EXAMPLE

M.165 MATH SIGNS & OPERATIONS

$X \perp\!\!\!\perp Y$

NAME IT / GIVE AN EXAMPLE

M.166 MATH SIGNS & OPERATIONS

$\hat{\theta}$

NAME IT / GIVE AN EXAMPLE

M.167 MATH SIGNS & OPERATIONS

$\mathcal{L}(\theta)$

NAME IT / GIVE AN EXAMPLE

M.168 MATH SIGNS & OPERATIONS

$H_0 \quad H_1$

NAME IT / GIVE AN EXAMPLE

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x bar (sample mean)

Average of the observed data: $\frac{1}{n}\sum x_i$ S.1 §13 Statistics.

EXAMPLE

Data $2, 4, 9$: $\bar{x} = 5$

M.159

Mu (population mean)

The true mean of a population or distribution S.1 §13 Statistics.

EXAMPLE

Fair die: $\mu = 3.5$

M.158

Variance of X

$\mathbb{E}[(X-\mu)^2]$: average squared spread S.2 §13 Statistics. $\mathrm{Cov}(X, Y)$ is the two-variable version S.38 §13 Statistics.

EXAMPLE

Coin $X \in \{0, 1\}$: $\mathrm{Var}(X) = \frac{1}{4}$

M.157

Distributed as

Names the distribution a random variable follows S.20 §13 Statistics.

EXAMPLE

$X \sim \mathrm{Bin}(10, 0.5)$: 10 fair coin flips

M.162

Correlation

Strength of linear relationship, from -1 to 1; r for a sample, ρ for a population S.38 §13 Statistics.

EXAMPLE

Points $(1, 2), (2, 4), (3, 6)$: $r = 1$

M.161

Standard deviation

σ: population SD (σ^2 is the variance); s: sample SD, divides by $n-1$ S.2 §13 Statistics.

EXAMPLE

Data $2, 4, 4, 4, 5, 5, 7, 9$: $\sigma = 2$

M.160

Independent

$P(X, Y) = P(X)P(Y)$: knowing one tells nothing about the other S.9 §13 Statistics.

EXAMPLE

Two dice: $P(6, 6) = \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36}$

M.165

Normal distribution

Bell curve with mean μ and *variance* σ^2 (check: some software takes σ) S.20 §13 Statistics.

EXAMPLE

$\mathcal{N}(5, 4)$: mean 5, SD 2

M.164

i.i.d.

Independent and identically distributed: every draw from the same distribution, no influence between them S.24 §13 Statistics.

EXAMPLE

$X_1, \ldots, X_5 \overset{\text{iid}}{\sim} \mathcal{N}(0, 1)$

M.163

Null / alternative hypothesis

H_0: the no-effect claim being tested; H_1 (or H_a): what you accept if H_0 is rejected S.28 §13 Statistics.

EXAMPLE

$H_0 : \mu = 5$ vs $H_1 : \mu \neq 5$

M.168

Likelihood

Probability of the observed data viewed as a function of the parameter S.26 §13 Statistics.

EXAMPLE

2 heads in 3 flips: $\mathcal{L}(p) = 3p^2(1-p)$

M.167

Theta hat (estimate)

A hat marks an estimate or prediction of the un-hatted quantity: $\hat{y}$, $\hat{\beta}$, $\hat{p}$ S.26 §13 Statistics.

EXAMPLE

3 heads in 10 flips: $\hat{p} = 0.3$

M.166

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M.169 MATH SIGNS & OPERATIONS



NAME IT / GIVE AN EXAMPLE

M.170 MATH SIGNS & OPERATIONS



NAME IT / GIVE AN EXAMPLE

M.171 MATH SIGNS & OPERATIONS

rad

NAME IT / GIVE AN EXAMPLE

M.172 MATH SIGNS & OPERATIONS

$\triangle ABC$

NAME IT / GIVE AN EXAMPLE

M.173 MATH SIGNS & OPERATIONS



NAME IT / GIVE AN EXAMPLE

M.174 MATH SIGNS & OPERATIONS



NAME IT / GIVE AN EXAMPLE

M.175 MATH SIGNS & OPERATIONS

$\overline{AB}$

NAME IT / GIVE AN EXAMPLE

M.176 MATH SIGNS & OPERATIONS

$\overrightarrow{AB}$

NAME IT / GIVE AN EXAMPLE

M.177 MATH SIGNS & OPERATIONS

$\overset{\frown}{AB}$

NAME IT / GIVE AN EXAMPLE

M.178 MATH SIGNS & OPERATIONS

NAME IT / GIVE AN EXAMPLE

M.179 MATH SIGNS & OPERATIONS

NAME IT / GIVE AN EXAMPLE

M.180 MATH SIGNS & OPERATIONS



NAME IT / GIVE AN EXAMPLE

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Radian

Angle whose arc equals the radius; 2π rad $= 360°$ T.2 §06 Trigonometry. Angles with no unit are radians.

EXAMPLE

$\frac{\pi}{2}$ rad $= 90°$; $\quad 1$ rad $\approx 57.3°$

M.171

Degree

$\frac{1}{360}$ of a full turn; $1° = 60'$ (minutes) $= 3600''$ (seconds) T.2 §06 Trigonometry.

EXAMPLE

$180° = \pi$ rad; $\quad 30' = 0.5°$

M.170

Angle

The angle at the middle vertex; $m\angle$ is its measure G.3 §05 Geometry.

EXAMPLE

$\angle ABC = 90°$ in a square corner

M.169

Is perpendicular to

Meet at a right angle; slopes multiply to -1 A1.8 §03 Algebra I. For vectors: orthogonal.

EXAMPLE

$y = 2x \perp y = -\frac{1}{2}x; \quad (1,2) \perp (2,-1)$

M.174

Is parallel to

Same direction, never meet; equal slopes A1.8 §03 Algebra I.

EXAMPLE

$y = 2x + 1 \parallel y = 2x + 5$

M.173

Triangle ABC

The triangle with vertices A, B, C G.5 §05 Geometry.

EXAMPLE

$\triangle ABC$ with sides $3, 4, 5$ is a right triangle

M.172

Arc AB

The arc of a circle from A to B; length $r\theta$ G.17 §05 Geometry.

EXAMPLE

$90°$ arc, $r = 2$: length $\pi \approx 3.14$

M.177

Ray / vector AB

Starts at A through B; as a vector, $B - A$. ($\overleftrightarrow{AB}$: the whole line.)

EXAMPLE

$A = (1,1)$, $B = (4,5)$: $\overrightarrow{AB} = (3,4)$

M.176

Segment AB

The segment from A to B; often its length AB too G.12 §05 Geometry.

EXAMPLE

$A = (0,0)$, $B = (3,4)$: $AB = 5$

M.175

Equal by definition

Same as $:=$; also written $\stackrel{\text{def}}{=}$.

EXAMPLE

$\sigma \triangleq \sqrt{\text{Var}(X)}$

M.180

Centered dots

"And so on" between operators. Vertical and diagonal dots continue the columns and diagonal of a matrix.

EXAMPLE

$1 + 2 + \cdots + 10 = 55$

M.179

Ellipsis (low dots)

"And so on" in a list separated by commas.

EXAMPLE

$1, 2, \ldots, 10$

M.178

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M.181 MATH SIGNS & OPERATIONS



NAME IT / GIVE AN EXAMPLE

M.182 MATH SIGNS & OPERATIONS

s.t.

NAME IT / GIVE AN EXAMPLE

M.K1 MATH SIGNS & OPERATIONS

When is $p \to q$ false?

RECALL IT

M.K1 MATH SIGNS & OPERATIONS

Converse, inverse and contrapositive of $p \to q$

RECALL IT

M.K1 MATH SIGNS & OPERATIONS

Negate $\forall x\, P(x)$ and $\exists x\, P(x)$

RECALL IT

M.K1 MATH SIGNS & OPERATIONS

Truth table of XOR and iff

RECALL IT

M.K2 MATH SIGNS & OPERATIONS

Precedence: -3^2 and 2^{3^2}

RECALL IT

M.K2 MATH SIGNS & OPERATIONS

Bound vs free index

RECALL IT

M.K2 MATH SIGNS & OPERATIONS

$\sin^2 x$ vs $\sin^{-1} x$

RECALL IT

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Only when p is true and q is false.
$p \to q \equiv \neg p \vee q$.

M.K1

Such that / subject to

Introduces a condition; in optimization, the constraints
O.1 §14 Optimization.

EXAMPLE

$\min x^2$ s.t. $x \geq 1$: minimum 1 at $x = 1$

M.182

QED (end of proof)

Quod erat demonstrandum, "which was to be shown"; also □.

EXAMPLE

"…so n^2 is even. ■"

M.181

$p \oplus q$: true when $p \neq q$ (TF, FT).
$p \leftrightarrow q$: true when $p = q$ (TT, FF).

M.K1

$\neg\forall x\, P(x) \equiv \exists x\, \neg P(x)$
$\neg\exists x\, P(x) \equiv \forall x\, \neg P(x)$

M.K1

Converse $q \to p$; inverse $\neg p \to \neg q$; contrapositive $\neg q \to \neg p$ ($\equiv p \to q$).
Converse ≡ inverse.

M.K1

$\sin^2 x = (\sin x)^2$; $\sin^{-1} x = \arcsin x$ (inverse function), not $1/\sin x$.

M.K2

Bound: summed over, can be renamed ($\sum_i a_i = \sum_k a_k$).
Free: the equation holds for each value ($y_i = 2x_i$).

M.K2

$-3^2 = -(3^2) = -9$ (exponent before unary minus).
$2^{3^2} = 2^9 = 512$ (exponents group right to left).

M.K2

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GR.1 GREEK ALPHABET

Α α

NAME THE LETTER

GR.2 GREEK ALPHABET

Β β

NAME THE LETTER

GR.3 GREEK ALPHABET

Γ γ

NAME THE LETTER

GR.4 GREEK ALPHABET

Δ δ

NAME THE LETTER

GR.5 GREEK ALPHABET

Ε ϵ ε

NAME THE LETTER

GR.6 GREEK ALPHABET

Ζ ζ

NAME THE LETTER

GR.7 GREEK ALPHABET

Η η

NAME THE LETTER

GR.8 GREEK ALPHABET

Θ θ ϑ

NAME THE LETTER

GR.9 GREEK ALPHABET

Ι ι

NAME THE LETTER

GR.10 GREEK ALPHABET

Κ κ ϰ

NAME THE LETTER

GR.11 GREEK ALPHABET

Λ λ

NAME THE LETTER

GR.12 GREEK ALPHABET

Μ μ

NAME THE LETTER

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Gamma

GAM-uh

Math: $\Gamma(n) = (n-1)!$, the gamma function D.9 §18 Graph Gallery; Euler's constant $\gamma \approx 0.577$. **ML:** RL discount factor O.21 §14 Optimization; RBF kernel width ML.15 §15 Machine Learning. **Physics:** Lorentz factor, heat-capacity ratio, photon. **Finance:** option gamma FN.28 §17 Finance.

GR.3

Beta

BAY-tuh (UK: BEE-tuh)

Stats: regression coefficients β_0, β_1 S.40 §13 Statistics; Type II error rate, power $= 1 - \beta$ S.28 §13 Statistics; Beta distribution D.10 §18 Graph Gallery. **ML:** Adam momentum β_1, β_2 ML.27 §15 Machine Learning. **Finance:** market beta FN.14 §17 Finance. **EE:** transistor current gain.

GR.2

Alpha

AL-fuh

Stats: significance level / Type I error rate S.28 §13 Statistics. **ML:** learning rate; elastic-net mix ML.8 §15 Machine Learning; Beta/Dirichlet shape D.10 §18 Graph Gallery. **Physics:** angle, angular acceleration, thermal diffusivity P.31 §12 Physics. **Finance:** excess return (Jensen's alpha) FN.14 §17 Finance.

GR.1

Zeta

ZAY-tuh (UK: ZEE-tuh)

Math: Riemann zeta function $\zeta(s) = \sum n^{-s}$, with $\zeta(2) = \frac{\pi^2}{6}$ C2.16 §08 Calculus II. **Engineering:** damping ratio of a second-order system: $\zeta < 1$ underdamped, $\zeta = 1$ critical, $\zeta > 1$ overdamped EE.16 §16 Electrical Eng., P.15 §12 Physics.

GR.6

Epsilon

EP-sih-lon

Both forms (ϵ, ε) are the same letter. **Math:** an arbitrarily small positive number C1.1 §07 Calculus I. **Stats:** error term $\varepsilon \sim \mathcal{N}(0, \sigma^2)$ S.40 §13 Statistics. **ML:** stability constant (`1e-8`) ML.27 §15 Machine Learning; ε-greedy. **Physics/EE:** permittivity ε_0 EE.18 §16 Electrical Eng.; emissivity P.28 §12 Physics; strain.

GR.5

Delta

DEL-tuh

Δ: a finite change Δx A1.6 §03 Algebra I; discriminant A1.15 §03 Algebra I; Laplacian $\Delta = \nabla^2$. δ: a small change; ε–δ limits C1.1 §07 Calculus I; Kronecker δ_{ij} and Dirac $\delta(x)$; backprop error signal ML.22 §15 Machine Learning. **Finance:** option delta $\partial V / \partial S$ FN.28 §17 Finance.

GR.4

Iota

eye-OH-tuh

Rarely used because it looks like i (and has no dot). Inclusion map $\iota\colon A \hookrightarrow B$; index generator in APL. "Not one iota": not the smallest amount.

GR.9

Theta

THAY-tuh, THEE-tuh

θ: an angle T.1 §06 Trigonometry, polar angle T.19 §06 Trigonometry; generic model parameters, estimate $\hat{\theta}$ S.26 §13 Statistics. Θ: parameter space; tight bound $\Theta(n)$ O.10 §14 Optimization. **Finance:** option theta (time decay). ϑ: variant, used for a second angle.

GR.8

Eta

AY-tuh (UK: EE-tuh)

ML: learning rate (step size) of gradient descent C1.26 §07 Calculus I, ML.27 §15 Machine Learning. **Engineering:** efficiency $\eta = P_{\text{out}}/P_{\text{in}}$ P.6 §12 Physics. **Physics:** dynamic viscosity. **Stats:** linear predictor $\eta = X\beta$ in GLMs; effect size η^2.

GR.7

Mu

MYOO

Stats: population mean $\mu = \mathbb{E}[X]$ S.1 §13 Statistics. **Units:** micro-, 10^{-6} (μm, μF) P.1 §12 Physics. **Physics:** friction coefficient P.4 §12 Physics; dynamic viscosity P.21 §12 Physics; permeability μ_0 EE.18 §16 Electrical Eng.. **Finance:** drift of a price process FN.27 §17 Finance.

GR.12

Lambda

LAM-duh

λ: eigenvalue LA.23 §09 Linear Algebra (Λ = diagonal eigenvalue matrix); Poisson / exponential rate S.18 §13 Statistics; regularization strength ML.8 §15 Machine Learning; Lagrange multiplier O.17 §14 Optimization; wavelength $\lambda = v/f$ P.15 §12 Physics. Programming: anonymous function.

GR.11

Kappa

KAP-uh

Linear algebra: condition number $\kappa(A)$ LA.29 §09 Linear Algebra. **Geometry:** curvature. **Stats/ML:** Cohen's κ, agreement beyond chance ML.17 §15 Machine Learning. **Physics:** thermal conductivity (k or κ) P.24 §12 Physics; dielectric constant. $\varkappa$: variant form.

GR.10

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GR.13 GREEK ALPHABET

N ν

NAME THE LETTER

GR.14 GREEK ALPHABET

Ξ ξ

NAME THE LETTER

GR.15 GREEK ALPHABET

O ο

NAME THE LETTER

GR.16 GREEK ALPHABET

Π π ϖ

NAME THE LETTER

GR.17 GREEK ALPHABET

P ρ ϱ

NAME THE LETTER

GR.18 GREEK ALPHABET

Σ σ ς

NAME THE LETTER

GR.19 GREEK ALPHABET

T τ

NAME THE LETTER

GR.20 GREEK ALPHABET

Υ υ

NAME THE LETTER

GR.21 GREEK ALPHABET

Φ ϕ φ

NAME THE LETTER

GR.22 GREEK ALPHABET

Χ χ

NAME THE LETTER

GR.23 GREEK ALPHABET

Ψ ψ

NAME THE LETTER

GR.24 GREEK ALPHABET

Ω ω

NAME THE LETTER

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Omicron

OM-ih-kron

Identical to Latin *o*, so almost never used as a symbol. The asymptotic symbols big-O and little-o O.10 §14 Optimization are Latin letters ("order of").

GR.15

Xi

KSY, ZY or KSEE

ML: SVM slack variables ξ_i ML.15 §15 Machine Learning. **Math:** an unspecified intermediate point (mean value theorem, Taylor remainder) C1.27 §07 Calculus I; a generic random variable. **Physics:** correlation length; damping ratio in some texts.

GR.14

Nu

NOO (or NYOO)

Looks like a Latin *v*. **Stats:** degrees of freedom S.21 §13 Statistics of the t and χ^2 distributions S.30 §13 Statistics. **Physics:** frequency ($\omega = 2\pi\nu$) P.15 §12 Physics; kinematic viscosity $\nu = \mu/\rho$ P.21 §12 Physics; Poisson's ratio.

GR.13

Sigma

SIG-muh

Σ: summation M.53 §01 Math Signs; covariance matrix. σ: standard deviation, σ^2 variance S.2 §13 Statistics; sigmoid $\sigma(z)$ ML.9 §15 Machine Learning; singular values LA.28 §09 Linear Algebra; Stefan–Boltzmann constant P.28 §12 Physics; volatility (finance). ς: word-final form, not used in math.

GR.18

Rho

ROH

Stats: population correlation S.38 §13 Statistics. **Physics:** density (mass per volume) P.19 §12 Physics; resistivity $R = \rho L/A$ P.33 §12 Physics. **Math:** spherical radius C3.21 §10 Calculus III; spectral radius $\rho(A)$. **Finance:** option rho (rate sensitivity). ϱ: variant form.

GR.17

Pi

PIE

$\pi \approx 3.14159$, circumference over diameter G.17 §05 Geometry. Π: product operator M.54 §01 Math Signs. **ML/Stats:** mixing weights π_k ML.37 §15 Machine Learning; RL policy $\pi(a \mid s)$; prior $\pi(\theta)$. **Number theory:** prime-counting $\pi(x)$. ϖ: rare variant.

GR.16

Phi

FY (or FEE)

Φ: standard normal CDF S.20 §13 Statistics; magnetic or electric flux EE.18 §16 Electrical Eng.. ϕ: standard normal pdf; golden ratio $\varphi \approx 1.618$; phase angle EE.14 §16 Electrical Eng.; azimuth C3.21 §10 Calculus III; feature map $\phi(\mathbf{x})$ ML.15 §15 Machine Learning; Euler's totient. ϕ, φ: same letter. Not $\varnothing$.

GR.21

Upsilon

OOP-sih-lon

Rare because υ looks like u and v; occasionally a velocity. Particle physics: the Υ meson.

GR.20

Tau

TOW (rhymes with cow)

EE: time constant $\tau = RC$ EE.11 §16 Electrical Eng.. **Physics:** torque $\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}$; shear stress P.21 §12 Physics. **Stats:** Kendall's rank correlation τ. **ML:** softmax temperature ML.24 §15 Machine Learning; a threshold. **Math:** $\tau = 2\pi$ (one full turn).

GR.19

Omega

oh-MAY-guh

Ω: the ohm, unit of resistance EE.1 §16 Electrical Eng.; sample space S.7 §13 Statistics; asymptotic lower bound $\Omega(n)$ O.10 §14 Optimization; solid angle. ω: angular frequency $\omega = 2\pi f$ EE.13 §16 Electrical Eng.; angular velocity T.2 §06 Trigonometry; an outcome $\omega \in \Omega$.

GR.24

Psi

SY (or PSY)

Physics: quantum wave function ψ ($|\psi|^2$ = probability density); stream function in fluid flow. **Math:** digamma function $\psi = \Gamma'/\Gamma$ D.9 §18 Graph Gallery. **Engineering:** yaw angle; magnetic flux linkage Ψ.

GR.23

Chi

KY (rhymes with sky)

Stats: χ^2 distribution and tests of fit and independence S.32 §13 Statistics, D.11 §18 Graph Gallery. **Math:** indicator (characteristic) function χ_A; Euler characteristic $\chi = V - E + F$. **Physics:** magnetic or electric susceptibility. Not the Latin x.

GR.22

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GR.K1

GREEK ALPHABET

Which letters are “learning rate”?

RECALL IT

GR.K1

GREEK ALPHABET

σ vs Σ in statistics

RECALL IT

GR.K1

GREEK ALPHABET

The option “Greeks”

RECALL IT

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Delta $\Delta = \partial V/\partial S$; gamma $\Gamma = \partial^2 V/\partial S^2$; theta $\Theta = \partial V/\partial t$; rho $\rho = \partial V/\partial r$; vega $= \partial V/\partial \sigma$ (not a Greek letter).

GR.K1

σ: standard deviation (σ^2 variance). Σ: covariance matrix, or summation.

GR.K1

η (most common) or α. Momentum: β; discount: γ; regularization: λ.

GR.K1

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A1.1 (1/3) ALGEBRA I

Properties of Real Numbers & Order of Operations

PART 1 OF 3

1. Commutative
2. Associative
3. Distributive

STATE EACH NUMBERED FORMULA

A1.1 (2/3) ALGEBRA I

Properties of Real Numbers & Order of Operations

PART 2 OF 3

1. Identities
2. Inverses
3. Zero property

STATE EACH NUMBERED FORMULA

A1.1 (3/3) ALGEBRA I

Properties of Real Numbers & Order of Operations

PART 3 OF 3

1. Order of operations

STATE EACH NUMBERED FORMULA

A1.K1 ALGEBRA I

Canceling rule: what may cancel?

RECALL IT

A1.K1 ALGEBRA I

Cancel these:

$$\frac{x^2-9}{x^2+x-6}, \quad \frac{2-x}{x-2}$$

RECALL IT

A1.K1 ALGEBRA I

Common algebra traps

RECALL IT

A1.2 (1/2) ALGEBRA I

Fractions & Rational Arithmetic

PART 1 OF 2

1. Add / subtract
2. Multiply
3. Divide

STATE EACH NUMBERED FORMULA

A1.2 (2/2) ALGEBRA I

Fractions & Rational Arithmetic

PART 2 OF 2

1. Simplify
2. Mixed number

STATE EACH NUMBERED FORMULA

A1.3 (1/3) ALGEBRA I

Exponent Rules

PART 1 OF 3

1. Product of powers
2. Quotient of powers
3. Power of a power

STATE EACH NUMBERED FORMULA

A1.3 (2/3) ALGEBRA I

Exponent Rules

PART 2 OF 3

1. Power of a product
2. Power of a quotient

STATE EACH NUMBERED FORMULA

A1.3 (3/3) ALGEBRA I

Exponent Rules

PART 3 OF 3

1. Zero exponent
2. Negative exponent

STATE EACH NUMBERED FORMULA

A1.4 (1/2) ALGEBRA I

Radicals & Rational Exponents

PART 1 OF 2

1. Rational exponent
2. Product rule
3. Quotient rule

STATE EACH NUMBERED FORMULA

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1. $\underbrace{(\,)}_{\text{Parentheses}} \to \underbrace{x^n}_{\text{Exponents}} \to \underbrace{\times \div}_{\text{left to right}} \to \underbrace{+ -}_{\text{left to right}}$

TERMS a, b, c real numbers; identity 0 (for $+$), 1 (for $\times$)

PEMDAS: "Please Excuse My Dear Aunt Sally" (Parentheses, Exponents, Multiply/Divide, Add/Subtract).

1. $a + 0 = a, \quad a \cdot 1 = a$
2. $a + (-a) = 0, \quad a \cdot \frac{1}{a} = 1 \ (a \neq 0)$
3. $a \cdot 0 = 0; \quad ab = 0 \Rightarrow a = 0 \text{ or } b = 0$

TERMS a, b, c real numbers; identity 0 (for $+$), 1 (for $\times$)

PEMDAS: "Please Excuse My Dear Aunt Sally" (Parentheses, Exponents, Multiply/Divide, Add/Subtract).

1. $a + b = b + a, \quad ab = ba$
2. $(a + b) + c = a + (b + c), \quad (ab)c = a(bc)$
3. $a(b + c) = ab + ac$

TERMS a, b, c real numbers; identity 0 (for $+$), 1 (for $\times$)

PEMDAS: "Please Excuse My Dear Aunt Sally" (Parentheses, Exponents, Multiply/Divide, Add/Subtract).

$(a+b)^2 \neq a^2 + b^2$; $\sqrt{a^2 + b^2} \neq a + b$

$\frac{1}{a+b} \neq \frac{1}{a} + \frac{1}{b}$; $\ln(a+b) \neq \ln a + \ln b$

$-3^2 = -9$; never divide by a variable that may be 0

$\frac{x-3}{x-2} \ (x \neq -3, 2)$

-1 (opposite factors)

Only common *factors* (multiplied), never terms:

$\frac{(x+3)(x-2)}{(x+3)(x+1)} = \frac{x-2}{x+1}, x \neq -3$

$\frac{x+3}{3} \neq x$

1. $a^m \cdot a^n = a^{m+n}$
2. $\frac{a^m}{a^n} = a^{m-n}$
3. $(a^m)^n = a^{mn}$

TERMS a, b bases ($\neq 0$ for exponents ≤ 0); m, n exponents

Multiply → add exponents; power of a power → multiply exponents.

1. $\frac{ka}{kb} = \frac{a}{b} \ (k \neq 0)$
2. $3\frac{1}{2} = \frac{3 \cdot 2 + 1}{2} = \frac{7}{2}$

TERMS $b, d \neq 0$; LCD least common denominator

Adding: "same bottom first". Dividing: "keep, change, flip".

1. $\frac{a}{b} \pm \frac{c}{d} = \frac{ad \pm bc}{bd}$
2. $\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}$
3. $\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \cdot \frac{d}{c}$

TERMS $b, d \neq 0$; LCD least common denominator

Adding: "same bottom first". Dividing: "keep, change, flip".

1. $a^{m/n} = \sqrt[n]{a^m} = \left(\sqrt[n]{a}\right)^m$
2. $\sqrt{ab} = \sqrt{a}\sqrt{b}$
3. $\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$

TERMS n index; a radicand (≥ 0 for even n)

"Power over root": in $a^{m/n}$ the top is the power, the bottom is the root.

1. $a^0 = 1 \ (a \neq 0)$
2. $a^{-n} = \frac{1}{a^n}$

TERMS a, b bases ($\neq 0$ for exponents ≤ 0); m, n exponents

Multiply → add exponents; power of a power → multiply exponents.

1. $(ab)^n = a^n b^n$
2. $\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$

TERMS a, b bases ($\neq 0$ for exponents ≤ 0); m, n exponents

Multiply → add exponents; power of a power → multiply exponents.

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A1.4 (2/2) ALGEBRA I

Radicals & Rational Exponents

PART 2 OF 2

1. Rationalize
2. Conjugate
3. Even root of a square

STATE EACH NUMBERED FORMULA

A1.5 (1/2) ALGEBRA I

Scientific Notation & Significant Figures

PART 1 OF 2

1. Scientific notation
2. Multiply / divide
3. Add / subtract

STATE EACH NUMBERED FORMULA

A1.5 (2/2) ALGEBRA I

Scientific Notation & Significant Figures

PART 2 OF 2

1. Significant figures ($\times, \div$)
2. Decimal places ($+, -$)

STATE EACH NUMBERED FORMULA

A1.6 ALGEBRA I

Slope

1. Slope (two points)
2. Sign of slope

STATE EACH NUMBERED FORMULA

A1.7 ALGEBRA I

Equations of a Line

1. Slope–intercept
2. Point–slope
3. Standard form

STATE EACH NUMBERED FORMULA

A1.8 ALGEBRA I

Parallel & Perpendicular Lines

1. Parallel
2. Perpendicular
3. Distance point to line

STATE EACH NUMBERED FORMULA

A1.9 ALGEBRA I

Functions, Domain & Range

1. Function notation
2. Vertical line test
3. Domain restrictions

STATE EACH NUMBERED FORMULA

A1.10 ALGEBRA I

Solving Linear & Literal Equations

1. Balance principle
2. Linear solution
3. Literal equation

STATE EACH NUMBERED FORMULA

A1.11 (1/2) ALGEBRA I

Absolute Value Equations & Inequalities

PART 1 OF 2

1. Definition
2. Equation
3. "Less than"

STATE EACH NUMBERED FORMULA

A1.11 (2/2) ALGEBRA I

Absolute Value Equations & Inequalities

PART 2 OF 2

1. "Greater than"
2. Triangle inequality

STATE EACH NUMBERED FORMULA

A1.12 ALGEBRA I

Systems of Linear Equations

1. 2×2 system
2. Cramer's rule (2×2)
3. Solution count test

STATE EACH NUMBERED FORMULA

A1.13 (1/2) ALGEBRA I

Polynomial Products & Special Products

PART 1 OF 2

1. Square of a sum
2. Square of a difference
3. Difference of squares

STATE EACH NUMBERED FORMULA

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1. keep the fewest sig. figs. of the inputs
2. keep the fewest decimal places

TERMS c coefficient $1 \le |c| < 10$; k integer exponent

Big number → positive exponent; tiny number → negative exponent.

1. $x = c \times 10^k, \quad 1 \le |c| < 10$
2. $(c_1 \times 10^a)(c_2 \times 10^b) = c_1 c_2 \times 10^{a+b}$
3. $c_1 \times 10^k + c_2 \times 10^k = (c_1 + c_2) \times 10^k$

TERMS c coefficient $1 \le |c| < 10$; k integer exponent

Big number → positive exponent; tiny number → negative exponent.

1. $\dfrac{c}{\sqrt{a}} = \dfrac{c\sqrt{a}}{a}$
2. $\dfrac{1}{\sqrt{a}+\sqrt{b}} = \dfrac{\sqrt{a}-\sqrt{b}}{a-b}$
3. $\sqrt{x^2} = |x|$

TERMS n index; a radicand (≥ 0 for even n)

"Power over root": in $a^{m/n}$ the top is the power, the bottom is the root.

1. $m_1 = m_2 \ (b_1 \neq b_2)$
2. $m_1 \cdot m_2 = -1 \iff m_2 = -\dfrac{1}{m_1}$
3. $d = \dfrac{|Ax_0 + By_0 - C|}{\sqrt{A^2 + B^2}}$

TERMS m_1, m_2 slopes; (x_0, y_0) point; A, B, C standard-form coefficients

Perpendicular slopes: "flip it and switch the sign."

1. $y = mx + b$
2. $y - y_1 = m(x - x_1)$
3. $Ax + By = C$

TERMS m slope; b y-intercept; (x_1, y_1) known point

1. $m = \dfrac{y_2 - y_1}{x_2 - x_1} = \dfrac{\Delta y}{\Delta x}$
2. $m > 0 \ \nearrow, \quad m < 0 \ \searrow, \quad m = 0 \ \rightarrow$

TERMS $(x_1, y_1), (x_2, y_2)$ points on the line; Δ "change in"

"Rise over run" — y changes on top, like rising up.

1. $|x| = \begin{cases} x, & x \ge 0 \\ -x, & x < 0 \end{cases}$
2. $|x - a| = r \iff x = a \pm r \ (r \ge 0)$
3. $|x - a| < r \iff a - r < x < a + r$

TERMS a center; r radius / tolerance

"Less than" is a sandwich (and); "greater than" goes two ways (or).

1. $a = b \Rightarrow a + c = b + c, \quad ac = bc$
2. $ax + b = c \Rightarrow x = \dfrac{c - b}{a}$
3. $A = \frac{1}{2}bh \Rightarrow h = \dfrac{2A}{b}$

TERMS $a \neq 0$; check the solution by substitution

"Undo in reverse order": last thing done to x is the first thing undone.

1. $y = f(x)$
2. each $x \mapsto$ exactly one y
3. $\frac{1}{g(x)} : \ g \neq 0; \quad \sqrt{g(x)} : \ g \ge 0; \quad \ln g(x) : \ g > 0$

TERMS x input; $f(x)$ output; domain = allowed inputs; range = outputs

Domain = "what can go in"; range = "what comes out".

1. $(a + b)^2 = a^2 + 2ab + b^2$
2. $(a - b)^2 = a^2 - 2ab + b^2$
3. $(a + b)(a - b) = a^2 - b^2$

TERMS a, b, c, d terms; degree = largest exponent

FOIL: First, Outer, Inner, Last.

1. $\begin{cases} a_1 x + b_1 y = c_1 \\ a_2 x + b_2 y = c_2 \end{cases}$
2. $x = \dfrac{c_1 b_2 - b_1 c_2}{a_1 b_2 - b_1 a_2}, \quad y = \dfrac{a_1 c_2 - c_1 a_2}{a_1 b_2 - b_1 a_2}$
3. $a_1 b_2 - b_1 a_2 \neq 0 \iff$ unique solution

TERMS a_i, b_i coefficients; c_i constants; determinant $a_1 b_2 - b_1 a_2$

1. $|x - a| > r \iff x < a - r$ or $x > a + r$
2. $|a + b| \le |a| + |b|$

TERMS a center; r radius / tolerance

"Less than" is a sandwich (and); "greater than" goes two ways (or).

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A1.13 (2/2) ALGEBRA I

Polynomial Products & Special Products

PART 2 OF 2

1. FOIL
2. Cube of a sum

STATE EACH NUMBERED FORMULA

A1.14 (1/2) ALGEBRA I

Factoring

PART 1 OF 2

1. Greatest common factor
2. Trinomial ($a = 1$)
3. Difference of squares

STATE EACH NUMBERED FORMULA

A1.14 (2/2) ALGEBRA I

Factoring

PART 2 OF 2

1. Sum of cubes
2. Difference of cubes
3. AC method

STATE EACH NUMBERED FORMULA

A1.K2 ALGEBRA I

Factoring order (decision tree)

RECALL IT

A1.K2 ALGEBRA I

$x^2 + bx + c = (x + p)(x + q)$: what are p and q?

RECALL IT

A1.K2 ALGEBRA I

Quadratic letters: $a, b, c, h, k, r_1, r_2, \Delta$

RECALL IT

A1.15 ALGEBRA I

Quadratic Formula & Discriminant

1. Quadratic formula
2. Discriminant
3. Vieta's formulas

STATE EACH NUMBERED FORMULA

A1.16 ALGEBRA I

Vertex Form & Completing the Square

1. Vertex form
2. Vertex from standard form
3. Completing the square

STATE EACH NUMBERED FORMULA

A1.K3 ALGEBRA I

Completing the square: $x^2 + bx = ?$

RECALL IT

A1.K3 ALGEBRA I

Completing the square when $a \neq 1$

RECALL IT

A1.17 ALGEBRA I

Quadratic & Polynomial Inequalities

1. Sign rule
2. Upward parabola

STATE EACH NUMBERED FORMULA

A1.18 ALGEBRA I

Direct & Inverse Variation

1. Direct
2. Inverse
3. Joint

STATE EACH NUMBERED FORMULA

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1. $a^3 + b^3 = (a+b)(a^2 - ab + b^2)$
2. $a^3 - b^3 = (a-b)(a^2 + ab + b^2)$
3. $ax^2 + bx + c$: find $pq = ac,\ p+q = b$

TERMS GCF greatest common factor; pq = constant, $p+q$ = middle coefficient

Sum/difference of cubes signs: SOAP — Same, Opposite, Always Positive.

1. $ab + ac = a(b+c)$
2. $x^2 + (p+q)x + pq = (x+p)(x+q)$
3. $a^2 - b^2 = (a-b)(a+b)$

TERMS GCF greatest common factor; pq = constant, $p+q$ = middle coefficient

Sum/difference of cubes signs: SOAP — Same, Opposite, Always Positive.

1. $(a+b)(c+d) = ac + ad + bc + bd$
2. $(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$

TERMS a, b, c, d terms; degree = largest exponent

FOIL: First, Outer, Inner, Last.

$h = -\dfrac{b}{2a}$, $k = c - \dfrac{b^2}{4a}$, c = y-intercept

$r = h \pm \sqrt{-k/a}$; $r_1 + r_2 = -\dfrac{b}{a}$,

$r_1 r_2 = \dfrac{c}{a}$

$\Delta = b^2 - 4ac = -4ak$

$pq = c,\ p+q = b$; roots are $-p, -q$

$p+q = -2h,\ pq = h^2 + k$

$x^2 - x - 12 = (x+3)(x-4)$

1. GCF (and -1) 2. count terms: 2: squares/cubes; 3: square, $a = 1$, AC; 4: grouping
3. re-factor, check by expanding

$x^2 + bx = \left(x + \frac{b}{2}\right)^2 - \left(\frac{b}{2}\right)^2$

$x^2 + 6x + 5 = (x+3)^2 - 4$, vertex $(-3, -4)$

1. $y = a(x-h)^2 + k$
2. $h = -\dfrac{b}{2a}, \quad k = f(h)$
3. $x^2 + bx = \left(x + \frac{b}{2}\right)^2 - \frac{b^2}{4}$

TERMS (h, k) vertex; a leading coefficient

1. $ax^2 + bx + c = 0 \Rightarrow x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
2. $\Delta = b^2 - 4ac$
3. $x_1 + x_2 = -\dfrac{b}{a}, \quad x_1 x_2 = \dfrac{c}{a}$

TERMS $a \neq 0$, b, c coefficients; Δ discriminant

Sung to "Pop Goes the Weasel": "x equals negative b, plus or minus the square root, of b squared minus four a c, all over two a."

1. $y = kx$
2. $y = \dfrac{k}{x}$
3. $y = kxz$

TERMS $k \neq 0$ constant of variation

1. $p(x) = (x - r_1)(x - r_2)\cdots$
2. $a > 0$: $ax^2 + bx + c < 0 \iff r_1 < x < r_2$

TERMS r_i roots; test one point per interval

Factor a from the x-terms, complete inside, multiply the correction by a:

$2x^2 + 8x + 3 = 2(x+2)^2 - 5$

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A1.19

ALGEBRA I

Ratios, Proportions & Percent Change

1. Percent change
2. Proportion
3. Percent of

STATE EACH NUMBERED FORMULA

A1.20

ALGEBRA I

Linear Inequalities

1. Add / subtract
2. Multiply by positive
3. Multiply by negative

STATE EACH NUMBERED FORMULA

A1.21 (1/2)

ALGEBRA I

Sets, Set Operations & Interval Notation

PART 1 OF 2

1. Union / intersection
2. Complement / difference
3. Inclusion–exclusion

STATE EACH NUMBERED FORMULA

A1.21 (2/2)

ALGEBRA I

Sets, Set Operations & Interval Notation

PART 2 OF 2

1. De Morgan's laws
2. Interval notation

STATE EACH NUMBERED FORMULA

A1.K4

ALGEBRA I

Solving equations: which tool for which type?

RECALL IT

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1. $A \cup B$ (or), $A \cap B$ (and)
2. A^c (not), $A \setminus B$
3. $|A \cup B| = |A| + |B| - |A \cap B|$

TERMS A, B sets; U universal set; $|A|$ size of A

Square bracket = "included" (closed); parenthesis = "not included" (open). De Morgan: "break the bar, change the sign."

1. $a < b \implies a + c < b + c$
2. $a < b,\ c > 0 \implies ac < bc$
3. $a < b,\ c < 0 \implies ac > bc$

TERMS $c > 0$ keeps direction; $c < 0$ flips it

"Multiply or divide by a negative, flip the sign."

1. $\dfrac{\text{new} - \text{old}}{\text{old}} \times 100\%$
2. $\dfrac{a}{b} = \dfrac{c}{d} \iff ad = bc$
3. $p\%$ of $x = \dfrac{p}{100}\, x$

TERMS old $\neq 0$ reference value; $b, d \neq 0$

Percent change: "difference over original."

Linear: isolate x. $|\cdot|$: split $\pm$.
Quadratic: factor or formula.
Rational: $\times$ LCD, exclude zeros.
Radical: power both sides, check.
Exponential: take logs. Log: exponentiate.

1. $(A \cup B)^c = A^c \cap B^c,\ \ (A \cap B)^c = A^c \cup B^c$
2. $[a, b) = \{x : a \leq x < b\}$

TERMS A, B sets; U universal set; $|A|$ size of A

Square bracket = "included" (closed); parenthesis = "not included" (open). De Morgan: "break the bar, change the sign."

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A2.1 (1/2) ALGEBRA II

Functions: Composition & Inverses

PART 1 OF 2

1. Function arithmetic
2. Composition
3. Inverse identity

STATE EACH NUMBERED FORMULA

A2.1 (2/2) ALGEBRA II

Functions: Composition & Inverses

PART 2 OF 2

1. Horizontal line test
2. Inverse graph reflection
3. Inverse of a composition

STATE EACH NUMBERED FORMULA

A2.2 (1/3) ALGEBRA II

Parent Functions & Function Families

PART 1 OF 3

1. Linear (identity)
2. Quadratic
3. Cubic

STATE EACH NUMBERED FORMULA

A2.2 (2/3) ALGEBRA II

Parent Functions & Function Families

PART 2 OF 3

1. Absolute value
2. Square root
3. Reciprocal

STATE EACH NUMBERED FORMULA

A2.2 (3/3) ALGEBRA II

Parent Functions & Function Families

PART 3 OF 3

1. Exponential
2. Logarithmic

STATE EACH NUMBERED FORMULA

A2.3 (1/2) ALGEBRA II

Even & Odd Functions

PART 1 OF 2

1. Even function
2. Odd function
3. Polynomial rule

STATE EACH NUMBERED FORMULA

A2.3 (2/2) ALGEBRA II

Even & Odd Functions

PART 2 OF 2

1. Products
2. Even–odd decomposition
3. Symmetric integrals

STATE EACH NUMBERED FORMULA

A2.4 (1/3) ALGEBRA II

Piecewise & Step Functions

PART 1 OF 3

1. Piecewise definition
2. Floor
3. Ceiling

STATE EACH NUMBERED FORMULA

A2.4 (2/3) ALGEBRA II

Piecewise & Step Functions

PART 2 OF 3

1. Sign function
2. Heaviside step
3. ReLU

STATE EACH NUMBERED FORMULA

A2.4 (3/3) ALGEBRA II

Piecewise & Step Functions

PART 3 OF 3

1. Continuity at a break

STATE EACH NUMBERED FORMULA

A2.5 (1/2) ALGEBRA II

Transformations of Functions

PART 1 OF 2

1. General form
2. Vertical shift
3. Horizontal shift

STATE EACH NUMBERED FORMULA

A2.5 (2/2) ALGEBRA II

Transformations of Functions

PART 2 OF 2

1. Vertical stretch / reflection
2. Horizontal scale / reflection
3. Point mapping

STATE EACH NUMBERED FORMULA

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1 $y = x$
2 $y = x^2$
3 $y = x^3$

TERMS key points: $(0,0), (1,1)$; b^x via $(0,1)$; $\log_b x$ via $(1,0)$

One key point each: x, x^2, x^3, $|x|$, $\sqrt{x}$ pass through $(0,0)$ and $(1,1)$; $\frac{1}{x}$ through $(1,1)$; b^x through $(0,1)$; $\log_b x$ through $(1,0)$.

A2.2 (1/3)

1 $f(a) = f(b) \Rightarrow a = b$
2 (a,b) on $f \iff (b,a)$ on f^{-1}
3 $(f \circ g)^{-1} = g^{-1} \circ f^{-1}$

TERMS $f \circ g$: g first; f^{-1} inverse (not $1/f$); one-to-one

Composition works inside-out: apply g first, then f. Inverses: "socks and shoes" — undo the last step first.

A2.1 (2/2)

1 $(f \pm g)(x) = f(x) \pm g(x), \quad (fg)(x) = f(x)\,g(x)$
2 $(f \circ g)(x) = f\big(g(x)\big)$
3 $f^{-1}\big(f(x)\big) = f\big(f^{-1}(x)\big) = x$

TERMS $f \circ g$: g first; f^{-1} inverse (not $1/f$); one-to-one

Composition works inside-out: apply g first, then f. Inverses: "socks and shoes" — undo the last step first.

A2.1 (1/2)

1 $f(-x) = f(x)$
2 $f(-x) = -f(x)$
3 even powers only $\Rightarrow$ even; odd powers only $\Rightarrow$ odd

TERMS E even, O odd; test by computing $f(-x)$

Even $\leftrightarrow$ even powers $\leftrightarrow$ mirror in the y-axis. Odd $\leftrightarrow$ odd powers $\leftrightarrow$ spin 180° about the origin.

A2.3 (1/2)

1 $y = b^x \quad (b > 1)$
2 $y = \log_b x \quad (b > 1)$

TERMS key points: $(0,0), (1,1)$; b^x via $(0,1)$; $\log_b x$ via $(1,0)$

One key point each: x, x^2, x^3, $|x|$, $\sqrt{x}$ pass through $(0,0)$ and $(1,1)$; $\frac{1}{x}$ through $(1,1)$; b^x through $(0,1)$; $\log_b x$ through $(1,0)$.

A2.2 (3/3)

1 $y = |x|$
2 $y = \sqrt{x}$
3 $y = \dfrac{1}{x}$

TERMS key points: $(0,0), (1,1)$; b^x via $(0,1)$; $\log_b x$ via $(1,0)$

One key point each: x, x^2, x^3, $|x|$, $\sqrt{x}$ pass through $(0,0)$ and $(1,1)$; $\frac{1}{x}$ through $(1,1)$; b^x through $(0,1)$; $\log_b x$ through $(1,0)$.

A2.2 (2/3)

1 $\operatorname{sgn} x = \begin{cases} -1, & x < 0 \\ 0, & x = 0 \\ 1, & x > 0 \end{cases}$
2 $H(x) = \begin{cases} 0, & x < 0 \\ 1, & x \geq 0 \end{cases}$
3 $\operatorname{ReLU}(x) = \max(0, x)$

TERMS c breakpoint; • included, ○ excluded endpoint

Floor is the floor of the room (down); ceiling is up. Closed dot • = endpoint included; open dot ○ = excluded.

A2.4 (2/3)

1 $f(x) = \begin{cases} f_1(x), & x < c \\ f_2(x), & x \geq c \end{cases}$
2 $\lfloor x \rfloor$ = greatest integer $\leq x$
3 $\lceil x \rceil$ = least integer $\geq x$

TERMS c breakpoint; • included, ○ excluded endpoint

Floor is the floor of the room (down); ceiling is up. Closed dot • = endpoint included; open dot ○ = excluded.

A2.4 (1/3)

1 $\mathrm{E} \cdot \mathrm{E} = \mathrm{O} \cdot \mathrm{O} = \mathrm{E}, \quad \mathrm{E} \cdot \mathrm{O} = \mathrm{O}$
2 $f(x) = \underbrace{\frac{f(x)+f(-x)}{2}}_{\text{even}} + \underbrace{\frac{f(x)-f(-x)}{2}}_{\text{odd}}$
3 $\int_{-a}^{a} f_{\text{odd}}\,dx = 0, \quad \int_{-a}^{a} f_{\text{even}}\,dx = 2\int_{0}^{a} f_{\text{even}}\,dx$

TERMS E even, O odd; test by computing $f(-x)$

Even $\leftrightarrow$ even powers $\leftrightarrow$ mirror in the y-axis. Odd $\leftrightarrow$ odd powers $\leftrightarrow$ spin 180° about the origin.

A2.3 (2/2)

1 $y = a\,f(x)$
2 $y = f(bx)$
3 $(x, y) \mapsto \left(\dfrac{x}{b} + h,\ a\,y + k\right)$

TERMS h right, k up; $|a|$ vertical stretch; $1/|b|$ horizontal scale

Inside the parentheses acts on x and does the opposite (minus h shifts right); outside acts on y and does what it says.

A2.5 (2/2)

1 $y = a\,f\big(b\,(x - h)\big) + k$
2 $y = f(x) + k$
3 $y = f(x - h)$

TERMS h right, k up; $|a|$ vertical stretch; $1/|b|$ horizontal scale

Inside the parentheses acts on x and does the opposite (minus h shifts right); outside acts on y and does what it says.

A2.5 (1/2)

1 $\lim_{x \to c^-} f(x) = \lim_{x \to c^+} f(x) = f(c)$

TERMS c breakpoint; • included, ○ excluded endpoint

Floor is the floor of the room (down); ceiling is up. Closed dot • = endpoint included; open dot ○ = excluded.

A2.4 (3/3)

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A2.K1

ALGEBRA II

Transformations: $y = a\,f\big(b(x-h)\big)+k$ what does each constant do?

RECALL IT

A2.K1

ALGEBRA II

Transformations: minus sign inside vs. outside?

RECALL IT

A2.K1

ALGEBRA II

Transformations: what shift does $f(2x-6)$ give?

RECALL IT

A2.6 (1/3)

ALGEBRA II

Complex Numbers

PART 1 OF 3

1. Imaginary unit
2. Standard form
3. Add / subtract

STATE EACH NUMBERED FORMULA

A2.6 (2/3)

ALGEBRA II

Complex Numbers

PART 2 OF 3

1. Multiply
2. Conjugate & modulus
3. Divide

STATE EACH NUMBERED FORMULA

A2.6 (3/3)

ALGEBRA II

Complex Numbers

PART 3 OF 3

1. Polar form
2. Conjugate root theorem

STATE EACH NUMBERED FORMULA

A2.7

ALGEBRA II

Polynomial Long & Synthetic Division

1. Division algorithm
2. As a fraction
3. Synthetic division

STATE EACH NUMBERED FORMULA

A2.8 (1/2)

ALGEBRA II

Remainder, Factor & Rational Root Theorems

PART 1 OF 2

1. Remainder theorem
2. Factor theorem
3. Rational root theorem

STATE EACH NUMBERED FORMULA

A2.8 (2/2)

ALGEBRA II

Remainder, Factor & Rational Root Theorems

PART 2 OF 2

1. Fundamental theorem of algebra
2. Descartes' rule of signs
3. Vieta's formulas

STATE EACH NUMBERED FORMULA

A2.9 (1/2)

ALGEBRA II

Polynomial Graphs: End Behavior & Multiplicity

PART 1 OF 2

1. Standard form
2. End behavior
3. Multiplicity

STATE EACH NUMBERED FORMULA

A2.9 (2/2)

ALGEBRA II

Polynomial Graphs: End Behavior & Multiplicity

PART 2 OF 2

1. Turning points
2. Real zeros
3. y-intercept

STATE EACH NUMBERED FORMULA

A2.10 (1/3)

ALGEBRA II

Rational Functions & Asymptotes

PART 1 OF 3

1. Rational function
2. Vertical asymptote
3. Hole

STATE EACH NUMBERED FORMULA

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Factor first: $f(2(x-3))$
compress by $\frac{1}{2}$, then shift **right 3** (not 6).

A2.K1

$-f(x)$: reflect in the x-axis (upside down)
$f(-x)$: reflect in the y-axis (left–right)
$f^{-1}(x)$: reflect in $y = x$

A2.K1

h: shift right ($f(x+2)$ is left)
k: shift up
a: vertical stretch $|a|$; $a<0$ flips over x-axis
b: horizontal scale $\frac{1}{|b|}$; $b<0$ flips over y-axis
$(x,y) \mapsto \left(\frac{x}{b}+h,\ ay+k\right)$

A2.K1

1. $z = r(\cos\theta + i\sin\theta) = re^{i\theta}$
2. $p(z)=0 \Rightarrow p(\bar{z})=0$

TERMS $a=\operatorname{Re} z$, $b=\operatorname{Im} z$; $\bar{z}$ conjugate; $|z|$ modulus

Powers of i cycle $i,-1,-i,1$: divide the exponent by 4 and use the remainder ($i^7=i^3=-i$).

A2.6 (3/3)

1. $(a+bi)(c+di) = (ac-bd)+(ad+bc)\,i$
2. $\bar{z}=a-bi,\quad |z|=\sqrt{a^2+b^2},\quad z\bar{z}=|z|^2$
3. $\dfrac{z}{w}=\dfrac{z\,\bar{w}}{w\,\bar{w}}$

TERMS $a=\operatorname{Re} z$, $b=\operatorname{Im} z$; $\bar{z}$ conjugate; $|z|$ modulus

Powers of i cycle $i,-1,-i,1$: divide the exponent by 4 and use the remainder ($i^7=i^3=-i$).

A2.6 (2/3)

1. $i^2=-1,\qquad \sqrt{-n}=i\sqrt{n}\ \ (n>0)$
2. $z=a+bi$
3. $(a+bi)\pm(c+di)=(a\pm c)+(b\pm d)\,i$

TERMS $a=\operatorname{Re} z$, $b=\operatorname{Im} z$; $\bar{z}$ conjugate; $|z|$ modulus

Powers of i cycle $i,-1,-i,1$: divide the exponent by 4 and use the remainder ($i^7=i^3=-i$).

A2.6 (1/3)

1. $P(x)=a_n(x-r_1)(x-r_2)\cdots(x-r_n)$
2. # positive roots $= V,\ V-2,\ V-4,\ldots$
3. $\sum r_i = -\dfrac{a_{n-1}}{a_n},\qquad \prod r_i=(-1)^n\dfrac{a_0}{a_n}$

TERMS a_n leading coeff.; a_0 constant; $p \mid a_0$: p divides a_0

Rational roots: "factors of the last *over factors of the* first.*"*

A2.8 (2/2)

1. $P(x)=(x-c)\,Q(x)+P(c)$
2. $P(c)=0 \iff (x-c)$ is a factor
3. $x=\pm\dfrac{p}{q},\ \ p\mid a_0,\ \ q\mid a_n$

TERMS a_n leading coeff.; a_0 constant; $p \mid a_0$: p divides a_0

Rational roots: "factors of the last *over factors of the* first.*"*

A2.8 (1/2)

1. $P(x)=D(x)\,Q(x)+R(x),\quad \deg R<\deg D$
2. $\dfrac{P(x)}{D(x)}=Q(x)+\dfrac{R(x)}{D(x)}$
3. divide by $x-c$: bring down, $\times c$, add

TERMS P dividend; D divisor; Q quotient; R remainder ($\deg R<\deg D$)

Long-division loop: Divide, Multiply, Subtract, Bring down ("Does McDonald's Sell Burgers?").

A2.7

1. $f(x)=\dfrac{P(x)}{Q(x)}$
2. $x=c$ where $Q(c)=0,\ P(c)\neq 0$
3. $(x-c)$ cancels from P and Q

TERMS VA vertical asymptote; HA horizontal; a_n, b_n leading coefficients

Degrees: "Bigger on Bottom, $y=0$; Exactly Even, divide the leads; Bigger on Top by one, slant."

A2.10 (1/3)

1. # turns $\le n-1$
2. # x-intercepts $\le n$
3. $P(0)=a_0$

TERMS n degree; a_n leading coefficient; m multiplicity of root r

The leading term rules the ends; multiplicity rules the middle: odd crosses, even bounces.

A2.9 (2/2)

1. $P(x)=a_nx^n+a_{n-1}x^{n-1}+\cdots+a_0$
2. $P(x)\approx a_nx^n$ as $|x|\to\infty$
3. $(x-r)^m$ a factor of P

TERMS n degree; a_n leading coefficient; m multiplicity of root r

The leading term rules the ends; multiplicity rules the middle: odd crosses, even bounces.

A2.9 (1/2)

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A2.10 (2/3) ALGEBRA II

Rational Functions & Asymptotes

PART 2 OF 3

1. HA, bottom-heavy
2. HA, equal degrees
3. Slant (oblique) asymptote

STATE EACH NUMBERED FORMULA

A2.10 (3/3) ALGEBRA II

Rational Functions & Asymptotes

PART 3 OF 3

1. Intercepts

STATE EACH NUMBERED FORMULA

A2.11 (1/2) ALGEBRA II

Rational Equations & Inequalities

PART 1 OF 2

1. Clear denominators
2. Excluded values
3. Rational inequality

STATE EACH NUMBERED FORMULA

A2.11 (2/2) ALGEBRA II

Rational Equations & Inequalities

PART 2 OF 2

1. Work-rate equation
2. Partial-fraction split

STATE EACH NUMBERED FORMULA

A2.12 (1/2) ALGEBRA II

Radical Equations & Extraneous Roots

PART 1 OF 2

1. Power both sides
2. Why extra roots appear
3. Sign condition

STATE EACH NUMBERED FORMULA

A2.12 (2/2) ALGEBRA II

Radical Equations & Extraneous Roots

PART 2 OF 2

1. Two radicals
2. Rational exponent

STATE EACH NUMBERED FORMULA

A2.13 (1/2) ALGEBRA II

Exponential Growth & Decay

PART 1 OF 2

1. Discrete growth
2. Compound interest
3. Continuous compounding

STATE EACH NUMBERED FORMULA

A2.13 (2/2) ALGEBRA II

Exponential Growth & Decay

PART 2 OF 2

1. Natural growth / decay
2. Half-life / doubling time
3. Half-life form

STATE EACH NUMBERED FORMULA

A2.14 (1/2) ALGEBRA II

Logarithms: Definition & Properties

PART 1 OF 2

1. Definition
2. Product rule
3. Quotient rule

STATE EACH NUMBERED FORMULA

A2.14 (2/2) ALGEBRA II

Logarithms: Definition & Properties

PART 2 OF 2

1. Power rule
2. Special values
3. Inverse identities

STATE EACH NUMBERED FORMULA

A2.K2 ALGEBRA II

Exponents & logs: convert $\log_b x = y$

RECALL IT

A2.K2 ALGEBRA II

Exponents & logs: graphs of b^x and $\log_b x$

RECALL IT

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1. $\dfrac{P}{Q} = \dfrac{R}{S} \;\Rightarrow\; PS = QR \quad (Q, S \neq 0)$
2. $Q(x) = 0 \;\Rightarrow\; x \notin \text{domain}$
3. $\dfrac{P(x)}{Q(x)} > 0$

TERMS LCD least common denominator; excluded values: $Q(x) = 0$

Multiply by the LCD, then check every candidate against the "forbidden list".

1. x-int: $P(x) = 0$; $\quad y$-int: $f(0) = \dfrac{P(0)}{Q(0)}$

TERMS VA vertical asymptote; HA horizontal; a_n, b_n leading coefficients

Degrees: "Bigger on Bottom, $y = 0$; Exactly Even, divide the leads; Bigger on Top by one, slant."

1. $\deg P < \deg Q \;\Rightarrow\; y = 0$
2. $\deg P = \deg Q \;\Rightarrow\; y = \dfrac{a_n}{b_n}$
3. $\deg P = \deg Q + 1 \;\Rightarrow\; y = \text{quotient of } P \div Q$

TERMS VA vertical asymptote; HA horizontal; a_n, b_n leading coefficients

Degrees: "Bigger on Bottom, $y = 0$; Exactly Even, divide the leads; Bigger on Top by one, slant."

1. $\sqrt{u} + \sqrt{w} = c$
2. $x^{m/n} = k \;\Rightarrow\; x = k^{n/m}$

TERMS n index; extraneous root: fails the original equation

Isolate, power, solve, check.

1. $\sqrt[n]{u} = v \;\Rightarrow\; u = v^n$
2. $a = b \Rightarrow a^2 = b^2$, but $a^2 = b^2 \not\Rightarrow a = b$
3. $\sqrt{u} = v$ requires $v \geq 0$

TERMS n index; extraneous root: fails the original equation

Isolate, power, solve, check.

1. $\dfrac{1}{t_1} + \dfrac{1}{t_2} = \dfrac{1}{T}$
2. $\dfrac{1}{(x-a)(x-b)} = \dfrac{1}{a-b}\left(\dfrac{1}{x-a} - \dfrac{1}{x-b}\right)$

TERMS LCD least common denominator; excluded values: $Q(x) = 0$

Multiply by the LCD, then check every candidate against the "forbidden list".

1. $\log_b x = y \iff b^y = x$
2. $\log_b(xy) = \log_b x + \log_b y$
3. $\log_b \dfrac{x}{y} = \log_b x - \log_b y$

TERMS $b > 0$, $b \neq 1$ base; $x > 0$ argument; $\ln = \log_e$

"A log is an exponent": $\log_b x$ *asks "b to what power gives x?"*

1. $y = y_0\, e^{kt}$
2. $t_{1/2} = \dfrac{\ln 2}{|k|}$
3. $y = y_0 \left(\frac{1}{2}\right)^{t/t_{1/2}}$

TERMS P principal; r rate; n periods/yr; k growth constant

"Pert": continuous compounding is $A = Pe^{rt}$. Rule of 72: doubling time $\approx 72/(\text{percent rate})$ FN.3 *§17 Finance.*

1. $y = y_0(1+r)^t$
2. $A = P\left(1 + \dfrac{r}{n}\right)^{nt}$
3. $A = Pe^{rt}$

TERMS P principal; r rate; n periods/yr; k growth constant

"Pert": continuous compounding is $A = Pe^{rt}$. Rule of 72: doubling time $\approx 72/(\text{percent rate})$ FN.3 *§17 Finance.*

b^x: through $(0,1)$, asymptote $y = 0$, range $(0, \infty)$
$\log_b x$: through $(1,0)$, asymptote $x = 0$, domain $(0, \infty)$
mirror images in $y = x$

$$b^y = x$$

A log is an exponent.
$\log_2 8 = 3$ since $2^3 = 8$.

1. $\log_b x^p = p \log_b x$
2. $\log_b 1 = 0, \qquad \log_b b = 1$
3. $b^{\log_b x} = x, \qquad \log_b(b^x) = x$

TERMS $b > 0$, $b \neq 1$ base; $x > 0$ argument; $\ln = \log_e$

"A log is an exponent": $\log_b x$ *asks "b to what power gives x?"*

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A2.K2

ALGEBRA II

Exponents & logs: the three log rules

RECALL IT

A2.15 (1/3)

ALGEBRA II

Graphs of Exponential & Logarithmic Functions

PART 1 OF 3

1. Exponential function
2. Growth vs. decay
3. Logarithmic function

STATE EACH NUMBERED FORMULA

A2.15 (2/3)

ALGEBRA II

Graphs of Exponential & Logarithmic Functions

PART 2 OF 3

1. Base-e form
2. Transformed exponential
3. Transformed logarithm

STATE EACH NUMBERED FORMULA

A2.15 (3/3)

ALGEBRA II

Graphs of Exponential & Logarithmic Functions

PART 3 OF 3

1. Growth ranking

STATE EACH NUMBERED FORMULA

A2.16 (1/2)

ALGEBRA II

Change of Base & Exponential Equations

PART 1 OF 2

1. Change of base
2. Exponential equation
3. Same base

STATE EACH NUMBERED FORMULA

A2.16 (2/2)

ALGEBRA II

Change of Base & Exponential Equations

PART 2 OF 2

1. Doubling time
2. Quadratic type

STATE EACH NUMBERED FORMULA

A2.17 (1/2)

ALGEBRA II

Logarithmic Equations

PART 1 OF 2

1. Exponentiate
2. One-to-one property
3. Domain check

STATE EACH NUMBERED FORMULA

A2.17 (2/2)

ALGEBRA II

Logarithmic Equations

PART 2 OF 2

1. Log inequality

STATE EACH NUMBERED FORMULA

A2.18 (1/2)

ALGEBRA II

Arithmetic Sequences & Series

PART 1 OF 2

1. nth term
2. Recursive form
3. Sum (first and last)

STATE EACH NUMBERED FORMULA

A2.18 (2/2)

ALGEBRA II

Arithmetic Sequences & Series

PART 2 OF 2

1. Sum (no last term)
2. Gauss sum

STATE EACH NUMBERED FORMULA

A2.19 (1/2)

ALGEBRA II

Geometric Sequences & Series

PART 1 OF 2

1. nth term
2. Recursive form
3. Finite sum

STATE EACH NUMBERED FORMULA

A2.19 (2/2)

ALGEBRA II

Geometric Sequences & Series

PART 2 OF 2

1. Infinite sum
2. Repeating decimal

STATE EACH NUMBERED FORMULA

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1. $b^x = e^{x \ln b}$
2. $y = a\,b^{x-h} + k$
3. $y = a \log_b(x-h) + k$

TERMS $b > 0$, $b \neq 1$; asymptotes $y = 0$ (b^x) and $x = 0$ ($\log_b x$)

Exponential: point $(0,1)$ and a floor at $y = 0$. Log: point $(1,0)$ and a wall at $x = 0$.

A2.15 (2/3)

1. $f(x) = b^x$
2. $b > 1$ increasing; $\quad 0 < b < 1$ decreasing
3. $g(x) = \log_b x$

TERMS $b > 0$, $b \neq 1$; asymptotes $y = 0$ (b^x) and $x = 0$ ($\log_b x$)

Exponential: point $(0,1)$ and a floor at $y = 0$. Log: point $(1,0)$ and a wall at $x = 0$.

A2.15 (1/3)

$$\log_b(xy) = \log_b x + \log_b y$$
$$\log_b \tfrac{x}{y} = \log_b x - \log_b y$$
$$\log_b x^p = p \log_b x$$

A2.K2

1. $t = \dfrac{\ln 2}{\ln(1+r)}$
2. $a\,b^{2x} + c\,b^x + d = 0$

TERMS $\ln = \log_e$; isolate b^x before taking logs

To get the exponent down, take a log.

A2.16 (2/2)

1. $\log_b x = \dfrac{\ln x}{\ln b} = \dfrac{\log_{10} x}{\log_{10} b}$
2. $b^x = c \implies x = \dfrac{\ln c}{\ln b}$
3. $b^u = b^v \iff u = v$

TERMS $\ln = \log_e$; isolate b^x before taking logs

To get the exponent down, take a log.

A2.16 (1/2)

1. $\log x \ll x^p \ll b^x \quad (x \to \infty,\ p > 0,\ b > 1)$

TERMS $b > 0$, $b \neq 1$; asymptotes $y = 0$ (b^x) and $x = 0$ ($\log_b x$)

Exponential: point $(0,1)$ and a floor at $y = 0$. Log: point $(1,0)$ and a wall at $x = 0$.

A2.15 (3/3)

1. $a_n = a_1 + (n-1)\,d$
2. $a_n = a_{n-1} + d$
3. $S_n = \dfrac{n\,(a_1 + a_n)}{2}$

TERMS a_1 first term; d common difference; n number of terms

Arithmetic sum: "average of first and last, times the number of terms" (Gauss's trick).

A2.18 (1/2)

1. $\log_b u < \log_b v \iff u < v \quad (b > 1)$

TERMS arguments must be > 0; $\log_b u = c \iff u = b^c$

Condense, exponentiate, check that every argument is positive.

A2.17 (2/2)

1. $\log_b u = c \iff u = b^c$
2. $\log_b u = \log_b v \iff u = v \quad (u, v > 0)$
3. $u > 0$ for every $\log_b u$ in the original

TERMS arguments must be > 0; $\log_b u = c \iff u = b^c$

Condense, exponentiate, check that every argument is positive.

A2.17 (1/2)

1. $S_\infty = \displaystyle\sum_{n=1}^{\infty} a_1 r^{n-1} = \frac{a_1}{1-r}, \quad |r| < 1$
2. $0.\overline{27} = \dfrac{0.27}{1-0.01} = \dfrac{27}{99} = \dfrac{3}{11}$

TERMS a_1 first term; r common ratio; converges iff $|r| < 1$

Infinite geometric sum: "first term over one minus the ratio."

A2.19 (2/2)

1. $a_n = a_1 r^{n-1}$
2. $a_n = r\,a_{n-1}$
3. $S_n = \dfrac{a_1\,(1-r^n)}{1-r} \quad (r \neq 1)$

TERMS a_1 first term; r common ratio; converges iff $|r| < 1$

Infinite geometric sum: "first term over one minus the ratio."

A2.19 (1/2)

1. $S_n = \dfrac{n}{2}\left[2a_1 + (n-1)\,d\right]$
2. $1 + 2 + \cdots + n = \dfrac{n(n+1)}{2}$

TERMS a_1 first term; d common difference; n number of terms

Arithmetic sum: "average of first and last, times the number of terms" (Gauss's trick).

A2.18 (2/2)

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A2.20 (1/2) ALGEBRA II

Arithmetic, Geometric & Harmonic Means

PART 1 OF 2

1. Arithmetic mean
2. Geometric mean
3. Harmonic mean

STATE EACH NUMBERED FORMULA

A2.20 (2/2) ALGEBRA II

Arithmetic, Geometric & Harmonic Means

PART 2 OF 2

1. AM–GM–HM inequality
2. Two numbers
3. Weighted mean

STATE EACH NUMBERED FORMULA

A2.21 (1/2) ALGEBRA II

Sigma Notation & Summation Formulas

PART 1 OF 2

1. Constant sum
2. Sum of integers
3. Sum of squares

STATE EACH NUMBERED FORMULA

A2.21 (2/2) ALGEBRA II

Sigma Notation & Summation Formulas

PART 2 OF 2

1. Sum of cubes
2. Linearity
3. Telescoping sum

STATE EACH NUMBERED FORMULA

A2.22 (1/2) ALGEBRA II

Mathematical Induction

PART 1 OF 2

1. Base case
2. Inductive step
3. Conclusion

STATE EACH NUMBERED FORMULA

A2.22 (2/2) ALGEBRA II

Mathematical Induction

PART 2 OF 2

1. Strong induction

STATE EACH NUMBERED FORMULA

A2.23 (1/2) ALGEBRA II

Binomial Theorem

PART 1 OF 2

1. Binomial theorem
2. Binomial coefficient
3. General term

STATE EACH NUMBERED FORMULA

A2.23 (2/2) ALGEBRA II

Binomial Theorem

PART 2 OF 2

1. Pascal's rule
2. Symmetry & row sum
3. Linear approximation

STATE EACH NUMBERED FORMULA

A2.24 (1/3) ALGEBRA II

Matrix Operations

PART 1 OF 3

1. Matrix product
2. Size rule
3. Add & scale

STATE EACH NUMBERED FORMULA

A2.24 (2/3) ALGEBRA II

Matrix Operations

PART 2 OF 3

1. 2×2 product

STATE EACH NUMBERED FORMULA

A2.24 (3/3) ALGEBRA II

Matrix Operations

PART 3 OF 3

1. Transpose
2. Identity & order

STATE EACH NUMBERED FORMULA

A2.25 (1/3) ALGEBRA II

Determinants, Inverses & Cramer's Rule

PART 1 OF 3

1. 2×2 determinant
2. 2×2 inverse
3. Cramer's rule

STATE EACH NUMBERED FORMULA

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1. $\displaystyle\sum_{i=1}^{n} c = cn$
2. $\displaystyle\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$
3. $\displaystyle\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$

TERMS i index; n upper limit; c constant

Sum of cubes = (sum of integers)^2: $1+8+27=36=6^2$.

A2.21 (1/2)

1. AM $\geq$ GM $\geq$ HM
2. $\dfrac{a+b}{2} \;\geq\; \sqrt{ab} \;\geq\; \dfrac{2ab}{a+b}$
3. $\bar{x}_w = \dfrac{\sum w_i x_i}{\sum w_i}$

TERMS $x_i > 0$ values; w_i weights; equality iff all x_i equal

Add → arithmetic; multiply → geometric; "per" rates → harmonic.

A2.20 (2/2)

1. $\text{AM} = \dfrac{x_1 + x_2 + \cdots + x_n}{n}$
2. $\text{GM} = \sqrt[n]{x_1 x_2 \cdots x_n}$
3. $\text{HM} = \dfrac{n}{\frac{1}{x_1} + \frac{1}{x_2} + \cdots + \frac{1}{x_n}}$

TERMS $x_i > 0$ values; w_i weights; equality iff all x_i equal

Add → arithmetic; multiply → geometric; "per" rates → harmonic.

A2.20 (1/2)

1. $P(n_0), \ldots, P(k) \Rightarrow P(k+1)$

TERMS $P(n)$ statement; n_0 base case; k arbitrary case

Dominoes: the first one falls, and each falling domino knocks over the next.

A2.22 (2/2)

1. $P(n_0)$ is true
2. $P(k) \Rightarrow P(k+1)$ for all $k \geq n_0$
3. $P(n)$ is true for all $n \geq n_0$

TERMS $P(n)$ statement; n_0 base case; k arbitrary case

Dominoes: the first one falls, and each falling domino knocks over the next.

A2.22 (1/2)

1. $\displaystyle\sum_{i=1}^{n} i^3 = \left[\frac{n(n+1)}{2}\right]^2$
2. $\displaystyle\sum_{i=1}^{n}(c\,a_i + b_i) = c\sum_{i=1}^{n} a_i + \sum_{i=1}^{n} b_i$
3. $\displaystyle\sum_{i=1}^{n}(a_{i+1} - a_i) = a_{n+1} - a_1$

TERMS i index; n upper limit; c constant

Sum of cubes = (sum of integers)^2: $1+8+27=36=6^2$.

A2.21 (2/2)

1. $(AB)_{ij} = \displaystyle\sum_{k=1}^{n} a_{ik}\, b_{kj}$
2. $(m \times n)(n \times p) = (m \times p)$
3. $(A+B)_{ij} = a_{ij} + b_{ij}, \quad (cA)_{ij} = c\,a_{ij}$

TERMS $m \times n$: m rows, n cols; a_{ij} row i, col j; I identity

"Row times column": slide across the row of A and down the column of B.

A2.24 (1/3)

1. $\displaystyle\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$
2. $\displaystyle\binom{n}{k} = \binom{n}{n-k}, \qquad \sum_{k=0}^{n}\binom{n}{k} = 2^n$
3. $(1+x)^n \approx 1 + nx \quad (|x| \ll 1)$

TERMS $\binom{n}{k}$ "n choose k"; $n!$ factorial, $0! = 1$

Coefficients come from Pascal's triangle: 1; 1 1; 1 2 1; 1 3 3 1; …

A2.23 (2/2)

1. $\displaystyle(a+b)^n = \sum_{k=0}^{n}\binom{n}{k} a^{n-k}\, b^k$
2. $\displaystyle\binom{n}{k} = \frac{n!}{k!\,(n-k)!}$
3. $\displaystyle T_{k+1} = \binom{n}{k} a^{n-k} b^k$

TERMS $\binom{n}{k}$ "n choose k"; $n!$ factorial, $0! = 1$

Coefficients come from Pascal's triangle: 1; 1 1; 1 2 1; 1 3 3 1; …

A2.23 (1/2)

1. $\det\begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$
2. $A^{-1} = \dfrac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$
3. $x = \dfrac{\det A_x}{\det A}, \qquad y = \dfrac{\det A_y}{\det A}$

TERMS $\det A = 0$: singular (no inverse); A_x: x-column replaced by $\mathbf{b}$

Two-by-two inverse: "swap the diagonal, negate the off-diagonal, divide by the determinant."

A2.25 (1/3)

1. $(A^{\mathsf{T}})_{ij} = a_{ji}, \qquad (AB)^{\mathsf{T}} = B^{\mathsf{T}} A^{\mathsf{T}}$
2. $AI = IA = A, \qquad AB \neq BA$ in general

TERMS $m \times n$: m rows, n cols; a_{ij} row i, col j; I identity

"Row times column": slide across the row of A and down the column of B.

A2.24 (3/3)

1. $\begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} e & f \\ g & h \end{bmatrix} = \begin{bmatrix} ae+bg & af+bh \\ ce+dg & cf+dh \end{bmatrix}$

TERMS $m \times n$: m rows, n cols; a_{ij} row i, col j; I identity

"Row times column": slide across the row of A and down the column of B.

A2.24 (2/3)

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A2.25 (2/3) ALGEBRA II

Determinants, Inverses & Cramer's Rule

PART 2 OF 3

1. 3×3 cofactor expansion

STATE EACH NUMBERED FORMULA

A2.25 (3/3) ALGEBRA II

Determinants, Inverses & Cramer's Rule

PART 3 OF 3

1. Product rule
2. Matrix solution

STATE EACH NUMBERED FORMULA

A2.26 (1/3) ALGEBRA II

Conic Sections

PART 1 OF 3

1. Circle
2. Parabola
3. Ellipse

STATE EACH NUMBERED FORMULA

A2.26 (2/3) ALGEBRA II

Conic Sections

PART 2 OF 3

1. Hyperbola
2. Hyperbola asymptotes
3. Eccentricity

STATE EACH NUMBERED FORMULA

A2.26 (3/3) ALGEBRA II

Conic Sections

PART 3 OF 3

1. General-form test

STATE EACH NUMBERED FORMULA

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1 $(x-h)^2 + (y-k)^2 = r^2$

2 $(x-h)^2 = 4p\,(y-k)$

3 $\dfrac{(x-h)^2}{a^2} + \dfrac{(y-k)^2}{b^2} = 1, \quad c^2 = a^2 - b^2$

TERMS (h,k) center; a, b semi-axes; c focal distance; p focus distance

Ellipse subtracts ($c^2 = a^2 - b^2$), hyperbola adds ($c^2 = a^2 + b^2$).

1 $\det(AB) = \det A \, \det B, \qquad \det(A^{-1}) = \dfrac{1}{\det A}$

2 $A\mathbf{x} = \mathbf{b} \;\Rightarrow\; \mathbf{x} = A^{-1}\mathbf{b}$

TERMS $\det A = 0$: singular (no inverse); A_x: x-column replaced by $\mathbf{b}$

Two-by-two inverse: "swap the diagonal, negate the off-diagonal, divide by the determinant."

1 $\det \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = a(ei - fh) - b(di - fg) + c(dh - eg)$

TERMS $\det A = 0$: singular (no inverse); A_x: x-column replaced by $\mathbf{b}$

Two-by-two inverse: "swap the diagonal, negate the off-diagonal, divide by the determinant."

1 $Ax^2 + Cy^2 + Dx + Ey + F = 0$

TERMS (h,k) center; a, b semi-axes; c focal distance; p focus distance

Ellipse subtracts ($c^2 = a^2 - b^2$), hyperbola adds ($c^2 = a^2 + b^2$).

1 $\dfrac{(x-h)^2}{a^2} - \dfrac{(y-k)^2}{b^2} = 1, \quad c^2 = a^2 + b^2$

2 $y - k = \pm\dfrac{b}{a}\,(x-h)$

3 $e = \dfrac{c}{a}$

TERMS (h,k) center; a, b semi-axes; c focal distance; p focus distance

Ellipse subtracts ($c^2 = a^2 - b^2$), hyperbola adds ($c^2 = a^2 + b^2$).

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G.1 (1/3) GEOMETRY

Points, Lines, Rays, Segments & Planes

PART 1 OF 3

1. Segment
2. Ray
3. Line

STATE EACH NUMBERED FORMULA

G.1 (2/3) GEOMETRY

Points, Lines, Rays, Segments & Planes

PART 2 OF 3

1. Segment addition postulate
2. Collinearity (coordinates)
3. Plane postulate

STATE EACH NUMBERED FORMULA

G.1 (3/3) GEOMETRY

Points, Lines, Rays, Segments & Planes

PART 3 OF 3

1. Intersections

STATE EACH NUMBERED FORMULA

G.2 (1/3) GEOMETRY

Angles: Measure & Types

PART 1 OF 3

1. Acute angle
2. Right angle
3. Obtuse angle

STATE EACH NUMBERED FORMULA

G.2 (2/3) GEOMETRY

Angles: Measure & Types

PART 2 OF 3

1. Straight angle
2. Reflex angle
3. Full turn

STATE EACH NUMBERED FORMULA

G.2 (3/3) GEOMETRY

Angles: Measure & Types

PART 3 OF 3

1. Angle addition postulate
2. Angle bisector

STATE EACH NUMBERED FORMULA

G.3 (1/3) GEOMETRY

Angle Relationships

PART 1 OF 3

1. Complementary
2. Supplementary
3. Linear pair

STATE EACH NUMBERED FORMULA

G.3 (2/3) GEOMETRY

Angle Relationships

PART 2 OF 3

1. Vertical angles
2. Corresponding angles
3. Alternate interior angles

STATE EACH NUMBERED FORMULA

G.3 (3/3) GEOMETRY

Angle Relationships

PART 3 OF 3

1. Alternate exterior angles
2. Same-side interior angles

STATE EACH NUMBERED FORMULA

G.4 (1/2) GEOMETRY

Classifying Triangles

PART 1 OF 2

1. Equilateral
2. Isosceles (base angles theorem)
3. Scalene

STATE EACH NUMBERED FORMULA

G.4 (2/2) GEOMETRY

Classifying Triangles

PART 2 OF 2

1. By largest angle
2. Side–angle order
3. Equilateral area

STATE EACH NUMBERED FORMULA

G.5 GEOMETRY

Triangle Angle Sum & Triangle Inequality

1. Angle sum
2. Exterior angle theorem
3. Triangle inequality

STATE EACH NUMBERED FORMULA

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1. $\ell_1 \cap \ell_2 =$ a point, $\varnothing$, or the whole line

TERMS bar: segment; one arrow: ray; two arrows: line; collinear = on one line

Bar = stops at both ends (segment); one arrow = one way (ray); two arrows = both ways (line).

G.1 (3/3)

1. $AB + BC = AC$
2. $m_{AB} = m_{BC}$
3. A, B, C noncollinear $\Rightarrow$ exactly one plane

TERMS bar: segment; one arrow: ray; two arrows: line; collinear = on one line

Bar = stops at both ends (segment); one arrow = one way (ray); two arrows = both ways (line).

G.1 (2/3)

1. $\overline{AB}$ (length AB)
2. $\overrightarrow{AB}$
3. $\overleftrightarrow{AB}$

TERMS bar: segment; one arrow: ray; two arrows: line; collinear = on one line

Bar = stops at both ends (segment); one arrow = one way (ray); two arrows = both ways (line).

G.1 (1/3)

1. $m\angle ABC = m\angle ABD + m\angle DBC$
2. $m\angle ABD = m\angle DBC = \frac{1}{2}\, m\angle ABC$

TERMS θ angle measure; $\angle ABC$ vertex B; $180^\circ = \pi$ rad

A-cute *angles are small; obtuse angles are wide; a straight angle is a straight line.*

G.2 (3/3)

1. $\theta = 180^\circ = \pi$
2. $180^\circ < \theta < 360^\circ$
3. $360^\circ = 2\pi$ rad

TERMS θ angle measure; $\angle ABC$ vertex B; $180^\circ = \pi$ rad

A-cute *angles are small; obtuse angles are wide; a straight angle is a straight line.*

G.2 (2/3)

1. $0^\circ < \theta < 90^\circ$
2. $\theta = 90^\circ = \frac{\pi}{2}$
3. $90^\circ < \theta < 180^\circ$

TERMS θ angle measure; $\angle ABC$ vertex B; $180^\circ = \pi$ rad

A-cute *angles are small; obtuse angles are wide; a straight angle is a straight line.*

G.2 (1/3)

1. $\angle 1 = \angle 7$
2. $\angle 4 + \angle 5 = 180^\circ$

TERMS $\ell_1 \parallel \ell_2$, transversal t; 1–4 top crossing, 5–8 bottom, CCW from upper right

Complementary makes a Corner (90°); Supplementary makes a Straight line (180°).

G.3 (3/3)

1. $\angle 1 = \angle 3$
2. $\angle 1 = \angle 5$
3. $\angle 3 = \angle 5$

TERMS $\ell_1 \parallel \ell_2$, transversal t; 1–4 top crossing, 5–8 bottom, CCW from upper right

Complementary makes a Corner (90°); Supplementary makes a Straight line (180°).

G.3 (2/3)

1. $\alpha + \beta = 90^\circ$
2. $\alpha + \beta = 180^\circ$
3. $\angle 1 + \angle 2 = 180^\circ$

TERMS $\ell_1 \parallel \ell_2$, transversal t; 1–4 top crossing, 5–8 bottom, CCW from upper right

Complementary makes a Corner (90°); Supplementary makes a Straight line (180°).

G.3 (1/3)

1. $A + B + C = 180^\circ$
2. $\angle_{\text{ext}} = \angle_{\text{remote}_1} + \angle_{\text{remote}_2}$
3. $|a - b| < c < a + b$

TERMS A, B, C angles opposite sides a, b, c; exterior angle on an extended side

G.5

1. $\max(A, B, C) < 90^\circ,\ = 90^\circ,\ > 90^\circ$
2. $a > b \iff A > B$
3. Area $= \dfrac{\sqrt{3}}{4}\, s^2$

TERMS a, b, c sides opposite angles A, B, C; s side length

Isosceles *is Greek for "equal legs"*; scalene *for "uneven".*

G.4 (2/2)

1. $a = b = c, \quad A = B = C = 60^\circ$
2. $a = b \iff A = B$
3. $a,\ b,\ c$ all different

TERMS a, b, c sides opposite angles A, B, C; s side length

Isosceles *is Greek for "equal legs"*; scalene *for "uneven".*

G.4 (1/2)

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G.6 (1/2) GEOMETRY

Triangle Congruence

PART 1 OF 2

1. SSS (side–side–side)
2. SAS (side–angle–side)
3. ASA (angle–side–angle)

STATE EACH NUMBERED FORMULA

G.6 (2/2) GEOMETRY

Triangle Congruence

PART 2 OF 2

1. AAS (angle–angle–side)
2. HL (hypotenuse–leg)
3. CPCTC

STATE EACH NUMBERED FORMULA

G.7 (1/2) GEOMETRY

Pythagorean Theorem

PART 1 OF 2

1. Pythagorean theorem
2. Missing side
3. Converse

STATE EACH NUMBERED FORMULA

G.7 (2/2) GEOMETRY

Pythagorean Theorem

PART 2 OF 2

1. Acute / obtuse test
2. Euclid's formula for triples

STATE EACH NUMBERED FORMULA

G.8 (1/2) GEOMETRY

Special Right Triangles

PART 1 OF 2

1. 45–45–90 triangle
2. 30–60–90 triangle

STATE EACH NUMBERED FORMULA

G.8 (2/2) GEOMETRY

Special Right Triangles

PART 2 OF 2

1. Square diagonal
2. Equilateral height

STATE EACH NUMBERED FORMULA

G.9 (1/2) GEOMETRY

Similar Triangles & Scale Factor

PART 1 OF 2

1. Similarity ratio
2. Perimeter ratio

STATE EACH NUMBERED FORMULA

G.9 (2/2) GEOMETRY

Similar Triangles & Scale Factor

PART 2 OF 2

1. Area ratio
2. Volume ratio

STATE EACH NUMBERED FORMULA

G.10 (1/2) GEOMETRY

Similarity Criteria & Proportional Segments

PART 1 OF 2

1. AA similarity
2. SSS similarity
3. SAS similarity

STATE EACH NUMBERED FORMULA

G.10 (2/2) GEOMETRY

Similarity Criteria & Proportional Segments

PART 2 OF 2

1. Side-splitter theorem
2. Angle bisector theorem
3. Geometric mean (altitude)

STATE EACH NUMBERED FORMULA

G.11 (1/2) GEOMETRY

Triangle Centers

PART 1 OF 2

1. Centroid
2. Circumcenter
3. Incenter

STATE EACH NUMBERED FORMULA

G.11 (2/2) GEOMETRY

Triangle Centers

PART 2 OF 2

1. Orthocenter
2. Euler line
3. Circumradius

STATE EACH NUMBERED FORMULA

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1. $a^2 + b^2 = c^2$
2. $c = \sqrt{a^2 + b^2}, \quad a = \sqrt{c^2 - b^2}$
3. $c^2 = a^2 + b^2 \Rightarrow$ right angle opposite c

TERMS a, b legs; c hypotenuse (longest side); $m > n > 0$ integers

Common triples: 3–4–5, 5–12–13, 8–15–17, 7–24–25.

1. $\angle A = \angle D,\ \angle B = \angle E,\ BC = EF$
2. $\angle C = \angle F = 90^\circ,\ AB = DE,\ BC = EF$
3. $\triangle ABC \cong \triangle DEF \Rightarrow AB = DE,\ \angle A = \angle D,\ \ldots$

TERMS $\cong$ congruent; vertex order gives correspondence; included = between

The valid tests: SSS, SAS, ASA, AAS, HL. SSA ("the donkey rule") and AAA are not *tests.*

1. $AB = DE,\ BC = EF,\ CA = FD$
2. $AB = DE,\ \angle B = \angle E,\ BC = EF$
3. $\angle A = \angle D,\ AB = DE,\ \angle B = \angle E$

TERMS $\cong$ congruent; vertex order gives correspondence; included = between

The valid tests: SSS, SAS, ASA, AAS, HL. SSA ("the donkey rule") and AAA are not *tests.*

1. $d = s\sqrt{2}$
2. $h = \frac{\sqrt{3}}{2}\, s$

TERMS x shortest side; s side, d diagonal, h height

30–60–90: "1, root 3, 2" — the short leg is half the hypotenuse.

1. $x : x : x\sqrt{2}$
2. $x : x\sqrt{3} : 2x$

TERMS x shortest side; s side, d diagonal, h height

30–60–90: "1, root 3, 2" — the short leg is half the hypotenuse.

1. $c^2 < a^2 + b^2 \Rightarrow$ acute, $\quad c^2 > a^2 + b^2 \Rightarrow$ obtuse
2. $(m^2 - n^2,\ 2mn,\ m^2 + n^2)$

TERMS a, b legs; c hypotenuse (longest side); $m > n > 0$ integers

Common triples: 3–4–5, 5–12–13, 8–15–17, 7–24–25.

1. $\angle A = \angle D,\ \angle B = \angle E \Rightarrow \triangle ABC \sim \triangle DEF$
2. $\dfrac{AB}{DE} = \dfrac{BC}{EF} = \dfrac{CA}{FD}$
3. $\dfrac{AB}{DE} = \dfrac{BC}{EF},\ \ \angle B = \angle E$

TERMS h altitude to hypotenuse; p, q hypotenuse pieces

1. $\dfrac{\text{Area}_1}{\text{Area}_2} = k^2$
2. $\dfrac{\text{Vol}_1}{\text{Vol}_2} = k^3$

TERMS $\sim$ similar; k scale factor; P perimeter

Length k, area k^2, volume k^3.

1. $\triangle ABC \sim \triangle DEF \implies \dfrac{AB}{DE} = \dfrac{BC}{EF} = \dfrac{AC}{DF} = k$
2. $\dfrac{P_1}{P_2} = k$

TERMS $\sim$ similar; k scale factor; P perimeter

Length k, area k^2, volume k^3.

1. $AH \perp BC,\ \ BH \perp CA,\ \ CH \perp AB$
2. $G = \frac{1}{3}H + \frac{2}{3}O$
3. $R = \dfrac{abc}{4 \cdot \text{Area}}$

TERMS G, O, I, H centers; R, r circum-, inradius; s semi-perimeter

Medians → Centroid, Perpendicular bisectors → Circumcenter, Angle bisectors → Incenter, Altitudes → Orthocenter: "MC–PC–AI–AO".

1. $G = \left(\dfrac{x_1 + x_2 + x_3}{3},\ \dfrac{y_1 + y_2 + y_3}{3}\right)$
2. $OA = OB = OC = R$
3. $r = \dfrac{\text{Area}}{s}$

TERMS G, O, I, H centers; R, r circum-, inradius; s semi-perimeter

Medians → Centroid, Perpendicular bisectors → Circumcenter, Angle bisectors → Incenter, Altitudes → Orthocenter: "MC–PC–AI–AO".

1. $DE \parallel BC \iff \dfrac{AD}{DB} = \dfrac{AE}{EC}$
2. $\dfrac{BD}{DC} = \dfrac{AB}{AC}$
3. $h^2 = pq$

TERMS h altitude to hypotenuse; p, q hypotenuse pieces

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G.12 (1/2) GEOMETRY

Distance & Midpoint Formulas

PART 1 OF 2

1. Distance formula
2. Midpoint formula
3. n-D Euclidean distance

STATE EACH NUMBERED FORMULA

G.12 (2/2) GEOMETRY

Distance & Midpoint Formulas

PART 2 OF 2

1. Section formula
2. Manhattan distance

STATE EACH NUMBERED FORMULA

G.13 (1/2) GEOMETRY

Polygon Angle Sums & Diagonals

PART 1 OF 2

1. Interior angle sum
2. Each interior angle (regular)
3. Exterior angle sum

STATE EACH NUMBERED FORMULA

G.13 (2/2) GEOMETRY

Polygon Angle Sums & Diagonals

PART 2 OF 2

1. Each exterior angle (regular)
2. Number of diagonals

STATE EACH NUMBERED FORMULA

G.14 (1/3) GEOMETRY

Quadrilateral Properties

PART 1 OF 3

1. Parallelogram
2. Rectangle
3. Rhombus

STATE EACH NUMBERED FORMULA

G.14 (2/3) GEOMETRY

Quadrilateral Properties

PART 2 OF 3

1. Square
2. Trapezoid midsegment
3. Kite

STATE EACH NUMBERED FORMULA

G.14 (3/3) GEOMETRY

Quadrilateral Properties

PART 3 OF 3

1. Angle sum
2. Area from diagonals

STATE EACH NUMBERED FORMULA

G.K1 GEOMETRY

Plane figures: area of triangle, trapezoid, rhombus, circle, sector

RECALL IT

G.K1 GEOMETRY

Solids: volume of prism, pyramid, cylinder, cone, sphere

RECALL IT

G.K1 GEOMETRY

Solids: surface area of cylinder, cone, sphere, hemisphere

RECALL IT

G.15 (1/2) GEOMETRY

Area of Plane Figures

PART 1 OF 2

1. Rectangle
2. Parallelogram
3. Triangle

STATE EACH NUMBERED FORMULA

G.15 (2/2) GEOMETRY

Area of Plane Figures

PART 2 OF 2

1. Trapezoid
2. Rhombus / kite
3. Regular polygon

STATE EACH NUMBERED FORMULA

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1. $\Sigma_{\text{int}} = (n-2)\cdot 180^\circ$
2. $\dfrac{(n-2)\cdot 180^\circ}{n}$
3. $\Sigma_{\text{ext}} = 360^\circ$

TERMS n number of sides; e exterior angle; regular = all equal

G.13 (1/2)

1. $P = \left(\dfrac{nx_1 + mx_2}{m+n},\ \dfrac{ny_1 + my_2}{m+n}\right)$
2. $d_1 = |x_2 - x_1| + |y_2 - y_1|$

TERMS $(x_1, y_1), (x_2, y_2)$ points; M midpoint; ratio $m : n$

G.12 (2/2)

1. $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
2. $M = \left(\dfrac{x_1 + x_2}{2},\ \dfrac{y_1 + y_2}{2}\right)$
3. $d(\mathbf{p}, \mathbf{q}) = \sqrt{\sum_{i=1}^{n} (p_i - q_i)^2}$

TERMS $(x_1, y_1), (x_2, y_2)$ points; M midpoint; ratio $m : n$

G.12 (1/2)

1. rectangle $\cap$ rhombus
2. $m = \frac{1}{2}(b_1 + b_2)$
3. $AB = AD,\ \ CB = CD,\ \ AC \perp BD$

TERMS $ABCD$ in order; AC, BD diagonals (d_1, d_2); b_1, b_2 bases

Every square is a rectangle and a rhombus, but not the other way round.

G.14 (2/3)

1. $AB \parallel CD,\ AB = CD,\ \angle A = \angle C$
2. $\angle A = 90^\circ,\ \ AC = BD$
3. $AB = BC = CD = DA,\ \ AC \perp BD$

TERMS $ABCD$ in order; AC, BD diagonals (d_1, d_2); b_1, b_2 bases

Every square is a rectangle and a rhombus, but not the other way round.

G.14 (1/3)

1. $e = \dfrac{360^\circ}{n}$
2. $\dfrac{n(n-3)}{2}$

TERMS n number of sides; e exterior angle; regular = all equal

G.13 (2/2)

Prism Bh Pyramid $\frac{1}{3}Bh$
Cylinder $\pi r^2 h$ Cone $\frac{1}{3}\pi r^2 h$
Sphere $\frac{4}{3}\pi r^3$

G.K1

Triangle $\frac{1}{2}bh$ Trapezoid $\frac{1}{2}(b_1 + b_2)h$
Rhombus/kite $\frac{1}{2}d_1 d_2$
Circle πr^2 Sector $\frac{1}{2}r^2\theta$ (θ in rad)

G.K1

1. $\angle A + \angle B + \angle C + \angle D = 360^\circ$
2. Area $= \frac{1}{2}\, d_1 d_2$

TERMS $ABCD$ in order; AC, BD diagonals (d_1, d_2); b_1, b_2 bases

Every square is a rectangle and a rhombus, but not the other way round.

G.14 (3/3)

1. $A = \frac{1}{2}\,(b_1 + b_2)\,h$
2. $A = \frac{1}{2}\, d_1 d_2$
3. $A = \frac{1}{2}\, aP$

TERMS b, h base, perpendicular height; d_1, d_2 diagonals; a apothem; P perimeter

G.15 (2/2)

1. $A = lw$
2. $A = bh$
3. $A = \frac{1}{2}\, bh$

TERMS b, h base, perpendicular height; d_1, d_2 diagonals; a apothem; P perimeter

G.15 (1/2)

Cylinder $2\pi r^2 + 2\pi rh$
Cone $\pi r^2 + \pi r\ell$
Sphere $4\pi r^2$ Hemisphere $3\pi r^2$

G.K1

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G.16 (1/3) GEOMETRY

Triangle Area: Heron's Formula & Alternatives

PART 1 OF 3

1. Base–height
2. SAS (trigonometric) area
3. Heron's formula

STATE EACH NUMBERED FORMULA

G.16 (2/3) GEOMETRY

Triangle Area: Heron's Formula & Alternatives

PART 2 OF 3

1. Inradius form
2. Circumradius form

STATE EACH NUMBERED FORMULA

G.16 (3/3) GEOMETRY

Triangle Area: Heron's Formula & Alternatives

PART 3 OF 3

1. Coordinate (shoelace) formula

STATE EACH NUMBERED FORMULA

G.17 (1/2) GEOMETRY

Circles: Circumference, Area, Arcs & Sectors

PART 1 OF 2

1. Circumference
2. Area
3. Arc length

STATE EACH NUMBERED FORMULA

G.17 (2/2) GEOMETRY

Circles: Circumference, Area, Arcs & Sectors

PART 2 OF 2

1. Sector area
2. Segment area
3. Chord length

STATE EACH NUMBERED FORMULA

G.18 (1/3) GEOMETRY

Circle Theorems

PART 1 OF 3

1. Inscribed angle theorem
2. Thales' theorem
3. Tangent–radius theorem

STATE EACH NUMBERED FORMULA

G.18 (2/3) GEOMETRY

Circle Theorems

PART 2 OF 3

1. Intersecting chords
2. Power of a point (outside)
3. Two tangents

STATE EACH NUMBERED FORMULA

G.18 (3/3) GEOMETRY

Circle Theorems

PART 3 OF 3

1. Cyclic quadrilateral

STATE EACH NUMBERED FORMULA

G.19 (1/2) GEOMETRY

Equation of a Circle

PART 1 OF 2

1. Standard form
2. Centered at the origin
3. General form

STATE EACH NUMBERED FORMULA

G.19 (2/2) GEOMETRY

Equation of a Circle

PART 2 OF 2

1. Point position test
2. Tangent line at a point
3. Parametric form

STATE EACH NUMBERED FORMULA

G.20 (1/2) GEOMETRY

Coordinate Geometry Proofs

PART 1 OF 2

1. Parallel sides
2. Perpendicular sides
3. Equal lengths

STATE EACH NUMBERED FORMULA

G.20 (2/2) GEOMETRY

Coordinate Geometry Proofs

PART 2 OF 2

1. Bisection
2. Collinear points
3. Right-angle (dot product) test

STATE EACH NUMBERED FORMULA

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1 Area $= \frac{1}{2}\,|x_1(y_2-y_3)+x_2(y_3-y_1)+x_3(y_1-y_2)|$

TERMS a, b, c sides; s semi-perimeter; r, R in-/circumradius

G.16 (3/3)

1 Area $= rs$

2 Area $= \dfrac{abc}{4R}$

TERMS a, b, c sides; s semi-perimeter; r, R in-/circumradius

G.16 (2/3)

1 Area $= \frac{1}{2}\,bh$

2 Area $= \frac{1}{2}\,ab\sin C$

3 Area $= \sqrt{s(s-a)(s-b)(s-c)}, \quad s = \dfrac{a+b+c}{2}$

TERMS a, b, c sides; s semi-perimeter; r, R in-/circumradius

G.16 (1/3)

1 $\angle_{\text{inscribed}} = \frac{1}{2}\angle_{\text{central}}$

2 angle in a semicircle $= 90°$

3 tangent $\perp$ radius

TERMS P outside point; T tangency point; A, B secant points

G.18 (1/3)

1 $A_{\text{sector}} = \frac{1}{2}\,r^2\theta = \dfrac{\theta^\circ}{360^\circ}\cdot\pi r^2$

2 $A_{\text{seg}} = \frac{1}{2}\,r^2(\theta - \sin\theta)$

3 $c = 2r\sin\dfrac{\theta}{2}$

TERMS r radius, d diameter; θ central angle (radians); s arc

"Cherry pie is delicious" ($C = \pi d$); "apple pies are too" ($A = \pi r^2$).

G.17 (2/2)

1 $C = 2\pi r = \pi d$

2 $A = \pi r^2$

3 $s = r\theta = \dfrac{\theta^\circ}{360^\circ}\cdot 2\pi r$

TERMS r radius, d diameter; θ central angle (radians); s arc

"Cherry pie is delicious" ($C = \pi d$); "apple pies are too" ($A = \pi r^2$).

G.17 (1/2)

1 $(x-h)^2 + (y-k)^2 = r^2$

2 $x^2 + y^2 = r^2$

3 $x^2 + y^2 + Dx + Ey + F = 0$

TERMS (h, k) center; r radius; D, E, F general-form constants

The signs flip: $(x-2)^2 + (y+3)^2 = 16$ has center $(2, -3)$.

G.19 (1/2)

1 $\angle A + \angle C = 180^\circ$

TERMS P outside point; T tangency point; A, B secant points

G.18 (3/3)

1 $ab = cd$

2 $PT^2 = PA \cdot PB$

3 $PT_1 = PT_2$

TERMS P outside point; T tangency point; A, B secant points

G.18 (2/3)

1 $M_{AC} = M_{BD}$

2 $m_{AB} = m_{BC}$

3 $(x_2 - x_1)(x_3 - x_1) + (y_2 - y_1)(y_3 - y_1) = 0$

TERMS m slope; d distance; M midpoint; a, b, c general constants

G.20 (2/2)

1 $m_1 = m_2$

2 $m_1 m_2 = -1$

3 $d_1 = d_2$

TERMS m slope; d distance; M midpoint; a, b, c general constants

G.20 (1/2)

1 $(x_0 - h)^2 + (y_0 - k)^2 \lessgtr r^2$

2 $(x_0 - h)(x - h) + (y_0 - k)(y - k) = r^2$

3 $x = h + r\cos t, \quad y = k + r\sin t$

TERMS (h, k) center; r radius; D, E, F general-form constants

The signs flip: $(x-2)^2 + (y+3)^2 = 16$ has center $(2, -3)$.

G.19 (2/2)

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G.21 (1/3) GEOMETRY

Prisms & Cylinders

PART 1 OF 3

1. Prism / cylinder volume
2. Rectangular prism volume
3. Rectangular prism surface area

STATE EACH NUMBERED FORMULA

G.21 (2/3) GEOMETRY

Prisms & Cylinders

PART 2 OF 3

1. Cylinder volume
2. Cylinder surface area

STATE EACH NUMBERED FORMULA

G.21 (3/3) GEOMETRY

Prisms & Cylinders

PART 3 OF 3

1. Cube
2. Space diagonal

STATE EACH NUMBERED FORMULA

G.22 (1/3) GEOMETRY

Pyramids, Cones & Spheres

PART 1 OF 3

1. Pyramid volume
2. Cone volume
3. Cone surface area

STATE EACH NUMBERED FORMULA

G.22 (2/3) GEOMETRY

Pyramids, Cones & Spheres

PART 2 OF 3

1. Sphere volume
2. Sphere surface area
3. Slant height

STATE EACH NUMBERED FORMULA

G.22 (3/3) GEOMETRY

Pyramids, Cones & Spheres

PART 3 OF 3

1. Frustum volume

STATE EACH NUMBERED FORMULA

G.23 (1/3) GEOMETRY

Surface Area of Solids

PART 1 OF 3

1. Prism
2. Cylinder lateral area
3. Regular pyramid

STATE EACH NUMBERED FORMULA

G.23 (2/3) GEOMETRY

Surface Area of Solids

PART 2 OF 3

1. Cone lateral area
2. Sphere
3. Hemisphere (closed)

STATE EACH NUMBERED FORMULA

G.23 (3/3) GEOMETRY

Surface Area of Solids

PART 3 OF 3

1. Euler's polyhedron formula

STATE EACH NUMBERED FORMULA

G.24 (1/4) GEOMETRY

Coordinate Transformations

PART 1 OF 4

1. Translation
2. Reflection in the x-axis
3. Reflection in the y-axis

STATE EACH NUMBERED FORMULA

G.24 (2/4) GEOMETRY

Coordinate Transformations

PART 2 OF 4

1. Reflection in the line $y = x$
2. Reflection in the line $y = -x$

STATE EACH NUMBERED FORMULA

G.24 (3/4) GEOMETRY

Coordinate Transformations

PART 3 OF 4

1. Rotation matrix

STATE EACH NUMBERED FORMULA

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1 $V = s^3, \quad SA = 6s^2$
2 $d = \sqrt{l^2 + w^2 + h^2}$

TERMS B base area; h height; l, w length, width; r radius

G.21 (3/3)

1 $V = \pi r^2 h$
2 $SA = 2\pi r^2 + 2\pi rh$

TERMS B base area; h height; l, w length, width; r radius

G.21 (2/3)

1 $V = Bh$
2 $V = lwh$
3 $SA = 2(lw + lh + wh)$

TERMS B base area; h height; l, w length, width; r radius

G.21 (1/3)

1 $V = \frac{1}{3}\pi h\left(R^2 + Rr + r^2\right)$

TERMS B base area; h height; r radius; ℓ slant height; R big radius

"A cone is one-third of the cylinder" with the same base and height.

G.22 (3/3)

1 $V = \frac{4}{3}\pi r^3$
2 $SA = 4\pi r^2$
3 $\ell = \sqrt{r^2 + h^2}$

TERMS B base area; h height; r radius; ℓ slant height; R big radius

"A cone is one-third of the cylinder" with the same base and height.

G.22 (2/3)

1 $V = \frac{1}{3}Bh$
2 $V = \frac{1}{3}\pi r^2 h$
3 $SA = \pi r^2 + \pi r\ell$

TERMS B base area; h height; r radius; ℓ slant height; R big radius

"A cone is one-third of the cylinder" with the same base and height.

G.22 (1/3)

1 $V - E + F = 2$

TERMS B, P base area, perimeter; ℓ slant height; L lateral area

G.23 (3/3)

1 $L = \pi r\ell$
2 $SA = 4\pi r^2$
3 $SA = 3\pi r^2$

TERMS B, P base area, perimeter; ℓ slant height; L lateral area

G.23 (2/3)

1 $SA = 2B + Ph$
2 $L = 2\pi rh$
3 $SA = B + \frac{1}{2}P\ell$

TERMS B, P base area, perimeter; ℓ slant height; L lateral area

G.23 (1/3)

1 $\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$

TERMS (a, b) shift; θ CCW angle; k scale factor; (p, q) center

G.24 (3/4)

1 $(x, y) \mapsto (y, x)$
2 $(x, y) \mapsto (-y, -x)$

TERMS (a, b) shift; θ CCW angle; k scale factor; (p, q) center

G.24 (2/4)

1 $(x, y) \mapsto (x + a,\ y + b)$
2 $(x, y) \mapsto (x, -y)$
3 $(x, y) \mapsto (-x, y)$

TERMS (a, b) shift; θ CCW angle; k scale factor; (p, q) center

G.24 (1/4)

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G.24 (4/4)

GEOMETRY

Coordinate Transformations

PART 4 OF 4

1. Dilation
2. Dilation about a center

STATE EACH NUMBERED FORMULA

G.25 (1/4)

GEOMETRY

Rotations, Rigid Motions & Compositions

PART 1 OF 4

1. Rotation 90° counterclockwise
2. Rotation 180°
3. Rotation 270° counterclockwise

STATE EACH NUMBERED FORMULA

G.25 (2/4)

GEOMETRY

Rotations, Rigid Motions & Compositions

PART 2 OF 4

1. Rotation about a point

STATE EACH NUMBERED FORMULA

G.25 (3/4)

GEOMETRY

Rotations, Rigid Motions & Compositions

PART 3 OF 4

1. Two reflections, parallel mirrors
2. Two reflections, crossing mirrors

STATE EACH NUMBERED FORMULA

G.25 (4/4)

GEOMETRY

Rotations, Rigid Motions & Compositions

PART 4 OF 4

1. Isometry
2. Similarity transformation

STATE EACH NUMBERED FORMULA

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1 $$\begin{bmatrix} x' \\ y' \end{bmatrix} = R_\theta \begin{bmatrix} x-a \\ y-b \end{bmatrix} + \begin{bmatrix} a \\ b \end{bmatrix}$$

TERMS R_θ rotation matrix; (a,b) center; d, φ mirror gap, angle

90° *counterclockwise: "swap and negate the first". Two mirrors double:* $2d$ *or* 2φ.

G.25 (2/4)

1 $(x,y) \mapsto (-y,\ x)$
2 $(x,y) \mapsto (-x,\ -y)$
3 $(x,y) \mapsto (y,\ -x)$

TERMS R_θ rotation matrix; (a,b) center; d, φ mirror gap, angle

90° *counterclockwise: "swap and negate the first". Two mirrors double:* $2d$ *or* 2φ.

G.25 (1/4)

1 $(x,y) \mapsto (kx,\ ky)$
2 $(x,y) \mapsto \big(p + k(x-p),\ q + k(y-q)\big)$

TERMS (a,b) shift; θ CCW angle; k scale factor; (p,q) center

G.24 (4/4)

1 $d(P',Q') = d(P,Q)$
2 $d(P',Q') = k\,d(P,Q)$

TERMS R_θ rotation matrix; (a,b) center; d, φ mirror gap, angle

90° *counterclockwise: "swap and negate the first". Two mirrors double:* $2d$ *or* 2φ.

G.25 (4/4)

1 translation by $2d$
2 rotation by 2φ

TERMS R_θ rotation matrix; (a,b) center; d, φ mirror gap, angle

90° *counterclockwise: "swap and negate the first". Two mirrors double:* $2d$ *or* 2φ.

G.25 (3/4)

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T.1 (1/2) TRIGONOMETRY

Right-Triangle Trigonometric Ratios

PART 1 OF 2

1. Sine (SOH)
2. Cosine (CAH)
3. Tangent (TOA)

STATE EACH NUMBERED FORMULA

T.1 (2/2) TRIGONOMETRY

Right-Triangle Trigonometric Ratios

PART 2 OF 2

1. Cosecant
2. Secant
3. Cotangent

STATE EACH NUMBERED FORMULA

T.2 (1/3) TRIGONOMETRY

Radians, Degrees & Angular Speed

PART 1 OF 3

1. Conversion constant
2. Degrees $\to$ radians
3. Radians $\to$ degrees

STATE EACH NUMBERED FORMULA

T.2 (2/3) TRIGONOMETRY

Radians, Degrees & Angular Speed

PART 2 OF 3

1. Arc length
2. Sector area

STATE EACH NUMBERED FORMULA

T.2 (3/3) TRIGONOMETRY

Radians, Degrees & Angular Speed

PART 3 OF 3

1. Angular speed
2. Linear speed

STATE EACH NUMBERED FORMULA

T.3 (1/2) TRIGONOMETRY

Coterminal & Reference Angles

PART 1 OF 2

1. Standard position
2. Coterminal angles

STATE EACH NUMBERED FORMULA

T.3 (2/2) TRIGONOMETRY

Coterminal & Reference Angles

PART 2 OF 2

1. Reference angle by quadrant
2. Reference-angle rule

STATE EACH NUMBERED FORMULA

T.K1 TRIGONOMETRY

Taylor polynomials of $\sin x$ and $\cos x$ at 0

RECALL IT

T.K1 TRIGONOMETRY

Why do Taylor terms of $\sin$ and $\cos$ alternate and skip?

RECALL IT

T.K1 TRIGONOMETRY

Unit circle: what is the point at angle θ?

RECALL IT

T.K1 TRIGONOMETRY

Exact $\sin, \cos, \tan$ at $\frac{\pi}{6}, \frac{\pi}{4}, \frac{\pi}{3}$

RECALL IT

T.K1 TRIGONOMETRY

Unit circle coordinates at $\frac{2\pi}{3}, \frac{5\pi}{4}, \frac{11\pi}{6}$

RECALL IT

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1. $\pi \text{ rad} = 180^\circ$
2. $\theta_{\text{rad}} = \theta^\circ \cdot \dfrac{\pi}{180^\circ}$
3. $\theta^\circ = \theta_{\text{rad}} \cdot \dfrac{180^\circ}{\pi}$

TERMS r radius; θ in radians; ω angular speed; f rev. per time

Degrees to radians: multiply by $\frac{\pi}{180}$ ("pi on top to go to radians").

T.2 (1/3)

1. $\csc\theta = \dfrac{1}{\sin\theta} = \dfrac{\text{hyp}}{\text{opp}}$
2. $\sec\theta = \dfrac{1}{\cos\theta} = \dfrac{\text{hyp}}{\text{adj}}$
3. $\cot\theta = \dfrac{1}{\tan\theta} = \dfrac{\text{adj}}{\text{opp}}$

TERMS opp, adj = legs relative to θ; hyp = hypotenuse

SOH-CAH-TOA: Sine = Opposite/Hypotenuse, Cosine = Adjacent/Hypotenuse, Tangent = Opposite/Adjacent. Each "co-" reciprocal pairs with a non-"co" function: csc↔sin, sec↔cos.

T.1 (2/2)

1. $\sin\theta = \dfrac{\text{opp}}{\text{hyp}}$
2. $\cos\theta = \dfrac{\text{adj}}{\text{hyp}}$
3. $\tan\theta = \dfrac{\text{opp}}{\text{adj}} = \dfrac{\sin\theta}{\cos\theta}$

TERMS opp, adj = legs relative to θ; hyp = hypotenuse

SOH-CAH-TOA: Sine = Opposite/Hypotenuse, Cosine = Adjacent/Hypotenuse, Tangent = Opposite/Adjacent. Each "co-" reciprocal pairs with a non-"co" function: csc↔sin, sec↔cos.

T.1 (1/2)

1. $\theta > 0 : \circlearrowleft, \qquad \theta < 0 : \circlearrowright$
2. $\theta + 2\pi k \;\; (\theta + 360^\circ k),\ k \in \mathbb{Z}$

TERMS $k \in \mathbb{Z}$; θ' reference angle; quadrant gives the sign

Reference angle = "how far to the nearest x-axis".

T.3 (1/2)

1. $\omega = \dfrac{\theta}{t} = 2\pi f$
2. $v = r\omega$

TERMS r radius; θ in radians; ω angular speed; f rev. per time

Degrees to radians: multiply by $\frac{\pi}{180}$ ("pi on top to go to radians").

T.2 (3/3)

1. $s = r\theta$
2. $A = \frac{1}{2}r^2\theta$

TERMS r radius; θ in radians; ω angular speed; f rev. per time

Degrees to radians: multiply by $\frac{\pi}{180}$ ("pi on top to go to radians").

T.2 (2/3)

Derivatives cycle
$\sin \to \cos \to -\sin \to -\cos$; at 0 they are $0, \pm 1$.
sin odd $\Rightarrow$ odd powers; cos even $\Rightarrow$ even powers.

T.K1

$$\sin x \approx x - \frac{x^3}{6} + \frac{x^5}{120}$$
$$\cos x \approx 1 - \frac{x^2}{2} + \frac{x^4}{24}$$
radians; error $\le$ first omitted term

T.K1

1. $\theta' = \begin{cases} \theta, & \text{QI} \\ \pi - \theta, & \text{QII} \end{cases} \qquad \theta' = \begin{cases} \theta - \pi, & \text{QIII} \\ 2\pi - \theta, & \text{QIV} \end{cases} \qquad (0 \le \theta < 2\pi)$
2. $f(\theta) = \pm f(\theta')$

TERMS $k \in \mathbb{Z}$; θ' reference angle; quadrant gives the sign

Reference angle = "how far to the nearest x-axis".

T.3 (2/2)

$\frac{2\pi}{3}$: $\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$
$\frac{5\pi}{4}$: $\left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$
$\frac{11\pi}{6}$: $\left(\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$

T.K1

$\frac{\pi}{6}$: $\frac{1}{2}$, $\frac{\sqrt{3}}{2}$, $\frac{\sqrt{3}}{3}$
$\frac{\pi}{4}$: $\frac{\sqrt{2}}{2}$, $\frac{\sqrt{2}}{2}$, 1
$\frac{\pi}{3}$: $\frac{\sqrt{3}}{2}$, $\frac{1}{2}$, $\sqrt{3}$

T.K1

$P(\theta) = (\cos\theta,\ \sin\theta)$ on $x^2 + y^2 = 1$;
$\tan\theta = \frac{y}{x}$.

T.K1

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T.K1 TRIGONOMETRY

Quadrant signs: which functions are positive where?

RECALL IT

T.K1 TRIGONOMETRY

Reference-angle method: the four steps

RECALL IT

T.4 (1/2) TRIGONOMETRY

Unit Circle & Exact Values

PART 1 OF 2

1. Unit-circle definition
2. Tangent as slope
3. Circle of radius r

STATE EACH NUMBERED FORMULA

T.4 (2/2) TRIGONOMETRY

Unit Circle & Exact Values

PART 2 OF 2

1. Exact values (first quadrant)
2. Quadrant signs (ASTC)

STATE EACH NUMBERED FORMULA

T.5 (1/2) TRIGONOMETRY

Graphs of the Six Trig Functions

PART 1 OF 2

1. Periodicity
2. Sine and cosine
3. Cosine is shifted sine

STATE EACH NUMBERED FORMULA

T.5 (2/2) TRIGONOMETRY

Graphs of the Six Trig Functions

PART 2 OF 2

1. Tangent asymptotes
2. Cotangent asymptotes
3. Secant and cosecant range

STATE EACH NUMBERED FORMULA

T.6 (1/2) TRIGONOMETRY

Pythagorean & Fundamental Identities

PART 1 OF 2

1. Pythagorean identity
2. Tangent–secant form
3. Cotangent–cosecant form

STATE EACH NUMBERED FORMULA

T.6 (2/2) TRIGONOMETRY

Pythagorean & Fundamental Identities

PART 2 OF 2

1. Even / odd (negative angle)
2. Periodicity
3. Supplement / half-turn

STATE EACH NUMBERED FORMULA

T.7 (1/2) TRIGONOMETRY

Reciprocal, Quotient & Cofunction Identities

PART 1 OF 2

1. Reciprocal identities
2. Quotient identities

STATE EACH NUMBERED FORMULA

T.7 (2/2) TRIGONOMETRY

Reciprocal, Quotient & Cofunction Identities

PART 2 OF 2

1. Cofunction: sine / cosine
2. Cofunction: tangent / cotangent
3. Cofunction: secant / cosecant

STATE EACH NUMBERED FORMULA

T.8 TRIGONOMETRY

Sum & Difference Formulas

1. Sine of a sum / difference
2. Cosine of a sum / difference
3. Tangent of a sum / difference

STATE EACH NUMBERED FORMULA

T.9 (1/4) TRIGONOMETRY

Double-Angle & Half-Angle Formulas

PART 1 OF 4

1. Sine double angle
2. Tangent double angle

STATE EACH NUMBERED FORMULA

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1. $P(\theta) = (\cos\theta,\ \sin\theta)$
2. $\tan\theta = \dfrac{y}{x} = \dfrac{\sin\theta}{\cos\theta}$
3. $\sin\theta = \dfrac{y}{r},\quad \cos\theta = \dfrac{x}{r},\quad \tan\theta = \dfrac{y}{x}$

TERMS $P(\theta)$ point on $x^2+y^2=1$; $r=\sqrt{x^2+y^2}$; ASTC signs

Sine values are $\frac{\sqrt{0}}{2}$, $\frac{\sqrt{1}}{2}$, $\frac{\sqrt{2}}{2}$, $\frac{\sqrt{3}}{2}$, $\frac{\sqrt{4}}{2}$; cosine is the reverse. Signs by quadrant: "All Students Take Calculus" (All, Sin, Tan, Cos positive).

1. Reduce to $[0, 2\pi)$. 2. Reference angle θ' to the x-axis.
3. Evaluate at θ'. 4. Attach the ASTC sign.

"All Students Take Calculus"
QI all; QII sin, csc; QIII tan, cot; QIV cos, sec.

1. $x = \dfrac{\pi}{2} + k\pi$
2. $x = k\pi$
3. $|\sec x| \geq 1, \quad |\csc x| \geq 1$

TERMS P period; $k \in \mathbb{Z}$; asymptotes where the denominator $= 0$

Asymptotes sit where the denominator *is zero: tan, sec $\to \cos x = 0$; cot, csc $\to \sin x = 0$.*

1. $f(x+P) = f(x)$ for all x
2. $-1 \leq \sin x,\ \cos x \leq 1$
3. $\cos x = \sin\left(x + \frac{\pi}{2}\right)$

TERMS P period; $k \in \mathbb{Z}$; asymptotes where the denominator $= 0$

Asymptotes sit where the denominator *is zero: tan, sec $\to \cos x = 0$; cot, csc $\to \sin x = 0$.*

	θ	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
1	$\sin\theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
	$\cos\theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0

2. QI: all QII: sin QIII: tan QIV: cos

TERMS $P(\theta)$ point on $x^2+y^2=1$; $r=\sqrt{x^2+y^2}$; ASTC signs

Sine values are $\frac{\sqrt{0}}{2}$, $\frac{\sqrt{1}}{2}$, $\frac{\sqrt{2}}{2}$, $\frac{\sqrt{3}}{2}$, $\frac{\sqrt{4}}{2}$; cosine is the reverse. Signs by quadrant: "All Students Take Calculus" (All, Sin, Tan, Cos positive).

1. $\csc\theta = \dfrac{1}{\sin\theta},\quad \sec\theta = \dfrac{1}{\cos\theta},\quad \cot\theta = \dfrac{1}{\tan\theta}$
2. $\tan\theta = \dfrac{\sin\theta}{\cos\theta}, \qquad \cot\theta = \dfrac{\cos\theta}{\sin\theta}$

TERMS $\frac{\pi}{2} - \theta$ complement; pairs sin/cos, tan/cot, sec/csc

"Co" means "complement's": cosine is the sine of the complementary angle ($\sin 30° = \cos 60°$).

1. $\sin(-\theta) = -\sin\theta,\ \cos(-\theta) = \cos\theta,\ \tan(-\theta) = -\tan\theta$
2. $\sin(\theta + 2\pi) = \sin\theta, \quad \tan(\theta+\pi) = \tan\theta$
3. $\sin(\pi-\theta) = \sin\theta,\ \cos(\pi-\theta) = -\cos\theta,\ \sin(\theta+\pi) = -\sin\theta$

TERMS $\sin^2\theta = (\sin\theta)^2$; odd: sin, tan; even: cos

Divide $\sin^2 + \cos^2 = 1$ by $\cos^2$ or $\sin^2$ to get the other two. Cosine is the only even one.

1. $\sin^2\theta + \cos^2\theta = 1$
2. $1 + \tan^2\theta = \sec^2\theta$
3. $1 + \cot^2\theta = \csc^2\theta$

TERMS $\sin^2\theta = (\sin\theta)^2$; odd: sin, tan; even: cos

Divide $\sin^2 + \cos^2 = 1$ by $\cos^2$ or $\sin^2$ to get the other two. Cosine is the only even one.

1. $\sin 2\theta = 2\sin\theta\cos\theta$
2. $\tan 2\theta = \dfrac{2\tan\theta}{1-\tan^2\theta}$

TERMS $\pm$ by the quadrant of $\frac{\theta}{2}$; $\sin^2\theta = (\sin\theta)^2$

Half-angle: "sine is minus, cosine is plus" under the root.

1. $\sin(\alpha \pm \beta) = \sin\alpha\cos\beta \pm \cos\alpha\sin\beta$
2. $\cos(\alpha \pm \beta) = \cos\alpha\cos\beta \mp \sin\alpha\sin\beta$
3. $\tan(\alpha \pm \beta) = \dfrac{\tan\alpha \pm \tan\beta}{1 \mp \tan\alpha\tan\beta}$

TERMS α, β any angles; take top or bottom signs together

Sine keeps the sign and mixes (sin cos + cos sin); cosine keeps the pairs and flips the sign.

1. $\sin\left(\frac{\pi}{2} - \theta\right) = \cos\theta, \quad \cos\left(\frac{\pi}{2} - \theta\right) = \sin\theta$
2. $\tan\left(\frac{\pi}{2} - \theta\right) = \cot\theta$
3. $\sec\left(\frac{\pi}{2} - \theta\right) = \csc\theta$

TERMS $\frac{\pi}{2} - \theta$ complement; pairs sin/cos, tan/cot, sec/csc

"Co" means "complement's": cosine is the sine of the complementary angle ($\sin 30° = \cos 60°$).

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T.9 (2/4) TRIGONOMETRY

Double-Angle & Half-Angle Formulas

PART 2 OF 4

1. Cosine double angle (three forms)

STATE EACH NUMBERED FORMULA

T.9 (3/4) TRIGONOMETRY

Double-Angle & Half-Angle Formulas

PART 3 OF 4

1. Sine half angle
2. Cosine half angle
3. Tangent half angle

STATE EACH NUMBERED FORMULA

T.9 (4/4) TRIGONOMETRY

Double-Angle & Half-Angle Formulas

PART 4 OF 4

1. Triple angle

STATE EACH NUMBERED FORMULA

T.10 (1/3) TRIGONOMETRY

Power-Reduction Formulas

PART 1 OF 3

1. Sine squared
2. Cosine squared
3. Tangent squared

STATE EACH NUMBERED FORMULA

T.10 (2/3) TRIGONOMETRY

Power-Reduction Formulas

PART 2 OF 3

1. Product of squares
2. Cosine to the fourth

STATE EACH NUMBERED FORMULA

T.10 (3/3) TRIGONOMETRY

Power-Reduction Formulas

PART 3 OF 3

1. Sine cubed
2. Cosine cubed

STATE EACH NUMBERED FORMULA

T.11 (1/3) TRIGONOMETRY

Product-to-Sum & Sum-to-Product

PART 1 OF 3

1. Sine times cosine
2. Cosine times cosine
3. Sine times sine

STATE EACH NUMBERED FORMULA

T.11 (2/3) TRIGONOMETRY

Product-to-Sum & Sum-to-Product

PART 2 OF 3

1. Sum of sines
2. Difference of sines

STATE EACH NUMBERED FORMULA

T.11 (3/3) TRIGONOMETRY

Product-to-Sum & Sum-to-Product

PART 3 OF 3

1. Sum of cosines
2. Difference of cosines

STATE EACH NUMBERED FORMULA

T.K2 TRIGONOMETRY

Identity hexagon: how to read it

RECALL IT

T.K2 TRIGONOMETRY

Proving an identity: the standard moves

RECALL IT

T.12 (1/2) TRIGONOMETRY

Graphs of Sinusoids

PART 1 OF 2

1. General sinusoid
2. Amplitude
3. Period

STATE EACH NUMBERED FORMULA

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1. $\sin 3\theta = 3\sin\theta - 4\sin^3\theta, \quad \cos 3\theta = 4\cos^3\theta - 3\cos\theta$

TERMS $\pm$ by the quadrant of $\frac{\theta}{2}$; $\sin^2\theta = (\sin\theta)^2$

Half-angle: "sine is minus, cosine is plus" under the root.

T.9 (4/4)

1. $\sin\dfrac{\theta}{2} = \pm\sqrt{\dfrac{1-\cos\theta}{2}}$
2. $\cos\dfrac{\theta}{2} = \pm\sqrt{\dfrac{1+\cos\theta}{2}}$
3. $\tan\dfrac{\theta}{2} = \dfrac{1-\cos\theta}{\sin\theta} = \dfrac{\sin\theta}{1+\cos\theta}$

TERMS $\pm$ by the quadrant of $\frac{\theta}{2}$; $\sin^2\theta = (\sin\theta)^2$

Half-angle: "sine is minus, cosine is plus" under the root.

T.9 (3/4)

1. $\cos 2\theta = \cos^2\theta - \sin^2\theta = 2\cos^2\theta - 1 = 1 - 2\sin^2\theta$

TERMS $\pm$ by the quadrant of $\frac{\theta}{2}$; $\sin^2\theta = (\sin\theta)^2$

Half-angle: "sine is minus, cosine is plus" under the root.

T.9 (2/4)

1. $\sin^3\theta = \dfrac{3\sin\theta - \sin 3\theta}{4}$
2. $\cos^3\theta = \dfrac{3\cos\theta + \cos 3\theta}{4}$

TERMS $\sin^n\theta = (\sin\theta)^n$; result has only first powers of $\cos k\theta$

"Squared becomes half, plus or minus half of the double": $\sin^2, \cos^2 = \frac{1}{2} \mp \frac{1}{2}\cos 2\theta$.

T.10 (3/3)

1. $\sin^2\theta\cos^2\theta = \dfrac{1-\cos 4\theta}{8}$
2. $\cos^4\theta = \dfrac{3 + 4\cos 2\theta + \cos 4\theta}{8}$

TERMS $\sin^n\theta = (\sin\theta)^n$; result has only first powers of $\cos k\theta$

"Squared becomes half, plus or minus half of the double": $\sin^2, \cos^2 = \frac{1}{2} \mp \frac{1}{2}\cos 2\theta$.

T.10 (2/3)

1. $\sin^2\theta = \dfrac{1-\cos 2\theta}{2}$
2. $\cos^2\theta = \dfrac{1+\cos 2\theta}{2}$
3. $\tan^2\theta = \dfrac{1-\cos 2\theta}{1+\cos 2\theta}$

TERMS $\sin^n\theta = (\sin\theta)^n$; result has only first powers of $\cos k\theta$

"Squared becomes half, plus or minus half of the double": $\sin^2, \cos^2 = \frac{1}{2} \mp \frac{1}{2}\cos 2\theta$.

T.10 (1/3)

1. $\cos A + \cos B = 2\cos\frac{A+B}{2}\cos\frac{A-B}{2}$
2. $\cos A - \cos B = -2\sin\frac{A+B}{2}\sin\frac{A-B}{2}$

TERMS A, B angles; half-sum $\frac{A+B}{2}$; half-difference $\frac{A-B}{2}$

Sum-to-product: "half-sum, half-difference".
Product-to-sum: "average the sum and difference angles".

T.11 (3/3)

1. $\sin A + \sin B = 2\sin\frac{A+B}{2}\cos\frac{A-B}{2}$
2. $\sin A - \sin B = 2\cos\frac{A+B}{2}\sin\frac{A-B}{2}$

TERMS A, B angles; half-sum $\frac{A+B}{2}$; half-difference $\frac{A-B}{2}$

Sum-to-product: "half-sum, half-difference".
Product-to-sum: "average the sum and difference angles".

T.11 (2/3)

1. $\sin A\cos B = \frac{1}{2}\left[\sin(A+B) + \sin(A-B)\right]$
2. $\cos A\cos B = \frac{1}{2}\left[\cos(A-B) + \cos(A+B)\right]$
3. $\sin A\sin B = \frac{1}{2}\left[\cos(A-B) - \cos(A+B)\right]$

TERMS A, B angles; half-sum $\frac{A+B}{2}$; half-difference $\frac{A-B}{2}$

Sum-to-product: "half-sum, half-difference".
Product-to-sum: "average the sum and difference angles".

T.11 (1/3)

1. $y = A\sin\big(B(x-C)\big) + D$
2. amplitude $= |A|$
3. period $= \dfrac{2\pi}{|B|}$

TERMS A amplitude; B angular freq.; C phase shift; D midline

"A-B-C-D": Amplitude, B for the period, C for the shift ("Copy the sign backwards": $x - C$ moves right), D for the Datum (midline).

T.12 (1/2)

One side only; convert to sin, cos; common denominators; factor; use $\sin^2 + \cos^2 = 1$; conjugates.

T.K2

Left column sin, tan, sec; right cos, cot, csc; center 1. Diagonals: reciprocals. Neighbors multiply to the middle one. Down-triangles: top^2 + top^2 = bottom^2.

T.K2

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T.12 (2/2) TRIGONOMETRY

Graphs of Sinusoids

PART 2 OF 2

1. Phase shift
2. Midline, max, min
3. Frequency

STATE EACH NUMBERED FORMULA

T.13 (1/2) TRIGONOMETRY

Harmonic Addition: Combining Sine & Cosine

PART 1 OF 2

1. Amplitude–phase form
2. Combined amplitude
3. Phase angle

STATE EACH NUMBERED FORMULA

T.13 (2/2) TRIGONOMETRY

Harmonic Addition: Combining Sine & Cosine

PART 2 OF 2

1. Cosine form
2. Maximum value

STATE EACH NUMBERED FORMULA

T.14 (1/3) TRIGONOMETRY

Inverse Trigonometric Functions

PART 1 OF 3

1. Arcsine
2. Arccosine
3. Arctangent

STATE EACH NUMBERED FORMULA

T.14 (2/3) TRIGONOMETRY

Inverse Trigonometric Functions

PART 2 OF 3

1. Cancellation
2. Complementary sum
3. Negative inputs

STATE EACH NUMBERED FORMULA

T.14 (3/3) TRIGONOMETRY

Inverse Trigonometric Functions

PART 3 OF 3

1. Triangle composition
2. Two-argument arctangent

STATE EACH NUMBERED FORMULA

T.15 (1/2) TRIGONOMETRY

Solving Trigonometric Equations

PART 1 OF 2

1. Sine equation
2. Cosine equation
3. Tangent equation

STATE EACH NUMBERED FORMULA

T.15 (2/2) TRIGONOMETRY

Solving Trigonometric Equations

PART 2 OF 2

1. Multiple angle
2. Factored form

STATE EACH NUMBERED FORMULA

T.16 (1/2) TRIGONOMETRY

Law of Sines & Triangle Area

PART 1 OF 2

1. Law of sines
2. Extended law of sines
3. Area (SAS)

STATE EACH NUMBERED FORMULA

T.16 (2/2) TRIGONOMETRY

Law of Sines & Triangle Area

PART 2 OF 2

1. Ambiguous-case height

STATE EACH NUMBERED FORMULA

T.17 (1/2) TRIGONOMETRY

Law of Cosines

PART 1 OF 2

1. Law of cosines
2. Angle form
3. Cyclic forms

STATE EACH NUMBERED FORMULA

T.17 (2/2) TRIGONOMETRY

Law of Cosines

PART 2 OF 2

1. Acute / obtuse test

STATE EACH NUMBERED FORMULA

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1. $a\sin x + b\cos x = R\cos(x-\delta),\quad \delta = \operatorname{atan2}(a,b)$
2. $\max_x (a\sin x + b\cos x) = \sqrt{a^2+b^2}$

TERMS $R = \sqrt{a^2+b^2}$ amplitude; φ phase angle

"Legs a and b, hypotenuse R, angle φ": draw the triangle.

T.13 (2/2)

1. $a\sin x + b\cos x = R\sin(x+\varphi)$
2. $R = \sqrt{a^2+b^2}$
3. $\cos\varphi = \dfrac{a}{R},\quad \sin\varphi = \dfrac{b}{R} \;\Rightarrow\; \varphi = \operatorname{atan2}(b,a)$

TERMS $R = \sqrt{a^2+b^2}$ amplitude; φ phase angle

"Legs a and b, hypotenuse R, angle φ": draw the triangle.

T.13 (1/2)

1. phase shift $= C \quad \left(= \frac{\varphi}{B} \text{ for } \sin(Bx-\varphi)\right)$
2. $y = D,\quad y_{\max} = D + |A|,\quad y_{\min} = D - |A|$
3. $f = \dfrac{|B|}{2\pi} = \dfrac{1}{\text{period}}$

TERMS A amplitude; B angular freq.; C phase shift; D midline

"A-B-C-D": Amplitude, B for the period, C for the shift ("Copy the sign backwards": $x - C$ moves right), D for the Datum (midline).

T.12 (2/2)

1. $\cos(\arcsin x) = \sqrt{1-x^2}$
2. $\theta = \operatorname{atan2}(y,x) \in (-\pi, \pi]$

TERMS principal ranges: arcsin, arctan in QI/QIV; arccos in QI/QII

Ranges: arcsin and arctan live on the right *half of the circle (QI, QIV); arccos on the* top *half (QI, QII).*

T.14 (3/3)

1. $\sin(\arcsin x) = x;\quad \arcsin(\sin\theta) = \theta$ only if $|\theta| \le \frac{\pi}{2}$
2. $\arcsin x + \arccos x = \dfrac{\pi}{2}$
3. $\arcsin(-x) = -\arcsin x,\quad \arccos(-x) = \pi - \arccos x$

TERMS principal ranges: arcsin, arctan in QI/QIV; arccos in QI/QII

Ranges: arcsin and arctan live on the right *half of the circle (QI, QIV); arccos on the* top *half (QI, QII).*

T.14 (2/3)

1. $y = \arcsin x \iff \sin y = x,\quad y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
2. $y = \arccos x \in [0,\pi],\quad x \in [-1,1]$
3. $y = \arctan x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right),\quad x \in \mathbb{R}$

TERMS principal ranges: arcsin, arctan in QI/QIV; arccos in QI/QII

Ranges: arcsin and arctan live on the right *half of the circle (QI, QIV); arccos on the* top *half (QI, QII).*

T.14 (1/3)

1. $\dfrac{a}{\sin A} = \dfrac{b}{\sin B} = \dfrac{c}{\sin C}$
2. $\dfrac{a}{\sin A} = 2R$
3. $K = \frac{1}{2}ab\sin C$

TERMS a, b, c opposite A, B, C; R circumradius; K area

"Side over sine of the angle across from it" is the same for all three pairs.

T.16 (1/2)

1. $\sin(kx) = a:$ solve for kx, then divide by k
2. $f(x)\,g(x) = 0 \Rightarrow f(x) = 0$ or $g(x) = 0$

TERMS $n \in \mathbb{Z}$; $|a| \le 1$ for sin, cos; period 2π (sin, cos), π (tan)

"Find them in one lap, then go around again": $+2\pi n$ (or $+\pi n$ for tan).

T.15 (2/2)

1. $\sin x = a \Rightarrow x = \arcsin a + 2\pi n$ or $x = \pi - \arcsin a + 2\pi n$
2. $\cos x = a \Rightarrow x = \pm\arccos a + 2\pi n$
3. $\tan x = a \Rightarrow x = \arctan a + \pi n$

TERMS $n \in \mathbb{Z}$; $|a| \le 1$ for sin, cos; period 2π (sin, cos), π (tan)

"Find them in one lap, then go around again": $+2\pi n$ (or $+\pi n$ for tan).

T.15 (1/2)

1. $c^2 < a^2 + b^2 \iff C < 90°$

TERMS C = angle between a and b, opposite c

It is the Pythagorean theorem with a correction: when $C = 90°$, $\cos C = 0$.

T.17 (2/2)

1. $c^2 = a^2 + b^2 - 2ab\cos C$
2. $\cos C = \dfrac{a^2+b^2-c^2}{2ab}$
3. $a^2 = b^2 + c^2 - 2bc\cos A,\quad b^2 = a^2 + c^2 - 2ac\cos B$

TERMS C = angle between a and b, opposite c

It is the Pythagorean theorem with a correction: when $C = 90°$, $\cos C = 0$.

T.17 (1/2)

1. $h = b\sin A$

TERMS a, b, c opposite A, B, C; R circumradius; K area

"Side over sine of the angle across from it" is the same for all three pairs.

T.16 (2/2)

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T.18 (1/3) TRIGONOMETRY

Triangle Area, Radii & Law of Tangents

PART 1 OF 3

1. Area (SAS)
2. Area (ASA / AAS)
3. Heron's formula (SSS)

STATE EACH NUMBERED FORMULA

T.18 (2/3) TRIGONOMETRY

Triangle Area, Radii & Law of Tangents

PART 2 OF 3

1. Circumradius
2. Inradius

STATE EACH NUMBERED FORMULA

T.18 (3/3) TRIGONOMETRY

Triangle Area, Radii & Law of Tangents

PART 3 OF 3

1. Law of tangents
2. Mollweide's check

STATE EACH NUMBERED FORMULA

T.19 (1/2) TRIGONOMETRY

Polar Coordinates

PART 1 OF 2

1. Polar to Cartesian
2. Radius
3. Angle

STATE EACH NUMBERED FORMULA

T.19 (2/2) TRIGONOMETRY

Polar Coordinates

PART 2 OF 2

1. Non-uniqueness
2. Basic polar curves

STATE EACH NUMBERED FORMULA

T.20 (1/2) TRIGONOMETRY

Vectors: Components, Magnitude & Direction

PART 1 OF 2

1. Components from polar
2. Magnitude
3. Direction angle

STATE EACH NUMBERED FORMULA

T.20 (2/2) TRIGONOMETRY

Vectors: Components, Magnitude & Direction

PART 2 OF 2

1. Unit vector
2. Resultant (sum)
3. Rotation by angle α

STATE EACH NUMBERED FORMULA

T.21 (1/2) TRIGONOMETRY

Vectors, Dot Product & Cosine Similarity

PART 1 OF 2

1. Norm (length)
2. Dot product
3. Cosine similarity

STATE EACH NUMBERED FORMULA

T.21 (2/2) TRIGONOMETRY

Vectors, Dot Product & Cosine Similarity

PART 2 OF 2

1. Orthogonality test
2. Scalar projection
3. Vector projection

STATE EACH NUMBERED FORMULA

T.22 (1/2) TRIGONOMETRY

Polar Form of Complex Numbers

PART 1 OF 2

1. Trigonometric form
2. Modulus and argument
3. Back to rectangular

STATE EACH NUMBERED FORMULA

T.22 (2/2) TRIGONOMETRY

Polar Form of Complex Numbers

PART 2 OF 2

1. Multiplication
2. Division
3. Conjugate

STATE EACH NUMBERED FORMULA

T.23 (1/3) TRIGONOMETRY

Euler's Formula & De Moivre's Theorem

PART 1 OF 3

1. Euler's formula
2. Exponential polar form
3. De Moivre's theorem

STATE EACH NUMBERED FORMULA

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1 $$\frac{a-b}{a+b} = \frac{\tan\frac{A-B}{2}}{\tan\frac{A+B}{2}}$$

2 $$\frac{a+b}{c} = \frac{\cos\frac{A-B}{2}}{\sin\frac{C}{2}}$$

TERMS K area; s semi-perimeter; R circumradius; r inradius

Check with the 3–4–5 triangle: $s = 6$, $K = 6$, $R = \frac{60}{24} = 2.5$ (half the hypotenuse), $r = 1$.

T.18 (3/3)

1 $$R = \frac{abc}{4K}$$

2 $$r = \frac{K}{s}$$

TERMS K area; s semi-perimeter; R circumradius; r inradius

Check with the 3–4–5 triangle: $s = 6$, $K = 6$, $R = \frac{60}{24} = 2.5$ (half the hypotenuse), $r = 1$.

T.18 (2/3)

1 $$K = \frac{1}{2}ab\sin C$$

2 $$K = \frac{a^2 \sin B \sin C}{2\sin A}$$

3 $$K = \sqrt{s(s-a)(s-b)(s-c)}, \quad s = \tfrac{a+b+c}{2}$$

TERMS K area; s semi-perimeter; R circumradius; r inradius

Check with the 3–4–5 triangle: $s = 6$, $K = 6$, $R = \frac{60}{24} = 2.5$ (half the hypotenuse), $r = 1$.

T.18 (1/3)

1 $$\mathbf{v} = \langle \|\mathbf{v}\|\cos\theta,\ \|\mathbf{v}\|\sin\theta \rangle$$

2 $$\|\mathbf{v}\| = \sqrt{v_x^2 + v_y^2}$$

3 $$\theta = \operatorname{atan2}(v_y, v_x)$$

TERMS v_x, v_y components; $\|\mathbf{v}\|$ length; θ direction; α rotation

"Cos goes with x, sin goes with y" (as on the unit circle).

T.20 (1/2)

1 $$(r,\theta) = (r,\theta+2\pi k) = (-r,\theta+\pi)$$

2 $$r = a \text{ (circle)}, \quad \theta = \alpha \text{ (line)}, \quad r = 2a\cos\theta$$

TERMS r distance from origin; θ angle from $+x$-axis

Polar is "how far, which way"; Cartesian is "how far across, how far up".

T.19 (2/2)

1 $$x = r\cos\theta, \qquad y = r\sin\theta$$

2 $$r = \sqrt{x^2 + y^2}$$

3 $$\tan\theta = \frac{y}{x}, \qquad \theta = \operatorname{atan2}(y, x)$$

TERMS r distance from origin; θ angle from $+x$-axis

Polar is "how far, which way"; Cartesian is "how far across, how far up".

T.19 (1/2)

1 $$\mathbf{u}\cdot\mathbf{v} = 0 \iff \theta = \tfrac{\pi}{2}$$

2 $$\operatorname{comp}_{\mathbf{v}}\mathbf{u} = \frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{v}\|} = \|\mathbf{u}\|\cos\theta$$

3 $$\operatorname{proj}_{\mathbf{v}}\mathbf{u} = \frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{v}\|^2}\mathbf{v}$$

TERMS θ angle between $\mathbf{u}, \mathbf{v}$; n dimension; $\|\cdot\|$ length

Positive dot product: acute angle (same-ish direction); zero: perpendicular; negative: obtuse.

T.21 (2/2)

1 $$\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}$$

2 $$\mathbf{u}\cdot\mathbf{v} = \sum_{i=1}^{n} u_i v_i = \|\mathbf{u}\|\,\|\mathbf{v}\|\cos\theta$$

3 $$\cos\theta = \frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{u}\|\,\|\mathbf{v}\|}$$

TERMS θ angle between $\mathbf{u}, \mathbf{v}$; n dimension; $\|\cdot\|$ length

Positive dot product: acute angle (same-ish direction); zero: perpendicular; negative: obtuse.

T.21 (1/2)

1 $$\hat{\mathbf{u}} = \frac{\mathbf{v}}{\|\mathbf{v}\|} = \langle \cos\theta, \sin\theta \rangle$$

2 $$\mathbf{u} + \mathbf{v} = \langle u_x + v_x,\ u_y + v_y \rangle$$

3 $$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

TERMS v_x, v_y components; $\|\mathbf{v}\|$ length; θ direction; α rotation

"Cos goes with x, sin goes with y" (as on the unit circle).

T.20 (2/2)

1 $$e^{i\theta} = \cos\theta + i\sin\theta$$

2 $$z = re^{i\theta} = r(\cos\theta + i\sin\theta)$$

3 $$z^n = r^n e^{in\theta} = r^n(\cos n\theta + i\sin n\theta)$$

TERMS r modulus; θ argument; $k = 0, \ldots, n-1$ root index

Powers: "multiply the angle". Roots: "divide the angle, then add $\frac{2\pi}{n}$ steps".

T.23 (1/3)

1 $$z_1 z_2 = r_1 r_2 \operatorname{cis}(\theta_1 + \theta_2)$$

2 $$\frac{z_1}{z_2} = \frac{r_1}{r_2}\operatorname{cis}(\theta_1 - \theta_2)$$

3 $$\bar{z} = r\operatorname{cis}(-\theta)$$

TERMS $r = |z|$ modulus; $\theta = \arg z$; $\operatorname{cis}\theta = \cos\theta + i\sin\theta$

"Multiply lengths, add angles."

T.22 (2/2)

1 $$z = r(\cos\theta + i\sin\theta) = r\operatorname{cis}\theta$$

2 $$r = |z| = \sqrt{a^2+b^2}, \qquad \theta = \arg z = \operatorname{atan2}(b, a)$$

3 $$a = r\cos\theta, \qquad b = r\sin\theta$$

TERMS $r = |z|$ modulus; $\theta = \arg z$; $\operatorname{cis}\theta = \cos\theta + i\sin\theta$

"Multiply lengths, add angles."

T.22 (1/2)

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T.23 (2/3)

TRIGONOMETRY

Euler's Formula & De Moivre's Theorem

PART 2 OF 3

1. nth roots

STATE EACH NUMBERED FORMULA

T.23 (3/3)

TRIGONOMETRY

Euler's Formula & De Moivre's Theorem

PART 3 OF 3

1. Cosine and sine from exponentials
2. Euler's identity

STATE EACH NUMBERED FORMULA

T.24 (1/3)

TRIGONOMETRY

Hyperbolic Functions

PART 1 OF 3

1. Hyperbolic sine
2. Hyperbolic cosine
3. Hyperbolic tangent

STATE EACH NUMBERED FORMULA

T.24 (2/3)

TRIGONOMETRY

Hyperbolic Functions

PART 2 OF 3

1. Fundamental identity
2. Exponential split
3. Addition and double angle

STATE EACH NUMBERED FORMULA

T.24 (3/3)

TRIGONOMETRY

Hyperbolic Functions

PART 3 OF 3

1. Derivatives
2. Inverse functions
3. Link to trigonometry

STATE EACH NUMBERED FORMULA

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1. $\sinh x = \dfrac{e^x - e^{-x}}{2}$
2. $\cosh x = \dfrac{e^x + e^{-x}}{2}$
3. $\tanh x = \dfrac{\sinh x}{\cosh x} = \dfrac{e^x - e^{-x}}{e^x + e^{-x}}$

TERMS sinh, cosh = odd, even parts of e^x; tanh = sinh / cosh

Osborn's rule: take a trig identity, change cos → cosh, sin → sinh, *and flip the sign of every product of two sines* ($\sin^2 \to -\sinh^2$).

1. $\cos\theta = \dfrac{e^{i\theta} + e^{-i\theta}}{2}, \quad \sin\theta = \dfrac{e^{i\theta} - e^{-i\theta}}{2i}$
2. $e^{i\pi} + 1 = 0$

TERMS r modulus; θ argument; $k = 0, \ldots, n-1$ root index

Powers: "multiply the angle". Roots: "divide the angle, then add $\frac{2\pi}{n}$ *steps".*

1. $z^{1/n} = r^{1/n}\, e^{i(\theta + 2\pi k)/n}, \qquad k = 0, 1, \ldots, n-1$

TERMS r modulus; θ argument; $k = 0, \ldots, n-1$ root index

Powers: "multiply the angle". Roots: "divide the angle, then add $\frac{2\pi}{n}$ *steps".*

1. $\dfrac{d}{dx}\sinh x = \cosh x, \quad \dfrac{d}{dx}\cosh x = \sinh x$
2. $\operatorname{arsinh} x = \ln\left(x + \sqrt{x^2+1}\right), \quad \operatorname{artanh} x = \frac{1}{2}\ln\frac{1+x}{1-x}$
3. $\cos(ix) = \cosh x, \qquad \sin(ix) = i\sinh x$

TERMS sinh, cosh = odd, even parts of e^x; tanh = sinh / cosh

Osborn's rule: take a trig identity, change cos → cosh, sin → sinh, *and flip the sign of every product of two sines* ($\sin^2 \to -\sinh^2$).

1. $\cosh^2 x - \sinh^2 x = 1$
2. $e^x = \cosh x + \sinh x$
3. $\sinh(x+y) = \sinh x \cosh y + \cosh x \sinh y, \quad \cosh 2x = \cosh^2 x + \sinh^2 x$

TERMS sinh, cosh = odd, even parts of e^x; tanh = sinh / cosh

Osborn's rule: take a trig identity, change cos → cosh, sin → sinh, *and flip the sign of every product of two sines* ($\sin^2 \to -\sinh^2$).

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C1.1 (1/3) CALCULUS I

Limits & Limit Laws

PART 1 OF 3

1. Limit
2. Sum & constant multiple laws
3. Product law

STATE EACH NUMBERED FORMULA

C1.1 (2/3) CALCULUS I

Limits & Limit Laws

PART 2 OF 3

1. Quotient law
2. Power & root laws
3. Direct substitution

STATE EACH NUMBERED FORMULA

C1.1 (3/3) CALCULUS I

Limits & Limit Laws

PART 3 OF 3

1. Existence (two-sided)
2. ε–δ definition

STATE EACH NUMBERED FORMULA

C1.2 (1/2) CALCULUS I

One-Sided & Infinite Limits

PART 1 OF 2

1. Left-hand limit
2. Right-hand limit
3. Infinite limit

STATE EACH NUMBERED FORMULA

C1.2 (2/2) CALCULUS I

One-Sided & Infinite Limits

PART 2 OF 2

1. Vertical asymptote
2. Nonzero over zero
3. Standard examples

STATE EACH NUMBERED FORMULA

C1.3 (1/3) CALCULUS I

Limits at Infinity & Dominant Terms

PART 1 OF 3

1. Reciprocal power
2. Divide by the highest power
3. Horizontal asymptote

STATE EACH NUMBERED FORMULA

C1.3 (2/3) CALCULUS I

Limits at Infinity & Dominant Terms

PART 2 OF 3

1. Degree rule (rational functions)
2. Growth hierarchy

STATE EACH NUMBERED FORMULA

C1.3 (3/3) CALCULUS I

Limits at Infinity & Dominant Terms

PART 3 OF 3

1. Standard limits at infinity
2. Roots at $-\infty$

STATE EACH NUMBERED FORMULA

C1.4 (1/3) CALCULUS I

Indeterminate Forms

PART 1 OF 3

1. Quotient forms
2. Product form
3. Difference form

STATE EACH NUMBERED FORMULA

C1.4 (2/3) CALCULUS I

Indeterminate Forms

PART 2 OF 3

1. Power forms
2. Exponential trick
3. 1^{∞} shortcut

STATE EACH NUMBERED FORMULA

C1.4 (3/3) CALCULUS I

Indeterminate Forms

PART 3 OF 3

1. Not indeterminate

STATE EACH NUMBERED FORMULA

C1.5 (1/2) CALCULUS I

Squeeze Theorem

PART 1 OF 2

1. Squeeze theorem
2. Bounded times vanishing
3. Absolute-value form

STATE EACH NUMBERED FORMULA

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1 $\lim_{x\to a} f(x)$ exists $\iff \lim_{x\to a^-} f(x) = \lim_{x\to a^+} f(x)$

2 $\lim_{x\to a} f(x) = L \iff \forall \varepsilon > 0\ \exists \delta > 0 : \quad 0 < |x-a| < \delta \implies |f(x) - L| < \varepsilon$

TERMS a point approached; L limit; $a^{\mp}$ from left/right; ε, δ tolerances

A limit asks where f is heading, *not where it* arrives.

1 $\lim_{x\to a} \dfrac{f}{g} = \dfrac{\lim f}{\lim g} \quad (\lim g \neq 0)$

2 $\lim_{x\to a}[f]^n = \left[\lim_{x\to a} f\right]^n, \quad \lim_{x\to a} \sqrt[n]{f} = \sqrt[n]{\lim_{x\to a} f}$

3 $\lim_{x\to a} f(x) = f(a)$ if f is continuous at a

TERMS a point approached; L limit; $a^{\mp}$ from left/right; ε, δ tolerances

A limit asks where f is heading, *not where it* arrives.

1 $\lim_{x\to a} f(x) = L$

2 $\lim_{x\to a}\big[c\,f \pm g\big] = c \lim_{x\to a} f \pm \lim_{x\to a} g$

3 $\lim_{x\to a}\big[f\,g\big] = \lim_{x\to a} f \cdot \lim_{x\to a} g$

TERMS a point approached; L limit; $a^{\mp}$ from left/right; ε, δ tolerances

A limit asks where f is heading, *not where it* arrives.

1 $\lim_{x\to\pm\infty} \dfrac{c}{x^p} = 0 \quad (p > 0)$

2 $\dfrac{3x^2+1}{5x^2-x} = \dfrac{3+1/x^2}{5-1/x} \xrightarrow[x\to\infty]{} \dfrac{3}{5}$

3 $y = L$ if $\lim_{x\to\infty} f(x) = L$ or $\lim_{x\to-\infty} f(x) = L$

TERMS n, m degrees; a_n, b_m leading coefficients; $\ll$ ratio $\to 0$

BOBO BOTN EATS DC: Bigger On Bottom $\to$ 0; Bigger On Top $\to$ None (infinite); Exponents Are The Same $\to$ Divide Coefficients.

1 $x = a$ if $\lim_{x\to a^{\pm}} f(x) = \pm\infty$

2 $\lim_{x\to a^{\pm}} \dfrac{c}{x-a} = \pm\infty \quad (c > 0)$

3 $\lim_{x\to 0^{\pm}} \dfrac{1}{x} = \pm\infty, \quad \lim_{x\to 0} \dfrac{1}{x^2} = +\infty, \quad \lim_{x\to 0^+} \ln x = -\infty$

TERMS $a^{\mp}$ from left/right; ∞ unbounded (not a number); $c \neq 0$

"Nonzero over zero" is never indeterminate: it is always $\pm\infty$ (or does not exist). Only the sign *needs work.*

1 $\lim_{x\to a^-} f(x) = L$

2 $\lim_{x\to a^+} f(x) = L$

3 $\lim_{x\to a} f(x) = +\infty$

TERMS $a^{\mp}$ from left/right; ∞ unbounded (not a number); $c \neq 0$

"Nonzero over zero" is never indeterminate: it is always $\pm\infty$ (or does not exist). Only the sign *needs work.*

1 $\dfrac{0}{0}, \qquad \dfrac{\infty}{\infty}$

2 $0 \cdot \infty \longrightarrow \dfrac{f}{1/g}$ or $\dfrac{g}{1/f}$

3 $\infty - \infty \longrightarrow$ one fraction

TERMS f, g functions; $\exp(u) = e^u$; $c \neq 0$ constant

Seven indeterminate forms:
$\frac{0}{0}$, $\frac{\infty}{\infty}$, $0\cdot\infty$, $\infty - \infty$, 1^∞, 0^0, ∞^0.

1 $e^{-x} \to 0, \quad \arctan x \to \pm\frac{\pi}{2}, \quad \dfrac{\ln x}{x} \to 0$

2 $\sqrt{x^2} = |x| = -x\ (x < 0) \implies \lim_{x\to-\infty} \dfrac{\sqrt{x^2+1}}{x} = -1$

TERMS n, m degrees; a_n, b_m leading coefficients; $\ll$ ratio $\to 0$

BOBO BOTN EATS DC: Bigger On Bottom $\to$ 0; Bigger On Top $\to$ None (infinite); Exponents Are The Same $\to$ Divide Coefficients.

1 $\lim_{x\to\pm\infty} \dfrac{a_n x^n + \cdots}{b_m x^m + \cdots} = \begin{cases} 0, & n < m \\ \frac{a_n}{b_m}, & n = m \\ \pm\infty, & n > m \end{cases}$

2 $\ln n \ll n^p \ll a^n \ll n! \ll n^n \quad (p > 0,\ a > 1)$

TERMS n, m degrees; a_n, b_m leading coefficients; $\ll$ ratio $\to 0$

BOBO BOTN EATS DC: Bigger On Bottom $\to$ 0; Bigger On Top $\to$ None (infinite); Exponents Are The Same $\to$ Divide Coefficients.

1 $g \le f \le h$ near a, $\lim_{x\to a} g = \lim_{x\to a} h = L \implies \lim_{x\to a} f = L$

2 $|f(x)| \le M$, $\lim g = 0 \implies \lim f\,g = 0$

3 $|f(x) - L| \le g(x) \to 0 \implies f(x) \to L$

TERMS $g \le f \le h$ near a; L common limit; M a bound

Two police officers walking a suspect to the same cell: wherever they end up, so does the suspect.

1 $\frac{c}{0}\ (c \neq 0) \to \pm\infty, \quad \frac{0}{\infty} \to 0, \quad \infty + \infty \to \infty, \quad 0^{+\infty} \to 0$

TERMS f, g functions; $\exp(u) = e^u$; $c \neq 0$ constant

Seven indeterminate forms:
$\frac{0}{0}$, $\frac{\infty}{\infty}$, $0\cdot\infty$, $\infty - \infty$, 1^∞, 0^0, ∞^0.

1 $1^\infty, \qquad 0^0, \qquad \infty^0$

2 $\lim f^{\,g} = \exp\Big(\lim g \ln f\Big)$

3 $f \to 1,\ g \to \infty : \quad \lim f^{\,g} = e^{\lim g\,(f-1)}$

TERMS f, g functions; $\exp(u) = e^u$; $c \neq 0$ constant

Seven indeterminate forms:
$\frac{0}{0}$, $\frac{\infty}{\infty}$, $0\cdot\infty$, $\infty - \infty$, 1^∞, 0^0, ∞^0.

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C1.5 (2/2) CALCULUS I

Squeeze Theorem

PART 2 OF 2

1. Classic oscillation
2. Sine limit (proof)

STATE EACH NUMBERED FORMULA

C1.6 (1/3) CALCULUS I

Special Limits & the Number e

PART 1 OF 3

1. Sine limit
2. Scaled sine
3. Cosine limits

STATE EACH NUMBERED FORMULA

C1.6 (2/3) CALCULUS I

Special Limits & the Number e

PART 2 OF 3

1. Exponential limit
2. Logarithm limit

STATE EACH NUMBERED FORMULA

C1.6 (3/3) CALCULUS I

Special Limits & the Number e

PART 3 OF 3

1. Definition of e
2. Continuous growth

STATE EACH NUMBERED FORMULA

C1.K1 CALCULUS I

Evaluating a limit: the six steps

RECALL IT

C1.K1 CALCULUS I

$\frac{0}{0}$ limit: which algebraic move?

RECALL IT

C1.K1 CALCULUS I

Limit at $\pm\infty$ of a rational function

RECALL IT

C1.K1 CALCULUS I

ε–δ definition of a limit

RECALL IT

C1.7 (1/2) CALCULUS I

Continuity & the Intermediate Value Theorem

PART 1 OF 2

1. Continuity at a point
2. Continuity on an interval
3. Combinations

STATE EACH NUMBERED FORMULA

C1.7 (2/2) CALCULUS I

Continuity & the Intermediate Value Theorem

PART 2 OF 2

1. Composition
2. Intermediate value theorem
3. Root location (Bolzano)

STATE EACH NUMBERED FORMULA

C1.8 (1/2) CALCULUS I

Types of Discontinuity

PART 1 OF 2

1. Removable
2. Jump
3. Infinite

STATE EACH NUMBERED FORMULA

C1.8 (2/2) CALCULUS I

Types of Discontinuity

PART 2 OF 2

1. Oscillating (essential)
2. Continuous extension
3. Piecewise boundary check

STATE EACH NUMBERED FORMULA

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1. $\displaystyle\lim_{x\to0}\frac{e^x-1}{x}=1$
2. $\displaystyle\lim_{x\to0}\frac{\ln(1+x)}{x}=1$

TERMS x in radians; $e\approx2.71828$; k,r constants

Small x (radians): $\sin x\approx x,\ 1-\cos x\approx\frac{x^2}{2}$, $e^x\approx1+x,\ \ln(1+x)\approx x$.

1. $\displaystyle\lim_{x\to0}\frac{\sin x}{x}=1$
2. $\displaystyle\lim_{x\to0}\frac{\sin kx}{x}=k$
3. $\displaystyle\lim_{x\to0}\frac{1-\cos x}{x}=0,\qquad\lim_{x\to0}\frac{1-\cos x}{x^2}=\frac{1}{2}$

TERMS x in radians; $e\approx2.71828$; k,r constants

Small x (radians): $\sin x\approx x,\ 1-\cos x\approx\frac{x^2}{2}$, $e^x\approx1+x,\ \ln(1+x)\approx x$.

1. $-|x|\le x\sin\frac{1}{x}\le|x|\ \Rightarrow\ \displaystyle\lim_{x\to0}x\sin\frac{1}{x}=0$
2. $\cos x\le\dfrac{\sin x}{x}\le1\ \Rightarrow\ \displaystyle\lim_{x\to0}\frac{\sin x}{x}=1$

TERMS $g\le f\le h$ near a; L common limit; M a bound

Two police officers walking a suspect to the same cell: wherever they end up, so does the suspect.

Polynomial zero at a: factor out $(x-a)$, cancel.
Root in a sum/difference: conjugate.
Nested fractions: common denominator.
$x\to\infty$: divide by highest power.
Trig/exp near 0: match a special limit.

1 Direct substitution.
2 Classify: number, $\frac{c}{0}$, or indeterminate.
3 Algebra: factor, conjugate, common denominator, divide by highest power.
4 Special limits / squeeze.
5 L'Hôpital ($\frac{0}{0}$, $\frac{\infty}{\infty}$ only).
6 One-sided check.

1. $\displaystyle\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n=e$
2. $\displaystyle\lim_{n\to\infty}\left(1+\frac{r}{n}\right)^n=e^r$

TERMS x in radians; $e\approx2.71828$; k,r constants

Small x (radians): $\sin x\approx x,\ 1-\cos x\approx\frac{x^2}{2}$, $e^x\approx1+x,\ \ln(1+x)\approx x$.

1. f continuous at $a\iff\displaystyle\lim_{x\to a}f(x)=f(a)$
2. $f\in C[a,b]:$ continuous on (a,b), $\lim_{x\to a^+}f=f(a)$, $\lim_{x\to b^-}f=f(b)$
3. f,g continuous at $a\implies f\pm g,\ fg,\ \frac{f}{g}\ (g(a)\neq0)$ continuous at a

TERMS $C[a,b]$ continuous on $[a,b]$; N target value; c point where $f(c)=N$

Continuous = you can draw the graph without lifting your pencil.

$\lim_{x\to a}f(x)=L$ iff for every $\varepsilon>0$ there is $\delta>0$ with $0<|x-a|<\delta\Rightarrow|f(x)-L|<\varepsilon$.

Output tolerance ε is met by input tolerance δ.

Degree top n, bottom m:
$n<m\to0$; $\quad n=m\to\frac{a_n}{b_m}$;
$n>m\to\pm\infty$.

Growth: $\ln x\ll x^p\ll a^x\ll x!\ll x^x$.

1. $\displaystyle\lim_{x\to0}\sin\frac{1}{x}$ does not exist
2. $\tilde f(x)=\begin{cases}f(x), & x\neq a\\ \lim_{x\to a}f(x), & x=a\end{cases}$
3. $\displaystyle\lim_{x\to a^-}f_{\text{left}}=\lim_{x\to a^+}f_{\text{right}}=f(a)$

TERMS L limit; $a^{\mp}$ left/right; $\tilde f$ extended function

Hole, step, wall, wiggle: only the hole can be patched.

1. $\displaystyle\lim_{x\to a}f(x)=L$ exists, but $f(a)\neq L$ or $f(a)$ undefined
2. $\displaystyle\lim_{x\to a^-}f(x)\neq\lim_{x\to a^+}f(x)\quad$ (both finite)
3. $\displaystyle\lim_{x\to a^{\pm}}f(x)=\pm\infty$

TERMS L limit; $a^{\mp}$ left/right; $\tilde f$ extended function

Hole, step, wall, wiggle: only the hole can be patched.

1. g cont. at a, f cont. at $g(a)\implies\displaystyle\lim_{x\to a}f(g(x))=f\Big(\lim_{x\to a}g(x)\Big)$
2. $f\in C[a,b]$, N between $f(a),f(b)\implies\exists\,c\in(a,b):\ f(c)=N$
3. $f\in C[a,b]$, $f(a)\,f(b)<0\implies\exists\,c\in(a,b):\ f(c)=0$

TERMS $C[a,b]$ continuous on $[a,b]$; N target value; c point where $f(c)=N$

Continuous = you can draw the graph without lifting your pencil.

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C1.9 (1/2) CALCULUS I

Definition of the Derivative

PART 1 OF 2

1. Derivative (limit definition)
2. Derivative at a point
3. Difference quotient

STATE EACH NUMBERED FORMULA

C1.9 (2/2) CALCULUS I

Definition of the Derivative

PART 2 OF 2

1. Notations
2. Forward difference
3. Central difference

STATE EACH NUMBERED FORMULA

C1.10 (1/2) CALCULUS I

Differentiability vs. Continuity

PART 1 OF 2

1. Differentiable implies continuous
2. Converse fails
3. One-sided derivatives

STATE EACH NUMBERED FORMULA

C1.10 (2/2) CALCULUS I

Differentiability vs. Continuity

PART 2 OF 2

1. Corner
2. Cusp
3. Vertical tangent

STATE EACH NUMBERED FORMULA

C1.11 (1/2) CALCULUS I

Power, Constant & Sum Rules

PART 1 OF 2

1. Constant rule
2. Power rule
3. Constant multiple rule

STATE EACH NUMBERED FORMULA

C1.11 (2/2) CALCULUS I

Power, Constant & Sum Rules

PART 2 OF 2

1. Sum & difference rule
2. Polynomial derivative
3. Linear function

STATE EACH NUMBERED FORMULA

C1.12 (1/2) CALCULUS I

Product & Quotient Rules

PART 1 OF 2

1. Product rule
2. Quotient rule
3. Triple product

STATE EACH NUMBERED FORMULA

C1.12 (2/2) CALCULUS I

Product & Quotient Rules

PART 2 OF 2

1. Reciprocal rule
2. Sigmoid derivative

STATE EACH NUMBERED FORMULA

C1.13 (1/2) CALCULUS I

Chain Rule

PART 1 OF 2

1. Chain rule (Lagrange)
2. Chain rule (Leibniz)
3. Longer chains

STATE EACH NUMBERED FORMULA

C1.13 (2/2) CALCULUS I

Chain Rule

PART 2 OF 2

1. General power rule
2. Exponential of a function
3. Log of a function

STATE EACH NUMBERED FORMULA

C1.14 (1/3) CALCULUS I

Derivatives of Trig, Exponential & Log Functions

PART 1 OF 3

1. Sine & cosine
2. Tangent & cotangent
3. Secant & cosecant

STATE EACH NUMBERED FORMULA

C1.14 (2/3) CALCULUS I

Derivatives of Trig, Exponential & Log Functions

PART 2 OF 3

1. Natural exponential
2. General exponential

STATE EACH NUMBERED FORMULA

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1. $f'(a)$ exists $\implies \lim_{x\to a} f(x) = f(a)$
2. f continuous at $a \not\implies f'(a)$ exists
3. $f'(a)$ exists $\iff f'_-(a) = f'_+(a)$

TERMS $f'_\mp$ one-sided derivatives; corner, cusp, vertical tangent

Smooth $\Rightarrow$ connected, but connected $\not\Rightarrow$ smooth: a crease or a spike is still continuous.

1. $f'(x) = \dfrac{dy}{dx} = \dfrac{d}{dx}f(x) = Df(x) = \dot{y}$
2. $f'(x) \approx \dfrac{f(x+h)-f(x)}{h}$, error $O(h)$
3. $f'(x) \approx \dfrac{f(x+h)-f(x-h)}{2h}$, error $O(h^2)$

TERMS $h \to 0$ step; $\Delta y/\Delta x$ secant slope; $O(h^k)$ error order

"Rise over run, with the run shrinking to zero."

1. $f'(x) = \dfrac{dy}{dx} = \lim_{h\to 0} \dfrac{f(x+h)-f(x)}{h}$
2. $f'(a) = \lim_{x\to a} \dfrac{f(x)-f(a)}{x-a}$
3. $\dfrac{\Delta y}{\Delta x} = \dfrac{f(x+h)-f(x)}{h}$

TERMS $h \to 0$ step; $\Delta y/\Delta x$ secant slope; $O(h^k)$ error order

"Rise over run, with the run shrinking to zero."

1. $\dfrac{d}{dx}\left[f(x) \pm g(x)\right] = f'(x) \pm g'(x)$
2. $\dfrac{d}{dx}\sum_{k=0}^{n} a_k x^k = \sum_{k=1}^{n} k\, a_k x^{k-1}$
3. $\dfrac{d}{dx}(mx+b) = m$

TERMS c constant; n any real exponent; a_k coefficients

Power rule: "bring the power down, subtract one."

1. $\dfrac{d}{dx}\, c = 0$
2. $\dfrac{d}{dx}\, x^n = n\, x^{n-1}$
3. $\dfrac{d}{dx}\left[c\, f(x)\right] = c\, f'(x)$

TERMS c constant; n any real exponent; a_k coefficients

Power rule: "bring the power down, subtract one."

1. $f(x) = |x|: \quad f'_-(0) = -1 \neq 1 = f'_+(0)$
2. $f(x) = x^{2/3}: \quad f'(x) = \frac{2}{3}x^{-1/3} \to \pm\infty$ as $x \to 0^{\pm}$
3. $f(x) = x^{1/3}: \quad f'(x) = \frac{1}{3}x^{-2/3} \to +\infty$ as $x \to 0$

TERMS $f'_\mp$ one-sided derivatives; corner, cusp, vertical tangent

Smooth $\Rightarrow$ connected, but connected $\not\Rightarrow$ smooth: a crease or a spike is still continuous.

1. $\dfrac{d}{dx} f\big(g(x)\big) = f'\big(g(x)\big) \cdot g'(x)$
2. $\dfrac{dy}{dx} = \dfrac{dy}{du} \cdot \dfrac{du}{dx}$
3. $\dfrac{d}{dx} f\Big(g\big(h(x)\big)\Big) = f'\big(g(h)\big)\, g'(h)\, h'(x)$

TERMS $u = g(x)$ inner; $y = f(u)$ outer; one factor per link

"Derivative of the outside (inside left alone) times derivative of the inside."

1. $\dfrac{d}{dx}\left[\dfrac{1}{g}\right] = -\dfrac{g'}{g^2}$
2. $\sigma(z) = \dfrac{1}{1+e^{-z}} \implies \sigma'(z) = \sigma(z)\big(1-\sigma(z)\big)$

TERMS f, g, h differentiable; $g \neq 0$; σ sigmoid

Quotient: "low d-high minus high d-low, over the square of what's below."

1. $\dfrac{d}{dx}\left[f\, g\right] = f'g + f\, g'$
2. $\dfrac{d}{dx}\left[\dfrac{f}{g}\right] = \dfrac{f'g - f\, g'}{g^2}$
3. $(fgh)' = f'gh + fg'h + fgh'$

TERMS f, g, h differentiable; $g \neq 0$; σ sigmoid

Quotient: "low d-high minus high d-low, over the square of what's below."

1. $\dfrac{d}{dx}e^x = e^x$
2. $\dfrac{d}{dx}a^x = a^x \ln a$

TERMS x in radians; $a > 0$ base; ln natural log

Derivatives of "co" functions (cos, cot, csc) get a minus sign. The derivatives of tan, sec *use* sec; *those of* cot, csc *use* csc.

1. $\dfrac{d}{dx}\sin x = \cos x, \qquad \dfrac{d}{dx}\cos x = -\sin x$
2. $\dfrac{d}{dx}\tan x = \sec^2 x, \qquad \dfrac{d}{dx}\cot x = -\csc^2 x$
3. $\dfrac{d}{dx}\sec x = \sec x \tan x, \qquad \dfrac{d}{dx}\csc x = -\csc x \cot x$

TERMS x in radians; $a > 0$ base; ln natural log

Derivatives of "co" functions (cos, cot, csc) get a minus sign. The derivatives of tan, sec *use* sec; *those of* cot, csc *use* csc.

1. $\dfrac{d}{dx}\left[g(x)\right]^n = n\left[g(x)\right]^{n-1} g'(x)$
2. $\dfrac{d}{dx}e^{g(x)} = e^{g(x)}\, g'(x)$
3. $\dfrac{d}{dx}\ln g(x) = \dfrac{g'(x)}{g(x)}$

TERMS $u = g(x)$ inner; $y = f(u)$ outer; one factor per link

"Derivative of the outside (inside left alone) times derivative of the inside."

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C1.14 (3/3) CALCULUS I

Derivatives of Trig, Exponential & Log Functions

PART 3 OF 3

1. Natural logarithm
2. Log of absolute value
3. General logarithm

STATE EACH NUMBERED FORMULA

C1.15 (1/2) CALCULUS I

Inverse Functions & Inverse Trig Derivatives

PART 1 OF 2

1. Inverse function derivative
2. Leibniz form
3. Arcsine

STATE EACH NUMBERED FORMULA

C1.15 (2/2) CALCULUS I

Inverse Functions & Inverse Trig Derivatives

PART 2 OF 2

1. Arccosine
2. Arctangent
3. Arcsecant

STATE EACH NUMBERED FORMULA

C1.16 (1/2) CALCULUS I

Hyperbolic Functions & Their Derivatives

PART 1 OF 2

1. Hyperbolic sine & cosine
2. Hyperbolic tangent
3. Fundamental identity

STATE EACH NUMBERED FORMULA

C1.16 (2/2) CALCULUS I

Hyperbolic Functions & Their Derivatives

PART 2 OF 2

1. Derivatives of sinh & cosh
2. Derivative of tanh
3. Inverse hyperbolic derivatives

STATE EACH NUMBERED FORMULA

C1.K2 CALCULUS I

The five derivative combination rules

RECALL IT

C1.K2 CALCULUS I

Which derivative rule do I apply first?

RECALL IT

C1.K2 CALCULUS I

Derivatives of the six trig functions

RECALL IT

C1.K2 CALCULUS I

Derivatives of arcsin, arccos, arctan

RECALL IT

C1.17 (1/2) CALCULUS I

Implicit Differentiation

PART 1 OF 2

1. Power of y
2. Product with y
3. Function of y

STATE EACH NUMBERED FORMULA

C1.17 (2/2) CALCULUS I

Implicit Differentiation

PART 2 OF 2

1. Implicit slope formula
2. Circle example
3. Second derivative

STATE EACH NUMBERED FORMULA

C1.18 (1/2) CALCULUS I

Logarithmic Differentiation

PART 1 OF 2

1. Logarithmic derivative
2. Recover the derivative
3. Variable base and exponent

STATE EACH NUMBERED FORMULA

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1. $\dfrac{d}{dx}\arccos x = -\dfrac{1}{\sqrt{1-x^2}}$
2. $\dfrac{d}{dx}\arctan x = \dfrac{1}{1+x^2}$
3. $\dfrac{d}{dx}\operatorname{arcsec} x = \dfrac{1}{|x|\sqrt{x^2-1}} \quad (|x|>1)$

TERMS f^{-1} inverse; $b=f(a)$; $|x|<1$ for arcsin

Inverse trig derivatives are algebraic: no trig left. "co" inverses get a minus sign.

1. $(f^{-1})'(b) = \dfrac{1}{f'(a)}, \quad b=f(a)$
2. $\dfrac{dx}{dy} = \dfrac{1}{dy/dx}$
3. $\dfrac{d}{dx}\arcsin x = \dfrac{1}{\sqrt{1-x^2}} \quad (|x|<1)$

TERMS f^{-1} inverse; $b=f(a)$; $|x|<1$ for arcsin

Inverse trig derivatives are algebraic: no trig left. "co" inverses get a minus sign.

1. $\dfrac{d}{dx}\ln x = \dfrac{1}{x} \quad (x>0)$
2. $\dfrac{d}{dx}\ln|x| = \dfrac{1}{x} \quad (x\neq 0)$
3. $\dfrac{d}{dx}\log_a x = \dfrac{1}{x\ln a}$

TERMS x in radians; $a>0$ base; ln natural log

Derivatives of "co" functions (cos, cot, csc) get a minus sign. The derivatives of tan, sec *use* sec; *those of* cot, csc *use* csc.

Linearity: $(af+bg)' = af'+bg'$.
Power: $(x^n)' = nx^{n-1}$.
Product: $(fg)' = f'g+fg'$.
Quotient: $(f/g)' = (f'g-fg')/g^2$.
Chain: $(f(g))' = f'(g)\,g'$.

1. $\dfrac{d}{dx}\sinh x = \cosh x, \qquad \dfrac{d}{dx}\cosh x = \sinh x$
2. $\dfrac{d}{dx}\tanh x = \operatorname{sech}^2 x = 1-\tanh^2 x$
3. $\dfrac{d}{dx}\operatorname{arsinh} x = \dfrac{1}{\sqrt{x^2+1}}, \quad \dfrac{d}{dx}\operatorname{artanh} x = \dfrac{1}{1-x^2}$

TERMS sinh, cosh, tanh hyperbolic; sech $=1/\cosh$

Hyperbolic = trig with the signs "fixed": $\cosh' = +\sinh$, *and the identity has a minus:* $\cosh^2 - \sinh^2 = 1$.

1. $\sinh x = \dfrac{e^x - e^{-x}}{2}, \qquad \cosh x = \dfrac{e^x + e^{-x}}{2}$
2. $\tanh x = \dfrac{\sinh x}{\cosh x} = \dfrac{e^x - e^{-x}}{e^x + e^{-x}}$
3. $\cosh^2 x - \sinh^2 x = 1$

TERMS sinh, cosh, tanh hyperbolic; sech $=1/\cosh$

Hyperbolic = trig with the signs "fixed": $\cosh' = +\sinh$, *and the identity has a minus:* $\cosh^2 - \sinh^2 = 1$.

$\arcsin x \to \frac{1}{\sqrt{1-x^2}}$,
$\arccos x \to -\frac{1}{\sqrt{1-x^2}}$,
$\arctan x \to \frac{1}{1+x^2}$. With $u=g(x)$, multiply by u'.

$\sin \to \cos$, $\cos \to -\sin$, $\tan \to \sec^2$,
$\cot \to -\csc^2$, $\sec \to \sec\tan$,
$\csc \to -\csc\cot$.
"co" functions take the minus sign.

The rule for the *last* operation you would do to evaluate the expression (outermost layer).
Simplify first; logs for many factors or x^x; implicit if y is not isolated.

1. $\dfrac{d}{dx}\ln f(x) = \dfrac{f'(x)}{f(x)}$
2. $f'(x) = f(x)\cdot\dfrac{d}{dx}\ln f(x)$
3. $\dfrac{d}{dx}\,x^x = x^x(\ln x + 1)$

TERMS f'/f relative rate; $f>0$; $f_1\cdots f_n$ factors

"Log it, differentiate, multiply back."

1. $F(x,y)=0 \implies \dfrac{dy}{dx} = -\dfrac{F_x}{F_y}$
2. $x^2+y^2=r^2 \implies \dfrac{dy}{dx} = -\dfrac{x}{y}$
3. $\dfrac{d^2y}{dx^2} = \dfrac{d}{dx}\left(-\dfrac{x}{y}\right) = -\dfrac{y - x\,y'}{y^2} = -\dfrac{r^2}{y^3}$

TERMS $y=y(x)$; $y'=dy/dx$; F_x, F_y partials

"Every y gets a y'."

1. $\dfrac{d}{dx}\left[y^n\right] = n\,y^{n-1}\dfrac{dy}{dx}$
2. $\dfrac{d}{dx}\left[x\,y\right] = y + x\dfrac{dy}{dx}$
3. $\dfrac{d}{dx}\,g(y) = g'(y)\,\dfrac{dy}{dx}$

TERMS $y=y(x)$; $y'=dy/dx$; F_x, F_y partials

"Every y gets a y'."

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C1.18 (2/2) CALCULUS I

Logarithmic Differentiation

PART 2 OF 2

1. General f^g
2. Products of many factors

STATE EACH NUMBERED FORMULA

C1.19 (1/2) CALCULUS I

Higher-Order Derivatives

PART 1 OF 2

1. Second derivative
2. nth derivative
3. Motion

STATE EACH NUMBERED FORMULA

C1.19 (2/2) CALCULUS I

Higher-Order Derivatives

PART 2 OF 2

1. Powers
2. Exponential & trig patterns
3. General Leibniz rule

STATE EACH NUMBERED FORMULA

C1.20 (1/2) CALCULUS I

Related Rates

PART 1 OF 2

1. Time chain rule
2. Expanding circle
3. Expanding sphere

STATE EACH NUMBERED FORMULA

C1.20 (2/2) CALCULUS I

Related Rates

PART 2 OF 2

1. Sliding ladder
2. Filling cone

STATE EACH NUMBERED FORMULA

C1.21 (1/2) CALCULUS I

Tangent Lines & Linear Approximation

PART 1 OF 2

1. Tangent line
2. Normal line
3. Linearization

STATE EACH NUMBERED FORMULA

C1.21 (2/2) CALCULUS I

Tangent Lines & Linear Approximation

PART 2 OF 2

1. Approximation error
2. Standard linearizations

STATE EACH NUMBERED FORMULA

C1.22 (1/2) CALCULUS I

Differentials & Error Propagation

PART 1 OF 2

1. Differential
2. Change estimate
3. Relative error

STATE EACH NUMBERED FORMULA

C1.22 (2/2) CALCULUS I

Differentials & Error Propagation

PART 2 OF 2

1. Power-law errors
2. Differential rules

STATE EACH NUMBERED FORMULA

C1.23 (1/2) CALCULUS I

Critical Points, Extrema & Concavity

PART 1 OF 2

1. Critical point
2. Fermat's theorem
3. Increasing / decreasing test

STATE EACH NUMBERED FORMULA

C1.23 (2/2) CALCULUS I

Critical Points, Extrema & Concavity

PART 2 OF 2

1. Concavity test
2. Inflection point
3. Second-derivative test

STATE EACH NUMBERED FORMULA

C1.24 CALCULUS I

Extreme Value Theorem & Closed-Interval Method

1. Extreme value theorem
2. Candidate set
3. Absolute vs. local

STATE EACH NUMBERED FORMULA

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1. $\dfrac{d^n}{dx^n}x^m = \dfrac{m!}{(m-n)!}x^{m-n}\ (n \le m), \qquad \dfrac{d^{m+1}}{dx^{m+1}}x^m = 0$
2. $\dfrac{d^n}{dx^n}e^{kx} = k^n e^{kx}, \qquad \dfrac{d^n}{dx^n}\sin x = \sin\left(x + \dfrac{n\pi}{2}\right)$
3. $(fg)^{(n)} = \displaystyle\sum_{k=0}^{n}\binom{n}{k}f^{(k)}g^{(n-k)}$

TERMS $f^{(n)}$ nth derivative; s, v, a position, velocity, accel.

Sine's derivatives: $\sin \to \cos \to -\sin \to -\cos \to \sin$, *a four-step cycle.*

1. $f''(x) = \dfrac{d^2y}{dx^2} = \dfrac{d}{dx}\left(\dfrac{dy}{dx}\right)$
2. $f^{(n)}(x) = \dfrac{d^ny}{dx^n}$
3. $s(t), \quad v = s'(t), \quad a = v' = s''(t), \quad j = s'''(t)$

TERMS $f^{(n)}$ nth derivative; s, v, a position, velocity, accel.

Sine's derivatives: $\sin \to \cos \to -\sin \to -\cos \to \sin$, *a four-step cycle.*

1. $\dfrac{d}{dx}f^g = f^g\left(g'\ln f + \dfrac{gf'}{f}\right)$
2. $\dfrac{(f_1f_2\cdots f_n)'}{f_1f_2\cdots f_n} = \dfrac{f_1'}{f_1} + \dfrac{f_2'}{f_2} + \cdots + \dfrac{f_n'}{f_n}$

TERMS f'/f relative rate; $f > 0$; $f_1 \cdots f_n$ factors

"Log it, differentiate, multiply back."

1. $y - f(a) = f'(a)\,(x-a)$
2. $y - f(a) = -\dfrac{1}{f'(a)}\,(x-a)$
3. $f(x) \approx L(x) = f(a) + f'(a)\,(x-a)$

TERMS a base point; $L(x)$ linearization; f'' curvature

"Near the point, the curve is the tangent line."

1. $x^2 + y^2 = \ell^2 \ \Rightarrow\ x\dfrac{dx}{dt} + y\dfrac{dy}{dt} = 0$
2. $V = \frac{1}{3}\pi r^2 h, \quad \dfrac{r}{h} = \dfrac{R}{H} \ \Rightarrow\ V = \dfrac{\pi R^2}{3H^2}h^3$

TERMS t time; $\frac{dx}{dt}$ rate; ℓ, R, H fixed sizes

"Differentiate first, substitute numbers after."

1. $\dfrac{d}{dt}f(x(t)) = f'(x)\,\dfrac{dx}{dt}$
2. $A = \pi r^2 \ \Rightarrow\ \dfrac{dA}{dt} = 2\pi r\dfrac{dr}{dt}$
3. $V = \frac{4}{3}\pi r^3 \ \Rightarrow\ \dfrac{dV}{dt} = 4\pi r^2\dfrac{dr}{dt}$

TERMS t time; $\frac{dx}{dt}$ rate; ℓ, R, H fixed sizes

"Differentiate first, substitute numbers after."

1. $y = kx^n \implies \dfrac{dy}{y} = n\,\dfrac{dx}{x}$
2. $d(u+v) = du + dv, \quad d(uv) = u\,dv + v\,du$

TERMS dx increment; dy tangent change; Δy true change

dy rides the tangent; Δy rides the curve.

1. $dy = f'(x)\,dx$
2. $\Delta y = f(x+dx) - f(x) \approx dy$
3. $\dfrac{dy}{y} = \dfrac{f'(x)}{f(x)}\,dx$

TERMS dx increment; dy tangent change; Δy true change

dy rides the tangent; Δy rides the curve.

1. $f(x) - L(x) \approx \frac{1}{2}f''(a)\,(x-a)^2$
2. $\sqrt{1+x} \approx 1 + \frac{x}{2}, \quad (1+x)^k \approx 1 + kx, \quad \sin x \approx x$

TERMS a base point; $L(x)$ linearization; f'' curvature

"Near the point, the curve is the tangent line."

1. $f \in C[a,b] \implies \exists c, d \in [a,b]:\ f(c) \le f(x) \le f(d)\ \ \forall x \in [a,b]$
2. abs. extrema $\in \{\text{critical points in } (a,b)\} \cup \{a, b\}$
3. $f(d) = \max_{[a,b]} f$ (global), $\qquad f(c) \ge f(x)$ near c (local)

TERMS $[a,b]$ closed interval; c, d abs. min/max locations

"Closed and continuous: the max and min must show up." Check the ends!

1. $f'' > 0 \Rightarrow$ concave up $(\cup)$; $\qquad f'' < 0 \Rightarrow$ concave down $(\cap)$
2. f'' changes sign at c
3. $f'(c) = 0:\ f''(c) > 0 \Rightarrow$ local min, $\ f''(c) < 0 \Rightarrow$ local max

TERMS c critical point; $\cup$ concave up; $\cap$ concave down

Concave up holds water like a cup (a min); concave down spills like a frown (a max).

1. c critical $\iff f'(c) = 0$ or $f'(c)$ undefined $(c \in \operatorname{dom} f)$
2. f has a local extremum at c, $f'(c)$ exists $\implies f'(c) = 0$
3. $f' > 0$ on $I \Rightarrow f \nearrow$; $\qquad f' < 0$ on $I \Rightarrow f \searrow$

TERMS c critical point; $\cup$ concave up; $\cap$ concave down

Concave up holds water like a cup (a min); concave down spills like a frown (a max).

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C1.25 (1/2) CALCULUS I

Curve Sketching: First- & Second-Derivative Tests

PART 1 OF 2

1. First-derivative test
2. Second-derivative test
3. Sign chart

STATE EACH NUMBERED FORMULA

C1.25 (2/2) CALCULUS I

Curve Sketching: First- & Second-Derivative Tests

PART 2 OF 2

1. Symmetry
2. Asymptotes

STATE EACH NUMBERED FORMULA

C1.26 (1/2) CALCULUS I

Optimization & Gradient Descent

PART 1 OF 2

1. Stationarity condition
2. Single critical point
3. Gradient descent (1-D)

STATE EACH NUMBERED FORMULA

C1.26 (2/2) CALCULUS I

Optimization & Gradient Descent

PART 2 OF 2

1. Gradient descent (vector)
2. Step-size condition

STATE EACH NUMBERED FORMULA

C1.27 (1/2) CALCULUS I

Mean Value Theorem

PART 1 OF 2

1. Mean value theorem
2. Rolle's theorem
3. Zero derivative

STATE EACH NUMBERED FORMULA

C1.27 (2/2) CALCULUS I

Mean Value Theorem

PART 2 OF 2

1. Lipschitz bound
2. Cauchy's mean value theorem

STATE EACH NUMBERED FORMULA

C1.28 (1/2) CALCULUS I

L'Hôpital's Rule

PART 1 OF 2

1. L'Hôpital's rule
2. Repeated application
3. Product form

STATE EACH NUMBERED FORMULA

C1.28 (2/2) CALCULUS I

L'Hôpital's Rule

PART 2 OF 2

1. Power forms
2. Growth comparisons

STATE EACH NUMBERED FORMULA

C1.29 (1/3) CALCULUS I

Antiderivatives & Basic Integrals

PART 1 OF 3

1. Antiderivative
2. Power rule
3. Reciprocal

STATE EACH NUMBERED FORMULA

C1.29 (2/3) CALCULUS I

Antiderivatives & Basic Integrals

PART 2 OF 3

1. Exponentials
2. Sine & cosine
3. Secant squared & arctangent

STATE EACH NUMBERED FORMULA

C1.29 (3/3) CALCULUS I

Antiderivatives & Basic Integrals

PART 3 OF 3

1. Linearity
2. Initial value problem

STATE EACH NUMBERED FORMULA

C1.K3 CALCULUS I

Antiderivatives of x^n, $\frac{1}{x}$, e^{kx}, a^x

RECALL IT

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1. $f'(x^*) = 0$
2. only one critical point on I, local min $\implies$ absolute min on I
3. $x_{k+1} = x_k - \eta\, f'(x_k)$

TERMS x^* optimum; η step size; ∇L gradient; M curvature bound

"Downhill is minus the gradient."

1. $f(-x) = f(x)$ (even), $\quad f(-x) = -f(x)$ (odd)
2. $\lim_{x\to a} f = \pm\infty$ (vertical), $\quad \lim_{x\to\pm\infty} f = L$ (horizontal)

TERMS c critical; $+\to-$ sign change of f'; ? inconclusive

f': up or down. f'': cup or cap.

1. f': $+\to-$ at $c \Rightarrow$ local max; $\quad -\to+ \Rightarrow$ local min
2. $f'(c) = 0$, $f''(c) > 0 \Rightarrow$ min; $\;f''(c) < 0 \Rightarrow$ max; $\;f''(c) = 0 \Rightarrow$?
3. sign of f' (or f'') on each interval between its zeros

TERMS c critical; $+\to-$ sign change of f'; ? inconclusive

f': up or down. f'': cup or cap.

1. $|f(x) - f(y)| \le M|x - y|, \quad M = \sup|f'|$
2. $\dfrac{f(b) - f(a)}{g(b) - g(a)} = \dfrac{f'(c)}{g'(c)}$

TERMS $c \in (a,b)$; secant slope $\frac{f(b)-f(a)}{b-a}$; $M = \sup|f'|$

"At some moment, your speedometer showed your average speed."

1. f continuous on $[a,b]$, differentiable on $(a,b) \implies \exists\, c \in (a,b) :\; f'(c) = \frac{f(b)-f(a)}{b-a}$
2. $f(a) = f(b) \implies \exists\, c \in (a,b) :\; f'(c) = 0$
3. $f' \equiv 0$ on $I \implies f$ constant on I

TERMS $c \in (a,b)$; secant slope $\frac{f(b)-f(a)}{b-a}$; $M = \sup|f'|$

"At some moment, your speedometer showed your average speed."

1. $\boldsymbol{\theta}_{k+1} = \boldsymbol{\theta}_k - \eta\, \nabla L(\boldsymbol{\theta}_k)$
2. $0 < \eta < \dfrac{2}{M}, \quad M = \max|f''|$

TERMS x^* optimum; η step size; ∇L gradient; M curvature bound

"Downhill is minus the gradient."

1. $\displaystyle\int f(x)\,dx = F(x) + C \iff F'(x) = f(x)$
2. $\displaystyle\int x^n\,dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$
3. $\displaystyle\int \frac{1}{x}\,dx = \ln|x| + C$

TERMS $F' = f$; C constant of integration; (x_0, y_0) initial value

"Add one to the power, divide by the new power"; don't forget $+C$.

1. $1^\infty, 0^0, \infty^0 :\; \lim f^{\,g} = e^{\lim g \ln f}$
2. $\displaystyle\lim_{x\to\infty} \frac{\ln x}{x^p} = 0, \qquad \lim_{x\to\infty} \frac{x^n}{e^x} = 0$

TERMS $\frac{0}{0}, \frac{\infty}{\infty}$ forms; f', g' separate derivatives

Differentiate top and bottom separately, not the quotient rule.

1. $\lim f = \lim g = 0$ or $\pm\infty \implies \lim \frac{f(x)}{g(x)} = \lim \frac{f'(x)}{g'(x)}$
2. $\displaystyle\lim \frac{f}{g} = \lim \frac{f'}{g'} = \lim \frac{f''}{g''} = \cdots$
3. $0 \cdot \infty :\; \lim f\, g = \lim \dfrac{f}{1/g}$

TERMS $\frac{0}{0}, \frac{\infty}{\infty}$ forms; f', g' separate derivatives

Differentiate top and bottom separately, not the quotient rule.

$\int x^n dx = \frac{x^{n+1}}{n+1} + C\ (n \neq -1)$
$\int \frac{1}{x} dx = \ln|x| + C$
$\int e^{kx} dx = \frac{1}{k} e^{kx} + C \quad \int a^x dx = \frac{a^x}{\ln a} + C$

1. $\displaystyle\int \big[af(x) + bg(x)\big]\,dx = a\int f\,dx + b\int g\,dx$
2. $F' = f,\ F(x_0) = y_0 \implies C = y_0 - F_0(x_0)$

TERMS $F' = f$; C constant of integration; (x_0, y_0) initial value

"Add one to the power, divide by the new power"; don't forget $+C$.

1. $\displaystyle\int e^x\,dx = e^x + C, \qquad \int a^x\,dx = \frac{a^x}{\ln a} + C$
2. $\displaystyle\int \sin x\,dx = -\cos x + C, \qquad \int \cos x\,dx = \sin x + C$
3. $\displaystyle\int \sec^2 x\,dx = \tan x + C, \qquad \int \frac{dx}{1+x^2} = \arctan x + C$

TERMS $F' = f$; C constant of integration; (x_0, y_0) initial value

"Add one to the power, divide by the new power"; don't forget $+C$.

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C1.K3

CALCULUS I

Antiderivatives of the trig functions

RECALL IT

C1.K3

CALCULUS I

Integrals giving inverse trig functions

RECALL IT

C1.30 (1/2)

CALCULUS I

Riemann Sums & the Definite Integral

PART 1 OF 2

1. Definite integral
2. Width and grid
3. Sample points

STATE EACH NUMBERED FORMULA

C1.30 (2/2)

CALCULUS I

Riemann Sums & the Definite Integral

PART 2 OF 2

1. Reversed limits
2. Summation formulas
3. Over/underestimates

STATE EACH NUMBERED FORMULA

C1.31 (1/3)

CALCULUS I

Properties of Definite Integrals

PART 1 OF 3

1. Zero width
2. Linearity
3. Additivity

STATE EACH NUMBERED FORMULA

C1.31 (2/3)

CALCULUS I

Properties of Definite Integrals

PART 2 OF 3

1. Comparison
2. Bounds
3. Absolute value

STATE EACH NUMBERED FORMULA

C1.31 (3/3)

CALCULUS I

Properties of Definite Integrals

PART 3 OF 3

1. Even & odd symmetry
2. Mean value theorem for integrals

STATE EACH NUMBERED FORMULA

C1.32 (1/2)

CALCULUS I

Fundamental Theorem of Calculus

PART 1 OF 2

1. Part 1 (accumulation)
2. Part 1 with chain rule
3. Variable limits on both ends

STATE EACH NUMBERED FORMULA

C1.32 (2/2)

CALCULUS I

Fundamental Theorem of Calculus

PART 2 OF 2

1. Part 2 (evaluation)
2. Every continuous f has an antiderivative

STATE EACH NUMBERED FORMULA

C1.33 (1/2)

CALCULUS I

Net Change Theorem

PART 1 OF 2

1. Net change theorem
2. Displacement
3. Total distance

STATE EACH NUMBERED FORMULA

C1.33 (2/2)

CALCULUS I

Net Change Theorem

PART 2 OF 2

1. Amount from a rate
2. Velocity from acceleration

STATE EACH NUMBERED FORMULA

C1.34 (1/2)

CALCULUS I

Integration by u-Substitution

PART 1 OF 2

1. Substitution rule
2. Differential
3. Definite integrals

STATE EACH NUMBERED FORMULA

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1. $$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\,\Delta x$$
2. $$\Delta x = \frac{b-a}{n}, \qquad x_i = a + i\,\Delta x$$
3. $x_i^* = x_{i-1}$ (left), x_i (right), $\frac{x_{i-1}+x_i}{2}$ (midpoint)

TERMS $\Delta x = \frac{b-a}{n}$ width; x_i^* sample point; L_n, R_n sums

"Height times width, added up, with the width shrinking to zero."

$$\int \frac{dx}{a^2+x^2} = \frac{1}{a}\arctan\frac{x}{a} + C$$
$$\int \frac{dx}{\sqrt{a^2-x^2}} = \arcsin\frac{x}{a} + C$$
$$\int \frac{dx}{x\sqrt{x^2-a^2}} = \frac{1}{a}\operatorname{arcsec}\frac{|x|}{a} + C$$

$$\int \sin = -\cos,\ \int \cos = \sin,\ \int \sec^2 = \tan,$$
$$\int \csc^2 = -\cot$$
$$\int \sec\tan = \sec,\ \int \csc\cot = -\csc$$
$$\int \tan = \ln|\sec x|,$$
$$\int \sec = \ln|\sec x + \tan x| \quad \text{(all } +C\text{)}$$

1. $f \le g$ on $[a,b] \implies \int_a^b f\,dx \le \int_a^b g\,dx$
2. $m \le f \le M \implies m(b-a) \le \int_a^b f\,dx \le M(b-a)$
3. $$\left|\int_a^b f\,dx\right| \le \int_a^b |f|\,dx$$

TERMS m, M min/max of f; even/odd symmetry; c intermediate point

Odd over symmetric limits: zero. Even: double the half.

1. $$\int_a^a f(x)\,dx = 0$$
2. $$\int_a^b [cf \pm g]\,dx = c\int_a^b f\,dx \pm \int_a^b g\,dx$$
3. $$\int_a^b f\,dx + \int_b^c f\,dx = \int_a^c f\,dx$$

TERMS m, M min/max of f; even/odd symmetry; c intermediate point

Odd over symmetric limits: zero. Even: double the half.

1. $$\int_a^b f\,dx = -\int_b^a f\,dx$$
2. $$\sum_{i=1}^{n} i = \frac{n(n+1)}{2}, \qquad \sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$$
3. $f \nearrow$: $L_n \le \int_a^b f \le R_n$

TERMS $\Delta x = \frac{b-a}{n}$ width; x_i^* sample point; L_n, R_n sums

"Height times width, added up, with the width shrinking to zero."

1. $$\int_a^b f(x)\,dx = F(b) - F(a) = \Big[F(x)\Big]_a^b$$
2. $$F(x) = \int_a^x f(t)\,dt$$

TERMS F antiderivative; t dummy variable; $[F]_a^b = F(b) - F(a)$

"Integrate then differentiate: you get back where you started."

1. $$\frac{d}{dx}\int_a^x f(t)\,dt = f(x)$$
2. $$\frac{d}{dx}\int_a^{g(x)} f(t)\,dt = f\big(g(x)\big)\,g'(x)$$
3. $$\frac{d}{dx}\int_{h(x)}^{g(x)} f(t)\,dt = f\big(g(x)\big)g'(x) - f\big(h(x)\big)h'(x)$$

TERMS F antiderivative; t dummy variable; $[F]_a^b = F(b) - F(a)$

"Integrate then differentiate: you get back where you started."

1. $\int_{-a}^{a} f\,dx = 2\int_0^a f\,dx$ (even), $\int_{-a}^{a} f\,dx = 0$ (odd)
2. $\exists\, c \in [a,b]: \int_a^b f\,dx = f(c)\,(b-a)$

TERMS m, M min/max of f; even/odd symmetry; c intermediate point

Odd over symmetric limits: zero. Even: double the half.

1. $$\int f\big(g(x)\big)\,g'(x)\,dx = \int f(u)\,du$$
2. $u = g(x), \qquad du = g'(x)\,dx$
3. $$\int_a^b f\big(g(x)\big)g'(x)\,dx = \int_{g(a)}^{g(b)} f(u)\,du$$

TERMS $u = g(x)$; $du = g'(x)\,dx$; new limits $g(a), g(b)$

Look for an "inside function" whose derivative is also present.

1. $$Q(b) = Q(a) + \int_a^b Q'(t)\,dt$$
2. $$v(t) = v(0) + \int_0^t a(\tau)\,d\tau$$

TERMS F' rate; s, v, a position, velocity, accel.; $|v|$ speed

Displacement keeps the signs; distance takes the absolute value.

1. $$\int_a^b F'(x)\,dx = F(b) - F(a)$$
2. $$s(t_2) - s(t_1) = \int_{t_1}^{t_2} v(t)\,dt$$
3. $$\text{distance} = \int_{t_1}^{t_2} |v(t)|\,dt$$

TERMS F' rate; s, v, a position, velocity, accel.; $|v|$ speed

Displacement keeps the signs; distance takes the absolute value.

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Integration by u-Substitution

PART 2 OF 2

1. Log pattern
2. Linear substitution

STATE EACH NUMBERED FORMULA

Newton's Method for Roots

PART 1 OF 2

1. Newton's iteration
2. Stopping rule
3. Square-root iteration

STATE EACH NUMBERED FORMULA

Newton's Method for Roots

PART 2 OF 2

1. Quadratic convergence
2. Optimization version

STATE EACH NUMBERED FORMULA

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1 $e_{n+1} \approx \dfrac{f''(r)}{2f'(r)}\, e_n^2$

2 $x_{n+1} = x_n - \dfrac{f'(x_n)}{f''(x_n)}$

TERMS x_n estimate; r root; e_n error; $f'(x_n) \neq 0$

"Follow the tangent line down to the axis, then repeat."

C1.35 (2/2)

1 $x_{n+1} = x_n - \dfrac{f(x_n)}{f'(x_n)}$

2 $|x_{n+1} - x_n| <$ tolerance or $|f(x_n)| <$ tolerance

3 $x_{n+1} = \dfrac{1}{2}\left(x_n + \dfrac{a}{x_n}\right) \to \sqrt{a}$

TERMS x_n estimate; r root; e_n error; $f'(x_n) \neq 0$

"Follow the tangent line down to the axis, then repeat."

C1.35 (1/2)

1 $\displaystyle\int \frac{g'(x)}{g(x)}\, dx = \ln|g(x)| + C$

2 $\displaystyle\int f(ax+b)\, dx = \frac{1}{a}F(ax+b) + C$

TERMS $u = g(x)$; $du = g'(x)\, dx$; new limits $g(a), g(b)$

Look for an "inside function" whose derivative is also present.

C1.34 (2/2)

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C2.1 (1/2) CALCULUS II

Integration by Parts

PART 1 OF 2

1. Integration by parts
2. Definite form
3. Tabular method

STATE EACH NUMBERED FORMULA

C2.1 (2/2) CALCULUS II

Integration by Parts

PART 2 OF 2

1. Cyclic (solve-for-I) case
2. Lone log / inverse trig
3. Reduction formula

STATE EACH NUMBERED FORMULA

C2.2 (1/3) CALCULUS II

Trigonometric Integrals

PART 1 OF 3

1. Power reduction for $\sin^2$
2. Power reduction for $\cos^2$
3. Odd power

STATE EACH NUMBERED FORMULA

C2.2 (2/3) CALCULUS II

Trigonometric Integrals

PART 2 OF 3

1. Tangent integral
2. Secant integral
3. Arctangent form

STATE EACH NUMBERED FORMULA

C2.2 (3/3) CALCULUS II

Trigonometric Integrals

PART 3 OF 3

1. Arcsine form
2. Product to sum

STATE EACH NUMBERED FORMULA

C2.3 (1/2) CALCULUS II

Trigonometric Substitution

PART 1 OF 2

1. Sine substitution
2. Tangent substitution
3. Secant substitution

STATE EACH NUMBERED FORMULA

C2.3 (2/2) CALCULUS II

Trigonometric Substitution

PART 2 OF 2

1. Back-substitution
2. Complete the square first

STATE EACH NUMBERED FORMULA

C2.4 (1/2) CALCULUS II

Partial Fractions

PART 1 OF 2

1. Distinct linear factors
2. Repeated linear factor
3. Irreducible quadratic factor

STATE EACH NUMBERED FORMULA

C2.4 (2/2) CALCULUS II

Partial Fractions

PART 2 OF 2

1. Log integral
2. Power integral
3. Heaviside cover-up

STATE EACH NUMBERED FORMULA

C2.5 (1/3) CALCULUS II

Numerical Integration: Trapezoid & Simpson's Rules

PART 1 OF 3

1. Step size
2. Midpoint rule
3. Trapezoid rule

STATE EACH NUMBERED FORMULA

C2.5 (2/3) CALCULUS II

Numerical Integration: Trapezoid & Simpson's Rules

PART 2 OF 3

1. Simpson's rule

STATE EACH NUMBERED FORMULA

C2.5 (3/3) CALCULUS II

Numerical Integration: Trapezoid & Simpson's Rules

PART 3 OF 3

1. Error bounds
2. Simpson as a blend

STATE EACH NUMBERED FORMULA

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1. $\sin^2 x = \dfrac{1-\cos 2x}{2}$
2. $\cos^2 x = \dfrac{1+\cos 2x}{2}$
3. $\displaystyle\int \sin^3 x\,dx = \int (1-\cos^2 x)\sin x\,dx$

TERMS m, n powers; $a > 0$ constant; $\sin^2 + \cos^2 = 1$

Odd power: "save one, convert the rest." Even powers: "reduce the power."

1. $I = \displaystyle\int e^x \sin x\,dx = \tfrac{1}{2}e^x(\sin x - \cos x) + C$
2. $\displaystyle\int \ln x\,dx = x\ln x - x + C$
3. $\displaystyle\int x^n e^x\,dx = x^n e^x - n\int x^{n-1}e^x\,dx$

TERMS u differentiate; dv integrate; LIATE order for choosing u

LIATE picks u: Log, Inverse trig, Algebraic, Trig, Exponential. "Ultraviolet voodoo": $uv - \int v\,du$.

1. $\displaystyle\int u\,dv = uv - \int v\,du$
2. $\displaystyle\int_a^b u\,dv = \Big[uv\Big]_a^b - \int_a^b v\,du$
3. $\displaystyle\int p\,g\,dx = p\,G_1 - p'G_2 + p''G_3 - \cdots$

TERMS u differentiate; dv integrate; LIATE order for choosing u

LIATE picks u: Log, Inverse trig, Algebraic, Trig, Exponential. "Ultraviolet voodoo": $uv - \int v\,du$.

1. $\sqrt{a^2-x^2}$: $x = a\sin\theta \Rightarrow \sqrt{a^2-x^2} = a\cos\theta$
2. $\sqrt{a^2+x^2}$: $x = a\tan\theta \Rightarrow \sqrt{a^2+x^2} = a\sec\theta$
3. $\sqrt{x^2-a^2}$: $x = a\sec\theta \Rightarrow \sqrt{x^2-a^2} = a\tan\theta$

TERMS $a > 0$; θ triangle angle; identities $1 - \sin^2 = \cos^2$, $1 + \tan^2 = \sec^2$

Match the root to the identity: $a^2 - x^2 \to \sin$, $a^2 + x^2 \to \tan$, $x^2 - a^2 \to \sec$.

1. $\displaystyle\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\frac{x}{a} + C$
2. $\sin A\cos B = \tfrac{1}{2}\big[\sin(A-B) + \sin(A+B)\big]$

TERMS m, n powers; $a > 0$ constant; $\sin^2 + \cos^2 = 1$

Odd power: "save one, convert the rest." Even powers: "reduce the power."

1. $\displaystyle\int \tan x\,dx = \ln|\sec x| + C$
2. $\displaystyle\int \sec x\,dx = \ln|\sec x + \tan x| + C$
3. $\displaystyle\int \frac{dx}{a^2+x^2} = \frac{1}{a}\tan^{-1}\frac{x}{a} + C$

TERMS m, n powers; $a > 0$ constant; $\sin^2 + \cos^2 = 1$

Odd power: "save one, convert the rest." Even powers: "reduce the power."

1. $\displaystyle\int \frac{dx}{x-a} = \ln|x-a| + C$
2. $\displaystyle\int \frac{dx}{(x-a)^k} = \frac{-1}{(k-1)(x-a)^{k-1}} + C \quad (k \geq 2)$
3. $A = \left.\dfrac{px+q}{x-b}\right|_{x=a}$

TERMS A, B constants; numerator degree < denominator degree

"One fraction per factor; each numerator one degree below its denominator piece."

1. $\dfrac{px+q}{(x-a)(x-b)} = \dfrac{A}{x-a} + \dfrac{B}{x-b}$
2. $\dfrac{r(x)}{(x-a)^2} = \dfrac{A}{x-a} + \dfrac{B}{(x-a)^2}$
3. $\dfrac{r(x)}{x^2+c} = \dfrac{Ax+B}{x^2+c} \quad (c > 0)$

TERMS A, B constants; numerator degree < denominator degree

"One fraction per factor; each numerator one degree below its denominator piece."

1. $\sin\theta = \dfrac{x}{a} \Rightarrow \cos\theta = \dfrac{\sqrt{a^2-x^2}}{a}$
2. $\sqrt{x^2+2x+5} = \sqrt{(x+1)^2+4}$

TERMS $a > 0$; θ triangle angle; identities $1 - \sin^2 = \cos^2$, $1 + \tan^2 = \sec^2$

Match the root to the identity: $a^2 - x^2 \to \sin$, $a^2 + x^2 \to \tan$, $x^2 - a^2 \to \sec$.

1. $|E_T| \leq \dfrac{K_2(b-a)^3}{12n^2}, \qquad |E_S| \leq \dfrac{K_4(b-a)^5}{180n^4}$
2. $S_{2n} = \dfrac{2M_n + T_n}{3}$

TERMS $h = \frac{b-a}{n}$ spacing; $f_i = f(a+ih)$; K derivative bounds

Simpson's weights: 1, 4, 2, 4, 2, . . . , 4, 1. Trapezoid: 1, 2, 2, . . . , 2, 1.

1. $S_n = \dfrac{h}{3}\Big[f_0 + 4f_1 + 2f_2 + 4f_3 + \cdots + 4f_{n-1} + f_n\Big] \quad (n \text{ even})$

TERMS $h = \frac{b-a}{n}$ spacing; $f_i = f(a+ih)$; K derivative bounds

Simpson's weights: 1, 4, 2, 4, 2, . . . , 4, 1. Trapezoid: 1, 2, 2, . . . , 2, 1.

1. $h = \dfrac{b-a}{n}, \qquad f_i = f(a+ih)$
2. $M_n = h\big[f(\bar{x}_1) + f(\bar{x}_2) + \cdots + f(\bar{x}_n)\big]$
3. $T_n = \dfrac{h}{2}\Big[f_0 + 2f_1 + 2f_2 + \cdots + 2f_{n-1} + f_n\Big]$

TERMS $h = \frac{b-a}{n}$ spacing; $f_i = f(a+ih)$; K derivative bounds

Simpson's weights: 1, 4, 2, 4, 2, . . . , 4, 1. Trapezoid: 1, 2, 2, . . . , 2, 1.

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C2.K1

CALCULUS II

Integration: which method? (in order)

RECALL IT

C2.K1

CALCULUS II

Parts or u-substitution?

RECALL IT

C2.6 (1/3)

CALCULUS II

Improper Integrals

PART 1 OF 3

1. Type 1: infinite interval
2. Type 2: unbounded integrand
3. Doubly infinite

STATE EACH NUMBERED FORMULA

C2.6 (2/3)

CALCULUS II

Improper Integrals

PART 2 OF 3

1. p-test at infinity
2. p-test at zero
3. Exponential tail

STATE EACH NUMBERED FORMULA

C2.6 (3/3)

CALCULUS II

Improper Integrals

PART 3 OF 3

1. Comparison test
2. Limit comparison

STATE EACH NUMBERED FORMULA

C2.7

CALCULUS II

Area Between Curves

1. Vertical slices
2. Horizontal slices
3. Crossing curves

STATE EACH NUMBERED FORMULA

C2.8 (1/2)

CALCULUS II

Volumes: Slicing, Disks & Washers

PART 1 OF 2

1. Slicing
2. Disk method
3. Washer method

STATE EACH NUMBERED FORMULA

C2.8 (2/2)

CALCULUS II

Volumes: Slicing, Disks & Washers

PART 2 OF 2

1. Other axis

STATE EACH NUMBERED FORMULA

C2.9

CALCULUS II

Shell Method

1. Shell method (y-axis)
2. Shell, general axis
3. Shell (x-axis)

STATE EACH NUMBERED FORMULA

C2.10 (1/2)

CALCULUS II

Arc Length

PART 1 OF 2

1. Arc-length element
2. Graph $y = f(x)$
3. Graph $x = g(y)$

STATE EACH NUMBERED FORMULA

C2.10 (2/2)

CALCULUS II

Arc Length

PART 2 OF 2

1. Parametric curve
2. Polar curve

STATE EACH NUMBERED FORMULA

C2.11 (1/2)

CALCULUS II

Surface Area of Revolution

PART 1 OF 2

1. About the x-axis
2. About the y-axis
3. Parametric curve

STATE EACH NUMBERED FORMULA

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1. $\displaystyle\int_a^{\infty} f(x)\,dx = \lim_{b\to\infty}\int_a^b f(x)\,dx$
2. $\displaystyle\int_a^b f(x)\,dx = \lim_{t\to a^+}\int_t^b f(x)\,dx$
3. $\displaystyle\int_{-\infty}^{\infty} f\,dx = \int_{-\infty}^{c} f\,dx + \int_c^{\infty} f\,dx$

TERMS p exponent; converges = finite limit; $f, g \geq 0$ for comparison

p-test: "p bigger than one converges" (at infinity); near zero it is the opposite.

u-sub when one factor is the derivative of the inside of the other: $\int 2x\,e^{x^2}dx$.

Parts when the factors are unrelated types: $\int x\,e^{x}dx$.

1 Simplify, basic table. 2 u-sub: inner function + its derivative.
3 Parts (LIATE): product of unlike types.
4 Trig powers: odd → save one; even → reduce.
5 $\sqrt{a^2 \pm x^2}$: trig sub. 6 P/Q: divide, partial fractions.
7 Otherwise: numerical or series.

1. $A = \displaystyle\int_a^b \big[f(x) - g(x)\big]\,dx \quad (f \geq g \text{ on } [a,b])$
2. $A = \displaystyle\int_c^d \big[x_{\text{right}}(y) - x_{\text{left}}(y)\big]\,dy$
3. $A = \displaystyle\int_a^b \big|f(x) - g(x)\big|\,dx$

TERMS f top, g bottom; a, b intersections

"Top minus bottom" (or "right minus left").

1. $0 \leq f \leq g:\quad \displaystyle\int g < \infty \Rightarrow \int f < \infty$
2. $\displaystyle\lim \frac{f(x)}{g(x)} = c \in (0,\infty) \Rightarrow$ same behavior

TERMS p exponent; converges = finite limit; $f, g \geq 0$ for comparison

p-test: "p bigger than one converges" (at infinity); near zero it is the opposite.

1. $\displaystyle\int_1^{\infty} \frac{dx}{x^p} = \begin{cases} \dfrac{1}{p-1}, & p > 1 \\ \text{diverges}, & p \leq 1 \end{cases}$
2. $\displaystyle\int_0^1 \frac{dx}{x^p}$ converges $\iff p < 1$
3. $\displaystyle\int_0^{\infty} e^{-ax}\,dx = \frac{1}{a} \quad (a > 0)$

TERMS p exponent; converges = finite limit; $f, g \geq 0$ for comparison

p-test: "p bigger than one converges" (at infinity); near zero it is the opposite.

1. $V = 2\pi\displaystyle\int_a^b x\,h(x)\,dx$
2. $V = 2\pi\displaystyle\int_a^b \rho(x)\,h(x)\,dx$
3. $V = 2\pi\displaystyle\int_c^d y\,w(y)\,dy$

TERMS ρ radius to axis; h strip height; dx thickness

Shell = "circumference × height × thickness": $2\pi\rho \cdot h \cdot dx$.

1. axis $y = k$: $\quad R(x) = |f(x) - k|$

TERMS R outer, r inner radius from the axis; $A(x)$ slice area

Washer = "big disk minus small disk" (not "disk of the difference").

1. $V = \displaystyle\int_a^b A(x)\,dx$
2. $V = \pi\displaystyle\int_a^b \big[R(x)\big]^2\,dx$
3. $V = \pi\displaystyle\int_a^b \Big(\big[R(x)\big]^2 - \big[r(x)\big]^2\Big)\,dx$

TERMS R outer, r inner radius from the axis; $A(x)$ slice area

Washer = "big disk minus small disk" (not "disk of the difference").

1. $S = 2\pi\displaystyle\int_a^b f(x)\sqrt{1 + \big[f'(x)\big]^2}\,dx$
2. $S = 2\pi\displaystyle\int_a^b x\sqrt{1 + \big[f'(x)\big]^2}\,dx$
3. $S = 2\pi\displaystyle\int_\alpha^\beta y(t)\sqrt{x'(t)^2 + y'(t)^2}\,dt$

TERMS ρ radius to axis; ds arc element

"Circumference × slant length": $2\pi\rho\,ds$.

1. $L = \displaystyle\int_\alpha^\beta \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\,dt$
2. $L = \displaystyle\int_\alpha^\beta \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\,d\theta$

TERMS ds arc element; f' slope; $x(t), y(t)$ parametric; $r(\theta)$ polar

"Pythagoras, then integrate": $ds = \sqrt{dx^2 + dy^2}$, factor out whichever differential you integrate in.

1. $ds = \sqrt{dx^2 + dy^2}$
2. $L = \displaystyle\int_a^b \sqrt{1 + \big[f'(x)\big]^2}\,dx$
3. $L = \displaystyle\int_c^d \sqrt{1 + \big[g'(y)\big]^2}\,dy$

TERMS ds arc element; f' slope; $x(t), y(t)$ parametric; $r(\theta)$ polar

"Pythagoras, then integrate": $ds = \sqrt{dx^2 + dy^2}$, factor out whichever differential you integrate in.

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C2.11 (2/2) CALCULUS II

Surface Area of Revolution

PART 2 OF 2

1. General form

STATE EACH NUMBERED FORMULA

C2.12 (1/2) CALCULUS II

Average Value of a Function

PART 1 OF 2

1. Average value
2. Mean Value Theorem for Integrals
3. Root mean square

STATE EACH NUMBERED FORMULA

C2.12 (2/2) CALCULUS II

Average Value of a Function

PART 2 OF 2

1. Weighted average

STATE EACH NUMBERED FORMULA

C2.13 (1/2) CALCULUS II

Work, Pumping & Hydrostatic Force

PART 1 OF 2

1. Work by a variable force
2. Spring (Hooke's law)
3. Pumping a liquid

STATE EACH NUMBERED FORMULA

C2.13 (2/2) CALCULUS II

Work, Pumping & Hydrostatic Force

PART 2 OF 2

1. Hydrostatic pressure
2. Hydrostatic force

STATE EACH NUMBERED FORMULA

C2.14 (1/2) CALCULUS II

Moments & Center of Mass

PART 1 OF 2

1. Discrete masses
2. Rod with density $\rho(x)$
3. Centroid $\bar{x}$ of a region

STATE EACH NUMBERED FORMULA

C2.14 (2/2) CALCULUS II

Moments & Center of Mass

PART 2 OF 2

1. Centroid $\bar{y}$ of a region
2. Theorem of Pappus

STATE EACH NUMBERED FORMULA

C2.15 (1/3) CALCULUS II

Sequences

PART 1 OF 3

1. Sequence
2. Limit of a sequence
3. Geometric limit

STATE EACH NUMBERED FORMULA

C2.15 (2/3) CALCULUS II

Sequences

PART 2 OF 3

1. Monotone convergence
2. Squeeze theorem
3. Function link

STATE EACH NUMBERED FORMULA

C2.15 (3/3) CALCULUS II

Sequences

PART 3 OF 3

1. Growth order

STATE EACH NUMBERED FORMULA

C2.16 (1/3) CALCULUS II

Series, Partial Sums & the Divergence Test

PART 1 OF 3

1. Partial sum
2. Sum of a series
3. Divergence (nth-term) test

STATE EACH NUMBERED FORMULA

C2.16 (2/3) CALCULUS II

Series, Partial Sums & the Divergence Test

PART 2 OF 3

1. Geometric series
2. Telescoping series
3. Harmonic series

STATE EACH NUMBERED FORMULA

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1 $\bar{f}_w = \dfrac{\int_a^b f(x)\,w(x)\,dx}{\int_a^b w(x)\,dx}$

TERMS $[a,b]$ interval; c point where $f(c) = f_{\text{avg}}$; w weight

"Average = total over length."

C2.12 (2/2)

1 $f_{\text{avg}} = \dfrac{1}{b-a}\displaystyle\int_a^b f(x)\,dx$

2 $\exists\, c \in [a,b] :\ f(c) = f_{\text{avg}}$

3 $f_{\text{rms}} = \sqrt{\dfrac{1}{b-a}\displaystyle\int_a^b f(x)^2\,dx}$

TERMS $[a,b]$ interval; c point where $f(c) = f_{\text{avg}}$; w weight

"Average = total over length."

C2.12 (1/2)

1 $S = \displaystyle\int 2\pi\rho\,ds$

TERMS ρ radius to axis; ds arc element

"Circumference × slant length": $2\pi\rho\,ds$.

C2.11 (2/2)

1 $\bar{x} = \dfrac{\sum m_i x_i}{\sum m_i}$

2 $\bar{x} = \dfrac{\int_a^b x\,\rho(x)\,dx}{\int_a^b \rho(x)\,dx}$

3 $\bar{x} = \dfrac{1}{A}\displaystyle\int_a^b x\big[f(x) - g(x)\big]\,dx$

TERMS $\bar{x}, \bar{y}$ center of mass; A area; ρ density

"Moment over mass."

C2.14 (1/2)

1 $P = \rho g h$

2 $F = \displaystyle\int_a^b \rho g\, h(y)\, w(y)\, dy$

TERMS ρ density; g gravity; h depth; w width; D lift distance

"Slab, weight, distance" for pumping; "depth times width" for walls.

C2.13 (2/2)

1 $W = \displaystyle\int_a^b F(x)\,dx$

2 $W = \displaystyle\int_0^d kx\,dx = \tfrac{1}{2}kd^2$

3 $W = \displaystyle\int_a^b \rho g\, A(y)\, D(y)\, dy$

TERMS ρ density; g gravity; h depth; w width; D lift distance

"Slab, weight, distance" for pumping; "depth times width" for walls.

C2.13 (1/2)

1 bounded and monotonic $\implies$ convergent

2 $b_n \le a_n \le c_n,\ \ b_n, c_n \to L \ \Rightarrow\ a_n \to L$

3 $f(x) \to L$ as $x \to \infty \ \Rightarrow\ f(n) \to L$

TERMS a_n nth term; L limit; $|r| < 1$; $p > 0$, $b > 1$

Growth order: logs $\ll$ powers $\ll$ exponentials $\ll$ factorials.

C2.15 (2/3)

1 $\{a_n\}_{n=1}^{\infty} = a_1, a_2, a_3, \ldots$

2 $\lim_{n\to\infty} a_n = L$

3 $\lim_{n\to\infty} r^n = 0 \ \ (|r| < 1)$

TERMS a_n nth term; L limit; $|r| < 1$; $p > 0$, $b > 1$

Growth order: logs $\ll$ powers $\ll$ exponentials $\ll$ factorials.

C2.15 (1/3)

1 $\bar{y} = \dfrac{1}{A}\displaystyle\int_a^b \tfrac{1}{2}\Big(\big[f(x)\big]^2 - \big[g(x)\big]^2\Big)\,dx$

2 $V = 2\pi\bar{\rho}\,A$

TERMS $\bar{x}, \bar{y}$ center of mass; A area; ρ density

"Moment over mass."

C2.14 (2/2)

1 $\displaystyle\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r} \ \ (|r| < 1)$

2 $\displaystyle\sum_{n=1}^{\infty} (b_n - b_{n+1}) = b_1 - \lim_{n\to\infty} b_n$

3 $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n} = \infty$

TERMS S_N partial sum; a first term; r ratio

"If the terms don't go to zero, the series won't go."

C2.16 (2/3)

1 $S_N = \displaystyle\sum_{n=1}^{N} a_n$

2 $\displaystyle\sum_{n=1}^{\infty} a_n = \lim_{N\to\infty} S_N$

3 $\lim_{n\to\infty} a_n \neq 0 \ \Rightarrow$ diverges

TERMS S_N partial sum; a first term; r ratio

"If the terms don't go to zero, the series won't go."

C2.16 (1/3)

1 $\ln n \ll n^p \ll b^n \ll n! \ll n^n$

TERMS a_n nth term; L limit; $|r| < 1$; $p > 0$, $b > 1$

Growth order: logs $\ll$ powers $\ll$ exponentials $\ll$ factorials.

C2.15 (3/3)

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C2.16 (3/3) CALCULUS II

Series, Partial Sums & the Divergence Test

PART 3 OF 3

1. Linearity

STATE EACH NUMBERED FORMULA

C2.17 CALCULUS II

Integral Test & p-Series

1. Integral test
2. p-series
3. Remainder estimate

STATE EACH NUMBERED FORMULA

C2.18 (1/2) CALCULUS II

Comparison & Limit Comparison Tests

PART 1 OF 2

1. Comparison (convergence)
2. Comparison (divergence)
3. Limit comparison

STATE EACH NUMBERED FORMULA

C2.18 (2/2) CALCULUS II

Comparison & Limit Comparison Tests

PART 2 OF 2

1. Dominant-term rule

STATE EACH NUMBERED FORMULA

C2.19 (1/2) CALCULUS II

Ratio, Root & Alternating Series Tests

PART 1 OF 2

1. Ratio test
2. Root test
3. Conclusion (both tests)

STATE EACH NUMBERED FORMULA

C2.19 (2/2) CALCULUS II

Ratio, Root & Alternating Series Tests

PART 2 OF 2

1. Alternating series test
2. Alternating error bound

STATE EACH NUMBERED FORMULA

C2.20 (1/2) CALCULUS II

Absolute vs. Conditional Convergence

PART 1 OF 2

1. Absolute convergence
2. Absolute implies convergent
3. Conditional convergence

STATE EACH NUMBERED FORMULA

C2.20 (2/2) CALCULUS II

Absolute vs. Conditional Convergence

PART 2 OF 2

1. Riemann rearrangement

STATE EACH NUMBERED FORMULA

C2.21 (1/3) CALCULUS II

Power Series & Radius of Convergence

PART 1 OF 3

1. Power series
2. Radius (ratio form)
3. Radius (root form)

STATE EACH NUMBERED FORMULA

C2.21 (2/3) CALCULUS II

Power Series & Radius of Convergence

PART 2 OF 3

1. Interval of convergence
2. Term-by-term derivative
3. Term-by-term integral

STATE EACH NUMBERED FORMULA

C2.21 (3/3) CALCULUS II

Power Series & Radius of Convergence

PART 3 OF 3

1. Geometric source series

STATE EACH NUMBERED FORMULA

C2.22 (1/3) CALCULUS II

Taylor & Maclaurin Series

PART 1 OF 3

1. Taylor series

STATE EACH NUMBERED FORMULA

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1. $0 \le a_n \le b_n,\ \sum b_n < \infty \ \Rightarrow\ \sum a_n < \infty$
2. $0 \le b_n \le a_n,\ \sum b_n = \infty \ \Rightarrow\ \sum a_n = \infty$
3. $\lim_{n\to\infty} \frac{a_n}{b_n} = c,\ \ 0 < c < \infty \ \Rightarrow$ same behavior

TERMS $a_n, b_n \ge 0$; $c \in (0,\infty)$ limit of a_n/b_n

"Smaller than convergent converges; bigger than divergent diverges."

C2.18 (1/2)

1. $\sum_{n=1}^{\infty} f(n)$ and $\int_1^{\infty} f(x)\,dx$ converge together
2. $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges $\iff p > 1$
3. $\int_{N+1}^{\infty} f\,dx \le \sum_{n=N+1}^{\infty} a_n \le \int_N^{\infty} f\,dx$

TERMS $f(n) = a_n$ positive, decreasing; p exponent

p-series: "p bigger than one converges" (same rule as the p-integral).

C2.17

1. $\sum(ca_n + b_n) = c\sum a_n + \sum b_n$

TERMS S_N partial sum; a first term; r ratio

"If the terms don't go to zero, the series won't go."

C2.16 (3/3)

1. $\sum(-1)^n b_n$ converges if $b_{n+1} \le b_n$ and $b_n \to 0$
2. $|S - S_N| \le b_{N+1}$

TERMS L limit ratio/root; $b_n > 0$ decreasing to 0

Ratio for factorials, root for nth powers; alternating: "decreasing to zero is enough."

C2.19 (2/2)

1. $L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$
2. $L = \lim_{n\to\infty} \sqrt[n]{|a_n|}$
3. $\begin{cases} L < 1 & \text{converges absolutely} \\ L > 1 & \text{diverges} \\ L = 1 & \text{inconclusive} \end{cases}$

TERMS L limit ratio/root; $b_n > 0$ decreasing to 0

Ratio for factorials, root for nth powers; alternating: "decreasing to zero is enough."

C2.19 (1/2)

1. $\frac{3n+1}{n^3+2} \sim \frac{3n}{n^3} = \frac{3}{n^2}$

TERMS $a_n, b_n \ge 0$; $c \in (0,\infty)$ limit of a_n/b_n

"Smaller than convergent converges; bigger than divergent diverges."

C2.18 (2/2)

1. $\sum_{n=0}^{\infty} c_n (x-a)^n$
2. $R = \lim_{n\to\infty} \left|\frac{c_n}{c_{n+1}}\right|$
3. $\frac{1}{R} = \limsup_{n\to\infty} \sqrt[n]{|c_n|}$

TERMS c_n coefficients; a center; R radius

"Ratio test on the whole term, solve $|x-a| < R$, then check both ends."

C2.21 (1/3)

1. conditional $\Rightarrow$ rearranging can give any sum

TERMS $\sum |a_n|$ absolute series; conditional = converges, not absolutely

"Absolute is robust, conditional is fragile."

C2.20 (2/2)

1. $\sum |a_n| < \infty$
2. $\sum |a_n| < \infty \ \Rightarrow\ \sum a_n$ converges
3. $\sum a_n$ converges, $\sum |a_n| = \infty$

TERMS $\sum |a_n|$ absolute series; conditional = converges, not absolutely

"Absolute is robust, conditional is fragile."

C2.20 (1/2)

1. $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots$

TERMS a center; $f^{(n)}(a)$ nth derivative; $n!$ factorial

sin uses odd powers (sine is odd), cos uses even powers (cosine is even), e^x uses all.

C2.22 (1/3)

1. $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n \quad (|x| < 1)$

TERMS c_n coefficients; a center; R radius

"Ratio test on the whole term, solve $|x-a| < R$, then check both ends."

C2.21 (3/3)

1. $|x-a| < R$ converges, $|x-a| > R$ diverges
2. $f'(x) = \sum_{n=1}^{\infty} n\,c_n (x-a)^{n-1}$
3. $\int f\,dx = C + \sum_{n=0}^{\infty} \frac{c_n (x-a)^{n+1}}{n+1}$

TERMS c_n coefficients; a center; R radius

"Ratio test on the whole term, solve $|x-a| < R$, then check both ends."

C2.21 (2/3)

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C2.22 (2/3) CALCULUS II

Taylor & Maclaurin Series

PART 2 OF 3

1. Taylor polynomial
2. Maclaurin series

STATE EACH NUMBERED FORMULA

C2.22 (3/3) CALCULUS II

Taylor & Maclaurin Series

PART 3 OF 3

1. Exponential
2. Sine
3. Cosine

STATE EACH NUMBERED FORMULA

C2.23 (1/2) CALCULUS II

More Maclaurin Series & Building New Ones

PART 1 OF 2

1. Geometric
2. Natural log
3. Arctangent

STATE EACH NUMBERED FORMULA

C2.23 (2/2) CALCULUS II

More Maclaurin Series & Building New Ones

PART 2 OF 2

1. Binomial series
2. Generalized binomial coefficient
3. Build by substitution

STATE EACH NUMBERED FORMULA

C2.24 (1/2) CALCULUS II

Taylor's Theorem & the Lagrange Remainder

PART 1 OF 2

1. Taylor's theorem
2. Lagrange remainder
3. Remainder bound

STATE EACH NUMBERED FORMULA

C2.24 (2/2) CALCULUS II

Taylor's Theorem & the Lagrange Remainder

PART 2 OF 2

1. Big-O form
2. Series equals f

STATE EACH NUMBERED FORMULA

C2.K2 CALCULUS II

Series convergence: which test, in order?

RECALL IT

C2.K2 CALCULUS II

Ratio test gives $L = 1$: what now?

RECALL IT

C2.K2 CALCULUS II

Maclaurin series with intervals: $\frac{1}{1-x}$, $\ln(1+x)$, $\tan^{-1} x$, $(1+x)^k$

RECALL IT

C2.K2 CALCULUS II

Maclaurin series: e^x, $\sin x$, $\cos x$

RECALL IT

C2.25 (1/3) CALCULUS II

Parametric Equations

PART 1 OF 3

1. Parametric curve
2. Eliminating the parameter
3. Parametric slope

STATE EACH NUMBERED FORMULA

C2.25 (2/3) CALCULUS II

Parametric Equations

PART 2 OF 3

1. Second derivative
2. Horizontal / vertical tangents
3. Area under a parametric curve

STATE EACH NUMBERED FORMULA

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1. $\dfrac{1}{1-x} = \displaystyle\sum_{n=0}^{\infty} x^n = 1 + x + x^2 + \cdots$
2. $\ln(1+x) = \displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n+1}x^n}{n} = x - \frac{x^2}{2} + \cdots$
3. $\tan^{-1} x = \displaystyle\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1} = x - \frac{x^3}{3} + \cdots$

TERMS $|x| < 1$ (geometric, binomial); $-1 < x \le 1$ (ln); $|x| \le 1$ ($\tan^{-1}$)

$\ln(1+x)$ and $\tan^{-1} x$ are "integrated geometric series": divide by the new power.

1. $e^x = \displaystyle\sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \cdots$
2. $\sin x = \displaystyle\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \cdots$
3. $\cos x = \displaystyle\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \cdots$

TERMS a center; $f^{(n)}(a)$ nth derivative; $n!$ factorial

sin uses odd powers (sine is odd), cos uses even powers (cosine is even), e^x uses all.

1. $P_n(x) = \displaystyle\sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k$
2. $f(x) = \displaystyle\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n$

TERMS a center; $f^{(n)}(a)$ nth derivative; $n!$ factorial

sin uses odd powers (sine is odd), cos uses even powers (cosine is even), e^x uses all.

1. $f(x) = P_n(x) + O\big((x-a)^{n+1}\big)$
2. $R_n(x) \to 0 \;\Rightarrow\; f(x) = \displaystyle\sum \frac{f^{(n)}(a)}{n!}(x-a)^n$

TERMS P_n Taylor polynomial; c between a and x; $M \ge |f^{(n+1)}|$

"The error is the first term you left out, with the derivative taken at some c."

1. $f(x) = P_n(x) + R_n(x)$
2. $R_n(x) = \dfrac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}$
3. $|R_n(x)| \le \dfrac{M\,|x-a|^{n+1}}{(n+1)!}$

TERMS P_n Taylor polynomial; c between a and x; $M \ge |f^{(n+1)}|$

"The error is the first term you left out, with the derivative taken at some c."

1. $(1+x)^k = \displaystyle\sum_{n=0}^{\infty} \binom{k}{n} x^n = 1 + kx + \frac{k(k-1)}{2!}x^2 + \cdots$
2. $\displaystyle\binom{k}{n} = \frac{k(k-1)\cdots(k-n+1)}{n!}$
3. $e^{-x^2} = \displaystyle\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{n!}$

TERMS $|x| < 1$ (geometric, binomial); $-1 < x \le 1$ (ln); $|x| \le 1$ ($\tan^{-1}$)

$\ln(1+x)$ and $\tan^{-1} x$ are "integrated geometric series": divide by the new power.

$\sum x^n$, $|x| < 1$
$x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots$, $-1 < x \le 1$
$x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots$, $|x| \le 1$
$\sum \binom{k}{n} x^n$, $|x| < 1$

Inconclusive. Typical for rational/algebraic terms: use (limit) comparison with $\sum \frac{1}{n^p}$, or the integral test.

1 Divergence: $\lim a_n \neq 0 \Rightarrow$ div.
2 Geometric ($|r| < 1$) / p-series ($p > 1$).
3 Comparison / limit comparison.
4 Ratio: $n!$, c^n. 5 Root: $(b_n)^n$.
6 Alternating: $b_n \downarrow 0$. 7 Integral test.

1. $\dfrac{d^2y}{dx^2} = \dfrac{\frac{d}{dt}\left(\frac{dy}{dx}\right)}{dx/dt}$
2. $\frac{dy}{dt} = 0$ (horiz.), $\quad \frac{dx}{dt} = 0$ (vert.)
3. $A = \displaystyle\int_{\alpha}^{\beta} y(t)\,x'(t)\,dt$

TERMS t parameter; $x'(t), y'(t)$ derivatives; $\alpha \le t \le \beta$

"Everything over dx/dt": the slope is y'/x', and the second derivative is (slope)$'$ over x' once more.

1. $x = f(t), \quad y = g(t), \quad a \le t \le b$
2. $x = r\cos t,\ y = r\sin t \;\Rightarrow\; x^2 + y^2 = r^2$
3. $\dfrac{dy}{dx} = \dfrac{dy/dt}{dx/dt} \quad \left(\dfrac{dx}{dt} \neq 0\right)$

TERMS t parameter; $x'(t), y'(t)$ derivatives; $\alpha \le t \le \beta$

"Everything over dx/dt": the slope is y'/x', and the second derivative is (slope)$'$ over x' once more.

$\sum \frac{x^n}{n!}$, $\sum \frac{(-1)^n x^{2n+1}}{(2n+1)!}$, $\sum \frac{(-1)^n x^{2n}}{(2n)!}$
all converge for every real x

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C2.25 (3/3) CALCULUS II

Parametric Equations

PART 3 OF 3

1. Speed & arc length
2. Line segment

STATE EACH NUMBERED FORMULA

C2.26 (1/3) CALCULUS II

Polar Area & Calculus

PART 1 OF 3

1. Polar to Cartesian
2. Polar area
3. Area between polar curves

STATE EACH NUMBERED FORMULA

C2.26 (2/3) CALCULUS II

Polar Area & Calculus

PART 2 OF 3

1. Polar slope

STATE EACH NUMBERED FORMULA

C2.26 (3/3) CALCULUS II

Polar Area & Calculus

PART 3 OF 3

1. Polar arc length
2. Tangent at the pole

STATE EACH NUMBERED FORMULA

C2.27 (1/2) CALCULUS II

Differential Equations: Terms, Slope Fields & Equilibria

PART 1 OF 2

1. Order
2. Initial value problem
3. Slope field

STATE EACH NUMBERED FORMULA

C2.27 (2/2) CALCULUS II

Differential Equations: Terms, Slope Fields & Equilibria

PART 2 OF 2

1. Equilibrium solution
2. Stability test
3. Existence & uniqueness

STATE EACH NUMBERED FORMULA

C2.28 (1/3) CALCULUS II

Differential Equations: Separable, Growth & Logistic

PART 1 OF 3

1. Separation of variables
2. Exponential growth / decay
3. Doubling time / half-life

STATE EACH NUMBERED FORMULA

C2.28 (2/3) CALCULUS II

Differential Equations: Separable, Growth & Logistic

PART 2 OF 3

1. Logistic equation
2. Logistic solution
3. Fastest growth

STATE EACH NUMBERED FORMULA

C2.28 (3/3) CALCULUS II

Differential Equations: Separable, Growth & Logistic

PART 3 OF 3

1. Newton's law of cooling

STATE EACH NUMBERED FORMULA

C2.29 (1/3) CALCULUS II

Euler's Method for Differential Equations

PART 1 OF 3

1. Initial value problem
2. Euler step
3. Global error

STATE EACH NUMBERED FORMULA

C2.29 (2/3) CALCULUS II

Euler's Method for Differential Equations

PART 2 OF 3

1. Improved Euler (Heun)

STATE EACH NUMBERED FORMULA

C2.29 (3/3) CALCULUS II

Euler's Method for Differential Equations

PART 3 OF 3

1. Runge–Kutta 4 (RK4)
2. RK4 stages

STATE EACH NUMBERED FORMULA

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1 $$\frac{dy}{dx} = \frac{\frac{dr}{d\theta}\sin\theta + r\cos\theta}{\frac{dr}{d\theta}\cos\theta - r\sin\theta}$$

TERMS $r(\theta)$ polar curve; α, β angle limits; $r' = \frac{dr}{d\theta}$

Polar area is a sum of thin pizza slices: $\frac{1}{2}r^2\,d\theta$.

C2.26 (2/3)

1 $x = r\cos\theta, \quad y = r\sin\theta, \quad r^2 = x^2 + y^2$

2 $$A = \frac{1}{2}\int_\alpha^\beta r^2\,d\theta$$

3 $$A = \frac{1}{2}\int_\alpha^\beta \left(r_{\text{out}}^2 - r_{\text{in}}^2\right)d\theta$$

TERMS $r(\theta)$ polar curve; α, β angle limits; $r' = \frac{dr}{d\theta}$

Polar area is a sum of thin pizza slices: $\frac{1}{2}r^2\,d\theta$.

C2.26 (1/3)

1 $$L = \int_\alpha^\beta \sqrt{x'(t)^2 + y'(t)^2}\,dt$$

2 $x = x_0 + t\,\Delta x, \quad y = y_0 + t\,\Delta y, \quad 0 \le t \le 1$

TERMS t parameter; $x'(t), y'(t)$ derivatives; $\alpha \le t \le \beta$

"Everything over dx/dt": the slope is y'/x', and the second derivative is (slope)′ over x' once more.

C2.25 (3/3)

1 $y' = f(y), \quad f(y^*) = 0 \;\Rightarrow\; y \equiv y^*$

2 $f'(y^*) < 0$ stable, $\quad f'(y^*) > 0$ unstable

3 f, $\frac{\partial f}{\partial y}$ continuous near $(x_0, y_0) \Rightarrow$ unique solution

TERMS n order; (x_0, y_0) initial point; y^* equilibrium; f slope function

Phase line: arrows point up where $f > 0$, down where $f < 0$; a stable equilibrium has arrows pointing into *it.*

C2.27 (2/2)

1 $F(x, y, y', \ldots, y^{(n)}) = 0$

2 $y' = f(x, y), \quad y(x_0) = y_0$

3 segment of slope $f(x, y)$ at each (x, y)

TERMS n order; (x_0, y_0) initial point; y^* equilibrium; f slope function

Phase line: arrows point up where $f > 0$, down where $f < 0$; a stable equilibrium has arrows pointing into *it.*

C2.27 (1/2)

1 $$L = \int_\alpha^\beta \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\,d\theta$$

2 $r(\theta_0) = 0,\ r'(\theta_0) \neq 0 \;\Rightarrow$ tangent line $\theta = \theta_0$

TERMS $r(\theta)$ polar curve; α, β angle limits; $r' = \frac{dr}{d\theta}$

Polar area is a sum of thin pizza slices: $\frac{1}{2}r^2\,d\theta$.

C2.26 (3/3)

1 $$\frac{dT}{dt} = -k(T - T_a) \;\Rightarrow\; T = T_a + (T_0 - T_a)e^{-kt}$$

TERMS k rate; y_0, P_0 initial values; r growth rate; K capacity; T_a ambient

Separable: "get all the y's with dy, all the x's with dx, then integrate both sides."

C2.28 (3/3)

1 $$\frac{dP}{dt} = rP\left(1 - \frac{P}{K}\right)$$

2 $$P(t) = \frac{K}{1 + Ae^{-rt}}, \qquad A = \frac{K - P_0}{P_0}$$

3 $$P = \frac{K}{2}, \qquad \left(\frac{dP}{dt}\right)_{\max} = \frac{rK}{4}$$

TERMS k rate; y_0, P_0 initial values; r growth rate; K capacity; T_a ambient

Separable: "get all the y's with dy, all the x's with dx, then integrate both sides."

C2.28 (2/3)

1 $$\frac{dy}{dx} = g(x)\,h(y) \;\Rightarrow\; \int \frac{dy}{h(y)} = \int g(x)\,dx$$

2 $$\frac{dy}{dt} = ky \;\Rightarrow\; y = y_0 e^{kt}$$

3 $$T = \frac{\ln 2}{|k|}$$

TERMS k rate; y_0, P_0 initial values; r growth rate; K capacity; T_a ambient

Separable: "get all the y's with dy, all the x's with dx, then integrate both sides."

C2.28 (1/3)

1 $$y_{n+1} = y_n + \frac{h}{6}\left(k_1 + 2k_2 + 2k_3 + k_4\right)$$

2 $k_1 = f(t_n, y_n), \qquad k_2 = f\left(t_n + \frac{h}{2}, y_n + \frac{h}{2}k_1\right)$
$k_3 = f\left(t_n + \frac{h}{2}, y_n + \frac{h}{2}k_2\right), \qquad k_4 = f(t_n + h, y_n + hk_3)$

TERMS h step; $f(t, y)$ slope; k_i RK4 slopes; $O(h)$ first order

"New value = old value + step × slope."

C2.29 (3/3)

1 $$y_{n+1} = y_n + \frac{h}{2}\left(k_1 + k_2\right)$$
$$k_1 = f(t_n, y_n), \quad k_2 = f\left(t_n + h, y_n + hk_1\right)$$

TERMS h step; $f(t, y)$ slope; k_i RK4 slopes; $O(h)$ first order

"New value = old value + step × slope."

C2.29 (2/3)

1 $$\frac{dy}{dt} = f(t, y), \qquad y(t_0) = y_0$$

2 $$y_{n+1} = y_n + h\,f(t_n, y_n), \qquad t_{n+1} = t_n + h$$

3 $$|y(t_N) - y_N| = O(h)$$

TERMS h step; $f(t, y)$ slope; k_i RK4 slopes; $O(h)$ first order

"New value = old value + step × slope."

C2.29 (1/3)

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C2.30 (1/3) CALCULUS II

First-Order Linear ODEs & Integrating Factors

PART 1 OF 3

1. Standard form
2. Integrating factor
3. Product collapse

STATE EACH NUMBERED FORMULA

C2.30 (2/3) CALCULUS II

First-Order Linear ODEs & Integrating Factors

PART 2 OF 3

1. General solution
2. Constant coefficients
3. Mixing problem

STATE EACH NUMBERED FORMULA

C2.30 (3/3) CALCULUS II

First-Order Linear ODEs & Integrating Factors

PART 3 OF 3

1. Bernoulli equation

STATE EACH NUMBERED FORMULA

C2.31 (1/3) CALCULUS II

Second-Order Linear ODEs: Constant Coefficients

PART 1 OF 3

1. Homogeneous equation
2. Characteristic equation
3. Superposition

STATE EACH NUMBERED FORMULA

C2.31 (2/3) CALCULUS II

Second-Order Linear ODEs: Constant Coefficients

PART 2 OF 3

1. Distinct real roots ($\Delta > 0$)
2. Repeated root ($\Delta = 0$)
3. Complex roots $\alpha \pm \beta i$ ($\Delta < 0$)

STATE EACH NUMBERED FORMULA

C2.31 (3/3) CALCULUS II

Second-Order Linear ODEs: Constant Coefficients

PART 3 OF 3

1. Damped spring
2. Natural frequency

STATE EACH NUMBERED FORMULA

C2.32 (1/4) CALCULUS II

Nonhomogeneous ODEs: Undetermined Coefficients & Resonance

PART 1 OF 4

1. Nonhomogeneous equation
2. Solution structure
3. Polynomial forcing

STATE EACH NUMBERED FORMULA

C2.32 (2/4) CALCULUS II

Nonhomogeneous ODEs: Undetermined Coefficients & Resonance

PART 2 OF 4

1. Exponential forcing
2. Sinusoidal forcing
3. Modification rule

STATE EACH NUMBERED FORMULA

C2.32 (3/4) CALCULUS II

Nonhomogeneous ODEs: Undetermined Coefficients & Resonance

PART 3 OF 4

1. Resonance

STATE EACH NUMBERED FORMULA

C2.32 (4/4) CALCULUS II

Nonhomogeneous ODEs: Undetermined Coefficients & Resonance

PART 4 OF 4

1. Variation of parameters

STATE EACH NUMBERED FORMULA

C2.33 (1/2) CALCULUS II

Laplace Transform Basics

PART 1 OF 2

1. Definition
2. Linearity
3. Derivative rule

STATE EACH NUMBERED FORMULA

C2.33 (2/2) CALCULUS II

Laplace Transform Basics

PART 2 OF 2

1. First shift (s-shift)
2. Second shift (t-shift)
3. Convolution

STATE EACH NUMBERED FORMULA

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1 $$y' + Py = Qy^n, \quad v = y^{1-n}$$
$$\Rightarrow\ v' + (1-n)Pv = (1-n)Q$$

TERMS P, Q coefficient functions; μ integrating factor; C constant

"e to the integral of P", then the left side is $(\mu y)'$.

C2.30 (3/3)

1 $$y = \frac{1}{\mu(x)}\left[\int \mu(x)Q(x)\,dx + C\right]$$
2 $$y' + ay = b \ \Rightarrow\ y = \frac{b}{a} + Ce^{-ax}$$
3 $$\frac{dA}{dt} = r_{\text{in}}c_{\text{in}} - \frac{r_{\text{out}}}{V(t)}A$$

TERMS P, Q coefficient functions; μ integrating factor; C constant

"e to the integral of P", then the left side is $(\mu y)'$.

C2.30 (2/3)

1 $$y' + P(x)\,y = Q(x)$$
2 $$\mu(x) = e^{\int P(x)\,dx}$$
3 $$\mu y' + \mu P y = (\mu y)' = \mu Q$$

TERMS P, Q coefficient functions; μ integrating factor; C constant

"e to the integral of P", then the left side is $(\mu y)'$.

C2.30 (1/3)

1 $$mx'' + cx' + kx = 0$$
2 $$\omega_0 = \sqrt{\frac{k}{m}}, \qquad \beta = \sqrt{\omega_0^2 - \left(\frac{c}{2m}\right)^2}$$

TERMS $\Delta = b^2 - 4ac$; r roots; c_1, c_2 constants; m, c, k mass, damping, stiffness

"Guess e^{rx}; the quadratic decides the shape." Repeated root: "multiply by x".

C2.31 (3/3)

1 $$y = c_1 e^{r_1 x} + c_2 e^{r_2 x}$$
2 $$y = (c_1 + c_2 x)e^{rx}$$
3 $$y = e^{\alpha x}(c_1 \cos\beta x + c_2 \sin\beta x)$$

TERMS $\Delta = b^2 - 4ac$; r roots; c_1, c_2 constants; m, c, k mass, damping, stiffness

"Guess e^{rx}; the quadratic decides the shape." Repeated root: "multiply by x".

C2.31 (2/3)

1 $$ay'' + by' + cy = 0$$
2 $$ar^2 + br + c = 0$$
3 $$y = c_1 y_1 + c_2 y_2$$

TERMS $\Delta = b^2 - 4ac$; r roots; c_1, c_2 constants; m, c, k mass, damping, stiffness

"Guess e^{rx}; the quadratic decides the shape." Repeated root: "multiply by x".

C2.31 (1/3)

$$x'' + \omega_0^2 x = \frac{F_0}{m}\cos\omega_0 t$$
1 $$\Rightarrow\ x_p = \frac{F_0}{2m\omega_0}\,t\sin\omega_0 t$$

TERMS y_h homogeneous, y_p particular; ω_0 natural frequency; W Wronskian

"Guess what you see" (same family as g); if the guess is already in y_h, "bump it by x".

C2.32 (3/4)

1 $$g = e^{kx} \ \rightarrow\ y_p = Ae^{kx}$$
2 $$g = \cos\omega x,\ \sin\omega x \ \rightarrow\ y_p = A\cos\omega x + B\sin\omega x$$
3 $$\text{trial} \subset y_h \ \Rightarrow\ \text{use } x\cdot\text{trial (or } x^2\cdot\text{trial)}$$

TERMS y_h homogeneous, y_p particular; ω_0 natural frequency; W Wronskian

"Guess what you see" (same family as g); if the guess is already in y_h, "bump it by x".

C2.32 (2/4)

1 $$ay'' + by' + cy = g(x)$$
2 $$y = y_h + y_p$$
3 $$g = p_n(x) \ \rightarrow\ y_p = A_n x^n + \cdots + A_1 x + A_0$$

TERMS y_h homogeneous, y_p particular; ω_0 natural frequency; W Wronskian

"Guess what you see" (same family as g); if the guess is already in y_h, "bump it by x".

C2.32 (1/4)

1 $$\mathcal{L}\{e^{at}f(t)\} = F(s-a)$$
2 $$\mathcal{L}\{u(t-c)f(t-c)\} = e^{-cs}F(s)$$
$$\mathcal{L}\{f * g\} = F(s)\,G(s)$$
3 $$(f*g)(t) = \int_0^t f(\tau)g(t-\tau)\,d\tau$$

TERMS $F(s)$ transform; s variable; u step; δ impulse; $f(0), f'(0)$ initial values

"Derivative in $t \rightarrow$ times s in s" (minus the initial values).

C2.33 (2/2)

1 $$\mathcal{L}\{f\}(s) = F(s) = \int_0^\infty e^{-st}f(t)\,dt$$
2 $$\mathcal{L}\{af + bg\} = aF + bG$$
$$\mathcal{L}\{f'\} = sF - f(0)$$
3 $$\mathcal{L}\{f''\} = s^2F - sf(0) - f'(0)$$

TERMS $F(s)$ transform; s variable; u step; δ impulse; $f(0), f'(0)$ initial values

"Derivative in $t \rightarrow$ times s in s" (minus the initial values).

C2.33 (1/2)

1 $$y_p = -y_1\int \frac{y_2\,g}{W}\,dx + y_2\int \frac{y_1\,g}{W}\,dx$$
$$W = y_1y_2' - y_1'y_2$$

TERMS y_h homogeneous, y_p particular; ω_0 natural frequency; W Wronskian

"Guess what you see" (same family as g); if the guess is already in y_h, "bump it by x".

C2.32 (4/4)

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LA.1 (1/3) LINEAR ALGEBRA

Vectors: Components, Magnitude & Unit Vectors

PART 1 OF 3

1. Vector between two points
2. Component (basis) form
3. Magnitude (Euclidean norm)

STATE EACH NUMBERED FORMULA

LA.1 (2/3) LINEAR ALGEBRA

Vectors: Components, Magnitude & Unit Vectors

PART 2 OF 3

1. Unit vector (normalizing)
2. Magnitude–direction form (2-D)
3. Length of a multiple

STATE EACH NUMBERED FORMULA

LA.1 (3/3) LINEAR ALGEBRA

Vectors: Components, Magnitude & Unit Vectors

PART 3 OF 3

1. Distance between points

STATE EACH NUMBERED FORMULA

LA.2 (1/2) LINEAR ALGEBRA

Vectors & Linear Combinations

PART 1 OF 2

1. Vector addition
2. Scalar multiplication
3. Linear combination

STATE EACH NUMBERED FORMULA

LA.2 (2/2) LINEAR ALGEBRA

Vectors & Linear Combinations

PART 2 OF 2

1. Subtraction
2. Matrix form of a combination
3. Algebra rules

STATE EACH NUMBERED FORMULA

LA.K1 LINEAR ALGEBRA

Point, segment, ray, line, vector: notation & meaning

RECALL IT

LA.K1 LINEAR ALGEBRA

Vector subtraction $\mathbf{a} - \mathbf{b}$ geometrically

RECALL IT

LA.K1 LINEAR ALGEBRA

Dot product vs. cross product

RECALL IT

LA.3 (1/3) LINEAR ALGEBRA

Dot Product & the Angle Between Vectors

PART 1 OF 3

1. Algebraic dot product
2. Geometric dot product
3. Angle between vectors

STATE EACH NUMBERED FORMULA

LA.3 (2/3) LINEAR ALGEBRA

Dot Product & the Angle Between Vectors

PART 2 OF 3

1. Sign test
2. Length from the dot product
3. Algebraic properties

STATE EACH NUMBERED FORMULA

LA.3 (3/3) LINEAR ALGEBRA

Dot Product & the Angle Between Vectors

PART 3 OF 3

1. Cauchy–Schwarz inequality
2. Triangle inequality
3. Cosine similarity

STATE EACH NUMBERED FORMULA

LA.4 (1/2) LINEAR ALGEBRA

Scalar & Vector Projection

PART 1 OF 2

1. Scalar projection (component)
2. Vector projection
3. Orthogonal component

STATE EACH NUMBERED FORMULA

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1 $d(P,Q) = \|Q - P\|$

TERMS $\mathbf{v}$ vector; v_i components; $\|\mathbf{v}\|$ length; $\hat{\mathbf{v}}$ unit vector

"Tip minus tail." A hat on a vector means "length 1".

1 $\hat{\mathbf{v}} = \dfrac{\mathbf{v}}{\|\mathbf{v}\|}$

2 $\mathbf{v} = \|\mathbf{v}\| \langle \cos\theta,\ \sin\theta \rangle$

3 $\|c\mathbf{v}\| = |c|\,\|\mathbf{v}\|$

TERMS $\mathbf{v}$ vector; v_i components; $\|\mathbf{v}\|$ length; $\hat{\mathbf{v}}$ unit vector

"Tip minus tail." A hat on a vector means "length 1".

1 $\overrightarrow{PQ} = Q - P = \langle q_1 - p_1,\ q_2 - p_2,\ q_3 - p_3 \rangle$

2 $\mathbf{v} = \langle v_1, v_2, v_3 \rangle = v_1\mathbf{i} + v_2\mathbf{j} + v_3\mathbf{k}$

3 $\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}$

TERMS $\mathbf{v}$ vector; v_i components; $\|\mathbf{v}\|$ length; $\hat{\mathbf{v}}$ unit vector

"Tip minus tail." A hat on a vector means "length 1".

Point $A = (x, y)$: location.
Segment $\overline{AB}$: two endpoints.
Ray $\overrightarrow{AB}$: from A through B, forever.
Line $\overleftrightarrow{AB}$: infinite both ways.
Vector $\mathbf{v} = \langle v_1, v_2 \rangle$, $\overrightarrow{AB} = B - A$: length + direction, no fixed position.

1 $\mathbf{u} - \mathbf{v} = \mathbf{u} + (-1)\mathbf{v}$

2 $c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k = V\mathbf{c}$

3 $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}, \quad c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$

TERMS $\mathbf{v}_i$ vectors; c_i scalar weights; V columns $\mathbf{v}_i$

Add tip-to-tail; scaling stretches (and a negative scalar flips).

1 $\mathbf{u} + \mathbf{v} = (u_1 + v_1,\ \ldots,\ u_n + v_n)$

2 $c\,\mathbf{v} = (cv_1,\ \ldots,\ cv_n)$

3 $\mathbf{w} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_k\mathbf{v}_k$

TERMS $\mathbf{v}_i$ vectors; c_i scalar weights; V columns $\mathbf{v}_i$

Add tip-to-tail; scaling stretches (and a negative scalar flips).

1 $\mathbf{u} \cdot \mathbf{v} = \displaystyle\sum_{i=1}^{n} u_i v_i = u_1 v_1 + \cdots + u_n v_n$

2 $\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\|\,\|\mathbf{v}\| \cos\theta$

3 $\theta = \arccos \dfrac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\|\,\|\mathbf{v}\|}$

TERMS θ angle between $\mathbf{u}, \mathbf{v}$; $\hat{\mathbf{u}}$ unit vector

"Dot" gives a dot-*sized answer: a single number. Zero dot product = right angle.*

$\mathbf{u} \cdot \mathbf{v} = \sum u_i v_i = \|\mathbf{u}\|\|\mathbf{v}\| \cos\theta$: a scalar; sign gives the angle type.
$\mathbf{a} \times \mathbf{b}$: a 3-D vector $\perp$ both, length $\|\mathbf{a}\|\|\mathbf{b}\| \sin\theta$ (area), right-hand rule.

$\mathbf{a} - \mathbf{b} = \mathbf{a} + (-\mathbf{b})$ is the arrow from the tip of $\mathbf{b}$ to the tip of $\mathbf{a}$.
$(4,3) - (1,2) = (3,1)$.

1 $\operatorname{comp}_{\mathbf{a}} \mathbf{b} = \dfrac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\|} = \|\mathbf{b}\| \cos\theta$

2 $\operatorname{proj}_{\mathbf{a}} \mathbf{b} = \dfrac{\mathbf{a} \cdot \mathbf{b}}{\mathbf{a} \cdot \mathbf{a}}\,\mathbf{a}$

3 $\mathbf{b}_\perp = \mathbf{b} - \operatorname{proj}_{\mathbf{a}} \mathbf{b}$

TERMS $\mathbf{a}$ direction ($\neq \mathbf{0}$); θ angle; $\mathbf{b}_\perp \perp \mathbf{a}$

"Dot over dot, times the direction": $\frac{\mathbf{a}\cdot\mathbf{b}}{\mathbf{a}\cdot\mathbf{a}}\,\mathbf{a}$*; the vector you project* onto *appears three times.*

1 $|\mathbf{u} \cdot \mathbf{v}| \le \|\mathbf{u}\|\,\|\mathbf{v}\|$

2 $\|\mathbf{u} + \mathbf{v}\| \le \|\mathbf{u}\| + \|\mathbf{v}\|$

3 $\cos\theta = \hat{\mathbf{u}} \cdot \hat{\mathbf{v}}$

TERMS θ angle between $\mathbf{u}, \mathbf{v}$; $\hat{\mathbf{u}}$ unit vector

"Dot" gives a dot-*sized answer: a single number. Zero dot product = right angle.*

1 $\mathbf{u} \cdot \mathbf{v}\ > 0,\ = 0,\ < 0$

2 $\mathbf{v} \cdot \mathbf{v} = \|\mathbf{v}\|^2$

3 $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}, \quad \mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}$

TERMS θ angle between $\mathbf{u}, \mathbf{v}$; $\hat{\mathbf{u}}$ unit vector

"Dot" gives a dot-*sized answer: a single number. Zero dot product = right angle.*

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LA.4 (2/2) LINEAR ALGEBRA

Scalar & Vector Projection

PART 2 OF 2

1. Unit-vector form
2. Projection matrix onto a line

STATE EACH NUMBERED FORMULA

LA.5 (1/3) LINEAR ALGEBRA

Cross Product & Triple Products

PART 1 OF 3

1. Determinant formula
2. Component formula
3. Magnitude = area

STATE EACH NUMBERED FORMULA

LA.5 (2/3) LINEAR ALGEBRA

Cross Product & Triple Products

PART 2 OF 3

1. Orthogonality (right-hand rule)
2. Anticommutativity
3. Basis products

STATE EACH NUMBERED FORMULA

LA.5 (3/3) LINEAR ALGEBRA

Cross Product & Triple Products

PART 3 OF 3

1. Scalar triple product
2. Vector triple product
3. Distributive law

STATE EACH NUMBERED FORMULA

LA.6 (1/2) LINEAR ALGEBRA

Vector Spaces & Subspaces

PART 1 OF 2

1. Subspace test
2. Span is a subspace
3. Homogeneous solutions

STATE EACH NUMBERED FORMULA

LA.6 (2/2) LINEAR ALGEBRA

Vector Spaces & Subspaces

PART 2 OF 2

1. Subspaces of $\mathbb{R}^3$
2. Dimension of a sum
3. Orthogonal complement

STATE EACH NUMBERED FORMULA

LA.7 (1/2) LINEAR ALGEBRA

Span, Linear Independence, Basis & Dimension

PART 1 OF 2

1. Span
2. Linear independence
3. Dimension

STATE EACH NUMBERED FORMULA

LA.7 (2/2) LINEAR ALGEBRA

Span, Linear Independence, Basis & Dimension

PART 2 OF 2

1. Square determinant test
2. Counting rule
3. Unique coordinates

STATE EACH NUMBERED FORMULA

LA.8 (1/2) LINEAR ALGEBRA

Change of Basis

PART 1 OF 2

1. Change-of-basis matrix
2. New $\to$ standard
3. Standard $\to$ new

STATE EACH NUMBERED FORMULA

LA.8 (2/2) LINEAR ALGEBRA

Change of Basis

PART 2 OF 2

1. Similar matrix
2. Similarity invariants

STATE EACH NUMBERED FORMULA

LA.9 (1/3) LINEAR ALGEBRA

Linear Transformations & Their Matrices

PART 1 OF 3

1. Linearity
2. Matrix representation
3. Columns rule

STATE EACH NUMBERED FORMULA

LA.9 (2/3) LINEAR ALGEBRA

Linear Transformations & Their Matrices

PART 2 OF 3

1. Composition
2. Standard 2-D maps
3. Area scaling

STATE EACH NUMBERED FORMULA

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1. $(\mathbf{a}\times\mathbf{b})\cdot\mathbf{a} = (\mathbf{a}\times\mathbf{b})\cdot\mathbf{b} = 0$
2. $\mathbf{b}\times\mathbf{a} = -(\mathbf{a}\times\mathbf{b}), \quad \mathbf{a}\times\mathbf{a} = \mathbf{0}$
3. $\mathbf{i}\times\mathbf{j} = \mathbf{k}, \quad \mathbf{j}\times\mathbf{k} = \mathbf{i}, \quad \mathbf{k}\times\mathbf{i} = \mathbf{j}$

TERMS $\mathbf{a},\mathbf{b},\mathbf{c}\in\mathbb{R}^3$; θ angle; $\mathbf{i},\mathbf{j},\mathbf{k}$ unit axes

Right hand: fingers along $\mathbf{a}$*, curl to* $\mathbf{b}$*, thumb* $= \mathbf{a}\times\mathbf{b}$*. Triple product: "BAC − CAB".*

1. $\mathbf{a}\times\mathbf{b} = \begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\ a_1 & a_2 & a_3\\ b_1 & b_2 & b_3\end{vmatrix}$
2. $\mathbf{a}\times\mathbf{b} = \langle a_2b_3 - a_3b_2,\ a_3b_1 - a_1b_3,\ a_1b_2 - a_2b_1\rangle$
3. $\|\mathbf{a}\times\mathbf{b}\| = \|\mathbf{a}\|\,\|\mathbf{b}\|\sin\theta$

TERMS $\mathbf{a},\mathbf{b},\mathbf{c}\in\mathbb{R}^3$; θ angle; $\mathbf{i},\mathbf{j},\mathbf{k}$ unit axes

Right hand: fingers along $\mathbf{a}$*, curl to* $\mathbf{b}$*, thumb* $= \mathbf{a}\times\mathbf{b}$*. Triple product: "BAC − CAB".*

1. $\operatorname{proj}_{\mathbf{a}}\mathbf{b} = (\hat{\mathbf{a}}\cdot\mathbf{b})\,\hat{\mathbf{a}}$
2. $P = \dfrac{\mathbf{a}\mathbf{a}^\top}{\mathbf{a}^\top\mathbf{a}}$

TERMS $\mathbf{a}$ direction ($\neq\mathbf{0}$); θ angle; $\mathbf{b}_\perp\perp\mathbf{a}$

"Dot over dot, times the direction": $\frac{\mathbf{a}\cdot\mathbf{b}}{\mathbf{a}\cdot\mathbf{a}}\,\mathbf{a}$*; the vector you project* onto *appears three times.*

1. $\{\mathbf{0}\}$, lines through $\mathbf{0}$, planes through $\mathbf{0}$, $\mathbb{R}^3$
2. $\dim(W_1+W_2) = \dim W_1 + \dim W_2 - \dim(W_1\cap W_2)$
3. $W^\perp = \{\mathbf{x} : \mathbf{x}\cdot\mathbf{w} = 0 \ \ \forall\mathbf{w}\in W\}$

TERMS W subspace; $\mathbf{0}$ zero vector; $W^\perp$ orthogonal complement

"Through the origin and flat." If $\mathbf{0}$ *is missing, it is not a subspace.*

1. $\mathbf{0}\in W; \quad \mathbf{u},\mathbf{v}\in W\Rightarrow\mathbf{u}+\mathbf{v}\in W; \quad c\mathbf{u}\in W$
2. $\operatorname{span}\{\mathbf{v}_1,\dots,\mathbf{v}_k\}\subseteq V$
3. $\{\mathbf{x} : A\mathbf{x} = \mathbf{0}\}$

TERMS W subspace; $\mathbf{0}$ zero vector; $W^\perp$ orthogonal complement

"Through the origin and flat." If $\mathbf{0}$ *is missing, it is not a subspace.*

1. $\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c}) = \det\begin{bmatrix}a_1 & a_2 & a_3\\ b_1 & b_2 & b_3\\ c_1 & c_2 & c_3\end{bmatrix}$
2. $\mathbf{a}\times(\mathbf{b}\times\mathbf{c}) = \mathbf{b}(\mathbf{a}\cdot\mathbf{c}) - \mathbf{c}(\mathbf{a}\cdot\mathbf{b})$
3. $\mathbf{a}\times(\mathbf{b}+\mathbf{c}) = \mathbf{a}\times\mathbf{b} + \mathbf{a}\times\mathbf{c}$

TERMS $\mathbf{a},\mathbf{b},\mathbf{c}\in\mathbb{R}^3$; θ angle; $\mathbf{i},\mathbf{j},\mathbf{k}$ unit axes

Right hand: fingers along $\mathbf{a}$*, curl to* $\mathbf{b}$*, thumb* $= \mathbf{a}\times\mathbf{b}$*. Triple product: "BAC − CAB".*

1. $P = \begin{bmatrix}\mathbf{b}_1 & \mathbf{b}_2 & \cdots & \mathbf{b}_n\end{bmatrix}$
2. $\mathbf{v} = P\,[\mathbf{v}]_B$
3. $[\mathbf{v}]_B = P^{-1}\mathbf{v}$

TERMS P columns = new basis; $[\mathbf{v}]_B$ coordinates in B

"P *takes you from the new basis to the old (standard) one;* P^{-1} *takes you back."*

1. $\mathbf{v}_1,\dots,\mathbf{v}_n$ basis of $\mathbb{R}^n \iff \det[\mathbf{v}_1\cdots\mathbf{v}_n]\neq 0$
2. $k>n\Rightarrow$ dependent; $\quad k<n\Rightarrow$ cannot span $\mathbb{R}^n$
3. $\mathbf{v} = c_1\mathbf{b}_1+\cdots+c_n\mathbf{b}_n$ in exactly one way

TERMS $\mathbf{v}_i$ vectors; c_i weights; basis = independent + spanning

A basis is a "minimal complete set": enough vectors to reach everything, none redundant. n *vectors in* $\mathbb{R}^n$ *form a basis* $\iff \det\neq 0$.

1. $\operatorname{span}\{\mathbf{v}_1,\dots,\mathbf{v}_k\} = \{c_1\mathbf{v}_1+\cdots+c_k\mathbf{v}_k\}$
2. $c_1\mathbf{v}_1+\cdots+c_k\mathbf{v}_k = \mathbf{0}\Rightarrow c_1=\cdots=c_k=0$
3. $\dim V =$ # vectors in any basis

TERMS $\mathbf{v}_i$ vectors; c_i weights; basis = independent + spanning

A basis is a "minimal complete set": enough vectors to reach everything, none redundant. n *vectors in* $\mathbb{R}^n$ *form a basis* $\iff \det\neq 0$.

1. $(S\circ T)(\mathbf{x}) = BA\,\mathbf{x}$
2. $\underbrace{\begin{bmatrix}s_x & 0\\ 0 & s_y\end{bmatrix}}_{\text{scale}}\ \underbrace{\begin{bmatrix}1 & k\\ 0 & 1\end{bmatrix}}_{\text{shear}}\ \underbrace{\begin{bmatrix}1 & 0\\ 0 & -1\end{bmatrix}}_{\text{reflect in } x\text{-axis}}$
3. $\operatorname{area}(T(S)) = |\det A|\cdot\operatorname{area}(S)$

TERMS $\mathbf{e}_i$ standard basis; A matrix of T; $|\det A|$ area factor

"Columns are where the basis vectors land."

1. $T(a\mathbf{u}+b\mathbf{v}) = a\,T(\mathbf{u}) + b\,T(\mathbf{v})$
2. $T(\mathbf{x}) = A\mathbf{x}$
3. $A = \begin{bmatrix}T(\mathbf{e}_1) & T(\mathbf{e}_2) & \cdots & T(\mathbf{e}_n)\end{bmatrix}$

TERMS $\mathbf{e}_i$ standard basis; A matrix of T; $|\det A|$ area factor

"Columns are where the basis vectors land."

1. $[A]_B = P^{-1}AP$
2. det, tr, eigenvalues and rank of $P^{-1}AP$ = those of A

TERMS P columns = new basis; $[\mathbf{v}]_B$ coordinates in B

"P *takes you from the new basis to the old (standard) one;* P^{-1} *takes you back."*

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LA.9 (3/3) LINEAR ALGEBRA

Linear Transformations & Their Matrices

PART 3 OF 3

1. Kernel and image

STATE EACH NUMBERED FORMULA

LA.10 (1/3) LINEAR ALGEBRA

Matrix Multiplication & the Dimension Rule

PART 1 OF 3

1. Entry formula
2. Dimension rule
3. Matrix–vector, column view

STATE EACH NUMBERED FORMULA

LA.10 (2/3) LINEAR ALGEBRA

Matrix Multiplication & the Dimension Rule

PART 2 OF 3

1. Matrix–vector, row view
2. Column-by-column
3. Identity

STATE EACH NUMBERED FORMULA

LA.10 (3/3) LINEAR ALGEBRA

Matrix Multiplication & the Dimension Rule

PART 3 OF 3

1. Non-commutativity
2. Matrix powers
3. Cost

STATE EACH NUMBERED FORMULA

LA.K2 LINEAR ALGEBRA

Matrix multiplication: entry rule & dimension rule

RECALL IT

LA.K2 LINEAR ALGEBRA

Matrix times vector: two views

RECALL IT

LA.11 (1/4) LINEAR ALGEBRA

Matrix Algebra Rules & Identities

PART 1 OF 4

1. Identity & inverse
2. Associative & distributive
3. Not commutative

STATE EACH NUMBERED FORMULA

LA.11 (2/4) LINEAR ALGEBRA

Matrix Algebra Rules & Identities

PART 2 OF 4

1. Transpose
2. Transpose of sum & multiple
3. Transpose of a product

STATE EACH NUMBERED FORMULA

LA.11 (3/4) LINEAR ALGEBRA

Matrix Algebra Rules & Identities

PART 3 OF 4

1. Inverse of a product
2. Inverse of transpose, multiple, inverse
3. Trace rules

STATE EACH NUMBERED FORMULA

LA.11 (4/4) LINEAR ALGEBRA

Matrix Algebra Rules & Identities

PART 4 OF 4

1. Gram matrix is symmetric

STATE EACH NUMBERED FORMULA

LA.12 (1/3) LINEAR ALGEBRA

Inner, Outer, Hadamard, Kronecker & Block Products

PART 1 OF 3

1. Inner product
2. Outer product
3. Hadamard (elementwise) product

STATE EACH NUMBERED FORMULA

LA.12 (2/3) LINEAR ALGEBRA

Inner, Outer, Hadamard, Kronecker & Block Products

PART 2 OF 3

1. Kronecker product
2. Column view of Ax
3. Outer-product view of AB

STATE EACH NUMBERED FORMULA

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1. $(A\mathbf{x})_i = \mathbf{r}_i \cdot \mathbf{x}$
2. $AB = \begin{bmatrix} A\mathbf{b}_1 & A\mathbf{b}_2 & \cdots & A\mathbf{b}_p \end{bmatrix}$
3. $I_m A = A I_n = A$

TERMS a_{ik} entry row i col k; $\mathbf{a}_j$ column j; I identity

"Row times column": slide across the row of A and down the column of B. "Inner must match, outer is the answer."

1. $(AB)_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}$
2. $(m \times n)(n \times p) = m \times p$
3. $A\mathbf{x} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n$

TERMS a_{ik} entry row i col k; $\mathbf{a}_j$ column j; I identity

"Row times column": slide across the row of A and down the column of B. "Inner must match, outer is the answer."

1. $\ker T = \text{Null}\, A, \quad \text{im}\, T = \text{Col}\, A$

TERMS $\mathbf{e}_i$ standard basis; A matrix of T; $|\det A|$ area factor

"Columns are where the basis vectors land."

Row view: $(A\mathbf{x})_i = \text{row}_i \cdot \mathbf{x}$.
Column view: $A\mathbf{x} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n$.
$\left[\begin{smallmatrix} 2 & 1 \\ 1 & 3 \end{smallmatrix}\right]\left[\begin{smallmatrix} 2 \\ 1 \end{smallmatrix}\right] = 2\left[\begin{smallmatrix} 2 \\ 1 \end{smallmatrix}\right] + 1\left[\begin{smallmatrix} 1 \\ 3 \end{smallmatrix}\right] = \left[\begin{smallmatrix} 5 \\ 5 \end{smallmatrix}\right]$

$(AB)_{ij} = \sum_k a_{ik} b_{kj}$: row i of A · column j of B.
$(m \times n)(n \times p) = m \times p$: inner sizes match, outer sizes give the result.
$\left[\begin{smallmatrix} 1 & 2 \\ 3 & 4 \end{smallmatrix}\right]\left[\begin{smallmatrix} 5 & 6 \\ 7 & 8 \end{smallmatrix}\right] = \left[\begin{smallmatrix} 19 & 22 \\ 43 & 50 \end{smallmatrix}\right]$

1. $AB \neq BA$ in general
2. $A^k = \underbrace{AA\cdots A}_{k}, \quad A^0 = I$
3. $m \cdot n \cdot p$ multiplications

TERMS a_{ik} entry row i col k; $\mathbf{a}_j$ column j; I identity

"Row times column": slide across the row of A and down the column of B. "Inner must match, outer is the answer."

1. $(AB)^{-1} = B^{-1}A^{-1}$
2. $(A^\top)^{-1} = (A^{-1})^\top, \quad (cA)^{-1} = \frac{1}{c}A^{-1}, \quad (A^{-1})^{-1} = A$
3. $\text{tr}(A + B) = \text{tr}\, A + \text{tr}\, B, \qquad \text{tr}(AB) = \text{tr}(BA)$

TERMS I identity; $A^\top$ transpose; A^{-1} inverse; tr trace

"Socks and shoes": to undo (or transpose) a product, reverse the order, $(AB)^{-1} = B^{-1}A^{-1}$.

1. $(A^\top)_{ij} = a_{ji}, \qquad (A^\top)^\top = A$
2. $(A + B)^\top = A^\top + B^\top, \qquad (cA)^\top = cA^\top$
3. $(AB)^\top = B^\top A^\top$

TERMS I identity; $A^\top$ transpose; A^{-1} inverse; tr trace

"Socks and shoes": to undo (or transpose) a product, reverse the order, $(AB)^{-1} = B^{-1}A^{-1}$.

1. $AI = IA = A, \qquad AA^{-1} = A^{-1}A = I$
2. $A(BC) = (AB)C, \qquad A(B + C) = AB + AC$
3. $AB \neq BA$ (in general)

TERMS I identity; $A^\top$ transpose; A^{-1} inverse; tr trace

"Socks and shoes": to undo (or transpose) a product, reverse the order, $(AB)^{-1} = B^{-1}A^{-1}$.

1. $A \otimes B = \begin{bmatrix} a_{11}B & \cdots & a_{1n}B \\ \vdots & \ddots & \vdots \\ a_{m1}B & \cdots & a_{mn}B \end{bmatrix}$
2. $A\mathbf{x} = \sum_j x_j \mathbf{a}_j$
3. $AB = \sum_k \mathbf{a}_k \mathbf{b}_k^\top$

TERMS $\odot$ elementwise; $\otimes$ Kronecker; $\mathbf{a}_k$ columns of A

Inner product "collapses" ($1 \times n$ times $n \times 1$ is 1×1); outer product "expands" ($n \times 1$ times $1 \times m$ is $n \times m$).

1. $\mathbf{u}^\top \mathbf{v} = \sum_i u_i v_i$
2. $\mathbf{u}\mathbf{v}^\top = \begin{bmatrix} u_i v_j \end{bmatrix}$
3. $(A \odot B)_{ij} = a_{ij} b_{ij}$

TERMS $\odot$ elementwise; $\otimes$ Kronecker; $\mathbf{a}_k$ columns of A

Inner product "collapses" ($1 \times n$ times $n \times 1$ is 1×1); outer product "expands" ($n \times 1$ times $1 \times m$ is $n \times m$).

1. $(A^\top A)^\top = A^\top A$

TERMS I identity; $A^\top$ transpose; A^{-1} inverse; tr trace

"Socks and shoes": to undo (or transpose) a product, reverse the order, $(AB)^{-1} = B^{-1}A^{-1}$.

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LA.12 (3/3) LINEAR ALGEBRA

Inner, Outer, Hadamard, Kronecker & Block Products

PART 3 OF 3

1. Block multiplication

STATE EACH NUMBERED FORMULA

LA.13 (1/3) LINEAR ALGEBRA

Determinants: Cofactor Expansion & Geometry

PART 1 OF 3

1. 2×2 determinant
2. Cofactor (Laplace) expansion
3. Checkerboard of signs

STATE EACH NUMBERED FORMULA

LA.13 (2/3) LINEAR ALGEBRA

Determinants: Cofactor Expansion & Geometry

PART 2 OF 3

1. 3×3 rule of Sarrus

STATE EACH NUMBERED FORMULA

LA.13 (3/3) LINEAR ALGEBRA

Determinants: Cofactor Expansion & Geometry

PART 3 OF 3

1. Triangular matrix
2. Geometric meaning
3. Invertibility test

STATE EACH NUMBERED FORMULA

LA.14 (1/3) LINEAR ALGEBRA

Determinant Properties & Row Operations

PART 1 OF 3

1. Product rule
2. Transpose
3. Inverse

STATE EACH NUMBERED FORMULA

LA.14 (2/3) LINEAR ALGEBRA

Determinant Properties & Row Operations

PART 2 OF 3

1. Scalar multiple
2. Row swap
3. Row scaling

STATE EACH NUMBERED FORMULA

LA.14 (3/3) LINEAR ALGEBRA

Determinant Properties & Row Operations

PART 3 OF 3

1. Row replacement
2. Zero or repeated row
3. Cramer's rule ($n \times n$)

STATE EACH NUMBERED FORMULA

LA.15 (1/3) LINEAR ALGEBRA

Matrix Inverse by Gauss–Jordan

PART 1 OF 3

1. Definition
2. Gauss–Jordan method
3. 2×2 inverse

STATE EACH NUMBERED FORMULA

LA.15 (2/3) LINEAR ALGEBRA

Matrix Inverse by Gauss–Jordan

PART 2 OF 3

1. Adjugate formula
2. Solving with the inverse

STATE EACH NUMBERED FORMULA

LA.15 (3/3) LINEAR ALGEBRA

Matrix Inverse by Gauss–Jordan

PART 3 OF 3

1. Invertible matrix theorem

STATE EACH NUMBERED FORMULA

LA.16 (1/3) LINEAR ALGEBRA

Special Matrices

PART 1 OF 3

1. Diagonal
2. Triangular
3. Symmetric & skew-symmetric

STATE EACH NUMBERED FORMULA

LA.16 (2/3) LINEAR ALGEBRA

Special Matrices

PART 2 OF 3

1. Orthogonal
2. Idempotent
3. Nilpotent

STATE EACH NUMBERED FORMULA

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1. $\det A = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33}$

TERMS M_{ij} minor; $(-1)^{i+j}$ sign; $|\det A|$ area factor

Expand along the row or column with the most zeros.
"$\det = 0 \iff$ *flat* $\iff$ *no inverse.*"

1. $\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$
2. $\det A = \sum_{j=1}^{n} (-1)^{i+j} a_{ij} M_{ij}$
3. $(-1)^{i+j}$: $\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$

TERMS M_{ij} minor; $(-1)^{i+j}$ sign; $|\det A|$ area factor

Expand along the row or column with the most zeros.
"$\det = 0 \iff$ *flat* $\iff$ *no inverse.*"

1. $\begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} E \\ F \end{bmatrix} = \begin{bmatrix} AE + BF \\ CE + DF \end{bmatrix}$

TERMS $\odot$ elementwise; $\otimes$ Kronecker; $\mathbf{a}_k$ columns of A

Inner product "collapses" ($1 \times n$ times $n \times 1$ is 1×1); outer product "expands" ($n \times 1$ times $1 \times m$ is $n \times m$).

1. $\det(cA) = c^n \det A$
2. $R_i \leftrightarrow R_j$: $\det \to -\det$
3. $R_i \to cR_i$: $\det \to c \det$

TERMS c scalar; n matrix size; A_i column i replaced by $\mathbf{b}$

"Swap flips, scale scales, replace keeps."

1. $\det(AB) = \det A \, \det B$
2. $\det\left(A^\top\right) = \det A$
3. $\det\left(A^{-1}\right) = \dfrac{1}{\det A}$

TERMS c scalar; n matrix size; A_i column i replaced by $\mathbf{b}$

"Swap flips, scale scales, replace keeps."

1. $\det A = a_{11}a_{22}\cdots a_{nn}$
2. $|\det A|$ = area / volume scale factor
3. $\det A \neq 0 \iff A^{-1}$ exists

TERMS M_{ij} minor; $(-1)^{i+j}$ sign; $|\det A|$ area factor

Expand along the row or column with the most zeros.
"$\det = 0 \iff$ *flat* $\iff$ *no inverse.*"

1. $A^{-1} = \dfrac{1}{\det A} \operatorname{adj} A, \qquad \operatorname{adj} A = C^\top$
2. $A\mathbf{x} = \mathbf{b} \Rightarrow \mathbf{x} = A^{-1}\mathbf{b}$

TERMS I identity; C cofactors; $\operatorname{adj} A = C^\top$

"Whatever turns A into I turns I into A^{-1}."

1. $AA^{-1} = A^{-1}A = I$
2. $\left[\, A \mid I \,\right] \xrightarrow{\text{row ops}} \left[\, I \mid A^{-1} \,\right]$
3. $\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \dfrac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

TERMS I identity; C cofactors; $\operatorname{adj} A = C^\top$

"Whatever turns A into I turns I into A^{-1}."

1. $R_i \to R_i + kR_j$: det unchanged
2. zero row or two equal rows $\Rightarrow \det A = 0$
3. $x_i = \dfrac{\det A_i}{\det A}$

TERMS c scalar; n matrix size; A_i column i replaced by $\mathbf{b}$

"Swap flips, scale scales, replace keeps."

1. $Q^\top Q = I$
2. $P^2 = P$
3. $N^k = 0$ for some k

TERMS d_i diagonal entries; I identity; k a power

Orthogonal matrices are "rigid": they rotate or reflect without stretching, so their inverse is just the transpose.

1. $D = \operatorname{diag}(d_1, \ldots, d_n)$
2. upper/lower; $\det = \prod a_{ii}$
3. $A = A^\top, \qquad A^\top = -A$

TERMS d_i diagonal entries; I identity; k a power

Orthogonal matrices are "rigid": they rotate or reflect without stretching, so their inverse is just the transpose.

1. A^{-1} exists $\iff \det A \neq 0 \iff \operatorname{rank} A = n \iff \operatorname{Null} A = \{\mathbf{0}\} \iff$ columns form a basis $\iff 0$ is not an eigenvalue

TERMS I identity; C cofactors; $\operatorname{adj} A = C^\top$

"Whatever turns A into I turns I into A^{-1}."

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LA.16 (3/3) LINEAR ALGEBRA

Special Matrices

PART 3 OF 3

1. Permutation
2. Positive (semi)definite

STATE EACH NUMBERED FORMULA

LA.17 (1/2) LINEAR ALGEBRA

Gaussian Elimination & Row Reduction

PART 1 OF 2

1. Augmented matrix
2. Elementary row operations
3. Row echelon form (REF)

STATE EACH NUMBERED FORMULA

LA.17 (2/2) LINEAR ALGEBRA

Gaussian Elimination & Row Reduction

PART 2 OF 2

1. Reduced row echelon form (RREF)
2. Number of solutions
3. Cost

STATE EACH NUMBERED FORMULA

LA.18 (1/3) LINEAR ALGEBRA

LU & Cholesky Decompositions

PART 1 OF 3

1. LU factorization
2. With pivoting
3. Solving with LU

STATE EACH NUMBERED FORMULA

LA.18 (2/3) LINEAR ALGEBRA

LU & Cholesky Decompositions

PART 2 OF 3

1. Cholesky factorization
2. Cholesky entries

STATE EACH NUMBERED FORMULA

LA.18 (3/3) LINEAR ALGEBRA

LU & Cholesky Decompositions

PART 3 OF 3

1. Determinant via LU
2. Cost

STATE EACH NUMBERED FORMULA

LA.19 (1/2) LINEAR ALGEBRA

Rank, Column Space, Null Space & Rank–Nullity

PART 1 OF 2

1. Column space
2. Null space
3. Rank

STATE EACH NUMBERED FORMULA

LA.19 (2/2) LINEAR ALGEBRA

Rank, Column Space, Null Space & Rank–Nullity

PART 2 OF 2

1. Rank–nullity theorem
2. Four fundamental subspaces
3. General solution

STATE EACH NUMBERED FORMULA

LA.20 (1/3) LINEAR ALGEBRA

Orthogonality, Projections & Least Squares

PART 1 OF 3

1. Orthogonality test
2. Projection onto a line
3. Least-squares problem

STATE EACH NUMBERED FORMULA

LA.20 (2/3) LINEAR ALGEBRA

Orthogonality, Projections & Least Squares

PART 2 OF 3

1. Normal equations
2. Projection onto Col A
3. Projection matrix

STATE EACH NUMBERED FORMULA

LA.20 (3/3) LINEAR ALGEBRA

Orthogonality, Projections & Least Squares

PART 3 OF 3

1. Orthonormal columns
2. Projection with orthonormal basis

STATE EACH NUMBERED FORMULA

LA.21 (1/2) LINEAR ALGEBRA

Gram–Schmidt & QR Factorization

PART 1 OF 2

1. First vector
2. Subtract the shadows
3. Normalize

STATE EACH NUMBERED FORMULA

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1 pivots $= 1$, zeros above and below
2 unique / none / infinitely many
3 $\approx \frac{2}{3}n^3$ operations

TERMS $[A \mid \mathbf{b}]$ augmented; pivot = leading entry; free = no pivot

Solutions: a pivot in every column → unique; a row $[0 \cdots 0 \mid c \neq 0]$ → none; a free variable → infinitely many.

LA.17 (2/2)

1 $A\mathbf{x} = \mathbf{b} \longleftrightarrow \left[\, A \mid \mathbf{b} \,\right]$
2 $R_i \leftrightarrow R_j, \quad R_i \to cR_i \ (c \neq 0), \quad R_i \to R_i + cR_j$
3 zeros below each pivot; pivots step right

TERMS $[A \mid \mathbf{b}]$ augmented; pivot = leading entry; free = no pivot

Solutions: a pivot in every column → unique; a row $[0 \cdots 0 \mid c \neq 0]$ → none; a free variable → infinitely many.

LA.17 (1/2)

1 $P\mathbf{x} =$ reordered $\mathbf{x}$
2 $\mathbf{x}^\top A\mathbf{x} > 0 \ (\geq 0)$

TERMS d_i diagonal entries; I identity; k a power

Orthogonal matrices are "rigid": they rotate or reflect without stretching, so their inverse is just the transpose.

LA.16 (3/3)

1 $\det A = \pm \prod_i u_{ii}$
2 $LU \approx \frac{2}{3}n^3, \quad$ Cholesky $\approx \frac{1}{3}n^3$

TERMS L lower-, U upper-triangular; P permutation; SPD for Cholesky

LU = "elimination with a memory". Cholesky = "square root of a matrix".

LA.18 (3/3)

1 $A = LL^\top$
2 $l_{jj} = \sqrt{a_{jj} - \sum_{k<j} l_{jk}^2}, \qquad l_{ij} = \frac{1}{l_{jj}}\left(a_{ij} - \sum_{k<j} l_{ik}l_{jk}\right) \ (i > j)$

TERMS L lower-, U upper-triangular; P permutation; SPD for Cholesky

LU = "elimination with a memory". Cholesky = "square root of a matrix".

LA.18 (2/3)

1 $A = LU$
2 $PA = LU$
3 $L\mathbf{y} = \mathbf{b}$ (forward), $\quad U\mathbf{x} = \mathbf{y}$ (back)

TERMS L lower-, U upper-triangular; P permutation; SPD for Cholesky

LU = "elimination with a memory". Cholesky = "square root of a matrix".

LA.18 (1/3)

1 $\mathbf{u} \perp \mathbf{v} \iff \mathbf{u} \cdot \mathbf{v} = 0$
2 $\text{proj}_{\mathbf{a}}\, \mathbf{b} = \dfrac{\mathbf{a} \cdot \mathbf{b}}{\mathbf{a} \cdot \mathbf{a}}\, \mathbf{a}$
3 $\hat{\mathbf{x}} = \arg\min_{\mathbf{x}} \|A\mathbf{x} - \mathbf{b}\|^2$

TERMS $\hat{\mathbf{x}}$ best fit; P projection matrix; Q orthonormal columns

Least squares: the error $\mathbf{b} - A\hat{\mathbf{x}}$ is perpendicular to every column of A.

LA.20 (1/3)

1 $\text{rank}\, A + \dim \text{Null}\, A = n$
2 $\text{Row}\, A \perp \text{Null}\, A, \qquad \text{Col}\, A \perp \text{Null}\, A^\top$
3 $\mathbf{x} = \mathbf{x}_p + \mathbf{x}_n$

TERMS n columns; rank = pivots; nullity = free variables

"Rank = pivots; nullity = free variables; together = columns."

LA.19 (2/2)

1 $\text{Col}\, A = \text{span}\{\text{columns}\} = \{A\mathbf{x}\}$
2 $\text{Null}\, A = \{\mathbf{x} : A\mathbf{x} = \mathbf{0}\}$
3 $\text{rank}\, A = \dim \text{Col}\, A = \dim \text{Row}\, A$

TERMS n columns; rank = pivots; nullity = free variables

"Rank = pivots; nullity = free variables; together = columns."

LA.19 (1/2)

1 $\mathbf{u}_1 = \mathbf{v}_1$
2 $\mathbf{u}_k = \mathbf{v}_k - \sum_{j=1}^{k-1} \dfrac{\mathbf{u}_j \cdot \mathbf{v}_k}{\mathbf{u}_j \cdot \mathbf{u}_j}\, \mathbf{u}_j$
3 $\mathbf{q}_k = \dfrac{\mathbf{u}_k}{\|\mathbf{u}_k\|}$

TERMS $\mathbf{v}_k$ input; $\mathbf{u}_k$ orthogonal; $\mathbf{q}_k$ orthonormal

"Subtract the shadows, then normalize."

LA.21 (1/2)

1 $Q^\top Q = I$
2 $\hat{\mathbf{b}} = QQ^\top \mathbf{b} = \sum_i (\mathbf{q}_i \cdot \mathbf{b})\, \mathbf{q}_i$

TERMS $\hat{\mathbf{x}}$ best fit; P projection matrix; Q orthonormal columns

Least squares: the error $\mathbf{b} - A\hat{\mathbf{x}}$ is perpendicular to every column of A.

LA.20 (3/3)

1 $A^\top A\, \hat{\mathbf{x}} = A^\top \mathbf{b}$
2 $\hat{\mathbf{b}} = A(A^\top A)^{-1} A^\top \mathbf{b}$
3 $P = A(A^\top A)^{-1} A^\top, \quad P^2 = P = P^\top$

TERMS $\hat{\mathbf{x}}$ best fit; P projection matrix; Q orthonormal columns

Least squares: the error $\mathbf{b} - A\hat{\mathbf{x}}$ is perpendicular to every column of A.

LA.20 (2/3)

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LA.21 (2/2) LINEAR ALGEBRA

Gram–Schmidt & QR Factorization

PART 2 OF 2

1. QR factorization
2. Entries of R
3. Least squares via QR

STATE EACH NUMBERED FORMULA

LA.22 (1/3) LINEAR ALGEBRA

Orthogonal Matrices, Rotations & Reflections

PART 1 OF 3

1. Orthogonal matrix
2. Preserves geometry
3. Determinant

STATE EACH NUMBERED FORMULA

LA.22 (2/3) LINEAR ALGEBRA

Orthogonal Matrices, Rotations & Reflections

PART 2 OF 3

1. 2-D rotation
2. Composing rotations
3. 2-D reflection

STATE EACH NUMBERED FORMULA

LA.22 (3/3) LINEAR ALGEBRA

Orthogonal Matrices, Rotations & Reflections

PART 3 OF 3

1. 3-D rotation about z
2. Householder reflection

STATE EACH NUMBERED FORMULA

LA.23 (1/2) LINEAR ALGEBRA

Eigenvalues & Eigenvectors

PART 1 OF 2

1. Eigen-equation
2. Characteristic equation
3. Eigenvectors

STATE EACH NUMBERED FORMULA

LA.23 (2/2) LINEAR ALGEBRA

Eigenvalues & Eigenvectors

PART 2 OF 2

1. Trace = sum
2. Determinant = product
3. 2×2 shortcut

STATE EACH NUMBERED FORMULA

LA.24 (1/3) LINEAR ALGEBRA

Eigenvalue Properties & Power Iteration

PART 1 OF 3

1. Powers
2. Inverse
3. Shift

STATE EACH NUMBERED FORMULA

LA.24 (2/3) LINEAR ALGEBRA

Eigenvalue Properties & Power Iteration

PART 2 OF 3

1. Shared spectrum
2. Multiplicities
3. Cayley–Hamilton

STATE EACH NUMBERED FORMULA

LA.24 (3/3) LINEAR ALGEBRA

Eigenvalue Properties & Power Iteration

PART 3 OF 3

1. Power iteration
2. Rayleigh quotient

STATE EACH NUMBERED FORMULA

LA.25 (1/2) LINEAR ALGEBRA

Diagonalization & Matrix Powers

PART 1 OF 2

1. Diagonalization
2. Eigenvalue matrix
3. Matrix powers

STATE EACH NUMBERED FORMULA

LA.25 (2/2) LINEAR ALGEBRA

Diagonalization & Matrix Powers

PART 2 OF 2

1. Spectral theorem
2. When possible
3. Matrix exponential

STATE EACH NUMBERED FORMULA

LA.26 (1/2) LINEAR ALGEBRA

Markov Chains & Stochastic Matrices

PART 1 OF 2

1. Transition matrix
2. Distribution update
3. Stationary distribution

STATE EACH NUMBERED FORMULA

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1 $R(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$

2 $R(\alpha)R(\beta) = R(\alpha+\beta), \quad R(\theta)^{-1} = R(-\theta) = R(\theta)^\top$

3 $F(\theta) = \begin{bmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{bmatrix}$

TERMS $Q^{-1} = Q^\top$; θ counterclockwise angle; $\hat{\mathbf{u}}$ unit normal

Rotation matrix: "first column is where $\mathbf{e}_1$ lands: $(\cos\theta, \sin\theta)$."

1 $Q^\top Q = QQ^\top = I, \qquad Q^{-1} = Q^\top$

2 $\|Q\mathbf{x}\| = \|\mathbf{x}\|, \qquad (Q\mathbf{x}) \cdot (Q\mathbf{y}) = \mathbf{x} \cdot \mathbf{y}$

3 $\det Q = \pm 1$

TERMS $Q^{-1} = Q^\top$; θ counterclockwise angle; $\hat{\mathbf{u}}$ unit normal

Rotation matrix: "first column is where $\mathbf{e}_1$ lands: $(\cos\theta, \sin\theta)$."

1 $A = QR$

2 $r_{ij} = \mathbf{q}_i \cdot \mathbf{a}_j \ (i \le j)$

3 $R\,\hat{\mathbf{x}} = Q^\top \mathbf{b}$

TERMS $\mathbf{v}_k$ input; $\mathbf{u}_k$ orthogonal; $\mathbf{q}_k$ orthonormal

"Subtract the shadows, then normalize."

1 $\operatorname{tr} A = \sum_i \lambda_i$

2 $\det A = \prod_i \lambda_i$

3 $\lambda^2 - (\operatorname{tr} A)\lambda + \det A = 0$

TERMS λ eigenvalue; $\mathbf{v} \neq \mathbf{0}$ eigenvector; tr trace

"Eigen" is German for "own": an eigenvector keeps its own direction; λ is the stretch factor.

1 $A\mathbf{v} = \lambda\mathbf{v} \quad (\mathbf{v} \neq \mathbf{0})$

2 $\det(A - \lambda I) = 0$

3 $(A - \lambda I)\mathbf{v} = \mathbf{0}$

TERMS λ eigenvalue; $\mathbf{v} \neq \mathbf{0}$ eigenvector; tr trace

"Eigen" is German for "own": an eigenvector keeps its own direction; λ is the stretch factor.

1 $R_z(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$

2 $H = I - 2\hat{\mathbf{u}}\hat{\mathbf{u}}^\top$

TERMS $Q^{-1} = Q^\top$; θ counterclockwise angle; $\hat{\mathbf{u}}$ unit normal

Rotation matrix: "first column is where $\mathbf{e}_1$ lands: $(\cos\theta, \sin\theta)$."

1 $\mathbf{x}_{k+1} = \dfrac{A\mathbf{x}_k}{\|A\mathbf{x}_k\|}$

2 $R(\mathbf{x}) = \dfrac{\mathbf{x}^\top A\mathbf{x}}{\mathbf{x}^\top\mathbf{x}}, \quad \lambda_{\min} \le R \le \lambda_{\max}$

TERMS λ eigenvalue; $\mathbf{v}$ eigenvector; p characteristic polynomial

"Whatever you do to A, do to λ": $A^k \to \lambda^k$, $A^{-1} \to 1/\lambda$, $A + cI \to \lambda + c$.

1 A, $A^\top$, $P^{-1}AP$ have the same eigenvalues

2 $1 \le$ geometric $\le$ algebraic

3 $p(A) = 0$

TERMS λ eigenvalue; $\mathbf{v}$ eigenvector; p characteristic polynomial

"Whatever you do to A, do to λ": $A^k \to \lambda^k$, $A^{-1} \to 1/\lambda$, $A + cI \to \lambda + c$.

1 $A^k\mathbf{v} = \lambda^k\mathbf{v}$

2 $A^{-1}\mathbf{v} = \frac{1}{\lambda}\mathbf{v}$

3 $(A + cI)\mathbf{v} = (\lambda + c)\mathbf{v}$

TERMS λ eigenvalue; $\mathbf{v}$ eigenvector; p characteristic polynomial

"Whatever you do to A, do to λ": $A^k \to \lambda^k$, $A^{-1} \to 1/\lambda$, $A + cI \to \lambda + c$.

1 $P_{ij} = \Pr(\text{next} = j \mid \text{now} = i), \quad \sum_j P_{ij} = 1$

2 $\boldsymbol{\pi}_{k+1} = \boldsymbol{\pi}_k P, \qquad \boldsymbol{\pi}_k = \boldsymbol{\pi}_0 P^k$

3 $\boldsymbol{\pi} P = \boldsymbol{\pi}, \quad \sum_i \pi_i = 1$

TERMS P row-stochastic; π stationary; a, b switch probabilities

"Rows sum to one; the steady state is the eigenvector for 1."

1 $A = A^\top \ \Rightarrow \ A = Q\Lambda Q^\top$

2 n independent eigenvectors

3 $e^{At} = Pe^{Dt}P^{-1}$

TERMS P eigenvectors; D eigenvalues; Q orthonormal (symmetric case)

In the eigenbasis, a matrix just stretches each axis; powers become powers of numbers.

1 $A = PDP^{-1}$

2 $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$

3 $A^k = PD^kP^{-1}$

TERMS P eigenvectors; D eigenvalues; Q orthonormal (symmetric case)

In the eigenbasis, a matrix just stretches each axis; powers become powers of numbers.

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LA.26 (2/2) LINEAR ALGEBRA

Markov Chains & Stochastic Matrices

PART 2 OF 2

1. Eigenvalue bound
2. Convergence
3. Two-state chain

STATE EACH NUMBERED FORMULA

LA.27 (1/2) LINEAR ALGEBRA

Symmetric, Positive Definite Matrices & Quadratic Forms

PART 1 OF 2

1. Quadratic form
2. Positive definite
3. Eigenvalue test

STATE EACH NUMBERED FORMULA

LA.27 (2/2) LINEAR ALGEBRA

Symmetric, Positive Definite Matrices & Quadratic Forms

PART 2 OF 2

1. Sylvester's criterion
2. 2×2 test
3. Semidefinite & Gram matrices

STATE EACH NUMBERED FORMULA

LA.28 (1/2) LINEAR ALGEBRA

Singular Value Decomposition (SVD)

PART 1 OF 2

1. SVD
2. Singular values
3. Singular vectors

STATE EACH NUMBERED FORMULA

LA.28 (2/2) LINEAR ALGEBRA

Singular Value Decomposition (SVD)

PART 2 OF 2

1. Best rank-k approximation
2. Spectral norm
3. Rank

STATE EACH NUMBERED FORMULA

LA.29 (1/3) LINEAR ALGEBRA

Norms, Trace & Condition Number

PART 1 OF 3

1. L_1 norm
2. L_2 norm
3. L_∞ norm

STATE EACH NUMBERED FORMULA

LA.29 (2/3) LINEAR ALGEBRA

Norms, Trace & Condition Number

PART 2 OF 3

1. L_p norm
2. Frobenius norm
3. Induced (operator) norm

STATE EACH NUMBERED FORMULA

LA.29 (3/3) LINEAR ALGEBRA

Norms, Trace & Condition Number

PART 3 OF 3

1. Condition number
2. Trace

STATE EACH NUMBERED FORMULA

LA.30 (1/2) LINEAR ALGEBRA

Matrix Calculus Identities

PART 1 OF 2

1. Linear function
2. Quadratic form
3. Squared norm

STATE EACH NUMBERED FORMULA

LA.30 (2/2) LINEAR ALGEBRA

Matrix Calculus Identities

PART 2 OF 2

1. Least-squares gradient
2. Trace derivative
3. Log-determinant derivative

STATE EACH NUMBERED FORMULA

LA.31 (1/3) LINEAR ALGEBRA

Pseudoinverse & Inverse-Update Identities

PART 1 OF 3

1. Pseudoinverse (full column rank)
2. Pseudoinverse via SVD

STATE EACH NUMBERED FORMULA

LA.31 (2/3) LINEAR ALGEBRA

Pseudoinverse & Inverse-Update Identities

PART 2 OF 3

1. Sherman–Morrison formula

STATE EACH NUMBERED FORMULA

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1. $A \succ 0 \iff$ all leading principal minors > 0
2. $\begin{bmatrix} a & b \\ b & c \end{bmatrix} \succ 0 \iff a > 0,\ ac - b^2 > 0$
3. $A \succeq 0 \iff \lambda_i \geq 0; \qquad X^\top X \succeq 0$

TERMS $\succ 0$ PD; $\succeq 0$ PSD; minors = top-left determinants

Positive definite = a bowl (unique minimum); indefinite = a saddle C3.16 *§10 Calculus III.*

1. $q(\mathbf{x}) = \mathbf{x}^\top A \mathbf{x} = \sum_{i,j} a_{ij} x_i x_j$
2. $A \succ 0 \iff \mathbf{x}^\top A \mathbf{x} > 0 \ \ \forall \mathbf{x} \neq \mathbf{0}$
3. $A \succ 0 \iff$ all $\lambda_i > 0$

TERMS $\succ 0$ PD; $\succeq 0$ PSD; minors = top-left determinants

Positive definite = a bowl (unique minimum); indefinite = a saddle C3.16 *§10 Calculus III.*

1. $\lambda = 1$ always; $\quad |\lambda| \leq 1$
2. P^k has all entries $> 0 \Rightarrow \boldsymbol{\pi}_k \to \boldsymbol{\pi}$
3. $P = \begin{bmatrix} 1-a & a \\ b & 1-b \end{bmatrix} \Rightarrow \boldsymbol{\pi} = \left(\frac{b}{a+b}, \frac{a}{a+b} \right)$

TERMS P row-stochastic; π stationary; a, b switch probabilities

"Rows sum to one; the steady state is the eigenvector for 1."

1. $\|\mathbf{x}\|_1 = \sum_i |x_i|$
2. $\|\mathbf{x}\|_2 = \sqrt{\sum_i x_i^2}$
3. $\|\mathbf{x}\|_\infty = \max_i |x_i|$

TERMS σ singular values; κ condition number; tr trace

Large κ = "wobbly": tiny changes in the data cause big changes in the answer.

1. $A_k = \sum_{i=1}^{k} \sigma_i \, \mathbf{u}_i \mathbf{v}_i^\top$
2. $\|A\|_2 = \sigma_1$
3. $\operatorname{rank} A = \#\{\sigma_i > 0\}$

TERMS U, V orthogonal; Σ singular values $\sigma_1 \geq \sigma_2 \geq \cdots$

Every matrix is "rotate, stretch, rotate" ($V^\top$, Σ, U).

1. $A = U \Sigma V^\top$
2. $\sigma_i = \sqrt{\lambda_i (A^\top A)}$
3. $A \mathbf{v}_i = \sigma_i \mathbf{u}_i, \qquad A^\top A = V \Sigma^\top \Sigma V^\top$

TERMS U, V orthogonal; Σ singular values $\sigma_1 \geq \sigma_2 \geq \cdots$

Every matrix is "rotate, stretch, rotate" ($V^\top$, Σ, U).

1. $\nabla_{\mathbf{x}} (\mathbf{a}^\top \mathbf{x}) = \mathbf{a}$
2. $\nabla_{\mathbf{x}} (\mathbf{x}^\top A \mathbf{x}) = (A + A^\top) \mathbf{x}$
3. $\nabla_{\mathbf{x}} \|\mathbf{x}\|^2 = 2\mathbf{x}$

TERMS ∇ gradient; $X^{-\top} = (X^{-1})^\top$; A symmetric $\Rightarrow 2A\mathbf{x}$

Matrix calculus mirrors scalar calculus: $\frac{d}{dx}(ax) = a$, $\frac{d}{dx}(ax^2) = 2ax$; just watch transposes and shapes.

1. $\kappa(A) = \dfrac{\sigma_{\max}}{\sigma_{\min}} = \|A\| \, \|A^{-1}\|$
2. $\operatorname{tr} A = \sum_i a_{ii} = \sum_i \lambda_i$

TERMS σ singular values; κ condition number; tr trace

Large κ = "wobbly": tiny changes in the data cause big changes in the answer.

1. $\|\mathbf{x}\|_p = \left(\sum_i |x_i|^p \right)^{1/p}$
2. $\|A\|_F = \sqrt{\sum_{i,j} a_{ij}^2} = \sqrt{\operatorname{tr}\left(A^\top A\right)}$
3. $\|A\|_2 = \max_{\mathbf{x} \neq \mathbf{0}} \dfrac{\|A\mathbf{x}\|_2}{\|\mathbf{x}\|_2} = \sigma_{\max}$

TERMS σ singular values; κ condition number; tr trace

Large κ = "wobbly": tiny changes in the data cause big changes in the answer.

1. $\left(A + \mathbf{u}\mathbf{v}^\top\right)^{-1} = A^{-1} - \dfrac{A^{-1}\mathbf{u}\mathbf{v}^\top A^{-1}}{1 + \mathbf{v}^\top A^{-1} \mathbf{u}}$

TERMS A^+ pseudoinverse; Σ^+ inverted singular values; $\mathbf{u}\mathbf{v}^\top$ rank 1

A rank-1 change to a matrix needs only a rank-1 correction to its inverse: $O(n^2)$ instead of $O(n^3)$.

1. $A^+ = (A^\top A)^{-1} A^\top$
2. $A^+ = V \Sigma^+ U^\top$

TERMS A^+ pseudoinverse; Σ^+ inverted singular values; $\mathbf{u}\mathbf{v}^\top$ rank 1

A rank-1 change to a matrix needs only a rank-1 correction to its inverse: $O(n^2)$ instead of $O(n^3)$.

1. $\nabla_\beta \|\mathbf{y} - X\beta\|^2 = -2X^\top (\mathbf{y} - X\beta)$
2. $\dfrac{\partial \operatorname{tr}(AB)}{\partial A} = B^\top$
3. $\dfrac{\partial \ln \det X}{\partial X} = X^{-\top}$

TERMS ∇ gradient; $X^{-\top} = (X^{-1})^\top$; A symmetric $\Rightarrow 2A\mathbf{x}$

Matrix calculus mirrors scalar calculus: $\frac{d}{dx}(ax) = a$, $\frac{d}{dx}(ax^2) = 2ax$; just watch transposes and shapes.

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Pseudoinverse & Inverse-Update Identities

PART 3 OF 3

1 Woodbury identity

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1 $(A+UCV)^{-1} = A^{-1} - A^{-1}U\left(C^{-1}+VA^{-1}U\right)^{-1}VA^{-1}$

TERMS A^{+} pseudoinverse; Σ^{+} inverted singular values; $\mathbf{u}\mathbf{v}^{\top}$ rank 1

A rank-1 change to a matrix needs only a rank-1 correction to its inverse: $O(n^2)$ instead of $O(n^3)$.

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C3.1 (1/2) CALCULUS III

3-D Coordinates, Distance & Spheres

PART 1 OF 2

1. Distance formula
2. Midpoint
3. Sphere (standard form)

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C3.1 (2/2) CALCULUS III

3-D Coordinates, Distance & Spheres

PART 2 OF 2

1. Sphere (general form)
2. Coordinate planes

STATE EACH NUMBERED FORMULA

C3.2 (1/3) CALCULUS III

Vectors in Space: Dot & Cross Products

PART 1 OF 3

1. Magnitude & unit vector
2. Dot product
3. Vector projection

STATE EACH NUMBERED FORMULA

C3.2 (2/3) CALCULUS III

Vectors in Space: Dot & Cross Products

PART 2 OF 3

1. Cross product
2. Length of the cross product
3. Cross-product algebra

STATE EACH NUMBERED FORMULA

C3.2 (3/3) CALCULUS III

Vectors in Space: Dot & Cross Products

PART 3 OF 3

1. Scalar triple product
2. Triangle area

STATE EACH NUMBERED FORMULA

C3.3 (1/3) CALCULUS III

Lines & Planes in Space

PART 1 OF 3

1. Line (vector / parametric)
2. Line (symmetric)
3. Plane (point–normal)

STATE EACH NUMBERED FORMULA

C3.3 (2/3) CALCULUS III

Lines & Planes in Space

PART 2 OF 3

1. Plane (general)
2. Plane through 3 points
3. Distance point to plane

STATE EACH NUMBERED FORMULA

C3.3 (3/3) CALCULUS III

Lines & Planes in Space

PART 3 OF 3

1. Angle between planes
2. Distance point to line
3. Distance between skew lines

STATE EACH NUMBERED FORMULA

C3.4 (1/3) CALCULUS III

Cylinders & Quadric Surfaces

PART 1 OF 3

1. Ellipsoid
2. Elliptic paraboloid
3. Hyperbolic paraboloid

STATE EACH NUMBERED FORMULA

C3.4 (2/3) CALCULUS III

Cylinders & Quadric Surfaces

PART 2 OF 3

1. Hyperboloid of one sheet
2. Hyperboloid of two sheets
3. Elliptic cone

STATE EACH NUMBERED FORMULA

C3.4 (3/3) CALCULUS III

Cylinders & Quadric Surfaces

PART 3 OF 3

1. Cylinder

STATE EACH NUMBERED FORMULA

C3.5 (1/3) CALCULUS III

Vector-Valued Functions & Motion

PART 1 OF 3

1. Vector function
2. Derivative
3. Integral

STATE EACH NUMBERED FORMULA

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1 $\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2 + v_3^2}, \qquad \hat{\mathbf{v}} = \dfrac{\mathbf{v}}{\|\mathbf{v}\|}$

2 $\mathbf{u} \cdot \mathbf{v} = u_1v_1 + u_2v_2 + u_3v_3 = \|\mathbf{u}\|\|\mathbf{v}\| \cos\theta$

3 $\text{proj}_{\mathbf{v}}\, \mathbf{u} = \dfrac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{v}\|^2}\, \mathbf{v}$

TERMS θ angle between; $\mathbf{i}, \mathbf{j}, \mathbf{k}$ unit axis vectors; $\perp$ perpendicular

Cycle $\mathbf{i} \to \mathbf{j} \to \mathbf{k} \to \mathbf{i}$: forward gives + ($\mathbf{j} \times \mathbf{k} = \mathbf{i}$), backward gives − ($\mathbf{k} \times \mathbf{j} = -\mathbf{i}$). Dot → cosine (alignment); cross → sine (area).

1 $x^2 + y^2 + z^2 + Gx + Hy + Iz + J = 0$

2 $z = 0\ (xy), \quad x = 0\ (yz), \quad y = 0\ (xz)$

TERMS (h, k, l) center; r radius; $\Delta x, \Delta y, \Delta z$ coordinate differences

In 3-D: one equation → a surface; two equations → a curve; three → points.

1 $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$

2 $M = \left(\frac{x_1+x_2}{2},\ \frac{y_1+y_2}{2},\ \frac{z_1+z_2}{2}\right)$

3 $(x - h)^2 + (y - k)^2 + (z - l)^2 = r^2$

TERMS (h, k, l) center; r radius; $\Delta x, \Delta y, \Delta z$ coordinate differences

In 3-D: one equation → a surface; two equations → a curve; three → points.

1 $\mathbf{r}(t) = \mathbf{r}_0 + t\mathbf{v}: \quad x = x_0 + at,\ y = y_0 + bt,\ z = z_0 + ct$

2 $\dfrac{x - x_0}{a} = \dfrac{y - y_0}{b} = \dfrac{z - z_0}{c}$

3 $a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$

TERMS P_0 point; $\mathbf{v}$ direction; $\mathbf{n} = (a, b, c)$ normal; t parameter

Line = point + direction; plane = point + normal.

1 $\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w}) = \det \begin{bmatrix} \mathbf{u} \\ \mathbf{v} \\ \mathbf{w} \end{bmatrix}$

2 $A = \frac{1}{2}\left\|\overrightarrow{PQ} \times \overrightarrow{PR}\right\|$

TERMS θ angle between; $\mathbf{i}, \mathbf{j}, \mathbf{k}$ unit axis vectors; $\perp$ perpendicular

Cycle $\mathbf{i} \to \mathbf{j} \to \mathbf{k} \to \mathbf{i}$: forward gives + ($\mathbf{j} \times \mathbf{k} = \mathbf{i}$), backward gives − ($\mathbf{k} \times \mathbf{j} = -\mathbf{i}$). Dot → cosine (alignment); cross → sine (area).

1 $\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} = (u_2v_3 - u_3v_2,\ u_3v_1 - u_1v_3,\ u_1v_2 - u_2v_1)$

2 $\|\mathbf{u} \times \mathbf{v}\| = \|\mathbf{u}\|\|\mathbf{v}\| \sin\theta$

3 $\mathbf{v} \times \mathbf{u} = -\mathbf{u} \times \mathbf{v}, \qquad \mathbf{i} \times \mathbf{j} = \mathbf{k}$

TERMS θ angle between; $\mathbf{i}, \mathbf{j}, \mathbf{k}$ unit axis vectors; $\perp$ perpendicular

Cycle $\mathbf{i} \to \mathbf{j} \to \mathbf{k} \to \mathbf{i}$: forward gives + ($\mathbf{j} \times \mathbf{k} = \mathbf{i}$), backward gives − ($\mathbf{k} \times \mathbf{j} = -\mathbf{i}$). Dot → cosine (alignment); cross → sine (area).

1 $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} + \dfrac{z^2}{c^2} = 1$

2 $z = \dfrac{x^2}{a^2} + \dfrac{y^2}{b^2}$

3 $z = \dfrac{x^2}{a^2} - \dfrac{y^2}{b^2}$

TERMS a, b, c semi-axes; trace = slice by x, y or $z = k$

In "= 1" form count the minus signs: 0 → ellipsoid, 1 → one sheet, 2 → two sheets; "= 0" → cone. One variable to the first power → paraboloid (same signs: bowl; opposite: saddle).

1 $\cos\theta = \dfrac{|\mathbf{n}_1 \cdot \mathbf{n}_2|}{\|\mathbf{n}_1\|\|\mathbf{n}_2\|}$

2 $D = \dfrac{\|\overrightarrow{P_0P_1} \times \mathbf{v}\|}{\|\mathbf{v}\|}$

3 $D = \dfrac{|(\mathbf{r}_2 - \mathbf{r}_1) \cdot (\mathbf{v}_1 \times \mathbf{v}_2)|}{\|\mathbf{v}_1 \times \mathbf{v}_2\|}$

TERMS P_0 point; $\mathbf{v}$ direction; $\mathbf{n} = (a, b, c)$ normal; t parameter

Line = point + direction; plane = point + normal.

1 $ax + by + cz = d$

2 $\mathbf{n} = \overrightarrow{PQ} \times \overrightarrow{PR}$

3 $D = \dfrac{|ax_1 + by_1 + cz_1 - d|}{\sqrt{a^2 + b^2 + c^2}}$

TERMS P_0 point; $\mathbf{v}$ direction; $\mathbf{n} = (a, b, c)$ normal; t parameter

Line = point + direction; plane = point + normal.

1 $\mathbf{r}(t) = \big(x(t),\ y(t),\ z(t)\big)$

2 $\mathbf{r}'(t) = \lim_{h \to 0} \dfrac{\mathbf{r}(t + h) - \mathbf{r}(t)}{h} = (x', y', z')$

3 $\displaystyle\int_a^b \mathbf{r}(t)\, dt = \left(\int_a^b x\, dt,\ \int_a^b y\, dt,\ \int_a^b z\, dt\right)$

TERMS $\mathbf{r}(t)$ position; $\mathbf{v} = \mathbf{r}'$; $\mathbf{a} = \mathbf{r}''$; g gravity

Everything happens one component at a time.

1 $f(x, y) = 0 \quad (z \text{ absent})$

TERMS a, b, c semi-axes; trace = slice by x, y or $z = k$

In "= 1" form count the minus signs: 0 → ellipsoid, 1 → one sheet, 2 → two sheets; "= 0" → cone. One variable to the first power → paraboloid (same signs: bowl; opposite: saddle).

1 $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} - \dfrac{z^2}{c^2} = 1$

2 $\dfrac{z^2}{c^2} - \dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = 1$

3 $\dfrac{z^2}{c^2} = \dfrac{x^2}{a^2} + \dfrac{y^2}{b^2}$

TERMS a, b, c semi-axes; trace = slice by x, y or $z = k$

In "= 1" form count the minus signs: 0 → ellipsoid, 1 → one sheet, 2 → two sheets; "= 0" → cone. One variable to the first power → paraboloid (same signs: bowl; opposite: saddle).

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C3.5 (2/3) CALCULUS III

Vector-Valued Functions & Motion

PART 2 OF 3

1. Velocity, speed, acceleration
2. Product rules
3. Constant length

STATE EACH NUMBERED FORMULA

C3.5 (3/3) CALCULUS III

Vector-Valued Functions & Motion

PART 3 OF 3

1. Projectile motion
2. Tangent line

STATE EACH NUMBERED FORMULA

C3.6 (1/3) CALCULUS III

Arc Length, Curvature & the TNB Frame

PART 1 OF 3

1. Arc length
2. Arc-length function
3. Unit tangent

STATE EACH NUMBERED FORMULA

C3.6 (2/3) CALCULUS III

Arc Length, Curvature & the TNB Frame

PART 2 OF 3

1. Curvature (definition)
2. Curvature (cross-product formula)
3. Curvature of a graph

STATE EACH NUMBERED FORMULA

C3.6 (3/3) CALCULUS III

Arc Length, Curvature & the TNB Frame

PART 3 OF 3

1. Normal & binormal
2. Acceleration components
3. Frenet–Serret formulas

STATE EACH NUMBERED FORMULA

C3.7 (1/2) CALCULUS III

Functions of Several Variables & Level Curves

PART 1 OF 2

1. Function of two variables
2. Graph
3. Level curve

STATE EACH NUMBERED FORMULA

C3.7 (2/2) CALCULUS III

Functions of Several Variables & Level Curves

PART 2 OF 2

1. Vertical traces
2. Level surface
3. Function of n variables

STATE EACH NUMBERED FORMULA

C3.8 (1/3) CALCULUS III

Limits & Continuity in Several Variables

PART 1 OF 3

1. ε–δ definition
2. Two-path test
3. Polar substitution

STATE EACH NUMBERED FORMULA

C3.8 (2/3) CALCULUS III

Limits & Continuity in Several Variables

PART 2 OF 3

1. Squeeze theorem
2. Continuity
3. Direction-dependent limit

STATE EACH NUMBERED FORMULA

C3.8 (3/3) CALCULUS III

Limits & Continuity in Several Variables

PART 3 OF 3

1. Lines are not enough

STATE EACH NUMBERED FORMULA

C3.K1 CALCULUS III

Partial derivative f_x: definition and meaning?

RECALL IT

C3.K1 CALCULUS III

Gradient: three geometric facts?

RECALL IT

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1 $L = \displaystyle\int_a^b \|\mathbf{r}'(t)\|\,dt$

2 $s(t) = \displaystyle\int_a^t \|\mathbf{r}'(u)\|\,du, \qquad \frac{ds}{dt} = \|\mathbf{r}'(t)\|$

3 $\mathbf{T}(t) = \dfrac{\mathbf{r}'(t)}{\|\mathbf{r}'(t)\|}$

TERMS s arc length; κ curvature; τ torsion; $\mathbf{T}, \mathbf{N}, \mathbf{B}$ frame

Tight turn = big curvature: $\kappa = 1/R$, the reciprocal of the radius of the best-fitting (osculating) circle.

1 $\mathbf{r}(t) = \mathbf{r}_0 + \left(v_0 \cos\alpha\ t,\ \ v_0 \sin\alpha\ t - \frac{1}{2} g t^2\right)$

2 $\mathbf{L}(s) = \mathbf{r}(t_0) + s\,\mathbf{r}'(t_0)$

TERMS $\mathbf{r}(t)$ position; $\mathbf{v} = \mathbf{r}'$; $\mathbf{a} = \mathbf{r}''$; g gravity

Everything happens one component at a time.

1 $\mathbf{v} = \mathbf{r}', \qquad \text{speed} = \|\mathbf{r}'\|, \qquad \mathbf{a} = \mathbf{r}''$

2 $(\mathbf{u}\cdot\mathbf{w})' = \mathbf{u}'\cdot\mathbf{w} + \mathbf{u}\cdot\mathbf{w}', \quad (\mathbf{u}\times\mathbf{w})' = \mathbf{u}'\times\mathbf{w} + \mathbf{u}\times\mathbf{w}'$

3 $\|\mathbf{r}(t)\| = c \ \Rightarrow\ \mathbf{r}\cdot\mathbf{r}' = 0$

TERMS $\mathbf{r}(t)$ position; $\mathbf{v} = \mathbf{r}'$; $\mathbf{a} = \mathbf{r}''$; g gravity

Everything happens one component at a time.

1 $z = f(x,y), \quad (x,y) \in D \subseteq \mathbb{R}^2$

2 $\{(x,y,z) :\ z = f(x,y),\ (x,y) \in D\}$

3 $f(x,y) = c$

TERMS D domain; c constant level; a, b fixed coordinates

Graph = landscape; level curves = the hiking map of that landscape.

1 $\mathbf{N} = \dfrac{\mathbf{T}'}{\|\mathbf{T}'\|}, \qquad \mathbf{B} = \mathbf{T}\times\mathbf{N}$

2 $\mathbf{a} = a_T\mathbf{T} + a_N\mathbf{N}, \quad a_T = \dfrac{d}{dt}\|\mathbf{v}\|, \quad a_N = \kappa\|\mathbf{v}\|^2$

3 $\dfrac{d\mathbf{T}}{ds} = \kappa\mathbf{N}, \quad \dfrac{d\mathbf{N}}{ds} = -\kappa\mathbf{T} + \tau\mathbf{B}, \quad \dfrac{d\mathbf{B}}{ds} = -\tau\mathbf{N}$

TERMS s arc length; κ curvature; τ torsion; $\mathbf{T}, \mathbf{N}, \mathbf{B}$ frame

Tight turn = big curvature: $\kappa = 1/R$, the reciprocal of the radius of the best-fitting (osculating) circle.

1 $\kappa = \left\|\dfrac{d\mathbf{T}}{ds}\right\| = \dfrac{\|\mathbf{T}'(t)\|}{\|\mathbf{r}'(t)\|}$

2 $\kappa = \dfrac{\|\mathbf{r}'\times\mathbf{r}''\|}{\|\mathbf{r}'\|^3}$

3 $\kappa = \dfrac{|f''(x)|}{\left(1 + f'(x)^2\right)^{3/2}}$

TERMS s arc length; κ curvature; τ torsion; $\mathbf{T}, \mathbf{N}, \mathbf{B}$ frame

Tight turn = big curvature: $\kappa = 1/R$, the reciprocal of the radius of the best-fitting (osculating) circle.

1 $|f(x,y) - L| \le g(x,y), \ \ g \to 0 \ \Rightarrow\ f \to L$

2 $\displaystyle\lim_{(x,y)\to(a,b)} f(x,y) = f(a,b)$

3 $f = \dfrac{xy}{x^2+y^2}$ on $y = mx$: $\quad f = \dfrac{m}{1+m^2}$

TERMS (a,b) point approached; L limit; ε, δ tolerances; m slope

Paths can only disprove *a limit; to* prove *one, squeeze.*

1 $0 < \|(x,y) - (a,b)\| < \delta \ \Rightarrow\ |f(x,y) - L| < \varepsilon$

2 $f \xrightarrow{C_1} L_1, \ \ f \xrightarrow{C_2} L_2, \ \ L_1 \neq L_2 \ \Rightarrow$ no limit

3 $x = r\cos\theta,\ y = r\sin\theta$: $\quad |f - L| \le g(r) \to 0$

TERMS (a,b) point approached; L limit; ε, δ tolerances; m slope

Paths can only disprove *a limit; to* prove *one, squeeze.*

1 $z = f(x,b) \quad \text{and} \quad z = f(a,y)$

2 $F(x,y,z) = c$

3 $f : \mathbb{R}^n \to \mathbb{R}, \quad \mathbf{x} = (x_1, \ldots, x_n) \mapsto f(\mathbf{x})$

TERMS D domain; c constant level; a, b fixed coordinates

Graph = landscape; level curves = the hiking map of that landscape.

$\nabla f = (f_x, f_y)$ points uphill (steepest ascent),
$\|\nabla f\|$ is the steepest slope,
$\nabla f \perp$ level curves.

$$f_x = \lim_{h\to 0} \frac{f(x+h,y) - f(x,y)}{h}$$

Freeze y, differentiate in x: slope of the trace in the plane $y = b$.

1 $f = \dfrac{x^2 y}{x^4 + y^2}$: $\quad y = mx \to 0, \quad y = x^2 \to \frac{1}{2}$

TERMS (a,b) point approached; L limit; ε, δ tolerances; m slope

Paths can only disprove *a limit; to* prove *one, squeeze.*

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C3.K1 CALCULUS III

$f = x^2y + 3y$: **find** f_x, f_y, $\nabla f(1,2)$

RECALL IT

C3.K1 CALCULUS III

Chain rule tree: how to read it?

RECALL IT

C3.9 (1/2) CALCULUS III

Partial Derivatives

PART 1 OF 2

1. Partial derivative in x
2. Partial derivative in y
3. Geometric meaning

STATE EACH NUMBERED FORMULA

C3.9 (2/2) CALCULUS III

Partial Derivatives

PART 2 OF 2

1. Notation
2. Mixed partial notation
3. Numerical partial (central difference)

STATE EACH NUMBERED FORMULA

C3.10 (1/2) CALCULUS III

Higher-Order Partials & Clairaut's Theorem

PART 1 OF 2

1. Pure second partials
2. Mixed second partial
3. Clairaut's theorem

STATE EACH NUMBERED FORMULA

C3.10 (2/2) CALCULUS III

Higher-Order Partials & Clairaut's Theorem

PART 2 OF 2

1. Higher orders
2. Laplace's equation
3. Clairaut counterexample

STATE EACH NUMBERED FORMULA

C3.11 (1/2) CALCULUS III

Tangent Planes, Linear Approximation & Differentials

PART 1 OF 2

1. Tangent plane
2. Linearization
3. Total differential

STATE EACH NUMBERED FORMULA

C3.11 (2/2) CALCULUS III

Tangent Planes, Linear Approximation & Differentials

PART 2 OF 2

1. Differentiability
2. Normal vector to a graph
3. Error propagation

STATE EACH NUMBERED FORMULA

C3.12 (1/3) CALCULUS III

Gradient & Directional Derivative

PART 1 OF 3

1. Gradient
2. Directional derivative
3. Steepest ascent

STATE EACH NUMBERED FORMULA

C3.12 (2/3) CALCULUS III

Gradient & Directional Derivative

PART 2 OF 3

1. Orthogonal to level sets
2. Tangent plane to a level surface
3. Normal line

STATE EACH NUMBERED FORMULA

C3.12 (3/3) CALCULUS III

Gradient & Directional Derivative

PART 3 OF 3

1. Gradient rules

STATE EACH NUMBERED FORMULA

C3.13 (1/2) CALCULUS III

Multivariable Chain Rule

PART 1 OF 2

1. One parameter
2. Two parameters
3. Vector form

STATE EACH NUMBERED FORMULA

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1 $$\frac{\partial f}{\partial x} = f_x = \lim_{h\to 0} \frac{f(x+h,y)-f(x,y)}{h}$$

2 $$\frac{\partial f}{\partial y} = f_y = \lim_{h\to 0} \frac{f(x,y+h)-f(x,y)}{h}$$

3 $$f_x(a,b) = \frac{d}{dx} f(x,b)\Big|_{x=a}$$

TERMS ∂ partial (others held fixed); h step; f_{xy}: x first, then y

"Freeze the other variables, then differentiate as usual."

Multiply the partial derivatives along each branch from output to input; add over all branches.

$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}$$

$$f_x = 2xy, \quad f_y = x^2 + 3$$

$$\nabla f(1,2) = (4,4)$$

1 $$f_{xxy} = f_{xyx} = f_{yxx}$$

2 $$f_{xx} + f_{yy} = 0$$

3 $$f = \frac{xy(x^2-y^2)}{x^2+y^2}: \quad f_{xy}(0,0) = -1 \neq 1 = f_{yx}(0,0)$$

TERMS subscript order = order taken; continuity $\Rightarrow f_{xy} = f_{yx}$

Smooth functions forget the order: "x then y" = "y then x".

1 $$f_{xx} = \frac{\partial^2 f}{\partial x^2}, \qquad f_{yy} = \frac{\partial^2 f}{\partial y^2}$$

2 $$f_{xy} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right)$$

3 $$f_{xy} = f_{yx} \quad \text{if both are continuous near } (a,b)$$

TERMS subscript order = order taken; continuity $\Rightarrow f_{xy} = f_{yx}$

Smooth functions forget the order: "x then y" = "y then x".

1 $$\frac{\partial f}{\partial x} = f_x = \partial_x f = D_1 f$$

2 $$\frac{\partial^2 f}{\partial y\,\partial x} = (f_x)_y = f_{xy}$$

3 $$f_x \approx \frac{f(x+h,y)-f(x-h,y)}{2h}$$

TERMS ∂ partial (others held fixed); h step; f_{xy}: x first, then y

"Freeze the other variables, then differentiate as usual."

1 $$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}\right)$$

2 $$D_{\mathbf{u}} f = \nabla f \cdot \mathbf{u} = \|\nabla f\| \cos\theta \quad (\|\mathbf{u}\| = 1)$$

3 $$\max_{\mathbf{u}} D_{\mathbf{u}} f = \|\nabla f\| \quad \text{at} \quad \mathbf{u} = \frac{\nabla f}{\|\nabla f\|}$$

TERMS ∇ del; $\mathbf{u}$ unit direction; θ angle between ∇f and $\mathbf{u}$

The gradient points straight uphill; go the opposite way to descend C1.26 §07 Calculus I.

1 $$\lim_{(h,k)\to(0,0)} \frac{f(a+h,b+k)-L(a+h,b+k)}{\sqrt{h^2+k^2}} = 0$$

2 $$\mathbf{n} = (f_x,\ f_y,\ -1)$$

3 $$|\Delta f| \lesssim |f_x|\,|\Delta x| + |f_y|\,|\Delta y|$$

TERMS (a,b) base point; $\Delta x, \Delta y$ small steps; dz total differential

Zoom in far enough on a smooth surface and it is flat.

1 $$z = f(a,b) + f_x(a,b)\,(x-a) + f_y(a,b)\,(y-b)$$

2 $$L(x,y) = f(a,b) + f_x\,\Delta x + f_y\,\Delta y \approx f(x,y)$$

3 $$dz = f_x\,dx + f_y\,dy$$

TERMS (a,b) base point; $\Delta x, \Delta y$ small steps; dz total differential

Zoom in far enough on a smooth surface and it is flat.

1 $$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}$$

2 $$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial s}$$

3 $$\frac{d}{dt} f(\mathbf{x}(t)) = \nabla f \cdot \mathbf{x}'(t)$$

TERMS x, y (or x_i) intermediate variables; s, t independent variables

Sum over every path from the input to the output: multiply along each path, add the paths.

1 $$\nabla(fg) = f\nabla g + g\nabla f, \qquad \nabla(\mathbf{a}\cdot\mathbf{x}) = \mathbf{a}, \qquad \nabla\|\mathbf{x}\|^2 = 2\mathbf{x}$$

TERMS ∇ del; $\mathbf{u}$ unit direction; θ angle between ∇f and $\mathbf{u}$

The gradient points straight uphill; go the opposite way to descend C1.26 §07 Calculus I.

1 $$\nabla f \perp \{f = c\}$$

2 $$\nabla F(P_0)\cdot(\mathbf{x}-\mathbf{x}_0) = 0$$

3 $$\mathbf{r}(t) = \mathbf{x}_0 + t\,\nabla F(P_0)$$

TERMS ∇ del; $\mathbf{u}$ unit direction; θ angle between ∇f and $\mathbf{u}$

The gradient points straight uphill; go the opposite way to descend C1.26 §07 Calculus I.

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C3.13 (2/2) CALCULUS III

Multivariable Chain Rule

PART 2 OF 2

1. General form
2. Matrix form

STATE EACH NUMBERED FORMULA

C3.14 (1/2) CALCULUS III

Implicit Differentiation with Partials

PART 1 OF 2

1. Implicit curve
2. Implicit surface
3. Implicit function theorem

STATE EACH NUMBERED FORMULA

C3.14 (2/2) CALCULUS III

Implicit Differentiation with Partials

PART 2 OF 2

1. Cyclic (triple product) rule

STATE EACH NUMBERED FORMULA

C3.15 (1/2) CALCULUS III

The Jacobian

PART 1 OF 2

1. Jacobian matrix
2. Linear approximation
3. Jacobian determinant

STATE EACH NUMBERED FORMULA

C3.15 (2/2) CALCULUS III

The Jacobian

PART 2 OF 2

1. Area scaling
2. Chain rule
3. Inverse function theorem

STATE EACH NUMBERED FORMULA

C3.16 (1/2) CALCULUS III

Hessian & the Second-Derivative Test

PART 1 OF 2

1. Critical point
2. Hessian matrix
3. Discriminant

STATE EACH NUMBERED FORMULA

C3.16 (2/2) CALCULUS III

Hessian & the Second-Derivative Test

PART 2 OF 2

1. Second-derivative test
2. Second-order Taylor
3. Eigenvalue test (n variables)

STATE EACH NUMBERED FORMULA

C3.17 CALCULUS III

Absolute Extrema on Closed, Bounded Regions

1. Extreme value theorem
2. Candidate list
3. Boundary reduction

STATE EACH NUMBERED FORMULA

C3.18 (1/2) CALCULUS III

Lagrange Multipliers

PART 1 OF 2

1. Tangency condition
2. Lagrangian
3. Two constraints

STATE EACH NUMBERED FORMULA

C3.18 (2/2) CALCULUS III

Lagrange Multipliers

PART 2 OF 2

1. Meaning of λ

STATE EACH NUMBERED FORMULA

C3.19 (1/3) CALCULUS III

Double & Triple Integrals

PART 1 OF 3

1. Definition (Riemann sum)
2. Iterated integral (rectangle)
3. Type I region

STATE EACH NUMBERED FORMULA

C3.19 (2/3) CALCULUS III

Double & Triple Integrals

PART 2 OF 3

1. Type II region
2. Area & volume
3. Triple integral

STATE EACH NUMBERED FORMULA

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1. $$\left(\frac{\partial x}{\partial y}\right)_z \left(\frac{\partial y}{\partial z}\right)_x \left(\frac{\partial z}{\partial x}\right)_y = -1$$

TERMS $F = 0$ defining equation; F_x, F_y, F_z partials (denominator $\neq 0$)

"Minus the ratio, with the dependent variable's partial on the bottom."

C3.14 (2/2)

1. $F(x,y) = 0 \;\Rightarrow\; \dfrac{dy}{dx} = -\dfrac{F_x}{F_y} \quad (F_y \neq 0)$
2. $F(x,y,z) = 0 \;\Rightarrow\; \dfrac{\partial z}{\partial x} = -\dfrac{F_x}{F_z},\; \dfrac{\partial z}{\partial y} = -\dfrac{F_y}{F_z}$
3. $F(a,b) = 0,\; F_y(a,b) \neq 0 \;\Rightarrow\; y = g(x)$ near a

TERMS $F = 0$ defining equation; F_x, F_y, F_z partials (denominator $\neq 0$)

"Minus the ratio, with the dependent variable's partial on the bottom."

C3.14 (1/2)

1. $$\frac{\partial u}{\partial t_j} = \sum_{i=1}^{n} \frac{\partial u}{\partial x_i}\frac{\partial x_i}{\partial t_j}$$
2. $$J_{\mathbf{g}\circ\mathbf{f}}(\mathbf{x}) = J_{\mathbf{g}}\big(\mathbf{f}(\mathbf{x})\big)\, J_{\mathbf{f}}(\mathbf{x})$$

TERMS x, y (or x_i) intermediate variables; s, t independent variables

Sum over every path from the input to the output: multiply along each path, add the paths.

C3.13 (2/2)

1. $\nabla f(a,b) = \mathbf{0}$ (or ∇f undefined)
2. $$H = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix}$$
3. $D = \det H = f_{xx}f_{yy} - f_{xy}^2$

TERMS f_{xx}, f_{yy}, f_{xy} second partials; $D = \det H$; λ_i eigenvalues

Hessian positive definite → bowl (min); negative definite → dome (max); mixed → saddle LA.27 *§09 Linear Algebra.*

C3.16 (1/2)

1. $dx\,dy = \left|\dfrac{\partial(x,y)}{\partial(u,v)}\right| du\,dv$
2. $J_{\mathbf{g}\circ\mathbf{f}} = J_{\mathbf{g}}\, J_{\mathbf{f}}$
3. $\det J_{\mathbf{f}}(\mathbf{a}) \neq 0 \;\Rightarrow\; J_{\mathbf{f}^{-1}} = \big(J_{\mathbf{f}}\big)^{-1}$

TERMS rows = outputs f_i, columns = inputs x_j; $|\det J|$ area scale

Rows are outputs, columns are inputs; $|\det J|$ is the local area-stretch factor.

C3.15 (2/2)

1. $$J = \begin{bmatrix} \dfrac{\partial f_1}{\partial x_1} & \cdots & \dfrac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \dfrac{\partial f_m}{\partial x_1} & \cdots & \dfrac{\partial f_m}{\partial x_n} \end{bmatrix}$$
2. $\mathbf{f}(\mathbf{x}+\mathbf{h}) \approx \mathbf{f}(\mathbf{x}) + J\mathbf{h}$
3. $$\frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} x_u & x_v \\ y_u & y_v \end{vmatrix} = x_u y_v - x_v y_u$$

TERMS rows = outputs f_i, columns = inputs x_j; $|\det J|$ area scale

Rows are outputs, columns are inputs; $|\det J|$ is the local area-stretch factor.

C3.15 (1/2)

1. $\nabla f = \lambda\, \nabla g, \qquad g(x,y) = c$
2. $\mathcal{L}(x,y,\lambda) = f(x,y) - \lambda\big(g(x,y) - c\big), \quad \nabla\mathcal{L} = \mathbf{0}$
3. $\nabla f = \lambda\, \nabla g + \mu\, \nabla h, \qquad g = c_1,\; h = c_2$

TERMS $g = c$ constraint; λ multiplier (shadow price); $\mathcal{L}$ Lagrangian

At the optimum the level curve of f kisses the constraint: parallel gradients.

C3.18 (1/2)

1. f continuous on closed, bounded $D \;\Rightarrow\; \max_D f,\; \min_D f$ exist
2. $\{\nabla f = \mathbf{0}$ in $D\} \cup \{$critical points on $\partial D\} \cup \{$corners$\}$
3. $\partial D:\; \mathbf{r}(t),\;\; g(t) = f\big(\mathbf{r}(t)\big),\;\; g'(t) = 0$

TERMS D closed, bounded region; ∂D boundary; corners = edge endpoints

"Inside, edges, corners — then compare."

C3.17

1. $$\begin{cases} D > 0,\; f_{xx} > 0 & \text{local min} \\ D > 0,\; f_{xx} < 0 & \text{local max} \\ D < 0 & \text{saddle point} \\ D = 0 & \text{inconclusive} \end{cases}$$
2. $f(\mathbf{a}+\mathbf{h}) \approx f(\mathbf{a}) + \nabla f \cdot \mathbf{h} + \frac{1}{2}\,\mathbf{h}^{\mathsf{T}} H\,\mathbf{h}$
3. all $\lambda_i > 0 \Rightarrow$ min; all $\lambda_i < 0 \Rightarrow$ max; mixed signs $\Rightarrow$ saddle

TERMS f_{xx}, f_{yy}, f_{xy} second partials; $D = \det H$; λ_i eigenvalues

Hessian positive definite → bowl (min); negative definite → dome (max); mixed → saddle LA.27 *§09 Linear Algebra.*

C3.16 (2/2)

1. $$\iint_D f\,dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x,y)\,dx\,dy$$
2. $\displaystyle\iint_D 1\,dA = \text{Area}(D), \qquad \iiint_E 1\,dV = \text{Vol}(E)$
3. $$\iiint_E f\,dV = \iint_D \left[\int_{u_1(x,y)}^{u_2(x,y)} f\,dz\right] dA$$

TERMS dA area element; inner limits may depend on the outer variable

Integrate the inside first (its limits may depend on the outer variable), treating the outer variable as a constant.

C3.19 (2/3)

1. $$\iint_R f\,dA = \lim_{\|P\|\to 0} \sum_{i,j} f(x_i^*, y_j^*)\,\Delta A$$
2. $$\iint_R f\,dA = \int_a^b \int_c^d f(x,y)\,dy\,dx$$
3. $$\iint_D f\,dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x,y)\,dy\,dx$$

TERMS dA area element; inner limits may depend on the outer variable

Integrate the inside first (its limits may depend on the outer variable), treating the outer variable as a constant.

C3.19 (1/3)

1. $$\lambda = \frac{df^*}{dc}$$

TERMS $g = c$ constraint; λ multiplier (shadow price); $\mathcal{L}$ Lagrangian

At the optimum the level curve of f kisses the constraint: parallel gradients.

C3.18 (2/2)

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C3.19 (3/3) CALCULUS III

Double & Triple Integrals

PART 3 OF 3

1. Average value
2. Separable integrand

STATE EACH NUMBERED FORMULA

C3.20 CALCULUS III

Fubini's Theorem & Changing the Order

1. Fubini's theorem
2. Tonelli's theorem
3. Reversing the order

STATE EACH NUMBERED FORMULA

C3.21 (1/2) CALCULUS III

Polar, Cylindrical & Spherical Coordinates

PART 1 OF 2

1. Polar conversion
2. Polar double integral
3. Cylindrical coordinates

STATE EACH NUMBERED FORMULA

C3.21 (2/2) CALCULUS III

Polar, Cylindrical & Spherical Coordinates

PART 2 OF 2

1. Spherical conversion
2. Spherical volume element
3. Gaussian integral

STATE EACH NUMBERED FORMULA

C3.22 (1/2) CALCULUS III

Change of Variables in Multiple Integrals

PART 1 OF 2

1. Change of variables (2-D)
2. Change of variables (3-D)
3. Inverse Jacobian

STATE EACH NUMBERED FORMULA

C3.22 (2/2) CALCULUS III

Change of Variables in Multiple Integrals

PART 2 OF 2

1. Linear map
2. Density transformation

STATE EACH NUMBERED FORMULA

C3.23 (1/2) CALCULUS III

Mass, Moments & Center of Mass

PART 1 OF 2

1. Mass
2. First moments
3. Center of mass

STATE EACH NUMBERED FORMULA

C3.23 (2/2) CALCULUS III

Mass, Moments & Center of Mass

PART 2 OF 2

1. Moments of inertia
2. Centroid
3. Probability analog

STATE EACH NUMBERED FORMULA

C3.24 CALCULUS III

Surface Area

1. Area of a graph
2. Area of a parametric surface
3. Sphere check

STATE EACH NUMBERED FORMULA

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1 $x = r\cos\theta, \quad y = r\sin\theta, \quad r^2 = x^2 + y^2$

2 $\iint_D f\,dA = \int_\alpha^\beta \int_{r_1(\theta)}^{r_2(\theta)} f(r\cos\theta, r\sin\theta)\, r\,dr\,d\theta$

3 $(r,\theta,z): \quad dV = r\,dz\,dr\,d\theta$

TERMS r,θ polar; z height; ρ,ϕ,θ spherical; extra factor = Jacobian

"Don't forget the extra r": it is the Jacobian
C3.15 §10 Calculus III.

1 $\int_a^b \int_c^d f\,dy\,dx = \int_c^d \int_a^b f\,dx\,dy$

2 $f \geq 0 \Rightarrow$ order can always be swapped

3 $\int_0^1 \int_x^1 f\,dy\,dx = \int_0^1 \int_0^y f\,dx\,dy$

TERMS $dy\,dx$ vertical strips; $dx\,dy$ horizontal strips; same region

Same region, new strips: the limits change, the region does not.

1 $f_{\text{avg}} = \dfrac{1}{\text{Area}(D)} \iint_D f\,dA$

2 $\iint_{[a,b]\times[c,d]} g(x)h(y)\,dA = \int_a^b g\,dx \cdot \int_c^d h\,dy$

TERMS dA area element; inner limits may depend on the outer variable

Integrate the inside first (its limits may depend on the outer variable), treating the outer variable as a constant.

1 $\begin{pmatrix} x \\ y \end{pmatrix} = A \begin{pmatrix} u \\ v \end{pmatrix} \Rightarrow |\det J| = |\det A|$

2 $f_Y(\mathbf{y}) = f_X(\mathbf{x}(\mathbf{y})) \left|\det \dfrac{\partial \mathbf{x}}{\partial \mathbf{y}}\right|$

TERMS S region in u,v; $|\det J|$ area scale factor

New variables, new area element: always multiply by $|\det J|$.

1 $\iint_R f(x,y)\,dx\,dy = \iint_S f(x(u,v), y(u,v)) \left|\dfrac{\partial(x,y)}{\partial(u,v)}\right| du\,dv$

2 $\iiint_E f\,dV = \iiint_S f \left|\dfrac{\partial(x,y,z)}{\partial(u,v,w)}\right| du\,dv\,dw$

3 $\dfrac{\partial(x,y)}{\partial(u,v)} = \left(\dfrac{\partial(u,v)}{\partial(x,y)}\right)^{-1}$

TERMS S region in u,v; $|\det J|$ area scale factor

New variables, new area element: always multiply by $|\det J|$.

1 $x = \rho\sin\phi\cos\theta, \quad y = \rho\sin\phi\sin\theta, \quad z = \rho\cos\phi$

2 $dV = \rho^2 \sin\phi\,d\rho\,d\phi\,d\theta$

3 $\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$

TERMS r,θ polar; z height; ρ,ϕ,θ spherical; extra factor = Jacobian

"Don't forget the extra r": it is the Jacobian
C3.15 §10 Calculus III.

1 $A = \iint_D \sqrt{1 + f_x^2 + f_y^2}\,dA$

2 $A = \iint_D \|\mathbf{r}_u \times \mathbf{r}_v\|\,du\,dv$

3 $A = \int_0^{2\pi} \int_0^{\pi} R^2 \sin\phi\,d\phi\,d\theta = 4\pi R^2$

TERMS D domain; f_x, f_y slopes; $\mathbf{r}_u \times \mathbf{r}_v$ normal

Steeper surface, bigger stretch factor; flat ($f_x = f_y = 0$) gives back the plain area.

1 $I_x = \iint_D y^2\rho\,dA, \quad I_y = \iint_D x^2\rho\,dA, \quad I_0 = I_x + I_y$

2 $\bar{x} = \dfrac{1}{A}\iint_D x\,dA, \qquad \bar{y} = \dfrac{1}{A}\iint_D y\,dA$

3 $P((X,Y) \in D) = \iint_D f\,dA, \qquad \mathbb{E}[X] = \iint x\,f\,dA$

TERMS ρ density; m mass; M_x, M_y moments; $(\bar{x},\bar{y})$ center of mass

Moment about an axis uses the other *coordinate: M_x has y, M_y has x.*

1 $m = \iint_D \rho(x,y)\,dA \qquad \left(\text{solid: } m = \iiint_E \rho\,dV\right)$

2 $M_x = \iint_D y\,\rho\,dA, \qquad M_y = \iint_D x\,\rho\,dA$

3 $\bar{x} = \dfrac{M_y}{m}, \qquad \bar{y} = \dfrac{M_x}{m}$

TERMS ρ density; m mass; M_x, M_y moments; $(\bar{x},\bar{y})$ center of mass

Moment about an axis uses the other *coordinate: M_x has y, M_y has x.*

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V.1 (1/3) VECTOR CALCULUS

Vector Fields & Line Integrals

PART 1 OF 3

1. Vector field
2. Gradient field
3. Flow line

STATE EACH NUMBERED FORMULA

V.1 (2/3) VECTOR CALCULUS

Vector Fields & Line Integrals

PART 2 OF 3

1. Scalar line integral
2. Vector line integral
3. Component form

STATE EACH NUMBERED FORMULA

V.1 (3/3) VECTOR CALCULUS

Vector Fields & Line Integrals

PART 3 OF 3

1. Orientation
2. Additivity

STATE EACH NUMBERED FORMULA

V.2 (1/2) VECTOR CALCULUS

Work, Circulation & Flux Across a Curve

PART 1 OF 2

1. Work
2. Circulation
3. Unit tangent & normal (2-D)

STATE EACH NUMBERED FORMULA

V.2 (2/2) VECTOR CALCULUS

Work, Circulation & Flux Across a Curve

PART 2 OF 2

1. Flux across a plane curve
2. Mass of a wire
3. Center of mass of a wire

STATE EACH NUMBERED FORMULA

V.3 (1/2) VECTOR CALCULUS

Fundamental Theorem for Line Integrals

PART 1 OF 2

1. FT for line integrals
2. Path independence
3. Closed loops

STATE EACH NUMBERED FORMULA

V.3 (2/2) VECTOR CALCULUS

Fundamental Theorem for Line Integrals

PART 2 OF 2

1. Equivalent conditions
2. Conservation of energy

STATE EACH NUMBERED FORMULA

V.4 (1/2) VECTOR CALCULUS

Conservative Fields & Potentials

PART 1 OF 2

1. Conservative field
2. Path independence
3. 2-D test

STATE EACH NUMBERED FORMULA

V.4 (2/2) VECTOR CALCULUS

Conservative Fields & Potentials

PART 2 OF 2

1. 3-D test
2. Why the test works
3. The hole counterexample

STATE EACH NUMBERED FORMULA

V.5 (1/4) VECTOR CALCULUS

Divergence & Curl

PART 1 OF 4

1. Del operator
2. Divergence

STATE EACH NUMBERED FORMULA

V.5 (2/4) VECTOR CALCULUS

Divergence & Curl

PART 2 OF 4

1. Curl

STATE EACH NUMBERED FORMULA

V.5 (3/4) VECTOR CALCULUS

Divergence & Curl

PART 3 OF 4

1. 2-D (scalar) curl
2. Divergence as flux density
3. Curl as circulation density

STATE EACH NUMBERED FORMULA

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1 $\displaystyle\int_{-C} \mathbf{F}\cdot d\mathbf{r} = -\int_C \mathbf{F}\cdot d\mathbf{r}, \qquad \int_{-C} f\,ds = \int_C f\,ds$

2 $\displaystyle\int_{C_1\cup C_2} = \int_{C_1} + \int_{C_2}$

TERMS $\mathbf{F}=(P,Q,R)$ field; $\mathbf{r}(t)$ path; ds arc length; C curve

$d\mathbf{r} = \mathbf{r}'(t)\,dt$ (direction matters); $ds = \|\mathbf{r}'(t)\|\,dt$ (only length matters).

1 $\displaystyle\int_C f\,ds = \int_a^b f\big(\mathbf{r}(t)\big)\,\|\mathbf{r}'(t)\|\,dt$

2 $\displaystyle\int_C \mathbf{F}\cdot d\mathbf{r} = \int_a^b \mathbf{F}\big(\mathbf{r}(t)\big)\cdot\mathbf{r}'(t)\,dt$

3 $\displaystyle\int_C \mathbf{F}\cdot d\mathbf{r} = \int_C P\,dx + Q\,dy + R\,dz$

TERMS $\mathbf{F}=(P,Q,R)$ field; $\mathbf{r}(t)$ path; ds arc length; C curve

$d\mathbf{r} = \mathbf{r}'(t)\,dt$ (direction matters); $ds = \|\mathbf{r}'(t)\|\,dt$ (only length matters).

1 $\mathbf{F}(x,y,z) = \langle P,\ Q,\ R\rangle$

2 $\mathbf{F} = \nabla f$

3 $\mathbf{r}'(t) = \mathbf{F}\big(\mathbf{r}(t)\big)$

TERMS $\mathbf{F}=(P,Q,R)$ field; $\mathbf{r}(t)$ path; ds arc length; C curve

$d\mathbf{r} = \mathbf{r}'(t)\,dt$ (direction matters); $ds = \|\mathbf{r}'(t)\|\,dt$ (only length matters).

1 $\displaystyle\int_C \nabla f\cdot d\mathbf{r} = f\big(\mathbf{r}(b)\big) - f\big(\mathbf{r}(a)\big)$

2 $\displaystyle\int_{C_1} \nabla f\cdot d\mathbf{r} = \int_{C_2} \nabla f\cdot d\mathbf{r}$

3 $\displaystyle\oint_C \nabla f\cdot d\mathbf{r} = 0$

TERMS f potential; $A=\mathbf{r}(a)$ start, $B=\mathbf{r}(b)$ end; U potential energy

The Fundamental Theorem of Calculus for line integrals: only the endpoints matter.

1 $\displaystyle\int_C \mathbf{F}\cdot\mathbf{n}\,ds = \int_C P\,dy - Q\,dx$

2 $\displaystyle m = \int_C \rho\,ds$

3 $\displaystyle\bar{x} = \frac{1}{m}\int_C x\,\rho\,ds, \qquad \bar{y} = \frac{1}{m}\int_C y\,\rho\,ds$

TERMS $\mathbf{T}$ tangent, $\mathbf{n}$ normal; Γ circulation; ρ density

Work goes along *($\mathbf{T}$), flux goes* across *($\mathbf{n}$). Flux: "$P\,dy - Q\,dx$" (swap and negate).*

1 $\displaystyle W = \int_C \mathbf{F}\cdot\mathbf{T}\,ds = \int_C \mathbf{F}\cdot d\mathbf{r}$

2 $\displaystyle\Gamma = \oint_C \mathbf{F}\cdot\mathbf{T}\,ds$

3 $\displaystyle\mathbf{T} = \frac{(x',y')}{\|\mathbf{r}'\|}, \qquad \mathbf{n} = \frac{(y',-x')}{\|\mathbf{r}'\|}$

TERMS $\mathbf{T}$ tangent, $\mathbf{n}$ normal; Γ circulation; ρ density

Work goes along *($\mathbf{T}$), flux goes* across *($\mathbf{n}$). Flux: "$P\,dy - Q\,dx$" (swap and negate).*

1 $\nabla\times\mathbf{F} = \mathbf{0}$

2 $P_y = \varphi_{xy} = \varphi_{yx} = Q_x$

$\mathbf{F} = \left(\dfrac{-y}{x^2+y^2}, \dfrac{x}{x^2+y^2}\right)$

3 $P_y = Q_x, \quad \displaystyle\oint_{x^2+y^2=1} \mathbf{F}\cdot d\mathbf{r} = 2\pi$

TERMS φ potential; P,Q,R components; simply connected = no holes

The Fundamental Theorem of Calculus for line integrals: only the endpoints matter C1.32 *§07 Calculus I.*

1 $\mathbf{F} = \nabla\varphi$

2 $\displaystyle\int_C \mathbf{F}\cdot d\mathbf{r} = \varphi(B) - \varphi(A), \qquad \oint_C \mathbf{F}\cdot d\mathbf{r} = 0$

3 $\displaystyle\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$

TERMS φ potential; P,Q,R components; simply connected = no holes

The Fundamental Theorem of Calculus for line integrals: only the endpoints matter C1.32 *§07 Calculus I.*

$\mathbf{F} = \nabla f \iff$ path independent

1 $\iff \displaystyle\oint_C \mathbf{F}\cdot d\mathbf{r} = 0$ for all closed C

2 $\mathbf{F} = -\nabla U \ \Rightarrow\ K_A + U_A = K_B + U_B$

TERMS f potential; $A=\mathbf{r}(a)$ start, $B=\mathbf{r}(b)$ end; U potential energy

The Fundamental Theorem of Calculus for line integrals: only the endpoints matter.

1 $(\nabla\times\mathbf{F})\cdot\mathbf{k} = Q_x - P_y$

2 $\nabla\cdot\mathbf{F}(p) = \displaystyle\lim_{V\to 0}\frac{1}{V}\iint_{\partial E}\mathbf{F}\cdot\mathbf{n}\,dS$

3 $(\nabla\times\mathbf{F})\cdot\mathbf{n} = \displaystyle\lim_{A\to 0}\frac{1}{A}\oint_C \mathbf{F}\cdot d\mathbf{r}$

TERMS ∇ del; P,Q,R components; $\mathbf{i},\mathbf{j},\mathbf{k}$ unit vectors

Divergence = "spreading out" (source >0, sink <0); curl = "spinning" (right-hand rule).

1 $\displaystyle\nabla\times\mathbf{F} = \begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\ \partial_x & \partial_y & \partial_z\\ P & Q & R\end{vmatrix}$

$= \langle R_y - Q_z,\ P_z - R_x,\ Q_x - P_y\rangle$

TERMS ∇ del; P,Q,R components; $\mathbf{i},\mathbf{j},\mathbf{k}$ unit vectors

Divergence = "spreading out" (source >0, sink <0); curl = "spinning" (right-hand rule).

1 $\displaystyle\nabla = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right)$

2 $\displaystyle\nabla\cdot\mathbf{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}$

TERMS ∇ del; P,Q,R components; $\mathbf{i},\mathbf{j},\mathbf{k}$ unit vectors

Divergence = "spreading out" (source >0, sink <0); curl = "spinning" (right-hand rule).

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V.5 (4/4) VECTOR CALCULUS

Divergence & Curl

PART 4 OF 4

1. Incompressible / irrotational

STATE EACH NUMBERED FORMULA

V.6 (1/3) VECTOR CALCULUS

Vector Identities: Product Rules, curl grad & div curl

PART 1 OF 3

1. Curl of a gradient
2. Divergence of a curl
3. Divergence of a gradient

STATE EACH NUMBERED FORMULA

V.6 (2/3) VECTOR CALCULUS

Vector Identities: Product Rules, curl grad & div curl

PART 2 OF 3

1. Gradient of a product
2. Divergence of $f\mathbf{F}$
3. Curl of $f\mathbf{F}$

STATE EACH NUMBERED FORMULA

V.6 (3/3) VECTOR CALCULUS

Vector Identities: Product Rules, curl grad & div curl

PART 3 OF 3

1. Divergence of a cross product
2. Curl of a curl

STATE EACH NUMBERED FORMULA

V.7 (1/2) VECTOR CALCULUS

Green's Theorem

PART 1 OF 2

1. Circulation (tangential) form
2. Flux (normal) form

STATE EACH NUMBERED FORMULA

V.7 (2/2) VECTOR CALCULUS

Green's Theorem

PART 2 OF 2

1. Area formulas
2. Region with holes
3. Shoelace formula

STATE EACH NUMBERED FORMULA

V.8 (1/3) VECTOR CALCULUS

Parametrized Surfaces & Surface Area

PART 1 OF 3

1. Parametric surface
2. Tangent vectors
3. Normal vector

STATE EACH NUMBERED FORMULA

V.8 (2/3) VECTOR CALCULUS

Parametrized Surfaces & Surface Area

PART 2 OF 3

1. Tangent plane
2. Surface area element
3. Surface area

STATE EACH NUMBERED FORMULA

V.8 (3/3) VECTOR CALCULUS

Parametrized Surfaces & Surface Area

PART 3 OF 3

1. Graph $z = g(x, y)$

STATE EACH NUMBERED FORMULA

V.9 (1/2) VECTOR CALCULUS

Surface Integrals of Scalar Functions

PART 1 OF 2

1. Surface integral
2. Over a graph $z = g(x, y)$
3. Over a sphere

STATE EACH NUMBERED FORMULA

V.9 (2/2) VECTOR CALCULUS

Surface Integrals of Scalar Functions

PART 2 OF 2

1. Mass of a shell
2. Center of mass of a shell
3. Average over a surface

STATE EACH NUMBERED FORMULA

V.10 (1/2) VECTOR CALCULUS

Surface Integrals of Vector Fields (Flux)

PART 1 OF 2

1. Unit normal (orientation)
2. Flux integral
3. Parametric formula

STATE EACH NUMBERED FORMULA

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1 $\nabla(fg) = f\,\nabla g + g\,\nabla f$

2 $\nabla \cdot (f\mathbf{F}) = f\,\nabla \cdot \mathbf{F} + \nabla f \cdot \mathbf{F}$

3 $\nabla \times (f\mathbf{F}) = f\,\nabla \times \mathbf{F} + \nabla f \times \mathbf{F}$

TERMS f, g scalars; $\mathbf{F}, \mathbf{G}$ fields; ∇^2 Laplacian

grad → curl → div: two steps in a row give zero ("the boundary of a boundary is empty").

V.6 (2/3)

1 $\nabla \times (\nabla f) = \mathbf{0}$

2 $\nabla \cdot (\nabla \times \mathbf{F}) = 0$

3 $\nabla \cdot (\nabla f) = \nabla^2 f$

TERMS f, g scalars; $\mathbf{F}, \mathbf{G}$ fields; ∇^2 Laplacian

grad → curl → div: two steps in a row give zero ("the boundary of a boundary is empty").

V.6 (1/3)

1 $\nabla \cdot \mathbf{F} = 0 \qquad / \qquad \nabla \times \mathbf{F} = \mathbf{0}$

TERMS ∇ del; P, Q, R components; $\mathbf{i}, \mathbf{j}, \mathbf{k}$ unit vectors

Divergence = "spreading out" (source > 0, sink < 0); curl = "spinning" (right-hand rule).

V.5 (4/4)

1 $\text{Area}(D) = \frac{1}{2}\oint_C (x\,dy - y\,dx) = \oint_C x\,dy = -\oint_C y\,dx$

2 $\partial D = C_{\text{outer}}\ (\circlearrowleft)\ \cup\ C_{\text{inner}}\ (\circlearrowright)$

3 $A = \frac{1}{2}\left|\sum_{i=1}^{n} \left(x_i y_{i+1} - x_{i+1} y_i\right)\right|$

TERMS C boundary (ccw); D region; P, Q components; $\mathbf{n}$ outward normal

Circulation around the boundary = total spin inside. Counterclockwise is positive.

V.7 (2/2)

1 $\oint_C P\,dx + Q\,dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA$

2 $\oint_C \mathbf{F} \cdot \mathbf{n}\,ds = \oint_C P\,dy - Q\,dx = \iint_D \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y}\right) dA$

TERMS C boundary (ccw); D region; P, Q components; $\mathbf{n}$ outward normal

Circulation around the boundary = total spin inside. Counterclockwise is positive.

V.7 (1/2)

1 $\nabla \cdot (\mathbf{F} \times \mathbf{G}) = \mathbf{G} \cdot (\nabla \times \mathbf{F}) - \mathbf{F} \cdot (\nabla \times \mathbf{G})$

2 $\nabla \times (\nabla \times \mathbf{F}) = \nabla(\nabla \cdot \mathbf{F}) - \nabla^2 \mathbf{F}$

TERMS f, g scalars; $\mathbf{F}, \mathbf{G}$ fields; ∇^2 Laplacian

grad → curl → div: two steps in a row give zero ("the boundary of a boundary is empty").

V.6 (3/3)

1 $\mathbf{r}_x \times \mathbf{r}_y = (-g_x, -g_y, 1), \quad dS = \sqrt{1 + g_x^2 + g_y^2}\,dA$

TERMS $\mathbf{r}(u,v)$ surface; $\mathbf{r}_u, \mathbf{r}_v$ tangents; D parameter domain

"Cross the tangents, take the length": $\|\mathbf{r}_u \times \mathbf{r}_v\|$ is the area stretch.

V.8 (3/3)

1 $\mathbf{N} \cdot \big(\mathbf{x} - \mathbf{r}(u_0, v_0)\big) = 0$

2 $dS = \|\mathbf{r}_u \times \mathbf{r}_v\|\,du\,dv$

3 $A(S) = \iint_D \|\mathbf{r}_u \times \mathbf{r}_v\|\,dA$

TERMS $\mathbf{r}(u,v)$ surface; $\mathbf{r}_u, \mathbf{r}_v$ tangents; D parameter domain

"Cross the tangents, take the length": $\|\mathbf{r}_u \times \mathbf{r}_v\|$ is the area stretch.

V.8 (2/3)

1 $\mathbf{r}(u,v) = \big(x(u,v),\, y(u,v),\, z(u,v)\big), \quad (u,v) \in D$

2 $\mathbf{r}_u = \dfrac{\partial \mathbf{r}}{\partial u}, \qquad \mathbf{r}_v = \dfrac{\partial \mathbf{r}}{\partial v}$

3 $\mathbf{N} = \mathbf{r}_u \times \mathbf{r}_v$

TERMS $\mathbf{r}(u,v)$ surface; $\mathbf{r}_u, \mathbf{r}_v$ tangents; D parameter domain

"Cross the tangents, take the length": $\|\mathbf{r}_u \times \mathbf{r}_v\|$ is the area stretch.

V.8 (1/3)

1 $\mathbf{n} = \dfrac{\mathbf{r}_u \times \mathbf{r}_v}{\|\mathbf{r}_u \times \mathbf{r}_v\|}$

2 $\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_S \mathbf{F} \cdot \mathbf{n}\,dS$

3 $\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_D \mathbf{F}\big(\mathbf{r}(u,v)\big) \cdot \big(\mathbf{r}_u \times \mathbf{r}_v\big)\,dA$

TERMS $\mathbf{n}$ unit normal; $d\mathbf{S} = \mathbf{n}\,dS$; $\mathbf{r}_u \times \mathbf{r}_v$ normal; D domain

"Only what goes through *counts": $\mathbf{F} \cdot \mathbf{n}$. No $\|\cdot\|$ in the parametric formula.*

V.10 (1/2)

1 $m = \iint_S \rho\,dS$

2 $\bar{x} = \dfrac{1}{m}\iint_S x\,\rho\,dS \quad$ (same for $\bar{y}, \bar{z}$)

3 $\bar{f} = \dfrac{1}{A(S)}\iint_S f\,dS$

TERMS f integrand; dS surface element; ρ surface density; m mass

dS is surface *area, never just dA: always include the stretch factor.*

V.9 (2/2)

1 $\iint_S f\,dS = \iint_D f\big(\mathbf{r}(u,v)\big)\,\|\mathbf{r}_u \times \mathbf{r}_v\|\,dA$

2 $\iint_S f\,dS = \iint_D f(x,y,g(x,y))\sqrt{1 + g_x^2 + g_y^2}\,dA$

3 $\iint_S f\,dS = \int_0^{2\pi}\int_0^{\pi} f\,a^2 \sin\phi\,d\phi\,d\theta$

TERMS f integrand; dS surface element; ρ surface density; m mass

dS is surface *area, never just dA: always include the stretch factor.*

V.9 (1/2)

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V.10 (2/2) VECTOR CALCULUS

Surface Integrals of Vector Fields (Flux)

PART 2 OF 2

1. Graph $z = g(x, y)$, upward
2. Sphere, outward
3. Mass flow rate

STATE EACH NUMBERED FORMULA

V.11 (1/2) VECTOR CALCULUS

Flux & the Divergence Theorem

PART 1 OF 2

1. Flux
2. Divergence theorem
3. Gauss's law

STATE EACH NUMBERED FORMULA

V.11 (2/2) VECTOR CALCULUS

Flux & the Divergence Theorem

PART 2 OF 2

1. Continuity equation
2. Solid with a cavity
3. Volume from the boundary

STATE EACH NUMBERED FORMULA

V.12 (1/2) VECTOR CALCULUS

Stokes' Theorem

PART 1 OF 2

1. Stokes' theorem
2. Green as a special case
3. Surface independence

STATE EACH NUMBERED FORMULA

V.12 (2/2) VECTOR CALCULUS

Stokes' Theorem

PART 2 OF 2

1. Closed surface
2. Faraday's law
3. Ampère's law

STATE EACH NUMBERED FORMULA

V.K1 VECTOR CALCULUS

The big theorems: the one-sentence pattern?

RECALL IT

V.K1 VECTOR CALCULUS

Which derivative goes with FTC, line integrals, Green, Stokes, divergence?

RECALL IT

V.K1 VECTOR CALCULUS

Orientation conventions: Green, Stokes, divergence?

RECALL IT

V.K1 VECTOR CALCULUS

Which theorem for: closed plane curve; closed space curve; closed surface; gradient field?

RECALL IT

V.K1 VECTOR CALCULUS

Flux of $\mathbf{F} = (x, y, z)$ out of the unit sphere, two ways?

RECALL IT

V.13 (1/3) VECTOR CALCULUS

The Laplacian & Classic PDEs

PART 1 OF 3

1. Laplacian
2. Heat (diffusion) equation
3. Wave equation

STATE EACH NUMBERED FORMULA

V.13 (2/3) VECTOR CALCULUS

The Laplacian & Classic PDEs

PART 2 OF 3

1. Laplace's equation
2. Poisson's equation
3. Mean-value property

STATE EACH NUMBERED FORMULA

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1 $$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0$$

2 $\partial E = S_{\text{outer}} \cup S_{\text{inner}}$, $\mathbf{n}$ pointing out of E

3 $$V = \frac{1}{3} \iint_{\partial E} \mathbf{r} \cdot \mathbf{n} \, dS$$

TERMS $S = \partial E$ closed surface; $\mathbf{n}$ outward normal; E solid; ρ density

What flows out through the skin = what is produced inside.

V.11 (2/2)

1 $$\text{flux} = \iint_S \mathbf{F} \cdot \mathbf{n} \, dS$$

2 $$\iint_{S=\partial E} \mathbf{F} \cdot \mathbf{n} \, dS = \iiint_E \nabla \cdot \mathbf{F} \, dV$$

3 $$\iint_{\partial E} \mathbf{E} \cdot \mathbf{n} \, dS = \frac{Q_{\text{enc}}}{\varepsilon_0}$$

TERMS $S = \partial E$ closed surface; $\mathbf{n}$ outward normal; E solid; ρ density

What flows out through the skin = what is produced inside.

V.11 (1/2)

1 $$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_D \left(-Pg_x - Qg_y + R \right) dA$$

2 $$\mathbf{n} = \frac{\mathbf{r}}{a}, \qquad dS = a^2 \sin\phi \, d\phi \, d\theta$$

3 $$\dot{m} = \iint_S \rho \, \mathbf{v} \cdot \mathbf{n} \, dS$$

TERMS $\mathbf{n}$ unit normal; $d\mathbf{S} = \mathbf{n} \, dS$; $\mathbf{r}_u \times \mathbf{r}_v$ normal; D domain

"Only what goes through *counts": $\mathbf{F} \cdot \mathbf{n}$. No $\| \cdot \|$ in the parametric formula.*

V.10 (2/2)

Integral of a derivative over a region = integral of the function over the oriented boundary:
$\int_{\partial\Omega}$ field $= \int_{\Omega}$ derivative.

V.K1

1 $$\iint_{\partial E} (\nabla \times \mathbf{F}) \cdot \mathbf{n} \, dS = 0$$

2 $$\oint_C \mathbf{E} \cdot d\mathbf{r} = -\frac{d}{dt} \iint_S \mathbf{B} \cdot \mathbf{n} \, dS$$

3 $$\oint_C \mathbf{B} \cdot d\mathbf{r} = \mu_0 I_{\text{enc}}$$

TERMS $C = \partial S$ boundary; $\mathbf{n}$ unit normal; $\nabla \times \mathbf{F}$ curl

Right-hand rule: curl your fingers along C, your thumb gives $\mathbf{n}$.

V.12 (2/2)

1 $$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} \, dS$$

2 $S \subset xy$-plane, $\mathbf{n} = \mathbf{k}$: $(\nabla \times \mathbf{F}) \cdot \mathbf{k} = Q_x - P_y$

3 $$\iint_{S_1} (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = \iint_{S_2} (\nabla \times \mathbf{F}) \cdot d\mathbf{S}$$

TERMS $C = \partial S$ boundary; $\mathbf{n}$ unit normal; $\nabla \times \mathbf{F}$ curl

Right-hand rule: curl your fingers along C, your thumb gives $\mathbf{n}$.

V.12 (1/2)

Green; Stokes; divergence theorem; FT for line integrals (endpoints only).

V.K1

Green: counterclockwise (region on the left, holes clockwise).
Stokes: right-hand rule (fingers along C, thumb $\mathbf{n}$).
Divergence: outward normal.

V.K1

f'; ∇f; $Q_x - P_y$; $\nabla \times \mathbf{F}$; $\nabla \cdot \mathbf{F}$

V.K1

1 $$\nabla^2 u = 0$$

2 $$\nabla^2 u = f$$

3 $$u(p) = \frac{1}{4\pi r^2} \iint_{\text{sphere}(p,r)} u \, dS$$

TERMS ∇^2 Laplacian; α diffusivity; c wave speed; h grid step

The Laplacian compares a point to the average of its neighbors: positive means the point is lower than its surroundings.

V.13 (2/3)

1 $$\nabla^2 f = \nabla \cdot \nabla f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$$

2 $$u_t = \alpha \nabla^2 u$$

3 $$u_{tt} = c^2 \nabla^2 u$$

TERMS ∇^2 Laplacian; α diffusivity; c wave speed; h grid step

The Laplacian compares a point to the average of its neighbors: positive means the point is lower than its surroundings.

V.13 (1/3)

Direct: $\mathbf{F} \cdot \mathbf{n} = 1$, area 4π.
Divergence: $\nabla \cdot \mathbf{F} = 3$, $3 \cdot \frac{4}{3}\pi = 4\pi$.

V.K1

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The Laplacian & Classic PDEs

PART 3 OF 3

1. Discrete Laplacian
2. Laplacian in polar coordinates

STATE EACH NUMBERED FORMULA

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1 $$\nabla^2 u \approx \frac{u_E + u_W + u_N + u_S - 4u_C}{h^2}$$

2 $$\nabla^2 u = u_{rr} + \frac{1}{r}u_r + \frac{1}{r^2}u_{\theta\theta}$$

TERMS ∇^2 Laplacian; α diffusivity; c wave speed; h grid step

The Laplacian compares a point to the average of its neighbors: positive means the point is lower than its surroundings.

V.13 (3/3)

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P.1 (1/2) PHYSICS & NATURAL PHENOMENA

Units & Dimensional Analysis

PART 1 OF 2

1. Unit of force (newton)
2. Unit of energy (joule)
3. Unit of power (watt)

STATE EACH NUMBERED FORMULA

P.1 (2/2) PHYSICS & NATURAL PHENOMENA

Units & Dimensional Analysis

PART 2 OF 2

1. Dimensional homogeneity
2. Power-law ansatz
3. Buckingham π theorem

STATE EACH NUMBERED FORMULA

P.K1 PHYSICS & NATURAL PHENOMENA

SI prefixes: symbols and powers of ten, from pico to tera

RECALL IT

P.K1 PHYSICS & NATURAL PHENOMENA

The seven SI base units

RECALL IT

P.K1 PHYSICS & NATURAL PHENOMENA

N, J, W, Pa in base SI units

RECALL IT

P.K1 PHYSICS & NATURAL PHENOMENA

Factor-label conversion: how, and the squared-unit trap

RECALL IT

P.2 (1/2) PHYSICS & NATURAL PHENOMENA

Kinematics: Constant Acceleration

PART 1 OF 2

1. Velocity
2. Acceleration
3. Velocity–time

STATE EACH NUMBERED FORMULA

P.2 (2/2) PHYSICS & NATURAL PHENOMENA

Kinematics: Constant Acceleration

PART 2 OF 2

1. Position–time
2. Timeless equation
3. Average-velocity form

STATE EACH NUMBERED FORMULA

P.3 (1/3) PHYSICS & NATURAL PHENOMENA

Projectile Motion

PART 1 OF 3

1. Launch components
2. Horizontal motion
3. Vertical motion

STATE EACH NUMBERED FORMULA

P.3 (2/3) PHYSICS & NATURAL PHENOMENA

Projectile Motion

PART 2 OF 3

1. Time of flight (level)
2. Maximum height
3. Range (level ground)

STATE EACH NUMBERED FORMULA

P.3 (3/3) PHYSICS & NATURAL PHENOMENA

Projectile Motion

PART 3 OF 3

1. Trajectory equation

STATE EACH NUMBERED FORMULA

P.4 (1/3) PHYSICS & NATURAL PHENOMENA

Newton's Laws & Friction

PART 1 OF 3

1. First law (inertia)
2. Second law
3. Third law

STATE EACH NUMBERED FORMULA

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p 10^{-12}, n 10^{-9}, μ 10^{-6}, m 10^{-3}, c 10^{-2}
k 10^{3}, M 10^{6}, G 10^{9}, T 10^{12}

1 [left side] = [right side]
2 $Q = C\,x_1^a x_2^b x_3^c$
3 # dimensionless groups $= n - k$

TERMS N = kg·m/s^2; n variables; k base dimensions (M, L, T)

"Units first": if the units of both sides do not match, the equation is wrong.

1 $[\text{force}] = \dfrac{\text{kg}\cdot\text{m}}{\text{s}^2} = \text{N}$
2 $[\text{energy}] = \text{N}\cdot\text{m} = \text{J}$
3 $[\text{power}] = \dfrac{\text{J}}{\text{s}} = \text{W}$

TERMS N = kg·m/s^2; n variables; k base dimensions (M, L, T)

"Units first": if the units of both sides do not match, the equation is wrong.

Multiply by fractions equal to 1 so unwanted units cancel. Square/cube the factor for areas/volumes:
1 cm^2 = 10^{-4} m^2, 1 km^2 = 10^6 m^2

N = kg·m/s^2 J = N·m = kg·m^2/s^2
W = J/s = kg·m^2/s^3
Pa = N/m^2 = kg/(m·s^2)

m (length), kg (mass), s (time),
A (current), K (temperature),
mol (amount), cd (luminous intensity)

1 $v_{0x} = v_0\cos\theta, \quad v_{0y} = v_0\sin\theta$
2 $x = v_{0x}\,t$
3 $y = y_0 + v_{0y}t - \frac{1}{2}gt^2, \quad v_y = v_{0y} - gt$

TERMS v_0 launch speed; θ launch angle; $g = 9.81$ m/s^2; H height; R range

"Horizontal: cruise. Vertical: free fall." The clock is the only thing the two directions share.

1 $x = x_0 + v_0 t + \frac{1}{2}at^2$
2 $v^2 = v_0^2 + 2a\,\Delta x$
3 $\Delta x = \frac{1}{2}(v_0 + v)\,t$

TERMS x position (m); v velocity (m/s); a acceleration (m/s^2); t time (s)

"Velocity is the slope of position; acceleration is the slope of velocity" (and area under v–t is distance).

1 $v = \dfrac{dx}{dt}$
2 $a = \dfrac{dv}{dt} = \dfrac{d^2x}{dt^2}$
3 $v = v_0 + at$

TERMS x position (m); v velocity (m/s); a acceleration (m/s^2); t time (s)

"Velocity is the slope of position; acceleration is the slope of velocity" (and area under v–t is distance).

1 $\sum \mathbf{F} = \mathbf{0} \iff \mathbf{v} = \text{const}$
2 $\sum \mathbf{F} = m\mathbf{a}$
3 $\mathbf{F}_{AB} = -\mathbf{F}_{BA}$

TERMS m mass (kg); N normal force; μ_k, μ_s friction coefficients; θ incline

"F equals m a" — every problem starts with a free-body diagram: draw all forces, then sum them.

1 $y = x\tan\theta - \dfrac{g\,x^2}{2v_0^2\cos^2\theta}$

TERMS v_0 launch speed; θ launch angle; $g = 9.81$ m/s^2; H height; R range

"Horizontal: cruise. Vertical: free fall." The clock is the only thing the two directions share.

1 $t_f = \dfrac{2v_0\sin\theta}{g}$
2 $H = \dfrac{v_0^2\sin^2\theta}{2g}$
3 $R = \dfrac{v_0^2\sin 2\theta}{g}$

TERMS v_0 launch speed; θ launch angle; $g = 9.81$ m/s^2; H height; R range

"Horizontal: cruise. Vertical: free fall." The clock is the only thing the two directions share.

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P.4 (2/3) PHYSICS & NATURAL PHENOMENA

Newton's Laws & Friction

PART 2 OF 3

1. Weight
2. Kinetic friction
3. Static friction

STATE EACH NUMBERED FORMULA

P.4 (3/3) PHYSICS & NATURAL PHENOMENA

Newton's Laws & Friction

PART 3 OF 3

1. Incline components

STATE EACH NUMBERED FORMULA

P.5 (1/2) PHYSICS & NATURAL PHENOMENA

Hooke's Law & Springs

PART 1 OF 2

1. Hooke's law
2. Spring potential energy
3. Springs in series

STATE EACH NUMBERED FORMULA

P.5 (2/2) PHYSICS & NATURAL PHENOMENA

Hooke's Law & Springs

PART 2 OF 2

1. Springs in parallel
2. Stress–strain (Young's modulus)

STATE EACH NUMBERED FORMULA

P.6 (1/3) PHYSICS & NATURAL PHENOMENA

Work, Energy & Power

PART 1 OF 3

1. Work
2. Kinetic energy
3. Gravitational potential energy

STATE EACH NUMBERED FORMULA

P.6 (2/3) PHYSICS & NATURAL PHENOMENA

Work, Energy & Power

PART 2 OF 3

1. Work–energy theorem
2. Conservation of mechanical energy
3. Energy with losses

STATE EACH NUMBERED FORMULA

P.6 (3/3) PHYSICS & NATURAL PHENOMENA

Work, Energy & Power

PART 3 OF 3

1. Power
2. Efficiency

STATE EACH NUMBERED FORMULA

P.7 (1/3) PHYSICS & NATURAL PHENOMENA

Momentum & Impulse

PART 1 OF 3

1. Momentum
2. Second law (momentum form)
3. Impulse–momentum theorem

STATE EACH NUMBERED FORMULA

P.7 (2/3) PHYSICS & NATURAL PHENOMENA

Momentum & Impulse

PART 2 OF 3

1. Conservation of momentum
2. Perfectly inelastic collision
3. Elastic collision (1-D)

STATE EACH NUMBERED FORMULA

P.7 (3/3) PHYSICS & NATURAL PHENOMENA

Momentum & Impulse

PART 3 OF 3

1. Center of mass

STATE EACH NUMBERED FORMULA

P.8 (1/2) PHYSICS & NATURAL PHENOMENA

Circular Motion & Gravitation

PART 1 OF 2

1. Centripetal acceleration
2. Centripetal force
3. Speed and period

STATE EACH NUMBERED FORMULA

P.8 (2/2) PHYSICS & NATURAL PHENOMENA

Circular Motion & Gravitation

PART 2 OF 2

1. Newton's law of gravitation
2. Surface gravity
3. Orbital speed

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1. $F = -kx$
2. $U_s = \frac{1}{2}kx^2$
3. $\dfrac{1}{k_{\text{eq}}} = \dfrac{1}{k_1} + \dfrac{1}{k_2}$

TERMS k spring constant (N/m); x displacement (m); E Young's modulus (Pa)

Hooke's anagram "ut tensio, sic vis": as the extension, so the force. Springs combine the opposite *way to resistors* P.33 §12 Physics.

1. $F_{\parallel} = mg\sin\theta, \quad N = mg\cos\theta$

TERMS m mass (kg); N normal force; μ_k, μ_s friction coefficients; θ incline

"F equals m a" — every problem starts with a free-body diagram: draw all forces, then sum them.

1. $W = mg$
2. $f_k = \mu_k N$
3. $f_s \le \mu_s N$

TERMS m mass (kg); N normal force; μ_k, μ_s friction coefficients; θ incline

"F equals m a" — every problem starts with a free-body diagram: draw all forces, then sum them.

1. $W_{\text{net}} = \Delta KE$
2. $KE_1 + PE_1 = KE_2 + PE_2$
3. $KE_1 + PE_1 = KE_2 + PE_2 + E_{\text{loss}}$

TERMS m mass; v speed; h height; θ angle between $\mathbf{F}$ and motion; η efficiency

1. $W = Fd\cos\theta = \displaystyle\int \mathbf{F}\cdot d\mathbf{x}$
2. $KE = \frac{1}{2}mv^2$
3. $PE = mgh$

TERMS m mass; v speed; h height; θ angle between $\mathbf{F}$ and motion; η efficiency

1. $k_{\text{eq}} = k_1 + k_2$
2. $\dfrac{F}{A} = E\,\dfrac{\Delta L}{L_0}$

TERMS k spring constant (N/m); x displacement (m); E Young's modulus (Pa)

Hooke's anagram "ut tensio, sic vis": as the extension, so the force. Springs combine the opposite *way to resistors* P.33 §12 Physics.

1. $m_1\mathbf{v}_1 + m_2\mathbf{v}_2 = m_1\mathbf{v}_1' + m_2\mathbf{v}_2'$
2. $v' = \dfrac{m_1 v_1 + m_2 v_2}{m_1 + m_2}$
3. $v_1' - v_2' = -(v_1 - v_2)$

TERMS m mass (kg); $\mathbf{v}$ velocity; primes = after; Δt contact time

1. $\mathbf{p} = m\mathbf{v}$
2. $\sum \mathbf{F} = \dfrac{d\mathbf{p}}{dt}$
3. $\mathbf{J} = \displaystyle\int \mathbf{F}\,dt = \mathbf{F}_{\text{avg}}\,\Delta t = \Delta\mathbf{p}$

TERMS m mass (kg); $\mathbf{v}$ velocity; primes = after; Δt contact time

1. $P = \dfrac{W}{t} = Fv$
2. $\eta = \dfrac{P_{\text{out}}}{P_{\text{in}}}$

TERMS m mass; v speed; h height; θ angle between $\mathbf{F}$ and motion; η efficiency

1. $F = \dfrac{G\,m_1 m_2}{r^2}$
2. $g = \dfrac{GM}{R^2}$
3. $v_{\text{orbit}} = \sqrt{\dfrac{GM}{r}}$

TERMS r radius; ω angular speed; G gravitational constant; M, R planet mass, radius

Gravity is an inverse-square law: double the distance, one quarter the force.

1. $a_c = \dfrac{v^2}{r} = \omega^2 r$
2. $F_c = \dfrac{mv^2}{r}$
3. $v = \omega r = \dfrac{2\pi r}{T}$

TERMS r radius; ω angular speed; G gravitational constant; M, R planet mass, radius

Gravity is an inverse-square law: double the distance, one quarter the force.

1. $x_{\text{cm}} = \dfrac{\sum m_i x_i}{\sum m_i}$

TERMS m mass (kg); $\mathbf{v}$ velocity; primes = after; Δt contact time

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Orbits, Escape Velocity & Kepler's Laws

PART 1 OF 2

1. Gravitational potential energy
2. Escape velocity
3. Field strength at distance r

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Orbits, Escape Velocity & Kepler's Laws

PART 2 OF 2

1. Kepler's first & second laws
2. Kepler's third law
3. Orbital energy

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Rotational Kinematics

PART 1 OF 3

1. Arc length
2. Angular velocity & acceleration
3. Linear–angular link

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Rotational Kinematics

PART 2 OF 3

1. Angular velocity–time
2. Angular position–time
3. Rotational timeless equation

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Rotational Kinematics

PART 3 OF 3

1. Frequency & rpm

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Torque & Static Equilibrium

PART 1 OF 2

1. Torque (magnitude)
2. Torque (vector)
3. Law of the lever

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Torque & Static Equilibrium

PART 2 OF 2

1. Translational equilibrium
2. Rotational equilibrium

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Moment of Inertia

PART 1 OF 2

1. Point masses
2. Continuous body
3. Parallel-axis theorem

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Moment of Inertia

PART 2 OF 2

1. Perpendicular-axis theorem
2. Radius of gyration

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Rotational Dynamics & Angular Momentum

PART 1 OF 2

1. Newton's second law for rotation
2. Angular momentum (rigid body)
3. Angular momentum (particle)

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Rotational Dynamics & Angular Momentum

PART 2 OF 2

1. Torque–angular momentum
2. Conservation of angular momentum
3. Rotational work & power

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Rotational Kinetic Energy & Rolling

PART 1 OF 2

1. Rotational kinetic energy
2. Rolling without slipping
3. Total kinetic energy (rolling)

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1 $s = r\theta$

2 $\omega = \dfrac{d\theta}{dt}, \quad \alpha = \dfrac{d\omega}{dt}$

3 $v_t = r\omega, \quad a_t = r\alpha$

TERMS θ angle (rad); ω angular velocity; α angular acceleration; r radius

Swap $x \to \theta$, $v \to \omega$, $a \to \alpha$ and every kinematics equation still works.

1 ellipse, star at a focus; $\dfrac{dA}{dt} = \text{const}$

2 $T^2 = \dfrac{4\pi^2}{GM} a^3$

3 $E = KE + U = -\dfrac{GMm}{2a}$

TERMS r distance from center; a semi-major axis; T period; M central mass

Kepler III: "square the period, cube the distance."

1 $U = -\dfrac{GMm}{r}$

2 $v_{\text{esc}} = \sqrt{\dfrac{2GM}{r}} = \sqrt{2}\, v_{\text{orbit}}$

3 $g(r) = \dfrac{GM}{r^2} = g\left(\dfrac{R}{r}\right)^2$

TERMS r distance from center; a semi-major axis; T period; M central mass

Kepler III: "square the period, cube the distance."

1 $\tau = rF\sin\phi = r_\perp F$

2 $\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}$

3 $F_1 d_1 = F_2 d_2$

TERMS τ torque (N·m); $r_\perp$ lever arm; ϕ angle between $\mathbf{r}$ and $\mathbf{F}$

Door handles sit far from the hinges: a long lever arm needs only a small force.

1 $\omega = 2\pi f = \dfrac{2\pi \cdot \text{rpm}}{60}$

TERMS θ angle (rad); ω angular velocity; α angular acceleration; r radius

Swap $x \to \theta$, $v \to \omega$, $a \to \alpha$ and every kinematics equation still works.

1 $\omega = \omega_0 + \alpha t$

2 $\theta = \theta_0 + \omega_0 t + \frac{1}{2}\alpha t^2$

3 $\omega^2 = \omega_0^2 + 2\alpha\,\Delta\theta$

TERMS θ angle (rad); ω angular velocity; α angular acceleration; r radius

Swap $x \to \theta$, $v \to \omega$, $a \to \alpha$ and every kinematics equation still works.

1 $I_z = I_x + I_y$

2 $I = MR_g^2$

TERMS m_i, r_i mass and distance to axis; M total mass; d axis offset

Mass far from the axis counts quadratically: a hoop (MR^2) is twice as hard to spin as a disk ($\frac{1}{2}MR^2$) of equal mass and radius.

1 $I = \sum m_i r_i^2$

2 $I = \displaystyle\int r^2\, dm$

3 $I = I_{\text{cm}} + Md^2$

TERMS m_i, r_i mass and distance to axis; M total mass; d axis offset

Mass far from the axis counts quadratically: a hoop (MR^2) is twice as hard to spin as a disk ($\frac{1}{2}MR^2$) of equal mass and radius.

1 $\sum \mathbf{F} = \mathbf{0}$

2 $\sum \tau = 0$ about any point

TERMS τ torque (N·m); $r_\perp$ lever arm; ϕ angle between $\mathbf{r}$ and $\mathbf{F}$

Door handles sit far from the hinges: a long lever arm needs only a small force.

1 $KE_{\text{rot}} = \frac{1}{2} I\omega^2$

2 $v_{\text{cm}} = R\omega, \quad a_{\text{cm}} = R\alpha$

3 $KE = \frac{1}{2}Mv_{\text{cm}}^2 + \frac{1}{2}I_{\text{cm}}\omega^2$

TERMS I moment of inertia; ω angular speed; R radius; v_{cm} center speed

Race down a ramp: sphere (0.4) beats disk (0.5) beats hoop (1), whatever their mass or size. The smaller I/MR^2, the faster.

1 $\sum \boldsymbol{\tau} = \dfrac{d\mathbf{L}}{dt}$

2 $I_1\omega_1 = I_2\omega_2$

3 $W = \tau\,\Delta\theta, \quad P = \tau\omega$

TERMS τ torque; I moment of inertia; ω, α angular velocity, acceleration; L angular momentum

"Pull in, spin up": L fixed, so smaller I means larger ω.

1 $\sum \tau = I\alpha$

2 $L = I\omega$

3 $\mathbf{L} = \mathbf{r} \times \mathbf{p}, \quad L = mvr_\perp$

TERMS τ torque; I moment of inertia; ω, α angular velocity, acceleration; L angular momentum

"Pull in, spin up": L fixed, so smaller I means larger ω.

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Rotational Kinetic Energy & Rolling

PART 2 OF 2

1. Speed after descending h
2. Acceleration down an incline

STATE EACH NUMBERED FORMULA

Oscillations & Simple Harmonic Motion

PART 1 OF 3

1. SHM equation
2. Solution
3. Angular frequency (spring–mass)

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Oscillations & Simple Harmonic Motion

PART 2 OF 3

1. Period & frequency
2. Maximum speed & acceleration
3. Energy

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Oscillations & Simple Harmonic Motion

PART 3 OF 3

1. Damped oscillation

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Pendulums

PART 1 OF 2

1. Restoring force
2. Simple pendulum period
3. Physical pendulum period

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Pendulums

PART 2 OF 2

1. Large-amplitude correction
2. Measuring g

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Waves: Speed, Frequency & Wavelength

PART 1 OF 3

1. Wave speed
2. Traveling sinusoidal wave
3. Wave number & angular frequency

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Waves: Speed, Frequency & Wavelength

PART 2 OF 3

1. Wave speed on a string
2. Standing waves (both ends fixed)
3. Superposition & interference

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Waves: Speed, Frequency & Wavelength

PART 3 OF 3

1. Electromagnetic waves

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Sound: Intensity, Decibels & the Doppler Effect

PART 1 OF 2

1. Speed of sound in air
2. Intensity of a point source
3. Sound intensity level

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Sound: Intensity, Decibels & the Doppler Effect

PART 2 OF 2

1. Doppler effect
2. Beat frequency
3. Doppler for light (slow source)

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Pressure, Hydrostatics & Buoyancy

PART 1 OF 2

1. Pressure
2. Hydrostatic pressure
3. Pascal's principle (hydraulics)

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1 $T = \dfrac{2\pi}{\omega} = 2\pi\sqrt{\dfrac{m}{k}}, \quad f = \dfrac{1}{T}$

2 $v_{\max} = A\omega, \quad a_{\max} = A\omega^2$

3 $E = \frac{1}{2}kA^2 = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$

TERMS A amplitude; ϕ phase; k spring constant; m mass; ω angular frequency

SHM = "a spring and a cosine"; the period depends on the system (m, k), never on how hard you pluck it.

1 $m\ddot{x} = -kx \iff \ddot{x} = -\omega^2 x$

2 $x(t) = A\cos(\omega t + \phi)$

3 $\omega = \sqrt{\dfrac{k}{m}}$

TERMS A amplitude; ϕ phase; k spring constant; m mass; ω angular frequency

SHM = "a spring and a cosine"; the period depends on the system (m, k), never on how hard you pluck it.

1 $v = \sqrt{\dfrac{2gh}{1 + I_{\text{cm}}/(MR^2)}}$

2 $a = \dfrac{g\sin\theta}{1 + I_{\text{cm}}/(MR^2)}$

TERMS I moment of inertia; ω angular speed; R radius; v_{cm} center speed

Race down a ramp: sphere (0.4) beats disk (0.5) beats hoop (1), whatever their mass or size. The smaller I/MR^2, the faster.

1 $T \approx 2\pi\sqrt{\dfrac{L}{g}}\left(1 + \dfrac{\theta_0^2}{16}\right)$

2 $g = \dfrac{4\pi^2 L}{T^2}$

TERMS L length; θ angle (rad); I moment of inertia; d pivot to center of mass

"Period depends on length, not on mass": a 1 m pendulum ticks once per second.

1 $F_t = -mg\sin\theta \approx -mg\,\theta$

2 $T \approx 2\pi\sqrt{\dfrac{L}{g}}$

3 $T = 2\pi\sqrt{\dfrac{I}{mgd}}$

TERMS L length; θ angle (rad); I moment of inertia; d pivot to center of mass

"Period depends on length, not on mass": a 1 m pendulum ticks once per second.

1 $x(t) = A\,e^{-\gamma t}\cos(\omega_d t + \phi), \qquad \gamma = \dfrac{b}{2m}, \quad \omega_d = \sqrt{\omega^2 - \gamma^2}$

TERMS A amplitude; ϕ phase; k spring constant; m mass; ω angular frequency

SHM = "a spring and a cosine"; the period depends on the system (m, k), never on how hard you pluck it.

1 $c = f\lambda, \quad c = 3.00 \times 10^8 \text{ m/s}$

TERMS f frequency; λ wavelength; $k = 2\pi/\lambda$; μ mass per length; L string length

"The medium sets the speed, the source sets the frequency."

1 $v = \sqrt{\dfrac{F_T}{\mu}}$

2 $\lambda_n = \dfrac{2L}{n}, \quad f_n = \dfrac{nv}{2L}$

3 $y = y_1 + y_2; \quad \Delta r = m\lambda$ (constructive)

TERMS f frequency; λ wavelength; $k = 2\pi/\lambda$; μ mass per length; L string length

"The medium sets the speed, the source sets the frequency."

1 $v = f\lambda = \dfrac{\lambda}{T}$

2 $y(x,t) = A\sin(kx - \omega t)$

3 $k = \dfrac{2\pi}{\lambda}, \quad \omega = 2\pi f$

TERMS f frequency; λ wavelength; $k = 2\pi/\lambda$; μ mass per length; L string length

"The medium sets the speed, the source sets the frequency."

1 $P = \dfrac{F}{A}$

2 $P = P_0 + \rho g h$

3 $\dfrac{F_1}{A_1} = \dfrac{F_2}{A_2}$

TERMS ρ density; h depth; P_0 surface pressure; V volume

Archimedes: "the fluid pushes up with the weight of the fluid you push aside."

1 $f' = f\,\dfrac{v \pm v_o}{v \mp v_s}$ (upper signs: moving toward each other)

2 $f_{\text{beat}} = |f_1 - f_2|$

3 $\dfrac{\Delta f}{f} \approx \dfrac{v_r}{c}$

TERMS I intensity; I_0 threshold; v_o, v_s observer, source speeds; v sound speed

Doppler signs: pick the sign that makes f' bigger when the two approach. Decibels: $+10$ dB $= \times 10$ intensity, $+3$ dB $\approx \times 2$.

1 $v \approx 331 + 0.6\,T_C \text{ m/s}$

2 $I = \dfrac{P}{4\pi r^2}$

3 $\beta = 10\log_{10}\dfrac{I}{I_0} \text{ dB}$

TERMS I intensity; I_0 threshold; v_o, v_s observer, source speeds; v sound speed

Doppler signs: pick the sign that makes f' bigger when the two approach. Decibels: $+10$ dB $= \times 10$ intensity, $+3$ dB $\approx \times 2$.

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Pressure, Hydrostatics & Buoyancy

PART 2 OF 2

1. Archimedes' principle
2. Floating fraction

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Continuity & Bernoulli's Equation

PART 1 OF 2

1. Volumetric flow rate
2. Continuity equation
3. Bernoulli's equation

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Continuity & Bernoulli's Equation

PART 2 OF 2

1. Torricelli's law

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Viscosity & the Reynolds Number

PART 1 OF 2

1. Newton's law of viscosity
2. Reynolds number
3. Pipe flow regimes

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Viscosity & the Reynolds Number

PART 2 OF 2

1. Poiseuille's law

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Drag & Terminal Velocity

PART 1 OF 2

1. Quadratic drag
2. Terminal velocity
3. Stokes drag ($\mathrm{Re} \ll 1$)

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Drag & Terminal Velocity

PART 2 OF 2

1. Power against drag

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Heat, Temperature & Phase Change

PART 1 OF 2

1. Sensible heat
2. Latent heat
3. Calorimetry balance

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Heat, Temperature & Phase Change

PART 2 OF 2

1. Kelvin
2. Fahrenheit

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Conduction: Fourier's Law & Thermal Resistance

PART 1 OF 2

1. Fourier's law
2. Heat flow through a slab
3. Conduction resistance

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Conduction: Fourier's Law & Thermal Resistance

PART 2 OF 2

1. Thermal circuit

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Convection & Newton's Law of Cooling

1. Convective heat flow
2. Newton's law of cooling
3. Time constant

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1 $v_{\text{out}} = \sqrt{2gh}$

TERMS A area; v speed; P pressure; ρ density; h height

"Fast fluid, low pressure" — squeeze the pipe, speed goes up, pressure goes down.

P.20 (2/2)

1 $Q = Av$

2 $A_1 v_1 = A_2 v_2$

3 $P_1 + \frac{1}{2}\rho v_1^2 + \rho g h_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho g h_2$

TERMS A area; v speed; P pressure; ρ density; h height

"Fast fluid, low pressure" — squeeze the pipe, speed goes up, pressure goes down.

P.20 (1/2)

1 $F_B = \rho_{\text{fluid}}\, g\, V_{\text{displaced}}$

2 $\frac{V_{\text{sub}}}{V} = \frac{\rho_{\text{object}}}{\rho_{\text{fluid}}}$

TERMS ρ density; h depth; P_0 surface pressure; V volume

Archimedes: "the fluid pushes up with the weight of the fluid you push aside."

P.19 (2/2)

1 $F_D = \frac{1}{2}\rho\, C_D\, A\, v^2$

2 $v_t = \sqrt{\frac{2mg}{\rho\, C_D A}}$

3 $F_D = 6\pi\mu r v$

TERMS ρ fluid density; C_D drag coefficient; A frontal area; μ viscosity

Drag goes as v^2, drag power as v^3.

P.22 (1/2)

1 $Q = \frac{\pi r^4 \Delta P}{8\mu L}$

TERMS μ viscosity; $\nu = \mu/\rho$; L length scale; r pipe radius; ΔP pressure drop

"Re is big, flow goes wild" — high Reynolds means turbulence; low means smooth, syrupy flow.

P.21 (2/2)

1 $\tau = \mu \frac{du}{dy}$

2 $\text{Re} = \frac{\rho v L}{\mu} = \frac{vL}{\nu} = \frac{\text{inertial forces}}{\text{viscous forces}}$

3 $\text{Re} < 2300$ laminar, $\text{Re} > 4000$ turbulent

TERMS μ viscosity; $\nu = \mu/\rho$; L length scale; r pipe radius; ΔP pressure drop

"Re is big, flow goes wild" — high Reynolds means turbulence; low means smooth, syrupy flow.

P.21 (1/2)

1 $T_K = T_C + 273.15$

2 $T_F = \frac{9}{5} T_C + 32$

TERMS m mass; c specific heat; L latent heat; ΔT temperature change

Water: $c \approx 4186$ J/(kg·K); melting ice takes as much heat as warming the water by about 80°C.

P.23 (2/2)

1 $Q = mc\,\Delta T$

2 $Q = mL$

3 $\sum Q_i = 0$

TERMS m mass; c specific heat; L latent heat; ΔT temperature change

Water: $c \approx 4186$ J/(kg·K); melting ice takes as much heat as warming the water by about 80°C.

P.23 (1/2)

1 $P_{\text{drag}} = F_D v \propto v^3$

TERMS ρ fluid density; C_D drag coefficient; A frontal area; μ viscosity

Drag goes as v^2, drag power as v^3.

P.22 (2/2)

1 $\dot{Q} = hA\,(T_s - T_\infty)$

2 $T(t) = T_\infty + (T_0 - T_\infty)\, e^{-t/\tau}$

3 $\tau = \frac{\rho V c}{hA} = \frac{mc}{hA}$

TERMS h convection coefficient; T_∞ surroundings; T_0 initial; τ time constant

"Hot things cool fastest when they are hottest": the rate is proportional to the temperature gap.

P.25

1 $\dot{Q} = \frac{\Delta T}{R_{\text{total}}}, \quad R_{\text{series}} = R_1 + R_2 + \cdots$

TERMS k conductivity; A area; L thickness; R thermal resistance

Heat flows like current: ΔT is the "voltage," R the "resistor" P.33 §12 Physics.

P.24 (2/2)

1 $q = -k \frac{dT}{dx}$

2 $\dot{Q} = \frac{kA\,\Delta T}{L}$

3 $R_{\text{cond}} = \frac{L}{kA}$

TERMS k conductivity; A area; L thickness; R thermal resistance

Heat flows like current: ΔT is the "voltage," R the "resistor" P.33 §12 Physics.

P.24 (1/2)

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Biot Number & Lumped Capacitance

PART 1 OF 2

1. Biot number
2. Characteristic length
3. Lumped-model criterion

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Biot Number & Lumped Capacitance

PART 2 OF 2

1. Fourier number

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Natural Convection: Nu, Pr, Gr & Ra

PART 1 OF 2

1. Nusselt number
2. Prandtl number
3. Grashof number

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Natural Convection: Nu, Pr, Gr & Ra

PART 2 OF 2

1. Rayleigh number
2. Onset of convection
3. Empirical correlation

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Thermal Radiation

PART 1 OF 2

1. Stefan–Boltzmann law
2. Net radiative exchange
3. Wien's displacement law

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Thermal Radiation

PART 2 OF 2

1. Stefan–Boltzmann constant

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Ideal Gas Law & Thermodynamics

PART 1 OF 3

1. Ideal gas law
2. First law of thermodynamics
3. Work by a gas

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Ideal Gas Law & Thermodynamics

PART 2 OF 3

1. Isothermal work
2. Molecular kinetic energy
3. Carnot efficiency

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Ideal Gas Law & Thermodynamics

PART 3 OF 3

1. Second law (entropy)

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Entropy & the Second Law

PART 1 OF 2

1. Clausius entropy change
2. Second law
3. Boltzmann entropy

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Entropy & the Second Law

PART 2 OF 2

1. Heat-engine efficiency
2. Refrigerator COP
3. Heat-pump COP

STATE EACH NUMBERED FORMULA

Diffusion & the Heat Equation

PART 1 OF 2

1. Fick's first law
2. Fick's second law (diffusion equation)
3. Heat equation

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1 $$\mathrm{Nu} = \frac{hL}{k_{\text{fluid}}}$$
2 $$\mathrm{Pr} = \frac{\nu}{\alpha}$$
3 $$\mathrm{Gr} = \frac{g\beta\,\Delta T\,L^3}{\nu^2}$$

TERMS h convection coeff.; k_f fluid conductivity; β expansion coeff.; ν, α diffusivities

Ra is the "Reynolds number of buoyancy": above ≈ 1708, a heated layer starts to roll into convection cells.

1 $$\mathrm{Fo} = \frac{\alpha t}{L_c^2}$$

TERMS h convection coeff.; k solid conductivity; $L_c = V/A_s$; α diffusivity

"Biot small, same temperature all" — below 0.1, the inside keeps up with the surface.

1 $$\mathrm{Bi} = \frac{hL_c}{k_{\text{solid}}} = \frac{\text{internal conduction resistance}}{\text{surface convection resistance}}$$
2 $$L_c = \frac{V}{A_s}$$
3 $\mathrm{Bi} < 0.1 \Rightarrow$ lumped model valid

TERMS h convection coeff.; k solid conductivity; $L_c = V/A_s$; α diffusivity

"Biot small, same temperature all" — below 0.1, the inside keeps up with the surface.

1 $$\sigma = 5.670 \times 10^{-8}\ \frac{\mathrm{W}}{\mathrm{m^2K^4}}$$

TERMS ε emissivity; σ Stefan–Boltzmann const.; T in kelvin; A area

"Fourth power": double the absolute temperature, sixteen times the radiation. Always use kelvin.

1 $$P = \varepsilon\sigma A T^4$$
2 $$P_{\text{net}} = \varepsilon\sigma A\left(T^4 - T_{\text{surr}}^4\right)$$
3 $$\lambda_{\max} T = 2.898 \times 10^{-3}\ \mathrm{m \cdot K}$$

TERMS ε emissivity; σ Stefan–Boltzmann const.; T in kelvin; A area

"Fourth power": double the absolute temperature, sixteen times the radiation. Always use kelvin.

1 $$\mathrm{Ra} = \mathrm{Gr} \cdot \mathrm{Pr} = \frac{g\beta\,\Delta T\,L^3}{\nu\,\alpha}$$
2 $$\mathrm{Ra}_c \approx 1708$$
3 $$\mathrm{Nu} = C\,\mathrm{Ra}^n$$

TERMS h convection coeff.; k_f fluid conductivity; β expansion coeff.; ν, α diffusivities

Ra is the "Reynolds number of buoyancy": above ≈ 1708, a heated layer starts to roll into convection cells.

1 $\Delta S \geq 0$ (isolated system)

TERMS n moles; $R = 8.314$ J/(mol·K); T in K; Q heat in; W work by gas

"You can't win, you can't break even, you can't get out of the game" (first, second, third laws).

1 $$W = nRT \ln \frac{V_2}{V_1}$$
2 $$\langle KE \rangle = \tfrac{3}{2} k_B T$$
3 $$\eta_{\text{Carnot}} = 1 - \frac{T_C}{T_H}$$

TERMS n moles; $R = 8.314$ J/(mol·K); T in K; Q heat in; W work by gas

"You can't win, you can't break even, you can't get out of the game" (first, second, third laws).

1 $$PV = nRT$$
2 $$\Delta U = Q - W$$
3 $$W = \int P\,dV$$

TERMS n moles; $R = 8.314$ J/(mol·K); T in K; Q heat in; W work by gas

"You can't win, you can't break even, you can't get out of the game" (first, second, third laws).

1 $$J = -D\frac{dC}{dx}$$
2 $$\frac{\partial C}{\partial t} = D\frac{\partial^2 C}{\partial x^2}$$
3 $$\frac{\partial T}{\partial t} = \alpha\frac{\partial^2 T}{\partial x^2}$$

TERMS D diffusion coeff. ($\mathrm{m^2/s}$); C concentration; α thermal diffusivity

Diffusion distance grows like $\sqrt{t}$: to go twice as far takes four times as long.

1 $$\eta = \frac{W}{Q_H} = 1 - \frac{Q_C}{Q_H} \leq 1 - \frac{T_C}{T_H}$$
2 $$\mathrm{COP}_R = \frac{Q_C}{W} \leq \frac{T_C}{T_H - T_C}$$
3 $$\mathrm{COP}_{HP} = \frac{Q_H}{W} \leq \frac{T_H}{T_H - T_C}$$

TERMS S entropy; Ω microstates; Q_H, Q_C hot/cold heat; T_H, T_C in K

Heat flows downhill in T because that is the direction that raises total entropy: $\frac{Q}{T_C} > \frac{Q}{T_H}$.

1 $$\Delta S = \int \frac{dQ_{\text{rev}}}{T} \quad \left(= \frac{Q}{T} \text{ at constant } T\right)$$
2 $$\Delta S_{\text{total}} \geq 0$$
3 $$S = k_B \ln \Omega$$

TERMS S entropy; Ω microstates; Q_H, Q_C hot/cold heat; T_H, T_C in K

Heat flows downhill in T because that is the direction that raises total entropy: $\frac{Q}{T_C} > \frac{Q}{T_H}$.

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Diffusion & the Heat Equation

PART 2 OF 2

1. Diffusion length

STATE EACH NUMBERED FORMULA

Coulomb's Law, Electric Field & Potential

PART 1 OF 3

1. Coulomb's law
2. Electric field
3. Field of a point charge

STATE EACH NUMBERED FORMULA

Coulomb's Law, Electric Field & Potential

PART 2 OF 3

1. Potential of a point charge
2. Electric potential energy
3. Uniform field

STATE EACH NUMBERED FORMULA

Coulomb's Law, Electric Field & Potential

PART 3 OF 3

1. Superposition
2. Lorentz force

STATE EACH NUMBERED FORMULA

Electric Circuits: Ohm's Law & Power

PART 1 OF 2

1. Ohm's law
2. Electric power
3. Resistivity

STATE EACH NUMBERED FORMULA

Electric Circuits: Ohm's Law & Power

PART 2 OF 2

1. Series resistors
2. Parallel resistors
3. Electrical energy

STATE EACH NUMBERED FORMULA

Reflection, Refraction & Total Internal Reflection

PART 1 OF 2

1. Law of reflection
2. Index of refraction
3. Snell's law

STATE EACH NUMBERED FORMULA

Reflection, Refraction & Total Internal Reflection

PART 2 OF 2

1. Critical angle
2. Wavelength in a medium

STATE EACH NUMBERED FORMULA

Thin Lenses & Mirrors

PART 1 OF 2

1. Thin lens / mirror equation
2. Magnification
3. Spherical mirror focal length

STATE EACH NUMBERED FORMULA

Thin Lenses & Mirrors

PART 2 OF 2

1. Lensmaker's equation
2. Optical power

STATE EACH NUMBERED FORMULA

Special Relativity Basics

PART 1 OF 3

1. Lorentz factor
2. Time dilation
3. Length contraction

STATE EACH NUMBERED FORMULA

Special Relativity Basics

PART 2 OF 3

1. Mass–energy equivalence
2. Relativistic energy & momentum
3. Energy–momentum relation

STATE EACH NUMBERED FORMULA

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1 $V = \dfrac{kq}{r}$

2 $U = qV, \quad U_{12} = \dfrac{kq_1q_2}{r}$

3 $E = \dfrac{\Delta V}{d}$

TERMS q charge (C); k Coulomb constant; r distance; E field (N/C); V potential (V)

Field is force per charge; potential is energy per charge. "Field $\sim 1/r^2$, potential $\sim 1/r$."

1 $F = k\dfrac{|q_1q_2|}{r^2}, \quad k = \dfrac{1}{4\pi\varepsilon_0}$

2 $\mathbf{E} = \dfrac{\mathbf{F}}{q}$

3 $E = \dfrac{kq}{r^2}$

TERMS q charge (C); k Coulomb constant; r distance; E field (N/C); V potential (V)

Field is force per charge; potential is energy per charge. "Field $\sim 1/r^2$, potential $\sim 1/r$."

1 $L_{\text{diff}} \approx \sqrt{2Dt}$

TERMS D diffusion coeff. (m^2/s); C concentration; α thermal diffusivity

Diffusion distance grows like $\sqrt{t}$: to go twice as far takes four times as long.

1 $R_{\text{series}} = R_1 + R_2 + \cdots$

2 $\dfrac{1}{R_{\text{parallel}}} = \dfrac{1}{R_1} + \dfrac{1}{R_2} + \cdots$

3 $E = Pt$

TERMS V voltage; I current; R resistance; ρ_e resistivity; P power

"VIR" triangle: cover the one you want — $V = IR$, $I = V/R$, $R = V/I$.

1 $V = IR$

2 $P = IV = I^2R = \dfrac{V^2}{R}$

3 $R = \dfrac{\rho_e L}{A}$

TERMS V voltage; I current; R resistance; ρ_e resistivity; P power

"VIR" triangle: cover the one you want — $V = IR$, $I = V/R$, $R = V/I$.

1 $\mathbf{E} = \sum_i \mathbf{E}_i$

2 $\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$

TERMS q charge (C); k Coulomb constant; r distance; E field (N/C); V potential (V)

Field is force per charge; potential is energy per charge. "Field $\sim 1/r^2$, potential $\sim 1/r$."

1 $\dfrac{1}{f} = \dfrac{1}{d_o} + \dfrac{1}{d_i}$

2 $m = \dfrac{h_i}{h_o} = -\dfrac{d_i}{d_o}$

3 $f = \dfrac{R}{2}$

TERMS f focal length; d_o, d_i object, image distances; m magnification

Object inside $f \rightarrow$ magnifying glass: virtual, upright, bigger.

1 $\sin\theta_c = \dfrac{n_2}{n_1} \quad (n_1 > n_2)$

2 $\lambda_n = \dfrac{\lambda_0}{n}$

TERMS n refractive index; θ from the normal; θ_c critical angle; c light speed

"Slow down, bend toward the normal" (like a car whose wheels hit sand at an angle).

1 $\theta_r = \theta_i$

2 $n = \dfrac{c}{v}$

3 $n_1 \sin\theta_1 = n_2 \sin\theta_2$

TERMS n refractive index; θ from the normal; θ_c critical angle; c light speed

"Slow down, bend toward the normal" (like a car whose wheels hit sand at an angle).

1 $E_0 = mc^2$

2 $E = \gamma mc^2, \quad p = \gamma mv$

3 $E^2 = (pc)^2 + (mc^2)^2$

TERMS v speed; c speed of light; γ Lorentz factor; subscript 0 = rest frame

"Moving clocks run slow, moving rulers shrink."

1 $\gamma = \dfrac{1}{\sqrt{1 - v^2/c^2}}$

2 $\Delta t = \gamma\, \Delta t_0$

3 $L = \dfrac{L_0}{\gamma}$

TERMS v speed; c speed of light; γ Lorentz factor; subscript 0 = rest frame

"Moving clocks run slow, moving rulers shrink."

1 $\dfrac{1}{f} = (n-1)\left(\dfrac{1}{R_1} - \dfrac{1}{R_2}\right)$

2 $P = \dfrac{1}{f}, \quad P_{\text{total}} = P_1 + P_2$

TERMS f focal length; d_o, d_i object, image distances; m magnification

Object inside $f \rightarrow$ magnifying glass: virtual, upright, bigger.

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Special Relativity Basics

PART 3 OF 3

1. Velocity addition

STATE EACH NUMBERED FORMULA

Photons & Matter Waves

PART 1 OF 2

1. Photon energy
2. Photon momentum
3. de Broglie wavelength

STATE EACH NUMBERED FORMULA

Photons & Matter Waves

PART 2 OF 2

1. Photoelectric effect
2. Heisenberg uncertainty

STATE EACH NUMBERED FORMULA

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1 $KE_{\max} = hf - \phi$

2 $\Delta x \, \Delta p \geq \dfrac{\hbar}{2}$

TERMS h Planck's constant; f frequency; λ wavelength; ϕ work function

$hc \approx 1240$ eV·nm: a 500 nm photon carries $\frac{1240}{500} \approx 2.5$ eV.

1 $E = hf = \dfrac{hc}{\lambda}$

2 $p = \dfrac{h}{\lambda}$

3 $\lambda = \dfrac{h}{mv}$

TERMS h Planck's constant; f frequency; λ wavelength; ϕ work function

$hc \approx 1240$ eV·nm: a 500 nm photon carries $\frac{1240}{500} \approx 2.5$ eV.

1 $u = \dfrac{u' + v}{1 + u'v/c^2}$

TERMS v speed; c speed of light; γ Lorentz factor; subscript 0 = rest frame

"Moving clocks run slow, moving rulers shrink."

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S.1 (1/2) STATISTICS & PROBABILITY

Measures of Center

PART 1 OF 2

1. Sample mean
2. Population mean
3. Weighted mean

STATE EACH NUMBERED FORMULA

S.1 (2/2) STATISTICS & PROBABILITY

Measures of Center

PART 2 OF 2

1. Median
2. Mode
3. Trimmed mean

STATE EACH NUMBERED FORMULA

S.2 (1/2) STATISTICS & PROBABILITY

Variance & Standard Deviation

PART 1 OF 2

1. Population variance
2. Sample variance
3. Standard deviation

STATE EACH NUMBERED FORMULA

S.2 (2/2) STATISTICS & PROBABILITY

Variance & Standard Deviation

PART 2 OF 2

1. Computational formula
2. Coefficient of variation
3. Range

STATE EACH NUMBERED FORMULA

S.3 (1/2) STATISTICS & PROBABILITY

Quartiles, IQR & Outliers

PART 1 OF 2

1. Interquartile range
2. Tukey's fences
3. Five-number summary

STATE EACH NUMBERED FORMULA

S.3 (2/2) STATISTICS & PROBABILITY

Quartiles, IQR & Outliers

PART 2 OF 2

1. Percentile position
2. Percentile rank

STATE EACH NUMBERED FORMULA

S.4 (1/2) STATISTICS & PROBABILITY

z-Scores & Standardization

PART 1 OF 2

1. Population z-score
2. Sample z-score
3. Back-transform

STATE EACH NUMBERED FORMULA

S.4 (2/2) STATISTICS & PROBABILITY

z-Scores & Standardization

PART 2 OF 2

1. Standardized data
2. Min–max scaling

STATE EACH NUMBERED FORMULA

S.5 STATISTICS & PROBABILITY

Shape: Skewness & Kurtosis

1. Sample skewness
2. Excess kurtosis
3. Skew and the center

STATE EACH NUMBERED FORMULA

S.6 (1/2) STATISTICS & PROBABILITY

Sample Spaces, Events & the Axioms

PART 1 OF 2

1. Sample space and event
2. Axiom 1: non-negativity
3. Axiom 2: normalization

STATE EACH NUMBERED FORMULA

S.6 (2/2) STATISTICS & PROBABILITY

Sample Spaces, Events & the Axioms

PART 2 OF 2

1. Axiom 3: additivity
2. Classical probability
3. Frequentist probability

STATE EACH NUMBERED FORMULA

S.7 (1/2) STATISTICS & PROBABILITY

Basic Probability Rules

PART 1 OF 2

1. Bounds
2. Complement rule
3. Addition rule

STATE EACH NUMBERED FORMULA

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1 $\sigma^2 = \dfrac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2$

2 $s^2 = \dfrac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$

3 $\sigma = \sqrt{\sigma^2}, \qquad s = \sqrt{s^2}$

TERMS $\mu, \bar{x}$ means; N, n sizes; $n-1$ Bessel's correction

Divide by $n-1$ for a Sample ("s for sample, subtract one").

1 $\tilde{x} = \begin{cases} x_{\left(\frac{n+1}{2}\right)}, & n \text{ odd} \\ \frac{1}{2}\left(x_{(n/2)} + x_{(n/2+1)}\right), & n \text{ even} \end{cases}$

2 $\text{mode} = \arg\max_x \ \text{freq}(x)$

3 $\bar{x}_{\text{tr}} = \dfrac{1}{n-2g}\sum_{i=g+1}^{n-g} x_{(i)}$

TERMS n, N sizes; w_i weights; $x_{(k)}$ kth sorted value; g trimmed per end

Mean = balance point; median = halfway point; mode = most popular.

1 $\bar{x} = \dfrac{1}{n}\sum_{i=1}^{n} x_i$

2 $\mu = \dfrac{1}{N}\sum_{i=1}^{N} x_i$

3 $\bar{x}_w = \dfrac{\sum_i w_i x_i}{\sum_i w_i}$

TERMS n, N sizes; w_i weights; $x_{(k)}$ kth sorted value; g trimmed per end

Mean = balance point; median = halfway point; mode = most popular.

1 $L = \frac{p}{100}(n-1) + 1$

2 $\text{PR}(x) = \dfrac{\#\{x_i \le x\}}{n} \times 100\%$

TERMS Q_1, Q_3 quartiles; $\tilde{x}$ median; p percentile; n sample size

The box holds the middle half; whiskers reach to 1.5 boxes.

1 $\text{IQR} = Q_3 - Q_1$

2 $x < Q_1 - 1.5\,\text{IQR} \ \text{ or } \ x > Q_3 + 1.5\,\text{IQR}$

3 $\min,\ Q_1,\ \tilde{x},\ Q_3,\ \max$

TERMS Q_1, Q_3 quartiles; $\tilde{x}$ median; p percentile; n sample size

The box holds the middle half; whiskers reach to 1.5 boxes.

1 $\sigma^2 = \dfrac{1}{N}\sum_{i=1}^{N} x_i^2 - \mu^2$

2 $CV = \dfrac{s}{\bar{x}}$

3 $R = x_{\max} - x_{\min}$

TERMS $\mu, \bar{x}$ means; N, n sizes; $n-1$ Bessel's correction

Divide by $n-1$ for a Sample ("s for sample, subtract one").

1 $g_1 = \dfrac{\frac{1}{n}\sum(x_i - \bar{x})^3}{\left(\frac{1}{n}\sum(x_i - \bar{x})^2\right)^{3/2}}$

2 $g_2 = \dfrac{\frac{1}{n}\sum(x_i - \bar{x})^4}{\left(\frac{1}{n}\sum(x_i - \bar{x})^2\right)^{2}} - 3$

3 right skew: $\text{mode} < \tilde{x} < \bar{x}$

TERMS $\bar{x}$ mean; $\tilde{x}$ median; n sample size

Skew points to the tail: the tail is where the mean goes.

1 $\bar{z} = 0, \qquad s_z = 1$

2 $x' = \dfrac{x - x_{\min}}{x_{\max} - x_{\min}}$

TERMS μ, σ population mean, SD; $\bar{x}, s$ sample mean, SD

"How many SDs from the mean?"

1 $z = \dfrac{x - \mu}{\sigma}$

2 $z = \dfrac{x - \bar{x}}{s}$

3 $x = \mu + z\,\sigma$

TERMS μ, σ population mean, SD; $\bar{x}, s$ sample mean, SD

"How many SDs from the mean?"

1 $0 \le P(A) \le 1$

2 $P(A^c) = 1 - P(A)$

3 $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

TERMS A^c not A; $\cup$ or; $\cap$ and; p_i event probabilities

"Or" adds (minus the overlap); "at least one" = 1− "none."

1 $P\left(\bigcup_i A_i\right) = \sum_i P(A_i) \ \ (A_i \text{ disjoint})$

2 $P(A) = \dfrac{|A|}{|S|}$

3 $P(A) = \lim_{n\to\infty} \dfrac{n_A}{n}$

TERMS S sample space; A event; $|A|$ count; n_A times A occurred

Kolmogorov's three: never negative, total one, disjoint pieces add.

1 $S = \{\text{all outcomes}\}, \qquad A \subseteq S$

2 $P(A) \ge 0$

3 $P(S) = 1$

TERMS S sample space; A event; $|A|$ count; n_A times A occurred

Kolmogorov's three: never negative, total one, disjoint pieces add.

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S.7 (2/2) STATISTICS & PROBABILITY

Basic Probability Rules

PART 2 OF 2

1. Disjoint events
2. At least one
3. Three-event inclusion–exclusion

STATE EACH NUMBERED FORMULA

S.8 (1/3) STATISTICS & PROBABILITY

Counting: Permutations & Combinations

PART 1 OF 3

1. Multiplication principle
2. Factorial
3. Permutations

STATE EACH NUMBERED FORMULA

S.8 (2/3) STATISTICS & PROBABILITY

Counting: Permutations & Combinations

PART 2 OF 3

1. Combinations
2. With repetition, ordered
3. With repetition, unordered

STATE EACH NUMBERED FORMULA

S.8 (3/3) STATISTICS & PROBABILITY

Counting: Permutations & Combinations

PART 3 OF 3

1. Arrangements with repeats

STATE EACH NUMBERED FORMULA

S.9 (1/2) STATISTICS & PROBABILITY

Conditional Probability & Independence

PART 1 OF 2

1. Conditional probability
2. Multiplication rule
3. Chain rule

STATE EACH NUMBERED FORMULA

S.9 (2/2) STATISTICS & PROBABILITY

Conditional Probability & Independence

PART 2 OF 2

1. Independence
2. Conditional independence
3. Disjoint $\Rightarrow$ dependent

STATE EACH NUMBERED FORMULA

S.10 STATISTICS & PROBABILITY

Law of Total Probability

1. Partition
2. Law of total probability
3. Two-case form

STATE EACH NUMBERED FORMULA

S.11 (1/2) STATISTICS & PROBABILITY

Bayes' Theorem

PART 1 OF 2

1. Bayes' theorem
2. Evidence (two cases)
3. Bayes over a partition

STATE EACH NUMBERED FORMULA

S.11 (2/2) STATISTICS & PROBABILITY

Bayes' Theorem

PART 2 OF 2

1. Odds form
2. Proportional form

STATE EACH NUMBERED FORMULA

S.12 (1/2) STATISTICS & PROBABILITY

Random Variables, PMFs, PDFs & CDFs

PART 1 OF 2

1. PMF (discrete)
2. PDF (continuous)
3. CDF

STATE EACH NUMBERED FORMULA

S.12 (2/2) STATISTICS & PROBABILITY

Random Variables, PMFs, PDFs & CDFs

PART 2 OF 2

1. Density from CDF
2. Interval from CDF
3. Quantile function

STATE EACH NUMBERED FORMULA

S.13 (1/3) STATISTICS & PROBABILITY

Expected Value & Variance of a Random Variable

PART 1 OF 3

1. Expected value (discrete)
2. Expected value (continuous)
3. LOTUS

STATE EACH NUMBERED FORMULA

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1. $\binom{n}{r} = {}_nC_r = \dfrac{n!}{r!\,(n-r)!}$
2. n^r
3. $\binom{n+r-1}{r}$

TERMS n items; r chosen; n_j identical copies of type j

Permutation: Position matters. Combination: Committee (order doesn't matter).

S.8 (2/3)

1. m ways then n ways $= m \cdot n$ ways
2. $n! = n(n-1)(n-2)\cdots 1, \quad 0! = 1$
3. ${}_nP_r = \dfrac{n!}{(n-r)!}$

TERMS n items; r chosen; n_j identical copies of type j

Permutation: Position matters. Combination: Committee (order doesn't matter).

S.8 (1/3)

1. $P(A \cup B) = P(A) + P(B)$
2. $P(\text{at least one}) = 1 - \prod_{i=1}^{k}(1-p_i)$
3. $P(A \cup B \cup C) = \sum P(A) - \sum P(A \cap B) + P(A \cap B \cap C)$

TERMS A^c not A; $\cup$ or; $\cap$ and; p_i event probabilities

"Or" adds (minus the overlap); "at least one" = 1− "none."

S.7 (2/2)

1. $P(A \cap B) = P(A)\,P(B) \iff P(A \mid B) = P(A)$
2. $P(A \cap B \mid C) = P(A \mid C)\,P(B \mid C)$
3. $A \cap B = \varnothing,\ P(A), P(B) > 0 \Rightarrow P(A \mid B) = 0 \neq P(A)$

TERMS $P(A \mid B)$ A given B; $\cap$ and; C conditioning event

"Given" shrinks the sample space to B. Disjoint means "can't both happen"; independent means "doesn't matter if the other happened".

S.9 (2/2)

1. $P(A \mid B) = \dfrac{P(A \cap B)}{P(B)}$
2. $P(A \cap B) = P(A \mid B)\,P(B)$
3. $P(A \cap B \cap C) = P(A)\,P(B \mid A)\,P(C \mid A \cap B)$

TERMS $P(A \mid B)$ A given B; $\cap$ and; C conditioning event

"Given" shrinks the sample space to B. Disjoint means "can't both happen"; independent means "doesn't matter if the other happened".

S.9 (1/2)

1. $\dfrac{n!}{n_1!\,n_2!\cdots n_k!}$

TERMS n items; r chosen; n_j identical copies of type j

Permutation: Position matters. Combination: Committee (order doesn't matter).

S.8 (3/3)

1. $\dfrac{P(A \mid B)}{P(A^c \mid B)} = \dfrac{P(B \mid A)}{P(B \mid A^c)} \cdot \dfrac{P(A)}{P(A^c)}$
2. $P(A \mid B) \propto P(B \mid A)\,P(A)$

TERMS $P(A)$ prior; $P(B \mid A)$ likelihood; $P(B)$ evidence; A^c not A

Posterior $\propto$ likelihood $\times$ prior.

S.11 (2/2)

1. $P(A \mid B) = \dfrac{P(B \mid A)\,P(A)}{P(B)}$
2. $P(B) = P(B \mid A)P(A) + P(B \mid A^c)P(A^c)$
3. $P(A_j \mid B) = \dfrac{P(B \mid A_j)P(A_j)}{\sum_i P(B \mid A_i)P(A_i)}$

TERMS $P(A)$ prior; $P(B \mid A)$ likelihood; $P(B)$ evidence; A^c not A

Posterior $\propto$ likelihood $\times$ prior.

S.11 (1/2)

1. $A_i \cap A_j = \varnothing\ (i \neq j), \quad \bigcup_i A_i = S$
2. $P(B) = \sum_i P(B \mid A_i)\,P(A_i)$
3. $P(B) = P(B \mid A)P(A) + P(B \mid A^c)P(A^c)$

TERMS A_i partition of S; B event of interest; A^c not A

"Divide and conquer": sum over cases of (chance of case) $\times$ (chance of B in that case).

S.10

1. $E[X] = \mu = \sum_x x\,p(x)$
2. $E[X] = \int_{-\infty}^{\infty} x\,f(x)\,dx$
3. $E[g(X)] = \sum_x g(x)\,p(x)$

TERMS X, Y random variables; p, f PMF, PDF; a, b constants; g function

Adding a constant shifts the mean but not the spread; scaling by a scales variance by a^2.

S.13 (1/3)

1. $f(x) = F'(x)$
2. $P(a < X \le b) = F(b) - F(a)$
3. $Q(p) = F^{-1}(p) = \min\{x : F(x) \ge p\}$

TERMS X random variable; p PMF; f PDF; F CDF; Q quantile

PMF = point masses; PDF = area; CDF = "so far" total; quantile = CDF read backwards.

S.12 (2/2)

1. $p(x) = P(X = x), \qquad \sum_x p(x) = 1$
2. $P(a \le X \le b) = \int_a^b f(x)\,dx, \quad \int_{-\infty}^{\infty} f = 1$
3. $F(x) = P(X \le x)$

TERMS X random variable; p PMF; f PDF; F CDF; Q quantile

PMF = point masses; PDF = area; CDF = "so far" total; quantile = CDF read backwards.

S.12 (1/2)

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S.13 (2/3) STATISTICS & PROBABILITY

Expected Value & Variance of a Random Variable

PART 2 OF 3

1. Variance
2. Linear transformation
3. Linearity of expectation

STATE EACH NUMBERED FORMULA

S.13 (3/3) STATISTICS & PROBABILITY

Expected Value & Variance of a Random Variable

PART 3 OF 3

1. Variance of a sum
2. Product of independents

STATE EACH NUMBERED FORMULA

S.14 (1/3) STATISTICS & PROBABILITY

Joint, Marginal & Conditional Distributions

PART 1 OF 3

1. Joint PMF
2. Marginal
3. Conditional

STATE EACH NUMBERED FORMULA

S.14 (2/3) STATISTICS & PROBABILITY

Joint, Marginal & Conditional Distributions

PART 2 OF 3

1. Independence of variables
2. Conditional expectation
3. Law of total expectation

STATE EACH NUMBERED FORMULA

S.14 (3/3) STATISTICS & PROBABILITY

Joint, Marginal & Conditional Distributions

PART 3 OF 3

1. Law of total variance

STATE EACH NUMBERED FORMULA

S.15 (1/2) STATISTICS & PROBABILITY

Bernoulli, Geometric & Negative Binomial

PART 1 OF 2

1. Bernoulli
2. Geometric PMF
3. Geometric mean, variance

STATE EACH NUMBERED FORMULA

S.15 (2/2) STATISTICS & PROBABILITY

Bernoulli, Geometric & Negative Binomial

PART 2 OF 2

1. Geometric tail
2. Negative binomial PMF
3. Negative binomial mean, variance

STATE EACH NUMBERED FORMULA

S.16 (1/2) STATISTICS & PROBABILITY

Binomial Distribution

PART 1 OF 2

1. Binomial PMF
2. Mean
3. Variance

STATE EACH NUMBERED FORMULA

S.16 (2/2) STATISTICS & PROBABILITY

Binomial Distribution

PART 2 OF 2

1. Sum of Bernoullis
2. CDF
3. Normal approximation

STATE EACH NUMBERED FORMULA

S.17 (1/2) STATISTICS & PROBABILITY

Hypergeometric & Discrete Uniform

PART 1 OF 2

1. Hypergeometric PMF
2. Hypergeometric mean
3. Hypergeometric variance

STATE EACH NUMBERED FORMULA

S.17 (2/2) STATISTICS & PROBABILITY

Hypergeometric & Discrete Uniform

PART 2 OF 2

1. Discrete uniform PMF
2. Discrete uniform mean, variance

STATE EACH NUMBERED FORMULA

S.18 (1/2) STATISTICS & PROBABILITY

Poisson Distribution

PART 1 OF 2

1. Poisson PMF
2. Mean and variance
3. Rate scaling

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1 $p(x,y) = P(X = x,\ Y = y)$

2 $p_X(x) = \sum_y p(x,y), \qquad f_X(x) = \int f(x,y)\,dy$

3 $p(y \mid x) = \dfrac{p(x,y)}{p_X(x)}$

TERMS $p(x,y)$ joint; p_X marginal; $p(y \mid x)$ conditional

Joint = the whole table; marginal = the margins; conditional = one row, rescaled to sum to 1.

1 $\mathrm{Var}(X+Y) = \mathrm{Var}\,X + \mathrm{Var}\,Y + 2\,\mathrm{Cov}(X,Y)$

2 $E[XY] = E[X]\,E[Y]$

TERMS X, Y random variables; p, f PMF, PDF; a, b constants; g function

Adding a constant shifts the mean but not the spread; scaling by a scales variance by a^2.

1 $\mathrm{Var}(X) = E\big[(X-\mu)^2\big] = E[X^2] - \big(E[X]\big)^2$

2 $E[aX+b] = aE[X]+b, \quad \mathrm{Var}(aX+b) = a^2\,\mathrm{Var}(X)$

3 $E[X+Y] = E[X] + E[Y]$

TERMS X, Y random variables; p, f PMF, PDF; a, b constants; g function

Adding a constant shifts the mean but not the spread; scaling by a scales variance by a^2.

1 $P(X=1) = p,\ P(X=0) = 1-p; \quad \mu = p,\ \sigma^2 = p(1-p)$

2 $P(X=k) = (1-p)^{k-1}p, \quad k = 1,2,\dots$

3 $\mu = \dfrac{1}{p}, \qquad \sigma^2 = \dfrac{1-p}{p^2}$

TERMS p success probability; k trial number; r successes needed

Geometric: "how long until the first win?" $\Rightarrow 1/p$ tries on average.

1 $\mathrm{Var}(Y) = E\big[\mathrm{Var}(Y \mid X)\big] + \mathrm{Var}\big(E[Y \mid X]\big)$

TERMS $p(x,y)$ joint; p_X marginal; $p(y \mid x)$ conditional

Joint = the whole table; marginal = the margins; conditional = one row, rescaled to sum to 1.

1 $p(x,y) = p_X(x)\,p_Y(y) \ \text{ for all } x, y$

2 $E[Y \mid X = x] = \sum_y y\,p(y \mid x)$

3 $E[Y] = E\big[E[Y \mid X]\big]$

TERMS $p(x,y)$ joint; p_X marginal; $p(y \mid x)$ conditional

Joint = the whole table; marginal = the margins; conditional = one row, rescaled to sum to 1.

1 $X = B_1 + B_2 + \cdots + B_n$

2 $P(X \le k) = \sum_{j=0}^{k} \binom{n}{j} p^j (1-p)^{n-j}$

3 $X \approx N\big(np,\ np(1-p)\big)$

TERMS n trials; k successes; p success probability

BINS conditions: Binary outcomes, Independent trials, fixed Number, Same p.

1 $P(X=k) = \binom{n}{k} p^k (1-p)^{n-k}, \ \ k = 0,\dots,n$

2 $\mu = np$

3 $\sigma^2 = np(1-p)$

TERMS n trials; k successes; p success probability

BINS conditions: Binary outcomes, Independent trials, fixed Number, Same p.

1 $P(X > k) = (1-p)^k$

2 $P(X=k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}, \ k \ge r$

3 $\mu = \dfrac{r}{p}, \qquad \sigma^2 = \dfrac{r(1-p)}{p^2}$

TERMS p success probability; k trial number; r successes needed

Geometric: "how long until the first win?" $\Rightarrow 1/p$ tries on average.

1 $P(X=k) = \dfrac{\lambda^k e^{-\lambda}}{k!}, \ \ k = 0,1,2,\dots$

2 $\mu = \sigma^2 = \lambda$

3 $\lambda = r\,t$

TERMS λ expected count; k observed count; r rate; t interval

Poisson: mean = variance = λ.

1 $P(X=x) = \dfrac{1}{m}, \ \ x = a,\dots,b, \ m = b-a+1$

2 $\mu = \dfrac{a+b}{2}, \qquad \sigma^2 = \dfrac{m^2-1}{12}$

TERMS N population; K successes in it; n sample; k successes in sample

Hypergeometric = "draws without putting back".

1 $P(X=k) = \dfrac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}$

2 $\mu = n\,\dfrac{K}{N}$

3 $\sigma^2 = n\,\dfrac{K}{N}\left(1 - \dfrac{K}{N}\right)\dfrac{N-n}{N-1}$

TERMS N population; K successes in it; n sample; k successes in sample

Hypergeometric = "draws without putting back".

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S.18 (2/2) STATISTICS & PROBABILITY

Poisson Distribution

PART 2 OF 2

1. Sum of Poissons
2. Binomial limit

STATE EACH NUMBERED FORMULA

S.19 (1/3) STATISTICS & PROBABILITY

Continuous Distributions: Uniform & Exponential

PART 1 OF 3

1. Uniform PDF
2. Uniform CDF
3. Uniform mean, variance

STATE EACH NUMBERED FORMULA

S.19 (2/3) STATISTICS & PROBABILITY

Continuous Distributions: Uniform & Exponential

PART 2 OF 3

1. Exponential PDF
2. Exponential CDF
3. Exponential mean, variance

STATE EACH NUMBERED FORMULA

S.19 (3/3) STATISTICS & PROBABILITY

Continuous Distributions: Uniform & Exponential

PART 3 OF 3

1. Memoryless property

STATE EACH NUMBERED FORMULA

S.20 (1/3) STATISTICS & PROBABILITY

Normal Distribution & the CDF

PART 1 OF 3

1. Normal PDF
2. Notation
3. Normal CDF

STATE EACH NUMBERED FORMULA

S.20 (2/3) STATISTICS & PROBABILITY

Normal Distribution & the CDF

PART 2 OF 3

1. Standardizing
2. Interval probability
3. Symmetry

STATE EACH NUMBERED FORMULA

S.20 (3/3) STATISTICS & PROBABILITY

Normal Distribution & the CDF

PART 3 OF 3

1. Empirical rule
2. Sum of independent normals

STATE EACH NUMBERED FORMULA

S.21 (1/3) STATISTICS & PROBABILITY

Degrees of Freedom

PART 1 OF 3

1. General rule
2. Sample variance
3. One-sample t

STATE EACH NUMBERED FORMULA

S.21 (2/3) STATISTICS & PROBABILITY

Degrees of Freedom

PART 2 OF 3

1. Two-sample t (pooled)
2. χ^2 goodness of fit
3. χ^2 independence

STATE EACH NUMBERED FORMULA

S.21 (3/3) STATISTICS & PROBABILITY

Degrees of Freedom

PART 3 OF 3

1. Regression residuals
2. ANOVA

STATE EACH NUMBERED FORMULA

S.22 (1/2) STATISTICS & PROBABILITY

Sampling Distributions: χ^2, t & F

PART 1 OF 2

1. Chi-square
2. Sample variance
3. Student's t

STATE EACH NUMBERED FORMULA

S.22 (2/2) STATISTICS & PROBABILITY

Sampling Distributions: χ^2, t & F

PART 2 OF 2

1. t-statistic
2. F distribution
3. Relationships

STATE EACH NUMBERED FORMULA

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1 $f(x) = \lambda e^{-\lambda x}, \quad x \geq 0$

2 $F(x) = 1 - e^{-\lambda x}$

3 $\mu = \dfrac{1}{\lambda}, \qquad \sigma^2 = \dfrac{1}{\lambda^2}$

TERMS a, b endpoints; λ rate; s, t elapsed and extra time

Exponential: mean $= 1/\lambda$; "memoryless" — the past doesn't matter.

1 $f(x) = \dfrac{1}{b-a}, \quad a \leq x \leq b$

2 $F(x) = \dfrac{x-a}{b-a}$

3 $\mu = \dfrac{a+b}{2}, \qquad \sigma^2 = \dfrac{(b-a)^2}{12}$

TERMS a, b endpoints; λ rate; s, t elapsed and extra time

Exponential: mean $= 1/\lambda$; "memoryless" — the past doesn't matter.

1 $X_1 + X_2 \sim \text{Pois}(\lambda_1 + \lambda_2)$

2 $\binom{n}{k} p^k (1-p)^{n-k} \to \dfrac{\lambda^k e^{-\lambda}}{k!}$

TERMS λ expected count; k observed count; r rate; t interval

Poisson: mean $=$ variance $= \lambda$.

1 $Z = \dfrac{X - \mu}{\sigma} \sim N(0,1)$

2 $P(a \leq X \leq b) = \Phi(z_b) - \Phi(z_a)$

3 $\Phi(-z) = 1 - \Phi(z)$

TERMS μ mean; σ SD; Φ standard normal CDF; z standard score

68–95–99.7: within 1, 2, 3 SDs.

1 $f(x) = \dfrac{1}{\sigma\sqrt{2\pi}}\, e^{-\frac{(x-\mu)^2}{2\sigma^2}}$

2 $X \sim N(\mu, \sigma^2)$

3 $F(x) = P(X \leq x) = \displaystyle\int_{-\infty}^{x} f(t)\, dt$

TERMS μ mean; σ SD; Φ standard normal CDF; z standard score

68–95–99.7: within 1, 2, 3 SDs.

1 $P(X > s + t \mid X > s) = P(X > t)$

TERMS a, b endpoints; λ rate; s, t elapsed and extra time

Exponential: mean $= 1/\lambda$; "memoryless" — the past doesn't matter.

1 $df = n_1 + n_2 - 2$

2 $df = k - 1 - m$

3 $df = (r-1)(c-1)$

TERMS n observations; k categories; r, c table size; p predictors; m fitted parameters

"One estimate, one freedom lost."

1 $df = n -$ (number of estimated parameters / constraints)

2 $s^2 = \dfrac{1}{n-1} \displaystyle\sum_{i=1}^{n} (x_i - \bar{x})^2$

3 $df = n - 1$

TERMS n observations; k categories; r, c table size; p predictors; m fitted parameters

"One estimate, one freedom lost."

1 $P(|Z| \leq 1, 2, 3) \approx 0.68,\ 0.95,\ 0.997$

2 $X + Y \sim N(\mu_X + \mu_Y,\ \sigma_X^2 + \sigma_Y^2)$

TERMS μ mean; σ SD; Φ standard normal CDF; z standard score

68–95–99.7: within 1, 2, 3 SDs.

1 $\dfrac{\bar{X} - \mu}{s/\sqrt{n}} \sim t_{n-1}$

2 $F = \dfrac{V_1/d_1}{V_2/d_2} \sim F_{d_1,d_2}$

3 $t_k^2 = F_{1,k}, \qquad t_k \to N(0,1)\ (k \to \infty)$

TERMS Z standard normal; k, d_1, d_2 degrees of freedom; V chi-square

χ^2 = squares; t = mean with estimated SD; F = ratio of variances.

1 $Q = \displaystyle\sum_{i=1}^{k} Z_i^2 \sim \chi_k^2, \quad \mu = k,\ \sigma^2 = 2k$

2 $\dfrac{(n-1)s^2}{\sigma^2} \sim \chi_{n-1}^2$

3 $T = \dfrac{Z}{\sqrt{V/k}} \sim t_k$

TERMS Z standard normal; k, d_1, d_2 degrees of freedom; V chi-square

χ^2 = squares; t = mean with estimated SD; F = ratio of variances.

1 $df = n - p - 1$

2 $df_{\text{between}} = k - 1, \quad df_{\text{within}} = N - k$

TERMS n observations; k categories; r, c table size; p predictors; m fitted parameters

"One estimate, one freedom lost."

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S.23 (1/2) STATISTICS & PROBABILITY

Law of Large Numbers & Probability Inequalities

PART 1 OF 2

1. Mean of the sample mean
2. Variance of the sample mean
3. Weak law of large numbers

STATE EACH NUMBERED FORMULA

S.23 (2/2) STATISTICS & PROBABILITY

Law of Large Numbers & Probability Inequalities

PART 2 OF 2

1. Markov's inequality
2. Chebyshev's inequality
3. Monte Carlo estimate

STATE EACH NUMBERED FORMULA

S.24 (1/2) STATISTICS & PROBABILITY

Central Limit Theorem & Standard Error

PART 1 OF 2

1. Central limit theorem
2. Standardized mean
3. SE of a mean

STATE EACH NUMBERED FORMULA

S.24 (2/2) STATISTICS & PROBABILITY

Central Limit Theorem & Standard Error

PART 2 OF 2

1. SE of a proportion
2. SE of a difference of means
3. SE of a difference of proportions

STATE EACH NUMBERED FORMULA

S.25 (1/2) STATISTICS & PROBABILITY

Point Estimators: Bias, Variance & MSE

PART 1 OF 2

1. Bias
2. Mean squared error
3. Standard error

STATE EACH NUMBERED FORMULA

S.25 (2/2) STATISTICS & PROBABILITY

Point Estimators: Bias, Variance & MSE

PART 2 OF 2

1. Consistency
2. Cramér–Rao bound
3. Fisher information

STATE EACH NUMBERED FORMULA

S.26 (1/2) STATISTICS & PROBABILITY

Maximum Likelihood Estimation (MLE)

PART 1 OF 2

1. Likelihood
2. Log-likelihood
3. MLE condition

STATE EACH NUMBERED FORMULA

S.26 (2/2) STATISTICS & PROBABILITY

Maximum Likelihood Estimation (MLE)

PART 2 OF 2

1. Known MLEs
2. Normal variance MLE
3. Asymptotic normality

STATE EACH NUMBERED FORMULA

S.27 (1/2) STATISTICS & PROBABILITY

Confidence Intervals

PART 1 OF 2

1. z-interval for a mean
2. t-interval for a mean
3. Interval for a proportion

STATE EACH NUMBERED FORMULA

S.27 (2/2) STATISTICS & PROBABILITY

Confidence Intervals

PART 2 OF 2

1. Difference of two means
2. Sample size for a mean
3. Sample size for a proportion

STATE EACH NUMBERED FORMULA

S.28 (1/2) STATISTICS & PROBABILITY

Hypothesis Testing, p-Values & Errors

PART 1 OF 2

1. Hypotheses
2. Test statistic (z)
3. Two-sided p-value

STATE EACH NUMBERED FORMULA

S.28 (2/2) STATISTICS & PROBABILITY

Hypothesis Testing, p-Values & Errors

PART 2 OF 2

1. One-sided p-value
2. Decision rule
3. Error rates

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1. $\bar{X} \approx N\left(\mu, \dfrac{\sigma^2}{n}\right) \quad (n \text{ large})$
2. $z = \dfrac{\bar{x} - \mu}{\sigma/\sqrt{n}}$
3. $SE_{\bar{x}} = \dfrac{\sigma}{\sqrt{n}} \approx \dfrac{s}{\sqrt{n}}$

TERMS μ, σ population mean, SD; n sample size; $\hat{p}$ sample proportion

"Root-n law": four times the data halves the error.

1. $P(X \geq a) \leq \dfrac{E[X]}{a} \quad (X \geq 0)$
2. $P\big(|X - \mu| \geq k\sigma\big) \leq \dfrac{1}{k^2}$
3. $\dfrac{1}{n}\sum_{i=1}^{n} g(X_i) \to E[g(X)]$

TERMS $\bar{X}_n$ mean of n draws; ε tolerance; k number of SDs

LLN: the average converges. CLT S.24 §13 Statistics *: and here is the shape of its error.*

1. $E[\bar{X}] = \mu$
2. $\mathrm{Var}(\bar{X}) = \dfrac{\sigma^2}{n}$
3. $P\big(|\bar{X}_n - \mu| > \varepsilon\big) \to 0 \quad (n \to \infty)$

TERMS $\bar{X}_n$ mean of n draws; ε tolerance; k number of SDs

LLN: the average converges. CLT S.24 §13 Statistics *: and here is the shape of its error.*

1. $\hat{\theta}_n \to \theta \ \text{ as } n \to \infty$
2. $\mathrm{Var}(\hat{\theta}) \geq \dfrac{1}{n\, I(\theta)}$
3. $I(\theta) = E\left[\left(\frac{\partial}{\partial\theta} \ln f(X \mid \theta)\right)^2\right]$

TERMS θ true parameter; $\hat{\theta}$ estimator; $I(\theta)$ Fisher information

MSE = variance + bias²: the dartboard picture (spread vs off-center).

1. $\mathrm{Bias}(\hat{\theta}) = E[\hat{\theta}] - \theta$
2. $\mathrm{MSE}(\hat{\theta}) = E\big[(\hat{\theta} - \theta)^2\big] = \mathrm{Var}(\hat{\theta}) + \mathrm{Bias}^2$
3. $SE(\hat{\theta}) = \sqrt{\mathrm{Var}(\hat{\theta})}$

TERMS θ true parameter; $\hat{\theta}$ estimator; $I(\theta)$ Fisher information

MSE = variance + bias²: the dartboard picture (spread vs off-center).

1. $SE_{\hat{p}} = \sqrt{\dfrac{p(1-p)}{n}}$
2. $SE_{\bar{x}_1 - \bar{x}_2} = \sqrt{\dfrac{s_1^2}{n_1} + \dfrac{s_2^2}{n_2}}$
3. $SE_{\hat{p}_1 - \hat{p}_2} = \sqrt{\dfrac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \dfrac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$

TERMS μ, σ population mean, SD; n sample size; $\hat{p}$ sample proportion

"Root-n law": four times the data halves the error.

1. $\bar{x} \pm z^* \dfrac{\sigma}{\sqrt{n}}$
2. $\bar{x} \pm t^*_{n-1} \dfrac{s}{\sqrt{n}}$
3. $\hat{p} \pm z^* \sqrt{\dfrac{\hat{p}(1-\hat{p})}{n}}$

TERMS z^*, t^* critical values; E margin of error; $\hat{p}$ sample proportion

Estimate ± (critical value)×(standard error).

1. $\hat{p} = \bar{x}, \quad \hat{\lambda}_{\text{Pois}} = \bar{x}, \quad \hat{\lambda}_{\text{Exp}} = \dfrac{1}{\bar{x}}, \quad \hat{\mu} = \bar{x}$
2. $\hat{\sigma}^2 = \dfrac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2$
3. $\hat{\theta} \approx N\left(\theta, \dfrac{1}{n\, I(\theta)}\right)$

TERMS θ parameter; $f(x \mid \theta)$ model; ℓ log-likelihood; n data points

"Which parameter makes the data I saw most probable?" Take logs to turn products into sums.

1. $L(\theta) = \prod_{i=1}^{n} f(x_i \mid \theta)$
2. $\ell(\theta) = \ln L(\theta) = \sum_{i=1}^{n} \ln f(x_i \mid \theta)$
3. $\hat{\theta}_{\text{MLE}} = \arg\max_{\theta} \ell(\theta) : \ \dfrac{d\ell}{d\theta} = 0$

TERMS θ parameter; $f(x \mid \theta)$ model; ℓ log-likelihood; n data points

"Which parameter makes the data I saw most probable?" Take logs to turn products into sums.

1. $p = P(Z \geq z \mid H_0) = 1 - \Phi(z)$
2. reject H_0 if $p < \alpha \iff |z| > z^*_{\alpha/2}$
3. $\alpha = P(\text{reject } H_0 \mid H_0), \quad \beta = P(\text{keep } H_0 \mid H_a), \quad \text{power} = 1 - \beta$

TERMS H_0 null; H_a alternative; μ_0 null value; α level; β miss rate

"If p is low, the null must go." Type I = false alarm; Type II = missed detection.

1. $H_0 : \ \mu = \mu_0 \quad \text{vs} \quad H_a : \ \mu \neq \mu_0 \ (\text{or } >, <)$
2. $z = \dfrac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$
3. $p = P\big(|Z| \geq |z| \ \big|\ H_0\big) = 2\big(1 - \Phi(|z|)\big)$

TERMS H_0 null; H_a alternative; μ_0 null value; α level; β miss rate

"If p is low, the null must go." Type I = false alarm; Type II = missed detection.

1. $(\bar{x}_1 - \bar{x}_2) \pm t^* \sqrt{\dfrac{s_1^2}{n_1} + \dfrac{s_2^2}{n_2}}$
2. $n = \left(\dfrac{z^* \sigma}{E}\right)^2$
3. $n = \dfrac{(z^*)^2\, p(1-p)}{E^2}$

TERMS z^*, t^* critical values; E margin of error; $\hat{p}$ sample proportion

Estimate ± (critical value)×(standard error).

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S.K1 STATISTICS & PROBABILITY

Which test compares the means of two independent groups?

RECALL IT

S.K1 STATISTICS & PROBABILITY

Which test for 3 or more group means?

RECALL IT

S.K1 STATISTICS & PROBABILITY

Which test for categorical data?

RECALL IT

S.K1 STATISTICS & PROBABILITY

Which distribution? Counts, successes, waits

RECALL IT

S.K1 STATISTICS & PROBABILITY

The five steps of any hypothesis test

RECALL IT

S.K1 STATISTICS & PROBABILITY

One- vs two-tailed; z vs t

RECALL IT

S.29 (1/2) STATISTICS & PROBABILITY

Errors, Power, Sample Size & Effect Size

PART 1 OF 2

1. Power of a z-test
2. Sample size (one sample)
3. Sample size per group (two groups)

STATE EACH NUMBERED FORMULA

S.29 (2/2) STATISTICS & PROBABILITY

Errors, Power, Sample Size & Effect Size

PART 2 OF 2

1. Cohen's d
2. Pooled SD

STATE EACH NUMBERED FORMULA

S.30 (1/2) STATISTICS & PROBABILITY

t-Tests

PART 1 OF 2

1. One-sample t
2. Welch two-sample t
3. Welch degrees of freedom

STATE EACH NUMBERED FORMULA

S.30 (2/2) STATISTICS & PROBABILITY

t-Tests

PART 2 OF 2

1. Pooled two-sample t
2. Paired t

STATE EACH NUMBERED FORMULA

S.31 (1/2) STATISTICS & PROBABILITY

z-Tests for Proportions

PART 1 OF 2

1. One-proportion z
2. Pooled proportion
3. Two-proportion z

STATE EACH NUMBERED FORMULA

S.31 (2/2) STATISTICS & PROBABILITY

z-Tests for Proportions

PART 2 OF 2

1. Interval for $p_1 - p_2$
2. Relative lift

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1 proportion: one-proportion z.
2 proportions: two-proportion z ($= \chi^2$ on 2×2).
$r \times c$ table: χ^2 independence.
Shares vs claim: χ^2 goodness of fit.
Small expected counts: Fisher's exact.

S.K1

One-way ANOVA (F-test).
Not normal / ordinal: Kruskal–Wallis.
Then post-hoc pairwise tests with a multiple-testing correction.

S.K1

Welch two-sample t-test.
Skewed or small n: Mann–Whitney U.
Paired data: paired t (or Wilcoxon signed-rank).

S.K1

Two-tailed unless direction fixed in advance.
z: proportions or known σ.
t: means with σ estimated by s.

S.K1

1 H_0, H_a, α, sidedness.
2 Check conditions.
3 Statistic = (estimate − null)/SE.
4 p-value from null distribution.
5 Decide; report effect size + CI.

S.K1

Successes in n trials: Binomial.
Without replacement: Hypergeometric.
Events per interval: Poisson.
Wait between events: Exponential.
Trials to first success: Geometric.
Sum of many effects: Normal.

S.K1

1 $$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}, \quad df = n - 1$$

2 $$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

3 $$df \approx \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}}$$

TERMS $\bar{x}$ mean; s SD; n size; df degrees of freedom; $\bar{d}$ mean difference

Paired = one-sample test on differences.

S.30 (1/2)

1 $$d = \frac{\bar{x}_1 - \bar{x}_2}{s_p}$$

2 $$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

TERMS δ effect; σ SD; α, β error rates; s_p pooled SD

Power rises with: bigger effect, bigger n, less noise, larger α. "16 over d^2" per group.

S.29 (2/2)

1 $$1 - \beta = \Phi\left(\frac{|\delta|\sqrt{n}}{\sigma} - z^*_{\alpha/2}\right)$$

2 $$n = \left(\frac{(z^*_{\alpha/2} + z^*_{\beta})\,\sigma}{\delta}\right)^2$$

3 $$n = \frac{2\,(z^*_{\alpha/2} + z^*_{\beta})^2 \sigma^2}{\delta^2}$$

TERMS δ effect; σ SD; α, β error rates; s_p pooled SD

Power rises with: bigger effect, bigger n, less noise, larger α. "16 over d^2" per group.

S.29 (1/2)

1 $$(\hat{p}_1 - \hat{p}_2) \pm z^* \sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}}$$

2 $$\text{lift} = \frac{\hat{p}_2 - \hat{p}_1}{\hat{p}_1}$$

TERMS $\hat{p}$ sample proportion; p_0 null value; x successes; n trials

Test with the null (pooled) SE; estimate with the unpooled SE.

S.31 (2/2)

1 $$z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}}$$

2 $$\hat{p} = \frac{x_1 + x_2}{n_1 + n_2}$$

3 $$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1 - \hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

TERMS $\hat{p}$ sample proportion; p_0 null value; x successes; n trials

Test with the null (pooled) SE; estimate with the unpooled SE.

S.31 (1/2)

1 $$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}, \; df = n_1 + n_2 - 2$$

2 $$t = \frac{\bar{d}}{s_d/\sqrt{n}}, \quad df = n - 1$$

TERMS $\bar{x}$ mean; s SD; n size; df degrees of freedom; $\bar{d}$ mean difference

Paired = one-sample test on differences.

S.30 (2/2)

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S.32 (1/2) STATISTICS & PROBABILITY

Chi-Square Tests

PART 1 OF 2

1. Chi-square statistic
2. Goodness of fit
3. Expected count (two-way table)

STATE EACH NUMBERED FORMULA

S.32 (2/2) STATISTICS & PROBABILITY

Chi-Square Tests

PART 2 OF 2

1. Independence / homogeneity
2. Cramér's V

STATE EACH NUMBERED FORMULA

S.33 (1/2) STATISTICS & PROBABILITY

One-Way ANOVA & the F-Test

PART 1 OF 2

1. Sum-of-squares split
2. Between groups
3. Within groups

STATE EACH NUMBERED FORMULA

S.33 (2/2) STATISTICS & PROBABILITY

One-Way ANOVA & the F-Test

PART 2 OF 2

1. F-statistic
2. Effect size

STATE EACH NUMBERED FORMULA

S.34 (1/2) STATISTICS & PROBABILITY

Nonparametric (Rank) Tests

PART 1 OF 2

1. Mann–Whitney U
2. U normal approximation
3. Wilcoxon signed-rank

STATE EACH NUMBERED FORMULA

S.34 (2/2) STATISTICS & PROBABILITY

Nonparametric (Rank) Tests

PART 2 OF 2

1. W^+ normal approximation
2. Kruskal–Wallis H
3. Sign test

STATE EACH NUMBERED FORMULA

S.35 (1/2) STATISTICS & PROBABILITY

Multiple Testing: Bonferroni & Benjamini–Hochberg

PART 1 OF 2

1. Family-wise error rate
2. Bonferroni
3. Holm step-down

STATE EACH NUMBERED FORMULA

S.35 (2/2) STATISTICS & PROBABILITY

Multiple Testing: Bonferroni & Benjamini–Hochberg

PART 2 OF 2

1. False discovery rate
2. Benjamini–Hochberg

STATE EACH NUMBERED FORMULA

S.36 (1/2) STATISTICS & PROBABILITY

Bootstrap & Permutation Tests

PART 1 OF 2

1. Bootstrap resample
2. Bootstrap standard error
3. Percentile interval

STATE EACH NUMBERED FORMULA

S.36 (2/2) STATISTICS & PROBABILITY

Bootstrap & Permutation Tests

PART 2 OF 2

1. Permutation p-value
2. Resampling probability

STATE EACH NUMBERED FORMULA

S.37 (1/3) STATISTICS & PROBABILITY

Bayesian Inference & Conjugate Priors

PART 1 OF 3

1. Posterior
2. Proportional form
3. Beta prior

STATE EACH NUMBERED FORMULA

S.37 (2/3) STATISTICS & PROBABILITY

Bayesian Inference & Conjugate Priors

PART 2 OF 3

1. Beta–Binomial update
2. Posterior mean
3. Normal–normal update

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1. $SS_T = SS_B + SS_W$
2. $SS_B = \sum_{j=1}^{k} n_j(\bar{y}_j - \bar{y})^2$
3. $SS_W = \sum_{j=1}^{k} \sum_{i=1}^{n_j} (y_{ij} - \bar{y}_j)^2$

TERMS k groups; N total; $\bar{y}_j$ group mean; $\bar{y}$ grand mean

F = signal (between) / noise (within).

1. $df = (r-1)(c-1)$
2. $V = \sqrt{\dfrac{\chi^2}{n\,(\min(r,c)-1)}}$

TERMS O observed; E expected; k categories; r, c rows, columns

"Observed minus expected, squared, over expected."

1. $\chi^2 = \sum \dfrac{(O-E)^2}{E}$
2. $E_i = n\,p_{0i}, \quad df = k-1$
3. $E = \dfrac{(\text{row total})(\text{column total})}{\text{grand total}}$

TERMS O observed; E expected; k categories; r, c rows, columns

"Observed minus expected, squared, over expected."

1. $\mu_W = \dfrac{n(n+1)}{4}, \quad \sigma_W^2 = \dfrac{n(n+1)(2n+1)}{24}$
2. $H = \dfrac{12}{N(N+1)} \sum_{j=1}^{k} \dfrac{R_j^2}{n_j} - 3(N+1)$
3. $S = \#\{d_i > 0\} \sim \text{Bin}\left(n, \tfrac{1}{2}\right)$

TERMS R rank sum; n_1, n_2 group sizes; d_i paired differences

Rank first, then test: Mann–Whitney (2 independent), Wilcoxon (paired), Kruskal–Wallis (3+).

1. $U_1 = R_1 - \dfrac{n_1(n_1+1)}{2}$
2. $\mu_U = \dfrac{n_1 n_2}{2}, \quad \sigma_U^2 = \dfrac{n_1 n_2 (n_1 + n_2 + 1)}{12}$
3. $W^+ = \sum_{d_i > 0} \text{rank}(|d_i|)$

TERMS R rank sum; n_1, n_2 group sizes; d_i paired differences

Rank first, then test: Mann–Whitney (2 independent), Wilcoxon (paired), Kruskal–Wallis (3+).

1. $F = \dfrac{MS_B}{MS_W} = \dfrac{SS_B/(k-1)}{SS_W/(N-k)}$
2. $\eta^2 = \dfrac{SS_B}{SS_T}$

TERMS k groups; N total; $\bar{y}_j$ group mean; $\bar{y}$ grand mean

F = signal (between) / noise (within).

1. $x_1^*, \ldots, x_n^*$ drawn with replacement from $x_1, \ldots, x_n$
2. $SE^* = \sqrt{\dfrac{1}{B-1} \sum_{b=1}^{B} \left(\hat{\theta}_b^* - \bar{\theta}^*\right)^2}$
3. $\left[\hat{\theta}^*_{(\alpha/2)},\ \hat{\theta}^*_{(1-\alpha/2)}\right]$

TERMS x^* resample; B number of resamples; $\hat{\theta}_b^*$ resampled statistic

Bootstrap: "the sample is my population". Permutation: "if labels don't matter, shuffle them".

1. $\text{FDR} = E\left[\dfrac{V}{\max(R, 1)}\right]$
2. $k = \max\left\{i : \ p_{(i)} \le \dfrac{i}{m} q\right\}$; reject $H_{(1)}, \ldots, H_{(k)}$

TERMS m tests; $p_{(i)}$ sorted p-values; q target FDR; V false rejections

Bonferroni: "divide α by m". BH: "line through the origin with slope q/m".

1. $\text{FWER} = P(\ge 1 \text{ false positive}) = 1 - (1-\alpha)^m$
2. reject H_i if $p_i \le \dfrac{\alpha}{m}$
3. $p_{(i)} \le \dfrac{\alpha}{m - i + 1}$ for $i = 1, 2, \ldots$

TERMS m tests; $p_{(i)}$ sorted p-values; q target FDR; V false rejections

Bonferroni: "divide α by m". BH: "line through the origin with slope q/m".

1. $\text{Beta}(\alpha, \beta) \xrightarrow{k \text{ of } n} \text{Beta}(\alpha + k,\ \beta + n - k)$
2. $E[\theta \mid x] = \dfrac{\alpha + k}{\alpha + \beta + n}$
3. $\mu_{\text{post}} = \dfrac{\mu_0/\tau^2 + n\bar{x}/\sigma^2}{1/\tau^2 + n/\sigma^2}$

TERMS θ parameter; α, β Beta shapes; k successes; n trials

Prior + data → posterior; today's posterior is tomorrow's prior.

1. $p(\theta \mid x) = \dfrac{p(x \mid \theta)\, p(\theta)}{\int p(x \mid \theta')\, p(\theta')\, d\theta'}$
2. $p(\theta \mid x) \propto p(x \mid \theta)\, p(\theta)$
3. $p(\theta) = \dfrac{\theta^{\alpha-1}(1-\theta)^{\beta-1}}{B(\alpha, \beta)}, \quad 0 \le \theta \le 1$

TERMS θ parameter; α, β Beta shapes; k successes; n trials

Prior + data → posterior; today's posterior is tomorrow's prior.

1. $p = \dfrac{1 + \#\{|T_b^*| \ge |T_{\text{obs}}|\}}{B + 1}$
2. $P(x_i \notin \text{resample}) = \left(1 - \dfrac{1}{n}\right)^n \to e^{-1}$

TERMS x^* resample; B number of resamples; $\hat{\theta}_b^*$ resampled statistic

Bootstrap: "the sample is my population". Permutation: "if labels don't matter, shuffle them".

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S.37 (3/3) STATISTICS & PROBABILITY

Bayesian Inference & Conjugate Priors

PART 3 OF 3

1. Credible interval
2. MAP estimate

STATE EACH NUMBERED FORMULA

S.38 (1/2) STATISTICS & PROBABILITY

Covariance & Correlation

PART 1 OF 2

1. Sample covariance
2. Pearson correlation
3. Range of r

STATE EACH NUMBERED FORMULA

S.38 (2/2) STATISTICS & PROBABILITY

Covariance & Correlation

PART 2 OF 2

1. Population covariance
2. Spearman rank correlation
3. Test for correlation

STATE EACH NUMBERED FORMULA

S.39 (1/3) STATISTICS & PROBABILITY

Covariance Matrix & Multivariate Normal

PART 1 OF 3

1. Covariance matrix
2. Sample covariance matrix
3. Variance of a combination

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S.39 (2/3) STATISTICS & PROBABILITY

Covariance Matrix & Multivariate Normal

PART 2 OF 3

1. Linear transformation
2. Correlation matrix

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S.39 (3/3) STATISTICS & PROBABILITY

Covariance Matrix & Multivariate Normal

PART 3 OF 3

1. Multivariate normal density

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S.40 (1/3) STATISTICS & PROBABILITY

Least-Squares Linear Regression & R^2

PART 1 OF 3

1. Fitted line
2. Slope
3. Intercept

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S.40 (2/3) STATISTICS & PROBABILITY

Least-Squares Linear Regression & R^2

PART 2 OF 3

1. Coefficient of determination
2. Normal equations
3. Residual standard error

STATE EACH NUMBERED FORMULA

S.40 (3/3) STATISTICS & PROBABILITY

Least-Squares Linear Regression & R^2

PART 3 OF 3

1. Slope t-test

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S.41 (1/2) STATISTICS & PROBABILITY

Multiple Regression & Adjusted R^2

PART 1 OF 2

1. Model
2. Coefficient covariance
3. Coefficient t-test

STATE EACH NUMBERED FORMULA

S.41 (2/2) STATISTICS & PROBABILITY

Multiple Regression & Adjusted R^2

PART 2 OF 2

1. Adjusted R^2
2. Overall F-test
3. Variance inflation factor

STATE EACH NUMBERED FORMULA

S.42 (1/2) STATISTICS & PROBABILITY

Logistic Regression (Statistical View)

PART 1 OF 2

1. Logit link
2. Sigmoid (inverse link)
3. Odds ratio

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1 $\mathrm{Cov}(X,Y) = E\big[(X-\mu_X)(Y-\mu_Y)\big] = E[XY] - \mu_X\mu_Y$

2 $\rho_s = 1 - \dfrac{6\sum d_i^2}{n(n^2-1)}$

3 $t = \dfrac{r\sqrt{n-2}}{\sqrt{1-r^2}}, \quad df = n-2$

TERMS s_x, s_y SDs; z z-scores; d_i rank differences; n pairs

Correlation is not causation S.43 *§13 Statistics.*

1 $\mathrm{Cov}(X,Y) = \dfrac{1}{n-1}\sum_{i=1}^{n}(x_i-\bar{x})(y_i-\bar{y})$

2 $r = \dfrac{\mathrm{Cov}(X,Y)}{s_x\, s_y} = \dfrac{1}{n-1}\sum_{i=1}^{n} z_{x_i} z_{y_i}$

3 $-1 \le r \le 1$

TERMS s_x, s_y SDs; z z-scores; d_i rank differences; n pairs

Correlation is not causation S.43 *§13 Statistics.*

1 $P(a \le \theta \le b \mid x) = 0.95$

2 $\hat{\theta}_{\text{MAP}} = \arg\max_{\theta}\, p(\theta \mid x)$

TERMS θ parameter; α, β Beta shapes; k successes; n trials

Prior + data → posterior; today's posterior is tomorrow's prior.

1 $f(\mathbf{x}) = \dfrac{1}{(2\pi)^{p/2}|\Sigma|^{1/2}} \exp\!\left(-\tfrac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x}-\boldsymbol{\mu})\right)$

TERMS Σ covariance matrix; $\boldsymbol{\mu}$ mean vector; p variables; A matrix

Diagonal = variances, off-diagonal = covariances; eigenvectors = ellipse axes.

1 $\mathrm{Cov}(A\mathbf{X}+\mathbf{b}) = A\,\Sigma A^\top$

2 $R = D^{-1/2}\,\Sigma\, D^{-1/2}$

TERMS Σ covariance matrix; $\boldsymbol{\mu}$ mean vector; p variables; A matrix

Diagonal = variances, off-diagonal = covariances; eigenvectors = ellipse axes.

1 $\Sigma = E\big[(\mathbf{X}-\boldsymbol{\mu})(\mathbf{X}-\boldsymbol{\mu})^\top\big]$

2 $S = \dfrac{1}{n-1} X_c^\top X_c$

3 $\mathrm{Var}(\mathbf{a}^\top \mathbf{X}) = \mathbf{a}^\top \Sigma\, \mathbf{a}$

TERMS Σ covariance matrix; $\boldsymbol{\mu}$ mean vector; p variables; A matrix

Diagonal = variances, off-diagonal = covariances; eigenvectors = ellipse axes.

1 $t = \dfrac{b_1}{SE(b_1)}, \quad SE(b_1) = \dfrac{s_e}{\sqrt{\sum(x_i-\bar{x})^2}}$

TERMS b_0, b_1 intercept, slope; r correlation; e_i residual; s_e residual SE

The regression line always passes through $(\bar{x}, \bar{y})$.

1 $R^2 = 1 - \dfrac{SS_{\text{res}}}{SS_{\text{tot}}} = 1 - \dfrac{\sum(y_i-\hat{y}_i)^2}{\sum(y_i-\bar{y})^2}$

2 $\hat{\boldsymbol{\beta}} = (X^\top X)^{-1} X^\top \mathbf{y}$

3 $s_e = \sqrt{\dfrac{SS_{\text{res}}}{n-2}}$

TERMS b_0, b_1 intercept, slope; r correlation; e_i residual; s_e residual SE

The regression line always passes through $(\bar{x}, \bar{y})$.

1 $\hat{y} = b_0 + b_1 x$

2 $b_1 = \dfrac{\sum(x_i-\bar{x})(y_i-\bar{y})}{\sum(x_i-\bar{x})^2} = r\,\dfrac{s_y}{s_x}$

3 $b_0 = \bar{y} - b_1\bar{x}$

TERMS b_0, b_1 intercept, slope; r correlation; e_i residual; s_e residual SE

The regression line always passes through $(\bar{x}, \bar{y})$.

1 $\ln\dfrac{p}{1-p} = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$

2 $p = \sigma(z) = \dfrac{1}{1+e^{-z}}, \; z = \mathbf{x}^\top\boldsymbol{\beta}$

3 $\mathrm{OR}_j = e^{\beta_j}$

TERMS p probability of $Y=1$; β_j coefficients; z linear predictor

Logistic: linear in the log-odds; exponentiate a coefficient to get an odds ratio.

1 $R^2_{\text{adj}} = 1 - (1-R^2)\dfrac{n-1}{n-p-1}$

2 $F = \dfrac{R^2/p}{(1-R^2)/(n-p-1)}$

3 $\mathrm{VIF}_j = \dfrac{1}{1-R_j^2}$

TERMS X design matrix; $\boldsymbol{\beta}$ coefficients; p predictors; n rows

Assumptions LINE: Linear, Independent errors, Normal errors, Equal variance.

1 $\mathbf{y} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad \varepsilon_i \sim N(0,\sigma^2)$

2 $\mathrm{Cov}(\hat{\boldsymbol{\beta}}) = \hat{\sigma}^2 (X^\top X)^{-1}, \quad \hat{\sigma}^2 = \dfrac{SS_{\text{res}}}{n-p-1}$

3 $t_j = \dfrac{\hat{\beta}_j}{SE(\hat{\beta}_j)}, \quad df = n-p-1$

TERMS X design matrix; $\boldsymbol{\beta}$ coefficients; p predictors; n rows

Assumptions LINE: Linear, Independent errors, Normal errors, Equal variance.

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Logistic Regression (Statistical View)

PART 2 OF 2

1. Log-likelihood
2. Wald test

STATE EACH NUMBERED FORMULA

Correlation vs Causation

PART 1 OF 2

1. Average treatment effect
2. Randomized experiment
3. Adjustment for a confounder

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Correlation vs Causation

PART 2 OF 2

1. Simpson's paradox

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1. $P(A \mid B, z) > P(A \mid B^c, z)\ \forall z \not\Rightarrow P(A \mid B) > P(A \mid B^c)$

TERMS $Y(1), Y(0)$ potential outcomes; T treatment; Z confounder

Correlation is not causation: look for the confounder, reverse causation, and selection.

S.43 (2/2)

1. $\text{ATE} = E\big[Y(1) - Y(0)\big]$
2. $\text{ATE} = E[Y \mid T = 1] - E[Y \mid T = 0]$
3. $\text{ATE} = \sum_z \big(E[Y \mid T{=}1, z] - E[Y \mid T{=}0, z]\big) P(z)$

TERMS $Y(1), Y(0)$ potential outcomes; T treatment; Z confounder

Correlation is not causation: look for the confounder, reverse causation, and selection.

S.43 (1/2)

1. $\ell(\boldsymbol{\beta}) = \sum_{i=1}^{n} \big[y_i \ln p_i + (1 - y_i) \ln(1 - p_i)\big]$
2. $z_j = \dfrac{\hat{\beta}_j}{SE(\hat{\beta}_j)}$

TERMS p probability of $Y = 1$; β_j coefficients; z linear predictor

Logistic: linear in the log-odds; exponentiate a coefficient to get an odds ratio.

S.42 (2/2)

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O.1 (1/2) OPTIMIZATION & SIMULATION

Anatomy of an Optimization Problem

PART 1 OF 2

1. Standard form
2. Max–min equivalence
3. Feasible set

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O.1 (2/2) OPTIMIZATION & SIMULATION

Anatomy of an Optimization Problem

PART 2 OF 2

1. Global minimum
2. Local minimum
3. Constraint rewrites

STATE EACH NUMBERED FORMULA

O.K1 OPTIMIZATION & SIMULATION

Optimization problem types: form and solver

RECALL IT

O.K1 OPTIMIZATION & SIMULATION

Zeroth-, first- and second-order methods: what do they use?

RECALL IT

O.2 (1/2) OPTIMIZATION & SIMULATION

Convex Sets & Convex Functions

PART 1 OF 2

1. Convex set
2. Convex function (chord)
3. First-order condition

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O.2 (2/2) OPTIMIZATION & SIMULATION

Convex Sets & Convex Functions

PART 2 OF 2

1. Second-order condition
2. Strong convexity
3. Local = global

STATE EACH NUMBERED FORMULA

O.3 (1/2) OPTIMIZATION & SIMULATION

Linear Programming (LP)

PART 1 OF 2

1. Canonical form
2. Standard (equality) form

STATE EACH NUMBERED FORMULA

O.3 (2/2) OPTIMIZATION & SIMULATION

Linear Programming (LP)

PART 2 OF 2

1. Corner-point theorem
2. Iso-profit line

STATE EACH NUMBERED FORMULA

O.4 (1/3) OPTIMIZATION & SIMULATION

Simplex Method & LP Duality

PART 1 OF 3

1. Ratio test
2. Optimality test

STATE EACH NUMBERED FORMULA

O.4 (2/3) OPTIMIZATION & SIMULATION

Simplex Method & LP Duality

PART 2 OF 3

1. Primal–dual pair

STATE EACH NUMBERED FORMULA

O.4 (3/3) OPTIMIZATION & SIMULATION

Simplex Method & LP Duality

PART 3 OF 3

1. Weak duality
2. Strong duality
3. Complementary slackness

STATE EACH NUMBERED FORMULA

O.5 (1/2) OPTIMIZATION & SIMULATION

Quadratic Programming (QP)

PART 1 OF 2

1. QP form
2. Convexity test
3. Unconstrained solution

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LP: linear; simplex / IPM, vertex optimum.
MIP: integers; branch & bound.
QP: quadratic f, linear constr.
Convex: local = global.
NLP: Newton / BFGS, local only.
Combinatorial: DP or heuristics.

O.K1

1. $f(\mathbf{x}^*) \le f(\mathbf{x}) \quad \forall\, \mathbf{x} \in \mathcal{F}$
2. $f(\mathbf{x}^*) \le f(\mathbf{x}) \quad \forall\, \mathbf{x} \in \mathcal{F},\ \|\mathbf{x} - \mathbf{x}^*\| < \varepsilon$
3. $g \ge 0 \iff -g \le 0; \quad h = 0 \iff h \le 0,\ -h \le 0$

TERMS $\mathbf{x}$ decisions; f objective; $g_i \le 0$, $h_j = 0$ constraints; $\mathcal{F}$ feasible set

Three parts: decisions, objective, constraints.

O.1 (2/2)

1. $\min\limits_{\mathbf{x}\in\mathbb{R}^n} f(\mathbf{x}) \quad \text{subject to} \quad g_i(\mathbf{x}) \le 0 \ \ (i = 1,\dots,m), \qquad h_j(\mathbf{x}) = 0 \ \ (j = 1,\dots,p)$
2. $\max\limits_{\mathbf{x}} f(\mathbf{x}) = -\min\limits_{\mathbf{x}} \big(-f(\mathbf{x})\big)$
3. $\mathcal{F} = \{\mathbf{x} :\ g_i(\mathbf{x}) \le 0,\ h_j(\mathbf{x}) = 0\}$

TERMS $\mathbf{x}$ decisions; f objective; $g_i \le 0$, $h_j = 0$ constraints; $\mathcal{F}$ feasible set

Three parts: decisions, objective, constraints.

O.1 (1/2)

1. $f''(x) \ge 0 \quad \text{or} \quad \nabla^2 f(\mathbf{x}) \succeq 0$
2. $\nabla^2 f(\mathbf{x}) \succeq mI, \quad m > 0$
3. convex f, convex $\mathcal{F} \ \Rightarrow$ local min = global min

TERMS $\lambda \in [0,1]$ mixing weight; $\nabla^2 f$ Hessian; $\succeq 0$ positive semidefinite

Convex looks like a bowl: any chord lies above the curve, any tangent below it.

O.2 (2/2)

1. $\lambda\mathbf{x} + (1-\lambda)\mathbf{y} \in C \quad \forall\, \mathbf{x}, \mathbf{y} \in C,\ \lambda \in [0,1]$
2. $f(\lambda\mathbf{x} + (1-\lambda)\mathbf{y}) \le \lambda f(\mathbf{x}) + (1-\lambda) f(\mathbf{y})$
3. $f(\mathbf{y}) \ge f(\mathbf{x}) + \nabla f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x})$

TERMS $\lambda \in [0,1]$ mixing weight; $\nabla^2 f$ Hessian; $\succeq 0$ positive semidefinite

Convex looks like a bowl: any chord lies above the curve, any tangent below it.

O.2 (1/2)

Zeroth: function values only (SA, GA, Nelder–Mead).
First: gradients (GD, SGD, Adam).
Second: Hessian or its approximation (Newton, BFGS); fewer, costlier steps.

O.K1

1. leave: $\arg\min\limits_i \left\{ \dfrac{b_i}{a_{ij}} :\ a_{ij} > 0 \right\}$
2. $\bar{c}_j = c_j - \mathbf{c}_B^\top B^{-1} A_j \le 0 \ \ \forall j$

TERMS $\bar{c}_j$ reduced cost; $\mathbf{y}$ shadow prices; s_i slack

Each primal constraint becomes a dual variable; max becomes min.

O.4 (1/3)

1. optimum (if any) is attained at a vertex
2. $\mathbf{c}^\top \mathbf{x} = k$

TERMS $\mathbf{c}$ objective coeffs; $A, \mathbf{b}$ constraints; $\mathbf{s}$ slacks

"Check the corners."

O.3 (2/2)

1. $\max\limits_{\mathbf{x}} \ z = \mathbf{c}^\top \mathbf{x} \ \text{ s.t. } \ A\mathbf{x} \le \mathbf{b}, \ \ \mathbf{x} \ge \mathbf{0}$
2. $A\mathbf{x} + \mathbf{s} = \mathbf{b}, \quad \mathbf{x}, \mathbf{s} \ge \mathbf{0}$

TERMS $\mathbf{c}$ objective coeffs; $A, \mathbf{b}$ constraints; $\mathbf{s}$ slacks

"Check the corners."

O.3 (1/2)

1. $\min\limits_{\mathbf{x}} \ \frac{1}{2}\mathbf{x}^\top Q\mathbf{x} + \mathbf{c}^\top \mathbf{x} \quad \text{s.t.} \quad A\mathbf{x} \le \mathbf{b}, \quad E\mathbf{x} = \mathbf{d}$
2. $Q \succeq 0$
3. $\nabla f = Q\mathbf{x} + \mathbf{c} = \mathbf{0} \Rightarrow \mathbf{x}^* = -Q^{-1}\mathbf{c}$

TERMS Q quadratic (Hessian) matrix; $\mathbf{c}$ linear cost; ν multipliers

LP with a bowl on top: the bottom of the bowl may sit on a face, not a corner.

O.5 (1/2)

1. $\mathbf{c}^\top \mathbf{x} \le \mathbf{b}^\top \mathbf{y}$
2. $\mathbf{c}^\top \mathbf{x}^* = \mathbf{b}^\top \mathbf{y}^*$
3. $y_i^* \big(b_i - (A\mathbf{x}^*)_i\big) = 0$

TERMS $\bar{c}_j$ reduced cost; $\mathbf{y}$ shadow prices; s_i slack

Each primal constraint becomes a dual variable; max becomes min.

O.4 (3/3)

1. Primal: $\max \mathbf{c}^\top \mathbf{x}$ s.t. $A\mathbf{x} \le \mathbf{b}, \mathbf{x} \ge \mathbf{0}$ $\iff$ Dual: $\min \mathbf{b}^\top \mathbf{y}$ s.t. $A^\top \mathbf{y} \ge \mathbf{c}, \mathbf{y} \ge \mathbf{0}$

TERMS $\bar{c}_j$ reduced cost; $\mathbf{y}$ shadow prices; s_i slack

Each primal constraint becomes a dual variable; max becomes min.

O.4 (2/3)

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O.5 (2/2) OPTIMIZATION & SIMULATION

Quadratic Programming (QP)

PART 2 OF 2

1. Equality-constrained KKT system
2. Markowitz portfolio

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O.6 (1/2) OPTIMIZATION & SIMULATION

Integer & Mixed-Integer Programming (MIP)

PART 1 OF 2

1. MIP form
2. LP-relaxation bound
3. Big-M switch

STATE EACH NUMBERED FORMULA

O.6 (2/2) OPTIMIZATION & SIMULATION

Integer & Mixed-Integer Programming (MIP)

PART 2 OF 2

1. Branching
2. Logical constraints
3. Optimality gap

STATE EACH NUMBERED FORMULA

O.7 (1/2) OPTIMIZATION & SIMULATION

Knapsack Problem

PART 1 OF 2

1. 0/1 knapsack model
2. DP recurrence
3. Base case

STATE EACH NUMBERED FORMULA

O.7 (2/2) OPTIMIZATION & SIMULATION

Knapsack Problem

PART 2 OF 2

1. Fractional (greedy) bound
2. Complexity

STATE EACH NUMBERED FORMULA

O.8 (1/2) OPTIMIZATION & SIMULATION

Traveling Salesman Problem (TSP)

PART 1 OF 2

1. Objective
2. Degree constraints
3. Subtour elimination

STATE EACH NUMBERED FORMULA

O.8 (2/2) OPTIMIZATION & SIMULATION

Traveling Salesman Problem (TSP)

PART 2 OF 2

1. Number of tours
2. 2-opt move gain

STATE EACH NUMBERED FORMULA

O.9 (1/2) OPTIMIZATION & SIMULATION

Shortest Paths & Network Flow

PART 1 OF 2

1. Bellman optimality
2. Edge relaxation
3. Complexities

STATE EACH NUMBERED FORMULA

O.9 (2/2) OPTIMIZATION & SIMULATION

Shortest Paths & Network Flow

PART 2 OF 2

1. Maximum flow
2. Max-flow min-cut

STATE EACH NUMBERED FORMULA

O.10 (1/2) OPTIMIZATION & SIMULATION

Algorithmic Complexity (Big-O)

PART 1 OF 2

1. Big-O (upper bound)
2. Big-Ω and Big-Θ
3. Simplification rule

STATE EACH NUMBERED FORMULA

O.10 (2/2) OPTIMIZATION & SIMULATION

Algorithmic Complexity (Big-O)

PART 2 OF 2

1. Growth hierarchy
2. P and NP

STATE EACH NUMBERED FORMULA

O.11 (1/3) OPTIMIZATION & SIMULATION

Gradient Descent & Convergence

PART 1 OF 3

1. Gradient descent update
2. Step-size rule
3. Convex rate

STATE EACH NUMBERED FORMULA

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1. $x_j \le \lfloor v \rfloor$ or $x_j \ge \lceil v \rceil$
2. $y_1 + y_2 \le 1; \quad y_1 \le y_2; \quad \sum_k y_k \ge K$
3. $\text{gap} = \dfrac{z_{\text{bound}} - z_{\text{incumbent}}}{|z_{\text{incumbent}}|}$

TERMS I integer indices; y binary; M big constant; v fractional LP value

Relax, bound, branch, prune.

1. $\max \ \mathbf{c}^\top \mathbf{x} \ \text{ s.t. } A\mathbf{x} \le \mathbf{b}, \quad x_j \in \mathbb{Z} \ (j \in I)$
2. $z_{\text{MIP}} \le z_{\text{LP relax}}$
3. $x \le M y, \quad y \in \{0,1\}$

TERMS I integer indices; y binary; M big constant; v fractional LP value

Relax, bound, branch, prune.

1. $\begin{bmatrix} Q & E^\top \\ E & 0 \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \boldsymbol{\nu} \end{bmatrix} = \begin{bmatrix} -\mathbf{c} \\ \mathbf{d} \end{bmatrix}$
2. $\min_{\mathbf{w}} \ \mathbf{w}^\top \Sigma \mathbf{w} \ \text{ s.t. } \boldsymbol{\mu}^\top \mathbf{w} \ge r, \ \mathbf{1}^\top \mathbf{w} = 1$

TERMS Q quadratic (Hessian) matrix; $\mathbf{c}$ linear cost; ν multipliers

LP with a bowl on top: the bottom of the bowl may sit on a face, not a corner.

1. $\min \sum_{i \ne j} c_{ij} x_{ij}, \quad x_{ij} \in \{0,1\}$
2. $\sum_j x_{ij} = 1 \ \ \forall i, \qquad \sum_i x_{ij} = 1 \ \ \forall j$
3. $\sum_{i,j \in S} x_{ij} \le |S| - 1 \quad \forall S \subset V, \ 2 \le |S| \le n-1$

TERMS c_{ij} edge cost; x_{ij} binary edge choice; S subset of cities

$(n-1)!/2$: *fix the start, and a tour and its reverse are the same.*

1. sort by $\dfrac{v_i}{w_i}$, fill, split the last item
2. $O(nW)$ table vs 2^n subsets

TERMS v_i, w_i value, weight; W capacity; $V(i,w)$ best value table

Each item: "skip it or take it" — keep the better.

1. $\max \sum_{i=1}^{n} v_i x_i \ \text{ s.t. } \sum_{i=1}^{n} w_i x_i \le W, \quad x_i \in \{0,1\}$
2. $V(i,w) = \max\{V(i-1,w), \ v_i + V(i-1, w-w_i)\}$
3. $V(0,w) = 0, \qquad V(i,w) = V(i-1,w) \ \text{ if } w_i > w$

TERMS v_i, w_i value, weight; W capacity; $V(i,w)$ best value table

Each item: "skip it or take it" — keep the better.

1. $\max \sum_v f_{sv} \ \text{ s.t. } \ 0 \le f_{uv} \le c_{uv}, \qquad \sum_u f_{uv} = \sum_w f_{vw} \ \ (v \ne s,t)$
2. max flow = min cut capacity

TERMS $d(v)$ distance; w edge weight; f, c flow, capacity; s, t source, sink

A chain is only as strong as its weakest link: max flow = min cut.

1. $d(s) = 0, \quad d(v) = \min_{(u,v) \in E} \big[d(u) + w(u,v) \big]$
2. $d(v) \leftarrow \min\{d(v), \ d(u) + w(u,v)\}$
3. Dijkstra $O((V+E)\log V)$; Bellman–Ford $O(VE)$

TERMS $d(v)$ distance; w edge weight; f, c flow, capacity; s, t source, sink

A chain is only as strong as its weakest link: max flow = min cut.

1. $\dfrac{(n-1)!}{2}$
2. $\Delta = c_{ac} + c_{bd} - c_{ab} - c_{cd}$

TERMS c_{ij} edge cost; x_{ij} binary edge choice; S subset of cities

$(n-1)!/2$: *fix the start, and a tour and its reverse are the same.*

1. $\mathbf{x}_{k+1} = \mathbf{x}_k - \eta \nabla f(\mathbf{x}_k)$
2. $0 < \eta < \dfrac{2}{L}, \qquad \eta = \dfrac{1}{L}$ safe
3. $f(\mathbf{x}_k) - f^* \le \dfrac{\|\mathbf{x}_0 - \mathbf{x}^*\|^2}{2\eta k}$

TERMS η step size; L, m max/min curvature; $\kappa = L/m$; β momentum

"Walk downhill; take smaller steps where it is steep." Bigger κ, narrower valley, slower descent.

1. $O(1) < O(\log n) < O(n) < O(n \log n) < O(n^2) < O(n^3) < O(2^n) < O(n!)$
2. P: solvable in poly time; NP: checkable in poly time

TERMS n input size; c, n_0 constants; P poly-time, NP poly-checkable

Drop constants and lower-order terms: $3n^2 + 5n + 7 = O(n^2)$.

1. $f(n) = O(g(n)) \iff \exists c > 0, n_0 : \ f(n) \le c\, g(n) \ \ \forall n \ge n_0$
2. $f = \Omega(g) \iff g = O(f); \quad f = \Theta(g) \iff$ both
3. $3n^2 + 5n + 7 = O(n^2)$

TERMS n input size; c, n_0 constants; P poly-time, NP poly-checkable

Drop constants and lower-order terms: $3n^2 + 5n + 7 = O(n^2)$.

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O.11 (2/3) OPTIMIZATION & SIMULATION

Gradient Descent & Convergence

PART 2 OF 3

1. Strongly convex rate
2. Condition number

STATE EACH NUMBERED FORMULA

O.11 (3/3) OPTIMIZATION & SIMULATION

Gradient Descent & Convergence

PART 3 OF 3

1. Momentum (heavy ball)
2. Nesterov acceleration

STATE EACH NUMBERED FORMULA

O.12 (1/2) OPTIMIZATION & SIMULATION

Line Search: Armijo & Wolfe Conditions

PART 1 OF 2

1. Descent direction
2. Exact line search

STATE EACH NUMBERED FORMULA

O.12 (2/2) OPTIMIZATION & SIMULATION

Line Search: Armijo & Wolfe Conditions

PART 2 OF 2

1. Armijo (sufficient decrease)
2. Curvature (Wolfe)
3. Backtracking

STATE EACH NUMBERED FORMULA

O.13 (1/2) OPTIMIZATION & SIMULATION

Nonlinear Programming & Newton's Method

PART 1 OF 2

1. First-order condition
2. Second-order condition
3. Newton (1-D)

STATE EACH NUMBERED FORMULA

O.13 (2/2) OPTIMIZATION & SIMULATION

Nonlinear Programming & Newton's Method

PART 2 OF 2

1. Newton step (n-D)
2. Quadratic convergence
3. Gauss–Newton

STATE EACH NUMBERED FORMULA

O.14 (1/3) OPTIMIZATION & SIMULATION

Quasi-Newton Methods (BFGS, L-BFGS)

PART 1 OF 3

1. Search direction
2. Step and gradient change
3. Secant condition

STATE EACH NUMBERED FORMULA

O.14 (2/3) OPTIMIZATION & SIMULATION

Quasi-Newton Methods (BFGS, L-BFGS)

PART 2 OF 3

1. BFGS inverse update

STATE EACH NUMBERED FORMULA

O.14 (3/3) OPTIMIZATION & SIMULATION

Quasi-Newton Methods (BFGS, L-BFGS)

PART 3 OF 3

1. Curvature condition
2. L-BFGS cost

STATE EACH NUMBERED FORMULA

O.15 (1/2) OPTIMIZATION & SIMULATION

Coordinate Descent

PART 1 OF 2

1. Exact coordinate update
2. Coordinate gradient step

STATE EACH NUMBERED FORMULA

O.15 (2/2) OPTIMIZATION & SIMULATION

Coordinate Descent

PART 2 OF 2

1. Soft-thresholding
2. Lasso coordinate update

STATE EACH NUMBERED FORMULA

O.16 (1/2) OPTIMIZATION & SIMULATION

Stochastic Optimization & SGD

PART 1 OF 2

1. Stochastic program
2. SGD update
3. Minibatch variance

STATE EACH NUMBERED FORMULA

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1 $\nabla f(\mathbf{x})^\top \mathbf{p} < 0$

2 $\alpha^* = \arg\min_{\alpha>0} \ \phi(\alpha), \quad \phi(\alpha) = f(\mathbf{x} + \alpha \mathbf{p})$

TERMS $\mathbf{p}$ direction; α step; c_1, c_2 constants; ρ shrink factor

Armijo: "not too long". Wolfe curvature: "not too short".

O.12 (1/2)

1 $\mathbf{x}_{k+1} = \mathbf{x}_k - \eta \nabla f(\mathbf{x}_k) + \beta(\mathbf{x}_k - \mathbf{x}_{k-1})$

2 $O(1/k^2)$ convex; $\sim \sqrt{\kappa} \ln \frac{1}{\varepsilon}$ steps

TERMS η step size; L, m max/min curvature; $\kappa = L/m$; β momentum

"Walk downhill; take smaller steps where it is steep." Bigger κ, narrower valley, slower descent.

O.11 (3/3)

1 $f(\mathbf{x}_k) - f^* \leq \left(1 - \frac{1}{\kappa}\right)^k (f(\mathbf{x}_0) - f^*)$

2 $\kappa = \frac{L}{m} = \frac{\lambda_{\max}(\nabla^2 f)}{\lambda_{\min}(\nabla^2 f)}$

TERMS η step size; L, m max/min curvature; $\kappa = L/m$; β momentum

"Walk downhill; take smaller steps where it is steep." Bigger κ, narrower valley, slower descent.

O.11 (2/3)

1 $\mathbf{x}_{k+1} = \mathbf{x}_k - \left[\nabla^2 f(\mathbf{x}_k)\right]^{-1} \nabla f(\mathbf{x}_k)$

2 $\|\mathbf{x}_{k+1} - \mathbf{x}^*\| \leq C \|\mathbf{x}_k - \mathbf{x}^*\|^2$

3 $\mathbf{p} = -(J^\top J)^{-1} J^\top \mathbf{r}$

TERMS ∇f gradient; $\nabla^2 f$ Hessian; J Jacobian; $\mathbf{r}$ residuals

Newton fits a parabola and jumps to its bottom.

O.13 (2/2)

1 $\nabla f(\mathbf{x}^*) = \mathbf{0}$

2 $\nabla^2 f(\mathbf{x}^*) \succ 0$

3 $x_{k+1} = x_k - \frac{f'(x_k)}{f''(x_k)}$

TERMS ∇f gradient; $\nabla^2 f$ Hessian; J Jacobian; $\mathbf{r}$ residuals

Newton fits a parabola and jumps to its bottom.

O.13 (1/2)

1 $\phi(\alpha) \leq \phi(0) + c_1 \alpha \nabla f^\top \mathbf{p}$

2 $\nabla f(\mathbf{x} + \alpha \mathbf{p})^\top \mathbf{p} \geq c_2 \nabla f(\mathbf{x})^\top \mathbf{p}$

3 $\alpha \leftarrow \rho \alpha$ until Armijo holds

TERMS $\mathbf{p}$ direction; α step; c_1, c_2 constants; ρ shrink factor

Armijo: "not too long". Wolfe curvature: "not too short".

O.12 (2/2)

1 $\mathbf{y}_k^\top \mathbf{s}_k > 0$

2 $O(mn)$ time and memory per step

TERMS H_k inverse-Hessian approx.; $\mathbf{s}_k$ step; $\mathbf{y}_k$ gradient change

"Learn the curvature from how the gradient changes."

O.14 (3/3)

1 $H_{k+1} = (I - \rho_k \mathbf{s}_k \mathbf{y}_k^\top) H_k (I - \rho_k \mathbf{y}_k \mathbf{s}_k^\top) + \rho_k \mathbf{s}_k \mathbf{s}_k^\top, \qquad \rho_k = \frac{1}{\mathbf{y}_k^\top \mathbf{s}_k}$

TERMS H_k inverse-Hessian approx.; $\mathbf{s}_k$ step; $\mathbf{y}_k$ gradient change

"Learn the curvature from how the gradient changes."

O.14 (2/3)

1 $\mathbf{p}_k = -H_k \nabla f(\mathbf{x}_k)$

2 $\mathbf{s}_k = \mathbf{x}_{k+1} - \mathbf{x}_k, \quad \mathbf{y}_k = \nabla f_{k+1} - \nabla f_k$

3 $H_{k+1} \mathbf{y}_k = \mathbf{s}_k$

TERMS H_k inverse-Hessian approx.; $\mathbf{s}_k$ step; $\mathbf{y}_k$ gradient change

"Learn the curvature from how the gradient changes."

O.14 (1/3)

1 $\min_{\mathbf{x}} \ f(\mathbf{x}) = E_\xi[F(\mathbf{x}, \xi)]$

2 $\mathbf{x}_{k+1} = \mathbf{x}_k - \eta_k \mathbf{g}_k, \quad E[\mathbf{g}_k] = \nabla f(\mathbf{x}_k)$

3 $\mathrm{Var}(\mathbf{g}_k) = \frac{\sigma^2}{B}$

TERMS $\mathbf{g}_k$ noisy gradient; η_k step schedule; B batch size; ξ random data

"Many cheap noisy steps beat a few expensive exact ones."

O.16 (1/2)

1 $S_\lambda(z) = \mathrm{sign}(z) \max(|z| - \lambda, 0)$

2 $\beta_j \leftarrow \frac{S_\lambda(\mathbf{x}_j^\top \mathbf{r}_{(j)})}{\mathbf{x}_j^\top \mathbf{x}_j}$

TERMS x_i coordinate; L_i coordinate curvature; S_λ soft-threshold

"One knob at a time."

O.15 (2/2)

1 $x_i \leftarrow \arg\min_t \ f(x_1, \ldots, x_{i-1}, t, x_{i+1}, \ldots, x_n)$

2 $x_i \leftarrow x_i - \frac{1}{L_i} \frac{\partial f}{\partial x_i}$

TERMS x_i coordinate; L_i coordinate curvature; S_λ soft-threshold

"One knob at a time."

O.15 (1/2)

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O.16 (2/2) OPTIMIZATION & SIMULATION

Stochastic Optimization & SGD

PART 2 OF 2

1. Robbins–Monro steps
2. Convergence rates
3. Sample average approximation

STATE EACH NUMBERED FORMULA

O.17 (1/2) OPTIMIZATION & SIMULATION

Lagrange Multipliers

PART 1 OF 2

1. Lagrangian
2. Tangency (stationarity)
3. Constraint

STATE EACH NUMBERED FORMULA

O.17 (2/2) OPTIMIZATION & SIMULATION

Lagrange Multipliers

PART 2 OF 2

1. Sensitivity (shadow price)
2. Several constraints

STATE EACH NUMBERED FORMULA

O.18 (1/3) OPTIMIZATION & SIMULATION

KKT Conditions

PART 1 OF 3

1. Stationarity
2. Primal feasibility
3. Dual feasibility

STATE EACH NUMBERED FORMULA

O.18 (2/3) OPTIMIZATION & SIMULATION

KKT Conditions

PART 2 OF 3

1. Complementary slackness

STATE EACH NUMBERED FORMULA

O.18 (3/3) OPTIMIZATION & SIMULATION

KKT Conditions

PART 3 OF 3

1. Sufficiency (convex case)

STATE EACH NUMBERED FORMULA

O.19 (1/2) OPTIMIZATION & SIMULATION

Lagrangian Duality

PART 1 OF 2

1. Lagrangian
2. Dual function
3. Dual problem

STATE EACH NUMBERED FORMULA

O.19 (2/2) OPTIMIZATION & SIMULATION

Lagrangian Duality

PART 2 OF 2

1. Weak duality
2. Strong duality (Slater)
3. Saddle point

STATE EACH NUMBERED FORMULA

O.20 (1/3) OPTIMIZATION & SIMULATION

Penalty, Barrier & Projection Methods

PART 1 OF 3

1. Quadratic penalty
2. Log barrier
3. Central path gap

STATE EACH NUMBERED FORMULA

O.20 (2/3) OPTIMIZATION & SIMULATION

Penalty, Barrier & Projection Methods

PART 2 OF 3

1. Augmented Lagrangian

STATE EACH NUMBERED FORMULA

O.20 (3/3) OPTIMIZATION & SIMULATION

Penalty, Barrier & Projection Methods

PART 3 OF 3

1. Projected gradient

STATE EACH NUMBERED FORMULA

O.21 (1/2) OPTIMIZATION & SIMULATION

Dynamic Programming & the Bellman Equation

PART 1 OF 2

1. Bellman optimality equation
2. Finite-horizon recursion
3. Requirements

STATE EACH NUMBERED FORMULA

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1 $\lambda = \dfrac{\partial f^*}{\partial c}$

2 $\nabla f = \sum_{i=1}^{k} \lambda_i \nabla g_i, \quad g_i(\mathbf{x}) = c_i$

TERMS $\mathcal{L}$ Lagrangian; λ multiplier (shadow price); $g = c$ constraint

At the optimum the level curve kisses (is tangent to) the constraint.

1 $\mathcal{L}(\mathbf{x}, \lambda) = f(\mathbf{x}) - \lambda\big(g(\mathbf{x}) - c\big)$

2 $\nabla f(\mathbf{x}) = \lambda\, \nabla g(\mathbf{x})$

3 $g(\mathbf{x}) = c$

TERMS $\mathcal{L}$ Lagrangian; λ multiplier (shadow price); $g = c$ constraint

At the optimum the level curve kisses (is tangent to) the constraint.

1 $\sum_k \eta_k = \infty, \qquad \sum_k \eta_k^2 < \infty$

2 $O(1/\sqrt{k})$ convex; $\quad O(1/k)$ strongly convex

3 $\min_{\mathbf{x}} \; \dfrac{1}{N} \sum_{i=1}^{N} F(\mathbf{x}, \xi_i)$

TERMS $\mathbf{g}_k$ noisy gradient; η_k step schedule; B batch size; ξ random data

"Many cheap noisy steps beat a few expensive exact ones."

1 f, g_i convex, h_j affine $\Rightarrow$ KKT = global optimal

TERMS $\mu_i \geq 0$ inequality multipliers; ν_j equality multipliers; active: $g_i = 0$

Complementary slackness: either the constraint is tight or its multiplier is zero.

1 $\mu_i \, g_i(\mathbf{x}^*) = 0$

TERMS $\mu_i \geq 0$ inequality multipliers; ν_j equality multipliers; active: $g_i = 0$

Complementary slackness: either the constraint is tight or its multiplier is zero.

1 $\nabla f(\mathbf{x}^*) + \sum_i \mu_i \nabla g_i(\mathbf{x}^*) + \sum_j \nu_j \nabla h_j(\mathbf{x}^*) = \mathbf{0}$

2 $g_i(\mathbf{x}^*) \leq 0, \quad h_j(\mathbf{x}^*) = 0$

3 $\mu_i \geq 0$

TERMS $\mu_i \geq 0$ inequality multipliers; ν_j equality multipliers; active: $g_i = 0$

Complementary slackness: either the constraint is tight or its multiplier is zero.

1 $\min_{\mathbf{x}} \; f(\mathbf{x}) + \dfrac{\rho}{2}\Big[\sum_i \max\big(0, g_i(\mathbf{x})\big)^2 + \sum_j h_j(\mathbf{x})^2\Big]$

2 $\min_{\mathbf{x}} \; f(\mathbf{x}) - \mu \sum_i \ln\big(-g_i(\mathbf{x})\big)$

3 $f(\mathbf{x}^*(\mu)) - p^* \leq m\,\mu$

TERMS ρ penalty weight; μ barrier parameter; Π_C projection

Penalty: "a fine for trespassing". Barrier: "an electric fence". Projection: "step, then snap back".

1 $d^* \leq p^*$

2 convex $+ \; \exists\, \tilde{\mathbf{x}} : \; g_i(\tilde{\mathbf{x}}) < 0 \;\Rightarrow\; d^* = p^*$

3 $\min_{\mathbf{x}} \max_{\boldsymbol{\mu} \geq 0} \mathcal{L} = \max_{\boldsymbol{\mu} \geq 0} \min_{\mathbf{x}} \mathcal{L}$

TERMS q dual function; $\mu \geq 0$; $d^* \leq p^*$; gap $p^* - d^*$

The dual is a "price the constraints, then let x respond" game; its best price never beats the true optimum.

1 $\mathcal{L}(\mathbf{x}, \boldsymbol{\mu}, \boldsymbol{\nu}) = f(\mathbf{x}) + \sum_i \mu_i g_i(\mathbf{x}) + \sum_j \nu_j h_j(\mathbf{x})$

2 $q(\boldsymbol{\mu}, \boldsymbol{\nu}) = \inf_{\mathbf{x}} \; \mathcal{L}(\mathbf{x}, \boldsymbol{\mu}, \boldsymbol{\nu})$

3 $d^* = \max_{\boldsymbol{\mu} \geq \mathbf{0},\ \boldsymbol{\nu}} \; q(\boldsymbol{\mu}, \boldsymbol{\nu})$

TERMS q dual function; $\mu \geq 0$; $d^* \leq p^*$; gap $p^* - d^*$

The dual is a "price the constraints, then let x respond" game; its best price never beats the true optimum.

1 $V(s) = \max_a \Big[r(s,a) + \gamma \sum_{s'} P(s' \mid s, a)\, V(s') \Big]$

2 $V_t(s) = \max_a \big[r(s,a) + V_{t+1}(s') \big], \quad V_T \equiv 0$

3 optimal substructure + overlapping subproblems

TERMS V value; r reward; γ discount; $P(s' \mid s, a)$ transitions; Q action value

"Solve each subproblem once and remember it."

1 $\mathbf{x}_{k+1} = \Pi_C\big(\mathbf{x}_k - \eta \nabla f(\mathbf{x}_k)\big)$

TERMS ρ penalty weight; μ barrier parameter; Π_C projection

Penalty: "a fine for trespassing". Barrier: "an electric fence". Projection: "step, then snap back".

1 $\mathcal{L}_\rho = f + \boldsymbol{\nu}^\top \mathbf{h} + \dfrac{\rho}{2}\|\mathbf{h}\|^2, \qquad \boldsymbol{\nu} \leftarrow \boldsymbol{\nu} + \rho\, \mathbf{h}(\mathbf{x})$

TERMS ρ penalty weight; μ barrier parameter; Π_C projection

Penalty: "a fine for trespassing". Barrier: "an electric fence". Projection: "step, then snap back".

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O.21 (2/2) OPTIMIZATION & SIMULATION

Dynamic Programming & the Bellman Equation

PART 2 OF 2

1. Q-function
2. Value iteration
3. Discounted constant reward

STATE EACH NUMBERED FORMULA

O.22 (1/2) OPTIMIZATION & SIMULATION

Heuristics & Simulated Annealing

PART 1 OF 2

1. Metropolis acceptance
2. Cost change

STATE EACH NUMBERED FORMULA

O.22 (2/2) OPTIMIZATION & SIMULATION

Heuristics & Simulated Annealing

PART 2 OF 2

1. Geometric cooling
2. Logarithmic cooling

STATE EACH NUMBERED FORMULA

O.23 (1/2) OPTIMIZATION & SIMULATION

Local Search, Greedy & Tabu Search

PART 1 OF 2

1. Neighborhood
2. Hill climbing (descent)
3. Greedy construction

STATE EACH NUMBERED FORMULA

O.23 (2/2) OPTIMIZATION & SIMULATION

Local Search, Greedy & Tabu Search

PART 2 OF 2

1. Tabu move
2. Aspiration criterion
3. Iterated local search

STATE EACH NUMBERED FORMULA

O.24 (1/2) OPTIMIZATION & SIMULATION

Genetic Algorithms & Population Methods

PART 1 OF 2

1. Fitness-proportional selection
2. Tournament selection
3. One-point crossover

STATE EACH NUMBERED FORMULA

O.24 (2/2) OPTIMIZATION & SIMULATION

Genetic Algorithms & Population Methods

PART 2 OF 2

1. Mutation
2. Elitism
3. Particle swarm velocity

STATE EACH NUMBERED FORMULA

O.25 (1/2) OPTIMIZATION & SIMULATION

Pseudo-Random Number Generation

PART 1 OF 2

1. Linear congruential generator
2. Uniform output
3. Full period (Hull–Dobell)

STATE EACH NUMBERED FORMULA

O.25 (2/2) OPTIMIZATION & SIMULATION

Pseudo-Random Number Generation

PART 2 OF 2

1. Box–Muller
2. Lattice structure

STATE EACH NUMBERED FORMULA

O.26 (1/2) OPTIMIZATION & SIMULATION

Random Sampling Methods

PART 1 OF 2

1. Inverse transform
2. Exponential case
3. Discrete inverse

STATE EACH NUMBERED FORMULA

O.26 (2/2) OPTIMIZATION & SIMULATION

Random Sampling Methods

PART 2 OF 2

1. Rejection sampling
2. Importance sampling
3. Uniform on $[a, b]$

STATE EACH NUMBERED FORMULA

O.27 (1/2) OPTIMIZATION & SIMULATION

Monte Carlo Estimation & Integration

PART 1 OF 2

1. Monte Carlo estimator
2. Monte Carlo integration
3. Standard error

STATE EACH NUMBERED FORMULA

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1 $T_k = T_0\,\alpha^k \quad (0 < \alpha < 1)$

2 $T_k = \dfrac{c}{\ln(1+k)}$

TERMS Δ cost change; T temperature; α cooling rate

Hot → explore; cool → settle.

O.22 (2/2)

1 $P(\text{accept}) = \begin{cases} 1, & \Delta \le 0 \\ e^{-\Delta/T}, & \Delta > 0 \end{cases}$

2 $\Delta = f(\mathbf{x}_{\text{new}}) - f(\mathbf{x}_{\text{current}})$

TERMS Δ cost change; T temperature; α cooling rate

Hot → explore; cool → settle.

O.22 (1/2)

1 $Q(s,a) = r(s,a) + \gamma \sum_{s'} P(s' \mid s,a) \max_{a'} Q(s',a')$

2 $V_{k+1} = \mathcal{T} V_k, \quad \|V_k - V^*\| \le \gamma^k \|V_0 - V^*\|$

3 $\displaystyle\sum_{t=0}^{\infty} \gamma^t r = \frac{r}{1-\gamma}$

TERMS V value; r reward; γ discount; $P(s' \mid s,a)$ transitions; Q action value

"Solve each subproblem once and remember it."

O.21 (2/2)

1 $p_i = \dfrac{F_i}{\sum_j F_j}$

2 pick k at random, keep the best

3 $(a_{1:c}, b_{c+1:L}),\ (b_{1:c}, a_{c+1:L})$

TERMS F_i fitness; p_m mutation rate; L length; $\mathbf{p}, \mathbf{g}$ best positions

Survival of the fittest, mixing, and mutation: select, cross, mutate.

O.24 (1/2)

1 $\mathbf{x} \leftarrow \arg\min_{\mathbf{y} \in N(\mathbf{x}) \setminus \mathcal{T}} f(\mathbf{y})$

2 allow tabu $\mathbf{y}$ if $f(\mathbf{y}) < f_{\text{best}}$

3 $\mathbf{x} \leftarrow \text{LocalSearch}(\text{Perturb}(\mathbf{x}))$

TERMS $N(\mathbf{x})$ neighborhood; $\mathcal{T}$ tabu list; k tenure; aspiration

Tabu: "don't undo what you just did."

O.23 (2/2)

1 $N(\mathbf{x}) = \{\text{solutions one move from } \mathbf{x}\}$

2 $\mathbf{x} \leftarrow \arg\min_{\mathbf{y} \in N(\mathbf{x})} f(\mathbf{y})$ if $f(\mathbf{y}) < f(\mathbf{x})$

3 add the element with best immediate gain

TERMS $N(\mathbf{x})$ neighborhood; $\mathcal{T}$ tabu list; k tenure; aspiration

Tabu: "don't undo what you just did."

O.23 (1/2)

1 $Z = \sqrt{-2 \ln U_1}\,\cos(2\pi U_2)$

2 $(U_n, U_{n+1}, \dots)$ lie on few hyperplanes

TERMS X_0 seed; a, c, m multiplier, increment, modulus; $U = X/m$

Same seed, same "random" numbers: reproducible by design.

O.25 (2/2)

1 $X_{n+1} = (aX_n + c) \bmod m$

2 $U_n = \dfrac{X_n}{m} \in [0,1)$

3 $\gcd(c,m) = 1;\ \ p \mid m \Rightarrow p \mid a-1;\ \ 4 \mid m \Rightarrow 4 \mid a-1$

TERMS X_0 seed; a, c, m multiplier, increment, modulus; $U = X/m$

Same seed, same "random" numbers: reproducible by design.

O.25 (1/2)

1 flip each bit w.p. $p_m \approx \dfrac{1}{L}$

2 copy the best e individuals unchanged

3 $\mathbf{v} \leftarrow \omega\mathbf{v} + c_1 r_1 (\mathbf{p} - \mathbf{x}) + c_2 r_2 (\mathbf{g} - \mathbf{x}), \qquad \mathbf{x} \leftarrow \mathbf{x} + \mathbf{v}$

TERMS F_i fitness; p_m mutation rate; L length; $\mathbf{p}, \mathbf{g}$ best positions

Survival of the fittest, mixing, and mutation: select, cross, mutate.

O.24 (2/2)

1 $\hat{\theta}_N = \dfrac{1}{N} \displaystyle\sum_{i=1}^{N} f(X_i) \ \longrightarrow\ E\big[f(X)\big]$

2 $\displaystyle\int_a^b f(x)\,dx \approx \frac{b-a}{N} \sum_{i=1}^{N} f(U_i)$

3 $SE = \dfrac{\hat{\sigma}}{\sqrt{N}}$

TERMS N samples; $\hat{\sigma}$ sample SD; $U_i \sim \text{Unif}(a,b)$; z critical value

Error shrinks like $1/\sqrt{N}$: 100 times the samples for one more digit.

O.27 (1/2)

1 accept $x \sim g$ if $U \le \dfrac{f(x)}{M\,g(x)}$

2 $E_p\big[h(X)\big] = E_q\Big[h(X)\dfrac{p(X)}{q(X)}\Big]$

3 $X = a + (b-a)U$

TERMS $U \sim \text{Unif}(0,1)$; F^{-1} inverse CDF; M envelope bound; p/q weights

"Read the CDF backwards": pick a height U, find the x where the CDF reaches it.

O.26 (2/2)

1 $X = F^{-1}(U), \quad U \sim \text{Unif}(0,1)$

2 $X = -\dfrac{\ln(1-U)}{\lambda}$

3 $X = x_k$ where $F(x_{k-1}) < U \le F(x_k)$

TERMS $U \sim \text{Unif}(0,1)$; F^{-1} inverse CDF; M envelope bound; p/q weights

"Read the CDF backwards": pick a height U, find the x where the CDF reaches it.

O.26 (1/2)

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O.27 (2/2) OPTIMIZATION & SIMULATION

Monte Carlo Estimation & Integration

PART 2 OF 2

1. Confidence interval
2. Sample size
3. Estimating π

STATE EACH NUMBERED FORMULA

O.28 (1/3) OPTIMIZATION & SIMULATION

Variance Reduction Techniques

PART 1 OF 3

1. Antithetic variates
2. Antithetic variance

STATE EACH NUMBERED FORMULA

O.28 (2/3) OPTIMIZATION & SIMULATION

Variance Reduction Techniques

PART 2 OF 3

1. Control variate
2. Optimal coefficient

STATE EACH NUMBERED FORMULA

O.28 (3/3) OPTIMIZATION & SIMULATION

Variance Reduction Techniques

PART 3 OF 3

1. Importance sampling
2. Optimal proposal
3. Stratified sampling

STATE EACH NUMBERED FORMULA

O.29 (1/2) OPTIMIZATION & SIMULATION

Markov Chain Monte Carlo (Metropolis–Hastings)

PART 1 OF 2

1. Metropolis–Hastings acceptance
2. Symmetric proposal (Metropolis)
3. Detailed balance

STATE EACH NUMBERED FORMULA

O.29 (2/2) OPTIMIZATION & SIMULATION

Markov Chain Monte Carlo (Metropolis–Hastings)

PART 2 OF 2

1. Gibbs sampling
2. Effective sample size

STATE EACH NUMBERED FORMULA

O.30 (1/2) OPTIMIZATION & SIMULATION

Queueing Basics: Little's Law & M/M/1

PART 1 OF 2

1. Little's law
2. Utilization
3. M/M/1 state probabilities

STATE EACH NUMBERED FORMULA

O.30 (2/2) OPTIMIZATION & SIMULATION

Queueing Basics: Little's Law & M/M/1

PART 2 OF 2

1. M/M/1 number in system
2. M/M/1 time in system
3. Kendall notation

STATE EACH NUMBERED FORMULA

O.31 (1/2) OPTIMIZATION & SIMULATION

Discrete-Event Simulation

PART 1 OF 2

1. Next-event time advance
2. Arrival times
3. Lindley recursion

STATE EACH NUMBERED FORMULA

O.31 (2/2) OPTIMIZATION & SIMULATION

Discrete-Event Simulation

PART 2 OF 2

1. Time-average estimator
2. Replication interval

STATE EACH NUMBERED FORMULA

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1 $\hat{\theta}_c = \bar{Y} - c\,(\bar{X} - \mu_X)$

2 $c^* = \dfrac{\mathrm{Cov}(Y,X)}{\mathrm{Var}(X)}, \quad \mathrm{Var} = (1-\rho_{XY}^2)\dfrac{\sigma_Y^2}{N}$

TERMS ρ correlation; c^* control coeff.; p/q weights; w_s stratum weight

Same answer, less noise: mirror it, correct it, focus it, or split it.

1 $\hat{\theta} = \dfrac{1}{N}\sum_{i=1}^{N} \dfrac{f(U_i) + f(1-U_i)}{2}$

2 $\mathrm{Var} = \dfrac{\sigma^2}{2N}\,(1+\rho)$

TERMS ρ correlation; c^* control coeff.; p/q weights; w_s stratum weight

Same answer, less noise: mirror it, correct it, focus it, or split it.

1 $\hat{\theta}_N \pm 1.96\,\dfrac{\hat{\sigma}}{\sqrt{N}}$

2 $N = \left(\dfrac{z\,\hat{\sigma}}{\varepsilon}\right)^2$

3 $\hat{\pi} = 4 \cdot \dfrac{\#\{U_1^2 + U_2^2 \le 1\}}{N}$

TERMS N samples; $\hat{\sigma}$ sample SD; $U_i \sim \mathrm{Unif}(a,b)$; z critical value

Error shrinks like $1/\sqrt{N}$: 100 times the samples for one more digit.

1 $x_i \sim p(x_i \mid \mathbf{x}_{-i})$

2 $\mathrm{ESS} = \dfrac{N}{1 + 2\sum_{k\ge 1}\rho_k}$

TERMS p target (unnormalized); q proposal; α acceptance prob.; ESS

Always accept uphill in probability; accept downhill sometimes, with probability equal to the ratio.

1 $\alpha(x \to x') = \min\left\{1,\ \dfrac{p(x')\,q(x \mid x')}{p(x)\,q(x' \mid x)}\right\}, \quad$ accept if $U \le \alpha$

2 $\alpha = \min\left\{1,\ \dfrac{p(x')}{p(x)}\right\}$

3 $p(x)\,P(x \to x') = p(x')\,P(x' \to x)$

TERMS p target (unnormalized); q proposal; α acceptance prob.; ESS

Always accept uphill in probability; accept downhill sometimes, with probability equal to the ratio.

1 $\hat{\theta}_{IS} = \dfrac{1}{N}\sum_{i=1}^{N} h(X_i)\dfrac{p(X_i)}{q(X_i)}, \quad X_i \sim q$

2 $q^*(x) \propto |h(x)|\,p(x)$

3 $\hat{\theta}_{st} = \sum_s w_s\,\bar{Y}_s$

TERMS ρ correlation; c^* control coeff.; p/q weights; w_s stratum weight

Same answer, less noise: mirror it, correct it, focus it, or split it.

1 $t \leftarrow \min\{\text{scheduled event times}\}$

2 $A_n = A_{n-1} + T_n$

3 $W_{n+1} = \max\left(0,\ W_n + S_n - T_{n+1}\right)$

TERMS t clock; A_n, T_n arrival, gap; W_n, S_n wait, service; R replications

"Jump from event to event; nothing happens in between."

1 $L = \dfrac{\rho}{1-\rho}, \qquad L_q = \dfrac{\rho^2}{1-\rho}$

2 $W = \dfrac{1}{\mu-\lambda}, \qquad W_q = \dfrac{\rho}{\mu-\lambda}$

3 $A/S/c$

TERMS λ arrival rate; μ service rate; $\rho = \lambda/\mu$; L number; W time

"L equals lambda W": the line is as long as the rate times the wait. Near 100% busy, waits explode.

1 $L = \lambda W$

2 $\rho = \dfrac{\lambda}{\mu} < 1$

3 $P(N = n) = (1-\rho)\rho^n$

TERMS λ arrival rate; μ service rate; $\rho = \lambda/\mu$; L number; W time

"L equals lambda W": the line is as long as the rate times the wait. Near 100% busy, waits explode.

1 $\bar{L} = \dfrac{1}{T_{\text{end}}}\displaystyle\int_0^{T_{\text{end}}} N(t)\,dt$

2 $\bar{x} \pm t_{R-1}\,\dfrac{s}{\sqrt{R}}$

TERMS t clock; A_n, T_n arrival, gap; W_n, S_n wait, service; R replications

"Jump from event to event; nothing happens in between."

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ML.1 (1/2)

MACHINE LEARNING

The Learning Problem & Empirical Risk

PART 1 OF 2

1. Regularized empirical risk
2. Expected (true) risk
3. Generalization gap

STATE EACH NUMBERED FORMULA

ML.1 (2/2)

MACHINE LEARNING

The Learning Problem & Empirical Risk

PART 2 OF 2

1. Likelihood as a loss

STATE EACH NUMBERED FORMULA

ML.K1

MACHINE LEARNING

Choosing a model: first family by task & data

RECALL IT

ML.K1

MACHINE LEARNING

Choosing a metric: which for which situation?

RECALL IT

ML.2 (1/2)

MACHINE LEARNING

Loss Functions

PART 1 OF 2

1. Mean squared error
2. Mean absolute error
3. Huber loss

STATE EACH NUMBERED FORMULA

ML..2 (2/2)

MACHINE LEARNING

Loss Functions

PART 2 OF 2

1. Log-loss (binary cross-entropy)
2. Hinge loss
3. Quantile (pinball) loss

STATE EACH NUMBERED FORMULA

ML.3 (1/2)

MACHINE LEARNING

Entropy, Cross-Entropy & KL Divergence

PART 1 OF 2

1. Entropy
2. Cross-entropy
3. KL divergence

STATE EACH NUMBERED FORMULA

ML.3 (2/2)

MACHINE LEARNING

Entropy, Cross-Entropy & KL Divergence

PART 2 OF 2

1. Decomposition
2. Mutual information

STATE EACH NUMBERED FORMULA

ML.4

MACHINE LEARNING

Bias–Variance Trade-off

1. Bias–variance decomposition
2. Bias
3. Variance

STATE EACH NUMBERED FORMULA

ML.5 (1/2)

MACHINE LEARNING

Cross-Validation

PART 1 OF 2

1. k-fold CV estimate
2. Fold sizes
3. Leave-one-out

STATE EACH NUMBERED FORMULA

ML.5 (2/2)

MACHINE LEARNING

Cross-Validation

PART 2 OF 2

1. Standard error of CV

STATE EACH NUMBERED FORMULA

ML.6 (1/2)

MACHINE LEARNING

Feature Scaling & Encoding

PART 1 OF 2

1. Standardization
2. Min–max scaling
3. Robust scaling

STATE EACH NUMBERED FORMULA

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Tabular: gradient-boosted trees (+ linear baseline).
Images: CNN / pretrained. Text: transformer.
Small n or must explain: regularized linear, tree.
Clusters: k-means/GMM. Compress: PCA.

1 $L = -\log p_\theta(y \mid \mathbf{x}) \;\Rightarrow\; \text{ERM} = \text{MLE}$

TERMS θ params; L loss; λR penalty; $\mathcal{D}$ data distribution

Train to fit, validate to choose, test once to report.

1 $\hat{\theta} = \arg\min_\theta \underbrace{\frac{1}{n}\sum_{i=1}^{n} L\big(y_i,\ f_\theta(\mathbf{x}_i)\big)}_{\text{empirical risk } \hat{R}_n(\theta)} + \underbrace{\lambda\, R(\theta)}_{\text{regularization}}$

2 $R(\theta) = E_{(\mathbf{x},y)\sim\mathcal{D}}\Big[L\big(y, f_\theta(\mathbf{x})\big)\Big]$

3 $R(\hat{\theta}) - \hat{R}_n(\hat{\theta})$

TERMS θ params; L loss; λR penalty; $\mathcal{D}$ data distribution

Train to fit, validate to choose, test once to report.

1 $-\big[y \ln p + (1-y)\ln(1-p)\big]$

2 $\max\big(0,\ 1 - y\, f(\mathbf{x})\big)$

3 $L_\tau(r) = \max\big(\tau r,\ (\tau - 1)r\big)$

TERMS $r = y - \hat{y}$; p prob.; δ Huber cutoff; τ quantile

MSE → mean; MAE → median; pinball → quantile.

1 $\text{MSE} = \frac{1}{n}\sum_i (y_i - \hat{y}_i)^2$

2 $\text{MAE} = \frac{1}{n}\sum_i |y_i - \hat{y}_i|$

3 $L_\delta(r) = \begin{cases} \frac{1}{2}r^2, & |r| \le \delta \\ \delta\big(|r| - \frac{1}{2}\delta\big), & |r| > \delta \end{cases}$

TERMS $r = y - \hat{y}$; p prob.; δ Huber cutoff; τ quantile

MSE → mean; MAE → median; pinball → quantile.

Regression: MAE (robust), RMSE (big misses).
Rare positives: PR AUC, F_1. Ranking: ROC AUC.
Probabilities: log-loss, Brier, calibration.
Always compare to the trivial baseline.

1 $E\Big[\big(y - \hat{f}(\mathbf{x})\big)^2\Big] = \underbrace{\big(E[\hat{f}(\mathbf{x})] - f(\mathbf{x})\big)^2}_{\text{Bias}^2} + \underbrace{\text{Var}\big(\hat{f}(\mathbf{x})\big)}_{\text{Variance}} + \underbrace{\sigma^2}_{\text{irreducible noise}}$

2 $\text{Bias} = E[\hat{f}(\mathbf{x})] - f(\mathbf{x})$

3 $\text{Var}(\hat{f}) = E\Big[\big(\hat{f} - E[\hat{f}]\big)^2\Big]$

TERMS f truth; $\hat{f}$ fitted model; σ^2 noise variance

Too simple → bias (underfit); too complex → variance (overfit).

1 $H(p, q) = H(p) + D_{\text{KL}}(p \,\|\, q)$

2 $I(X;Y) = H(Y) - H(Y \mid X)$

TERMS p true dist.; q model dist.; log base 2 = bits, e = nats

KL is not symmetric: $D(p\|q) \neq D(q\|p)$.

1 $H(p) = -\sum_k p_k \log p_k$

2 $H(p, q) = -\sum_k p_k \log q_k$

3 $D_{\text{KL}}(p \,\|\, q) = \sum_k p_k \log \frac{p_k}{q_k} \ge 0$

TERMS p true dist.; q model dist.; log base 2 = bits, e = nats

KL is not symmetric: $D(p\|q) \neq D(q\|p)$.

1 $z = \frac{x - \mu}{\sigma}$

2 $x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \in [0, 1]$

3 $x' = \frac{x - \text{median}}{\text{IQR}}$

TERMS μ, σ train mean/SD; IQR $Q_3 - Q_1$; n_c category count; m smoothing

Fit the scaler on train, apply it to test.

1 $SE_{CV} \approx \frac{s_{\text{Err}}}{\sqrt{k}}$

TERMS k folds; Err_j fold-j error; n rows; s SD of fold errors

Every row gets one turn as the test set.

1 $\text{CV}_{(k)} = \frac{1}{k}\sum_{j=1}^{k} \text{Err}_j$

2 train on $\frac{k-1}{k}n$, validate on $\frac{n}{k}$

3 LOOCV: $k = n$

TERMS k folds; Err_j fold-j error; n rows; s SD of fold errors

Every row gets one turn as the test set.

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ML.6 (2/2) MACHINE LEARNING

Feature Scaling & Encoding

PART 2 OF 2

1. Log transform
2. One-hot encoding
3. Target (mean) encoding

STATE EACH NUMBERED FORMULA

ML.7 (1/3) MACHINE LEARNING

Linear Regression: Closed Form & Gradient

PART 1 OF 3

1. Model
2. Least-squares objective
3. Normal equations

STATE EACH NUMBERED FORMULA

ML.7 (2/3) MACHINE LEARNING

Linear Regression: Closed Form & Gradient

PART 2 OF 3

1. Closed form (OLS)
2. Gradient
3. Gradient-descent step

STATE EACH NUMBERED FORMULA

ML.7 (3/3) MACHINE LEARNING

Linear Regression: Closed Form & Gradient

PART 3 OF 3

1. Simple regression

STATE EACH NUMBERED FORMULA

ML.8 (1/2) MACHINE LEARNING

Regularization: Ridge, Lasso & Elastic Net

PART 1 OF 2

1. Ridge objective
2. Ridge closed form
3. Lasso objective

STATE EACH NUMBERED FORMULA

ML.8 (2/2) MACHINE LEARNING

Regularization: Ridge, Lasso & Elastic Net

PART 2 OF 2

1. Norms
2. Elastic net
3. Weight decay (gradient)

STATE EACH NUMBERED FORMULA

ML.9 (1/2) MACHINE LEARNING

Logistic Regression

PART 1 OF 2

1. Model
2. Log-odds (logit)
3. Odds ratio

STATE EACH NUMBERED FORMULA

ML.9 (2/2) MACHINE LEARNING

Logistic Regression

PART 2 OF 2

1. Log-loss gradient
2. Decision boundary

STATE EACH NUMBERED FORMULA

ML.10 (1/2) MACHINE LEARNING

Naive Bayes Classifier

PART 1 OF 2

1. Posterior
2. Decision rule (logs)
3. Laplace smoothing

STATE EACH NUMBERED FORMULA

ML.10 (2/2) MACHINE LEARNING

Naive Bayes Classifier

PART 2 OF 2

1. Gaussian likelihood

STATE EACH NUMBERED FORMULA

ML.11 (1/2) MACHINE LEARNING

k-Nearest Neighbors & Distance Metrics

PART 1 OF 2

1. Euclidean distance
2. Manhattan distance
3. Minkowski distance

STATE EACH NUMBERED FORMULA

ML.11 (2/2) MACHINE LEARNING

k-Nearest Neighbors & Distance Metrics

PART 2 OF 2

1. Cosine similarity
2. k-NN prediction

STATE EACH NUMBERED FORMULA

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1 $\hat{\boldsymbol{\beta}} = (X^\top X)^{-1} X^\top \mathbf{y}$

2 $\nabla_{\boldsymbol{\beta}} J = -\dfrac{2}{n} X^\top (\mathbf{y} - X\boldsymbol{\beta})$

3 $\boldsymbol{\beta} \leftarrow \boldsymbol{\beta} + \eta \dfrac{2}{n} X^\top (\mathbf{y} - X\boldsymbol{\beta})$

TERMS X design matrix; $\boldsymbol{\beta}$ coefficients; η learning rate

Normal equations: residuals are normal *(perpendicular) to the features.*

1 $\hat{\mathbf{y}} = X\boldsymbol{\beta}$

2 $J(\boldsymbol{\beta}) = \dfrac{1}{n} \|\mathbf{y} - X\boldsymbol{\beta}\|_2^2$

3 $X^\top X\, \boldsymbol{\beta} = X^\top \mathbf{y}$

TERMS X design matrix; $\boldsymbol{\beta}$ coefficients; η learning rate

Normal equations: residuals are normal *(perpendicular) to the features.*

1 $x' = \log(1 + x)$

2 color $\in \{r, g, b\} \mapsto (1,0,0), (0,1,0), (0,0,1)$

3 $x'_c = \dfrac{n_c \bar{y}_c + m\, \bar{y}}{n_c + m}$

TERMS μ, σ train mean/SD; IQR $Q_3 - Q_1$; n_c category count; m smoothing

Fit the scaler on train, apply it to test.

1 $\|\boldsymbol{\beta}\|_1 = \sum_j |\beta_j|, \quad \|\boldsymbol{\beta}\|_2^2 = \sum_j \beta_j^2$

2 $\lambda\left[\alpha \|\boldsymbol{\beta}\|_1 + (1-\alpha)\|\boldsymbol{\beta}\|_2^2\right]$

3 $\nabla\left(\lambda \|\boldsymbol{\beta}\|_2^2\right) = 2\lambda\boldsymbol{\beta}$

TERMS λ strength; α L1 mix; I identity; $\|\cdot\|_1, \|\cdot\|_2$ norms

Lasso has a Lasso that ropes coefficients to exactly zero; ridge just shrinks.

1 $\min_{\boldsymbol{\beta}} \|\mathbf{y} - X\boldsymbol{\beta}\|_2^2 + \lambda \|\boldsymbol{\beta}\|_2^2$

2 $\hat{\boldsymbol{\beta}} = \left(X^\top X + \lambda I\right)^{-1} X^\top \mathbf{y}$

3 $\min_{\boldsymbol{\beta}} \|\mathbf{y} - X\boldsymbol{\beta}\|_2^2 + \lambda \|\boldsymbol{\beta}\|_1$

TERMS λ strength; α L1 mix; I identity; $\|\cdot\|_1, \|\cdot\|_2$ norms

Lasso has a Lasso that ropes coefficients to exactly zero; ridge just shrinks.

1 $\hat{\beta}_1 = \dfrac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$

TERMS X design matrix; $\boldsymbol{\beta}$ coefficients; η learning rate

Normal equations: residuals are normal *(perpendicular) to the features.*

1 $P(y \mid \mathbf{x}) \;\propto\; P(y) \prod_{j=1}^{d} P(x_j \mid y)$

2 $\hat{y} = \arg\max_y \left[\log P(y) + \sum_j \log P(x_j \mid y) \right]$

3 $\hat{P}(x_j \mid y) = \dfrac{n_{jy} + 1}{n_y + V}$

TERMS $P(y)$ prior; n_{jy} counts; V vocabulary size; d features

Naive = pretend the features don't talk to each other.

1 $\nabla_{\boldsymbol{\beta}} L = \dfrac{1}{n} X^\top (\mathbf{p} - \mathbf{y})$

2 $p = 0.5 \iff \beta_0 + \boldsymbol{\beta}^\top \mathbf{x} = 0$

TERMS z linear score; σ sigmoid; β_j coefficient; p probability

Logistic regression is linear in the log-odds.

1 $p = P(y = 1 \mid \mathbf{x}) = \sigma(z) = \dfrac{1}{1 + e^{-z}}, \quad z = \beta_0 + \boldsymbol{\beta}^\top \mathbf{x}$

2 $\ln \dfrac{p}{1-p} = \beta_0 + \boldsymbol{\beta}^\top \mathbf{x}$

3 $\mathrm{OR}_j = e^{\beta_j}$

TERMS z linear score; σ sigmoid; β_j coefficient; p probability

Logistic regression is linear in the log-odds.

1 $\cos\theta = \dfrac{\mathbf{x} \cdot \mathbf{z}}{\|\mathbf{x}\|\, \|\mathbf{z}\|}$

2 $\hat{y} =$ majority (or mean) of $N_k(\mathbf{x})$

TERMS $\mathbf{x}, \mathbf{z}$ vectors; N_k k nearest; p Minkowski order

"You are the company you keep."

1 $d_2(\mathbf{x}, \mathbf{z}) = \sqrt{\sum_j (x_j - z_j)^2}$

2 $d_1(\mathbf{x}, \mathbf{z}) = \sum_j |x_j - z_j|$

3 $d_p(\mathbf{x}, \mathbf{z}) = \left(\sum_j |x_j - z_j|^p \right)^{1/p}$

TERMS $\mathbf{x}, \mathbf{z}$ vectors; N_k k nearest; p Minkowski order

"You are the company you keep."

1 $P(x_j \mid y) = \dfrac{1}{\sqrt{2\pi\sigma_{jy}^2}} e^{-(x_j - \mu_{jy})^2 / 2\sigma_{jy}^2}$

TERMS $P(y)$ prior; n_{jy} counts; V vocabulary size; d features

Naive = pretend the features don't talk to each other.

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ML.12 (1/2) MACHINE LEARNING

Decision Trees: Gini, Entropy & Information Gain

PART 1 OF 2

1. Gini impurity
2. Entropy
3. Information gain

STATE EACH NUMBERED FORMULA

ML.12 (2/2) MACHINE LEARNING

Decision Trees: Gini, Entropy & Information Gain

PART 2 OF 2

1. Variance reduction (regression)
2. Cost-complexity pruning

STATE EACH NUMBERED FORMULA

ML.13 (1/2) MACHINE LEARNING

Ensembles: Bagging, Random Forests & Boosting

PART 1 OF 2

1. Bagging average
2. Variance of an average
3. Out-of-bag fraction

STATE EACH NUMBERED FORMULA

ML.13 (2/2) MACHINE LEARNING

Ensembles: Bagging, Random Forests & Boosting

PART 2 OF 2

1. Boosting update
2. AdaBoost learner weight
3. AdaBoost reweighting

STATE EACH NUMBERED FORMULA

ML.14 (1/2) MACHINE LEARNING

Gradient Boosting

PART 1 OF 2

1. Additive model
2. Initial model
3. Pseudo-residuals

STATE EACH NUMBERED FORMULA

ML.14 (2/2) MACHINE LEARNING

Gradient Boosting

PART 2 OF 2

1. Squared-error residuals
2. Log-loss residuals
3. Second-order leaf weight

STATE EACH NUMBERED FORMULA

ML.15 (1/2) MACHINE LEARNING

Support Vector Machines

PART 1 OF 2

1. Soft-margin primal
2. Margin width
3. Hinge-loss form

STATE EACH NUMBERED FORMULA

ML.15 (2/2) MACHINE LEARNING

Support Vector Machines

PART 2 OF 2

1. Dual decision function
2. RBF kernel

STATE EACH NUMBERED FORMULA

ML.16 (1/2) MACHINE LEARNING

Kernels & the Kernel Trick

PART 1 OF 2

1. Kernel definition
2. Polynomial kernel
3. Worked identity

STATE EACH NUMBERED FORMULA

ML.16 (2/2) MACHINE LEARNING

Kernels & the Kernel Trick

PART 2 OF 2

1. Gram matrix (Mercer)
2. Representer theorem
3. Kernel ridge regression

STATE EACH NUMBERED FORMULA

ML.17 (1/2) MACHINE LEARNING

Classification Metrics

PART 1 OF 2

1. Precision
2. Recall (TPR, sensitivity)
3. F_1 score

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ML.17 (2/2) MACHINE LEARNING

Classification Metrics

PART 2 OF 2

1. Accuracy
2. False positive rate
3. F_β score

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1. $\bar{f} = \dfrac{1}{B}\sum_{b=1}^{B} \hat{f}_b$
2. $\mathrm{Var}(\bar{f}) = \rho\,\sigma^2 + \dfrac{1-\rho}{B}\,\sigma^2$
3. $\left(1 - \dfrac{1}{n}\right)^n \to e^{-1} \approx 0.368$

TERMS B models; ρ correlation; ν shrinkage; ε_m weighted error

Bagging fights variance; boosting fights bias.

1. $\Delta = \mathrm{Var}(\text{parent}) - \sum_c \dfrac{n_c}{n}\mathrm{Var}(c)$
2. $R_\alpha(T) = R(T) + \alpha\,|T|$

TERMS p_k class share; n_c/n child weight; $|T|$ leaves; α penalty

Split to purify; prune to generalize.

1. $\mathrm{Gini} = 1 - \sum_k p_k^2$
2. $H = -\sum_k p_k \log_2 p_k$
3. $IG = H(\text{parent}) - \sum_{c\,\in\,\text{children}} \dfrac{n_c}{n} H(c)$

TERMS p_k class share; n_c/n child weight; $|T|$ leaves; α penalty

Split to purify; prune to generalize.

1. $L = \frac{1}{2}(y - F)^2 \Rightarrow r_{im} = y_i - F_{m-1}(\mathbf{x}_i)$
2. $r_{im} = y_i - p_{m-1}(\mathbf{x}_i)$
3. $w_j^* = -\dfrac{\sum_{i\in j} g_i}{\sum_{i\in j} h_i + \lambda}$

TERMS F_m ensemble; h_m tree; ν shrinkage; r_{im} pseudo-residual

Each tree fixes what the previous trees got wrong.

1. $F_M(\mathbf{x}) = F_0 + \nu \sum_{m=1}^{M} h_m(\mathbf{x})$
2. $F_0 = \arg\min_c \sum_i L(y_i, c)$
3. $r_{im} = -\left.\dfrac{\partial L(y_i, F)}{\partial F}\right|_{F=F_{m-1}(\mathbf{x}_i)}$

TERMS F_m ensemble; h_m tree; ν shrinkage; r_{im} pseudo-residual

Each tree fixes what the previous trees got wrong.

1. $F_m(\mathbf{x}) = F_{m-1}(\mathbf{x}) + \nu\, h_m(\mathbf{x})$
2. $\alpha_m = \dfrac{1}{2}\ln\dfrac{1-\varepsilon_m}{\varepsilon_m}$
3. $w_i \leftarrow w_i\, e^{-\alpha_m y_i h_m(\mathbf{x}_i)}$

TERMS B models; ρ correlation; ν shrinkage; ε_m weighted error

Bagging fights variance; boosting fights bias.

1. $K(\mathbf{x}, \mathbf{z}) = \boldsymbol{\phi}(\mathbf{x})^\top \boldsymbol{\phi}(\mathbf{z})$
2. $K(\mathbf{x}, \mathbf{z}) = (\mathbf{x}^\top \mathbf{z} + c)^p$
3. $(\mathbf{x}^\top \mathbf{z})^2 = \boldsymbol{\phi}(\mathbf{x})^\top \boldsymbol{\phi}(\mathbf{z}),\ \boldsymbol{\phi}(\mathbf{x}) = (x_1^2, \sqrt{2}x_1x_2, x_2^2)$

TERMS $\boldsymbol{\phi}$ feature map; K kernel/Gram; α_i weights; λ penalty

Never compute $\boldsymbol{\phi}$; only compute K.

1. $f(\mathbf{x}) = \sum_i \alpha_i y_i K(\mathbf{x}_i, \mathbf{x}) + b$
2. $K(\mathbf{x}, \mathbf{z}) = e^{-\gamma\|\mathbf{x}-\mathbf{z}\|^2}$

TERMS $\mathbf{w}, b$ hyperplane; ξ_i slack; C penalty; α_i dual; γ RBF width

Maximize the street width; only the points on the curb (support vectors) matter.

1. $\min_{\mathbf{w},b,\boldsymbol{\xi}} \frac{1}{2}\|\mathbf{w}\|^2 + C\sum_i \xi_i \quad \text{s.t.} \quad y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1 - \xi_i,\ \ \xi_i \geq 0$
2. $\text{margin} = \dfrac{2}{\|\mathbf{w}\|}$
3. $\min\ \dfrac{1}{2}\|\mathbf{w}\|^2 + C\sum_i \max\left(0, 1 - y_i f(\mathbf{x}_i)\right)$

TERMS $\mathbf{w}, b$ hyperplane; ξ_i slack; C penalty; α_i dual; γ RBF width

Maximize the street width; only the points on the curb (support vectors) matter.

1. $\text{Accuracy} = \dfrac{TP + TN}{N}$
2. $\text{FPR} = \dfrac{FP}{FP + TN} = 1 - \text{Specificity}$
3. $F_\beta = \dfrac{(1+\beta^2)\,PR}{\beta^2 P + R}$

TERMS TP, FP, TN, FN confusion counts; N total; β recall weight

Precision: "of those I flagged, how many were right?"
Recall: "of the real ones, how many did I catch?"

1. $\text{Precision} = \dfrac{TP}{TP + FP}$
2. $\text{Recall} = \dfrac{TP}{TP + FN}$
3. $F_1 = \dfrac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$

TERMS TP, FP, TN, FN confusion counts; N total; β recall weight

Precision: "of those I flagged, how many were right?"
Recall: "of the real ones, how many did I catch?"

1. $K_{ij} = K(\mathbf{x}_i, \mathbf{x}_j), \quad K \succeq 0$
2. $f(\mathbf{x}) = \sum_{i=1}^{n} \alpha_i K(\mathbf{x}_i, \mathbf{x})$
3. $\boldsymbol{\alpha} = (K + \lambda I)^{-1}\mathbf{y}$

TERMS $\boldsymbol{\phi}$ feature map; K kernel/Gram; α_i weights; λ penalty

Never compute $\boldsymbol{\phi}$; only compute K.

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ML.18 (1/2) MACHINE LEARNING

ROC, AUC & Precision–Recall Curves

PART 1 OF 2

1. ROC curve
2. AUC

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ML.18 (2/2) MACHINE LEARNING

ROC, AUC & Precision–Recall Curves

PART 2 OF 2

1. PR curve
2. PR baseline
3. Youden's J

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ML.19 (1/2) MACHINE LEARNING

Regression Metrics

PART 1 OF 2

1. RMSE
2. MAE
3. MAPE

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ML.19 (2/2) MACHINE LEARNING

Regression Metrics

PART 2 OF 2

1. Coefficient of determination
2. Adjusted R^2

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ML.20 MACHINE LEARNING

Reading Metrics: Bad, Good & Perfect

1. Matthews correlation coefficient
2. Brier score
3. Baseline log-loss (balanced)

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ML.21 (1/2) MACHINE LEARNING

Probability Calibration

PART 1 OF 2

1. Perfect calibration
2. Expected calibration error
3. Platt scaling

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ML.21 (2/2) MACHINE LEARNING

Probability Calibration

PART 2 OF 2

1. Temperature scaling
2. Isotonic regression

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ML.22 (1/2) MACHINE LEARNING

Neural Networks & Backpropagation

PART 1 OF 2

1. Layer (forward pass)
2. ReLU
3. Sigmoid derivative

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ML.22 (2/2) MACHINE LEARNING

Neural Networks & Backpropagation

PART 2 OF 2

1. Output error
2. Backpropagation
3. Weight & bias gradients

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ML.23 (1/2) MACHINE LEARNING

Activation Functions & Their Derivatives

PART 1 OF 2

1. Sigmoid
2. Hyperbolic tangent
3. ReLU

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ML.23 (2/2) MACHINE LEARNING

Activation Functions & Their Derivatives

PART 2 OF 2

1. Leaky ReLU
2. GELU
3. Softmax & Jacobian

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ML.24 (1/2) MACHINE LEARNING

Softmax & Cross-Entropy Loss

PART 1 OF 2

1. Softmax
2. Cross-entropy loss
3. Gradient

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1. $\text{RMSE} = \sqrt{\frac{1}{n}\sum_i (y_i - \hat{y}_i)^2}$
2. $\text{MAE} = \frac{1}{n}\sum_i |y_i - \hat{y}_i|$
3. $\text{MAPE} = \frac{100\%}{n}\sum_i \left|\frac{y_i - \hat{y}_i}{y_i}\right|$

TERMS $y, \hat{y}$ actual/predicted; $\bar{y}$ mean; n rows; p predictors

RMSE punishes big misses; MAE treats all misses equally; both are in the units of y.

1. $\big(\text{Recall}(\tau),\ \text{Precision}(\tau)\big)$
2. $\text{precision}_{\text{random}} = \pi_+$
3. $J = \max_\tau \big(\text{TPR} - \text{FPR}\big)$

TERMS τ threshold; s score; π_+ prevalence; TPR, FPR rates

Lower the threshold ⇒ flag more ⇒ recall up, precision (usually) down. Raise it ⇒ the reverse.

1. $\big(\text{FPR}(\tau),\ \text{TPR}(\tau)\big),\ \tau \in \mathbb{R}$
2. $\text{AUC} = \int_0^1 \text{TPR}\, d(\text{FPR}) = P\big(s(\mathbf{x}_+) > s(\mathbf{x}_-)\big)$

TERMS τ threshold; s score; π_+ prevalence; TPR, FPR rates

Lower the threshold ⇒ flag more ⇒ recall up, precision (usually) down. Raise it ⇒ the reverse.

1. $P\big(y = 1 \mid \hat{p} = p\big) = p \quad \forall p$
2. $\text{ECE} = \sum_{b=1}^{B} \frac{n_b}{n} |\bar{y}_b - \bar{p}_b|$
3. $\hat{p} = \sigma(a\, s + b)$

TERMS $\hat{p}$ predicted prob.; n_b bin count; s score; T temperature

"70% should mean 70%."

1. $\text{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}}$
2. $\text{Brier} = \frac{1}{n}\sum_i (p_i - y_i)^2$
3. $-\ln \frac{1}{2} = \ln 2 \approx 0.693$

TERMS TP, TN, FP, FN counts; p_i prob.; $y_i \in \{0, 1\}$

A metric means nothing without its baseline: always ask "compared to what?"

1. $R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$
2. $R^2_{\text{adj}} = 1 - \big(1 - R^2\big)\frac{n-1}{n-p-1}$

TERMS $y, \hat{y}$ actual/predicted; $\bar{y}$ mean; n rows; p predictors

RMSE punishes big misses; MAE treats all misses equally; both are in the units of y.

1. $\delta^{(L)} = \nabla_{\mathbf{a}^{(L)}} L \odot \phi'\big(\mathbf{z}^{(L)}\big)$
2. $\delta^{(l)} = \Big(W^{(l+1)\top} \delta^{(l+1)}\Big) \odot \phi'\big(\mathbf{z}^{(l)}\big)$
3. $\frac{\partial L}{\partial W^{(l)}} = \delta^{(l)}\, \mathbf{a}^{(l-1)\top},\quad \frac{\partial L}{\partial \mathbf{b}^{(l)}} = \delta^{(l)}$

TERMS $W, \mathbf{b}$ weights/bias; ϕ activation; δ error; $\odot$ elementwise

Forward pass computes; backward pass blames (chain rule, layer by layer).

1. $\mathbf{z}^{(l)} = W^{(l)}\mathbf{a}^{(l-1)} + \mathbf{b}^{(l)},\ \ \mathbf{a}^{(l)} = \phi\big(\mathbf{z}^{(l)}\big)$
2. $\text{ReLU}(z) = \max(0, z)$
3. $\sigma'(z) = \sigma(z)\big(1 - \sigma(z)\big)$

TERMS $W, \mathbf{b}$ weights/bias; ϕ activation; δ error; $\odot$ elementwise

Forward pass computes; backward pass blames (chain rule, layer by layer).

1. $\hat{\mathbf{p}} = \text{softmax}(\mathbf{z}/T)$
2. $\hat{p} = g(s),\ g$ non-decreasing

TERMS $\hat{p}$ predicted prob.; n_b bin count; s score; T temperature

"70% should mean 70%."

1. $\hat{p}_k = \text{softmax}(\mathbf{z})_k = \frac{e^{z_k}}{\sum_{j=1}^{K} e^{z_j}}$
2. $L = -\sum_{k=1}^{K} y_k \ln \hat{p}_k = -\ln \hat{p}_{\text{true}}$
3. $\frac{\partial L}{\partial \mathbf{z}} = \hat{\mathbf{p}} - \mathbf{y}$

TERMS $\mathbf{z}$ logits; $\hat{\mathbf{p}}$ probs; $\mathbf{y}$ one-hot; T temperature

Gradient is "prediction minus truth."

1. $\max(\alpha z, z),\ \ \alpha \approx 0.01$
2. $\text{GELU}(z) = z\,\Phi(z) \approx \frac{z}{2}\Big[1 + \tanh\Big(\sqrt{\frac{2}{\pi}}\big(z + 0.0447 z^3\big)\Big)\Big]$
3. $s_k = \frac{e^{z_k}}{\sum_j e^{z_j}},\quad \frac{\partial s_k}{\partial z_j} = s_k(\delta_{kj} - s_j)$

TERMS z pre-activation; Φ normal CDF; α leak; δ_{kj} Kronecker

Hidden layers: ReLU/GELU. Output: identity (regression), sigmoid (binary), softmax (multi-class).

1. $\sigma(z) = \frac{1}{1 + e^{-z}},\ \ \sigma' = \sigma(1 - \sigma)$
2. $\tanh z = \frac{e^z - e^{-z}}{e^z + e^{-z}},\ \ \tanh' = 1 - \tanh^2$
3. $\max(0, z),\ \ \text{ReLU}' = \begin{cases} 1, & z > 0 \\ 0, & z < 0 \end{cases}$

TERMS z pre-activation; Φ normal CDF; α leak; δ_{kj} Kronecker

Hidden layers: ReLU/GELU. Output: identity (regression), sigmoid (binary), softmax (multi-class).

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ML.24 (2/2) MACHINE LEARNING

Softmax & Cross-Entropy Loss

PART 2 OF 2

1. Log-sum-exp trick
2. Temperature

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ML.25 (1/2) MACHINE LEARNING

Weight Initialization

PART 1 OF 2

1. Variance propagation
2. Xavier / Glorot
3. He (Kaiming)

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ML.25 (2/2) MACHINE LEARNING

Weight Initialization

PART 2 OF 2

1. Uniform version
2. Biases

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ML.26 (1/2) MACHINE LEARNING

Dropout, Batch & Layer Normalization

PART 1 OF 2

1. Dropout mask
2. Inverted dropout
3. Batch statistics

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ML.26 (2/2) MACHINE LEARNING

Dropout, Batch & Layer Normalization

PART 2 OF 2

1. Batch norm
2. Layer norm
3. RMSNorm

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ML.27 (1/2) MACHINE LEARNING

SGD, Momentum & Adam

PART 1 OF 2

1. SGD
2. Momentum
3. Adam moments

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ML.27 (2/2) MACHINE LEARNING

SGD, Momentum & Adam

PART 2 OF 2

1. Adam bias correction
2. Adam update
3. AdamW (decoupled decay)

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ML.28 (1/2) MACHINE LEARNING

Training Loop, LR Schedules & Gradient Clipping

PART 1 OF 2

1. Step decay
2. Exponential decay
3. Linear warmup

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ML.28 (2/2) MACHINE LEARNING

Training Loop, LR Schedules & Gradient Clipping

PART 2 OF 2

1. Cosine annealing
2. Gradient clipping (norm)
3. Early stopping

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ML.29 (1/3) MACHINE LEARNING

Convolutional Networks: Stride, Padding & Pooling

PART 1 OF 3

1. 2-D convolution
2. Output size
3. “Same” padding

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ML.29 (2/3) MACHINE LEARNING

Convolutional Networks: Stride, Padding & Pooling

PART 2 OF 3

1. Parameter count

STATE EACH NUMBERED FORMULA

ML.29 (3/3) MACHINE LEARNING

Convolutional Networks: Stride, Padding & Pooling

PART 3 OF 3

1. Max pooling
2. Receptive field (stacked)

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1 $w \sim \mathcal{U}\big[-a, a\big], \quad a = \sqrt{3\,\mathrm{Var}(w)}$

2 $\mathbf{b} = \mathbf{0}$

TERMS $n_{\text{in}}, n_{\text{out}}$ fan-in/out; w weight; $\mathbf{b}$ bias

ReLU → He ($2/n_{in}$); tanh → Xavier ($2/(n_{in}+n_{out})$).

1 $\mathrm{Var}(z) = n_{\text{in}}\,\mathrm{Var}(w)\,\mathrm{Var}(x)$

2 $\mathrm{Var}(w) = \dfrac{2}{n_{\text{in}} + n_{\text{out}}}$

3 $\mathrm{Var}(w) = \dfrac{2}{n_{\text{in}}}$

TERMS $n_{\text{in}}, n_{\text{out}}$ fan-in/out; w weight; $\mathbf{b}$ bias

ReLU → He ($2/n_{in}$); tanh → Xavier ($2/(n_{in}+n_{out})$).

1 $\ln \sum_j e^{z_j} = m + \ln \sum_j e^{z_j - m}, \; m = \max_j z_j$

2 $\hat{p}_k \propto e^{z_k/T}$

TERMS $\mathbf{z}$ logits; $\hat{\mathbf{p}}$ probs; $\mathbf{y}$ one-hot; T temperature

Gradient is "prediction minus truth."

1 $\theta \leftarrow \theta - \eta\,\nabla_\theta L_{\text{batch}}$

2 $\mathbf{v} \leftarrow \beta \mathbf{v} + \nabla L, \quad \theta \leftarrow \theta - \eta \mathbf{v}$

3 $\mathbf{m} \leftarrow \beta_1 \mathbf{m} + (1-\beta_1)\mathbf{g}, \quad \mathbf{v} \leftarrow \beta_2 \mathbf{v} + (1-\beta_2)\mathbf{g}^2$

TERMS η LR; $\mathbf{g}$ gradient; $\mathbf{m}, \mathbf{v}$ moments; t step; λ decay

Adam = momentum (m) + per-parameter step sizes (v).

1 $\hat{x}_i = \dfrac{x_i - \mu_B}{\sqrt{\sigma_B^2 + \epsilon}}, \quad y_i = \gamma \hat{x}_i + \beta$

2 $\hat{\mathbf{x}} = \dfrac{\mathbf{x} - \mu}{\sqrt{\sigma^2 + \epsilon}}, \; \mu, \sigma^2$ over features

3 $\hat{\mathbf{x}} = \dfrac{\mathbf{x}}{\sqrt{\frac{1}{d}\sum_j x_j^2 + \epsilon}} \odot \gamma$

TERMS p drop prob.; μ_B, σ_B^2 batch stats; γ, β scale/shift

Batch norm: across the batch; layer norm: across the features.

1 $m_j \sim \text{Bernoulli}(1-p)$

2 $\tilde{\mathbf{h}} = \dfrac{\mathbf{m} \odot \mathbf{h}}{1-p}$

3 $\mu_B = \dfrac{1}{m}\sum_{i=1}^{m} x_i, \quad \sigma_B^2 = \dfrac{1}{m}\sum_{i=1}^{m}(x_i - \mu_B)^2$

TERMS p drop prob.; μ_B, σ_B^2 batch stats; γ, β scale/shift

Batch norm: across the batch; layer norm: across the features.

1 $\eta_t = \eta_{\min} + \frac{1}{2}(\eta_{\max} - \eta_{\min})\Big(1 + \cos\dfrac{\pi t}{T}\Big)$

2 $\mathbf{g} \leftarrow \mathbf{g} \cdot \min\Big(1, \dfrac{c}{\|\mathbf{g}\|}\Big)$

3 $t^* = \arg\min_t \; L_{\text{val}}(t)$

TERMS η_t LR at t; T_w warmup; T length; c clip norm

Warm up, cruise, cool down.

1 $\eta_t = \eta_0\,\gamma^{\lfloor t/s \rfloor}$

2 $\eta_t = \eta_0\, e^{-kt}$

3 $\eta_t = \eta_{\max}\dfrac{t}{T_w}, \quad t \le T_w$

TERMS η_t LR at t; T_w warmup; T length; c clip norm

Warm up, cruise, cool down.

1 $\hat{\mathbf{m}} = \dfrac{\mathbf{m}}{1-\beta_1^t}, \quad \hat{\mathbf{v}} = \dfrac{\mathbf{v}}{1-\beta_2^t}$

2 $\theta \leftarrow \theta - \eta\,\dfrac{\hat{\mathbf{m}}}{\sqrt{\hat{\mathbf{v}}} + \epsilon}$

3 $\theta \leftarrow \theta - \eta\Big(\dfrac{\hat{\mathbf{m}}}{\sqrt{\hat{\mathbf{v}}} + \epsilon} + \lambda\theta\Big)$

TERMS η LR; $\mathbf{g}$ gradient; $\mathbf{m}, \mathbf{v}$ moments; t step; λ decay

Adam = momentum (m) + per-parameter step sizes (v).

1 $Y[i,j] = \max_{(u,v)\in\text{window}} X[iS+u,\, jS+v]$

2 $r_l = r_{l-1} + (F_l - 1)\prod_{k<l} S_k$

TERMS W input; F kernel; P pad; S stride; C channels

"W minus F plus 2P, over S, plus 1."

1 $(F^2 C_{\text{in}} + 1)\,C_{\text{out}}$

TERMS W input; F kernel; P pad; S stride; C channels

"W minus F plus 2P, over S, plus 1."

1 $Y[i,j] = \sum_{c=1}^{C_{\text{in}}}\sum_{u=0}^{F-1}\sum_{v=0}^{F-1} K[c,u,v]\; X[c,\, iS+u,\, jS+v] + b$

2 $O = \left\lfloor \dfrac{W - F + 2P}{S} \right\rfloor + 1$

3 $P = \dfrac{F-1}{2} \quad (S = 1)$

TERMS W input; F kernel; P pad; S stride; C channels

"W minus F plus 2P, over S, plus 1."

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ML.30 (1/2) MACHINE LEARNING

Recurrent Neural Networks

PART 1 OF 2

1. Hidden-state update
2. Output
3. Sequence loss

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ML.30 (2/2) MACHINE LEARNING

Recurrent Neural Networks

PART 2 OF 2

1. BPTT gradient chain
2. Vanishing / exploding test

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ML.31 (1/2) MACHINE LEARNING

LSTM & GRU Gates

PART 1 OF 2

1. Forget gate
2. Input gate & candidate
3. Cell update

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ML.31 (2/2) MACHINE LEARNING

LSTM & GRU Gates

PART 2 OF 2

1. Output gate & hidden state
2. GRU

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ML.32 (1/2) MACHINE LEARNING

Embeddings & Cosine Similarity

PART 1 OF 2

1. Embedding lookup
2. Cosine similarity
3. Cosine and Euclidean

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ML.32 (2/2) MACHINE LEARNING

Embeddings & Cosine Similarity

PART 2 OF 2

1. Skip-gram probability
2. Negative sampling
3. Analogy arithmetic

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ML.33 (1/2) MACHINE LEARNING

Scaled Dot-Product Attention

PART 1 OF 2

1. Scaled dot-product attention
2. Projections
3. Why $\sqrt{d_k}$

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ML.33 (2/2) MACHINE LEARNING

Scaled Dot-Product Attention

PART 2 OF 2

1. Causal mask

STATE EACH NUMBERED FORMULA

ML.34 (1/2) MACHINE LEARNING

Transformers: Multi-Head Attention & Positional Encoding

PART 1 OF 2

1. Attention head
2. Multi-head attention
3. Feed-forward sublayer

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ML.34 (2/2) MACHINE LEARNING

Transformers: Multi-Head Attention & Positional Encoding

PART 2 OF 2

1. Pre-LN block
2. Sinusoidal encoding (even)
3. Sinusoidal encoding (odd)

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ML.35 (1/2) MACHINE LEARNING

k-Means Clustering

PART 1 OF 2

1. Objective (inertia)
2. Assignment step
3. Update step

STATE EACH NUMBERED FORMULA

ML.35 (2/2) MACHINE LEARNING

k-Means Clustering

PART 2 OF 2

1. k-means++ seeding

STATE EACH NUMBERED FORMULA

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1 $\mathbf{f}_t = \sigma\big(W_f[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_f\big)$

2 $\mathbf{i}_t = \sigma\big(W_i[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_i\big), \quad \tilde{\mathbf{c}}_t = \tanh\big(W_c[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_c\big)$

3 $\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{c}}_t$

TERMS $\mathbf{f}, \mathbf{i}, \mathbf{o}$ gates; $\mathbf{c}_t$ cell; $\mathbf{h}_t$ hidden; $\odot$ elementwise

Forget, write, output: the cell state is a conveyor belt.

ML.31 (1/2)

1 $\dfrac{\partial \mathbf{h}_T}{\partial \mathbf{h}_k} = \prod_{t=k+1}^{T} \operatorname{diag}\big(1 - \mathbf{h}_t^2\big)\, W_h$

2 $\|\partial \mathbf{h}_T / \partial \mathbf{h}_k\| \sim \rho(W_h)^{T-k}$

TERMS $\mathbf{h}_t$ hidden state; $\mathbf{x}_t$ input; W_h recurrent weights; T length

Same cell, every time step; the hidden state is the memory.

ML.30 (2/2)

1 $\mathbf{h}_t = \tanh\big(W_h \mathbf{h}_{t-1} + W_x \mathbf{x}_t + \mathbf{b}\big)$

2 $\hat{\mathbf{y}}_t = \operatorname{softmax}\big(W_y \mathbf{h}_t + \mathbf{c}\big)$

3 $L = \sum_{t=1}^{T} L_t(\mathbf{y}_t, \hat{\mathbf{y}}_t)$

TERMS $\mathbf{h}_t$ hidden state; $\mathbf{x}_t$ input; W_h recurrent weights; T length

Same cell, every time step; the hidden state is the memory.

ML.30 (1/2)

1 $P(o \mid c) = \dfrac{\exp(\mathbf{u}_o^\top \mathbf{v}_c)}{\sum_{w=1}^{V} \exp(\mathbf{u}_w^\top \mathbf{v}_c)}$

2 $\log \sigma(\mathbf{u}_o^\top \mathbf{v}_c) + \sum_{k=1}^{K} \log \sigma(-\mathbf{u}_{n_k}^\top \mathbf{v}_c)$

3 $\mathbf{e}_{\text{king}} - \mathbf{e}_{\text{man}} + \mathbf{e}_{\text{woman}} \approx \mathbf{e}_{\text{queen}}$

TERMS E table; V vocab size; d dimension; $\mathbf{u}, \mathbf{v}$ vectors

Similar meaning → small angle.

ML.32 (2/2)

1 $\mathbf{e}_w = E^\top \mathbf{o}_w$

2 $\cos(\mathbf{u}, \mathbf{v}) = \dfrac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\|\,\|\mathbf{v}\|}$

3 $\|\hat{\mathbf{u}} - \hat{\mathbf{v}}\|^2 = 2 - 2\cos(\mathbf{u}, \mathbf{v})$

TERMS E table; V vocab size; d dimension; $\mathbf{u}, \mathbf{v}$ vectors

Similar meaning → small angle.

ML.32 (1/2)

1 $\mathbf{o}_t = \sigma\big(W_o[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_o\big), \quad \mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t)$

2 $\mathbf{h}_t = (1 - \mathbf{z}_t) \odot \mathbf{h}_{t-1} + \mathbf{z}_t \odot \tilde{\mathbf{h}}_t$

TERMS $\mathbf{f}, \mathbf{i}, \mathbf{o}$ gates; $\mathbf{c}_t$ cell; $\mathbf{h}_t$ hidden; $\odot$ elementwise

Forget, write, output: the cell state is a conveyor belt.

ML.31 (2/2)

1 $\text{head}_h = \text{Attention}\big(XW_Q^{(h)}, XW_K^{(h)}, XW_V^{(h)}\big)$

2 $\text{MHA}(X) = \big[\text{head}_1, \ldots, \text{head}_H\big] W_O$

3 $\text{FFN}(\mathbf{x}) = W_2\, \text{GELU}(W_1 \mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2$

TERMS H heads; d width; W_O output proj.; p position; LN layer norm

Attention mixes across tokens; the FFN thinks within each token.

ML.34 (1/2)

1 $S_{ij} = -\infty \;\; \text{for } j > i$

TERMS Q, K, V query/key/value; d_k key dim.; X input; n length

Query asks, key answers "how relevant?", value is what gets passed along.

ML.33 (2/2)

1 $\text{Attention}(Q, K, V) = \operatorname{softmax}\left(\dfrac{QK^\top}{\sqrt{d_k}}\right) V$

2 $Q = XW_Q, \quad K = XW_K, \quad V = XW_V$

3 $\text{Var}(\mathbf{q}^\top \mathbf{k}) = d_k$ for unit-variance entries

TERMS Q, K, V query/key/value; d_k key dim.; X input; n length

Query asks, key answers "how relevant?", value is what gets passed along.

ML.33 (1/2)

1 $P(\mathbf{x}_i \text{ chosen}) = \dfrac{D(\mathbf{x}_i)^2}{\sum_j D(\mathbf{x}_j)^2}$

TERMS K clusters; C_k cluster set; μ_k centroid; J inertia

Assign, update, repeat.

ML.35 (2/2)

1 $J = \sum_{k=1}^{K} \sum_{\mathbf{x}_i \in C_k} \|\mathbf{x}_i - \boldsymbol{\mu}_k\|^2$

2 $c_i = \arg\min_k \|\mathbf{x}_i - \boldsymbol{\mu}_k\|^2$

3 $\boldsymbol{\mu}_k = \dfrac{1}{|C_k|} \sum_{\mathbf{x}_i \in C_k} \mathbf{x}_i$

TERMS K clusters; C_k cluster set; μ_k centroid; J inertia

Assign, update, repeat.

ML.35 (1/2)

1 $\mathbf{x} \leftarrow \mathbf{x} + \text{MHA}(\text{LN}(\mathbf{x})), \quad \mathbf{x} \leftarrow \mathbf{x} + \text{FFN}(\text{LN}(\mathbf{x}))$

2 $PE_{(p,2i)} = \sin\big(p / 10000^{2i/d}\big)$

3 $PE_{(p,2i+1)} = \cos\big(p / 10000^{2i/d}\big)$

TERMS H heads; d width; W_O output proj.; p position; LN layer norm

Attention mixes across tokens; the FFN thinks within each token.

ML.34 (2/2)

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ML.36 (1/2) MACHINE LEARNING

Hierarchical Clustering & Silhouette

PART 1 OF 2

1. Silhouette
2. Single linkage
3. Complete linkage

STATE EACH NUMBERED FORMULA

ML.36 (2/2) MACHINE LEARNING

Hierarchical Clustering & Silhouette

PART 2 OF 2

1. Average linkage
2. Ward linkage

STATE EACH NUMBERED FORMULA

ML.37 (1/2) MACHINE LEARNING

Gaussian Mixtures & the EM Algorithm

PART 1 OF 2

1. Mixture density
2. Log-likelihood
3. E-step (responsibilities)

STATE EACH NUMBERED FORMULA

ML.37 (2/2) MACHINE LEARNING

Gaussian Mixtures & the EM Algorithm

PART 2 OF 2

1. M-step: means
2. M-step: variances
3. M-step: weights

STATE EACH NUMBERED FORMULA

ML.38 (1/2) MACHINE LEARNING

Principal Component Analysis (PCA)

PART 1 OF 2

1. Covariance matrix
2. Eigen-decomposition
3. Explained variance ratio

STATE EACH NUMBERED FORMULA

ML.38 (2/2) MACHINE LEARNING

Principal Component Analysis (PCA)

PART 2 OF 2

1. Scores (projection)
2. SVD route
3. Reconstruction

STATE EACH NUMBERED FORMULA

ML.39 (1/2) MACHINE LEARNING

t-SNE & UMAP

PART 1 OF 2

1. High-dim affinities
2. Symmetrized
3. Low-dim affinities

STATE EACH NUMBERED FORMULA

ML.39 (2/2) MACHINE LEARNING

t-SNE & UMAP

PART 2 OF 2

1. t-SNE objective
2. UMAP objective

STATE EACH NUMBERED FORMULA

ML.40 (1/2) MACHINE LEARNING

Autoencoders & Variational Autoencoders

PART 1 OF 2

1. Autoencoder loss
2. Evidence lower bound (ELBO)

STATE EACH NUMBERED FORMULA

ML.40 (2/2) MACHINE LEARNING

Autoencoders & Variational Autoencoders

PART 2 OF 2

1. Reparameterization
2. Gaussian KL (closed form)

STATE EACH NUMBERED FORMULA

ML.41 (1/2) MACHINE LEARNING

Generative Adversarial Networks

PART 1 OF 2

1. GAN minimax objective
2. Optimal discriminator
3. Value at the optimum

STATE EACH NUMBERED FORMULA

ML.41 (2/2) MACHINE LEARNING

Generative Adversarial Networks

PART 2 OF 2

1. Non-saturating generator loss
2. Jensen–Shannon divergence

STATE EACH NUMBERED FORMULA

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1 $p(x) = \sum_{k=1}^{K} \pi_k \mathcal{N}(x \mid \mu_k, \sigma_k^2)$

2 $\ell = \sum_{i=1}^{n} \ln \sum_{k=1}^{K} \pi_k \mathcal{N}(x_i \mid \mu_k, \sigma_k^2)$

3 $\gamma_{ik} = \dfrac{\pi_k \mathcal{N}(x_i \mid \mu_k, \sigma_k^2)}{\sum_j \pi_j \mathcal{N}(x_i \mid \mu_j, \sigma_j^2)}$

TERMS π_k weights; μ_k, σ_k^2 comp. params; γ_{ik} responsibility

E-step: guess memberships; M-step: refit the Gaussians.

1 $d_{\text{avg}} = \dfrac{1}{|A||B|} \sum_{\mathbf{x} \in A} \sum_{\mathbf{z} \in B} d(\mathbf{x}, \mathbf{z})$

2 $\Delta J = \dfrac{|A||B|}{|A| + |B|} \|\boldsymbol{\mu}_A - \boldsymbol{\mu}_B\|^2$

TERMS $a(i)$ own-cluster dist.; $b(i)$ nearest-other dist.; A, B clusters

Bottom-up: merge the closest pair, record the height, repeat.

1 $s(i) = \dfrac{b(i) - a(i)}{\max\{a(i),\, b(i)\}} \in [-1, 1]$

2 $d_{\text{single}} = \min_{\mathbf{x} \in A, \mathbf{z} \in B} d(\mathbf{x}, \mathbf{z})$

3 $d_{\text{complete}} = \max_{\mathbf{x} \in A, \mathbf{z} \in B} d(\mathbf{x}, \mathbf{z})$

TERMS $a(i)$ own-cluster dist.; $b(i)$ nearest-other dist.; A, B clusters

Bottom-up: merge the closest pair, record the height, repeat.

1 $\mathbf{z} = V_k^\top (\mathbf{x} - \bar{\mathbf{x}})$

2 $X_c = U \Sigma V^\top, \quad \lambda_j = \dfrac{\sigma_j^2}{n-1}$

3 $\hat{\mathbf{x}} = \bar{\mathbf{x}} + V_k \mathbf{z}$

TERMS X_c centered data; S covariance; $\mathbf{v}_k, \lambda_k$ eigenpairs

PC1 points along the longest direction of the data cloud.

1 $S = \dfrac{1}{n-1} X_c^\top X_c$

2 $S\,\mathbf{v}_k = \lambda_k \mathbf{v}_k, \quad \lambda_1 \geq \lambda_2 \geq \cdots$

3 $\dfrac{\lambda_k}{\sum_j \lambda_j}$

TERMS X_c centered data; S covariance; $\mathbf{v}_k, \lambda_k$ eigenpairs

PC1 points along the longest direction of the data cloud.

1 $\mu_k = \dfrac{\sum_i \gamma_{ik} x_i}{\sum_i \gamma_{ik}}$

2 $\sigma_k^2 = \dfrac{\sum_i \gamma_{ik} (x_i - \mu_k)^2}{\sum_i \gamma_{ik}}$

3 $\pi_k = \dfrac{N_k}{n}, \quad N_k = \sum_i \gamma_{ik}$

TERMS π_k weights; μ_k, σ_k^2 comp. params; γ_{ik} responsibility

E-step: guess memberships; M-step: refit the Gaussians.

1 $L = \left\| \mathbf{x} - g_\phi\big(f_\theta(\mathbf{x})\big) \right\|^2$

2 $\log p(\mathbf{x}) \geq \underbrace{E_{q_\phi(\mathbf{z}|\mathbf{x})}\big[\log p_\theta(\mathbf{x} \mid \mathbf{z})\big]}_{\text{reconstruction}} - \underbrace{D_{\text{KL}}\big(q_\phi(\mathbf{z} \mid \mathbf{x}) \,\|\, p(\mathbf{z})\big)}_{\text{regularization}}$

TERMS f, g enc./dec.; $\mathbf{z}$ code; q encoder dist.; $p(\mathbf{z})$ prior

Reconstruct well, but keep the codes close to the prior.

1 $C = D_{\text{KL}}(P \,\|\, Q) = \sum_{i \neq j} p_{ij} \log \dfrac{p_{ij}}{q_{ij}}$

2 $\sum_{i \neq j} \left[w_{ij} \log \dfrac{w_{ij}}{v_{ij}} + (1 - w_{ij}) \log \dfrac{1 - w_{ij}}{1 - v_{ij}} \right]$

TERMS p_{ij}, q_{ij} affinities; σ_i width; perplexity = eff. neighbors

Distances between clusters in a t-SNE plot mean nothing.

1 $p_{j|i} = \dfrac{\exp(-\|\mathbf{x}_i - \mathbf{x}_j\|^2 / 2\sigma_i^2)}{\sum_{k \neq i} \exp(-\|\mathbf{x}_i - \mathbf{x}_k\|^2 / 2\sigma_i^2)}$

2 $p_{ij} = \dfrac{p_{j|i} + p_{i|j}}{2n}$

3 $q_{ij} = \dfrac{(1 + \|\mathbf{y}_i - \mathbf{y}_j\|^2)^{-1}}{\sum_{k \neq l} (1 + \|\mathbf{y}_k - \mathbf{y}_l\|^2)^{-1}}$

TERMS p_{ij}, q_{ij} affinities; σ_i width; perplexity = eff. neighbors

Distances between clusters in a t-SNE plot mean nothing.

1 $L_G = -E_{\mathbf{z}}\big[\log D(G(\mathbf{z}))\big]$

2 $\text{JS}(p \| q) = \frac{1}{2} D_{\text{KL}}(p \| m) + \frac{1}{2} D_{\text{KL}}(q \| m), \; m = \frac{p+q}{2}$

TERMS G generator; D discriminator; p_g model dist.; JS divergence

Counterfeiter vs. detective: both improve until fakes are indistinguishable.

1 $\min_G \max_D \; V(D,G) = E_{\mathbf{x} \sim p_{\text{data}}}\big[\log D(\mathbf{x})\big] + E_{\mathbf{z} \sim p(\mathbf{z})}\big[\log\big(1 - D(G(\mathbf{z}))\big)\big]$

2 $D^*(\mathbf{x}) = \dfrac{p_{\text{data}}(\mathbf{x})}{p_{\text{data}}(\mathbf{x}) + p_g(\mathbf{x})}$

3 $V(D^*, G) = -\log 4 + 2\,\text{JS}(p_{\text{data}} \,\|\, p_g)$

TERMS G generator; D discriminator; p_g model dist.; JS divergence

Counterfeiter vs. detective: both improve until fakes are indistinguishable.

1 $\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, I)$

2 $D_{\text{KL}} = \frac{1}{2} \sum_j \left(\mu_j^2 + \sigma_j^2 - \ln \sigma_j^2 - 1\right)$

TERMS f, g enc./dec.; $\mathbf{z}$ code; q encoder dist.; $p(\mathbf{z})$ prior

Reconstruct well, but keep the codes close to the prior.

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ML.42 (1/2)

MACHINE LEARNING

Markov Decision Processes & the Bellman Equation

PART 1 OF 2

1. Discounted return
2. State value
3. Action value

STATE EACH NUMBERED FORMULA

ML.42 (2/2)

MACHINE LEARNING

Markov Decision Processes & the Bellman Equation

PART 2 OF 2

1. Bellman expectation equation
2. Bellman optimality equation

STATE EACH NUMBERED FORMULA

ML.43 (1/2)

MACHINE LEARNING

Q-Learning & Policy Gradients

PART 1 OF 2

1. TD error
2. Q-learning update
3. ε-greedy

STATE EACH NUMBERED FORMULA

ML.43 (2/2)

MACHINE LEARNING

Q-Learning & Policy Gradients

PART 2 OF 2

1. Policy gradient theorem
2. Advantage

STATE EACH NUMBERED FORMULA

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1 $\delta_t = r_{t+1} + \gamma \max_{a'} Q(s_{t+1}, a') - Q(s_t, a_t)$

2 $Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha\, \delta_t$

3 $a_t = \begin{cases} \text{random action}, & \text{prob. } \varepsilon \\ \arg\max_a Q(s_t, a), & \text{prob. } 1 - \varepsilon \end{cases}$

TERMS Q action value; α step; γ discount; ε explore; b baseline

Q-learning: "learn a guess from a guess." Policy gradient: "do more of what worked."

ML.43 (1/2)

1 $V^\pi(s) = \sum_a \pi(a \mid s) \sum_{s'} P(s' \mid s, a)\left[R(s, a, s') + \gamma V^\pi(s')\right]$

2 $Q^*(s, a) = \sum_{s'} P(s' \mid s, a)\left[R(s, a, s') + \gamma \max_{a'} Q^*(s', a')\right]$

TERMS s, a state/action; γ discount; π policy; P transitions; R reward

Value now = reward now + $\gamma\times$ value next.

ML.42 (2/2)

1 $G_t = \sum_{k=0}^{\infty} \gamma^k r_{t+k+1}$

2 $V^\pi(s) = E_\pi\left[G_t \mid s_t = s\right]$

3 $Q^\pi(s, a) = E_\pi\left[G_t \mid s_t = s, a_t = a\right]$

TERMS s, a state/action; γ discount; π policy; P transitions; R reward

Value now = reward now + $\gamma\times$ value next.

ML.42 (1/2)

1 $\nabla_\theta J(\theta) = E_{\pi_\theta}\left[\sum_t \nabla_\theta \log \pi_\theta(a_t \mid s_t)\left(G_t - b(s_t)\right)\right]$

2 $A(s, a) = Q(s, a) - V(s)$

TERMS Q action value; α step; γ discount; ε explore; b baseline

Q-learning: "learn a guess from a guess." Policy gradient: "do more of what worked."

ML.43 (2/2)

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EE.1 (1/2)

ELECTRICAL ENGINEERING

Charge, Current, Voltage, Power & Energy

PART 1 OF 2

1. Current
2. Voltage
3. Electric power

STATE EACH NUMBERED FORMULA

EE.1 (2/2)

ELECTRICAL ENGINEERING

Charge, Current, Voltage, Power & Energy

PART 2 OF 2

1. Energy
2. Kilowatt-hour
3. Charge quantization

STATE EACH NUMBERED FORMULA

EE.2 (1/2)

ELECTRICAL ENGINEERING

Resistance, Ohm's Law & Resistivity

PART 1 OF 2

1. Ohm's law
2. Conductance
3. Joule heating

STATE EACH NUMBERED FORMULA

EE.2 (2/2)

ELECTRICAL ENGINEERING

Resistance, Ohm's Law & Resistivity

PART 2 OF 2

1. Wire resistance
2. Conductivity
3. Temperature coefficient

STATE EACH NUMBERED FORMULA

EE.3 (1/2)

ELECTRICAL ENGINEERING

Series & Parallel Combinations

PART 1 OF 2

1. Resistors in series
2. Resistors in parallel
3. Product over sum

STATE EACH NUMBERED FORMULA

EE.3 (2/2)

ELECTRICAL ENGINEERING

Series & Parallel Combinations

PART 2 OF 2

1. Inductors
2. Capacitors
3. Impedances

STATE EACH NUMBERED FORMULA

EE.4

ELECTRICAL ENGINEERING

Kirchhoff's Laws

1. Kirchhoff's current law (KCL)
2. Kirchhoff's voltage law (KVL)
3. Single-loop current

STATE EACH NUMBERED FORMULA

EE.5 (1/2)

ELECTRICAL ENGINEERING

Voltage & Current Dividers

PART 1 OF 2

1. Voltage divider
2. General divider
3. Current divider (two)

STATE EACH NUMBERED FORMULA

EE.5 (2/2)

ELECTRICAL ENGINEERING

Voltage & Current Dividers

PART 2 OF 2

1. General current divider
2. Loaded divider

STATE EACH NUMBERED FORMULA

EE.6 (1/2)

ELECTRICAL ENGINEERING

Independent & Dependent Sources; Source Transformation

PART 1 OF 2

1. VCVS
2. VCCS
3. CCVS

STATE EACH NUMBERED FORMULA

EE.6 (2/2)

ELECTRICAL ENGINEERING

Independent & Dependent Sources; Source Transformation

PART 2 OF 2

1. CCCS
2. Source transformation
3. Practical source

STATE EACH NUMBERED FORMULA

EE.7

ELECTRICAL ENGINEERING

Nodal & Mesh Analysis

1. Nodal equation
2. Nodal matrix form
3. Mesh equation

STATE EACH NUMBERED FORMULA

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1. $V = IR$
2. $G = \dfrac{1}{R}$
3. $P = I^2R = \dfrac{V^2}{R}$

TERMS ρ resistivity (Ω·m); L length; A area; G conductance (S); α temp. coeff.

"Long and thin, resistance wins"; "twice the current, four times the heat" (I^2R).

1. $E = \displaystyle\int P\,dt = Pt \quad (\text{constant } P)$
2. $1\ \text{kWh} = 3.6 \times 10^6\ \text{J}$
3. $Q = ne, \quad e = 1.602 \times 10^{-19}\ \text{C}$

TERMS Q charge (C); I current (A); V voltage (V); P power (W); E energy (J)

Water analogy: voltage = pressure, current = flow rate, resistance = pipe narrowness, power = pressure × flow.

1. $I = \dfrac{dQ}{dt}$
2. $V = \dfrac{dW}{dQ}$
3. $P = VI$

TERMS Q charge (C); I current (A); V voltage (V); P power (W); E energy (J)

Water analogy: voltage = pressure, current = flow rate, resistance = pipe narrowness, power = pressure × flow.

1. $L_{\text{ser}} = L_1 + L_2, \qquad \dfrac{1}{L_{\text{par}}} = \dfrac{1}{L_1} + \dfrac{1}{L_2}$
2. $C_{\text{par}} = C_1 + C_2, \qquad \dfrac{1}{C_{\text{ser}}} = \dfrac{1}{C_1} + \dfrac{1}{C_2}$
3. $Z_{\text{ser}} = \displaystyle\sum_k Z_k, \qquad \dfrac{1}{Z_{\text{par}}} = \sum_k \dfrac{1}{Z_k}$

TERMS R_{eq} equivalent; $\|$ in parallel; L (H); C (F); Z impedance

"R and L add in series; C adds in parallel."

1. $R_{\text{eq}} = R_1 + R_2 + \cdots + R_n$
2. $\dfrac{1}{R_{\text{eq}}} = \dfrac{1}{R_1} + \dfrac{1}{R_2} + \cdots + \dfrac{1}{R_n}$
3. $R_1 \parallel R_2 = \dfrac{R_1 R_2}{R_1 + R_2}$

TERMS R_{eq} equivalent; $\|$ in parallel; L (H); C (F); Z impedance

"R and L add in series; C adds in parallel."

1. $R = \rho\,\dfrac{L}{A}$
2. $\sigma = \dfrac{1}{\rho}$
3. $R(T) \approx R_0\left[1 + \alpha\,(T - T_0)\right]$

TERMS ρ resistivity (Ω·m); L length; A area; G conductance (S); α temp. coeff.

"Long and thin, resistance wins"; "twice the current, four times the heat" (I^2R).

1. $I_k = I\,\dfrac{G_k}{G_1 + \cdots + G_n}$
2. $V_{\text{out}} = V_{\text{in}}\,\dfrac{R_2 \parallel R_L}{R_1 + R_2 \parallel R_L}$

TERMS R_1, R_2 resistors; G conductance; R_L load; $\|$ parallel

Voltage divider: "my resistor on top"; current divider: "the other resistor on top."

1. $V_{\text{out}} = V_{\text{in}}\,\dfrac{R_2}{R_1 + R_2}$
2. $V_k = V\,\dfrac{R_k}{R_1 + \cdots + R_n}$
3. $I_1 = I_{\text{total}}\,\dfrac{R_2}{R_1 + R_2}$

TERMS R_1, R_2 resistors; G conductance; R_L load; $\|$ parallel

Voltage divider: "my resistor on top"; current divider: "the other resistor on top."

1. $\displaystyle\sum_k I_k = 0 \iff \sum I_{\text{in}} = \sum I_{\text{out}}$
2. $\displaystyle\sum_{\text{loop}} V_k = 0$
3. $I = \dfrac{\sum V_{\text{src}}}{\sum R}$

TERMS I_k currents at a node (A); V_k voltages around a loop (V)

KCL: "what flows in must flow out" (conservation of charge). KVL: "every round trip ends where it started" (conservation of energy).

1. $\displaystyle\sum_{\text{branches}} \frac{V_n - V_m}{R_{nm}} = I_{\text{src},n}$
2. $G\,\mathbf{v} = \mathbf{i}$
3. $\displaystyle\sum_{\text{loop}} R\,I = \sum V_{\text{source}} \;\Rightarrow\; R\,\mathbf{i} = \mathbf{v}$

TERMS V_n node voltage; G conductance matrix; $\mathbf{i}$ source currents; mesh currents

Nodal: "voltages unknown, KCL equations". Mesh: "currents unknown, KVL equations".

1. $i = \beta\, i_x$
2. V_s in series with $R \iff I_s = \dfrac{V_s}{R}$ in parallel with R
3. $v = V_s - i\,R_s$

TERMS μ, β gains; g_m (S); r_m (Ω); v_x, i_x controlling variables

Diamond = dependent ("it depends"). Never switch off a dependent source when finding R_{th} or using superposition.

1. $v = \mu\, v_x$
2. $i = g_m\, v_x$
3. $v = r_m\, i_x$

TERMS μ, β gains; g_m (S); r_m (Ω); v_x, i_x controlling variables

Diamond = dependent ("it depends"). Never switch off a dependent source when finding R_{th} or using superposition.

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EE.8 ELECTRICAL ENGINEERING

Superposition Theorem

1. Superposition
2. Switching a source off
3. Power is not superposable

STATE EACH NUMBERED FORMULA

EE.9 (1/2) ELECTRICAL ENGINEERING

Thévenin & Norton Equivalents; Maximum Power Transfer

PART 1 OF 2

1. Thévenin voltage
2. Thévenin resistance
3. Norton current

STATE EACH NUMBERED FORMULA

EE.9 (2/2) ELECTRICAL ENGINEERING

Thévenin & Norton Equivalents; Maximum Power Transfer

PART 2 OF 2

1. Load voltage
2. Maximum power transfer
3. Efficiency

STATE EACH NUMBERED FORMULA

EE.K1 ELECTRICAL ENGINEERING

Circuit analysis: which method when?

RECALL IT

EE.K1 ELECTRICAL ENGINEERING

Superposition: how to switch a source off?

RECALL IT

EE.10 (1/3) ELECTRICAL ENGINEERING

Capacitors & Inductors

PART 1 OF 3

1. Capacitance
2. Capacitor law
3. Inductor law

STATE EACH NUMBERED FORMULA

EE.10 (2/3) ELECTRICAL ENGINEERING

Capacitors & Inductors

PART 2 OF 3

1. Stored energy
2. Combination
3. Parallel-plate capacitor

STATE EACH NUMBERED FORMULA

EE.10 (3/3) ELECTRICAL ENGINEERING

Capacitors & Inductors

PART 3 OF 3

1. Continuity

STATE EACH NUMBERED FORMULA

EE.11 (1/2) ELECTRICAL ENGINEERING

RC & RL Transients

PART 1 OF 2

1. Capacitor charging
2. Capacitor discharging
3. Time constants

STATE EACH NUMBERED FORMULA

EE.11 (2/2) ELECTRICAL ENGINEERING

RC & RL Transients

PART 2 OF 2

1. RL current rise
2. General first-order solution
3. Governing ODE

STATE EACH NUMBERED FORMULA

EE.12 (1/3) ELECTRICAL ENGINEERING

Second-Order (RLC) Transients & Damping

PART 1 OF 3

1. Standard form
2. Characteristic roots
3. Damping parameters

STATE EACH NUMBERED FORMULA

EE.12 (2/3) ELECTRICAL ENGINEERING

Second-Order (RLC) Transients & Damping

PART 2 OF 3

1. Overdamped ($\zeta > 1$)
2. Critically damped ($\zeta = 1$)
3. Underdamped ($\zeta < 1$)

STATE EACH NUMBERED FORMULA

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1. $V_L = V_{\text{th}} \dfrac{R_L}{R_{\text{th}} + R_L}$

2. $R_L = R_{\text{th}} \;\Rightarrow\; P_{\max} = \dfrac{V_{\text{th}}^2}{4R_{\text{th}}}$

3. $\eta = \dfrac{R_L}{R_{\text{th}} + R_L}$

TERMS V_{oc} open-circuit V; I_{sc} short-circuit I; R_L load; η efficiency

Any linear circuit, seen from two terminals, is "one battery plus one resistor."

1. $V_{\text{th}} = V_{\text{oc}}$

2. $R_{\text{th}} = \dfrac{V_{\text{oc}}}{I_{\text{sc}}}$

3. $I_N = I_{\text{sc}} = \dfrac{V_{\text{th}}}{R_{\text{th}}}$

TERMS V_{oc} open-circuit V; I_{sc} short-circuit I; R_L load; η efficiency

Any linear circuit, seen from two terminals, is "one battery plus one resistor."

1. $x = \displaystyle\sum_k x_k, \quad x_k = \text{response to source } k \text{ alone}$

2. $V_s = 0 \to \text{short}, \qquad I_s = 0 \to \text{open}$

3. $P = R\Big(\displaystyle\sum_k i_k\Big)^2 \neq \sum_k R\, i_k^2$

TERMS x_k response to source k; dead $V_s \to$ short; dead $I_s \to$ open

"One at a time, then add"; "dead voltage source = wire, dead current source = gap."

1. $Q = CV$

2. $i = C\,\dfrac{dv}{dt}$

3. $v = L\,\dfrac{di}{dt}$

TERMS C (F); L (H); ε permittivity; A plate area; d gap

Capacitors combine like resistors "upside down." A capacitor resists sudden voltage changes; an inductor resists sudden current changes.

Voltage source → short circuit.
Current source → open circuit.
Dependent sources stay on.
Never superpose power.

Fewest unknowns: nodal (nodes$-1-V_s$) vs. mesh (meshes$-I_s$).
One quantity: divider, superposition, Thévenin.
Varying load: Thévenin/Norton.
Mixed DC + AC: superposition.

1. $v_C(t) = V_s\big(1 - e^{-t/\tau}\big)$

2. $v_C(t) = V_0\, e^{-t/\tau}$

3. $\tau_{RC} = RC, \qquad \tau_{RL} = \dfrac{L}{R}$

TERMS τ time constant (s); $x(0^+)$ initial; $x(\infty)$ final; V_s source

After 1τ: 63% of the way; after 5τ: over 99% ("five tau and done").

1. $v_C(0^+) = v_C(0^-), \qquad i_L(0^+) = i_L(0^-)$

TERMS C (F); L (H); ε permittivity; A plate area; d gap

Capacitors combine like resistors "upside down." A capacitor resists sudden voltage changes; an inductor resists sudden current changes.

1. $E_C = \tfrac{1}{2}CV^2, \qquad E_L = \tfrac{1}{2}LI^2$

2. $C_{\parallel} = C_1 + C_2, \qquad \dfrac{1}{C_{\text{series}}} = \dfrac{1}{C_1} + \dfrac{1}{C_2}$

3. $C = \varepsilon_r \varepsilon_0\, \dfrac{A}{d}$

TERMS C (F); L (H); ε permittivity; A plate area; d gap

Capacitors combine like resistors "upside down." A capacitor resists sudden voltage changes; an inductor resists sudden current changes.

1. $x_n = A_1 e^{s_1 t} + A_2 e^{s_2 t}$

2. $x_n = (A_1 + A_2 t)\, e^{-\alpha t}$

3. $x_n = e^{-\alpha t}\big(B_1 \cos \omega_d t + B_2 \sin \omega_d t\big)$

TERMS α decay rate; $\omega_0 = 1/\sqrt{LC}$; $\zeta = \alpha/\omega_0$; ω_d damped freq.

$\zeta > 1$ "sluggish", $\zeta = 1$ "just right", $\zeta < 1$ "ringing". Series RLC: more $R \Rightarrow$ more damping; parallel RLC: more $R \Rightarrow$ less.

1. $\ddot{x} + 2\alpha\, \dot{x} + \omega_0^2\, x = \omega_0^2\, x(\infty)$

2. $s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$

3. $\alpha_{\text{ser}} = \dfrac{R}{2L}, \quad \alpha_{\text{par}} = \dfrac{1}{2RC}, \quad \omega_0 = \dfrac{1}{\sqrt{LC}}, \quad \zeta = \dfrac{\alpha}{\omega_0}$

TERMS α decay rate; $\omega_0 = 1/\sqrt{LC}$; $\zeta = \alpha/\omega_0$; ω_d damped freq.

$\zeta > 1$ "sluggish", $\zeta = 1$ "just right", $\zeta < 1$ "ringing". Series RLC: more $R \Rightarrow$ more damping; parallel RLC: more $R \Rightarrow$ less.

1. $i_L(t) = \dfrac{V_s}{R}\big(1 - e^{-t/\tau}\big)$

2. $x(t) = x(\infty) + \big[x(0^+) - x(\infty)\big] e^{-t/\tau}$

3. $RC\,\dfrac{dv}{dt} + v = V_s$

TERMS τ time constant (s); $x(0^+)$ initial; $x(\infty)$ final; V_s source

After 1τ: 63% of the way; after 5τ: over 99% ("five tau and done").

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EE.12 (3/3) ELECTRICAL ENGINEERING

Second-Order (RLC) Transients & Damping

PART 3 OF 3

1. Overshoot & settling

STATE EACH NUMBERED FORMULA

EE.13 (1/2) ELECTRICAL ENGINEERING

AC Signals & RMS Values

PART 1 OF 2

1. Sinusoid
2. Frequency and period
3. RMS value

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EE.13 (2/2) ELECTRICAL ENGINEERING

AC Signals & RMS Values

PART 2 OF 2

1. RMS of a sine
2. Other waveforms
3. Average power to a resistor

STATE EACH NUMBERED FORMULA

EE.14 (1/2) ELECTRICAL ENGINEERING

Phasors & Impedance

PART 1 OF 2

1. Phasor
2. Impedances
3. Ohm's law for AC

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EE.14 (2/2) ELECTRICAL ENGINEERING

Phasors & Impedance

PART 2 OF 2

1. Rectangular and polar
2. Admittance
3. Derivative rule

STATE EACH NUMBERED FORMULA

EE.15 (1/2) ELECTRICAL ENGINEERING

AC Power & Power Factor

PART 1 OF 2

1. Real power
2. Reactive power
3. Complex power

STATE EACH NUMBERED FORMULA

EE.15 (2/2) ELECTRICAL ENGINEERING

AC Power & Power Factor

PART 2 OF 2

1. Power factor
2. Power-factor correction

STATE EACH NUMBERED FORMULA

EE.16 (1/2) ELECTRICAL ENGINEERING

RLC Circuits & Resonance

PART 1 OF 2

1. Resonant frequency
2. Series impedance
3. Quality factor (series)

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EE.16 (2/2) ELECTRICAL ENGINEERING

RLC Circuits & Resonance

PART 2 OF 2

1. Quality factor (parallel)
2. Bandwidth
3. Half-power frequencies

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EE.17 (1/2) ELECTRICAL ENGINEERING

Filters, Frequency Response & Decibels

PART 1 OF 2

1. RC low-pass
2. Magnitude
3. Cutoff frequency

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EE.17 (2/2) ELECTRICAL ENGINEERING

Filters, Frequency Response & Decibels

PART 2 OF 2

1. Gain in decibels
2. Phase
3. RC high-pass

STATE EACH NUMBERED FORMULA

EE.18 (1/2) ELECTRICAL ENGINEERING

Electric & Magnetic Fields

PART 1 OF 2

1. Coulomb's law
2. Electric field
3. Field of a long wire

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1. $V_{\text{rms}} = \dfrac{V_m}{\sqrt{2}} \approx 0.707\, V_m$
2. square: V_m, triangle: $\dfrac{V_m}{\sqrt{3}}$
3. $P = \dfrac{V_{\text{rms}}^2}{R} = I_{\text{rms}}^2 R$

TERMS V_m peak (V); f Hz; ω rad/s; T period (s); ϕ phase

RMS = Root of the Mean of the Square; for sine waves, peak ≈ 1.414× RMS.

1. $v(t) = V_m \sin(\omega t + \phi)$
2. $\omega = 2\pi f, \qquad T = \dfrac{1}{f}$
3. $V_{\text{rms}} = \sqrt{\dfrac{1}{T}\displaystyle\int_0^T v^2\, dt}$

TERMS V_m peak (V); f Hz; ω rad/s; T period (s); ϕ phase

RMS = Root of the Mean of the Square; for sine waves, peak ≈ 1.414× RMS.

1. $\%OS = 100\, e^{-\pi\zeta/\sqrt{1-\zeta^2}}, \qquad t_s \approx \dfrac{4}{\alpha}$

TERMS α decay rate; $\omega_0 = 1/\sqrt{LC}$; $\zeta = \alpha/\omega_0$; ω_d damped freq.

$\zeta > 1$ "sluggish", $\zeta = 1$ "just right", $\zeta < 1$ "ringing". Series RLC: more $R \Rightarrow$ more damping; parallel RLC: more $R \Rightarrow$ less.

1. $P = V_{\text{rms}} I_{\text{rms}} \cos\theta$
2. $Q = V_{\text{rms}} I_{\text{rms}} \sin\theta$
3. $S = \tilde{V}\tilde{I}^* = P + jQ, \qquad |S|^2 = P^2 + Q^2$

TERMS P (W); Q (var); $|S|$ (VA); θ phase angle; pf power factor

The power triangle: P is the "beer" (useful), Q the "foam," $|S|$ the whole glass.

1. $Z = R + jX, \quad |Z| = \sqrt{R^2 + X^2}, \quad \theta = \tan^{-1}\dfrac{X}{R}$
2. $Y = \dfrac{1}{Z} = G + jB$
3. $\dfrac{d}{dt} \;\leftrightarrow\; j\omega$

TERMS $j^2 = -1$; $Z = R + jX$ (Ω); X reactance; $Y = 1/Z$ (S); $\angle$ phase

ELI the ICE man: in an inductor (L), voltage (E) leads current (I); in a capacitor (C), current (I) leads voltage (E).

1. $V_m \cos(\omega t + \phi) \;\leftrightarrow\; \tilde{V} = V_m \angle\phi = V_m e^{j\phi}$
2. $Z_R = R, \qquad Z_L = j\omega L, \qquad Z_C = \dfrac{1}{j\omega C} = -\dfrac{j}{\omega C}$
3. $\tilde{V} = \tilde{I}\, Z$

TERMS $j^2 = -1$; $Z = R + jX$ (Ω); X reactance; $Y = 1/Z$ (S); $\angle$ phase

ELI the ICE man: in an inductor (L), voltage (E) leads current (I); in a capacitor (C), current (I) leads voltage (E).

1. $Q_{\text{p}} = R\sqrt{\dfrac{C}{L}}$
2. $\text{BW} = \omega_2 - \omega_1 = \dfrac{\omega_0}{Q}$
3. $\omega_{1,2} \approx \omega_0 \mp \dfrac{\text{BW}}{2} \quad (Q \gtrsim 5)$

TERMS ω_0 (rad/s); Q quality factor; BW bandwidth; $\omega_{1,2}$ half-power

At resonance the inductor and capacitor reactances cancel: $\omega L = \frac{1}{\omega C}$.

1. $\omega_0 = \dfrac{1}{\sqrt{LC}}, \qquad f_0 = \dfrac{1}{2\pi\sqrt{LC}}$
2. $Z = R + j\left(\omega L - \dfrac{1}{\omega C}\right)$
3. $Q_{\text{s}} = \dfrac{1}{R}\sqrt{\dfrac{L}{C}} = \dfrac{\omega_0 L}{R}$

TERMS ω_0 (rad/s); Q quality factor; BW bandwidth; $\omega_{1,2}$ half-power

At resonance the inductor and capacitor reactances cancel: $\omega L = \frac{1}{\omega C}$.

1. $\text{pf} = \cos\theta = \dfrac{P}{|S|}$
2. $Q_C = P\,(\tan\theta_1 - \tan\theta_2), \qquad C = \dfrac{Q_C}{\omega V_{\text{rms}}^2}$

TERMS P (W); Q (var); $|S|$ (VA); θ phase angle; pf power factor

The power triangle: P is the "beer" (useful), Q the "foam," $|S|$ the whole glass.

1. $F = k\,\dfrac{q_1 q_2}{r^2}$
2. $\mathbf{E} = \dfrac{\mathbf{F}}{q}, \qquad E = \dfrac{V}{d}$ (uniform)
3. $B = \dfrac{\mu_0 I}{2\pi r}$

TERMS k Coulomb const.; μ_0; Φ_B flux (Wb); N turns; n turns/m; B (T)

Faraday: "change the flux, get a voltage"; the minus sign (Lenz) means the induced current opposes the change.

1. $|H|_{\text{dB}} = 20\log_{10}|H|$
2. $\angle H = -\tan^{-1}\dfrac{f}{f_c}$
3. $H(j\omega) = \dfrac{j\omega RC}{1 + j\omega RC}$

TERMS f_c cutoff (Hz); RC time constant (s); dB decibels; $\angle H$ phase

At the cutoff, $|H| = \frac{1}{\sqrt{2}}$, i.e. -3 dB (half power). Past it, -20 dB per decade for each RC stage.

1. $H(j\omega) = \dfrac{V_{\text{out}}}{V_{\text{in}}} = \dfrac{1}{1 + j\omega RC}$
2. $|H| = \dfrac{1}{\sqrt{1 + (f/f_c)^2}}$
3. $f_c = \dfrac{1}{2\pi RC}$

TERMS f_c cutoff (Hz); RC time constant (s); dB decibels; $\angle H$ phase

At the cutoff, $|H| = \frac{1}{\sqrt{2}}$, i.e. -3 dB (half power). Past it, -20 dB per decade for each RC stage.

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EE.18 (2/2) ELECTRICAL ENGINEERING

Electric & Magnetic Fields

PART 2 OF 2

1. Solenoid
2. Faraday's law
3. Lorentz force

STATE EACH NUMBERED FORMULA

EE.19 (1/2) ELECTRICAL ENGINEERING

Transformers & Three-Phase Power

PART 1 OF 2

1. Turns ratio
2. Ideal power balance
3. Reflected impedance

STATE EACH NUMBERED FORMULA

EE.19 (2/2) ELECTRICAL ENGINEERING

Transformers & Three-Phase Power

PART 2 OF 2

1. Three-phase power
2. Line loss

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EE.20 (1/2) ELECTRICAL ENGINEERING

Three-Phase Systems: Wye & Delta

PART 1 OF 2

1. Balanced phase voltages
2. Wye (Y) connection
3. Delta (Δ) connection

STATE EACH NUMBERED FORMULA

EE.20 (2/2) ELECTRICAL ENGINEERING

Three-Phase Systems: Wye & Delta

PART 2 OF 2

1. Total real power
2. Apparent and reactive
3. Y–Δ load conversion

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EE.21 (1/2) ELECTRICAL ENGINEERING

Batteries & Energy Storage

PART 1 OF 2

1. Energy
2. Runtime
3. C-rate

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EE.21 (2/2) ELECTRICAL ENGINEERING

Batteries & Energy Storage

PART 2 OF 2

1. Round-trip efficiency
2. Pack wiring
3. Terminal voltage

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EE.22 (1/3) ELECTRICAL ENGINEERING

LTI Systems, Impulse Response & Convolution

PART 1 OF 3

1. Linearity
2. Time invariance
3. Convolution (continuous)

STATE EACH NUMBERED FORMULA

EE.22 (2/3) ELECTRICAL ENGINEERING

LTI Systems, Impulse Response & Convolution

PART 2 OF 3

1. Convolution (discrete)
2. Convolution theorem
3. Sinusoidal steady state

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EE.22 (3/3) ELECTRICAL ENGINEERING

LTI Systems, Impulse Response & Convolution

PART 3 OF 3

1. BIBO stability

STATE EACH NUMBERED FORMULA

EE.23 (1/2) ELECTRICAL ENGINEERING

Laplace Transform & Transfer Functions

PART 1 OF 2

1. Laplace transform
2. Derivative and integral rules
3. Core pairs

STATE EACH NUMBERED FORMULA

EE.23 (2/2) ELECTRICAL ENGINEERING

Laplace Transform & Transfer Functions

PART 2 OF 2

1. *s*-domain impedances
2. Transfer function
3. Final and initial value theorems

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1. $P_{3\phi} = \sqrt{3}\, V_L I_L \cos\theta$
2. $P_{\text{loss}} = I^2 R = \left(\dfrac{P}{V}\right)^2 R$

TERMS N_p, N_s turns; p primary, s secondary; V_L, I_L line values

"Step up voltage, step down current": power in = power out (ideal).

1. $\dfrac{V_s}{V_p} = \dfrac{N_s}{N_p} = \dfrac{I_p}{I_s}$
2. $V_p I_p = V_s I_s$
3. $Z_{\text{in}} = \left(\dfrac{N_p}{N_s}\right)^2 Z_L$

TERMS N_p, N_s turns; p primary, s secondary; V_L, I_L line values

"Step up voltage, step down current": power in = power out (ideal).

1. $B = \mu_0 n I$
2. $\mathcal{E} = -N\dfrac{d\Phi_B}{dt}$
3. $\mathbf{F} = q\left(\mathbf{E} + \mathbf{v}\times\mathbf{B}\right)$

TERMS k Coulomb const.; μ_0; Φ_B flux (Wb); N turns; n turns/m; B (T)

Faraday: "change the flux, get a voltage"; the minus sign (Lenz) means the induced current opposes the change.

1. $E\,(\text{Wh}) = V_{\text{nominal}} \times \text{capacity (Ah)}$
2. $t_{\text{run}} \approx \dfrac{\text{capacity (Ah)}}{I\,(\text{A})}$
3. $\text{C-rate} = \dfrac{I}{\text{capacity}}$

TERMS Ah charge; Wh energy; C-rate; n_s, n_p series/parallel cells; η

"mAh is charge, Wh is energy": multiply by voltage to compare batteries fairly.

1. $P = 3V_{\text{ph}} I_{\text{ph}} \cos\theta = \sqrt{3}\, V_L I_L \cos\theta$
2. $|S| = \sqrt{3}\, V_L I_L, \qquad Q = \sqrt{3}\, V_L I_L \sin\theta$
3. $Z_Y = \dfrac{Z_\Delta}{3}$

TERMS ph phase quantities; L line quantities; $\sqrt{3}$ from 120° spacing

"Wye: voltage gets the root three; Delta: current gets the root three."

1. $\tilde{V}_a = V_p\angle 0^\circ, \quad \tilde{V}_b = V_p\angle -120^\circ, \quad \tilde{V}_c = V_p\angle 120^\circ$
2. $V_L = \sqrt{3}\, V_{\text{ph}}, \qquad I_L = I_{\text{ph}}$
3. $V_L = V_{\text{ph}}, \qquad I_L = \sqrt{3}\, I_{\text{ph}}$

TERMS ph phase quantities; L line quantities; $\sqrt{3}$ from 120° spacing

"Wye: voltage gets the root three; Delta: current gets the root three."

1. $y[n] = \displaystyle\sum_k x[k]\, h[n-k]$
2. $y = x * h \iff Y = X \cdot H$
3. $\cos\omega t \;\mapsto\; |H(j\omega)|\cos\big(\omega t + \angle H(j\omega)\big)$

TERMS h impulse response; $*$ convolution; δ impulse; H transfer function

"Flip and slide": convolution flips h and slides it across x.

1. $ax_1 + bx_2 \;\mapsto\; ay_1 + by_2$
2. $x(t - t_0) \;\mapsto\; y(t - t_0)$
3. $y(t) = (x * h)(t) = \displaystyle\int_{-\infty}^{\infty} x(\tau)\, h(t-\tau)\, d\tau$

TERMS h impulse response; $*$ convolution; δ impulse; H transfer function

"Flip and slide": convolution flips h and slides it across x.

1. $\eta_{\text{round trip}} = \dfrac{E_{\text{out}}}{E_{\text{in}}}$
2. $V_{\text{pack}} = n_s V_{\text{cell}}, \qquad \text{Ah}_{\text{pack}} = n_p\, \text{Ah}_{\text{cell}}$
3. $V = V_{\text{oc}} - I R_{\text{int}}$

TERMS Ah charge; Wh energy; C-rate; n_s, n_p series/parallel cells; η

"mAh is charge, Wh is energy": multiply by voltage to compare batteries fairly.

1. $Z_R = R, \qquad Z_L = sL, \qquad Z_C = \dfrac{1}{sC}$
2. $H(s) = \dfrac{Y(s)}{X(s)}\Big|_{\text{zero ICs}} = \mathcal{L}\{h(t)\}$
3. $f(\infty) = \lim_{s\to 0} sF(s), \qquad f(0^+) = \lim_{s\to\infty} sF(s)$

TERMS $s = \sigma + j\omega$; u step; δ impulse; $H(s)$ transfer function

"s is a derivative, 1/s is an integral."

1. $F(s) = \displaystyle\int_{0^-}^{\infty} f(t)\, e^{-st}\, dt$
2. $\mathcal{L}\{f'\} = sF(s) - f(0^-), \qquad \mathcal{L}\Big\{\displaystyle\int_0^t f\Big\} = \dfrac{F(s)}{s}$
3. $u(t) \to \dfrac{1}{s}, \quad e^{-at} \to \dfrac{1}{s+a}, \quad \sin\omega t \to \dfrac{\omega}{s^2+\omega^2}$

TERMS $s = \sigma + j\omega$; u step; δ impulse; $H(s)$ transfer function

"s is a derivative, 1/s is an integral."

1. $\displaystyle\int_{-\infty}^{\infty} |h(t)|\, dt < \infty, \qquad \sum_n |h[n]| < \infty$

TERMS h impulse response; $*$ convolution; δ impulse; H transfer function

"Flip and slide": convolution flips h and slides it across x.

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EE.24 (1/2)

ELECTRICAL ENGINEERING

Poles, Zeros & Stability

PART 1 OF 2

1. Pole–zero form
2. Natural response
3. Stability

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EE.24 (2/2)

ELECTRICAL ENGINEERING

Poles, Zeros & Stability

PART 2 OF 2

1. Standard second-order
2. DC gain
3. Order and properness

STATE EACH NUMBERED FORMULA

EE.25 (1/2)

ELECTRICAL ENGINEERING

Bode Plots

PART 1 OF 2

1. Bode (time-constant) form
2. Gain adds in dB
3. Real pole at ω_p

STATE EACH NUMBERED FORMULA

EE.25 (2/2)

ELECTRICAL ENGINEERING

Bode Plots

PART 2 OF 2

1. Real zero at ω_z
2. Pole at the origin
3. Complex pole pair

STATE EACH NUMBERED FORMULA

EE.26 (1/3)

ELECTRICAL ENGINEERING

Fourier Series, the DFT & FFT

PART 1 OF 3

1. Complex Fourier series
2. Fourier coefficient
3. Fourier transform

STATE EACH NUMBERED FORMULA

EE.26 (2/3)

ELECTRICAL ENGINEERING

Fourier Series, the DFT & FFT

PART 2 OF 3

1. Discrete Fourier transform
2. Frequency resolution
3. FFT speed-up

STATE EACH NUMBERED FORMULA

EE.26 (3/3)

ELECTRICAL ENGINEERING

Fourier Series, the DFT & FFT

PART 3 OF 3

1. Parseval's theorem

STATE EACH NUMBERED FORMULA

EE.27 (1/2)

ELECTRICAL ENGINEERING

Sampling, Nyquist & Quantization

PART 1 OF 2

1. Nyquist criterion
2. Alias frequency
3. ADC resolution (LSB)

STATE EACH NUMBERED FORMULA

EE.27 (2/2)

ELECTRICAL ENGINEERING

Sampling, Nyquist & Quantization

PART 2 OF 2

1. ADC output code
2. Quantization noise
3. SQNR

STATE EACH NUMBERED FORMULA

EE.28 (1/2)

ELECTRICAL ENGINEERING

The Z-Transform & Digital Filters

PART 1 OF 2

1. Z-transform
2. Delay property
3. Core pairs

STATE EACH NUMBERED FORMULA

EE.28 (2/2)

ELECTRICAL ENGINEERING

The Z-Transform & Digital Filters

PART 2 OF 2

1. Transfer function of a difference equation
2. Stability
3. Frequency response

STATE EACH NUMBERED FORMULA

EE.29 (1/2)

ELECTRICAL ENGINEERING

Decibels, SNR & Channel Capacity

PART 1 OF 2

1. Decibel (power)
2. Decibel (voltage)
3. Absolute levels

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1. $H(j\omega) = K\,\dfrac{\prod(1+j\omega/\omega_z)}{\prod(1+j\omega/\omega_p)}$

2. $|H|_{\text{dB}} = 20\log_{10}K + \sum 20\log_{10}|\text{zero factors}| - \sum 20\log_{10}|\text{pole factors}|$

3. -20 dB/dec above ω_p, -3 dB and -45° at ω_p

TERMS ω_p pole; ω_z zero; dB/dec slope; K DC gain; phase in degrees

"Pole: down 20 and back 90; zero: up 20 and forward 90." Phase changes over two decades centered on the corner.

1. $H(s) = \dfrac{\omega_0^2}{s^2 + 2\zeta\omega_0 s + \omega_0^2}$

2. $H(0) = K\,\dfrac{\prod(-z_i)}{\prod(-p_k)}$

3. $n = \deg(\text{den}) \;\geq\; m = \deg(\text{num})$

TERMS z_i zeros; p_k poles; K gain; stable $\Leftrightarrow \operatorname{Re} p_k < 0$

"Left is safe, right is unstable, the $j\omega$ axis rings forever"; distance from the axis = decay rate, height = oscillation frequency.

1. $H(s) = K\,\dfrac{(s-z_1)\cdots(s-z_m)}{(s-p_1)\cdots(s-p_n)}$

2. $y_n(t) = \displaystyle\sum_k A_k\, e^{p_k t}$

3. $\operatorname{Re}(p_k) < 0 \;\;\text{for all } k$

TERMS z_i zeros; p_k poles; K gain; stable $\Leftrightarrow \operatorname{Re} p_k < 0$

"Left is safe, right is unstable, the $j\omega$ axis rings forever"; distance from the axis = decay rate, height = oscillation frequency.

1. $X_k = \displaystyle\sum_{n=0}^{N-1} x_n\, e^{-j2\pi kn/N}$

2. $\Delta f = \dfrac{f_s}{N}$

3. DFT $O(N^2) \;\rightarrow\;$ FFT $O(N \log N)$

TERMS c_n coefficients; ω_0 fundamental; X_k bin k; f_s rate; N samples

Every signal is a sum of spinning arrows (sinusoids) T.23 *§06 Trigonometry; the FFT finds how much of each.*

1. $x(t) = \displaystyle\sum_{n=-\infty}^{\infty} c_n\, e^{jn\omega_0 t}$

2. $c_n = \dfrac{1}{T}\displaystyle\int_0^T x(t)\, e^{-jn\omega_0 t}\, dt$

3. $X(f) = \displaystyle\int_{-\infty}^{\infty} x(t)\, e^{-j2\pi ft}\, dt$

TERMS c_n coefficients; ω_0 fundamental; X_k bin k; f_s rate; N samples

Every signal is a sum of spinning arrows (sinusoids) T.23 *§06 Trigonometry; the FFT finds how much of each.*

1. $+20$ dB/dec, $\quad 0^\circ \rightarrow +90^\circ$

2. $\dfrac{1}{j\omega}$: -20 dB/dec through 0 dB at $\omega = 1$, $\;-90^\circ$

3. -40 dB/dec above ω_0, $\quad$ peak $\approx \dfrac{1}{2\zeta}$ $(\zeta \ll 1)$

TERMS ω_p pole; ω_z zero; dB/dec slope; K DC gain; phase in degrees

"Pole: down 20 and back 90; zero: up 20 and forward 90." Phase changes over two decades centered on the corner.

1. $D = \left\lfloor \dfrac{V_{\text{in}}}{V_{\text{ref}}}\, 2^N \right\rfloor$

2. $e_{\text{rms}} = \dfrac{\Delta}{\sqrt{12}}$

3. $\text{SQNR} \approx 6.02N + 1.76$ dB

TERMS f_s sampling rate; N bits; V_{ref} full scale; Δ LSB; D code

"Sample at least twice as fast as the fastest wiggle"; each extra bit buys about 6 dB.

1. $f_s > 2f_{\max}$

2. $f_{\text{alias}} = |f - kf_s|$

3. $\Delta = \dfrac{V_{\text{ref}}}{2^N}$

TERMS f_s sampling rate; N bits; V_{ref} full scale; Δ LSB; D code

"Sample at least twice as fast as the fastest wiggle"; each extra bit buys about 6 dB.

1. $\displaystyle\sum_n |x_n|^2 = \frac{1}{N}\sum_k |X_k|^2$

TERMS c_n coefficients; ω_0 fundamental; X_k bin k; f_s rate; N samples

Every signal is a sum of spinning arrows (sinusoids) T.23 *§06 Trigonometry; the FFT finds how much of each.*

1. $\text{dB} = 10\log_{10}\dfrac{P_2}{P_1}$

2. $\text{dB} = 20\log_{10}\dfrac{V_2}{V_1}$

3. $\text{dBm} = 10\log_{10}\dfrac{P}{1\text{ mW}}$

TERMS dB ratio; dBm vs. 1 mW; SNR linear in C; B bandwidth (Hz)

$+3$ dB $\approx 2\times$ power, $+10$ dB $= 10\times$ power, $+20$ dB $= 100\times$ power ($10\times$ voltage).

1. $y[n] = \sum_k b_k x[n-k] - \sum_{k\geq 1} a_k y[n-k] \;\Rightarrow\; H(z) = \dfrac{\sum b_k z^{-k}}{1 + \sum a_k z^{-k}}$

2. $|p_k| < 1 \;\;\text{for all poles}$

3. $H(e^{j\Omega}), \qquad \Omega = 2\pi\dfrac{f}{f_s}$

TERMS z^{-1} unit delay; b_k feed-forward; a_k feedback; poles inside $|z| = 1$

"s-plane left half $\leftrightarrow$ z-plane unit disk." z^{-1} = "one sample ago".

1. $X(z) = \displaystyle\sum_{n=0}^{\infty} x[n]\, z^{-n}$

2. $x[n-k] \;\leftrightarrow\; z^{-k}X(z)$

3. $\delta[n] \rightarrow 1, \qquad a^n u[n] \rightarrow \dfrac{z}{z-a} = \dfrac{1}{1-az^{-1}}$

TERMS z^{-1} unit delay; b_k feed-forward; a_k feedback; poles inside $|z| = 1$

"s-plane left half $\leftrightarrow$ z-plane unit disk." z^{-1} = "one sample ago".

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EE.29 (2/2) ELECTRICAL ENGINEERING

Decibels, SNR & Channel Capacity

PART 2 OF 2

1. Signal-to-noise ratio
2. Cascaded gains
3. Shannon–Hartley capacity

STATE EACH NUMBERED FORMULA

EE.30 (1/2) ELECTRICAL ENGINEERING

Feedback & Stability

PART 1 OF 2

1. Closed-loop transfer function
2. Loop gain
3. Sensitivity

STATE EACH NUMBERED FORMULA

EE.30 (2/2) ELECTRICAL ENGINEERING

Feedback & Stability

PART 2 OF 2

1. Steady-state error (step)
2. Characteristic equation
3. Stability margins

STATE EACH NUMBERED FORMULA

EE.31 (1/2) ELECTRICAL ENGINEERING

PID Control

PART 1 OF 2

1. PID law
2. Transfer function
3. Standard (ISA) form

STATE EACH NUMBERED FORMULA

EE.31 (2/2) ELECTRICAL ENGINEERING

PID Control

PART 2 OF 2

1. Discrete PID
2. Ziegler–Nichols tuning

STATE EACH NUMBERED FORMULA

EE.32 (1/2) ELECTRICAL ENGINEERING

Diodes, LEDs & Transistors

PART 1 OF 2

1. Shockley diode equation
2. Thermal voltage
3. LED series resistor

STATE EACH NUMBERED FORMULA

EE.32 (2/2) ELECTRICAL ENGINEERING

Diodes, LEDs & Transistors

PART 2 OF 2

1. Constant-drop model
2. Half-wave rectifier
3. Ripple with filter capacitor

STATE EACH NUMBERED FORMULA

EE.33 (1/2) ELECTRICAL ENGINEERING

BJT & MOSFET Operating Regions

PART 1 OF 2

1. BJT currents
2. BJT regions
3. MOSFET cutoff

STATE EACH NUMBERED FORMULA

EE.33 (2/2) ELECTRICAL ENGINEERING

BJT & MOSFET Operating Regions

PART 2 OF 2

1. MOSFET triode (linear)
2. MOSFET saturation
3. Transconductance

STATE EACH NUMBERED FORMULA

EE.34 (1/2) ELECTRICAL ENGINEERING

Operational Amplifiers

PART 1 OF 2

1. Open-loop gain
2. Ideal rules (negative feedback)
3. Inverting amplifier

STATE EACH NUMBERED FORMULA

EE.34 (2/2) ELECTRICAL ENGINEERING

Operational Amplifiers

PART 2 OF 2

1. Non-inverting amplifier
2. Voltage follower
3. Gain–bandwidth product

STATE EACH NUMBERED FORMULA

EE.35 (1/3) ELECTRICAL ENGINEERING

Op-Amp Configurations

PART 1 OF 3

1. Generalized inverting
2. Summing amplifier
3. Difference amplifier

STATE EACH NUMBERED FORMULA

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1 $e_{ss} = \dfrac{1}{1+K_p}, \qquad K_p = \lim_{s\to 0} L(s)$

2 $1 + L(s) = 0$

3 $\mathrm{GM} = -20\log_{10}|L(j\omega_{180})|, \qquad \mathrm{PM} = 180^\circ + \angle L(j\omega_c)$

TERMS G forward; H feedback; $L = GH$ loop gain; GM, PM margins

"Big loop gain, small error"; but "gain + lag = wobble." Aim for phase margin $\geq 45°$.

1 $T(s) = \dfrac{Y}{R} = \dfrac{G(s)}{1 + G(s)H(s)}$

2 $L(s) = G(s)H(s), \qquad |L| \gg 1 \Rightarrow T \approx \dfrac{1}{H}$

3 $S(s) = \dfrac{1}{1+L(s)}, \qquad E = S\,R \;\; (H = 1)$

TERMS G forward; H feedback; $L = GH$ loop gain; GM, PM margins

"Big loop gain, small error"; but "gain + lag = wobble." Aim for phase margin $\geq 45°$.

1 $\mathrm{SNR} = \dfrac{P_{\text{signal}}}{P_{\text{noise}}}, \quad \mathrm{SNR}_{\text{dB}} = 10\log_{10}\mathrm{SNR}$

2 $G_{\text{total,dB}} = G_{1,\text{dB}} + G_{2,\text{dB}} + \cdots$

3 $C = B\log_2(1 + \mathrm{SNR})$

TERMS dB ratio; dBm vs. 1 mW; SNR linear in C; B bandwidth (Hz)

+3 dB ≈ 2× power, +10 dB = 10× power, +20 dB = 100× power (10× voltage).

1 $I = I_s\left(e^{V/(nV_T)} - 1\right)$

2 $V_T = \dfrac{kT}{q} \approx 25.9\ \text{mV at 300 K}$

3 $R_{\text{LED}} = \dfrac{V_s - V_f}{I_{\text{LED}}}$

TERMS I_s saturation current; n ideality; $V_T = kT/q$; V_f forward voltage

Silicon diode "turns on" near 0.7 V; the current grows about 10× for every 60 mV more (with $n = 1$).

1 $u_k = K_p e_k + K_i T_s \sum_{j\leq k} e_j + K_d \dfrac{e_k - e_{k-1}}{T_s}$

2 $K_p = 0.6K_u, \quad T_i = \dfrac{T_u}{2}, \quad T_d = \dfrac{T_u}{8}$

TERMS K_p, K_i, K_d gains; $e = r - y$ error; u control; T_i, T_d times

P = present, I = past, D = future. "P reacts, I removes offset, D damps."

1 $u(t) = K_p e(t) + K_i \int_0^t e(\tau)\,d\tau + K_d \dfrac{de}{dt}$

2 $C(s) = K_p + \dfrac{K_i}{s} + K_d s$

3 $C(s) = K_p\left(1 + \dfrac{1}{T_i s} + T_d s\right)$

TERMS K_p, K_i, K_d gains; $e = r - y$ error; u control; T_i, T_d times

P = present, I = past, D = future. "P reacts, I removes offset, D damps."

1 $I_D = k_n\left[(V_{GS} - V_t)V_{DS} - \tfrac{1}{2}V_{DS}^2\right], \;\; V_{DS} < V_{GS} - V_t$

2 $I_D = \tfrac{1}{2}k_n(V_{GS} - V_t)^2, \;\; V_{DS} \geq V_{GS} - V_t$

3 $g_m^{\text{MOS}} = \dfrac{2I_D}{V_{GS} - V_t}, \qquad g_m^{\text{BJT}} = \dfrac{I_C}{V_T}$

TERMS β gain; V_t threshold; k_n (A/V^2); $V_{GS} - V_t$ overdrive; g_m (S)

The names clash: BJT "saturation" = switch fully on; MOSFET "saturation" = amplifier region. MOSFET switch on = triode.

1 $I_E = I_B + I_C, \qquad I_C = \beta I_B \;\; \text{(active)}$

2 cutoff: $V_{BE} < 0.7$; active: $V_{CE} > 0.2$; saturation: $V_{CE} \approx 0.2$ V

3 $V_{GS} < V_t \;\Rightarrow\; I_D = 0$

TERMS β gain; V_t threshold; k_n (A/V^2); $V_{GS} - V_t$ overdrive; g_m (S)

The names clash: BJT "saturation" = switch fully on; MOSFET "saturation" = amplifier region. MOSFET switch on = triode.

1 $V_D \approx 0.7$ V (Si) when on, $\qquad I = 0$ when off

2 $V_{\text{out,pk}} \approx V_m - 0.7\ \text{V}$

3 $V_r \approx \dfrac{I_{\text{load}}}{fC}$

TERMS I_s saturation current; n ideality; $V_T = kT/q$; V_f forward voltage

Silicon diode "turns on" near 0.7 V; the current grows about 10× for every 60 mV more (with $n = 1$).

1 $\dfrac{V_{\text{out}}}{V_{\text{in}}} = -\dfrac{Z_f}{Z_{\text{in}}}$

2 $V_{\text{out}} = -R_f\left(\dfrac{V_1}{R_1} + \dfrac{V_2}{R_2} + \cdots\right)$

3 $V_{\text{out}} = \dfrac{R_2}{R_1}(V_2 - V_1)$

TERMS Z_f feedback; Z_{in} input; RC (s); V_{sat} rail level

"Capacitor in feedback integrates; capacitor at the input differentiates."

1 $\dfrac{V_{\text{out}}}{V_{\text{in}}} = 1 + \dfrac{R_f}{R_g}$

2 $V_{\text{out}} = V_{\text{in}}$

3 $A_{\text{cl}} \cdot f_{-3\text{dB}} \approx \text{GBW}$

TERMS V_+, V_- inputs; R_f feedback; R_{in}, R_g; A open-loop gain; GBW

The two golden rules: "the inputs are at the same voltage" and "no current enters the inputs."

1 $V_{\text{out}} = A\,(V_+ - V_-)$

2 $V_+ = V_-, \qquad I_+ = I_- = 0$

3 $\dfrac{V_{\text{out}}}{V_{\text{in}}} = -\dfrac{R_f}{R_{\text{in}}}$

TERMS V_+, V_- inputs; R_f feedback; R_{in}, R_g; A open-loop gain; GBW

The two golden rules: "the inputs are at the same voltage" and "no current enters the inputs."

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EE.35 (2/3) ELECTRICAL ENGINEERING

Op-Amp Configurations

PART 2 OF 3

1. Integrator
2. Differentiator
3. Buffer (follower)

STATE EACH NUMBERED FORMULA

EE.35 (3/3) ELECTRICAL ENGINEERING

Op-Amp Configurations

PART 3 OF 3

1. Comparator

STATE EACH NUMBERED FORMULA

EE.36 (1/2) ELECTRICAL ENGINEERING

Digital Logic, Binary & Boolean Algebra

PART 1 OF 2

1. Binary value
2. Number of values
3. Basic operations

STATE EACH NUMBERED FORMULA

EE.36 (2/2) ELECTRICAL ENGINEERING

Digital Logic, Binary & Boolean Algebra

PART 2 OF 2

1. De Morgan's laws
2. Key identities
3. Sum of products

STATE EACH NUMBERED FORMULA

EE.37 (1/3) ELECTRICAL ENGINEERING

Number Systems & Two's Complement

PART 1 OF 3

1. Positional value (base b)
2. Hex and octal grouping
3. Unsigned range

STATE EACH NUMBERED FORMULA

EE.37 (2/3) ELECTRICAL ENGINEERING

Number Systems & Two's Complement

PART 2 OF 3

1. Two's complement value
2. Signed range
3. Negation

STATE EACH NUMBERED FORMULA

EE.37 (3/3) ELECTRICAL ENGINEERING

Number Systems & Two's Complement

PART 3 OF 3

1. Overflow rule
2. IEEE 754 floating point

STATE EACH NUMBERED FORMULA

EE.38 (1/2) ELECTRICAL ENGINEERING

Karnaugh Maps & Logic Minimization

PART 1 OF 2

1. Combining rule
2. Gray-code order
3. Group size

STATE EACH NUMBERED FORMULA

EE.38 (2/2) ELECTRICAL ENGINEERING

Karnaugh Maps & Logic Minimization

PART 2 OF 2

1. Minterm list
2. Don't cares
3. Product of sums

STATE EACH NUMBERED FORMULA

EE.39 (1/2) ELECTRICAL ENGINEERING

Flip-Flops, Registers & Counters

PART 1 OF 2

1. D flip-flop
2. JK flip-flop
3. T flip-flop

STATE EACH NUMBERED FORMULA

EE.39 (2/2) ELECTRICAL ENGINEERING

Flip-Flops, Registers & Counters

PART 2 OF 2

1. SR latch
2. Binary counter
3. Maximum clock frequency

STATE EACH NUMBERED FORMULA

EE.40 (1/2) ELECTRICAL ENGINEERING

Schematic Symbols: Passive Components & Switches

PART 1 OF 2

1. Resistor
2. Capacitor
3. Inductor

STATE EACH NUMBERED FORMULA

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1 $N = \sum_{i=0}^{n-1} b_i\, 2^i$

2 n bits $\to 2^n$ values

3 $A \cdot B$ (AND), $A + B$ (OR), $\bar{A}$ (NOT)

TERMS b_i bits; $\cdot$ AND; $+$ OR; bar NOT; minterm = one true row

De Morgan: "break the bar, change the operator" (same as for sets A1.21 *§03 Algebra I).*

1 $V_{\text{out}} = \begin{cases} +V_{\text{sat}}, & V_+ > V_- \\ -V_{\text{sat}}, & V_+ < V_- \end{cases}$

TERMS Z_f feedback; Z_{in} input; RC (s); V_{sat} rail level

"Capacitor in feedback integrates; capacitor at the input differentiates."

1 $V_{\text{out}} = -\dfrac{1}{RC}\int V_{\text{in}}\, dt, \qquad H(s) = -\dfrac{1}{sRC}$

2 $V_{\text{out}} = -RC\,\dfrac{dV_{\text{in}}}{dt}, \qquad H(s) = -sRC$

3 $V_{\text{out}} = V_{\text{in}}, \quad Z_{\text{in}} \to \infty,\ Z_{\text{out}} \to 0$

TERMS Z_f feedback; Z_{in} input; RC (s); V_{sat} rail level

"Capacitor in feedback integrates; capacitor at the input differentiates."

1 $N = -b_{n-1}2^{n-1} + \sum_{i=0}^{n-2} b_i 2^i$

2 -2^{n-1} to $2^{n-1} - 1$

3 $-x = \bar{x} + 1$

TERMS b base; n bits; sign bit b_{n-1}; $\bar{x}$ invert; s, f, e float fields

"Flip and add one." The most negative number has no positive partner (-128 in 8 bits).

1 $N = \sum_i d_i\, b^i$

2 1 hex digit = 4 bits, 1 octal digit = 3 bits

3 0 to $2^n - 1$

TERMS b base; n bits; sign bit b_{n-1}; $\bar{x}$ invert; s, f, e float fields

"Flip and add one." The most negative number has no positive partner (-128 in 8 bits).

1 $\overline{A \cdot B} = \bar{A} + \bar{B}, \qquad \overline{A + B} = \bar{A} \cdot \bar{B}$

2 $A + \bar{A} = 1, \quad A \cdot \bar{A} = 0, \quad A + AB = A$

3 $f = \sum(\text{minterms where } f = 1)$

TERMS b_i bits; $\cdot$ AND; $+$ OR; bar NOT; minterm = one true row

De Morgan: "break the bar, change the operator" (same as for sets A1.21 *§03 Algebra I).*

1 $f(A, B, C) = \sum m(\dots)$

2 $X = 0$ or 1 as convenient

3 group the 0s $\Rightarrow$ $f = \prod(\text{sum terms})$

TERMS m_i minterms; Gray code order; 2^k group drops k vars; X don't care

"Biggest boxes, fewest boxes, powers of two, wrap the edges."

1 $AB + A\bar{B} = A$

2 00, 01, 11, 10

3 2^k cells $\Rightarrow$ k variables eliminated

TERMS m_i minterms; Gray code order; 2^k group drops k vars; X don't care

"Biggest boxes, fewest boxes, powers of two, wrap the edges."

1 same-sign operands $\to$ opposite-sign result

2 $x = (-1)^s \times 1.f \times 2^{e-\text{bias}}$

TERMS b base; n bits; sign bit b_{n-1}; $\bar{x}$ invert; s, f, e float fields

"Flip and add one." The most negative number has no positive partner (-128 in 8 bits).

1 $V = IR$

2 $i = C\,\dfrac{dv}{dt}$

3 $v = L\,\dfrac{di}{dt}$

TERMS R (Ω); C (F); L (H); N_p, N_s turns; I_{rated} fuse rating

Zigzag = resistor (friction); two parallel plates = capacitor (a gap that stores charge); coils = inductor (a wound wire).

1 $Q^+ = S + \bar{R}Q \quad (SR = 0)$

2 n flip-flops $\to$ mod-2^n, $\qquad f_{Q_i} = \dfrac{f_{\text{clk}}}{2^{i+1}}$

3 $f_{\max} = \dfrac{1}{t_{cq} + t_{\text{logic}} + t_{\text{setup}}}$

TERMS Q^+ next state; $\oplus$ XOR; f_{clk} clock; t_{cq}, t_{setup} delays

"D for Data (copy), T for Toggle, JK: J jumps up, K knocks down."

1 $Q^+ = D$

2 $Q^+ = J\bar{Q} + \bar{K}Q$

3 $Q^+ = T \oplus Q$

TERMS Q^+ next state; $\oplus$ XOR; f_{clk} clock; t_{cq}, t_{setup} delays

"D for Data (copy), T for Toggle, JK: J jumps up, K knocks down."

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EE.40 (2/2)

ELECTRICAL ENGINEERING

Schematic Symbols: Passive Components & Switches

PART 2 OF 2

1. Transformer
2. Fuse
3. Lamp

STATE EACH NUMBERED FORMULA

SY.1

EE SYMBOLS



NAME IT / STATE ITS FORMULA

SY.2

EE SYMBOLS



NAME IT / STATE ITS FORMULA

SY.3

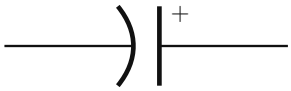
EE SYMBOLS



NAME IT / STATE ITS FORMULA

SY.4

EE SYMBOLS



NAME IT / STATE ITS FORMULA

SY.5

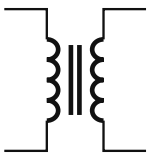
EE SYMBOLS



NAME IT / STATE ITS FORMULA

SY.6

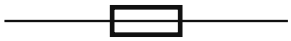
EE SYMBOLS



NAME IT / STATE ITS FORMULA

SY.7

EE SYMBOLS



NAME IT / STATE ITS FORMULA

SY.8

EE SYMBOLS



NAME IT / STATE ITS FORMULA

SY.9

EE SYMBOLS



NAME IT / STATE ITS FORMULA

EE.41 (1/2)

ELECTRICAL ENGINEERING

Schematic Symbols: Sources, Meters & Ground

PART 1 OF 2

1. Ideal voltage source
2. Ideal current source
3. AC source

STATE EACH NUMBERED FORMULA

EE.41 (2/2)

ELECTRICAL ENGINEERING

Schematic Symbols: Sources, Meters & Ground

PART 2 OF 2

1. Voltmeter
2. Ammeter
3. Ground

STATE EACH NUMBERED FORMULA

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Potentiometer

Adjustable resistor with a wiper
(adjustable voltage divider)

SY.2

Resistor

Limits current
$V = IR$, unit Ω

SY.1

1. $\dfrac{V_s}{V_p} = \dfrac{N_s}{N_p}$
2. opens when $I > I_{\text{rated}}$
3. $P = \dfrac{V^2}{R}$

TERMS R (Ω); C (F); L (H); N_p, N_s turns; I_{rated} fuse rating

Zigzag = resistor (friction); two parallel plates = capacitor (a gap that stores charge); coils = inductor (a wound wire).

EE.40 (2/2)

Inductor

Stores magnetic energy
$v = L\dfrac{di}{dt}, \quad E = \frac{1}{2}LI^2$

SY.5

Polarized Capacitor

Electrolytic capacitor
curved plate = negative; observe polarity

SY.4

Capacitor

Stores charge / electric energy
$Q = CV, \quad i = C\dfrac{dv}{dt}$

SY.3

Switch (SPST)

Single-pole single-throw switch
open = no current, closed = short

SY.8

Fuse

Melts to open the circuit
when current exceeds its rating

SY.7

Transformer

Two coupled coils on a core
$\dfrac{V_s}{V_p} = \dfrac{N_s}{N_p}$

SY.6

1. $R_{\text{in}} \to \infty$, in parallel
2. $R_{\text{in}} \to 0$, in series
3. $V_{\text{gnd}} = 0$ V

TERMS R_{in} meter resistance; ground = 0 V; V_m amplitude; ω (rad/s)

"Volts across, amps through": voltmeter across the part, ammeter in the path.

EE.41 (2/2)

1. $V =$ const for any I
2. $I =$ const for any V
3. $v(t) = V_m \sin(\omega t + \phi)$

TERMS R_{in} meter resistance; ground = 0 V; V_m amplitude; ω (rad/s)

"Volts across, amps through": voltmeter across the part, ammeter in the path.

EE.41 (1/2)

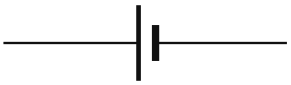
Lamp

Incandescent lamp / bulb
a resistive load: $P = \dfrac{V^2}{R}$

SY.9

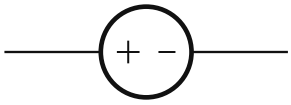
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SY.10 EE SYMBOLS



NAME IT / STATE ITS FORMULA

SY.11 EE SYMBOLS



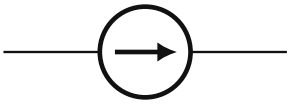
NAME IT / STATE ITS FORMULA

SY.12 EE SYMBOLS



NAME IT / STATE ITS FORMULA

SY.13 EE SYMBOLS



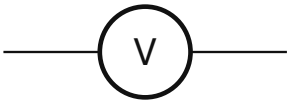
NAME IT / STATE ITS FORMULA

SY.14 EE SYMBOLS



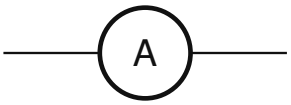
NAME IT / STATE ITS FORMULA

SY.15 EE SYMBOLS



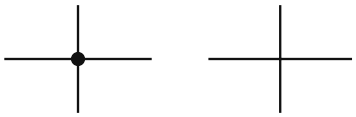
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SY.16 EE SYMBOLS



NAME IT / STATE ITS FORMULA

SY.17 EE SYMBOLS



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EE.42 (1/2) ELECTRICAL ENGINEERING

Schematic Symbols: Semiconductors & Op-Amps

PART 1 OF 2

1. Diode
2. Zener diode
3. BJT

STATE EACH NUMBERED FORMULA

EE.42 (2/2) ELECTRICAL ENGINEERING

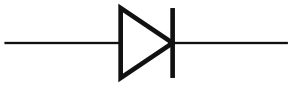
Schematic Symbols: Semiconductors & Op-Amps

PART 2 OF 2

1. MOSFET
2. Op-amp

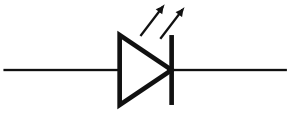
STATE EACH NUMBERED FORMULA

SY.18 EE SYMBOLS



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SY.19 EE SYMBOLS



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AC Voltage Source

Sinusoidal source
$v(t) = V_m \sin(\omega t + \phi)$

SY.12

DC Voltage Source

Ideal independent voltage source
$V =$ constant

SY.11

Battery

DC source (cell)
long plate $= +$, short plate $= -$

SY.10

Voltmeter

Measures voltage
connect in parallel; $R_{\text{in}} \to \infty$

SY.15

Ground

Reference node $= 0\,\text{V}$
(earth or common return)

SY.14

Current Source

Ideal independent current source
$I =$ constant, arrow gives direction

SY.13

1. $I = I_s\left(e^{V/(nV_T)} - 1\right)$
2. $V \approx V_Z$ in reverse breakdown
3. $I_C \approx \beta I_B$

TERMS anode $\to$ cathode; V_Z Zener voltage; β current gain; A open-loop gain

The diode's arrow points the way conventional current flows (anode $\to$ cathode; the bar is the cathode).

EE.42 (1/2)

Junction vs. Crossing

Dot $=$ wires connected (a node)
no dot $=$ wires cross, not connected

SY.17

Ammeter

Measures current
connect in series; $R_{\text{in}} \to 0$

SY.16

LED

Light-emitting diode
needs a series resistor: $R = \dfrac{V_s - V_f}{I}$

SY.19

Diode

One-way valve for current
anode $\to$ cathode (bar); $\approx 0.7\,\text{V}$ drop

SY.18

1. $I_D = f(V_{GS})$
2. $V_{\text{out}} = A\,(V_+ - V_-)$

TERMS anode $\to$ cathode; V_Z Zener voltage; β current gain; A open-loop gain

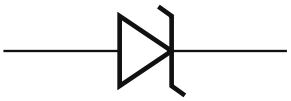
The diode's arrow points the way conventional current flows (anode $\to$ cathode; the bar is the cathode).

EE.42 (2/2)

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SY.20

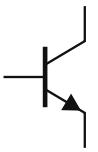
EE SYMBOLS



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SY.21

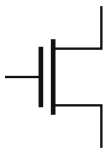
EE SYMBOLS



NAME IT / STATE ITS FORMULA

SY.22

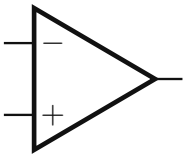
EE SYMBOLS



NAME IT / STATE ITS FORMULA

SY.23

EE SYMBOLS



NAME IT / STATE ITS FORMULA

EE.43 (1/2)

ELECTRICAL ENGINEERING

Schematic Symbols: Logic Gates

PART 1 OF 2

1. AND
2. OR
3. NOT

STATE EACH NUMBERED FORMULA

EE.43 (2/2)

ELECTRICAL ENGINEERING

Schematic Symbols: Logic Gates

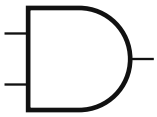
PART 2 OF 2

1. NAND
2. XOR

STATE EACH NUMBERED FORMULA

SY.24

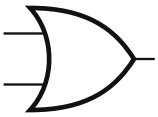
EE SYMBOLS



NAME IT / STATE ITS FORMULA

SY.25

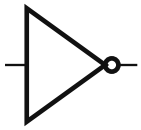
EE SYMBOLS



NAME IT / STATE ITS FORMULA

SY.26

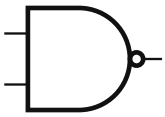
EE SYMBOLS



NAME IT / STATE ITS FORMULA

SY.27

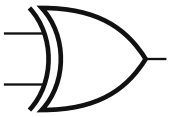
EE SYMBOLS



NAME IT / STATE ITS FORMULA

SY.28

EE SYMBOLS



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MOSFET (n-channel)

Voltage-controlled switch / amplifier
gate, drain, source; basis of CPUs and GPUs

SY.22

NPN Transistor (BJT)

Bipolar junction transistor
$I_C \approx \beta I_B$; emitter arrow points out

SY.21

Zener Diode

Conducts in reverse at V_Z
used as a voltage reference / regulator

SY.20

1. $Y = \overline{A \cdot B}$
2. $Y = A \oplus B = A\bar{B} + \bar{A}B$

TERMS A, B inputs; Y output; bubble = NOT; $\oplus$ XOR

A small circle ("bubble") on an output means NOT. AND has a flat back, OR has a curved back, XOR adds an extra curved line.

EE.43 (2/2)

1. $Y = A \cdot B$
2. $Y = A + B$
3. $Y = \bar{A}$

TERMS A, B inputs; Y output; bubble = NOT; $\oplus$ XOR

A small circle ("bubble") on an output means NOT. AND has a flat back, OR has a curved back, XOR adds an extra curved line.

EE.43 (1/2)

Op-Amp

Operational amplifier
$V_{\text{out}} = A(V_+ - V_-)$; ideal: $V_+ = V_-$

SY.23

NOT Gate (Inverter)

$Y = \bar{A}$
flips the input

SY.26

OR Gate

$Y = A + B$
1 if any input is 1

SY.25

AND Gate

$Y = A \cdot B$
1 only if all inputs are 1

SY.24

XOR Gate

$Y = A \oplus B$
1 if the inputs differ

SY.28

NAND Gate

$Y = \overline{A \cdot B}$
universal: builds every other gate

SY.27

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FN.1 (1/2) FINANCE

Simple vs. Compound Interest

PART 1 OF 2

1. Simple interest
2. Compound interest
3. Continuous compounding

STATE EACH NUMBERED FORMULA

FN.1 (2/2) FINANCE

Simple vs. Compound Interest

PART 2 OF 2

1. Solving for time
2. Solving for the rate

STATE EACH NUMBERED FORMULA

FN.2 (1/2) FINANCE

APR vs. Effective Annual Rate

PART 1 OF 2

1. Effective annual rate
2. Continuous EAR
3. Nominal from effective

STATE EACH NUMBERED FORMULA

FN.2 (2/2) FINANCE

APR vs. Effective Annual Rate

PART 2 OF 2

1. Continuous equivalent
2. Periodic rate conversion

STATE EACH NUMBERED FORMULA

FN.3 (1/2) FINANCE

Doubling Time & the Rule of 72

PART 1 OF 2

1. Exact doubling time
2. Rule of 72
3. Continuous doubling

STATE EACH NUMBERED FORMULA

FN.3 (2/2) FINANCE

Doubling Time & the Rule of 72

PART 2 OF 2

1. Tripling (rule of 114)

STATE EACH NUMBERED FORMULA

FN.4 (1/2) FINANCE

Inflation & Real Returns

PART 1 OF 2

1. Fisher equation
2. Exact real rate
3. Approximate real rate

STATE EACH NUMBERED FORMULA

FN.4 (2/2) FINANCE

Inflation & Real Returns

PART 2 OF 2

1. Real value (deflating)
2. Inflation rate from CPI

STATE EACH NUMBERED FORMULA

FN.5 (1/2) FINANCE

Present Value & Discounting

PART 1 OF 2

1. Present value
2. Future value
3. Discount factor

STATE EACH NUMBERED FORMULA

FN.5 (2/2) FINANCE

Present Value & Discounting

PART 2 OF 2

1. PV of a cash-flow stream

STATE EACH NUMBERED FORMULA

FN.6 (1/2) FINANCE

Annuities & Perpetuities

PART 1 OF 2

1. PV of an ordinary annuity
2. FV of an ordinary annuity

STATE EACH NUMBERED FORMULA

FN.6 (2/2) FINANCE

Annuities & Perpetuities

PART 2 OF 2

1. Perpetuity
2. Growing perpetuity

STATE EACH NUMBERED FORMULA

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1. $\mathrm{EAR} = \left(1 + \dfrac{\mathrm{APR}}{n}\right)^{n} - 1$
2. $\mathrm{EAR} = e^{r} - 1$
3. $\mathrm{APR}_n = n\left[(1+\mathrm{EAR})^{1/n} - 1\right]$

TERMS APR nominal; EAR effective; n, m periods/yr; r_c continuous rate

APR is the "sticker" rate; EAR is what you actually pay or earn. EAR $\geq$ APR whenever $n > 1$.

1. $t = \dfrac{\ln(A/P)}{n\ln\left(1 + \frac{r}{n}\right)}$
2. $r = n\left[\left(\dfrac{A}{P}\right)^{1/(nt)} - 1\right]$

TERMS P principal; A amount; r nominal rate; n periods/yr; t years

Compounding is "interest on interest"; "Pert" for continuous.

1. $I = Prt, \qquad A = P(1+rt)$
2. $A = P\left(1 + \dfrac{r}{n}\right)^{nt}$
3. $A = Pe^{rt}$

TERMS P principal; A amount; r nominal rate; n periods/yr; t years

Compounding is "interest on interest"; "Pert" for continuous.

1. $t_{\text{triple}} = \dfrac{\ln 3}{\ln(1+r)} \approx \dfrac{114}{r\,(\%)}$

TERMS r growth rate per period (% in the rules); $\ln 2 \approx 0.693$

Rule of 72 (use 69.3 for continuous compounding, since $\ln 2 \approx 0.693$).

1. $t_{\text{double}} = \dfrac{\ln 2}{\ln(1+r)}$
2. $t_{\text{double}} \approx \dfrac{72}{r\ (\text{in }\%)}$
3. $t_{\text{double}} = \dfrac{\ln 2}{r} \approx \dfrac{69.3}{r\,(\%)}$

TERMS r growth rate per period (% in the rules); $\ln 2 \approx 0.693$

Rule of 72 (use 69.3 for continuous compounding, since $\ln 2 \approx 0.693$).

1. $r_c = \ln(1+\mathrm{EAR}) = n\ln\left(1 + \dfrac{\mathrm{APR}}{n}\right)$
2. $i_m = (1+i_n)^{n/m} - 1$

TERMS APR nominal; EAR effective; n, m periods/yr; r_c continuous rate

APR is the "sticker" rate; EAR is what you actually pay or earn. EAR $\geq$ APR whenever $n > 1$.

1. $PV = \dfrac{FV}{(1+r)^t}$
2. $FV = PV\,(1+r)^t$
3. $DF_t = \dfrac{1}{(1+r)^t} = e^{-r_c t}$ (continuous)

TERMS PV today; FV at t; r rate/period; CF_t cash flow; DF_t discount factor

"A dollar today is worth more than a dollar tomorrow."

1. $\text{real value} = \dfrac{\text{nominal}}{(1+\pi)^t}$
2. $\pi_t = \dfrac{\mathrm{CPI}_t}{\mathrm{CPI}_{t-1}} - 1$

TERMS i nominal; r_{real} real rate; π inflation; CPI price index

"Real $\approx$ nominal minus inflation."

1. $(1+i) = (1+r_{\text{real}})(1+\pi)$
2. $r_{\text{real}} = \dfrac{1+i}{1+\pi} - 1$
3. $r_{\text{real}} \approx i - \pi$

TERMS i nominal; r_{real} real rate; π inflation; CPI price index

"Real $\approx$ nominal minus inflation."

1. $PV = \dfrac{C}{r}$
2. $PV = \dfrac{C_1}{r-g} \quad (r > g)$

TERMS C payment; n payments; r rate/period; g growth, $r > g$

A perpetuity is an annuity with $n \to \infty$: the $(1+r)^{-n}$ term vanishes.

1. $PV = C\,\dfrac{1-(1+r)^{-n}}{r}$
2. $FV = C\,\dfrac{(1+r)^n - 1}{r}$

TERMS C payment; n payments; r rate/period; g growth, $r > g$

A perpetuity is an annuity with $n \to \infty$: the $(1+r)^{-n}$ term vanishes.

1. $PV = \displaystyle\sum_{t=1}^{T} \frac{CF_t}{(1+r)^t}$

TERMS PV today; FV at t; r rate/period; CF_t cash flow; DF_t discount factor

"A dollar today is worth more than a dollar tomorrow."

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FN.7 (1/2)

FINANCE

Annuities Due & Growing Annuities

PART 1 OF 2

1. PV of an annuity due
2. FV of an annuity due

STATE EACH NUMBERED FORMULA

FN.7 (2/2)

FINANCE

Annuities Due & Growing Annuities

PART 2 OF 2

1. PV of a growing annuity
2. FV of a growing annuity
3. Equal-rate case

STATE EACH NUMBERED FORMULA

FN.K1

FINANCE

Time value of money: PV of an annuity, perpetuity, growing perpetuity, annuity due

RECALL IT

FN.K1

FINANCE

Time value of money: three checks before using any formula

RECALL IT

FN.8 (1/2)

FINANCE

Net Present Value & Internal Rate of Return

PART 1 OF 2

1. Net present value
2. Internal rate of return
3. Decision rule

STATE EACH NUMBERED FORMULA

FN.8 (2/2)

FINANCE

Net Present Value & Internal Rate of Return

PART 2 OF 2

1. Profitability index
2. Payback period
3. Modified IRR

STATE EACH NUMBERED FORMULA

FN.9 (1/2)

FINANCE

Loan Amortization & Mortgages

PART 1 OF 2

1. Level payment
2. Monthly conventions
3. Interest / principal split

STATE EACH NUMBERED FORMULA

FN.9 (2/2)

FINANCE

Loan Amortization & Mortgages

PART 2 OF 2

1. Remaining balance
2. Total interest

STATE EACH NUMBERED FORMULA

FN.10 (1/2)

FINANCE

Returns: Simple, Log & CAGR

PART 1 OF 2

1. Simple return
2. Log return
3. Multi-period return

STATE EACH NUMBERED FORMULA

FN.10 (2/2)

FINANCE

Returns: Simple, Log & CAGR

PART 2 OF 2

1. CAGR
2. Geometric mean return
3. Volatility drag

STATE EACH NUMBERED FORMULA

FN.11 (1/2)

FINANCE

Volatility & Risk

PART 1 OF 2

1. Sample volatility
2. Annualizing volatility
3. Square-root-of-time

STATE EACH NUMBERED FORMULA

FN.11 (2/2)

FINANCE

Volatility & Risk

PART 2 OF 2

1. Annualizing the mean
2. EWMA volatility

STATE EACH NUMBERED FORMULA

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$$PV_{\text{ann}} = C\frac{1-(1+r)^{-n}}{r} \quad PV_{\text{perp}} = \frac{C}{r}$$
$$PV_{\text{grow}} = \frac{C_1}{r-g} \quad \text{due: } \times(1+r)$$

FN.K1

1 $$PV = \frac{C_1}{r-g}\left[1-\left(\frac{1+g}{1+r}\right)^n\right]$$
2 $$FV = C_1\frac{(1+r)^n-(1+g)^n}{r-g}$$
3 $$g = r: \quad PV = \frac{nC_1}{1+r}$$

TERMS C level payment; C_1 first payment; g growth; r rate; n payments

"Due" = "do it now": payments at the start, one extra factor $(1+r)$.

FN.7 (2/2)

1 $$PV_{\text{due}} = C\frac{1-(1+r)^{-n}}{r}(1+r)$$
2 $$FV_{\text{due}} = C\frac{(1+r)^n-1}{r}(1+r)$$

TERMS C level payment; C_1 first payment; g growth; r rate; n payments

"Due" = "do it now": payments at the start, one extra factor $(1+r)$.

FN.7 (1/2)

1 $$PI = \frac{PV(\text{future cash flows})}{\text{initial investment}}$$
2 $$\min\left\{T : \textstyle\sum_{t=0}^{T} CF_t \geq 0\right\}$$
3 $$\text{MIRR} = \left(\frac{FV(\text{inflows at } r_{\text{re}})}{PV(\text{outflows at } r_{\text{fin}})}\right)^{1/T} - 1$$

TERMS CF_t cash flow ($CF_0 < 0$); r hurdle rate; IRR makes NPV $= 0$

NPV answers "how much value?"; IRR answers "what rate?" When they disagree, trust NPV.

FN.8 (2/2)

1 $$NPV = \sum_{t=0}^{T}\frac{CF_t}{(1+r)^t}$$
2 $$\sum_{t=0}^{T}\frac{CF_t}{(1+\text{IRR})^t} = 0$$
3 accept if $NPV > 0$ ($\Leftrightarrow$ IRR $> r$ for normal projects)

TERMS CF_t cash flow ($CF_0 < 0$); r hurdle rate; IRR makes NPV $= 0$

NPV answers "how much value?"; IRR answers "what rate?" When they disagree, trust NPV.

FN.8 (1/2)

1. Rate *per period*.
2. Number of periods.
3. When is the first payment (now $\rightarrow$ due; one period out $\rightarrow$ ordinary)?

FN.K1

1 $$R_t = \frac{P_t - P_{t-1} + D_t}{P_{t-1}}$$
2 $$r_t = \ln\frac{P_t}{P_{t-1}} = \ln(1+R_t)$$
3 $$1+R_{0,T} = \prod_{t=1}^{T}(1+R_t), \quad r_{0,T} = \sum_{t=1}^{T} r_t$$

TERMS P_t price; D_t dividend; V_0, V_T start/end value; T years

Log returns add over time; simple returns add across assets. +50% then −50% loses 25%.

FN.10 (1/2)

1 $$B_k = P(1+i)^k - M\frac{(1+i)^k-1}{i}$$
2 $$nM - P$$

TERMS P loan; i rate/period; n payments; B_k balance after k

Early payments are mostly interest; late payments are mostly principal.

FN.9 (2/2)

1 $$M = P\frac{i(1+i)^n}{(1+i)^n-1}$$
2 $$i = \frac{\text{APR}}{12}, \quad n = 12\times\text{years}$$
3 $$I_k = i\,B_{k-1}, \qquad \text{Pr}_k = M - I_k$$

TERMS P loan; i rate/period; n payments; B_k balance after k

Early payments are mostly interest; late payments are mostly principal.

FN.9 (1/2)

1 $$\mu_{\text{annual}} = 252\,\mu_{\text{daily}}$$
2 $$\sigma_t^2 = \lambda\sigma_{t-1}^2 + (1-\lambda)r_{t-1}^2$$

TERMS r_t returns; σ volatility; T horizon; λ decay; 252 days

"Risk grows with the square root of time; expected return grows with time."

FN.11 (2/2)

1 $$\sigma = \sqrt{\frac{1}{n-1}\sum_t (r_t - \bar{r})^2}$$
2 $$\sigma_{\text{annual}} = \sigma_{\text{daily}}\sqrt{252}$$
3 $$\sigma_T = \sigma\sqrt{T}$$

TERMS r_t returns; σ volatility; T horizon; λ decay; 252 days

"Risk grows with the square root of time; expected return grows with time."

FN.11 (1/2)

1 $$\text{CAGR} = \left(\frac{V_T}{V_0}\right)^{1/T} - 1$$
2 $$\bar{R}_{\text{geo}} = \left[\prod(1+R_t)\right]^{1/T} - 1 \leq \bar{R}_{\text{arith}}$$
3 $$\bar{R}_{\text{geo}} \approx \bar{R}_{\text{arith}} - \tfrac{1}{2}\sigma^2$$

TERMS P_t price; D_t dividend; V_0, V_T start/end value; T years

Log returns add over time; simple returns add across assets. +50% then −50% loses 25%.

FN.10 (2/2)

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FN.12 (1/2)

FINANCE

Portfolio Return, Variance & Diversification

PART 1 OF 2

1. Expected portfolio return
2. Portfolio variance
3. Two-asset variance

STATE EACH NUMBERED FORMULA

FN.12 (2/2)

FINANCE

Portfolio Return, Variance & Diversification

PART 2 OF 2

1. Diversification limit

STATE EACH NUMBERED FORMULA

FN.13 (1/2)

FINANCE

Efficient Frontier & Minimum-Variance Portfolio

PART 1 OF 2

1. Mean–variance problem
2. Two-asset minimum variance
3. Global minimum-variance portfolio

STATE EACH NUMBERED FORMULA

FN.13 (2/2)

FINANCE

Efficient Frontier & Minimum-Variance Portfolio

PART 2 OF 2

1. Tangency portfolio
2. Capital market line

STATE EACH NUMBERED FORMULA

FN.14 (1/2)

FINANCE

CAPM & Beta

PART 1 OF 2

1. CAPM (security market line)
2. Beta
3. Jensen's alpha

STATE EACH NUMBERED FORMULA

FN.14 (2/2)

FINANCE

CAPM & Beta

PART 2 OF 2

1. Portfolio beta
2. Risk decomposition

STATE EACH NUMBERED FORMULA

FN.15 (1/2)

FINANCE

Estimating Beta; Levered & Unlevered Beta

PART 1 OF 2

1. Market model
2. OLS beta
3. Systematic share

STATE EACH NUMBERED FORMULA

FN.15 (2/2)

FINANCE

Estimating Beta; Levered & Unlevered Beta

PART 2 OF 2

1. Adjusted (Blume) beta
2. Unlevering (Hamada)
3. Relevering

STATE EACH NUMBERED FORMULA

FN.16 (1/2)

FINANCE

Sharpe Ratio & Risk-Adjusted Return

PART 1 OF 2

1. Sharpe ratio
2. Sortino ratio
3. Downside deviation

STATE EACH NUMBERED FORMULA

FN.16 (2/2)

FINANCE

Sharpe Ratio & Risk-Adjusted Return

PART 2 OF 2

1. Treynor ratio
2. Information ratio
3. Annualizing Sharpe

STATE EACH NUMBERED FORMULA

FN.17 (1/2)

FINANCE

Weighted Average Cost of Capital

PART 1 OF 2

1. WACC
2. Firm value
3. Cost of equity (CAPM)

STATE EACH NUMBERED FORMULA

FN.17 (2/2)

FINANCE

Weighted Average Cost of Capital

PART 2 OF 2

1. Cost of debt

STATE EACH NUMBERED FORMULA

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1 $\min_{\mathbf{w}} \ \mathbf{w}^\top \Sigma \mathbf{w} \ \text{ s.t. } \mathbf{w}^\top \boldsymbol{\mu} = \mu_p, \ \mathbf{w}^\top \mathbf{1} = 1$

2 $w_1^* = \dfrac{\sigma_2^2 - \rho\sigma_1\sigma_2}{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2}$

3 $\mathbf{w}_{\text{GMV}} = \dfrac{\Sigma^{-1}\mathbf{1}}{\mathbf{1}^\top\Sigma^{-1}\mathbf{1}}$

TERMS $\mathbf{w}$ weights; $\boldsymbol{\mu}, \Sigma$ means/covariance; R_f risk-free; $\mathbf{1}$ ones

Two-fund separation: every efficient choice is "some cash + the tangency portfolio".

FN.13 (1/2)

1 $\sigma_p^2 = \dfrac{\overline{\sigma^2}}{N} + \left(1 - \dfrac{1}{N}\right)\overline{\text{cov}}$

TERMS w_i weights; Σ covariance matrix; ρ correlation; N assets

"Diversification is the only free lunch": with $\rho < 1$, portfolio risk is less than the weighted average risk.

FN.12 (2/2)

1 $E[R_p] = \sum_i w_i E[R_i] = \mathbf{w}^\top \boldsymbol{\mu}$

2 $\sigma_p^2 = \mathbf{w}^\top \Sigma \, \mathbf{w}$

3 $\sigma_p^2 = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\,\rho\,\sigma_1\sigma_2$

TERMS w_i weights; Σ covariance matrix; ρ correlation; N assets

"Diversification is the only free lunch": with $\rho < 1$, portfolio risk is less than the weighted average risk.

FN.12 (1/2)

1 $\beta_p = \sum_i w_i \beta_i$

2 $\sigma_i^2 = \beta_i^2\sigma_m^2 + \sigma_\varepsilon^2$

TERMS R_f risk-free; R_m market; β market sensitivity; α excess return

$\beta > 1$: amplifies the market; $\beta < 1$: dampens it; β is just a regression slope.

FN.14 (2/2)

1 $E[R_i] = R_f + \beta_i\big(E[R_m] - R_f\big)$

2 $\beta_i = \dfrac{\text{Cov}(R_i, R_m)}{\text{Var}(R_m)} = \rho_{im}\dfrac{\sigma_i}{\sigma_m}$

3 $\alpha_i = R_i - \big[R_f + \beta_i(R_m - R_f)\big]$

TERMS R_f risk-free; R_m market; β market sensitivity; α excess return

$\beta > 1$: amplifies the market; $\beta < 1$: dampens it; β is just a regression slope.

FN.14 (1/2)

1 $\mathbf{w}_T = \dfrac{\Sigma^{-1}(\boldsymbol{\mu} - R_f\mathbf{1})}{\mathbf{1}^\top\Sigma^{-1}(\boldsymbol{\mu} - R_f\mathbf{1})}$

2 $E[R_p] = R_f + \dfrac{E[R_T] - R_f}{\sigma_T}\,\sigma_p$

TERMS $\mathbf{w}$ weights; $\boldsymbol{\mu}, \Sigma$ means/covariance; R_f risk-free; $\mathbf{1}$ ones

Two-fund separation: every efficient choice is "some cash + the tangency portfolio".

FN.13 (2/2)

1 $S = \dfrac{E[R_p] - R_f}{\sigma_p}$

2 $\text{Sortino} = \dfrac{E[R_p] - R_f}{\sigma_{\text{down}}}$

3 $\sigma_{\text{down}} = \sqrt{\frac{1}{n}\sum_t \min(R_t - \tau, 0)^2}$

TERMS R_p, R_f, R_b portfolio/risk-free/benchmark; σ volatility; β_p beta; τ target

Sharpe = "excess return per unit of risk." Rough guide: < 0.5 poor, ≈ 1 good, > 2 excellent (and suspicious).

FN.16 (1/2)

1 $\beta_{\text{adj}} = 0.67\,\hat{\beta} + 0.33$

2 $\beta_U = \dfrac{\beta_L}{1 + (1 - T_c)\,D/E}$

3 $\beta_L = \beta_U\left[1 + (1 - T_c)\dfrac{D}{E}\right]$

TERMS $\hat{\beta}$ slope; D/E debt/equity; T_c tax rate; β_L, β_U levered/unlevered

Debt makes equity "louder": $\beta_L > \beta_U$ whenever $D > 0$.

FN.15 (2/2)

1 $R_{i,t} - R_f = \alpha_i + \beta_i(R_{m,t} - R_f) + \varepsilon_t$

2 $\hat{\beta} = \dfrac{\sum_t (x_t - \bar{x})(y_t - \bar{y})}{\sum_t (x_t - \bar{x})^2} = \dfrac{s_{xy}}{s_x^2}$

3 $R^2 = \dfrac{\hat{\beta}^2\sigma_m^2}{\sigma_i^2}$

TERMS $\hat{\beta}$ slope; D/E debt/equity; T_c tax rate; β_L, β_U levered/unlevered

Debt makes equity "louder": $\beta_L > \beta_U$ whenever $D > 0$.

FN.15 (1/2)

1 $r_D \approx$ YTM of the firm's bonds

TERMS E, D equity/debt values; $V = E + D$; r_E, r_D costs; T_c tax rate

"Pay each investor their price, in proportion to their share, and let the tax shield discount the debt."

FN.17 (2/2)

1 $\text{WACC} = \dfrac{E}{V}\,r_E + \dfrac{D}{V}\,r_D\,(1 - T_c)$

2 $V = E + D$

3 $r_E = R_f + \beta_L\big(E[R_m] - R_f\big)$

TERMS E, D equity/debt values; $V = E + D$; r_E, r_D costs; T_c tax rate

"Pay each investor their price, in proportion to their share, and let the tax shield discount the debt."

FN.17 (1/2)

1 $\text{Treynor} = \dfrac{E[R_p] - R_f}{\beta_p}$

2 $IR = \dfrac{E[R_p - R_b]}{\sigma(R_p - R_b)}$

3 $S_{\text{annual}} = S_{\text{daily}}\sqrt{252}$

TERMS R_p, R_f, R_b portfolio/risk-free/benchmark; σ volatility; β_p beta; τ target

Sharpe = "excess return per unit of risk." Rough guide: < 0.5 poor, ≈ 1 good, > 2 excellent (and suspicious).

FN.16 (2/2)

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FN.18 (1/2) FINANCE

Stock Valuation & Multiples

PART 1 OF 2

1. Gordon growth model
2. Implied required return
3. Sustainable growth

STATE EACH NUMBERED FORMULA

FN.18 (2/2) FINANCE

Stock Valuation & Multiples

PART 2 OF 2

1. Price–earnings ratio
2. Enterprise value

STATE EACH NUMBERED FORMULA

FN.19 (1/2) FINANCE

Discounted Cash Flow & Terminal Value

PART 1 OF 2

1. Free cash flow to the firm
2. DCF value
3. Gordon terminal value

STATE EACH NUMBERED FORMULA

FN.19 (2/2) FINANCE

Discounted Cash Flow & Terminal Value

PART 2 OF 2

1. Exit-multiple terminal value
2. Equity value per share

STATE EACH NUMBERED FORMULA

FN.20 (1/2) FINANCE

Financial Ratios & DuPont Analysis

PART 1 OF 2

1. Return on equity
2. DuPont decomposition

STATE EACH NUMBERED FORMULA

FN.20 (2/2) FINANCE

Financial Ratios & DuPont Analysis

PART 2 OF 2

1. Current ratio
2. Debt-to-equity

STATE EACH NUMBERED FORMULA

FN.21 (1/3) FINANCE

More Financial Statement Ratios

PART 1 OF 3

1. Gross / operating margin
2. Return on assets
3. Quick ratio

STATE EACH NUMBERED FORMULA

FN.21 (2/3) FINANCE

More Financial Statement Ratios

PART 2 OF 3

1. Interest coverage
2. Inventory turnover
3. Days sales outstanding

STATE EACH NUMBERED FORMULA

FN.21 (3/3) FINANCE

More Financial Statement Ratios

PART 3 OF 3

1. EV/EBITDA
2. Price-to-book
3. Dividend yield

STATE EACH NUMBERED FORMULA

FN.22 (1/2) FINANCE

Bond Pricing & Yield

PART 1 OF 2

1. Bond price
2. Zero-coupon bond
3. Premium / discount rule

STATE EACH NUMBERED FORMULA

FN.22 (2/2) FINANCE

Bond Pricing & Yield

PART 2 OF 2

1. Dirty price

STATE EACH NUMBERED FORMULA

FN.23 (1/2) FINANCE

Bond Yield Measures

PART 1 OF 2

1. Current yield
2. Yield to maturity
3. Approximate YTM

STATE EACH NUMBERED FORMULA

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1. $FCF = \text{EBIT}(1 - T_c) + \text{D\&A} - \text{CapEx} - \Delta\text{NWC}$
2. $V_0 = \sum_{t=1}^{T} \frac{FCF_t}{(1+\text{WACC})^t} + \frac{TV_T}{(1+\text{WACC})^T}$
3. $TV_T = \frac{FCF_T(1+g)}{\text{WACC} - g}$

TERMS FCF free cash flow; WACC discount rate; g terminal growth; T forecast years

In most DCFs the terminal value is 60–80% of the total: the answer lives in WACC and g.

1. $\frac{P}{E} = \frac{\text{price}}{\text{earnings per share}}$
2. $\text{EV} = \text{market cap} + \text{debt} - \text{cash}$

TERMS D_1 next dividend; r required return; g growth $< r$; b retention

Price is the present value of all future cash; growth near the discount rate makes values explode.

1. $P_0 = \frac{D_1}{r - g}$
2. $r = \frac{D_1}{P_0} + g$
3. $g = \text{ROE} \times b$

TERMS D_1 next dividend; r required return; g growth $< r$; b retention

Price is the present value of all future cash; growth near the discount rate makes values explode.

1. $\frac{\text{current assets}}{\text{current liabilities}}$
2. $\frac{D}{E} = \frac{\text{debt}}{\text{equity}}$

TERMS NI net income; equity = assets − liabilities; D debt; E equity

Accounting identity: assets = liabilities + equity.

1. $\text{ROE} = \frac{\text{net income}}{\text{equity}}$
2. $\text{ROE} = \underbrace{\frac{\text{NI}}{\text{sales}}}_{\text{margin}} \cdot \underbrace{\frac{\text{sales}}{\text{assets}}}_{\text{turnover}} \cdot \underbrace{\frac{\text{assets}}{\text{equity}}}_{\text{leverage}}$

TERMS NI net income; equity = assets − liabilities; D debt; E equity

Accounting identity: assets = liabilities + equity.

1. $TV_T = \text{multiple} \times \text{EBITDA}_T$
2. $\frac{V_0 - \text{debt} + \text{cash}}{\text{shares outstanding}}$

TERMS FCF free cash flow; WACC discount rate; g terminal growth; T forecast years

In most DCFs the terminal value is 60–80% of the total: the answer lives in WACC and g.

1. $\frac{\text{EV}}{\text{EBITDA}}$
2. $\frac{P}{B} = \frac{\text{market cap}}{\text{book equity}}$
3. $\frac{\text{annual dividend per share}}{\text{price}}$

TERMS COGS cost of goods sold; EBIT operating profit; EV enterprise value; B book equity

Five questions: profitable? liquid? solvent? efficient? cheap?

1. $\frac{\text{EBIT}}{\text{interest expense}}$
2. $\frac{\text{COGS}}{\text{average inventory}}$
3. $\text{DSO} = \frac{\text{receivables}}{\text{revenue}} \times 365$

TERMS COGS cost of goods sold; EBIT operating profit; EV enterprise value; B book equity

Five questions: profitable? liquid? solvent? efficient? cheap?

1. $\frac{\text{revenue} - \text{COGS}}{\text{revenue}}, \quad \frac{\text{EBIT}}{\text{revenue}}$
2. $\text{ROA} = \frac{\text{net income}}{\text{total assets}}$
3. $\frac{\text{current assets} - \text{inventory}}{\text{current liabilities}}$

TERMS COGS cost of goods sold; EBIT operating profit; EV enterprise value; B book equity

Five questions: profitable? liquid? solvent? efficient? cheap?

1. $\text{CY} = \frac{\text{annual coupon}}{P}$
2. $P = \sum_{t=1}^{n} \frac{C}{(1+y)^t} + \frac{F}{(1+y)^n}$
3. $y \approx \frac{C + (F - P)/n}{(F + P)/2}$

TERMS P price; C coupon; F face; n periods; s_n spot rate; f forward rate

Current yield "now"; YTM "to the end"; YTC "if they call"; worst of them for callable bonds.

1. $P_{\text{dirty}} = P_{\text{clean}} + C \cdot \frac{\text{days since last coupon}}{\text{days in period}}$

TERMS C coupon; F face; y yield per period; n periods; c coupon rate

Bond prices and yields move in opposite directions ("see-saw"). Coupon > yield ⇒ premium; coupon < yield ⇒ discount.

1. $P = \sum_{t=1}^{n} \frac{C}{(1+y)^t} + \frac{F}{(1+y)^n} = C\,\frac{1 - (1+y)^{-n}}{y} + \frac{F}{(1+y)^n}$
2. $P = \frac{F}{(1+y)^n}$
3. $c > y \Rightarrow P > F, \quad c < y \Rightarrow P < F$

TERMS C coupon; F face; y yield per period; n periods; c coupon rate

Bond prices and yields move in opposite directions ("see-saw"). Coupon > yield ⇒ premium; coupon < yield ⇒ discount.

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FN.23 (2/2) FINANCE

Bond Yield Measures

PART 2 OF 2

1. Bond-equivalent & effective yield
2. Yield to call
3. Spot & forward rates

STATE EACH NUMBERED FORMULA

FN.24 (1/2) FINANCE

Duration & Convexity

PART 1 OF 2

1. Macaulay duration
2. Modified duration
3. Convexity

STATE EACH NUMBERED FORMULA

FN.24 (2/2) FINANCE

Duration & Convexity

PART 2 OF 2

1. Price change approximation
2. DV01
3. Zero-coupon duration

STATE EACH NUMBERED FORMULA

FN.25 (1/2) FINANCE

Options: Payoffs & Put–Call Parity

PART 1 OF 2

1. Call payoff
2. Put payoff
3. Profit

STATE EACH NUMBERED FORMULA

FN.25 (2/2) FINANCE

Options: Payoffs & Put–Call Parity

PART 2 OF 2

1. Put–call parity
2. Break-even price

STATE EACH NUMBERED FORMULA

FN.26 (1/2) FINANCE

Binomial Option Pricing

PART 1 OF 2

1. Up / down factors (CRR)
2. Risk-neutral probability
3. One-step value

STATE EACH NUMBERED FORMULA

FN.26 (2/2) FINANCE

Binomial Option Pricing

PART 2 OF 2

1. Hedge ratio
2. n-step European option
3. American early exercise

STATE EACH NUMBERED FORMULA

FN.27 (1/2) FINANCE

Geometric Brownian Motion

PART 1 OF 2

1. Stochastic differential equation
2. Closed-form solution
3. Log-return distribution

STATE EACH NUMBERED FORMULA

FN.27 (2/2) FINANCE

Geometric Brownian Motion

PART 2 OF 2

1. Mean and median
2. Exact simulation step

STATE EACH NUMBERED FORMULA

FN.28 (1/2) FINANCE

Black–Scholes Option Pricing

PART 1 OF 2

1. Call price
2. d_1 and d_2
3. Put price

STATE EACH NUMBERED FORMULA

FN.28 (2/2) FINANCE

Black–Scholes Option Pricing

PART 2 OF 2

1. Black–Scholes PDE
2. Implied volatility

STATE EACH NUMBERED FORMULA

FN.29 (1/2) FINANCE

Option Greeks

PART 1 OF 2

1. Delta
2. Gamma
3. Vega

STATE EACH NUMBERED FORMULA

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1 $$\frac{\Delta P}{P} \approx -D_{\text{mod}}\,\Delta y + \frac{1}{2}\,\text{Cx}\,(\Delta y)^2$$

2 $$\text{DV01} = D_{\text{mod}} \times P \times 0.0001$$

3 $$D_{\text{Mac}} = n$$

TERMS P price; y yield; CF_t cash flow at t; Cx convexity; 1 bp = 0.01%

Duration ≈ "the percent price drop for a 1-point rise in rates."

FN.24 (2/2)

1 $$D_{\text{Mac}} = \frac{1}{P}\sum_t t\,\frac{CF_t}{(1+y)^t}$$

2 $$D_{\text{mod}} = \frac{D_{\text{Mac}}}{1+y} = -\frac{1}{P}\frac{dP}{dy}$$

3 $$\text{Cx} = \frac{1}{P}\frac{d^2P}{dy^2} = \frac{1}{P}\sum_t \frac{t(t+1)\,CF_t}{(1+y)^{t+2}}$$

TERMS P price; y yield; CF_t cash flow at t; Cx convexity; 1 bp = 0.01%

Duration ≈ "the percent price drop for a 1-point rise in rates."

FN.24 (1/2)

1 $$\text{BEY} = 2\,y_{\text{semi}}, \qquad \text{EAY} = \left(1+\tfrac{\text{BEY}}{2}\right)^2 - 1$$

2 $$P = \sum_{t=1}^{n_c}\frac{C}{(1+y_c)^t} + \frac{\text{call price}}{(1+y_c)^{n_c}}$$

3 $$s_n = \left(\frac{F}{P_n}\right)^{1/n} - 1, \quad (1+s_2)^2 = (1+s_1)(1+f_{1,2})$$

TERMS P price; C coupon; F face; n periods; s_n spot rate; f forward rate

Current yield "now"; YTM "to the end"; YTC "if they call"; worst of them for callable bonds.

FN.23 (2/2)

1 $$u = e^{\sigma\sqrt{\Delta t}}, \qquad d = \frac{1}{u}$$

2 $$p = \frac{e^{r\Delta t} - d}{u - d}$$

3 $$V_0 = e^{-r\Delta t}\left[p\,V_u + (1-p)\,V_d\right]$$

TERMS u, d factors; p risk-neutral prob.; Δt step; V_u, V_d next values

"Replicate, don't predict": the real-world probability of an up move never enters the price.

FN.26 (1/2)

1 $$C - P = S_0 - Ke^{-rT}$$

2 $$S_T = K + C \;\text{ (call)}, \qquad S_T = K - P \;\text{ (put)}$$

TERMS S_T price at expiry; K strike; C, P premiums; r rate; T expiry

"Call up, put down": a call profits when price goes up, a put when it goes down. The call payoff is a ReLU

ML.22 *§15 Machine Learning!*

FN.25 (2/2)

1 $$\max(S_T - K, 0)$$

2 $$\max(K - S_T, 0)$$

3 $$\text{profit} = \text{payoff} - \text{premium}$$

TERMS S_T price at expiry; K strike; C, P premiums; r rate; T expiry

"Call up, put down": a call profits when price goes up, a put when it goes down. The call payoff is a ReLU

ML.22 *§15 Machine Learning!*

FN.25 (1/2)

1 $$E[S_T] = S_0 e^{\mu T}, \quad \text{median}(S_T) = S_0 e^{(\mu - \sigma^2/2)T}$$

2 $$S_{t+\Delta t} = S_t \exp\left[\left(\mu - \tfrac{\sigma^2}{2}\right)\Delta t + \sigma\sqrt{\Delta t}\,Z\right], \;\; Z \sim N(0,1)$$

TERMS μ drift; σ volatility; W_t Brownian motion; $Z \sim N(0,1)$

Mean > median: a few huge winners pull the average up (log-normal skew).

FN.27 (2/2)

1 $$dS = \mu S\,dt + \sigma S\,dW_t$$

2 $$S_T = S_0 \exp\left[\left(\mu - \frac{\sigma^2}{2}\right)T + \sigma W_T\right]$$

3 $$\ln\frac{S_T}{S_0} \sim N\left(\left(\mu - \tfrac{\sigma^2}{2}\right)T,\; \sigma^2 T\right)$$

TERMS μ drift; σ volatility; W_t Brownian motion; $Z \sim N(0,1)$

Mean > median: a few huge winners pull the average up (log-normal skew).

FN.27 (1/2)

1 $$\Delta = \frac{V_u - V_d}{S_u - S_d}$$

2 $$V_0 = e^{-rT}\sum_{k=0}^{n}\binom{n}{k}p^k(1-p)^{n-k}\,\text{payoff}\left(S_0 u^k d^{n-k}\right)$$

3 $$V = \max\left(\text{exercise value},\; \text{continuation value}\right)$$

TERMS u, d factors; p risk-neutral prob.; Δt step; V_u, V_d next values

"Replicate, don't predict": the real-world probability of an up move never enters the price.

FN.26 (2/2)

1 $$\Delta = \frac{\partial V}{\partial S} = N(d_1)\;\text{(call)}, \quad N(d_1) - 1\;\text{(put)}$$

2 $$\Gamma = \frac{\partial^2 V}{\partial S^2} = \frac{\varphi(d_1)}{S\sigma\sqrt{T}}$$

3 $$\mathcal{V} = \frac{\partial V}{\partial \sigma} = S\,\varphi(d_1)\sqrt{T}$$

TERMS V option value; S price; φ normal pdf; N normal CDF; d_1, d_2

Delta = slope, Gamma = curvature, Vega = volatility, Theta = time, Rho = rate.

FN.29 (1/2)

1 $$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2\frac{\partial^2 V}{\partial S^2} + rS\frac{\partial V}{\partial S} - rV = 0$$

2 $$\sigma_{\text{imp}}: \;\; C_{\text{BS}}(\sigma_{\text{imp}}) = C_{\text{market}}$$

TERMS S_0 price; K strike; r rate; σ vol; T years; N normal CDF

$N(d_2)$ ≈ *risk-neutral probability the call finishes in the money;* $N(d_1)$ *is the hedge ratio (delta).*

FN.28 (2/2)

1 $$C = S_0\,N(d_1) - Ke^{-rT}N(d_2)$$

2 $$d_1 = \frac{\ln(S_0/K) + \left(r + \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}$$

3 $$P = Ke^{-rT}N(-d_2) - S_0 N(-d_1)$$

TERMS S_0 price; K strike; r rate; σ vol; T years; N normal CDF

$N(d_2)$ ≈ *risk-neutral probability the call finishes in the money;* $N(d_1)$ *is the hedge ratio (delta).*

FN.28 (1/2)

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FN.29 (2/2) FINANCE

Option Greeks

PART 2 OF 2

1. Theta (call)
2. Rho (call)
3. Delta–gamma approximation

STATE EACH NUMBERED FORMULA

FN.30 (1/2) FINANCE

Value at Risk (VaR) & Expected Shortfall

PART 1 OF 2

1. Parametric VaR
2. Expected shortfall
3. Normal quantiles

STATE EACH NUMBERED FORMULA

FN.30 (2/2) FINANCE

Value at Risk (VaR) & Expected Shortfall

PART 2 OF 2

1. Normal ES
2. Historical VaR

STATE EACH NUMBERED FORMULA

FN.31 (1/2) FINANCE

Expected Value of Bets & the Kelly Criterion

PART 1 OF 2

1. Expected value
2. Kelly fraction
3. Growth rate

STATE EACH NUMBERED FORMULA

FN.31 (2/2) FINANCE

Expected Value of Bets & the Kelly Criterion

PART 2 OF 2

1. Continuous Kelly

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1 $\mathrm{ES}_\alpha = \dfrac{\varphi(z_\alpha)}{1-\alpha}\,\sigma\sqrt{t}\,V$

2 $\mathrm{VaR}_\alpha = -\mathrm{quantile}_{1-\alpha}(\text{past P\&L})$

TERMS α confidence; z_α quantile; $\sigma\sqrt{t}$ horizon vol; V value

VaR: "the loss you won't exceed α of the time"; ES: "how bad it gets when you do."

FN.30 (2/2)

1 $\mathrm{VaR}_\alpha = z_\alpha\,\sigma\sqrt{t}\,V$

2 $\mathrm{ES}_\alpha = E\big[\text{loss} \mid \text{loss} > \mathrm{VaR}_\alpha\big]$

3 $z_{0.95} = 1.645, \quad z_{0.99} = 2.326$

TERMS α confidence; z_α quantile; $\sigma\sqrt{t}$ horizon vol; V value

VaR: "the loss you won't exceed α of the time"; ES: "how bad it gets when you do."

FN.30 (1/2)

1 $\Theta = \dfrac{\partial V}{\partial t} = -\dfrac{S\,\varphi(d_1)\,\sigma}{2\sqrt{T}} - rKe^{-rT}N(d_2)$

2 $\rho = \dfrac{\partial V}{\partial r} = KTe^{-rT}N(d_2)$

3 $\Delta V \approx \Delta\cdot\Delta S + \frac{1}{2}\Gamma(\Delta S)^2$

TERMS V option value; S price; φ normal pdf; N normal CDF; d_1, d_2

Delta = slope, Gamma = curvature, Vega = volatility, Theta = time, Rho = rate.

FN.29 (2/2)

1 $f^* = \dfrac{\mu - r}{\sigma^2}$

TERMS p, q win/loss prob.; b net odds; f^* bankroll fraction

Never bet on negative EV; even with an edge, bet a fraction (many use "half Kelly").

FN.31 (2/2)

1 $EV = p\cdot W - q\cdot L$

2 $f^* = \dfrac{bp-q}{b} = p - \dfrac{q}{b}, \quad q = 1-p$

3 $G(f) = p\ln(1+bf) + q\ln(1-f)$

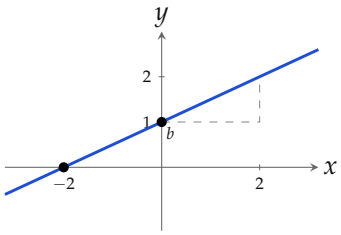
TERMS p, q win/loss prob.; b net odds; f^* bankroll fraction

Never bet on negative EV; even with an edge, bet a fraction (many use "half Kelly").

FN.31 (1/2)

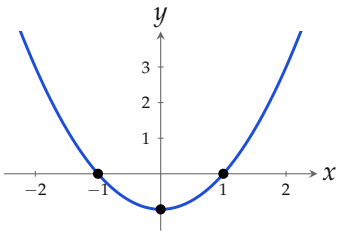
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F.1 FUNCTIONS



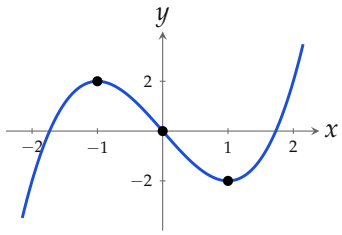
NAME IT / STATE ITS FORMULA

F.2 FUNCTIONS



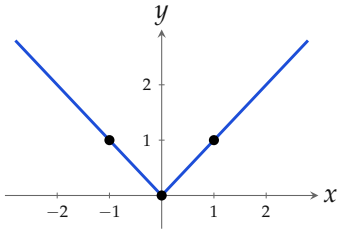
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F.3 FUNCTIONS



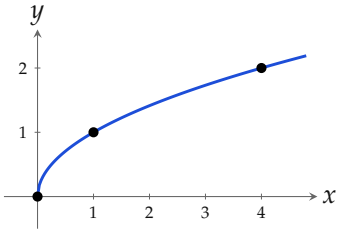
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F.4 FUNCTIONS



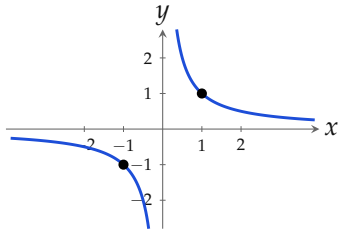
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F.5 FUNCTIONS



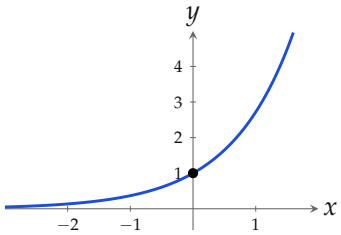
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F.6 FUNCTIONS



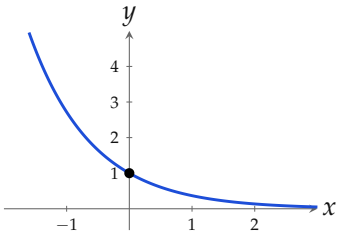
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F.7 FUNCTIONS



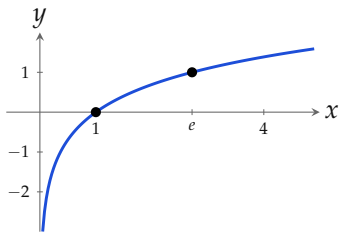
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F.8 FUNCTIONS



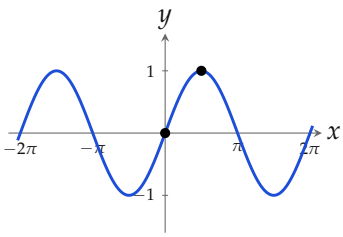
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F.9 FUNCTIONS



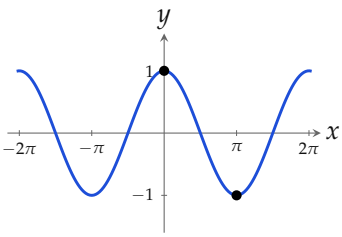
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F.10 FUNCTIONS



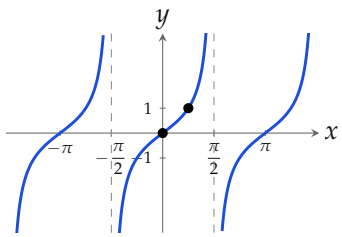
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F.11 FUNCTIONS



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F.12 FUNCTIONS



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Cubic

$y = ax^3 + bx^2 + cx + d$

odd end behavior; up to 2 turns

F.3

Quadratic

$y = ax^2 + bx + c$

parabola; vertex at $x = -\dfrac{b}{2a}$

F.2

Linear

$y = mx + b$

slope m, intercept b

F.1

Reciprocal

$y = \dfrac{1}{x}$

asymptotes $x = 0,\ y = 0$

F.6

Square Root

$y = \sqrt{x}$

domain $x \geq 0$

F.5

Absolute Value

$y = |x|$

V shape, vertex at $(0, 0)$

F.4

Natural Logarithm

$y = \ln x$

domain $x > 0$; VA $x = 0$; $\ln 1 = 0$

F.9

Exponential Decay

$y = e^{-x}$ $(b^x,\ 0 < b < 1)$

half-life $\dfrac{\ln 2}{k}$

F.8

Exponential Growth

$y = e^x$ $(b^x,\ b > 1)$

y-intercept 1; HA $y = 0$

F.7

Tangent

$y = \tan x = \dfrac{\sin x}{\cos x}$

period π; VA at $x = \pm\dfrac{\pi}{2}$

F.12

Cosine

$y = \cos x$

period 2π; $\cos 0 = 1$

F.11

Sine

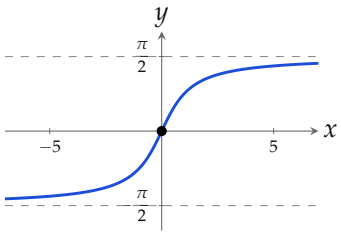
$y = \sin x$

period 2π; $\sin 0 = 0$

F.10

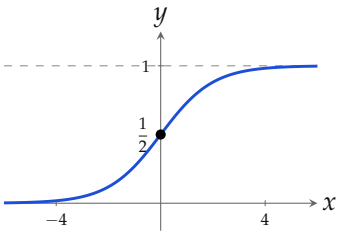
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F.13 FUNCTIONS



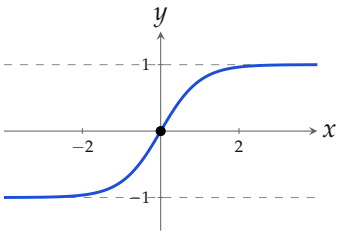
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F.14 FUNCTIONS



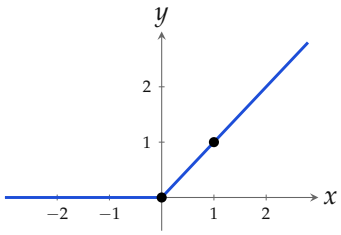
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F.15 FUNCTIONS



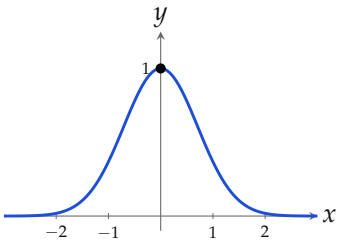
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F.16 FUNCTIONS



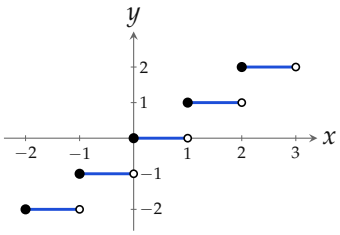
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F.17 FUNCTIONS



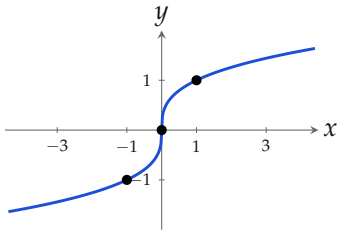
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F.18 FUNCTIONS



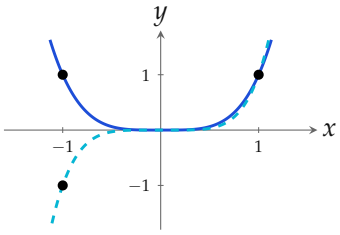
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F.19 FUNCTIONS



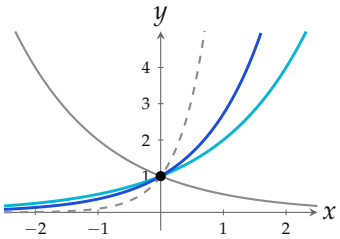
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F.20 FUNCTIONS



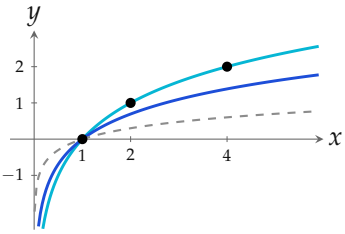
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F.21 FUNCTIONS



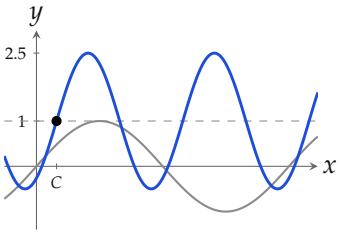
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F.22 FUNCTIONS



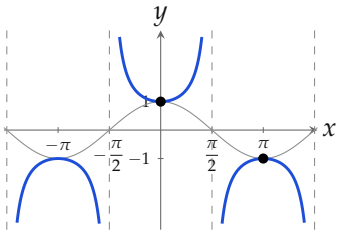
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F.23 FUNCTIONS



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F.24 FUNCTIONS



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Hyperbolic Tangent

$$\tanh x = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

range $(-1, 1)$; odd

F.15

Logistic Sigmoid

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

range $(0, 1)$; $\sigma(0) = \frac{1}{2}$

F.14

Arctangent

$$y = \arctan x$$

HA $y = \pm\frac{\pi}{2}$

F.13

Floor (Step) Function

$$y = \lfloor x \rfloor$$

greatest integer $\leq x$

F.18

Gaussian (Bell Curve)

$$y = e^{-x^2}$$

even; maximum 1 at $x = 0$

F.17

ReLU

$$\mathrm{ReLU}(x) = \max(0, x)$$

kink at 0

F.16

Exponential Family

$y = b^x \quad (b = \frac{1}{2}, 2, e, 10)$

all pass $(0, 1)$; larger b steeper; $b < 1$ decays

F.21

Even vs. Odd Powers

$y = x^4$ vs. $y = x^5$

even n: ends both up; odd n: ends opposite

F.20

Cube Root

$$y = \sqrt[3]{x} = x^{1/3}$$

all reals; odd; vertical tangent at 0

F.19

Secant

$$y = \sec x = \frac{1}{\cos x}$$

period 2π; $|y| \geq 1$; VA where $\cos x = 0$

F.24

Sinusoid with Phase Shift

$$y = A \sin\big(B(x - C)\big) + D$$

amplitude $|A|$, period $\dfrac{2\pi}{|B|}$, shift C, midline D

F.23

Logarithm Bases

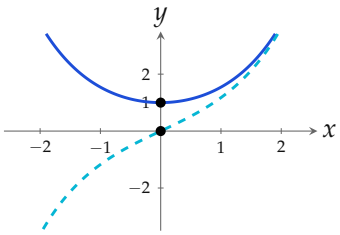
$$y = \log_b x = \frac{\ln x}{\ln b}$$

all pass $(1, 0)$; $\log_2 > \ln > \log_{10}$ for $x > 1$

F.22

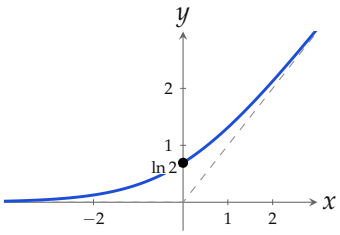
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F.25 FUNCTIONS



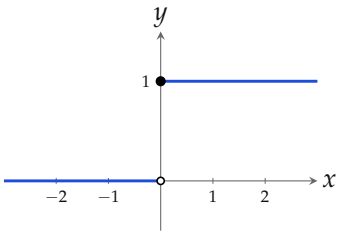
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F.26 FUNCTIONS



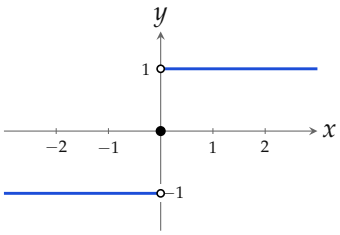
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F.27 FUNCTIONS



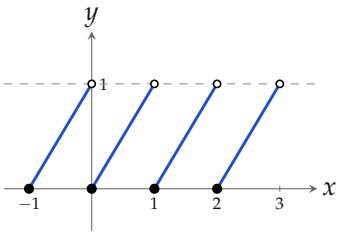
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F.28 FUNCTIONS



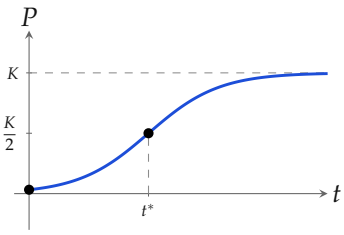
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F.29 FUNCTIONS



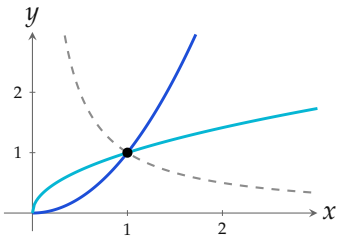
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F.30 FUNCTIONS



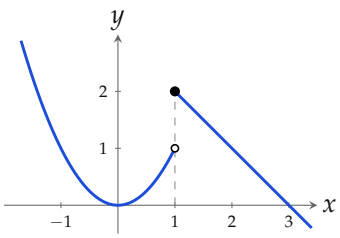
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F.31 FUNCTIONS



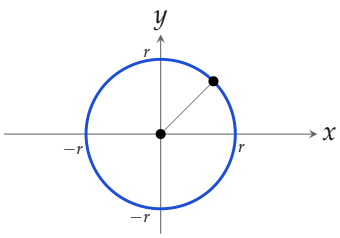
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F.32 FUNCTIONS



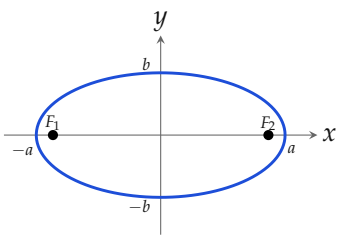
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F.33 FUNCTIONS



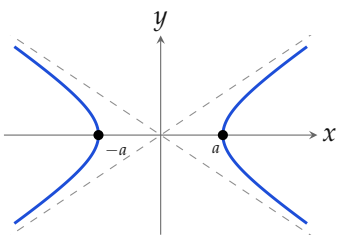
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F.34 FUNCTIONS



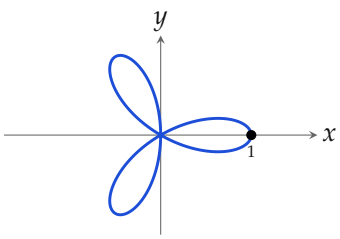
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F.35 FUNCTIONS



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F.36 FUNCTIONS



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Heaviside Step

$$H(x) = \begin{cases} 0, & x < 0 \\ 1, & x \geq 0 \end{cases}$$

$H'(x) = \delta(x)$; $H(0)$ is a convention

F.27

Softplus

$$\operatorname{softplus}(x) = \ln(1 + e^x)$$

smooth ReLU; derivative is $\sigma(x)$

F.26

Hyperbolic Sine & Cosine

$$\cosh x = \frac{e^x + e^{-x}}{2}, \quad \sinh x = \frac{e^x - e^{-x}}{2}$$

cosh even (catenary), sinh odd; $\cosh^2 x - \sinh^2 x = 1$

F.25

Logistic Growth

$$P(t) = \frac{K}{1 + Ae^{-rt}}, \quad A = \frac{K - P_0}{P_0}$$

fastest growth at $P = \frac{K}{2}$; capacity K

F.30

Sawtooth (Fractional Part)

$$y = x - \lfloor x \rfloor$$

period 1; range $[0, 1)$; jumps at integers

F.29

Sign Function

$$\operatorname{sgn} x = \frac{x}{|x|} \ (x \neq 0), \quad \operatorname{sgn} 0 = 0$$

$\frac{d}{dx}|x| = \operatorname{sgn} x$ for $x \neq 0$

F.28

Circle

$$x^2 + y^2 = r^2$$

center (h, k): $(x - h)^2 + (y - k)^2 = r^2$; not a function

F.33

Piecewise Function

$$f(x) = \begin{cases} x^2, & x < 1 \\ 3 - x, & x \geq 1 \end{cases}$$

use the branch whose condition holds; dots show which end is included

F.32

Power Law

$$y = Cx^k$$

$k > 1$ convex, $0 < k < 1$ concave, $k < 0$ decays; line on log–log axes

F.31

Polar Rose

$$r = \cos(n\theta)$$

n odd: n petals; n even: $2n$ petals; length 1

F.36

Hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

asymptotes $y = \pm\frac{b}{a}x$; $c^2 = a^2 + b^2$; $|d_1 - d_2| = 2a$

F.35

Ellipse

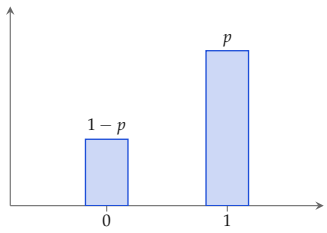
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

foci $(\pm c, 0)$, $c^2 = a^2 - b^2$; $d_1 + d_2 = 2a$

F.34

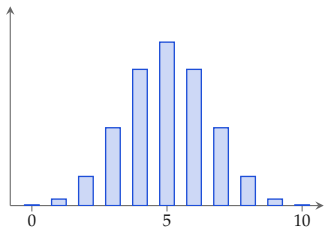
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D.1 PROBABILITY DISTRIBUTIONS



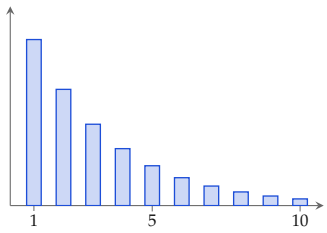
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D.2 PROBABILITY DISTRIBUTIONS



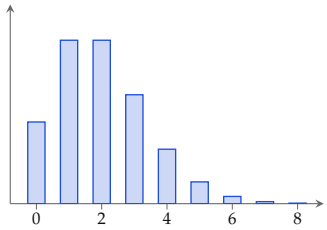
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D.3 PROBABILITY DISTRIBUTIONS



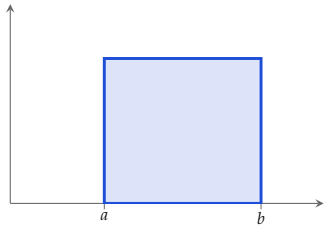
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D.4 PROBABILITY DISTRIBUTIONS



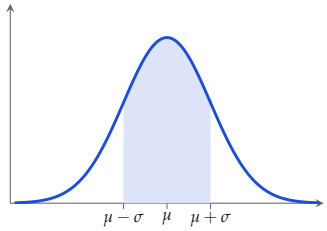
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D.5 PROBABILITY DISTRIBUTIONS



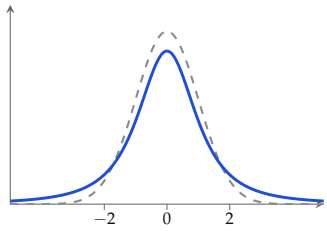
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D.6 PROBABILITY DISTRIBUTIONS



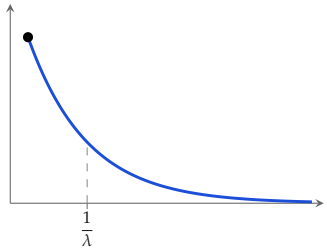
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D.7 PROBABILITY DISTRIBUTIONS



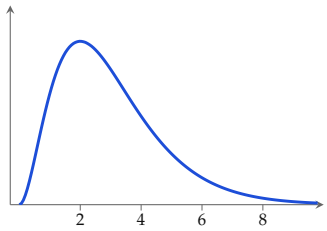
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D.8 PROBABILITY DISTRIBUTIONS



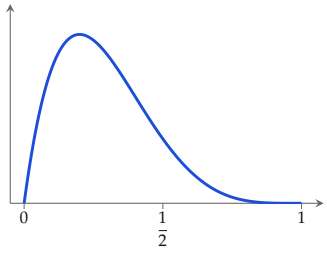
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D.9 PROBABILITY DISTRIBUTIONS



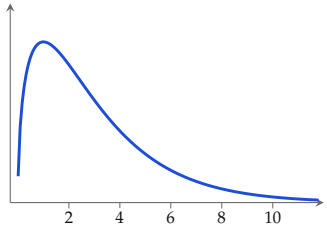
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D.10 PROBABILITY DISTRIBUTIONS



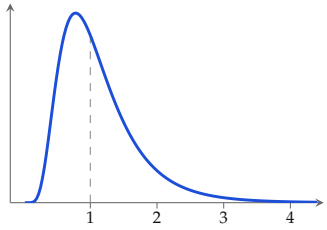
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D.11 PROBABILITY DISTRIBUTIONS



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D.12 PROBABILITY DISTRIBUTIONS



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Geometric

$$P(X=k) = (1-p)^{k-1}p, \quad k = 1, 2, \ldots$$

trials to first success;

$$\mu = \frac{1}{p}, \ \sigma^2 = \frac{1-p}{p^2}$$

D.3

Binomial

$$P(X=k) = \binom{n}{k} p^k (1-p)^{n-k}$$

$$\mu = np, \quad \sigma^2 = np(1-p)$$

D.2

Bernoulli

$$P(X=1) = p, \quad P(X=0) = 1-p$$

$$\mu = p, \quad \sigma^2 = p(1-p)$$

D.1

Normal

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

symmetric bell; 68–95–99.7

D.6

Uniform (Continuous)

$$f(x) = \frac{1}{b-a}, \quad a \le x \le b$$

$$\mu = \frac{a+b}{2}, \quad \sigma^2 = \frac{(b-a)^2}{12}$$

D.5

Poisson

$$P(X=k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

$$\mu = \sigma^2 = \lambda$$

D.4

Gamma

$$f(x) = \frac{x^{k-1} e^{-x/\theta}}{\Gamma(k)\,\theta^k}$$

$$\mu = k\theta, \quad \sigma^2 = k\theta^2$$

D.9

Exponential

$$f(x) = \lambda e^{-\lambda x}, \quad x \ge 0$$

$$\mu = \frac{1}{\lambda}$$; memoryless

D.8

Student's t (vs. normal)

$$f(t) \propto \left(1 + \frac{t^2}{\nu}\right)^{-\frac{\nu+1}{2}}$$

heavier tails; $t \to N(0,1)$ as $\nu \to \infty$

D.7

Log-Normal

$$\ln X \sim N(\mu, \sigma^2)$$

$$f(x) = \frac{1}{x\sigma\sqrt{2\pi}} e^{-\frac{(\ln x - \mu)^2}{2\sigma^2}}$$

D.12

Chi-Square

$$\chi^2_k = \sum_{i=1}^{k} Z_i^2$$

$$\mu = k, \quad \sigma^2 = 2k$$; right-skewed

D.11

Beta

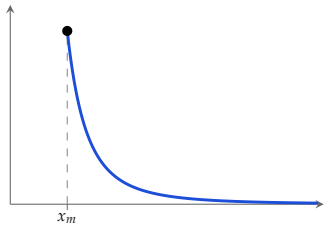
$$f(x) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha,\beta)}, \quad 0 \le x \le 1$$

$$\mu = \frac{\alpha}{\alpha+\beta}$$

D.10

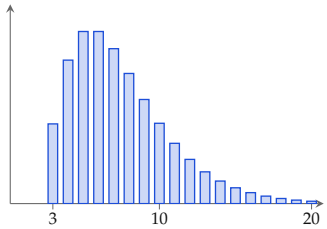
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D.13 PROBABILITY DISTRIBUTIONS



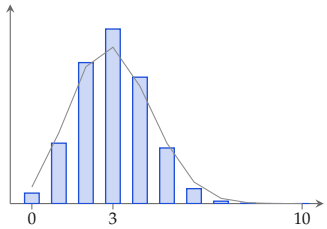
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D.14 PROBABILITY DISTRIBUTIONS



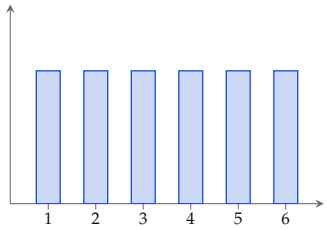
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D.15 PROBABILITY DISTRIBUTIONS



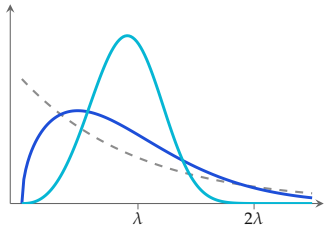
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D.16 PROBABILITY DISTRIBUTIONS



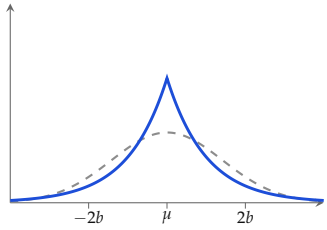
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D.17 PROBABILITY DISTRIBUTIONS



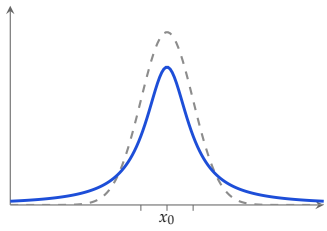
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D.18 PROBABILITY DISTRIBUTIONS



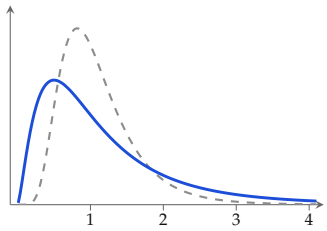
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D.19 PROBABILITY DISTRIBUTIONS



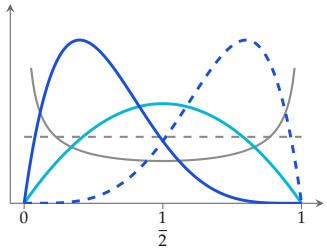
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D.20 PROBABILITY DISTRIBUTIONS



NAME IT / STATE ITS FORMULA

D.21 PROBABILITY DISTRIBUTIONS



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Hypergeometric

$$P(X=k)=\frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}$$

draws without replacement; $\mu = n\frac{K}{N}$

D.15

Negative Binomial

$$P(X=k)=\binom{k-1}{r-1}p^r(1-p)^{k-r}$$

trials to r-th success;
$\mu=\frac{r}{p}$, $\sigma^2=\frac{r(1-p)}{p^2}$

D.14

Pareto (Power Law)

$$f(x)=\frac{\alpha\, x_m^\alpha}{x^{\alpha+1}},\quad x\geq x_m$$

heavy tail; mean finite iff $\alpha > 1$

D.13

Laplace (vs. normal)

$$f(x)=\frac{1}{2b}e^{-|x-\mu|/b}$$

$\sigma^2 = 2b^2$; L1 loss / lasso prior

D.18

Weibull

$$f(x)=\frac{k}{\lambda}\left(\frac{x}{\lambda}\right)^{k-1}e^{-(x/\lambda)^k}$$

shape $k < 1, = 1, > 1$: failure rate falls, flat, rises

D.17

Discrete Uniform

$$P(X=k)=\frac{1}{n},\quad k=1,\ldots,n$$

fair die; $\mu=\frac{n+1}{2}$, $\sigma^2=\frac{n^2-1}{12}$

D.16

Beta Shapes

$$\text{mode}=\frac{\alpha-1}{\alpha+\beta-2}\quad(\alpha,\beta>1)$$

$\alpha=\beta<1$: U; $=1$: flat; $\alpha<\beta$: right-skewed

D.21

F-Distribution

$$F=\frac{\chi^2_{d_1}/d_1}{\chi^2_{d_2}/d_2}$$

ANOVA, nested models; $\mu=\frac{d_2}{d_2-2}$

D.20

Cauchy (vs. normal)

$$f(x)=\frac{1}{\pi\gamma\left[1+\left(\frac{x-x_0}{\gamma}\right)^2\right]}$$

no mean or variance; ratio of two normals

D.19

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AL.1 ALGORITHMS

Linear Regression

DEFINE IT

AL.2 ALGORITHMS

Logistic Regression

DEFINE IT

AL.3 ALGORITHMS

Ridge / Lasso

DEFINE IT

AL.4 ALGORITHMS

k-Nearest Neighbors

DEFINE IT

AL.5 ALGORITHMS

Naive Bayes

DEFINE IT

AL.6 ALGORITHMS

Decision Tree

DEFINE IT

AL.7 ALGORITHMS

Random Forest

DEFINE IT

AL.8 ALGORITHMS

Gradient Boosting (XGBoost, LightGBM)

DEFINE IT

AL.9 ALGORITHMS

Support Vector Machine

DEFINE IT

AL.10 ALGORITHMS

Neural Network (MLP)

DEFINE IT

AL.11 ALGORITHMS

CNN

DEFINE IT

AL.12 ALGORITHMS

RNN / LSTM

DEFINE IT

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Supervised, regularized regression. L2 / L1 penalties. Used for: many correlated features, feature selection.

AL.3 Algorithms

Supervised, classification. Sigmoid of a linear score. Used for: churn, credit risk, conversion.

AL.2 Algorithms

Supervised, regression. Fits $\hat{y} = X\beta$ by least squares. Used for: pricing, forecasting, effect sizes.

AL.1 Algorithms

Supervised. Splits data by if–then rules. Used for: interpretable rules, triage.

AL.6 Algorithms

Supervised, classification. Bayes' rule with independent features. Used for: spam filtering, text tagging.

AL.5 Algorithms

Supervised, classification or regression. Vote of the k closest points. Used for: recommendations, simple baselines, imputation.

AL.4 Algorithms

Supervised, classification. Maximum-margin boundary, kernels. Used for: small high-dimensional data, text.

AL.9 Algorithms

Supervised. Adds trees that fix previous errors. Used for: top accuracy on tabular data (fraud, pricing, ranking).

AL.8 Algorithms

Supervised. Averages many decorrelated trees (bagging). Used for: robust tabular baselines, feature importance.

AL.7 Algorithms

Supervised (deep learning). Recurrent state over a sequence. Used for: time series, speech (largely replaced by transformers).

AL.12 Algorithms

Supervised (deep learning). Convolutional filters share weights across space. Used for: images, medical scans, audio spectrograms.

AL.11 Algorithms

Supervised. Layers of linear maps and nonlinearities. Used for: flexible function fitting, embeddings.

AL.10 Algorithms

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AL.13 ALGORITHMS

Transformer

DEFINE IT

AL.14 ALGORITHMS

k-Means

DEFINE IT

AL.15 ALGORITHMS

DBSCAN

DEFINE IT

AL.16 ALGORITHMS

Hierarchical Clustering

DEFINE IT

AL.17 ALGORITHMS

Gaussian Mixture Model

DEFINE IT

AL.18 ALGORITHMS

PCA

DEFINE IT

AL.19 ALGORITHMS

t-SNE / UMAP

DEFINE IT

AL.20 ALGORITHMS

Isolation Forest

DEFINE IT

AL.21 ALGORITHMS

Autoencoder

DEFINE IT

AL.22 ALGORITHMS

Apriori / Association Rules

DEFINE IT

AL.23 ALGORITHMS

Matrix Factorization

DEFINE IT

AL.24 ALGORITHMS

LDA (Topic Model)

DEFINE IT

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Unsupervised, clustering. Groups dense regions and labels sparse points as noise; no K needed. Used for: geospatial hot spots, anomaly detection, irregular-shaped clusters.

AL.15 Algorithms

Unsupervised, clustering. Minimizes squared distance to K centroids. Used for: customer segmentation, quantization.

AL.14 Algorithms

Self-supervised / supervised. Attention over all tokens. Used for: LLMs, translation, vision transformers.

AL.13 Algorithms

Unsupervised, dimension reduction. Projects onto directions of maximum variance. Used for: compression, decorrelation, visualization.

AL.18 Algorithms

Unsupervised. Soft clustering with EM. Used for: density estimation, soft segments.

AL.17 Algorithms

Unsupervised. Merges clusters into a dendrogram. Used for: taxonomies, gene expression.

AL.16 Algorithms

Unsupervised (self-supervised). Network compresses and reconstructs input. Used for: denoising, anomaly detection by reconstruction error.

AL.21 Algorithms

Unsupervised, anomaly detection. Anomalies are isolated in few random splits. Used for: fraud, sensor faults, intrusion detection.

AL.20 Algorithms

Unsupervised, nonlinear embedding. Preserves local neighborhoods in 2-D. Used for: visualizing embeddings and single-cell data (not for distances).

AL.19 Algorithms

Unsupervised. Latent Dirichlet Allocation: documents as mixtures of topics. Used for: topic discovery in text.

AL.24 Algorithms

Unsupervised / collaborative filtering. Low-rank user $\times$ item factors. Used for: recommender systems.

AL.23 Algorithms

Unsupervised. Finds frequent itemsets with support, confidence, lift. Used for: market-basket analysis.

AL.22 Algorithms

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AL.25 ALGORITHMS

ARIMA

DEFINE IT

AL.26 ALGORITHMS

Q-Learning

DEFINE IT

AL.27 ALGORITHMS

Spectral Clustering

DEFINE IT

AL.28 ALGORITHMS

GAN

DEFINE IT

AL.29 ALGORITHMS

k-Medoids (PAM)

DEFINE IT

AL.30 ALGORITHMS

HDBSCAN

DEFINE IT

AL.31 ALGORITHMS

Mean Shift

DEFINE IT

AL.32 ALGORITHMS

Elastic Net

DEFINE IT

AL.33 ALGORITHMS

Generalized Linear Model (GLM)

DEFINE IT

AL.34 ALGORITHMS

Generalized Additive Model (GAM)

DEFINE IT

AL.35 ALGORITHMS

Quantile Regression

DEFINE IT

AL.36 ALGORITHMS

Gaussian Process Regression

DEFINE IT

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Unsupervised. Clusters using eigenvectors of a graph Laplacian. Used for: non-convex clusters, community detection.

AL.27 Algorithms

Reinforcement learning. Learns action values via the Bellman equation. Used for: games, control, recommendations over time.

AL.26 Algorithms

Supervised time series. AutoRegressive Integrated Moving Average. Used for: demand and sales forecasting.

AL.25 Algorithms

Unsupervised, clustering. Hierarchical DBSCAN: keeps the most stable clusters across all density levels, so it handles clusters of varying density; only a minimum cluster size is needed. Used for: embeddings, geospatial and log data.

AL.30 Algorithms

Unsupervised, clustering. Like k-means ML.35 §15 Machine Learning but each center is an actual data point, so any distance works and outliers matter less. Used for: clustering with Manhattan, edit or Gower distance.

AL.29 Algorithms

Unsupervised generative. Generator vs. discriminator game. Used for: synthetic images and data augmentation.

AL.28 Algorithms

Supervised. A linear score passed through a link function with an exponential-family outcome (logistic, Poisson, gamma). Used for: insurance claims, count data, rates.

AL.33 Algorithms

Supervised, regularized regression. Mixes L1 and L2 penalties ML.8 §15 Machine Learning: sparse like lasso but stable with correlated features. Used for: genomics, wide data ($p \gg n$).

AL.32 Algorithms

Unsupervised, clustering. Moves points uphill on a kernel density estimate until they reach a mode; the number of modes is the number of clusters. Used for: image segmentation, tracking.

AL.31 Algorithms

Supervised, Bayesian. A distribution over functions set by a kernel; gives predictions with uncertainty bands. Used for: small data, spatial kriging, Bayesian optimization.

AL.36 Algorithms

Supervised, regression. Minimizes the pinball loss to predict a chosen quantile instead of the mean. Used for: prediction intervals, delivery-time and demand ranges.

AL.35 Algorithms

Supervised. Sum of smooth one-feature curves (splines) inside a GLM: flexible yet interpretable. Used for: risk curves, dose–response, ecology.

AL.34 Algorithms

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AL.37 ALGORITHMS

Bayesian Optimization

DEFINE IT

AL.38 ALGORITHMS

Survival Analysis (Kaplan–Meier, Cox)

DEFINE IT

AL.39 ALGORITHMS

Exponential Smoothing (Holt–Winters)

DEFINE IT

AL.40 ALGORITHMS

Prophet

DEFINE IT

AL.41 ALGORITHMS

Hidden Markov Model / Viterbi

DEFINE IT

AL.42 ALGORITHMS

Kalman Filter

DEFINE IT

AL.43 ALGORITHMS

Word2Vec

DEFINE IT

AL.44 ALGORITHMS

Contrastive Learning (SimCLR, CLIP)

DEFINE IT

AL.45 ALGORITHMS

Diffusion Model

DEFINE IT

AL.46 ALGORITHMS

Graph Neural Network (GNN)

DEFINE IT

AL.47 ALGORITHMS

PageRank

DEFINE IT

AL.48 ALGORITHMS

Approximate Nearest Neighbors (HNSW, FAISS)

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Time series. Weighted averages that decay geometrically into the past, with optional trend and seasonal terms. Used for: inventory and short-horizon demand forecasts.

AL.39 Algorithms

Supervised, time-to-event. Handles censored data; Kaplan–Meier estimates $S(t)$, Cox regression models hazard ratios. Used for: churn timing, clinical trials, failure analysis.

AL.38 Algorithms

Optimization. Fits a surrogate (often a GP) to an expensive objective and picks the next point by an acquisition function. Used for: hyperparameter tuning, experiment design.

AL.37 Algorithms

State estimation. Recursive predict–update for a linear system with Gaussian noise; optimal blend of model and measurement. Used for: GPS/IMU tracking, navigation, smoothing noisy sensors.

AL.42 Algorithms

Unsupervised / sequence. Hidden states follow a Markov chain and emit observations; Viterbi finds the most likely state path by dynamic programming O.21 §14 Optimization. Used for: speech, gene finding, regime detection.

AL.41 Algorithms

Time series. Additive model: piecewise trend + Fourier seasonalities + holiday effects, fitted by regression. Used for: business forecasts with strong seasonality and missing data.

AL.40 Algorithms

Generative. Learns to reverse a gradual noising process, generating data by iterative denoising from pure noise. Used for: image, audio and video generation.

AL.45 Algorithms

Self-supervised. Pulls embeddings of matching pairs together and pushes non-matching pairs apart. Used for: image–text search, pre-training without labels.

AL.44 Algorithms

Self-supervised, embeddings. A shallow network predicts a word from its context (CBOW) or the context from a word (skip-gram), producing word vectors where similar words are close. Used for: text features, similarity, analogies.

AL.43 Algorithms

Search. Index structures that find near neighbors in sublinear time with small recall loss. Used for: vector databases, semantic search, RAG retrieval.

AL.48 Algorithms

Graph ranking. Stationary distribution of a random surfer on a link graph: the dominant eigenvector LA.23 §09 Linear Algebra of the transition matrix. Used for: web search, influence and citation ranking.

AL.47 Algorithms

Supervised / self-supervised. Each node updates its vector by aggregating messages from its neighbors. Used for: molecules, fraud rings, recommendation graphs.

AL.46 Algorithms

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AL.49 ALGORITHMS

LoRA & Fine-Tuning

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AL.50 ALGORITHMS

RLHF / DPO

DEFINE IT

AL.51 ALGORITHMS

Multi-Armed Bandit (Thompson Sampling, UCB)

DEFINE IT

AL.52 ALGORITHMS

Bootstrap

DEFINE IT

AL.53 ALGORITHMS

SMOTE

DEFINE IT

AL.54 ALGORITHMS

SHAP

DEFINE IT

AC.1 ACRONYMS

CNN

DEFINE IT

AC.2 ACRONYMS

RNN

DEFINE IT

AC.3 ACRONYMS

LSTM

DEFINE IT

AC.4 ACRONYMS

GRU

DEFINE IT

AC.5 ACRONYMS

GAN

DEFINE IT

AC.6 ACRONYMS

VAE

DEFINE IT

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Online learning. Balances exploring uncertain options and exploiting the best so far. Used for: adaptive A/B tests, ad and recommendation selection.

AL.51 Algorithms

Alignment. Reinforcement Learning from Human Feedback trains a reward model on human preferences and optimizes the LLM against it (PPO); DPO optimizes the preferences directly. Used for: chat assistants.

AL.50 Algorithms

Transfer learning. Fine-tuning updates a pre-trained model on task data; LoRA freezes the weights W and learns a low-rank update $W + BA$, cutting trainable parameters by orders of magnitude. Used for: adapting LLMs and diffusion models cheaply.

AL.49 Algorithms

Interpretability. SHapley Additive exPlanations: splits a prediction into per-feature contributions using game-theoretic Shapley values. Used for: explaining tree models and credit decisions.

AL.54 Algorithms

Preprocessing. Synthetic Minority Over-sampling: creates new minority-class points by interpolating between neighbors. Used for: imbalanced classification (apply to training folds only).

AL.53 Algorithms

Resampling. Resample the data with replacement many times and recompute a statistic to estimate its sampling distribution. Used for: confidence intervals when no formula exists S.27 §13 Statistics.

AL.52 Algorithms

Long Short-Term Memory: an RNN with gates that preserve long-range information.

AC.3 Acronyms

Recurrent Neural Network: processes sequences with a hidden state carried forward.

AC.2 Acronyms

Convolutional Neural Network: learns local filters (convolutions) shared across an image or signal.

AC.1 Acronyms

Variational Autoencoder: probabilistic autoencoder that learns a latent distribution.

AC.6 Acronyms

Generative Adversarial Network: a generator and a discriminator trained against each other.

AC.5 Acronyms

Gated Recurrent Unit: a simpler gated RNN.

AC.4 Acronyms

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AC.7 ACRONYMS

LLM

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AC.8 ACRONYMS

NLP

DEFINE IT

AC.9 ACRONYMS

RAG

DEFINE IT

AC.10 ACRONYMS

MLP

DEFINE IT

AC.11 ACRONYMS

SGD

DEFINE IT

AC.12 ACRONYMS

ReLU

DEFINE IT

AC.13 ACRONYMS

ROC / AUC

DEFINE IT

AC.14 ACRONYMS

MSE / RMSE / MAE

DEFINE IT

AC.15 ACRONYMS

PCA

DEFINE IT

AC.16 ACRONYMS

SVD

DEFINE IT

AC.17 ACRONYMS

SVM

DEFINE IT

AC.18 ACRONYMS

k-NN

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Retrieval-Augmented Generation: an LLM answers using documents retrieved by search.

AC.9 Acronyms

Natural Language Processing: computing with human language.

AC.8 Acronyms

Large Language Model: a very large transformer trained to predict text.

AC.7 Acronyms

Rectified Linear Unit: activation $\max(0, x)$.

AC.12 Acronyms

Stochastic Gradient Descent: gradient steps on random mini-batches.

AC.11 Acronyms

Multilayer Perceptron: a fully connected feed-forward neural network.

AC.10 Acronyms

Principal Component Analysis.

AC.15 Acronyms

Mean Squared Error / Root Mean Squared Error / Mean Absolute Error.

AC.14 Acronyms

Receiver Operating Characteristic curve / Area Under the Curve.

AC.13 Acronyms

k-Nearest Neighbors.

AC.18 Acronyms

Support Vector Machine.

AC.17 Acronyms

Singular Value Decomposition: $A = U\Sigma V^\top$.

AC.16 Acronyms

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AC.19 ACRONYMS

EDA

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AC.20 ACRONYMS

ETL

DEFINE IT

AC.21 ACRONYMS

TF-IDF

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AC.22 ACRONYMS

BERT

DEFINE IT

AC.23 ACRONYMS

GPT

DEFINE IT

AC.24 ACRONYMS

OLS

DEFINE IT

AC.25 ACRONYMS

MLE / MAP

DEFINE IT

AC.26 ACRONYMS

MCMC

DEFINE IT

AC.27 ACRONYMS

KL

DEFINE IT

AC.28 ACRONYMS

i.i.d.

DEFINE IT

AC.29 ACRONYMS

PDF / PMF / CDF

DEFINE IT

AC.30 ACRONYMS

IQR

DEFINE IT

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Term Frequency–Inverse Document Frequency: weights words that are frequent in a document but rare overall.

AC.21 Acronyms

Extract, Transform, Load: moving and reshaping data into a warehouse.

AC.20 Acronyms

Exploratory Data Analysis: summarizing and plotting data before modeling.

AC.19 Acronyms

Ordinary Least Squares.

AC.24 Acronyms

Generative Pre-trained Transformer: a decoder language model.

AC.23 Acronyms

Bidirectional Encoder Representations from Transformers: an encoder language model.

AC.22 Acronyms

Kullback–Leibler divergence.

AC.27 Acronyms

Markov Chain Monte Carlo.

AC.26 Acronyms

Maximum Likelihood Estimate / Maximum A Posteriori estimate (likelihood $\times$ prior).

AC.25 Acronyms

Interquartile Range, $Q_3 - Q_1$.

AC.30 Acronyms

Probability Density Function / Probability Mass Function / Cumulative Distribution Function.

AC.29 Acronyms

Independent and Identically Distributed.

AC.28 Acronyms

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AC.31 ACRONYMS

CI

DEFINE IT

AC.32 ACRONYMS

RL

DEFINE IT

AC.33 ACRONYMS

LP / MIP / NLP

DEFINE IT

AC.34 ACRONYMS

ODE / PDE

DEFINE IT

AC.35 ACRONYMS

GPU

DEFINE IT

AC.36 ACRONYMS

MLOps

DEFINE IT

AC.37 ACRONYMS

GLM / GAM

DEFINE IT

AC.38 ACRONYMS

ANOVA

DEFINE IT

AC.39 ACRONYMS

AIC / BIC

DEFINE IT

AC.40 ACRONYMS

KDE

DEFINE IT

AC.41 ACRONYMS

EM

DEFINE IT

AC.42 ACRONYMS

HMM

DEFINE IT

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Linear / Mixed-Integer / Nonlinear Programming (in optimization, NLP means nonlinear programming).

AC.33 Acronyms

Reinforcement Learning: learning actions from rewards.

AC.32 Acronyms

Confidence Interval (also Continuous Integration in software).

AC.31 Acronyms

Machine Learning Operations: deploying, monitoring, and maintaining models.

AC.36 Acronyms

Graphics Processing Unit: parallel hardware used to train neural networks.

AC.35 Acronyms

Ordinary / Partial Differential Equation.

AC.34 Acronyms

Akaike / Bayesian Information Criterion: fit minus a complexity penalty; lower is better.

AC.39 Acronyms

Analysis of Variance: tests whether several group means differ using an F statistic.

AC.38 Acronyms

Generalized Linear Model / Generalized Additive Model.

AC.37 Acronyms

Hidden Markov Model.

AC.42 Acronyms

Expectation–Maximization: alternates soft assignment and parameter updates ML.37 §15 Machine Learning.

AC.41 Acronyms

Kernel Density Estimation: a smooth density built from a bump on each data point.

AC.40 Acronyms

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AC.43 ACRONYMS

GP

DEFINE IT

AC.44 ACRONYMS

MDP

DEFINE IT

AC.45 ACRONYMS

TD

DEFINE IT

AC.46 ACRONYMS

PPO

DEFINE IT

AC.47 ACRONYMS

RLHF

DEFINE IT

AC.48 ACRONYMS

LoRA / PEFT

DEFINE IT

AC.49 ACRONYMS

MoE

DEFINE IT

AC.50 ACRONYMS

ViT

DEFINE IT

AC.51 ACRONYMS

GNN

DEFINE IT

AC.52 ACRONYMS

NER

DEFINE IT

AC.53 ACRONYMS

ANN

DEFINE IT

AC.54 ACRONYMS

TPR / FPR

DEFINE IT

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Temporal Difference learning: updates value estimates from the next estimate (bootstrapping).

AC.45 Acronyms

Markov Decision Process: states, actions, transition probabilities, rewards; the setting of RL.

AC.44 Acronyms

Gaussian Process.

AC.43 Acronyms

Low-Rank Adaptation / Parameter-Efficient Fine-Tuning.

AC.48 Acronyms

Reinforcement Learning from Human Feedback.

AC.47 Acronyms

Proximal Policy Optimization: a stable policy-gradient RL method with clipped updates.

AC.46 Acronyms

Graph Neural Network.

AC.51 Acronyms

Vision Transformer: a transformer applied to image patches.

AC.50 Acronyms

Mixture of Experts: a router sends each token to a few expert sub-networks.

AC.49 Acronyms

True Positive Rate (recall) / False Positive Rate: the axes of the ROC curve ML.18 §15 Machine Learning.

AC.54 Acronyms

Approximate Nearest Neighbor (search), or Artificial Neural Network; check context.

AC.53 Acronyms

Named Entity Recognition: tagging people, places, organizations in text.

AC.52 Acronyms

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AC.55 ACRONYMS

F1

DEFINE IT

AC.56 ACRONYMS

MAPE

DEFINE IT

AC.57 ACRONYMS

R^2

DEFINE IT

AC.58 ACRONYMS

FDR

DEFINE IT

AC.59 ACRONYMS

VIF

DEFINE IT

AC.60 ACRONYMS

RCT

DEFINE IT

AC.61 ACRONYMS

ATE

DEFINE IT

AC.62 ACRONYMS

DAG

DEFINE IT

AC.63 ACRONYMS

SQL

DEFINE IT

AC.64 ACRONYMS

API

DEFINE IT

AC.65 ACRONYMS

FLOPs

DEFINE IT

AC.66 ACRONYMS

TPU

DEFINE IT

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Coefficient of determination: share of variance explained S.40 §13 Statistics.

AC.57 Acronyms

Mean Absolute Percentage Error; undefined when actual values are 0.

AC.56 Acronyms

Harmonic mean of precision and recall ML.17 §15 Machine Learning.

AC.55 Acronyms

Randomized Controlled Trial: random assignment makes groups comparable.

AC.60 Acronyms

Variance Inflation Factor, $1/(1-R_j^2)$: flags multicollinearity (above 5–10).

AC.59 Acronyms

False Discovery Rate: expected share of false positives among discoveries (Benjamini–Hochberg).

AC.58 Acronyms

Structured Query Language: querying relational databases.

AC.63 Acronyms

Directed Acyclic Graph: used for causal diagrams, pipelines and Bayesian networks.

AC.62 Acronyms

Average Treatment Effect: mean outcome difference if everyone vs. no one were treated.

AC.61 Acronyms

Tensor Processing Unit: Google's accelerator for matrix math.

AC.66 Acronyms

Floating-Point Operations: the unit of compute for training and inference.

AC.65 Acronyms

Application Programming Interface: a defined way for programs to call each other.

AC.64 Acronyms

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AC.67 ACRONYMS

NPV / IRR

DEFINE IT

AC.68 ACRONYMS

APR / EAR

DEFINE IT

AC.69 ACRONYMS

CAGR

DEFINE IT

AC.70 ACRONYMS

VaR / ES

DEFINE IT

AC.71 ACRONYMS

CAPM

DEFINE IT

AC.72 ACRONYMS

P/E

DEFINE IT

AC.73 ACRONYMS

ROE / ROA

DEFINE IT

AC.74 ACRONYMS

KCL / KVL

DEFINE IT

AC.75 ACRONYMS

RMS

DEFINE IT

AC.76 ACRONYMS

SNR

DEFINE IT

AC.77 ACRONYMS

DFT / FFT

DEFINE IT

KT.1 KEY TERMS

Domain & Range

DEFINE IT

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Compound Annual Growth Rate
FN.10 §17 Finance.

AC.69 Acronyms

Annual Percentage Rate / Effective Annual Rate FN.2 §17 Finance.

AC.68 Acronyms

Net Present Value / Internal Rate of Return FN.8 §17 Finance.

AC.67 Acronyms

Price-to-Earnings ratio: share price divided by earnings per share.

AC.72 Acronyms

Capital Asset Pricing Model
FN.14 §17 Finance.

AC.71 Acronyms

Value at Risk / Expected Shortfall
FN.30 §17 Finance.

AC.70 Acronyms

Root Mean Square: the DC-equivalent value of an AC signal EE.13 §16 Electrical Eng..

AC.75 Acronyms

Kirchhoff's Current / Voltage Law EE.4 §16 Electrical Eng..

AC.74 Acronyms

Return on Equity / Return on Assets FN.20 §17 Finance.

AC.73 Acronyms

Domain: all allowed inputs of a function. Range: all outputs it produces.

KT.1 Key Terms

Discrete Fourier Transform / Fast Fourier Transform EE.26 §16 Electrical Eng..

AC.77 Acronyms

Signal-to-Noise Ratio, usually in dB EE.29 §16 Electrical Eng..

AC.76 Acronyms

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KT.2 KEY TERMS

Monotonic

DEFINE IT

KT.3 KEY TERMS

One-to-One (Injective)

DEFINE IT

KT.4 KEY TERMS

Even & Odd Function

DEFINE IT

KT.5 KEY TERMS

Asymptote

DEFINE IT

KT.6 KEY TERMS

Identity vs. Equation

DEFINE IT

KT.7 KEY TERMS

Theorem, Lemma, Corollary

DEFINE IT

KT.8 KEY TERMS

Necessary vs. Sufficient

DEFINE IT

KT.9 KEY TERMS

Population vs. Sample

DEFINE IT

KT.10 KEY TERMS

Parameter vs. Statistic

DEFINE IT

KT.11 KEY TERMS

Bias (Statistical)

DEFINE IT

KT.12 KEY TERMS

Confounder

DEFINE IT

KT.13 KEY TERMS

Stationarity

DEFINE IT

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Even: $f(-x) = f(x)$ (symmetric about the y-axis). Odd: $f(-x) = -f(x)$ (symmetric about the origin).

KT.4 Key Terms

Different inputs give different outputs; needed for an inverse.

KT.3 Key Terms

Always increasing or always decreasing.

KT.2 Key Terms

Theorem: major proven result. Lemma: helper result. Corollary: follows directly from a theorem.

KT.7 Key Terms

An identity is true for all values; an equation is true only for its solutions.

KT.6 Key Terms

A line a graph approaches but does not reach (vertical, horizontal, or slant).

KT.5 Key Terms

Parameter describes a population (μ); statistic describes a sample ($\bar{x}$).

KT.10 Key Terms

Population: every unit of interest. Sample: the subset you observe.

KT.9 Key Terms

A is necessary for B if $B \Rightarrow A$; sufficient if $A \Rightarrow B$.

KT.8 Key Terms

A time series whose mean, variance and autocorrelation do not change over time.

KT.13 Key Terms

A variable that affects both the cause and the outcome, creating a spurious association.

KT.12 Key Terms

Systematic difference between an estimator's average and the truth.

KT.11 Key Terms

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KT.14 KEY TERMS

Heteroscedasticity

DEFINE IT

KT.15 KEY TERMS

Multicollinearity

DEFINE IT

KT.16 KEY TERMS

Feature & Label

DEFINE IT

KT.17 KEY TERMS

Training, Validation, Test Sets

DEFINE IT

KT.18 KEY TERMS

Overfitting & Underfitting

DEFINE IT

KT.19 KEY TERMS

Hyperparameter

DEFINE IT

KT.20 KEY TERMS

Epoch & Batch

DEFINE IT

KT.21 KEY TERMS

Data Leakage

DEFINE IT

KT.22 KEY TERMS

One-Hot Encoding

DEFINE IT

KT.23 KEY TERMS

Normalization vs. Standardization

DEFINE IT

KT.24 KEY TERMS

Imputation

DEFINE IT

KT.25 KEY TERMS

Embedding

DEFINE IT

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Feature: an input variable. Label (target): the value to predict.

KT.16 Key Terms

Predictors that are highly correlated, making coefficients unstable.

KT.15 Key Terms

Residual variance that changes with the predictors.

KT.14 Key Terms

A setting chosen before training (learning rate, depth, k), not learned from data.

KT.19 Key Terms

Overfitting: memorizes noise (great train, poor test). Underfitting: too simple for both.

KT.18 Key Terms

Fit on training; tune on validation; report once on the untouched test set.

KT.17 Key Terms

Representing a category as a vector with a single 1.

KT.22 Key Terms

Information from outside the training data (e.g. the future or the test set) sneaks into training.

KT.21 Key Terms

Epoch: one full pass over the training data. Batch: the subset used for one gradient step.

KT.20 Key Terms

A dense vector representing an item (word, user, image) so that similar items are close.

KT.25 Key Terms

Filling in missing values (mean, median, model-based).

KT.24 Key Terms

Normalization rescales to $[0, 1]$; standardization gives mean 0, SD 1.

KT.23 Key Terms

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KT.26 KEY TERMS

Tensor

DEFINE IT

KT.27 KEY TERMS

Loss vs. Metric

DEFINE IT

KT.28 KEY TERMS

Class Imbalance

DEFINE IT

KT.29 KEY TERMS

Calibration

DEFINE IT

KT.30 KEY TERMS

Ensemble

DEFINE IT

KT.31 KEY TERMS

Transfer Learning & Fine-Tuning

DEFINE IT

KT.32 KEY TERMS

Attention

DEFINE IT

KT.33 KEY TERMS

Supervised vs. Unsupervised

DEFINE IT

KT.34 KEY TERMS

Self-Supervised Learning

DEFINE IT

KT.35 KEY TERMS

Latent Variable

DEFINE IT

KT.36 KEY TERMS

Convex vs. Non-Convex

DEFINE IT

KT.37 KEY TERMS

Vectorization

DEFINE IT

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One class is much rarer; use PR curves, resampling, or class weights.

KT.28 Key Terms

Loss: what training minimizes. Metric: what you report to judge usefulness.

KT.27 Key Terms

A multi-dimensional array (scalar, vector, matrix, and higher).

KT.26 Key Terms

Reusing a pre-trained model and adapting it to a new task.

KT.31 Key Terms

Combining several models to improve accuracy or stability.

KT.30 Key Terms

Predicted probabilities match observed frequencies.

KT.29 Key Terms

Labels are created from the data itself (e.g. predict the next word).

KT.34 Key Terms

Supervised learns from labeled examples; unsupervised finds structure without labels.

KT.33 Key Terms

Weighting inputs by relevance: $\text{softmax}\left(QK^\top/\sqrt{d}\right)V$.

KT.32 Key Terms

Replacing explicit loops with whole-array operations (NumPy, GPU kernels); often 10–100$\times$ faster.

KT.37 Key Terms

Convex problems have one global minimum; non-convex ones can have many local minima.

KT.36 Key Terms

A hidden quantity inferred from data, such as a cluster or topic.

KT.35 Key Terms

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KT.38 KEY TERMS

Broadcasting

DEFINE IT

KT.39 KEY TERMS

Sparsity

DEFINE IT

KT.40 KEY TERMS

Curse of Dimensionality

DEFINE IT

KT.41 KEY TERMS

Regularization

DEFINE IT

KT.42 KEY TERMS

Early Stopping

DEFINE IT

KT.43 KEY TERMS

Dropout

DEFINE IT

KT.44 KEY TERMS

Batch / Layer Normalization

DEFINE IT

KT.45 KEY TERMS

Vanishing & Exploding Gradients

DEFINE IT

KT.46 KEY TERMS

Learning Rate & Schedule

DEFINE IT

KT.47 KEY TERMS

Logits & Temperature

DEFINE IT

KT.48 KEY TERMS

Tokenization

DEFINE IT

KT.49 KEY TERMS

Context Window

DEFINE IT

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In high dimensions volume concentrates at the edges and distances become nearly equal, so data are sparse and nearest-neighbor methods degrade.

KT.40 Key Terms

Most entries are zero (one-hot data, text counts, graphs); store and compute only the nonzeros. Also: a model with many zero weights (lasso).

KT.39 Key Terms

Automatically stretching arrays of compatible shapes so element-wise operations work, e.g. subtracting a row of column means from every row of a matrix.

KT.38 Key Terms

Randomly zeroing a fraction of activations during training so the network cannot rely on any single unit.

KT.43 Key Terms

Stop training when validation loss stops improving; a cheap regularizer.

KT.42 Key Terms

Any constraint or penalty that reduces variance at the cost of a little bias: L1/L2 penalties ML.8 §15 Machine Learning, dropout, early stopping, data augmentation.

KT.41 Key Terms

Step size of gradient descent ML.27 §15 Machine Learning; too big diverges, too small crawls. Schedules (warm-up, cosine decay) change it during training.

KT.46 Key Terms

Gradients shrink or blow up as they are multiplied back through many layers; fixed by ReLU, residual connections, normalization, gradient clipping.

KT.45 Key Terms

Rescaling activations to mean 0, variance 1 (over the batch or over features), then learning a scale and shift; stabilizes training.

KT.44 Key Terms

The maximum number of tokens a language model can attend to at once.

KT.49 Key Terms

Splitting text into units (words, subwords such as BPE) that a model maps to IDs; costs and context limits are counted in tokens.

KT.48 Key Terms

Logits: raw scores before softmax ML.24 §15 Machine Learning. Temperature T divides logits: $T < 1$ sharpens, $T > 1$ flattens the distribution.

KT.47 Key Terms

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KT.50 KEY TERMS

Hallucination

DEFINE IT

KT.51 KEY TERMS

In-Context Learning

DEFINE IT

KT.52 KEY TERMS

Inductive Bias

DEFINE IT

KT.53 KEY TERMS

No Free Lunch

DEFINE IT

KT.54 KEY TERMS

Feature Engineering

DEFINE IT

KT.55 KEY TERMS

Distribution Shift (Drift)

DEFINE IT

KT.56 KEY TERMS

Target Leakage

DEFINE IT

KT.57 KEY TERMS

A/B Test

DEFINE IT

KT.58 KEY TERMS

Statistical Power

DEFINE IT

KT.59 KEY TERMS

Effect Size

DEFINE IT

KT.60 KEY TERMS

Statistical vs. Practical Significance

DEFINE IT

KT.61 KEY TERMS

p-Hacking

DEFINE IT

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The assumptions a model builds in (locality in CNNs, linearity in regression) that let it generalize.

KT.52 Key Terms

A large model performs a new task from instructions or examples in the prompt, without weight updates.

KT.51 Key Terms

A fluent but false output from a generative model; reduced by retrieval (RAG) and verification.

KT.50 Key Terms

Production data differ from training data: covariate shift (inputs change) or concept drift (input–label relation changes). Monitor and retrain.

KT.55 Key Terms

Creating informative inputs: ratios, lags, interactions, aggregates, encodings.

KT.54 Key Terms

Averaged over all possible problems, no learning algorithm beats another; good performance comes from matching assumptions to the data.

KT.53 Key Terms

$1 - \beta$: the probability a test detects a real effect of a given size S.28 §13 Statistics; rises with sample size and effect size.

KT.58 Key Terms

A randomized experiment comparing a control (A) and a variant (B) on a pre-chosen metric, with sample size set in advance from the desired power.

KT.57 Key Terms

A feature that is only known after the outcome (or is a proxy for it); gives great validation scores and a useless model. A form of leakage KT.21.

KT.56 Key Terms

Trying many analyses, subsets or stopping points until $p < 0.05$; produces false discoveries. Prevented by pre-registration and correction.

KT.61 Key Terms

With huge n, tiny useless effects become "significant"; always report effect size and a confidence interval.

KT.60 Key Terms

The magnitude of a difference (e.g. Cohen's $d = \Delta\mu/\sigma$), as opposed to whether it is statistically significant.

KT.59 Key Terms

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KT.62 KEY TERMS

Multiple Comparisons

DEFINE IT

KT.63 KEY TERMS

Simpson’s Paradox

DEFINE IT

KT.64 KEY TERMS

Selection & Survivorship Bias

DEFINE IT

KT.65 KEY TERMS

Correlation vs. Causation

DEFINE IT

KT.66 KEY TERMS

Heavy Tail

DEFINE IT

KT.67 KEY TERMS

Outlier vs. Anomaly

DEFINE IT

KT.68 KEY TERMS

Markov Property

DEFINE IT

KT.69 KEY TERMS

Numerical Stability

DEFINE IT

KT.70 KEY TERMS

Ill-Conditioned Problem

DEFINE IT

KT.71 KEY TERMS

Reproducibility

DEFINE IT

KT.72 KEY TERMS

Tidy (Long) vs. Wide Data

DEFINE IT

KT.73 KEY TERMS

Interpolation vs. Extrapolation

DEFINE IT

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The sample is not representative because of how it was chosen, e.g. studying only the funds or companies that survived.

KT.64 Key Terms

A trend in every subgroup reverses when the groups are combined, because group sizes differ; look for a lurking variable.

KT.63 Key Terms

Running m tests at level α gives about $m\alpha$ false positives; correct with Bonferroni (α/m) or FDR control.

KT.62 Key Terms

Outlier: a point far from the rest (1.5 IQR rule S.3 §13 Statistics). Anomaly: an outlier that signals a different process (fraud, fault).

KT.67 Key Terms

A distribution whose tail decays slower than exponential (Pareto, Cauchy); extremes dominate sums and averages converge slowly.

KT.66 Key Terms

Association alone does not show cause; causal claims need randomization or explicit assumptions (confounders KT.12).

KT.65 Key Terms

Tiny input changes cause large output changes (large condition number LA.29 §09 Linear Algebra); solve with QR/SVD or regularization.

KT.70 Key Terms

Avoiding overflow and cancellation, e.g. the log-sum-exp trick $\log \sum e^{x_i} = m + \log \sum e^{x_i - m}$ with $m = \max x_i$.

KT.69 Key Terms

The future depends only on the present state, not on the path that led there.

KT.68 Key Terms

Predicting inside the range of the training data vs. beyond it; extrapolation is far less reliable.

KT.73 Key Terms

Tidy: one row per observation, one column per variable. Wide: repeated measures spread across columns. Pivot to convert.

KT.72 Key Terms

Getting the same result again: fixed random seeds, versioned data and code, recorded environment.

KT.71 Key Terms

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KT.74 KEY TERMS

Big-$\mathcal{O}$ Notation

DEFINE IT

ET.1 EE TERMS

AC vs. DC

DEFINE IT

ET.2 EE TERMS

Ground

DEFINE IT

ET.3 EE TERMS

Short Circuit

DEFINE IT

ET.4 EE TERMS

Open Circuit

DEFINE IT

ET.5 EE TERMS

Series vs. Parallel

DEFINE IT

ET.6 EE TERMS

Impedance

DEFINE IT

ET.7 EE TERMS

Reactance

DEFINE IT

ET.8 EE TERMS

Conductance

DEFINE IT

ET.9 EE TERMS

Bandwidth

DEFINE IT

ET.10 EE TERMS

Gain

DEFINE IT

ET.11 EE TERMS

Feedback

DEFINE IT

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The reference node defined as 0 V; also a safety connection to earth.

ET.2 EE Terms

Alternating current reverses direction periodically (grid power); direct current flows one way (batteries, electronics).

ET.1 EE Terms

How cost grows with input size n, ignoring constants: $O(n)$, $O(n \log n)$, $O(n^2)$ O.10 §14 Optimization.

KT.74 Key Terms

Series: same current through each element. Parallel: same voltage across each element.

ET.5 EE Terms

A break in the path; no current flows.

ET.4 EE Terms

A near-zero-resistance path; current can become dangerously large.

ET.3 EE Terms

Reciprocal of resistance, $G = 1/R$, in siemens.

ET.8 EE Terms

The imaginary part of impedance from inductors (+) and capacitors (−).

ET.7 EE Terms

AC generalization of resistance, $Z = R + jX$, in ohms.

ET.6 EE Terms

Feeding part of the output back to the input; negative feedback stabilizes, positive feedback oscillates.

ET.11 EE Terms

Output-to-input ratio of an amplifier, often in dB.

ET.10 EE Terms

The range of frequencies a system passes (Hz); in data links, the achievable data rate.

ET.9 EE Terms

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ET.12 EE TERMS

ADC / DAC

DEFINE IT

ET.13 EE TERMS

PWM

DEFINE IT

ET.14 EE TERMS

Semiconductor

DEFINE IT

ET.15 EE TERMS

MOSFET

DEFINE IT

ET.16 EE TERMS

Breaker & Fuse

DEFINE IT

ET.17 EE TERMS

Duty Cycle

DEFINE IT

ET.18 EE TERMS

Harmonics

DEFINE IT

ET.19 EE TERMS

Microcontroller

DEFINE IT

ET.20 EE TERMS

Kilowatt-hour

DEFINE IT

ET.21 EE TERMS

Capacitance

DEFINE IT

ET.22 EE TERMS

Inductance

DEFINE IT

ET.23 EE TERMS

Time Constant

DEFINE IT

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A material (e.g. silicon) whose conductivity can be controlled by doping; basis of diodes and transistors.

ET.14 EE Terms

Pulse-Width Modulation: controlling average power by switching on and off at a varying duty cycle.

ET.13 EE Terms

Analog-to-Digital / Digital-to-Analog Converter.

ET.12 EE Terms

Fraction of each period that a signal is on.

ET.17 EE Terms

Devices that interrupt current when it exceeds a safe level.

ET.16 EE Terms

Metal-Oxide-Semiconductor Field-Effect Transistor: the voltage-controlled switch in all modern chips.

ET.15 EE Terms

Energy of 1 kW for 1 hour (3.6 MJ); the unit on electricity bills.

ET.20 EE Terms

A small computer on one chip with CPU, memory and I/O, used in embedded devices.

ET.19 EE Terms

Frequency components at integer multiples of the fundamental frequency.

ET.18 EE Terms

$\tau = RC$ or L/R: time to reach 63% of the final change; about 5τ to settle EE.11 §16 Electrical Eng..

ET.23 EE Terms

Flux linkage per ampere, L, in henries; an inductor resists changes in current and passes DC

EE.10 §16 Electrical Eng..

ET.22 EE Terms

Charge stored per volt, $C = Q/V$, in farads; a capacitor blocks DC and passes high frequencies

EE.10 §16 Electrical Eng..

ET.21 EE Terms

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ET.24 EE TERMS

RMS Value

DEFINE IT

ET.25 EE TERMS

Phasor

DEFINE IT

ET.26 EE TERMS

Power Factor

DEFINE IT

ET.27 EE TERMS

Resonance

DEFINE IT

ET.28 EE TERMS

Cutoff Frequency

DEFINE IT

ET.29 EE TERMS

Bode Plot

DEFINE IT

ET.30 EE TERMS

Decibel (dB)

DEFINE IT

ET.31 EE TERMS

Aliasing

DEFINE IT

ET.32 EE TERMS

Sampling Rate & Resolution

DEFINE IT

ET.33 EE TERMS

Pull-Up / Pull-Down Resistor

DEFINE IT

ET.34 EE TERMS

Debouncing

DEFINE IT

ET.35 EE TERMS

Decoupling Capacitor

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$\cos\phi$ = real power / apparent power; low values waste line capacity EE.15 §16 Electrical Eng..

ET.26 EE Terms

A complex number holding a sinusoid's amplitude and phase, turning AC circuit ODEs into algebra EE.14 §16 Electrical Eng..

ET.25 EE Terms

The DC value that delivers the same power; $V_{\text{peak}}/\sqrt{2}$ for a sine EE.13 §16 Electrical Eng..

ET.24 EE Terms

Gain (dB) and phase vs. log frequency; shows a system's frequency response at a glance.

ET.29 EE Terms

The -3 dB point of a filter, where output power is half; $f_c = \frac{1}{2\pi RC}$ for an RC filter EE.17 §16 Electrical Eng..

ET.28 EE Terms

Frequency where inductive and capacitive reactances cancel, $f_0 = \frac{1}{2\pi\sqrt{LC}}$ EE.16 §16 Electrical Eng..

ET.27 EE Terms

Samples per second (Hz) and bits per sample of an ADC; n bits give 2^n levels.

ET.32 EE Terms

Sampling below twice the highest frequency folds high frequencies onto low ones EE.27 §16 Electrical Eng.; prevent with an anti-aliasing low-pass filter.

ET.31 EE Terms

Logarithmic ratio: $10\log_{10}(P_2/P_1)$ for power, $20\log_{10}(V_2/V_1)$ for voltage; $+3$ dB $\approx$ double power.

ET.30 EE Terms

A capacitor placed next to a chip's supply pins to absorb fast current spikes and keep the voltage steady.

ET.35 EE Terms

Filtering the rapid on–off chatter of a mechanical switch (RC filter or software delay).

ET.34 EE Terms

Holds a digital input at a defined high or low level when nothing drives it.

ET.33 EE Terms

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ET.36 EE TERMS

Impedance Matching

DEFINE IT

ET.37 EE TERMS

Transformer

DEFINE IT

ET.38 EE TERMS

Three-Phase Power

DEFINE IT

ET.39 EE TERMS

Ripple

DEFINE IT

ET.40 EE TERMS

Noise Floor

DEFINE IT

ET.41 EE TERMS

Clock

DEFINE IT

ET.42 EE TERMS

Integrated Circuit (IC) & PCB

DEFINE IT

FT.1 FINANCE TERMS

Time Value of Money

DEFINE IT

FT.2 FINANCE TERMS

Liquidity

DEFINE IT

FT.3 FINANCE TERMS

Arbitrage

DEFINE IT

FT.4 FINANCE TERMS

Hedge

DEFINE IT

FT.5 FINANCE TERMS

Leverage

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Three sinusoids 120° apart; delivers constant total power with less conductor EE.19 §16 Electrical Eng..

ET.38 EE Terms

Two coupled coils that step AC voltage up or down by the turns ratio, $V_2/V_1 = N_2/N_1$ EE.19 §16 Electrical Eng..

ET.37 EE Terms

Making source and load impedances equal (or conjugate) to maximize power transfer EE.9 §16 Electrical Eng. or prevent reflections.

ET.36 EE Terms

A periodic square wave that synchronizes digital logic; its frequency limits how fast operations occur.

ET.41 EE Terms

The level of background noise below which signals cannot be distinguished; sets the useful dynamic range.

ET.40 EE Terms

Residual AC variation on a DC supply after rectification and filtering.

ET.39 EE Terms

How quickly an asset can be sold near its fair price.

FT.2 Finance Terms

Money now is worth more than the same amount later, because it can earn a return.

FT.1 Finance Terms

IC: many components fabricated on one chip. PCB: printed circuit board with copper traces connecting components.

ET.42 EE Terms

Using borrowed money to magnify returns (and losses).

FT.5 Finance Terms

A position taken to reduce the risk of another position.

FT.4 Finance Terms

A riskless profit from price differences; its absence underlies pricing models.

FT.3 Finance Terms

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FT.6 FINANCE TERMS

Bid–Ask Spread

DEFINE IT

FT.7 FINANCE TERMS

Market Capitalization

DEFINE IT

FT.8 FINANCE TERMS

Dividend

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FT.9 FINANCE TERMS

EBITDA

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FT.10 FINANCE TERMS

Free Cash Flow

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FT.11 FINANCE TERMS

Balance Sheet

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FT.12 FINANCE TERMS

Income Statement

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FT.13 FINANCE TERMS

Yield Curve

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FT.14 FINANCE TERMS

Basis Point

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FT.15 FINANCE TERMS

Bull vs. Bear Market

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FT.16 FINANCE TERMS

Index Fund / ETF

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FT.17 FINANCE TERMS

Short Selling

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A cash payment from a company’s profits to shareholders.

FT.8 Finance Terms

Share price times shares outstanding.

FT.7 Finance Terms

Difference between the highest buy and lowest sell price; a trading cost.

FT.6 Finance Terms

Snapshot of assets, liabilities and equity at a point in time.

FT.11 Finance Terms

Operating cash flow minus capital expenditures.

FT.10 Finance Terms

Earnings Before Interest, Taxes, Depreciation and Amortization; a proxy for operating cash profit.

FT.9 Finance Terms

One hundredth of a percent (0.01%).

FT.14 Finance Terms

Plot of bond yields vs. maturity; an inverted curve often precedes recessions.

FT.13 Finance Terms

Revenue, expenses and profit over a period.

FT.12 Finance Terms

Selling borrowed shares to profit if the price falls.

FT.17 Finance Terms

A fund that tracks a market index; ETFs trade like stocks.

FT.16 Finance Terms

Rising market (bull) vs. falling market, typically −20% or more (bear).

FT.15 Finance Terms

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FT.18 FINANCE TERMS

Drawdown

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FT.19 FINANCE TERMS

Implied Volatility

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FT.20 FINANCE TERMS

Credit Score / Default

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FT.21 FINANCE TERMS

Efficient Market Hypothesis

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FT.22 FINANCE TERMS

Backtesting

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FT.23 FINANCE TERMS

Compounding

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FT.24 FINANCE TERMS

Discount Rate

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FT.25 FINANCE TERMS

Opportunity Cost

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FT.26 FINANCE TERMS

Nominal vs. Real

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FT.27 FINANCE TERMS

Risk Premium

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FT.28 FINANCE TERMS

Risk-Free Rate

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FT.29 FINANCE TERMS

Alpha & Beta

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Credit score: estimate of a borrower's reliability. Default: failure to make required payments.

FT.20 Finance Terms

The volatility that makes an option-pricing model match the market price.

FT.19 Finance Terms

Decline from a peak to a trough in portfolio value.

FT.18 Finance Terms

Earning returns on previous returns, so growth is exponential FN.1 §17 Finance.

FT.23 Finance Terms

Testing a strategy on historical data; prone to overfitting and look-ahead bias.

FT.22 Finance Terms

Prices already reflect available information, so consistent excess returns are hard to earn.

FT.21 Finance Terms

Nominal: in current money. Real: adjusted for inflation, $1 + r_{\text{real}} = \frac{1+r_{\text{nom}}}{1+\pi}$ FN.4 §17 Finance.

FT.26 Finance Terms

The return given up on the best alternative use of the money or time.

FT.25 Finance Terms

The rate used to convert future cash flows to present value; reflects time value and risk FN.5 §17 Finance.

FT.24 Finance Terms

Beta: sensitivity to market moves FN.14 §17 Finance. Alpha: return beyond what beta explains.

FT.29 Finance Terms

Return on a (nearly) default-free asset such as short-term government bills.

FT.28 Finance Terms

Extra expected return demanded for bearing risk, e.g. equity return minus the risk-free rate.

FT.27 Finance Terms

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FT.30 FINANCE TERMS

Volatility

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FT.31 FINANCE TERMS

Diversification

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FT.32 FINANCE TERMS

Systematic vs. Idiosyncratic Risk

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FT.33 FINANCE TERMS

Coupon & Face Value

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FT.34 FINANCE TERMS

Yield to Maturity

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FT.35 FINANCE TERMS

Duration

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FT.36 FINANCE TERMS

Credit Spread

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FT.37 FINANCE TERMS

Call & Put Option

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FT.38 FINANCE TERMS

Futures & Forwards

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FT.39 FINANCE TERMS

Swap

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FT.40 FINANCE TERMS

Market vs. Limit Order

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FT.41 FINANCE TERMS

Order Book & Market Maker

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Market-wide risk that cannot be diversified away vs. company-specific risk that can.

FT.32 Finance Terms

Combining imperfectly correlated assets lowers portfolio risk without lowering expected return FN.12 §17 Finance.

FT.31 Finance Terms

Standard deviation of returns, usually annualized ($\sigma_{\text{annual}} = \sigma_{\text{daily}}\sqrt{252}$) FN.11 §17 Finance.

FT.30 Finance Terms

Weighted-average time of a bond's cash flows; approximate % price change per 1% yield change FN.24 §17 Finance.

FT.35 Finance Terms

The single discount rate that makes a bond's cash flows equal its price FN.22 §17 Finance.

FT.34 Finance Terms

Coupon: periodic interest a bond pays. Face (par) value: amount repaid at maturity.

FT.33 Finance Terms

Agreements to trade an asset at a set price on a future date; futures are standardized and exchange-traded.

FT.38 Finance Terms

The right (not obligation) to buy (call) or sell (put) at the strike price by expiry FN.25 §17 Finance.

FT.37 Finance Terms

Yield of a risky bond minus the yield of a comparable government bond; compensation for default risk.

FT.36 Finance Terms

Order book: all resting buy and sell limit orders. Market maker: firm quoting both sides, earning the spread.

FT.41 Finance Terms

Market: execute now at the best available price. Limit: execute only at the stated price or better.

FT.40 Finance Terms

Exchange of cash-flow streams, e.g. fixed for floating interest payments.

FT.39 Finance Terms

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FT.42 FINANCE TERMS

Margin & Margin Call

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FT.43 FINANCE TERMS

Amortization

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FT.44 FINANCE TERMS

Total Return

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FT.45 FINANCE TERMS

Look-Ahead Bias

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FT.46 FINANCE TERMS

Mean Reversion vs. Momentum

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Price change plus dividends or coupons, as a fraction of the starting value.

FT.44 Finance Terms

Repaying a loan through scheduled payments that cover interest and principal FN.9 §17 Finance.

FT.43 Finance Terms

Margin: collateral posted for a leveraged position. Margin call: demand for more collateral after losses.

FT.42 Finance Terms

Mean reversion: prices drift back toward an average. Momentum: recent winners keep winning for a while.

FT.46 Finance Terms

Using information in a backtest that would not have been available at decision time.

FT.45 Finance Terms

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